

Prepladder Version X Qbank

Subject: Pharmacology

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Pharmacokinetics

1. Which of the following drugs is not metabolised by the CYP2C19 enzyme?

- A. Clopidogrel
 - B. Omeprazole
 - C. Voriconazole
 - D. Warfarin
-

2. Which of the following is true about the elimination kinetics of phenytoin?

- A. The rate of elimination is proportional to plasma concentration
 - B. Half-life remains constant irrespective of plasma concentration
 - C. The clearance is independent of plasma concentration throughout
 - D. Kinetics changes from first order to zero order at higher doses
-

3. A patient is given a 300 mg dose of a drug. After 1 hour, 75 mg of the drug is excreted. Assuming the drug is eliminated from the system by following first-order kinetics, how much of the drug will remain after 4 hours of starting the drug?

- A. 72 mg
 - B. 96 mg
 - C. 122 mg
 - D. 148 mg
-

4. A drug given to a patient is known to follow a capacity-limited elimination pattern. Which of the following is incorrect about this mode of drug elimination?

- A. The amount of the drug eliminated per unit of time will remain constant
 - B. Clearance of the drug will be less at higher concentrations.
 - C. The half-life of the drug will be longer at lower concentrations
 - D. The rate of elimination of the drug is independent of plasma concentration.
-

5. The target concentration of a drug given to a patient should be 5 mg/L. Assuming the drug's bioavailability is 0.5 and it gets cleared out of the system at a rate of 2L/hr, what is the maintenance dose to be given twice daily?

- A. 190 mg
 - B. 240 mg
 - C. 300 mg
 - D. 330 mg
-

6. A 70 kg patient is administered 2 mg of warfarin. If its plasma concentration is 200 mcg/L, calculate the apparent volume of distribution of the drug.

- A. 0.25 L/kg
 - B. 0.10 L/kg
 - C. 0.15 L/kg
 - D. 0.20 L/kg
-

7. Which of the following statements regarding elimination characteristics of "hit and run" drugs are incorrect? These drugs are eliminated quickly and have short half-lives. These drugs produce a prolonged effect even after they are eliminated

from the body. These drugs require continuous administration to maintain their therapeutic effect. Omeprazole and reserpine are classified under these drugs. Regular blood samples are needed to monitor plasma concentration.

- A. 1, 3 and 5
 - B. 2, 3 and 5
 - C. 3 and 4
 - D. 1,2 and 3
-

8. A 32-year-old female on oral contraceptives containing estrogen is prescribed a course of ampicillin for a urinary tract infection. A few weeks later, she experiences an unexpected breakthrough bleeding and is concerned about the efficacy of her contraceptive. Which of the following describes the potential failure of her oral contraceptive?

- A. Antibiotics induce hepatic enzymes that increase estrogen metabolism
 - B. Antibiotics inhibit the absorption of estrogen in the stomach
 - C. Antibiotics destroy gut flora, reducing enterohepatic circulation of estrogen
 - D. Antibiotics increase renal excretion of estrogen
-

9. A 45-year-old male is undergoing treatment for latent tuberculosis with a standard regimen. After a few weeks of therapy, he develops peripheral neuropathy. His doctor explains that this side effect is related to the metabolism of isoniazid. Which type of phase II metabolism is responsible for this patient's symptoms?

- A. Slow glucuronidation
- B. Fast glucuronidation
- C. Fast acetylation
- D. Slow acetylation

10. Match the following CYP450 enzymes with the drugs they primarily metabolise:
CYP450 Enzyme Drug 1. CYP1A2 A. Phenytoin 2. CYP2C9 B. Midazolam 3. CYP2C19 C. Clopazine 4. CYP2D6 D. Pantoprazole 5. CYP3A4 E. Codeine

CYP450 Enzyme	Drug
1. CYP1A2	A. Phenytoin
2. CYP2C9	B. Midazolam
3. CYP2C19	C. Clopazine
4. CYP2D6	D. Pantoprazole
5. CYP3A4	E. Codeine

- A. 1:D, 2:C, 3:A, 4:E, 5:B
 - B. 1:C, 2:A, 3:D, 4:E, 5:B
 - C. 1:C, 2:A, 3:E, 4:B, 5:D
 - D. 1:A, 2:C, 3:B, 4:E, 5:D
-

11. In drug mechanisms observed in cancer cells, which of the following accurately describes the role of the P-glycoprotein pump (Pgp)?

- A. Pgp facilitates drug uptake into tumour cells
 - B. Pgp reduces drug elimination from tumour cells
 - C. Pgp is involved in drug efflux from tumour cells
 - D. Pgp has no significant impact on drug transport within tumour cells
-

12. Diazepam undergoes which of the following metabolic reactions?

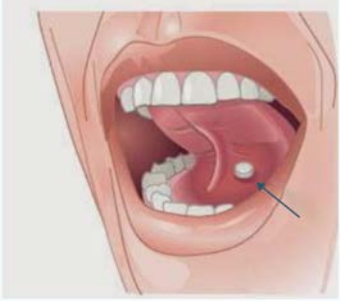
- A. Glycination
 - B. Glucuronide conjugation
 - C. Sulfation
 - D. Methylation
-

13. All of the following are phase 1 reactions except?

- A. Oxidation
 - B. Reduction
 - C. Methylation
 - D. Hydrolysis
-

14. Match the following: Route Percentage Bioavailability 1) a) 100 2) b) 30-90 3) c) 75-100 4) d) 5-100

Route	Percentage Bioavailability
 ... (content truncated)	a) 100

 Figure 1: Sublingual drug delivery.[12]	b) 30-90
2)  ... (content truncated)	c) 75-100
4) 	d) 5-100

- A. 1-d, 2-b, 3-c, 4-a
- B. 1-d, 2-c, 3-b, 4-a
- C. 1-a, 2-b, 3-c, 4-d
- D. 1-b, 2-a, 3-c, 4-d

15. What is the effect of hypoalbuminemia on drugs with high plasma protein binding?

- A. Decreased drug efficacy
 - B. Increased drug efficacy
 - C. Increased risk of drug toxicity
 - D. Decreased risk of drug toxicity
-

16. Which statement regarding pharmacokinetics is incorrect?

- A. The volume of distribution (V_d) is a measure of drug distribution in the body
 - B. Clearance (Cl) represents the rate at which a drug is eliminated from the body
 - C. Bioavailability (F) is the fraction of an administered dose that reaches systemic circulation unchanged
 - D. The half-life ($T_{1/2}$) is unaffected by liver metabolism
-

17. A 25-year-old patient presents with agitation, confusion, hyperthermia, and mydriasis. Urine toxicology confirms amphetamine use. What is the most appropriate management strategy for this patient?

- A. Benzodiazepines
 - B. Beta-blockers
 - C. Ammonium chloride
 - D. Sodium bicarbonate
-

18. Which of the following statements is correct regarding drug absorption and excretion?

- A. Highly polar drugs are readily absorbed through the lipid membranes
 - B. Non-polar drugs are readily absorbed
 - C. Maximum unionisation of an acidic drug occurs in the small intestine
 - D. Non-ionized forms of drugs are less likely to be absorbed than ionised forms.
-

19. A 55-year-old male with chronic obstructive pulmonary disease (COPD) is being treated with theophylline. He is prescribed erythromycin for a respiratory tract infection. Which of the following outcomes is most likely due to the interaction between these two drugs?

- A. Decreased plasma concentration of theophylline
 - B. Increased risk of theophylline toxicity
 - C. No significant interaction
 - D. Decreased efficacy of erythromycin
-

20. A patient with epilepsy is prescribed both Phenytoin and lamotrigine for seizure control. The patient reports worsening seizures despite adhering to the prescribed regimen. Which of the following best explains the potential reason for the failure of the regimen?

- A. Increased absorption of lamotrigine
 - B. Toxicity of lamotrigine
 - C. Increased metabolism of lamotrigine
 - D. Inhibition of transport of lamotrigine
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	2
Question 4	3
Question 5	2
Question 6	3
Question 7	1
Question 8	3
Question 9	4
Question 10	2
Question 11	3
Question 12	2
Question 13	3
Question 14	1
Question 15	3
Question 16	4
Question 17	3
Question 18	2
Question 19	2
Question 20	3

Solution for Question 1:

Correct answer : D) Warfarin

Explanation :

Warfarin is metabolised by CYP2C9.

Drugs metabolised by CYP2C19:

- PPI: Omeprazole (Option B) & lansoprazole.
- Amitriptyline.
- Citalopram.
- Phenytoin.
- Diazepam.
- Voriconazole and other Antifungal Azoles (Option C).
- Clopidogrel (Option A).

CYP-450 enzymes:

- Essential for phase 1 reactions (oxidation and reduction).
- Genetic polymorphism is present.
- Nomenclature of a CYP enzyme: Eg, CYP3A4. CYP: Cytochrome P450 superfamily 3: Family of CYP (number of the gene making CYP enzyme) A: Isoform/Subfamily of CYP 4: Individual enzyme
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- 3: Family of CYP (number of the gene making CYP enzyme)
- A: Isoform/Subfamily of CYP
- 4: Individual enzyme
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- 4: Individual enzyme
- CYP: Cytochrome P450 superfamily

- 3: Family of CYP (number of the gene making CYP enzyme)
 - A: Isoform/Subfamily of CYP
 - 4: Individual enzyme
-

Solution for Question 2:

Correct answer: D) Kinetics changes from first order to zero order at higher doses.

Explanation:

Phenytoin follows a different elimination pattern, where it initially follows first-order kinetics but the metabolising enzymes get saturated above therapeutic concentration and the pattern changes from first to zero order kinetics. Such pattern shows a curved plot and is called non-linear elimination.

Common drugs that follow a non-linear elimination:

- Phenytoin.
- Tolbutamide.
- Theophylline.
- Warfarin.

The rate of elimination is proportional to plasma concentration (Options A) & half-life remains constant irrespective of plasma concentration (Option B) : They are properties of drugs that purely follow first-order kinetics.

The clearance is independent of plasma concentration throughout (Option C) : It purely follow zero-order kinetics.

Solution for Question 3:

Correct answer: B) 96 mg

Explanation:

The given drug follows a first-order kinetic elimination; hence, its clearance rate remains constant and proportional to the plasma concentration per unit of time.

- Since 75 mg out of 300 mg (which is $\frac{1}{4}$) got excreted by the end of 1 hour, 225 mg will remain in the body (which is $\frac{3}{4}$). Hence, per first-order kinetics, $\frac{3}{4}$ of the drug will remain in the body at the end of each hour.
- This means at the end of 4 hours.

$$300 * (\frac{1}{4})(\frac{1}{4})(\frac{1}{4})(\frac{1}{4}) = 96 \text{ mg will remain in the body}$$

Solution for Question 4:

Correct answer: C) The half-life of the drug will be longer at lower concentrations.

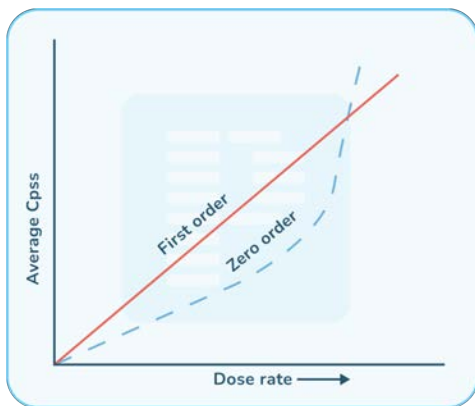
Explanation:

The half-life of a drug that follows a capacity-limited elimination/Michaelis Menton elimination/zero order kinetics will be less at lower concentrations and more at higher concentrations.

The key differences between drugs following zero and first-order kinetics are below.

	First-order kinetics	Zero-order kinetics
Description	A constant fraction of the drug is eliminated per unit of time.	A constant amount of the drug is eliminated per unit of time.
Rate of clearance	Constant	More at lower concentrations and less at higher concentrations.
Half-life	Constant	Less at lower concentrations and more at higher concentrations.
Rate of elimination	Proportional to its plasma concentration.	Independent of plasma concentration.
Examples	Most of the drugs.	Alcohol, aspirin, theophylline, tolbutamide, and warfarin.

(Options A, B, and D ruled out)



Solution for Question 5:

Correct answer: B) 240 mg

Explanation:

The maintenance dose of a drug is the dose that has to be repeated at specified intervals to maintain the target plasma concentration of a drug, and it's given by the equation :

Maintenance dose = Target plasma concentration (Cp) * Clearance (CL) * Dosing interval
Bioavailability (F)

Here, Cp = 5 mg/L ; CL = 2 L/hr ; F = 0.5 ; dosing interval is twice daily (BD).

Hence, $5 * 2 / 0.5 = 20 \text{ mg/hr} * 12 \text{ hours} = 240 \text{ mg BD}$

Solution for Question 6:

Correct answer: C) 0.15 L/kg

Explanation:

The apparent volume of distribution (V d) of a drug is calculated by the formula below :

$V_d = \text{Dose of drug administered} / \text{plasma concentration}$

Here, the total dose administered is 2mg, i.e., 2000 mcg, and the plasma concentration is 200 mcg/L.

Hence, $2000/200 = 10$

Since the patient weighs 70 kg, $10/70 = 0.143$, approximating 0.15 L/kg.

The volume of distribution refers to the apparent volume that would accommodate the entire drug in the body if the concentration is the same as in plasma.

Low V_d	High V_d
A larger portion of a drug is in the intravascular compartment.	A larger portion of a drug is in the extravascular compartment.
It is easily removed by haemodialysis.	It can be difficult to remove by haemodialysis.
Factors responsible: Extensive plasma protein binding Hydrophilic nature	Factors responsible: Extensive body tissue binding Lipophilic nature
Examples: warfarin, diclofenac	Examples: Digoxin, propranolol, morphine

Solution for Question 7:

Correct answer : A) 1, 3 and 5

Explanation:

Hit-and-run drugs are not defined by their rate of elimination, do not require continuous administration, and monitoring of plasma concentration has no value.

Hit and Run drugs:

- Create lasting effects by irreversibly binding to targets, initiating a cascade that persists even after the drug is eliminated (Statement 2).
- The effect lasts much longer than the drug itself.
- Do not require continuous administration due to prolonged effect.
- Plasma concentration monitoring is of no value.
- Examples include reserpine, guanethidine, MAO inhibitors, and omeprazole (Statement 4).

Solution for Question 8:

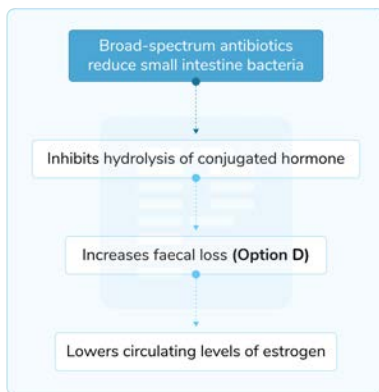
Correct answer : C) Antibiotics destroy gut flora, reducing enterohepatic circulation of estrogen

Explanation:

Broad-spectrum antibiotics can destroy gut flora , essential for the enterohepatic recirculation of oestrogen. This can potentially lead to decreased estrogen levels and contraceptive failure.

Metabolism of estrogen/oral contraceptives:

- CYP3A4 Enzyme for Estrogen Hydroxylation.
- Absorption occurs in the small intestine (Option B)
- Mechanism: Induction CYP450 enzymes in the liver: (Option A) Rifampicin induces CYP3A4, resulting in increased estrogen metabolism, leading to contraceptive failure.
- Induction CYP450 enzymes in the liver: (Option A) Rifampicin induces CYP3A4, resulting in increased estrogen metabolism, leading to contraceptive failure.
- Rifampicin induces CYP3A4, resulting in increased estrogen metabolism, leading to contraceptive failure.
- Interruption with enterohepatic cycling: Seen with penicillins , cephalosporins, tetracyclines, macrolides, antifungals, and sulphonamides.
- Seen with penicillins , cephalosporins, tetracyclines, macrolides, antifungals, and sulphonamides.
- Induction CYP450 enzymes in the liver: (Option A) Rifampicin induces CYP3A4, resulting in increased estrogen metabolism, leading to contraceptive failure.
- Rifampicin induces CYP3A4, resulting in increased estrogen metabolism, leading to contraceptive failure.
- Rifampicin induces CYP3A4, resulting in increased estrogen metabolism, leading to contraceptive failure.
- Seen with penicillins , cephalosporins, tetracyclines, macrolides, antifungals, and sulphonamides.



Solution for Question 9:

Correct answer : D) Slow acetylation

Explanation :

Isoniazid is metabolised primarily in the liver through acetylation , a phase II metabolic process . The rate at which isoniazid is acetylated depends on the genetic polymorphism of the NAT2 enzyme , leading to two major phenotypes: slow acetylators and fast acetylators (Option C ruled out)

	Fast Acetylators	Slow Acetylators
NAT2 enzyme activity	High	Low
Drug concentrations	Low	High
Increased risk for	Reduced microbial cure. Increased relapse. Acquired drug resistance.	Hepatotoxicity. Peripheral neuropathy/neuritis.

Acetylation:

- Compounds having amino or hydrazine residues are conjugated with the help of acetyl coenzyme A.
- Drugs undergoing acetylation (Mnemonic is C-SHIPPeD) : C Isoniazid. S ulphonamides. H ydralazine. I soniazid. P rocainamide. P ara-amino-salicylic acid. D apsone.
- C Isoniazid.
- S ulphonamides.

- Hydralazine.
- Isoniazid.
- Procainamide.
- Para-aminosalicylic acid.
- Dapsone.
- Clonazepam.
- Sulfonamides.
- Hydralazine.
- Isoniazid.
- Procainamide.
- Para-aminosalicylic acid.
- Dapsone.

Slow glucuronidation (Option A) and fast glucuronidation (Option B): Glucuronidation is not the primary phase II pathway for isoniazid. It is seen in clonazepam, diazepam, amitriptyline etc.

Solution for Question 10:

Correct answer : B) 1:C, 2:A, 3:D, 4:E, 5:B

Explanation :

Types of CYP enzymes and their substrates.

CYP Enzymes	Substrates/Drugs
CYP1A2	Theophylline Clozapine
CYP2C9 (earlier CYP2C8) (Saturation kinetics is seen)	Phenytoin Warfarin
CYP2C19	Clopidogrel (see note below) Azoles: Voriconazole, Ketoconazole PPI: Omeprazole, Pantoprazole

CYP2D6	Propranolol Quinidine Tamoxifen TCA SSRIs Opioids (Codeine) Antipsychotics (all except clozapine)
CYP2E1	Paracetamol
CYP3A4	Astemizole Midazolam Cyclosporine/Tacrolimus Diltiazem/ verapamil Oral contraceptive pills Fexofenadine/Terfenadine

Note: Clopidogrel metabolism and antiplatelet effects.

- Bioactivated through phase 2 conjugation by CYP2C19 enzyme to 2-oxo-clopidogrel, metabolised to an irreversible inhibitor of platelet ADP P2Y₁₂ receptors.
- Omeprazole inhibits CYP2C19, reducing the activation of clopidogrel and potentially decreasing its antiplatelet effect.

Solution for Question 11:

Correct answer: C) Pgp is involved in drug efflux from tumour cells

Explanation:

P-glycoprotein (P-gp), also known as ATP-binding cassette sub-family B member 1 (ABCB1) or multidrug resistance protein 1 (MDR1), is an efflux transporter protein.

- Its primary function is to pump out a wide range of substances, including drugs, from inside the cell to outside the cell (drug efflux). (Options: A, B, and D ruled out).
- When Pgp is overexpressed or hyperactive in cancer cells, it leads to increased efflux of chemotherapeutic drugs out of the cells (leads to drug resistance).
- Location of Pgp in various body tissues: GIT: Decreases absorption of the drug. Kidney: Increases urinary excretion of the drug. Liver: Increases bile excretion of the drug. Blood-brain-barrier/Placenta: Decreases drug entry into the brain/placental cells.
- GIT: Decreases absorption of the drug.

- Kidney: Increases urinary excretion of the drug.
- Liver: Increases bile excretion of the drug.
- GIT: Decreases absorption of the drug.
- Kidney: Increases urinary excretion of the drug.
- Liver: Increases bile excretion of the drug.
- Substrate of p-glycoprotein transporter: E.g. Digoxin.

	P-glycoprotein Inducer	P-glycoprotein Inhibitor
Activity	Increases the activity of Pgp	Decreases the activity of Pgp
Drugs	Rifampicin. Phenobarbitone.	Mnemonic: QVACKER Quinidine Verapamil Amiodarone Clarithromycin Ketoconazole Erythromycin Ritonavir
Combined with Substrate of p-glycoprotein (Digoxin)	Increases the activity of P-gp. Increased urinary excretion. Digoxin is excreted.	Reduces the activity of P-gp. Reduced urinary excretion. Increased plasma concentration of the drug. Toxicity of digoxin: Arrhythmias.

Solution for Question 12:

Correct answer: B) Glucuronide conjugation

Explanation:

Diazepam undergoes glucuronide conjugation.

Phase 2 reactions:

- These reactions involve the conjugation of the drug or its phase I metabolite with an endogenous substrate, usually derived from carbohydrates or amino acids, to form a polar, highly ionised organic acid that is easily excreted in urine or bile.
- Conjugation reactions have high energy requirements and are generally faster than phase I reactions.

- Glucuronide conjugation: This is the most important phase II reaction carried out by a group of UDP-glucuronyl transferases. Compounds with a hydroxyl or carboxylic acid group are easily conjugated with glucuronic acid derived from glucose. Substances undergoing glucuronide conjugation:

- This is the most important phase II reaction carried out by a group of UDP-glucuronyl transferases.

- Compounds with a hydroxyl or carboxylic acid group are easily conjugated with glucuronic acid derived from glucose.

- Substances undergoing glucuronide conjugation:

- This is the most important phase II reaction carried out by a group of UDP-glucuronyl transferases.

- Compounds with a hydroxyl or carboxylic acid group are easily conjugated with glucuronic acid derived from glucose.

- Substances undergoing glucuronide conjugation:

Drugs	Endogenous substances
Chloramphenicol. Paracetamol. Aspirin. Benzodiazepines: Diazepam, lorazepam. Morphine. Metronidazole.	Bilirubin. Thyroxine. Steroidal hormones.

Options	Reaction	Examples
Glycination (Option A)	Glycination: The drugs are conjugated with glycine.	Salicylates. Nicotinic acid.
Sulfation (Option C)	Sulfation: The phenolic compounds and steroids are sulfated by sulfotransferases.	Chloramphenicol. Methyldopa. Adrenal & sex steroids.
Methylation(Option D)	Methylation: The amines and phenols are methylated by methyl transferases.	Chloramphenicol. Methyldopa. Captopril. Mercaptopurine.

Solution for Question 13:

Correct answer: C) Methylation

Explanation:

Methylation is a phase 2 reaction.

Drugs undergo two types of reactions in the liver: Phase 1 and Phase 2.

	Phase 1 (Exposes Functional group on Drug)	Phase 2 (Makes drug Water Soluble)
The activity of the drug	Drugs convert into active & inactive forms.	The drug converts into active form only, except morphine & minoxidil. Enzymes transfer polar groups.
Types of reaction and their site	Microsomal reactions: Oxidation (Most Common) & Reduction (mediated by CYP450 enzymes). Hydrolysis. Deamination. Dehydrogenation. Cyclisation. Decyclisation (Options A, B and D ruled out).	Microsomal reaction: Glucuronide conjugation. Non-microsomal reactions: Acetylation. Methylation. Sulfation. Glycination. Glutathionation. Ribonucleoside/nucleotide synthesis.
M/C reaction	Oxidation & reduction.	Glucuronide conjugation.

(Microsomes- Smooth endoplasmic reticulum).

Solution for Question 14:

Correct answer: A) 1-d, 2-b, 3-c, 4-a

Explanation:

Route of administration	Bioavailability in %
1) Oral	5-100
2) Sublingual	30-90
3) Intramuscular	75-100
4) Intravenous	100

Bioavailability:

- It measures the fraction (F) of the administered dose of a drug that reaches the systemic circulation in an unchanged form.
- Bioavailability of a drug injected i.v. is 100%, but is frequently lower after oral ingestion because: The drug may be incompletely absorbed. The absorbed drug may undergo first-pass metabolism in the intestinal wall/liver or be excreted in bile.
- The drug may be incompletely absorbed.
- The absorbed drug may undergo first-pass metabolism in the intestinal wall/liver or be excreted in bile.
- The drug may be incompletely absorbed.
- The absorbed drug may undergo first-pass metabolism in the intestinal wall/liver or be excreted in bile.

Incomplete bioavailability after s.c. or i.m. injection is less common but may occur due to the drug's local binding.

First-pass (Pre-systemic) Metabolism:

- Metabolism of a drug during its passage from the site of absorption into the systemic circulation.
- All orally administered drugs undergo this process.
- By using sublingual, transdermal, or parenteral routes, drugs can bypass significant presystemic metabolism.
- However, some metabolism may still occur in the skin (for transdermal drugs) and in the lungs (for drugs entering venous blood through any route). (Options B, C, and D ruled out)

Route of Administration	Bioavailability Percentage
Oral	5-100
Sublingual	30-90
Rectal	30-60
Intramuscular (IM)	75-100
Intravenous (IV)	100
Subcutaneous (SC)	75 - 100
Intranasal	20-50
Transdermal	80-100

Solution for Question 15:

Correct answer: C) Increased risk of drug toxicity

Explanation:

Drugs with high plasma protein binding rely heavily on albumin for transportation in the bloodstream.

Hypoalbuminemia (low levels of albumin in the blood) reduces the available binding sites for these drugs, leading to higher concentrations of free, unbound drugs in the bloodstream.

This can result in increased drug potency and a higher risk of adverse effects or toxicity (Options A, B and D ruled out).

Examples include phenytoin and furosemide.

The clinically significant implications of plasma protein binding are:

- Highly plasma protein-bound drugs are restricted to the vascular compartment as they do not cross membranes → smaller volumes of distribution.
- The drug's bound fraction is not available for action but is in equilibrium with the free drug in plasma, which is available for metabolism and elimination, causing the bound fraction to dissociate as free drug concentration decreases.
- High protein binding generally prolongs the drug's action as the bound fraction is not readily metabolised or excreted.
- Glomerular filtration does not significantly reduce the concentration of free drugs in the efferent vessels.
- Highly protein-bound drugs are not removed by haemodialysis.
- A single drug can bind to multiple sites on the albumin molecule.
- Multiple drugs can bind to the same site on albumin, leading to potential displacement interactions.

Solution for Question 16:

Correct answer: D) The half-life ($T_{1/2}$) is unaffected by liver metabolism.

Explanation:

The Plasma half-life ($t_{1/2}$) of a drug is the time taken for its plasma concentration to be reduced to half of its original value. It is influenced by various factors, including metabolism.

- Drugs metabolised by the liver can have their half-life altered by changes in liver function or enzyme activity.
- Liver metabolism can either increase or decrease the half-life of a drug, depending on whether the metabolites are active or inactive and how rapidly they are eliminated from the body.

The volume of distribution (V_d) is a measure of drug distribution in the body (Option A):

- Once a drug has gained access to the bloodstream, it gets distributed to other tissues that initially had no drug, concentration gradient being in the direction of plasma to tissues.
- The extent of distribution of a drug and its pattern of tissue distribution depends on its: Lipid solubility. Ionisation at physiological pH (a function of its pK_a). The extent of binding to plasma and tissue proteins. The presence of tissue-specific transporters. Differences in regional blood flow.
- Lipid solubility.
- Ionisation at physiological pH (a function of its pK_a).
- The extent of binding to plasma and tissue proteins.
- The presence of tissue-specific transporters.
- Differences in regional blood flow.
- Lipid solubility.
- Ionisation at physiological pH (a function of its pK_a).
- The extent of binding to plasma and tissue proteins.
- The presence of tissue-specific transporters.
- Differences in regional blood flow.

Clearance (CL) represents the rate at which a drug is eliminated from the body (Option B):

- The clearance of a drug is the theoretical volume of plasma from which the drug is completely removed in a unit of time.
- It can be calculated as:

$$CL = \text{Rate of elimination} / C$$

Where C is the plasma concentration.

Bioavailability (F) is the fraction of an administered dose that reaches systemic circulation unchanged (Option C):

- Bioavailability refers to the rate and extent of a drug's absorption from a dosage form administered by any route, as determined by its concentration-time curve in blood or

excretion in urine.

- It measures the fraction (F) of the administered drug dose that reaches the systemic circulation in the unchanged form.

Solution for Question 17:

Correct answer: C) Ammonium chloride Explanation:

- Acidification of urine increases the renal elimination of amphetamines, aiding in their clearance from the body.
- Amphetamines are weak bases, meaning they exist in both ionized and non-ionized forms depending on the pH of their environment.
- In acidic conditions, such as an acidified urine pH, weak bases like amphetamines tend to remain in their non-ionized form, thus enhancing elimination.
- Acidification of urine can be achieved through the administration of ammonium chloride or ascorbic acid (vitamin C).
- These substances acidify urine by increasing the excretion of hydrogen ions (H^+) into the urine (Options: A, C, and D ruled out).

Solution for Question 18:

Correct answer: B) Non-polar drugs are readily absorbed

Explanation:

- Cell membranes are primarily composed of lipids.
- Lipophilic or non-polar drugs can dissolve in lipids and cross these lipid-rich barriers.
- Due to their affinity for lipid components, lipophilic substances can passively diffuse through cell membranes.
- This passive diffusion is a common mechanism by which drugs are absorbed.

Highly polar drugs are readily absorbed through the lipid membranes (Option A) :

- Highly polar drugs are less likely to pass through the lipid membranes due to their low solubility.
- Polar or hydrophilic drugs are water-soluble, lack lipid solubility, and require specialised transporters or channels for absorption across cell membranes.

Maximum unionisation of an acidic drug occurs in the small intestine (Option C):

- A drug's degree of ionisation is influenced by its pKa (the pH at which it is 50% ionised) and the pH of its environment.
- Acidic drugs tend to be ionised in basic environments (higher pH) and non-ionised in acidic environments (lower pH) and vice versa.
- Accordingly, the maximum unionisation of acidic drugs occurs in the stomach, not the small intestine.

Non-ionized forms of drugs are less likely to be absorbed than ionised forms (Option D):

- Non-ionized drugs are more lipid-soluble and more readily absorbed through lipid membranes.

Solution for Question 19:

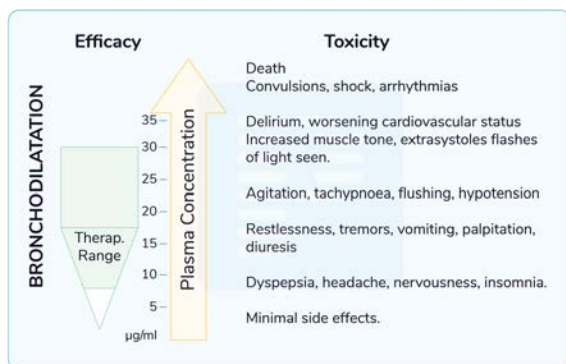
Correct answer: B) Increased risk of theophylline toxicity

Explanation:

Erythromycin is an inhibitor of the cytochrome P450 enzyme CYP1A2, which metabolises theophylline. Inhibition of this enzyme leads to higher plasma levels of theophylline, increasing the risk of toxicity.

Theophylline toxicity:

It has a narrow margin of safety, and dose-dependent toxicity starts from the upper half of the therapeutic concentration range. The image below shows the effects of theophylline toxicity.



Enzyme inhibitors: (leads to drug toxicity)

Drugs (Mnemonic- COKE i n G allon C ontain A rtificial P reser v atives):

C imetidine.	G rapefruit juice (Furano coumarins).
O meprazole.	C larithromycin & C iprofloxacin.
K etoconazole.	A cute alcoholism.
E rythromycin.	P rotease inhibitor (Ritonavir).
I soniazid (inhibits all CYP enzymes except CYP2E1).	V alproate.

Solution for Question 20:

Correct answer: C) Increased metabolism of lamotrigine

Explanation:

Phenytoin is known to induce hepatic enzymes, including cytochrome P450 enzymes, which play a crucial role in drug metabolism.

- One of the drugs affected by this induction is Carbamazepine.
- When Carbamazepine is metabolised more rapidly due to phenytoin's enzyme-inducing effects, its plasma concentrations decrease, potentially leading to reduced efficacy and worsening seizures despite adhering to the prescribed regimen.

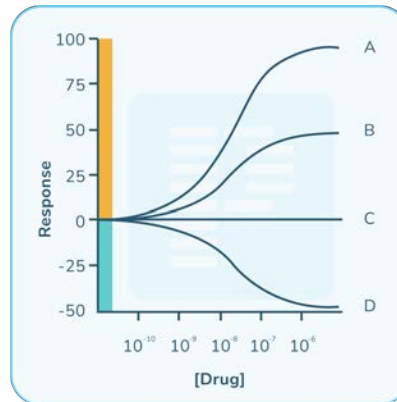
Enzyme inducers:

- They increase the activity of drug-metabolizing enzymes, usually within the cytochrome P450 (CYP) family.

- This leads to enhanced drug metabolism, which can reduce their therapeutic effects.
 - Drugs (Mnemonic- GRAASS) G riseofulvin. R ifampicin. A ntiepileptics: Phenytoin. Carbamazepine. Phenobarbital. A lcoholism (Chronic). S moking. S t. John's wart.
 - G riseofulvin.
 - R ifampicin.
 - A ntiepileptics: Phenytoin. Carbamazepine. Phenobarbital.
 - Phenytoin.
 - Carbamazepine.
 - Phenobarbital.
 - A lcoholism (Chronic).
 - S moking.
 - S t. John's wart.
 - G riseofulvin.
 - R ifampicin.
 - A ntiepileptics: Phenytoin. Carbamazepine. Phenobarbital.
 - Phenytoin.
 - Carbamazepine.
 - Phenobarbital.
 - A lcoholism (Chronic).
 - S moking.
 - S t. John's wart.
 - Phenytoin.
 - Carbamazepine.
 - Phenobarbital.
-

Pharmacodynamics

1. Which of the following agents inhibits the action of an agonist on a receptor or the subsequent response but does not have any effect of its own?



- A. Drug A
- B. Drug B
- C. Drug C
- D. Drug D

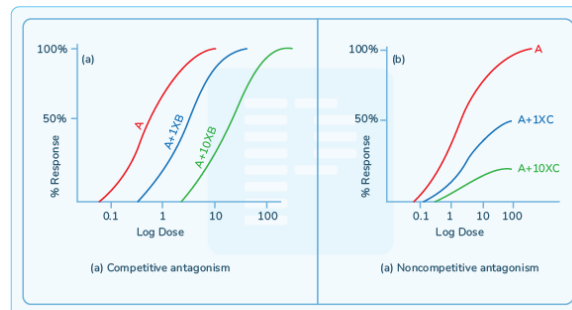
2. A patient with an overdose of a toxic substance is treated with a drug that directly binds to and inactivates the toxic substance. Which type of antagonist is being used in this treatment?

- A. Physical Antagonist
- B. Physiological Antagonist
- C. Chemical Antagonist
- D. Pharmacological Antagonist

3. Which type of antagonist acts on different receptors or mechanisms to produce opposite effects on the same physiological function?

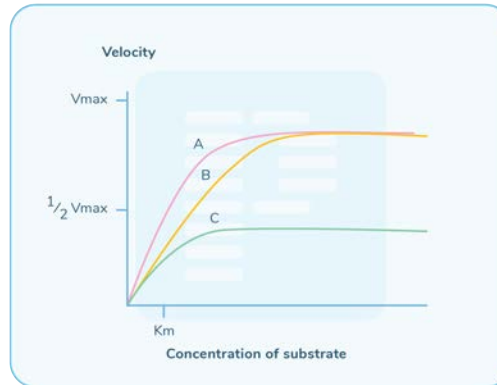
- A. Competitive antagonist
 - B. Non-competitive antagonist
 - C. Physiological antagonist
 - D. Inverse agonist
-

4. In a pharmacology experiment, you observe that adding a competitive antagonist increases the EC₅₀ without changing the maximum response. Adding a non-competitive antagonist decreases the maximum response without changing the EC₅₀. Which of the following best describes the dose-response curve changes for these scenarios?



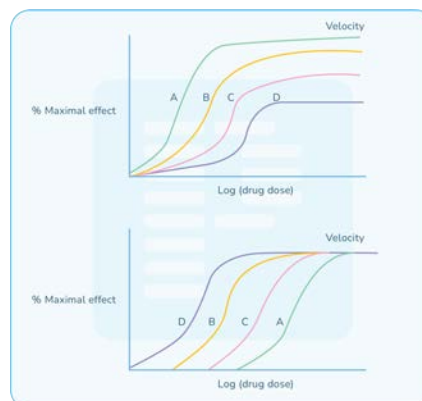
- A. Competitive antagonist shifts left, non-competitive antagonist shifts down.
 - B. Competitive antagonist shifts right, and non-competitive antagonist shifts down.
 - C. Competitive antagonist shifts down, and non-competitive antagonist shifts right.
 - D. Competitive antagonist shifts right, non-competitive antagonist shifts left.
-

5. Identify the correct statement based on the following Lineweaver Burke plot showing the effect of different types of inhibitors on a reaction.



- A. B denotes competitive inhibition
- B. C denotes non-competitive inhibition
- C. V_{max} is directly proportional to the enzyme concentration
- D. All of the above

6. A clinical trial of 4 new drugs was conducted, and the following graphs were obtained. Based on these graphs, which of the following statements is/are correct?
a) Drug A has the highest efficacy as well as the highest potency. b) Drug A has the highest efficacy but the lowest potency. c) Drug D has the highest efficacy as well as the highest potency. d) Drug B is more potent than drug C.



- A. Statements a and d are correct
 - B. Statements b and d are correct
 - C. Only statement b is correct
 - D. None of the statements are correct
-

7. Which of the following statements is true about dose-response curves (DRCs)?

- A. Graded DRCs measure the all-or-none response to different drug doses within a population.
 - B. Quantal DRCs plot the response to a drug continuously within an individual.
 - C. U-shaped DRCs are rare and can indicate toxicity at both low and high doses.
 - D. Graded DRCs are used to calculate the median effective dose (ED50) and the median lethal dose (LD50) in a population.
-

8. Lithium was prescribed to a newly diagnosed patient with bipolar disorder. Frequent monitoring is needed while using Lithium as it has a lower therapeutic index. How is the therapeutic index of a drug calculated?

- A. The ratio between the median toxic dose and median effective dose of lithium in the population.
 - B. The ratio between the median effective dose and median toxic dose of lithium in the population.
 - C. Product of median effective dose and median toxic dose of lithium.
 - D. Average of the sum of the median effective dose and median toxic dose of lithium.
-

9. A 35-year-old man with a history of epilepsy presents for a medication review. He has been taking phenytoin for the past year and wants to ensure his medication

dosage is appropriate. Which of the following is the therapeutic range of phenytoin?

- A. 10-20 mcg/ml
 - B. 40-50 mcg/ml
 - C. 50-100 mcg/ml
 - D. 100-120 mcg/ml
-

10. A 65-year-old man with a history of chronic heart failure presents for a follow-up appointment. He has been on digoxin therapy for several years. During the consultation, the cardiologist discusses the importance of therapeutic drug monitoring (TDM) to ensure optimal digoxin levels for effective treatment. TDM is NOT done in which of the following?

- A. Detect drug levels in case of poisoning levels.
 - B. Use of aminoglycosides in patients with renal failure.
 - C. Drugs with a narrow therapeutic index.
 - D. Drugs with irreversible action.
-

11. Which of the following types of receptors is the fastest acting?

- A. Enzyme-linked receptors
 - B. G-protein coupled receptors
 - C. Intracellular receptors
 - D. Ligand-gated ion channel receptors
-

12. Which is the active subunit of G-protein coupled receptors (GPCR)?

- A. Alpha subunit (■)
 - B. Beta subunit (β)
 - C. Gamma subunit (■)
 - D. Delta subunit (δ)
-

13. Which second messenger is associated with phospholipase C (PLC) activation by Gq protein?

- A. Cyclic AMP (cAMP)
 - B. Inositol trisphosphate (IP3) and diacylglycerol (DAG)
 - C. Cyclic GMP (cGMP)
 - D. Nitric oxide (NO)
-

14. When bound to its receptor, which ligand activates Gs proteins, leading to increased adenylyl cyclase activity and elevated cAMP levels?

- A. Serotonin at 5-HT1 receptors
 - B. Glucagon
 - C. Somatostatin
 - D. Thromboxane
-

15. Which of the following acts on enzyme-linked receptors?

- A. Epinephrine
- B. Insulin
- C. Acetylcholine
- D. Dopamine

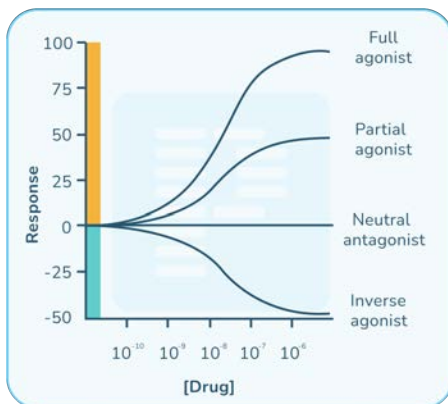
16. Which statement is not true about intracellular receptors?

- A. Intracytoplasmic receptors are attached to heat shock proteins (HSP) when inactive.
 - B. Intracellular receptors interact with lipid-soluble ligands that can pass through the cell membrane.
 - C. Intracellular receptors typically require second messengers to relay the signal inside the cell.
 - D. They interact with co-activators or co-repressors in the nucleus
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	2
Question 5	4
Question 6	2
Question 7	3
Question 8	1
Question 9	1
Question 10	4
Question 11	4
Question 12	1
Question 13	2
Question 14	2
Question 15	2
Question 16	3

Solution for Question 1:



Drug C (Option C) : Antagonist	It prevents the action of an agonist on a receptor or the subsequent response but does not have any effect of its own. Naloxone, an opioid receptor antagonist, is commonly used to reverse opioid overdose.
Drug A (Option A) : Agonist	It activates a receptor to produce an effect similar to the physiological signal molecule (e.g., morphine at opioid receptors).
Drug D (Option D) : Inverse Agonist	It activates a receptor to produce an effect opposite to that of an agonist (e.g., Rimonabant at cannabinoid receptors).
Drug B (Option B) : Partial Agonist	It activates a receptor to produce a submaximal effect and can block the action of a full agonist (e.g., Buprenorphine at opioid receptors).
Super Agonist	Not directly relevant to opioid overdose treatment but produces a maximum response greater than that of the endogenous agonist (e.g., BMS-986121 at GLP-1 receptors).

Solution for Question 2:

Type of Antagonists	Examples
Chemical Antagonist (Option C): Direct interaction between two drugs to form an inactive product.	KMnO ₄ and Alkaloids: Potassium permanganate oxidises alkaloids and is used in gastric lavage for poisoning. Tannins and Alkaloids: Tannins react with alkaloids to form insoluble alkaloidal tannate. Chelating Agents (BAL, Calcium Disodium EDTA): These agents are complex with toxic metals like arsenic

	(As) and lead (Pb). Nitrites and Cyanide: Nitrites form methemoglobin, which reacts with cyanide radicals.
Physical Antagonist (Option A) Physical methods are used to counteract the effects of a substance, often not involving receptor interactions.	Activated charcoal is used to physically adsorb toxins in the gastrointestinal tract to prevent their absorption.
Physiological Antagonist (Option B) Involves two agents that produce opposite effects on the same physiological function without interacting directly with each other or the same receptor.	Insulin and glucagon are physiological antagonists. Insulin lowers blood glucose levels, while glucagon raises them. Histamine and adrenaline on bronchial muscles and blood pressure. Hydrochlorothiazide and amiloride on urinary potassium excretion.
Pharmacological Antagonist (Option D) Drugs acting on the same receptor as the agonist but producing 0 overall effects and preventing the action of agonist on receptor.	Morphine and naloxone Acetylcholine and atropine

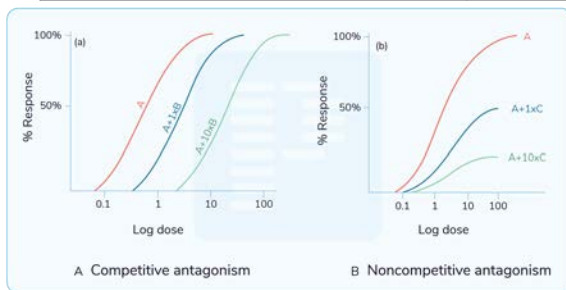
Solution for Question 3:

Type of Antagonist	Description	Mechanism	Example
Competitive Antagonist (Option A)	Competes with agonists for the same binding site, reducing the potency of the agonist.	Reversible binding to the same receptor site as the agonist.	Propranolol (β -blocker) binds to β -adrenergic receptors, reducing the effect of adrenaline (β -agonist)
Non-Competitive Antagonist (Option B)	Prevents the action of an agonist without competing with it for the binding site.	Irreversible binding to the receptor reduces the maximum effect of the agonist.	Phenoxybenzamine binds to α -adrenergic receptors, reducing the effect of adrenaline.
Physiological Antagonist (Option C)	Acts on different receptors or mechanisms to produce opposite effects on the same physiological function.	Does not directly interact with the same receptor or signaling pathway as the agonist.	Calcium channel blocker reduces the effect of adrenaline on the heart.

Inverse Agonist (Option D)	Selectively binds to the inactive conformation of a constitutively active receptor, reducing the basal activity of the receptor.	Binds to the inactive conformation of the receptor, reducing the basal activity.	Propranolol (β -blocker) can act as an inverse agonist on β -adrenergic receptors, reducing the basal activity of the receptor.
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Solution for Question 4:

	Competitive Antagonism	Non-Competitive Antagonism
Description	Competes with agonists for the same receptor site	Binds to a different site on the receptor, alters receptor function
Effect on Dose-Response Curve	Shifts to the right, no change in maximal response	Flattens the curve, reduces the maximal response
Reversibility	Reversible	Irreversible
Maximal Response	Unaffected	Reduced
Effective Concentration for 50% of population/ EC50 (Km)	Decreased	Unchanged



Solution for Question 5:

Correct Answer: D) All of the above.

According to Michaelis Menten kinetics:

- $V = (V_{\max} * [S]) / (K_m + [S])$

where $V_{\max}$ – maximum velocity of a reaction

K_m - Concentration of substrate at which velocity is $\frac{1}{2}$ of $V_{\max}$

- $V_{\max}$ is directly proportional to the enzyme concentration (Option C)
- K_m is inversely proportional to the affinity of the enzyme for its substrate.

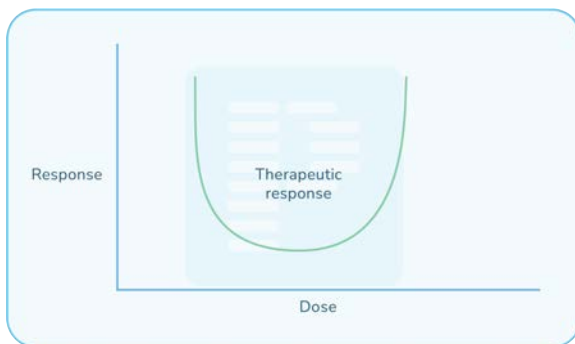
	COMPETITIVE INHIBITOR	NON-COMPETITIVE INHIBITOR	Uncompetitive inhibitor
Substrate resemblance	Resembles substrate.	It does not resemble the substrate.	It does not resemble the substrate.
Effect on increasing [S]	Inhibition can be reversed.	Inhibition cannot be reversed.	Inhibition can be reversed.
Binding site	Active site.	Allosteric site.	Enzyme substrate complex
$V_{\max}$	Unchanged. (Option A)	Decreases (as active enzyme decreases). (Option B)	Decreases (as active enzyme decreases).
K_m	Increases (more substrate needed to overcome inhibited enzyme).	Unchanged (increasing substrate does not overcome the inhibition).	Increases (more substrate needed to overcome inhibited enzyme).

Solution for Question 6:

- Drug A has the highest efficacy with the lowest potency.
- Drug D has the highest potency with the lowest efficacy .
- Efficacy is the maximal effect a drug can produce, represented by the $V_{\max}$ /y-axis value. The higher the $V_{\max}$, the higher the efficacy.
- Potency is the amount of drug required to produce a given effect. The lower the drug dose needed, the higher the potency.
- Both are independent of each other; efficacious drugs can have high or low potency and vice versa.
- The first graph helps us estimate the efficacy of the drugs (the higher the $V_{\max}$, the higher the efficacy, Thus Drug $A > B > C > D$).

- The second graph helps us estimate the drugs' potency (The lower the drug dose needed, the higher the potency). The left shift of the graph indicates an increase in potency.
- Thus Drug D > B > C > A.

Solution for Question 7:



Very few drugs form a U-shaped DRC; it has 3 parts

- 1st part: Less dose gives Adverse effect (deficiency)
- 2nd part: Therapeutic response (optimum dose)
- 3rd part: High dose gives Adverse effect (excess)
- Examples:

Drug /hormone/vitamin	Deficiency	Excess
Insulin	Diabetes	Hypoglycemia
Iron	Anemia	Hemochromatosis
Vitamin A	Night blindness	Liver damage, blurred vision.

Graded DRCs measure the all-or-none response to different drug doses within a population (Option A) and graded DRCs are used to calculate the median effective dose (ED50) and the median lethal dose (LD50) in a population (Option D):

Graded DRC is used for the determination of response in 1 individual.

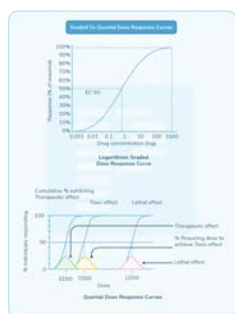
It measures the continuous response to different drug concentrations within an individual.

- When the dose is increased gradually, the response also increases
- Example: Antihypertensives

- Helpful in finding out efficacy and potency.

Quantal DRCs plot the response to a drug continuously within an individual (Option B):

- Quantal DRC is used to determine the response in a population (%).
- It measures an all-or-none response within a population, plotting the cumulative percentage of subjects that respond at each dose.
- They are used to Calculate ED50 and LD50. Therapeutic index of a drug■.
- Calculate ED50 and LD50.
- Therapeutic index of a drug■.
- Calculate ED50 and LD50.
- Therapeutic index of a drug■.

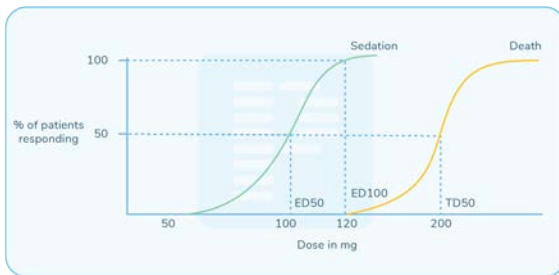


Solution for Question 8:

- The Therapeutic Index (TI) is a measure used to assess a drug's safety . It is defined as the ratio of the dose that produces toxicity to the dose that produces a therapeutic effect.

TI= median toxic dose /median effective dose ; (TD50/ED50)

- The therapeutic window is the range of drug concentrations that can safely and effectively treat the disease.
- If the TI window for a drug is narrow, it requires frequent monitoring.
- E.g., Warfarin, Theophylline, Digoxin, Lithium (Warning These Drugs are Lethal)



Solution for Question 9:

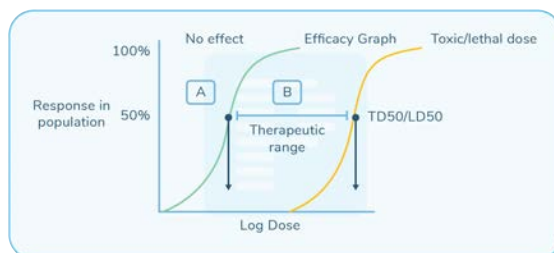
Correct Answer: A) 10-20 mcg/ml

The median effective dose (ED50) refers to the dose of a drug or biologic that produces a specific beneficial therapeutic response in 50% of the population that takes that dose.

LD stands for "Lethal Dose". LD 50 is the amount of a material, given all at once, which causes the death of 50% (one-half) of a group of test animals.

The Therapeutic Range is the range between the minimal therapeutic effect and the maximal acceptable adverse effect (i.e., the range between ED50 and TD50).

- The drug doesn't work below ED50 (Area A) and is lethal above TD50.
- 10-20 mcg/ml is the therapeutic range of phenytoin, which implies that the drug doesn't work below 10 mcg/ml and is lethal above 20 mcg/ml.
- The aim is to maintain the plasma concentration between this range for maximal therapeutic efficacy and safety.



Therapeutic ranges of other drugs:

- Valproate: 50-100 mcg/ml (Option C)
- Phenobarbitone: 15-40 mcg/ml

- Digoxin: 0.6-2.0 ng/ml
 - Theophylline: 10-20 mcg/ml
 - Lithium: 0.5-1.5 meq/ml
 - Vancomycin: 5-15 mcg/ml
-

Solution for Question 10:

Therapeutic drug monitoring (TDM) is the measurement of plasma drug concentration.

It is expensive and is only done in:

- Drugs with low safety margin, e.g., Digoxin, Anticonvulsants, Antiarrhythmics, theophylline, aminoglycoside antibiotics, lithium, and tricyclic antidepressants. (Option C)
- If individual variations are large, e.g., Antidepressants, Lithium.
- Nephrotoxic drugs used in renal failure, e.g., Aminoglycoside antibiotics and vancomycin. (Option B)
- In case of poisoning, detect drug levels and initiate treatment. (Option A)
- In case of failure of response, e.g., Antimicrobials.
- To check patient compliance, e.g., Phenytoin.

TDM is not done in

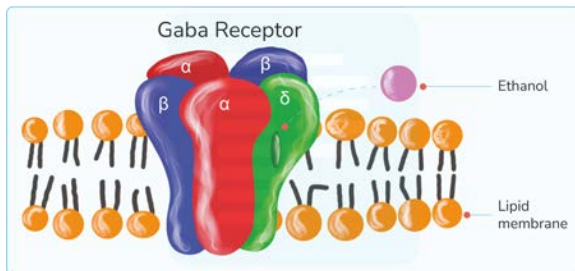
- Drugs whose response is easily measurable, e.g., Antihypertensives, hypoglycemics, diuretics, oral anticoagulants, and inhalational general anaesthetics.
 - Drugs activated in the body, e.g., Levodopa.
 - 'Hit and run drugs' (whose effect lasts much longer than the drug itself), e.g., reserpine, guanethidine, MAO inhibitors, omeprazole.
 - Drugs with irreversible action, e.g., Organophosphate anticholinesterases, phenoxybenzamine.
-

Solution for Question 11:

- Ligand-gated ion channels are cell surface receptors that enclose ion-selective channels (for Na⁺, K⁺, Ca²⁺, or Cl⁻) within their molecules.
- Agonist binding opens the channel and causes depolarisation/hyperpolarisation.
- Examples:

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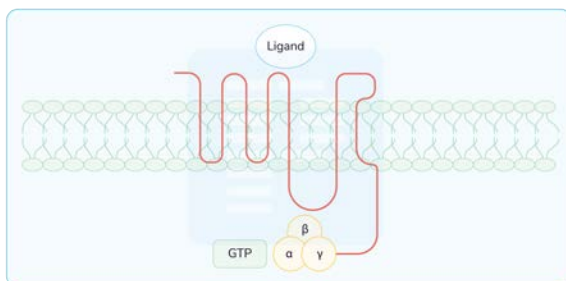


- The receptor is a pentameric protein with 5 subunits.
- The agonist directly operates ion channels in these receptors without any coupling protein or second messenger intervention.
- The onset and offset of responses are the fastest (in milliseconds).

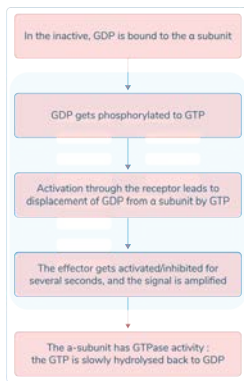
Solution for Question 12:

Correct Answer: A) Alpha subunit (■)

Alpha subunit (■) is the active subunit of G-protein coupled receptors (GPCR).



- G-protein coupled receptors (GPCR) are cell membrane receptors
- The molecule has 7 α-helical domains, which run in a serpentine manner into 3 extracellular and 3 intracellular loops.



Solution for Question 13:

- When ligand-bound G protein-coupled receptors (GPCRs) are activated, Gq protein stimulates phospholipase C (PLC) .
- PLC hydrolyses phosphatidylinositol 4,5-bisphosphate (PIP2) into inositol trisphosphate (IP3) and diacylglycerol (DAG) which act as second messengers.
- IP3 binds to receptors on the endoplasmic reticulum, releasing calcium ions (Ca²⁺) , which activate calcium-dependent enzymes and protein kinases.
- DAG remains in the plasma membrane, activating protein kinase C (PKC) to phosphorylate target proteins, initiating cellular responses like gene expression and cell growth

G Protein Type	Effector	Second Messenger(s)	Pathway	Gs	Adenylyl cyclase	Cyclic AMP	Increases cAMP, activates protein kinase A (PKA)	Gq	Adenylyl cyclase	Reduces cyclic AMP (cAMP)	Decreases cAMP, inhibits protein kinase A (PKA)	Gq	Phospholipase C (PLC)	Inositol trisphosphate (IP3) and DAG	IP3 releases Ca ²⁺ from ER; DAG activates protein kinase C (PKC)
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tent trun cat ed)																
G P rote in T ype	Eff ec tor	Sec ond Me sse nge r(s)	Pat hw ay													
Gs	Ad en yly l c ycl as e	Cyc lic AM P (c AM P)	Incr eas es cA MP, act ivat es prot ein kin ase A (PK A)													
Gi	Ad en yly l c ycl as e	Re duc es cycl ic A MP (cA MP)	Dec rea ses cA MP, inh ibits pro tein kin ase A (PK A)													
Gq	Ph os ph oli pa se C (PL C)	Ino sitol tris pho sph ate (IP 3) a nd DA G	IP3 rele ase s C a ²⁺ fro m E R; DA G a ctiv ate													

[illegible]

Solution for Question 14:

Protein	Ligands and Receptors
Gs subtype	Glucagon at its receptor

Epinephrine at β -adrenergic receptors	
Histamine at H2 receptors	
Gi subtype	Acetylcholine at M2 receptors
Dopamine at D2 receptors	
Serotonin at 5-HT1 receptors	
Gq subtype	Serotonin at 5-HT2 receptors
Acetylcholine at M1 receptors	
Histamine at H1 receptors	
Thromboxane	

Solution for Question 15:

- Insulin primarily acts on enzyme-linked receptors, specifically receptor tyrosine kinases (RTKs), to regulate glucose metabolism and other cellular processes.
- Enzyme-linked receptors consist of a large extracellular ligand-binding domain connected through a single transmembrane helical peptide chain to an intracellular subunit with enzymatic properties.

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- Epinephrine primarily acts on G protein-coupled receptors (GPCRs) and ligand-gated ion channels.
- Acetylcholine primarily acts on muscarinic and nicotinic receptors, which are G protein-coupled (GPCRs) and ligand-gated ion channels.
- Dopamine primarily acts on dopamine receptors, which belong to the G protein-coupled receptor (GPCR)

Solution for Question 16:

Intracellular receptors :

- They do not typically require second messengers to relay the signal inside the cell. Instead, these receptors, upon binding with their ligands, influence gene transcription, and protein synthesis directly through DNA.
- They directly act through DNA, influencing gene transcription.
- These receptors can increase or decrease gene transcription through coactivators and corepressors and are known for their slow and sustained effects.
- Based on their location, intracellular receptors are classified into intranuclear receptors, which are already inside the nucleus, and intracytoplasmic receptors, which translocate to the nucleus after ligand binding.

Table rendering failed

Clinical Trials and Miscellaneous

1. Which of the following is the recommended temperature for drugs which need to be stored in a refrigerator but not in the freezer ?

- A. 1-7°C
 - B. 2-8°C
 - C. 4-12°C
 - D. 7-15°C
-

2. Arrange the following steps in the correct order as they occur in the process of drug development: Filing of an Investigational New Drug (IND) application. Start toxicity and safety testing based on GLP guidelines. Approval from Central Drugs Standard Control Organization (CDSCO). Start pharmacokinetics and pharmacodynamics testing based on GCP guidelines. Filing of an NDA application to CDSCO. Approval from the Committee for Control and Supervision of Experiments on Animals (CPCSEA)

- A. 6, 2, 1, 3, 4, 5
 - B. 6, 2, 4, 1, 3, 5
 - C. 3, 2, 6, 1, 4, 5
 - D. 6, 2, 1, 4, 5, 3
-

3. Which of the following statements is true regarding the source of a drug?

- A. Atropine and vaccines are developed through rational drug design.
- B. HIT refers to a synthetic chemical with the potential to become a drug.
- C. Preclinical trials are not necessary for potential medications.

D. The most common way to find a potential drug is by high throughput screening of all available chemicals.

4. Match the phases of drug trials with their correct descriptions: Phase Description
1. Phase 0 a. Determines drug efficacy and side effects on a heterogeneous population
2. Phase 1 b. Monitor long-term side effects and drug interaction in hidden populations.
3. Phase 2 c. Human testing to assess drug safety and maximum tolerable dose (MDT).
4. Phase 3 d. Randomized trials to assess drug efficacy and optimal dosing
5. Phase 4 e. Low dosing trials to check drug-target binding.

Phase	Description
1. Phase 0	a. Determines drug efficacy and side effects on a heterogeneous population
2. Phase 1	b. Monitor long-term side effects and drug interaction in hidden populations.
3. Phase 2	c. Human testing to assess drug safety and maximum tolerable dose (MDT).
4. Phase 3	d. Randomized trials to assess drug efficacy and optimal dosing
5. Phase 4	e. Low dosing trials to check drug-target binding.

- A. 1: e, 2: c, 3: d, 4: a, 5: b
B. 1: c, 2: e, 3: a, 4: b, 5: a
C. 1: e, 2: c, 3: a, 4: b, 5: d
D. 1: c, 2: d, 3: e, 4: a, 5: b
-

5. Match the phases of clinical trials with their target populations: Phase Target Population
1. Phase 0 a. Patients with the disease of interest
2. Phase 1 b. General patient population post-marketing
3. Phase 2 c. Healthy volunteers
4. Phase 3 d. Small number of patients with the disease
5. Phase 4 e. Very small number of

healthy or diseased individuals

Phase	Target Population
1. Phase 0	a. Patients with the disease of interest
2. Phase 1	b. General patient population post-marketing
3. Phase 2	c. Healthy volunteers
4. Phase 3	d. Small number of patients with the disease
5. Phase 4	e. Very small number of healthy or diseased individuals

- A. 1: b, 2: c, 3: d, 4: e, 5: a
 - B. 1: e, 2: c, 3: d, 4: a, 5: b
 - C. 1: a, 2: d, 3: c, 4: e, 5: b
 - D. 1: e, 2: c, 3: a, 4: b, 5: d
-

6. A 68-year-old female with a history of atrial fibrillation and congestive heart failure presents to the emergency department with symptoms of nausea, vomiting, blurred vision, and severe bradycardia. On examination, there is evidence of cardiac dysrhythmias. Which of the following categories does this specific antidote belong to in the management?

- A. Counterfeit drugs
 - B. Essential drugs
 - C. Orphan drugs
 - D. Over-the-counter drugs
-

7. Which of the following statements are true about counterfeit drugs? Counterfeit drugs may contain incorrect ingredients. These drugs often have the same therapeutic effect as genuine drugs. Counterfeit drugs are usually subject to rigorous quality control. They can lead to treatment failure and contribute to drug

resistance. Counterfeit drugs may be labelled with fake packaging to resemble genuine products.

- A. 1, 2, and 4
 - B. 1, 3, and 5
 - C. 1, 4, and 5
 - D. 2, 3, and 5
-

8. A 28-year-old pregnant woman comes to the antenatal clinic for regular check up. An ultrasound of her abdomen showed craniofacial abnormalities, hypoplasia of the phalanges, and cleft palate. Which of the following medications is most likely to cause this condition?

- A. Valproate
 - B. Phenytoin
 - C. Isotretinoin
 - D. Warfarin
-

9. Which of the following is the primary aim of pharmacovigilance?

- A. To conduct clinical trials for new drugs
 - B. To ensure the quality and efficacy of pharmaceutical products
 - C. To reduce the risk of drug-related harm to the patient
 - D. To control the marketing and distribution of medications
-

10. All of the following are true about Placebo drugs except:

- A. It is used as a dummy medication

- B. It contains the active moiety
 - C. It can induce physiological responses
 - D. Placebo helps in blinding
-

11. A diabetic patient on regular insulin therapy accidentally injects himself with a higher dose than prescribed. He experiences sweating, trembling, confusion, and dizziness. Which type of adverse drug reaction (ADR) can be seen in this patient?

- A. Type A
 - B. Type B
 - C. Type C
 - D. Type D
-

12. A patient who was admitted with acute myocardial infarction is due for discharge. His previous medical records show a defective CYP2C19. Which of his following long-term medications should be avoided in this case?

- A. Ticagrelor
 - B. Ramipril
 - C. Clopidogrel
 - D. Atenolol
-

**13. Match the following adverse reactions with their causative agents. IMAGE
DRUG RESPONSIBLE Sulfonamides Carbamazepine Tetracyclines Griseofulvin**

IMAGE	DRUG RESPONSIBLE
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		Sulfonamides
		Carbamazepine
		Tetracyclines

- A. 1-a, 2-d, 3-b, 4-c
- B. 1-c, 2-b, 3-a, 4-d
- C. 1-b, 2-c, 3-d, 4-a
- D. 1-d, 2-b, 3-d, 4-c

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	2
Question 4	1
Question 5	2
Question 6	3
Question 7	3
Question 8	2
Question 9	3
Question 10	2
Question 11	1
Question 12	3
Question 13	2

Solution for Question 1:

The recommended temperature for drugs which need to be stored in a refrigerator but not in the freezer is 2-8°C.

The United States Pharmacopeia recommends appropriate storage temperatures for drugs based on criteria that should not affect their efficacy.

- Room temperature (controlled): 20 to 25°C.
- Cool storage: 8 to 15°C.
- Refrigerator storage: 2 to 8°C.

- Freezer storage: -25 to -10°C.

Apart from temperature regulations, other recommendations like “store in a cool, dry place or protect from light” should also be followed because those drugs can be susceptible to damage by humidity or light, respectively.

Solution for Question 2:

Steps in drug development:

Solution for Question 3:

Drug Sources:

- All chemicals that have the potential to become drugs are called HIT. (Option B)
- Pre-clinical trials are required for HIT/potential medicines. (Option C)
- Sources:

Natural	High throughput screening	Rational drug designing
Plants: Morphine, atropine (Option A) Animals: Vaccines, monoclonal antibody Microorganism: Penicillin, streptomycin	Screening all available chemicals with the potential to become a drug is the M/C way. (Option D)	Developed to bind at target receptors E.g. PPIs.

Solution for Question 4:

Phases of clinical trials:

Phases	Description
Phase 0	Design: An open-label trial Purpose: Check the target binding of a drug Check the pharmacokinetics of the drug Called a micro-dosing trial (100 micrograms dose) Non-mandatory clinical trial
Phase 1	Design: An open-label trial Purpose: Evaluate drug safety- the most important Evaluate MTD (maximum tolerable dose) , loading dose and drug half-life. Evaluate drug pharmacokinetics (absorption, distribution, metabolism and excretion) Called the first human trial
Phase 2	Design: Randomized clinical trial (RCT) Purpose: Explore drug efficacy Called a therapeutic exploratory trial or proof of concept trial
Phase 3	Design: Double-blinded RCT Purpose: Find the optimal drug dose Confirm efficacy Safety is priority Called a therapeutic confirmatory trial Called multi-site trial (done on a heterogeneous population) Drugs get market approval
Phase 4	Design: Open-label Purpose: Find hidden/missed/rare adverse effects. Evaluate compliance: long-term side effects Drug interactions Food interactions Called post-marketing surveillance

Solution for Question 5:

Target Population in Clinical Trials:

Phases	Number of Volunteers	Type of population
Phase 0	2-5	Healthy
Phase 1	20-50	Healthy
Phase 2	200-500	Patients
Phase 3	2000-5000	Patients
Phase 4	Larger population (cannot be predicted)	Patients on the approved drug

Solution for Question 6:

History of atrial fibrillation and congestive heart failure with symptoms of nausea, vomiting, blurred vision, and severe bradycardia along with evidence of cardiac dysrhythmias suggests Digoxin toxicity. Digoxin-specific antibody fragments (Digibind or DigiFab) are the specific antidote for digoxin toxicity, which is an orphan drug.

Orphan drugs:

These are:

- Drugs or biological agents for diagnosis/treatment/prevention of a rare disease or condition.

(or)

- A more common disease (endemic only in resource-poor countries) for which there is no reasonable expectation that the cost of developing and marketing it will be recovered from the sales of that drug.
- Rare disease: In India, Rare disease - the prevalence of the disease is <1 per 5 lakh population. In the USA, Rare disease - the prevalence of the disease is <1 per 2 lakh population.
- In India, Rare disease - the prevalence of the disease is <1 per 5 lakh population.
- In the USA, Rare disease - the prevalence of the disease is <1 per 2 lakh population.
- In India, the government gives incentives/money to pharmaceutical companies to make orphan drugs.
- Some examples of orphan drugs:
- In India, Rare disease - the prevalence of the disease is <1 per 5 lakh population.

- In the USA, Rare disease - the prevalence of the disease is <1 per 2 lakh population.

Desferrioxamine	Clofazimine	Sodium stibogluconate
Digibind	Colchicine	Sodium thiosulfate
Fomepizole	Eltrombopag	Sodium nitrite
Felbamate	Fomivirsen	ThioTEPA
Adenosine deaminase	Icatibant	Succimer
Azacitidine	Nilotinib	Rifabutin
Bevacizumab	Paromomycin	Somatropin
Bortezomib	Rifaximin	Liothyronine
Carboprost	Rituximab	Miltefosine

Orphan receptors :

- These are receptors whose natural ligands are unknown.
- They are discovered based on their structural similarity to known receptors.
- Types: GPCRs & nuclear receptors.
- Importance: To understand their function. Potential targets for new therapeutics, especially for diseases with Undiscovered treatment
- To understand their function.
- Potential targets for new therapeutics, especially for diseases with Undiscovered treatment
- Examples: GPR40 and GPR55 were once orphan receptors, but they are now known to interact with specific ligands, revealing new physiological roles.
- To understand their function.
- Potential targets for new therapeutics, especially for diseases with Undiscovered treatment

Counterfeit drugs (Option A): Fake medications that are illegally produced and sold to represent their origin, authenticity, or effectiveness deceptively.

Essential medicines (Option B):

- WHO defines essential medicines as “those that satisfy the priority healthcare needs of the population.”
- The drugs should be Available at all times. Available in all doses. Available in all formulations. Affordable and cost-effective.
- Available at all times.
- Available in all doses.
- Available in all formulations.

- Affordable and cost-effective.
- Available at all times.
- Available in all doses.
- Available in all formulations.
- Affordable and cost-effective.

Over-the-counter drugs (Option D):

- Medications available without a prescription, intended for self-medication and usually for less serious conditions.
- Examples: Paracetamol, antacids, laxatives, vitamins etc.

Solution for Question 7:

Counterfeit drugs:

- Illegal.
- Given to cheat or harm a person.
- These drugs do not have the same therapeutic effect as genuine drugs and may lead to treatment failure.
- Counterfeit drugs are not subjected to proper quality control.
- People making illegal drugs are punished or jailed.

Types	Description	Example
Spurious Drugs	Have no active ingredient	500 mg paracetamol tablet → when checked, it contains no paracetamol in it. Made by illegal people other than the manufacturer to copy the drug.
Misbranded Drugs	Low-quality drugs	500 mg tablet → when checked, it contains only 200 mg. Made by the same manufacturer to increase profit.
Adulterated Drugs	Addition of harmful or unwanted excipients	Cough syrup made in India containing ethylene glycol → when checked, GMP guidelines do not approve these substances. Ethylene glycol is harmful and not approved as an excipient by GMP.

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Solution for Question 8:

The ultrasound findings in a 28-year-old pregnant woman showing craniofacial abnormalities, hypoplasia of the phalanges, and cleft palate in the fetus are suggestive of fetal hydantoin syndrome caused by phenytoin intake.

Fetal hydantoin syndrome:

- It is associated with the use of phenytoin during pregnancy.
- The features include Hypertelorism. Low nasal bridge. Hypoplasia of the distal phalanges and nails. Growth impairment. Intellectual disability. Cleft lip/palate. Microcephaly.
- Hypertelorism.
- Low nasal bridge.
- Hypoplasia of the distal phalanges and nails.
- Growth impairment.
- Intellectual disability.
- Cleft lip/palate.
- Microcephaly.
- Hypertelorism.
- Low nasal bridge.
- Hypoplasia of the distal phalanges and nails.
- Growth impairment.
- Intellectual disability.
- Cleft lip/palate.
- Microcephaly.

Teratogenic drugs:

Teratogenic drugs are substances that can cause developmental abnormalities (teratogenesis) in a developing fetus when a pregnant woman is exposed to them.

Drug	Teratogenic Disorder
Acitretin	Craniofacial anomalies , heart defects, CNS malformations.

Alcohol	Fetal Alcohol Syndrome (growth retardation, facial anomalies, CNS dysfunction, ASD, VSD).
ACE inhibitors & Angiotensin-receptor blockers	Renal dysgenesis, oligohydramnios, lung hypoplasia, skull hypoplasia.
Carbamazepine	Neural tube defects (spina bifida), craniofacial defects.
Isotretinoin (Option B)	Craniofacial anomalies, heart defects, CNS malformations.
Lithium	Ebstein's anomaly (tricuspid valve malformation).
Thalidomide	Severe limb defects (phocomelia), ear, cardiac, and genitourinary malformations.
Carbimazole & methimazole	Choanal atresia, aplasia cutis, and fetal goitre.
Penicillamine	Cutis laxa syndrome.
Statins	Vertebral, anal, cardiac, tracheo-oesophageal fistula, renal, arterial, and limb defects (VACTERAL defects).
SSRIs	Persistent pulmonary hypertension of the newborn.
Valproic acid (Option A)	Neural tube defects (spina bifida), craniofacial defects, cardiovascular malformations, and developmental delay.
Warfarin (Option D)	Fetal warfarin syndrome: Nasal hypoplasia, stippled epiphyses, limb hypoplasia, and central nervous system abnormalities.

Solution for Question 9:

The WHO defines pharmacovigilance as “the science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or other drug-related problems”. The main aim of pharmacovigilance is to reduce the risk of drug-related harm to the patient.

It has an important role in educating doctors about adverse effects, official regulation of drug use, and the rational use of medicines, as it provides the basis for assessing the safety of medicines.

Activities involved in pharmacovigilance:

- Postmarketing surveillance and other methods of ADR monitoring, such as voluntary reporting by doctors, prescription event monitoring, computerised medical record linkage,

and other cohort/case-control studies, as well as anecdotal case reports by doctors.

- Dissemination of ADR data through drug alerts, medical letters, and advisories sent to doctors by pharmaceuticals and regulatory agencies.
- Changes in the labelling of medicines indicate restrictions in use, statutory warnings, precautions, or even drug withdrawal by the regulatory decision-making authority.
- The Central Drugs Standard Control Organisation (CDSCO) and the National Pharmacovigilance advisory committee coordinate pharmacovigilance programmes in India.
- Algorithms such as the Naranjo algorithm (casualty assessment) and modified Hartwig scale (severity grading) are used to assess ADR casualty and severity.

Solution for Question 10:

- A placebo is an inert substance given in a medicine's garb. A placebo is similar to the actual drug. It does not contain any active moiety (Option B) . It contains inert substances with excipients.
- A placebo is similar to the actual drug.
- It does not contain any active moiety (Option B) .
- It contains inert substances with excipients.
- It works by psychodynamic rather than pharmacodynamic means and often produces responses equivalent to the active drug (Option C) . Placebos are used in two situations: As a control device in clinical trials of drugs (dummy medication & blinding) (Options A & D) . To treat a patient who, in the physician's opinion, does not require an active drug.
- As a control device in clinical trials of drugs (dummy medication & blinding) (Options A & D) .
- To treat a patient who, in the physician's opinion, does not require an active drug.
- A placebo is similar to the actual drug.
- It does not contain any active moiety (Option B) .
- It contains inert substances with excipients.
- As a control device in clinical trials of drugs (dummy medication & blinding) (Options A & D) .
- To treat a patient who, in the physician's opinion, does not require an active drug.
- Placebos can induce physiological responses (e.g, releasing endorphins in the brain, causing analgesia).

- The administration of an active drug produces effects due to both pharmacological and psychodynamic effects; a placebo can supplement the pharmacological effects of active medicines.
- Examples: Lactose tablets/capsules, distilled water injection, multi-vitamin injections, and sham surgery.

Solution for Question 11:

A diabetic patient taking a higher dose of insulin and developing hypoglycemia (symptoms such as sweating, trembling, confusion, and dizziness) is suggestive of a type A adverse drug reaction.

Types of drug reactions (mnemonic-ABCDEF):

Type of reaction	Description	Example
A ugmented	Dose-dependent, predictable, and more common.	↑ Insulin → hypoglycemia.
B izarre	Dose-independent, non-predictable, less common, and more serious.	Allergy & idiosyncrasy.
C hronic	Dose & duration dependent	Suppression of hypothalamic-pituitary axis by chronic steroid use.
D elayed	Duration dependent	Kidney disease from long-term NSAID use.
E nd of use	Withdrawal symptoms	Opioid withdrawal.
F ailure	Failure of therapy	Enzyme inducers → OCP failure.

Solution for Question 12:

Clopidogrel, once enters the blood, is converted into its active form by the CYP2C19 enzyme and hence, people with genetic defects in this enzyme will be “poor metabolisers” of clopidogrel and hence will be ineffective.

Pharmacogenetics	Pharmacogenomics
The study of the genetic basis for variability in drug response.	It uses genetic information to guide the choice of drug and dose individually.
It deals with genetic influences on drug action as well as on drug handling of the body.	It tends to identify individuals who are either more likely or less likely to respond to a drug, as well as those who require altered doses of certain drugs.

In this patient's case, the defective CYP2C19 gene means they cannot properly metabolise clopidogrel. Pharmacogenomics takes this information to guide the choice and dose of drugs, ensuring that medications are tailored to the patient's genetic profile for optimal effectiveness and safety.

Patients can be divided into four types based on the ability to bio-activate clopidogrel.

- Optimal metabolisers : The usual dose of clopidogrel can be completely effective.
- Intermediate metabolisers : Therapeutic effect can be achieved by increasing the dose.
- Poor metabolisers : Clopidogrel will almost always be ineffective.
- Over metabolisers : They require excessive dosing and can lead to adverse effects.

Ticagrelor, Ramipril, and Atenolol are not metabolised by CYP2C19 (Options A, B, and D ruled out).

Solution for Question 13:

The images are numbered as:

1. Tetracyclines can commonly cause phototoxic reactions in the right leg. The image below shows phototoxic reactions on the chest.



2. Steven Johnson syndrome is common with many drugs, especially carbamazepine.



3. Erythema nodosum is common with sulfonamides .



4. Erythema multiforme can be caused by griseofulvin.



Adverse effects of a drug can manifest in different ways:

- Toxic effects.
 - Intolerance.
 - Idiosyncrasy.
 - Hypersensitivity.
 - Phototoxicity.
 - Drug dependency and addiction.
 - Teratogenicity.
 - Mutagenicity & carcinogenicity.
 - Withdrawal reactions.
-

Previous Year Questions

1. Loading dose of drug is dependent on:

- A. Clearance
 - B. Route of administration
 - C. Volume of distribution
 - D. Bioavailability
-

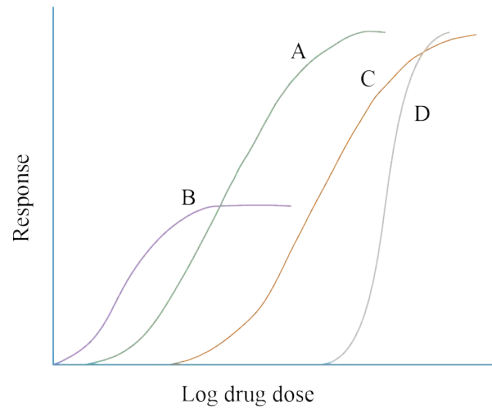
2. What is the mechanism of action of dapsone, considering that it is used in the treatment of leprosy, and fungal infections and, also for immunomodulatory actions?

- A. Cell wall synthesis inhibitor
 - B. Folic acid synthesis inhibitor
 - C. Protein synthesis inhibitor
 - D. Cell membrane inhibitor
-

3. A 35-year-old patient with a history of epilepsy is prescribed a combination of carbamazepine and valproate for seizure control. Several weeks into the treatment, the patient presents with abdominal pain, jaundice, and elevated liver enzymes. What is the most likely adverse effect associated with this medication combination?

- A. Pancreatitis
- B. Thrombocytopenia
- C. Hepatotoxicity
- D. Hyperammonemia

4. Which of the following statements is true about the given dose-response curve?



- A. Drug B is the most potent
- B. Drug B is the most efficacious
- C. Drug C is more efficacious than drug D
- D. Drug A is more potent than drug B

5. What is the extent of drug absorption when it is given intravenously?

- A. 10%
- B. b. 100%
- C. 50%
- D. Variable

6. Which of the following is not a prodrug?

- A. Proguanil

- B. Sulindac
 - C. Primidone
 - D. Lisdexamfetamine:
-

7. Which of the following causes an increase in ACh release from the presynaptic neuron?

- A. Potassium channel blocker
 - B. Sodium channel blocker
 - C. Calcium channel blocker
 - D. Chlorine channel blocker
-

8. A drug label suggests storage in a 'cool' place. What is the temperature at which this drug is to be stored?

(or)

A drug label suggests storage in a 'cool' place. What is the temperature at which this drug is to be stored?

- A. -2 degree Celsius
 - B. 0 degree Celsius
 - C. 2-8 degree Celsius
 - D. 8-15 degrees Celsius
-

9. Which drug's action is prolonged by the administration of cilastatin?

- A. Meropenem

- B. Imipenem
 - C. Cefazolin
 - D. Piperacillin
-

10. Which parameter is utilized to evaluate the level of drug absorption in a graph demonstrating plasma concentration versus time?

- A. T_{max}
 - B. C_{max}
 - C. T_{1/2}
 - D. Area under the curve
-

11. Which of the following describes LD 50?

- A. 50% animal mortality is an experiment after exposure to the drug
 - B. 50% reduction of infection in the given population of experimental animals
 - C. Dose of the drug that causes adverse effects in 50% of the population
 - D. Range of concentration at which the likelihood of efficacy is high and the probability of adverse effects is low
-

12. Which of the following is an example of a GPCR?

- A. Muscarinic
- B. Nicotinic
- C. GABA-A
- D. Insulin

13. In relation to drug absorption, what does the term "extent" signify?

- A. T_{max}
 - B. Area under the curve
 - C. C_{max}
 - D. t_{1/2}
-

14. Identify the drug categorized as schedule X from the options provided.

- A. Halothane
 - B. Thalidomide
 - C. Ketamine
 - D. Colistin
-

15. Which of the following acts by suicide inhibition? 1) Spironolactone 2) Ethinyl estradiol 3) Benzbromarone 4) Ritonavir

- A. 1, 2, and 3
 - B. 1 and 2
 - C. 3 and 4
 - D. 1, 3, and 4
-

16. A female patient complains of swelling and pain in her great toe. Her uric acid level is found to be high. A drug that reduces uric acid levels is prescribed. Which

of the following enzyme should the drug inhibit?

- A. Urease
 - B. Homogentisate oxidase
 - C. Lysyl oxidase
 - D. Xanthine oxidase
-

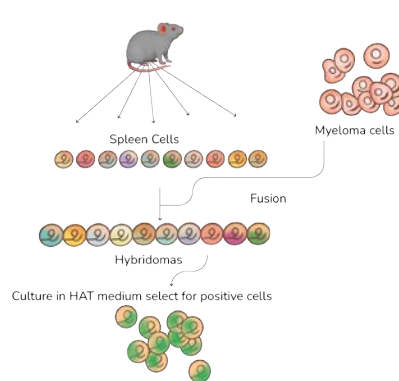
17. The therapeutic index is used for assessing which of the following factors?

- A. Safety
 - B. Efficacy
 - C. Patency
 - D. Toxicity
-

18. You, being an ardent researcher in the field of vaccine production, are the in-charge of preparing samples to make a vaccine against the new variant of SARS-CoV 2. After getting satisfactory results in preclinical studies, you seek approval to do clinical studies, which requires healthy volunteers. Which of the following phases does this refer to?

- A. Phase 1
 - B. Phase 2
 - C. Phase 3
 - D. Phase 4
-

19. The technique given in the picture is used for:



- A. Synthesis of monoclonal antibodies
- B. Preparing cell lines for viral culture
- C. Synthesis of vaccines
- D. Process of genetic engineering

20. Which of the following drugs does not increase the risk of bleeding in patients undergoing warfarin therapy?

- A. Cimetidine
- B. Carbamazepine
- C. Isoniazid
- D. Amiodarone

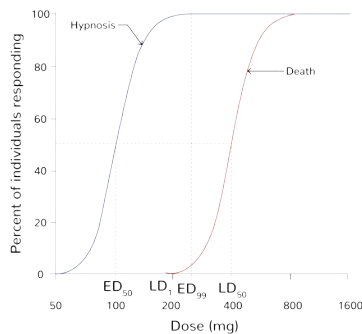
21. The loading dose of a drug is governed primarily by:

- A. Clearance
- B. Volume of distribution
- C. Plasma half-life
- D. Bioavailability

22. Which of the following routes of administration will not bypass hepatic first-pass metabolism?

- A. Sublingual
 - B. Oral
 - C. Intravenous
 - D. Transdermal
-

23. Please determine the therapeutic index of a medication based on the depicted dose-response curves.:



- A. 1
 - B. 2
 - C. 4
 - D. 8
-

24. Pharmacodynamics deals with:

- A. Mode of excretion of a drug
 - B. Mechanism of action of a drug
 - C. Transport of drug across the biological membranes
 - D. Latency of onset
-

25. The therapeutic index of a drug serves as a measure for:

- A. Potency
 - B. Safety
 - C. Efficacy
 - D. All of these
-

26. Which of the following is not a use of enteric-coated drugs?

- A. Protect the acid labile drug from gastric acidity
 - B. Protect the gastric mucosa from the irritant effects of certain drugs
 - C. Prolongs the half-life of the drug
 - D. Deliver drugs distal to the stomach
-

27. The normality of 2M HCL when compare to 1M H₂SO₄ is:

- A. Half
- B. Twice
- C. Same
- D. 4 times

28. Which statement is accurate regarding a patient receiving a high dosage of both penicillin and gentamicin?

- A. TDM is done for penicillin because of the narrow therapeutic index
 - B. TDM is done for gentamicin
 - C. TDM is done with both because of dangerous side effects
 - D. Both drugs do not need monitoring
-

29. Acetazolamide is given to a patient with glaucoma. It is a non-competitive inhibitor of the carbonic anhydrase enzyme. Which of the following would be the effect of this drug?

- A. Decrease in V_{max}
 - B. Decrease in K_m
 - C. Decrease in both K_m and V_{max}
 - D. No change in V_{max}
-

30. What does a low apparent volume of distribution of a drug signify?

- A. Low $t_{1/2}$
 - B. Low bioavailability
 - C. Low efficacy
 - D. Not extensively distributed to tissues
-

31. After which phase of the clinical trials is a new drug introduced into the market?

- A. Phase I
 - B. Phase II
 - C. Phase III
 - D. Phase IV
-

32. Aspirin and phenobarbitone are acidic drugs whereas diazepam is a basic drug, mark the false statement about these drugs:

- A. Aspirin is present in mainly non – ionized form in stomach, hence can be easily absorbed
 - B. Diazepam is mostly absorbed from intestine
 - C. Phenobarbitone can be absorbed from the stomach.
 - D. Diseases decreasing the transit time of drugs like diarrhea will increase the drug absorption in the small intestine
-

33. Match the following: Column A Column B 1. Omalizumab A. Anti-CD6 2. Itolizumab B. Anti-CD25 3. Daclizumab C. Anti-IgE 4. Belimumab D. Anti-IgG1 E. Anti-B lymphocyte stimulator

Column A	Column B
1. Omalizumab	A. Anti-CD6
2. Itolizumab	B. Anti-CD25
3. Daclizumab	C. Anti-IgE
4. Belimumab	D. Anti-IgG1
	E. Anti-B lymphocyte stimulator

- A. 1-C; 2-A; 3-B; 4-D

- B. 1-C; 2-D; 3-B; 4-E
 - C. 1-C; 2-A; 3-D; 4-B
 - D. 1-C; 2-A; 3-B; 4-E
-

34. Which of the following drug is not metabolized by acetylation?

- A. Isoniazid
 - B. Hydralazine
 - C. Phenytoin
 - D. Procainamide
-

35. Identify the false statement regarding suicide inhibition:

- A. The inhibitor can bind with any site resulting in suicidal inhibition
 - B. The binding of the enzyme to the substrate analogue is irreversible
 - C. They are enzyme specific and used in rational drug design
 - D. The inhibitor forms a product with the enzyme and the product inhibits it
-

36. A 65-year-old patient with hypertension has been prescribed Drug X, which has a bioavailability of 30%. Which of the following is the most appropriate statement for this patient's drug prescription?

- A. 70% of drug reach systemic circulation in unchanged form
- B. 30% of the drug is pure
- C. 70% of drug is potent in giving therapeutic effect
- D. 70% of the drug undergoes 1st pass metabolism in the liver

37. Which of the following drugs is absolutely contraindicated in pregnancy?

- A. Captopril
 - B. Paracetamol
 - C. Ranitidine
 - D. Cefixime
-

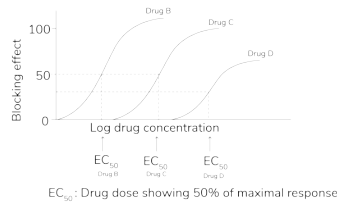
38. A patient was given an antipsychotic drug haloperidol and the patient developed acute dystonia. Which is the next best step?

- A. Give Benztropine
 - B. Change to clozapine
 - C. Give Fluphenazine
 - D. Increase dose of haloperidol
-

39. Which medications exhibit zero order kinetics among the options provided?

- A. Phenytoin
 - B. Epinephrine
 - C. Morphine
 - D. Propranolol
-

40. Identify the true statement about the given dose-response curve?



- A. Drug C is the most potent
- B. Drug B is the most efficacious
- C. Drug C is as efficacious as drug D
- D. Drug D is more potent than drug C

41. Which of the following drugs increases uric acid in urine?

- A. Colchicine
- B. Febuxostat
- C. Probenecid
- D. Allopurinol

42. Which medications should be excluded when treating postpartum hemorrhage in a pregnant woman who has a background of bronchial asthma during the third stage of labor?

- A. Carboprost
- B. Oxytocin
- C. Dinoprostone
- D. Methyl ergometrine

43. A patient given digoxin started having side effects like nausea and vomiting. The serum concentration of digoxin was 4 mg/dL. The plasma therapeutic range is below 1 mg/dL. If the half-life of digoxin is 40 hours, how long should one wait before resuming the treatment?

- A. 40 hours
 - B. 80 hours
 - C. 120 hours
 - D. 140-180 hours
-

44. Match the following drugs in Column A with their contraindications in Column B. Column A Column B A. Morphine B. Amiodarone C. Vigabatrin D. Estrogen preparations 1. QT prolongation 2. Thromboembolism 3. Pregnancy 4. Head injury

Column A	Column B
A. Morphine B. Amiodarone C. Vigabatrin D. Estrogen preparations	1. QT prolongation 2. Thromboembolism 3. Pregnancy 4. Head injury

- A. A-1, B-3, C-2, D-4
 - B. A-4, B-1, C-3, D-2
 - C. A-2, B-4, C-1, D-3
 - D. A-3, B-2, C-4, D-1
-

45. A young man sustained a scorpion sting 2 hours ago. On examination, his blood pressure is 168/100 mm Hg; HR, 72 beats / minute and respiratory rate, 32

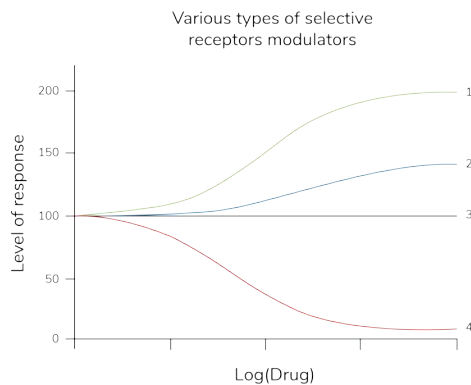
breaths/ minute. He also complains of breathlessness. How do you manage this patient?

- A. Metoprolol
- B. Ivabradine
- C. Prazosin
- D. Frusemide

46. Choose the correct options regarding the route of administration and bioavailability.: A. Intravenous = 1 B. $0.75 < \text{Oral} < 1$ C. $0.75 < \text{IM} \leq 1$ D. $0.75 < \text{SC} \leq 1$
IM - Intramuscular SC - Subcutaneous

- A. A, B, D
- B. A, C, D
- C. A and C
- D. A and D

47. Which of the following statement is correct regarding the given graph?



- A. Drug 1 represents agonist and drug 2 represents inverse agonist
- B. Drug 3 represents agonist and drug 4 represents inverse agonist

- C. Drug 1 represents agonist and drug 4 represent inverse agonist
 - D. Drug 2 represents partial agonist and drug 3 represents inverse agonist
-

48. Dorzolamide is a:

- A. Topically applied sulfonamide antibacterial
 - B. Topical ocular carbonic anhydrase inhibitor
 - C. Second – generation sulfonylurea
 - D. Opioid receptor agonist
-

49. Phospholipase C acts via which of the following G protein – coupled receptors?

- A. Gs
 - B. Gi
 - C. Gq
 - D. Go
-

50. Which of the following statements is correct regarding clinical trials?

- A. Phase 3 aim is to increase pharmacovigilance
 - B. Phase 4 is also called post marketing surveillance
 - C. Phase 3 is never double blinded
 - D. Phase 2 is done on healthy individuals
-

**51. Match the following teratogenic drug with their teratogenic effects.: Column A
Column B 1. Lithium 2. Chloramphenicol 3. Warfarin 4. Thalidomide a. Phocomelia
b. Depressed nasal bridge c. Gray baby syndrome d. Ebstein anomaly**

Table rendering failed

- A. 1-a, 2-b, 3-c, 4-d
 - B. 1-d, 2-c, 3-b, 4-a
 - C. 1-d, 2-a, 3-d, 4-c
 - D. 1-c, 2-d, 3-a, 4-b
-

52. In your role as a medical professional in India, how would you go about prescribing alprazolam to a patient?

- A. 1 mg OD before bedtime for 7 days
 - B. 1/2 mg HS for 7 days
 - C. 0.5 mg BD for 7 days
 - D. 0.5 mg daily before bedtime for 7 days
-

53. What formula is the dosing rate of a drug based on?

- A. Clearance/desired peak plasma concentration
 - B. Clearance of drug x desired peak plasma concentration
 - C. Clearance x loading dose
 - D. Clearance/ loading dose
-

54. Which drugs are subjected to therapeutic drug monitoring?

- A. Phenytoin
 - B. Heparin
 - C. Aspirin
 - D. Metformin
-

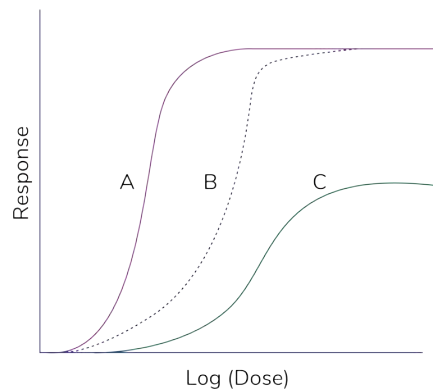
55. Adrenaline acting via beta 2 receptors cause bronchodilation whereas histamine acts through H1 receptors to cause bronchoconstriction. Which type of antagonism it is?

- A. Physiological antagonism
 - B. Chemical antagonism
 - C. Reversible antagonism
 - D. Competitive antagonism
-

56. All of the following can result in gynaecomastia except?

- A. Liver failure
 - B. Spironolactone
 - C. Digoxin
 - D. Aromatase inhibitors
-

57. The graph below shows the relationship between the log (dose) and the response of three drugs, A, B, and C. Which of these drugs has the highest potency?



- A. A
 - B. B
 - C. Both A and B
 - D. C
-

**58. Find therapeutic index of a drug from the information given below: LD 50: 400
ED 50: 100**

- A. 1
 - B. 2
 - C. 4
 - D. 8
-

59. True about non-competitive inhibition is:

- A. K_m remains same, V_{max} decreases
- B. K_m increases, V_{max} remains same
- C. K_m decreases, V_{max} increases
- D. K_m increases, V_{max} increases

60. Which of the following statements is incorrect regarding the selection of essential drugs?

- A. An adequate safety profile needs to be established
 - B. Cost to benefit has to be considered
 - C. Fixed drug combination is preferred over single drugs
 - D. Disease prevalence is considered
-

61. What is the half-life of a drug with an elimination rate constant of 0.05 per hour?

- A. 20.8 hours
 - B. 6.9 hours
 - C. 13.8 hours
 - D. 9.8 hours
-

62. Administration of which of the following drugs would increase the bioavailability of saquinavir?

- A. Ritonavir
 - B. Vitamin C
 - C. Cimetidine
 - D. Ganciclovir
-

63. If the dosage of a drug following first-order kinetics is doubled, how would it affect the half-life and plasma concentration?

- A. Half – life doubles and plasma concentration remains the same
 - B. Half – life remains the same and plasma concentration doubles
 - C. Half – life and plasma concentration remains the same
 - D. Half life and plasma concentration doubles
-

64. A patient is on phenytoin for a seizure disorder. He was prescribed sucralfate 4 times a day for peptic ulcers. What should be the minimum duration of time between consumption of these drugs?

- A. 30 minutes
 - B. 60 minutes
 - C. 90 minutes
 - D. 120 minutes
-

65. How does cyclosporine exert its mechanism of action?

- A. Inhibition of calcineurin
 - B. AMP kinase stimulation
 - C. IMP dehydrogenase inhibition
 - D. Dihydro-orotate dehydrogenase inhibition
-

66. Apparent volume of distribution of a drug is very high (6L/kg). Which of the following is true regarding the distribution of that drug?

- A. Highly bound to plasma proteins

- B. Confined to vascular compartment
 - C. Sequestered in body tissues
 - D. Both A and B
-

67. Which phase of a clinical trial is primarily focused on assessing the preliminary efficacy and determining the optimal dose of a drug?

- A. Phase 1
 - B. Phase 2
 - C. Phase 3
 - D. Phase 4
-

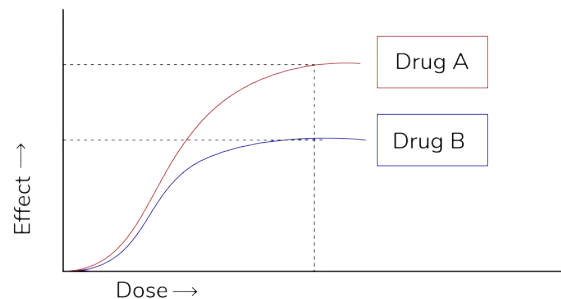
68. A dermatologist wants to prescribe acitretin to a patient with psoriasis. Which of the following statements is false regarding the drug?

- A. Contraindicated in persons with hyperlipidemia
 - B. It can cause increased sensitivity to sunlight
 - C. Female patients should avoid getting pregnant for 6 months after therapy
 - D. Acitretin has a half life of about 50 hours
-

69. When the pH = pKa of a drug, which of the following statements is true?

- A. Concentration of the drug is 50% ionic and 50% non-ionic
- B. Concentration of the drug is 90% ionic and 10% non-ionic
- C. Concentration of the drug is 75% ionic and 25% non-ionic
- D. Concentration of the drug is 25% ionic and 75% non-ionic

70. In a clinical trial evaluating the efficacy of a new drug, a normal efficacy potency graph is used. Which of the following statements regarding the graph is true?



- A. The graph shows the relationship between drug potency and its efficacy.
- B. The graph demonstrates that higher potency always leads to higher efficacy.
- C. The graph represents the dose-response curve for the drug.
- D. The graph indicates that drugs with low potency can never be effective.

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	3
Question 4	1
Question 5	2
Question 6	3
Question 7	1
Question 8	3
Question 9	2
Question 10	4
Question 11	1
Question 12	1
Question 13	2
Question 14	3
Question 15	3
Question 16	4
Question 17	1
Question 18	1
Question 19	1
Question 20	2
Question 21	2
Question 22	2

Question 23	3
Question 24	2
Question 25	2
Question 26	3
Question 27	3
Question 28	2
Question 29	1
Question 30	4
Question 31	3
Question 32	4
Question 33	4
Question 34	3
Question 35	1
Question 36	4
Question 37	1
Question 38	1
Question 39	1
Question 40	2
Question 41	3
Question 42	1
Question 43	2
Question 44	2
Question 45	3
Question 46	2
Question 47	3
Question 48	2

Question 49	3
Question 50	2
Question 51	2
Question 52	4
Question 53	2
Question 54	1
Question 55	1
Question 56	4
Question 57	1
Question 58	3
Question 59	1
Question 60	3
Question 61	3
Question 62	1
Question 63	2
Question 64	4
Question 65	1
Question 66	3
Question 67	2
Question 68	3
Question 69	1
Question 70	3

Solution for Question 1:

Correct Option C - Volume of distribution:

- The loading dose of a drug is calculated to rapidly achieve the desired plasma concentration of the drug. It is primarily dependent on the volume of distribution (Vd), which reflects the extent to which a drug is distributed in the body tissues relative to the plasma.

The formula for loading dose is:

$$\text{Loading Dose} = \frac{\text{Desired Plasma Concentration} \times \text{Volume of Distribution (Vd)}}{\text{Bioavailability (F)}}$$

- This equation shows that the loading dose is directly proportional to the volume of distribution.

Incorrect Options:

Option A - Clearance:

- Clearance is related to the maintenance dose rather than the loading dose. It represents the rate at which the drug is removed from the body and is important for determining the dosing rate needed to maintain a steady-state concentration, not for calculating the loading dose.

Option B - Route of administration:

- The route of administration can affect the bioavailability of a drug, but it does not directly determine the loading dose. The loading dose is based on the volume of distribution and desired plasma concentration, though the bioavailability factor must be considered if the drug is not administered intravenously.

Option D - Bioavailability:

- Bioavailability (F) does play a role in the calculation of the loading dose, particularly for non-intravenous routes where F is less than 1. However, the loading dose is fundamentally based on the volume of distribution, with bioavailability adjusting the dose for non-IV routes.

Solution for Question 2:

Correct Option B - Folic acid synthesis inhibitor:

- Dapsone inhibits dihydropteroate synthetase, disrupting dihydrofolic acid synthesis.
- Without folic acid production, microbial growth and survival are impaired.

Incorrect Options:

Option A - Cell wall synthesis inhibitor: Dapsone doesn't target cell wall synthesis, unlike beta-lactam antibiotics.

Option C - Protein synthesis inhibitor: Dapsone doesn't interfere with protein synthesis like macrolides or aminoglycosides.

Option D - Cell membrane inhibitor: Dapsone doesn't act on the cell membrane like polymyxins or polyenes.

Solution for Question 3:

Correct Option C - Hepatotoxicity:

- Liver damage or dysfunction is termed hepatotoxicity.
- Combining carbamazepine and valproate increases hepatotoxicity risk.
- Both drugs are metabolized in the liver, potentially producing toxic metabolites.
- Regular liver function monitoring is vital for prompt detection and management.

Incorrect Options:

Option A - Pancreatitis:

- Valproate use is associated with rare pancreatitis occurrences.
- Elevated liver enzymes suggest liver-related issues rather than pancreatitis.
- The addition of carbamazepine does not significantly increase pancreatitis risk.

Option B - Thrombocytopenia:

- Carbamazepine can rarely cause thrombocytopenia.
- Valproate typically does not lead to thrombocytopenia.
- Combined use doesn't substantially elevate thrombocytopenia risk.

Option D - Hyperammonemia:

- Valproate carries a higher risk of hyperammonemia, especially in metabolic disorders.
 - Carbamazepine usage is not typically linked to hyperammonemia.
 - The combination does not significantly increase hyperammonemia risk over valproate alone.
-

Solution for Question 4:

Correct Option A - Drug B is the most potent:

- Drug potency refers to the amount of drug required to produce a specific effect.
- Drug B is the most potent because it achieves the therapeutic effect at a lower concentration or dose compared to other drugs.
- Potency and efficacy are independent characteristics of drugs.

Incorrect Options:

Option B - Drug B is the most efficacious: Drug B is not the most efficacious as its maximum response is lower than others.

Option C - Drug C is more efficacious than drug D: Drug D is more efficacious than Drug C due to its higher maximum response.

Option D - Drug A is more potent than drug B: Drug B is more potent than Drug A, while Drug A is more efficacious.

Solution for Question 5:

Correct Option B - 100%:

- Bioavailability of 100% indicates that the entire administered dose reaches systemic circulation unchanged after intravenous administration, bypassing the absorption process.

Incorrect Options:

Option A, C & D are incorrect.

Solution for Question 6:

Correct Option C - Primidone:

- Primidone is not a prodrug.
- It's an antiepileptic medication converted into active metabolites, phenobarbital, and phenylethylmalonamide (PEMA), within the body.
- These metabolites are responsible for its antiepileptic effects.

Incorrect Options:

Option A - Proguanil: Proguanil is a prodrug converted to cycloguanil, its active metabolite, for antimalarial activity.

Option B - Sulindac: Sulindac is a prodrug converted to its active form, sulindac sulfide, providing anti-inflammatory and analgesic effects.

Option D - Lisdexamfetamine: Lisdexamfetamine, a prodrug, converts into dextroamphetamine, a CNS stimulant used for ADHD and binge eating disorder.

Solution for Question 7:

Correct Option A - Potassium channel blocker:

- Potassium channel blockers inhibit potassium efflux, prolonging action potential duration.
- This increases calcium influx, enhancing acetylcholine (ACh) release.

Incorrect Options:

Option B, C & D: Sodium channel blocker, Calcium channel blocker & Chlorine channel blocker:

- Sodium and Chlorine channel blockers, don't directly affect ACh release.
- Calcium channel blockers reduce calcium influx, potentially decreasing ACh release.

Solution for Question 8:

Correct Option C - 2-8 degrees Celsius:

- 'Cool' storage typically refers to a temperature range of 2-8 degrees Celsius (36-46 degrees Fahrenheit).
- This range is recommended for medications requiring refrigeration to maintain stability and effectiveness.
- Storing medications within this range prevents degradation, maintains potency, and ensures safety.

Incorrect Options:

Option A: -2 degrees Celsius: Too cold, risking freezing and medication damage.

Option B: 0 degrees Celsius: Freezing point of water, still too cold for most medications.

Option D: 8-15 degrees Celsius: Considered "cool" but might be warmer than ideal, closer to room temperature.

Solution for Question 9:

Correct Option B - Imipenem:

- Cilastatin inhibits renal enzyme dehydropeptidase I, preventing breakdown of imipenem in renal tubules.
- This inhibition increases imipenem's half-life and plasma levels, enhancing its effectiveness.

Incorrect Options:

Option A - Meropenem: Meropenem is similar to imipenem but not typically co-administered with cilastatin.

Option C - Cefazolin: Cefazolin is unrelated to imipenem and not affected by cilastatin.

Option D - Piperacillin: Piperacillin's action is not extended by cilastatin administration.

Solution for Question 10:

Correct Option D - AUC (Area under the curve):

- AUC measures total drug exposure in the bloodstream over time.
- It integrates plasma concentration-time curve, indicating both rate and extent of drug absorption.
- Larger AUC suggests greater drug exposure, providing comprehensive pharmacokinetic assessment.

Incorrect Options:

Option A - T_{max} (Time to reach maximum concentration): Time taken to reach C_{max} after drug administration, reflecting absorption rate.

Option B - C_{max} (Maximum concentration): Peak concentration achieved in bloodstream after administration, influencing efficacy and side effects.

Option C - T_{1/2} (Half-life): Time taken for drug concentration to decrease by half, reflecting elimination rate and dosing interval determination.

Solution for Question 11:

Correct Option A - 50% animal mortality is an experiment after exposure to the drug:

- LD₅₀ stands for "lethal dose 50," indicating the dose or concentration of a substance lethal to 50% of the exposed population in animal testing. It measures acute toxicity.

Incorrect Options:

Option B - 50% reduction of infection in experimental animals: Unrelated to LD₅₀.

Option C - Dose causing adverse effects in 50% of the population: Refers to TD₅₀, not LD₅₀.

Option D - Range where efficacy is high and adverse effects are low: Pertains to therapeutic window, not LD₅₀.

Solution for Question 12:

Correct Option A - Muscarinic receptors:

- Muscarinic receptors are a type of acetylcholine receptor that are GPCRs.
- They mediate parasympathetic nervous system effects and are involved in various physiological functions.
- They regulate heart rate, smooth muscle contraction, and glandular secretion.

Incorrect Options:

Option B - Nicotinic: Nicotinic receptors are ligand-gated ion channels, not GPCRs. They are also acetylcholine receptors but function differently by allowing ion flow across the membrane.

Option C - GABA-A: GABA-A receptors are ligand-gated ion channels that mediate the effects of the neurotransmitter GABA by allowing chloride ions to flow into the neuron, leading to hyperpolarization.

Option D - Insulin: The insulin receptor is a receptor tyrosine kinase, not a GPCR. It is involved in the regulation of glucose metabolism.

Solution for Question 13:

Correct Option B - Area under the curve:

- Area under the curve (AUC) measures total drug absorption over time, aiding in comparing drug formulations or different drugs.

Incorrect Options:

Option A - T_{max}: T_{max} is time to reach maximum concentration (C_{max}).

Option C - C_{max}: C_{max} is peak concentration after administration.

Option D - t_{1/2}: t_{1/2} is drug elimination half-life.

Solution for Question 14:

Correct Option is C: Ketamine: Ketamine is classified as a Schedule X drug in some countries. Schedule X is a category used to regulate drugs with a high potential for abuse and dependence. These drugs are subject to stricter controls and regulations due to their potential risks and misuse.

Incorrect Options:

Option A: Halothane: Halothane is a volatile inhalation anaesthetic and is not classified as a Schedule X drug. It is commonly used for general anaesthesia in surgical procedures.

Option B: Thalidomide: Thalidomide is not classified as a Schedule X drug. It is a medication with various uses, including treating certain cancers and skin conditions, but it is also known for its historical association with birth defects when used during pregnancy.

Option D: Colistin: Colistin is an antibiotic primarily used for the treatment of multidrug-resistant bacterial infections. It is not classified as a Schedule X drug.

Solution for Question 15:

Correct Option C - 3 and 4:

- Suicide inhibition is a type of enzyme inhibition in which the inhibitor forms a covalent bond with the enzyme, essentially inactivating it irreversibly.

Suicide inhibition is a type of enzyme inhibition in which the inhibitor forms a covalent bond with the enzyme, essentially inactivating it irreversibly.

1) Spironolactone: Drugs such as spironolactone, eplerenone, and finerenone competitively inhibit the binding of aldosterone to the mineralocorticoid receptors. 2) Ethinyl estradiol: Ethinyl estradiol is a synthetic estrogen used in contraceptive pills. It does not act by suicide inhibition. 3) Benzbromarone: Benzbromarone is a uricosuric agent used to treat gout. It inhibits xanthine oxidase irreversibly, making it a suicide inhibitor. 4) Ritonavir: Ritonavir is a protease inhibitor used to treat HIV/AIDS. It inhibits the HIV protease enzyme irreversibly, making it a suicide inhibitor.

Incorrect Options:

- Options A, B, and D are incorrect.

Options A, B, and D are incorrect.

Solution for Question 16:

Correct Option D - Xanthine oxidase:

- Xanthine oxidase is involved in purine metabolism, converting purines to uric acid in the liver. Inhibiting this enzyme decreases uric acid levels, making it a target for gout and hyperuricemia treatment with drugs like allopurinol and febuxostat.

Incorrect Options:

Option A - Urease: Urease breaks down urea, unrelated to uric acid metabolism.

Option B - Homogentisate oxidase: Homogentisate oxidase relates to phenylalanine and tyrosine metabolism, not uric acid.

Option C - Lysyl oxidase: Lysyl oxidase is involved in collagen and elastin cross-linking, unrelated to uric acid metabolism.

Solution for Question 17:

Correct Option A - Safety:

- Safety refers to the extent to which a drug can be used without causing harm or adverse effects to the patient. Assessing safety involves evaluating the potential risks and side effects associated with the use of a drug and determining the appropriate dose and duration of treatment to minimize any potential harm.

Incorrect Options:

Option B - Efficacy: Efficacy refers to the ability of a drug to produce the desired therapeutic effect or clinical benefit in treating a specific condition. It measures how well a drug works in achieving its intended purpose, such as relieving symptoms, improving health outcomes, or

curing a disease.

Option C - Patency: Patency, in the context of drugs, usually refers to the openness or unobstructed flow of blood vessels or other bodily passages. It may not be directly related to the assessment of drug effects or safety.

Option D - Toxicity: Toxicity refers to the potential of a drug to cause harmful or adverse effects on the body. It assesses the degree or extent to which a drug can be toxic or poisonous and cause damage to organs, tissues, or physiological processes. Evaluating toxicity helps determine the safe dosage range and monitor for any signs or symptoms of drug-induced harm. While therapeutic index can give an idea of increased risk of toxicity, option A is the better option.

Solution for Question 18:

Correct Option A - Phase 1:

- Seeking approval for clinical studies with healthy volunteers refers to Phase 1 of vaccine development, which involves initial safety and dosage evaluations.

Incorrect Options:

Option B - Phase 2: Phase 2 involves expanded safety studies and further evaluation of the vaccine in a larger group of people including patients.

Option C - Phase 3: Phase 3 involves large-scale efficacy studies in a broader population and comparing effect in patients when compared to current standard of treatment/placebo.

Option D - Phase 4: Phase 4 refers to post-marketing surveillance and monitoring of the vaccine after it has been approved and released to the market.

Solution for Question 19:

Correct Option A - Synthesis of monoclonal antibodies:

- Monoclonal antibodies are antibodies that are produced by identical immune cells derived from a single parent cell. They can be synthesized using hybridoma technology, which involves fusing specific antibody-producing cells with immortalized cells to create hybrid cell lines capable of producing a large quantity of identical antibodies. This technique is widely used in medical research, diagnostics, and therapeutic applications.

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Incorrect Options:

Option B - Preparing cell lines for viral culture: Viral culture involves growing and replicating viruses in laboratory settings. Cell lines derived from various organisms, including human or animal cells, are commonly used as host cells for viral replication. These cell lines are prepared and maintained to support viral growth and study viral behavior.

Option C - Synthesis of vaccines: Vaccines are created by different methods depending on the type of vaccine. Common techniques for vaccine production include using weakened or inactivated viruses, recombinant DNA technology, or subunit vaccines. These processes involve the synthesis or manipulation of specific antigens or genetic material to stimulate an immune response and provide protection against specific diseases.

Option D - Process of genetic engineering: Genetic engineering involves the manipulation and modification of an organism's genetic material to introduce new traits or characteristics. Techniques such as gene cloning, DNA sequencing, and gene editing (e.g., CRISPR-Cas9) are used in genetic engineering to alter the genetic code of organisms, including plants, animals, and microorganisms.

Solution for Question 20:

Correct Option B - Carbamazepine:

- Carbamazepine can cause a decrease in the level of Warfarin. This is due to it inhibiting CYP2C9/10 enzyme which is involved in warfarin metabolism. This would reduce the bleeding risk with warfarin.

Carbamazepine can cause a decrease in the level of Warfarin. This is due to it inhibiting CYP2C9/10 enzyme which is involved in warfarin metabolism. This would reduce the bleeding risk with warfarin.

Incorrect Options:

Option A - Cimetidine: Cimetidine is an H₂ receptor antagonist commonly used to reduce stomach acid production. It has the potential to interact with warfarin, leading to increased levels of warfarin in the blood and an increased risk of bleeding. Therefore, caution is advised, and close monitoring of the patient's international normalized ratio (INR) is necessary when cimetidine and warfarin are used together.

Option C - Isoniazid: Isoniazid reduces metabolism of R-enantiomer of warfarin. This is by inhibition of CYP3A4. Reduced metabolism results in higher levels of warfarin and higher bleeding risk. This requires close INR monitoring if both drugs need to be given simultaneously

Option D - Amiodarone: Amiodarone is an antiarrhythmic medication used to treat certain heart rhythm disorders. It can interact with warfarin and increase the risk of bleeding. Amiodarone can inhibit enzymes involved in warfarin metabolism, leading to increased levels of warfarin in the blood. Frequent monitoring of the INR is crucial when amiodarone and warfarin are used concomitantly.

Solution for Question 21:

Correct Option B - Volume of distribution:

- The volume of distribution (V_d) is a pharmacokinetic parameter that represents the apparent space in the body available to contain the drug. It is a theoretical volume that indicates how extensively a drug is distributed throughout the body relative to its concentration in the plasma. A higher volume of distribution suggests that the drug is extensively distributed in the tissues, while a lower volume of distribution indicates more confinement to the plasma. The volume of distribution influences the loading dose of a drug, as it helps determine the initial dose required to achieve a desired concentration in the body.

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distribution influences the loading dose of a drug, as it helps determine the initial dose required to achieve a desired concentration in the body.

Incorrect Options:

Option A - Clearance: Clearance refers to the rate at which a drug is eliminated from the body. It is a pharmacokinetic parameter that measures how efficiently the body removes a drug from the systemic circulation. Clearance is influenced by various factors, including liver function, kidney function, and metabolic processes. A higher clearance indicates faster elimination of the drug from the body, while a lower clearance means slower elimination. Clearance plays a role in determining the dosing frequency and adjustment of a drug.

Option C - Plasma half-life: The plasma half-life is the time it takes for the concentration of a drug in the plasma to decrease by half. It is a pharmacokinetic parameter that reflects the rate of elimination of the drug from the systemic circulation. The half-life is influenced by factors such as clearance, metabolism, and excretion. A longer half-life means the drug remains in the body for a longer duration, while a shorter half-life indicates faster elimination. The plasma half-life is used to determine the dosing interval for a drug, as it indicates how frequently the drug needs to be administered to maintain a therapeutic concentration.

Option D - Bioavailability: Bioavailability is the fraction or percentage of an administered drug that reaches systemic circulation in an unchanged form. It represents the extent and rate at which the drug is absorbed and becomes available at the site of action. Bioavailability is influenced by various factors, including the route of administration, drug formulation, and first-pass metabolism. A high bioavailability indicates efficient absorption and availability of the drug, while a low bioavailability means that a significant portion of the drug is lost or metabolized before reaching the systemic circulation. Bioavailability affects the dose adjustment and selection of the appropriate route of administration for a drug.

Solution for Question 22:

Correct Option B - Oral:

- Oral administration refers to taking a drug by mouth, where it is swallowed and absorbed through the gastrointestinal tract. After absorption, the drug enters the portal vein and is transported to the liver before reaching the systemic circulation. In the liver, the drug may undergo extensive metabolism by enzymes before it reaches the systemic circulation. This process is known as hepatic first-pass metabolism. Therefore, oral administration can result

in significant metabolism of the drug, leading to lower bioavailability compared to other routes that bypass hepatic first-pass metabolism.

Oral administration refers to taking a drug by mouth, where it is swallowed and absorbed through the gastrointestinal tract. After absorption, the drug enters the portal vein and is transported to the liver before reaching the systemic circulation. In the liver, the drug may undergo extensive metabolism by enzymes before it reaches the systemic circulation. This process is known as hepatic first-pass metabolism. Therefore, oral administration can result in significant metabolism of the drug, leading to lower bioavailability compared to other routes that bypass hepatic first-pass metabolism.

Incorrect Options:

Option A - Sublingual: Sublingual administration involves placing a drug under the tongue, where it is absorbed directly into the bloodstream through the sublingual mucosa. This route allows the drug to bypass the gastrointestinal tract and liver to some extent, leading to faster onset of action and higher bioavailability compared to oral administration. There can still be some metabolism of the drug by the liver during its passage through the systemic circulation but this is not termed as first pass metabolism

Option C - Intravenous: Intravenous administration involves directly injecting a drug into a vein, allowing it to enter the systemic circulation immediately without passing through the gastrointestinal tract or liver. Since the drug bypasses hepatic first-pass metabolism, it reaches the systemic circulation in its active form with maximum bioavailability. IV administration provides rapid and precise drug delivery, making it suitable for drugs that require immediate effects or when precise control over drug levels is necessary.

Option D - Transdermal: Transdermal administration involves applying a drug topically to the skin, from which it is absorbed into the systemic circulation. The drug passes through the skin layers and enters the systemic bloodstream. As it doesn't pass through the liver before entering systemic circulation, it is considered to bypass first pass metabolism by the liver

Solution for Question 23:

Correct Option C - 4:

- This option is correct because the calculated TI is indeed 4.

- To calculate the therapeutic index (TI) of a drug, you need to divide the median lethal dose (LD50) by the median effective dose (ED50).
- Given that LD50 is 400 and ED50 is 100, we can calculate the TI as follows:
- $TI = LD50 / ED50 = 400 / 100 = 4$

This option is correct because the calculated TI is indeed 4.

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Given that LD50 is 400 and ED50 is 100, we can calculate the TI as follows:

$$TI = LD50 / ED50 = 400 / 100 = 4$$

Incorrect Options:

Option A - 1: The calculated TI is 4, not 1.

Option B - 2: the calculated TI is 4, not 2.

Option D - 8: The calculated TI is 4, not 8.

Solution for Question 24:

Correct Option B - Mechanism of action of a drug:

- This is the primary focus of pharmacodynamics. It refers to the specific biochemical and physiological interactions between a drug and its target molecules in the body. Understanding the mechanism of action helps in elucidating how a drug produces its therapeutic effects or exerts its influence on biological systems. Pharmacokinetics: What the body does to the drug(Absorption, metabolism, excretion etc) Pharmacodynamics: What the drug does to the body (mechanism of action, effects, change to receptors etc)
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Incorrect Options:

Option A - Mode of excretion of a drug: This refers to the process by which a drug is eliminated from the body. It involves the removal of drugs or their metabolites from the bloodstream through various routes, such as renal excretion (via urine), hepatic excretion (via bile), or pulmonary excretion (via breath). This falls under pharmacokinetics

Option C - Transport of drug across the biological membranes: This aspect is more closely related to pharmacokinetics, which deals with the absorption, distribution, metabolism, and excretion of drugs (often referred to as ADME).

Option D - Latency of onset: The latency of onset refers to the time it takes for a drug to produce its pharmacological effect after administration. This can vary depending on various factors such as the drug's formulation, route of administration, absorption rate, and distribution throughout the body. While pharmacodynamics may indirectly influence the onset of drug action, the specific timing is not the primary focus of this discipline.

Solution for Question 25:

Correct Option B - Safety:

- The therapeutic index of a drug is an indicator of its safety. The therapeutic index is defined as the ratio of Lethal/Toxic dose 50 to Effective Dose 50. It reflects the margin of safety for a drug. A higher therapeutic index indicates a wider margin of safety, as the effective dose is significantly lower than the toxic dose.

Incorrect Options:

Option A - Potency: Potency refers to the measure of how much of a drug is needed to produce a specific effect. It is related to the drug's affinity for its target and its ability to activate or inhibit the target at a given concentration. Potency does not directly indicate the safety or efficacy of a

drug but rather focuses on the dose-response relationship.

Option C - Efficacy: Efficacy refers to the ability of a drug to produce the desired therapeutic effect. It measures the extent to which a drug can produce a maximum response irrespective of dose. Efficacy is not directly related to the therapeutic index but rather focuses on the effectiveness of a drug in achieving the desired clinical outcome.

Solution for Question 26:

Correct Option C - Prolongs the half-life of the drug:

- Enteric coating does not have a direct impact on the half-life of a drug. The half-life of a drug is determined by its pharmacokinetics, specifically its rate of elimination from the body. Enteric coating primarily affects drug release and absorption in the gastrointestinal tract, but it does not alter the drug's elimination rate or prolong its half-life.

Incorrect Options:

Option A - Protecting the acid-labile drug from gastric acidity: Some drugs are sensitive to the low pH of the stomach, which can degrade their effectiveness. Enteric coating provides a protective barrier that prevents the drug from being exposed to gastric acid, thus maintaining its stability until it reaches the intestine, where the pH is higher.

Option B - Protecting the gastric mucosa from irritant effects of certain drugs: Certain drugs, such as non-steroidal anti-inflammatory drugs (NSAIDs), can cause irritation and damage to the gastric mucosa, leading to adverse effects like gastric ulcers. Enteric coating helps prevent direct contact of the drug with the gastric mucosa, reducing the risk of irritation and protecting the stomach lining.

Option D - Delivering drugs distal to the stomach: Enteric-coated drugs are designed to resist dissolution in the stomach and instead release the active ingredient in the intestine. This can be useful when the drug needs to be absorbed or exert its effects specifically in the intestines, bypassing the stomach.

Solution for Question 27:

Correct Option C - Same:

- To compare the normality of 2M HCl to 1M H₂SO₄, consider their acid properties: 2M HCl: Since it's monoprotic, its normality is equal to its molarity, so it's 2N. 1M H₂SO₄: Being diprotic, its normality is twice its molarity, making it 2N as well.
- 2M HCl: Since it's monoprotic, its normality is equal to its molarity, so it's 2N.
- 1M H₂SO₄: Being diprotic, its normality is twice its molarity, making it 2N as well.
- Thus, both solutions have the same normality, which is 2N.
- 2M HCl: Since it's monoprotic, its normality is equal to its molarity, so it's 2N.
- 1M H₂SO₄: Being diprotic, its normality is twice its molarity, making it 2N as well.

Incorrect Options:

- Option A, B and D are incorrect. Refer to the explanation for the correct answer.
-

Solution for Question 28:

Correct Option B - TDM is done for gentamicin:

- TDM (Therapeutic Drug Monitoring) is done with gentamicin for a patient on a high dose of penicillin and gentamicin.
- Therapeutic Drug Monitoring (TDM) involves measuring drug concentrations in the blood to optimize dosing and ensure therapeutic efficacy while minimizing the risk of toxicity. It is particularly important for drugs with a narrow therapeutic index, where small changes in drug concentration can significantly affect the therapeutic response.
- In the given scenario, penicillin is not typically monitored through TDM because it has a wide therapeutic index, meaning that the difference between the effective and toxic doses is relatively large. Therefore, monitoring penicillin levels is not necessary for most patients.
- On the other hand, gentamicin, an aminoglycoside antibiotic, has a narrow therapeutic index, and its levels should be monitored through TDM. Aminoglycosides, including gentamicin, can cause dose-related nephrotoxicity and ototoxicity. By monitoring gentamicin levels, healthcare providers can adjust the dosage to maintain therapeutic concentrations within a safe range and reduce the risk of toxicity.

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Incorrect Options:

Option A - TDM is done with penicillin because of a narrow therapeutic index: As mentioned earlier, penicillin has a wide therapeutic index, so routine TDM is not necessary for penicillin.

Option C - TDM is done with both because of dangerous side effects: While gentamicin has the potential for dangerous side effects, routine TDM is not done for penicillin due to its wide therapeutic index. Therefore, TDM is primarily performed for gentamicin in this scenario.

Option D - Both drugs do not need monitoring: As explained, gentamicin, due to its narrow therapeutic index and potential for toxicity, should be monitored through TDM. Penicillin, however, does not typically require routine TDM due to its wide therapeutic index.

Solution for Question 29:

Correct Option A - Decrease in V_{max} :

- Acetazolamide inhibits carbonic anhydrase, reducing its activity and leading to a decreased V_{max} . This means the enzyme's maximum rate of catalyzing the reaction is lowered.

Incorrect Options:

Option B - Decrease in K_m : Diamox, as a non-competitive inhibitor, does not directly affect K_m . K_m remains unchanged because substrate-binding affinity is unaffected.

Option C - Decrease in both K_m and V_{max} : Diamox only decreases V_{max} , not K_m .

Option D - No change in V_{max} : Diamox, as a non-competitive inhibitor, decreases V_{max} .

Solution for Question 30:

Correct Option D - Not extensively distributed to tissues:

- A low apparent volume of distribution indicates that the drug is primarily confined to the plasma or blood and is not extensively distributed into the tissues.
- This could be due to limited penetration into tissues or high plasma protein binding.

Incorrect Options:

Option A - Low $t_{1/2}$ (half-life): Half-life is the time taken for the drug concentration to decrease by half and is not directly related to volume of distribution.

Option B - Low bioavailability: Bioavailability is the fraction of a drug reaching systemic circulation after administration and is not directly linked to volume of distribution.

Option C - Low efficacy: Efficacy refers to a drug's ability to produce the desired effect and is not determined by volume of distribution

Solution for Question 31:

Correct Option C - Phase III:

- Phase III is also called a therapeutic confirmatory trial, after which the drug is released into the market.

Incorrect Options:

Option A - Phase I: Phase I is the first stage of clinical trials, where the safety of the drug is assessed.

Option B - Phase II: Phase II is the second phase, the therapeutic exploratory phase, efficacy of the drug is established.

Option D - Phase IV: Phase IV is the post-marketing surveillance phase after the drug is in the market.

Solution for Question 32:

Correct Option D - Diseases decreasing the transit time of drugs like diarrhoea will increase the drug absorption in the small intestine.:

- The statement that diseases decreasing the transit time of drugs like diarrhoea will increase drug absorption in the small intestine is incorrect. Diarrhea, which is characterized by increased intestinal motility and decreased transit time, can actually decrease drug absorption as the drug has less time to be absorbed in the intestine.

Incorrect Options:

Option A - Aspirin is present in mainly non-ionized form in the stomach, which allows it to be easily absorbed: Non-ionized forms of drugs are generally more lipid-soluble and can readily cross biological membranes.

Option B - Diazepam is mostly absorbed from the intestine: Diazepam is a basic drug and is well-absorbed from the gastrointestinal tract, primarily in the small intestine.

Option C - Phenobarbital can be absorbed from the stomach. Phenobarbital is an acidic drug. Therefore, its absorption is primarily facilitated in the stomach and upper small intestine, where the pH is relatively lower.

Solution for Question 33:

Correct Option D - 1-C; 2-A; 3-B; 4-E:

Column A	Column B
1. Omalizumab	C. Anti-IgE
2. Itolizumab	A. Anti-CD6

3. Daclizumab	B. Anti-CD25
4. Belimumab	E. Anti-B lymphocyte stimulator

Incorrect Options:

- Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 34:

Correct Option C - Phenytoin:

- Phenytoin is primarily metabolized by hepatic cytochrome P450 enzymes, particularly CYP2C9 and CYP2C19.
- The main metabolic pathway involves the conversion of phenytoin to its inactive metabolite hydroxyphenytoin (p-HPPH), which accounts for up to 90% of its metabolism.
- p-HPPH undergoes further glucuronidation before excretion in the urine.

Incorrect Options:

Option A - Isoniazid: Metabolized by acetylation in the liver.

Option B - Hydralazine: Metabolized by N-acetylation in the liver.

Option D - Procainamide: Metabolized by acetylation in the liver.

Solution for Question 35:

Correct Option A - The inhibitor can bind with any site resulting in suicidal inhibition:

- Suicide inhibition involves irreversible enzyme inhibition.
- Inhibitor binds covalently to enzyme's active site, forming stable complex.
- Specific binding to active site allows for irreversible inhibition.
- Inhibitor mimics enzyme's substrate structure, interacting with active site residues.

- Results in covalent bond formation through catalytic activity of enzyme.

Incorrect Options:

Option B - The binding of the enzyme to the substrate analogue is irreversible: Suicide inhibitors form irreversible binding with enzyme's active site, unlike reversible inhibitors.

Option C - They are enzyme specific and used in rational drug design: Suicide inhibitors are enzyme specific and used in rational drug design.

Option D - The inhibitor forms a product with the enzyme and the product inhibits it: In suicide inhibition, inhibitor forms product that further inhibits enzyme's activity.

Solution for Question 36:

Correct Option D - 70% of the drug undergoes 1st pass metabolism in the liver:

- Bioavailability refers to the fraction of an administered dose of unchanged drug that reaches the systemic circulation. If a drug has a bioavailability of 30%, this means that 70% of the drug is metabolized and inactivated during its first pass through the liver before it reaches systemic circulation.

Incorrect Options:

Option A. 70% of drug reach systemic circulation in unchanged form

- Incorrect: This statement is the opposite of the actual situation. Only 30% of the drug reaches the systemic circulation in unchanged form, not 70%.

Option B. 30% of the drug is pure

- Incorrect: Bioavailability is not a measure of the purity of the drug. It indicates how much of the drug reaches the systemic circulation in an active form.

Option C. 70% of drug is potent in giving therapeutic effect

- Incorrect: This statement misinterprets bioavailability. Potency refers to the drug's ability to produce a therapeutic effect at a given dose, not the percentage that reaches systemic circulation.
-

Solution for Question 37:

Correct Option A - Captopril:

- Captopril is absolutely contraindicated in pregnancy as it causes fetal growth retardation, hypoplasia of organs and fetal death especially if given in the later half of the pregnancy.
- There is a 2.7 fold higher malformation rate in fetuses when exposed to ACE inhibitors in the first trimester and hence ACE inhibitors are stopped when the woman conceives.
- Some other drugs contraindicated in pregnancy: Thalidomide Phenytoin Warfarin Lithium Carbamazepine Tetracyclines
- Thalidomide
- Phenytoin
- Warfarin
- Lithium
- Carbamazepine
- Tetracyclines
- Thalidomide
- Phenytoin
- Warfarin
- Lithium
- Carbamazepine
- Tetracyclines

Incorrect Options:

Option B - Paracetamol: Paracetamol is one of the safest drugs in pregnancy.

Option C - Ranitidine: Ranitidine is an H₂ blocker used in peptic ulcer disease and safe in pregnancy.

Option D - Cefixime: Cefixime is cephalosporin which is safe in pregnancy.

Solution for Question 38:

Correct Option A - Give Benztropine:

- Acute muscle dystonia is the 1st extrapyramidal symptom caused by antipsychotics and is characterized by torticollis and uprolling of the eyes.
- Benztropine is an anticholinergic drug that is used to treat extrapyramidal symptoms.
- Other drugs used in treating EPS: Promethazine Diphenhydramine Benzhexol
- Promethazine
- Diphenhydramine
- Benzhexol
- Promethazine
- Diphenhydramine
- Benzhexol

Incorrect Options:

Option B - Change to clozapine:

- Clozapine can be given once the extrapyramidal symptoms are treated as it causes the least extrapyramidal symptoms.

Option C - Give Fluphenazine:

- Fluphenazine is a typical anti-psychotic that can worsen extrapyramidal symptoms.

Option D - Increase dose of haloperidol:

- Increasing the dose of haloperidol can also worsen extrapyramidal symptoms.

Solution for Question 39:

Correct Option A - Phenytoin:

- Phenytoin exhibits zero-order kinetics, meaning a constant amount of drug is eliminated per unit of time, irrespective of its concentration.
- This occurs due to saturation of the enzymes responsible for its metabolism.

Incorrect Options:

Option B, C & D: Epinephrine, Morphine & Propranolol :

- Follows first-order kinetics, where elimination is proportional to its concentration.

Solution for Question 40:

Correct option B - Drug B is the most efficacious:

- Drug B is the most efficacious because it exhibits the highest plateau on the dose-response curve.

Incorrect Options:

Option A: Drug C is the most potent - Potency determined by EC50 value, inversely proportional, so Drug C is not the most potent.

Option C: Drug C is as efficacious as Drug D - EC50 value affects potency, not efficacy, so Drug C is not equally efficacious as Drug D.

Option D: Drug D is more potent than Drug C - Potency determined by EC50 value, so Drug D is not necessarily more potent than Drug C.

Solution for Question 41:

Correct Option C - Probenecid:

- Probenecid is a uricosuric agent that increases the excretion of uric acid in urine by blocking the reabsorption of uric acid in the renal tubules.
- This is how it lowers serum uric acid levels

Probenecid is a uricosuric agent that increases the excretion of uric acid in urine by blocking the reabsorption of uric acid in the renal tubules.

This is how it lowers serum uric acid levels

Incorrect Options:

Option A - Colchicine: Colchicine is used to treat gout by reducing inflammation and pain. It doesn't increase uric acid in the urine.

Option B - Febuxostat: Febuxostat is a xanthine oxidase inhibitor that decreases the production of uric acid.

Option D - Allopurinol: Allopurinol is also a xanthine oxidase inhibitor that decreases the production of uric acid.

Solution for Question 42:

Correct Option A - Carboprost:

- Carboprost tromethamine, a synthetic prostaglandin, is contraindicated in pregnant women with bronchial asthma history due to its strong uterine contractions exacerbating asthma symptoms.

Incorrect Options:

Option B - Oxytocin: Oxytocin, a hormone, is safe for PPH management without significant respiratory effects.

Option C - Dinoprostone: Dinoprostone is not commonly used for PPH.

Option D - Methyl ergometrine: Methyl ergometrine, an ergot alkaloid, is safe for PPH management despite contraindications in hypertension/preeclampsia.

Solution for Question 43:

Correct Option B - 80 hours:

- In the scenario, with a digoxin serum concentration of 4 mg/dL, significantly above the therapeutic range, the patient should wait at least 80 hours before resuming treatment, considering digoxin's half-life of 40 hours.

Incorrect Options:

Option A - 40 hours: 40 hours is the half-life, but it takes longer for the serum concentration to decrease to a safe level.

Option C - 120 hours: 120 hours may not suffice for serum concentration to decrease adequately.

Option D - 140-180 hours: Range of 140-180 hours lacks scientific basis and explanation.

Solution for Question 44:

Correct Option B - A-4, B-1, C-3, D-2:

- Morphine: Contraindicated in head injury due to increased intracranial pressure.
- Amiodarone: Prolongs QT interval, potentially leading to Torsades de pointes.
- Vigabatrin: Contraindicated in pregnancy, increases risk of fetal malformations.
- Estrogen preparations: Increase risk of thromboembolism.

Incorrect Options:

- Option A, C & D are incorrect. Refer to the explanation for the correct answer.
-

Solution for Question 45:

Correct Option C - Prazosin:

- Prazosin, an alpha-blocker, dilates blood vessels and stabilizes blood pressure.
- Effective in managing hypertension and relieving respiratory distress caused by scorpion sting.
- Suitable treatment option for the young man presenting with scorpion sting, respiratory distress, and hypertension.

Incorrect Options:

Option A - Metoprolol: Beta-blocker, may worsen hypertension in scorpion sting.

Option B - Ivabradine: Reduces heart rate, not effective for blood pressure or respiratory distress.

Option D - Furosemide: Diuretic, increases urine output, but doesn't directly impact blood pressure or respiratory distress.

Solution for Question 46:

Correct Option B - A, C, D:

- A (Intravenous): Bioavailability is 100% (1) since the drug is directly injected into the bloodstream.
- C (Intramuscular) and D (Subcutaneous): Both have bioavailability greater than 0.75, typically ranging between 0.75 and 1, due to efficient absorption into systemic circulation.

Incorrect Options:

- Option A, C and D: B (Oral) The statement " $0.75 < \text{Oral} < 1$ " is too broad. Oral bioavailability varies widely among drugs and can be significantly less than 0.75 for some drugs due to factors like first-pass metabolism and poor absorption. Therefore, this option is incorrect.

Solution for Question 47:

Correct Option C - Drug 1 represents agonist and drug 4 represent inverse agonist:

- We can see that Drug 1 produces a higher response than Drug 4 at all concentrations and above the baseline.

We can see that Drug 1 produces a higher response than Drug 4 at all concentrations and above the baseline.

Incorrect Options:

Option A - Drug 1 represents agonist and drug 2 represents inverse agonist: Drug 1 is full agonist and 2 is partial agonist.

Option B - Drug 3 represents agonist and drug 4 represents inverse agonist: Drug 3 is antagonist while Drug 4 is an inverse agonist as it produces a response below the baseline.

Option D - Drug 2 represents partial agonist and drug 3 represents inverse agonist: This statement is also incorrect because even though Drug 2 is a partial agonist. Drug 3 represents an antagonist since it has zero efficacy with increasing drug concentrations.

Solution for Question 48:

Correct Option B - Topical ocular carbonic anhydrase inhibitor:

- Dorzolamide is an ophthalmic medication that is used to treat glaucoma by lowering intraocular pressure.
- It is a carbonic anhydrase inhibitor that is involved in the production of aqueous humor in the eye.
- By inhibiting the activity of this enzyme, dorzolamide reduces the production of aqueous humor, which helps to lower intraocular pressure.

Incorrect Options:

- Options A, C and D are incorrect.
-

Solution for Question 49:

Correct Option C - Gq:

- Phospholipase C (PLC) catalyzes the hydrolysis of phosphatidylinositol 4, 5-bisphosphate (PIP₂) to produce inositol 1,4,5-trisphosphate (IP₃) and diacylglycerol (DAG), acting as second messengers in cell signaling.
- Activated by Gq-coupled receptors, PLC is triggered by the exchange of GDP for GTP on the Gq alpha subunit.
- This activation pathway leads to the generation of IP₃ and DAG, initiating downstream signaling pathways like calcium release and PKC activation.

Incorrect Options:

Option A - Gs: Gs is not correct because these G proteins do not activate PLC. Gs stimulates the production of cyclic AMP (cAMP) by adenylate cyclase.

Option B - Gi: Gi is not correct because these G proteins do not activate PLC. Gi inhibits adenylate cyclase activity.

Option D - Go: Go is not correct because Go is involved in the inhibition of voltage-gated calcium channels and the regulation of potassium channels, but it does not activate PLC.

Solution for Question 50:

Correct Option B - Phase 4 is also called post-marketing surveillance:

- Phase 4 trials, also known as post-marketing surveillance, occur after a drug has been approved by regulatory agencies like the FDA/CDSCO.
- These trials aim to monitor the long-term safety and effectiveness of the drug in a larger patient population.
- Phase 4 helps identify any rare or long-term side effects that may not have been detected during earlier phases.

Incorrect Options:

Option A - Phase 3 aim is to increase pharmacovigilance: Phase 3 doesn't aim to increase pharmacovigilance; that's the role of Phase 4.

Option C - Phase 3 is never double blinded: Phase 3 trials are often double-blinded to confirm a drug's effectiveness and safety.

Option D - Phase 2 is done on healthy individuals: Phase 2 trials involve patients with the intended condition, not healthy individuals.

Solution for Question 51:

Correct Option B - 1-d, 2-c, 3-b, 4-a:

- Lithium: Associated with Ebstein anomaly, a heart defect where the tricuspid valve is displaced.
- Chloramphenicol: Linked to Gray baby syndrome, characterized by symptoms like abdominal distension and cyanosis.
- Warfarin: Prolonged exposure during pregnancy can cause fetal warfarin syndrome, leading to developmental abnormalities.
- Thalidomide: Known for causing phocomelia, a severe birth defect resulting in underdeveloped or absent limbs.

Incorrect Options:

- Option A, C & D are incorrect. Refer to the explanation for the correct answer.
-

Solution for Question 52:

Correct Option D - 0.5 mg daily before bedtime for 7 days:

- Prescribe 0.5 milligrams of the medication once daily before bedtime for a duration of 7 days.
- The specific dosing regimen and duration of treatment should be determined by the healthcare professional based on the patient's condition, response to treatment, and any potential contraindications or interactions with other medications

Incorrect Options:

- Option A, B & C are incorrect options.
-

Solution for Question 53:

Correct Option B - Clearance of drug x desired peak plasma concentration:

The dosing rate of a drug is typically based on the formula:

- The dosing rate of a drug is calculated by multiplying its clearance by the desired peak plasma concentration.
- This formula helps determine how quickly the drug should be administered to achieve the desired concentration in the plasma.

Incorrect Options:

- Option A, C & D are incorrect options.

Solution for Question 54:

Correct Option A - Phenytoin:

- Phenytoin, an antiepileptic drug, requires therapeutic drug monitoring (TDM) due to its narrow therapeutic range.
- TDM helps ensure its blood levels stay within the therapeutic range for optimal seizure control and to avoid adverse effects.

Incorrect Options:

Option B - Heparin: TDM is generally unnecessary for heparin because its effects can be evaluated using other tests like aPTT or anti-Xa activity.

Option C - Aspirin: Aspirin's effects are well understood, so TDM is not typically required.

Option D - Metformin: TDM is not commonly used for metformin as its therapeutic effects relate more to glucose metabolism than precise blood levels.

Solution for Question 55:

Correct Option A - Physiological antagonism:

- The type of antagonism described in the scenario is Physiological antagonism.
- Physiological antagonism refers to the interaction between two substances or pathways that have opposing physiological effects. In this case, adrenaline (also known as

epinephrine) acting through beta 2 receptors causes bronchodilation, while histamine acting through H1 receptors causes bronchoconstriction.

- The effects of these two substances on bronchial smooth muscle are opposite and counteract each other. Adrenaline stimulates beta 2 receptors, leading to relaxation of bronchial smooth muscle and bronchodilation, which helps in opening up the airways and facilitating easier breathing. On the other hand, histamine binding to H1 receptors triggers bronchoconstriction, causing narrowing of the airways and difficulty in breathing.
- The term physiological antagonism is used to describe this situation because the antagonism occurs at the physiological level, where two substances with different actions counterbalance each other's effects.
- Therefore, the correct answer is physiological antagonism, as it accurately describes the opposing physiological effects of adrenaline and histamine on bronchial smooth muscle.

The type of antagonism described in the scenario is Physiological antagonism.

Physiological antagonism refers to the interaction between two substances or pathways that have opposing physiological effects. In this case, adrenaline (also known as epinephrine) acting through beta 2 receptors causes bronchodilation, while histamine acting through H1 receptors causes bronchoconstriction.

The effects of these two substances on bronchial smooth muscle are opposite and counteract each other. Adrenaline stimulates beta 2 receptors, leading to relaxation of bronchial smooth muscle and bronchodilation, which helps in opening up the airways and facilitating easier breathing. On the other hand, histamine binding to H1 receptors triggers bronchoconstriction, causing narrowing of the airways and difficulty in breathing.

The term physiological antagonism is used to describe this situation because the antagonism occurs at the physiological level, where two substances with different actions counterbalance each other's effects.

Therefore, the correct answer is physiological antagonism, as it accurately describes the opposing physiological effects of adrenaline and histamine on bronchial smooth muscle.

Incorrect Options:

Option B - Chemical antagonism: Chemical antagonism refers to the direct chemical interaction between two substances that renders one of them inactive. It does not accurately describe the interaction between adrenaline and histamine.

Option C - Reversible antagonism: Reversible antagonism refers to the competitive binding of an antagonist to a receptor, which can be overcome by increasing the concentration of the agonist. It does not accurately describe the interaction between adrenaline and histamine, as they act through different receptors and their effects are not directly reversible.

Option D - Competitive antagonism: Competitive antagonism occurs when an antagonist competes with an agonist for binding to the same receptor site, thereby inhibiting the agonist's action. This term does not accurately describe the interaction between adrenaline and histamine since they act through different receptor types (beta 2 and H1 receptors, respectively) and their effects are not directly competitive.

Solution for Question 56:

Correct Option D - Aromatase inhibitors:

- Aromatase inhibitors are medications used to treat hormone receptor-positive breast cancer in postmenopausal women.
- They work by reducing estrogen levels in the body through inhibition of the enzyme aromatase, which converts androgens to estrogens.
- Aromatase inhibitors are not typically associated with gynecomastia.

Incorrect Options:

Option A - Liver failure: Liver failure can disrupt hormone metabolism, leading to hormonal imbalances such as increased estrogen levels, which may contribute to gynecomastia.

Option B - Spironolactone: Spironolactone, an aldosterone receptor antagonist, can interfere with androgen activity, potentially leading to gynecomastia.

Option C - Digoxin: While the mechanism is not well understood, digoxin, a cardiac medication, has been associated with gynecomastia in some cases.

Solution for Question 57:

Correct Option A - A:

- Potency is related to dose
- To produce a similar response if a drug is required in a lesser dose, it is more potent.

- Here A has more potency

Incorrect Options:

Option B - B: B has less potency than A

Option C - Both A and B: A has more potency than B

Option D - C: C here is least potent

Solution for Question 58:

Correct Option C - 4:

- Therapeutic index = LD_{50}/ED_{50}
- Therefore therapeutic index = $400/100 = 4$

Incorrect Options:

- Option A, B & D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 59:

Correct Option A - K_m remains same, V_{max} decreases:

- In non-competitive inhibition: K_m remains the same. V_{max} decreases. Binds to both free enzymes and enzyme-substrate complexes with the same affinity.
- K_m remains the same.
- V_{max} decreases.
- Binds to both free enzymes and enzyme-substrate complexes with the same affinity.
- K_m remains the same.
- V_{max} decreases.
- Binds to both free enzymes and enzyme-substrate complexes with the same affinity.

Incorrect Options:

- Option B, C and D are incorrect.
-

Solution for Question 60:

Correct Option C - Fixed drug combination is preferred over single drugs:

This statement is incorrect.

- Essential medicines meet priority healthcare needs based on efficacy, safety, and cost-effectiveness, as defined by WHO.
- Governments prioritize these medicines to optimize resource utilization.
- Selection criteria include efficacy, safety, availability, cost-benefit analysis, and consideration of prevalent diseases and resources.
- Fixed-dose combinations are not always preferred over single drugs due to flexibility limitations.

Incorrect Options:

The following statements are correct regarding Essential Drugs

Option A - An adequate safety profile needs to be established: Safety matters: Essential drugs must have acceptable safety profiles, with benefits outweighing risks.

Option B - Cost to benefit has to be considered: Cost matters too. Essential drugs should be cost-effective, delivering benefits that exceed their costs.

Option D - Disease prevalence is considered: Prevalence counts. Essential drugs are chosen based on the prevalence of the diseases they treat, ensuring they meet population needs.

Solution for Question 61:

Correct Option C - 13.8 hours:

- Drug half-life: Time for drug plasma concentration to decrease by half.
- Measures rate of drug elimination from body.

- Determines medication dosing interval and frequency.

Drug half-life: Time for drug plasma concentration to decrease by half.

Measures rate of drug elimination from body.

Determines medication dosing interval and frequency.

The formula for calculating the half-life of a drug is:

- Half-life = $0.693 / \text{elimination rate constant}$
- Putting in the given elimination rate constant of 0.05/hr, we get:
- Half-life = $0.693 / 0.05 = 13.86$ hours
- Therefore, the Correct Answer is C - 13.8 hours.

Half-life = $0.693 / \text{elimination rate constant}$

Putting in the given elimination rate constant of 0.05/hr, we get:

Half-life = $0.693 / 0.05 = 13.86$ hours

Therefore, the Correct Answer is C - 13.8 hours.

Incorrect Options:

- Options A, B & D are incorrect.

Options A, B & D are incorrect.

Solution for Question 62:

Correct Option A - Ritonavir:

- It's a protease inhibitor used to treat HIV. When combined with saquinavir, it boosts saquinavir levels by inhibiting cytochrome P450, increasing saquinavir's bioavailability. This enhances its antiviral effectiveness.

Incorrect Options:

Option B - Vitamin C: An essential nutrient, not known to affect saquinavir bioavailability. It does enhance oral iron preparation bioavailability.

Option C - Cimetidine: Reduces stomach acid, but its effect on saquinavir bioavailability is modest. It inhibits cytochrome P450.

Option D - Ganciclovir: Used for herpes and CMV infections, it doesn't significantly impact saquinavir bioavailability.

Solution for Question 63:

Correct Option B - Half-life remains the same, plasma concentration doubles:

- In first-order kinetics, doubling the dose of a drug leads to a doubling of its plasma concentration. However, the drug's half-life remains unchanged.
- The half-life represents the time taken for the plasma concentration of the drug to decrease by half.
- In this scenario, the rate of elimination increases proportionally with the increased dose, balancing the elevated plasma concentration.

Incorrect Options:

Option A - Half – life doubles and plasma concentration remains the same: Doubling the dose would result in a longer half-life, not a constant one.

Option C - Half – life and plasma concentration remains the same: Doubling the dose affects both half-life and plasma concentration.

Option D - Half life and plasma concentration doubles: Neither the half-life nor the plasma concentration doubles; doubling the dose leads to an increased plasma concentration, but the half-life remains the same.

Solution for Question 64:

Correct Option D - 120 minutes:

- Phenytoin and sucralfate are medications used for different purposes—phenytoin for seizure disorders and sucralfate for peptic ulcers.

- When taken together, sucralfate can hinder the absorption of phenytoin in the gastrointestinal tract, affecting its effectiveness.
- To avoid this interaction, it's recommended to space the intake of these medications by a minimum of 2 hours (120 minutes).
- This duration allows sucralfate to form a protective layer in the gastrointestinal tract, minimizing its interference with phenytoin absorption.

Incorrect Options:

Option A - 30 minutes: Too short a gap to mitigate the interaction adequately.

Option B - 60 minutes: Still insufficient time for sucralfate to exert its protective effect.

Option C - 90 minutes: Longer than the previous options but still falls short of the recommended 2-hour interval.

Solution for Question 65:

Correct Option A - Inhibition of calcineurin:

- Cyclosporine inhibits calcineurin activity.
- Calcineurin is a phosphatase enzyme involved in T-cell activation.
- Cyclosporine binds to cyclophilin, forming a complex that inhibits calcineurin.
- Reduction in cytokine production and T-cell proliferation leads to immunosuppression.
- Clinical use includes preventing organ transplant rejection and treating autoimmune diseases.

Cyclosporine inhibits calcineurin activity.

Calcineurin is a phosphatase enzyme involved in T-cell activation.

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Reduction in cytokine production and T-cell proliferation leads to immunosuppression.

Clinical use includes preventing organ transplant rejection and treating autoimmune diseases.

Incorrect Options:

Option B - AMP kinase stimulation: Cyclosporine does not stimulate AMP kinase activity.

Option C - IMP dehydrogenase inhibition: Cyclosporine does not inhibit IMP dehydrogenase; this is a property of mycophenolate mofetil.

Option D - Dihydro-orotate dehydrogenase inhibition: Cyclosporine does not inhibit dihydro-orotate dehydrogenase.

Solution for Question 66:

Correct Option C - Sequestered in body tissues:

- A high apparent volume of distribution suggests that the drug is extensively distributed beyond the vascular compartment and is sequestered in body tissues.
- It indicates that the drug has a tendency to distribute widely throughout the body, beyond the plasma and blood vessels.

A high apparent volume of distribution suggests that the drug is extensively distributed beyond the vascular compartment and is sequestered in body tissues.

It indicates that the drug has a tendency to distribute widely throughout the body, beyond the plasma and blood vessels.

Incorrect Options:

Option A - Highly bound to plasma proteins: If a drug is highly bound to plasma proteins, it tends to remain in the bloodstream and has a low apparent volume of distribution. This means that the drug is confined to the vascular compartment and has limited distribution to body tissues.

Option B - Confined to vascular compartment: If a drug is confined to the vascular compartment, it has limited distribution to body tissues. This implies a low apparent volume of distribution rather than a high one.

Option D - Both A and B: As explained above, neither option A nor option B is correct.

Solution for Question 67:

Correct Option B - Phase 2:

- Concentrates on assessing the preliminary efficacy of the drug and determining the optimal dose.
- It involves a larger group of patients (usually several hundred) and provides initial data on how well the drug works and its side effects.

Phases of clinical trials

Phase 0	• Tested on healthy human volunteers. • Done in only upto 10 participants.	• Micro dosing: 1/100th of a dose because high doses may be fateful.
Phase 1	• Also tested on healthy human volunteers. • Extends from months to years (10-100).	• Safety and nontoxicity profile of drugs. • Maximum tolerated dose of a drug (beyond this adverse effects of a drug occurs).
Phase 2	• Tested on patients. • 50-500 participants. • Lasts for 1-2 years.	• Effectiveness and efficacy. • Maximum drug failure seen.
Phase 3	• Patients can be 500-3000 in number. • This phase lasts 3-5 years.	• Comparison with existing drugs. • Randomized Control Trial.
Drug is launched after phase 3 trials		
Phase 4	• Also tested on patients. • Study is unending till the drug removal. • Longest phase (at least till 10-25 years).	• Monitor Long-term and rare side effects. • Post marketing surveillance.

Incorrect Options:

Option A - Phase 1: Focuses on evaluating the safety, tolerability, pharmacokinetics, and pharmacodynamics of a drug in a small group of healthy volunteers or patients.

Option C - Phase 3: Designed to confirm the drug's efficacy and safety in a much larger patient population (several hundred to several thousand), providing comprehensive evidence required for regulatory approval.

Option D - Phase 4: Conducted after the drug has been approved to monitor its long-term effectiveness and safety in the general population, and may explore additional uses or benefits.

Solution for Question 68:

Correct Option C - Female patients should avoid getting pregnant for 6 months after therapy:

- Female patients of childbearing potential should avoid pregnancy for at least 3 years after stopping acitretin therapy due to its teratogenic effects.
- Effective contraception should be used during treatment and for an extended period after discontinuation.

Incorrect Options:

Option A - Contraindicated in persons with hyperlipidemia: Acitretin is contraindicated in those with hyperlipidemia due to its potential to exacerbate lipid levels, requiring close lipid monitoring during therapy.

Option B - It can cause increased sensitivity to sunlight: Acitretin increases skin sensitivity to sunlight, necessitating sun protection measures like sunscreen and protective clothing to prevent sunburn.

Option D - Acitretin has a half life of about 50 hours: Acitretin has a long half-life of approximately 50 hours, emphasizing the need for extended contraceptive use and pregnancy avoidance post-therapy.

Solution for Question 69:

Correct Option A - Concentration of the drug is 50% ionic and 50% non-ionic:

- The pKa of a drug is the pH at which the drug exists in equal concentrations of its ionized (ionic) and non-ionized (non-ionic) forms.
- There for when $\text{pH} = \text{pKa}$, the drug is at its half-ionization point, meaning that it is 50% ionized and 50% non-ionized.

The pKa of a drug is the pH at which the drug exists in equal concentrations of its ionized (ionic) and non-ionized (non-ionic) forms.

There for when $\text{pH} = \text{pKa}$, the drug is at its half-ionization point, meaning that it is 50% ionized and 50% non-ionized.

Incorrect Options:

- Options B, C & D are incorrect.

Solution for Question 70:

Correct Option C - The graph represents the dose-response curve for the drug:

- The graph represents the dose-response curve for the drug, showing the relationship between drug dose and its observed response or effect.

Incorrect Options:

Option A - The graph shows the relationship between drug potency and its efficacy: The graph does not specifically represent the relationship between drug potency and efficacy; its primary focus is on illustrating the dose-response curve.

Option B - The graph demonstrates that higher potency always leads to higher efficacy: The graph does not necessarily indicate that higher potency always leads to higher efficacy; efficacy depends on various factors beyond potency alone.

Option D - The graph indicates that drugs with low potency can never be effective: The graph does not suggest that drugs with low potency can never be effective; efficacy can still be achieved with lower potency drugs if administered appropriately within the therapeutic range.

ANS Introduction and Cholinergic Drugs

1. A 62-year-old woman is brought to the emergency department because of acute chest pain with complaints of diaphoresis and lightheadedness. An ECG reveals ST-segment elevation in the inferior leads. Her blood pressure is 90/60 mm Hg, and her pulse is 42/min. The appropriate therapy is initiated, including medication to treat her bradycardia. The medication improved her vitals, but she now complains of severe right eye pain. Which of the following drugs can be used to treat this patient?

- A. Phenylephrine
- B. Pyridostigmine
- C. Pilocarpine
- D. Bethanechol

2. A 68 y/o man with a history of Parkinson's disease is scheduled for a laparoscopic cholecystectomy. He is experiencing excessive salivation and drooling due to his Parkinson's disease. Which of the following medications would be most appropriate to administer before surgery to reduce his salivary secretions?

- A. Glycopyrrolate
- B. Neostigmine
- C. Pilocarpine
- D. Bethanechol

3. A 30 y/o woman presents to the emergency department with acute angle-closure glaucoma. Her intraocular pressure is markedly elevated. Which of the following medications is most appropriate for lowering her intraocular pressure?

- A. Ipratropium
 - B. Atropine
 - C. Carbachol
 - D. Scopolamine
-

4. A 68 y/o woman with a long history of diabetes presents with complaints of frequent urinary tract infections and difficulty emptying her bladder. She reports feeling like she needs to urinate urgently, but when she tries to go, very little urine comes out. After ruling out a urinary tract obstruction, which medication would be most appropriate to address her symptoms?

- A. Bethanechol
 - B. Pilocarpine
 - C. Atropine
 - D. Edrophonium
-

5. Which of the following would possibly confirm the diagnosis of myasthenia gravis following an Edrophonium test?

- A. No change in muscle strength after administration of edrophonium
 - B. Worsening of muscle weakness after administration of edrophonium
 - C. Temporary improvement in muscle strength after administration of edrophonium
 - D. Development of severe muscle cramps after administration of edrophonium
-

6. A 45-year-old woman presents with generalised weakness, ptosis, diplopia, and difficulty swallowing. She reports that her symptoms worsen with exertion and improve with rest. After a thorough evaluation, the physician suspects a

neuromuscular junction disorder and prescribes a medication to enhance neuromuscular transmission. Which of the following medications is most appropriate for managing the patient's symptoms?

- A. Edrophonium
 - B. Pyridostigmine
 - C. IV-Ig
 - D. Atropine
-

7. Neostigmine is commonly used in clinical practice for several medical conditions. Which of the following is NOT an appropriate use of Neostigmine?

- A. Treatment of myasthenia gravis
 - B. Reversal of non-depolarizing neuromuscular blockade after surgery
 - C. Management of urinary retention
 - D. Treatment of Alzheimer's disease
-

8. A 65-year-old patient is being treated for myasthenia gravis and requires a medication that acts peripherally without affecting the central nervous system. Which of the following carbamate drugs cannot cross the blood-brain barrier and is appropriate for this patient?

- A. Physostigmine
 - B. Neostigmine
 - C. Pyridostigmine
 - D. Rivastigmine
-

9. Which of the following is the first-line drug of choice for managing mild to moderate symptoms of Alzheimer's disease?

- A. Memantine
 - B. Donepezil
 - C. Rivastigmine
 - D. Galantamine
-

10. Which of the following is a reversible anticholinesterase?

- A. Sarin
 - B. Malathion
 - C. Tabun
 - D. Tacrine
-

11. A 30-year-old woman presents with severe nausea, vomiting, abdominal pain, and diarrhea. She reports foraging for wild mushrooms earlier in the day and consuming them in a meal. Upon further questioning, it is discovered that she is experiencing visual disturbances and confusion. Physical examination reveals jaundice and signs of dehydration. Which of the following toxins is the likely culprit?

- A. Amatoxin
 - B. Bungarotoxin
 - C. Muscarine
 - D. Muscimol
-

12. A 45-year-old male presents with a chronic cough and shortness of breath. Upon further inquiry, it is revealed that he has been a smoker for 20 years, averaging a pack a day. His doctor discusses the risks of smoking and recommends nicotine cessation drugs to help him quit. Which of the following may be prescribed first-line?

- A. Topiramate
 - B. Clonidine
 - C. Nortriptyline
 - D. Bupropion
-

13. Match the drugs with their mechanism of action: Drug Action 1. Neostigmine a) Inhibits acetylcholinesterase in the CNS 2. Galantamine b) Inhibits acetylcholinesterase at the neuromuscular junction 3. Bethanechol c) Inhibits acetylcholinesterase at both the central and peripheral nervous system 4- Physostigmine d) Direct-acting cholinergic agonist at muscarinic receptors

Drug	Action
1. Neostigmine	a) Inhibits acetylcholinesterase in the CNS
2. Galantamine	b) Inhibits acetylcholinesterase at the neuromuscular junction
3. Bethanechol	c) Inhibits acetylcholinesterase at both the central and peripheral nervous system
4- Physostigmine	d) Direct-acting cholinergic agonist at muscarinic receptors

- A. 1-a, 2-b, 3-c, 4-d
 - B. 1-d, 2-a, 3-d, 4-c
 - C. 1-c, 2-a, 3-d, 4-b
 - D. 1-b, 2-d, 3-c, 4-a
-

14. A 30-year-old female with no significant medical history presents to the clinic with complaints of intermittent shortness of breath and wheezing over the past few months. Her physical examination and baseline spirometry are unremarkable. The physician decides to perform a methacholine challenge test. Which of the following best describes the purpose of the methacholine challenge test?

- A. To confirm the presence of chronic obstructive pulmonary disease (COPD)
 - B. To diagnose pulmonary embolism
 - C. To identify airway hyperreactivity consistent with asthma
 - D. To assess for restrictive lung disease
-

15. A 6-year-old girl is brought to the emergency department by her parents after they found her unresponsive. They report that her younger sister sprayed the patient with a bottle filled with pesticide. On examination, the patient is lethargic and salivating. The patient's blood pressure is 110/60 mm Hg, pulse is 58/min, and respirations are 42/min. Which of the following medications should be administered first during treatment of this patient's current condition?

- A. Neostigmine
 - B. Pralidoxime
 - C. Naloxone
 - D. Atropine
-

16. A 25-year-old male is undergoing a surgical procedure requiring general anesthesia. The anesthetic is administered. Shortly after administration, the patient experiences prolonged muscle paralysis, necessitating mechanical ventilation. Laboratory tests reveal a deficiency in pseudocholinesterase activity. Which of the following drugs is not responsible for the patient's prolonged muscle paralysis?

- A. Mivacurium
- B. Succinylcholine

- C. Rocuronium
 - D. Butyrylcholine
-

17. Which of the following symptoms is seen in Atropine poisoning?

- A. Miosis
 - B. Mydriasis
 - C. Bronchospasm
 - D. Sweating
-

18. A 45-year-old woman presents to her healthcare provider with complaints of severe dry mouth following radiation therapy for head and neck cancer. The physician prescribes a directly acting cholinergic drug to stimulate her salivary glands. Which of the following drugs is most appropriate for treating her condition?

- A. Carbachol
 - B. Bethanechol
 - C. Pilocarpine
 - D. Methacholine
-

19. A 60-year-old male patient presents with urinary retention following abdominal surgery. The physician prescribes a drug that will stimulate bladder contraction by directly activating muscarinic receptors. Which of the following drugs is most appropriate for this condition?

- A. Bethanechol - Direct-acting cholinergic agonist
- B. Neostigmine - Direct-acting anticholinergic
- C. Atropine - Indirect-acting cholinergic agonist

D. Pilocarpine - Indirect-acting anticholinergic

20. A 45 y/o male presents with severe abdominal pain, excessive salivation, sweating, bradycardia (heart rate 48 bpm), and miosis after ingesting an unknown substance. His blood pressure is 100/60 mmHg. Which muscarinic receptor type is primarily responsible for the observed bradycardia?

- A. M1 receptor
 - B. M2 receptor
 - C. M3 receptor
 - D. M4 receptor
-

21. A 35-year-old man is admitted to the hospital with symptoms of muscle weakness, fasciculations, and myotonic jerks. The patient has a history of smoking and has been taking a cholinesterase inhibitor for a recent injury. The treating physician suspects that the patient's symptoms are due to the effects of the cholinesterase inhibitor. Which of the following receptors is most likely involved in the patient's symptoms?

- A. NM Receptors
 - B. M3 Receptors
 - C. Nn Receptors
 - D. Adrenergic receptors
-

22. Which of the following statements regarding botulinum toxin (botox) is correct?

- A. Botox stimulates the release of Ach
- B. Botox type B is more effective than type A

- C. Botox injection causes muscle relaxation
 - D. Onabotulinumtoxin A is used to treat dry mouth
-

23. A 72-year-old patient presents with symptoms of memory loss and confusion. After a thorough evaluation, the patient is diagnosed with Alzheimer's disease. The physician prescribes a medication that inhibits the enzyme acetylcholinesterase, available as a transdermal patch. Which of the following drugs is most likely to be prescribed?

- A. Tacrine
 - B. Donepezil
 - C. Rivastigmine
 - D. Memantine
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	3
Question 4	1
Question 5	3
Question 6	2
Question 7	4
Question 8	2
Question 9	2
Question 10	4
Question 11	1
Question 12	4
Question 13	4
Question 14	3
Question 15	4
Question 16	3
Question 17	2
Question 18	3
Question 19	1
Question 20	2
Question 21	1
Question 22	3

Solution for Question 1:

Correct Answer: C) Pilocarpine

Explanation:

Given the clinical scenario, this patient had an inferior wall MI associated with bradycardia. Atropine, an antimuscarinic drug, blocks vagal effects on the nodes of the heart, effectively increasing heart rate. In the eye, it induces mydriasis, which narrows the anterior chamber angle and reduces aqueous humor outflow, potentially triggering acute angle-closure glaucoma.

The drug of choice for this condition is pilocarpine. When applied topically to the eye, pilocarpine rapidly penetrates the cornea, causing miosis, ciliary muscle contraction, and reduced intraocular pressure.

Therapeutic uses of pilocarpine:

- Third-line treatment for open-angle glaucoma, either alone or in combination with other medications.
- Counteracting the effects of mydriatics and preventing or breaking adhesions of the iris with the lens or cornea.
- Treating xerostomia (dry mouth) associated with Sjögren's syndrome or following head and neck radiation therapy.

Side effects:

- Topical use: Stinging, accommodation spasm, and systemic effects such as nausea, diarrhea, and bronchospasm at higher doses. Pilocarpine is associated with cataract formation after chronic use.
- Systemic use: Cholinergic effects like sweating, nausea, and diarrhea.

Phenylephrine (Option A) is used as a mydriatic and nasal decongestant.

Pyridostigmine (Option B) is used to treat myasthenia gravis.

Bethanechol (Option D) is used to treat urinary retention and stimulate bladder function.

Solution for Question 2:

Correct Answer: A) Glycopyrrolate

Explanation:

- Glycopyrrolate is a muscarinic antagonist that reduces salivary secretions.
- Glycopyrrolate is frequently used in pre-anesthetic medication and during surgery.

Glycopyrrolate:

- Glycopyrrolate is a potent and rapidly acting antimuscarinic drug that lacks central effects.
- MOA: Works by blocking muscarinic receptors and inhibits the effects of acetylcholine.
- Uses: Pre-anesthetic medication and during anesthesia - to reduce salivary and tracheobronchial secretions. Reduce drooling - Parkinson's disease. Reduce GI tone and motility. Treat primary axillary hyperhidrosis. Treat obstructive airway disease.
- Pre-anesthetic medication and during anesthesia - to reduce salivary and tracheobronchial secretions.
- Reduce drooling - Parkinson's disease.
- Reduce GI tone and motility.
- Treat primary axillary hyperhidrosis.
- Treat obstructive airway disease.
- Adverse Effects: Dry mouth, Constipation, Blurred vision, Gastrointestinal discomfort, Cognitive impairment.
- Pre-anesthetic medication and during anesthesia - to reduce salivary and tracheobronchial secretions.
- Reduce drooling - Parkinson's disease.
- Reduce GI tone and motility.
- Treat primary axillary hyperhidrosis.
- Treat obstructive airway disease.

Neostigmine (Option B) is a cholinergic agonist that increases salivary secretions, worsening the patient's drooling.

Pilocarpine (Option C) is a cholinergic agonist that increases salivary secretions. It is used to treat xerostomia (dry mouth)

Bethanechol (Option D) is a cholinergic agonist that increases salivary secretions. It is primarily used to treat urinary retention.

Solution for Question 3:

Correct Answer: C) Carbachol

Explanation:

- Carbachol is a cholinergic agonist that acts on both muscarinic and nicotinic receptors.
- When applied topically to the eye, carbachol produces miosis (pupillary constriction) and facilitates aqueous outflow, reducing intraocular pressure.

Carbachol:

- Carbachol is a cholinergic agonist, that acts by directly stimulating cholinergic receptors.
- It is a long-acting drug that is resistant to degradation by acetylcholinesterase and butyrylcholinesterase.
- Uses: Glaucoma, causes the ciliary muscle to contract, reducing intraocular pressure.
- Side effects: DUMBBELSS = Diarrhea, Urination, Miosis, Bronchospasm, Bradycardia, Excitation/seizures, Lacrimation, Salivation, and Sweating.

Ipratropium (Option A): is a muscarinic antagonist that acts selectively on bronchial muscles.

Atropine and scopolamine (Options B & D): are muscarinic receptor antagonists. They produce mydriasis (pupillary dilation), which can worsen angle closure and increase intraocular pressure.

Solution for Question 4:

Correct Answer: A) Bethanechol

Explanation:

- This patient, with a long history of diabetes and urinary symptoms, is most likely to have a neurogenic bladder, causing inadequate bladder emptying.
- Bethanechol is the most appropriate medication for this patient's neurogenic bladder.

Bethanechol:

- Bethanechol is a cholinergic agonist that stimulates muscarinic receptors, which are found primarily in the urinary and gastrointestinal tracts.
- MOA: Mimics the effects of acetylcholine, which plays a crucial role in the parasympathetic nervous system, at muscarinic receptors. Upon stimulating muscarinic receptors, it promotes bladder muscle contraction, aiding in urine expulsion and reducing urinary retention.
- Mimics the effects of acetylcholine, which plays a crucial role in the parasympathetic nervous system, at muscarinic receptors.
- Upon stimulating muscarinic receptors, it promotes bladder muscle contraction, aiding in urine expulsion and reducing urinary retention.
- Clinical Usage: Postoperative and Postpartum urinary retention with no actual obstruction. Neurogenic bladder. Symptomatic relief in congenital megacolon. Gastroesophageal reflux (Rare, due to its side effects).
- Postoperative and Postpartum urinary retention with no actual obstruction.
- Neurogenic bladder.
- Symptomatic relief in congenital megacolon.
- Gastroesophageal reflux (Rare, due to its side effects).
- Side Effects: Gastrointestinal: Belching, involuntary defecation. Urinary: Involuntary urination. Other: Flushing, sweating, hypotension.
- Gastrointestinal: Belching, involuntary defecation.
- Urinary: Involuntary urination.
- Other: Flushing, sweating, hypotension.
- Contraindications: Mechanical obstruction of the gastrointestinal or urinary tract. Asthma. Hypotension.
- Mechanical obstruction of the gastrointestinal or urinary tract.
- Asthma.
- Hypotension.
- Mimics the effects of acetylcholine, which plays a crucial role in the parasympathetic nervous system, at muscarinic receptors.
- Upon stimulating muscarinic receptors, it promotes bladder muscle contraction, aiding in urine expulsion and reducing urinary retention.
- Postoperative and Postpartum urinary retention with no actual obstruction.
- Neurogenic bladder.
- Symptomatic relief in congenital megacolon.
- Gastroesophageal reflux (Rare, due to its side effects).
- Gastrointestinal: Belching, involuntary defecation.

- Urinary: Involuntary urination.
- Other: Flushing, sweating, hypotension.
- Mechanical obstruction of the gastrointestinal or urinary tract.
- Asthma.
- Hypotension.

Pilocarpine (Option B): While pilocarpine is a muscarinic agonist, it's clinically used in topical applications for glaucoma and xerostomia (dry mouth).

Atropine (Option C): is a muscarinic receptor antagonist, working in direct opposition to bethanechol.

Edrophonium (Option D): short-acting cholinesterase inhibitor, is primarily utilized in diagnosing myasthenia gravis.

Solution for Question 5:

Correct Answer: C) Temporary improvement in muscle strength after administration of edrophonium

Explanation:

The edrophonium (Tensilon) test helps differentiate between myasthenic and cholinergic crises, which present with similar symptoms but require opposite treatments.

- Myasthenic crisis arises from inadequate acetylcholine at the neuromuscular junction.
- The cholinergic crisis results from excessive acetylcholine stimulation due to anticholinesterase medication overdose.

The tensilon test involves the injection of 10 mg i.v. Edrophonium, a short-acting anticholinesterase agent.

Steps: 2 mg i.v. Edrophonium → Observe for any adverse reactions → If no adverse reactions occur after 30–60 seconds, the remaining 8 mg is injected.

Results:

- Myasthenia Gravis: Short-lasting improvement in muscle strength.
- Other Muscular Dystrophies: No improvement. (Option A ruled out)
- Cholinergic crisis: No improvement or a worsening of their condition. (Option B ruled out)

Alternative Test: If edrophonium is unavailable, 1.5 mg of i.v. Neostigmine can be used alternatively. Atropine pretreatment can be administered to counteract the muscarinic effects of neostigmine.

Provocative test (d-Tubocurarine test): Myasthenia gravis patients are highly sensitive to d-tubocurarine (a non-depolarising muscle blocker); a 0.5 mg IV dose causes significant weakness in them but does not affect non-myasthenic individuals.

However, this test is hazardous and not commonly performed.

Solution for Question 6:

Correct Answer: B) Pyridostigmine

Explanation:

Based on the scenario, the likely diagnosis is Myasthenia gravis, which is managed with Pyridostigmine, neostigmine, and ambenonium. They are anti-cholinesterase agents that inhibit the breakdown of ACh by acetylcholinesterase and increase presynaptic levels of ACh, allowing for muscle contraction.

Edrophonium (Option A): short-acting acetylcholinesterase inhibitor, is used as a diagnostic tool in suspected cases of myasthenia gravis. This involves intravenous administration to individuals with a deficiency of acetylcholine receptors at the neuromuscular junction. The test typically results in a temporary improvement in muscle strength, confirming the diagnosis of myasthenia gravis by demonstrating the underlying defect in neuromuscular transmission.

IV- Ig (Option C): or intravenous immunoglobulins are used for exacerbations known as myasthenic crisis.

Atropine (Option D): is an anticholinergic drug given along with pyridostigmine to counteract the cholinergic side effects of pyridostigmine.

Solution for Question 7:

Correct Answer: D) Treatment of Alzheimer's disease

Explanation:

Neostigmine is a cholinesterase inhibitor that is commonly used for

- Treatment of myasthenia gravis, (Option A) as it increases the availability of acetylcholine at neuromuscular junctions, thereby improving muscle strength.
- To reverse the effects of non-depolarizing neuromuscular blocking agents after surgical procedures and (Option B)
- In managing urinary retention due to its ability to stimulate bladder muscle contractions. (Option C)
- Cobra bites because cobra toxin disrupts the neuromuscular junction, resulting in respiratory paralysis.

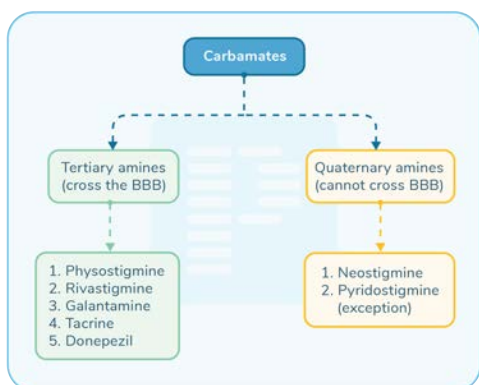
However, Neostigmine is not used for the treatment of Alzheimer's disease because it does not cross the blood-brain barrier and thus cannot affect the central nervous system.

Solution for Question 8:

Correct Answer: B) Neostigmine

Explanation:

Neostigmine is a carbamate drug that cannot cross the blood-brain barrier because it is a quaternary amine, which makes it more polar and less able to penetrate the lipid-rich environment of the BBB.



On the other hand, Physostigmine and Rivastigmine (Options A & D) are tertiary amines, allowing them to cross the BBB and exert their effects within the central nervous system.

Pyridostigmine (Option C), although used similarly to Neostigmine, has limited ability to cross the BBB due to its quaternary structure. Pyridostigmine can pass across the blood-brain barrier (BBB) and enter the brain when the BBB is disrupted.

Solution for Question 9:

Correct Answer: B) Donepezil

Explanation:

Donepezil is a cholinesterase inhibitor commonly used as a first-line treatment for managing symptoms of mild to moderate Alzheimer's disease. It works by increasing the levels of acetylcholine in the brain, which can help improve cognitive function and slow the progression of symptoms.

- Side effects: Nasopharyngitis, diarrhoea, nausea, and vomiting, which are linked to an overabundance of peripheral cholinergic stimulation. In rare cases, rhabdomyolysis has been reported, necessitating discontinuation of the medication.

Rivastigmine (Options C) and Galantamine (Options D) are also cholinesterase inhibitors and can be used, but Donepezil is often preferred due to its once-daily dosing and favourable side-effect profile.

Memantine (Option A), on the other hand, is typically used for moderate to severe Alzheimer's disease and works differently as an NMDA receptor antagonist.

Solution for Question 10:

Correct Answer: D) Tacrine

Explanation:

Tacrine is a reversible anticholinesterase used to manage symptoms of Alzheimer's disease.

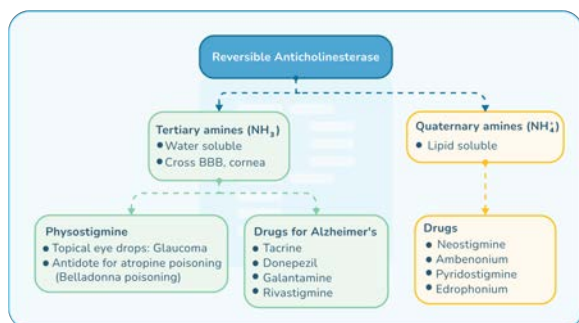
Sarin (Option A ruled out) and Tabun (Option C ruled out) are nerve agents, while Malathion (Option B ruled out) is a pesticide. All three are organophosphates and act as irreversible anticholinesterases.

Reversible anticholinesterases function by inhibiting acetylcholinesterase (AChE), preventing the breakdown of acetylcholine (ACh) and increasing its concentration in the synaptic cleft. This mechanism improves neurotransmission in conditions like myasthenia gravis and Alzheimer's disease.

- Myasthenia Gravis: These drugs enhance muscle strength by increasing ACh availability at neuromuscular junctions.
- Alzheimer's Disease: They temporarily improve cognitive function by boosting cholinergic transmission in the brain.

Unlike irreversible anticholinesterases, which form long-lasting bonds with AChE, reversible anticholinesterases bind transiently. This reverses their effects as the medication is metabolised or cleared from the body.

Reversible anticholinesterases can lead to side effects related to excessive cholinergic stimulation. These side effects may include nausea, vomiting, diarrhea, bradycardia, and bronchoconstriction.



Solution for Question 11:

Correct Answer: A) Amatoxin

Explanation:

The clinical scenario shows a patient with a history of consuming wild mushrooms, confusion, visual disturbance, and jaundice, which is a sign of hepatotoxicity. Amatoxin from Amanita phalloides mushroom can cause hepatotoxicity when ingested.

	Early mushroom poisoning	Late mushroom poisoning	Hallucinogenic mushroom poisoning
Species	Inocybe	Amanita phalloides	Amanita muscaria
Toxin	Muscarine (Option C)	Amatoxin (Option A)	Muscimol
Symptoms	Cholinergic symptoms (DUMBBELLS)	Hepatotoxicity	Hallucinations, seizures
Treatment	Atropine	Thioctic acid Silibinin (hepatoprotective)	Supportive treatment ATROPINE CONTRAINDICATED

Atropine's anticholinergic properties can worsen hallucinations and increase agitation, confusion, and delirium. It can also elevate heart rate and blood pressure, potentially exacerbating cardiovascular symptoms from hallucinogenic mushroom ingestion.

Bungarotoxin (Option B) is a toxin produced by venomous snakes that can lead to bleeding and paralysis.

Solution for Question 12:

Correct Answer: D) Bupropion

Explanation:

Bupropion is a first-line drug used for smoking cessation.

Smoking de-addiction drugs:

	Mechanism of Action	Effect on Smoking Cessation
First-Line Drugs		
Bupropion (Option D)	Blocks reuptake of dopamine and norepinephrine in the brain	Reduces nicotine withdrawal symptoms and cravings
Nicotine Replacement Therapy (NRT)	Provides controlled nicotine dose (patches, gum, lozenges, inhalers). Never given orally, sublingual due to its fast absorption	Alleviates withdrawal symptoms and cravings, gradually reducing nicotine dependence

Varenicline	Binds to nicotine receptors in the brain, partially stimulating them.	Most effective drug. Reduces cravings and pleasurable effects of smoking. Used in tobacco chewing.
All first-line drugs are C/I in seizures and heart diseases (MI, angina, hypertension)		
Second-Line Drugs		
Clonidine (Option B)	Alpha-2 adrenergic agonist, reduces sympathetic nervous system activity	Alleviates withdrawal symptoms such as anxiety, irritability, and restlessness. Safe in patients with heart disease/seizure.
Nortriptyline (Option C)	Tricyclic antidepressant, affecting norepinephrine and serotonin levels	Reduces withdrawal symptoms and cravings
Topiramate (Option A)	Blocks glutamate receptors and facilitates GABA neurotransmission	Potentially facilitates smoking cessation
Mecamylamine	Blocks nicotine receptors in the brain	Reduces the rewarding effects of nicotine, making smoking less appealing. Used in nicotine poisoning.

Solution for Question 13:

Correct Answer: D) 1-b, 2-d, 3-c, 4-a

Explanation:

Drug	Action
Neostigmine	Inhibits acetylcholinesterase at the neuromuscular junction
Galantamine	Inhibits acetylcholinesterase in the CNS
Bethanechol	Direct-acting cholinergic agonist at muscarinic receptors, promoting smooth muscle contraction and increasing gastrointestinal motility
Physostigmine	Inhibits acetylcholinesterase at both the central and peripheral nervous system, leading to increased

acetylcholine levels and enhanced cholinergic transmission
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Solution for Question 14:

Correct Answer: C) To identify airway hyperreactivity consistent with asthma

Explanation:

The methacholine challenge test is used to assess airway hypersensitivity, which is a hallmark of asthma.

Methacholine is a synthetic choline ester that acts as a muscarinic agonist.

The methacholine challenge test involves administering nebulised methacholine in increasing concentrations to the patient and measuring the resulting changes in lung function.

- A positive test is indicated by a 20% decrease in FEV1 (PC20) at a concentration ≤ 4 mg/mL, suggesting asthma.
- Intermediate results (PC20 between 4-16 mg/mL) indicate mild hyperreactivity,
- A PC20 > 16 mg/mL is negative. A negative test can help rule out asthma but should be interpreted cautiously if the patient has relevant symptoms.

Contraindications to methacholine testing: Severe airflow limitation, recent MI or stroke, uncontrolled hypertension, or pregnancy.

- COPD is typically diagnosed through clinical history, risk factors (like smoking), and spirometry showing irreversible airflow obstruction. (Option A ruled out)
- Pulmonary embolism is diagnosed using imaging studies such as a CT pulmonary angiogram or V/Q scan and blood tests like D-dimer. (Option B ruled out)
- Restrictive lung diseases are characterised by reduced lung volumes and are diagnosed through spirometry, which shows decreased total lung capacity (TLC) and other lung volume measurements. (Option D ruled out)

Solution for Question 15:

Correct Answer: D) Atropine

Explanation:

Organophosphate poisoning	
Common exposures	Pesticide: farmer/field worker, pediatric ingestion, suicide attempt Nerve agent: multiple patients presenting with similar symptoms
Manifestations	Muscarinic (DUMBBELSS) : D-Diarrhea U-Urination M-Miosis B-Bronchospasms, bronchorrhea, bradycardia E-Excitation, Emesis L-Lacrimation S-Salivation, sweating Nicotinic: Muscle weakness, paralysis, fasciculations.
Management	Decontamination Atropine reverses muscarinic symptoms Pralidoxime reverses nicotinic and muscarinic symptoms (reactivates cholinesterase)

This patient developed bradycardia, tachypnea, and excessive salivation after exposure to a pesticide, indicating acute organophosphate toxicity. Here are the key points:

Mechanism: Organophosphates are irreversible acetylcholinesterase inhibitors causing cholinergic excess by overstimulating muscarinic (DUMBBELSS) and nicotinic receptors (muscle weakness, fasciculations, paralysis).

Initial Management:

Atropine (option D):

Pralidoxime (Option B):

Neostigmine (Option A), another acetylcholinesterase inhibitor used to treat myasthenia gravis, would worsen this patient's cholinergic symptoms.

Naloxone (Option C) is used to treat opioid intoxication, which causes sedation and miosis, but typically suppresses the respiratory rate rather than increasing it. Opioid intoxication usually results from injection or ingestion, not from contact with aerosolised agents.

Solution for Question 16:

Correct Answer: C) Rocuronium

Explanation:

Rocuronium is not metabolised by pseudocholinesterase. Given the clinical scenario, the patient probably has pseudocholinesterase deficiency.

Anesthetic drugs metabolised by pseudocholinesterase (also known as butyrylcholinesterase) include:

Patients with pseudocholinesterase deficiency should avoid these drugs to prevent prolonged paralysis and other complications. Other nondepolarising neuromuscular blockers, such as atracurium, rocuronium, and vecuronium, are safe for use. These drugs are primarily eliminated through hepatic metabolism and excreted via bile and urine. Their action and duration are not dependent on pseudocholinesterase activity, making them suitable muscle relaxants for patients with this enzyme deficiency.

Other drugs metabolised by pseudocholinesterase:

Solution for Question 17:

Correct Answer: B) Mydriasis

Explanation:

Atropine is an anticholinergic medication that blocks acetylcholine muscarinic receptors, reversing the effects of DUMBELSS.

Anticholinergic vs Cholinergic symptoms:

Cholinergic symptoms (DUMBELS)	Anticholinergic symptoms
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Diarrhea Urination Miosis (Option A) Bronchospasm (Option C), bradycardia, bronchorrhea Excitation/ seizures, Emesis Lacrimation Salivation, Sweating (Option D)	Constipation Urinary retention Mydriasis (Option B) Tachycardia Hallucinations Delirium Seizures Dry eyes Dry mouth Decreased sweating Vasodilation (flushed skin)
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Famous poem about atropine:

- Hot as hare (hyperthermia)
- Red as a beat (flushes)
- Dry as bone (reduced secretions)
- Blind as bat (dilated pupil- blurred vision)
- Mad as hatter (hallucinations)

Solution for Question 18:

Correct Answer: C) Pilocarpine

Explanation: After undergoing radiation therapy for her head and neck, this 45-year-old woman is now complaining of xerostomia. One popular medication used to promote salivation is pilocarpine.

Muscarinic cholinergic receptor agonists can be classified into two types:

- The naturally occurring cholinomimetic alkaloids/ amides (including pilocarpine, muscarine, and arecoline) and their synthetic congeners
- Choline esters, including ACh and many synthetic esters

Amides/Alkaloids	Clinical significance
Pilocarpine (Option C)	Pilocarpine hydrochloride is used to treat xerostomia that develops after head and neck radiation treatments or that is linked to Sjögren syndrome and impairs their ability to secrete secretions (saliva and lacrimal), used for treatment of angle closure glaucoma.
Cevimeline	Cevimeline primarily stimulates the M1 and M3 receptors.

	Compared to pilocarpine, cevimeline may have fewer adverse effects and a long-lasting sialogogic impact on the salivary and lacrimal glands.
Choline esters	Clinical significance
Acetylcholine	ACh is used topically as a 1% solution into the eye to induce miosis during surgery, while it is rarely administered systemically.
Methacholine (Option D)	Inhaling methacholine can diagnose bronchial airway hyperreactivity in patients without asthma symptoms.
Bethanechol (Option B)	GI and urinary tracts are the main organs affected by bethanechol. Bethanechol is useful for treating postop urine retention, diabetic autonomic neuropathy, and other chronic hypotonic, myogenic, or neurogenic bladder conditions when the organic blockage is absent. It can also be used to treat insufficient bladder emptying.
Carbachol (Option A)	Carbachol is injected into the eye as a 0.01% to 3% solution and is used topically in ophthalmology to treat glaucoma and induce miosis during surgery.

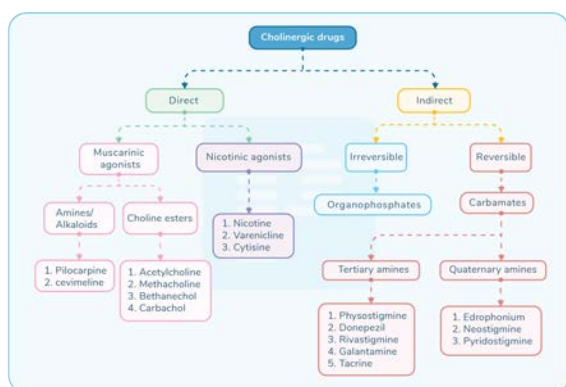
Solution for Question 19:

Correct Answer: A) Bethanechol - Direct-acting cholinergic agonist

Explanation: Following abdominal surgery, the 60-year-old male patient is complaining of urinary retention. A cholinergic agonist with direct action, Bethanechol, is a treatment for urinary retention.

Cholinergic drugs can be generally divided into:

1. Direct acting: Bonds to the muscarinic or nicotinic receptor directly.
2. Indirect acting: Increases acetylcholine (ACh) by blocking acetylcholine esterase.
 - Nicotic agonists are used to treat addiction to smoking.
 - Amides are lipid-soluble and can pass across the blood-brain barrier. They can be used to treat CNS disorders, but they may also cause typical CNS side effects.
 - Because choline esters are lipid insoluble, they cannot pass across the blood-brain barrier. They are used to treat peripheral disorders and have no central side effects.



Neostigmine (Option B) is an indirect-acting cholinergic agonist (not anticholinergic) as it inhibits acetylcholinesterase, thereby increasing acetylcholine levels at the synapse.

Atropine (Option C) is a direct-acting anticholinergic (muscarinic antagonist), blocking the action of acetylcholine at muscarinic receptors.

Pilocarpine (Option D) is a direct-acting cholinergic agonist, not an indirect-acting anticholinergic.

Solution for Question 20:

Correct Answer: B) M2 receptor

Explanation:

- Given his symptoms, he is likely to be exposed to a substance that acts on muscarinic receptors.
- M2 receptors are predominantly found in the heart.
- They mediate the parasympathetic effects on the heart, leading to a decrease in heart rate (bradycardia) and a decrease in the force of contraction.

Location	Receptor	Action
CNS	M1, M2, M3, M4, M5	Effects on cognitive function, motor control, appetite regulation, nociception.
Autonomic ganglia	M1	Modulates nicotinic actions of acetylcholine; may induce a long-lasting late EPSP.

Eye	M3	Iris: Pupillary constriction (miosis). Ciliary muscle: Contraction (accommodation for near vision).
Glands	M3	Increased secretion from salivary, tracheobronchial, lacrimal, and sweat glands.
Heart	M2	SA node: Decreases heart rate (negative chronotropic effect). AV node: Decreases conduction velocity (negative dromotropic effect). Atrium: Shortening of action potential duration, decreased contractility. Ventricle: Decreased contractility.
Blood vessels	M3	Vasodilation (mediated through the release of nitric oxide from endothelial cells)
Smooth muscle	M3	Gastrointestinal tract: Increased tone and peristalsis, sphincter relaxation. Urinary tract: Detrusor muscle contraction, bladder trigone and sphincter relaxation. Bronchial muscles: Constriction.
Gastric glands	M1	Stimulates histamine release, acid secretion

M4 receptors (Option D) are involved in modulating neurotransmitter release in the brain.

M5 receptors are involved in modulating dopaminergic pathways and vascular tone.

Solution for Question 21:

Correct Answer: A) NM Receptors

Explanation:

Nm receptors, found at neuromuscular junctions of skeletal muscles, transmit signals from motor neurons to muscle fibers, facilitating muscle contraction. Symptoms like muscle weakness, fasciculations, and myotonic jerks suggest nicotinic receptor overstimulation, implicating Nm

receptors in the patient's condition.

Characteristics of subtypes of nicotinic receptor:

	NM (Option A)	NN (Option C)
Location and function subserved	Neuromuscular Junction: Depolarization of muscle end plate-contraction of skeletal muscle	Autonomic ganglia: Depolarization -postganglionic impulse Adrenal medulla: catecholamine release CNS: site specific excitation or inhibition
Nature	Has Intrinsic ion channel, pentamer of α_2 , α , β or γ and δ subunits	Has intrinsic ion channel, pentamer of only α or β subunits
Transducer mechanism	Opening of cation (Na^+ K^+) channels	Opening of cation (Na^+ , K^+ , Ca^{2+}) channels
Agonists	Phenyl trimethyl ammonium (PTMA), Nicotine	Dimethylphenylpiperazinium (DMPP), Nicotine
Antagonists	Tubocurarine, α -Bungarotoxin, Atracurium, Vecuronium	Hexamethonium, Trimethaphan, Mecamylamine

M3 receptors (Option B) are muscarinic receptors found in the peripheral nervous system, including the neuromuscular junctions of skeletal muscles, involved in the transmission of signals from parasympathetic neurons to smooth muscles and cardiac muscles.

Adrenergic receptors (Option D) are receptors for the neurotransmitter norepinephrine and transmit signals from sympathetic neurons to smooth muscles and cardiac muscles. They are not typically associated with muscle weakness, fasciculations, and myotonic jerks.

Solution for Question 22:

Correct Answer: C) Botox injection causes muscle relaxation

Explanation:

Botulinum toxin injections cause muscle relaxation by blocking the release of acetylcholine at the neuromuscular junction, leading to temporary paralysis of the injected muscles. It is used to treat blepharospasm, achalasia, torticollis, squint, migraine pain, hyperhidrosis, and cosmetic procedures to reduce wrinkles.

- Botulinum toxin (botox) is derived from *Clostridium botulinum*, a bacterial toxin. It inhibits the release of acetylcholine (ACh) from nerve endings by blocking calcium channels, specifically calcium-dependent Snare protein, leading to muscle paralysis rather than stimulation. (Option A ruled out)
- Botulinum toxin is classified into types A to H. In humans, type A botulinum toxin is more commonly used therapeutically and is considered more effective than other types for various conditions. (Option B)
- Onabotulinumtoxin A can cause dry mouth as a side effect but is not used to treat it. Other side effects of Botox include muscle weakness, difficulty swallowing, and drooping eyelids. Additionally, botulinum toxin type A preparations, such as Onabotulinum Toxin A, Abobotulinum Toxin A, and Incobotulinum Toxin A, are made longer-acting by adding specific lipid molecules or moieties. RimabotulinumtoxinB is approved for the treatment of cervical dystonia. (Option D)

Solution for Question 23:

Correct Answer: C) Rivastigmine

Explanation:

- Alzheimer's Disease (AD) is a neurodegenerative disorder that primarily affects cholinergic neurons in the brain.
- Rivastigmine is a cholinesterase inhibitor that is available as a transdermal patch for AD.
- It can be applied for over 24 hours.
- Transdermal delivery can reduce adverse effects associated with oral formulations.

Drugs for Alzheimer's disease:

CHOLINERGICS	Anticholinesterases are the first-line drugs. Mechanism of action: They are tertiary amines, inhibiting acetylcholine esterase, increasing acetylcholine levels, and improving memory. Drugs: Tacrine (Option A): Not preferred/obsolete due to		
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	hepatotoxicity. Other drugs:		
Drugs	Donepezil (Option B)	Rivastigmine	Galantamine
Enzymes inhibited	AChE	AChE, BuChE	AChE
Mechanism	Noncompetitive	Noncompetitive	Competitive
Typical maintenance dose	10 mg once daily	9.5 mg/24 h (transdermal) 3-6 mg twice daily (oral)	8-12 mg twice daily (immediate release) 16-24 mg/d (extended-release)
FDA-approved indications	Mild-severe AD	Mild-moderate AD	Mild-moderate AD
Metabolism	CYP2D6	Esterases	CYP2D6, CYP3A4
Note: Acetylcholinesterase (AChE) is the major cholinesterase in the brain; butyrylcholinesterase (BuChE) is a serum and hepatic cholinesterase that is upregulated in AD brain. Typical starting doses are one-half of the maintenance dose and are given for the first month of therapy. Donepezil: DOC for AD as it is the longest-acting and most effective.			
NMDA ANTAGONIST	Memantine (Option D). For severe dementia (uncontrolled by anticholinesterases). Increases memory.		

ANTI- ■ -AMYLOID MONOCLONAL ANTIBODIES	For severe dementia (uncontrolled by anticholinesterases). Aducanumab: Blocks fibrils. Lecanemab: Blocks protofibrils.		
MISCELLANEOUS	Not FDA approved. Memory enhancers. Piribedil. Piracetam. Pyritinol (Pyritroxine). Citicoline. Ginkgo biloba.		

Anticholinergic Drugs

1. A 34-year-old male presents with symptoms of acute onset blurred vision, dry mouth, tachycardia, and urinary retention. His medical history is unremarkable, and he denies any recent illnesses or medication use. On examination, his pupils are dilated and unreactive to light. Which of the following is the most likely cause of his presentation?

- A. Cocaine toxicity
 - B. Opioid overdose
 - C. Benzodiazepine intoxication
 - D. Atropine poisoning
-

2. A 30-year-old woman presents with nausea, dizziness, and vomiting after travelling by boat during a recent vacation. She reports these symptoms occur whenever she is in motion, particularly when sailing or driving on winding roads. The patient states that her symptoms significantly interfere with her ability to enjoy travel and have become increasingly bothersome over time. She is prescribed a patch for her condition. Which is the drug of choice?

- A. Hyoscine
 - B. Acetazolamide
 - C. Meclizine
 - D. Diphenhydramine
-

3. Which of the following statements is false?

- A. Atropine causes delirium

- B. Hyoscine is used as a truth serum
 - C. Hyoscine does not cause sedation
 - D. Atropine is used to treat AV node block
-

4. A 65-year-old woman with schizophrenia, treated with haloperidol, develops worsening tremors and stiffness. She exhibits bradykinesia, rigidity, and resting tremor, suggestive of parkinsonism. To manage these symptoms, she is prescribed an anticholinergic. Which of the following is not typically used for this purpose?

- A. Benzhexol
 - B. Benztropine
 - C. Bipyridine
 - D. Promethazine
-

5. A 67 Y/o male with a history of COPD and advanced Parkinson's disease presents with severe sialorrhea, which significantly impacts his quality of life. The patient has tried multiple treatments, including atropine and scopolamine, with limited success. He is started on a trial of glycopyrrolate. Which of the following is true regarding glycopyrrolate?

- A. Glycopyrrolate crosses BBB and worsens Parkinson's disease
 - B. It is contraindicated in COPD as it may cause bronchospasm
 - C. It primarily acts on peripheral muscarinic receptors
 - D. It is short-acting, requiring multiple doses
-

6. A 68-year-old male patient with COPD (GOLD stage 2) comes in for a regular check-up. His current medications include a short-acting beta2-agonist (SABA) as needed. Despite this treatment, he complained of frequent symptoms and two exacerbations in the last year that required oral corticosteroids. His spirometry shows an FEV1 of 60% predicted. Which of the following drugs can be added to help the patient further?

- A. Tiotropium
 - B. Atropine
 - C. Hyoscine
 - D. Trimipramine
-

7. A 45-year-old woman presents to the ophthalmology clinic for a routine eye examination. The ophthalmologist performs a detailed fundoscopic examination to assess the retina. To facilitate this, the ophthalmologist administers an eye drop that causes mydriasis. Which of the following medications is most likely used for this purpose?

- A. Atropine
 - B. Tropicamide
 - C. Pilocarpine
 - D. Timolol
-

8. A 78-year-old female patient with a known diagnosis of Alzheimer's disease presents to the geriatric clinic with complaints of urinary urgency, frequency, and urge incontinence for the past six months. She is currently taking donepezil for Alzheimer's disease. Which of the following medications is most appropriate for managing this patient's urinary symptoms?

- A. Oxybutynin
- B. Trospium

- C. Solifenacin
 - D. Mirabegron
-

9. A 36-year-old woman presents with severe cramping abdominal pain and nausea. She reports that the pain started abruptly a few hours ago and is associated with bloating and diarrhea. She has a history of irritable bowel syndrome (IBS), and similar episodes have occurred in the past. Which of the following medications is most appropriate for relieving her symptoms?

- A. Telenzepine
 - B. Scopolamine
 - C. Hexamethonium
 - D. Thiopentone
-

10. A 9-year-old girl is brought to the pediatrician for frequent, forceful eye blinking and throat clearing, especially when anxious or excited. These brief episodes occur multiple times per hour. She was diagnosed and started on a ganglionic blocker to manage tics. Which of the following could be the possible side effects of the drug?

- A. Bradycardia
 - B. Urinary incontinence
 - C. Increased salivation
 - D. Orthostatic hypotension
-

11. A 43-year-old Man presents with increased mouth secretions, mild respiratory difficulty, and a heart rate of 52 bpm after eating sushi. While being examined, the patient loses consciousness. The patient is given atropine. Which of the following

describes the action of atropine on heart rate?

- A. Tachycardia
 - B. Initial tachycardia followed by bradycardia
 - C. Initial bradycardia followed by tachycardia
 - D. Worsens bradycardia
-

12. A 67-year-old male patient with a known history of primary open-angle glaucoma presents to the clinic for a routine eye examination. He reports no new symptoms, and his intraocular pressure (IOP) has been well controlled. Which of the following medications is preferred in this patient for dilating the pupils?

- A. Atropine
 - B. Phenylephrine
 - C. Homatropine
 - D. Cyclopentolate
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	3
Question 4	4
Question 5	3
Question 6	1
Question 7	2
Question 8	2
Question 9	2
Question 10	4
Question 11	3
Question 12	2

Solution for Question 1:

Correct Answer: D) Atropine poisoning

Explanation:

Atropine poisoning (Belladonna poisoning):

- Atropine is an anticholinergic medication that blocks the action of acetylcholine in the body by binding to muscarinic receptors.
- In cases of overdose or poisoning, it can cause dry mouth, blurred vision, tachycardia, urinary retention, and dilated pupils, among others.

- It is treated with physostigmine, which increases ACh levels; it can even cross the blood-brain barrier (BBB) to reverse the CNS effects of atropine poisoning.

Cocaine toxicity (Option A) causes symptoms such as euphoria, agitation, increased heart rate, elevated blood pressure, dilated pupils, and potentially seizures or cardiac arrhythmias. It typically presents with signs of sympathetic nervous system stimulation, such as tachycardia and hypertension, rather than the anticholinergic symptoms seen in atropine poisoning.

Opioids are a class of drugs that target opioid receptors in the brain and spinal cord to alleviate pain and induce feelings of euphoria. In an overdose, it produces symptoms of CNS depression. (Option B)

Benzodiazepines are commonly used in treating anxiety, insomnia, and certain seizure disorders. In cases of intoxication, benzodiazepines can cause sedation, drowsiness, confusion, impaired coordination, and respiratory depression. (Option C)

Solution for Question 2:

Correct Answer: A) Hyoscine

Explanation:

The patient's symptoms suggest she is experiencing motion sickness, characterized by nausea, dizziness, and vomiting triggered by motion, such as sailing or driving on winding roads. The drug of choice for her condition is Hyoscine.

- Hyoscine is the most effective drug used against motion sickness. It is an antimuscarinic that inhibits signal to vestibular nuclei, decreasing motion sickness. It is also the first choice because it is less sedating than other anti-motion sickness drugs like antihistamines. It can be taken orally or as a transdermal patch applied to the mastoid at least 4 hours before travel. Effects can last for 72 hours.
- It is an antimuscarinic that inhibits signal to vestibular nuclei, decreasing motion sickness.
- It is also the first choice because it is less sedating than other anti-motion sickness drugs like antihistamines.
- It can be taken orally or as a transdermal patch applied to the mastoid at least 4 hours before travel.
- Effects can last for 72 hours.
- It is an antimuscarinic that inhibits signal to vestibular nuclei, decreasing motion sickness.

- It is also the first choice because it is less sedating than other anti-motion sickness drugs like antihistamines.
 - It can be taken orally or as a transdermal patch applied to the mastoid at least 4 hours before travel.
 - Effects can last for 72 hours.
 - Dicyclomine is another anticholinergic that can be used for motion sickness.
 - Meclizine, diphenhydramine, cinnarazine, and promethazine are all antihistamines that treat motion sickness. They are not as effective as hyoscine. (Options A, B, and C ruled out)
 - Ephedrine may be added to the anti-motion sickness regimen to counteract sedative properties.
 - Acetazolamide, a carbonic anhydrase inhibitor, is used as the DOC for prophylaxis of acute mountain sickness. (Option D)
-

Solution for Question 3:

Correct Answer: C) Hyoscine does not cause sedation

Explanation:

- Hyoscine, also known as scopolamine, is a medication with anticholinergic properties. Its sedative effects are primarily due to its ability to block acetylcholine action, a neurotransmitter in the brain. (Option C).
 - By inhibiting the effects of acetylcholine, hyoscine disrupts certain neuronal pathways responsible for arousal and wakefulness. For this reason, hyoscine was one of the first drugs to be used as a truth serum (Option B ruled out).
 - Atropine causes delirium by blocking the action of acetylcholine, a neurotransmitter crucial for cognitive function and memory. (Option A ruled out).
 - Atropine decreases the parasympathetic influence on the heart by reducing the effects of ACh, leading to an increase in heart rate and improving conduction through the AV node by reducing its refractory period (Option D ruled out).
-

Solution for Question 4:

Correct Answer: D) Promethazine

Explanation:

Drug-induced parkinsonism can result from medications blocking D2 receptors in the basal ganglia (Eg. Haloperidol, Risperidone, Olanzapine, Metoclopramide, Cinnarizine). In the treatment of drug-induced parkinsonism, levodopa is not suitable. Instead, reducing acetylcholine action with anticholinergic and antihistaminic drugs is preferred.

- Promethazine and diphenhydramine are first-generation antihistamines that act on the basal ganglia and have anticholinergic properties. However, they are still classified as antihistamine agents. They are used for their antiemetic and anti-motion sickness properties. (Option D)
 - Benzhexol (trihexyphenidyl), benzotropine, and bipyridine are anticholinergic agents acting on the basal ganglia used to treat Parkinson's and drug-induced extrapyramidal symptoms. (Options A, B, and C ruled out)
-

Solution for Question 5:

Correct Answer: C) It primarily acts on peripheral muscarinic receptors

Explanation:

Glycopyrrolate is an anticholinergic drug that does not cross the blood-brain barrier as it is a quaternary amine (Option A ruled out). This means it has minimal penetration into the central nervous system (CNS), resulting in fewer CNS side effects.

- It primarily exerts its effects peripherally, targeting receptors in organs like the salivary glands, gastrointestinal tract, and respiratory system (Option C).
- By selectively acting on peripheral muscarinic receptors, glycopyrrolate can reduce secretions without causing significant CNS effects like sedation or cognitive impairment. This helps in symptom control in patients with COPD (Option B ruled out) and is useful in procedures such as endotracheal intubation.
- It is a long-acting agent requiring less number of doses to reach therapeutic effect (Option D ruled out)

Solution for Question 6:

Correct Answer: A) Tiotropium

Explanation:

This patient with moderate COPD (GOLD stage 2) who is experiencing frequent symptoms and exacerbations despite the use of a short-acting beta agonist (SABA) would benefit from the addition of a long-acting bronchodilator like Tiotropium, a long-acting muscarinic antagonist (LAMA).

- Anticholinergics are important in the treatment of chronic obstructive pulmonary disease (COPD), but they are less effective in most patients with asthma.
- Ipratropium, tiotropium, aclidinium, and umeclidinium are quaternary ammonium LAMAs used for their effects on the respiratory tract. They block acetylcholine action on muscarinic receptors, relaxing airways and improving airflow.

Drug	Dosage frequency	Route
Ipratropium	Four times daily	Metered-dose inhaler or nebuliser
Aclidinium	Twice daily	Dry powder inhaler
Tiotropium	Once-daily	Dry powder inhaler (for maintenance therapy of moderate-to-severe disease)
Umeclidinium		

- In COPD treatment, muscarinic antagonists are often used with inhaled long-acting β_2 adrenergic receptor agonists. Olodaterol (very long-acting β_2 adrenergic receptor agonist) is combined with tiotropium bromide to treat COPD.

Atropine (Option B) treats bradycardia and OPC poisoning to counteract muscarinic symptoms.

Hyoscine (Scopolamine) (Option C) is the preferred treatment for motion sickness. It is administered as hyoscine hydrobromide via a transdermal patch, which delivers approximately 0.5 mg of scopolamine over 72 hours.

Trimipramine (Option D) is a tricyclic antidepressant with anticholinergic properties, but it is not indicated for COPD management.

Solution for Question 7:

Correct Answer: B) Tropicamide

Explanation:

Tropicamide is a short-acting anticholinergic drug with a fast onset. While it is a less reliable cycloplegic, it is suitable for refraction testing in adults and as a short-acting mydriatic for fundoscopy.

Anticholinergic drugs block the effects of acetylcholine on the pupillary sphincter muscle, causing mydriasis, and the ciliary muscle, thereby paralyzing accommodation (cycloplegia), fixing the lens for far vision and blurring near vision.

Anticholinergic Drug	Characteristics	Therapeutic Uses
Atropine (Option A ruled out)	Long-lasting mydriasis and cycloplegia (3-5 days); hence not used in adults	Used in treating iritis, iridocyclitis, choroiditis, keratitis, and corneal ulcer Helps to decrease intraocular muscles & reduce spasms
Homatropine	Semi-synthetic atropine derivative, shorter duration than atropine	Used in the treatment of uveitis
Cyclopentolate Hydrochloride	Potent, rapid-acting	Used for refractive error testing Used in the treatment of uveitis Preferred cycloplegic in children (may cause transient behavioral changes in children due to systemic absorption)
Tropicamide (Option B)	Short-acting, fast onset, less reliable cycloplegic	Preferred cycloplegic in adults Used as a short-acting mydriatic for fundoscopy
Hyoscine	Similar cycloplegic action to atropine, more potent mydriatic, long duration but shorter than atropine	Useful for relieving pain in iridocyclitis due to ciliary muscle spasm

Pilocarpine (Option C): is a cholinergic agent that causes miosis (constriction of the pupil), not mydriasis.

Timolol (Option D): is used to lower intraocular pressure and is not used for pupil dilation.

Solution for Question 8:

Correct Answer: B) Trospium

Explanation:

Trospium, an anticholinergic medication, is commonly used to manage overactive bladder (OAB) symptoms in Alzheimer's patients. This medication is particularly suitable for those with Alzheimer's disease and the elderly due to its reduced CNS penetration, which minimizes the risk of cognitive side effects compared to other anticholinergics like oxybutynin.

ANS drugs acting on the urinary bladder:

i) Parasympathetic Drugs:

CHOLINERGIC DRUGS	ANTICHOLINERGIC DRUGS
Action: Detrusor contraction	Action: Block M3 receptors → Detrusor relaxation
Bethanechol Neostigmine	Non Selective M3 blockers: Trospium Oxybutynin (Option A) Fesoterodine Flavoxate Selective M3 Blockers: Solifenacin (Option C) Darifenacin Tolterodine
Uses: Bladder atony (overflow incontinence)	Uses: OAB (urge incontinence) Detrusor overactivity Increased bladder capacity Reduced frequency of bladder contractions Suppression of the urge to urinate

ii) Sympathetic drugs:

Beta 3 agonists	Alpha 1 activator
Site: Detrusor Action: Relaxation	Site: Internal urinary sphincter Action: Sphincter contraction
Mirabegron (Option D) Vibegron	Duloxetine (SNRI): Increased epinephrine availability → ■1 stimulation

Uses: Urge incontinence	Uses: Stress incontinence Urethral hypermobility or intrinsic sphincter dysfunction causes leaks with increased intra-abdominal pressure
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Solution for Question 9:

Correct Answer: B) Scopolamine

Explanation:

Scopolamine is an anticholinergic medication with antispasmodic effects, making it suitable for treating abdominal cramping and spasms linked to conditions like irritable bowel syndrome (IBS).

Scopolamine, by blocking muscarinic receptors in the GI tract, causes:

- Reduced Tone and Motility: This slows down the passage of gastrointestinal contents.
- Antispasmodic Effect: Helps to alleviate spasms and cramps by reducing the tone and contractions of the gastrointestinal tract.

Other antispasmodic anticholinergic drugs:

Telenzepine (Option A) and pirenzepine are selective M1 receptor antagonists used to reduce HCl production in peptic ulcer disease.

Hexamethonium (Option C) is a ganglionic blocker that inhibits the transmission of parasympathetic nerve impulses.

Thiopentone (Option D) is an ultra-short-acting barbiturate used as a general anesthetic. It is used in the management of refractory status epilepticus and is the drug of choice for narco analysis.

Solution for Question 10:

Correct Answer: D) Orthostatic hypotension

Explanation:

Based on the clinical scenario, the patient likely has Tourette syndrome, characterized by repetitive tics starting in childhood. Mecamylamine, a ganglionic blocker used for managing tics, can cause side effects such as orthostatic hypotension, tachycardia, urinary retention, dry mouth, and constipation.

Ganglion blockers inhibit the transmission of nerve impulses through autonomic ganglia and act primarily by blocking nicotinic acetylcholine receptors (nAChRs), which are ligand-gated ion channels found on the surface of ganglion cells.

GANGLION BLOCKERS	
Persistent depolarising blockers	Nicotine (Large doses) Anticholinesterases (Large doses)
Competitive blockers	Trimethaphan: Competes with Ach at NMJ. Analogous to the MOA of curare at the NMJ Used in hypertension related to a dissecting aortic aneurysm, as it helps block the sympathetically mediated increase in ejection velocity, which is beneficial
Channel blockers (Blocks Nn receptor after the channel opens, shortening the current flow duration by occluding or closing the channel)	Hexamethonium: Used to treat hypertensive emergencies (no longer in use) Mecamylamine: Used in nicotine dependence to reduce withdrawal symptoms and cravings (Used in adjunct with nicotine) Used in Mx of Tourette syndrome As an antihypertensive for moderate to severe HTN

Ganglion blockers inhibit both sympathetic and parasympathetic ganglia, affecting organs based on the dominant control of the autonomic nervous system.

EFFECTS ON GANGLION BLOCKADE ON ORGAN SYSTEMS		
Organ System	Effect of blockade	Blockade of the dominant tone
CVS (Option A ruled out) (Option D)	Heart: Tachycardia	Parasympathetic blockade
Blood vessels: Vasodilatation, postural hypotension due to decreased venous return	Sympathetic blockade	
GIT (Option C ruled out)	Decreased motility, constipation, and dry mouth	Parasympathetic blockade
Urinary bladder (Option B ruled out)	Urinary retention	Parasympathetic blockade
Eye	Mydriasis (pupil dilation) and cycloplegia (paralysis of accommodation).	Parasympathetic blockade

Sweat glands	Anhydrosis (decreased sweating)	Sympathetic blockade
Male sexual function	Inability to maintain erection Inhibition of ejaculation	Sympathetic and Parasympathetic blockade

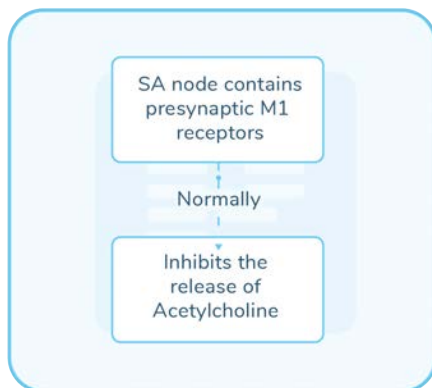
Solution for Question 11:

Correct Answer: C) Initial bradycardia followed by tachycardia

Explanation:

Effects of atropine on heart rate:

- Atropine when administered parenterally causes initial bradycardia followed by tachycardia.
(Option A,B and D ruled out)



- Therefore, stimulation of M1 receptors decreases the release of ACh and results in tachycardia, and inhibition of M1 receptors increases the release of ACh and results in bradycardia.
- Atropine initially blocks the M1 receptors causing an increase in ACh which in turn causes initial bradycardia.

As the atropine concentration increases, it blocks M2 receptors on the SA node which leads to a decrease in ACh causing tachycardia.

Solution for Question 12:

Correct Answer: B) Phenylephrine

Explanation:

For the patient mentioned above with a past medical history of open-angle glaucoma, the mydriatic of choice is phenylephrine.

Mydriatics and cycloplegics:

Drug	Indications	Ocular side effects
Atropine	Cycloplegia, mydriasis, iritis, uveitis	Photosensitivity, blurred vision, open-angle glaucoma
Homatropine		
Cyclopentolate	Cycloplegia, mydriasis	Photosensitivity, blurred vision, open-angle glaucoma
Phenylephrine	Mydriasis, vasoconstriction, uveitis, glaucoma	Acts for several hours

- Phenylephrine is an alpha-1 adrenergic agonist, causing pupil dilation without increasing the intraocular pressure (Option B).
 - Anticholinergic drugs such as atropine, homatropine, and cyclopentolate are potent mydriatics and cycloplegics that significantly raise the IOP (Options A, C & D ruled out).
 - Therefore, in patients with Open-angle glaucoma, anticholinergics are contraindicated.
-

Sympathomimetics

1. Which of the following is not classified as a vasopressor among the sympathomimetic drugs?

- A. Midodrine
 - B. Ephedrine
 - C. Phenylephrine
 - D. Dobutamine
-

2. A 45-year-old female with a BMI of 33 kg/m², currently on olanzapine, seeks treatment for obesity. The physician discusses various pharmacological options. Which newer approved anti-obesity drug might be effective for this patient, considering her olanzapine therapy?

- A. Rimonabant
 - B. Sibutramine
 - C. Samidorphan
 - D. Fenfluramine
-

3. Which anti-obesity drugs work primarily by inhibiting pancreatic lipase, thereby reducing fat absorption in the gastrointestinal tract?

- A. Orlistat
- B. Phentermine
- C. Liraglutide
- D. Bupropion

4. A 55-year-old male patient with a history of depression is being treated with phenelzine, a monoamine oxidase inhibitor (MAOI). He presents to the emergency department with a sudden onset of severe headache, palpitations, hypertension (blood pressure of 200/110 mmHg), and sweating. He reports consuming a large amount of aged cheese at a party earlier that evening. Which is the drug of choice for treating this patient's condition?

- A. Phentolamine
- B. Labetalol
- C. Hydralazine
- D. Captopril

5. A 32-week pregnant female is suspected of having premature labour in the next few days. Which of the following drugs can be administered to prevent the same?

- A. Phenylephrine
- B. Ritodrine
- C. Isoprenaline
- D. Methoxamine

6. A 7-year-old child is brought to the clinician due to difficulties in concentrating, impulsivity, and excessive fidgeting at home and in school. After a thorough assessment, the child is diagnosed with Attention Deficit Hyperactivity Disorder (ADHD). Which of the following medications would not be appropriate for the management of this child's symptoms?

- A. Methylphenidate

- B. Atomoxetine
 - C. Clonidine
 - D. Lorazepam
-

7. A 19-year-old college student is brought to the emergency after ingesting unknown pills night before his final exam. On examination, he presents with agitation, hyperthermia, and severe hypertension. His pupils are dilated, and urine catheterisation reveals dark urine. Which of the following is next best step for this patient?

- A. Administer intravenous sodium bicarbonate
 - B. Administer flumazenil
 - C. Administer ammonium chloride
 - D. Administer naloxone
-

8. A 25-year-old male presents to the clinic with complaints of excessive daytime sleepiness with sudden episodes of muscle weakness triggered by emotions. He reports that these symptoms have been significantly impacting his work performance. Which of the following is the drug of choice for treating this condition?

- A. Zolpidem
 - B. Modafinil
 - C. Lorazepam
 - D. Propranolol
-

9. A 60-year-old female presents to the clinic with complaints of frequent urination and incontinence. She has been diagnosed with overactive bladder and has tried lifestyle modifications with limited success. Her physician decides to start medication for her symptoms; it will act on which of the following receptors?

- A. Beta-3-agonist- Mirabegron
 - B. Beta-2-agonist- Terbutaline
 - C. Alpha-1-agonist- Phenylephrine
 - D. Alpha-1-antagonist- Tamsulosin
-

10. A 45-year-old male with a history of chronic sinusitis presents with complaints of persistent anosmia and nasal discharge. He used an over-the-counter nasal decongestant daily for the past three months. Which of the following statement is false regarding the treatment of the this condition? It results from abruptly stopping the use of oxymetazoline nasal spray. Long-term use of nasal decongestants is recommended to prevent this condition. It is caused by ischemic atrophy of the nasal mucosa and olfactory nerve injury. These drugs act on the alpha-adrenergic receptor, inducing vasodilation in the nose. Oxymetazoline can also be used to treat cases of adult rosacea.

- A. 2, 3, and 4
 - B. 2, 3 and 5
 - C. 2, 3, 4 and 5
 - D. 2 and 3
-

11. A 5-year-old child is brought to the ophthalmologist for a routine eye examination. Which of the following drugs is most appropriate for pupillary dilation to assess the retina in this child?

- A. Phenylephrine
- B. Atropine

- C. Pilocarpine
 - D. Cyclopentolate
-

12. Which of the following diseases can be treated with a drug that inhibits vesicular monoamine transporters (VMAT)?

- A. Huntington's chorea
 - B. Mononeuritis multiplex
 - C. Parkinson's disease
 - D. Lewy body dementia
-

13. Which of the following statements about Dobutamine is correct?

- A. It is administered as an I.M injection over the lateral aspect of the thigh usually.
 - B. It acts mainly on α adrenergic receptors, and is a weak β agonist.
 - C. Is an inotropic agent that increases cardiac output and is used in stress tests.
 - D. Cannot be used in cardiogenic shock without hypotension.
-

14. You are an intern assisting in an operating theatre. During surgery, the patient suddenly goes into shock, showing signs of refractory hypotension. The surgeon asks you to prepare dopamine for infusion. Which of the following dosage ranges would be appropriate for the situation?

- A. 0-2 mcg/kg/min
- B. 50-100 mcg/kg/min
- C. 10-50 mcg/kg/min
- D. 2-10 mcg/kg/min

15. Which of the following is a false statement regarding adrenaline?

- A. Adrenaline can be given in dilution with an anaesthetic in distal phalanges.
 - B. Adrenaline is contraindicated in patients with severe hypertension.
 - C. Adrenaline is commonly used in the management of anaphylactic shock.
 - D. Adrenaline is recommended for use in patients with Ventricular fibrillation.
-

16. A 45-year-old male is brought to the emergency department by paramedics after a severe anaphylactic reaction to a bee sting. The patient is in obvious distress, with signs of airway swelling, difficulty breathing, and low blood pressure. The emergency team prepares to administer adrenaline to stabilise the patient. What is the correct dilution and route of administration for adrenaline in this situation?

- A. 1:1,00,000 dilution, subcutaneous injection
 - B. 1:10,000 dilution, intravenous infusion
 - C. 1:1000 dilution, intramuscular injection
 - D. 1:100 dilution, intradermal injection
-

17. A 60-year-old male patient is brought to the emergency department after a car accident. He presents with hypotension, bradycardia, and signs of shock. The medical team decides to administer norepinephrine (NE) to stabilize his blood pressure. You are an intern monitoring the patient's blood pressure and heart rate continuously. Which of the following tracings most accurately represents the effect of norepinephrine on this patient's cardiovascular system?

- A. Increased systolic blood pressure with minimal change in diastolic blood pressure.

- B. Increased systolic and diastolic blood pressure with increased heart rate.
- C. Increased systolic and diastolic blood pressure with decreased heart rate.
- D. Decreased systolic and diastolic blood pressure with increased heart rate.

18. Match the following conditions with the most appropriate drug of choice.
Condition Drug of choice
1 Septic shock i Epinephrine
2 Malignant hypertension with end-organ damage ii Dopamine
3 Acute decompensated heart failure with oliguria iii Fenodolpam
4 Cardiac arrest iv Norepinephrine

	Condition		Drug of choice
1	Septic shock	i	Epinephrine
2	Malignant hypertension with end-organ damage	ii	Dopamine
3	Acute decompensated heart failure with oliguria	iii	Fenodolpam
4	Cardiac arrest	iv	Norepinephrine

- A. 1-iv, 2-iii, 3-ii, 4-i
- B. 1-iv, 2-iii, 3-i, 4-ii
- C. 1-iii, 2-i, 3-iv, 4-ii
- D. 1-i, 2-ii, 3-iii, 4-iv

19. β_2 receptors are present in all of the following organs except?

- A. Blood vessels.
- B. Uterus.
- C. Urinary bladder.
- D. Bronchi.

20. A patient is diagnosed with pheochromocytoma. Which drug can help improve the patient's symptoms by inhibiting tyrosine hydroxylase?

- A. Methyldopa
 - B. Clonidine
 - C. Metyrosine
 - D. Phenoxybenzamine
-

21. A patient is diagnosed with benign prostatic hyperplasia (BPH). He is prescribed tamsulosin. Which of the following receptors does this drug act on?

- A. $\alpha 1A$
 - B. $\alpha 1B$
 - C. $\alpha 2A$
 - D. $\alpha 2C$
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	1
Question 4	1
Question 5	2
Question 6	4
Question 7	3
Question 8	2
Question 9	1
Question 10	1
Question 11	2
Question 12	1
Question 13	3
Question 14	3
Question 15	1
Question 16	3
Question 17	3
Question 18	1
Question 19	3
Question 20	3
Question 21	1

Solution for Question 1:

CORRECT ANSWER: D) Dobutamine

EXPLANATION:

Dobutamine is a sympathomimetic drug that primarily acts on β_1 adrenergic receptors, enhancing myocardial contractility and leading to increased cardiac output. It is primarily used as an inotrope to support cardiac function in heart failure rather than as a vasopressor.

Vasopressors:

Vasopressors	Uses
Selective α_1 agonist	
Phenylephrine	DOC: Hypotension in aortic dissection surgeries To treat hypotension as a vasopressor in spinal anesthesia Mydriasis without cycloplegia
Midodrine	DOC: Postural hypotension
$\alpha+\beta$ receptors	
Ephedrine	Hypotension in pregnancy. Phenylephrine is the preferred agent for hypotension in pregnancy.

Solution for Question 2:

Correct Answer: C) Samidorphan

Explanation:

- Olanzapine is an atypical antipsychotic that can cause weight gain, partly due to its interaction with opioid receptors in the brain, leading to increased appetite and food intake.
- Samidorphan acts as an opioid receptor antagonist, blocking the effects of olanzapine on these receptors.
- By counteracting the appetite-stimulating effects of olanzapine, samidorphan can potentially mitigate or reduce the weight gain associated with olanzapine therapy.

DRUG	MOA	STATUS
Liraglutide	GLP-1 receptor agonist that enhances satiety and slows gastric emptying, leading to weight loss also used to treat type 2 diabetes	Approved (Newer anti-obesity drug)
Samidorphan (Option C)	Opioid Receptor antagonist blocks the effects of olanzapine on the receptors and hence reduce the weight gain associated with Olanzapine	Approved (Newer anti-obesity drug)
Setmelanotide	Selective agonist (activator) of the MC4R Decreases appetite and increases the feeling of fullness (satiety) Increases energy expenditure and metabolism Promotes weight loss	(Newer anti-obesity drug)
Rimonabant (Option A)	CB1 receptor antagonist; withdrawn due to psychiatric side effects.	Banned
Sibutramine (Option B)	Norepinephrine and serotonin reuptake inhibitor Increase sympathetic activity by increasing amine levels in the brain.	Banned (due to cardiotoxicity) (Withdrawn due to heart valve damage)
Fenfluramine (Option D)		

Solution for Question 3:

Correct answer: A) Orlistat

Explanation:

Category	Drugs
Inhibitors of Lipase	Orlistat (Option A)
Glucagon-like peptide-1 (GLP-1) receptor agonist (Suppress appetite)	Liraglutide (Option C),
Sympathomimetic amine (Suppress appetite)	Amphetamine, Phentermine (Option B)
Stimulators of Lipase	Mirabegron

Opioid antagonists (Early satiety)	Topiramate, Naltrexone
Norepinephrine-dopamine reuptake inhibitor (NDRI)	Bupropion (Option D)

Solution for Question 4:

Correct answer: A) Phentolamine

Explanation: The patient is experiencing a hypertensive crisis, which is a known potential adverse effect of consuming foods containing tyramine (aged cheese) while taking MAOI (phenelzine).

CHEESE REACTION	
Cause and Mechanism	The "cheese reaction" occurs when patients on MAOIs, such as phenelzine, consume foods high in tyramine (e.g., aged cheese). Tyramine is normally metabolized by monoamine oxidase in the gut. However, MAOIs inhibit this enzyme, leading to high levels of tyramine. Tyramine then enters the circulation and displaces norepinephrine from storage vesicles in adrenergic neurons, resulting in a sudden and excessive release of norepinephrine. This leads to severe hypertension, headache, palpitations, and other symptoms.
Treatment	Phentolamine, a non-specific reversible alpha-adrenergic blocker, is used to treat hypertensive emergencies caused by the cheese reaction. It counteracts the excessive effects of norepinephrine by blocking alpha-adrenergic receptors, thereby lowering blood pressure.

Solution for Question 5:

CORRECT ANSWER: B) Ritodrine

EXPLANATION:

Beta-2 adrenergic receptors are present in the uterus and smooth muscles, which relax them upon stimulation. Ritodrine (option B) is a selective beta-2 receptor agonist; hence, it can be used in this case to prevent premature labor by relaxing the uterus.

Options A, C, and D: Phenylephrine and methoxamine are alpha-adrenergic agonists and, hence, have no therapeutic benefit in premature labor. Isoprenaline is a non-specific beta-agonist, which causes side effects like tachycardia and palpitations when used as a tocolytic. It is less commonly employed now due to the availability of more selective beta-2 agonists that have fewer systemic side effects and are more effective.

Solution for Question 6:

CORRECT ANSWER: D) Lorazepam

EXPLANATION:

Lorazepam is a short-acting benzodiazepine that is not used to treat ADHD in children. It is used in insomnia, anxiety, and status epilepticus.

ADHD (Attention Deficit Hyperkinetic Disorder):

- Mostly seen in children < 12 years.
- Symptoms: Excessive motor activity, poor attention span (often fail at school), and impulsiveness.

Treatment of ADHD	
Treatment of Choice	Cognitive behavioral therapy (CBT)
Medical Management (if CBT is not effective)	
Stimulants	Non-stimulants
Increase sympathetic activity. Increase attention span and academic performance. Cause hyperkinesia/tics. Rapid onset and short duration.	Do not produce excessive movements/tics. Delayed onset. Less abuse potential.
Methylphenidate: (Option A) Drug of choice Increase brain catecholamines.	Clonidine, Guanfacine: (Option C) Treatment in ADHD with tics Atomoxetine-SNRI

Dexamphetamine Lisdexamfetamine Dextroamphetamine	CNS STIMULANTS(methylphenidate)- not preferred in children less than 7 years of age instead atomoxetine is preferred here(antidepressant) (Option B)
Side effects: Insomnia, abdominal pain, and weight loss.	Side effects: Insomnia, dry mouth and decreased appetite.

Solution for Question 7:

CORRECT ANSWER: C) Administer ammonium chloride

EXPLANATION:

This young student with signs of impulsiveness, hypertension, psychosis, and dark urine suggests amphetamine abuse due to exam performance pressure. Ammonium chloride is used to promote urinary drug excretion.

Amphetamine Poisoning:

- Abused as “performance-enhancing drugs” among students and sportspersons.
- Also called “rave party drugs”: Ecstasy/crystal/ methamphetamine/MDMA.
- Sympathomimetic toxidrome: Increase heart rate, respiration, temperature, bowel sounds, sweating and pupil dilatation.
- Triad: Sudden psychosis + Hypertension + Golden coloured urine.
- Treatment: Ammonium chloride turns urine pH acidic and causes forced diuresis of basic drugs like amphetamines. Chlorpromazine: treats psychosis and hypertension through alpha-blocking action.
- Ammonium chloride turns urine pH acidic and causes forced diuresis of basic drugs like amphetamines.
- Chlorpromazine: treats psychosis and hypertension through alpha-blocking action.
- Ammonium chloride turns urine pH acidic and causes forced diuresis of basic drugs like amphetamines.
- Chlorpromazine: treats psychosis and hypertension through alpha-blocking action.

Intravenous sodium bicarbonate (Option A): is used for urinary alkalization in cases of salicylate poisoning (weak acid), which manifests as tinnitus, nausea, and vomiting.

Flumazenil (Option B): is used to reverse the effects of benzodiazepine poisoning, which appears with symptoms of CNS depression and impaired mental status.

Naloxone (Option D): is an opioid antagonist used to treat opioid overdoses, which present as pinpoint pupils and respiratory depression.

Solution for Question 8:

CORRECT ANSWER: B) Modafinil

EXPLANATION:

Modafinil:

- Mechanism of action: Sympathetic CNS stimulant.
- DOC for daytime sleepiness. Occurs in: OSA (obstructive sleep apnea): AR-modafinil is used Narcolepsy.
- Occurs in: OSA (obstructive sleep apnea): AR-modafinil is used Narcolepsy.
- OSA (obstructive sleep apnea): AR-modafinil is used
- Narcolepsy.
- Occurs in: OSA (obstructive sleep apnea): AR-modafinil is used Narcolepsy.
- OSA (obstructive sleep apnea): AR-modafinil is used
- Narcolepsy.
- OSA (obstructive sleep apnea): AR-modafinil is used
- Narcolepsy.

Drugs approved for treatment of narcolepsy:

Zolpidem (Option A): Atypical benzodiazepine, sedative-hypnotic used for insomnia, not for treating narcolepsy.

Lorazepam (Option C): ■■■■benzodiazepine used for anxiety and sleep disorders but not specifically for narcolepsy.

Propranolol (Option D): beta-blocker used to manage hypertension and anxiety.

Solution for Question 9:

CORRECT ANSWER: A) Beta-3-agonist- Mirabegron

EXPLANATION:

Beta-3 agonists (mirabegron) are the preferred medication for people with urine incontinence due to overactive bladder.

Beta-3-agonists:

Beta-3 receptors are located in the urinary bladder and white & brown adipose tissues.

Medication	Sibutramine	Mirabegron/Vebegron
Mechanism	Promote lipolysis	Relax detrusor muscles (increase nitric oxide). Inhibits micturition
Use	Anti-obesity drug	Overactive bladder Urinary incontinence
Side effects	Cardiotoxic	Increase blood pressure UTI CYP2D6 inhibitor

Beta-2-agonist- Terbutaline (Option B) is used as a tocolytic to relax uterine smooth muscle and delay premature labor.

Alpha-1-agonist- Phenylephrine (Option A) is used as a nasal decongestant, mydriatic agent for eye exams, and as a vasopressor to treat hypotension.

Alpha-1-antagonist- Tamsulosin (Option D) is primarily used for BPH. It is a selective alpha-1A antagonist, which helps improve urinary symptoms with minimal effect on blood pressure.

Solution for Question 10:

CORRECT ANSWER: A) 2, 3, and 4

EXPLANATION: This patient's symptoms suggest rebound rhinorrhea as a result of Rhinitis Medicamentosa due to chronic nasal decongestant use.

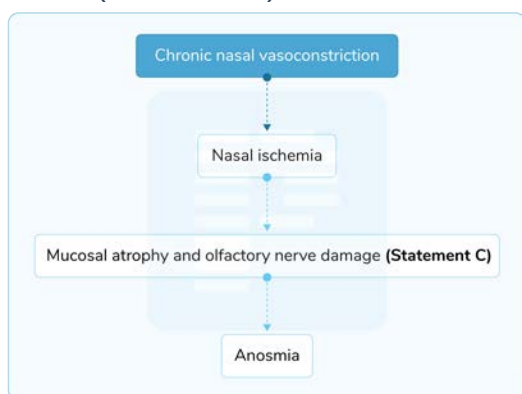
Nasal decongestants:

- MOA: Reduce rhinorrhea through α_1 -receptor-mediated vasoconstriction in the nose. (Statement D)

- Types:

Nasal drops	Oral medication
Xylometazoline Naphazoline Phenylephrine Oxymetazoline (otrivin) Also used to treat adult rosacea (Statement E)	Ephedrine Pseudoephedrine Phenylephrine

- Rhinitis Medicamentosa: Rebound rhinorrhea due to sudden stoppage after long-term use of nasal decongestants. (Statement A)
- Rebound rhinorrhea due to sudden stoppage after long-term use of nasal decongestants. (Statement A)
- Rebound rhinorrhea due to sudden stoppage after long-term use of nasal decongestants. (Statement A)



Treatment:

- Short course of Glucocorticoids nasal spray
- Avoid chronic nasal decongestant use (Statement B)

Solution for Question 11:

Correct answer: B) Atropine

Explanation:

Atropine is the drug of choice for fundoscopy in children.

Mydriatics (dilatation of pupil):

Drug	Action	Duration and side effects
Atropine 1% (Option B)	Drug of choice in children Blocks muscarinic M3 receptors (overactive in children) Cause circular muscle paralysis (Passive mydriasis) Use: mydriasis, cycloplegia, keratitis, iridocyclitis	Long-acting (>1 week) Photosensitivity
Phenylephrine 2.5-10% (Option A)	Drug of choice in adults Stimulate alpha-1 receptors Cause contraction of radial muscles (Active mydriasis) Not used as cycloplegic	Last several hours Increases blood pressure
Cyclopentolate 0.5-2% (Option D)	Same as atropine Use: iritis, uveitis, cycloplegia	Lasts one day Transient behavioural abnormalities in children
Tropicamide 0.5-1%	Commonly used for office exams	Effect for several hours

Note: Pilocarpine (Option C) cause active miosis through stimulation of M3 receptors.

Solution for Question 12:

Correct answer: A) Huntington's chorea

EXPLANATION:

Movement disorders like Huntington's chorea and tardive dyskinesia can be treated with drugs that inhibit vesicular monoamine transporters (VMAT).

- As the name suggests, VMATs are transporters of monoamines, especially catecholamines (adrenaline, noradrenaline, and dopamine), within storage vesicles.
- They are mainly of three types - VMAT 1, VMAT 2 & VAcHT.

Centrally, this process mainly involves the release of dopamine, which is abnormal in movement disorders involving excessive dopaminergic activity.	Peripherally, this process occurs at the synaptic terminals, which are responsible for maintaining blood pressure; hence, these drugs were once used to manage hypertension.
Inhibitors: Tetrabenazine, deutetrabenazine, valbenazine	Inhibitor: Reserpine
These drugs selectively inhibit VMAT-1	This drug inhibits both VMAT-1 & VMAT-2

Solution for Question 13:

Correct answer: C) Is an inotropic agent that increases cardiac output and is used in stress tests.

Explanation:

Dobutamine:

Mechanism	It is a sympathomimetic amine. It is mainly on β_1 adrenergic receptors and is a weak α agonist. (Option B) Administered at 2- 8 $\mu\text{g/kg/min}$ i. v. infusion. (Option A)
Uses	Acute heart failure. Cardiogenic shock without hypotension (DOC). (Option D) Stress Echocardiography. (Option C) Bridge therapy for heart transplant patients.

Solution for Question 14:

CORRECT ANSWER: C) 10-50 mcg/kg/min

EXPLANATION:

Dosage Range	Clinical Use	Mechanism of Action
0-2 mcg/kg /min	Renal perfusion improvement (oliguria)	Stimulates D1 receptors primarily in renal vasculature, leading to

		vasodilation and increased renal blood flow
2-10 mcg/kg/min	Acute Congestive heart failure	Stimulates β_1 receptors, leading to increased cardiac output and blood pressure
10-50 mcg/kg/min	Moderate to severe hypotension/shock	Predominantly acts on α_1 receptors, causing potent vasoconstriction

Solution for Question 15:

CORRECT ANSWER: A) Adrenaline can be given in dilution with an anesthetic in distal phalanges.

EXPLANATION:

Adrenaline cannot be given in dilution with an anesthetic in distal phalanges because, when used in high concentration or in combination with local anesthetics, it can cause vasoconstriction, leading to decreased blood supply, which may result in tissue necrosis, especially in areas with limited blood flow like the distal phalanges.

Adrenaline is contraindicated in patients with severe hypertension (Option B)

- It can exacerbate hypertension by increasing heart rate and constricting blood vessels.
- Therefore, it is generally contraindicated in patients with severe hypertension.

Adrenaline is commonly used in the management of anaphylactic shock. The dose is 1:1,000 dilution, intramuscular injection (Option C)

- It is the first-line treatment for anaphylactic shock because it can rapidly reverse severe allergic reactions by relieving airway constriction, increasing blood pressure, and reducing swelling and hives.

Adrenaline is recommended for use in patients with Ventricular fibrillation (Option D)

- Adrenaline is recommended for use in patients with ventricular fibrillation as part of advanced cardiac life support (ACLS) protocols.
- It helps to increase myocardial contractility and coronary perfusion pressure, thereby improving the chances of successful defibrillation.

Solution for Question 16:

CORRECT ANSWER: C) 1:1,000 dilution, intramuscular injection

EXPLANATION:

- Adrenaline (epinephrine) is the first-line treatment for severe anaphylactic reactions. In this scenario, the correct dilution of adrenaline for administration is 1:1,000.
- In severe cases, the preferred route is an intramuscular injection (on the anterolateral aspect of the thigh); it ensures rapid absorption and onset of action, crucial in managing anaphylaxis.

Uses of epinephrine:

Condition	Use	Dilution/Concentration	Route of Administration
Anaphylaxis	First-line treatment for severe allergic reactions	1:1000 (1 mg/mL)	Intramuscular (IM)
Cardiac arrest	To restore cardiac rhythm and blood pressure	1:10,000 (0.1 mg/mL)	Intravenous (IV) Intraosseous (IO)
Asthma	Treatment of acute severe asthma attacks	1:1000 (1 mg/mL)	Inhalation via nebulizer Intramuscular (IM) subcutaneous (SC)
Hypotension	To increase blood pressure	1:100,000 (0.01 mg/mL)	Intravenous (IV) infusion
Local anesthesia	Enhance duration and intensity of local anesthetic effect	1:200,000 to 1:100,000 (0.005 to 0.01 mg/mL)	Local injection

Solution for Question 17:

CORRECT ANSWER: C) Increased systolic and diastolic blood pressure with decreased heart rate

EXPLANATION:

NE stimulates α_1 receptors, leading to vasoconstriction and a subsequent increase in both systolic and diastolic BP, causing vasoconstriction and increased peripheral resistance, leading to increased diastolic BP.

- NE also acts on β_1 -receptors in the heart, which can increase heart rate and contractility, increasing systolic BP.
- However, the baroreceptor reflex may counteract this effect, resulting in a potential decrease in heart rate despite the direct β_1 action.
- NE has minimal to zero action on β_2 -receptors, primarily responsible for vasodilation.

Norepinephrine:

Description	Norepinephrine, or noradrenaline, is a sympathomimetic drug used primarily as a vasopressor.
Mechanism of Action (MOA)	It acts on α_1 , β_1 adrenergic receptors and has some affinity for alpha-2 receptors. Activation of α_1 receptors causes vasoconstriction, increasing peripheral resistance and blood pressure. Activation of β_1 receptors increases heart rate and myocardial contractility.
Uses	Treatment of acute hypotension (e.g., septic shock, cardiogenic shock). Cardiac arrest.
Side effects	Cardiovascular: Arrhythmias, bradycardia, hypertension. CNS: Anxiety, headache, tremor. Gastrointestinal: Nausea, vomiting. Respiratory: Respiratory difficulty, dyspnea. Metabolic: Hyperglycemia, lactic acidosis.
Contraindications	Peripheral vascular thrombosis (risk of ischemia). During anesthesia with cyclopropane or halothane (due to the risk of arrhythmias). Hypotension is due to blood volume deficits (hypovolemia) until an adequate fluid replacement.

Solution for Question 18:

Correct answer: A) 1-iv, 2-iii, 3-ii, 4-i

EXPLANATION:

Condition	Drug of choice
Septic shock	Norepinephrine
Malignant hypertension with end-organ damage	Fenodolpam
Acute renal failure	Dopamine
Cardiac arrest	Epinephrine

Uses: Norepinephrine acts on alpha1, alpha2, beta1

Epinephrine acts on alpha1, alpha2, beta1, beta2

Norepinephrine	Fenodolpam
It acts on both alpha and beta receptors. DOC for cardiogenic shock with hypotension. Stimulates α_1 receptors and increases the force of contraction of the heart. Stimulates α_1 receptors and causes vasoconstriction (increase in BP).	It is a partial D1 agonist. It is a rapidly acting vasodilator that should not be used for >48 hours. It can be given for control of severe hypertension (malignant hypertension with end-organ damage).
Dopamine	Epinephrine
It acts on D1 and D2 receptors. It acts on both alpha and beta receptors except β_2 . Dose-dependent action. Acute decompensated heart failure with oliguria. Cardiogenic and septic shock.	It acts on all receptors with a weak action on β_3 receptors. Uses: Anaphylactic shock. Brittle asthma. Cardiac arrest. Along with local anesthetics.

Dopamine exhibits dose-dependent actions, its effects vary based on the concentration administered.

Solution for Question 19:

Correct answer: C) Urinary bladder.

Explanation:

The Urinary bladder has β_3 receptors.

β -receptors:

- They are Gs-type G-protein-coupled receptors that activate adenylyl cyclase, increasing the levels of cyclic AMP.
- The table below shows the various types of receptors.

	β_1	β_2	β_3
Location	Heart and JG cells.	Bronchi, blood vessels, uterus, liver, GIT, urinary tract, and eye.	Adipose tissue and detrusor muscle of the urinary bladder.
Effects on stimulation	$\uparrow$ Heart rate. $\uparrow$ Conduction rate. $\uparrow$ Force of contraction. $\uparrow$ Renin release ($\uparrow$ BP).	Tremors. Hyperglycemia. Bronchodilation. Vasodilation. Tocolysis.	Lipolysis. Relaxation of the detrusor muscle.

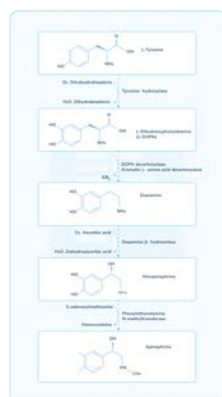
Solution for Question 20:

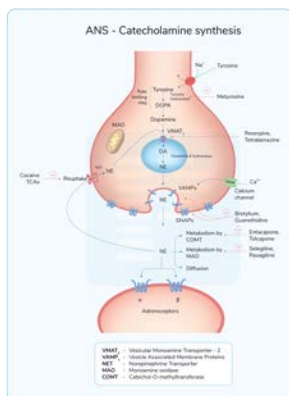
Correct answer: C) Metyrosine

Explanation:

Metyrosine inhibits tyrosine hydroxylase, the rate-limiting enzyme in synthesizing catecholamines.

Synthesis of catecholamines:





Methyldopa (Option A): Methyl DOPA is a centrally acting sympatholytic and alpha 2 agonist. It acts by inhibiting DOPA decarboxylase, which reduces dopamine synthesis.

Clonidine (Option B): acts by stimulating the presynaptic alpha-2 adrenergic receptors thus inhibiting the release of catecholamines.

Phenoxybenzamine (Option D): Phenoxybenzamine is an irreversible non-selective alpha blocker, Phentolamine and Tolazoline are reversible non-selective alpha-blockers. They inhibit the alpha-adrenergic receptors by acting on the postsynaptic terminals in smooth muscles.

Solution for Question 21:

Correct answer: A) $\alpha 1A$

EXPLANATION :

Tamsulosin is an $\alpha 1A$ blocker used in the treatment of BPH. Tamsulosin addresses the dynamic component of BPH, whereas finasteride, a 5- alpha reductase inhibitor addresses the static component of the disease.

α -blockers can be classified as:

- Non-selective blockers (used in the treatment of pheochromocytoma).
- Selective $\alpha 1$ blockers: DOC in BPH with hypertension. $\alpha 1A$ blockers: DOC in BPH without hypertension. $\alpha 1B$ blockers: Used in the treatment of hypertension.
- $\alpha 1A$ blockers: DOC in BPH without hypertension.
- $\alpha 1B$ blockers: Used in the treatment of hypertension.

- Selective α_2 blockers.
- α_1A blockers: DOC in BPH without hypertension.
- α_1B blockers: Used in the treatment of hypertension.

Alpha receptors:

They are a type of adrenergic receptor (membrane-bound G-protein coupled receptors) that function primarily by increasing or decreasing the production of second messengers cAMP or IP3/DAG.

They are present in various locations, and the table below describes them.

Table rendering failed

Mnemonic: A-C(ALPHA1-VASOCONSTRICTION) B-D(BETA2-VASODILATATION)

Sympatholytics

1. A 45-year-old hypertensive patient presents to the emergency department with severe chest pain and a rapid heartbeat. His blood pressure is 180/120mmHg. To manage his condition, a Phentolamine is administered. Later, he experiences an allergic reaction, and epinephrine is given to treat this condition. Which of the following best describes the Vasomotor reversal of Dale?

- A. Epinephrine administration after phentolamine causes an increased blood pressure
- B. Epinephrine administration after phentolamine causes decreased blood pressure.
- C. Epinephrine administration after phentolamine causes an increased heart rate.
- D. Epinephrine administration after phentolamine causes decrease in heart rate.

2. A 58-year-old patient presents with poorly controlled hypertension despite being on multiple antihypertensive medications. The physician initiates Clonidine therapy to help manage the patient's blood pressure. Which of the following is TRUE regarding the mechanism of action of Clonidine?

- A. Antagonises α_2 -adrenergic receptors.
- B. Stimulates α_1 -adrenergic receptors.
- C. Inhibits RAAS.
- D. Stimulates α_2 -adrenergic receptors.

3. What is the preferred medication for managing a hypertensive crisis caused by the sudden withdrawal of Clonidine?

- A. Intravenous labetalol
- B. Phentolamine

- C. Nitroglycerin
 - D. Amlodipine
-

4. A 32-year-old pregnant woman with chronic hypertension presents to her obstetrician's office for a routine prenatal visit. She has been managed with methyldopa for hypertension control. Which adverse effect should the physician monitor for during her treatment?

- A. Hyperkalemia
 - B. Hypertensive crisis
 - C. Hyperglycemia
 - D. Hemolytic anemia
-

5. A 52-year-old male with severe episodic hypertension, headaches, palpitations, and elevated urinary metanephrines is diagnosed with pheochromocytoma. What is the drug of choice to manage hypertension and symptoms before surgery?

- A. Propranolol
 - B. Phenoxybenzamine
 - C. Metoprolol
 - D. Prazosin
-

6. During the surgical excision of a pheochromocytoma, a continuous IV infusion of an alpha blocker was used to prevent the intraoperative rise of blood pressure due to catecholamine spillage into the bloodstream from the tumor. Match the alpha-blockers with their appropriate class. a. Phenoxybenzamine 1. Alpha 1 selective b. Prazosin 2. Alpha Non-selective and reversible c. Phentolamine 3. Alpha 2 selective d. Yohimbine 4. Alpha Non-selective and irreversible

a. Phenoxybenzamine	1. Alpha 1 selective
b. Prazosin	2. Alpha Non-selective and reversible
c. Phentolamine	3. Alpha 2 selective
d. Yohimbine	4. Alpha Non-selective and irreversible

- A. a-2, b-3, c-1, d-4
- B. a-4, b-1, c-2, d-3
- C. a-1, b-3, c-2, d-4
- D. a-3, b-2, c-4, d-1

7. A 58-year-old man with benign prostate hypertrophy was recently prescribed Prazosin. Which side effects should the patient be informed about before starting the medication? Postural hypotension Retrograde ejaculation Floppy iris syndrome Reflex bradycardia

- A. 1 and 2
- B. 1 and 3
- C. 1, 2 and 4
- D. 1, 2 and 3

8. A 40-year-old man with a recent history of myocardial infarction was started on Beta blocker as a secondary prophylaxis of MI. Which of the following beta blockers is most cardioselective?

- A. Metoprolol
- B. Propranolol
- C. Labetalol
- D. Tilisolol.

9. A 38-year-old female patient who was on propranolol developed bradycardia and was admitted to the cardiac ICU. On discharge, her medication was changed from propranolol to another beta-blocker, which has intrinsic sympathomimetic activity and membrane-stabilizing properties. Which of the following drugs was most likely prescribed?

- A. Carvedilol
 - B. Metoprolol
 - C. Pindolol
 - D. Propranolol
-

10. A 63-year-old male has complained of tremors in his hands for the past two years. Physical examination reveals bilateral symmetric tremors predominantly in his hands, which worsen with intentional movements and are relieved by rest. Which of the following is the drug of choice in essential tremors?

- A. Propranolol
 - B. Levodopa
 - C. Diazepam
 - D. Gabapentin
-

11. A 45-year-old female with a history of hypertension on metoprolol presents for a routine follow-up. She mentions experiencing fatigue and occasional dizziness. She also notes episodes of constipation. Which of the following is NOT an adverse effect of beta blockers?

- A. Hyperkalemia

- B. Hyperglycemia
 - C. Bronchospasm
 - D. Retrograde ejaculation
-

12. Which of the following calcium channel blockers can cause heart blockage when combined with beta-blockers in patients with heart failure?

- A. Nimodipine
 - B. Nifedipine
 - C. Amlodipine
 - D. Diltiazem
-

13. Which of the following symptoms of hypoglycemia is NOT masked by propranolol?

- A. Tremors
 - B. Palpitations
 - C. Tachycardia
 - D. Sweating
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	2
Question 4	4
Question 5	2
Question 6	2
Question 7	4
Question 8	1
Question 9	3
Question 10	1
Question 11	4
Question 12	4
Question 13	4

Solution for Question 1:

Correct answer: B) Epinephrine administration after phentolamine causes decreased blood pressure.

Explanation:

Epinephrine administration after phentolamine (an α -blocker) causes Vasodilation and decreases blood pressure.

- The Vasomotor reversal of Dale is a phenomenon in which the effect of adrenaline (epinephrine) is reversed from vasoconstriction to vasodilation due to the blockade of α

receptors by an α -blocker

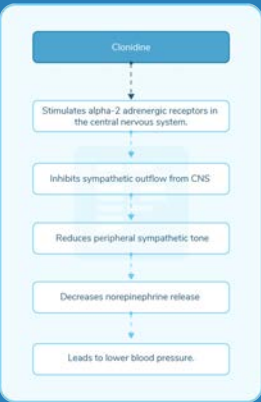
- In this scenario, the patient is given an α -blocker, which blocks the α receptors, and then epinephrine is administered to treat the allergic reaction. As a result, the epinephrine causes vasodilation and decreases blood pressure, which is the Vasomotor reversal of Dale

Solution for Question 2:

Correct answer: D) Stimulates α_2 -adrenergic receptors.

Explanation:

Clonidine reduces sympathetic outflow from the CNS by agonising α_2 -adrenergic receptors.

Mechanism of action	 <pre>graph TD; A[Clonidine] --> B[Stimulates alpha-2 adrenergic receptors in the central nervous system.]; B --> C[Inhibits sympathetic outflow from CNS]; C --> D[Reduces peripheral sympathetic tone]; D --> E[Decreases norepinephrine release]; E --> F[Leads to lower blood pressure.];</pre>	
Uses	Hypertension Diabetic diarrhoea Addiction withdrawal Menopausal hot flashes Diagnosing hypertension Suspected pheochromocytoma	Atrial fibrillation ADHD Tourette syndrome Hyperhidrosis

Solution for Question 3:

Correct Answer: B) Phentolamine

Explanation:

- Clonidine is commonly used to manage hypertension
- Sudden discontinuation of clonidine can lead to a withdrawal syndrome characterized by various symptoms, including headache, apprehension, tremors, abdominal pain, sweating, and tachycardia
- This syndrome typically occurs 18 to 36 hours after stopping the medication and is associated with increased sympathetic discharge, as evidenced by elevated plasma and urine concentrations of catecholamines and metabolites
- In the management of a hypertensive crisis, promptly resume Clonidine administration, and Phentolamine is considered the drug of choice for a hypertensive crisis
- Common side effects include dry eyes and mouth due to its antimuscarinic properties, sedation, hypotension, and erectile dysfunction

Solution for Question 4:

Correct answer: D) Hemolytic anemia

Explanation:

- Methyldopa, a centrally acting sympatholytic agent, is utilized in hypertension management, particularly during pregnancy and in patients with renal insufficiency
- Damage to red blood cells (Warm Antibody Hemolytic Anemia) occurs when alpha-methyldopa attaches to the RBC surface, rendering the RBC antigenic
- This attachment triggers the production of antibodies (IgG), which target and destroy the RBCs and is classified as a Type II Hypersensitivity reaction
- Regular monitoring, including Coombs' test for hemolytic anemia and liver function tests, is imperative for the early detection of adverse events

Category	Details
Mechanism of Action	Centrally converted to methyl norepinephrine, exerting alpha-2 agonistic action in the CNS, reducing adrenergic outflow.
Uses	Management of hypertension, particularly in pregnant women or those with renal insufficiency.
Side Effects	Nausea, Diarrhea, Headache, Dizziness, Sedation, Dry mouth, Rash. Rare complications include

Coombs-positive hemolytic anaemia, lupus-like syndrome, myocarditis, pancreatitis, hepatotoxicity, and immune thrombocytopenia.

Solution for Question 5:

Correct answer: B) Phenoxybenzamine

Explanation:

- Phenoxybenzamine is utilized to manage paroxysmal hypertension and sweating caused by conditions like pheochromocytoma.

Class	Mechanism of Action	Side Effects
Nonselective, irreversible alpha-adrenergic receptor antagonists	Forms covalent bonds, likely with cysteine residues, within alpha-adrenergic receptors, reducing sympathetic effects induced by catecholamines. Results in decreased vasoconstriction and systemic vascular resistance.	Postural hypotension, palpitations, nasal congestion, miosis, inhibition of ejaculation.

- Prazosin can be used as an alternative, but the drug of choice for pheochromocytoma is Phenoxybenzamine.

Solution for Question 6:

Correct answer: B) a-4, b-1, c-2, d-3

Explanation:

Classification of Alpha-blockers based on their receptor binding:

Table rendering failed

Solution for Question 7:

Correct answer: D) 1, 2 and 3

Explanation:

Prazosin is a highly selective α_1 blocker that has been prescribed to this patient for BPH.

It blocks the α_1 receptors in the bladder trigone and the prostatic smooth muscle, thus improving urine flow and reducing residual urine volume.

Mechanism of action	Uses	Side effects
Alpha 1a receptor blockade Receptors located on the internal urethral sphincter and prostatic smooth muscle Located on the iris	Benign prostatic hyperplasia Improves urine flow, thus reducing residual urine volume in the bladder. No clinical use	Retrograde ejaculation IUS sphincter opens Behind this sphincter, there is the opening of seminal vesicles Opening of this sphincter causes retrograde ejaculation. Floppy iris syndrome Miosis occurs due to paralysis of the iris
Alpha 1b receptor blockade Located on vascular smooth muscles	Hypertension Raynaud's disease Causes vasodilation, thus reducing peripheral resistance. Scorpion bite Reduced preload, thus preventing myocardial damage and pulmonary oedema due to scorpion bite.	Postural hypotension (first dose effect). Avoided by using low-dose prazosin at bedtime Severe postural hypotension can be treated with Midodrine(alpha 1 agonist) Hypotension also leads to reflex tachycardia.

Solution for Question 8:

Correct answer: A) Metoprolol

Explanation:

- Metoprolol is a second-generation/cardioprotective β -blocker used as a secondary prophylaxis for myocardial infarction. It helps prevent reinfarction and arrhythmias.

- It is more potent in blocking **β**₁-receptor (cardiac muscle) than **β**₂-receptor (bronchial muscles).

Classification of **β**-blockers:

First generation(Older / non selective)	Second generation (Cardioselective)	Third generation
Propranolol (Option B) Sotalol Pindolol (100% bioavailability) Timolol Nadolol (Maximum t _{1/2})	Metoprolol (Option A) Esmolol (Minimum t _{1/2}) Atenolol Bisoprolol Acebutolol	With alpha-blocking activity Labetalol (Option C) Celiprolol Nebivolol (NO-releasing property) Betaxolol
With calcium channel blockade Carvedilol (maximum plasma protein binding)		
With beta 2 agonism Bopindolol		
With K ⁺ channel opening action Tilisolol (Option D)		

Solution for Question 9:

Correct Answer: C) Pindolol

Explanation:

- Pindolol is a potent, non-selective beta blocker with Intrinsic sympathomimetic activity
Membrane stabilizing property
Lower risk for rebound hypertension when medication is stopped
- Intrinsic sympathomimetic activity
- Membrane stabilizing property
- Lower risk for rebound hypertension when medication is stopped
- Intrinsic sympathomimetic activity
- Membrane stabilizing property
- Lower risk for rebound hypertension when medication is stopped
- As it has a sympathetic activity, it does not cause bradycardia; thus, it is safe for cardiovascular conditions

- It also blocks sodium channels on the heart muscle's membrane, hence exhibiting a membrane stabilizing action, and thus, it can be used in Atrial fibrillation Ventricular fibrillation
- Atrial fibrillation
- Ventricular fibrillation
- Atrial fibrillation
- Ventricular fibrillation
- It also blocks sodium channels in neurons, leading to decreased nerve conduction.

Non selective ■1 + ■2 blocker	Cardioselective ■1 blocker		
Without ISA	With ISA	With ■ blocking activity	Metoprolol (Option B) Esmolol Atenolol Bisoprolol Acebutolol Betaxolol Nebivolol
Propranolol (Option D) Sotalol Timolol	Pindolol Celiprolol Oxprenolol Penbutolol Pindolol Bopindolol	Labetalol Carvedilol (Option A)	

ISA: intrinsic sympathomimetic action

Beta-blockers with membrane stabilizing activity (MSA) are:

- Propranolol (Maximum MSA)
- Metoprolol
- Sotalol
- Labetalol
- Acebutolol
- Pindolol

Carvedilol (Option A): It has multiple mechanisms of action, namely, ■1 + ■2 + ■1 blockade and Calcium channel blocker. It is used as a cardioprotective agent in congestive cardiac failure.

Metoprolol (Option B): It is the prototype cardioselective ■1 blocker, which may be used in diabetic patients receiving oral hypoglycemia (beta blockers are usually contraindicated). It is also used in people intolerant to propranolol, which causes cold hands and feet.

Propranolol (Option D): It is a non-selective beta blocker without intrinsic sympathomimetic activity, and membrane stabilizing property is present only at high doses.

Solution for Question 10:

Correct Answer: A) Propranolol

Explanation:

Essential tremor is seen in the hands and arms bilaterally and is symmetric. It can be elicited with the arms suspended against gravity in a fixed posture and during goal-directed activity. It is relieved by alcohol. Propranolol(beta-blocker) is the drug of choice.

Uses of beta-blockers:

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Solution for Question 11:

Correct Answer: D) Retrograde ejaculation

Explanation:

Retrograde ejaculation is a side effect of alpha-blockers

The cardiac side effects of beta blockers include

- Beta-blocker withdrawal can lead to the precipitation of an acute MI.
- Negative chronotropic effects include the development of sinus bradycardia and heart block.
- Heart failure may exacerbate symptoms in patients with decompensated heart failure or preexisting myocardial dysfunction.

The extracardiac side effects include

- Increased airway resistance; hence, they are contraindicated in patients with asthma due to bronchospasm.
- Exacerbation of peripheral artery disease and worsen claudication and precipitate Raynaud's disease.

- In Hypoglycemia, Beta-blockers mask the signs seen in a hypoglycemic patient, such as tachycardia.
- Hyperkalemia
- Depression, fatigue, and sexual dysfunction such as erectile dysfunction.
- Weight gain

Solution for Question 12:

Correct Answer: D) Diltiazem

Explanation:

Diltiazem, which is a non-dihydropyridine calcium channel blocker, can cause bradycardia and heart block when used in combination with beta-blockers in patients with heart failure.

- A non-dihydropyridine calcium channel blocker (i.e., diltiazem, verapamil) should not be simultaneously prescribed with a beta blocker since both classes have negative inotropic and chronotropic effects and can lead to bradycardia and heart block.
- Dihydropyridine calcium channel blockers (e.g., Nifedipine, Nimodipine, and Amlodipine), when added to beta-blockers, increase heart rate and decrease blood pressure, resulting in an augmented symptom benefit compared with either drug class alone.

Feature	DHP Calcium Channel Blockers	Non-DHP Calcium Channel Blockers
Examples	Amlodipine, Nifedipine, Felodipine	Diltiazem, Verapamil
Mechanism of Action	Primarily block L-type calcium channels in vascular smooth muscle	Block L-type calcium channels in the heart
Uses	Hypertension	Angina, rate control in arrhythmias
Side Effects	Peripheral edema, flushing, headache	Bradycardia, heart block, dizziness

Empty table

Solution for Question 13:

Correct Answer: D) Sweating

Explanation:

Sweating, a symptom of hypoglycemia, is caused by cholinergic receptors and is not inhibited by beta blockers.

- Prolonged propranolol therapy may impair carbohydrate tolerance in prediabetics by decreasing insulin release, leading to hyperglycemia and diabetes.
 - In diabetics, propranolol is avoided because of the risk of hypoglycemic episodes, as glycogenolysis in the heart, skeletal muscles, and liver that occurs due to sympathetic stimulation is attenuated.
 - In diabetic patients, beta blockers mask the signs of hypoglycemia mediated through sympathetic stimulation (tachycardia, tremor, palpitations) (Option A, B, C ruled out). Patients fail to recognise key symptoms of a hypoglycemic state and neglect getting appropriate treatment.
-

Previous Year Questions

1. A 10-year-old child is diagnosed with Attention Deficit Hyperactivity Disorder (ADHD) and is started on medication to increase norepinephrine levels as part of the treatment plan. Which of the following drugs is used for this purpose, and what is its mechanism of action?

- A. Methylphenidate
 - B. Reboxetine
 - C. Amitriptyline
 - D. Fluoxetine
-

2. Which of the following can be given for long term management of obesity?

- A. Liraglutide
 - B. Metformin
 - C. Sibutramine
 - D. Fenfluramine
-

3. Apart from mu-opioid receptors, on which other receptor does tramadol act?

- A. Anticholinergic
 - B. 5HT 1a inhibition
 - C. Serotonergic and norepinephrine reuptake inhibition
 - D. Antihistaminic
-

4. Which is the following drug that causes postural hypotension commonly?

- A. Beta blocker
 - B. Alpha blocker
 - C. ACE inhibitor
 - D. Angiotensin receptor blockers
-

5. Osimertinib is used in NSCLC with mutation due to:

- A. ROS1 mutation
 - B. T890M mutation
 - C. T790M mutation
 - D. M790T mutation
-

6. Which drug is used to test for Myasthenia Gravis?

- A. Pyridostigmine
 - B. Neostigmine
 - C. Edrophonium
 - D. Atropine
-

7. A female patient with a history of bronchial asthma has developed an acute attack of migraine. Which drug should be avoided?

- A. Sumatriptan

- B. Propranolol
 - C. Ergotamine
 - D. Lasmiditan
-

8. Chelator contraindicated in Iron poisoning?

- A. Calcium disodium EDTA
 - B. Deferoxamine
 - C. British anti lewisite (BAL)
 - D. Penicillamine
-

9. Which of the following medications is most commonly used for maintenance therapy in opioid withdrawal?

- A. Buprenorphine
 - B. Naloxone
 - C. Naltrexone
 - D. Acamprosate
-

10. Example of SNRI antidepressant drug

- A. Reboxatine
 - B. Tramadol
 - C. Paroxetine
 - D. Bupropion
-

11. Mechanism of action of atropine in treatment of organophosphate poisoning is?

- A. It inhibits secretion of acetylcholine
 - B. It is reactivation of acetylcholinesterase enzyme
 - C. It has antimuscarinic activity
 - D. It is agonist of acetylcholine receptors
-

12. Match the following dilution of adrenaline and its use" A. 1:1000 1. Cardiac arrest B. 1:10000 2. Anaphylaxis C. 1:100000 3. Local anesthesia

A. 1:1000	1. Cardiac arrest
B. 1:10000	2. Anaphylaxis
C. 1:100000	3. Local anesthesia

- A. A-1, B-2, C-3
 - B. A-2, B-1, C-3
 - C. A-3, B-2, C-1
 - D. A-1, B-3, C-2
-

13. Which of the following side effects are likely to be observed when a patient already taking metoprolol is prescribed verapamil?

- A. Bradycardia with AV block
 - B. Torsades de points
 - C. Tachycardia
 - D. Ventricular fibrillation
-

14. Beta-blockers should be avoided in all except:

- A. Asthma
 - B. Glaucoma
 - C. Bradycardia
 - D. Hypothyroidism
-

15. Which alpha-blocker would you choose to administer to a 72-year-old male patient who is experiencing a weak urinary stream, frequent urination in small amounts, and is currently taking prazosin? The patient also reports feeling dizzy frequently and having low blood pressure.

- A. Alfuzosin
 - B. Terazosin
 - C. Tamsulosin
 - D. Doxazosin
-

16. What dose of epinephrine is used anaphylactic shock?

- A. 1:10000 I.V. - 10 ml
 - B. 1:1000 I.M. - 0.5 ml
 - C. 1:1000 I.M. - 1 ml
 - D. 1:1000 I.V.- 0.5 ml
-

17. The response on stimulation of muscarinic receptor is:

- A. Erection
 - B. Ejaculation
 - C. Bronchodilation
 - D. Increased myocardial contractility
-

18. Which of the following cannot be used to decrease intraocular pressure in glaucoma?

- A. Dexamethasone
 - B. Acetazolamide
 - C. Apraclonidine
 - D. Mannitol
-

19. What is the mechanism by which acetylcholine causes a decrease in the heart rate?

- A. Delayed diastolic depolarization
 - B. Increase in plateau
 - C. Decrease in preload
 - D. Increase in afterload
-

20. A child, upon disturbing a bee hive, had several stings. After a few minutes, the child developed breathing difficulty and hypotension. What is the drug of choice for managing this child?

- A. Dobutamine
- B. Adrenaline

- C. Noradrenaline
 - D. Dopamine
-

21. The dose at which dopamine acts on beta 1 receptors is _____.

- A. 2-10 mcg/kg/min
 - B. <2 mcg/kg/min
 - C. 10-20 mcg/kg/min
 - D. >20 mcg/kg/min
-

22. Immediate treatment of anaphylactic shock is:

- A. Hydrocortisone
 - B. Atropine
 - C. Adrenaline
 - D. Benzylpenicillin
-

23. In the scenario of a patient having open angle glaucoma and bronchial asthma, which of the options below could be recommended for treatment?

- A. Latanoprost
 - B. Timolol
 - C. Gemeprost
 - D. Mannitol infusion
-

24. Which of the following is the most effective drug in smoking cessation?

- A. Varenicline
 - B. Buspirone SR
 - C. Nicotine gum
 - D. Rimonabant
-

25. A patient after consuming alcohol started experiencing vomiting, abdominal pain, and chest pain on examination. The patient was hypotensive. What is the drug causing these symptoms?

- A. Cefoperazone
 - B. Propranolol
 - C. Calcium channel blockers
 - D. Pantoprazole
-

26. Which of the options below acts as a selective alpha-1 receptor antagonist in medical practice?

- A. Phenoxybenzamine
 - B. Phentolamine
 - C. Prazosin
 - D. Idazoxan
-

27. A patient with diabetes developed postoperative urinary retention. Which of the following drugs can be used for short-term treatment of the condition?

- A. Tamsulosin

- B. Bethanechol
 - C. Terazosin
 - D. Methacholine
-

28. Which of the following statements about the treatment of a farmer presenting with symptoms of diarrhea, sweating, excessive tearing, and salivation in the emergency department is not accurate?

- A. Carbamates bind to acetylcholinesterase irreversibly.
 - B. Atropine is used as an antidote for OP poisoning
 - C. Pralidoxime has a role in reversing the enzyme in carbamate poisoning
 - D. Succinylcholine use is avoided in rapid sequence intubation
-

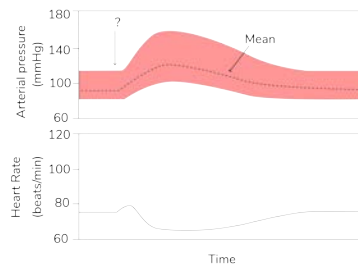
29. Which among the following is a short-acting beta blocker?

- A. Nadolol
 - B. Propranolol
 - C. Esmolol
 - D. Atenolol
-

30. A patient came to the casualty with acute bronchial asthma after treatment for glaucoma. Which of the following is a most probable causative agent?

- A. Timolol
- B. Betaxolol
- C. Latanoprost
- D. Bimatoprost

31. Which drug elicits the following response on blood pressure and heart rate, as shown in the image?



- A. Norepinephrine
- B. Isoproterenol
- C. Epinephrine
- D. Dopamine

32. The most appropriate description of the vasomotor reversal of Dale is that:

- A. Repeated administration of ephedrine decreases its effect on blood pressure
- B. A patient pretreated with phentolamine develops severe hypotension on the administration of adrenaline
- C. High dose of acetylcholine after
- D. An increase in pulse pressure is produced by the intravenous administration of isoprenaline

33. A patient comes to the casualty with organophosphate poisoning. He was started on atropine infusion and pralidoxime. After 2 hours, the patient had a sudden rise in temperature. What is the likely cause of fever?

- A. Atropine toxicity
 - B. A side effect of pralidoxime
 - C. Due to organophosphate poisoning
 - D. Idiopathic
-

34. A male patient was admitted to the emergency department following ingestion of an unidentified toxic substance. The medical team suspected muscarinic poisoning and initiated appropriate treatment. What was the potential symptom or characteristic that aided in making this diagnosis?

- A. Mydriasis
 - B. Diuresis
 - C. Muscle fasciculations
 - D. Bradycardia
-

35. Which of the following is the most effective oral drug for smoking cessation?

- A. Varenicline
 - B. Cevimeline
 - C. Muscarine
 - D. Atropine
-

36. Match the following: a. Beta 1 1. Mirabegron b. Beta 2 2. Betaxolol c. Beta 3 3. Isoprenaline

a. Beta 1	1. Mirabegron
b. Beta 2	2. Betaxolol
c. Beta 3	3. Isoprenaline

- A. a-2, b-1, c-3
 - B. a-3, b-1, c-2
 - C. a-2, b-3, c-1
 - D. a-3, b-2, c-1
-

37. A patient arrived with a past record of several instances of vomiting. Following the administration of an intravenous medication, the patient subsequently experienced unusual movements. Which of the options listed below is suitable for managing this condition?

- A. Hyoscine
 - B. Cyproheptadine
 - C. Levodopa
 - D. Benzhexol
-

38. What is the possible cause of poisoning in a farmer who has a pin-point pupil and an increase in secretions?

- A. Organophosphate poisoning
 - B. Dhatura poisoning
 - C. Opioid poisoning
 - D. Atropine poisoning
-

39. Pilocarpine produces:

- A. Active miosis
 - B. Active mydriasis
 - C. Passive miosis
 - D. Passive mydriasis
-

40. What can be administered to manage a patient who arrives at the emergency department experiencing symptoms such as diarrhea, sweating, excessive tearing, and increased salivation, with a pulse rate of 55 beats per minute and a plasma cholinesterase level of 50U/mL?

- A. Neostigmine
 - B. Atropine
 - C. Epinephrine
 - D. Benztropine
-

41. What could be the likely diagnosis for a 45-year-old female patient who visited the medicine OPD complaining of muscle weakness and blurred vision after exertion, mainly in the evening? Upon examination, the patient displayed ptosis, and their symptoms improved after undergoing the edrophonium test.

- A. Lambert Eaton syndrome
 - B. Myasthenia gravis
 - C. Gullian barre syndrome
 - D. Bronchial asthma
-

42. Which of the following antiglaucoma medications is unsafe in infants?

- A. Timolol
 - B. Brimonidine
 - C. Latanoprost
 - D. Dorzolamide
-

43. What is the drug used to differentiate myasthenia gravis from cholinergic crisis?

- A. Obidoxime
 - B. Edrophonium
 - C. Donepezil
 - D. Atropine
-

44. Intravenous mannitol is used in _____?

- A. Pulmonary edema
 - B. Congestive cardiac failure
 - C. Acute kidney injury with anuria
 - D. Glaucoma
-

45. Which drug is most likely to cause tremors with prolonged use?

- A. Salbutamol
 - B. Propofol
 - C. Betaxolol
 - D. Timolol
-

46. Which of the following is a long acting bronchodilator?

- A. Salbutamol
 - B. Terbutaline
 - C. Adrenaline
 - D. Formoterol
-

47. What is the preferred medication for managing Alzheimer's disease?

- A. Donepezil
 - B. Atropine
 - C. Physostigmine
 - D. Fluoxetine
-

48. An elderly woman received a botox injection for the treatment of wrinkles. Which of the following is the mechanism of action of botulinum toxin?

- A. Inhibits the release of acetylcholine
 - B. Release of noradrenaline at synaptic cleft
 - C. Selectively and irreversibly inhibits nicotinic receptors
 - D. Stimulates muscarinic and nicotinic receptors
-

49. A 5-year-old male child is brought to the ophthalmology clinic by his mother with complaints of blurring of vision for one week. The patient further explains that he only has blurring of the far vision and has no problem in his near vision. The

patient gives a history of no other abnormalities. For a detailed examination of the eye, the doctor uses a short-acting topical medication which also has a cycloplegic effect. Which of the following drug was used by the doctor in this case?

- A. Atropine oral
 - B. Atropine eye ointment
 - C. Tropicamide eye drops
 - D. Phenylephrine eye drops
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	3
Question 4	2
Question 5	3
Question 6	3
Question 7	2
Question 8	3
Question 9	1
Question 10	1
Question 11	3
Question 12	2
Question 13	1
Question 14	2
Question 15	3
Question 16	2
Question 17	1
Question 18	1
Question 19	1
Question 20	2
Question 21	1
Question 22	3

Question 23	1
Question 24	1
Question 25	1
Question 26	3
Question 27	2
Question 28	3
Question 29	3
Question 30	1
Question 31	1
Question 32	2
Question 33	1
Question 34	4
Question 35	1
Question 36	3
Question 37	4
Question 38	1
Question 39	1
Question 40	2
Question 41	2
Question 42	2
Question 43	2
Question 44	4
Question 45	1
Question 46	4
Question 47	1
Question 48	1

Solution for Question 1:

Correct Option A - Methylphenidate:

- Methylphenidate is a central nervous system stimulant commonly used in the treatment of ADHD. It works by inhibiting the reuptake of both norepinephrine and dopamine, increasing their levels in the brain. This helps improve attention, focus, and impulse control in individuals with ADHD. Although Methylphenidate acts on both norepinephrine and dopamine, it is the increased norepinephrine that contributes significantly to its therapeutic effects in managing ADHD symptoms.

Incorrect Options:

Option B - Reboxetine:

- Reboxetine is a selective norepinephrine reuptake inhibitor (NRI) that increases norepinephrine levels by inhibiting its reuptake. While it does increase norepinephrine levels and could theoretically be used in ADHD, it is not commonly used for this purpose. Methylphenidate is preferred due to its established efficacy in ADHD treatment.

Option C - Amitriptyline:

- Amitriptyline is a tricyclic antidepressant that inhibits the reuptake of both norepinephrine and serotonin. It is not typically used to treat ADHD, particularly in children, due to its side effect profile and the availability of more targeted medications like Methylphenidate.

Option D - Fluoxetine:

- Fluoxetine is a selective serotonin reuptake inhibitor (SSRI) that primarily increases serotonin levels. It does not significantly affect norepinephrine levels and is not used to treat ADHD. Fluoxetine is mainly used to treat depression, anxiety, and other mood disorders.

Solution for Question 2:

Correct Option A - Liraglutide:

- Liraglutide is a medication that is designed to help manage type 2 diabetes, but it has also been shown to promote weight loss. It works by mimicking the effects of a hormone called glucagon-like peptide-1 (GLP-1), which increases insulin secretion, slows down digestion, and reduces appetite. Liraglutide is typically administered as a daily injection and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Liraglutide include nausea, vomiting, diarrhea, and constipation. More serious side effects include pancreatitis and thyroid tumors, although these are rare.

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Common side effects of Liraglutide include nausea, vomiting, diarrhea, and constipation. More serious side effects include pancreatitis and thyroid tumors, although these are rare.

Incorrect Option:

Option B - Metformin:

- Metformin is a medication that is commonly used to manage type 2 diabetes, but it has also been shown to promote weight loss. It works by reducing glucose production in the liver and improving insulin sensitivity. Metformin is usually taken orally and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Metformin include nausea, vomiting, diarrhea, and stomach pain. More serious side effects include lactic acidosis, although this is rare.

Metformin is a medication that is commonly used to manage type 2 diabetes, but it has also been shown to promote weight loss. It works by reducing glucose production in the liver and improving insulin sensitivity. Metformin is usually taken orally and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.

Common side effects of Metformin include nausea, vomiting, diarrhea, and stomach pain. More serious side effects include lactic acidosis, although this is rare.

Option C - Sibutramine:

- Sibutramine is a medication that was previously used to manage obesity, but it has been discontinued due to safety concerns. It works by increasing the levels of certain neurotransmitters in the brain, which helps to reduce appetite and promote weight loss. Sibutramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Sibutramine included dry mouth, headache, constipation, and insomnia. More serious side effects included heart attacks and strokes, which is why it was discontinued.

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Common side effects of Sibutramine included dry mouth, headache, constipation, and insomnia. More serious side effects included heart attacks and strokes, which is why it was taken off the market.

Option D - Fenfluramine:

- Fenfluramine is a medication that was previously used to manage obesity, but it has also been discontinued due to safety concerns. It works by increasing serotonin levels in the brain, which helps reduce appetite and promote weight loss. Fenfluramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Fenfluramine include dry mouth, headache, constipation, and insomnia. More serious side effects included heart valve damage, which is why it was taken off the market.

Fenfluramine is a medication that was previously used to manage obesity, but it has also been discontinued due to safety concerns. It works by increasing serotonin levels in the brain, which helps reduce appetite and promote weight loss. Fenfluramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.

Common side effects of Fenfluramine include dry mouth, headache, constipation, and insomnia. More serious side effects included heart valve damage, which is why it was taken off the market.

Solution for Question 3:

Correct Option C - Serotonergic and norepinephrine reuptake inhibition:

- Tramadol acts as a serotonergic and norepinephrine reuptake inhibitor. Tramadol is a centrally acting analgesic that primarily binds to mu-opioid receptors to exert its pain-relieving effects. However, in addition to its opioid activity, tramadol also has the ability to inhibit the reuptake of serotonin and norepinephrine. By inhibiting the reuptake of these

neurotransmitters, tramadol increases their availability in the synaptic cleft, leading to enhanced analgesic effects. This dual mechanism of action contributes to tramadol's unique pharmacological profile.

Tramadol acts as a serotonergic and norepinephrine reuptake inhibitor. Tramadol is a centrally acting analgesic that primarily binds to mu-opioid receptors to exert its pain-relieving effects. However, in addition to its opioid activity, tramadol also has the ability to inhibit the reuptake of serotonin and norepinephrine. By inhibiting the reuptake of these neurotransmitters, tramadol increases their availability in the synaptic cleft, leading to enhanced analgesic effects. This dual mechanism of action contributes to tramadol's unique pharmacological profile.

Incorrect Options:

Option A - Anticholinergic: Tramadol does not have anticholinergic activity. Anticholinergic drugs act by blocking the effects of acetylcholine at muscarinic receptors and are not relevant to tramadol's mechanism of action.

Option B - 5HT 1a inhibition: Tramadol does not directly inhibit 5HT_{1a} receptors. While it does affect serotonin levels through its reuptake inhibition, it is not specific to 5HT_{1a} receptors.

Option D - Antihistaminic: Tramadol does not possess significant antihistaminic activity. Antihistaminic drugs act by blocking the effects of histamine at histamine receptors and are not relevant to tramadol's mechanism of action.

Solution for Question 4:

Correct Option B - Alpha blocker:

- Alpha-blockers inhibit vasoconstriction; Blockade of vasoconstrictors for Alpha receptors reduces peripheral resistance and causes blood pooling in capacitance vessels; venous return and cardiac output are reduced fall in BP.
- Postural reflex interferes with marked hypotension on standing, followed by dizziness and syncope. Hypovolemia accentuates the hypotension.

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Postural reflex interferes with marked hypotension on standing, followed by dizziness and syncope. Hypovolemia accentuates the hypotension.

Incorrect Options:

Option A - Beta blockers: Beta 1 blocker cause Hypotension while Beta 2 blockers increase BP

Option C - ACE inhibitor: Side effects of ACE inhibitor Are Hypotension especially common when starting the medication or when the dosage is increased, Dry cough, Angioedema, Hyperkalemia and skin rashes.

Option D - Angiotensin receptor blockers: Side effects of Angiotensin receptor blockers Are Hypotension ,Angioedema, Hyperkalemia ,skin rashes and They may reduce blood supply to the kidneys and can cause a temporary increase in serum creatinine levels.

Solution for Question 5:

Correct Option C - T790M mutation:

- Osimertinib is a third-generation epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor (TKI) primarily used in NSCLC.
- It is specifically effective against NSCLC that harbors the T790M mutation in the EGFR gene.
- The T790M mutation is commonly associated with resistance to first-generation EGFR TKIs like gefitinib and erlotinib.
- Osimertinib works by selectively inhibiting mutant forms of EGFR, including those with the T790M mutation, while sparing wild-type EGFR, which helps reduce toxicity.
- Clinical trials have demonstrated the efficacy of osimertinib in patients with NSCLC who have developed resistance to previous EGFR TKIs due to the T790M mutation.

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Clinical trials have demonstrated the efficacy of osimertinib in patients with NSCLC who have developed resistance to previous EGFR TKIs due to the T790M mutation.

Incorrect Options:

Option A - ROS1 mutation: Osimertinib is not indicated for NSCLC with ROS1 mutations. ROS1 rearrangements are a distinct molecular subtype of NSCLC, and targeted therapies like crizotinib, ceritinib, and entrectinib are more commonly used in this case.

Option B - T890M mutation: There is no known mutation referred to as T890M. The correct mutation associated with osimertinib efficacy is T790M.

Option D - M790T mutation: Similarly, there is no known mutation termed M790T. The correct mutation is T790M.

Solution for Question 6:

Correct Option C – Edrophonium:

- Edrophonium is a short-acting acetylcholinesterase inhibitor used in the diagnosis of Myasthenia Gravis.
- The drug works by temporarily increasing the levels of acetylcholine at the neuromuscular junction, which improves muscle strength if the patient has Myasthenia Gravis. This is known as the Tensilon test.
- If muscle strength improves after the administration of edrophonium, it indicates the presence of Myasthenia Gravis.

Incorrect Options

Options A, B, D:

- Pyridostigmine, Neostigmine and Atropine are not used to test for Myasthenia gravis

Solution for Question 7:

Correct Option B - Propranolol:

- Propranolol is a non-selective beta-blocker that blocks both beta-1 (found in the heart) and beta-2 receptors (found in the bronchial smooth muscle).
- Blocking beta-2 receptors can lead to bronchoconstriction, which can exacerbate symptoms in patients with bronchial asthma or lead to severe bronchospasm.
- Propranolol is contraindicated in patients with asthma due to its potential to worsen asthma symptoms, making it a poor choice in this scenario.

Incorrect Options:

- Option A - Sumatriptan: Sumatriptan is a serotonin receptor agonist (5-HT_{1B/1D}) used to treat acute migraines. It is not contraindicated in asthma and can be used for migraine treatment.
- Option C - Ergotamine: Ergotamine is an ergot alkaloid that is used to treat migraines by causing vasoconstriction. It is also not specifically contraindicated in asthma.
- Option D - Lasmiditan: Lasmiditan is a serotonin 5-HT_{1F} receptor agonist used to treat migraines. Like the other drugs mentioned, it is not contraindicated in asthma.

Solution for Question 8:

Correct Option C - British Anti-Lewisite (BAL):

- Mechanism of Action: BAL (dimercaprol) is a chelator primarily used for the treatment of heavy metal poisoning, specifically arsenic, mercury, and lead.
- Use in Iron Poisoning: BAL is contraindicated in iron poisoning because it is not effective for iron and can potentially displace iron from its chelation sites, worsening the situation. It does not effectively bind iron and could potentially complicate the clinical picture in iron toxicity.

Incorrect Options:

Option A - Calcium Disodium EDTA:

- Mechanism of Action: Calcium disodium EDTA (ethylenediaminetetraacetic acid) is a chelator used for lead poisoning and other heavy metals. While it is not the first choice for iron poisoning, it is used in some cases of iron overload to a limited extent.
- Use in Iron Poisoning: Calcium disodium EDTA is not typically used for iron poisoning, but it is not contraindicated. It is more commonly used for lead and other heavy metal poisonings.

Option B - Deferoxamine:

- Mechanism of Action: Deferoxamine is a specific iron chelator that binds to free iron in the bloodstream and facilitates its excretion.
- Use in Iron Poisoning: Deferoxamine is the drug of choice for treating iron poisoning. It is effective in removing excess iron from the body and is specifically indicated for iron overload conditions.

Option D - Penicillamine:

- Mechanism of Action: Penicillamine is a chelator used primarily for copper poisoning (e.g., Wilson's disease) and can also be used for lead poisoning.
- Use in Iron Poisoning: Penicillamine is generally not used for iron poisoning, but it is not contraindicated. It is less effective compared to deferoxamine for iron overload and is not the first choice.

Solution for Question 9:

Correct Option A - Buprenorphine:

- Buprenorphine is a partial opioid agonist that is commonly used in maintenance therapy for opioid dependence. It helps manage withdrawal symptoms and cravings by binding to the same opioid receptors in the brain that full agonists like heroin or methadone do but with a ceiling effect that reduces the risk of misuse, respiratory depression, and overdose. Buprenorphine can be used alone or in combination with naloxone (as in Suboxone) to further deter misuse. It is an effective and widely used medication for maintenance therapy in opioid use disorder.

Incorrect Options:

Option B - Naloxone:

- Naloxone is an opioid antagonist that is primarily used as a rescue medication in cases of opioid overdose. It rapidly reverses the effects of opioids, particularly respiratory depression. However, naloxone is not used for maintenance therapy in opioid withdrawal, as it would precipitate withdrawal symptoms in individuals dependent on opioids.

Option C - Naltrexone:

- Naltrexone is an opioid antagonist that is used in the maintenance treatment of opioid and alcohol dependence. For opioid dependence, naltrexone is used after detoxification to prevent relapse. It works by blocking the euphoric and sedative effects of opioids. While

effective, it is not typically used during the initial withdrawal phase or for managing withdrawal symptoms but rather for relapse prevention in those who have already undergone detoxification.

Option D - Acamprosate:

- Acamprosate is a medication used to help maintain abstinence in individuals with alcohol dependence. It works by modulating glutamatergic neurotransmission and is not used for opioid withdrawal or maintenance therapy. Acamprosate does not have any effect on opioid receptors and is specific to alcohol use disorder.

Solution for Question 10:

Correct Option A - Reboxetine:

- Reboxetine is a selective norepinephrine reuptake inhibitor (NRI) and is used as an antidepressant. It affects norepinephrine reuptake exclusively and does not significantly impact serotonin levels.

Incorrect Options

Option B - Tramadol:

- Tramadol is an analgesic with weak serotonin and norepinephrine reuptake inhibition properties but is not classified as a primary antidepressant or SNRI. It is mainly used for pain management, and its SNRI activity is a secondary effect rather than its primary indication.

Option C - Paroxetine:

- Paroxetine is a selective serotonin reuptake inhibitor (SSRI). It primarily affects serotonin reuptake and does not significantly impact norepinephrine levels. Therefore, it is not classified as an SNRI.

Option D - Bupropion:

- Bupropion is a norepinephrine-dopamine reuptake inhibitor (NDRI). It primarily affects norepinephrine and dopamine reuptake, but does not significantly impact serotonin levels, so it is not classified as an SNRI.

Solution for Question 11:

Correct Option C - It has antimuscarinic activity:

- It has antimuscarinic activity: Atropine, in the context of organophosphate poisoning, is used as an antidote to counteract the excessive stimulation of acetylcholine receptors caused by organophosphate insecticides or nerve agents.
- Atropine acts as a competitive antagonist at muscarinic acetylcholine receptors, blocking the effects of excess acetylcholine and reversing the toxic symptoms.

It has antimuscarinic activity: Atropine, in the context of organophosphate poisoning, is used as an antidote to counteract the excessive stimulation of acetylcholine receptors caused by organophosphate insecticides or nerve agents.

Atropine acts as a competitive antagonist at muscarinic acetylcholine receptors, blocking the effects of excess acetylcholine and reversing the toxic symptoms.

Incorrect Options:

Option A - It inhibits secretion of acetylcholine: Atropine does not directly inhibit the secretion of acetylcholine but rather acts as an antagonist at acetylcholine receptors.

Option B - It is a reactivator of acetylcholinesterase enzyme: Atropine does not reactivate acetylcholinesterase, the enzyme that breaks down acetylcholine. It works by blocking the effects of excessive acetylcholine stimulation.

Option D - It is an agonist of acetylcholine receptors: Atropine is not an agonist of acetylcholine receptors. It acts as an antagonist, blocking the binding of acetylcholine to these receptors.

Solution for Question 12:

Correct Option B - A-2, B-1, C-3:

- Dilution 1:1000: This dilution is used for the treatment of anaphylaxis, which is a severe allergic reaction. Adrenaline at a concentration of 1:1000 is administered intramuscularly or subcutaneously to rapidly alleviate symptoms and counteract the life-threatening effects of anaphylaxis.

- Dilution 1:10000: This dilution is used for cardiac arrest. Adrenaline at a concentration of 1:10000 is commonly administered during cardiopulmonary resuscitation (CPR) to restore circulation and improve the chances of successful resuscitation in cases of cardiac arrest.
- Dilution 1:100000: This dilution is used for local anesthesia. Adrenaline at a concentration of 1:10 is added to local anesthetic solutions to enhance their effectiveness by causing vasoconstriction at the site of injection. This helps to prolong the duration of action and reduce bleeding, thus improving the local anesthetic effect.

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Dilution 1:100000: This dilution is used for local anesthesia. Adrenaline at a concentration of 1:10 is added to local anesthetic solutions to enhance their effectiveness by causing vasoconstriction at the site of injection. This helps to prolong the duration of action and reduce bleeding, thus improving the local anesthetic effect.

Incorrect Options:

Option A - A-1, B-2, C-3: This option incorrectly matches dilutions to its use.

Option C - A-3, B-2, C-1: This option incorrectly matches dilutions to its use

Option D - A-1, B-3, C-2: This option incorrectly matches dilutions to its use

Solution for Question 13:

Correct Option A - Bradycardia with AV block:

- Verapamil, as a calcium channel blocker, primarily acts by inhibiting calcium influx into cardiac cells, thereby reducing contractility and slowing down the electrical conduction through the AV node. Metoprolol, as a beta-blocker, blocks the effects of beta-adrenergic stimulation, leading to decreased heart rate and AV node conduction.

- When these two medications are used together, there is an increased risk of excessive slowing of cardiac conduction, which can manifest as bradycardia (slow heart rate) and AV block (impaired conduction through the AV node). This can result in symptoms such as dizziness, fainting, or even heart block requiring intervention.

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When these two medications are used together, there is an increased risk of excessive slowing of cardiac conduction, which can manifest as bradycardia (slow heart rate) and AV block (impaired conduction through the AV node). This can result in symptoms such as dizziness, fainting, or even heart block requiring intervention.

Incorrect Options:

Option B - Torsades de pointes: Torsades de pointes is a specific type of ventricular tachycardia characterized by a unique pattern on an electrocardiogram (ECG). While both metoprolol and verapamil can affect cardiac conduction, the combination of these medications is not specifically associated with the development of torsades de pointes.

Option C - Tachycardia: Both metoprolol and verapamil have negative chronotropic effects, meaning they decrease heart rate. Therefore, the combination of these medications would not typically result in tachycardia (increased heart rate).

Option D - Ventricular fibrillation: Ventricular fibrillation is a life-threatening arrhythmia characterized by the rapid and chaotic contraction of the ventricles. While both metoprolol and verapamil can affect cardiac conduction, the combination of these medications is not typically associated with ventricular fibrillation.

Solution for Question 14:

Correct Option B - Glaucoma:

- Glaucoma is a condition of increased Intraocular pressure within the eyeball, causing gradual loss of sight. Betaxolol is the beta blocker used in glaucoma. Beta-blockers block Beta receptors, particularly B₂, in the ciliary body, and decrease cAMP, which in turn decreases aqueous humor production.

Glaucoma is a condition of increased Intraocular pressure within the eyeball, causing gradual loss of sight. Betaxolol is the beta blocker used in glaucoma. Beta-blockers block Beta receptors, particularly B₂, in the ciliary body, and decrease cAMP, which in turn decreases aqueous humor production.

Incorrect Options:

Option A - Asthma: Beta-blockers cause bronchoconstriction, so contraindicated in bronchial asthma.

Option C - Bradycardia: Beta-blockers decrease heart rate, therefore contraindicated in bradycardia.

Option D - Hypothyroidism: Beta blockers are used in Thyrotoxicosis.

Solution for Question 15:

Correct Option C - Tamsulosin:

- The above clinical features suggest that the patient is having benign prostatic hyperplasia. Prazosin is a selective α_1 antagonist, used in treating hypertension and for benign prostatic hyperplasia. Postural hypotension, stress incontinence, and impotence are unwanted effects. Tamsulosin is α_{1A} selective and acts mainly on the urogenital tract. It is used to treat benign prostatic hyperplasia and may cause less postural hypotension than other α_1 antagonists.

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Incorrect Options:

Option A - Alfuzosin: Alfuzosin block α_{1B} receptors decrease peripheral vascular resistance and lower arterial blood pressure by causing relaxation of both arterial and venous smooth muscle.

Option B - Terazosin: Terazosin block α_{1B} receptors decrease peripheral vascular resistance and lower arterial blood pressure by causing relaxation of both arterial and venous smooth muscle.

Option D - Doxazosin: Doxazosin block α_1B receptors decrease peripheral vascular resistance and lower arterial blood pressure by causing relaxation of both arterial and venous smooth muscle.

Solution for Question 16:

Correct Option B - 1:1000 I.M. - 0.5 ml:

- In the management of anaphylactic shock, epinephrine is the drug of choice as it helps reverse the severe allergic reaction and relieve symptoms. The recommended dose of epinephrine for anaphylactic shock is 1:1000 concentration, which means 1 mg of epinephrine in 1 mL of solution. It is administered intramuscularly (I.M.) in the mid-anterolateral aspect of the thigh.

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Incorrect Options:

Option A - 1:10000 I.V. - 10 mL: Represents a more diluted concentration and a higher volume, which is not the standard dose for anaphylactic shock.

Option C - 1:1000 I.M. - 1 mL: Represents a higher volume of epinephrine than the recommended dose.

Option D - 1:1000 I.V. - 0.5 mL: Represents the correct concentration but the incorrect route of administration. Intravenous (I.V.) administration of epinephrine is reserved for severe cases and should be administered by healthcare professionals.

Solution for Question 17:

Correct Option A - Erection:

- Stimulation of muscarinic receptors can have various effects on different organs and tissues in the body. Muscarinic receptors are a type of acetylcholine receptor that respond to the neurotransmitter acetylcholine.
- One of the responses of muscarinic receptor stimulation is the promotion of erection in the male reproductive system. Muscarinic receptors are involved in the parasympathetic control of penile erection. Stimulation of these receptors leads to the release of nitric oxide (NO) in the penile arteries and smooth muscle relaxation, resulting in increased blood flow to the erectile tissue and the subsequent achievement and maintenance of an erection.

Stimulation of muscarinic receptors can have various effects on different organs and tissues in the body. Muscarinic receptors are a type of acetylcholine receptor that respond to the neurotransmitter acetylcholine.

One of the responses of muscarinic receptor stimulation is the promotion of erection in the male reproductive system. Muscarinic receptors are involved in the parasympathetic control of penile erection. Stimulation of these receptors leads to the release of nitric oxide (NO) in the penile arteries and smooth muscle relaxation, resulting in increased blood flow to the erectile tissue and the subsequent achievement and maintenance of an erection.

Incorrect Options:

Option B - Ejaculation: Ejaculation is not primarily mediated by muscarinic receptor stimulation. It is a complex process involving the sympathetic and somatic nervous systems, and it is not directly influenced by muscarinic receptor activity.

Option C - Bronchodilation: Muscarinic receptor stimulation primarily leads to bronchoconstriction, not bronchodilation. Activation of muscarinic receptors in the airways causes smooth muscle contraction, narrowing the airway diameter and leading to bronchoconstriction.

Option D - Increased myocardial contractility: Muscarinic receptor stimulation primarily has negative chronotropic and inotropic effects on the heart. Activation of muscarinic receptors in the heart leads to decreased heart rate (chronotropic effect) and reduced myocardial contractility (inotropic effect), resulting in a decrease in the force of cardiac contraction.

Solution for Question 18:

Correct Option A - Dexamethasone:

- Dexamethasone is a corticosteroid medication that is not typically used to decrease intraocular pressure in glaucoma. While corticosteroids can have anti-inflammatory effects and may be used in certain ocular conditions, they are not commonly prescribed as primary treatments for reducing intraocular pressure in glaucoma.

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Incorrect Options:

Option B - Acetazolamide: Acetazolamide is a carbonic anhydrase inhibitor that can effectively lower intraocular pressure in glaucoma. It reduces the production of aqueous humour, thereby decreasing the pressure inside the eye.

Option C - Apraclonidine: Apraclonidine is an alpha-adrenergic agonist that can be used to reduce intraocular pressure in glaucoma. It works by decreasing the production of aqueous humour and increasing the outflow of fluid from the eye.

Option D - Mannitol: Mannitol is an osmotic diuretic that is sometimes used in acute glaucoma episodes to rapidly reduce intraocular pressure. It works by drawing fluid out of the eye, thereby lowering the pressure.

Solution for Question 19:

Correct Option A - Delayed diastolic depolarization:

- This mechanism refers to the delayed depolarization of the sinoatrial (SA) node during the relaxation phase of the cardiac cycle (diastole). The gradual depolarization of the SA node is responsible for initiating each heartbeat. When acetylcholine binds to muscarinic receptors in the SA node, it activates a signaling pathway that increases potassium efflux from the cells, hyperpolarizing the cell membrane. This hyperpolarization delays the diastolic depolarization, leading to a decrease in heart rate.

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hyperpolarizing the cell membrane. This hyperpolarization delays the diastolic depolarization, leading to a decrease in heart rate.

Incorrect Options:

Option B - Increase in plateau: Acetylcholine does not cause an increase in the plateau phase. Instead, it primarily affects the pacemaker activity of the SA node and the conduction through the atrioventricular (AV) node, leading to a decrease in heart rate.

Option C - Decrease in preload: Acetylcholine does not directly cause a decrease in preload. Instead, it primarily affects heart rate through its influence on the SA node and AV node. However, a decrease in heart rate may indirectly lead to a decrease in preload over time due to a longer diastolic filling period.

Option D - Increase in afterload: Acetylcholine does not directly cause an increase in afterload. Its primary effect on heart rate is related to its action on the SA node and AV node. Changes in afterload are more influenced by factors such as arterial resistance, vascular tone, and conditions like hypertension.

Solution for Question 20:

Correct Option B - Adrenaline:

- Adrenaline acts on all the receptors, it is used in diluted form. A ratio of 1:1000 (1 gram adrenaline in 1000ml of water) is used in anaphylactic shock. The intramuscular route of administration of adrenaline is preferred because, in shock conditions, blood flow is decreased to the periphery.
- The subcutaneous route has less blood flow than the intramuscular route. If the patient does not respond to the IM route in anaphylactic shock then the IV route can be preferred.

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The subcutaneous route has less blood flow than the intramuscular route. If the patient does not respond to the IM route in anaphylactic shock then the IV route can be preferred.

Incorrect Options:

Option A - Dobutamine: This is an inotropic agent used in pump failure accompanying myocardial infarction and cardiac surgery. It is used as a short-term management of severe congestive heart failure.

Option C - Noradrenaline: It acts on α_1 , α_2 , and β_1 receptors (no/weak action on β_2 receptor). It is not used in anaphylactic shock because it has no action on the β_2 receptor. It causes vasoconstriction which increases total peripheral resistance and diastolic blood pressure.

Option D - Dopamine: It is used in cardiogenic shock or septic shock with oliguria because it increases GFR to correct oliguria and the force of contraction. Dopamine does not cause tachycardia because it majorly acts on muscle tissues more than nodal tissue.

Solution for Question 21:

Correct Option A - 2-10 mcg/kg/min:

- Dopamine is a medication that can act on different receptors in the body at different doses. At low doses (2-10 mcg/kg/min), dopamine acts on dopamine receptors, leading to increased renal and mesenteric blood flow. At higher doses (>10 mcg/kg/min), dopamine can also act on beta 1 receptors in the heart to increase cardiac output and blood pressure, as well as on alpha receptors to increase peripheral vascular resistance.
- Therefore, the dose range of 2-10 mcg/kg/min is the range at which dopamine acts on beta 1 receptors, and is commonly used to treat hypotension and shock.

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Therefore, the dose range of 2-10 mcg/kg/min is the range at which dopamine acts on beta 1 receptors, and is commonly used to treat hypotension and shock.

Incorrect Options:

- Option B, C & D are Incorrect Choice.

Solution for Question 22:

Correct Option C - Adrenaline:

- Adrenaline (Epinephrine) is the immediate treatment of choice for anaphylactic shock. It acts rapidly to counteract the severe symptoms of anaphylaxis. Adrenaline works by stimulating alpha and beta-adrenergic receptors, which helps to reverse systemic vasodilation, increase blood pressure, improve bronchial airflow, and relieve swelling and hives. It is typically administered via intramuscular injection into the thigh as soon as anaphylaxis is suspected.

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Incorrect Options:

Option A - Hydrocortisone: Hydrocortisone is a corticosteroid medication with anti-inflammatory properties. While corticosteroids like hydrocortisone may be used as part of the overall treatment plan for anaphylaxis, they are not considered the immediate treatment. Corticosteroids help reduce inflammation and prevent delayed allergic reactions but act more slowly than adrenaline.

Option B - Atropine: Atropine is an anticholinergic medication that can help counteract certain effects of anaphylaxis, such as bronchoconstriction. However, it is not considered the first-line treatment for anaphylactic shock. Atropine may be used in specific cases when there is severe bronchospasm or other specific indications.

Option D - Benzylpenicillin: Benzylpenicillin, also known as penicillin G, is an antibiotic and is not indicated for the immediate treatment of anaphylactic shock. Anaphylaxis is typically caused by an allergic reaction to specific allergens, such as certain foods, medications, or insect stings, and requires immediate management with adrenaline.

Solution for Question 23:

Correct Option A - Latanoprost:

- Latanoprost is a prostaglandin analog that is commonly used to lower intraocular pressure in glaucoma. It works by increasing the outflow of aqueous humor from the eye, thus reducing intraocular pressure. Latanoprost is generally well-tolerated and has a favorable side effect profile. It is often considered a first-line treatment option for open-angle glaucoma.

Latanoprost is a prostaglandin analog that is commonly used to lower intraocular pressure in glaucoma. It works by increasing the outflow of aqueous humor from the eye, thus reducing intraocular pressure. Latanoprost is generally well-tolerated and has a favorable side effect profile. It is often considered a first-line treatment option for open-angle glaucoma.

Incorrect Options:

Option B - Timolol: Timolol is a non-selective beta-blocker that can also be used to lower intraocular pressure in glaucoma. However, it is contraindicated or should be used with caution in patients with bronchial asthma. Beta-blockers can cause bronchospasm and worsen respiratory symptoms in individuals with asthma.

Option C - Gemeprost: Gemeprost is a medication used in obstetrics and gynecology for various purposes, such as cervical ripening and termination of pregnancy. It is not indicated for the treatment of glaucoma and is not relevant to this scenario.

Option D - Mannitol infusion: Mannitol is an osmotic diuretic used in various medical conditions, including glaucoma. It can help reduce intraocular pressure by decreasing fluid volume in the eye. However, it is not a preferred long-term treatment for open-angle glaucoma and is typically used in acute situations or as a temporary measure.

Solution for Question 24:

Correct Option A - Varenicline:

- Varenicline is a medication that is used to help people quit smoking. It works by binding to the same receptors in the brain that nicotine binds to, reducing cravings and withdrawal symptoms. Varenicline is more effective than other smoking cessation medications, including nicotine replacement therapy and bupropion, and has been shown to double the likelihood of successfully quitting smoking compared to placebo.
- However, varenicline has been associated with some side effects, including nausea, headache, and sleep disturbances, and may not be suitable for everyone.

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However, varenicline has been associated with some side effects, including nausea, headache, and sleep disturbances, and may not be suitable for everyone.

Incorrect Options:

Option B - Buspirone SR: Buspirone SR hydrochloride extended-release tablets (SR) are a medication that belongs to the class of drugs known as anxiolytics. While buspirone hydrochloride is not indicated specifically for smoking cessation, it has been studied as an off-label treatment for smoking cessation due to its ability to reduce anxiety and stress, which can be triggers for smoking. Buspirone hydrochloride extended-release tablets (SR) are formulations of the drug that are designed to be released into the bloodstream slowly over time, allowing for once-daily dosing. This can be more convenient for patients who have difficulty adhering to multiple daily dosing regimens.

Option C - Nicotine gum: Nicotine gum is a form of nicotine replacement therapy that is designed to help people quit smoking by reducing the intensity of nicotine withdrawal symptoms. The gum allows for the absorption of nicotine through the lining of the mouth, which can help to reduce cravings and withdrawal symptoms. While nicotine replacement therapy can be effective in smoking cessation, it is usually less effective than varenicline. Nicotine gum may also have side effects, including mouth irritation and hiccups, and should not be used by people who have dental problems or jaw pain.

Option D - Rimonabant: Rimonabant is a medication that was developed as an anti-obesity drug and was also studied for its potential in smoking cessation. It works by blocking the CB1 receptor in the brain, which is involved in regulating appetite and addictive behaviors.

Solution for Question 25:

Correct Option A - Cefoperazone:

- Cefoperzone is a cephalosporin and responsible for the symptoms in the patient after consuming alcohol which indicates disulfiram-like action.

- Cephalosporins are contraindicated in active drinkers due to the presence of methyl tetrathiol ring (MTT ring) which causes disulfiram-like action.
- Drugs causing disulfiram like action and contraindicated in alcoholics: Chlorpropamide
Cephalosporins (Cefoperazone, cefamandole, cefotetan, ceftiofur)
Metronidazole Procarbazine
- Chlorpropamide
- Cephalosporins (Cefoperazone, cefamandole, cefotetan, ceftiofur)
- Griseofulvin
- Metronidazole
- Procarbazine
- Chlorpropamide
- Cephalosporins (Cefoperazone, cefamandole, cefotetan, ceftiofur)
- Griseofulvin
- Metronidazole
- Procarbazine

Incorrect Options:

Option B - Propranolol: Propranolol does not show disulfiram-like action.

Option C - Calcium channel blockers: Calcium channel blockers do not show disulfiram-like action.

Option D - Pantoprazole: Pantoprazole does not show disulfiram-like action.

Solution for Question 26:

Correct Option C - Prazosin:

- Prazosin is a selective alpha-1 receptor antagonist.

Incorrect Options:

Option A - Phenoxybenzamine: Phenoxybenzamine is a non-selective alpha receptor antagonist. It blocks both alpha-1 and alpha-2 adrenergic receptors.

Option B - Phentolamine: Phentolamine is a non-selective alpha receptor antagonist that blocks both alpha-1 and alpha-2 adrenergic receptors. It is used in the management of conditions like

pheochromocytoma, prevention of tissue necrosis following extravasation of certain vasoconstrictive drugs, and diagnosis of pheochromocytoma.

Option D - Idazoxan: Idazoxan is an alpha-2 receptor antagonist. It selectively blocks alpha-2 adrenergic receptors.

Solution for Question 27:

Correct Option B - Bethanechol:

- Bethanechol can be used for short-term treatment of postoperative urinary retention in a patient with diabetes
- Bethanechol is a cholinergic agonist that stimulates the muscarinic receptors in the bladder, promoting bladder contractions and increasing bladder tone. It is commonly used to treat urinary retention by helping to empty the bladder.
- In the given scenario, the patient has developed postoperative urinary retention, which refers to the inability to empty the bladder completely after surgery. Bethanechol can be administered to promote bladder emptying and relieve urinary retention in this case.

Bethanechol can be used for short-term treatment of postoperative urinary retention in a patient with diabetes

Bethanechol is a cholinergic agonist that stimulates the muscarinic receptors in the bladder, promoting bladder contractions and increasing bladder tone. It is commonly used to treat urinary retention by helping to empty the bladder.

In the given scenario, the patient has developed postoperative urinary retention, which refers to the inability to empty the bladder completely after surgery. Bethanechol can be administered to promote bladder emptying and relieve urinary retention in this case.

Incorrect Options:

Option A and C - Tamsulosin and Terazosin: Tamsulosin and terazosin are alpha-1 adrenergic blockers primarily used to treat symptoms of benign prostatic hyperplasia (BPH), such as urinary hesitancy, urgency, and weak urinary flow. They relax smooth muscles in the prostate and bladder neck to facilitate urine flow. While these medications may be beneficial for BPH-related urinary symptoms, they may not be as effective in treating postoperative urinary retention.

Option D - Methacholine: Methacholine is not used for urinary retention. It is a cholinergic agonist primarily used in diagnostic tests to assess airway hyperreactivity, such as in the diagnosis of asthma.

Solution for Question 28:

Correct Option C - Pralidoxime has a role in reversing the enzyme in carbamate poisoning:

- This statement is incorrect. Pralidoxime is an antidote for organophosphate poisoning, but it does not reverse the inhibition of acetylcholinesterase by carbamates. Pralidoxime is used for nicotinic effects of organophosphates.
- The scenario described in the question is consistent with organophosphate or carbamate poisoning. The symptoms are caused by the buildup of acetylcholine, a neurotransmitter, in the body.

Incorrect Options:

Option A - Carbamates bind to acetylcholinesterase irreversibly: This statement is true. Carbamates and organophosphates inhibit acetylcholinesterase, which leads to an accumulation of acetylcholine at nerve synapses and neuromuscular junctions. Carbamates bind to the enzyme irreversibly, whereas organophosphates bind reversibly.

Option B - Atropine is used as an antidote for OP poisoning: This statement is true. Atropine is an antidote for organophosphate poisoning. It blocks the effects of excessive acetylcholine in the body but does not reverse the inhibition of acetylcholinesterase.

Option D - Succinylcholine use is avoided in rapid sequence intubation: As Succinylcholine is a depolarizing neuromuscular blocking agent that can cause a rapid rise in potassium levels, leading to cardiac arrhythmias or cardiac arrest. Its use is generally avoided in patients with known or suspected hyperkalemia, severe burns, or skeletal muscle injury.

Solution for Question 29:

Correct Option C - Esmolol:

- Esmolol is a short-acting beta blocker used in hospitals to control rapid heart rates or high blood pressure during surgery or other procedures. It has a half-life of only about 9 minutes, meaning its effects wear off quickly.

Incorrect Options:

Option A - Nadolol: Nadolol is a long-acting beta blocker taken once a day. It has a half-life of about 20-24 hours, meaning it stays in the body long. Nadolol treats high blood pressure, angina, and heart failure.

Option B - Propranolol: Propranolol is a non-selective beta blocker used to treat high blood pressure, angina, migraines, and other conditions. It has a half-life of about 4-6 hours, which means it is a relatively short-acting agent.

Option D - Atenolol: Atenolol is a selective beta-blocker for treating high blood pressure, angina, and other conditions. It has a half-life of about 6-7 hours, which means it is a relatively short-acting agent.

Solution for Question 30:

Correct Option A - Timolol:

- The patient's history of glaucoma treatment suggests the possibility of medication-induced bronchial asthma. Certain drugs used to treat glaucoma can cause bronchospasm in susceptible individuals.
- Timolol is a non-selective beta blocker commonly used to treat glaucoma by reducing intraocular pressure. However, it can also cause bronchospasm in patients with a history of asthma or chronic obstructive pulmonary disease (COPD). This is because beta blockers can block the beta-2 receptors in the lungs, which can lead to constriction of the airways.

The patient's history of glaucoma treatment suggests the possibility of medication-induced bronchial asthma. Certain drugs used to treat glaucoma can cause bronchospasm in susceptible individuals.

Timolol is a non-selective beta blocker commonly used to treat glaucoma by reducing intraocular pressure. However, it can also cause bronchospasm in patients with a history of asthma or chronic obstructive pulmonary disease (COPD). This is because beta blockers can block the beta-2 receptors in the lungs, which can lead to constriction of the airways.

Incorrect Options:

Option B - Betaxolol: Betaxolol is a beta-1 selective blocker that is used to treat glaucoma. It is less likely to cause bronchospasm than non-selective beta blockers like timolol.

Option C - Latanoprost: Latanoprost is a prostaglandin analog used to treat glaucoma by increasing the outflow of aqueous humor from the eye. It does not directly affect the airways and is unlikely to cause bronchospasm.

Option D - Bimatoprost: Bimatoprost is another prostaglandin analog used to treat glaucoma. Like latanoprost, it is unlikely to cause bronchospasm.

Solution for Question 31:

Correct Option A - Norepinephrine:

- The image exhibits an increase in arterial blood pressure and decrease in heart rate as an effect of a drug.
- norepinephrine increases systolic , diastolic and mean arterial pressure and causes reflex bradycardia
- When released, norepinephrine causes the arteries and veins to constrict, increasing peripheral vascular resistance and raising blood pressure.
- This response is due to the activation of alpha-adrenergic receptors in the smooth muscle of blood vessels.

The image exhibits an increase in arterial blood pressure and decrease in heart rate as an effect of a drug.

norepinephrine increases systolic , diastolic and mean arterial pressure and causes reflex bradycardia

When released, norepinephrine causes the arteries and veins to constrict, increasing peripheral vascular resistance and raising blood pressure.

This response is due to the activation of alpha-adrenergic receptors in the smooth muscle of blood vessels.

Incorrect Options:

Option B - Isoproterenol: Beta 2 mediated vasodilation and leads to decreases mean blood pressure and increases heart rate.

Option C - Epinephrine: Increases heart rate and blood pressure at lower doses by stimulating beta-1 adrenergic receptors in the heart but at higher doses rise of blood pressure but later beta 2 action predominates and leads to decrease in blood pressure. This is a biphasic reaction of Epinephrine.

Option D - Dopamine: Dopamine can increase or decrease blood pressure and heart rate depending on the dose and the type of receptors it stimulates. At low doses, dopamine stimulates mainly dopaminergic receptors in the kidneys, causing vasodilation and increasing renal blood flow. At the same time, at higher doses, it can also stimulate alpha-1 adrenergic receptors, increasing blood pressure.

Solution for Question 32:

Correct Option B - A patient pretreated with phentolamine develops severe hypotension on the administration of adrenaline:

- Vasomotor reversal of Dale after administration of adrenaline after alpha blockade (phentolamine) leads to decreases in blood pressure.
- Because of alpha blockade there is no alpha induced increase in BP
- Only beta receptor induced vasodilation - decrease in BP

Vasomotor reversal of Dale after administration of adrenaline after alpha blockade (phentolamine) leads to decreases in blood pressure.

Because of alpha blockade there is no alpha induced increase in BP

Only beta receptor induced vasodilation - decrease in BP

Incorrect Options:

Option A - Repeated administration of ephedrine decreases its effect on blood pressure:

- Phentolamine is an alpha-blocker and would not cause a vasomotor reversal of Dale.

Option C - High dose of acetylcholine after:

- Pretreatment with atropine, which is a muscarinic receptor antagonist, would not cause a vasomotor reversal of Dale.

Option D - An increase in pulse pressure is produced by the intravenous administration of isoprenaline:

- Intravenous administration of isoprenaline, which is a non-selective beta agonist, would not cause a vasomotor reversal of Dale.

Solution for Question 33:

Correct Option A - Atropine toxicity:

- In this case, the likely cause of fever is atropine toxicity. Atropine is a medication used to counteract the effects of organophosphate poisoning. It works by blocking the action of acetylcholine at muscarinic receptors in the body. However, atropine also has other effects on the body, such as increased heart rate, pupil dilation, and secretions drying up. These effects can lead to a rise in body temperature, which is known as atropine fever.

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Incorrect Options:

Option B - A side effect of pralidoxime: Pralidoxime, on the other hand, is used to reactivate acetylcholinesterase, the enzyme that is inhibited by organophosphate poisoning. It can cause side effects such as headache, dizziness, and blurred vision, but it is unlikely to cause fever.

Option C - Due to organophosphate poisoning: Organophosphate poisoning can cause a range of symptoms such as muscle weakness, respiratory distress, and seizures, but it is not typically associated with fever.

Option D - Idiopathic: Idiopathic fever refers to a fever of unknown cause, which is not the case in this scenario as the cause of the fever is likely related to atropine toxicity. Therefore, option d is also incorrect.

Solution for Question 34:

Correct Option D - Bradycardia:

- Muscarinic receptors are a type of acetylcholine receptor that are found in various organs of the body, including the heart, lungs, and digestive system.
- Activation of these receptors by acetylcholine causes a wide range of physiological responses, including decreased heart rate, increased secretions, smooth muscle contraction, and pupil constriction.
- Muscarinic poisoning can occur due to exposure to organophosphate or carbamate insecticides, which inhibit the activity of the enzyme acetylcholinesterase. This leads to the accumulation of acetylcholine in the synaptic cleft, which can cause excessive stimulation of muscarinic receptors and lead to symptoms such as bradycardia, excessive secretions, bronchoconstriction, and gastrointestinal distress.

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Incorrect Options:

Option A - Mydriasis: Mydriasis, or pupil dilation, is typically associated with activation of the sympathetic nervous system, whereas muscarinic receptors are part of the parasympathetic nervous system.

Option B - Diuresis: Diuresis, or increased urine production, can be caused by a variety of factors and is not a specific symptom of muscarinic poisoning.

Option C - Muscle fasciculations: Muscle fasciculations can occur in cases of acetylcholinesterase inhibitor poisoning, but they are not specific to muscarinic receptors and can also be seen with nicotinic receptor activation.

Solution for Question 35:

Correct Option A - Varenicline:

- The most effective oral drug for smoking cessation among the options provided is Varenicline.
- Varenicline is a partial agonist of the nicotine receptor in the brain and works by both reducing cravings for nicotine and reducing the pleasurable effects of smoking.

Incorrect Options:

Option B - Cevimeline: Cevimeline is not a medication used for smoking cessation. It is a muscarinic agonist used to treat symptoms of dry mouth in patients with Sjogren's syndrome, a chronic autoimmune disease that affects the body's moisture-producing glands.

Option C - Muscarine: Muscarine is a naturally occurring compound found in certain mushrooms. It is not used as a medication for smoking cessation.

Option D - Atropine: Atropine is a medication used to treat certain heart and eye conditions but is not typically used for smoking cessation. In fact, atropine can have the opposite effect and may increase the urge to smoke.

Solution for Question 36:

Correct Option C - a-2, b-3, c-1:

The correct matching is as follows:

- a. Beta 1 - 2. Betaxolol
- b. Beta 2 - 3. Isoprenaline
- c. Beta 3 - 1. Mirabegron

Explanation:

- (1) Mirabegron is a selective beta-3 adrenergic receptor agonist used primarily for the treatment of overactive bladder. The beta-3 adrenergic receptors are found primarily in the bladder smooth muscle.

- (2) Betaxolol is a selective beta-1 adrenergic receptor antagonist. It is commonly used as an anti-hypertensive agent and also for the treatment of glaucoma. Beta-1 adrenergic receptors are found mainly in the heart.
- (3) Isoprenaline (also known as isoproterenol) is a non-selective beta adrenergic receptor agonist. It acts on both beta-1 and beta-2 adrenergic receptors. It is primarily used in emergencies to increase heart rate and cardiac output.

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(2) Betaxolol is a selective beta-1 adrenergic receptor antagonist. It is commonly used as an anti-hypertensive agent and also for the treatment of glaucoma. Beta-1 adrenergic receptors are found mainly in the heart.

(3) Isoprenaline (also known as isoproterenol) is a non-selective beta adrenergic receptor agonist. It acts on both beta-1 and beta-2 adrenergic receptors. It is primarily used in emergencies to increase heart rate and cardiac output.

Incorrect Options:

- Option A, B & D are incorrect choices.

Solution for Question 37:

Correct Option D - Benzhexol:

- Based on the presented scenario, the drug that can be used to treat the patient's abnormal movements is Benzhexol (also known as trihexyphenidyl).
- Benzhexol is an anticholinergic medication commonly used to treat drug-induced extrapyramidal symptoms, including abnormal movements such as dystonia, parkinsonism, and drug-induced tremors. It works by blocking the effects of excessive acetylcholine activity in the brain.
- In the case of the patient who developed abnormal movements after receiving an intravenous drug, it is possible that the drug caused extrapyramidal side effects. Benzhexol can be prescribed to alleviate these symptoms.

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In the case of the patient who developed abnormal movements after receiving an intravenous drug, it is possible that the drug caused extrapyramidal side effects. Benzhexol can be prescribed to alleviate these symptoms.

Incorrect Options:

Option A - Hyoscine: Hyoscine, also known as scopolamine, is primarily used for the prevention and treatment of motion sickness and nausea. It does not specifically target or treat abnormal movements.

Option B - Cyproheptadine: Cyproheptadine is an antihistamine often used for allergic conditions and appetite stimulation. It can sometimes be used off-label to treat serotonin syndrome, but it is not typically used for drug-induced abnormal movements.

Option C - Levodopa: Levodopa is a medication commonly used in the treatment of Parkinson's disease. It helps replenish dopamine levels in the brain. However, levodopa is not typically used to treat drug-induced abnormal movements caused by other medications.

Solution for Question 38:

Correct Option A - Organophosphate poisoning:

- Increased Secretion and pinpoint pupil is common presentation of Organophosphate poisoning.
- Anticholinesterases (anti-ChEs) or cholinesterase inhibitors suppress the cholinesterase enzyme. They primarily inhibit both acetyl ChE and butyryl ChE. Phosphoric acid (organophosphates) or carbamic acid (carbamates) esters are the two anticholinesterases.
- In addition to causing symptoms like nausea, vomiting, stomach cramps, diarrhea, miosis, sweating, increased salivary, tracheobronchial, and gastric secretions, muscarinic, nicotinic, and central effects can also result in symptoms like bronchospasm, hypotension, muscular twitching, weakness, convulsions, and coma. The death-causing paralysis of the respiratory system.

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Incorrect Options:

Option B - Datura poisoning: Hallucinations, delirium, agitation, confusion, and disorientation can all result from datura toxicity. There may also be seizures. They also prevent acetylcholine from working, which causes symptoms like constipation, urine retention, dry mouth, hazy vision, and dilated pupils (mydriasis). Tachycardia, arrhythmias, and variations in blood pressure (both high and low blood pressure) can all result from datura intoxication.

Option C - Opioid poisoning: The central nervous system can be suppressed by opioids, which can result in shallow breathing, respiratory stoppage, and lethargy. symptoms are constipation, nausea, vomiting, bradycardia, hypotension, and bradycardia are all typical signs of opioid intoxication. Skin that feels clammy and cool is common.

Option D - Atropine poisoning: Dry mouth, lessened perspiration, dilated pupils (mydriasis), hazy eyesight, and urine retention are symptoms of atropine toxicity. Tachycardia, agitation, disorientation, delirium, hallucinations, constipation, discomfort in the abdomen, and flushed, dry, and heated skin are also present.

Solution for Question 39:

Correct Option A - Active miosis:

- Pilocarpine is a cholinergic agonist that acts on the muscarinic receptors in the eye. When applied topically to the eye, pilocarpine causes the contraction of the iris sphincter muscle, leading to active miosis, which is the constriction of the pupil.

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Incorrect Options:

Option B - Active mydriasis: Mydriasis refers to the dilation of the pupil. Pilocarpine actually has the opposite effect of mydriasis. It causes miosis, not mydriasis.

Option C - Passive miosis: The term "passive miosis" is not commonly used in medical terminology. Miosis itself refers to the constriction of the pupil, which can be actively induced by pilocarpine.

Option D - Passive mydriasis: Similar to option B, "passive mydriasis" is not a recognized term in medical practice. Mydriasis refers to the dilation of the pupil, which is not produced by pilocarpine.

Solution for Question 40:

Correct Option B - Atropine:

- The patient presents with diarrhea, sweating, excessive lacrimation and salivation. These features are suggestive of organophosphorus poisoning and the treatment of choice is IV atropine.

The patient presents with diarrhea, sweating, excessive lacrimation and salivation. These features are suggestive of organophosphorus poisoning and the treatment of choice is IV atropine.

Incorrect Options:

Option A - Neostigmine: Neostigmine is not used in organophosphorus poisoning.

Option C - Epinephrine: Epinephrine is not used for organophosphorus poisoning.

Option D - Benztropine: Benztropine is not used for organophosphorus poisoning.

Solution for Question 41:

Correct Option B - Myasthenia gravis:

- Myasthenia gravis is an autoimmune disorder. One in every 10,000 people is affected by this. Antibodies are produced against the Nm receptor. It is a nicotinic receptor present at neuromuscular junctions.
- The function of Acetylcholine gets limited in this condition. The muscles are unable to work effectively and efficiently. In particular, the eyelids, facial muscles, and pharyngeal muscles are affected. As the disease progresses, other muscles get involved.

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The function of Acetylcholine gets limited in this condition. The muscles are unable to work effectively and efficiently. In particular, the eyelids, facial muscles, and pharyngeal muscles are affected. As the disease progresses, other muscles get involved.

Incorrect Options:

Option A - Lambert Eaton syndrome: In Lambert Eaton syndrome (LEMS), eye muscle weakness, if present, tends to be mild. But on the contrary, unlike Myasthenia Gravis, eye muscle weakness was never the only symptom of the disease. Edrophonium is used to differentiate myasthenia gravis from cholinergic crisis and Lambert-Eaton syndrome.

Option C - Gullian barre syndrome: Guillain-Barré syndrome (GBS) is characterized by ascending paralysis and areflexia. It appears often secondary to an infection.

Option D - Bronchial asthma: The edrophonium test is contraindicated in patients with bronchial asthma.

Solution for Question 42:

Correct Option B - Brimonidine:

- Brimonidine is an alpha-2 adrenergic agonist used in the treatment of glaucoma. However, it has been associated with serious adverse effects, such as central nervous system depression, hypotension, and apnea in infants. Due to these risks, brimonidine is generally considered unsafe in infants.

Brimonidine is an alpha-2 adrenergic agonist used in the treatment of glaucoma. However, it has been associated with serious adverse effects, such as central nervous system depression, hypotension, and apnea in infants. Due to these risks, brimonidine is generally considered

unsafe in infants.

Incorrect Options:

Option A - Timolol: Timolol is a beta-blocker commonly used in the treatment of glaucoma. While it is generally considered safe and effective for adults and older children, it can be used with caution in infants.

Option C - Latanoprost: Latanoprost is a prostaglandin analog commonly used in the treatment of glaucoma. While it is generally safe and well-tolerated, it is important to use caution and follow appropriate dosage adjustments in infants. However, compared to brimonidine, latanoprost is considered safer in infants.

Option D - Dorzolamide: Dorzolamide is a carbonic anhydrase inhibitor used in the treatment of glaucoma. It is generally well-tolerated and safe for use in infants.

Solution for Question 43:

Correct Option B - Edrophonium:

- Edrophonium is a short-acting reversible acetylcholinesterase inhibitor used in the diagnosis of myasthenia gravis (MG) and to differentiate between MG and cholinergic crisis.
- In myasthenia gravis, there is a deficiency or dysfunction of acetylcholine receptors at the neuromuscular junction, leading to muscle weakness and fatigue. Edrophonium temporarily increases the concentration of acetylcholine in the synaptic cleft by inhibiting acetylcholinesterase, the enzyme responsible for its breakdown. If a patient with myasthenia gravis receives edrophonium and experiences a temporary improvement in muscle strength, it suggests a diagnosis of myasthenia gravis.
- On the other hand, in cholinergic crisis, there is an excess of acetylcholine due to overstimulation of cholinergic receptors. Administering edrophonium in cholinergic crisis can worsen symptoms and lead to further muscle weakness and respiratory compromise.

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gravis receives edrophonium and experiences a temporary improvement in muscle strength, it suggests a diagnosis of myasthenia gravis.

On the other hand, in cholinergic crisis, there is an excess of acetylcholine due to overstimulation of cholinergic receptors. Administering edrophonium in cholinergic crisis can worsen symptoms and lead to further muscle weakness and respiratory compromise.

Incorrect Option:

Option A - Obidoxime: Obidoxime is an antidote used for the treatment of organophosphate poisoning, not for differentiating myasthenia gravis from cholinergic crisis.

Option C - Donepezil: Donepezil is a long-acting acetylcholinesterase inhibitor used in the treatment of Alzheimer's disease and is not specifically used to differentiate myasthenia gravis from cholinergic crisis.

Option D - Atropine: Atropine is an anticholinergic medication that blocks the effects of acetylcholine. It is used for the treatment of cholinergic crisis but does not help differentiate myasthenia gravis from cholinergic crisis

Solution for Question 44:

Correct Option D - Glaucoma:

- Intravenous mannitol is commonly used as an osmotic diuretic in the treatment of acute episodes of increased intraocular pressure associated with glaucoma. It works by reducing the intraocular fluid volume and lowering intraocular pressure, providing temporary relief and protecting the optic nerve.

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Incorrect Options:

Option A - Pulmonary edema: While mannitol may be used in the management of certain types of pulmonary edema, such as in cases of increased intracranial pressure or cerebral edema. Other interventions, such as diuretics and vasodilators, are typically used in the management of pulmonary edema.

Option B - Congestive cardiac failure: Mannitol is not a standard treatment for congestive heart failure. The primary treatment for congestive heart failure involves medications that help improve cardiac function, reduce fluid retention, and manage symptoms.

Option C - Acute kidney injury with anuria: Mannitol may be used in some cases of acute kidney injury to promote diuresis and maintain urine flow. However, its use in anuric kidney injury is limited, as there is no urine output to enhance. Other interventions, such as fluid management, hemodialysis, or other renal replacement therapies, are typically employed in the treatment of anuric kidney injury.

Solution for Question 45:

Correct Option A - Salbutamol:

- Long-term use of Salbutamol, a bronchodilator commonly used for asthma and other respiratory conditions, can be associated with the development of tremors as a side effect.

Long-term use of Salbutamol, a bronchodilator commonly used for asthma and other respiratory conditions, can be associated with the development of tremors as a side effect.

Incorrect Options:

Option B - Propofol: Propofol is a sedative-hypnotic agent used for anesthesia. Long-term use of Propofol is not typically associated with the development of tremors.

Option C - Betaxolol: Betaxolol is a beta-blocker medication primarily used for the treatment of high blood pressure and certain heart conditions. Tremors are not a commonly reported side effect of Betaxolol.

Option D - Timolol: Timolol is also a beta-blocker used to treat conditions such as high blood pressure and glaucoma. While it can cause side effects, tremors are not commonly associated with its use.

Solution for Question 46:

Correct Option D - Formoterol:

- Formoterol is a long-acting bronchodilator commonly used to treat asthma and chronic obstructive pulmonary disease (COPD). It has a longer duration of action compared to short-acting bronchodilators like Salbutamol.

Formoterol is a long-acting bronchodilator commonly used to treat asthma and chronic obstructive pulmonary disease (COPD). It has a longer duration of action compared to short-acting bronchodilators like Salbutamol.

Incorrect Options:

Option A - Salbutamol: Salbutamol is a short-acting bronchodilator commonly used for acute relief of bronchospasm. It has a shorter duration of action compared to Formoterol.

Option B - Terbutaline: Terbutaline is another short-acting bronchodilator similar to Salbutamol. It is not a long-acting bronchodilator like Formoterol.

Option C - Adrenaline: Adrenaline, also known as epinephrine, is a bronchodilator that can provide immediate relief in acute asthma attacks but is not considered a long-acting bronchodilator.

Solution for Question 47:

Correct Option A - Donepezil:

- Donepezil is a cholinesterase inhibitor and is considered the drug of choice for the treatment of Alzheimer's disease. It helps improve cognitive function and slows down the progression of symptoms in patients with Alzheimer's disease.

Incorrect Options:

Option B - Atropine: Atropine is an anticholinergic medication that blocks the effects of acetylcholine. It is not used for the treatment of Alzheimer's disease and may worsen cognitive function in this population.

Option C - Physostigmine: Physostigmine is a cholinesterase inhibitor used in the treatment of anticholinergic toxicity and certain other conditions. It is not the drug of choice for Alzheimer's disease.

Option D - Fluoxetine: Fluoxetine is a selective serotonin reuptake inhibitor (SSRI) primarily used for the treatment of depression and other psychiatric disorders. It is not indicated for the

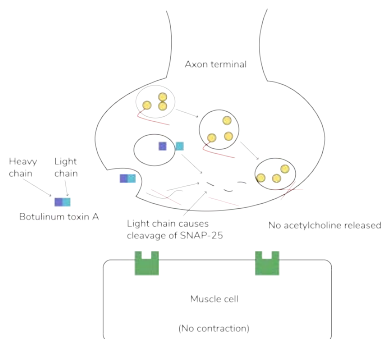
treatment of Alzheimer's disease.

Solution for Question 48:

Correct Option A - Inhibits the release of acetylcholine:

- Botulinum toxin works by selectively and irreversibly inhibiting the release of acetylcholine at the neuromuscular junction, leading to muscle relaxation. This property of botulinum toxin makes it useful in a variety of medical conditions such as dystonias, spasticity, and hyperhidrosis.

Botulinum toxin works by selectively and irreversibly inhibiting the release of acetylcholine at the neuromuscular junction, leading to muscle relaxation. This property of botulinum toxin makes it useful in a variety of medical conditions such as dystonias, spasticity, and hyperhidrosis.



Incorrect Options:

Option B - Release of noradrenaline at synaptic cleft: Botulinum toxin does not release noradrenaline at the synaptic cleft.

Option C - Selectively and irreversibly inhibits nicotinic receptors: Botulinum toxin does bind to nicotinic receptors, but its effect is not irreversible.

Option D - Stimulates muscarinic and nicotinic receptors: Botulinum toxin does not stimulate muscarinic and nicotinic receptors, but rather it inhibits the release of acetylcholine from presynaptic nerve terminals, leading to temporary paralysis of the muscles.

Solution for Question 49:

Correct Option C - Tropicamide eye drops:

- In this case, the doctor used a short-acting topical medication with cycloplegic effect for a detailed examination of the child's eyes. Tropicamide is a commonly used cycloplegic and mydriatic agent that is often employed in ophthalmology clinics for diagnostic purposes. Tropicamide eye drops cause temporary paralysis of the ciliary muscle of the eye, resulting in cycloplegia (loss of accommodation) and mydriasis (pupil dilation). These effects allow for a more accurate examination of the patient's refractive error and the posterior segment of the eye.

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Incorrect Options:

Option A - Atropine oral: Atropine is a potent anticholinergic medication with cycloplegic and mydriatic effects. However, in this case, the doctor used a topical medication for examination, not an oral formulation.

Option B - Atropine eye ointment: Atropine eye ointment can be used as a topical cycloplegic agent, but it has a slower onset of action compared to eye drops. It is typically used for prolonged dilation and cycloplegia, such as in cases of uveitis or amblyopia treatment. In this case, where a detailed examination is needed, a short-acting eye drop would be more appropriate.

Option D - Phenylephrine eye drops: Phenylephrine is a sympathomimetic medication that acts as a mydriatic (pupil dilation agent) by stimulating alpha-adrenergic receptors in the iris dilator muscle. While it can be used for pupillary dilation, it does not have cycloplegic effects. In this case, where cycloplegia is desired for a detailed examination, tropicamide eye drops would be a more suitable choice.

Non Steroidal Anti Inflammatory Drugs

1. Match the following drugs to their mechanism of action: a) Aspirin i) Inhibits COX 3 b) Meloxicam ii) Inhibits COX 1 and 2 c) Parecoxib iii) Inhibits COX 2 > COX1 d) Paracetamol iv) Inhibits only COX 2

a) Aspirin	i) Inhibits COX 3
b) Meloxicam	ii) Inhibits COX 1 and 2
c) Parecoxib	iii) Inhibits COX 2 > COX1
d) Paracetamol	iv) Inhibits only COX 2

- A. a-ii, b-iii, c-iv, d-i
- B. a-ii, b-iv, c-i, d-iii
- C. a-i, b-iii, c-iv, d-ii
- D. a-ii, b-i, c-iii, d-iv

2. Which of the following statements about NSAIDs is false?

- A. Used in IBD to control pain
- B. Not used to treat neuropathic pain
- C. Used for patent ductus arteriosus closure
- D. Exerts anti-platelet action

3. A 65-year-old male patient with a history of hypertension and congestive heart failure presents with worsening lower extremity edema and shortness of breath. He has been on a regimen of furosemide for managing his fluid overload and NSAIDs for chronic arthritis pain. He has recently increased his dose of NSAIDs due to

worsening pain. Which of the following is not a side effect of the drug responsible for failure of furosemide

- A. Gastric ulcer
 - B. Edema
 - C. Hypokalemia
 - D. Urticaria
-

4. A 40-year-old male with a family history of colorectal cancer presents for a screening colonoscopy due to his increased risk. Colonoscopy confirms the presence of multiple adenomatous polyps in the patient's colon. Which drug is prescribed as chemoprevention?

- A. Mycophenolate mofetil
 - B. Aspirin
 - C. Methotrexate
 - D. All trans-retinoic acid
-

5. A 9-year-old child presents with fever, myalgia and cough persisting for the past three days. The cough is non-productive and accompanied by mild nasal congestion. The child has a past medical history of eczema. She had a similar episode a year before and was treated with ibuprofen. Her mother has a known allergy to nonsteroidal anti-inflammatory drugs (NSAIDs). What is the reason this patient will be advised against taking aspirin?

- A. Cough
 - B. Patient age
 - C. Mother's allergy
 - D. Eczema
-

6. A 45-year-old woman presents with confusion, nausea, vomiting, and ringing in her ears. She has a history of chronic pain and admits to taking more aspirin than prescribed over the past few days. On examination, she is hyperventilating, her breath sounds are clear, and her vital signs include a BP of 110/70 mmHg, HR of 105 bpm, respiratory rate of 30/min, and temperature of 38.5°C (101.3°F). ABG analysis shows the following: pH: 7.45 | PaCO₂: 25 mmHg | HCO₃⁻: 15 mEq/L. Which of the following best explains the mechanism behind the patient's acid-base disorder?

- A. Central respiratory stimulation and disruption of oxidative phosphorylation
 - B. Renal tubular damage and acid-base disorder
 - C. Increased renal bicarbonate production
 - D. Carbonic anhydrase inhibition
-

7. Which of the following is the drug of choice used in the closure of a PDA in premature infants?

- A. Indomethacin
 - B. Ibuprofen
 - C. Nimesulide
 - D. Mefenamic acid
-

8. Which of the following NSAIDs is associated with a decreased risk of developing colon cancer?

- A. Sulindac
- B. Diclofenac
- C. Piroxicam

D. Ketorolac

9. A 65-year-old man with a history of chronic osteoarthritis presents to the emergency department with chest pain and shortness of breath. He has been taking an NSAID for pain management for the past few months. On examination, he appears diaphoretic and in distress. The ECG reveals ST-segment elevation in the anterior leads. Troponin levels are elevated. What is the most likely drug causing this patient's symptoms?

- A. Aspirin
 - B. Naproxen
 - C. Nefopam
 - D. Rofecoxib
-

10. A 25-year-old woman is brought to the emergency department by her roommate after intentionally ingesting a large quantity of paracetamol tablets. She reports feeling depressed and wanting to harm herself. On examination, she appears lethargic and complains of right upper quadrant abdominal pain. Laboratory tests reveal elevated liver enzymes. What is the primary pharmacological drug used in the treatment of paracetamol poisoning?

- A. N-Acetyl cysteine
 - B. Naloxone
 - C. Diazepam
 - D. Activated Charcoal
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	3
Question 4	2
Question 5	2
Question 6	1
Question 7	2
Question 8	1
Question 9	4
Question 10	1

Solution for Question 1:

Correct answer: A) a-ii, b-iii, c-iv, d-i

Explanation:

a) Aspirin	ii) Inhibits COX 1 and 2
b) Meloxicam	iii) Inhibits COX 2 > COX1 (Preferential COX 2 inhibitors)
c) Parecoxib	iv) Inhibits only COX 2 (Selective COX 2 inhibitors)
d) Paracetamol	i) Inhibits COX 3

- Cyclooxygenase (COX) inhibitors are used in various conditions and diseases for their analgesic, anti-inflammatory, and antipyretic properties.

- The COX enzyme catalyzes the conversion of arachidonic acid into prostaglandin.
- The enzyme has 2 known isoforms, cyclooxygenase-1 (COX-1) and cyclooxygenase-2 (COX-2).
- The COX-1 enzyme regulates many cellular processes, including platelet aggregation, kidney afferent arteriole vasodilation, and gastric mucosa acid protection. The COX-2 enzyme is an inducible enzyme and increases during inflammatory processes.

NSAIDs classification:



Empty table

Solution for Question 2:

Correct answer: A) Used in IBD to control pain

Explanation:

NSAIDs are contraindicated in patients with IBD (ulcerative colitis or Crohn's disease) because they can exacerbate inflammation and worsen symptoms due to their potential to disrupt the intestinal mucosal barrier, and they also reduce the acid secretion in the stomach, leading to increased risk of GI bleeding, ulcers, and perforation. COX 1 and 2 inhibition by NSAIDs decrease prostaglandin levels necessary for maintaining gastrointestinal mucosal defense.

NSAIDs (Option B) are not helpful for the treatment of neuropathic pain. NSAIDs primarily target inflammation and do not effectively address the underlying mechanisms of neuropathic pain, which involve dysfunction in the nervous system.

Prostaglandins (Option C) keep the ductus arteriosus open during fetal development, but in PDA, this opening remains after birth, causing abnormal blood flow. NSAIDs help close the ductus arteriosus by blocking prostaglandin production, thereby restoring normal blood flow.

Aspirin (Option D) irreversibly inhibits COX activity, which in turn blocks thromboxane A2 production and inhibits platelet function. Aspirin, hence, increases bleeding time and reduces clot formation.

Solution for Question 3:

Correct answer: C) Hypokalemia

Explanation:

The patient's increased NSAID dose and worsening edema and shortness of breath indicate fluid retention due to decreased efficacy of the loop diuretic (furosemide).

- NSAIDs can cause hyperkalemia by decreasing renin secretion and reducing aldosterone release, which lowers potassium excretion
- Combined use of NSAIDs and furosemide impairs renal function by reducing renal blood flow and prostaglandin-mediated vasodilation. This results in decreased diuretic efficacy, fluid retention, and exacerbation of heart failure symptoms

Side effects of NSAIDs:

System	Side effects	Additional information
Gastrointestinal	Nausea and vomiting	Selective COX-2 inhibitors: Less likely to cause GI side effects
GI hemorrhage, erosion, ulcers (Option A)	Due to ↓prostaglandin synthesis, leading to loss of mucosal protection	
Hematological	Platelet dysfunction → Increased bruising and hemorrhage risk	From reduced thromboxane A2 activity

Renal	Renal dysfunction/failure Reduction in efficacy of loop diuretics & antihypertensives Salt and water retention Hyperkalemia (Option C) Acute tubular necrosis Renal papillary necrosis Interstitial nephritis Analgesic nephropathy	Analgesic nephropathy leads to renal failure with papillary necrosis and chronic interstitial nephritis, which is less common since the ban of phenacetin.
Cardiovascular	Precipitate heart failure Edema (Option B)	
Hepatic	Elevated liver enzymes Hepatic failure	
Hypersensitivity	Urticaria (Option D) Flushing Anaphylaxis	
Dermatological	Steven-Johnson syndrome Photosensitivity	

Solution for Question 4:

Correct answer: B) Aspirin

Explanation:

- Aspirin is used as chemoprophylaxis for colorectal cancer for those who are at high risk, such as in Lynch syndrome or Familial Adenomatous Polyposis by suppressing the polyp formation and providing symptomatic relief.
- Colonic tumors express large quantities of COX-2. Aspirin inhibits COX-2, reducing inflammation, and potentially slowing down the growth of precancerous and cancerous cells.

Dose	Action
Low dose aspirin (40-300 mg/day)	Antiplatelet action
Medium dose aspirin (300-2400 mg/day)	Analgesic and Antipyretic action
High dose aspirin (2400-4000 mg/day)	Anti-inflammatory action

Uses of Aspirin:

- DOC for niacin-induced flushing
- Rheumatic fever
- Rheumatoid arthritis
- Osteoarthritis
- Kawasaki disease (Along with IVIG)
- Essential thrombocytopenia
- ACE inhibitor-induced dry cough
- Reduce the risk of colorectal cancer (FAP)
- APLA syndrome

Mycophenolate mofetil (Option A): is an immunosuppressant used in cases of organ transplant to reduce risk of rejection.

Methotrexate (Option C): is used in chemotherapy, immunosuppressant and disease modifying antirheumatic agent.

All trans-retinoic acid (Option D): is used to treat acute promyelocytic leukemia.

Solution for Question 5:

Correct answer: B) Patient age

Explanation:

The patient is a 9-year-old child. Aspirin is typically avoided in children due to the risk of Reye syndrome.

- Reye syndrome is a life-threatening condition that develops in children and those below the age of 20 years who have taken aspirin for a viral illness
- The mechanism of this is unknown. It is characterized by liver dysfunction and encephalopathy
- It is treated by withdrawing aspirin and supportive measures. Its incidence has decreased because aspirin is no longer used except in Kawasaki disease which is the only current indication for using aspirin in children.

Contraindications of aspirin

NSAID allergy (Due to cross-reactivity with ibuprofen allergy)
Asthma (Due to bronchospasm from NSAIDs)
Children (Risk of Reye syndrome if they have a viral infection)
Gout ($\uparrow$ uric acid levels)
Chronic liver disease (Hepatic injury at high doses)
Peptic ulcer disease or gastritis ($\uparrow$ risk of GI bleed)
Bleeding disorder ($\uparrow$ risk of bleeding)

Cough and past history of eczema (Options A and B) are not contraindications to aspirin use.

Mother's allergy (Option C) to NSAIDs is not significant, as the patient has tolerated ibuprofen in the past.

Solution for Question 6:

Correct answer: A) Central respiratory stimulation and disruption of oxidative phosphorylation

Explanation:

The given scenario is consistent with aspirin poisoning. A low PaCO_2 (normal: 35-45 mm Hg) and a low HCO_3^- (normal: 22-26 mEq/L), despite a near-normal pH and a respiratory rate of 30/min, suggest that it's a mixed acid-base disorder, respiratory alkalosis with high anion gap metabolic acidosis.

- Aspirin stimulates the brain's respiratory center, causing hyperventilation ($\downarrow \text{CO}_2$ levels) and respiratory alkalosis
- The kidneys excrete more bicarbonate to counteract respiratory alkalosis, resulting in metabolic acidosis (Compensatory metabolic acidosis). High doses of aspirin can disrupt oxidative phosphorylation and kidney function, exacerbating metabolic acidosis

Dosage and Toxicity:

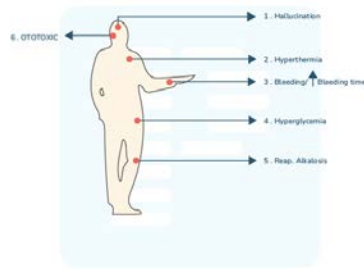
- Therapeutic Dose: 3-5 g/day can cause mild chronic toxicity (salicylism)
- Toxic Dose: Acute ingestion of 10-30 g in adults can be lethal, with much lower doses (4 mL of methyl salicylate) posing a significant risk in children

Clinical presentation:

- Salicylism: Headache, dizziness, tinnitus, reversible hearing and vision impairment, drowsiness, vomiting, hyperventilation, electrolyte imbalance

- Severe toxicity: Dehydration, high tone deafness, confusion, petechial haemorrhages, hyperpyrexia, seizures, hypoglycemia, psychosis, coma

Management: Treatment is supportive and includes gastric lavage, correction of metabolic acidosis, vit K supplements and urinary alkalinization with intravenous sodium bicarbonate to enhance salicylate excretion. In severe cases, hemodialysis may be necessary.



Solution for Question 7:

Correct answer: B) Ibuprofen

Explanation:

- Ibuprofen, a nonselective COX inhibitor, is the drug of choice for closing a patent ductus arteriosus (PDA) in premature infants by inhibiting prostaglandin synthesis, primarily PGE₂ (maintains ductus arteriosus patency during fetal development) through COX enzyme inhibition
- Due to its lower risk of renal toxicity, it is considered a safer option than indomethacin for PDA closure in neonates.

Indomethacin (Option A): is the drug of choice for treating ankylosing spondylitis, acute gout, and rheumatoid arthritis. It is also used to close PDA in premature infants, but it carries a higher risk of renal toxicity.

Nimesulide (Option C): is a preferential COX 2 inhibitor no longer in use as it causes Steven Johnson syndrome and is highly hepatotoxic. It is banned in India for use in children under 12 years of age.

Mefenamic acid (Option D): is indicated primarily as an analgesic in muscle, joint, and soft tissue pain where strong anti-inflammatory action is not needed. It is also a drug of choice in treating dysmenorrhea.

(Note: Ibuprofen is generally not recommended to be given with aspirin as both are inhibitors of COX-1, thereby reducing each other's efficacy).

Solution for Question 8:

Correct answer: A) Sulindac

Explanation:

Sulindac is a prodrug converted to an active sulfide metabolite in the body. It decreases the quantity and size of colorectal polyps, especially in individuals with familial adenomatous polyposis (FAP), reducing the risk of colorectal cancer.

Sulindac is prescribed to treat various inflammatory diseases like arthritis.

Diclofenac (Option B): is an NSAID used in arthritis pain and sprain. The liver metabolises diclofenac primarily via CYP2C9. It is more hepatotoxic.

Piroxicam (Option C): a non-selective and reversible COX inhibitor, reduces synovial fluid prostaglandin levels and inhibits platelet aggregation. Liver metabolism involves hydroxylation and glucuronide conjugation, with elimination via urine and bile, and enterohepatic cycling prolongs its half-life to approximately 2 days.

Ketorolac (Option D): a potent analgesic, is used for short-term management of acute pain (≤ 5 days) and as an injectable for managing pain following surgery, dental procedures.

Solution for Question 9:

Correct answer: D) Rofecoxib

Explanation:

Rofecoxib and Valdecoxib, selective COX-2 inhibitors, were withdrawn from the market due to their association with a higher risk of atherothrombotic events.

- COX-2 inhibitors reduce endothelial PGI₂ production without affecting platelet TXA₂ synthesis → Prothrombotic state → Increased cardiovascular risk.
- Hence, they should be reserved for high-risk peptic ulcer patients at the lowest dose and shortest duration and avoided in patients with a risk of heart disease.

Aspirin and Naproxen (Options A and B): are safer for cardiovascular patients due to their antiplatelet effects. Aspirin irreversibly inhibits platelet aggregation primarily by inhibiting COX-1, which reduces the risk of cardiovascular events. Similarly, Naproxen's longer half-life and COX inhibition contribute to cardiovascular safety.

Nefopam (Option C): doesn't act on the COX enzyme. It is a serotonin-norepinephrine dopamine reuptake inhibitor (SNDRI). It is used to treat migraines and post-anesthetic shivering. It is not known to cause cardiovascular events like STEMI.

Solution for Question 10:

Correct answer: A) N-Acetyl cysteine

Explanation:

N-acetylcysteine (NAC) is the antidote for paracetamol overdose.

Paracetamol Poisoning :

Mechanism of Toxicity: Paracetamol (acetaminophen) overdose → Increased production of toxic metabolite NAPQI (catalysed by CYP2E1) → Depletes glutathione (which normally neutralises NAPQI) → Leads to NAPQI binding to liver cells, causing Zonal necrosis/ centrilobular necrosis with periportal sparing.

Risk Factors:

- Factors Depleting GSH: Alcohol, fasting, or malnutrition.
- Children (due to lower glucuronidation capacity)
- On Isoniazid (Induces CYP2E1 enzyme)

Clinical presentation:

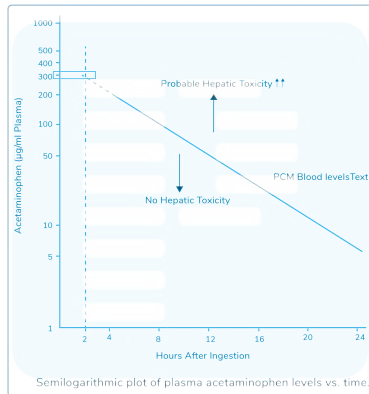
Early Stage (12–36 hours post-ingestion)

Later Stage (2–4 days post-ingestion)

Nausea, vomiting & abdominal pain
Liver tenderness
↑ plasma transaminases
No impairment of consciousness

Right subcostal pain with tender hepatomegaly
Jaundice
Coagulopathy
Renal impairment or failure
Hypoglycemia
Hepatic encephalopathy

Rupert Matthew Nomogram: A clinical tool used to assess the risk of hepatotoxicity following an acute paracetamol (acetaminophen) overdose.



Treatment:

- Activated charcoal (Option D): Within 4 hrs of ingestion (↓ paracetamol absorption).
- N-acetylcysteine (NAC): Antidote for paracetamol overdose. It replenishes GSH stores and detoxifies NAPQI. Early administration of NAC (within 8-16 hrs) improves outcome.

Naloxone (Option B): is used to reverse opioid overdoses.

Diazepam (Option C): is used to treat anxiety, muscle spasms, and seizures.

Histamine, Serotonin and Lipid Autocoids

1. A 32-year-old patient presents with symptoms of seasonal allergies, including sneezing, rhinorrhea, and itchy eyes. The patient has a history of performing tasks requiring alertness throughout the day. Which antihistamines would be the most suitable choice for this patient?

- A. Diphenhydramine
 - B. Fexofenadine
 - C. Hydroxyzine
 - D. Loratadine
-

2. Which prostaglandin analogue is not used to conduct MTP in the first trimester of pregnancy?

- A. Misoprostol
 - B. Dinoprostone
 - C. Carboprost
 - D. Latanoprost
-

3. In a 35-year-old woman at 40 weeks gestation presenting with an unripe cervix and scheduled for induction, what is the preferred subtype of prostaglandin for cervical ripening in preparation for labour?

- A. PGE2
- B. PGE1
- C. PGF2■
- D. PGI2

4. Match the following anti-inflammatory drugs with their mechanism of action. 1) Corticosteroid a) Cyclooxygenase inhibition 2) NSAIDs b) Lipoxygenase inhibition 3) Zileuton c) Phospholipase A 2 inhibition 4) Licofelone d) Platelet-activating receptor blocker 5) Apafant e) COX + LOX inhibition

1) Corticosteroid	a) Cyclooxygenase inhibition
2) NSAIDs	b) Lipoxygenase inhibition
3) Zileuton	c) Phospholipase A 2 inhibition
4) Licofelone	d) Platelet-activating receptor blocker
5) Apafant	e) COX + LOX inhibition

- A. 1-e, 2-a, 3-c, 4-b, 5-d
B. 1-c, 2-a, 3-b, 4-e, 5-d
C. 1-c, 2-e, 3-a, 4-d, 5-b
D. 1-b, 2-c, 3-d, 4-a, 5-e
-

5. A 32-year-old female with a known diagnosis of hereditary angioedema (HAE) presents to the ER with facial swelling and difficulty breathing. She reports that the symptoms started suddenly and have progressively worsened. Which drug can be used as a bradykinin B2 receptor blocker to treat the patient?

- A. Icatibant
B. Lanadelumab
C. Berotralstat
D. Ecallantide
-

6. Cimetidine can reduce the effectiveness of ketoconazole. Which of the following mechanisms explain this interaction? Cimetidine decreases gastric acid production, leading to reduced absorption of ketoconazole. Cimetidine delays gastric emptying, reducing the absorption rate of ketoconazole. Cimetidine increases gastric pH, leading to reduced solubility and absorption of ketoconazole. Cimetidine induces liver enzymes, leading to increased clearance of ketoconazole.

- A. 1, 2
 - B. 1, 3
 - C. 2, 3
 - D. 2, 4
-

7. A 55-year-old male patient is being treated for peptic ulcer disease and presents with symptoms of breast tenderness and decreased libido. His current medication includes an H2 blocker. Which of the following H2 blockers is most likely responsible for these antiandrogenic effects?

- A. Ranitidine
 - B. Famotidine
 - C. Cimetidine
 - D. Nizatidine
-

8. Which of the following drugs is known for its maximum antipruritic effect?

- A. Doxylamine
 - B. Meclizine
 - C. Hydroxyzine
 - D. Chlorpheniramine
-

9. A 50-year-old patient with a history of cardiac arrhythmias presents with symptoms of allergic rhinitis. The physician considers prescribing either terfenadine or fexofenadine for symptom relief. Which differentiating factors should be considered when selecting the appropriate antihistamine for this patient?

- A. Fexofenadine is more likely to cause sedation.
 - B. Fexofenadine is more potent than terfenadine.
 - C. Fexofenadine is less likely to cause cardiac toxicity.
 - D. Fexofenadine is more expensive than terfenadine.
-

10. Which of the following is an ionotropic receptor?

- A. 5-HT1
 - B. 5-HT2
 - C. 5-HT3
 - D. 5-HT4
-

11. A 35-year-old woman is experiencing severe allergic reactions after eating shellfish. Her doctor prescribes a medication to help manage these symptoms. Which histamine receptors are primarily involved in regulating the allergic response?

- A. H1 receptor
 - B. H2 receptor
 - C. H3 receptor
 - D. H4 receptor
-

12. A 35-year-old woman is experiencing severe nausea and vomiting after chemotherapy. Her doctor prescribes a medication to help manage these symptoms. Which of the following medications is most likely to be prescribed?

- A. Ondansetron
 - B. Metoclopramide
 - C. Cisapride
 - D. Tegaserod
-

13. A 35-year-old woman has been experiencing frequent and severe migraines. Her doctor recommends a medication that blocks 5HT-2 receptors to help prevent these attacks. Which of the following medications is most likely to be prescribed?

- A. Ketanserin
 - B. Ritanserin
 - C. Quetiapine
 - D. Cyproheptadine
-

14. A 28-year-old woman with a history of migraine without aura, currently on preventive medication, is planning to conceive. She seeks advice on safe medications for treating acute migraine attacks during pregnancy. Which of the following drugs is recommended?

- A. Sumatriptan
 - B. Acetaminophen
 - C. Flunarizine
 - D. Propranolol
-

15. A 35-year-old woman presents to the emergency department with a severe migraine headache that has been ongoing for several hours. She reports that her headache is accompanied by nausea, photophobia, and phonophobia. She has tried over-the-counter pain relievers without relief. What medication would you prescribe to help alleviate her symptoms?

- A. Sumatriptan
 - B. Paracetamol
 - C. Telotristat
 - D. Morphine
-

16. Which of the following statements best describes the interaction between ergotamine and triptans in migraine treatment?

- A. Ergotamine and triptans have a synergistic effect, leading to enhanced vasoconstriction.
 - B. Ergotamine and triptans compete for the same receptor, leading to decreased efficacy of both drugs.
 - C. Ergotamine and triptans have no interaction and can be safely used together for migraine relief.
 - D. Concurrent use of ergotamine and triptans can lead to serotonin syndrome due to additive serotonin receptor agonism.
-

17. A 35-year-old woman presents with a history of recurrent headaches. She describes a severe throbbing headache on the right side of her head associated with nausea, vomiting, and sensitivity to light and sound. She has tried over-the-counter pain relievers without relief. Triptans are being considered. Which of the following is the fastest-acting oral triptan?

- A. Sumatriptan

- B. Rizatriptan
 - C. Almotriptan
 - D. Zolmitriptan
-

18. Which medication among the following acts via the mechanism of action of antagonising calcitonin gene-related peptide (CGRP) receptors?

- A. Sumatriptan
 - B. Rimegepant
 - C. Lasmiditan
 - D. Metoclopramide
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	1
Question 4	2
Question 5	1
Question 6	2
Question 7	3
Question 8	3
Question 9	3
Question 10	3
Question 11	1
Question 12	1
Question 13	4
Question 14	2
Question 15	1
Question 16	1
Question 17	2
Question 18	2

Solution for Question 1:

Correct answer: B) Fexofenadine

Explanation:

- Fexofenadine, a second-generation antihistamine, has minimal sedative effects due to its limited ability to cross the blood-brain barrier. Thus, it is helpful in patients performing tasks requiring alertness during the day.
- Sedation and drowsiness are common side effects of first-generation antihistamines, which can impair cognitive function and alertness, affecting the patient's ability to perform tasks requiring mental acuity throughout the day.

Drugs that act by blocking Histamine H1 receptors:

Generation	Drug	Side Effects	Generalised Uses
1st Generation	Diphenhydramine (Option A)	Sedation, drowsiness, dry mouth, blurred vision	Allergy symptoms, Insomnia, Motion sickness
Chlorpheniramine	Sedation, drowsiness, dry mouth	Allergy symptoms	
Hydroxyzine (Option C)	Anxiety, Allergy symptoms, Pruritus		
2nd Generation	Loratadine (Option D)	Minimal sedation, dry mouth	Allergy symptoms, Urticaria
Cetirizine	Allergy symptoms, Itchiness, Urticaria		
Fexofenadine (Option B)	Non-sedating, minimal side effects	Allergy symptoms, Chronic idiopathic urticaria	

Solution for Question 2:

Correct answer: D) Latanoprost - PGE2

Explanation:

- Latanoprost is a PGE2 analogue that is used as a topical drug in the treatment of glaucoma.
- It reduces intraocular pressure by increasing the uveoscleral outflow of aqueous humour.
- It plays no role in the medical termination of pregnancy.

Medical Termination of Pregnancy:

DOC: Misoprostol 400µg, 48 hours after 200 mg of Mifepristone administration.

- Mifepristone is an anti-progesterone drug that makes the fetus non-viable.
- Misoprostol is a PGE1 analogue that causes uterine contraction, thus expelling the abortus.

Table rendering failed

Solution for Question 3:

Correct answer: A) PGE2

Explanation: PGE2 (Dinoprostone) is the drug of choice for cervical ripening. It only makes the cervix soft and ripe without affecting the uterine tone and motility.

Prostaglandin	Preparation	Indications
PGE1 (Option B)	Misoprostol	NSAID-induced peptic ulcer 1st trimester MTP abortion
Alprostadil	Erectile dysfunction (cavernosal injection) Critical limb ischemia Open ductus arteriosus	
Lubiprostone	Irritable bowel syndrome	
Gemeprost	1st trimester MTP abortion	
PGE2 (Option A)	Dinoprostone	DOC for cervical ripening (Vaginal) Induction of labour Medical abortion
PGF2■ (Option C)	Dinoprost	Medical abortion
Carboprost	Resistant postpartum hemorrhage Medical abortion	
Travoprost Latanoprost Bimatoprost Tafluprost	Topical eye drops for treatment of Glaucoma	
PGI2 (Option D)	Iloprost Epoprostenol Treprostinil	Pulmonary artery hypertension (via inhalation route) As an antiplatelet during hemodialysis

Solution for Question 4:

Correct answer: B) 1-c, 2-a, 3-b, 4-e, 5-d

Explanation:



Drug	Mechanism of Action	Indications
Corticosteroids	Blocks phospholipase A2 enzyme, thus reducing the production of arachidonic acid.	Most effective anti-inflammatory. Long-term usage causes Cushing syndrome.
NSAIDs	It blocks the cyclooxygenase enzyme and inhibits the conversion of arachidonic acid to prostaglandins.	Used in Gout, arthritis, musculoskeletal pain, dysmenorrhea etc Long-term use can cause gastritis and renal failure.
Zileuton	Blocks 5-lipoxygenase enzyme.	Asthma.
Licofelone	Blocks both COX and LOX enzymes.	Controls inflammation in osteoarthritis.
Apafant	Blocks platelet-activating receptor.	Controls inflammation in acute pancreatitis

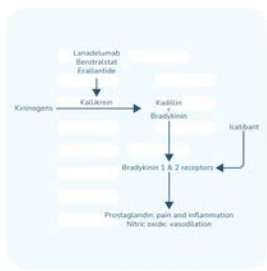
Solution for Question 5:

Correct answer: A) Icatibant

Explanation:

- Hereditary angioedema occurs due to defective/deficient C1 esterase inhibitor protein. This leads to unregulated activation of the complement system and kallikrein pathway, leading to excessive bradykinin production. Thus, exogenous C1 esterase inhibitors are used as first-line treatment to replace the deficient protein.
- Icatibant is a bradykinin B2 receptor antagonist resistant to kinin enzymatic degradation and can be used as an alternative to C1 esterase inhibitor for treating hereditary angioedema.

Drugs acting on kinin-kallikrein system:



Drugs used for the management of hereditary angioedema include the following:

Prophylaxis	Drug of choice: Danazol (increases C1 esterase production) Alternatives: Kallikrein antagonists Lanadelumab: Monoclonal antibody against plasma kallikrein (Option B) Berotralstat: Plasma enzyme blocker (Option C)
Treatment	Drug of choice: C1 esterase inhibitor. Alternatives: Kallikrein antagonist Ecallantide: Potent enzyme inhibitor (Option D) Bradykinin B2 receptor antagonist Icatibant

Solution for Question 6:

Correct answer: B) 1, 3

Explanation:

- Cimetidine, a H₂ receptor antagonist, reduces gastric acid secretion, leading to decreased absorption of ketoconazole when administered concurrently.
 - This interaction can result in decreased efficacy of ketoconazole.
 - Cimetidine is also a potent inhibitor of cytochrome P450 enzymes, leading to decreased metabolism of several drugs and subsequent elevation of their serum levels.
 - Theophylline, phenytoin, and warfarin are primarily metabolised by cytochrome P450 enzymes. Therefore, concurrent administration of cimetidine can lead to decreased metabolism and increased levels of these drugs, potentially causing toxicity due to elevated plasma concentrations.
-

Solution for Question 7:

Correct answer: C) Cimetidine

Explanation:

- Cimetidine is an H₂ receptor antagonist that reduces gastric acid secretion in conditions such as peptic ulcer disease and GERD.
- Notable side effects include CNS disturbances and antiandrogenic effects like gynecomastia.
- Cimetidine is known for its antiandrogenic effects, including gynecomastia, breast tenderness, and decreased libido, due to its ability to inhibit estradiol metabolism and displace dihydrotestosterone from its receptor.

Ranitidine (Option A):

It has no significant antiandrogenic effects and is generally well-tolerated, with a lower risk of CNS effects and fewer drug interactions compared to cimetidine.

Famotidine (Option B):

It also lacks significant antiandrogenic effects and has minimal drug interactions. It is more potent and has a longer action duration than ranitidine and cimetidine.

Nizatidine (Option D):

It has no significant antiandrogenic effects and minimal drug interactions, with a similar efficacy and side effect profile to ranitidine, although it is used less frequently.

Solution for Question 8:

Correct answer: C) Hydroxyzine

Explanation: Hydroxyzine is a first-generation anti-histamine with a maximum anti-pruritic effect.

Drug Name	Primary Use	Antipruritic Effect	Mechanism of Action	Other Characteristics
Doxylamine (Option A)	DOC for morning sickness	Moderate	Blocks H1 histamine receptors	Treat allergy symptoms and used as a short-term sleep aid due to its sedative effects
Meclizine (Option B)	Seasickness	Mild to moderate	Blocks H1 histamine receptors and muscarinic receptors	Primarily used to prevent and treat nausea, vomiting, and dizziness caused by motion.
Hydroxyzine (Option C)	Antipruritic, anxiolytic	Maximum	Blocks H1 histamine receptors and has central nervous system depressant activity.	Strong antihistamine used for treating itching, anxiety and as a sedative.
Chlorpheniramine (Option D)	Allergic reactions	Mild to moderate	Blocks H1 histamine receptors	Commonly used for allergic reactions like hay fever and hives. It is the only first-generation antihistamine that can be given during the day.

Solution for Question 9:

Correct answer: C) Fexofenadine is less likely to cause cardiac toxicity.

Explanation:

- Terfenadine, a first-generation antihistamine, is associated with a risk of cardiac arrhythmias, including QT interval prolongation and torsades de pointes, due to its effects on cardiac potassium channels.
- In contrast, fexofenadine, the active metabolite of terfenadine, has a lower propensity for such cardiac adverse effects.

Feature	Terfenadine	Fexofenadine
Type	First-generation antihistamine	Second-generation antihistamine
Cardiac toxicity	Withdrawn due to risk of cardiac arrhythmias (e.g., Torsades de Pointes) when co-administered with CYP3A4 inhibitors. (Note: CAT drugs - Cisapride, Astemizole, and Terfenadine have been withdrawn due to cardiotoxicity).	Generally considered safer with minimal risk of cardiac arrhythmias.
Drug Interactions	Prone to interactions with CYP3A4 inhibitors.	Less prone to interactions with CYP3A4 inhibitors.
CNS Penetration	Penetrates blood-brain barrier.	Poor penetration of the blood-brain barrier.
Pharmacokinetics	Metabolism leads to a longer half-life and duration of action.	Minimal metabolism, shorter half-life, and shorter duration of action.
Sedative Effects	It may cause sedation and impair psychomotor performance.	Minimal sedation or impairment of psychomotor performance.

Solution for Question 10:

Correct Answer: C) 5HT3

Explanation: 5HT3 is an ionotropic receptor.

R e c e p t o r t y p e	5-HT1	5-HT2	5-HT3	5-HT4-7
S u b t y p e s	5HT-1A, 5HT-1B, 5HT-1D, 5HT-1E, 5HT-1F, 5HT-1P.			
T y p e o f r e c e p t o r	GPCR	GPCR	Ionotropic ligand-gat ed	GPCR
M e c h a n i s m	Inhibit serotonergic activity in the brain, and their agonist drugs are used in migraine prophylaxis and as anxiolytics.	Mediate vascular and visceral smooth muscle contraction, platelet aggregation and neuronal activation in the brain.	Elicit reflex effects like emesis, gut peristalsis, bradycardia, transient hypotensio n, apnoea, pain, and itch.	Mediate intesti nal secretion and augmenta tion of peristal sis.

Solution for Question 11:

Correct answer: A) H1 receptor

Explanation: H1 receptors are primarily involved in regulating allergic responses, including smooth muscle contraction, increased vascular permeability, and stimulation of sensory nerve endings. Antihistamines that target H1 receptors are commonly used to alleviate symptoms of

allergies and allergic reactions.

Receptor	Functions	Locations
H1 (OptionA)	Allergic responses Smooth muscle contraction Increased vascular permeability Stimulation of sensory nerve endings.	Smooth muscle cells Endothelial cells Neurons Immune cells
H2 (OptionB)	Gastric acid secretion Airway mucus production Vascular permeability	Parietal cells in the stomach lining.
H3 (OptionC)	Neurotransmitter release modulation	Central nervous system (CNS)
H4 (OptionD)	Immune responses Chemotaxis Cytokine production	Mast cells Eosinophils T cells

Solution for Question 12:

Correct answer: A) Ondansetron

Explanation: Ondansetron is a selective 5-HT₃ antagonist commonly used to prevent nausea and vomiting, particularly in patients undergoing chemotherapy or radiation therapy.

	5HT-3 Antagonists (Anti-motility drugs)	5HT-4 Agonists (Prokinetic drugs)
Action	Block 5HT-3 receptors in the gastrointestinal tract and central nervous system. Decrease intestinal motility and secretion. Inhibit the chemoreceptor trigger zone (CTZ) in the area postrema, reducing nausea and vomiting.	Activate 5HT-4 receptors in the gastrointestinal tract. Increase gastrointestinal motility and secretion. Enhances the release of acetylcholine from enteric neurons.
Drugs	Ondansetron (Option A) Granisetron Dolasetron Palonosetron Alosetron	Metoclopramide (Option B) Cisapride (withdrawn due to QT prolongation) (Option C) Tegaserod (restricted use due to cardiovascular risks) (Option D)

Uses	Chemotherapy-induced nausea and vomiting (CINV) Postoperative nausea and vomiting (PONV) Radiation-induced nausea and vomiting (RINV) Irritable bowel syndrome with diarrhoea (IBS-D): Alosetron	Gastroparesis Gastroesophageal reflux disease (GERD) Functional dyspepsia Chronic constipation Postoperative ileus
Adverse Effects	Headache Constipation Dizziness Fatigue Prolonged QT interval (rare)	Diarrhea Abdominal pain Nausea Headache Prolonged QT interval (Cisapride, Tegaserod)

Solution for Question 13:

Correct answer: D) Cyproheptadine

Explanation:

- Cyproheptadine primarily blocks 5-HT₂ receptors and has additional H₁ antihistaminic, anticholinergic, and sedative properties.
- It can be used in the prophylaxis of migraine.

Clinical Uses of Cyproheptadine include

- Migraine prophylaxis.
- Serotonin syndrome.
- Carcinoid syndrome.
- Cold urticaria.
- Promote weight gain in AIDS patients.

Drugs that act by blocking 5-HT₂ receptors (5HT-2 antagonists)

Drug	Uses
Cyproheptadine	Migraine prophylaxis, antihistamine
Methysergide	Migraine and cluster headache prophylaxis
Pizotifen	Migraine prophylaxis, antihistamine

Quetiapine (Option C)	Schizophrenia, bipolar disorder
Ketanserin (Option A)	Hypertension, peripheral vascular disease
Ritanserin (Option B)	Investigated for schizophrenia, depression, and sleep disorders

Solution for Question 14:

Correct Answer: B) Acetaminophen

Explanation: Acetaminophen or Paracetamol is DOC for migraine in pregnancy as well as DOC for Migraine prophylaxis.

Drugs Used for Migraine in Pregnancy:

The recommended treatment for migraine during pregnancy involves a step-by-step approach:

- Initial Treatment: For mild migraine attacks, use paracetamol. If this does not provide relief, consider adding: Codeine Caffeine (in moderation) Metoclopramide (although it may have some side effects)
- Codeine
- Caffeine (in moderation)
- Metoclopramide (although it may have some side effects)
- Codeine
- Caffeine (in moderation)
- Metoclopramide (although it may have some side effects)
- Triptans: If the above treatments fail, sumatriptan is the safest triptan option due to its extensive use and minimal risk of adverse effects.

Drugs used for Migraine Prophylaxis:

- Flunarizine (Calcium channel blocker)
- Cyproheptadine: (Antihistamine that can help alleviate migraine symptoms).
- Candesartan: (Angiotensin II receptor blocker).
- Clonidine
- Pizotifen (Serotonin antagonist that can help reduce migraine frequency).
- Propranolol (Beta-blocker, DOC for migraine prophylaxis).

- Drugs releasing GABA: Gabapentin, valproate, nortriptyline, and topiramate are medications that can help manage migraine symptoms by releasing the neurotransmitter GABA.
- Topiramate
- Methysergide (Manages migraine symptoms by blocking serotonin receptors).

Solution for Question 15:

Correct answer: A) Sumatriptan

Explanation: Sumatriptan is a more potent and specific migraine treatment that provides superior pain relief than paracetamol.

Mechanism of Action	It is a selective 5-HT _{1B/1D} receptor agonist that activates other subtypes of 5-HT ₁ receptors only at very high concentrations. Inhibits the release of calcitonin gene-related peptide (CGRP) and causes cranial vasoconstriction.
Administration and Efficacy	It is administered at the onset of a migraine attack and is as effective and better tolerated than ergotamine. About 3/4 of patients obtain complete or significant relief within 2-3 hours. Available in various formulations such as Oral tablets, Nasal sprays, Injections and Rectal suppositories.
Pharmacokinetics	It is absorbed rapidly and completely after subcutaneous injection, with an oral bioavailability of only 15%. It is rapidly metabolised by MAO-A isoenzyme, and metabolites are excreted in urine, with an elimination half-life of approximately 2 hours.
Side Effects	Common side effects include tightness in the head and chest, feeling of heat and other paresthesias in the limbs, dizziness, and weakness. These are usually mild and short-lasting but can be dose-related.

Paracetamol (Option B): It is not effective enough for treating moderate to severe migraine pain. Sumatriptan is a more potent and specific migraine treatment that provides superior pain relief compared to paracetamol.

Telotristat (Option C): It is not indicated for treating acute migraine attacks. It is used to treat carcinoid syndrome diarrhoea.

Morphine (Option D): It is not recommended for migraine due to its side effects and potential for abuse. Opioids like morphine are generally not considered first-line treatments for migraine.

Solution for Question 16:

Correct answer: A) Ergotamine and triptans have a synergistic effect, leading to enhanced vasoconstriction.

Explanation:

- Ergots have a mechanism of action similar to triptans. They stimulate the 5-HT_{1B/1D} receptors but are non-selective, decreasing the release of CGRP.
- They are more potent vasoconstrictors than triptans.
- Ergotamine and triptans have a synergistic effect, leading to enhanced vasoconstriction, and hence should never be combined.
- Side effects of Ergotamine
Coronary vasoconstriction (Angina and MI)
Gangrene of the extremities
Hypertension
Arrhythmias
Sweating
Bradycardia
- Coronary vasoconstriction (Angina and MI)
- Gangrene of the extremities
- Hypertension
- Arrhythmias
- Sweating
- Bradycardia
- Ergotamine is contraindicated in
Ischaemic heart disease
Hypertension
Epilepsy
Hepatic or renal impairment
Pregnancy
- Ischaemic heart disease
- Hypertension
- Epilepsy
- Hepatic or renal impairment
- Pregnancy
- Coronary vasoconstriction (Angina and MI)
- Gangrene of the extremities
- Hypertension

- Arrhythmias
- Sweating
- Bradycardia
- Ischaemic heart disease
- Hypertension
- Epilepsy
- Hepatic or renal impairment
- Pregnancy

Solution for Question 17:

Correct Answer: B) Rizatriptan

Explanation:

- Rizatriptan is the fastest-acting oral triptan and the second most effective drug for migraines, after sumatriptan, the most effective for acute migraines.
- Triptans act by decreasing the release of CGRP and cause vasoconstriction.

The types of triptans, along with their route of administration, include

Drug names	Route of administration	Bioavailability	Half-life
Rizatriptan (Option B)	Oral	45%	1.5 hours
Sumatriptan (Option A)	Oral, Nasal, Subcutaneous	15%	2.5 hours
Zolmitriptan (Option D)	Oral, Nasal	40%	2 hours
Almotriptan (Option C)	Oral	70%	3 hours
Eletriptan	Oral	50%	4 hours
Frovatriptan	Oral	25%	3 hours
Naratriptan	Oral	70%	3 hours

Solution for Question 18:

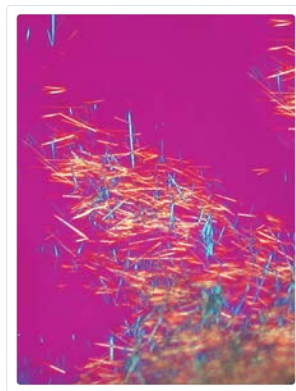
Correct answer: B) Rimegepant

Explanation:

Pathophysiology of Migraine	Migraine is characterised by a pulsating unilateral headache that comes in bouts lasting 4-48 hours and is associated with nausea, vomiting, sensitivity to light and sound, flashes of light, and vertigo. 5th cranial nerve stimulation (5-HT _{1F} receptor) ↓ Release of neurotransmitters like CGRP, neurokinin, Nitrous Oxide, 5-HT ↓ Dilation of meningeal vessels causing headache
Drugs used in acute treatment	Triptans: Decrease the release of CGRP and cause vasoconstriction. E.g., Sumatriptan, Rizatriptan. (Option A ruled out) CGRP ligand blockers: Eptinezumab, Fremanezumab CGRP receptor blockers: Rimegepant (Option B) 5-HT _{1F} agonist: Causes inhibition of neuronal firing. E.g., Lasmiditan (Option C ruled out) Ergot alkaloids: Vasoconstrictor. E.g., Ergotamine D ₂ Blockers: Metoclopramide (Option D ruled out) Opioids and NSAIDs

Gout and Rheumatoid Arthritis

1. A patient presents with a fever and a red, tender, swollen left knee, which started two days ago. He was recently diagnosed with hypertension and started on hydrochlorothiazide. Immediate joint aspiration is done, and the microscopic picture is given below. Which of the following is the most appropriate initial management?



- A. Probenecid
- B. Naproxen
- C. Sulfipyrazone
- D. Febuxostat

2. Which of the following drugs treats an acute episode of gout by binding to synovial tubulin thus inhibiting the migration of granulocytes into the joint ?

- A. Etoricoxib
- B. Piroxicam
- C. Prednisolone
- D. Colchicine

3. Which of the following drugs is not used in the management of chronic hyperuricemia?

- A. Etoricoxib
 - B. Sulfipyrazone
 - C. Probenecid
 - D. Febuxostat
-

4. A patient complains of persistent pain and stiffness over both his big toes and right knee over the past 2 years. X-ray of big toe showed the following changes. Which of the following is the most appropriate drug of choice?

- A. Piroxicam
 - B. Allopurinol
 - C. Hydrocortisone
 - D. Etoricoxib
-

5. All of the following about allopurinol are incorrect except?

- A. Allopurinol is long-acting than alloxanthine.
 - B. Allopurinol is largely bound to plasma proteins.
 - C. Allopurinol reduces the activity of azathioprine.
 - D. Allopurinol is contraindicated during lactation.
-

6. A 48-year-old man with a history of chronic gout presented with recurrent episodes of painful attacks despite being on allopurinol. His physician suggests a drug that inhibits the reabsorption of uric acid in the proximal tubules. Which of the following drugs uses this mechanism of action?

- A. Probenicid
 - B. Febuxostat
 - C. Rasburicase
 - D. Colchicine
-

7. A 75-year-old male with a history of gout presents to the clinic with increased frequency of gout attacks. His daily medications include rosuvastatin, low-dose aspirin and losartan for hypertension. He has been advised to avoid certain medications that can exacerbate his gout. Which of the following drugs should be avoided in this patient?

- A. Allopurinol
 - B. Losartan
 - C. Aspirin
 - D. Rosuvastatin
-

8. A 42-year-old female presented to the hospital with early morning stiffness in her joints for the last few weeks. She reports severe pain in her right wrist and left knee. Increased CRP levels are seen in laboratory testing. What is the most appropriate treatment for this condition?

- A. Methotrexate
- B. NSAIDs
- C. Hydroxychloroquine
- D. Colchicine

9. A 45-year-old female has been experiencing pain and stiffness in her wrists despite initial treatments. Her rheumatologist decides to start her on disease-modifying therapy (DMARDs). Which of the following statements incorrectly describes the mechanisms of action of the DMARDs? Methotrexate inhibits dihydrofolate reductase, reducing the production of DNA and RNA. Hydroxychloroquine inhibits T-cell activation and reduces the production of pro-inflammatory cytokines. Sulfasalazine is metabolised to 5-aminosalicylic acid (5-ASA), which reduces inflammation of joints in rheumatoid arthritis (RA) Leflunomide inhibits the synthesis of purines, leading to a reduction in lymphocyte proliferation. Infliximab is an oral DMARD that acts by inhibiting interleukin-6.

- A. 3, 4 and 5
- B. 1, 3, 4, and 5
- C. 1, 4 and 5
- D. 3 and 4 only

10. A 42-year-old female with a 3-year history of worsening joint pain presented to the clinic with early morning stiffness and swelling in her wrists. Her symptoms were not relieved with ibuprofen. A physical examination of the hands is shown. What is the most appropriate first-line treatment for her condition?



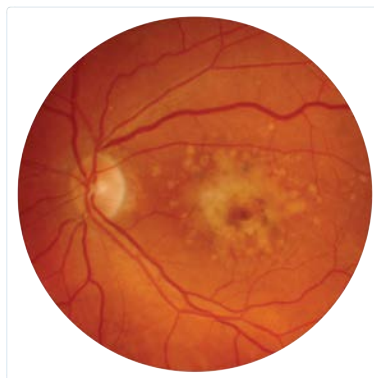
- A. Hydroxychloroquine
- B. Sulfasalazine

- C. Methotrexate
 - D. Colchicine
-

11. A 52-year-old female patient with a known case of rheumatoid arthritis for 5 years presents to the clinic with complaints of nausea, fatigue, mouth ulcers and an unusual rash on her legs. She has been on a stable regimen of DMARDs for the past few years. Laboratory investigations reveal a decreased Hb level, elevated liver enzymes, and mild leukopenia. Which drug is most likely responsible for her current symptoms?

- A. Hydroxychloroquine
 - B. Sulfasalazine
 - C. Methotrexate
 - D. Penicillamine
-

12. A 52-year-old female with a 10-year history of rheumatoid arthritis has been on certain drug therapy for the past 5 years. She recently underwent an ophthalmological examination, where the ophthalmologist noted the presence of the following condition. What is the maximum recommended daily dose of the drug (in mg/kg) that is considered safe to minimise the risk of developing this condition?



- A. 2.5 mg/kg
- B. 5 mg/kg
- C. 6.5 mg/kg
- D. 10 mg/kg

13. A 33-year-old male patient with rheumatoid arthritis (RA) has been undergoing treatment with methotrexate for the past two years. Despite the treatment, his disease activity has worsened, prompting the addition of a newer drug. Six months later, he got married, and subsequently, the couple faced difficulty conceiving, which was attributed to the new medication. Which of the following drugs is most likely responsible?

- A. Hydroxychloroquine
- B. Sulfasalazine
- C. Leflunomide
- D. Etanercept

14. Match the following Biologicals with their respective MOA. BIOLOGICALS MOA
a) Rituximab i) Anti-TNF α b) Anakinra ii) Anti-IL6 c) Infliximab iii) Anti-IL1 d) Tocilizumab iv) CD20

BIOLOGICALS	MOA
a) Rituximab	i) Anti-TNF α
b) Anakinra	ii) Anti-IL6
c) Infliximab	iii) Anti-IL1
d) Tocilizumab	iv) CD20

- A. a-iii, b-iv, c-ii, d-i
- B. a-iv, b-i, c-ii, d-iii

C. a-iv, b-iii, c-i, d-ii

D. a-iii, b-ii, c-i, d-iv

15. Which of the following is not a Janus kinase (JAK) inhibitor?

A. Baricitinib

B. Upadacitinib

C. Tofacitinib

D. Alectinib

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	1
Question 4	2
Question 5	4
Question 6	1
Question 7	3
Question 8	2
Question 9	1
Question 10	3
Question 11	3
Question 12	2
Question 13	2
Question 14	3
Question 15	4

Solution for Question 1:

Correct answer: B) Naproxen

Explanation:

The given clinical scenario and microscopy showing negatively birefringent urate crystals suggest acute gouty arthritis, and naproxen is the most appropriate initial drug for this patient.

- Gout is a metabolic disorder characterized by hyperuricemia (normal plasma urate 2-6mg/dl).
- Secondary hyperuricemia can be caused by diseases like leukemias, lymphomas, polycythemias; or drugs like thiazides, furosemide, pyrazinamide, ethambutol, levodopa, that reduce the uric acid excretion by kidney.
- Acute gouty arthritis: It is a sudden onset of severe inflammation in a small joint (M/C: metatarsophalangeal joint of the great toe) due to precipitation of urate crystals in the joint space. The joint becomes red, swollen, and extremely painful, requiring immediate treatment. Management: First-line: NSAIDs (all except aspirin and paracetamol). E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib. NSAID resistant cases: Colchicine. Colchicine resistant cases: Intra-articular corticosteroids.
- It is a sudden onset of severe inflammation in a small joint (M/C: metatarsophalangeal joint of the great toe) due to precipitation of urate crystals in the joint space.
- The joint becomes red, swollen, and extremely painful, requiring immediate treatment.
- Management: First-line: NSAIDs (all except aspirin and paracetamol). E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib. NSAID resistant cases: Colchicine. Colchicine resistant cases: Intra-articular corticosteroids.
- First-line: NSAIDs (all except aspirin and paracetamol). E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib.
- E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib.
- NSAID resistant cases: Colchicine.
- Colchicine resistant cases: Intra-articular corticosteroids.
- It is a sudden onset of severe inflammation in a small joint (M/C: metatarsophalangeal joint of the great toe) due to precipitation of urate crystals in the joint space.
- The joint becomes red, swollen, and extremely painful, requiring immediate treatment.
- Management: First-line: NSAIDs (all except aspirin and paracetamol). E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib. NSAID resistant cases: Colchicine. Colchicine resistant cases: Intra-articular corticosteroids.
- First-line: NSAIDs (all except aspirin and paracetamol). E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib.
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- First-line: NSAIDs (all except aspirin and paracetamol). E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib.
- E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib.

- NSAID resistant cases: Colchicine.
- Colchicine resistant cases: Intra-articular corticosteroids.
- E.g. Naproxen, piroxicam, diclofenac, indomethacin, or etoricoxib.

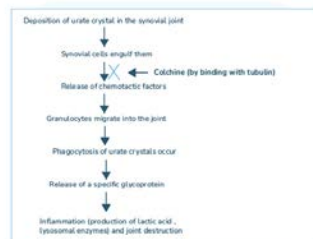
Solution for Question 2:

Correct answer: D) Colchicine

Explanation:

Colchicine is the drug that acts by binding to the fibrillary protein tubulin, thus inhibiting the migration of granulocytes into the joint (naproxen and piroxicam share the exact mechanism of action).

Mechanism of action of colchicine:



Solution for Question 3:

Correct answer: A) Etoricoxib

Explanation:

Etoricoxib, which is an NSAID, will not be used in chronic hyperuricemia/chronic gout. Instead, it can be used to treat acute gout attacks.

Chronic gout:

- Persistence of pain and stiffness in joints in between the attacks.
 - Other cardinal features are hyperuricemia, tophi, and urate stones in the kidney.
 - In most cases, hyperuricemia is due to the under-secretion of uric acid, while in a few, it is due to overproduction.
 - Chronic gouty arthritis may cause progressive deformities and permanent disabilities.
 - Drugs used in chronic gout: Uricosurics: They increase uric acid excretion. Drugs: Probenecid, sulfinpyrazone, lesinurad, benzbromarone. Uric acid synthesis inhibitors: They inhibit xanthine oxidase, inhibiting uric acid synthesis. Allopurinol, febuxostat.
 - Uricosurics: They increase uric acid excretion. Drugs: Probenecid, sulfinpyrazone, lesinurad, benzbromarone.
 - They increase uric acid excretion.
 - Drugs: Probenecid, sulfinpyrazone, lesinurad, benzbromarone.
 - Uric acid synthesis inhibitors: They inhibit xanthine oxidase, inhibiting uric acid synthesis. Allopurinol, febuxostat.
 - They inhibit xanthine oxidase, inhibiting uric acid synthesis.
 - Allopurinol, febuxostat.
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 - Uric acid synthesis inhibitors: They inhibit xanthine oxidase, inhibiting uric acid synthesis. Allopurinol, febuxostat.
 - They inhibit xanthine oxidase, inhibiting uric acid synthesis.
 - Allopurinol, febuxostat.
 - They increase uric acid excretion.
 - Drugs: Probenecid, sulfinpyrazone, lesinurad, benzbromarone.
 - They inhibit xanthine oxidase, inhibiting uric acid synthesis.
 - Allopurinol, febuxostat.
-

Solution for Question 4:

Correct answer: B) Allopurinol

Explanation:

The given clinical scenario along with the x-ray findings of Martel sign is suggestive of chronic gout and allopurinol is the most appropriate drug of choice among the given options. Rat bite erosions, also known as Martel sign or G sign, are seen in gout and refer to punched out erosions with sclerotic and overhanging margins.

Allopurinol:

- Allopurinol was initially synthesized as an antineoplastic agent, but was later found to lack anti-cancer property, but an inhibitor of xanthine oxidase (XO).
- Both allopurinol and its primary metabolite, oxypurinol (alloxanthine), reduce urate production by inhibiting XO, the enzyme that converts xanthine to uric acid.
- Allopurinol competitively inhibits XO at low concentrations and is a non-competitive inhibitor at high concentrations.
- It is the drug of choice for treatment of chronic gout, however there is a risk of acute attack in the beginning due to compensatory mobilisation of tissue stores into plasma.
- Hence, colchicine or NSAIDs are given along with in the beginning.
- Other uses of allopurinol: Tumour lysis syndrome. Lesch-Nyhan syndrome. Post-transplant patients.
- Tumour lysis syndrome.
- Lesch-Nyhan syndrome.
- Post-transplant patients.
- Tumour lysis syndrome.
- Lesch-Nyhan syndrome.
- Post-transplant patients.

Piroxicam, Etoricoxib (NSAIDS) and Hydrocortisone (steroids), are used in the management of acute gout attacks. (Options A, C and D ruled out)

Solution for Question 5:

Correct answer: D) Allopurinol is contraindicated during lactation.

Explanation:

Pharmacokinetics of allopurinol:

- Allopurinol (half-life of 2 hours) is shorter acting than its metabolite alloxanthine (half-life of 24 hours). (Option A)
- 80% of orally administered allopurinol is absorbed orally; it is not bound to plasma proteins and is metabolised to xanthine (Option B).
- Drug interactions: Allopurinol inhibits the degradation of 6-mercaptopurine and azathioprine, increasing their concentrations (Option C). Probenecid + allopurinol: Probenecid shortens the $t_{1/2}$ of allopurinol. Allopurinol prolongs the $t_{1/2}$ of probenecid. Allopurinol can potentiate warfarin and theophylline by inhibiting their metabolism. Allopurinol + ampicillin: Higher incidence of skin rashes.
- Allopurinol inhibits the degradation of 6-mercaptopurine and azathioprine, increasing their concentrations (Option C).
- Probenecid + allopurinol: Probenecid shortens the $t_{1/2}$ of allopurinol. Allopurinol prolongs the $t_{1/2}$ of probenecid.
- Probenecid shortens the $t_{1/2}$ of allopurinol.
- Allopurinol prolongs the $t_{1/2}$ of probenecid.
- Allopurinol can potentiate warfarin and theophylline by inhibiting their metabolism.
- Allopurinol + ampicillin: Higher incidence of skin rashes.
- Contraindications: Hypersensitivity, pregnancy, and lactation (Option D).
- Allopurinol inhibits the degradation of 6-mercaptopurine and azathioprine, increasing their concentrations (Option C).
- Probenecid + allopurinol: Probenecid shortens the $t_{1/2}$ of allopurinol. Allopurinol prolongs the $t_{1/2}$ of probenecid.
- Probenecid shortens the $t_{1/2}$ of allopurinol.
- Allopurinol prolongs the $t_{1/2}$ of probenecid.
- Allopurinol can potentiate warfarin and theophylline by inhibiting their metabolism.
- Allopurinol + ampicillin: Higher incidence of skin rashes.
- Probenecid shortens the $t_{1/2}$ of allopurinol.
- Allopurinol prolongs the $t_{1/2}$ of probenecid.
- Cautious use is recommended in Extremes of age. Patients with kidney and liver disease.
- Extremes of age.
- Patients with kidney and liver disease.

- Extremes of age.
- Patients with kidney and liver disease.

Solution for Question 6:

Correct answer: A) Probenecid

Explanation:

Probenecid inhibits the reabsorption of uric acid in the proximal tubules.

Uricosuric drugs	
Mechanism	Inhibit OAPT transporter/ URAT-1 in kidney ⇒ Inhibits reabsorption ⇒ Increase excretion of uric acid
Drugs	1. Probenecid: Side effects: diarrhoea, hypersensitivity, uric acid stones Contraindication: nephrolithiasis, renal failure Precaution: take plenty of oral fluids Concomitant colchicine or NSAIDs indicated in attack of gout 2. Sulphinpyrazole: Non-steroidal anti-inflammatory drug Recommended in both acute and chronic gout 3. Benzbromarone: Safe in renal failure ... (content truncated)

Rasburicase degrades uric acid in the blood into a harmless molecule of allantoin.

Solution for Question 7:

Correct answer: C) Aspirin

Explanation:

Aspirin affects renal handling of uric acid in two ways:

- High doses (>3 g/day): Uricosuric effect (blocking urate reabsorption).
- Low doses (<2 g/day): Uric acid retention (inhibiting urate secretion).

Gout triggers can include alcohol, red meat and certain medications.

Drugs Associated with Gout	
Drugs causing gout	Drugs beneficial in gout
Mnemonic – PAINED P: Pyrazinamide A: Aspirin (Normal dose <2 gms) I: Immunosuppressive: Cyclosporine N: Niacin E: Ethambutol D: Diuretic like loop diuretics, thiazides	Mnemonic – FALSE F: Fenofibrate A: Aspirin in toxic dose >5 gm L: Losartan (OPTION B) Se: Statins (Pravastatin, rosuvastatin) (OPTION D)
Treatment for drug-induced gout: Continue the drug + Allopurinol.	

Note: Allopurinol (Option A) benefits gout by reducing uric acid production by inhibiting xanthine oxidase.

Solution for Question 8:

Correct answer: B) NSAIDs

Explanation:

Rheumatoid arthritis (RA):

- Type 3 and 4 hypersensitivity common in a middle-aged female.
- Symptoms: Early morning stiffness, pain, redness and swelling.
- Joints involved: MCP, ICP and knee.
- Management:

Treatment of Rheumatoid Arthritis	
Acute arthritis Mild-moderate DOC: NSAIDs(OPTION B) All except paracetamol (Piroxicam, Indomethacin, Aspirin, Diclofenac, Celecoxib)	Long-term definitive treatment DMARDs (Disease-modifying antirheumatic drugs) DOC: Methotrexate (OPTION A) Alternate: Hydroxychloroquine, Sulfasalazine, Infliximab etc. (OPTION D)

2. Severe flare/ Resistant to NSAIDS
 1-2 joints: Intra-articular corticosteroids (Prednisolone, Triamcinolone)
 >2 joints: Oral prednisolone

Action begins after 1 month
 Not recommended for acute arthritis.
 Bridge therapy – Start with corticosteroid and maintain the patient on DMARD.

Solution for Question 9:

Correct answer: A) 3, 4 and 5

Explanation:

DMARD's		
No.	Medication	Mechanism of Action
1.	Methotrexate (1) (anchor drug)	Inhibition of cytokine production, chemotaxis and cell mediated immune reaction.
2.	Hydroxychloroquine	Block lymphocyte proliferation like chloroquine but with longer action.
3.	Sulfasalazine	The sulfapyridine component blocks lymphocyte proliferation.
4.	Leflunomide	Inhibits DHOLD (Dihydro orotate dehydrogenase) and thymidylate synthase.
5.	Infliximab (I/V) Certolizumab Adalimumab Golimumab Etanercept	Anti-TNF■
6.	Chloroquine	Inhibits IL1, IL6 TNF-Alpha in lymphocytes, reducing T cell activation.
7.	Tocilizumab Sarilumab	Anti-IL-6
8.	Rituximab	Anti-CD-20
9.	Anakinra	Anti-IL-1

10.	Abatacept	Recombinant CTLA4, inhibits CD80/86
11.	Tofacitinib	Janus kinase (JAK) inhibitors
12.	Penicillamine Gold (Au) Azathioprine	Inhibits macrophage and decreases IL-1
13.	Azathioprine	Cleaved to 6-mercaptopurine (6MP) and inhibit purine synthesis

Solution for Question 10:

Correct answer: C) Methotrexate

Explanation:

Methotrexate should be used as the first line of treatment for this adult female patient with chronic pain with early morning stiffness and swelling in her wrist joints, as well as hand deformity (swan-neck and boutonnière deformity) in the image.

Methotrexate (Anchor drug):

- Dose: Low dose (10-25 mg/week).
- Uses: Most effective drug for RA.
- Onset: action starts > 1 month.
- Contraindications: Pregnancy (neural tube defects) and lactation.
- Toxicity antidote: Folinic acid/ Leucovorin.
- Methotrexate resistance: Use anti-TNF-alpha or CD20 inhibitor.



- Hydroxychloroquine and Sulfasalazine are used in RA but are generally considered less effective than methotrexate for moderate to severe cases. (Options A and B ruled out)

Solution for Question 11:

Correct answer: C) Methotrexate

Explanation:

A patient with a known case of rheumatoid arthritis on DMARDs presenting with nausea, fatigue, mouth ulcers, rash on her legs, and labs showing anaemia, elevated liver enzymes, and mild leukopenia suggests methotrexate use. It is the first-line drug used for patients with chronic rheumatoid arthritis.

Side effects of methotrexate:

- Nausea, vomiting, and fatigue.
- Oral ulcers.
- Hair loss.
- Hematologic: Myelosuppression (decreased white blood cells, red blood cells, and platelets).
- Myelosuppression (decreased white blood cells, red blood cells, and platelets).
- Hepatic: Elevated liver enzymes. Hepatotoxicity.
- Elevated liver enzymes.
- Hepatotoxicity.
- Gastrointestinal: Stomatitis (inflammation of the mouth). Diarrhoea.
- Stomatitis (inflammation of the mouth).
- Diarrhoea.
- Other: Skin rashes. Increased incidence of chest infection.
- Skin rashes.
- Increased incidence of chest infection.
- Myelosuppression (decreased white blood cells, red blood cells, and platelets).
- Elevated liver enzymes.
- Hepatotoxicity.
- Stomatitis (inflammation of the mouth).
- Diarrhoea.
- Skin rashes.

- Increased incidence of chest infection.

Side effects of Hydroxychloroquine.(Option A):

- Gastrointestinal upset.
- Skin rash.
- Retinal toxicity.
- Corneal opacity.

Side effects of Sulfasalazine (Option B):

- Gastrointestinal upset.
- Rash.
- Liver enzyme abnormalities.
- Myelosuppression.

The combination of nausea, oral ulcers, significant hepatotoxicity, and a decrease in white blood cell count is more characteristic of methotrexate.

Side effects of Penicillamine (Option D):

- Diarrhoea.
- Dermatitis.
- Drug-induced lupus.
- Nephrotoxicity.

Solution for Question 12:

Correct answer: B) 5 mg/kg

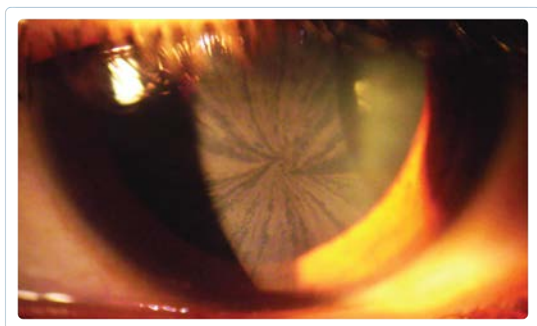
Explanation:

The given image shows the bull's eye maculopathy caused by hydroxychloroquine. The recommended dose to prevent this adverse effect should not be >5mg/kg.

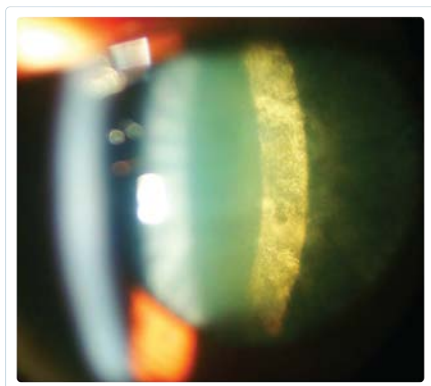
Hydroxychloroquine	
Mechanism of Action	Hydroxychloroquine ■ Lysosome Accumulation ■ ↑ Lysosomal pH ■ Inhibits Antigen Processing ■ ↓ Antigen-MHC Presentation ■ ↓ T-Cell Activation ■

	Inhibits TLRs ■ ↓ Pro-inflammatory Cytokines ■ ↓ Inflammation in RA.
Uses	Rheumatoid arthritis. Systemic lupus erythematosus (SLE). Prevention and treatment of malaria. Autoimmune diseases like Sjögren's syndrome
Side Effects	Gastrointestinal (nausea, vomiting, diarrhoea). Dermatological (rash, pigmentation changes). Ocular toxicity: Vortex keratopathy (cornea verticillata). Posterior subcapsular cataract. Bull's eye retinopathy (deposits like concentric rings in the retina). Cardiovascular (cardiomyopathy, QT prolongation). Neurological (headache, dizziness). Hematologic (anaemia, leukopenia).
Additional Info	Regular ophthalmologic examinations to monitor for retinal toxicity. Maximum safe dose of hydroxychloroquine to minimise the risk of bull's-eye maculopathy, is recommended to be 5 mg/kg of real body weight per day.

Cornea verticillata:



Posterior subcapsular cataract:



Solution for Question 13:

Correct answer: B) Sulfasalazine

Explanation:

Sulfasalazine	
Mechanism of Action	Sulfasalazine ■ Active Metabolites ■ Inhibits Cyclooxygenase/Lipoxygenase ■ ↓ Prostaglandins/Leukotrienes ■ ↓ Inflammation in GI Tract and Joints + Immunomodulation + Antimicrobial Effects.
Uses	IBD ulcerative colitis and Crohn's disease. Rheumatoid arthritis. Juvenile idiopathic arthritis.
Side Effects	Gastrointestinal(nausea, vomiting, diarrhea). Headache. Rash. Blood disorders like leukopenia and thrombocytopenia. Liver dysfunction. Reversible infertility in males (oligospermia).
Additional Info	Administered cautiously in patients with known sulfa allergies. Monitoring of blood counts and liver function is recommended during therapy.

Solution for Question 14:

Correct answer: C) a-iv, b-iii, c-i, d-ii

Explanation:

Mechanism of Action (MOA)	Biological Drug
---------------------------	-----------------

TNF-alpha inhibitor	Adalimumab Etanercept Infliximab
IL-6 receptor antagonist	Tocilizumab Sarilumab
CD20-directed cytolytic monoclonal antibody	Rituximab
CTLA-4 Ig fusion protein	Abatacept
IL-1 receptor antagonist	Anakinra

Solution for Question 15:

Correct answer: D) Alectinib

Explanation:

Alectinib is a tyrosine kinase and RET proto-oncogene used to treat ALK-positive non-small cell lung cancer.

Janus kinase (JAK) inhibitors	
Drugs	Baricitinib (Option A) Upadacitinib (Option B) Tofacitinib (Option C)
MOA	JAK Inhibitor ■ Inhibits JAKs ■ Blockade of Cytokine Signaling ■ ↓ Inflammatory Mediators ■ ↓ Inflammation in GI Tract and Joints + Immunomodulation.
Uses	Mnemonic: "RAJPUT" Rheumatoid Arthritis Ankylosing Spondylitis Juvenile Idiopathic Arthritis Psoriatic Arthritis Ulcerative Colitis Tofacitinib
Side Effects	Increased risk of infections Liver enzyme elevation Gastrointestinal disturbances Headache Fatigue Increased cholesterol levels

Previous Year Questions

1. What is pegloticase used for ?

- A. Chronic gout
 - B. Rheumatoid arthritis
 - C. Psoriatic arthritis
 - D. Paralytic ileus
-

2. How does colchicine work in the body?

- A. Inhibits synthesis of uric acid
 - B. Promotes excretion of uric acid
 - C. Inhibits xanthine oxidase enzyme
 - D. Binds to tubulin and inhibits granulocyte migration
-

3. What is the initial treatment of choice for chronic gout in a 52-year-old woman with hypertension, type 2 diabetes mellitus, severe pain, and swelling in her left great toe, sporadic similar episodes over the past 3 years, and a daily intake of 6 beers?

- A. Allopurinol
 - B. Febuxostat
 - C. Probenecid
 - D. Sulfapyrazone
-

4. What is the most suitable medication for alleviating the symptoms of a 56-year-old postmenopausal woman who has been experiencing worsening pain and swelling in her left knee for the past 6 months? She has a medical history of peptic ulcer disease and is currently taking cimetidine. On examination, palpable crepitus and restricted range of motion are observed in her left knee.

- A. Ketorolac
 - B. Etoricoxib
 - C. Piroxicam
 - D. Ibuprofen
-

5. Which of the following is the compound that prostaglandin, indicated for multiple conditions, is derived from?

- A. Stearic acid
 - B. Arachidonic acid
 - C. Linoleic acid
 - D. Linolenic acid
-

6. An elderly woman on treatment for migraine presents with chest pain. She has a previous history of angina/MI. Which of the following mechanism of the prescribed drug for migraine might have caused this presentation

- A. 5-HT antagonist
 - B. 5-HT agonist
 - C. Calcium Channel Blocker action
 - D. Nitric oxide-mediated vasodilation
-

7. Which of the following is correct about the drugs used in Gout? A. Aspirin is not used as it decreases uric acid excretion B. Diclofenac is used as it concentrates in the joint C. Indomethacin is used as it acts by multiple pathways D. Aspirin is used as it is most powerful NSAID

- A. A, B & C
 - B. A, B, C & D
 - C. A & B
 - D. C & D
-

8. Which of the following statements is true? A. Morphine used in pulmonary edema B. Pethidine used in biliary colic C. Indomethacin is used in headache D. Aspirin used in peptic ulcer disease

- A. A and C
 - B. A, B and C
 - C. A and D
 - D. B, C and D
-

9. Which medication can be administered to reduce the frequency of future headache episodes in a 26-year-old female patient who experiences one-sided pulsating headaches accompanied by nausea, vomiting, and sensitivity to light, and finds relief in a dimly lit environment?

- A. Propranolol
- B. Alprazolam
- C. Diazepam
- D. Fluoxetine

10. Why is cisatracurium preferred over atracurium?

- A. It has rapid onset of action
 - B. It has less depressant action on heart
 - C. It has short duration of action
 - D. It causes less release of histamine
-

11. Which of the following helps in maintaining the patency of ductus arteriosus?

- A. PGF2 α
 - B. PGE1
 - C. PGH2
 - D. PGI2
-

12. In the management of Thalamic Dejerine Syndrome, which of the following drugs is the least useful?

- A. Pregabalin
 - B. Carbamazepine
 - C. Gabapentin
 - D. NSAID analgesic
-

13. A female delivered a male baby developed PPH was treated with ergometrine but not able to stop the bleeding. Which of the following is a prostaglandin

derivative that can be used in the treatment of this patient?

- A. Misoprostol
 - B. Dinoprost
 - C. Carboprost
 - D. Latanoprost
-

14. A 55-year-old male with complaints of urine hesitancy and retention of urine comes to ER. On investigation, his USG shows residual urine of 130 ml. Which of the following drug is used in the treatment of this patient?

- A. Calcium channel blockers
 - B. 5 alpha reductase inhibitors
 - C. ACE inhibitors
 - D. Aldosterone antagonist
-

15. Which drug is known to cause hand-foot syndrome?

- A. Capecitabine
 - B. Cyclophosphamide
 - C. Gemcitabine
 - D. Paclitaxel
-

16. Drug for HIV induced lipodystrophy:

- A. Tesamorelin
- B. Somatropin

- C. Octreotide
 - D. Pegvisomant
-

17. Methotrexate usage causes reduced synthesis of:

- A. TMP
 - B. CMP
 - C. GMP
 - D. AMP
-

18. Which among the following is not preferred as an analgesic in rheumatoid arthritis?

- A. Tramadol
 - B. Ketorolac
 - C. Naproxen
 - D. Oxycodone
-

19. A student complains of developing urticaria following consumption of seafood. Considering the need for non-sedating medication due to an upcoming exam, which of the following drugs would you choose to prescribe?

- A. Pheniramine
 - B. Promethazine
 - C. Hydroxyzine
 - D. Fexofenadine
-

20. A patient presents with acute swelling in the big toe and his serum uric acid levels were increased. The doctor prescribed a medicine for the management of this acute condition. What is the mechanism of this drug?

- A. Pyrimidine antimetabolite
 - B. Uricosuric drug
 - C. Xanthine oxidase inhibitor
 - D. Neutrophil chemotaxis prevention
-

21. A 40 year old male presents with diffuse abdominal pain. He has a known history of chronic NSAID's usage. Physical examination reveals rigidity. X-ray abdomen is as shown below. What is the diagnosis?



- A. Intestinal obstruction
 - B. Pericarditis
 - C. Peptic ulcer perforation
 - D. Gallstones
-

22. Which of the following factors could be accountable for the exacerbation of the patient's symptoms after the intern administered an analgesic for biliary colic?

- A. Morphine
 - B. Diclofenac
 - C. Nefopam
 - D. Etoricoxib
-

23. Pegloticase is used in management of:

- A. Chronic gout
 - B. Rheumatoid arthritis
 - C. Psoriatic arthritis
 - D. Calcium pyrophosphate dehydrate crystal deposition (CPPD)
-

24. Which medication is employed for the management of motion sickness?

- A. Cetirizine
 - B. Promethazine
 - C. Misolastin
 - D. Loratidine
-

25. Which of the following drugs is not used for the prophylaxis of migraine?

- A. Propranolol
- B. Imipramine
- C. Flutamide

D. Flunarizine

26. Mechanism of action of allopurinol is?

- A. Xanthine oxidase inhibition
 - B. Recombinant uricase
 - C. Increase uric acid excretion
 - D. Decrease chemotaxis
-

27. Which of the subsequent options is an abortifacient, derived from synthetic PFG2 alpha?

- A. Dinoprost
 - B. Dinoprostone
 - C. Misoprostol
 - D. Carboprost
-

28. In a patient with chronic kidney disease who has undergone nephrectomy, which of the following medications can be given as a pain reliever after the surgery?

- A. Naproxen
 - B. Indomethacin
 - C. Diclofenac
 - D. Acetaminophen
-

29. What should be the subsequent course of action in the management of a young female who presents to the emergency department after ingesting 100 tablets of aspirin?

- A. N-acetyl cysteine to replenish glutathione stores
 - B. Pralidoxime
 - C. Glucagon to control bradycardia and hypoglycemia
 - D. Give sodium bicarbonate to alkalize urine
-

30. What medication was administered to a patient experiencing frequent episodes of gout, resulting in a decrease in symptoms by inhibiting the production of uric acid?

- A. Probenecid
 - B. Colchicine
 - C. Diclofenac
 - D. Allopurinol
-

31. NAPQI is the metabolite of which of the following drug toxicity?

- A. Aspirin
 - B. Paracetamol
 - C. Diclofenac
 - D. Ketorolac
-

32. Urinary alkalinization will be most useful in toxicity of which of the following drugs?

- A. Beta-blockers
 - B. Calcium channel blockers
 - C. Acetaminophen
 - D. Aspirin
-

33. A pregnant woman was prescribed indomethacin. Which of the following effects does the drug have on the fetus?

- A. Premature closure of ductus arteriosus
 - B. Keeps ductus arteriosus patent
 - C. Closure of fossa ovalis
 - D. Premature closure of ductus venosus
-

34. Which of the following drugs are used in the closure of patent ductus arteriosus?

- A. Magnesium sulfate
 - B. Alprostadil
 - C. Indomethacin
 - D. Paracetamol
-

35. A 56-year-old male comes to the outpatient department for a routine health check-up. The patient is obese and has smoked for 30 years. His past medical history includes hypertension for 10 years and diabetes mellitus for 8 years, for which he takes amlodipine and metformin pills regularly. On examination, his

blood pressure is 140/90 mmHg, and lab findings reveal a cholesterol level of 170 mg/dL. The physician prescribes aspirin to prevent ischemic heart disease. This drug prevents ischemic heart disease by which of the following mechanism of action?

- A. Inhibiting TXA2 synthesis by platelets
 - B. Inhibiting prostacyclin synthesis
 - C. Inhibiting the release of EDRF
 - D. Inhibiting the synthesis of endothelins
-

36. A patient on aspirin presented with occult blood in the stool. This is due to inhibition of _____ by aspirin:

- A. Phospholipase A2
 - B. Thromboxane
 - C. Phospholipase C
 - D. Phospholipase D
-

37. A patient presents with swelling in his big toe, and his serum uric acid levels were found to be elevated. After acute treatment, the patient was put on a maintenance regimen with a drug which is considered first-line management for chronic gout; What is the mechanism of action of this drug?

- A. Pyrimidine antimetabolite
 - B. Inhibitor of neutrophil recruitment
 - C. Uricosuric drug
 - D. Xanthine oxidase inhibitor
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	1
Question 4	2
Question 5	2
Question 6	2
Question 7	1
Question 8	2
Question 9	1
Question 10	4
Question 11	2
Question 12	4
Question 13	3
Question 14	2
Question 15	1
Question 16	1
Question 17	1
Question 18	2
Question 19	4
Question 20	4
Question 21	3
Question 22	1

Question 23	1
Question 24	2
Question 25	3
Question 26	1
Question 27	4
Question 28	4
Question 29	4
Question 30	4
Question 31	2
Question 32	4
Question 33	1
Question 34	3
Question 35	1
Question 36	2
Question 37	4

Solution for Question 1:

Correct Option is A: A Pegloticase is used for the treatment of chronic gout. It is specifically indicated for patients with severe gout who have not responded well to other treatments or who cannot tolerate other treatment options. Pegloticase works by breaking down uric acid, which helps to reduce the build-up of uric acid crystals in the joints and alleviate symptoms of gout.

Incorrect Options:

Option B: Rheumatoid arthritis is not a indicated use for pegloticase. Rheumatoid arthritis is a different autoimmune condition that requires different treatment approaches.

Option C: Psoriatic arthritis is also not an indicated use for pegloticase. Psoriatic arthritis is a type of arthritis associated with psoriasis and requires specific treatments targeting both the joint inflammation and skin manifestations.

Option D: Paralytic ileus is a condition characterized by the impairment of intestinal motility, and pegloticase is not indicated for its treatment. Paralytic ileus usually requires supportive care and addressing the underlying cause of the condition.

Solution for Question 2:

Correct Option D - Colchicine:

- Colchicine exerts its mechanism of action by binding to tubulin and inhibiting granulocyte migration, specifically targeting the microtubules.
- This action reduces the inflammatory response associated with conditions such as gout and familial Mediterranean fever.

Incorrect Options:

Option A - Inhibits synthesis of uric acid: This is not the mechanism of action of colchicine. Colchicine does not directly inhibit the synthesis of uric acid.

Option B - Promotes excretion of uric acid: Colchicine does not promote the excretion of uric acid. Its primary action is related to its anti-inflammatory properties.

Option C - Inhibits xanthine oxidase enzyme: Colchicine does not inhibit the xanthine oxidase enzyme. Xanthine oxidase inhibitors, such as allopurinol, are used to reduce uric acid production.

Solution for Question 3:

Correct Option A

- Allopurinol: Allopurinol is the first choice for chronic gout. It is a xanthine oxidase inhibitor that reduces the production of uric acid and helps prevent the formation of urate crystals that cause gout attacks.

Incorrect options:

Option B. Febuxostat: Febuxostat is another xanthine oxidase inhibitor used for the management of chronic gout. While it is effective, it is not the first choice in this scenario.

Option C. Probenecid: Probenecid is a uricosuric agent that increases the excretion of uric acid from the body. It is often used as an adjunctive therapy in patients with gout who do not achieve adequate control with xanthine oxidase inhibitors. However, it is not the first choice for initial treatment.

Option D. Sulfipyrazone: Sulfipyrazone is also a uricosuric agent used in the management of gout. Similar to Probenecid, it is not the first choice for initial treatment.

Solution for Question 4:

Correct Option B:

- Etoricoxib: Etoricoxib is a nonsteroidal anti-inflammatory drug (NSAID) that is commonly used for the symptomatic treatment of osteoarthritis and other painful conditions. It provides pain relief and helps reduce inflammation in the affected joints.

Incorrect options:

Incorrect A. Ketorolac: Ketorolac is another NSAID used for the short-term management of moderate to severe pain. However, its use is generally limited to a maximum of 5 days due to the risk of adverse effects, such as gastrointestinal bleeding.

Incorrect C. Piroxicam: Piroxicam is an NSAID indicated for the treatment of rheumatoid arthritis and osteoarthritis. While it can be effective for pain relief, it is associated with a higher risk of gastrointestinal side effects compared to some other NSAIDs.

Incorrect D. Ibuprofen: Ibuprofen is a commonly used NSAID for the relief of pain and inflammation. It can be effective for knee pain; however, the specific patient's history of peptic ulcer disease may require a more selective NSAID with a lower risk of gastrointestinal adverse effects, such as etoricoxib.

Solution for Question 5:

Correct Option: B

- Prostaglandins are a class of lipid compounds that play diverse roles in the body, including regulating inflammation, blood flow, and various physiological processes. They are derived from a fatty acid called arachidonic acid, which is an omega-6 polyunsaturated fatty acid.
- Arachidonic acid is obtained from the diet and is also synthesized in the body from essential fatty acids. When the body needs prostaglandins, arachidonic acid is metabolized by enzymes called cyclooxygenases (COX) to produce prostaglandins.
- These prostaglandins then act as local hormones, exerting their effects in the immediate vicinity of their synthesis. They are involved in a wide range of biological processes, including inflammation, pain signaling, blood vessel constriction and dilation, regulation of blood clotting, and control of gastrointestinal secretions.

Incorrect Option:

Option A) Stearic acid: Stearic acid is a saturated fatty acid found in various foods, including animal fats and chocolate. It is not directly involved in the synthesis of prostaglandins.

Option C) Linoleic acid: Linoleic acid is an omega-6 polyunsaturated fatty acid that is also involved in the synthesis of various lipid compounds in the body. However, it is not the direct precursor of prostaglandins.

Option D) Linolenic acid: Linolenic acid is an omega-3 polyunsaturated fatty acid, and while it is involved in the synthesis of other important lipid compounds in the body, it is not the precursor for prostaglandin synthesis.

Solution for Question 6:

Correct Option B - 5-HT agonist:

- The mechanism of the prescribed drug for migraine as a 5-HT agonist might have caused the presentation of chest pain in this elderly woman with a history of angina/MI.
- Migraine is a complex neurological condition that involves changes in the levels of neurotransmitters, including serotonin (5-HT), in the brain. Various drugs are used for the treatment of migraines, and one class of drugs includes 5-HT agonists, also known as triptans. These drugs work by binding to specific serotonin receptors and exerting their effects.

- In the case of this elderly woman with a history of angina/MI, the 5-HT agonist prescribed for migraine treatment can potentially cause chest pain as a side effect. 5-HT agonists have vasoactive properties and can cause constriction of blood vessels, including coronary arteries. This vasoconstriction can lead to reduced blood flow to the heart, which can trigger or worsen angina symptoms and potentially result in chest pain.

Incorrect Options:

Option A - 5-HT antagonist:

- 5-HT antagonists, also known as serotonin antagonists, work by blocking the effects of serotonin in the body. They are not typically associated with causing chest pain in individuals with a history of angina/MI. These drugs may have other side effects, but they are not directly related to chest pain in this context.

Option C -Calcium Channel Blocker action:

- Calcium channel blockers are commonly used for the treatment of angina and hypertension. They work by blocking calcium channels in blood vessels, leading to vasodilation and relaxation of smooth muscle. Although calcium channel blockers can have side effects, such as hypotension or dizziness, they are not typically associated with causing chest pain in individuals with a history of angina/MI.

Option D - Nitric oxide-mediated vasodilation:

- Nitric oxide-mediated vasodilation is a physiological process that involves the release of nitric oxide to promote relaxation and dilation of blood vessels. Nitric oxide donors or drugs that enhance nitric oxide signaling are often used to treat conditions associated with vascular dysfunction, such as angina. These drugs are not commonly associated with causing chest pain in individuals with a history of angina/MI.

Solution for Question 7:

Correct Option A - A, B & C:

A. Aspirin is not commonly used in the treatment of gout because it can interfere with the excretion of uric acid and potentially worsen the condition. Aspirin competes with uric acid for renal tubular secretion, leading to decreased uric acid excretion and potential accumulation.

B. Diclofenac is a nonsteroidal anti-inflammatory drug (NSAID) that can be used in the treatment of acute gout attacks. It is known to concentrate in inflamed joints, providing localized relief from

pain and inflammation.

C. Indomethacin is another NSAID that is commonly used in the treatment of gout. It acts by inhibiting the production of inflammatory mediators, such as prostaglandins, and provides relief from pain and inflammation associated with gout.

Incorrect Options:

Option B - A, B, C & D: This option is incorrect because option D (Aspirin is used as it is the most powerful NSAID) is not an accurate statement regarding the drugs used in gout.

Option C - A & B: This option is incorrect because option c is also accurate.

Option D - C & D: This option is incorrect because option D is not accurate.

Solution for Question 8:

Correct Option B - A, B and C:

- A. Morphine is used in the management of pulmonary edema. It helps relieve symptoms such as shortness of breath and anxiety by reducing preload and afterload on the heart and promoting vasodilation. B. Pethidine (also known as meperidine) is commonly used to manage the pain associated with biliary colic. It helps relax the smooth muscles in the bile ducts, reducing the intensity of the pain. C. Indomethacin is a nonsteroidal anti-inflammatory drug (NSAID) that is sometimes used to treat specific types of headaches, such as cluster headaches and hemicrania continua. It helps reduce inflammation and alleviate headache symptoms.

Incorrect Options:

Option A - A and C: This option is correct, as explained above.

Option C - A and D: Aspirin is not typically used in the management of peptic ulcer disease due to its potential to irritate the gastric mucosa and exacerbate ulcer symptoms. It is generally recommended to avoid aspirin in patients with peptic ulcers, as other medications like proton pump inhibitors (PPIs) are more commonly used for ulcer treatment.

Option D - B, C, and D: This option is incorrect because, as mentioned earlier, aspirin is not recommended for peptic ulcer disease.

Solution for Question 9:

Correct Option: A: Propranolol

- Based on the clinical history, the patient is suffering from a Migraine headache. Propranolol is the most commonly used drug to prevent future attacks of headaches. It reduces the frequency as well as the severity of attacks in up to 70% of patients. The effect is generally seen in 4 weeks and is sustained during prolonged therapy.

Incorrect Options:

Option B: Alprazolam, a benzodiazepine can be used in acute attacks of migraine to induce sleep.

Option C: Diazepam may be used in acute attacks of migraine.

Option D: Fluoxetine is an SSRI that may help treat migraine in some people.

Solution for Question 10:

Correct Option is d. It causes less release of histamine.

Option D. It causes less release of histamine: Cisatracurium is preferred over atracurium because it causes less release of histamine. Histamine release can lead to adverse effects such as bronchoconstriction, hypotension, and flushing. Cisatracurium is considered a non-histamine-releasing neuromuscular blocking agent, making it a favorable choice in patients who are susceptible to histamine-related reactions.

Incorrect Options

Option A. It has rapid onset of action: Cisatracurium and atracurium have similar onset of action. Both drugs are classified as intermediate-acting neuromuscular blocking agents and have comparable time to onset.

Option B. It has less depressant action on the heart: Cisatracurium and atracurium have similar cardiovascular effects. They do not have significant direct effects on the heart and do not cause substantial cardiovascular depression.

Option C. It has a short duration of action: Cisatracurium and atracurium have similar durations of action. The duration of effect for both drugs is moderate, allowing for adequate muscle relaxation during surgical procedures.

Solution for Question 11:

Correct Option B - PGE1:

- PGE1 (Prostaglandin E1) helps in maintaining the patency of the ductus arteriosus. It acts as a vasodilator and keeps the ductus arteriosus open, allowing blood to bypass the lungs in fetal circulation. PGE1 is naturally produced in the body and is commonly used as a medication (e.g., alprostadil) to maintain ductal patency in certain congenital heart defects.

PGE1 (Prostaglandin E1) helps in maintaining the patency of the ductus arteriosus. It acts as a vasodilator and keeps the ductus arteriosus open, allowing blood to bypass the lungs in fetal circulation. PGE1 is naturally produced in the body and is commonly used as a medication (e.g., alprostadil) to maintain ductal patency in certain congenital heart defects.

Incorrect Options:

Option A - PGF2 α : PGF2 α (Prostaglandin F2 α) is not involved in maintaining the patency of the ductus arteriosus. Instead, it plays a role in vasoconstriction and contraction of smooth muscle.

Option C - PGH2: PGH2 (Prostaglandin H2) is an intermediate in the biosynthesis of other prostaglandins and is not directly involved in maintaining the patency of the ductus arteriosus.

Option D - PGI2: While PGI2 (Prostacyclin) is a potent vasodilator and inhibitor of platelet aggregation, it does not specifically play a role in maintaining the patency of the ductus arteriosus.

Solution for Question 12:

Correct Option D - NSAID Analgesic:

- Nonsteroidal anti-inflammatory drugs (NSAIDs) are a class of medications commonly used to relieve pain and reduce inflammation. However, they are generally not considered as first-line treatment for neuropathic pain syndromes like Thalamal Dejerine Syndrome. Neuropathic pain is often not primarily driven by inflammation, and NSAIDs are less effective in addressing the underlying mechanisms of neuropathic pain. Therefore, among the options provided, NSAID analgesics are the least useful in managing Thalamal Dejerine Syndrome.

Incorrect Options:

Option A - Pregabalin: Pregabalin is an anticonvulsant and analgesic medication commonly used to treat neuropathic pain conditions, including Thalamal Dejerine Syndrome. It acts by modulating calcium channels in the nervous system, reducing the release of neurotransmitters involved in pain signaling. Pregabalin can be effective in managing neuropathic pain, including the pain associated with thalamic lesions. Therefore, pregabalin is considered useful in Thalamal Dejerine Syndrome.

Option B - Carbamazepine: Carbamazepine is an anticonvulsant medication often used to treat various types of neuropathic pain, including trigeminal neuralgia and diabetic neuropathy. It works by stabilizing sodium channels in neurons, reducing their excitability. Carbamazepine has been shown to provide pain relief in some patients with Thalamal Dejerine Syndrome, making it a useful option in managing the condition.

Option C - Gabapentin: Gabapentin is another anticonvulsant medication commonly used for neuropathic pain management. It modulates calcium channels and is believed to inhibit the release of certain neurotransmitters involved in pain transmission. Similar to pregabalin, gabapentin has been found to be effective in treating neuropathic pain conditions, including Thalamal Dejerine Syndrome. Therefore, gabapentin can be considered useful in managing this condition.

Solution for Question 13:

Correct Option C - Carboprost:

- Carboprost tromethamine is a prostaglandin F_{2α} analogue that is often used for the treatment of postpartum hemorrhage that has not responded to other interventions, such as uterine massage, oxytocin, and ergometrine. It causes uterine contractions and helps to control bleeding.

Incorrect Options:

Option A - Misoprostol: Misoprostol is a prostaglandin E1 analogue. While it can be used to prevent and treat postpartum hemorrhage (PPH), it is not the first-line option in cases of severe PPH that do not respond to other interventions.

Option B - Dinoprost: Dinoprost, also known as prostaglandin F2 α , is used for various medical purposes, including induction of labor and abortion, but it is not the primary choice for treating PPH.

Option D - Latanoprost: Latanoprost is not used to treat postpartum hemorrhage. It is an eye drop medication used to treat certain types of glaucoma and ocular hypertension by reducing intraocular pressure.

Solution for Question 14:

Correct Option B - 5 alpha reductase inhibitors:

- 5 alpha reductase inhibitors are commonly used to treat benign prostatic hyperplasia (BPH), which is a condition characterized by an enlarged prostate gland causing symptoms such as urine hesitancy, retention, frequent urination, etc. These medications work by reducing the size of the prostate gland, relieving the pressure on the urethra, and improving urinary flow.

Incorrect Options:

Option A - Calcium channel blockers: Calcium channel blockers are commonly used to treat conditions such as high blood pressure, angina, and certain heart rhythm disorders. They do not directly impact urine hesitancy or retention and are not the appropriate treatment for this patient's condition.

Option C - ACE inhibitors: ACE inhibitors are primarily used to treat high blood pressure and heart failure. They do not have a direct role in treating urine hesitancy or retention caused by an enlarged prostate.

Option D - Aldosterone antagonist: Aldosterone antagonists are typically used to treat conditions like heart failure, high blood pressure, and certain kidney diseases by affecting the balance of electrolytes and fluid in the body. They are not indicated for the treatment of urine hesitancy or retention caused by an enlarged prostate.

Solution for Question 15:

Correct Option A – Capecitabine:

- Capecitabine is known to cause hand-foot syndrome, also known as Palmoplantar erythrodysesthesia. It is characterized by erythema, edema, and dysesthesia, which can progress to blistering and ulceration on the hands and feet.

Incorrect Options:

Options B, C, D:

- Cyclophosphamide, Gemcitabine, and Paclitaxel do not cause hand-foot syndrome.
-

Solution for Question 16:

Correct Option A - Tesamorelin:

- Tesamorelin is a synthetic growth hormone-releasing hormone (GHRH) analog that is used specifically for reducing abdominal fat in patients with HIV-associated lipodystrophy. It works by stimulating the release of growth hormone, which helps to reduce visceral fat accumulation. It is the primary medication approved for this indication.

Incorrect Options:

Option B - Somatropin:

- Somatropin is a recombinant growth hormone used to treat growth hormone deficiencies and growth disorders. It is not specifically indicated for HIV-induced lipodystrophy, although it may have some effect on body composition. Its use is more general and not targeted at HIV-associated lipodystrophy.

Option C - Octreotide:

- Octreotide is a somatostatin analog used to treat conditions like acromegaly and certain types of tumors. It is not used for treating HIV-induced lipodystrophy. Its effects are related to inhibiting growth hormone release rather than addressing fat distribution issues related to HIV.

Option D - Pegvisomant:

- Pegvisomant is a growth hormone receptor antagonist used to treat acromegaly by blocking the action of growth hormone at its receptor. It is not used for treating HIV-induced lipodystrophy, as its mechanism does not target fat redistribution or accumulation associated with HIV.
-

Solution for Question 17:

Correct Option A - TMP (Thymidine Monophosphate):

- Methotrexate is a folate analog that inhibits the enzyme dihydrofolate reductase (DHFR). DHFR is crucial for the conversion of dihydrofolate to tetrahydrofolate, a cofactor required for the synthesis of thymidylate (TMP), which is necessary for DNA synthesis. By inhibiting this enzyme, methotrexate reduces the synthesis of TMP, leading to impaired DNA synthesis, particularly affecting rapidly dividing cells.

Incorrect Options:

Option B - CMP (Cytidine Monophosphate):

- CMP synthesis is not directly affected by methotrexate. Cytidine monophosphate is derived from the pyrimidine biosynthesis pathway, which is not directly influenced by the folate cycle inhibited by methotrexate.

Option C - GMP (Guanosine Monophosphate):

- GMP synthesis is part of the purine biosynthesis pathway. While folate metabolism plays a role in purine synthesis, methotrexate's primary effect is on the pyrimidine synthesis pathway, particularly affecting TMP synthesis.

Option D - AMP (Adenosine Monophosphate):

- Similar to GMP, AMP is a purine nucleotide. While methotrexate can indirectly affect purine synthesis due to folate pathway inhibition, the most direct and significant impact is on TMP synthesis in the pyrimidine pathway.
-

Solution for Question 18:

Correct Option B - Ketorolac:

- Rheumatoid arthritis is a chronic autoimmune disorder characterized by joint inflammation and pain. The goal of treatment in rheumatoid arthritis is to reduce pain, inflammation, and joint damage. Analgesics are medications that are used to relieve pain.
- Ketorolac is also an NSAID, but it is not preferred for use in rheumatoid arthritis due to its potential side effects. It is a potent analgesic that works by blocking the production of prostaglandins. However, it is associated with a higher risk of gastrointestinal bleeding, kidney damage, and other serious adverse effects. Ketorolac is typically used for short-term pain relief in conditions such as postoperative pain, renal colic, and migraines.

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Incorrect Options:

Option A - Tramadol: Tramadol is an opioid analgesic commonly used to manage pain in rheumatoid arthritis. It has both opioid and non-opioid mechanisms of action and is considered effective for moderate to severe pain.

Option C - Naproxen: Naproxen is a nonsteroidal anti-inflammatory drug (NSAID) commonly used to manage pain and inflammation in rheumatoid arthritis. It works by blocking the production of prostaglandins, which are responsible for pain and inflammation. However, long-term use of NSAIDs may increase the risk of gastrointestinal bleeding, cardiovascular events, and kidney damage.

Option D - Oxycodone: Oxycodone is another opioid analgesic sometimes used to manage pain in rheumatoid arthritis. It is more potent than tramadol and may be used for severe pain. However, tramadol may cause side effects such as nausea, dizziness, and constipation.

Solution for Question 19:

Correct choice: D (Fexofenadine)

- Fexofenadine is a second generation anti histaminic which act for 12–24 hours or more.
- Fexofenadine does not cross blood-brain barrier. It does not produce sedation or impair psychomotor performance. It is free of atropinic side effects. It is rapidly absorbed, excreted unchanged in urine and bile.
- It is the active metabolite of terfenadine, the first nonsedating SGA that was withdrawn because of several deaths due to polymorphic ventricular tachycardia (Torsades de pointes).

Second Generation Antihistaminics		
Fexofenadine	120-180 mg oral	Allegra, Altiva, Fexo 120, 180 mg tab
Loratadine	10 mg oral	lorfats, lordin, lormeg, 10 mg tab. 1 mg/ml susp.
Desloratadine	5 mg oral	Deslor, lorday, neoloridin 5 mg tab
Cetirizine	10 mg oral	alerid, cetzine, zirtin, sizon 10 mg tab levocet 5 mg tab, 2.5mg/5 ml syr.
levocetirizine	5-10 mg oral	levosiz, levorid, teczine 5, 10 mg tab levocet 5 mg tab, 2.5mg/5 ml syr
Azelastine	4 mg oral 0.28 mg intranasal	azep nasal spray 0.14 mg per puff nasal spray
Mizolastine	10 mg oral	Elina 10 mg tab
Ebastine	10 mg oral	Ebast 10 mg tab
Rupatadine	10 mg oral	rupahist 10 mg tab

Incorrect choices:

Option A: Pheniramine : Is a first generation anti histaminic used for treating pruritis and allergic rhinitis. It has sedative effect.

Option B: Promethazine: For treating allergy, motion sickness and also used as a sleep aid. It has anti histaminic, anti cholinergic and antidopaminergic properties.

Option C: Hydroxyzine: It is used to treat allergic skin diseases. It also has a sedative effect.

Solution for Question 20:

Correct Option : D Neutrophil chemotaxis prevention:

This is a case of acute gout.

Colchicine acts by preventing neutrophil chemotaxis.

It Inhibits microtubules and mitosis.

Adverse effect - Diarrhea.

Incorrect options

Option A- Pyrimidine antimetabolite-They are cytotoxic drugs used in the treatment of solid Tumors. Eg-5- Flurouracil, Citarabine .

Option B- Uricosuric Drugs- Acts by eliminating uric Acid through Urine. Eg-Losartan, Probenecid, Benzobromone.

They are used in treatment of chronic gout.

Option c : Xanthine based inhibitors are used in the treatment of Hyperurecemia, Tumor lysis syndrome , Lesh Nyhan Syndrome (rarely used), Chagas disease, Secondary hyperuricemia and Chronic Gout. The case here presented is of Chronic Gout.They are contraindicated in Acute Gout.

Solution for Question 21:

Correct Option C:

The above image of the X-ray shows free air under the diaphragm which is seen in peptic ulcer perforation and since there is usage history of NSAIDs, this peptic ulcer perforation is due to chronic usage of NSAIDs.

Incorrect Options:

Option A: Since multiple air fluids levels are absent, it is not a case of intestinal obstruction.

Option B: Pericarditis does not present with stomach pain but with chest pain.

Option D: In gallstone there is no air under the diaphragm.

Solution for Question 22:

Correct Option A: Morphine

- Morphine could be responsible for the aggravation of the patient's symptoms in biliary colic.
- Biliary colic is typically caused by the obstruction of the bile ducts by gallstones, leading to intermittent pain in the right upper abdomen. The pain is often triggered by the contraction of the gallbladder in response to a meal, which increases the pressure on the obstructed bile duct.
- Morphine, as an opioid analgesic, can cause smooth muscle contraction, including the contraction of the sphincter of Oddi, which is the muscular valve that regulates the flow of bile from the common bile duct into the small intestine. In patients with biliary colic, morphine-induced sphincter of Oddi contraction can worsen the obstruction and lead to increased pain or even the development of acute cholecystitis.
- Therefore, in patients with biliary colic, it is generally recommended to avoid or use opioids with caution

Incorrect Options:

Option B: Non-opioid analgesics, such as nonsteroidal anti-inflammatory drugs (NSAIDs) like diclofenac may be considered as alternative options for pain relief in these patients.

Option C: Nefopam is a centrally acting analgesic with muscle relaxant properties and is not known to cause sphincter of Oddi contraction. However, it is not commonly used for biliary colic and is not typically the first choice for managing this condition.

Option D: Selective COX-2 inhibitors like etoricoxib, may be considered as alternative options for pain relief in these patients.

Solution for Question 23:

Correct Option A: Chronic gout

- Pegloticase is used in the management of chronic gout.
- Pegloticase is a medication that is specifically indicated for the treatment of chronic gout in adults who have not responded well to conventional therapies or who cannot tolerate those therapies. It is an enzyme that converts uric acid into a soluble compound, effectively lowering the levels of uric acid in the body.

Incorrect Options:

Option B: Rheumatoid arthritis : Pegloticase is Not used in its management.

Option C: Psoriatic arthritis : Pegloticase is Not used in its management.

Option D: Calcium pyrophosphate dehydrate crystal deposition (CPPD): Pegloticase is Not used in its management.

Solution for Question 24:

Correct Option B: Promethazine

- Promethazine is the drug commonly used for the treatment of motion sickness.
- Promethazine is an antihistamine with antiemetic (anti-nausea) properties. It works by blocking histamine receptors in the brain, particularly in the vomiting center, reducing the sensation of nausea and preventing vomiting. It is effective in treating various types of nausea and vomiting, including motion sickness.

Incorrect options:

Option A, C & D: Cetirizine, loratadine, and mizolastine are also antihistamines, but they are primarily used for the treatment of allergic conditions such as hay fever and allergic rhinitis. While they may have some mild sedative effects, they are not typically considered the first-line treatment for motion sickness.

Solution for Question 25:

Correct option:

Option C. Flutamide

- Flutamide is not used for the prophylaxis of migraine. Flutamide is an antiandrogen medication primarily used in the treatment of certain hormone-dependent conditions, such as prostate cancer and hirsutism. It is not indicated or commonly prescribed for the prevention or prophylaxis of migraine headaches.

Incorrect options:

Option A: Propranolol: Propranolol is a beta-blocker that can be effective in reducing the frequency and severity of migraines. It works by blocking the action of adrenaline and other stress hormones, which can trigger migraines.

Option B: Imipramine: Imipramine is a tricyclic antidepressant that is sometimes used for migraine prophylaxis. It helps regulate the levels of certain neurotransmitters in the brain, which may help prevent migraine attacks.

Option D: Flunarizine: Flunarizine is a calcium channel blocker that has been used in some countries for the prophylaxis of migraines. It works by reducing the influx of calcium into cells, which can help prevent the excessive dilation of blood vessels in the brain that is associated with migraines.

Solution for Question 26:

Correct Option A - Xanthine oxidase inhibition:

- Allopurinol is a medication primarily used for the management of gout and certain types of kidney stones.

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Mechanism of Action

- Allopurinol is a xanthine oxidase inhibitor. It inhibits the enzyme xanthine oxidase, which is responsible for the conversion of hypoxanthine to xanthine and then to uric acid.

- By inhibiting xanthine oxidase, allopurinol reduces the production of uric acid, thereby lowering the uric acid levels in the blood and urine.

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By inhibiting xanthine oxidase, allopurinol reduces the production of uric acid, thereby lowering the uric acid levels in the blood and urine.

Incorrect Options:

Options B - Recombinant uricase: Recombinant uricase, also known as pegloticase, is a medication used for the treatment of chronic gout in adults who have not responded adequately to other therapies. Uricase catalyzes the conversion of uric acid, a byproduct of purine metabolism, into allantoin, a compound that is more soluble and can be easily eliminated by the kidneys. By converting uric acid to allantoin, pegloticase reduces the serum uric acid levels and helps dissolve urate crystals in tissues.

Option C - Increase uric acid excretion: Uricosuric agents work by inhibiting the reabsorption of uric acid in the renal tubules, thereby increasing its elimination in the urine. Eg Probenecid, Benzbromarone and Lesinurad.

Option D - Decrease chemotaxis : Colchicine inhibits granulocyte migration and is used in acute gout.

Solution for Question 27:

Correct Option d. Carboprost.

Option D. Carboprost: Carboprost is a synthetic prostaglandin F2 alpha analogue used as an abortifacient. It is primarily used for the termination of pregnancy in cases where other methods have failed or are contraindicated. It is administered intramuscularly and is particularly effective in the second trimester of pregnancy

Incorrect options

Option A. Dinoprost: Dinoprost, also known as prostaglandin F2 alpha, is a naturally occurring prostaglandin used for various medical purposes, including the induction of labor, treatment of postpartum hemorrhage, and control of uterine bleeding. While it is not commonly used as an abortifacient, it can have abortifacient effects in certain circumstances.

Option B. Dinoprostone: Dinoprostone is a synthetic prostaglandin E2 analogue used for cervical ripening, induction of labor, and the termination of pregnancy. It is not specifically used as an abortifacient but can be used for inducing labor or as part of the process for medical abortion.

Option C. Misoprostol: Misoprostol is a synthetic prostaglandin E1 analogue that is commonly used as an abortifacient. It is effective for medical abortion when used alone or in combination with other medications.

Solution for Question 28:

Correct Option D - Acetaminophen:

- Among the options listed, acetaminophen (paracetamol) is the most appropriate postoperative analgesic for a patient with chronic kidney disease who underwent a nephrectomy.
- Acetaminophen is generally considered safe and well-tolerated in patients with chronic kidney disease. It has analgesic and antipyretic properties.
- Patients with chronic kidney disease often have reduced kidney function and impaired clearance of certain medications, including NSAIDs. NSAIDs can cause further kidney damage and potentially worsen renal function, so their use is generally avoided in this patient population. However, acetaminophen is considered a safer alternative as it is primarily metabolized in the liver and excreted by the kidneys in a non-toxic form.

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Patients with chronic kidney disease often have reduced kidney function and impaired clearance of certain medications, including NSAIDs. NSAIDs can cause further kidney damage and potentially worsen renal function, so their use is generally avoided in this patient population. However, acetaminophen is considered a safer alternative as it is primarily metabolized in the liver and excreted by the kidneys in a non-toxic form.

Incorrect Options:

Option A - Naproxen: Naproxen is not safe in CKD

Option B - Indomethacin: Indomethacin is not safe in CKD

Option C - Diclofenac: Diclofenac is not safe in CKD

Solution for Question 29:

Correct Option D - Give sodium bicarbonate to alkalize urine:

- Aspirin, or acetylsalicylic acid, is a non-steroidal anti-inflammatory drug (NSAID) commonly used to relieve pain, reduce inflammation, and prevent blood clots. However, aspirin overdose can cause severe toxicity and potentially life-threatening complications such as metabolic acidosis, respiratory depression, and cardiovascular collapse.
- Administering sodium bicarbonate to alkalize urine is a recommended intervention in managing aspirin overdose. Aspirin overdose can cause metabolic acidosis, leading to severe complications such as cerebral edema and cardiovascular collapse. Alkalization of urine with sodium bicarbonate can help enhance the excretion of aspirin and prevent the accumulation of salicylates in the body, thereby reducing the risk of toxicity.

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Incorrect Options:

Option A - N-acetyl cysteine to replenish glutathione stores: N-acetyl cysteine is not indicated in the management of aspirin overdose. It is commonly used as an antidote for acetaminophen overdose, which can cause liver toxicity by depleting glutathione stores. Aspirin overdose does not cause glutathione depletion, so N-acetyl cysteine is not useful in this situation.

Option B - Pralidoxime: Pralidoxime is not indicated in the management of aspirin overdose. Pralidoxime is an antidote for organophosphate poisoning, which can cause cholinergic toxicity by inhibiting acetylcholinesterase activity. Aspirin overdose does not cause cholinergic toxicity,

so pralidoxime is useless in this situation.

Option C - Glucagon to control bradycardia and hypoglycemia: Glucagon is not indicated in the management of aspirin overdose. Glucagon is used as an antidote to Beta blocker toxicity.

Solution for Question 30:

Correct Option D - Allopurinol:

- Gout is a type of arthritis caused by the buildup of uric acid crystals in the joints, leading to inflammation and pain. One of the mainstays of gout treatment is reducing uric acid levels in the blood, which can be achieved through lifestyle modifications and medications.
- Allopurinol is a xanthine oxidase inhibitor, inhibiting the enzyme that converts xanthine to uric acid. By inhibiting uric acid synthesis, allopurinol helps lower uric acid levels in the blood, which can prevent the formation of urate crystals in the joints and reduce the frequency and severity of gout attacks.

Gout is a type of arthritis caused by the buildup of uric acid crystals in the joints, leading to inflammation and pain. One of the mainstays of gout treatment is reducing uric acid levels in the blood, which can be achieved through lifestyle modifications and medications.

Allopurinol is a xanthine oxidase inhibitor, inhibiting the enzyme that converts xanthine to uric acid. By inhibiting uric acid synthesis, allopurinol helps lower uric acid levels in the blood, which can prevent the formation of urate crystals in the joints and reduce the frequency and severity of gout attacks.

Incorrect Options:

Option A - Probenecid: Probenecid, conversely, is a uricosuric agent that increases uric acid's excretion in the urine.

Option B - Colchicine: Colchicine is an anti-inflammatory medication commonly used to treat acute gout attacks, but does not affect uric acid levels.

Option C - Diclofenac: Diclofenac is a nonsteroidal anti-inflammatory drug (NSAID) that can also be used to manage gout symptoms, but does not inhibit uric acid synthesis.

Solution for Question 31:

Correct Option B - Paracetamol:

- NAPQI (N-acetyl-p-benzoquinone imine) is a toxic metabolite that is formed when paracetamol (also known as acetaminophen) is metabolized by the liver.
- Under normal circumstances, paracetamol is metabolized into non-toxic compounds that are excreted in the urine.
- However, when the dose of paracetamol exceeds a certain threshold, the liver's metabolic pathways become overwhelmed, and NAPQI accumulates in the liver, where it can cause hepatotoxicity.
- Acetaminophen nomogram is used to interpret serum acetaminophen concentrations in relation to time since ingestion, in order to assess potential hepatotoxicity.

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Under normal circumstances, paracetamol is metabolized into non-toxic compounds that are excreted in the urine.

However, when the dose of paracetamol exceeds a certain threshold, the liver's metabolic pathways become overwhelmed, and NAPQI accumulates in the liver, where it can cause hepatotoxicity.

Acetaminophen nomogram is used to interpret serum acetaminophen concentrations in relation to time since ingestion, in order to assess potential hepatotoxicity.

Incorrect Options:

- Option A, C and D are not metabolized to NAPQI

Solution for Question 32:

Correct Option D - Aspirin:

- Urinary alkalinization is a technique used to increase the pH of urine, which can increase the excretion of weakly acidic drugs that are cleared by the kidneys such as aspirin.

- When the pH of urine is increased, more of aspirin is ionized and therefore, it is unable to be reabsorbed by the kidneys, leading to increased excretion of the drug.

Urinary alkalinization is a technique used to increase the pH of urine, which can increase the excretion of weakly acidic drugs that are cleared by the kidneys such as aspirin.

When the pH of urine is increased, more of aspirin is ionized and therefore, it is unable to be reabsorbed by the kidneys, leading to increased excretion of the drug.

Incorrect Options:

Options A, B and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 33:

Correct Option A - Premature closure of ductus arteriosus:

- Indomethacin is a nonsteroidal anti-inflammatory drug (NSAID) that can have specific effects on the fetus when used during pregnancy. One of the significant effects of indomethacin on the fetus is the premature closure of the ductus arteriosus.
- The ductus arteriosus is a blood vessel in a fetus that connects the pulmonary artery to the aorta, bypassing the lungs. This shunt is important for fetal circulation. However, after birth, the ductus arteriosus is supposed to close and become a ligament.
- Indomethacin, as a prostaglandin inhibitor, can cause the premature closure of the ductus arteriosus in a developing fetus. This can lead to impaired blood flow and oxygenation in the fetus, potentially resulting in complications such as fetal distress, fetal hypoxia, and even fetal death.

Indomethacin is a nonsteroidal anti-inflammatory drug (NSAID) that can have specific effects on the fetus when used during pregnancy. One of the significant effects of indomethacin on the fetus is the premature closure of the ductus arteriosus.

The ductus arteriosus is a blood vessel in a fetus that connects the pulmonary artery to the aorta, bypassing the lungs. This shunt is important for fetal circulation. However, after birth, the ductus arteriosus is supposed to close and become a ligament.

Indomethacin, as a prostaglandin inhibitor, can cause the premature closure of the ductus arteriosus in a developing fetus. This can lead to impaired blood flow and oxygenation in the fetus, potentially resulting in complications such as fetal distress, fetal hypoxia, and even fetal

death.

Incorrect Options :

Option B - Keeps ductus arteriosus patent: It is used to close a patent ductus arteriosus

Option C - Closure of fossa ovalis: Indomethacin does not have a direct effect on the closure of the fossa ovalis. The fossa ovalis is a small depression in the interatrial septum of the heart and plays a role in fetal circulation. Its closure occurs naturally after birth and is not influenced by indomethacin.

Option D - Premature closure of ductus venosus: Indomethacin primarily affects the closure of the ductus arteriosus, not the ductus venosus. The ductus venosus is a vessel that shunts oxygenated blood from the umbilical vein to the inferior vena cava in the fetus.

- It's important to note that indomethacin is sometimes used in specific cases during pregnancy, such as to manage certain maternal or fetal conditions. However, its use requires careful monitoring and consideration of the potential risks to the fetus.

Solution for Question 34:

Correct Option C - Indomethacin:

- To encourage PDA closure, NSAIDs like indomethacin and ibuprofen are frequently utilized. These medications function by decreasing prostaglandin production, which is what keeps the ductus arteriosus open.

To encourage PDA closure, NSAIDs like indomethacin and ibuprofen are frequently utilized. These medications function by decreasing prostaglandin production, which is what keeps the ductus arteriosus open.

Incorrect Options:

Option A - Magnesium sulfate: This drug is used in the treatment of preeclampsia, preterm labour, torsades de pointes, acute myocardial infarction, eclampsia, seizure control and migraine headaches. It has no effect in patency of PDA

Option B - Alprostadil: This drug is used for the treatment of erectile dysfunction, Patent Ductus Arteriosus and Pulmonary Arterial Hypertension. It is used primarily used to treat a number of heart-related and erectile dysfunction-related medical issues. It helps maintain the patency of the ductus arteriosus.

Option D - Paracetamol: This drug is used for pain relief, and anti-pyretic. It does not cause closure of PDA as it does not affect prostaglandin synthesis.

Solution for Question 35:

Correct Option A - Inhibiting TXA2 synthesis by platelets:

- Aspirin, also known as acetylsalicylic acid, prevents ischemic heart disease by inhibiting the synthesis of thromboxane A2 (TXA2) by platelets. TXA2 is a potent platelet aggregator and vasoconstrictor. By irreversibly inhibiting the enzyme cyclooxygenase (COX) in platelets, aspirin reduces the production of TXA2, which helps to prevent platelet aggregation and the formation of blood clots in coronary arteries. This mechanism of action makes aspirin an effective antiplatelet medication for reducing the risk of cardiovascular events, including heart attacks and strokes.

Aspirin, also known as acetylsalicylic acid, prevents ischemic heart disease by inhibiting the synthesis of thromboxane A2 (TXA2) by platelets. TXA2 is a potent platelet aggregator and vasoconstrictor. By irreversibly inhibiting the enzyme cyclooxygenase (COX) in platelets, aspirin reduces the production of TXA2, which helps to prevent platelet aggregation and the formation of blood clots in coronary arteries. This mechanism of action makes aspirin an effective antiplatelet medication for reducing the risk of cardiovascular events, including heart attacks and strokes.

Incorrect Options:

Option B - Inhibiting prostacyclin synthesis: Prostacyclin, also known as prostaglandin I2 (PGI2), is a potent vasodilator and inhibitor of platelet aggregation. In contrast to inhibiting prostacyclin synthesis, aspirin actually inhibits the synthesis of thromboxane A2, which is derived from arachidonic acid by platelets and has proaggregatory effects. The inhibition of prostacyclin synthesis by aspirin is not its primary mechanism of action in preventing ischemic heart disease.

Option C - Inhibiting the release of EDRF: Endothelium-derived relaxing factor (EDRF) refers to nitric oxide (NO), which is a potent vasodilator released by the endothelial cells lining blood vessels. While NO plays a crucial role in vascular health and regulation of blood flow, aspirin's mechanism of action in preventing ischemic heart disease is not directly related to inhibiting the release of EDRF.

Option D - Inhibiting the synthesis of endothelins: Endothelins are vasoconstrictor peptides produced by endothelial cells. They play a role in regulating vascular tone and blood pressure. However, aspirin's mechanism of action in preventing ischemic heart disease does not involve

inhibiting the synthesis of endothelins.

Solution for Question 36:

Correct Option B - Thromboxane:

- The inhibition of thromboxane by aspirin is responsible for the occurrence of occult blood in the stool.
- Aspirin is a nonsteroidal anti-inflammatory drug (NSAID) that works by inhibiting the activity of cyclooxygenase (COX) enzymes, particularly COX-1 and COX-2. COX-1 is involved in the production of prostaglandins and thromboxane.
- Thromboxane A₂ is a potent pro-inflammatory and pro-thrombotic molecule that promotes platelet aggregation and vasoconstriction. By inhibiting COX-1, aspirin reduces the production of thromboxane A₂, which leads to decreased platelet aggregation and a reduced tendency for blood clot formation.
- Occult blood in the stool refers to the presence of blood that is not visibly apparent, but can be detected through laboratory tests. Aspirin's inhibition of thromboxane A₂ interferes with platelet function and can contribute to gastrointestinal bleeding, including the presence of occult blood in the stool.
- It is important to note that the occurrence of occult blood in the stool should be evaluated by a healthcare professional, as it can have various causes, including gastrointestinal disorders and other factors unrelated to aspirin use.

The inhibition of thromboxane by aspirin is responsible for the occurrence of occult blood in the stool.

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function and can contribute to gastrointestinal bleeding, including the presence of occult blood in the stool.

It is important to note that the occurrence of occult blood in the stool should be evaluated by a healthcare professional, as it can have various causes, including gastrointestinal disorders and other factors unrelated to aspirin use.

Incorrect Options:

- Option A, C and D are incorrect. Refer to the explanation for the correct answer.

Solution for Question 37:

Correct Option D - Xanthine oxidase inhibitor:

This patient after initial treatment of acute gout flare should be treated with Xanthine oxidase inhibitors which are considered first-line in management of chronic gout, to prevent relapses.

- Gout is a type of arthritis caused by an accumulation of uric acid crystals in the joints, leading to inflammation and pain.
- Xanthine oxidase is an enzyme involved in the metabolism of purines, which are converted to uric acid.
- Xanthine oxidase inhibitors block the activity of this enzyme, resulting in reduced uric acid production and lower levels of uric acid in the blood.
- Allopurinol and febuxostat are two commonly used xanthine oxidase inhibitors in treating gout.
- These drugs are considered first-line agents for managing gout in chronic gout, because they effectively reduce serum uric acid levels and prevent gout flares.

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These drugs are considered first-line agents for managing gout in chronic gout, because they effectively reduce serum uric acid levels and prevent gout flares.

Incorrect Options:

Option A - Pyrimidine antimetabolites: Pyrimidine antimetabolites are drugs that interfere with the synthesis of nucleic acids, specifically pyrimidines, and are used in cancer treatment. They have no role in the treatment of gout.

Option B - Inhibitor of neutrophil recruitment: Inhibitors of neutrophil recruitment are drugs that prevent the migration of white blood cells called neutrophils to the site of inflammation. colchicine works by this mechanism of action, but it is not used in maintenance treatment of gout, instead it is used in acute gout flare.

Option C - Uricosuric drug: Uricosuric drugs are a class of medications that increase the excretion of uric acid by the kidneys, resulting in lower serum uric acid levels. They are used in patients with gout who have under-excretion of uric acid as the cause of their hyperuricemia. However, they are not considered first-line agents in gout treatment and are generally used in combination with xanthine oxidase inhibitors or as an alternative in patients who cannot tolerate xanthine oxidase inhibitors.

Diuretics and Antidiuretics

1. A 45-year-old male patient has hypertension and hypokalemia. His blood pressure is consistently elevated despite being on multiple antihypertensive medications. Laboratory tests reveal elevated plasma aldosterone concentration and suppressed plasma renin activity. A CT scan of the abdomen shows an adrenal adenoma. Which is the most appropriate pharmacologic treatment for this patient's condition?

- A. Lisinopril
- B. Hydrochlorothiazide
- C. Amlodipine
- D. Spironolactone

2. Which of the following drugs is not used in the treatment of central diabetes insipidus?

- A. Carbamazepine
- B. Desmopressin
- C. Chlorpropamide
- D. Indomethacin

3. A 68-year-old patient presents with symptoms of lethargy, nausea, and muscle cramps. Laboratory tests reveal hyponatremia and plasma hypoosmolality. He has a history of lung cancer and was recently started on selective serotonin reuptake inhibitors (SSRIs) for depression. He is diagnosed with the syndrome of inappropriate antidiuretic hormone (SIADH). Which of the following statements regarding the management of SIADH is incorrect?

- A. Fluid restriction to less than 800ml/day is the first-line treatment for Mild to Moderate SIADH.
 - B. Severe SIADH is corrected with administration of 100 ml iv bolus of 3% NaCl
 - C. Targeted sodium correction should be $>1\text{mEq/L}$ per hour in case of severe SIADH
 - D. Demeclocycline and lithium can be used in the treatment of mild to moderate SIADH.
-

4. Which of the following is the incorrect statement with respect to vasopressin inhibitors?

- A. Conivaptan is used only as an intravenous formulation for hospitalised patients of hypovolemic/euvolemic hyponatremia.
 - B. Tolvaptan is given a black box warning and is intended to be given under close monitoring of serum Na^+
 - C. Osmotic demyelination syndrome is one of the serious adverse effects of the usage of vasopressin antagonists.
 - D. Tolvaptan is a non-selective V_1/V_2 antagonist, while conivaptan is a selective V_2 antagonist
-

5. A 7-year-old child presents with a history of bed-wetting three to four times a week. The child has no significant medical history and physical examination is normal. Which of the following is an incorrect statement with respect to initial management for nocturnal enuresis in this child?

- A. The drug of choice for nocturnal enuresis is intranasal desmopressin, administered at bedtime.
- B. Primary nocturnal enuresis is the most common type where the child has never developed bladder control.
- C. Initial management, in this case, is reducing water intake in the evening and doing bladder exercises.
- D. Allergic reactions similar to urticaria or anaphylaxis can occur due to desmopressin administration.

6. A 42-year-old patient, previously operated on for pituitary adenoma, was diagnosed with post-operative central diabetes insipidus. All of the following are true regarding the drug of choice for this patient, except?

- A. The drug acts on V1 Receptor located in renal tubules
 - B. The preferred route is intranasal, as the drug is metabolised by trypsin in the gut
 - C. Cerebral oedema can occur as a serious adverse effect of this drug
 - D. The drug is contraindicated in patients with acute renal failure
-

7. A 55-year-old man with a history of alcoholic cirrhosis presents with progressively worsening abdominal distention, decreased urine output, and swelling in his legs. On examination, his abdomen is tense and distended with caput medusa and ascites, and his chest shows spider angioma. Laboratory tests reveal elevated serum creatinine levels. Which of the following is the drug of choice for this patient?

- A. Terlipressin
 - B. Spironolactone
 - C. Furosemide
 - D. Lactulose
-

8. A 7-year-old child presents with persistent bedwetting despite implementing various lifestyle modifications, such as reducing fluid intake before bedtime and establishing a regular toileting schedule. Which of the following would be the appropriate drug of choice for this child?

- A. Terlipressin

- B. Felypressin
 - C. Desmopressin
 - D. Octreotide
-

9. A 55-year-old patient presents following a severe head injury. On examination, the patient is unconscious with signs of increased intracranial pressure (ICP). A CT scan reveals significant cerebral oedema and brain mass effect. The patient is scheduled for an emergency neurosurgery. Which medication is most appropriate for reducing cerebral oedema before and after the surgery in this patient?

- A. Loop diuretics
 - B. Thiazide diuretics
 - C. Osmotic diuretics
 - D. Carbonic anhydrase inhibitors
-

10. A 70-year-old female with chronic heart failure is being treated with a diuretic to manage her symptoms of fluid overload. Recently, she has started to experience hearing loss and ringing in her ears. On reviewing her medications, which class of diuretics is most likely responsible for these symptoms?

- A. Thiazide diuretics
 - B. Osmotic diuretics
 - C. Carbonic anhydrase inhibitors
 - D. Loop diuretics
-

11. A patient on long-term lithium therapy visits with complaints of frequent urination and increased thirst. He was diagnosed with nephrogenic diabetes

insipidus (DI) based on urine analysis and plasma ADH levels. He was prescribed a combination of amiloride and hydrochlorothiazide. What is the reason for the use of thiazide diuretics in nephrogenic DI?

- A. Thiazides reduce urine output by increasing free water absorption in proximal tubules
 - B. Thiazides directly increase the sensitivity of the kidneys to ADH.
 - C. Thiazides help to increase lithium excretion, reducing its nephrotoxic effects.
 - D. Thiazides decrease the risk of hypokalemia when used with amiloride.
-

12. Which of the following statements is true with respect to thiazides and loop diuretics?

- A. Thiazides are preferred over loop diuretics in hypertensive patients with low GFR.
 - B. Thiazides cause indirect vasodilatation by prostaglandin release, unlike loop diuretics
 - C. Thiazides are preferred over loop diuretics in patients with osteoporosis
 - D. Thiazides will not cause hypokalemia, unlike loop diuretics
-

13. A 56-year-old chronic hypertensive who hasn't taken his medication in the past three days presents with an acute severe headache, blurry vision, palpitations, and diaphoresis. On examination, blood pressure is 240/140 mm Hg. Which of the following is the ideal drug of choice for this patient?

- A. Furosemide
 - B. Chlorthalidone
 - C. Acetazolamide
 - D. Mannitol
-

14. A patient with chronic renal failure and resistant oedema was treated with a drug that alleviated the oedema. However, increasing doses were needed over time. Although the drug remained effective, the patient developed involuntary hand twitching and perioral numbness. Which class of drug was given?

- A. Loop diuretics
 - B. Carbonic anhydrase inhibitors
 - C. Thiazide diuretics
 - D. Osmotic diuretics
-

15. A 32-year-old mountain climber presents to the base camp complaining of severe headache, nausea, and fatigue after ascending rapidly to a high altitude. On examination, his respiratory rate is increased, and arterial blood gas analysis shows decreased partial pressure of carbon dioxide (PaCO_2). Given the Acute Mountain Sickness (AMS) suspicion, acetazolamide is administered to alleviate symptoms. Which of the following best describes acetazolamide's primary mechanism of action in treating AMS?

- A. Metabolic Acidosis
 - B. Respiratory Alkalosis
 - C. Respiratory Acidosis
 - D. Metabolic Alkalosis
-

16. All of the following are preferred uses of acetazolamide, except?

- A. Uric acid kidney stones
 - B. Mountain sickness
 - C. Pulmonary hypertension
 - D. Glaucoma
-

17. A 45-year-old man who has been treated for glaucoma comes with complaints of muscle weakness, fatigue, and palpitations. Laboratory tests reveal significant hypokalemia and metabolic acidosis. Which class of diuretics is most likely responsible for these symptoms, considering the maximum loss of potassium?

- A. Loop diuretics
 - B. Thiazide diuretics
 - C. Osmotic diuretics
 - D. Carbonic anhydrase inhibitors
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	3
Question 4	4
Question 5	1
Question 6	1
Question 7	1
Question 8	3
Question 9	3
Question 10	4
Question 11	1
Question 12	3
Question 13	1
Question 14	1
Question 15	1
Question 16	3
Question 17	4

Solution for Question 1:

Correct Answer: D) Spironolactone

Explanation :

Primary hyperaldosteronism (Conn's syndrome) is characterised by excessive production of aldosterone from the adrenal glands, leading to hypertension and hypokalemia. Spironolactone, an aldosterone antagonist and a potassium-sparing diuretic, is the most appropriate drug of choice.

Spironolactone:

- Aldosterone combines with an intracellular mineralocorticoid (MR) receptor and induces the formation of aldosterone-induced proteins (AIPs), which promote Na⁺ reabsorption.
- Spironolactone acts as a competitive antagonist and inhibits the formation of AIP by combining with MR.
- This prevents aldosterone from promoting sodium reabsorption and potassium excretion, which are the primary mechanisms by which aldosterone increases blood pressure and causes hypokalemia.

Lisinopril (Option A) is an angiotensin-converting enzyme (ACE) inhibitor that reduces aldosterone levels by inhibiting the production of angiotensin II. However, in primary hyperaldosteronism, there is an autonomous overproduction of aldosterone. Thus, lisinopril will be ineffective.

Hydrochlorothiazide (Option B) is a thiazide diuretic that inhibits the reabsorption of sodium and potassium, thus hyperaldosteronism induced hypokalemia will be worsened.

Amlodipine (Option C) is a calcium channel blocker that works by relaxing the smooth muscle. While effective for managing hypertension, it does not influence aldosterone levels or activity.

Solution for Question 2:

Correct Answer: D) Indomethacin

Explanation :

Indomethacin, a prostaglandin synthase inhibitor, is used for the treatment of nephrogenic diabetes insipidus, especially lithium-induced nephrogenic DI. It works by attenuating prostaglandin synthesis in the renal tubules, thereby potentiating the effects of antidiuretic hormone (ADH).

Oral carbamazepine (Option A) has shown antidiuretic activity in patients with neurohypophyseal diabetes insipidus. It is suggested for initial use in neurohypophyseal diabetes insipidus if oral therapy is indicated, although its exact mechanism of antidiuretic action is unclear.

Antidiuretic peptides (Option B) are the primary treatment for central DI, with desmopressin being the peptide of choice.

Chlorpropamide (Option C), an oral sulfonylurea, potentiates the action of small or residual amounts of circulating vasopressin and reduces urine volume in more than half of all patients with central DI. If polyuria is not controlled satisfactorily with chlorpropamide alone, adding a thiazide diuretic to the regimen usually results in an adequate reduction in urine volume.

Solution for Question 3:

Correct Answer: C). Targeted sodium correction should be $>1\text{mEq/L}$ per hour in case of severe SIADH

Explanation:

Sodium correction should not increase $0.5\text{--}1\text{mEq/L}$ per hour, as it can cause severe consequences such as osmotic demyelination syndrome, which can even lead to quadriplegia.

Severity of SIADH	Symptoms	Management
Mild	Asymptomatic, Mild lethargy, nausea	Fluid Restriction
Moderate	Pronounced Lethargy, Confusion, Cramps, Nausea and Vomiting	Fluid Restriction, Loop Diuretics, Vasopressin Antagonists (VAPTANS), Demeclocycline, Lithium
Severe	Severe symptoms (e.g., severe confusion, seizures, coma)	Hypertonic saline, fluid restriction, Vasopressin Antagonist

Fluid restriction is the mainstay of treatment for mild to moderate SIADH, which is to be kept below 800ml/day . (Option A)

In cases of severe SIADH, 100ml i.v. bolus of 3% NaCl is used to correct hyponatremia. (Option B)

Demeclocycline (Tetracycline) and lithium (Option D) can be used in the treatment of mild to moderate SIADH but not routinely preferred due to nephrotoxicity and availability of better drugs such as loop diuretics and vasopressin antagonists.

Solution for Question 4:

Correct Answer : D) Tolvaptan is a non-selective V1/V2 antagonist, while conivaptan is a selective V2 antagonist

Explanation :

Tolvaptan is a selective V2 antagonist while conivaptan is non selective V1/V2 antagonist. Since ADH acts on V2 receptors which reabsorbs water, tolvaptan helps in improving volume overload status by blocking those receptors.

Conivaptan(Option A) is FDA-approved for the treatment of hospitalised patients diagnosed with hypervolemic/euvolemic hyponatremia. It is also used in cases of emergency treatment of SIADH. Tolvaptan and mozavaptan are oral vasopressin antagonists used for long term management of the same.

Tolvaptan(Option B) is given a Black box warning as it can cause serious consequences by rapidly correcting hyponatremia. It is also hepatotoxic, thus should be used for a maximum of 30 days and should not be used in patients with liver disease.

Osmotic demyelination syndrome(Option C) can occur with the usage of vasopressin antagonists due to the risk of rapid correction of hyponatremia.

Solution for Question 5:

Correct Answer: A) The drug of choice for nocturnal enuresis is intranasal desmopressin, administered at bedtime.

Explanation :

The drug of choice for nocturnal enuresis is oral desmopressin administered at bedtime that too only if behavioral and training methods fail. Intranasal desmopressin is no longer recommended because of the risk of hyponatremia in children. only if behavioral and training methods fail.

(Option B) Nocturnal Enuresis is of 2 types- Primary and Secondary.

- Primary nocturnal enuresis is 3-5/ week incidences of bedwetting in children aged more than 5 years, indicating the child has not developed bladder control.
- Secondary nocturnal enuresis is a bedwetting incident occurring after the child has been dry for a period of at least 6 months, indicating the child has developed bladder control before

Solution for Question 6:

Correct Answer: A) The drug acts on V1 Receptor located in renal tubules

Explanation:

Desmopressin, which is the drug of choice in central diabetes insipidus, acts on V2 receptors located in renal tubules, not on V1

Receptor Type	G Sub type	Location	Actions
V1 (V1a)	Gq	Vascular smooth muscle, liver, CNS	Vasoconstriction, glycogenolysis, platelet aggregation, GIT smooth muscle constriction
V2	Gs	Renal collecting ducts	Increase water reabsorption
V3 (V1b)	Gq	Anterior pituitary, Pancreas, Adrenal Medulla	Release of ACTH

Preferred route for Desmopressin (Option B) is nasal or intravenous, as oral route is less efficacious due to getting metabolized by trypsin in the gut. However, in patients of acute rhinitis, oral route can also be used.

Since desmopressin (Option C) helps in absorbing more water, rapid correction of hyponatremia might occur, leading to cerebral oedema

Desmopressin (Option D) is not used in patients of acute renal failure as rapid increase in extracellular water can cause volume overload due to reduced renal function. Also, it is to be used very cautiously in patients of angina, hypertension and heart failure.

Solution for Question 7:

Correct Answer: A) Terlipressin

Explanation:

Terlipressin is a vasopressin analogue that acts as a V1 receptor agonist, leading to vasoconstriction of splanchnic vessels and reducing portal hypertension. It is the drug of choice for treating hepatorenal syndrome, as it helps improve renal perfusion and function in patients with advanced liver disease.

Spironolactone (Option B) is a potassium-sparing diuretic commonly used in the management of cirrhotic ascites. It can help manage fluid overload but not the renal dysfunction seen in hepatorenal syndrome.

Furosemide (Option C) is a loop diuretic used to treat fluid overload in patients with cirrhosis and ascites. However, in hepatorenal syndrome, loop diuretics may worsen renal perfusion and should be avoided.

Lactulose (Option D) is a non-absorbable sugar used to treat hepatic encephalopathy by reducing ammonia levels in the colon. While hepatic encephalopathy can occur in patients with advanced liver disease, it is not the primary concern in hepatorenal syndrome, which involves renal dysfunction secondary to portal hypertension.

Solution for Question 8:

Correct Answer: C) Desmopressin

Explanation:

DESMOPRESSIN (ADH analogue) is the drug of choice in nocturnal enuresis. It acts on the V2 receptors and increases aquaporin II production, which causes increased water reabsorption from urine.

Vasopressin receptor type	G Protein	Second messenger	Effect	Location
---------------------------	-----------	------------------	--------	----------

	n			
V1 (V1A)	Gq type	Increase IP3/DAG	Vasoconstriction, Increase BP. It can be used in the treatment of shock.	Smooth muscle cells of blood vessels.
V2	Gs type	Increase cAMP	Increases vWF and Factor 8 (present in endothelial cells of blood vessels and nephron)	Increases aquaporin II on collecting ducts (Reabsorb water from urine)
V3 (V1B)	Gq type	Increase IP3/DAG	Increases ACTH and endorphins (stress hormone)	Present in brain neurons (a neuropeptide)

SYNTHETIC ADH ANALOGUES :

Property	Terlipressin	Felypressin	Desmopressin
Receptor	V1 agonist	V1 agonist	V2 agonist
Administration	Injections only	Injections only	Orally, Nasally
Uses	Control bleeding in esophageal varices	Add-on to local anaesthetic	Treatment of Hemophilia A.
	Hepatorenal syndrome (DOC)	Epinephrine alternative	Nocturnal enuresis (bed wetting).
	Resistant shock		Neurogenic diabetes insipidus (DOC).

Solution for Question 9:

Correct Answer: C) Osmotic diuretics

Explanation :

Osmotic diuretics(mannitol) elevate the extracellular fluid compartment osmolality and thereby shift water from the extravascular into the intravascular compartment. This mechanism allows this group of drugs to treat conditions that raise ICP and also in acute attacks of glaucoma.

Loop and Thiazide diuretics(Options A & B)on the contrary can lower blood pressure and can lead to electrolyte imbalances, which may further compromise cerebral perfusion and worsen

neurological outcomes in patients with increased ICP.

Carbonic anhydrase inhibitors (Option D) are not as potent as osmotic diuretics, which are more effective in rapidly reducing intracranial pressure. They have limited ability to penetrate the blood-brain barrier, which limits their efficacy in managing increased ICP.

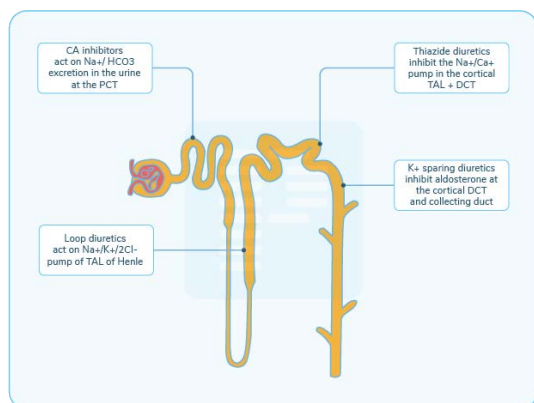
Solution for Question 10:

Correct Answer: D) Loop diuretics

Explanation:

This patient has chronic congestive heart failure. She has been treated with furosemide, which is a loop diuretic. Loop diuretics are known to cause ototoxicity, which can manifest as hearing loss and tinnitus.

The most ototoxic loop diuretic is ethacrynic acid



CLASSIFICATION OF DIURETICS :

Type	Mechanism of action	Examples	Clinical uses	Side effects
1. Carbonic anhydrase inhibitors (Option C)	Inhibit carbonic anhydrase enzyme in the proximal tubule, decreasing bicarbonate reabsorption	Acetazolamide, Methazolamide	Angle closure Glaucoma, Altitude sickness	Metabolic acidosis, Hypokalemia, hyperammonemia
2. Loop diuretics (Option D)	Inhibit the Na-K-2Cl symporter in the thick ascending	Furosemide, Bumetanide	Acute pulmonary edema, Chronic heart failure,	Hypokalemia, Ototoxicity, Metabolic alkalosis

	loop of Henle		Hypercalcemia	s, Hyperglycemia, Hyperuricemia
3. Thiazide diuretics (Option A)	Inhibit sodium reabsorption in the distal convoluted tubule	Hydrochloro thiazide, Chlorthalidone	Hypertension, Edema	Hypokalemia, Hyponatremia, Hyperglycemia , Hyperuricemia , metabolic alkalosis.
4. Potassium-sparing	Inhibit sodium channels in the distal nephron or block aldosterone receptors	Spironolactone, Amiloride	Heart failure, Hyperaldosteronism	Hyperkalemia, Gynecomastia (for Spironolactone) ,metabolic acidosis
5. Osmotic diuretics (Option B)	Increase osmolarity of the glomerular filtrate, inhibiting water reabsorption.	Mannitol, Glycerin	Cerebral oedema, Acute glaucoma	Dehydration, Electrolyte imbalance

Solution for Question 11:

Correct Answer: A) Thiazides reduce urine output by increasing free water absorption in proximal tubules

Explanation:

Thiazide, though being a diuretic, reduces urine output in patients of nephrogenic diabetes insipidus by increasing free water absorption in the proximal tubules.

Thiazides do not directly increase the sensitivity of the kidneys to ADH.(Option B)

Thiazides do not help to increase lithium excretion, whereas amiloride does(Option C)

In contrast, thiazides actually cause hypokalemia and amiloride is added as an adjunct since the latter is an antidiuretic agent(Option D)

Solution for Question 12:

Correct Answer: C) Thiazides are preferred over loop diuretics in patients with osteoporosis

Explanation:

Thiazide diuretics decrease urinary excretion of calcium ions and are hence preferred over loop diuretics (which increase calcium excretion) in potential hypocalcemic states like osteoporosis

Loop diuretics (Option A) are preferred in patients with low GFR (<30 ml/min/BSA).only thiazide which is effective in low GFR cases is metolazone.

Thiazides(Option B) cause direct vasodilatation by opening Ca^{2+} activated K^{+} channels on vascular smooth muscles.

Both loop diuretics (Option D)and thiazides can cause hypokalemia.

Solution for Question 13:

Correct Answer: A) Furosemide

Explanation :

A hypertensive emergency is defined as blood pressure more than 220/120 mm Hg. Furosemide is the drug of choice in hypertensive emergencies due to its rapid onset and potent diuretic effect, effectively reducing blood volume and controlling severe hypertension.

Solution for Question 14:

Correct Answer: A) Loop diuretics

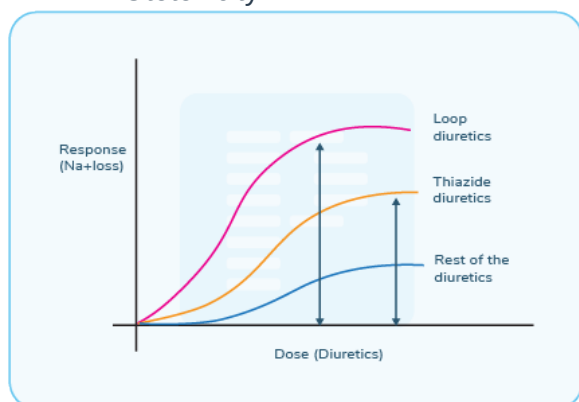
Explanation :

Involuntary hand twitching, along with perioral numbness, is suggestive of tetany due to hypocalcemia. High ceiling diuretics (loop diuretics), which are preferred in treating oedema (including resistant oedema and for rapid fluid mobilisation), impair the absorption of calcium ions, thus causing tetany.

- Loop diuretics like furosemide primarily inhibit the sodium-potassium-chloride cotransporter in the thick ascending limb of the loop of Henle.
- This inhibition disrupts the reabsorption of sodium, chloride, and water, which normally generates a positive potential gradient in the tubule's lumen.
- This gradient facilitates the reabsorption of calcium ions. However, when loop diuretics block this transporter, the potential gradient is disrupted, impairing calcium reabsorption and leading to increased urinary excretion, ultimately resulting in hypocalcemia.

Common Adverse Effects of loop diuretics (also seen with Thiazides, except for Calcium):

- Hyponatremia → Hypotension
- Hypokalemia → Muscle weakness increases the risk of digoxin toxicity
- Hypocalcemia → Tetany (Thiazides usually increase calcium levels)
- Hypomagnesemia → Pseudogout
- Decrease in H^+ ions → Metabolic alkalosis
- Hyperuricemia → in high doses
- Hyperglycemia and dyslipidemia
- Ototoxicity



Carbonic Anhydrase Inhibitors (Option B), such as acetazolamide, have a weak diuretic effect and are primarily used for conditions like glaucoma or altitude sickness, not for managing significant fluid overload in CHF or chronic renal failure.

Thiazide Diuretics (Option C) are less effective than loop diuretics for severe oedema, especially in patients with chronic renal failure. They are typically used for mild to moderate oedema and maintenance therapy, not for rapid fluid mobilisation. Osmotic Diuretics (Option D), like mannitol,

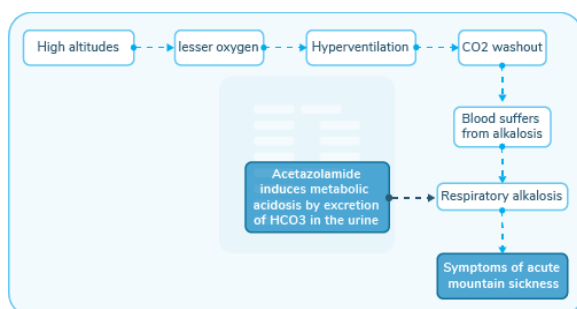
are used to reduce intracranial pressure or treat acute renal failure but are not suitable for the long-term management of chronic oedema.

Solution for Question 15:

Correct Answer: A) Metabolic acidosis

Explanation:

In the scenario described, the patient's symptoms of headache, nausea, fatigue, and increased respiratory rate suggest acute mountain sickness (AMS), which is further supported by the decreased PaCO₂ seen in the arterial blood gas analysis, which suggests respiratory alkalosis.



MOA OF ACETAZOLAMIDE IN TREATING AMS :

- Inhibits kidney carbonic anhydrase.
- Increases urine bicarbonate ions excretion → Metabolic acidosis.
- Counteracts respiratory alkalosis from high altitude hyperventilation.
- Improves oxygenation and alleviates AMS symptoms.

Solution for Question 16:

Correct Answer: C) Pulmonary hypertension

Explanation:

Acetazolamide cannot be used to treat pulmonary hypertension. The drug acts by inhibiting both membrane-bound and cytoplasmic forms of carbonic anhydrase. This results in nearly complete elimination of NaHCO_3 reabsorption in the proximal tubule

USES OF CARBONIC ANHYDRASE INHIBITORS :

Indications	Mechanism of action	Medication and uses
Glaucoma (Option D)	MOA: Reduces aqueous humor secretion from ciliary cells	Eye Drops: Brinzolamide, Dorzolamide Use: Primary open-angle glaucoma IV Drugs: Acetazolamide Use: Emergency treatment for acute congestive glaucoma
Alkalinisation of Urine	MOA: excretion of bicarbonate Side Effects: Higher risk of alkaline renal stones Increased risk of calcium phosphate stones Treatment: Thiazide diuretics: Reduce calcium excretion	Useful for acidic drug poisoning Effective against acidic kidney stones, including uric acid stones (Option A) and cystine stones (associated with homocystinuria)
Metabolic Acidosis in Blood (Option B)	Condition: Elevated levels of H^+ ions in the blood	Treatment: Acetazolamide is the drug of choice for mountain sickness (especially HAPE-High Altitude Pulmonary Edema)
Epilepsy	Suppresses the activity of neurons	Acetazolamide was earlier used for epilepsy
Periodic muscle paralysis	Imbalance of potassium ion in blood Occurs due to Hyperkalemia, Hypokalemia	Acetazolamide is used in periodic muscle paralysis.

Solution for Question 17:

Answer: D. Carbonic anhydrase inhibitor

Explanation :

This patient complains of muscle weakness, fatigue and palpitations after being started on a diuretic. These symptoms suggest hypokalemia, and his ABG shows metabolic acidosis, which suggests the use of a carbonic anhydrase inhibitor, which is commonly used in glaucoma.

Diuretics causing maximum potassium loss: Carbonic anhydrase inhibitors cause the most potassium loss, followed by thiazides (Option B), loop diuretics (Option A), and osmotic diuretics (Option C). metabolic alkalosis is seen with thiazides and loops

Diuretics causing maximum sodium loss: Loop diuretics lead to the greatest sodium loss, followed by thiazides, osmotic diuretics, carbonic anhydrase inhibitors, and potassium-sparing diuretics.

CLASSIFICATION OF DIURETICS :

Type	Mechanism of action	Examples	Clinical uses	Side effects
Carbonic anhydrase inhibitors (Option C)	Inhibit carbonic anhydrase enzyme in the proximal tubule, decreasing bicarbonate reabsorption	Acetazolamide, Methazolamide	Glaucoma, Altitude sickness	Metabolic acidosis, Hypokalemia
Loop diuretics (Option D)	Inhibit the Na-K-2Cl symporter in the thick ascending loop of Henle	Furosemide, Bumetanide	Acute pulmonary edema, Chronic heart failure, Hypercalcemia	Hypokalemia, Ototoxicity, Metabolic alkalosis, Hyperglycemia, Hyperuricemia
Thiazide diuretics (Option A)	Inhibit sodium reabsorption in the distal convoluted tubule	Hydrochlorothiazide, Chlorthalidone	Hypertension, Edema	Hypokalemia, Hyponatremia, Hyperglycemia, Hyperuricemia
Potassium-sparing	Inhibit sodium channels in the distal nephron or block aldosterone receptors	Spironolactone, Amiloride	Heart failure, Hyperaldosteronism	Hyperkalemia, Gynecomastia (for Spironolactone)
Osmotic diuretics (Option B)	Increase osmolarity of the glomerular filtrate, inhibiting water reabsorption.	Mannitol, Glycerin	Cerebral oedema, Acute glaucoma	Dehydration, Electrolyte imbalance

Previous Year Questions

1. A 35-year-old male experiences headache, nausea, and shortness of breath while trekking. The physician prescribes a drug to alleviate his symptoms. The drug administered primarily acts on which part of the nephron?

- A. Proximal convoluted tubule (PCT)
 - B. Loop of Henle
 - C. Distal convoluted tubule (DCT)
 - D. Collecting duct (CD)
-

2. Which of the following drugs is appropriate for treating a 75-year-old woman with urinary incontinence triggered by stress, such as coughing?

- A. Ipratropium
 - B. Tolterodine
 - C. Telenzepine
 - D. Tropicamide
-

3. Which of the following diuretics is matched correctly with its primary site of action? 1) Osmotic diuretics - Distal convoluted tubule. 2) Thiazide diuretics - Proximal convoluted tubule 3) Carbonic anhydrase inhibitors - Proximal convoluted tubule 4) Potassium-sparing diuretics - Collecting duct.

- A. 1, 2, and 3
- B. 3 and 4
- C. 2, 3, and 4
- D. 1 and 2.

4. A 30-year-old man joins a pilgrimage to a shrine at a mountain peak. At the summit, he develops tightness in the chest, associated with breathlessness and altered sensorium. Which of the following can be given for symptomatic relief?

- A. Mannitol
 - B. Digoxin
 - C. Diltiazem
 - D. Acetazolamide
-

5. What is the preferred medication for treating neurogenic diabetes insipidus?

- A. Conivaptan
 - B. Vasopressin
 - C. Terlipressin
 - D. Desmopressin
-

6. Which of the following drugs does not require dose reduction in patients with renal failure?

- A. Gentamicin
 - B. Vancomycin
 - C. Amphotericin B
 - D. Doxycycline
-

7. Which of the following is a true statement regarding Aliskiren?

- A. It is indicated in diabetes mellitus
 - B. It is used in the treatment of diabetic nephropathy
 - C. It increases renin levels
 - D. It increases aldosterone levels
-

8. Which drug is most suitable for treating edema in a patient with heart disease and presenting with breathlessness with rales?

- A. ACE inhibitor
 - B. Na-K-2Cl symporter inhibitor in loop of Henle
 - C. Na-Cl symporter inhibitor in DCT
 - D. Aldosterone blocker in DCT
-

9. Which of the subsequent medications can lead to the development of gynecomastia?

- A. Tamoxifen
 - B. Oxytocin
 - C. Spironolactone
 - D. Clomiphene citrate
-

10. Which of the following drugs can cause the shift of potassium back into the cells?

- A. Epinephrine
- B. Acetazolamide

- C. Atropine
 - D. Lactic acid
-

11. A patient was brought to the hospital following a road traffic accident. The patient was disoriented. CT head showed no evidence of hemorrhage, but features of cerebral edema and raised intracranial pressure was seen. Which of the following drugs can be used for the initial management of raised intracranial pressure?

- A. Furosemide
 - B. Mannitol
 - C. Acetazolamide
 - D. Glycerol
-

12. What is the probable reason behind the patient experiencing a dry mouth and increased thirst after being administered tolvaptan, following a diagnosis of ADPKD?

- A. Increased free water clearance
 - B. V2 receptor antagonism
 - C. Increased renal cAMP levels
 - D. Increase in urine osmolality
-

13. A patient on lithium developed hypertension. He was started on thiazides for hypertension. After few days, he developed coarse tremors and other symptoms suggestive of lithium toxicity. Explain the likely mechanism of this interaction:

- A. Thiazide inhibits the metabolism of lithium

- B. Thiazides act as an add on drug to lithium
 - C. Thiazides increase the tubular resorption of lithium
 - D. Thiazides cause water loss thereby increase lithium levels
-

14. Which one of the listed medications acts as a carbonic anhydrase inhibitor?

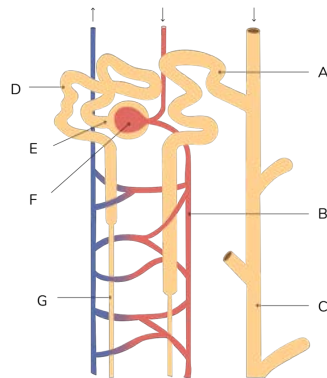
- A. Acetazolamide
 - B. Hydrochlorothiazide
 - C. Furosemide
 - D. Mannitol
-

15. Match the following in column A to their site of action in column B. Column A
Column B A. Acetazolamide B. Torsemide C. Chlorthalidone D. Vasopressin 1. Thick ascending loop of Henle 2. Collecting duct 3. Distal convoluted tubule 4. Proximal convoluted tubule

Column A	Column B
A. Acetazolamide B. Torsemide C. Chlorthalidone D. Vasopressin	1. Thick ascending loop of Henle 2. Collecting duct 3. Distal convoluted tubule 4. Proximal convoluted tubule

- A. A-2, B-3, C-1, D-4
 - B. A-3, B-4, C-2, D-1
 - C. A-1, B-2, C-4, D-3
 - D. A-4, B-1, C-3, D-2
-

16. A 50-year-old woman was prescribed a diuretic by a doctor to manage hypertension. Which of the following diuretic acts on the site 'A' is marked in the given image?



- A. Furosemide
 - B. Hydrochlorothiazide
 - C. Acetazolamide
 - D. Mannitol
-

17. Which of the following conditions is not treated with high-ceiling diuretics?

- A. Generalized edema
 - B. Cerebral edema
 - C. Acute pulmonary edema
 - D. Pulmonary hypertension
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	4
Question 5	4
Question 6	4
Question 7	2
Question 8	2
Question 9	3
Question 10	1
Question 11	2
Question 12	2
Question 13	3
Question 14	1
Question 15	4
Question 16	2
Question 17	4

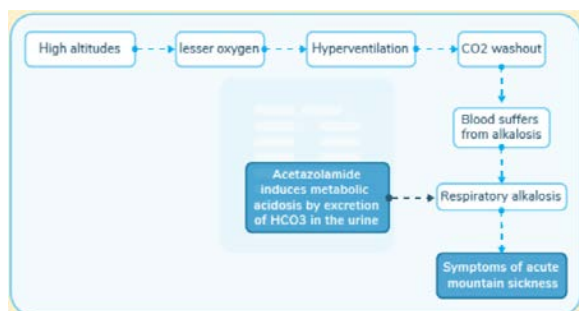
Solution for Question 1:

Answer: A) Proximal convoluted tubule (PCT)

Explanation:

The patient is experiencing symptoms of acute mountain sickness (AMS) due to hypoxia at high altitude. The drug commonly prescribed for AMS is acetazolamide, a carbonic anhydrase inhibitor, which primarily acts on the proximal convoluted tubule (PCT) of the nephron.

- Mechanism of Acetazolamide in AMS: Inhibits carbonic anhydrase in the PCT, leading to: Decreased bicarbonate (HCO_3^-) reabsorption → metabolic acidosis Increased renal excretion of bicarbonate → compensatory hyperventilation This enhances ventilation and improves oxygenation, helping alleviate AMS symptoms.
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- This enhances ventilation and improves oxygenation, helping alleviate AMS symptoms.
- Decreased bicarbonate (HCO_3^-) reabsorption → metabolic acidosis
- Increased renal excretion of bicarbonate → compensatory hyperventilation



Acetazolamide in AMS Symptom Alleviation

- Inhibits kidney carbonic anhydrase.
- Increases urine bicarbonate ions excretion → Metabolic acidosis.
- Counteracts respiratory alkalosis from high altitude hyperventilation.
- Improves oxygenation and alleviates AMS symptoms.

- Loop of Henle (Option B) is incorrect because loop diuretics (e.g., furosemide) act here by inhibiting Na⁺-K⁺-2Cl⁻ cotransport, but they are not used for AMS.
 - Distal Convoluted Tubule (DCT) (Option C) is incorrect because thiazide diuretics act here by inhibiting Na⁺-Cl⁻ symport, but they do not correct AMS-related hypoxia.
 - Collecting Duct (CD) (Option D) is incorrect because the collecting duct is the site of aldosterone and ADH regulation, affected by potassium-sparing diuretics (e.g., spironolactone), which do not treat AMS.
-

Solution for Question 2:

Correct Option B - Tolterodine:

- Mechanism of Action: Tolterodine is an antimuscarinic agent that specifically targets muscarinic receptors in the bladder. It reduces involuntary bladder contractions by blocking these receptors, which can help in managing symptoms of urinary incontinence, particularly urge incontinence.
- Use in Stress Incontinence: While tolterodine is primarily used to treat urge incontinence, it can sometimes be beneficial in mixed incontinence cases where urgency is also a factor. However, the first-line treatment for pure stress incontinence typically involves non-pharmacological approaches like pelvic floor exercises. Pharmacologically, stress incontinence is less commonly treated with medications, but if there is an urgency component, tolterodine may be considered.

Incorrect Options:

Option A - Ipratropium:

- Mechanism of Action: Ipratropium is an anticholinergic drug used mainly as a bronchodilator in the treatment of chronic obstructive pulmonary disease (COPD) and asthma. It has no role in the management of urinary incontinence.

Option C - Telenzepine:

- Mechanism of Action: Telenzepine is a selective M1 muscarinic receptor antagonist used mainly for reducing gastric acid secretion in conditions like peptic ulcer disease. It is not indicated for treating urinary incontinence.

Option D - Tropicamide:

- Mechanism of Action: Tropicamide is an antimuscarinic drug used primarily as a mydriatic agent to dilate the pupil for ophthalmic examinations. It has no effect on bladder function and is not used for urinary incontinence.
-

Solution for Question 3:

Correct Option B - 3 and 4:

- Osmotic diuretics - Proximal convoluted tubule and descending limb of the loop of Henle, not the distal convoluted tubule.
- Thiazide diuretics – It mainly acts on the Distal convoluted tubule; the proximal convoluted tubule may be a secondary site of action.
- Carbonic anhydrase inhibitors – Proximal convoluted tubule.
- Potassium-sparing diuretics - Collecting duct.

Incorrect Options:

- Options A, C, and D are incorrect.
-

Solution for Question 4:

Correct Option D - Acetazolamide:

- In the above case, there is chest tightness along with shortness of breath and altered sensorium at the top of a mountain which is suggestive of mountain sickness.
- The drug that provides relief is acetazolamide.
- Acetazolamide is a diuretic which inhibits carbonic acid anhydrase. It increases HCO_3 excretion in the kidneys causing a metabolic acidosis that counteracts the respiratory alkalosis (caused by hyperventilation due to hypoxic conditions).
- This allows for better respiratory response to hypoxic conditions.

In the above case, there is chest tightness along with shortness of breath and altered sensorium at the top of a mountain which is suggestive of mountain sickness.

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This allows for better respiratory response to hypoxic conditions.

Incorrect Options:

- Option A, B and C are not indicated in mountain sickness.

Solution for Question 5:

Correct Option D - Desmopressin:

- Desmopressin is the drug of choice for the treatment of neurogenic diabetes insipidus, a condition characterized by impaired water balance due to decreased vasopressin (ADH) secretion. Desmopressin is a synthetic analog of vasopressin.

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Incorrect Options:

Option A - Conivaptan: Conivaptan is a medication used for the treatment of hyponatremia but is not typically employed for neurogenic diabetes insipidus. Patients with DI are likely to develop hypernatremia.

Option B - Vasopressin: Vasopressin is not commonly used for the treatment of neurogenic diabetes insipidus. Desmopressin is the preferred first choice. Vasopressin may be considered as an alternative.

Option C - Terlipressin: Terlipressin is primarily used for the management of bleeding esophageal varices and hepatorenal syndrome. It is not the drug of choice for neurogenic diabetes insipidus.

Solution for Question 6:

Correct Option D - Doxycycline:

- Doxycycline is a broad-spectrum antibiotic from the tetracycline class. It is primarily eliminated via non-renal routes, such as hepatic metabolism and fecal excretion. As a result, renal impairment does not significantly affect its clearance. Therefore, dose reduction is not typically necessary in patients with renal failure.

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Incorrect Options:

Option A - Gentamicin: Gentamicin is an aminoglycoside antibiotic that is eliminated primarily by the kidneys. In patients with renal failure, the clearance of gentamicin is reduced, leading to an increased risk of toxicity. Therefore, dose adjustments are necessary to avoid excessive drug accumulation and potential nephrotoxicity.

Option B - Vancomycin: Vancomycin is a glycopeptide antibiotic that is primarily eliminated by the kidneys. In patients with renal impairment, the clearance of vancomycin is decreased, resulting in prolonged drug exposure. Dose adjustments are necessary to maintain therapeutic levels and prevent the risk of nephrotoxicity and other adverse effects.

Option C - Amphotericin B: Amphotericin B is an antifungal medication used to treat severe systemic fungal infections. It is primarily excreted by the kidneys, and renal impairment can lead to decreased clearance and increased drug levels. Dose reduction or alternative dosing strategies are often required in patients with renal failure to minimize the risk of nephrotoxicity.

Solution for Question 7:

Correct Option B - It is used in the treatment of diabetic nephropathy:

- Aliskiren is a direct renin inhibitor used for the management of hypertension. It works by inhibiting the enzyme renin, which is involved in the renin-angiotensin-aldosterone system (RAAS). By inhibiting renin, Aliskiren reduces the production of angiotensin I, which

subsequently decreases the formation of angiotensin II and the release of aldosterone.

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Incorrect Options:

Option A - It is indicated in diabetes mellitus: Aliskiren is not specifically indicated for diabetes mellitus. However, it may be prescribed to individuals with hypertension who also have diabetes, as it can effectively lower blood pressure in this population.

Option C - It increases renin levels: Aliskiren inhibits renin, leading to a decrease in renin levels rather than an increase.

Option D - It increases aldosterone levels: Aliskiren inhibits the production of angiotensin II, which, in turn, leads to a decrease in aldosterone levels.

Solution for Question 8:

Correct Option B - Na-K-2Cl symporter inhibitor in loop of Henle:

- The best drug for the treatment of edema in a patient with heart disease and lower limb edema would be a loop diuretic, which is a Na-K-2Cl symporter inhibitor in the loop of Henle.
- Loop diuretics, such as furosemide and bumetanide, act on the Na-K-2Cl symporter in the thick ascending limb of the loop of Henle to inhibit sodium reabsorption. By blocking sodium reabsorption, loop diuretics increase the excretion of sodium, chloride, and water, resulting in diuresis and reduction of edema.

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Incorrect Options:

Option A - ACE inhibitor: ACE inhibitors (angiotensin-converting enzyme inhibitors) are commonly used in the treatment of heart disease, but they primarily target the renin-angiotensin-aldosterone system (RAAS) to reduce blood pressure and improve cardiac function. While they can have some mild diuretic effects, they are not the first-line choice for managing edema and pulmonary edema.

Option C - Na-Cl symporter inhibitor in DCT: Na-Cl symporter inhibitors in the distal convoluted tubule (DCT), such as thiazide diuretics (e.g., hydrochlorothiazide), are effective in reducing blood pressure but have limited efficacy in treating edema. They primarily promote sodium and water excretion in the DCT but are less potent than loop diuretics. They are not preferred for acute management of pulmonary edema.

Option D - Aldosterone blocker in DCT: Aldosterone blockers, such as spironolactone and eplerenone, act on the aldosterone receptors in the distal tubules to inhibit the effects of aldosterone, a hormone that promotes sodium and water retention. While aldosterone blockers can be beneficial in some cases of heart failure, they are typically used as adjunct therapy rather than as the primary treatment for edema.

Solution for Question 9:

Correct Option C - Spironolactone:

- Amongst the above mentioned drugs, spironolactone is known present with gynaecomastia as a side effect with usage.
- Gynaecomastia is seen because spironolactone decreases testosterone production and increases peripheral conversion of testosterone to estradiol.
- It displaces estradiol from plasma binding protein sites.

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Gynaecomastia is seen because spironolactone decreases testosterone production and increases peripheral conversion of testosterone to estradiol.

It displaces estradiol from plasma binding protein sites.

Incorrect Options:

Option A - Tamoxifen: Tamoxifen does not cause gynaecomastia. BEing a SERM, it is inhibitory at the ER receptors in the breast. Therefore it can be used to decrease gynaecomastia.

Option B - Oxytocin: Oxytocin does not cause gynaecomastia. Oxytocin is involved in the milk let down reflex.

Option D - Clomiphene citrate: Clomiphene citrate stimulates ovulation and prevents unopposed estrogen activity in the menstrual cycle. It does not cause gynaecomastia.

Solution for Question 10:

Correct Option A - Epinephrine: Acts on beta-adrenergic receptors, particularly β_2 receptors, causing potassium to move from the extracellular space into cells, thereby reducing serum potassium levels.

Incorrect Options: Options B, C, and D do not cause a shift of potassium into cells.

Solution for Question 11:

Correct Option B - Mannitol:

- First line management in a case of cerebral oedema and raised intracranial pressure is administration of mannitol.
- Craniotomy and evacuation is required if there is bleeding. ICP varied between 4 and 14 mmHg which is the normal limit. Limits above 20 mmHg can cause brain injury. The aim is to maintain ICP <20 mmHg.

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Craniotomy and evacuation is required if there is bleeding. ICP varied between 4 and 14 mmHg which is the normal limit. Limits above 20 mmHg can cause brain injury. The aim is to maintain ICP <20 mmHg.

Incorrect Options:

Option A - Furosemide: Furosemide is not used as the initial line of management.

Option C - Acetazolamide: Acetazolamide is not used as the initial line of management.

Option D - Glycerol: Glycerol is an osmotic stool softener.

Solution for Question 12:

Correct Option B - V2 receptor antagonism

- The likely cause of the patient's dry mouth and increased thirst after taking tolvaptan is V2 receptor antagonist.
- Tolvaptan is a medication that belongs to a class of drugs known as vasopressin receptor antagonists, specifically targeting the V2 receptors. V2 receptors are located in the renal collecting ducts and are responsible for regulating water reabsorption. By antagonizing the V2 receptors, tolvaptan reduces water reabsorption and promotes the excretion of free water, leading to increased urine output.
- One of the common side effects of V2 receptor antagonism, such as with tolvaptan, is polyuria (increased urine output), which can result in dehydration. As a compensatory response, the body may experience a dry mouth and increased thirst to counterbalance the loss of water. This can be particularly noticeable during the initiation of tolvaptan therapy or with dose adjustment.

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Tolvaptan is a medication that belongs to a class of drugs known as vasopressin receptor antagonists, specifically targeting the V2 receptors. V2 receptors are located in the renal collecting ducts and are responsible for regulating water reabsorption. By antagonizing the V2 receptors, tolvaptan reduces water reabsorption and promotes the excretion of free water, leading to increased urine output.

One of the common side effects of V2 receptor antagonism, such as with tolvaptan, is polyuria (increased urine output), which can result in dehydration. As a compensatory response, the body may experience a dry mouth and increased thirst to counterbalance the loss of water. This can be particularly noticeable during the initiation of tolvaptan therapy or with dose adjustment.

Incorrect Options:

Option A - Increased free water clearance

Option C - Increased renal cAMP levels

Option D - Increase in urine osmolality

Solution for Question 13:

Correct Option C - Thiazides increase the tubular resorption of lithium:

- The likely mechanism of interaction between thiazides and lithium in this case is that thiazides increase the tubular resorption of lithium in the kidneys.
- Thiazide diuretics are commonly used to treat hypertension by promoting diuresis (increased urine production) and reducing fluid volume in the body. However, one of the side effects of thiazides is that they decrease the excretion of certain substances, including lithium, through the kidneys.
- Lithium is a medication commonly used to stabilize mood in conditions such as bipolar disorder. It is primarily eliminated from the body through the kidneys. When thiazides are administered concurrently with lithium, they can interfere with the renal clearance of lithium by increasing its tubular resorption.
- As a result, the elevated levels of lithium in the blood can lead to symptoms of lithium toxicity, which may include coarse tremors, gastrointestinal disturbances, confusion, and other neurological symptoms.

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As a result, the elevated levels of lithium in the blood can lead to symptoms of lithium toxicity, which may include coarse tremors, gastrointestinal disturbances, confusion, and other neurological symptoms.

Incorrect Options:

- Options A, B & D are incorrect choice.

Solution for Question 14:

Correct Option A - Acetazolamide:

- Acetazolamide is a sulfonamide derivative that inhibits carbonic anhydrase and improves salt, potassium, bicarbonate, and water excretion.

Acetazolamide is a sulfonamide derivative that inhibits carbonic anhydrase and improves salt, potassium, bicarbonate, and water excretion.

Incorrect Options:

Option B - Hydrochlorothiazide: Hydrochlorothiazide is a thiazide diuretic

Option C - Furosemide: Furosemide is a loop diuretic

Option D - Mannitol: Mannitol is an osmotic diuretic.

Solution for Question 15:

Correct Option D - A-4, B-1, C-3, D-2:

1. Acetazolamide: This is a carbonic anhydrase inhibitor that works in the proximal convoluted tubule of the nephron to inhibit the reabsorption of bicarbonate ions. This results in a mild diuretic effect and the excretion of excess fluids and electrolytes.

2. Torsemide: This is a loop diuretic that works in the thick ascending limb of the loop of Henle to inhibit the reabsorption of sodium, potassium, and chloride ions. Torsemide blocks the $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter, thus decreasing the amount of ions that are reabsorbed into the

bloodstream and increasing the amount of ions that are excreted in the urine.

3. Chlorthalidone: This is a thiazide-like diuretic that works in the distal convoluted tubule to inhibit the reabsorption of sodium ions. By blocking the Na^+/Cl^- cotransporter, chlorthalidone decreases the amount of sodium that is reabsorbed into the bloodstream and increases the amount of sodium that is excreted in the urine. This results in a moderate diuretic effect.

4. Vasopressin: This is not a diuretic, but rather an antidiuretic hormone (ADH) that works in the collecting duct to increase water reabsorption. Vasopressin acts on the V2 receptor in the collecting duct to stimulate the insertion of aquaporin-2 channels into the apical membrane, which allows for the reabsorption of water. This results in a decrease in urine volume and an increase in urine concentration.

Incorrect Options:

- Option A, B and C are incorrect. Refer to the explanation for the correct answer.

Solution for Question 16:

Correct Option B - Hydrochlorothiazide:

- Hydrochlorothiazide works by inhibiting the reabsorption of sodium and chloride in the distal convoluted tubule of the nephron. This leads to increased excretion of sodium, chloride, and water. Hydrochlorothiazide is a thiazide diuretic that works on the kidneys to increase salt and water excretion.

Hydrochlorothiazide works by inhibiting the reabsorption of sodium and chloride in the distal convoluted tubule of the nephron. This leads to increased excretion of sodium, chloride, and water. Hydrochlorothiazide is a thiazide diuretic that works on the kidneys to increase salt and water excretion.

Incorrect Options:

Option A - Furosemide: Furosemide is a loop diuretic that works by blocking the sodium-potassium-chloride (Na-K-2Cl) transporter in the thick ascending loop of Henle in the kidney's nephron. This leads to increased excretion of sodium, chloride, and water.

Option C - Acetazolamide: Acetazolamide is a carbonic anhydrase inhibitor. The mechanism of action involves the inhibition of carbonic anhydrase in the proximal convoluted tubule, leading to decreased reabsorption of bicarbonate and increased excretion of bicarbonate and other electrolytes.

Option D - Mannitol: Mannitol is an osmotic diuretic that may be used in emergency situations such as hypertensive crisis. Mannitol creates an osmotic gradient in the nephron that draws water into the urine, increasing water excretion and electrolytes.

Solution for Question 17:

Correct Option D - Pulmonary hypertension:

- High-ceiling diuretics, also known as loop diuretics, are medications commonly used to treat conditions characterized by fluid overload. Loop diuretics act on the ascending limb of loop of Henle, a section of the kidney tubule, to inhibit sodium and chloride reabsorption by the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$, leading to increased urinary output and fluid loss. They are considered the most potent diuretics available and are effective in rapidly reducing fluid volume in the body.
- Pulmonary hypertension, on the other hand, is a condition characterized by increased pulmonary vascular resistance in the lungs blood vessels, which can lead to symptoms such as shortness of breath and chest pain. PDE5 inhibitors like sildenafil, vardenafil or tadalafil are used for Pulmonary hypertension treatment. While loop diuretics may be used as adjunctive therapy in treating pulmonary hypertension, they are not typically used as the primary treatment for this condition.

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Incorrect Options:

Option A - Generalized edema: Generalized edema refers to fluid accumulation in the tissues throughout the body, typically due to conditions such as heart failure, liver disease, or kidney disease. Loop diuretics are commonly used in the treatment of generalized edema.

Option B - Cerebral edema: Cerebral edema is characterized by fluid accumulation in the brain, which can lead to increased intracranial pressure and neurological symptoms. Loop diuretics and osmotic diuretics are often used in conjunction with other treatments, such as corticosteroids or osmotic agents, to manage cerebral edema.

Option C - Acute pulmonary edema: Acute pulmonary edema is a condition characterized by the accumulation of fluid in the lungs, typically due to heart failure. Loop diuretics are commonly used in the treatment of acute pulmonary edema to reduce fluid accumulation and improve respiratory function.

Antihypertensives

1. Which of the following drugs can be used in the management of angina pectoris and hypertension and has anti-arrhythmic properties?

- A. Metoprolol
- B. Amlodipine
- C. Verapamil
- D. Hydrochlorothiazide

2. Three patients come to the clinic with hypertension: Patient A: A 52-year-old known hypertensive who has tried maximum doses of three different antihypertensive drugs without success; current BP 168/104 mm Hg. Patient B: A 30-year-old pregnant woman with a BP of 158/98 mm Hg. Patient C: A 65-year-old man with diabetes mellitus and a BP of 160/100 mm Hg. Choose the correct antihypertensive regimen for the above patients.

- A. Patient A: Existing regimen + Spironolactone Patient B: Labetalol Patient C: ACE inhibitors
- B. Patient A: Existing regimen + loop diuretics Patient B: ACE inhibitors Patient C: Beta-blockers
- C. Patient A: Sodium nitroprusside Patient B: Thiazides Patient C: Labetalol
- D. Patient A: Labetalol + Spironolactone Patient B: Calcium channel blockers Patient C: Beta-blockers

3. A 38-year-old woman comes to the emergency department due to a throbbing headache. On vital examination, BP is 210/140 mm Hg. She is started on a Nitroprusside IV drip. After 8 hours, develop lactic acidosis and neurological

symptoms. She is diagnosed with nitroprusside-induced cyanide toxicity. Which of the following drug combinations is used to treat this condition?

- A. Pralidoxime + atropine
 - B. Sodium nitrite + sodium thiosulfate + hydroxocobalamin
 - C. Activated charcoal + N-acetylcysteine
 - D. Sodium nitrite only
-

4. A 58-year-old man comes to the emergency department due to loss of consciousness and severe hypotension. He has a history of angina for which he takes nitroglycerin and metoprolol tablets. He was recently prescribed sildenafil to help him with his erectile dysfunction. Which of the following is true about the concomitant use of nitrates and Sildenafil?

- A. Sildenafil should be avoided for 24 hours before and after use of nitrates.
 - B. Sildenafil and nitrates can be taken together if doses are low.
 - C. Sildenafil are absolutely contraindicated with those taking nitrates.
 - D. None of the above.
-

5. A 68-year-old man with congestive heart failure who is on captopril and digoxin is being reviewed for worsening of his symptoms. A potassium channel opener is added along with nitrates to his regimen as a cardioprotective agent. Which of the following is the K⁺ channel opener most likely to be?

- A. Minoxidil
 - B. Diazoxide
 - C. Hydralazine
 - D. Nicorandil
-

6. A 68-year-old man comes to the ER with complaints of chest pain radiating to his left arm for the last 30 minutes. He has a history of diabetes, hyperlipidemia and hypertension. ECG showed features of stable angina. Sublingual glyceryl trinitrate was administered, which provided pain relief. What is the mechanism of action of this drug?

- A. It donates nitric oxide, which activates guanylyl cyclase, thus increasing cytosolic cGMP.
 - B. It hyperphosphorylates Myosin light chain kinase.
 - C. It blocks the L-type voltage-gated calcium channel in the vascular smooth muscle.
 - D. It inhibits the degradation of cytosolic cGMP.
-

7. Which of the following medications is contraindicated with metoprolol due to significant drug interactions?

- A. Amlodipine
 - B. Verapamil
 - C. Diltiazem
 - D. Nimodipine
-

8. A 65-year-old patient with hypertension and stable angina pectoris reports experiencing excessive swelling in the lower extremities, headaches, and overgrowth of gum tissue (gingival hyperplasia) after being on a calcium channel blocker for several months. Which of the following calcium channel blockers most likely cause these adverse effects?

- A. Verapamil
- B. Amlodipine
- C. Nimodipine

D. Diltiazem

9. Which of the following calcium channel blockers is cerebroselective, specifically for preventing and managing cerebral vasospasm following aneurysmal subarachnoid haemorrhage?

- A. Amlodipine
 - B. Verapamil
 - C. Nimodipine
 - D. Diltiazem
-

10. A 48-year-old woman with a 10-year history of poorly controlled hypertension visits your clinic for follow-up. Her blood pressure is 158/96 mmHg, despite being on lisinopril and hydrochlorothiazide. Her serum creatinine and potassium levels are within normal limits. Which calcium channel blockers would be the most appropriate addition to her long-term blood pressure control treatment regimen?

- A. Nifedipine
 - B. Nicardipine
 - C. Clevidipine
 - D. Amlodipine
-

11. Match the following
1. Aliskiren a) Central sympatholytic
2. Lisinopril b) Acts on JG cells of the kidney to reduce renin release
3. Losartan c) Direct renin inhibitor
4. Clonidine d) Prevents formation of Angiotensin 2
5. Beta-blockers e) Block Angiotensin 1 receptor

1. Aliskiren

a) Central sympatholytic

2. Lisinopril	b) Acts on JG cells of the kidney to reduce renin release
3. Losartan	c) Direct renin inhibitor
4. Clonidine	d) Prevents formation of Angiotensin 2
5. Beta-blockers	e) Block Angiotensin 1 receptor

- A. 1-a, 2-b, 3-c, 4-d, 5-e
 - B. 1-b, 2-c, 3-d, 4-e, 5-a
 - C. 1-c, 2-d, 3-e, 4-a, 5-b
 - D. 1-d, 2-e, 3-a, 4-b, 5-c
-

12. Which of the following drugs is classified as a non-dihydropyridine (non-DHP) calcium channel blocker?

- A. Amlodipine
 - B. Felodipine
 - C. Diltiazem
 - D. Nifedipine
-

13. Which of the following statements about the pharmacological properties of ARBs is correct? 1) Losartan blocks the thromboxane A2 receptor, resulting in an antiplatelet effect and uricosuric action. 2) Irbesartan blocks the SA node in the heart, making it effective in atrial fibrillation. 3) Telmisartan acts as a PPAR-alpha agonist, which improves insulin sensitivity.

- A. 1,2,3
- B. 1,2
- C. Only 3
- D. 1,3

14. Which angiotensin II receptor blocker (ARB) is considered the most potent and has the highest affinity for the AT-1 receptor?

- A. Losartan
 - B. Valsartan
 - C. Candesartan
 - D. Irbesartan
-

15. Which of the following adverse effects of ACE inhibitors is associated with increased bradykinin levels?

- A. Hypotension
 - B. Hyperkalemia
 - C. Cough
 - D. Dysgeusia
-

16. A 58-year-old female with hypertension and heart failure is prescribed an ACE inhibitor for blood pressure control. What is the most significant drug interaction to be cautious about when prescribing ACE inhibitors?

- A. Calcium channel blockers
 - B. Loop diuretics
 - C. Statins
 - D. Potassium-sparing diuretics
-

17. A 62-year-old female presents for management of her hypertension and heart failure. She has a past medical history of type 2 diabetes mellitus, hyperlipidemia, and chronic kidney disease (stage 3). The initiation of an ACE inhibitor is considered. Which of the following are the uses of ACE inhibitors in this scenario?

- A. Reduction of proteinuria
 - B. Reduction of preload and afterload
 - C. Prevention of diabetic nephropathy
 - D. All the above
-

18. Which of the following ACEI is not a prodrug?

- A. Captopril
 - B. Enalapril
 - C. Fosinopril
 - D. Ramipril
-

19. Which of the following ACEIs/ARBs are most safe in renal failure?

- A. Captopril and Losartan
 - B. Fosinopril and Telmisartan
 - C. Enalapril and Candesartan
 - D. Benazepril and Losartan
-

20. A 56-year-old male presents to the emergency department with severe headache, blurred vision, and chest pain. He reports a history of hypertension but admits to poor medication adherence. On examination, his blood pressure is 200/120 mmHg, and he appears diaphoretic and anxious. Which of the following drugs can be used to treat his condition?

- A. Lisinopril
 - B. Diltiazem
 - C. Losartan
 - D. Nicardipine
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	1
Question 5	4
Question 6	1
Question 7	2
Question 8	4
Question 9	3
Question 10	4
Question 11	3
Question 12	3
Question 13	1,2
Question 14	3
Question 15	3
Question 16	4
Question 17	4
Question 18	1
Question 19	2
Question 20	4

Solution for Question 1:

Correct Answer: C) Verapamil

Explanation:

Verapamil selectively blocks the entry of calcium ions through the L-type calcium channels in the cell membranes of cardiac muscle cells and vascular smooth muscle cells. This results in the following effects:

- Negative inotropic effect: It reduces the contractility of the heart muscle, leading to a decrease in myocardial oxygen demand, which is beneficial in angina pectoris.
- Vasodilatory effect: Verapamil relaxes vascular smooth muscle cells, leading to the dilation of coronary arteries and peripheral blood vessels. This increases blood flow to the myocardium and decreases peripheral vascular resistance, making it useful in treating hypertension and angina pectoris.
- Anti-arrhythmic effect: Verapamil has Class IV anti-arrhythmic properties, which can slow conduction through the atrioventricular (AV) node and prolong the refractory period of the AV node and atrial muscle fibres. This makes it useful in the management of certain supraventricular tachyarrhythmias, such as atrial fibrillation and atrial flutter.

Metoprolol(Option A):It is a beta-blocker and would also be an appropriate choice for managing unstable angina and hypertension. However, verapamil is preferred in this case as it is a cardio-specific CCB and has additional benefits in the management of angina pectoris.

Amlodipine(Option B):It is a dihydropyridine calcium channel blocker, which has a more potent vasodilatory effect but lacks the negative inotropic and anti-arrhythmic properties of verapamil. While it can be used for hypertension, it is not the preferred choice in the management of angina pectoris.

Hydrochlorothiazide(Option C):It is a thiazide diuretic commonly used in the treatment of hypertension, but it is not the most appropriate choice for managing acute angina pectoris or acute coronary syndrome.

Solution for Question 2:

Correct Answer: A) Patient A: Existing regimen + Spironolactone

Patient B: Labetalol

Patient C: ACE inhibitors

Explanation:

Patient A "Resistant hypertension" is not controlled by 3 different drugs (at maximum tolerable dose).	Drugs useful in this patient are Thiazides+ ACE Inhibitors/ARBs + Calcium Channel Blockers. Additionally, Potassium-sparing diuretics such as Spironolactone or eplerenone must be added in case of Resistant hypertension.
Patient B Pregnant woman with hypertension.	DOC: Labetalol (Beta-blocker) If chronic hypertension develops: Alpha methyl dopa.
Safe antihypertensives during pregnancy Labetalol, Hydralazine, Methyldopa, and Calcium channel blockers like Nifedipine. (LHMC- Lady Hardinge Medical College)	
Unsafe antihypertensive drugs during pregnancy: ACE inhibitors: Can lead to renal agenesis in the fetus, causing oligohydramnios and leading to POTTER sequence (Pulmonary hypoplasia, Oligohydramnios, Twisted/wrinkled skin, Twisted Potter faces, Extremity defects, Renal agenesis). Thiazides/Loop diuretics/Beta-blockers: Bear the risk of placental ischemia, which can be fatal to the fetus.	
Patient C Hypertensive patients with diabetes as a comorbidity.	ACE inhibitors are ideal for this patient because they are nephroprotective against diabetes-induced nephropathy. Beta-blockers should be avoided in diabetic patients as they can mask the sympathetic symptoms of hypoglycemia, which can be fatal.

Solution for Question 3:

Correct Answer: B) Sodium nitrite + sodium thiosulfate + hydroxocobalamin

Explanation:

Sodium nitrite + sodium thiosulfate + hydroxocobalamin is the combination of choice for the management of cyanide toxicity.

- Haemoglobin combines with sodium nitrite to form methemoglobin, which has a higher affinity for cyanide than haemoglobin.

- This results in the formation of cyanmethemoglobin. But this can easily dissociate into cyanide again, thus, sodium thiosulfate is given, which will result in the formation of sodium thiocyanate which is more stable and is excreted into the urine.
- Hydroxocobalamin also has a very high affinity for cyanide leading to formation of cyanocobalamin, which gets excreted into the urine.



Option A: Pralidoxime + atropine	This combination of drugs serves as an antidote for organophosphate poisoning. Atropine is used for atropinization of all the nicotinic and muscarinic receptors thus preventing their further stimulation, while pralidoxime is used to reverse the inhibited acetylcholinesterase and reactivate the enzyme .
Option C: Activated charcoal + N-acetyl cysteine	This combination of drugs is used to treat acetaminophen/paracetamol poisoning. Activated charcoal binds to the unabsorbed paracetamol present in the gut which can be flushed out using a lavage, while N-acetyl cysteine serves to replenish the glutathione stores which binds to the toxic metabolite of paracetamol N-acetyl-p-benzoquinoneimine.
Option D: Sodium nitrite only	If only sodium nitrite is used, the cyanmethemoglobin formed will dissociate back into cyanide and methemoglobin before being excreted out of the body.

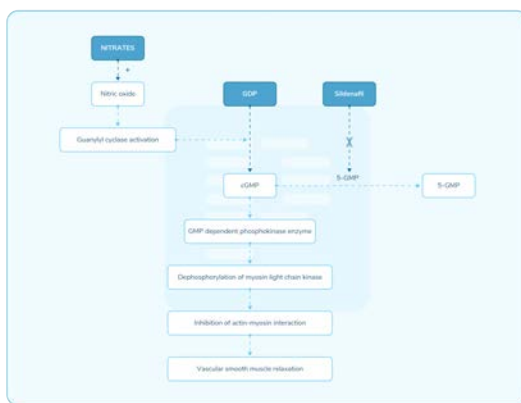
Solution for Question 4:

Correct Answer: A) Sildenafil should be avoided for 24 hours before and after use of nitrates.

Explanation:

- Sildenafil (PDE-5 inhibitor) should be avoided for 24 hours before and after use of nitrates to prevent severe hypotension.

- Nitrates are potent donors of nitric oxide, which in turn activates soluble guanylyl cyclase, thus increasing the production of cGMP. This results in the activation of cGMP-dependent protein kinase (PKG), which dephosphorylates myosin light chain kinase, thus interfering with actin-myosin binding, resulting in vascular smooth muscle relaxation
- PDE-5 inhibitors is responsible for destroying cGMP. This results in the accumulation of cytosolic cGMP, and hence, vasodilation is sustained.
- If the two drugs are used together, there will be potentiation of the activity of nitric oxide, hence causing marked hypotension and even precipitation of myocardial infarction.
- Patients on nitrates may use sildenafil, but there should be a gap of at least 24 hours before or after the use of nitrates.



Solution for Question 5:

Correct Answer: D) Nicorandil

Explanation:

Nicorandil has a dual mechanism of action:

- Nicorandil exhibits cardioprotective activity by stimulating ischemic preconditioning, a process where brief episodes of ischemia followed by reperfusion confer protection against subsequent prolonged ischemic events. This cardioprotective effect is mediated by the activation of mitochondrial ATP-sensitive potassium channels.
- The opening of these channels during ischemic preconditioning helps to protect the myocardium from the detrimental effects of prolonged ischemia and vascular occlusion.

Option A:
Minoxidil

Minoxidil is a potassium channel opener that causes vasodilation near hair follicles, improving

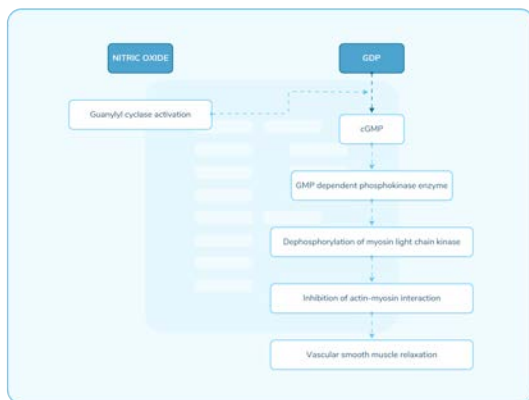
	microcirculation and stimulating hair growth, but it does not provide cardioprotection.
Option B: Diazoxide	Diazoxide blocks potassium channels in pancreatic beta cells, reducing insulin secretion. It is primarily used in insulinoma to prevent hyperinsulinemia but does not provide cardioprotection.
Option D: Hydralazine	Hydralazine is a potassium channel opener that dilates arteries, reducing afterload and cardiac workload. It has antioxidant properties that reduce oxidative stress. It increases renal blood flow, enhancing GFR and reducing circulating volume, thus decreasing preload. While it helps in heart failure and hypertension, it lacks specific cardioprotective actions related to ischemic heart conditions or dual nitrate-like effects seen with Nicorandil.

Solution for Question 6:

Correct Answer: A) It donates nitric oxide, which activates guanylyl cyclase, thus increasing cytosolic cGMP.

Explanation:

- Glyceryl trinitrate is a potent nitric oxide donor, which in turn activates soluble guanylyl cyclase thus increasing production of cGMP.
- This results in the activation of cGMP-dependent protein kinase (PKG), which dephosphorylates myosin light chain kinase (Option B), thus interfering with actin-myosin binding, resulting in vascular smooth muscle relaxation.
- This vasodilating action provides rapid angina relief by reducing the afterload and increasing blood flow to the ischemic myocardium.



Classification of vasodilators

Type	Mechanism of action	Drugs	Uses	Side effects
Arterial dilator	The K ⁺ channel opener causes K ⁺ to move out of the vascular smooth muscle, causing hyperpolarisation and thus leading to vasodilation.	Hydralazine	Hypertensive emergency; Preeclampsia; Chronic heart failure.	Drug-induced lupus. May precipitate angina in CAD patients.
Minoxidil	Alopecia (used in combination with aromatase inhibitor: Finasteride)	Hirsutism in females.		
Diazoxide (it blocks K ⁺ channels in the beta cells of the pancreas, thus reducing insulin secretion)	Insulinoma	Hyperglycemia Fluid retention. Hypotension (with IV injection)		
Calcium channel blockers blocks the L-type voltage-gated calcium channel in the vascular smooth muscle (Option C)	Dihydropyridine (vessels > heart) Nicardipine Nifedipine Felodipine	Hypertension and Angina.	Edema Flushing Headaches	
Non-dihydropyridine (heart > vessels) Verapamil Diltiazem	Arrhythmia and Angina	Bradycardia		

Phosphodiesterase inhibitors-5' It inhibits the enzyme responsible for the degradation of cGMP, thus keeping the cytosolic cGMP elevated	Sildenafil Tadalafil	Erectile dysfunction and Pulmonary artery hypertension.	Flushing Nasal congestion Dyspepsia Hypotension	
Venous dilator	NO donor It increases cytosolic cGMP, which activates protein kinase that dephosphorylates Myosin light chain kinase, thus inhibiting actin-myosin interaction.	Nitrates (Enzyme dependent)	Angina Cyanide poisoning	Flushing
Nitroprusside (Enzyme independent.)	Hypertensive emergencies (rapid action)	Cyanide toxicity		
Arterial + venous dilator	Has Dual action: NO donor + ATP sensitive K ⁺ channel opener.	Nicorandil	Angina	Flushing dizziness Aphthous ulcers

Solution for Question 7:

Correct Answer: B) Verapamil

Explanation:

Verapamil, a non-dihydropyridine calcium channel blocker, has the potential for significant drug interactions with medications metabolised by the CYP3A4 enzyme, such as metoprolol (beta-blockers), as verapamil can inhibit CYP3A4 and increase the plasma levels of this drug.

Contraindications of Verapamil:

Drug Interactions of Verapamil:

- Direct oral anticoagulants (DOACs) such as dabigatran, rivaroxaban, and apixaban increase bleeding risk when taken with calcium channel blockers like verapamil and

diltiazem. This is due to the calcium channel blockers' inhibition of P-glycoprotein (P-gp) and CYP3A4.

- Calcium channel blockers can enhance the effects of alpha- and beta-adrenergic blockers (increasing blood pressure and heart rate), raise digoxin levels significantly, and elevate the risk of toxicity from quinidine and disopyramide.

Solution for Question 8:

Correct Answer: D) Diltiazem

Explanation:

The side effects mentioned in the scenario, including peripheral oedema (excessive swelling in the lower extremities), headaches, and gingival hyperplasia (overgrowth of the gum tissue), are commonly associated with the long-term use of certain calcium channel blockers, particularly non-dihydropyridine CCBs like diltiazem.

Adverse effects of Calcium channel blockers:

Side Effect	Dihydropyridine CCBs (e.g., Amlodipine, Nimodipine)	Non-dihydropyridine CCBs (e.g., Verapamil, Diltiazem)
Peripheral Edema	Common	Common
Headache	Common	Less common
Flushing	Common	Less common
Gingival Hyperplasia	Rare	Common (especially with diltiazem)
Dizziness	Possible	Possible
Hypotension	Possible	Possible
Reflex Tachycardia	Rare	Common (due to negative inotropic effect)
Constipation	Possible	Possible
Nausea	Possible	Possible
Fatigue	Possible	Possible

Solution for Question 9:

Correct Answer: D) Nimodipine

Explanation:

- Nimodipine is a cerebroselective calcium channel blocker specifically used in managing cerebral vasospasm following aneurysmal subarachnoid haemorrhage (SAH).
- It is the most appropriate choice as a prophylactic treatment to prevent cerebral vasospasm and improve cerebral blood flow, thereby reducing the risk of ischemic stroke.

Mechanism of Action of Nimodipine: It selectively blocks the entry of calcium ions through the L-type calcium channels in the smooth muscle cells of cerebral arteries and arterioles. This results in the following effects:

- Vasodilation of cerebral blood vessels: Nimodipine causes vasodilation and improves cerebral blood flow by blocking calcium influx into the smooth muscle cells of cerebral arteries and arterioles.
- Prevention of cerebral vasospasm: Cerebral vasospasm, a common complication after SAH, is characterised by the narrowing of cerebral arteries, leading to decreased cerebral blood flow and an increased risk of ischemic stroke. Nimodipine helps prevent and alleviate cerebral vasospasm by promoting vasodilation and improving cerebral perfusion.
- Cerebroselective action: Nimodipine has a high affinity for the L-type calcium channels in cerebral blood vessels, making it selective for the cerebral vasculature. This selectivity minimises systemic side effects, such as hypotension and reflex tachycardia, commonly associated with non-selective calcium channel blockers.

Administration: Typically given orally or via nasogastric tube, starting within 4 days of the subarachnoid haemorrhage.

Solution for Question 10:

Correct Answer: D) Amlodipine

Explanation:

Amlodipine is a long-acting dihydropyridine CCB, and it would be an appropriate choice to achieve better blood pressure control in the long term. It can be used as an add-on therapy with other antihypertensive agents, and its once-daily dosing improves patient compliance.

Mechanism of action of CCB (Dihydropyridines):

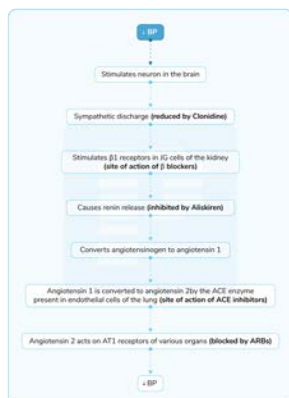
It blocks the entry of calcium ions through the L-type calcium channels in vascular smooth muscle cells, causing vasodilation and a reduction in peripheral vascular resistance.

Drug	Use	Rationale for Choice
Nifedipine (Option A) Short-acting	Primarily used for angina pectoris	Not recommended for long-term hypertension management due to its short duration of action and potential for reflex tachycardia.
Nicardipine (Option B) Short-acting	Commonly used in the management of hypertensive emergencies and perioperative hypertension or severe hypertension in the intensive care unit.	The short duration of action makes it less suitable for long-term hypertension management.
Clevidipine (Option C) Ultra-short-acting	Very short half-life makes it unsuitable for long-term hypertension management.	
Amlodipine (Option D) Long-acting.	Recommended as a first-line or add-on therapy for the long-term management of hypertension	Preferred choice due to its long duration of action (up to 24 hours), once-daily dosing, and effectiveness in lowering blood pressure with good tolerability.

Solution for Question 11:

Correct Answer: C) 1-c, 2-d, 3-e, 4-a, 5-b

Explanation:



Name of drug	Class	Mechanism of action
Aliskiren	Direct renin inhibitor	Blocks the formation of angiotensin 1, which decreases BP.
Lisinopril	ACE inhibitor	Blocks the conversion of angiotensin 1 to angiotensin 2
Losartan	ARB	Blocks angiotensin 1 receptor
Clonidine	Central sympat holytic	It acts on alpha 2A receptors, decreasing sympathetic discharge from the brain and reducing renin secretion.
Metoprolol, Atenolol	Beta-blockers	Act on JG cells of the kidney to reduce renin release.

Solution for Question 12:

Correct Answer: C) Diltiazem

Explanation:

Diltiazem is an example of non-dihydropyridines (non-DHPs) calcium channel blockers.

Calcium channel blockers (CCBs) can be classified into two main groups: dihydropyridines (DHPs) and non-dihydropyridines (non-DHPs).

Group	Examples	Main Effects
Non-dihydro pyridines (non-DHPs).	Diltiazem (Benzothiazepines)	Moderate effects on cardiac muscle and vascular smooth muscle.

Verapamil (Phenylalkylamines)	It primarily affects cardiac muscle, leading to negative inotropic effects and slow conduction through the AV node.	
Dihydropyridines	Amlodipine, Bepridil, Felodipine, Isradipine, Nicardipine, Nifedipine, Nisoldipine	It predominantly acts on the smooth vascular muscle, resulting in vasodilation and reduced peripheral vascular resistance; some also affect the cardiac muscle.

Solution for Question 13:

Correct Answer: B) 1,2

Explanation:

A R B' s	Mechanism of Action	Use	Special Features
L o s a r t a n	Acts as a competitive antagonist and inverse agonist at AT1 receptors, demonstrating selectivity for AT1 over AT2 receptors.	Hypertension, heart failure	It does not interfere with bradykinin degradation, reducing the risk of cough. It possesses an antiplatelet effect by blocking thromboxane A2 receptors. It exhibits uricosuric action, which is potentially beneficial for gout.
I r b e s a r t a n	Functions as a selective competitive antagonist of AT1 receptors.	Hypertension, atrial fibrillation	Effective in atrial fibrillation by blocking the sinoatrial (SA) node in the heart.

T e l m i s a r t a n	Acts as a selective competitive antagonist of AT1 receptors.	Hypertension, prevention of cardiovascular events, management of diabetic nephropathy	Acts as a peroxisome proliferator-activated receptor-gamma (PPAR-γ) agonist, similar to pioglitazone. Stimulation of PPAR-γ leads to improved insulin sensitivity and reduced risk of type 2 diabetes by decreasing insulin resistance.
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Solution for Question 14:

Correct Answer: C) Candesartan

Explanation:

- Candesartan is one of the most potent ARBs because of its high affinity for the AT-1 receptor.
- It has a relatively longer plasma half-life than other ARBs (about 9 hours).
- The recommended oral dosage of candesartan ranges from 4 to 32 mg once or twice daily.

Candesartan:

Mechanism of Action	Blocks angiotensin II receptors (AT-1), exerting its pharmacological effects.
Uses	It is commonly prescribed for various conditions, including hypertension, heart failure, and diabetic nephropathy, with off-label uses for conditions such as cerebrovascular accident (stroke), left ventricular hypertrophy, and migraines.
Side Effects	Dizziness, hypotension, hyperkalemia, cough, headache, back pain, angioedema, upper respiratory tract infections.
Specific Characteristics	Candesartan is associated with a risk of teratogenicity, evidenced by a boxed warning for fetal toxicity.

Solution for Question 15:

Correct Answer: C) Cough

Explanation:

- ACE inhibitors commonly cause a persistent dry cough, which is not dose-related.
- The mechanism underlying ACE inhibitor-induced cough involves the inhibition of bradykinin breakdown. ACE (angiotensin-converting enzyme) degrades bradykinin, which mediates vasodilation and inflammation.
- When ACE inhibitors block this enzyme, bradykinin levels increase, irritating the respiratory tract and triggering cough receptors. Additionally, the inhibition of substance P breakdown may contribute to cough.
- The onset of cough associated with ACE inhibitors can vary but often occurs within weeks to months after initiating therapy. However, it tends to resolve spontaneously within days to weeks after discontinuation of the ACE inhibitor.
- The management of cough involves switching to an alternative antihypertensive medication, such as an angiotensin II receptor blocker (ARB), which does not affect bradykinin metabolism and is less likely to induce cough.

Adverse effects of ACEI:

Adverse Effect	Description
Hypotension	Decrease in blood pressure, especially upon initiation of therapy or with dosage increases.
Hyperkalemia	Increased serum potassium levels due to reduced aldosterone secretion.
Cough	Persistent dry cough, possibly due to bradykinin buildup in the lungs.
Angioedema	Due to increased bradykinin levels, swelling of deeper layers of skin, particularly around the eyes and lips, can be life-threatening.
Rashes/Urticaria	Skin reactions like rashes or hives occur in a small percentage of users.
Dysgeusia	Reversible taste changes are more common with captopril.
Rare Effects	Granulocytopenia (low white blood cells) and proteinuria (excess protein in urine) are more concerning in patients with kidney disease.

Acute Renal Failure	Possible in patients with bilateral renal artery stenosis due to decreased kidney filtration.
Fetal Effects	ACE inhibitors are contraindicated during pregnancy due to potential harm to the fetus, including fetal renal damage due to inhibition of the fetal renin-angiotensin system.

Solution for Question 16:

Correct Answer: D) Potassium-sparing diuretics

Explanation:

When prescribing ACE inhibitors, certain drug interactions can pose significant risks, especially the interaction with potassium-sparing diuretics. Here are some key interactions and their clinical implications:

Drug Interaction	Mechanism	Clinical Implication
ACE Inhibitors + Potassium Supplements/K-sparing diuretics	Risk of hyperkalemia due to reduced aldosterone secretion by ACE inhibitors combined with potassium-retaining agents.	Monitoring serum potassium levels is crucial to avoid hyperkalemia, which can cause serious cardiac issues such as arrhythmias.
ACE Inhibitors + NSAIDs	NSAIDs can reduce the anti-hypertensive effects of ACE inhibitors and decrease renal blood flow, increasing the risk of acute renal failure (ARF) or oliguria.	Use caution when combining NSAIDs with ACE inhibitors, and closely monitor renal function, particularly in high-risk patients.
ACE Inhibitors + Lithium	ACE inhibitors can increase the reabsorption of lithium, potentially leading to lithium toxicity.	Monitor lithium levels carefully when starting or adjusting ACE inhibitors in patients taking lithium, and adjust the lithium dosage as needed.
ACE Inhibitors + Antacids	Reduced bioavailability of ACE inhibitors due to binding with antacids.	Administer ACE inhibitors at least two hours before or after antacids to maintain their efficacy.
ACE Inhibitors + Diuretics	Diuretics enhance the hypotensive effects of ACE inhibitors by depleting sodium and raising renin levels.	In patients already on diuretics, start ACE inhibitors at a low dose to minimise the risk of excessive blood pressure reduction initially.

Solution for Question 17:

Correct Answer: D) All the above

Explanation:

The patient has multiple comorbidities that would benefit from the use of ACE inhibitors (ACEIs), including:

- Reduction of proteinuria: ACEIs help decrease proteinuria, which slows the progression of kidney damage in Chronic Kidney disease (CKD).
- Reduction of preload and afterload: ACEIs improve symptoms and outcomes in heart failure patients by reducing both preload and afterload on the heart.
- Prevention of diabetic nephropathy: ACEIs help prevent the progression of diabetic nephropathy by lowering intraglomerular pressure and hyperfiltration, thereby preventing albuminuria.

Uses of ACEI:

- Hypertension: ACEIs are particularly effective in patients with comorbid conditions such as diabetes, CKD, and scleroderma.
- Myocardial Infarction (MI): ACEIs are used both in the acute management of MI and for long-term secondary prophylaxis.
- Congestive Heart Failure (CHF): ACEIs improve outcomes by reducing the workload on the heart.
- Prophylaxis in high cardiovascular-risk individuals: ACEIs protect both the myocardium and vasculature, possibly through improved endothelial function.
- Scleroderma: ACEIs can prevent the rise in blood pressure and renal function deterioration mediated by angiotensin II in patients with scleroderma.

Solution for Question 18:

Correct Answer: A) Captopril

Explanation:

- A prodrug is an inactive substance metabolised within the body to form an active drug. Most ACE inhibitors (ACEIs) are prodrugs that must be converted in the liver to their active forms. However, some ACEIs are administered in their active form.
 - Captopril and Lisinopril are not prodrugs; they are active in their administered forms.
 - The other ACEIs listed, including Enalapril, Fosinopril, and Ramipril, are prodrugs. They require metabolic conversion to their active forms, enalaprilat, fosinoprilat, and ramiprilat.
 - Enalaprat is the fastest acting ACE inhibitor and can be used intravenously
-

Solution for Question 19:

Correct Answer: B) Fosinopril and Telmisartan

Explanation:

Fosinopril and Telmisartan are considered to be safer in renal failure. ACE inhibitors (ACEIs) and angiotensin II receptor blockers (ARBs) are commonly used in the treatment of hypertension and chronic kidney disease. Their safety in patients with renal failure varies based on their excretion pathways:

- ACEIs: Most ACE inhibitors, such as enalapril, lisinopril, benazepril, and ramipril, are predominantly excreted by the kidneys. Fosinopril is an exception, as it is excreted by both the kidneys and the liver, making it safer for patients with impaired renal function.
- ARBs: Some ARBs, such as telmisartan, are largely excreted by the liver, which reduces the strain on the kidneys. In contrast, losartan is primarily excreted by the kidneys, and candesartan is excreted by both the kidneys and the liver.

Given these considerations, fosinopril and telmisartan are the safest choices for patients with renal failure due to their respective excretion mechanisms.

Solution for Question 20:

Correct Answer: D) Nicardipine

Explanation:

- The patient is most likely in a hypertensive emergency, an acute, marked elevation in blood pressure associated with signs of target-organ damage.
- These include pulmonary oedema, cardiac ischemia, neurologic deficits, acute renal failure, aortic dissection, and eclampsia.
- Nicardipine, a calcium channel blocker, is used to treat hypertensive emergencies.

Drugs used to treat hypertensive emergencies include

- Dihydropyridine calcium channel blockers: Intravenous Nicardipine (DOC), Clevidipine.
 - Nitrates: Sodium nitroprusside and Nitroglycerin.
 - Adrenergic blocking agents: Labetalol (DOC in pregnancy) and Esmolol.
 - Others: Enalaprilat and Hydralazine.
-

Anti Anginal drugs

1. A 60-year-old male with a history of hypertension presents to the emergency with a sudden onset of chest pain radiating to his jaw. The pain started 35 minutes ago while he was walking and has not improved with rest. On physical examination, ECG showed ST-segment depression in the anterior leads. What is the most appropriate initial treatment for managing this condition?

- A. Atenolol
- B. Nitroglycerin
- C. Aspirin
- D. Clopidogrel

2. Which of the following statements correctly describes the use of beta-blockers in the management of chronic angina?

- A. It dilates coronary blood vessels by blocking beta-2 receptors in the heart.
- B. It decreases myocardial oxygen demand by reducing heart rate and contractility.
- C. It dilates coronary arteries directly, improving blood flow to the myocardium.
- D. It increases the workload on the heart by increasing blood pressure.

3. A 40-year-old female presents to the emergency with recurrent chest pain over the past few days. The pain occurs at rest, often waking her from sleep and resolves spontaneously. Her resting ECG shows transient ST-segment elevation with a normal coronary angiogram. Which of the following most appropriately matches the treatment and mechanism of action of the drug of choice? Drug Mechanism of Action 1. Nitroglycerin a. Reduce myocardial oxygen demand by decreasing heart rate. 2. Propranolol b. Vasodilates coronary arteries by increasing

cGMP. 3. Nifedipine c. Inhibit calcium influx into vascular smooth muscle and cardiac cells. 4. Aspirin d. Inhibits platelet aggregation by irreversibly acetylating cyclooxygenase.

Drug	Mechanism of Action
1. Nitroglycerin	a. Reduce myocardial oxygen demand by decreasing heart rate.
2. Propranolol	b. Vasodilates coronary arteries by increasing cGMP.
3. Nifedipine	c. Inhibit calcium influx into vascular smooth muscle and cardiac cells.
4. Aspirin	d. Inhibits platelet aggregation by irreversibly acetylating cyclooxygenase.

- A. 3 and c
 - B. 3 and a
 - C. 2 and a
 - D. 1 and c
-

4. Which of the following statements about the pharmacokinetics of nitrates is incorrect?

- A. Nitrates undergo extensive first-pass metabolism in the liver
 - B. Nitrates are highly lipid-soluble drugs
 - C. Isosorbide mononitrate has 100% bioavailability when taken orally
 - D. Pentaerythritol tetranitrate is the shortest-acting nitrate when administered orally
-

5. A 50-year-old male presents to the hospital with an acute onset of chest pain that started 20 minutes ago while he was watching television. An ECG revealed ST-segment elevation in the lateral leads. His medical history includes hypertension and erectile dysfunction. Which of the following statements

regarding the management of his acute chest pain is true? Nitrates can be safely administered as they enhance vasodilation by increasing cyclic GMP. Nitrates are contraindicated in this patient due to the risk of hypotension. Nitrates are avoided as they cause bradycardia when combined with PDE-5 inhibitors. Nitrates are contraindicated as ECG findings suggest right ventricular dysfunction. Nitrates can be safely administered in this patient after a gap of at least 24 hours.

- A. 2 and 5
 - B. 2, 3 and 4
 - C. 1, 3, and 5
 - D. 4 and 5
-

6. Which of the following is a potassium channel opener used to manage resistant angina?

- A. Diazoxide
 - B. Nicorandil
 - C. Ranolazine
 - D. Verapamil
-

7. A 63-year-old male with a history of coronary artery disease presents to the clinic with worsening chest pain during physical exertion. The cardiologist performs a pharmacological stress test to evaluate his coronary blood flow. During the test, the patient's symptoms were exacerbated. Which of the following drugs is not associated with this phenomenon?

- A. Adenosine
- B. Dipyridamole
- C. Nitroglycerin
- D. Nitroprusside

8. A 65-year-old male taking propranolol presents to a cardiologist with worsening chest pain. The cardiologist discusses the option of second-line antianginal drugs to manage his symptoms better. Which of the following statements is correct? Ranolazine acts on sodium channels and is primarily used to treat acute angina. Fasudil acts as a Rho kinase inhibitor used in chronic angina. Trimetazidine inhibits thiolase to shift fuel metabolism from glucose to fatty acids. Ivabradine reduces heart rate by selectively inhibiting the funny current in the sinoatrial node. Ranolazine is contraindicated in patients with existing diabetes mellitus.

- A. 2 and 4
 - B. 1, 2 and 4
 - C. 1, 2 and 3
 - D. 2 only
-

9. Which of the following correctly describes the mechanism of action of cilnidipine?

- A. Blocks L-type calcium channels in vascular smooth muscle
 - B. Blocks T-type calcium channels in cardiac and vascular smooth muscle
 - C. Blocks both L-type and N-type calcium channels
 - D. Blocks only N-type calcium channels in the nervous system
-

10. Which of the following statements correctly matches with the medications? 1. Metoprolol A. Contraindicated with Non-DHP calcium channel blockers. 2. Ivabradine B. Increase risk of oral ulcers. 3. Trimetazidine C. Abrupt discontinuation can exacerbate angina. 4. Enoxaparin D. Increase risk of Parkinson's disease. 5. Nicorandil E. Contraindicated in renal insufficiency.

1. Metoprolol	A. Contraindicated with Non-DHP calcium channel blockers.
2. Ivabradine	B. Increase risk of oral ulcers.
3. Trimetazidine	C. Abrupt discontinuation can exacerbate angina.
4. Enoxaparin	D. Increase risk of Parkinson's disease.
5. Nicorandil	E. Contraindicated in renal insufficiency.

- A. 1:C, 2: A, 3:D, 4:E, 5: B
 B. 1:A, 2: C, 3:D, 4:B, 5: E
 C. 1:D, 2: C, 3:B, 4:A, 5: B
 D. 1:C, 2: B, 3:E, 4:A, 5: D
-

11. A 60-year-old male presents to the clinic complaining of chest pain. He describes the pain occurring with exertion and relief by rest. His baseline ECG is unremarkable. Which of the following drugs is most likely to worsen his condition?

- A. Nifedipine
 B. Felodipine
 C. Diltiazem
 D. Amlodipine
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	4
Question 5	1
Question 6	2
Question 7	3
Question 8	1
Question 9	3
Question 10	1
Question 11	1

Solution for Question 1:

- This patient with sudden onset chest pain radiating to the jaw and not improving on rest with ECG showing ST changes suggests acute angina with sublingual nitroglycerin as a drug of choice.
- Angina pectoris is characterised by chest pain caused by reduced blood flow and oxygen supply to the heart muscle. The pain is elicited by the metabolites that accumulate due to myocardial ischaemia.

Nitrates:

- They are best to abort an acute attack of angina.
- MOA: Venodilation → Reduce preload → Reduce myocardial contractility → Decreased O₂ demand

- Other beneficial effects: Dilates coronary artery (increase cGMP): effective in vasospastic angina. Redistribution of coronary blood flow to the ischemic area (Robinhood phenomenon).
- Dilates coronary artery (increase cGMP): effective in vasospastic angina.
- Redistribution of coronary blood flow to the ischemic area (Robinhood phenomenon).
- Dilates coronary artery (increase cGMP): effective in vasospastic angina.
- Redistribution of coronary blood flow to the ischemic area (Robinhood phenomenon).
- Uses: Angina (stable, vasospastic and chronic) Acute coronary syndrome or myocardial infarction (STEMI/NSTEMI) Acute pulmonary edema Abdominal colicky pain
- Angina (stable, vasospastic and chronic)
- Acute coronary syndrome or myocardial infarction (STEMI/NSTEMI)
- Acute pulmonary edema
- Abdominal colicky pain
- Angina (stable, vasospastic and chronic)
- Acute coronary syndrome or myocardial infarction (STEMI/NSTEMI)
- Acute pulmonary edema
- Abdominal colicky pain

Beta-blockers(OPTION A) like atenolol are important in the long-term management of angina; they are not the immediate treatment for acute pain relief.

Aspirin (OPTION C) prevents platelet aggregation and reduces the risk of future ischemic events but does not relieve acute chest pain.

Clopidogrel (OPTION D) is an antiplatelet agent used to prevent further ischemic events but does not provide immediate relief of acute angina symptoms.

Solution for Question 2:

Beta-blockers help manage angina by reducing heart rate and the force of heart contractions.

- This lowers the heart's oxygen demand, decreasing the likelihood of future angina episodes.
- They also reduce the heart's workload by lowering blood pressure and heart rate.

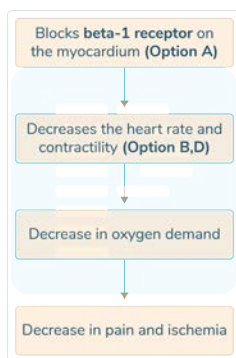
Chronic stable angina (Common form):

- Attacks are predictably provoked (stable angina) by exercise, emotion, eating or coitus.

- Pathogenesis: Atherosclerotic plaque → gradual → ↓ blood supply to myocardium → ↓ O₂ supply to myocardium → Anginal attack
- Atherosclerotic plaque → gradual → ↓ blood supply to myocardium → ↓ O₂ supply to myocardium → Anginal attack
- Atherosclerotic plaque → gradual → ↓ blood supply to myocardium → ↓ O₂ supply to myocardium → Anginal attack

Beta-blockers:

- DOC in chronic stable angina.
- MOA:



- Side effects: Risk of vasoconstriction (block beta-2 receptor in coronary arteries). (Option C)
- Risk of Vasospastic angina. (Option C)
- Risk of Vasospastic angina.
- Alternate treatment for chronic angina: Calcium channel blockers and nitrates.
- Risk of vasoconstriction (block beta-2 receptor in coronary arteries). (Option C)
- Risk of Vasospastic angina.

Cardioselective beta blocker	Non-Selective beta blocker
Blocks beta-1 > beta-2 Mnemonic- MAN B Metoprolol Atenolol Nebivolol Bisoprolol	Blocks beta-1 and beta-2 equally. Propranolol Contraindications- Vasospastic or Prinzmetal angina.

Solution for Question 3:

Correct Answer: A) 3 and c

Explanation:

This young female has recurring chest pain with ECG changes and a normal coronary artery angiogram, suggesting Prinzmetal angina as the most likely diagnosis, with calcium channel blockers (nifedipine) as the treatment of choice.

- Chronic Vasospastic angina (Prinzmetal angina) occurs due to recurrent localised (occasionally) coronary vasospasm, which may be superimposed on arteriosclerotic coronary artery disease.
- Attacks occur at rest or during sleep and are predictable.
- DOC- Calcium channel blockers (Dihydropyridine type)

Calcium channel blockers	
DHP	Non-DHP
MOA: Coronary artery dilator Drugs: Amlodipine, Nifedipine and Felodipine. Use: Chronic Vasospastic angina Combined with beta-blockers to reduce the risk of tachycardia.	MOA- Decrease heart rate and O ₂ demand (heart & coronary artery). Drugs: Verapamil, Diltiazem Uses: Stable angina, Coronary dilator and Vasospastic angina. Never combined with beta-blockers (marked bradycardia).

- Other treatment options are long-acting nitrates, oral nitrates and transdermal patches.

Propranolol(Options B and C) reduces myocardial oxygen demand by decreasing heart rate and contractility are not used in Prinzmetal angina.

Nitroglycerin (Option D) vasodilates coronary arteries and can relieve acute symptoms; it is not the first choice for Prinzmetal angina.

Solution for Question 4:

Pharmacokinetics of Nitrates:

- Highly lipid soluble. (OPTION B)

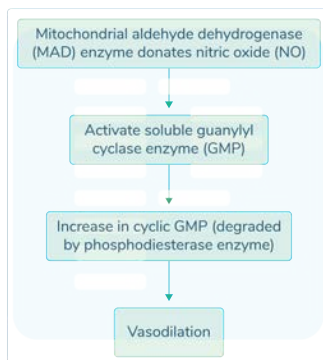
- High first-pass metabolism in the liver (given sublingual). (OPTION A) Isosorbide mononitrate doesn't have first-pass metabolism (given orally).
- Isosorbide mononitrate doesn't have first-pass metabolism (given orally).
- DOC for acute angina: Nitroglycerin or isosorbide dinitrate.
- Isosorbide mononitrate doesn't have first-pass metabolism (given orally).

Nitrates	
Glyceryl trinitrate (Nitroglycerin)	Shortest acting Given as oral, sublingual, transdermal, rectal, nasal spray and IV.
Isosorbide dinitrate	Prodrug; converted to active form (mononitrate).
Isosorbide mononitrate	Oral route 100% bioavailability (OPTION C)
Pentaerythritol tetranitrate	Longest acting (OPTION D)

Solution for Question 5:

Nitrates:

MOA:



Nitrates

Drug interaction	PDE 5 inhibitors (Sildenafil, Tadalafil): Causes massive vasodilation. Contraindicated as they can cause sudden hypotension and death. Prevent: 24-hour gap with sildenafil 72-hour gap with tadalafil
Side effects	Bradycardia and Hypotension (Bezold-Jarisch reflex) . Flushing and oedema. Reflex tachycardia Monday's disease: Down-regulation of MAD enzymes. Tachycardia, dizziness, headache with re-exposure after a drug-free interval.
Contraindications	SBP < 90 mmHg HR >100 BPM and <50 BPM Right heart failure

Solution for Question 6:

CORRECT ANSWER: B) Nicorandil

EXPLANATION:

Nicorandil:

- Potassium channel opener (KATP-stimulatory action).
- MOA: Causes coronary dilation and decreases O₂ demand.
- Other actions: Antioxidant action: protects the heart. Ischemic preconditioning makes the heart resistant to ischemia.
- Antioxidant action: protects the heart.
- Ischemic preconditioning makes the heart resistant to ischemia.
- Uses: Resistant angina
- Side effects- Aphthous ulcer and hypotension.
- Contraindicated with PDE-5 inhibitor (sildenafil).
- Antioxidant action: protects the heart.
- Ischemic preconditioning makes the heart resistant to ischemia.

Diazoxide (Option A), a potassium channel opener, primarily treats hypertensive circumstances rather than resistant angina.

Ranolazine (Option C) is a late sodium channel inhibitor for treating chronic angina and ventricular tachycardia.

Verapamil (Option D) is a calcium channel blocker that can treat vasospastic angina.

Drugs acting on potassium channels:

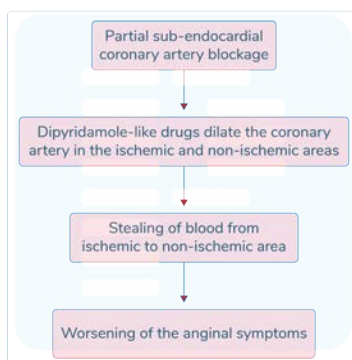
- Class III antiarrhythmics - block potassium channels
- Nicorandil- Potassium channel opener

Solution for Question 7:

Nitroglycerin does not cause coronary steal phenomenon and improves blood flow to ischemic areas.

Coronary steal phenomenon/syndrome:

- Used in pharmacological stress tests to diagnose partially blocked coronary arteries.
- MOA:



Drugs causing coronary steal syndrome

Mnemonic- HINDA

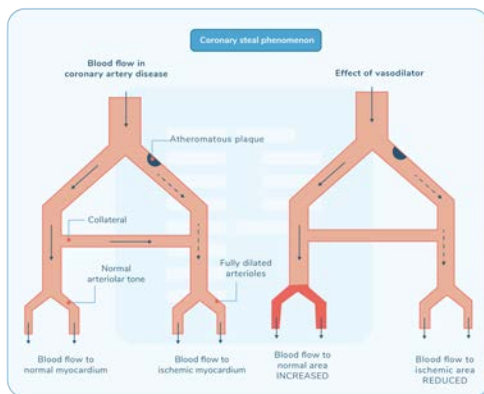
H: Hydralazine

I: Isoflurane (General anaesthetic)

N: Nitroprusside (OPTION D)

D: Dipyridamole (antiplatelet and vasodilator) (OPTION B)

A: Adenosine (OPTION A)



Solution for Question 8:

CORRECT ANSWER: A) 2 and 4

Second-line anti-anginal drugs (Mnemonic- No DRIFT):

- Nicorandil
- Dipyridamole
- Ranolazine
- Ivabradine
- Fasudil.
- Trimetazidine

Drugs	Ranolazine	Fasudil	Trimetazidine	Ivabradine
Target	Blocks late Na channels	Rho kinase inhibitor (Rock enzyme)	Partial fatty acid oxidation (PFOX) inhibitor	Inhibit funny current channels in the SA node.
MOA	↓ Myocardial contractility and O ₂ demand.	Coronary artery dilatation.	Inhibit thiolase enzyme and shift the fuel metabolism from fatty acids to glucose.	Reduce heart rate and O ₂ demand.

Use	1. Chronic stable angina 2. Ventricular tachycardia 3. Improves glycemic control (lower HbA1c and increase insulin)	Chronic vasospastic angina	Second-line anti-anginal drug	1. Chronic stable angina 2. Chronic heart failure
Side effects	1. QT prolongation 2. Liver failure		Parkinson disease	1. Luminous phenomenon 2. Atrial fibrillation 3. QT Prolongation

Solution for Question 9:

Correct Answer: C) Blocks both L-type and N-type calcium channels

Explanation:

- Cilnidipine: 4th generation Ca²⁺ channel blocker. Blocks both L-type and N-type calcium channels, providing antihypertensive effects and additional benefits by inhibiting sympathetic nerve activity.
- 4th generation Ca²⁺ channel blocker.
- Blocks both L-type and N-type calcium channels, providing antihypertensive effects and additional benefits by inhibiting sympathetic nerve activity.
- Dihydropyridine calcium channel blockers like amlodipine blocks L-type calcium channels in vascular smooth muscle.
- Mibefradil blocks T-type calcium channels in cardiac and vascular smooth muscle.
- 4th generation Ca²⁺ channel blocker.
- Blocks both L-type and N-type calcium channels, providing antihypertensive effects and additional benefits by inhibiting sympathetic nerve activity.

Feature	DHP Calcium Channel Blockers	Non-DHP Calcium Channel Blockers
Examples	Amlodipine, Nifedipine, Felodipine, Cilnidipine	Diltiazem, Verapamil
Mechanism of Action	Primarily block L-type calcium channels in vascular smooth muscle	Block L-type calcium channels in the heart

Uses	Hypertension	Angina, rate control in arrhythmias
Side Effects	Peripheral edema, flushing, headache	Bradycardia, heart block, dizziness

Solution for Question 10:

Correct Answer: A) 1:C, 2: A, 3:D, 4:E, 5: B

Explanation:

Medication	Adverse effects
1. Beta-blockers (e.g metoprolol)	Abrupt discontinuation can exacerbate angina. Bradycardia, AV block Bronchospasm Peripheral vasoconstriction Worsening of acute heart failure. Worsening of psoriasis.
2. Ivabradine	Contraindicated with diltiazem or verapamil. Bradycardia, heart block QT prolongation, atrial fibrillation Enhanced lightness in restricted areas of the visual field.
3. Trimetazidine	Increase incidence of Parkinson's disease. Agranulocytosis
4.Low-molecular-weight heparin (e.g., enoxaparin, dalteparin)	Accumulation in renal insufficiency. Heparin-induced thrombocytopenia. Raised liver enzymes.
5. Nicorandil	Hypotension Headache Buccal and GI ulcers Contraindicated with PDE5 inhibitor.

Solution for Question 11:

- Short-acting dihydropyridines CCBs such as nifedipine can cause hypotension, tachycardia, flushing, and dizziness, which may worsen myocardial ischemia and trigger anginal symptoms.
 - Dizziness and flushing are substantially less of an issue with long-release formulations, which have a longer half-life and provide more consistent plasma drug concentrations.
-

Myocardial Infarction & Drugs for Heart Failure

1. Which of the following statements regarding the mechanism of action of Digoxin is incorrect?

- A. Digoxin inhibits the Na⁺-K⁺ ATPase pump, leading to an increase in intracellular sodium concentration.
- B. Increased intracellular Ca²⁺ concentration due to Digoxin leads to enhanced myocardial contractility.
- C. Digoxin prolongs the (AV) node's refractory period, slowing conduction.
- D. Digoxin decreases vagal tone, resulting in increased heart rate.

2. A 58-year-old male patient presents with severe chest pain, radiating to his left arm and jaw. An electrocardiogram (ECG) confirms ST-segment elevation myocardial infarction (STEMI). The cardiologist recommends an urgent percutaneous coronary intervention (PCI) with the placement of an endovascular stent to restore blood flow to the affected coronary artery. What is the drug composition commonly used in drug-eluting endovascular stents?

- A. Paclitaxel
- B. Sirolimus
- C. Heparin
- D. Clopidogrel

3. A 65-year-old male presents with severe substernal chest pain radiating to his arm and back. ECG showed sinus rhythm with small Q waves in leads III and aVF, T-wave inversions in leads II and III, and aVF and ST depressions in V5 and V6. His vitals are normal. Cardiac markers are elevated. This patient can receive all of the

following except?

- A. Ramipril
 - B. Ticagrelor
 - C. Streptokinase
 - D. Enoxaparin
-

4. A 55-year-old male with a history of hypertension presents with increasing shortness of breath, fatigue, and swelling in his legs. On examination, he has elevated jugular venous pressure, bilateral pitting edema, and pulmonary crackles. Echocardiography reveals reduced ejection fraction. A new medication was recently started to manage his condition, but he developed severe angioedema, leading to the discontinuation of the drug. Identify the drug?

- A. Sacubitril/Valsartan
 - B. Ivabradine
 - C. Omapatrilat
 - D. Dapagliflozin
-

5. A 68-year-old female patient with a history of chronic heart failure and atrial fibrillation is being treated with digoxin. Recently, she has been prescribed additional medications for comorbid conditions. She presents to the emergency department with symptoms of nausea, vomiting, confusion and visual disturbances. Her serum digoxin level is found to be elevated. Which of the following medications is most likely to have precipitated digoxin toxicity in this patient?

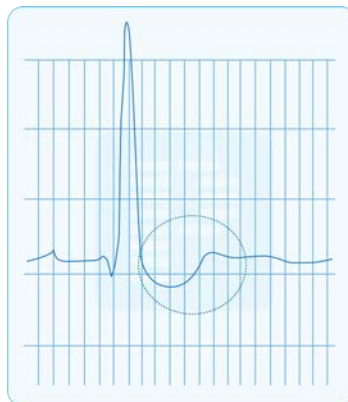
- A. Furosemide
- B. St. John's Wort
- C. Cholestyramine

D. Rifampin

6. Which ECG changes would not be seen in digoxin toxicity?

- A. Increased PR Interval
 - B. Widened QRS
 - C. Narrow QRS Complex
 - D. Inverted T Wave
-

7. A 72-year-old female with a history of CHF presents to the emergency department with nausea, vomiting, and visual disturbances. Her ECG reveals a sinus bradycardia with PR interval prolongation, inverted T waves and a particular finding below. Laboratory investigations show elevated serum digoxin levels. Which of the following drugs should not be used in the treatment of drug toxicity for this patient?



- A. Nifedipine
- B. Digibind
- C. Lidocaine
- D. KCL

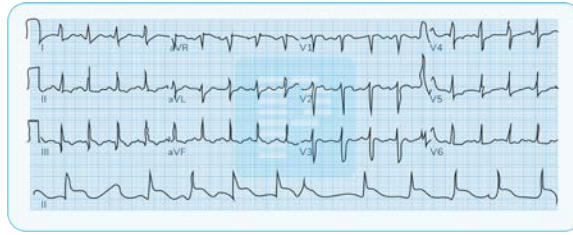
8. A 68-year-old male presents to the clinic with worsening dyspnea on exertion, fatigue, and ankle swelling. He has a past medical history of chronic heart failure, which was diagnosed 5 years ago. Despite being on several medications, he continues to experience symptoms. On examination, his heart rate is irregular and there is evidence of jugular venous distension. What is the drug of choice for managing his condition?

- A. Lisinopril
 - B. Carvedilol
 - C. Furosemide
 - D. Digoxin
-

9. All of the following statements regarding the pharmacokinetics of digoxin are true except?

- A. Half-life 36 to 48 hours
 - B. Narrow therapeutic index
 - C. Digoxin is primarily excreted unchanged by the kidney.
 - D. Digoxin is concentrated in the plasma more than in the heart.
-

10. A patient presents with left-sided chest pain, which started 5 hours ago, radiating to the left arm and doesn't get relieved on taking rest. An ECG is taken immediately which is given below. Which of the following is the drug for definitive management in this patient?



- A. Nitroglycerin
 - B. Reteplase
 - C. Aspirin & ticagrelor
 - D. Morphine
-

11. A 68-year-old male patient with a history of chronic heart failure has been experiencing worsening symptoms despite being on optimal medical therapy, including ACE inhibitors, beta-blockers, and diuretics. His cardiologist decides to add a cardiac glycoside to his treatment regimen to improve his symptoms. Which of the following medications is NOT a cardiac glycoside?

- A. Doxorubicin
 - B. Digitoxin
 - C. Digoxin
 - D. Ouabain
-

12. Which of the following is a PDE-3 inhibitor used in managing heart failure?

- A. Milrinone
 - B. Roflumilast
 - C. Cilostazol
 - D. Theophylline.
-

13. Which of the following is not a mechanism of action of levosimendan?

- A. It inhibits phosphodiesterase-3
 - B. It sensitizes cardiac troponin C to Ca^{2+} ions
 - C. It inhibits vasopressin-2 receptor
 - D. It opens ATP-sensitive K^{+} channels in vessels
-

14. Which of the following is true regarding natriuretic peptides and their recombinant forms?

- A. NT-proBNP remains characteristically normal in atrial fibrillation
 - B. Nesiritide acts by decreasing the level of cyclic GMP
 - C. There are no recombinant forms for ANP
 - D. C-type natriuretic peptide is synthesized by kidneys and endothelium
-

15. A known asthmatic is diagnosed with chronic heart failure. Which of the following drugs provides the most therapeutic benefit to this patient?

- A. Carvedilol
 - B. Sotalol
 - C. Nebivolol
 - D. Bisoprolol
-

16. All of the following are incorrect about sacubitril except?

- A. It inhibits the enzyme that is involved in the synthesis of natriuretic peptides.

- B. It is contraindicated to combine with ACE inhibitors.
 - C. The half-life of its active metabolite is 3 hours.
 - D. Nesiritide is generally preferred over ARNIs (sacubitril-valsartan).
-

17. A patient who had a history of myocardial infarction was diagnosed with heart failure. She was prescribed a drug “A” after monitoring renal function tests and potassium levels for a trial period of 4 weeks. What is the mechanism of action of the drug “A”?

- A. Inhibition of aldosterone receptors in kidneys.
 - B. Inhibits conversion of angiotensin 1 into angiotensin 2.
 - C. Inhibiting beta-adrenergic receptors of the heart.
 - D. Inhibits sodium glucose cotransporter-2 in kidneys.
-

18. A patient was admitted with chest pain, and the ECG is given in the image below. His blood report shows a troponin level of 210 ng/ml. Which of the following is less likely to be involved in managing this patient?



- A. Ticagrelor
 - B. Enoxaparin
 - C. Tenecteplase
 - D. Bivalirudin
-

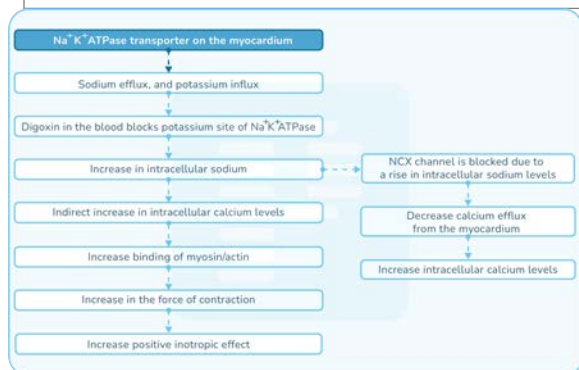
Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	3
Question 4	3
Question 5	1
Question 6	2
Question 7	1
Question 8	4
Question 9	4
Question 10	2
Question 11	1
Question 12	1
Question 13	3
Question 14	4
Question 15	3
Question 16	2
Question 17	2
Question 18	3

Solution for Question 1:

Mechanism of action of digoxin:

Direct action	Indirect action
Digoxin ■ Inhibition of Na ⁺ -K ⁺ -ATPase Pump ■ Intracellular Na ⁺ Accumulation ■ Intracellular Ca ²⁺ ■ Increase ■ Enhanced Myocardial Contractility ■ Positive Inotropic Effect ■ Increased Cardiac Output ■ Treatment of Heart Failure	Vagotonic effect ■ Increased parasympathetic/vagal tone ■ Blocking both S.A and A.V node ■ Prolongation of Refractory Period ■ Slowing of Conduction Through AV Node ■ Control of Rate in Arrhythmias



Solution for Question 2:

- Sirolimus, also known as rapamycin, is a commonly used drug in drug-eluting stents.
- It is an immunosuppressive agent that inhibits the mTOR (mammalian target of rapamycin) pathway, preventing smooth muscle cell proliferation and neointimal hyperplasia, thus reducing the risk of restenosis.
- Sirolimus and its analogs, such as everolimus and zotarolimus, are widely used in modern drug-eluting stents due to their effectiveness and favorable safety profiles.

Solution for Question 3:

- The clinical presentation and ECG findings suggest that the patient is experiencing an acute coronary syndrome (ACS), likely a non-ST-segment elevation myocardial infarction

(NSTEMI), given the presence of elevated cardiac markers and the ECG changes without ST-elevation.

- Streptokinase is a thrombolytic agent used to dissolve blood clots in the treatment of ST-segment elevation myocardial infarction (STEMI).
- It is contraindicated in NSTEMI because the risks of bleeding outweigh the benefits since NSTEMI is typically managed with antiplatelets, anticoagulants, and sometimes percutaneous coronary intervention (PCI), rather than thrombolysis.

Solution for Question 4:

- Omapatrilat is a dual inhibitor of Neprilysin and Angiotensin-Converting Enzyme (ACE).
- This dual inhibition leads to increased levels of bradykinin, a peptide that can cause vasodilation and increased vascular permeability.
- The accumulation of bradykinin is associated with a higher risk of severe angioedema, which is a serious and potentially life-threatening side effect.

Omapatrilat

Mechanism	Omapatrilat ■ Inhibition of Neprilysin and Angiotensin-Converting Enzyme (ACE) ■ Decreased Degradation of Natriuretic Peptides and Inhibition of Angiotensin II Production ■ Increased Levels of Natriuretic Peptides and Decreased Vasoconstriction ■ Vasodilation and Decreased Blood Pressure ■ Decreased Cardiac Afterload ■ Improved Cardiac Output ■ Treatment of Heart Failure
Uses	Primarily for the treatment of hypertension and heart failure.
Adverse Effects	Angioedema Hypotension Cough Dizziness Fatigue

Other newer drugs for treatment of heart failure

Drug Name	Type	Uses	Adverse Effects
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Sacubitril/Valsartan	ARNI (Angiotensin Receptor-Neprilysin Inhibitor)	Heart failure with reduced ejection fraction (HFrEF).	Hypotension Hyperkalemia Renal impairment Angioedema
Ivabradine	Funny channel (I _{Na+}) inhibitor	Heart failure with reduced ejection fraction (HFrEF) in patients with resting heart rate $\geq$ 70 bpm on maximally tolerated beta-blockers.	Bradycardia Hypertension Atrial fibrillation Luminous phenomena (visual brightness)
Vericiguat	Soluble guanylate cyclase (sGC) stimulator	Chronic heart failure with reduced ejection fraction (HFrEF).	Hypotension Anemia Dizziness
Dapagliflozin	SGLT2 inhibitor	Heart failure with reduced ejection fraction (HFrEF).	Urinary tract infections

Solution for Question 5:

Diuretics can lead to electrolyte imbalances, particularly hypokalemia, which increases the risk of digoxin toxicity.

Hypokalemia enhances digoxin's effects on the heart, leading to toxicity.

St. John's Wort(Option B)induces the metabolism of digoxin, which typically lowers its levels in the blood and decreases its efficacy rather than increasing the risk of toxicity.

Cholestyramine(Option C): It reduces the absorption of digoxin in the gut, leading to decreased efficacy rather than an increased risk of toxicity.

Rifampin(Option D): This P-gp inducer increases digoxin efflux from enterocytes and hepatocytes, resulting in lower digoxin levels and decreased efficacy.

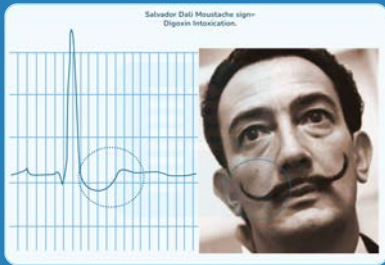
Drug interactions of digoxin	
Drug Class	Explanation
Diuretics (e.g., Furosemide, Hydrochlorothiazide)	Increased risk of digoxin toxicity due to electrolyte imbalance (hypokalemia).
ARBs and ACE Inhibitors	It can increase digoxin levels by reducing its renal clearance.
Beta-blockers (e.g., Metoprolol)	It can increase the risk of bradycardia and heart block.

Macrolide Antibiotics (e.g., Clarithromycin)	It can increase digoxin levels by inhibiting its metabolism.
Quinidine	Increase digoxin levels by inhibiting its renal clearance.
Herbal Supplements (e.g., St. John's Wort)	It can decrease digoxin levels by inducing its metabolism.
Cholestyramine and Antacids	It can reduce digoxin absorption in the gut, leading to decreased efficacy.
P-gp Inducers (e.g., Rifampin)	It can decrease digoxin levels by increasing its efflux from enterocytes and hepatocytes.
P-gp Inhibitors (e.g., Verapamil, Amiodarone)	It can increase digoxin levels by inhibiting its efflux from enterocytes and hepatocytes.

Condition	Reason for Contraindication
Severe ventricular arrhythmias	Digoxin can exacerbate these conditions.
Hypokalemia	Increases the risk of digoxin toxicity.
Hypercalcemia	Increases the risk of digoxin toxicity.
Hypomagnesemia	Increases the risk of digoxin toxicity.
Severe renal impairment	Digoxin is excreted by the kidneys; impaired function can lead to toxicity.
Acute myocardial infarction	Digoxin may increase oxygen demand and worsen myocardial damage.
Hypertrophic obstructive cardiomyopathy	Digoxin can increase outflow obstruction.
WPW (Wolff-Parkinson-White) syndrome with atrial fibrillation	Can increase ventricular rate.
Drugs	Reason for Contraindication
Amiodarone	Increases digoxin levels.
Quinidine	Increases digoxin levels.
Spironolactone	Can increase digoxin levels.
Diuretics (loop and thiazide)	Can cause electrolyte disturbances, increasing digoxin toxicity risk.
Succinylcholine	Can increase the risk of arrhythmias.
Sympathomimetics (e.g., epinephrine, norepinephrine)	Can increase the risk of arrhythmias.
Calcium channel blockers	Can decrease digoxin clearance and increase toxicity risk.

Solution for Question 6:

ECG Changes with Digoxin

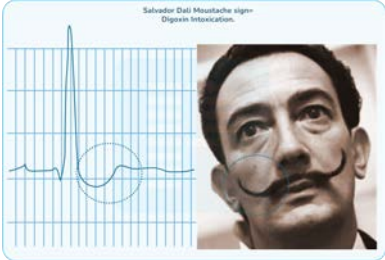


Salvador Dalí Moustache signs Digoxin Intoxication.

- ↑ PR Interval → AV node block
- ↑ RR Interval → SA node block
- Narrow QRS → Tells about Ventricular depolarisation - ↑ intracellular sodium
- Inverted T wave → ↓ intracellular potassium
- ... (content truncated)

Solution for Question 7:

Features Of Digoxin Toxicity	Treatment
Arrhythmias (e.g., atrial fibrillation, ventricular tachycardia)	Lidocaine Propranolol (for ventricular arrhythmias) Digibind (if refractory)
Hyperkalemia	Insulin-glucose therapy Potassium-lowering agents (Calcium gluconate)
Hypokalemia	Potassium supplementation (KCL)
Bradycardia	Atropine
Neurological symptoms (e.g., confusion, delirium)	Supportive care Correction of electrolyte imbalances.

Investigations for Digoxin Toxicity	 Salvador Dalí Moustache sign - Digoxin Intoxication. Serum Digoxin levels Electrolyte levels (potassium, calcium) ECG Changes with Digoxin ↑ PR Interval → AV node block ... (content truncated)
Antidote	Digoxin-specific antibody fragments (Digibind)

Option A: Nifedipine (calcium channel blockers) can slow atrioventricular conduction, which might worsen bradycardia or heart block associated with digoxin toxicity.

Solution for Question 8:

Digoxin, a cardiac glycoside, is commonly used to manage chronic heart failure with atrial fibrillation and rapid ventricular response. It helps control heart rate and improve symptoms in patients with atrial fibrillation.

- Digoxin at therapeutic concentrations mildly inhibits the cardiac Na^+/K^+ ATPase, causing an increase in intracellular Na^+ .
- Increased Na^+ inhibits Ca^{2+} extrusion via the NCX, resulting in higher intracellular Ca^{2+} and enhanced contractility.
- The increased contractility and cardiac output provide symptomatic relief (primarily in systolic dysfunction) in patients with heart failure.
- When the main trigger for neurohumoral activation is removed, sympathetic nerve tone, heart rate, and peripheral vascular resistance drop.
- The decrease in preload and afterload reduces chamber dilation and wall stress.
- Increased renal perfusion lowers renin production and increases diuresis, further decreasing preload.

Uses of digoxin:

- The most effective drug capable of relieving CHF symptoms and restoring cardiac compensation, especially in patients with dilated heart and low ejection fraction.
- Rate control in atrial fibrillation and atrial flutter.

Lisinopril(Option A) is an ACE inhibitor used in heart failure but does not directly address the patient's atrial fibrillation or rapid ventricular response.

Carvedilol(Option B) is a beta-blocker used in heart failure but does not specifically target atrial fibrillation.

Furosemide(Option C) is a loop diuretic that manages fluid overload in heart failure but does not address the patient's arrhythmia.

Solution for Question 9:

Pharmacokinetics	Absorption: Well-absorbed orally. Distribution: Large volume of distribution. Concentrated in the heart to a greater extent than in plasma, approximately 20 times higher. Crosses the blood-brain barrier and placenta. Metabolism: Primarily Hepatic, half-life 36 to 48 hours. Excretion: Renal, with 70-80% of the administered dose excreted unchanged in the urine. Narrow Therapeutic index Nonlinear pharmacokinetics (Clearance and Half-life can vary with changes in dosage)
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Solution for Question 10:

- The above ECG shows ST segment elevation in leads II, III and aVF, which suggests ST-elevation myocardial infarction (STEMI - inferior MI in this case). Reperfusion should be the definitive treatment; hence, reteplase (thrombolytic) should be ideal.
- Pathophysiology of MI:

Atherosclerotic plaque rupture → platelet aggregation and thrombosis → occlusion of vessels → ischaemia followed by infarction

• Management of MI:

Initial management	Oxygen Nitrates Morphine Loading dose (dual antiplatelets + statin)
Definitive management	Reperfusion therapy: Primary percutaneous coronary intervention: If presents within 12 hours. (door to balloon time <90 minutes) Thrombolysis: If presents more than 12 hours or no access to PCI within 2 hours of presentation. Recombinant tissue plasminogen activators: alteplase, reteplase and tenecteplase. (door to needle time <30 minutes)
Long-term management	Lifestyle modifications Dual antiplatelets ACE inhibitors Beta-blockers Statins Oral hypoglycemics (if diabetic)

Solution for Question 11:

Doxorubicin is an anthracycline antibiotic and an antineoplastic agent used to treat various malignancies, such as breast, leukaemia, and Hodgkin's lymphoma.

Cardiac glycosides are glycosidic drugs having cardiac inotropic properties. They increase myocardial contractility and output in a hypodynamic heart without a proportionate increase in O₂ consumption, increasing the efficiency of a failing heart.

Cardiac glycoside	Source
Digoxin	Digitalis lanata (foxglove plant)
Digitoxin	Digitalis purpurea (foxglove plant)
Ouabain	Strophanthus gratus (African plant)
Lanatoside C	Digitalis lanata
Acetyldigoxin	Synthetic derivative

Bufalin	Bufo gargarizans (Chinese toad)
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Solution for Question 12:

Milrinone is a phosphodiesterase-3 inhibitor used in the treatment of heart failure.

PDE-3 inhibitors:

- They are also called inodilators due to their positive inotropic and vasodilatory effects.
- The inotropic effect increases cardiac output, and vasodilation reduces pre- and afterload.
- Drugs: Inamrinone (amrinone) Milrinone More selective for PDE-3 10 times more potent than inamrinone Half-life: 40-80 minutes Enoximone Vesnarimone
- Inamrinone (amrinone)
- Milrinone More selective for PDE-3 10 times more potent than inamrinone Half-life: 40-80 minutes
- More selective for PDE-3
- 10 times more potent than inamrinone
- Half-life: 40-80 minutes
- Enoximone
- Vesnarimone
- Inamrinone (amrinone)
- Milrinone More selective for PDE-3 10 times more potent than inamrinone Half-life: 40-80 minutes
- More selective for PDE-3
- 10 times more potent than inamrinone
- Half-life: 40-80 minutes
- Enoximone
- Vesnarimone
- More selective for PDE-3
- 10 times more potent than inamrinone
- Half-life: 40-80 minutes
- Uses: Short-term circulation support in advanced chronic heart failure.

- Side effects: Thrombocytopenia (due to PDE-3 inhibition in platelets)

Roflumilast (Option B) is a selective PDE-4 inhibitor used to treat severe COPD.

Cilostazol (Option C) is a PDE-3 inhibitor similar to milrinone, but it affects platelet aggregation and vasodilation rather than myocardial contractility. It is used in the treatment of intermittent claudication.

Theophylline (Option D) is a non-selective PDE inhibitor used in the treatment of asthma and COPD (causes bronchodilation).

Solution for Question 13:

Inhibition of vasopressin-2 receptors is not one of the mechanisms of action of levosimendan, a drug used in patients with heart failure.

Levosimendan acts by the following mechanisms:

- Inotropism Inhibiting phosphodiesterase-3 (PDE-3), thereby causing elevated cAMP in cardiac myocytes and increased contractility (option A). Sensitising cardiac troponins to the intracellular Ca^{2+} ions, thus increasing its contractility (option B).
- Inhibiting phosphodiesterase-3 (PDE-3), thereby causing elevated cAMP in cardiac myocytes and increased contractility (option A).
- Sensitising cardiac troponins to the intracellular Ca^{2+} ions, thus increasing its contractility (option B).
- Vasodilation by opening ATP-sensitive K^{+} channels in the blood vessels, thus reducing preload and afterload (option D).
- Inhibiting phosphodiesterase-3 (PDE-3), thereby causing elevated cAMP in cardiac myocytes and increased contractility (option A).
- Sensitising cardiac troponins to the intracellular Ca^{2+} ions, thus increasing its contractility (option B).

Thus, levosimendan is called an “inodilator”

- Tolvaptan is a drug that inhibits the vasopressin V_2 receptors in the collecting ducts, where the hormone ADH acts; thus, tolvaptan improves water retention and hyponatremia.
- Hence, it is used predominantly in SIADH (syndrome of inappropriate antidiuretic hormone) and also in chronic heart failure.

Solution for Question 14:

- C-type natriuretic peptide is synthesised by the kidneys and endothelial cells.
- Natriuretic peptides are substances that increase the excretion of Na^+ along with water in urine. Hence, heart failure can be improved by using drugs that reduce their degradation or by using their recombinant forms.

NT-proBNP (OPTION A) levels are generally used in diagnosing heart failure, but the levels can also be increased in

- Atrial fibrillation
- Pulmonary embolism
- Pulmonary arterial hypertension

Nesiritide (OPTION B), a recombinant human BNP just like other natriuretic peptides, acts by increasing the levels of cyclic GMP, thus facilitating urinary Na^+ excretion.

Carperitide (OPTION C) is a human recombinant ANP used for the same therapeutic purpose.

Solution for Question 15:

- Nebivolol (Option C) is a cardioselective beta-blocker that possibly provides the most therapeutic benefit to this patient.
- The heart consists of beta-1 adrenergic receptors hence selective beta-1 blockers are termed cardioselective. These drugs are generally preferred over non-selective beta-blockers.
- In addition, this patient is asthmatic and hence non-selective beta blockade is contraindicated (bronchoconstriction due to beta-2 blockade).
- Among the given options, both nebivolol and bisoprolol are cardioselective beta blockers but nebivolol has an additional advantage by causing vasodilation. This helps the heart to function even better by reducing the preload and afterload.

Solution for Question 16:

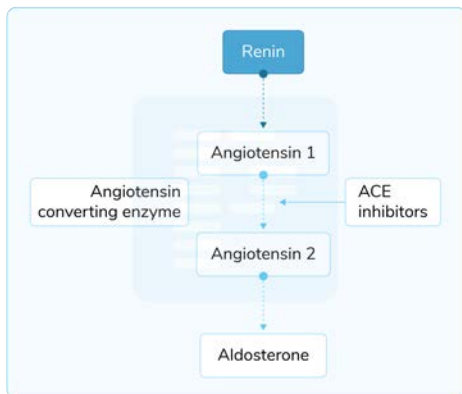
- Sacubitril is contraindicated to use with ACE inhibitors.(Option B)
- Sacubitril is a drug that inhibits neprilysin, an enzyme that degrades peptides(Option A) like ANP, BNP, and C-type natriuretic peptides. The resultant reduction in their degradation increases natriuresis, thus improving heart failure.
- Neprilysin is also associated with the degradation of substance P and bradykinin, which are also increased while using ACE inhibitors. This increases the cumulative risk of angioedema; hence, this combination is contraindicated.
- Hence it is generally combined with an angiotensin receptor blocker (ARB), called ARNIs (Angiotensin receptor - neprilysin inhibitors).

Nesiritide (OPTION C), a recombinant BNP, can also be used in heart failure, but ARNIs are preferred over nesiritide because the latter has the disadvantage of directly raising the BNP levels (being the recombinant form) in blood and tissue.

The active metabolite of sacubitril (OPTION D), LBQ657, has a longer half-life of about 11 hours.

Solution for Question 17:

Angiotensin-converting enzyme inhibitors (ACE inhibitors), which act by inhibiting the conversion of angiotensin 1 to angiotensin 2, are a group of drugs that require monitoring of kidney function tests and potassium levels over a trial period.



- Functions of Angiotensin II : Vasoconstriction Stimulation of aldosterone release from the adrenal glands. Direct hypertrophic and proliferative effects on cardiomyocytes and fibroblasts. Stimulation of NE release from sympathetic nerve endings and the adrenal medulla.
- Vasoconstriction
- Stimulation of aldosterone release from the adrenal glands.
- Direct hypertrophic and proliferative effects on cardiomyocytes and fibroblasts.
- Stimulation of NE release from sympathetic nerve endings and the adrenal medulla.
- ACE inhibitors benefit the cardiovascular system by blocking the synthesis of angiotensin II and preventing these actions.
- Vasoconstriction
- Stimulation of aldosterone release from the adrenal glands.
- Direct hypertrophic and proliferative effects on cardiomyocytes and fibroblasts.
- Stimulation of NE release from sympathetic nerve endings and the adrenal medulla.

Inhibition of aldosterone receptors (Option A) in kidneys- aldosterone antagonists (spironolactone, triamterine).

Inhibiting beta-adrenergic (Option C) of the heart- beta-blockers (propranolol, metoprolol, nebivolol).

Inhibits sodium glucose cotransporter-2(Option D) in kidneys- SGLT-2 inhibitors (dapagliflozin, sotagliflozin).

Solution for Question 18:

The above clinical vignette, supported by the presence of widespread T wave inversion and raised troponin levels (normal <4.5 ng/ml), is suggestive of non-ST elevation myocardial infarction (NSTEMI). Thrombolytics like tenecteplase will not be required in such cases.

Management of NSTEMI:

Initial management	Oxygen Nitrates Morphine Loading dose (dual antiplatelets + statin)
Definitive management	Anticoagulation - one of the below Unfractionated heparin Enoxaparin (low molecular weight heparin) Bivalirudin (direct thrombin inhibitor) Fondaparinux (factor Xa inhibitor)
Long-term management	Lifestyle modifications Dual antiplatelets ACE inhibitors Beta-blockers Statins Oral hypoglycemics (if diabetic)

Hypolipidemic Drugs and Anti-Arrhythmic Drugs

1. A 62-year-old male with a history of diabetes and hyperlipidemia presents for a routine follow-up. Despite lifestyle modifications and treatment with metformin, his lipid profile remains suboptimal. The physician decides to initiate combination therapy with a statin and a fibrate to improve his lipid profile further. Which of the following best explains the mechanism of action of the fibrate in this patient?

- A. Inhibits HMG-CoA reductase
 - B. Activates peroxisome proliferator-activated receptor alpha (PPAR- α)
 - C. Binds to bile acids in the intestine
 - D. Inhibits lipolysis in adipose tissue
-

2. A 55-year-old male presents with complaints of fatigue and exertional chest discomfort. His medical history is significant for hypertension and a family history of coronary artery disease. Physical examination reveals elevated blood pressure and mild obesity. Laboratory investigations show elevated LDL cholesterol levels. Based on his risk factors and lab results, he is prescribed statin therapy. Which of the following effects may not be seen after the commencement of statin therapy?

- A. Antiplatelet action
 - B. Anticoagulant action
 - C. Increase in uric acid
 - D. Reduced CRP
-

3. Which of the following statin is metabolised by CYP3A4 enzyme?

- A. Fluvastatin

- B. Rosuvastatin
 - C. Pravastatin
 - D. Atorvastatin
-

4. A 58-year-old male on treatment for hyperlipidemia, hypertension, and CAD presents with severe muscle pain and weakness. He reports starting a new medication for a respiratory infection a few days ago. On examination, he appears distressed, with generalized muscle tenderness and weakness. Laboratory studies reveal markedly elevated serum creatine kinase (CK) levels, along with myoglobinuria. Which of the following medications is most likely responsible for this patient's presentation?

- A. Aspirin
 - B. Ramipril
 - C. Metformin
 - D. Clarithromycin
-

5. Which of the following statements is false?

- A. Aspirin is used to counteract flushing from niacin
 - B. Bile acid sequestrants can be used in pregnancy
 - C. Colesevelam is approved for use in T2DM
 - D. Niacin reduces uric acid levels
-

6. A 56-year-old male with a history of hyperlipidemia is being managed on a lipid-lowering therapy regimen. His current medications include atorvastatin for cholesterol management. During a follow-up visit, the patient's lipid profile

indicates the need for additional triglyceride lowering, and the clinician considers adding a fibrate to his treatment plan. Which of the following fibrate drugs must be avoided in this patient?

- A. Fenofibrate
- B. Clofibrate
- C. Bezafibrate
- D. Gemfibrozil

7. Match the following newer hypolipidemic drugs with their corresponding Mechanisms of Action (MOAs): Drug Mechanisms of Action 1. Ezetimibe A. Inhibition of cholesterol absorption in the intestines 2. Alirocumab B. Inhibition of PCSK9 enzyme 3. Lomitapide C. Inhibition of microsomal triglyceride transfer protein (MTP) 4. Bempedoic acid D. Inhibition of ATP citrate lyase

Drug	Mechanisms of Action
1. Ezetimibe	A. Inhibition of cholesterol absorption in the intestines
2. Alirocumab	B. Inhibition of PCSK9 enzyme
3. Lomitapide	C. Inhibition of microsomal triglyceride transfer protein (MTP)
4. Bempedoic acid	D. Inhibition of ATP citrate lyase

- A. 1-D, 2-C, 3-B, 4-A
- B. 1-B, 2-A, 3-D, 4-C
- C. 1-C, 2-D, 3-B, 4-A
- D. 1-D, 2-B, 3-A, 4-C

8. Match the following Antiarrhythmic Drugs with their Mechanisms of Action: Antiarrhythmic Drugs Mechanisms of Action 1. Propranolol A. Beta-adrenergic receptor blocker (Class II) 2. Procainamide B. Sodium channel blocker (Class I) 3.

Sotalol C. Potassium channel blocker (Class III) 4.Verapamil D. Calcium channel blocker (Class IV)

Antiarrhythmic Drugs	Mechanisms of Action
1. Propranolol	A. Beta-adrenergic receptor blocker (Class II)
2. Procainamide	B. Sodium channel blocker (Class I)
3. Sotalol	C. Potassium channel blocker (Class III)
4.Verapamil	D. Calcium channel blocker (Class IV)

- A. 1-A, 2-B, 3-C, 4-D
- B. 1-B, 2-A, 3-C, 4-D
- C. 1-C, 2-B, 3-A, 4-D
- D. 1-D, 2-B, 3-A, 4-C

9. Which of the following Class 1 antiarrhythmic drugs have an indirect effect on Inward rectifier potassium (K+) channels?

- A. Quinidine
- B. Lidocaine
- C. Flecainide
- D. Propafenone

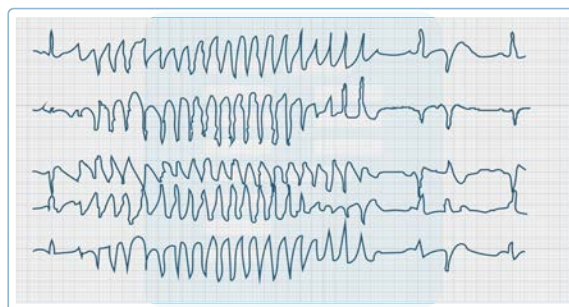
10. A 35-year-old man presents to the emergency department with palpitations and dizziness. His electrocardiogram (ECG) reveals a narrow, complex tachycardia with a rate of 200 beats per minute. The patient's history is significant for episodes of palpitations and syncope. On physical examination, his blood pressure is 110/70 mmHg, and he is diaphoretic but alert. A further review of his ECG shows a short PR interval and a delta wave. Which of the following is the most appropriate drug of choice to treat this patient?

- A. Lidocaine
 - B. Esmolol
 - C. Procainamide
 - D. Ajmaline
-

11. A 65-year-old patient with a history of hypertension and coronary artery disease presents to the emergency department with palpitations and shortness of breath. An ECG confirms atrial fibrillation with a rapid ventricular rate. A drug is administered to control the ventricular rate in this patient. Which of the following is not an AV nodal blocking drug?

- A. Adenosine
 - B. Atropine
 - C. Metoprolol
 - D. Digoxin
-

12. A 42-year-old man is brought to the ED by his wife after fainting, with recent episodes of dizziness and palpitations. His history includes hypertension and atrial fibrillation, for which he was recently started on a new medication. His vital signs are blood pressure: 100/60 mmHg , heart rate: 140 bpm. An ECG is obtained and shows the following pattern: Based on the clinical presentation and the ECG, which of the following interventions is most appropriate for the initial management of this patient?



- A. Administer intravenous amiodarone
 - B. Administer intravenous adenosine
 - C. Administer intravenous magnesium sulfate
 - D. Perform synchronised cardioversion
-

13. A 60-year-old woman presents to her primary care physician with complaints of fatigue, weight gain, and cold intolerance over the past few months. She has a history of atrial fibrillation and has been on amiodarone for the past year. On examination, she has a diffusely enlarged thyroid gland with no palpable nodules. Which of the following mechanisms best explains how amiodarone can cause thyroid dysfunction in this patient?

- A. Stimulation of thyroid hormone synthesis
 - B. Inhibition of thyroid hormone synthesis
 - C. Increased thyroid hormone release
 - D. Suppression of thyroid-stimulating hormone (TSH) secretion
-

14. A 35-year-old woman presents to the emergency department with sudden palpitations and lightheadedness. She has a previous history of MI and her medications include a beta blocker. On examination, her heart rate is 180 bpm, regular, and she appears anxious. An ECG shows a narrow complex tachycardia consistent with paroxysmal supraventricular tachycardia (PSVT). What is the most appropriate next step in the management of this patient's PSVT?

- A. Intravenous verapamil
- B. Intravenous adenosine
- C. Intravenous amiodarone
- D. Intravenous vernakalant

15. Match the following drug toxicities with their appropriate antidotes. Drug toxicity Antidote A. Lignocaine B. Amiodarone C. Verapamil D. Beta blockers 1. 20% Intralipid 2. IV Calcium gluconate 3. Glucagon hormone

Drug toxicity	Antidote
A. Lignocaine B. Amiodarone C. Verapamil D. Beta blockers	1. 20% Intralipid 2. IV Calcium gluconate 3. Glucagon hormone

- A. A-1,B-2,C-3,D-2
- B. A-2,B-3,C-2,D-1
- C. A-1,B-1,C-2,D-3
- D. A-2,B-3.C-3,D-1

16. Which of the following is the drug of choice for treating ventricular tachyarrhythmias induced by digoxin toxicity?

- A. Amiodarone
- B. Lidocaine
- C. Procainamide
- D. Verapamil

Correct Answers

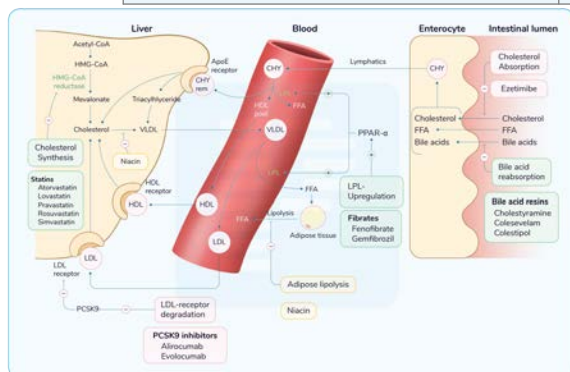
Question	Correct Answer
Question 1	2
Question 2	3
Question 3	4
Question 4	4
Question 5	4
Question 6	4
Question 7	1
Question 8	1
Question 9	2
Question 10	3
Question 11	2
Question 12	3
Question 13	2
Question 14	2
Question 15	3
Question 16	2

Solution for Question 1:

Drugs

Mechanism of action

Statins / HMG CoA reductase inhibitors (Option A) Atrovastatin Rosuvastatin Simvastatin Pitavastatin Pravastatin	Inhibits HMG CoA reductase → ■cholesterol production in liver (■LDL cholesterol)
Fibrates / Activates PPAR-α (Option B) Gemfibrozil Bezafibrate Fenofibrate	Activates PPAR-α → ■expression of genes involved responsible for triglycerides breakdown and VLDL clearance. Fibrates can ■triglycerides and ■HDL cholesterol levels.
Resins / Bile acid sequestrants (Option C) Cholestyramine Colesevelam Colestipol	Sequesters bile acids (■recycling of bile acids from the intestine to the liver) → To compensate this, the liver increases the conversion of cholesterol into bile acid → ■in hepatic cholesterol levels.
Lipolysis inhibitor/Triglyceride synthesis inhibitor (Option D) Niacin (Nicotinic acid)	Inhibits lipolysis in adipose tissue → ■hepatic synthesis of VLDL and LDL → ■LDL cholesterol and triglycerides in blood. Niacin also inhibits the breakdown of HDL → ■HDL levels.



Solution for Question 2:

Correct Answer: C) Increase in uric acid

Explanation:

Statins have been shown to be able to reduce levels of uric acid. They can be used in individuals with dyslipidemia who are at risk of CVD due to increased uric acid levels.

Pleiotropic effects of statins refer to their additional beneficial effects beyond simply lowering cholesterol levels. These pleiotropic effects contribute to the overall cardiovascular protective benefits of statin therapy.

- Direct action on atherosclerotic plaques- stabilises plaques preventing rupture and cardiovascular events
- Reduction in CRP (Option D) levels due to anti-inflammatory property
- Antiplatelet (Option A) and anti-coagulant (Option B) action
- Reduced levels of uric acid
- Improvement of endothelial function
- Increased nitric oxide bioavailability
- Antioxidant properties

Solution for Question 3:

Correct Answer: D) Atorvastatin

Explanation:

Fluvastatin (Option A), rosuvastatin (Option B), and pravastatin (Option C) are not considerably metabolized by CYP3A4 enzyme. CYP3A4 metabolises Atorvastatin (Option D).

CYP3A4 metabolises other drugs such as macrolides, azole antifungals, cyclosporine, and other drugs. This is why concurrent use of statin with these drugs can reduce statin catabolism and increase the risk of myopathy.

- Most potent statin: Rosuvastatin
- Most efficacious statin: Rosuvastatin > Atorvastatin
- Longest-acting statin: Rosuvastatin > Atorvastatin
- Safe in renal failure: Atorvastatin

Solution for Question 4:

Correct Answer: D) Clarithromycin

Explanation:

Clarithromycin is a macrolide antibiotic used to treat various bacterial infections, including respiratory tract infections. Rhabdomyolysis is a known rare side effect of clarithromycin, especially when used concomitantly with statin medications. This happens because clarithromycin inhibits the enzyme CYP3A4, which is responsible for metabolising statins. The interaction between these drugs increases statin levels in the blood, leading to muscle toxicity. This patient's symptoms, including severe muscle pain and weakness, along with markedly elevated CK levels and myoglobinuria, are highly suggestive of rhabdomyolysis secondary to clarithromycin and statin use.

Side effects of statins:

- Myopathy (mild to severe): It can present as muscle aches, weakness, or rhabdomyolysis. Reduced catabolism from drugs such as fibrates, cyclosporine, digoxin, and macrolides, and conditions such as liver disease, kidney disease, and hypothyroidism increase the risk of myopathy.
- Hepatotoxicity: Rare side effect and difficult to predict.

Contraindications to statin use:

- Pregnancy: Must be stopped before conception
- Children (Pravastatin can be used in children <8 years old)
- Used cautiously in renal impairment

Aspirin (Option A) is a COX inhibitor used for its cardioprotective and anti-platelet effects. Its most common side effects are hypersensitivity, increased bleeding risk, and Reye syndrome in children.

Ramipril (Option B) is an angiotensin-converting enzyme (ACE) inhibitor used to treat high blood pressure and heart failure. It is associated with adverse effects such as dry cough, hyperkalemia, angioedema, and postural hypotension.

Metformin (Option C) is an oral medication commonly used to treat type 2 diabetes. While metformin can rarely cause lactic acidosis as a severe side effect, it's not associated with rhabdomyolysis or severe muscle pain and weakness, which are the predominant symptoms in this case.

Solution for Question 5:

Correct Answer: D) Niacin reduces uric acid levels

Explanation:

Niacin increases uric levels and is contraindicated in gout for this very reason. Niacin increases uric acid levels primarily due to its inhibitory effect on the enzyme uricase. Uricase is responsible for breaking down uric acid into a more soluble form that can be easily excreted by the kidneys. When niacin inhibits uricase, uric acid accumulates in the blood, leading to higher levels.

Resins (Bile acid sequestrants):

Drugs include cholestyramine, cholestipol, and colesevelam. They are used in adjunct with statin therapy.

- It is safe to use in pregnancy. (Option B)
- Colesevelam is approved for use in T2DM. (Option C)
- They may be used off-label for hyperthyroidism treatment.
- They are used to reduce pruritis from liver disease

Niacin:

Niacin is Vitamin B3 or nicotinic acid. It inhibits lipolysis in adipose tissue and ■triglyceride production by the liver. It also ■HDL levels. Its use is limited by its side effects:

- Flushing- taking aspirin before niacin helps to reduce flushing (Option A)
- Dry skin
- Dyspepsia
- Hepatotoxicity- presents with fatigue and weakness
- Hyperglycemia- it can indicate insulin resistance and should be used cautiously in those with diabetes
- Increases uric acid levels- contraindicated in gout (Option D)

Solution for Question 6:

Correct Answer: D) Gemfibrozil

Explanation:

Gemfibrozil is a fibrate that significantly increases the risk of myopathy and rhabdomyolysis when combined with statins like atorvastatin. This is due to its strong inhibitory effect on the CYP2C8 and CYP3A4 enzymes, which are involved in the metabolism of many statins. As a result, it increases the plasma concentration of statins, leading to an increased risk of muscle toxicity.

Fibrates Mechanism: Activate peroxisome proliferator-activated receptor alpha (PPAR- α).

Effects:

Gemfibrozil (Option D)	Gemfibrozil effectively lowers plasma TG levels by enhancing breakdown and suppressing hepatic synthesis. Gemfibrozil appears to reduce VLDL secretion by the liver. Has CYP action, hence if combined with a statin increases the risk of myopathy
Bezafibrate (Option C)	This second-generation fibric acid derivative is an alternative to gemfibrozil in mixed hyperlipidemias Has the least risk of myopathy.
Fenofibrate (Option A)	This second-generation prodrug fibric acid derivative raises HDL-C and lowers LDL-C more than other fibrates. most suitable fibrate for combining with statins because it minimally affects statin metabolism as it has no CYP action (less risk of myopathy).
Clofibrate (Option B)	Clofibrate is an older fibrate and is less commonly used today. Associated with an increased risk of gallstone formation. All fibrates in general increase the lithogenicity of bile.
Additional information	Clofibrate, bezafibrate, and fenofibrate may increase the effects of warfarin. Fibrates are C/I in pregnancy and children. Hepatic and renal dysfunction are relative contraindications.

Solution for Question 7:

Correct Answer: A) 1-D, 2-C, 3-B, 4-A

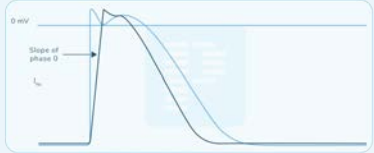
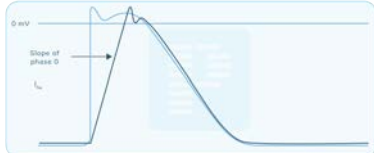
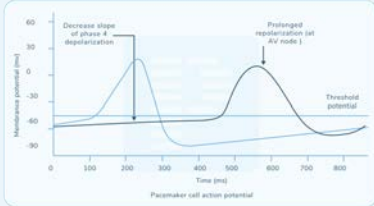
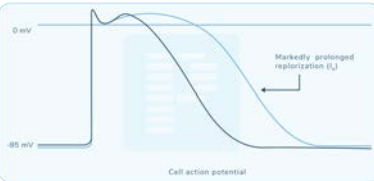
Explanation:

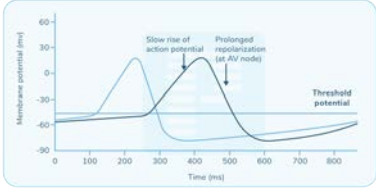
Ezetimibe	Disrupts cholesterol and phytosterol absorption in small intestines via PCIL1 transporter protein interference. Ezetimibe, with statins, enhances LDL-C reduction by inhibiting statin-induced cholesterol synthesis rise. Primarily used with statins for primary hypercholesterolemia. Avoid combining with bile acid sequestrants due to inhibited ezetimibe absorption.
PCSK9 inhibitors	PCSK9 inhibitors: Monoclonal antibodies block PCSK9 action, promoting LDL receptor recycling and degradation, enhancing LDL clearance. Alirocumab and evolocumab: Lower LDL-C in familial hypercholesterolemia or cardiovascular disease, given subcutaneously every 2 weeks or monthly, complement statins to reduce LDL-C. Inclisiran, a new drug which is a silencing RNA.
Lomitapide	An inhibitor of microsomal triglyceride transfer protein (MTP). Action on MTP hinders the production of VLDLs in liver cells and chylomicrons in the intestinal epithelial cells.
Omega 3 fatty acids	Present in fish Decreases FFA delivery to liver and decreases activity of TG synthesizing enzymes. Decrease VLDL production in liver Drugs: Eicosapentaenoic acid (Icosapent), Docosahexaenoic acid
Bempedoic acid	A prodrug, transforms into its active state, bempedoic acid-CoA which competitively blocks ATP Citrate Lyase, diminishing the production of hepatic cholesterol synthesis by limiting the flow of acetyl-CoA to HMG-CoA reductase.

Solution for Question 8:

Correct Answer: A) 1-A, 2-B, 3-C, 4-D

Explanation:

Sodium channel blockers (Class 1)	
Class 1A Quinidine Procainamide Disopyramide	 Blocks sodium channels (Na^+) during phase 0 of the cardiac action potential.
Class 1B Lidocaine Mexiletine	 Blocks sodium channels (Na^+) during phase 0 of the cardiac action potential, but with fast dissociation.
Class 1C Flecainide Propafenone	 Potent sodium channel blockers (Na^+) with slow dissociation from the channel.
Beta blockers (Class 2) Propranolol Metoprolol Esmolol Carvedilol	 Decrease SA and AV nodal activity by $\downarrow$ cAMP thereby $\downarrow$ Ca currents.
Potassium channel blockers (Class 3) Amiodarone Ibutilide Dofetilide Sotalol	 Delays repolarization, lengthens the cardiac action potential, and increases the refractory period.

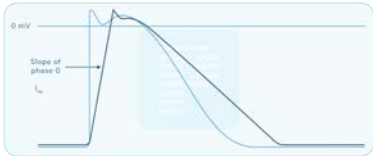
Calcium channel blockers (Class 4) Diltiazem Verapamil	 Reducing calcium influx during phase 2 of the cardiac action potential and decreases myocardial contractility, heart rate, and AV nodal conduction velocity.
Miscellaneous drugs (Class 5) Atropine Adenosine Digoxin Magnesium sulfate	Atropine: DOC in bradyarrhythmias Adenosine: used in acute attack of paroxysmal supraventricular tachycardia and to treat supra ventricular tachycardia in heart failure patients Digoxin: Treat atrial flutter/fibrillation Magnesium sulfate: DOC in Torsades de pointes

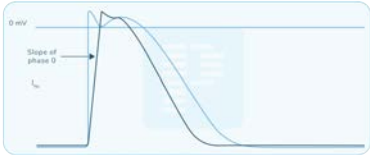
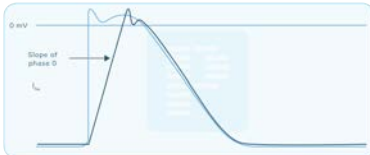
Solution for Question 9:

Correct Answer: B) Lidocaine

Explanation:

Class 1B antiarrhythmic drugs like lidocaine block sodium channels weakly, which means they primarily affect phase 0 of the cardiac action potential. In some studies, lidocaine increased current through inward rectifier potassium (K⁺) channels, which plays a role in maintaining the resting membrane potential of cardiac cells by allowing K⁺ ions to flow into the cell, thereby stabilising the negative charge inside the cell.

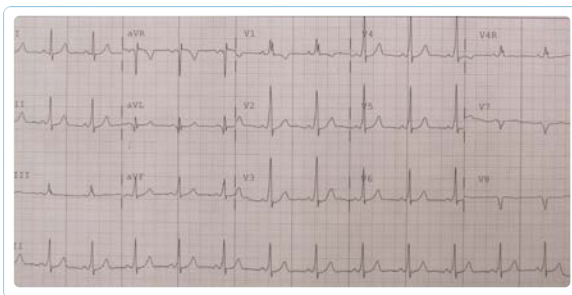
Sodium channel blockers	Description
Class 1A: Quinidine (Option A) Procainamide Disopyramide	 Blocks Na⁺ channels during phase 0 of the cardiac action potential (AP) ↑AP duration Closes inward rectifier K⁺ channels
Uses: Ventricular and atrial tachycardia	

Class 1B: Lidocaine (Option B) Mexiletine	 Blocks Na⁺ channels during phase 0 of the cardiac AP, but with fast dissociation ↓AP duration Opens inward rectifier K⁺ channels
Uses: Only ventricular tachycardia	
Class 1C: Flecainide (Option C) Propafenone (Option D)	 Potent Na⁺ channel blockers with slow dissociation from the channel No effect on AP duration and inward rectifier K⁺ channels
Uses: Resistant cases of ventricular and atrial tachycardia	

Solution for Question 10:

This patient has WPW syndrome, indicated by the ECG findings of a short PR interval and a delta wave, along with his symptoms of palpitations and dizziness.

- Procainamide is the most appropriate drug for treating WPW syndrome with atrial fibrillation or supraventricular tachycardia (SVT). It effectively slows conduction through both the accessory pathway and the atrioventricular (AV) node, aiding in the restoration of normal sinus rhythm.
- For WPW syndrome, procainamide is used in combination with propafenone to manage narrow QRS complex tachycardia and in combination with propafenone, verapamil, or propranolol to manage wide QRS complex tachycardia.



Lidocaine (Option A): Lidocaine is primarily used for ventricular arrhythmias and is not effective for atrial arrhythmias associated with WPW syndrome. It does not address the accessory pathway conduction issue.

Esmolol (Option B): While esmolol is a beta-blocker that can slow heart rate, it is not recommended in WPW syndrome as it may enhance conduction through the accessory pathway, potentially worsening the arrhythmia.

Ajmaline (Option D): Ajmaline, a new Class 1a drug, is not used for treating arrhythmias but is employed in diagnosing Brugada syndrome. This genetic condition features defective sodium channels that remain open for prolonged periods. Ajmaline is unable to block these sodium channels in patients with Brugada syndrome, making it a useful diagnostic tool rather than a treatment option.

Solution for Question 11:

Correct Answer: B) Atropine

Explanation:

Atropine is not an AV node blocker. Atropine blocks the vagal (parasympathetic) influences on the heart, and increases the heart rate, making it useful in treating bradyarrhythmias.

Drugs used as AV nodal blocks:

Adenosine
(Option A)

Administered as a rapid IV bolus.
DOC for acute termination of reentrant supraventricular arrhythmias.
■ AV nodal refractoriness, interrupting reentrant circuits relying on the AV node.

Beta Blockers (Option C)	Inhibit adrenergic effects contributing to arrhythmias. ■HR and ■intracellular Ca ²⁺ overload, inhibiting afterdepolarization-mediated automaticity. ■AV nodal refractoriness, useful in terminating reentrant arrhythmias involving the AV node. Control ventricular response in atrial fibrillation or atrial flutter.
Calcium Channel Blockers - Verapamil, Diltiazem	Block Ca ²⁺ channels in cardiac cells. ■AV nodal conduction velocity and ■AV nodal refractoriness. Treat reentrant arrhythmias involving the AV node. Approved for temporary control of rapid ventricular rate in atrial flutter or AF and rapid conversion of PSVT to sinus rhythm.
Digoxin (Option D)	■AV nodal refractoriness. Useful in terminating reentrant arrhythmias involving the AV node. Controls ventricular response in patients with AF.

Metoprolol (Option C) is a beta-blocker (Class 2 antiarrhythmic) that slows AV nodal conduction effectively. It is a preferred choice for rapid ventricular rate control in atrial fibrillation and atrial flutter, especially in patients with a history of hypertension and coronary artery disease. It provides both rate control and reduces myocardial oxygen demand.

Solution for Question 12:

Correct Answer: C) Administer intravenous magnesium sulfate

Explanation:

The patient started a new antiarrhythmic medication for atrial fibrillation, which triggered torsades de pointes (TdP). Antiarrhythmic drugs can prolong the QT interval by blocking cardiac potassium channels, leading to TdP. TdP is characterized by irregular polymorphic ventricular tachycardia with a "twisting" pattern on the EKG, caused by early afterdepolarisations (EADs) that disrupt the normal repolarisation process.

Magnesium sulphate is the drug of choice for TdP, regardless of serum magnesium levels. It blocks L-type calcium channels in the heart, preventing EADs and interrupting the arrhythmia, but it does not shorten the QT interval on an ECG.

Drugs causing QT prolongation

Cardioactive drugs	Non-cardioactive drugs
Quinidine Sotalol Procainamide Disopyramide Dofetilide Ibutilide Amiodarone	Fluoroquinolones and Macrolides Astemizole and Terfenadine (Antihistamines) Bedaquiline (anti-tuberculosis drug) Cisapride (drug for constipation) Tricyclic antidepressants and SSRI

Amiodarone (Option A) is an antiarrhythmic medication often used for various tachyarrhythmias. However, it is not the first-line treatment for torsades de pointes, which is specifically treated with magnesium sulfate.

Adenosine (Option B) is used to terminate certain types of supraventricular tachycardias by transiently blocking AV node conduction. It is ineffective for treating torsades de pointes and may be unsafe in this condition.

Synchronized (Option D) cardioversion is used for hemodynamically unstable tachyarrhythmias. While it can be used if the patient is unstable, the preferred initial treatment for a stable patient with torsades de pointes is magnesium sulfate.

Solution for Question 13:

Amiodarone can cause thyroid dysfunction by inhibiting thyroid hormone synthesis, known as the Wolff-Chaikoff effect. It contains iodine and can lead to excess iodine in the thyroid gland, suppressing thyroid hormone synthesis.

Mechanism of action of Amiodarone:

Adverse effects:

- Hypotension and Bradycardia (with i.v administration or after drug accumulation)
- Photosensitivity: Skin pigmentation - Purple color (purple man syndrome, smurf syndrome)
- Corneal microdeposits: Reversible upon drug discontinuation.
- Pulmonary alveolitis and fibrosis (On long-term use)
- Peripheral neuropathy
- Thyroid Dysfunction: Interfere with thyroid function as it acts as a source of iodine and can potentially lead to goitre, hypothyroidism (Wolff-Chaikoff effect), or, in rare cases, hyperthyroidism (Jod-Basedow effect).

Drug Interactions:

- Inhibits CYP3A4, CYP2C9, and P-glycoprotein, leading to potential interactions with drugs metabolised by this system.
- Increases plasma levels of digoxin and warfarin.
- Additive AV block with beta-blockers or calcium channel blockers.

Solution for Question 14:

Adenosine is the drug of choice for terminating PSVT. It works by transiently blocking AV node conduction, by activation of acetylcholine-sensitive K⁺ current in the atrium, sinus, and AV nodes, shortening of action potential duration and hyperpolarisation.

Adenosine and verapamil are both used to terminate PSVT. However, there are some key differences between the two drugs:

Feature	Adenosine (Option B)	Verapamil (Option A)
Efficacy	Equivalent to or better than verapamil for terminating PSVT	Effective for terminating PSVT
Duration of Action	Short-acting (< 1 minute)	Longer-acting than adenosine
Adverse Effects	Transient and generally mild (e.g., dyspnea, chest pain, flushing, hypotension, cardiac arrest)	C/I in patients with hypotension, CHF, or those receiving beta-blockers
Safety	Safe in patients with wide QRS tachycardia	C/I in patients with wide QRS tachycardia
Other Considerations	Requires rapid i.v injection, not suitable for prophylaxis of recurrent PSVT	Can be used for prophylaxis of recurrent PSVT
Use in Asthma	Can precipitate bronchospasm, making verapamil the preferred choice for these patients	DOC for PSVT in asthmatic patients

Amiodarone (Option C), a Class 3 antiarrhythmic drug with multiple mechanisms including beta-blocking properties, is often used for rate control in atrial fibrillation.

The main use of vernakalant (Option D), a class 3 antiarrhythmic drug, is for the rapid conversion of recent-onset atrial fibrillation (AF) to sinus rhythm in adults.

Solution for Question 15:

Drug toxicity	Antidote	Mechanism of antidote
Verapamil	IV Calcium gluconate	Raises blood calcium levels, facilitating cellular entry to restore normal heart function and blood pressure
Amiodarone	20% Intralipid	Creates an enlarged lipid phase, causing toxic drugs to move from tissue to plasma, reducing drug concentration in cardiac tissue → Enhances cardiac conduction, contractility, and coronary perfusion
Lignocaine		
Beta-blockers	Glucagon hormone	Bypasses β -receptors by acting on cardiac glucagon receptors to enhance cardiac contractility and AV conduction

Solution for Question 16:

MOA of lidocaine in treating digoxin induced VT :

- Sodium Channel Blockade: Digoxin toxicity → Excessive intracellular calcium → ■ automaticity and triggered activity in ventricular myocardium. Lidocaine → Blocks sodium channels in cardiac cells → Stabilizes the membrane potential and ■ automaticity, particularly during depolarisation.
- Decreased Ectopic Activity: Digoxin toxicity can cause ectopic pacemaker activity. Lidocaine suppresses ectopic pacemaker activity by decreasing cardiac cell excitability, and preventing abnormal electrical impulses.

- **Reduced Reentry:** By inhibiting reentrant pathways in the myocardium, lidocaine helps terminate sustained ventricular arrhythmias.
- **Fast Onset of Action:** Administered intravenously, lidocaine rapidly acts to manage acute ventricular arrhythmias induced by digoxin toxicity.

Amiodarone(Option A) is not the preferred choice for ventricular tachyarrhythmias induced by digoxin toxicity due to potential drug interactions and exacerbation of digoxin toxicity.

Procainamide(Option C) is an antiarrhythmic that can treat ventricular arrhythmias, but it is not preferred in digoxin toxicity due to its potential to exacerbate the condition and cause further complications.

Verapamil(Option D) is a calcium channel blocker for supraventricular arrhythmias and hypertension. It is contraindicated in digoxin toxicity as it can increase digoxin levels and worsen ventricular arrhythmias.

Previous Year Questions

1. Which antihypertensive drug is contraindicated in pregnancy?

- A. Methyldopa
 - B. Enalapril
 - C. Labetalol
 - D. Nifedipine
-

2. A young patient started to take a weight loss medication that acts by inhibiting fat absorption from food. After a few weeks, she developed easy bruising and increased menstrual bleeding. Deficiency of which of the following vitamins is responsible for her condition?

- A. Vitamin E
 - B. Vitamin K
 - C. Vitamin B6
 - D. Vitamin D
-

3. What is the mechanism of action of alteplase, a tissue plasminogen activator (tPA)?

- A. Inhibits Vit K dependent factors
 - B. Inhibits factor X
 - C. Promotes fibrinolysis
 - D. Activation of thrombin
-

4. Which of the following drugs acts by inhibiting ATP binding cassette?

- A. Tacrolimus
 - B. Nifedipine
 - C. Diltiazem
 - D. Verapamil
-

5. Which of the following is typically not given to a patient with elevated uric acid levels and the development of hypertension in the context of gout?

- A. Hydrochlorothiazide
 - B. Atenolol
 - C. Enalapril
 - D. Prazosin
-

6. A patient presented with exertional nocturnal dyspnoea, fatigue and cardiomegaly. Pedal oedema, gallop rhythm. On auscultation there were rales in both the lungs. Which of the following drugs can be used to prevent the progression of the disease?

- A. Verapamil
 - B. Digoxin
 - C. Captopril
 - D. Carvedilol
-

7. Which of the subsequent antihypertensive medications are contraindicated during pregnancy?

- A. Alpha methyldopa
 - B. Hydralazine
 - C. Propranolol
 - D. Labetalol
-

8. Identify the Pcsk9 inhibitor.

- A. Evolocumab
 - B. Lomitapide
 - C. Mipomersen
 - D. Torcetrapib
-

9. Which of the following is the best drug to reduce LDL cholesterol levels?

- A. Niacin
 - B. Fibrates
 - C. Statins
 - D. Resins
-

10. A 32-year-old pregnant female presents to the emergency department with a BP of 220/120mm Hg, along with headache, seizure, and blurred vision. Which of the following is the preferred intravenous medication for the management of hypertension in pregnancy?

- A. IV Labetalol

- B. IV nitroglycerine
 - C. IV hydralazine
 - D. IV sodium nitroprusside
-

11. Which of the following beta-adrenergic blockers used in the treatment of chronic heart failure (CHF) is known to prolong survival?

- A. Bisoprolol
 - B. Labetalol
 - C. Propranolol
 - D. Atenolol
-

12. Which anticoagulant is recommended for a pregnant female at 29 weeks of gestation who develops deep vein thrombosis (DVT) with calf tenderness and is diagnosed via Doppler ultrasound?

- A. Warfarin
 - B. Low molecular weight heparin (LMWH)
 - C. Apixaban
 - D. Rivaroxaban
-

13. A hypertensive patient is suffering from frequent attacks of renal colic. Preferred anti hypertensive drug is:

- A. Ethacrynic acid
- B. Acetazolamide
- C. Hydrochlorothiazide

D. Spironolactone

14. A hypertensive patient stopped antihypertensive medication due to dry eyes, dry mouth and constipation. After stopping the drug, he developed hypertensive emergency (BP > 180/120). Most likely drug is:

- A. Clonidine
 - B. Lisinopril
 - C. Amlodipine
 - D. Telmisartan
-

15. A patient with raised triglyceride started on medication. He soon developed flushing, gout and raised liver enzymes. Likely cause is:

- A. Nicotinamide
 - B. Atorvastatin
 - C. Fenofibrate
 - D. Cholestyramine
-

16. An unconscious patient found to have hot dry skin, dilated pupils, abnormal slurred speech. Most likely overdose is?

- A. Dhatura
 - B. Cannabis
 - C. Alcohol
 - D. Cocaine
-

17. Scientists administered norepinephrine to guinea pigs, resulting in an increase in systolic and diastolic blood pressure and a decrease in heart rate. What mechanism explains this response?

- A. Beta-1 receptor blockade
 - B. Alpha-1 receptor blockade
 - C. Baroreceptor stimulation
 - D. Baroreceptor inhibition
-

18. A patient with rheumatoid arthritis is taking indomethacin and an ACE inhibitor for hypertension. What potential side effect is likely to be seen?

- A. Hyperkalemia
 - B. Hypercalcemia
 - C. Hyponatremia
 - D. Hyperphosphatemia
-

19. A patient presents with hypertriglyceridemia. Which of the following is the best drug to reduce triglycerides (TG)?

- A. Fibrates
 - B. Niacin
 - C. Statins
 - D. Resins
-

20. Which medication is known to induce orthostatic (postural) hypotension and may lead to potentially severe blood pressure drops when patients transition from sitting or lying down to a standing position, potentially causing dizziness, lightheadedness, or even fainting?

- A. Metformin
 - B. Prazosin
 - C. Acarbose
 - D. Nor-adrenaline
-

21. Beta-blockers are ideally avoided in which of the following?

- A. Essential tremors
 - B. Diabetes Mellitus with hypertension
 - C. Angina pectoris
 - D. Migraine prophylaxis
-

22. Rosuvastatin should not be combined with which of the following drugs?

- A. Oxycodone
 - B. Clarithromycin
 - C. Adalimumab
 - D. Rivaroxaban
-

23. A 53 year old man is found to have stage 2 hypertension without any co morbidities. Which drug will you start ?

- A. Furosemide

- B. Chlorthalidone
 - C. Spironolactone
 - D. Triamterene
-

24. Which of the following is not used in heart failure?

- A. Trimetazidine
 - B. Sacubitril
 - C. Metoprolol
 - D. Nesiritide
-

25. Propranolol is drug of choice for:

- A. Ulcerated infantile hemangioma
 - B. Lymphangioma circumscriptum
 - C. Capillary malformation
 - D. Pyogenic granuloma
-

26. Which of the following GPCRs does not act via the K⁺ channel?

- A. Angiotensin 1 receptor
 - B. Muscarinic M2 receptor
 - C. Dopamine D2 receptor
 - D. Serotonin 5 HT 1 receptor
-

27. Which PDE-5 inhibitor has the longest duration of action among the options provided?

- A. Sildenafil
 - B. Tadalafil
 - C. Vardenafil
 - D. Avanafil
-

28. What is the mechanism of action of sodium nitroprusside?

- A. Increased cAMP
 - B. Stimulation of guanylyl cyclase
 - C. Calcium channel blockage
 - D. K⁺ channel opener
-

29. Dopamine at low doses:

- A. Increases in cardiac contractility
 - B. Vasoconstriction
 - C. Tachycardia
 - D. Increases renal blood flow
-

30. Which anti-hypertensive medication is suitable for administration to a patient suffering from renal insufficiency, exhibiting symptoms of significantly reduced urine output, pedal edema, headache, and a blood pressure reading of 160/90?

- A. Aliskiren
- B. Chlorthalidone

C. Amlodipine

D. Prazosin

31. A patient was brought to the emergency with a history of consumption of 8 tablets of digoxin. On examination he was unstable and his heart rate was 56 bpm. ECG showed a 3rd-degree heart block. What is the next step in the management of this patient?

A. Antibody against digoxin

B. Lidocaine

C. DC cardioversion

D. Phenytoin

32. Which of the following is a PCSK9 inhibitor?

A. Evolocumab

B. Ezetimibe

C. Bempedoic acid

D. Clofibrate

33. Which of the following drugs does not reduce mortality in patients with heart failure with reduced ejection fraction (HFrEF)?

A. Digoxin

B. Beta-blockers

C. ACE inhibitors

D. Hydralazine + isosorbide dinitrate

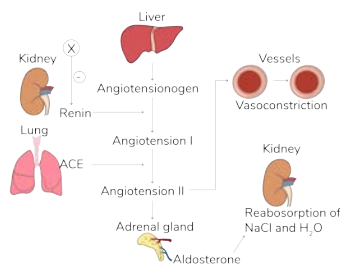
34. Which of the following statement is correct regarding the mechanism of action of labetalol?

- A. Directly acts as arterial vasodilator resulting in indirect effect tachycardia
- B. Blocks both alpha and beta adrenergic receptors
- C. Primarily acts as a vasodilator with little effect on arterioles
- D. Decreases peripheral vascular resistance by directly acting as a relaxant for vascular smooth muscles

35. Which of the following drugs reduces mortality in patients with heart failure with reduced ejection fraction (HFREF)?

- A. Metoprolol
- B. Furosemide
- C. Torsemide
- D. Prazosin

36. A patient with a history of hypertension is given the drug 'X'. Identify 'X'?



- A. Captopril
- B. Spironolactone
- C. Aliskiren
- D. Losartan

37. Which drug is responsible for the side effects mentioned above in a patient with a previous history of myocardial infarction and ventricular arrhythmia who has been on treatment for a few months and has developed fatigue, dyspnea, weight gain, and pulmonary fibrosis?

- A. Amiodarone
- B. Atenolol
- C. Aspirin
- D. Spironolactone

38. Why is a chronic smoker who was undergoing nicotine replacement therapy and taking clonidine tablets for smoking de-addiction experiencing headaches after discontinuing clonidine?

- A. Postural hypotension

- B. Receptor downregulation
 - C. Rebound hypertension
 - D. Receptor hypersensitivity
-

39. The true statement among the following is___:

- A. The dose of telmisartan should be reduced in renal failure but not in hepatic failure.
 - B. The dose of irbesartan need not be reduced in case of mild-moderate hepatic failure and renal failure.
 - C. The dose of candesartan should be reduced in renal failure but not in mild-moderate liver failure
 - D. Losartan acts as a thromboxane A2 antagonist and inhibits platelet aggregation.
-

40. Match the following antiarrhythmic drugs with their mechanism of action:

Mechanism of action Drug 1. Na⁺ channel blocker 2. K⁺ channel blocker 3. Na⁺, K⁺ ATPase inhibitor 4 Beta-blocker a. Quinidine b. Digoxin c. Esmolol d. Ibutilide

Mechanism of action	Drug
1. Na ⁺ channel blocker 2. K ⁺ channel blocker 3. Na ⁺ , K ⁺ ATPase inhibitor 4 Beta-blocker	a. Quinidine b. Digoxin c. Esmolol d. Ibutilide

- A. 1-d, 2-b, 3-a, 4-c
 - B. 1-a, 2-c, 3-d, 4-b
 - C. 1-a, 2-d, 3-b, 4-c
 - D. 1-d, 2-c, 3-a, 4-b
-

41. For preconception counseling, which antihypertensive medication should be stopped by a 35-year-old woman with a history of chronic hypertension visiting the outpatient department?

- A. Labetalol
 - B. Calcium channel blocker
 - C. Angiotensin – converting enzyme inhibitor
 - D. Alpha-methyldopa
-

42. A patient started on antihypertensive presented with the following picture. Which of the following drugs is the most likely cause?



- A. Lisinopril
 - B. Nifedipine
 - C. Atenolol
 - D. Clonidine
-

43. A hypertensive patient was started on some anti-hypertensive drug but he developed a dry cough. Which of the following drug can be responsible for this effect?

- A. Calcium channel blockers
 - B. Beta blockers
 - C. ACE inhibitors
 - D. Diuretics
-

44. What medications should be administered to manage a patient who is 34 weeks pregnant and experiencing seizures, with a blood pressure reading of 200/110 mmHg?

- A. Oral Labetalol
 - B. IV labetalol plus magnesium sulphate
 - C. IV Lorazepam
 - D. IV Enalapril
-

45. Which drug among the options works by blocking the late sodium current in the heart?

- A. Ivabradine
 - B. Ranolazine
 - C. Trimetazidine
 - D. Fasudil
-

46. A patient presents to the OPD with a chronic dry cough with no fever or other associated symptoms. He had recently been started on an antihypertensive for hypertension. Which among the following is most likely responsible to cause these symptoms?

- A. Bradykinin and renin

- B. Angiotensin II and bradykinin
 - C. Only bradykinin
 - D. Histamine
-

47. Which of the following drugs causes activation of K⁺-ATP channels?

- A. Nicorandil
 - B. Molsidomine
 - C. Isosorbide dinitrate
 - D. Nitroglycerin
-

48. A patient who is on treatment for angina, presents to you with erectile dysfunction. You decide against prescribing sildenafil because the patient is being treated with _____:

- A. Calcium channel blockers
 - B. Beta blockers
 - C. Nitrates
 - D. ACE inhibitors
-

49. Which of the following drugs is associated with QT prolongation?

- A. Lignocaine
- B. Quinidine
- C. Magnesium sulfate
- D. All of the above

50. What is the mechanism of action of sacubitril, a newly approved drug for the treatment of chronic heart failure?

- A. Angiotensin II inhibitor
 - B. ACE inhibitor
 - C. Renin inhibitor
 - D. Neutral endopeptidase inhibitor
-

51. Isosorbide dinitrate is indicated for congestive heart failure in combination with which of the following vasodilators?

- A. Minoxidil
 - B. Hydralazine
 - C. Nimodipine
 - D. Nitrendipine
-

52. A person presents with acute chest pain. Nitro-glycerine was given sublingually and pain relieved within a minute. What is the mechanism of this drug?

- A. Release of endothelin
 - B. Release of NO
 - C. Calcium channel blockade
 - D. Beta blockade
-

53. Preferred drug for the treatment of ventricular tachycardia is:

- A. Digoxin
 - B. Propranolol
 - C. Diltiazem
 - D. Lignocaine
-

54. Drug of choice for paroxysmal supraventricular tachycardia is:

- A. Digoxin
 - B. Propranolol
 - C. Adenosine
 - D. Diltiazem
-

55. A 44-year-old male was admitted to the ICU after being diagnosed with Subarachnoid hemorrhage secondary to a ruptured aneurysm. The patient is a known case of stage 2 hypertension and was taking Ramipril for the same. Which of the following calcium channel blockers can be used in this case, which might help relieve the cerebral vasospasm?

- A. Nimodipine
 - B. Felodipine
 - C. Amlodipine
 - D. Nitrendipine
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	4
Question 5	1
Question 6	3
Question 7	3
Question 8	1
Question 9	3
Question 10	1
Question 11	1
Question 12	2
Question 13	3
Question 14	1
Question 15	1
Question 16	1
Question 17	3
Question 18	1
Question 19	1
Question 20	2
Question 21	2
Question 22	2

Question 23	2
Question 24	1
Question 25	1
Question 26	1
Question 27	2
Question 28	2
Question 29	4
Question 30	3
Question 31	1
Question 32	1
Question 33	1
Question 34	2
Question 35	1
Question 36	3
Question 37	1
Question 38	3
Question 39	4
Question 40	3
Question 41	3
Question 42	1
Question 43	3
Question 44	2
Question 45	2
Question 46	3
Question 47	1
Question 48	3

Question 49	2
Question 50	4
Question 51	2
Question 52	2
Question 53	4
Question 54	3
Question 55	1

Solution for Question 1:

Explanation:

Enalapril, an angiotensin-converting enzyme (ACE) inhibitor, is contraindicated in pregnancy due to its teratogenic effects on fetal development.

- Risks in Pregnancy (Especially 2nd & 3rd Trimester): Fetal renal dysgenesis → Oligohydramnios → Pulmonary hypoplasia Hypocalvaria (skull hypoplasia) Fetal growth restriction Neonatal renal failure and anuria
- Fetal renal dysgenesis → Oligohydramnios → Pulmonary hypoplasia
- Hypocalvaria (skull hypoplasia)
- Fetal growth restriction
- Neonatal renal failure and anuria
- Why ACE Inhibitors Are Harmful in Pregnancy: ACE inhibitors block the renin-angiotensin-aldosterone system (RAAS), which is essential for fetal kidney development. This results in reduced amniotic fluid (oligohydramnios), leading to Potter sequence (lung hypoplasia, limb deformities).
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- This results in reduced amniotic fluid (oligohydramnios), leading to Potter sequence (lung hypoplasia, limb deformities).
- Fetal renal dysgenesis → Oligohydramnios → Pulmonary hypoplasia
- Hypocalvaria (skull hypoplasia)

- Fetal growth restriction
- Neonatal renal failure and anuria
- ACE inhibitors block the renin-angiotensin-aldosterone system (RAAS), which is essential for fetal kidney development.
- This results in reduced amniotic fluid (oligohydramnios), leading to Potter sequence (lung hypoplasia, limb deformities).

Adverse effects of ACEI:

Adverse Effect	Description
Hypotension	Decrease in blood pressure, especially upon initiation of therapy or with dosage increases.
Hyperkalemia	Increased serum potassium levels due to reduced aldosterone secretion.
Cough	Persistent dry cough, possibly due to bradykinin buildup in the lungs.
Angioedema	Due to increased bradykinin levels, swelling of deeper layers of skin, particularly around the eyes and lips, can be life-threatening.
Rashes/Urticaria	Skin reactions like rashes or hives occur in a small percentage of users.
Dysgeusia	Reversible taste changes are more common with captopril.
Rare Effects	Granulocytopenia (low white blood cells) and proteinuria (excess protein in urine) are more concerning in patients with kidney disease.
Acute Renal Failure	Possible in patients with bilateral renal artery stenosis due to decreased kidney filtration.
Fetal Effects	ACE inhibitors are contraindicated during pregnancy due to potential harm to the fetus, including fetal renal damage due to inhibition of the fetal renin-angiotensin system.

- Methyldopa (Option A): Preferred for chronic hypertension (centrally acting α_1 -agonist).
- Labetalol (Option C): Used in hypertensive emergencies and preeclampsia (combined α - & β -blocker).
- Nifedipine (Option D): Safe for hypertension and preeclampsia (calcium channel blocker).

Solution for Question 2:

Correct Option B - Vitamin K:

- A deficiency of vitamin K is responsible for easy bruising and increased menstrual bleeding in this scenario.
- The drug Being used here is Lipase Inhibitor like orlistat. Which acts by inhibiting fat absorption. Along with this absorption of fat soluble vitamins like A,E,D and k can be hindered.
- Vitamin K is an essential nutrient that plays a crucial role in blood clotting. It is needed for Clotting factor II,VII,IX and X .Hence deficiency of Vit K can cause increased risk of bleeding due to lack of clotting factor formation.
- Without enough vitamin K, blood can take longer to clot, leading to easy bruising and increased bleeding.

Incorrect Options:

Option A - Vitamin E: Vitamin E is an antioxidant that helps to protect cells from damage caused by free radicals. It is not directly involved in blood clotting and is therefore not likely to be responsible for the symptoms described in the scenario.

Option C - Vitamin B6: Vitamin B6 is involved in a variety of metabolic processes in the body, including the production of red blood cells. But it is not responsible for the above given scenario.

Option D - Vitamin D: Vitamin D is important for bone health and immune function. But it is not responsible for the above given scenario.

Solution for Question 3:

Correct Option C - Promotes fibrinolysis:

- Alteplase is a tissue plasminogen activator and it acts by promoting fibrinolysis. It activates plasminogen bound to fibrin, and catalyzes the conversion of plasminogen to plasmin which results in increased formation of plasmin. This leads to fibrinolysis.

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Incorrect Options:

Option A - Inhibits vit K dependent factors: Alteplase does not inhibit vit K dependent factors.

Option B - Inhibits factor X: Alteplase does not inhibit factor X.

Option D - Activation of thrombin: Alteplase does not act through activation of thrombin.

Solution for Question 4:

Correct Option D - Verapamil:

ATP-binding cassettes are also called p-glycoprotein. They are transporters of drugs and they primarily function as efflux pumps. Eg- Digoxin is thrown out by p-glycoprotein.

This P-glycoprotein can be inhibited by certain drugs (QVAC):

- Quinidine
- Verapamil
- Amiodarone
- Clarithromycin

Incorrect Options:

Option A - Tacrolimus:

- This option is incorrect as Tacrolimus does not inhibit ATP binding cassette.

Option B - Nifedipine:

- This option is incorrect as Nifedipine does not inhibit ATP binding cassette.

Option C - Diltiazem:

- This option is incorrect as Diltiazem does not inhibit ATP binding cassette.

Solution for Question 5:

Correct Option A - Hydrochlorothiazide:

- Hydrochlorothiazide is a thiazide diuretic that can increase uric acid levels by reducing its excretion through the kidneys. In patients with gout, caution should be exercised when using hydrochlorothiazide, and the dosage may need to be adjusted or alternative medications considered to minimize the risk of gout flares.

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Incorrect Options:

Option B - Atenolol: Atenolol is a beta-blocker that is commonly used in the management of hypertension. It does not directly affect uric acid levels and is generally considered safe in patients with gout.

Option C - Enalapril: Enalapril is an angiotensin-converting enzyme (ACE) inhibitor that is commonly used for the treatment of hypertension. It does not have a direct impact on uric acid levels and is generally considered safe in patients with gout.

Option D - Prazosin: Prazosin, on the other hand, is an alpha-1 adrenergic receptor blocker primarily used for the treatment of benign prostatic hyperplasia (BPH) and hypertension. It does not directly influence uric acid levels, but it is not typically used as a first-line antihypertensive medication in patients with gout and hypertension.

Solution for Question 6:

Correct Option C - Captopril:

The likely diagnosis in this patient is heart failure. Captopril is an ACE inhibitor that can be used to manage hypertension and heart failure.

The following drugs can be used to prevent the progression of heart failure:

- Spironolactone
- SGLT- 2 inhibitors
- ACE inhibitors

- ARBs
- ARNI(Combination of ARBs + sacubitril
- Beta Blockers
- The first drugs that a patient is started on as soon as a diagnosis of Heart failure is made, are ACE inhibitors, ARBs, or ARNI (Angiotensin Receptor/Neprilysin Inhibitor).
- There are many contraindications to beta-blockers so the patient is usually started on a very low dose.

Incorrect Options:

Option A - Verapamil:

- This is a Calcium channel blocker, that is absolutely contraindicated in CHF.

Option B - Digoxin:

- This option is incorrect as it has no role in decreasing mortality in heart failure. Digoxin only increases the force of contraction.

Option D - Carvedilol:

- This option is incorrect as there are many contraindications to beta blockers and they are not considered first line.

Solution for Question 7:

Correct Option C: Propranolol

- Propranolol is not typically used in pregnancy due to its potential adverse effects on fetal growth and development.
- Propranolol is generally avoided during pregnancy, particularly during the first trimester, due to concerns about its potential adverse effects on fetal growth and development.

Incorrect Options:

Option A: Alpha methyldopa- Alpha methyldopa is commonly used as an antihypertensive medication during pregnancy. It has a long history of safe use and is considered one of the first-line choices for managing hypertension in pregnant women.

Option B: Hydralazine- Hydralazine is also frequently used in pregnancy to control blood pressure. It is considered safe for use during pregnancy and is often used as an alternative if

other agents are not suitable or effective.

Option D: Labetalol- Labetalol is commonly used in pregnancy to manage hypertension. It is considered safe and effective for use during pregnancy and is often recommended as a first-line choice for treating hypertension in pregnant women.

Solution for Question 8:

Correct Option A - Evolocumab:

- Evolocumab is used for lowering LDL cholesterol in patients with hypercholesterolemia. It is a monoclonal antibody that inhibits Proprotein Convertase Subtilisin/Kexin Type 9 (PCSK9). By binding to PCSK9, it prevents PCSK9 from degrading the low-density lipoprotein (LDL) receptors on hepatocytes. This increases the number of LDL receptors available to clear LDL cholesterol from the blood, thus lowering LDL cholesterol levels.

Incorrect Options:

Option B - Lomitapide: Lomitapide inhibits the microsomal triglyceride transfer protein (MTP), which is essential for the assembly and secretion of very low-density lipoprotein (VLDL) in the liver. This reduces the levels of LDL cholesterol and triglycerides.

Option C - Mipomersen: Mipomersen is an antisense oligonucleotide that targets and inhibits the synthesis of apolipoprotein B-100 (apoB-100), the primary protein component of LDL and VLDL. This reduces LDL cholesterol.

Option D - Torcetrapib: Torcetrapib was a cholesteryl ester transfer protein (CETP) inhibitor intended to increase high-density lipoprotein (HDL) cholesterol. However, its development was terminated due to adverse effects in clinical trials, including increased mortality.

Solution for Question 9:

Correct Option C - Statins:

- Statins competitively inhibit the enzyme HMG-CoA reductase and upregulate LDL receptors on hepatocytes, which enhances the clearance of LDL cholesterol from the bloodstream.
- They also increase HDL cholesterol and decrease triglycerides.
- Hence, they are drug of choice for treating hypercholesterolemia.

Incorrect Options:

Option A - Niacin:

- Niacin is used to increase HDL level

Option B - Fibrates

- Fibrates are used to reduce triglyceride levels

Option D - Resins

- Resins is the drug of choice to reduce blood cholesterol in pregnancy

Solution for Question 10:

Correct Option A - IV Labetalol:

- The patient's presentation is suggestive of a hypertensive crisis. IV Labetalol is the preferred intravenous medication for the management of hypertension in pregnancy due to its efficacy and safety profile. It acts by blocking both alpha-1, beta-1, and beta-2 adrenergic receptors, leading to vasodilation and a subsequent reduction in blood pressure.

Incorrect Options:

Options B, C, and D are not the preferred drugs for managing hypertensive crisis in pregnancy.

Solution for Question 11:

Correct Option A - Bisoprolol:

- Mechanism of Action: Bisoprolol is a selective beta-1 adrenergic blocker. It works by decreasing the heart rate, reducing myocardial contractility, and lowering blood pressure, which reduces the workload on the heart.
- Prolonging Survival: Bisoprolol is one of the beta-blockers that has been proven to improve survival in patients with chronic heart failure (CHF). It is specifically indicated in the management of CHF due to its ability to reduce mortality and hospitalization by counteracting the harmful effects of excessive sympathetic stimulation on the heart.
- Clinical Trials: The CIBIS-II trial (Cardiac Insufficiency Bisoprolol Study II) demonstrated a significant reduction in mortality and hospitalization rates in patients with heart failure treated with bisoprolol.

Incorrect Options:

Option B - Labetalol:

- Mechanism of Action: Labetalol is a non-selective beta-blocker with additional alpha-1 adrenergic blocking activity.
- Use in Heart Failure: Labetalol is primarily used for the management of hypertension and hypertensive emergencies rather than for chronic heart failure. It is not known to prolong survival in CHF and is not routinely used for this purpose.

Option C - Propranolol:

- Mechanism of Action: Propranolol is a non-selective beta-blocker that affects both beta-1 and beta-2 receptors.
- Use in Heart Failure: Propranolol is generally used for conditions such as hypertension, angina, and arrhythmias but is not commonly used in the management of chronic heart failure. It does not have evidence supporting a survival benefit in CHF patients.

Option D - Atenolol:

- Mechanism of Action: Atenolol is a selective beta-1 adrenergic blocker, similar to bisoprolol.
- Use in Heart Failure: Atenolol is less commonly used in chronic heart failure and has not shown a survival benefit in this population. It is more frequently used for managing hypertension and angina pectoris.

Solution for Question 12:

Correct Option B - Low Molecular Weight Heparin (LMWH):

- LMWH, such as enoxaparin, is the preferred anticoagulant for treating DVT in pregnancy. It is safe, does not cross the placenta, and has a lower risk of causing heparin-induced thrombocytopenia (HIT) compared to unfractionated heparin. LMWH is also effective in preventing the extension of the thrombus and reducing the risk of pulmonary embolism, making it the standard treatment for DVT during pregnancy.

Incorrect Options:

Option A - Warfarin:

- Warfarin is contraindicated in pregnancy, especially during the first trimester and near term, due to its teratogenic effects and the risk of causing fetal bleeding. Warfarin crosses the placenta and can lead to fetal warfarin syndrome, characterized by congenital malformations. Therefore, it is not recommended for use in pregnant women.

Warfarin is contraindicated in pregnancy, especially during the first trimester and near term, due to its teratogenic effects and the risk of causing fetal bleeding. Warfarin crosses the placenta and can lead to fetal warfarin syndrome, characterized by congenital malformations. Therefore, it is not recommended for use in pregnant women.

Option C - Apixaban:

- Apixaban is a direct oral anticoagulant (DOAC) that is not recommended for use during pregnancy because its safety in pregnant women has not been established. DOACs, including apixaban, are generally avoided during pregnancy due to concerns about potential fetal exposure and lack of sufficient data on their effects.

Option D - Rivaroxaban:

- Like apixaban, rivaroxaban is a DOAC and is not recommended for use in pregnancy due to the lack of sufficient safety data. The potential for fetal exposure and unknown risks make DOACs less desirable compared to LMWH, which has a well-established safety profile in pregnancy.

Solution for Question 13:

Correct Option C - Hydrochlorothiazide:

- Hydrochlorothiazide is a thiazide diuretic commonly used to treat hypertension. It is particularly useful in patients with renal colic due to its ability to reduce calcium excretion in the urine, which can help prevent the formation of calcium-containing kidney stones.

Thiazide diuretics are effective in managing hypertension and have a protective effect against stone formation by decreasing calcium levels in the urine.

Incorrect Options:

Option A - Ethacrynic Acid:

- Ethacrynic acid is a loop diuretic and is not typically used for hypertension in the long term. It is used in situations where more potent diuresis is required or where thiazide diuretics are ineffective. Loop diuretics can increase calcium excretion in the urine, which could exacerbate kidney stone formation rather than prevent it.

Option B - Acetazolamide:

- Acetazolamide is a carbonic anhydrase inhibitor that works as a diuretic but is not commonly used for hypertension. It can lead to increased bicarbonate excretion and subsequent metabolic acidosis. It is more commonly used in conditions like glaucoma, altitude sickness, or metabolic alkalosis. Its effect on calcium excretion is not beneficial for preventing renal colic.

Option D - Spironolactone:

- Spironolactone is a potassium-sparing diuretic that works by antagonizing aldosterone. It is used to treat conditions like heart failure, primary aldosteronism, and as part of combination therapy for hypertension. However, it does not affect calcium excretion and is not typically used for managing renal colic or calcium-based kidney stones.

Solution for Question 14:

Correct Option A - Clonidine:

- Clonidine is an alpha-2 adrenergic agonist used to treat hypertension. It works by decreasing sympathetic outflow, which lowers blood pressure. When clonidine is stopped abruptly, it can lead to a rebound increase in blood pressure, potentially causing a hypertensive emergency. Symptoms like dry eyes, dry mouth, and constipation are common side effects of clonidine. Therefore, the development of a hypertensive emergency after stopping clonidine is consistent with its withdrawal effects.

Incorrect Options:

Option B - Lisinopril:

- Lisinopril is an ACE inhibitor used to treat hypertension and heart failure. It typically does not cause rebound hypertension if discontinued. Common side effects include cough and elevated potassium levels, but not dry eyes, dry mouth, or constipation. Discontinuation of lisinopril is unlikely to cause a hypertensive emergency.

Option C - Amlodipine:

- Amlodipine is a calcium channel blocker used to manage hypertension and angina. It generally does not cause rebound hypertension when stopped. Side effects can include swelling, dizziness, and flushing, but not dry eyes, dry mouth, or constipation. Discontinuation of amlodipine is less likely to result in a hypertensive emergency.

Option D - Telmisartan:

- Telmisartan is an angiotensin II receptor blocker (ARB). Similar to lisinopril, discontinuation of telmisartan typically does not cause rebound hypertension. Common side effects include dizziness and hyperkalemia, but not dry eyes, dry mouth, or constipation. A hypertensive emergency due to the abrupt discontinuation of telmisartan is unlikely.

Solution for Question 15:

Correct Option A - Nicotinamide (Niacin):

- Niacin (nicotinic acid, not to be confused with nicotinamide) is commonly used to lower triglycerides and raise HDL cholesterol. However, it is known to cause several side effects, including flushing, hyperuricemia (which can precipitate gout), and elevated liver enzymes (indicating hepatotoxicity). Flushing is a particularly well-known side effect due to the vasodilatory effect of niacin, which can be mitigated by taking aspirin before the dose.

Incorrect Options:

Option B - Atorvastatin:

- Atorvastatin, a statin, is used to lower LDL cholesterol and triglycerides. While it can cause elevated liver enzymes and muscle-related side effects (myopathy), it is not typically associated with flushing or gout.

Option C - Fenofibrate:

- Fenofibrate is a fibrate that is effective in lowering triglycerides. It can occasionally cause elevated liver enzymes and muscle-related side effects, but it is not typically associated with

flushing or gout.

Option D - Cholestyramine:

- Cholestyramine is a bile acid sequestrant used to lower LDL cholesterol. It can cause gastrointestinal side effects such as constipation but is not associated with flushing, gout, or elevated liver enzymes.

Solution for Question 16:

Correct Option A - Dhatura:

- Dhatura (Datura) contains anticholinergic alkaloids like atropine, scopolamine, and hyoscyamine. An overdose of Dhatura can lead to anticholinergic syndrome, characterized by hot, dry skin (due to inhibition of sweating), dilated pupils (mydriasis), and altered mental status, which may include slurred speech, confusion, and hallucinations. The combination of these symptoms is classic for anticholinergic toxicity.

Incorrect Options:

- Cannabis intoxication can lead to symptoms such as altered mental status, red eyes, increased appetite, and dry mouth. However, it is not typically associated with hot, dry skin or significant pupil dilation. The symptoms described are more consistent with anticholinergic toxicity rather than cannabis intoxication.
- Alcohol intoxication often presents with symptoms such as slurred speech, incoordination, and altered mental status. However, alcohol does not typically cause hot, dry skin or dilated pupils. In fact, alcohol may lead to hypothermia and constricted pupils in severe cases.
- Cocaine, a stimulant, typically causes symptoms such as dilated pupils, tachycardia, sweating, and agitation. However, it usually does not cause hot, dry skin, which is more characteristic of anticholinergic toxicity. The slurred speech and hot, dry skin are not typical of cocaine intoxication.

Solution for Question 17:

Answer: C) Baroreceptor stimulation

Explanation:

The administration of norepinephrine (NE) causes:

- $\uparrow$ Systolic and diastolic blood pressure (due to α -mediated vasoconstriction).
- $\downarrow$ Heart rate (bradycardia) despite norepinephrine's β -stimulatory effects.

This paradoxical bradycardia occurs due to the activation of the baroreceptor reflex in response to increased blood pressure.

Aspect	Details
Mechanism	Baroreceptors in carotid sinus and aortic arch detect BP changes Increased BP stretches receptors, sending signals to medulla via glossopharyngeal and vagus nerves Medulla inhibits vasoconstrictor center and stimulates vagal parasympathetic center
Actions	Vasodilation of blood vessels Decreased heart rate and contraction strength Reduced BP due to lower peripheral resistance and cardiac output
Importance	Maintains BP stability during activities like standing Fastest acting Tends to reset within 1-2 days for long-term regulation

- Beta-1 receptor blockade (Option A) is incorrect because norepinephrine stimulates β receptors, increasing contractility, but the reflex bradycardia is due to vagal activation, not β blockade.
- Alpha-1 receptor blockade (Option B) is incorrect because norepinephrine activates α receptors, leading to vasoconstriction and increased blood pressure. Blocking α receptors would cause vasodilation and hypotension, the opposite of the observed effect.
- Baroreceptor inhibition (Option D) is incorrect because if baroreceptors were inhibited, heart rate would increase (not decrease) due to unchecked sympathetic activation.

Solution for Question 18:

Answer: A) Hyperkalemia

Explanation:

The combination of indomethacin (a nonsteroidal anti-inflammatory drug, NSAID) and an ACE inhibitor increases the risk of hyperkalemia due to their combined effects on renal function and potassium homeostasis.

- Mechanism: ACE inhibitors (e.g., enalapril, lisinopril) block angiotensin-converting enzymes, reducing aldosterone secretion. Aldosterone promotes potassium excretion, so reduced aldosterone leads to potassium retention. Indomethacin (NSAID) further reduces renal perfusion by inhibiting prostaglandins, worsening potassium retention. This can lead to hyperkalemia, particularly in patients with chronic kidney disease (CKD), dehydration, or concurrent diuretic use.
- ACE inhibitors (e.g., enalapril, lisinopril) block angiotensin-converting enzymes, reducing aldosterone secretion.
- Aldosterone promotes potassium excretion, so reduced aldosterone leads to potassium retention.
- Indomethacin (NSAID) further reduces renal perfusion by inhibiting prostaglandins, worsening potassium retention.
- This can lead to hyperkalemia, particularly in patients with chronic kidney disease (CKD), dehydration, or concurrent diuretic use.
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- Aldosterone promotes potassium excretion, so reduced aldosterone leads to potassium retention.
- Indomethacin (NSAID) further reduces renal perfusion by inhibiting prostaglandins, worsening potassium retention.
- This can lead to hyperkalemia, particularly in patients with chronic kidney disease (CKD), dehydration, or concurrent diuretic use.

Drug Interaction	Mechanism	Clinical Implication
ACE Inhibitors + Potassium Supplements/K-sparing diuretics	Risk of hyperkalemia due to reduced aldosterone secretion by ACE inhibitors combined with potassium-retaining agents.	Monitoring serum potassium levels is crucial to avoid hyperkalemia, which can cause serious cardiac issues such as arrhythmias.
ACE Inhibitors + NSAIDs	NSAIDs can reduce the anti-hypertensive effects of ACE inhibitors and decrease renal blood flow, increasing the risk of acute renal failure (ARF) or oliguria.	Use caution when combining NSAIDs with ACE inhibitors, and closely monitor renal function, particularly in high-risk patients.
ACE Inhibitors + Lithium	ACE inhibitors can increase the reabsorption of lithium, potentially leading to lithium toxicity.	Monitor lithium levels carefully when starting or adjusting ACE inhibitors in patients taking lithium, and adjust the lithium dosage as needed.

ACE Inhibitors + Antacids	Reduced bioavailability of ACE inhibitors due to binding with antacids.	Administer ACE inhibitors at least two hours before or after antacids to maintain their efficacy.
ACE Inhibitors + Diuretics	Diuretics enhance the hypotensive effects of ACE inhibitors by depleting sodium and raising renin levels.	In patients already on diuretics, start ACE inhibitors at a low dose to minimise the risk of excessive blood pressure reduction initially.

- Hypercalcemia (Option B) is incorrect because neither NSAIDs nor ACE inhibitors cause significant calcium elevation. ACE inhibitors may slightly lower calcium levels by increasing urinary calcium excretion.
- Hyponatremia (Option C) is incorrect because ACE inhibitors do not cause sodium retention; instead, they promote mild natriuresis (sodium excretion).
- Hyperphosphatemia (Option D) is incorrect because NSAIDs and ACE inhibitors do not directly affect phosphate metabolism; hyperphosphatemia is more commonly seen in CKD due to decreased phosphate excretion.

Solution for Question 19:

Correct Option A – Fibrates:

- Fibrates are the drug of choice (DOC) for treating hypertriglyceridemia.
- They work by activating peroxisome proliferator-activated receptor alpha (PPAR α), which increases the activity of lipoprotein lipase.
- This enzyme breaks down triglycerides in the blood, thereby reducing their levels.

Incorrect Options:

Option B - Niacin

- Niacin is used to increase HDL levels.

Option C – Statins:

- Statins are used to reduce LDL cholesterol levels.

Option D – Resins:

- Resins are the drug of choice to reduce blood cholesterol in pregnancy.
-

Solution for Question 20:

Correct Option B - Prazosin:

- Prazosin is an alpha-1 adrenergic blocker that is commonly used to treat hypertension. However, it can also cause postural hypotension, which is a drop in blood pressure upon standing up from a sitting or lying position. This can lead to dizziness, lightheadedness, and an increased risk of falls, particularly in elderly patients who may already have compromised balance and coordination.
- Prazosin works by blocking the alpha-1 adrenergic receptors, which results in vasodilation and relaxation of smooth muscles in blood vessels, leading to a decrease in blood pressure. However, this vasodilatory effect can cause a sudden drop in blood pressure when changing positions, especially from lying down to standing.

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Incorrect Options:

Option A - Metformin: Metformin is an antidiabetic medication that does not typically cause postural hypotension. Its primary action is to decrease glucose production in the liver and increase insulin sensitivity, making it a commonly used medication for the management of type 2 diabetes.

Option C - Acarbose: Acarbose is an alpha-glucosidase inhibitor used in the treatment of type 2 diabetes. It does not have a direct effect on blood pressure and is not associated with postural hypotension.

Option D - Nor-adrenaline: Nor-adrenaline (norepinephrine) is a naturally occurring neurotransmitter and hormone that acts as a vasoconstrictor. It is not typically associated with postural hypotension. In fact, it can increase blood pressure by constricting blood vessels.

Solution for Question 21:

Correct Option B - Diabetes Mellitus with hypertension:

- Beta-blockers should ideally be avoided due to their potential to mask some symptoms of low blood sugar and possibly worsen glucose control.

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Incorrect Options:

Option A - Essential tremors: Essential tremors are involuntary rhythmic movements that typically affect the hands, arms, or other body parts. Beta-blockers can be used as a treatment option for essential tremors. They help reduce tremor severity by blocking the effects of adrenaline (epinephrine) on beta receptors in the body, thereby reducing sympathetic nervous system activity.

Option C - Angina pectoris: Angina pectoris is chest pain or discomfort that occurs when the heart muscle doesn't receive enough oxygen-rich blood. Beta-blockers are frequently prescribed for the management of angina pectoris. They reduce the workload on the heart by blocking the effects of adrenaline, leading to a decrease in heart rate and blood pressure, which can help alleviate anginal symptoms.

Option D - Migraine prophylaxis: Beta-blockers are commonly used for migraine prophylaxis, which involves the prevention or reduction of migraine attacks. They are particularly effective in patients with frequent migraines or those who have certain triggers. Beta-blockers help in the prevention of migraines by blocking the effects of adrenaline and reducing the excitability of the nervous system.

Solution for Question 22:

Correct Option B - Clarithromycin:

- Rosuvastatin is a statin medication used to lower cholesterol levels in the body. It works by inhibiting the enzyme HMG-CoA reductase, which is involved in cholesterol synthesis.
- Clarithromycin is an antibiotic commonly used to treat bacterial infections. It belongs to the macrolide class of antibiotics and can inhibit the metabolism of certain medications by affecting the liver enzyme CYP3A4, which can lead to increased levels of certain statins, including rosuvastatin, in the blood. This can increase the risk of statin-related side effects, particularly myopathy (muscle damage) and rhabdomyolysis (severe muscle breakdown).
- Therefore, it is generally recommended to avoid combining rosuvastatin with clarithromycin unless the potential benefits outweigh the risks. If necessary, alternative antibiotics that do not significantly inhibit CYP3A4, such as azithromycin, may be considered.

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Therefore, it is generally recommended to avoid combining rosuvastatin with clarithromycin unless the potential benefits outweigh the risks. If necessary, alternative antibiotics that do not significantly inhibit CYP3A4, such as azithromycin, may be considered.

Incorrect Options:

Option A - Oxycodone: There is no significant drug interaction between rosuvastatin and oxycodone. These medications can generally be used together if indicated, and no specific precautions or dosage adjustments are typically required.

Option C - Adalimumab: Adalimumab is a medication used to treat inflammatory conditions such as rheumatoid arthritis and Crohn's disease. There is no known significant interaction between adalimumab and rosuvastatin. These medications can generally be used together without any specific concerns.

Option D - Rivaroxaban: Rivaroxaban is an anticoagulant medication used to prevent blood clot formation. There is no known significant interaction between rivaroxaban and rosuvastatin. These medications can generally be used together without any specific concerns.

Solution for Question 23:

Correct Option B - Chlorthalidone:

- In a 53-year-old man with stage 2 hypertension and no co-morbidities, the preferred initial medication would be a thiazide-like diuretic. Chlorthalidone, which belongs to the thiazide class of diuretics, is an appropriate choice.
- Thiazide diuretics, such as, Chlorthalidone are considered first-line agents for treating hypertension due to their proven efficacy and safety profile. They help lower blood pressure by promoting diuresis (increased urine production) and reducing fluid volume in the body. This results in a decrease in systemic vascular resistance and blood pressure.

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Incorrect Options:

Option A - Furosemide: Furosemide is a loop diuretic, which is more potent than thiazide diuretics. They are not usually the first-line choice for uncomplicated hypertension.

Option C - Spironolactone: Spironolactone is a potassium-sparing diuretic and is often used as an adjunct therapy in cases where additional blood pressure control is needed. It is not typically used as monotherapy for initial treatment of hypertension.

Option D - Triamterene: Triamterene is another potassium-sparing diuretic commonly used in combination with other diuretics to reduce potassium loss. It is not typically used as monotherapy for initial treatment of hypertension.

Solution for Question 24:

Correct Option A - Trimetazidine:

- Trimetazidine is not commonly used in the management of heart failure. It is primarily used in the treatment of angina pectoris and ischemic heart disease. Trimetazidine acts by

improving myocardial glucose utilization and reducing fatty acid oxidation, which helps in preserving cardiac energy metabolism. However, it does not have direct effects on heart failure pathophysiology, such as reducing preload or afterload, improving cardiac contractility, or inhibiting neurohormonal activation, which are important therapeutic targets in heart failure.

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Incorrect Options:

Option B - Sacubitril: Sacubitril is a neprilysin inhibitor used in the management of heart failure with reduced ejection fraction (HFrEF). It is combined with the angiotensin receptor blocker (ARB) valsartan to form sacubitril/valsartan, which has been shown to reduce cardiovascular mortality and hospitalizations in HFrEF patients.

Option C - Metoprolol: Metoprolol is a beta-blocker commonly used in the treatment of heart failure. It helps in reducing heart rate, blood pressure, and cardiac workload, thereby improving symptoms and outcomes in heart failure patients.

Option D - Nesiritide: Nesiritide is a recombinant form of human B-type natriuretic peptide (BNP) used in acute decompensated heart failure. It has vasodilatory and diuretic effects, leading to decreased preload and afterload, improved cardiac output, and relief of symptoms in acute heart failure episodes. However, its long-term use in chronic heart failure is limited.

Solution for Question 25:

Correct Option A - Ulcerated infantile hemangioma:

- Propranolol is the drug of choice for ulcerated infantile hemangioma.
- Propranolol is a non-selective beta-blocker that has been found to be highly effective in treating ulcerated infantile hemangiomas, which are benign tumors of blood vessels that commonly occur in infants. Propranolol helps to reduce the size and improve the appearance of the hemangioma by causing vasoconstriction and reducing blood flow to the affected area.

It also promotes the regression of the abnormal blood vessels.

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Incorrect Options:

Option B - Lymphangioma circumscriptum: Lymphangiomas are malformations of the lymphatic vessels, and propranolol is not the drug of choice for this condition. Propranolol has no effect on lymphatic system. Surgical excision or other treatment modalities may be employed.

Option C - Capillary malformation: Propranolol is not the primary treatment for capillary malformations. Other treatment options, such as laser therapy, may be considered.

Option D - Pyogenic granuloma: Pyogenic granuloma is a benign vascular lesion that typically occurs on the skin or mucous membranes. Propranolol is not commonly used as a first-line treatment for this condition, and other treatments, such as surgical excision or cauterization, may be more appropriate.

Solution for Question 26:

Correct Option A - Angiotensin 1 receptor:

- The angiotensin 1 receptor is a G protein-coupled receptor (GPCR) that does not directly act via a K⁺ channel. Activation of the angiotensin 1 receptor by angiotensin II leads to intracellular signaling through G proteins and subsequent activation of downstream pathways, such as the phospholipase C (PLC) pathway and the mitogen-activated protein kinase (MAPK) pathway. These pathways result in various cellular responses, including vasoconstriction, aldosterone release, and increased blood pressure regulation. While these pathways can indirectly influence ion channels and cellular activity, they don't directly modulate K⁺ channels.

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Incorrect Options:

Option B - Muscarinic M2 receptor: The muscarinic M2 receptor is a GPCR that primarily acts via the activation of K⁺ channels. When activated by acetylcholine, the M2 receptor initiates a signaling cascade that involves the beta-gamma subunits of G proteins binding to and activating K⁺ channels, leading to hyperpolarization of the cell membrane and a decrease in heart rate.

Option C - Dopamine D2 receptor: The dopamine D2 receptor is also a GPCR that primarily acts by inhibiting adenylyl cyclase activity, reducing cyclic AMP (cAMP) levels, and indirectly modulating ion channels and cellular responses. While the D2 receptor's activation can have effects on ion channels, it doesn't directly act on K⁺ channels.

Option D - Serotonin 5-HT₁ receptor: The serotonin 5-HT₁ receptor is a GPCR that can have various effects, including modulation of ion channels and second messenger pathways. Similar to the other receptors mentioned, it doesn't directly act via K⁺ channels.

Solution for Question 27:

Correct Option B - Tadalafil:

- Tadalafil is a PDE-5 inhibitor used in the treatment of pulmonary hypertension. Tadalafil is the longest-acting PDE-5 inhibitor. Duration of action can last up to 36 hours.

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Incorrect Options:

Option A - Sildenafil: Belongs to phosphodiesterase (PDE) V inhibitors. Duration of action typically lasts 4-6 hours.

Option C - Vardenafil: Belongs to phosphodiesterase (PDE) V inhibitors. Duration of action typically lasts around 4-5 hours.

Option D - Avanafil: Avanafil's duration of action is typically around 4-6 hours.

Solution for Question 28:

Correct Option B - Stimulation of guanylyl cyclase:

- Sodium nitroprusside is a potent vasodilator used for the rapid reduction of blood pressure in hypertensive emergencies. Its mechanism of action involves the stimulation of guanylyl cyclase, leading to an increase in cyclic guanosine monophosphate (cGMP) levels within smooth muscle cells. This, in turn, leads to various intracellular effects that result in vasodilation and decreased systemic vascular resistance.

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Incorrect Options:

Option A - Increased cAMP: This option is not the mechanism of action of sodium nitroprusside. Increased levels of cyclic adenosine monophosphate (cAMP) are associated with the activation of certain receptors, such as beta-adrenergic receptors. cAMP generally leads to relaxation of smooth muscle, but sodium nitroprusside's mechanism is mediated through cGMP.

Option C - Calcium channel blockage: Sodium nitroprusside does not directly block calcium channels. Its mechanism is centered around the stimulation of guanylyl cyclase and the subsequent increase in cGMP levels, leading to vasodilation.

Option D - K⁺ channel opener: Sodium nitroprusside does not directly open potassium (K⁺) channels. Its primary effect is on guanylyl cyclase, which increases cGMP levels and promotes vasodilation.

Solution for Question 29:

Correct Option D - Increases renal blood flow:

- Low infusion rates (0.5 to 2 micrograms/kg per minute) produces vasodilation in the kidneys, resulting in increased urinary flow.
- Intermediate infusion rates (from 2 to 10 micrograms/kg/min) stimulate myocardial contractility and increased electrical conductivity leading to increased cardiac output.
- Higher doses cause vasoconstriction and increased blood pressure via the adrenergic receptors

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Higher doses cause vasoconstriction and increased blood pressure via the adrenergic receptors

Incorrect Options:

Option A - Increase in Cardiac contractility: Seen in intermediate infusion rates.

Option B - Vasoconstriction: Higher doses cause vasoconstriction leading to poor peripheral circulation. Dopamine is a peripheral vasostimulant

Option C - Tachycardia: Not seen at low doses

Solution for Question 30:

Correct Option C - Amlodipine:

- In the given scenario, with a patient presenting with renal insufficiency, very low urine output, pedal edema, headache, and high blood pressure (BP 160/90), the most appropriate anti-hypertensive option would be amlodipine.
- Amlodipine is a calcium channel blocker that is commonly used to treat hypertension. It works by relaxing and dilating the blood vessels, which leads to a reduction in blood pressure. Amlodipine is often well-tolerated and can be used in patients with renal insufficiency.

In the given scenario, with a patient presenting with renal insufficiency, very low urine output, pedal edema, headache, and high blood pressure (BP 160/90), the most appropriate anti-hypertensive option would be amlodipine.

Amlodipine is a calcium channel blocker that is commonly used to treat hypertension. It works by relaxing and dilating the blood vessels, which leads to a reduction in blood pressure. Amlodipine is often well-tolerated and can be used in patients with renal insufficiency.

Incorrect Options :

Option A - Aliskiren: Aliskiren is a direct renin inhibitor that can be used to treat hypertension. However, it is generally not recommended in patients with renal insufficiency, as it can further decrease kidney function.

Option B - Chlorthalidone: Chlorthalidone is a thiazide-like diuretic that is often used to treat hypertension. However, in a patient with very low urine output and pedal edema, adding a diuretic may not be the initial choice, as it may exacerbate fluid and electrolyte imbalances.

Option D - Prazosin: Prazosin is an alpha-1 adrenergic blocker that is used to treat high blood pressure. However, in this scenario, with symptoms of renal insufficiency and very low urine output, the focus should be on addressing the underlying renal issue and controlling blood pressure with other medications.

Solution for Question 31:

Correct Option A - Antibody against digoxin:

- The next step in the management of this patient with a history of consumption of 8 tablets of digoxin, unstable clinical condition, and 3rd-degree heart block is to administer antidote therapy for digoxin toxicity.
- The antidote for digoxin toxicity is digoxin immune Fab (also known as Digoxin-specific antibody fragments or Digoxin Immune Fab).
- This medication works by binding to the digoxin molecules in the bloodstream and preventing them from binding to the sodium-potassium ATPase pump in cardiac myocytes, thereby reversing the toxic effects of digoxin.

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This medication works by binding to the digoxin molecules in the bloodstream and preventing them from binding to the sodium-potassium ATPase pump in cardiac myocytes, thereby reversing the toxic effects of digoxin.

Incorrect Options:

Option B - Lidocaine: It is a local anesthetic and antiarrhythmic medication that is commonly used in the management of ventricular arrhythmias, such as ventricular tachycardia and ventricular fibrillation.

Option C - DC cardioversion: It is a medical procedure that is used to treat certain types of arrhythmias, such as atrial fibrillation, atrial flutter, and ventricular tachycardia.

Option D - Phenytoin: It is an antiepileptic drug that is used to treat seizures and epilepsy. It has no role here

Solution for Question 32:

Correct Option A - Evolocumab:

- Evolocumab is a monoclonal antibody and a type of PCSK9 inhibitor used in the treatment of high cholesterol levels.
- PCSK9 (Proprotein Convertase Subtilisin/Kexin type 9) is a protein that plays a crucial role in regulating the amount of LDL (low-density lipoprotein) cholesterol in the blood.
- LDL cholesterol is often referred to as "bad" cholesterol because high levels of LDL can contribute to the development of cardiovascular disease.
- PCSK9 works by binding to the LDL receptor on the surface of liver cells and preventing it from recycling back to the surface to continue removing LDL cholesterol from the bloodstream. This results in an accumulation of LDL cholesterol in the blood, which can lead to plaque formation and an increased risk of heart disease.
- Evolocumab is designed to bind to PCSK9 and prevent it from binding to the LDL receptor, allowing the receptor to continue removing LDL cholesterol from the blood. By reducing the amount of circulating PCSK9, Evolocumab can significantly lower LDL cholesterol levels and reduce the risk of cardiovascular events.
- Evolocumab is administered by subcutaneous injection every 2 to 4 weeks, depending on the dose and the patient's response. It is typically used in combination with other lipid-lowering therapies, such as statins, in patients with high cholesterol levels who have not

been able to achieve their target LDL cholesterol levels with diet and exercise alone.

Evolocumab is a monoclonal antibody and a type of PCSK9 inhibitor used in the treatment of high cholesterol levels.

PCSK9 (Proprotein Convertase Subtilisin/Kexin type 9) is a protein that plays a crucial role in regulating the amount of LDL (low-density lipoprotein) cholesterol in the blood.

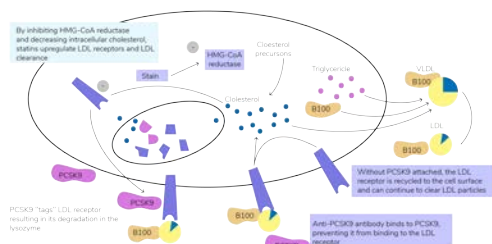
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Evolocumab is administered by subcutaneous injection every 2 to 4 weeks, depending on the dose and the patient's response. It is typically used in combination with other lipid-lowering therapies, such as statins, in patients with high cholesterol levels who have not been able to achieve their target LDL cholesterol levels with diet and exercise alone.

Action of PCSK9 Inhibitors



Incorrect Options:

Option B - Ezetimibe: Ezetimibe is a cholesterol absorption inhibitor and works by inhibiting the absorption of dietary and biliary cholesterol in the small intestine.

Option C - Bempedoic acid: Bempedoic acid is not a PCSK9 inhibitor, although it is a newer medication used to treat high cholesterol levels.

Option D - Clofibrate: Clofibrate is not a PCSK9 inhibitor. It belongs to a class of drugs called fibrates and is used to lower triglyceride levels and increase HDL cholesterol levels in patients with hyperlipidemia.

Solution for Question 33:

Correct Option A - Digoxin:

- Heart failure with reduced ejection fraction (HFrEF) is a common condition in which the heart cannot pump enough blood to meet the body's needs. Several drugs have been shown to improve symptoms and reduce mortality in patients with HFrEF.
- Digoxin is a medication used for many years to treat heart failure. While it can improve symptoms such as shortness of breath and fatigue, it has not been shown to reduce mortality in patients with HFrEF. However, digoxin may still have a role in managing symptoms in certain patients with HFrEF.

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Digoxin is a medication used for many years to treat heart failure. While it can improve symptoms such as shortness of breath and fatigue, it has not been shown to reduce mortality in patients with HFrEF. However, digoxin may still have a role in managing symptoms in certain patients with HFrEF.

Incorrect Options:

Option B - Beta-blockers: Beta-blockers such as metoprolol, carvedilol, and bisoprolol are among the most effective medications for treating HFrEF. They work by reducing the workload on the heart and improving its pumping ability. Beta-blockers have been shown to reduce mortality by about 30% in patients with HFrEF.

Option C - ACE inhibitors: ACE inhibitors such as lisinopril, enalapril, and ramipril are another class of effective drugs in treating HFrEF. They work by dilating blood vessels and reducing the workload on the heart. ACE inhibitors have been shown to reduce mortality by about 20-25% in patients with HFrEF.

Option D - Hydralazine + isosorbide dinitrate: Hydralazine + isosorbide dinitrate, sometimes used in patients with HFrEF who cannot tolerate ACE inhibitors or angiotensin receptor blockers (ARBs). Hydralazine is a vasodilator that relaxes blood vessels, while isosorbide dinitrate is a nitrate that also dilates blood vessels. This combination has reduced mortality by about 25% in African-American patients with HFrEF.

Solution for Question 34:

Correct Option B - Blocks both alpha and beta adrenergic receptors:

- Labetalol is a non-selective beta-blocker and an alpha-blocker. It blocks both beta-adrenergic receptors and alpha-adrenergic receptors. By blocking these receptors, labetalol reduces sympathetic nervous system activity, leading to decreased heart rate, decreased contractility, and vasodilation.
- Labetalol is a non-selective beta blocker.
- Blocking beta receptors reduces the effects of epinephrine and norepinephrine on the heart, decreasing heart rate and contractility.
- Blocking alpha-1 receptors reduces arterial resistance by causing dilation of the blood vessels. The net effect is a decrease in blood pressure.

Labetalol is a non-selective beta blocker.

Blocking beta receptors reduces the effects of epinephrine and norepinephrine on the heart, decreasing heart rate and contractility.

Blocking alpha-1 receptors reduces arterial resistance by causing dilation of the blood vessels. The net effect is a decrease in blood pressure.

Incorrect Options:

Option A - Directly acts as arterial vasodilator resulting in indirect effect tachycardia:

- Labetalol primarily acts as a non-selective beta-blocker and alpha-blocker. Its vasodilatory effect is mainly due to its blockade of alpha-adrenergic receptors, leading to relaxation of vascular smooth muscle. This vasodilation results in decreased systemic vascular resistance and blood pressure.

Option C - Primarily acts as a vasodilator with little effect on arterioles:

- Labetalol acts as both a beta-blocker and an alpha-blocker, causing vasodilation in both arteries and arterioles. Its alpha-blocking activity leads to arteriolar dilation, while its beta-blocking activity contributes to arterial vasodilation.

Labetalol acts as both a beta-blocker and an alpha-blocker, causing vasodilation in both arteries and arterioles. Its alpha-blocking activity leads to arteriolar dilation, while its beta-blocking activity contributes to arterial vasodilation.

Option D - Decreases peripheral vascular resistance by directly acting as a relaxant for vascular smooth muscles.:

- Labetalol decreases peripheral vascular resistance, but it does so indirectly by blocking both alpha-adrenergic receptors and beta-adrenergic receptors. By blocking these receptors, labetalol reduces the vasoconstrictor effects of catecholamines such as norepinephrine and epinephrine. This results in vasodilation and decreased peripheral vascular resistance. Labetalol does not directly act as a relaxant for vascular smooth muscles.

Solution for Question 35:

Correct Option A - Metoprolol:

- It is a beta-blocker that have been shown in multiple large-scale clinical trials to reduce mortality in patients with heart failure with reduced ejection fraction (HFREF).

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Incorrect Options:

Option B - Furosemide: Furosemide and torsemide are loop diuretics that are commonly used in the management of heart failure to reduce fluid buildup and relieve symptoms, but they do not have a mortality benefit. They however have symptomatic benefit. They are used extensively in acute decompensated heart failure.

Option C - Torsemide: Torsemide is a loop diuretic that is primarily used to reduce symptoms of fluid overload in patients with heart failure, but it has not been shown to reduce mortality in this population.

Option D - Prazosin: Prazosin is an alpha-blocker that is used primarily to treat high blood pressure and benign prostatic hyperplasia, but it is not recommended for the management of heart failure and has not been shown to reduce mortality in HFREF patients.

Solution for Question 36:

Correct Option C - Aliskiren:

- Aliskiren is a medication that belongs to a class of drugs called renin inhibitors. It is primarily used to treat hypertension
- The mode of action of aliskiren involves its inhibition of renin, an enzyme involved in the production of angiotensin I.

Aliskiren is a medication that belongs to a class of drugs called renin inhibitors. It is primarily used to treat hypertension

The mode of action of aliskiren involves its inhibition of renin, an enzyme involved in the production of angiotensin I.

Incorrect Options:

Option A - Captopril: Captopril is an angiotensin-converting enzyme (ACE) inhibitor. Its mode of action involves inhibiting the activity of ACE, an enzyme responsible for converting angiotensin I into angiotensin II. By inhibiting ACE, captopril reduces the production of angiotensin II, potent vasoconstrictor. This results in vasodilation (widening of blood vessels), leading to decreased peripheral resistance and lower blood pressure. Captopril is commonly used for the treatment of hypertension and heart failure.

Option B - Spironolactone: Spironolactone is a potassium-sparing diuretic and aldosterone receptor antagonist. Its mode of action involves binding to aldosterone receptors, which are located in the kidneys and other tissues. By binding to these receptors, spironolactone inhibits the effects of aldosterone, a hormone that promotes sodium and water retention while increasing potassium excretion. As a result, spironolactone promotes sodium and water excretion while retaining potassium, leading to diuresis (increased urine production) and the maintenance of potassium balance. Spironolactone treats conditions such as hypertension, heart failure, and certain hormonal disorders.

Option D - Losartan: Losartan is an angiotensin II receptor antagonist targeting the angiotensin II type 1 (AT1) receptors. Its mode of action involves selectively blocking these receptors, preventing the binding of angiotensin II. By blocking the actions of angiotensin II, losartan causes vasodilation and reduces the vasoconstrictive effects of angiotensin II. This leads to a decrease in peripheral resistance and blood pressure. Losartan is commonly prescribed for the management of hypertension and can also be used in certain cases of heart failure.

Solution for Question 37:

Correct Option A - Amiodarone:

- Amiodarone is an antiarrhythmic drug that is used to treat various types of arrhythmias, including ventricular arrhythmias. It works by prolonging cardiac cells' action potential and refractory period, thereby reducing the likelihood of abnormal electrical impulses and arrhythmias.
- However, amiodarone has several potential side effects, including pulmonary toxicity, thyroid dysfunction, liver dysfunction, corneal deposits, skin discoloration, and peripheral neuropathy. The side effects mentioned in the question, such as fatigue, dyspnea, weight gain, and pulmonary fibrosis, are consistent with amiodarone-induced pulmonary toxicity.
- Amiodarone-induced pulmonary toxicity can present in different forms, including interstitial pneumonitis, organizing pneumonia, and pulmonary fibrosis. It typically occurs after several months to years of treatment and can be irreversible in severe cases. The incidence of amiodarone-induced pulmonary toxicity is estimated to be around 5%, but it can be higher in elderly patients and those with pre-existing lung disease.

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Incorrect Options:

Option B - Atenolol: Atenolol is a beta-blocker that is used to treat hypertension and some types of arrhythmias. Its side effects can include fatigue, dizziness, bradycardia, and hypotension, but it is not associated with pulmonary toxicity.

Option C - Aspirin: Aspirin is an antiplatelet drug used to prevent thrombotic events, such as myocardial infarction and stroke. Its side effects include gastrointestinal bleeding, tinnitus, and increased bleeding risk, but it is not associated with pulmonary toxicity.

Option D - Spironolactone: Spironolactone is a potassium-sparing diuretic that is used to treat heart failure and hypertension. Its side effects can include hyperkalemia, gynecomastia, and

menstrual irregularities, but it is not associated with pulmonary toxicity.

Solution for Question 38:

Correct Option C - Rebound hypertension:

- Clonidine is a centrally acting alpha-2 adrenergic receptor agonist commonly used to manage hypertension, opioid withdrawal, and smoking cessation. When used for smoking cessation, clonidine helps to reduce the withdrawal symptoms associated with nicotine addiction, such as cravings, anxiety, and irritability.
- One of the side effects of clonidine therapy is rebound hypertension, which can occur when the drug is abruptly discontinued as alpha 1 receptors are upregulated. Rebound hypertension is characterized by a sudden and significant increase in blood pressure, which can cause symptoms such as headache, dizziness, and confusion.
- In this scenario, the patient stopped taking clonidine tablets and now presents with a headache, a common symptom of rebound hypertension. When clonidine is discontinued, the alpha-2 adrenergic receptors become less active, which can lead to increased sympathetic activity and a subsequent increase in blood pressure.

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One of the side effects of clonidine therapy is rebound hypertension, which can occur when the drug is abruptly discontinued as alpha 1 receptors are upregulated. Rebound hypertension is characterized by a sudden and significant increase in blood pressure, which can cause symptoms such as headache, dizziness, and confusion.

In this scenario, the patient stopped taking clonidine tablets and now presents with a headache, a common symptom of rebound hypertension. When clonidine is discontinued, the alpha-2 adrenergic receptors become less active, which can lead to increased sympathetic activity and a subsequent increase in blood pressure.

Incorrect Options:

Option A - Postural hypotension: Postural hypotension is a condition characterized by a sudden drop in blood pressure upon standing up, which can cause dizziness and fainting. This is not the

cause of the headache in this scenario, as the patient's symptoms are inconsistent with posturalhypotension.

Option B - Receptor Downregulation: Receptor Downregulation is a phenomenon in which the number of receptors on the cell surface decreases in response to an increase in the stimulation of the receptors. This is not the cause of the headache in this scenario, as upregulation of receptors is what leads to rebound hypetension.

Option D - Receptor hypersensitivity: Receptor hypersensitivity is a condition in which the receptors on the cell surface become more sensitive to the ligand, which can result in an exaggerated response to the drug. This is not the cause of the headache in this scenario, as hypersensitivity of the receptors would not result in rebound hypertension or headache.

Solution for Question 39:

Correct Option D - Losartan acts as a thromboxane A2 antagonist and inhibits platelet aggregation.:

- Losartan is an angiotensin II receptor antagonist that blocks the binding of angiotensin II to the AT1 receptor subtype, leading to the dilation of blood vessels and reduced blood pressure. Losartan blocks thromboxane A2 receptors and has some platelet antiaggregatory property.

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Incorrect Options:

Option A - The dose of telmisartan should be reduced in renal failure but not in hepatic failure.: Telmisartan is primarily eliminated by the liver and metabolized by the cytochrome P450 system. In case of hepatic impairment, the dose of telmisartan should be reduced. The dose does not need to be reduced in renal failure.

Option B - The dose of irbesartan need not be reduced in case of mild-moderate hepatic failure and renal failure.: Irbesartan is metabolized by the liver and excreted by the kidneys. In case of mild to moderate hepatic impairment or renal impairment, the dose of irbesartan should be reduced.

Option C - The dose of candesartan should be reduced in renal failure but not in mild-moderate liver failure.: Candesartan is mainly eliminated through the liver and excreted through the bile. In case of mild to moderate liver failure, the dose of candesartan should be reduced, but no dose adjustment is required in renal failure.

Solution for Question 40:

Correct Option C - 1-a, 2-d, 3-b, 4-c:

- (1) Na⁺ channel blocker - Quinidine: Quinidine is a class I antiarrhythmic drug that acts by blocking sodium channels in the cardiac cells. By blocking these channels, it decreases the influx of sodium ions during the depolarization phase of the cardiac action potential. This leads to a reduction in the conduction velocity and prolongation of the refractory period in cardiac tissues, helping to suppress abnormal electrical impulses and arrhythmias (1-a).
- (2) K⁺ channel blocker - Ibutilide: Ibutilide is a class III antiarrhythmic drug that primarily blocks potassium channels in the heart. By inhibiting these channels, it prolongs the repolarization phase of the cardiac action potential, resulting in an increase in the action potential duration and effective refractory period. This action helps to prevent re-entry arrhythmias and stabilize the heart's electrical activity (2-d).
- (3) Na⁺ K⁺ ATPase inhibitor - Digoxin: Digoxin is not primarily considered an antiarrhythmic drug but rather a cardiac glycoside. However, it does have some antiarrhythmic effects. Digoxin inhibits the activity of the Na⁺ K⁺ ATPase pump, and increases intracellular calcium levels, leading to increased contractility of the heart. This indirectly helps to control certain arrhythmias, such as atrial fibrillation (3-b).
- (4) Beta-blocker - Esmolol: Esmolol belongs to the class of beta-blockers, specifically a short-acting selective beta-1 adrenergic receptor blocker. It works by blocking the beta-1 adrenergic receptors in the heart, thereby reducing the effects of sympathetic stimulation. By blocking these receptors, esmolol decreases the heart rate, slows down the conduction of electrical signals through the atrioventricular node, and reduces the force of myocardial contraction. (4-c).

(1) Na⁺ channel blocker - Quinidine: Quinidine is a class I antiarrhythmic drug that acts by blocking sodium channels in the cardiac cells. By blocking these channels, it decreases the influx of sodium ions during the depolarization phase of the cardiac action potential. This leads to a reduction in the conduction velocity and prolongation of the refractory period in cardiac tissues, helping to suppress abnormal electrical impulses and arrhythmias (1-a).

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Incorrect Options:

- Options A, B and D are incorrect choice.

Solution for Question 41:

Correct Option C - Angiotensin – converting enzyme inhibitor:

- ACE inhibitors should be discontinued in women planning for pregnancy or as soon as pregnancy is confirmed. ACE inhibitors have been associated with an increased risk of fetal malformations, particularly in the first trimester. They can also cause complications such as reduced fetal growth, kidney problems, and low amniotic fluid levels.
- When considering preconception counseling for a chronic hypertensive woman, there are certain antihypertensive medications that should be discontinued due to potential risks to the developing fetus. Among the options provided:

ACE inhibitors should be discontinued in women planning for pregnancy or as soon as pregnancy is confirmed. ACE inhibitors have been associated with an increased risk of fetal malformations, particularly in the first trimester. They can also cause complications such as reduced fetal growth, kidney problems, and low amniotic fluid levels.

When considering preconception counseling for a chronic hypertensive woman, there are certain antihypertensive medications that should be discontinued due to potential risks to the developing fetus. Among the options provided:

Incorrect Options:

Option A - Labetalol: Labetalol is generally considered safe for use during pregnancy and is commonly used to manage hypertension in pregnant women. It does not need to be discontinued.

Option B - Calcium channel blockers: Calcium channel blockers are also considered safe for use during pregnancy and are commonly used to manage hypertension. They do not need to be discontinued.

Option D - Alpha methyldopa: Alpha methyldopa is generally considered safe for use during pregnancy and is commonly used to manage hypertension. It does not need to be discontinued. It has the most extensive safety data for use in pregnancy

Solution for Question 42:

Correct Option A - Lisinopril:

- Lisinopril is an angiotensin-converting enzyme (ACE) inhibitor used primarily for the treatment of hypertension (high blood pressure) and certain heart conditions. While Lisinopril is generally well-tolerated, it can cause several side effects.
- Angioedema: Although rare, Lisinopril can cause angioedema, which is swelling of the face, lips, tongue, throat, or extremities. This is a serious allergic reaction that requires immediate medical attention.
- Treated by Antihistamines + corticosteroids
- Sometimes involves larynx and causes laryngeal edema. SC injections of epinephrine are given in this case to produce bronchodilation.

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Treated by Antihistamines + corticosteroids

Sometimes involves larynx and causes laryngeal edema. SC injections of epinephrine are given in this case to produce bronchodilation.

Incorrect Options:

Option B - Nifedipine: Nifedipine works by blocking the influx of calcium ions into smooth muscle cells, specifically in the arterial walls. By blocking calcium channels, it inhibits the contractility of smooth muscle cells, leading to vasodilation and a decrease in peripheral vascular resistance. This results in a reduction in blood pressure and improved oxygen supply to the heart muscle.

Option C - Atenolol: Atenolol works by selectively blocking beta-1 adrenergic receptors in the heart, thereby reducing the effects of catecholamines (such as adrenaline or epinephrine) on the heart. By blocking these receptors, atenolol decreases the heart rate and cardiac output, leading to a decrease in blood pressure. It also reduces the contractility of the heart, which can be beneficial in certain cardiovascular conditions.

Option D - Clonidine: Clonidine works by stimulating alpha-2 adrenergic receptors in the brain, specifically in the central nervous system (CNS). Activation of these receptors leads to the inhibition of norepinephrine release from nerve terminals, resulting in a decrease in sympathetic outflow from the CNS. This reduction in sympathetic activity leads to a decrease in peripheral vascular resistance, heart rate, and blood pressure.

Solution for Question 43:

Correct Option C - ACE inhibitors:

- ACE inhibitors (angiotensin-converting enzyme inhibitors) are a class of medications commonly prescribed for hypertension. They work by blocking the action of angiotensin-converting enzyme, which converts angiotensin I to angiotensin II, a potent vasoconstrictor. Dry cough is a well-known and relatively common side effect of ACE inhibitors, occurring in up to 10% of patients. The cough is thought to be due to an increase in bradykinin levels caused by ACE inhibition.

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patients. The cough is thought to be due to an increase in bradykinin levels caused by ACEinhibition.

Incorrect Options:

Option A - Calcium channel blockers: Calcium channel blockers (CCBs) are a class of medications commonly used to treat hypertension. They work by blocking calcium channels in smooth muscle cells, leading to vasodilation and a reduction in blood pressure. Dry cough is not a common side effect associated with calcium channel blockers.

Option B - Beta-blockers: Beta-blockers are another class of medications used to treat hypertension. They work by blocking the effects of adrenaline (epinephrine) on beta receptors, resulting in decreased heart rate and blood pressure. While beta-blockers can cause side effects such as fatigue, dizziness, and sexual dysfunction, dry cough is not typically associated with this class of drugs.

Option D - Diuretics: Diuretics, also known as water pills, are frequently used as a first-line treatment for hypertension. They work by increasing the excretion of sodium and water from the body, resulting in reduced blood volume and lower blood pressure. Dry cough is not a commonly reported side effect of diuretics.

Solution for Question 44:

Correct Option B - IV labetalol plus magnesium sulphate:

- The combination of intravenous (IV) labetalol and magnesium sulfate is commonly used in the management of severe hypertension and seizures associated with preeclampsia or eclampsia.
- Labetalol is a beta-blocker that helps lower blood pressure by blocking the effects of certain stress hormones on the heart and blood vessels. Magnesium sulfate is an anticonvulsant medication that helps prevent and control seizures in women with preeclampsia or eclampsia.

The combination of intravenous (IV) labetalol and magnesium sulfate is commonly used in the management of severe hypertension and seizures associated with preeclampsia or eclampsia.

Labetalol is a beta-blocker that helps lower blood pressure by blocking the effects of certain stress hormones on the heart and blood vessels. Magnesium sulfate is an anticonvulsant medication that helps prevent and control seizures in women with preeclampsia or eclampsia.

Incorrect Options:

Option A - Oral Labetalol: Oral Labetalol alone may not be sufficient for immediate control of severe hypertension and seizure activity in this critical situation. IV administration allows for rapid onset of action.

Option C - IV Lorazepam: IV Lorazepam is a sedative and anti-anxiety medication but is not the first-line treatment for severe hypertension and seizures associated with preeclampsia or eclampsia.

Option D - IV Enalapril: IV Enalapril is an angiotensin-converting enzyme (ACE) inhibitor and is contraindicated during pregnancy due to potential harm to the fetus, especially in the presence of severe hypertension and organ dysfunction.

Solution for Question 45:

Correct Option B - Ranolazine:

- Among the options provided, ranolazine is the drug that acts by inhibiting the late sodium current in the heart.
- Ranolazine is an antianginal medication that is primarily used in the treatment of chronic stable angina. It works by inhibiting the late sodium current in cardiomyocytes, which helps to reduce intracellular sodium accumulation during ischemic conditions. By inhibiting the late sodium current, ranolazine indirectly reduces calcium overload in cardiac cells and improves myocardial relaxation, leading to an anti-ischemic effect.
- The inhibition of the late sodium current by ranolazine has several effects on cardiac function. It can help to decrease ventricular diastolic tension, reduce myocardial oxygen consumption, and improve myocardial perfusion. These actions contribute to its efficacy in reducing angina symptoms.

Among the options provided, ranolazine is the drug that acts by inhibiting the late sodium current in the heart.

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The inhibition of the late sodium current by ranolazine has several effects on cardiac function. It can help to decrease ventricular diastolic tension, reduce myocardial oxygen consumption, and improve myocardial perfusion. These actions contribute to its efficacy in reducing angina symptoms.

Incorrect Options:

Option A - Ivabradine: Ivabradine, on the other hand, acts as a selective inhibitor of the pacemaker I_f current in the sinoatrial node, leading to heart rate reduction. It is primarily used for the treatment of stable angina and heart failure with reduced ejection fraction.

Option C - Trimetazidine: Trimetazidine is a metabolic agent that improves myocardial energy metabolism by enhancing glucose oxidation and reducing fatty acid oxidation. It does not directly inhibit the late sodium current.

Option D - Fasudil: Fasudil is a calcium channel blocker that primarily acts by inhibiting Rho-kinase, an enzyme involved in regulating smooth muscle contraction. It is mainly used for the treatment of cerebral vasospasm after subarachnoid hemorrhage and is not directly related to inhibiting the late sodium current in the heart.

Solution for Question 46:

Correct Option C - Only bradykinin:

- The above case is that of a patient with chronic cough who has been on antihypertensives recently.
- This is because chronic dry cough is a common side effect of ACE inhibitors.
- This is so as there is accumulation of bradykinin, prostaglandins and substance P due to inhibition of Angiotensin-II.
- They cause a persistent brassy, dry cough.

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This is because chronic dry cough is a common side effect of ACE inhibitors.

This is so as there is accumulation of bradykinin, prostaglandins and substance P due to inhibition of Angiotensin-II.

They cause a persistent brassy, dry cough.

Incorrect Options:

Option A - Bradykinin and renin: The chronic dry cough is because of increased levels of bradykinin and renin is not involved.

Option B - Angiotensin II and bradykinin: Angiotensin II and bradykinin is not the correct choice as angiotensin is not involved.

Option D - Histamine: Histamine does not cause chronic dry cough.

Solution for Question 47:

Correct Option A - Nicorandil:

- Nicorandil is a medication that causes the activation of ATP-sensitive potassium (K⁺-ATP) channels. It is a hybrid compound that combines nitrate-like properties with potassium channel activation effects. By activating K⁺-ATP channels, Nicorandil promotes arterial and venous dilation, leading to decreased vascular resistance and increased blood flow. It is primarily used in the treatment of angina pectoris.

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Incorrect Options:

Option B - Molsidomine: Molsidomine is a nitric oxide donor that primarily works by releasing nitric oxide, leading to vasodilation. However, it does not directly activate K⁺-ATP channels.

Option C - Isosorbide dinitrate: Isosorbide dinitrate is a nitrate medication that acts as a vasodilator. It primarily works by releasing nitric oxide, which relaxes smooth muscles and dilates blood vessels.

Option D - Nitroglycerin: Nitroglycerin is also a nitrate medication that acts as a vasodilator by releasing nitric oxide.

Solution for Question 48:

Correct Option C - Nitrates:

- Nitrates: Nitrates are a class of medications commonly used in the treatment of angina.
- Sildenafil, the medication used for erectile dysfunction, should not be prescribed concomitantly with nitrates.
- Both medications can cause a significant drop in blood pressure when used together, leading to a potentially dangerous condition called hypotension.
- Therefore, nitrates are the correct option for the given scenario.

Nitrates: Nitrates are a class of medications commonly used in the treatment of angina.

Sildenafil, the medication used for erectile dysfunction, should not be prescribed concomitantly with nitrates.

Both medications can cause a significant drop in blood pressure when used together, leading to a potentially dangerous condition called hypotension.

Therefore, nitrates are the correct option for the given scenario.

Incorrect Options:

Option A - Calcium channel blockers: Calcium channel blockers are commonly used in the treatment of angina to relax and widen the blood vessels, reducing the workload on the heart and improving blood flow. They are generally not contraindicated with sildenafil, the medication used to treat erectile dysfunction. Therefore, calcium channel blockers are an incorrect option for the given scenario.

Option B - Beta blockers: Beta blockers are another class of medications commonly used to treat angina. They work by reducing the heart rate and blood pressure. While beta blockers can potentially contribute to erectile dysfunction, they are not an absolute contraindication for sildenafil use. In fact, sildenafil can be prescribed cautiously in patients taking beta blockers under appropriate medical supervision. Therefore, beta blockers are an incorrect option for the given scenario.

Option D - ACE inhibitors: ACE inhibitors are a class of medications commonly used to treat high blood pressure and heart failure. They do not typically cause significant interactions or contraindications with sildenafil. Therefore, ACE inhibitors are an incorrect option for the given scenario.

Solution for Question 49:

Correct Option B - Quinidine:

- Quinidine is an antiarrhythmic medication that can cause QT prolongation, along with other cardiac rhythm abnormalities. It is known to have a potential proarrhythmic effect.

Incorrect Options:

Option A - Lignocaine: Lignocaine (also known as Lidocaine) is a local anesthetic and is not typically associated with significant QT prolongation. It can shorten the QT interval.

Option C - Magnesium sulfate: Magnesium sulfate is commonly used in the treatment of certain cardiac arrhythmias and is not typically associated with significant QT prolongation.

Option D - All of the above: Lignocaine and magnesium sulfate are not strongly associated with QT prolongation. Only quinidine is known for its potential to prolong the QT interval.

Solution for Question 50:

Correct Option D - Neutral endopeptidase inhibitor:

- Sacubitril is a neutral endopeptidase inhibitor, also known as a neprilysin inhibitor. It is used in the treatment of chronic heart failure.
- The mechanism of action of sacubitril involves inhibiting the enzyme neprilysin, which is responsible for the degradation of several vasoactive peptides, including natriuretic peptides, bradykinin, and adrenomedullin.
- By inhibiting neprilysin, sacubitril increases the levels of these vasoactive peptides, leading to vasodilation, diuresis, and natriuresis.
- This helps to reduce the workload on the heart and improve cardiac function in patients with heart failure.

Sacubitril is a neutral endopeptidase inhibitor, also known as a neprilysin inhibitor. It is used in the treatment of chronic heart failure.

The mechanism of action of sacubitril involves inhibiting the enzyme neprilysin, which is responsible for the degradation of several vasoactive peptides, including natriuretic peptides, bradykinin, and adrenomedullin.

By inhibiting neprilysin, sacubitril increases the levels of these vasoactive peptides, leading to vasodilation, diuresis, and natriuresis.

This helps to reduce the workload on the heart and improve cardiac function in patients with heart failure.

Incorrect Options:

Option A - Angiotensin II inhibitor: Angiotensin II inhibitors, also known as angiotensin receptor blockers (ARBs), work by blocking the action of angiotensin II, a hormone that constricts blood vessels and promotes water and salt retention. They are commonly used in the treatment of hypertension and heart failure, but sacubitril is not an angiotensin II inhibitor.

Option B - ACE inhibitor: ACE inhibitors, or angiotensin-converting enzyme inhibitors, block the enzyme ACE, which converts angiotensin I to angiotensin II. Sacubitril is not an ACE inhibitor.

Option C - Renin inhibitor: Renin inhibitors target the enzyme renin, which plays a key role in the production of angiotensin II. By inhibiting renin, these drugs reduce the production of angiotensin II and help lower blood pressure. However, sacubitril is not a renin inhibitor.

Solution for Question 51:

Correct Option B - Hydralazine:

- Isosorbide dinitrate can be used in combination with hydralazine for the treatment of congestive heart failure.
- Hydralazine is a direct-acting vasodilator that works by relaxing the smooth muscles of the blood vessels, thereby reducing peripheral resistance and improving cardiac output.

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Hydralazine is a direct-acting vasodilator that works by relaxing the smooth muscles of the blood vessels, thereby reducing peripheral resistance and improving cardiac output.

Incorrect Options:

Option A - Minoxidil: Minoxidil is a potent vasodilator primarily used for the treatment of hypertension and male pattern baldness. It is not commonly used in combination with isosorbide dinitrate for congestive heart failure.

Option C - Nimodipine: These are calcium channel blockers primarily used for the treatment of cerebral vasospasm following subarachnoid hemorrhage. They are not typically used in combination with isosorbide dinitrate for congestive heart failure.

Option D - Nitrendipine: These are calcium channel blockers primarily used for the treatment of cerebral vasospasm following subarachnoid hemorrhage. They are not typically used in combination with isosorbide dinitrate for congestive heart failure.

Solution for Question 52:

Correct Option B - Release of NO:

- The likely mechanism of action for nitroglycerin in relieving acute chest pain is the release of nitric oxide (NO).
- Nitroglycerin is a medication that belongs to the class of drugs known as nitrates. When administered sublingually, nitroglycerin is rapidly absorbed into the bloodstream and undergoes biotransformation to release nitric oxide (NO). Nitric oxide acts as a potent vasodilator, causing relaxation of the smooth muscles in the blood vessels, particularly in the veins. This leads to venous dilation, which reduces preload and decreases the workload on the heart.
- By dilating the veins, nitroglycerin reduces the amount of blood returning to the heart, thereby reducing cardiac preload. This results in a decrease in ventricular filling pressure and myocardial oxygen demand, relieving the chest pain associated with conditions such as angina.

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By dilating the veins, nitroglycerin reduces the amount of blood returning to the heart, thereby reducing cardiac preload. This results in a decrease in ventricular filling pressure and myocardial oxygen demand, relieving the chest pain associated with conditions such as angina.

Incorrect Options:

Option A, C & D - Release of endothelin, Calcium channel blockade & Beta blockade are not mechanisms of action for nitroglycerin. Endothelin is a potent vasoconstrictor, and its release would have the opposite effect of nitroglycerin in causing vasoconstriction. Calcium channel blockade and beta blockade are mechanisms of action for other types of medications used to treat angina, but they are not the primary mechanisms of action for nitroglycerin.

Solution for Question 53:

Correct Option D - Lignocaine:

- Lignocaine (also known as Lidocaine): Class 1 Lignocaine is the preferred drug for the treatment of ventricular tachycardia, a condition characterized by rapid, abnormal heart rhythms originating from the ventricles. It is administered intravenously for its antiarrhythmic effects.

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Incorrect Options:

Option A - Digoxin: Digoxin is primarily used for the treatment of heart failure and certain arrhythmias but is not the preferred drug for ventricular tachycardia.

Option B - Propranolol: Propranolol is a beta-blocker medication used for various cardiovascular conditions, including arrhythmias. While it can be used for certain types of arrhythmias, other antiarrhythmic drugs like Lignocaine are preferred for ventricular tachycardia.

Option C - Diltiazem: Diltiazem is a calcium channel blocker used for the treatment of hypertension and certain heart rhythm disorders, but it is not typically the preferred drug for ventricular tachycardia.

Solution for Question 54:

Correct Option C - Adenosine:

- Adenosine is the drug of choice for acute termination of PSVT. It is a potent and rapidly acting medication that is administered intravenously during an episode of PSVT. Adenosine works by temporarily blocking the electrical conduction in the heart, thereby interrupting the abnormal circuit responsible for PSVT. The use of adenosine results in a brief pause in the heart rhythm, followed by restoration of normal sinus rhythm in most cases.

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Incorrect Options:

Option A - Digoxin: Digoxin is a medication commonly used to treat heart failure and certain arrhythmias. However, it is not typically the drug of choice for acute episodes of PSVT. Digoxin has a slower onset of action and is more commonly used for long-term management of certain types of arrhythmias.

Option B - Propranolol: Propranolol is a beta-blocker that can be effective in managing various types of tachycardia, including PSVT. It works by blocking the effects of adrenaline on the heart, thereby slowing down the heart rate. While propranolol may be used in some cases of PSVT, it is not considered the first-line drug of choice for acute termination of an episode.

Option D - Diltiazem: Diltiazem is a calcium channel blocker that can also be used in the management of PSVT. It works by reducing the influx of calcium into the heart muscle cells, leading to a decrease in heart rate and contractility. Diltiazem is commonly used in cases where adenosine is contraindicated or ineffective. It is particularly useful in patients with underlying cardiovascular conditions, such as hypertension or coronary artery disease.

Solution for Question 55:

Correct Option A - Nimodipine:

- Nimodipine is a calcium channel blocker that is specifically used for the prevention and treatment of cerebral vasospasm following subarachnoid hemorrhage (SAH). It has a

selective action on cerebral blood vessels, dilating them and improving blood flow in the brain. Nimodipine is the drug of choice for treating vasospasm in this context due to its specific action on cerebral arteries.

Nimodipine is a calcium channel blocker that is specifically used for the prevention and treatment of cerebral vasospasm following subarachnoid hemorrhage (SAH). It has a selective action on cerebral blood vessels, dilating them and improving blood flow in the brain. Nimodipine is the drug of choice for treating vasospasm in this context due to its specific action on cerebral arteries.

Incorrect Options:

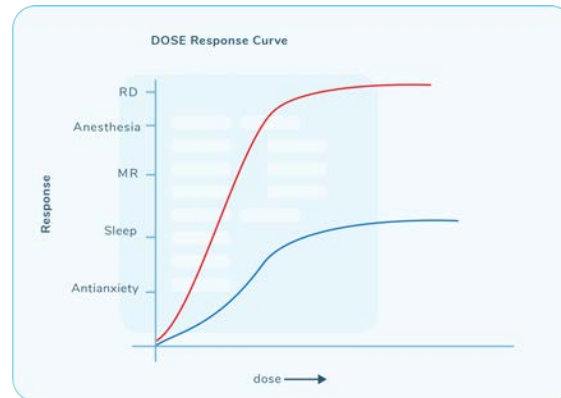
Option B - Felodipine: Felodipine is a calcium channel blocker that primarily acts on peripheral blood vessels, resulting in vasodilation and reduced systemic vascular resistance. It is mainly used for the treatment of hypertension. However, felodipine does not have a specific indication for the treatment of cerebral vasospasm following SAH.

Option C - Amlodipine: Amlodipine is a calcium channel blocker that acts primarily on peripheral blood vessels, similar to felodipine. It is widely used for the treatment of hypertension and angina pectoris. While amlodipine may have some vasodilatory effects on cerebral blood vessels, it does not have a specific indication for the treatment of cerebral vasospasm following SAH.

Option D - Nitrendipine: Nitrendipine is another calcium channel blocker that primarily acts on peripheral blood vessels, leading to vasodilation. It is commonly used for the treatment of hypertension. However, like felodipine and amlodipine, it does not have a specific indication for the treatment of cerebral vasospasm.

Sedatives Hypnotics, Drugs for Insomnia, and Alcohol

1. Which of the following statements regarding the dose-response curves for benzodiazepines and barbiturates is correct?



- A. Benzodiazepines demonstrate a steeper dose-response curve compared to barbiturates.
- B. Benzodiazepines have a narrower therapeutic index compared to Barbiturates
- C. Both benzodiazepines and barbiturates display a linear dose-response relationship.
- D. Benzodiazepines and barbiturates show similar potency but differ in their maximal efficacy on the dose-response curve.

2. Which of the following statements regarding GABA receptors is incorrect?
GABA-A receptors are dimeric metabotropic receptors and mediate inhibitory neurotransmission. GABA-B receptors are inhibitory G protein-coupled receptors and mediate slow inhibitory neurotransmission through the efflux of potassium. Activation of GABA-A receptors results in the efflux of chloride ions into the neuron. α and γ subunits of GABA-A receptors modulate drug efficacy. GABA-C receptors are structurally and functionally similar to GABA-A receptors

- A. 1 and 3

- B. 1, 3 and 5
 - C. 3 and 5
 - D. 2, 3 and 5
-

3. Arrange the following barbiturates based on their duration of action from shortest to longest: Secobarbital Phenobarbital Thiopental Pentobarbital

- A. 3, 1, 2, 4
 - B. 2, 1, 3, 4
 - C. 1, 2, 4, 3
 - D. 2, 1, 3, 4
-

4. A 42-year-old male with treatment-resistant major depressive disorder is scheduled for electroconvulsive therapy (ECT). The anesthesiologist plans to use a rapid-onset medication to minimize the side effects and facilitate quick recovery. Which of the following is the drug of choice for this patient?

- A. Thiopental
 - B. Propofol
 - C. Methohexital
 - D. Midazolam
-

5. A 2-day-old neonate is brought to the hospital with abnormal jerking movements in arms and legs. On examination, electroencephalogram (EEG) recording revealed abnormal electrical activity in the brain. What is the drug of choice for managing this patient condition?

- A. Phenobarbital

- B. Phenytoin
 - C. Levetiracetam
 - D. Midazolam
-

6. A 25-year-old student with a past history of suicide is brought to the emergency by his roommate, who found him unresponsive with an empty bottle of unknown pills. On examination, the patient is comatose with shallow breathing, hypotension, and decreased reflexes with mild pupillary dilatation. What is the most appropriate treatment for this patient's condition?

- A. Hemodialysis
 - B. Sodium bicarbonate
 - C. Flumazenil
 - D. Naloxone
-

7. Duration of action of which of the following benzodiazepines is determined by its rate of biotransformation?

- A. Lorazepam
 - B. Flurazepam
 - C. Diazepam
 - D. Clonazepam
-

8. A patient presents with a history of rigorous movements of the body associated with uprolling of eyes and wetting. Which of the following drugs will most likely not be suitable for this patient?

- A. Clonazepam

- B. Flurazepam
 - C. Clorazepate
 - D. Lorazepam
-

9. Which of the following drugs selectively helps improve insomnia without muscle relaxation?

- A. Clobazam
 - B. Zopiclone
 - C. Flurazepam
 - D. Carisoprodol
-

10. A 45-year-old blind patient presents with difficulty falling asleep and maintaining sleep. The patient reports a history of insomnia that has persisted despite trying various lifestyle modifications. Which of the following drugs is approved for the treatment of this condition?

- A. Tasimelteon
 - B. Zolpidem
 - C. Etizolam
 - D. Lorazepam
-

11. A patient with sleep disturbances is prescribed suvorexant. How does this drug help in improving insomnia?

- A. Inhibiting H1 histamine receptors
- B. Serotonergic 5HT2A/2C antagonism

- C. Orexin receptor antagonism
 - D. Inhibiting serotonergic 5HT_{2C} receptor alone
-

12. A 35-year-old male presents with severe headache, nausea, vomiting, blurred vision, disorientation, and fruity breath odour after consuming homemade liquor 12 hours ago. Blood gas analysis shows pH 7.25, bicarbonate 12 mmol/L, and anion gap metabolic acidosis. What is the drug of choice for his condition?

- A. N-acetylcysteine
 - B. Fomepizole
 - C. Atropine
 - D. Naloxone
-

13. A 42-year-old man with a history of chronic alcohol abuse is admitted to the hospital due to severe alcohol withdrawal symptoms. After stabilisation, he expresses his desire to overcome his alcohol dependence and seeks assistance from the healthcare team. Which of the following is NOT an anti-craving drug commonly used in the management of alcohol dependence?

- A. Naltrexone
 - B. Acamprosate
 - C. Disulfiram
 - D. Topiramate
-

14. Which of the following statements regarding the mechanism of action of disulfiram is incorrect?

- A. Disulfiram inhibits the enzyme aldehyde dehydrogenase, leading to an accumulation of acetaldehyde in the body.
 - B. Acetaldehyde accumulation results in unpleasant symptoms such as nausea, vomiting, and headache when alcohol is consumed.
 - C. Disulfiram increases the production of dopamine in the brain, reducing the desire to consume alcohol.
 - D. Disulfiram therapy is often used as an adjunct in the treatment of alcohol dependence.
-

15. A patient recovering from alcohol dependence on disulfiram developed a fungal infection and was given an antifungal medication. Shortly after, the patient developed symptoms reminiscent of a disulfiram-like reaction. Which of the following antifungal drugs is most likely to have caused this reaction?

- A. Fluconazole
 - B. Clotrimazole
 - C. Amphotericin B
 - D. Griseofulvin
-

Correct Answers

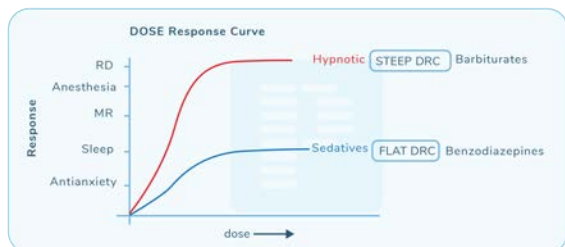
Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	3
Question 5	1
Question 6	1
Question 7	1
Question 8	2
Question 9	2
Question 10	1
Question 11	3
Question 12	2
Question 13	4
Question 14	3
Question 15	4

Solution for Question 1:

Correct Answer: A) Benzodiazepines demonstrate a steeper dose-response curve compared to barbiturates.

Explanation:

Benzodiazepines typically have a higher affinity for their receptor sites and thus exhibit a sharper increase in response with increasing doses compared to barbiturates. This means that even small increases in benzodiazepine dosage can lead to significant changes in the response.



Option B:

- Barbiturates have a narrower therapeutic index compared to benzodiazepines, meaning that the difference between the effective dose and toxic dose is smaller.
- This makes barbiturates more prone to causing overdose and toxicity with increasing doses.

Option C:

- A linear dose-response relationship implies that the response increases proportionally with increasing dosage.
- However, this is not typically observed with benzodiazepines and barbiturates, as their response tends to plateau or even decrease at higher doses due to receptor saturation.

Option D:

- Benzodiazepines and barbiturates may have similar potency in terms of the dose required to produce a certain effect, but they differ in their maximal efficacy.
- Benzodiazepines often have a lower maximal efficacy compared to barbiturates.

Solution for Question 2:

Correct Answer: A) 1 and 3

Explanation:

GABA receptors:

- Chief inhibitory receptors of the CNS.
- Termination of action by reuptake through GAT (GABA transporters) in neurons.

- Three types of GABA receptors:

Type s	MOA	Drugs
GABA A	Pentameric ionotropic ligand-gated Cl ⁻ channels 5 Subunits: 2 α , 2 β and 1 γ α and γ subunits affect the efficacy of benzodiazepine binding and channel modulation Fast inhibitory neurotransmission through hyperpolarisation of the neurons by Cl ⁻ ion influx	1. Benzodiazepines (diazepam etc.) Acts as a positive allosteric modulator Increase the frequency of opening of the GABA-A receptor. 2. Barbiturates Binds at multiple sites Increase mean open time with no effect on frequency. 3. Volatile anaesthetics 4. Brexanolone (neuroactive steroid) 5. Ethanol
GABA B	Metabotropic inhibitory G protein-coupled receptor Heterodimers with 2 subunits Gi type of GPCR Inhibit adenylyl cyclase, activate K ⁺ channels (K ⁺ efflux), and reduce Ca ²⁺ conductance Slow inhibitory neurotransmission	1. Baclofen (muscle relaxant) 2. gamma-hydroxybutyric acid (narcolepsy)
GABA C	Ionotropic receptor Structurally and functionally similar to GABA A receptors. Location: Retina	

Solution for Question 3:

Correct Answer: A) 3, 1, 2, 4

Explanation:

Classification of barbiturates:

Type	Duration	Drugs
Ultra short-acting	5-15 mins	Redistribution into fats Thiopentone: Sodium pentothal Uses: general anesthesia, narcoanalysis Side effect: Acute intermittent porphyria, thrombosis with arterial

		injection Methohexitone: Use: DOC in Electroconvulsive therapy Propofol Etomidate Ketamine Fentanyl (Opioid)
Short-acting	3-8 hours	Drugs ending with -barbitone or -barbital Butobarbitone Amobarbitone Secobarbital Pentobarbitone
Long-acting	Days	Phenobarbitone (Phenobarbital) Primidone: Used in essential tremors

Solution for Question 4:

Correct Answer: C) Methohexital

Explanation:

Electroconvulsive therapy is advised for treatment-resistant psychiatric disorders (severe depression, schizophrenia, and catatonia) and is performed under general anaesthesia with methohexital as the drug of choice. It involves delivering a small electric current into the brain, creating a brief seizure to modify brain chemistry.

Methohexital:

- Intravenous ultra-short-acting.
- Action: 5–7 min.
- Elimination: more rapid clearance than other barbiturates.
- Can increase ictal activity, and seizures.
- Uses: DOC for providing anaesthesia during Electroconvulsive therapy (ECT) to prevent seizure-induced trauma.
- Note- Other drug used for ECT: Succinylcholine.

Incorrect options:

Thiopental (Option A) is also an ultra-short-acting barbiturate, but methohexital is preferred for ECT due to its rapid clearance and ictal activity.

Propofol (Option B) is a commonly used anaesthetic but is not the first choice for ECT due to its potential for prolonged recovery and hypotension.

Midazolam (Option D) is a benzodiazepine used for sedation and treatment of status epilepticus. However, it is not recommended for ECT due to its longer recovery time.

Solution for Question 5:

Correct Answer: A) Phenobarbital

Explanation:

Neonatal seizures are described as sudden, paroxysmal, abnormal electrical activity in the brain that manifests as abnormal tonic-clonic movements, impaired attention, and changes in breathing patterns. Phenobarbital is the treatment of choice for neonatal seizures due to its efficacy and proven safety profile.

Phenobarbital (Phenobarbitone)	
Uses	DOC in Neonatal seizures DOC in Neonatal jaundice: Crigler Najjar syndrome type 2 DOC in Gilbert syndrome
MOA	Increase GABA-A receptor Cl ⁻ channel opening duration without affecting the frequency.
Pharmacokinetics	CYP3A4 Inducer (induces UDP-GT activity in Gilbert syndrome) pH-dependent renal excretion
Side effects	Paradoxical excitation: Seen in children Hyperkinesia Excessive sensation and pain perception Rapid metabolism of oral contraceptives
Contraindication	Acute intermittent porphyria (CYP enzyme inducers increase heme demand)

Phenytoin (Option B) is not the first-line treatment for neonatal seizures since it requires frequent blood level monitoring and has negative neurodevelopmental effects in newborns.

Levetiracetam (Option C) can be used as an anti-seizure medication in children over one month old, but it is not the first-line treatment for neonatal seizures.

Midazolam (Option D) is used to treat refractory seizures in neonates but is not recommended as a first-line epileptic treatment in neonates.

Solution for Question 6:

Correct Answer: B) Sodium bicarbonate

Explanation:

This young patient's decreased consciousness, hypotension, respiratory depression, and mildly dilated pupils suggest barbiturate poisoning, which should be managed by forced diuresis with intravenous sodium bicarbonate.

Barbiturate Poisoning:

- Symptoms: Unconsciousness, shallow breathing, bullous eruption on skin.
- Signs: Hypotension and bradycardia (decrease cardiac contractility) Severe respiratory depression (medullary respiratory centre) Renal failure Pulmonary (atelectasis, oedema, and bronchopneumonia)
- Hypotension and bradycardia (decrease cardiac contractility)
- Severe respiratory depression (medullary respiratory centre)
- Renal failure
- Pulmonary (atelectasis, oedema, and bronchopneumonia)
- Treatment: Supportive measures, no specific antidote. Activated charcoal (reduce absorption within few hours of ingestion) Forced alkaline diuresis by using sodium bicarbonate (increase renal excretion) Last option: Hemodialysis Alternate options: Respiratory stimulants/Analeptics Bemegride Doxapram Picrotoxin
- Supportive measures, no specific antidote.
- Activated charcoal (reduce absorption within few hours of ingestion)
- Forced alkaline diuresis by using sodium bicarbonate (increase renal excretion)
- Last option: Hemodialysis
- Alternate options: Respiratory stimulants/Analeptics Bemegride Doxapram Picrotoxin
- Bemegride

- Doxapram
- Picrotoxin
- Hypotension and bradycardia (decrease cardiac contractility)
- Severe respiratory depression (medullary respiratory centre)
- Renal failure
- Pulmonary (atelectasis, oedema, and bronchopneumonia)
- Supportive measures, no specific antidote.
- Activated charcoal (reduce absorption within few hours of ingestion)
- Forced alkaline diuresis by using sodium bicarbonate (increase renal excretion)
- Last option: Hemodialysis
- Alternate options: Respiratory stimulants/Analeptics Bemegride Doxapram Picrotoxin
- Bemegride
- Doxapram
- Picrotoxin
- Bemegride
- Doxapram
- Picrotoxin

Hemodialysis (Option A) is used as a last resort, not as the first line in managing barbiturate poisoning.

Flumazenil (Option C) is an antidote for benzodiazepine overdose; symptoms include dilated pupils and slurred speech. However, respiratory depression is mild, hypotension is uncommon, and reflexes are usually preserved when compared to barbiturate poisoning.

Naloxone (Option D) is an antidote for opioid overdose, which presents with altered mental status, respiratory depression, and pinpoint pupils.

Solution for Question 7:

Correct Answer: A) Lorazepam

Explanation:

Among the given drugs, the duration of action of lorazepam is determined by its rate of biotransformation/metabolism & excretion.

Benzodiazepines can be classified into class A and class B.

CLASS A	CLASS B
Drugs in this class need to be converted into their active metabolites	Drugs in this class are in their active forms
Active metabolites are biotransformed slower than parent drugs - duration of action of drugs has little to no relationship with rate of biotransformation of parent drugs	Duration of action is determined by the rate of biotransformation of the drug itself
All benzodiazepines except those in class B	Lorazepam Oxazepam Temazepam Triazolam Midazolam

Solution for Question 8:

Correct Answer: B) Flurazepam

Explanation:

The above clinical vignette shows an episode of seizure and flurazepam will not be beneficial in controlling the same.

Flurazepam is used for insomnia.

Benzodiazepines and their uses:

Uses	Drugs	Uses	Drugs
Seizure control	Clobazam Clonazepam Clorazepate Diazepam Lorazepam Midazolam	Pre-anaesthesia & anaesthesia	Diazepam Lorazepam Midazolam Remimazolam

Status epilepticus	Diazepam Lorazepam	Alcohol withdrawal	Chlordiazepoxide Clorazepate Diazepam Lorazepam Midazolam Oxazepam
Muscle relaxation	Diazepam	Insomnia	Estazolam Flurazepam Quazepam Temazepam Triazolam
Mood disorders (anxiety, mania)	Alprazolam Chlordiazepoxide Clobazam Clonazepam Clorazepate Diazepam Lorazepam Midazolam Oxazepam		

Solution for Question 9:

Correct Answer: B) Zopiclone

Explanation:

Zopiclone, which belongs to a group of drugs called “atypical benzodiazepines/Z compounds” is the drug which helps in improving insomnia without any muscle relaxant effect.

Atypical benzodiazepines (non-benzodiazepine hypnotics):

Binding of the inhibitory neurotransmitter GABA to its receptor GABA-A is responsible for various responses like sedative-hypnotism, muscle relaxation, anxiolysis, amnesia and anticonvulsant.

GABA-A receptor has two alpha subunits:

- Alpha-1: Hypnotism.
- Alpha-2: Muscle relaxation (seizure activities).

Z compounds are hypnotics that selectively act on the alpha-1 subunit and don't exhibit any muscle relaxation.

- Zolpidem.

- Zaleplon.
- Zopiclone.
- Eszopiclone.

Clobazam (Option A) is a typical benzodiazepine that is predominantly used for seizure control, mood disorders, etc.

Flurazepam (Option C) has hypnotic properties but will also cause some degree of muscle relaxation.

Carisoprodol (Option D) is a pure muscle relaxant.

Solution for Question 10:

Correct Answer: A) Tasimelteon

Explanation:

Tasimelteon (melatonin agonist) is approved for the treatment of insomnia in blind patients.

Role of melatonin in sleep:

- Melatonin is a hormone that maintains the circadian rhythm by raising and falling cyclically.
- It typically rises during the evening and night as a person prepares to sleep and falls during wakefulness.
- Melatonin receptors: MT1 and MT2.
- Drugs that act as MT1 & MT2 receptor agonists are called melatonin congeners: ramelteon and tasimelteon.

Ramelteon and Tasimelteon:

Ramelteon	Tasimelteon
MT1 & MT2 agonist. Half-life: 2 hours. Metabolised by CYP1A, CYP2C, and CYP3A4 (has high first-pass metabolism). 8mg is taken 30 minutes before bedtime. Uses: Insomnia in Night shift workers. Jet lag.	MT1 & MT2 agonist. Uses: Maintain circadian rhythm in blind people. Night-time sleep disturbances in Smith-Magenis syndrome (reversal of circadian rhythm).

Solution for Question 11:

Correct Answer: C) Orexin receptor antagonism

Explanation:

Suvorexant acts by inhibiting the orexin receptors 1 & 2.

Orexins are produced by neurons in the hypothalamus and play an important role in sleep regulation. These neurons are active during wakefulness and dormant during sleep. Hence, inhibiting orexins will improve insomnia.

The drugs include suvorexant and lemborexant. The advantage of orexin antagonists over benzodiazepines is that they don't cause anterograde amnesia. The most common side effect is daytime somnolence.

Solution for Question 12:

Correct Answer: B) Fomepizole

Explanation:

The clinical vignette of a patient presenting with severe headache, nausea, vomiting, blurred vision, disorientation, and fruity breath odour after consumption of homemade liquor and blood gas analysis showing anion gap metabolic acidosis is suggestive of methanol poisoning.

Fomepizole is the antidote of choice for this condition.

Fomepizole inhibits alcohol dehydrogenase, preventing the formation of toxic metabolites such as formaldehyde and formic acid.

Methanol:

Adulteration in Country liquor. Solvent in industrial processes. Antifreeze.	
Mechanism of Action	Methanol ■ Formaldehyde (oxidation by alcohol dehydrogenase). Formaldehyde ■ Formic acid (oxidation by aldehyde dehydrogenase). Formic Acid ■ Inhibition of Cytochrome c Oxidase ■ Cellular Hypoxia and Metabolic Acidosis.
Features of Toxicity	CNS: Headache, dizziness, confusion, ataxia, seizures Ocular: Blurred ("snowfield" vision). Gastrointestinal: Nausea, vomiting, fruity breath and abdominal pain Metabolic: Severe metabolic acidosis, elevated anion gap
Treatment	Mnemonic: SHAFE Sodium Bicarbonate to correct metabolic acidosis Hemodialysis to remove methanol and its metabolites Antidote: Fomepizole or Ethanol to inhibit ADH Folinic acid or folic acid to enhance formate metabolism.

N-acetylcysteine (Option A):

- Used as an antidote for acetaminophen (paracetamol) poisoning.
- Works by replenishing glutathione levels and preventing liver damage.

Atropine (Option C):

- Anticholinergic drugs used primarily in the treatment of poisoning (e.g., organophosphate poisoning)

Naloxone (Option D):

- It is an opioid antagonist used to reverse the effects of opioid overdose, such as respiratory depression and sedation.

Solution for Question 13:

Correct Answer: D) Topiramate

Explanation:

Topiramate is an anticonvulsant medication that has been found to have some efficacy in reducing alcohol consumption, possibly through its effects on neurotransmitter systems. It is used as an off-label drug for dependence (not FDA-approved).

Alcohol dependence is a chronic disease characterised by an inability to control alcohol consumption despite adverse social, occupational, or health consequences. Dependence occurs only in heavy and round-the-clock drinking.

Management of alcohol dependence:

- Both psychological and medical supportive measures should be provided.
- Anti-craving drugs:

	Drug Name	Mechanism of Action	Notes
FDA-approved anti-craving drugs	Disulfiram	Inhibits acetaldehyde dehydrogenase ADH	Reduces cravings by causing unpleasant reactions if alcohol is consumed
Naltrexone	Blocks opioid receptors	Reduces cravings and desire to drink	
Acamprosate	Modulates glutamate & GABA systems	Reduces cravings and helps maintain abstinence	
Not FDA-approved (off-label use)	Baclofen	GABA agonist	Helps with alcohol withdrawal symptoms
Gabapentin & Topiramate	Modulates GABA & glutamate systems	Helps with alcohol withdrawal symptoms	

Solution for Question 14:

Correct Answer: C) Disulfiram increases the production of dopamine in the brain, reducing the desire to consume alcohol.

Explanation:

Disulfiram works by inhibiting the enzyme aldehyde dehydrogenase, which is responsible for alcohol metabolism.

When alcohol is consumed while on disulfiram therapy, there is an accumulation of acetaldehyde, a toxic metabolite, leading to unpleasant symptoms such as nausea, vomiting,

and headache.

Disulfiram ■ Inhibition of Aldehyde Dehydrogenase (ALDH) ■ Accumulation of Acetaldehyde ■ Unpleasant symptoms including flushing, nausea, vomiting, headache, palpitations, sweating, shortness of breath, and hypotension ■ Creates a strong negative association between alcohol consumption and the discomfort.

Disulfiram therapy acts as a deterrent to alcohol consumption and is often used as an adjunct in the treatment of alcohol dependence. Disulfiram does not directly affect dopamine production in the brain.

Solution for Question 15:

Correct Answer: D) Griseofulvin

Explanation:

Griseofulvin inhibits aldehyde dehydrogenase, causing acetaldehyde accumulation and resulting in symptoms like flushing, headache, nausea, vomiting, and palpitations when alcohol is consumed.

Disulfiram-like reaction:

- A disulfiram-like reaction refers to an adverse reaction that mimics the effects of disulfiram (Antabuse) when certain substances are consumed together with alcohol.
- Supports the treatment of chronic alcoholism by producing an acute sensitivity to ethanol.
- When alcohol is ingested by someone taking disulfiram, it inhibits the enzyme aldehyde dehydrogenase, leading to the accumulation of acetaldehyde in the blood, which causes unpleasant symptoms. Flushing. Headache. Nausea & vomiting. Sweating. Palpitations, chest pain, and dizziness. Dyspnea. Hypotension and circulatory collapse.
- Flushing.
- Headache.
- Nausea & vomiting.
- Sweating.
- Palpitations, chest pain, and dizziness.
- Dyspnea.
- Hypotension and circulatory collapse.

- Flushing.
- Headache.
- Nausea & vomiting.
- Sweating.
- Palpitations, chest pain, and dizziness.
- Dyspnea.
- Hypotension and circulatory collapse.

Drugs causing disulfiram-like reaction: (Mnemonic- CGMP)

They are contraindicated in active drinkers.

- Chlorpropamide.
- Cephalosporins (mnemonic- PMTX): CefoPerazone. CefaMandole. CefoTetan. CefoXitin.
- CefoPerazone.
- CefaMandole.
- CefoTetan.
- CefoXitin.
- Griseofulvin.
- Metronidazole.
- Procarbazine.
- CefoPerazone.
- CefaMandole.
- CefoTetan.
- CefoXitin.

Other antifungals are not known to cause a disulfiram-like reaction (Options A, B and C).

Anti Epileptic Drugs

1. A 45-year-old man presents with complaints of recurrent episodes of sudden jerking movements involving his arms and legs. He has no other medical conditions and has been prescribed an antiepileptic drug. Which of the following medications acts by blocking sodium channels?

- A. Tiagabine
 - B. Levetiracetam
 - C. Phenytoin
 - D. Phenobarbitone
-

2. A 10-year-old child is brought by his parents due to episodes of brief staring spells associated with a loss of awareness, lasting 10 to 20 seconds, occurring multiple times a day, especially during stress or fatigue. Which of the following is the drug of choice for this condition?

- A. Valproate
 - B. Phenytoin
 - C. Carbamazepine
 - D. Ethosuximide
-

3. A 12-year-old with Lennox-Gastaut syndrome is on valproate for seizure control. During a follow-up, the neurologist reviews valproate's uses and potential side effects. Which of the following is NOT true regarding valproate?

- A. Used in the treatment of Dravet syndrome
- B. Drug of choice in atypical absence seizures

- C. Used in the treatment of mania
 - D. Causes purple glove syndrome
-

4. Which of the following is not a potential adverse effect of Phenytoin?

- A. Generalised lymphadenopathy
 - B. Osteomalacia
 - C. Megaloblastic anemia
 - D. Weight Loss
-

5. A 6-month-old infant is brought up due to concerns about episodes of sudden, repetitive muscle contractions involving his arms and legs. An EEG reveals a hypsarrhythmia pattern. Genetic testing reveals a mutation in the TSC1 gene. Which of the following is the drug of choice in this condition?

- A. Clonazepam
 - B. Valproate
 - C. Vigabatrin
 - D. Ethosuximide
-

6. A 60-year-old woman presents with severe, sharp, electric shock-like pain on the right side of her face. The pain, triggered by talking, chewing, and touching her face, lasts a few seconds to minutes and has become increasingly frequent and debilitating. Which of the following medications is the drug of choice for treating her condition?

- A. Carbamazepine
- B. Gabapentin

- C. Diazepam
 - D. Levetiracetam
-

7. Which of the following adverse effects is associated with carbamazepine use?

- A. Hypoglycemia
 - B. Hypertension
 - C. Weight gain
 - D. Agranulocytosis
-

8. A 28-year-old pregnant woman with a history of epilepsy presents for a prenatal visit. She is currently taking medication to control her seizures and is concerned about the safety of her medication during pregnancy. Which of the following drugs is considered the drug of choice for managing her condition?

- A. Carbamazepine
 - B. Lamotrigine
 - C. Gabapentin
 - D. Levetiracetam
-

9. A 28-year-old male with a history of epilepsy presents to the emergency department with ongoing tonic-clonic seizures that have lasted for the past 20 minutes. What is the initial drug of choice for the management of status epilepticus in this clinical scenario?

- A. Phenobarbital
- B. Lorazepam
- C. Lamotrigine

D. Valproate

10. What is the recommended therapeutic range for phenytoin?

- A. 4 to 12 mcg/ml
 - B. 5 to 40 mcg/ml
 - C. 10 to 20 mcg/ml
 - D. 50 to 100 mcg/ml
-

11. A 28-year-old woman with epilepsy and obesity presents for a follow-up visit. She has been experiencing increased seizure frequency and is concerned about her weight. Which anti-epileptic drug is most likely to cause weight loss and potentially benefit this patient?

- A. Carbamazepine
 - B. Gabapentin
 - C. Topiramate
 - D. Phenytoin
-

12. A 6-year-old boy with infantile spasms has been treated for 6 months, showing improved seizure control. The neurologist is concerned about the potential for permanent peripheral visual field defects with long-term use of his current medication. Which antiepileptic drug is most likely to cause these visual field defects, requiring regular ophthalmologic monitoring?

- A. Carbamazepine
- B. Valproic acid
- C. Vigabatrin

D. Levetiracetam

13. A 35-year-old woman with a history of anxiety and panic attacks seeks treatment for increased symptoms affecting her daily life. The physician considers prescribing a medication acting on GABA channels. Which medication would be most appropriate?

- A. Zolpidem
 - B. Gabapentin
 - C. Alprazolam
 - D. Topiramate
-

14. A 28-year-old man with a history of alcohol use disorder has been abstinent for the past 2 weeks. He now reports experiencing alcohol cravings, anxiety, and difficulty sleeping. His physician considers prescribing a medication that acts on the glutamate system to help maintain abstinence and reduce the risk of relapse. Which medication would be most appropriate for this patient's condition?

- A. Memantine
 - B. Riluzole
 - C. Acamprosate
 - D. Topiramate
-

15. A 12-year-old girl with treatment-resistant epilepsy has tried multiple antiepileptic drugs without adequate seizure control. Her parents are interested in exploring alternative treatment options, including cannabidiol (CBD). Which of the following statements about CBD is most accurate?

- A. CBD is a psychoactive compound that can cause intoxication and impairment.

- B. CBD has been FDA-approved for the treatment of seizures associated with Lennox-Gastaut syndrome and Dravet syndrome.
 - C. CBD is primarily used for the treatment of chronic pain and inflammation.
 - D. CBD has no known drug interactions and can be safely used with all other medications.
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	4
Question 4	4
Question 5	3
Question 6	1
Question 7	4
Question 8	4
Question 9	2
Question 10	3
Question 11	3
Question 12	3
Question 13	3
Question 14	3
Question 15	2

Solution for Question 1:

Correct Answer: C) Phenytoin

Explanation:

The patient is most likely suffering from recurrent seizures. Phenytoin blocks sodium channels and inhibits glutamate release.

Mechanism of seizures:

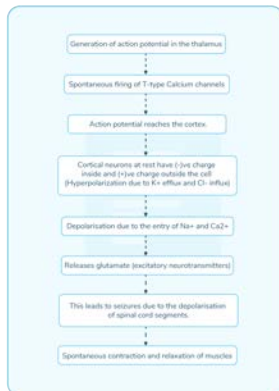


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Solution for Question 2:

Correct Answer: D) Ethosuximide

Explanation:

- The clinical scenario is characteristic of typical absence seizures, a type of generalised onset non-motor seizure with unresponsiveness and staring spells.
- They are generally seen in children aged 5 to 15 years.
- A classic EEG finding of a 3-Hz spike and wave discharges is found.
- Ethosuximide is the drug of choice in typical absence seizures. It blocks T-type calcium channels in the thalamus and prevents the firing of action potential to the cortex.
- Valproate (Option A) is the DOC in atypical absence seizures.
- Phenytoin (Option B) and Carbamazepine (Option C) are contraindicated in the Absence seizures.

Solution for Question 3:

Correct Answer: D) Causes purple glove syndrome

Explanation:

Purple glove syndrome is a complication associated with the intravenous use of phenytoin, not valproate.

- Valproate is widely used for several indications, including: Seizure Disorders: It is the DOC for atypical absence seizures, myoclonic seizures, and generalised tonic-clonic seizures (GTCS). Mood Disorders: Valproate is effective in the treatment of mania and bipolar disorder. Panic Attacks: It can be used off-label for the treatment of panic attacks. Migraine Prophylaxis: Valproate is considered a second-line treatment for preventing migraines. Mixed Seizure Syndromes: It is used in conditions such as Lennox-Gastaut syndrome. Severe Myoclonic Epilepsy of Infancy: Valproate is also used in Dravet syndrome.
- Seizure Disorders: It is the DOC for atypical absence seizures, myoclonic seizures, and generalised tonic-clonic seizures (GTCS).
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- Migraine Prophylaxis: Valproate is considered a second-line treatment for preventing migraines.
- Mixed Seizure Syndromes: It is used in conditions such as Lennox-Gastaut syndrome.
- Severe Myoclonic Epilepsy of Infancy: Valproate is also used in Dravet syndrome.
- Valproate is a CYP enzyme inhibitor that inhibits CYP2C9, which can lead to increased levels of drugs metabolised by this enzyme, such as phenytoin and warfarin.
- Adverse effects of Valproate: Gastrointestinal: Nausea, vomiting, and indigestion. Neurological: Tremor, sedation, dizziness, and ataxia. Hepatic: Hepatotoxicity (mainly in children <2 years of age). Hematologic: Thrombocytopenia and other blood dyscrasias. Metabolic: Weight gain, hyperammonemia, and pancreatitis. Teratogenicity: Risk of congenital malformations, such as neural tube defects (spina bifida occulta), if used during pregnancy. Others: Hair loss (alopecia) and polycystic ovary syndrome (PCOS).

- Gastrointestinal: Nausea, vomiting, and indigestion.
 - Neurological: Tremor, sedation, dizziness, and ataxia.
 - Hepatic: Hepatotoxicity (mainly in children <2 years of age).
 - Hematologic: Thrombocytopenia and other blood dyscrasias.
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 - Others: Hair loss (alopecia) and polycystic ovary syndrome (PCOS).
-

Solution for Question 4:

Correct Answer: D) Weight Loss

Explanation:

Weight Loss is not a side effect of phenytoin use.

- Phenytoin is an antiepileptic medication primarily used to manage and prevent seizures, particularly in the context of epilepsy.
- It is a sodium channel blocker that stabilises the neuronal membrane by inhibiting the repetitive firing of action potentials. This action helps to reduce the occurrence and severity of seizures in individuals with epilepsy.

Adverse effects of Phenytoin include:

Solution for Question 5:

Correct Answer: C) Vigabatrin

Explanation:

The infant most likely has Infantile spasms due to Tuberous sclerosis.

- Infantile spasms, characterised by intermittent muscle spasms and developmental regression, often present with a characteristic hypsarrhythmia pattern on EEG.
- These spasms can be associated with tuberous sclerosis, a genetic disorder caused by mutations in the TSC1 or TSC2 genes.

Treatment of Infantile Spasms:

- Vigabatrin It is the drug of choice for treating infantile spasms associated with tuberous sclerosis and drug of choice for infantile spasms without tuberous sclerosis is ACTH It ↑ GABA levels by inhibiting the enzyme GABA transaminase, thereby reducing seizure activity.
 - It is the drug of choice for treating infantile spasms associated with tuberous sclerosis and drug of choice for infantile spasms without tuberous sclerosis is ACTH
 - It ↑ GABA levels by inhibiting the enzyme GABA transaminase, thereby reducing seizure activity.
 - Corticosteroids: These drugs can offer symptomatic relief but are not suitable for long-term use due to their potential side effects.
 - Other antiepileptic drugs, Clonazepam (Option A) and Valproate (Option B), are sometimes used but are not the first-line treatments for this specific condition.
 - It is the drug of choice for treating infantile spasms associated with tuberous sclerosis and drug of choice for infantile spasms without tuberous sclerosis is ACTH
 - It ↑ GABA levels by inhibiting the enzyme GABA transaminase, thereby reducing seizure activity.
-

Solution for Question 6:

Correct Answer: A) Carbamazepine

Explanation:

- The patient's symptoms are indicative of trigeminal neuralgia, a neuropathic pain disorder characterised by severe stabbing pain along the trigeminal nerve, typically affecting one side of the face.
- Carbamazepine is the drug of choice for managing this condition due to its proven efficacy in reducing pain.

Drug	Mechanism of Action	Uses	Side Effects
Carbamazepine	Inhibits high-frequency neuronal discharges and modulates voltage-gated sodium channels.	Trigeminal neuralgia, GTCS, partial seizures, bipolar disorder.	Sedation, dizziness, ataxia, GI disturbances, bone marrow suppression, hypersensitivity reactions, water retention.
Gabapentin	Enhances GABA release modulates voltage-sensitive Ca ²⁺ channels.	Partial seizures, post-herpetic neuralgia, migraine prophylaxis.	Sedation, tiredness, dizziness, tremor, unsteadiness.
Diazepam	Enhances GABAergic neurotransmission.	Status epilepticus, Febrile seizure.	Thrombophlebitis, hypotension, respiratory depression.
Topiramate	Multiple mechanisms, including sodium channel inactivation, GABA potentiation, glutamate receptor antagonism, and potassium channel modulation.	GTCS, myoclonic epilepsy, and absence of seizures.	Impairment of attention, sedation, ataxia, word-finding difficulties, poor memory, weight loss, paresthesias, renal stones.

Solution for Question 7:

Correct Answer: D) Agranulocytosis

Explanation:

Agranulocytosis is a severe adverse effect of Carbamazepine, characterised by a dangerously low white blood cell count, which leaves the patient vulnerable to infections.

Adverse Effects	Description
-----------------	-------------

Neurotoxicity	Sedation, dizziness, vertigo, diplopia, and ataxia
Gastrointestinal Effects	Vomiting and diarrhoea occur with higher doses.
Acute Intoxication	Coma, convulsions, cardiovascular collapse.
Haematological	Agranulocytosis, aplastic anaemia.
Hypersensitivity Reactions	Rashes, photosensitivity, hepatitis, lupus-like syndrome.
Water Retention and Hyponatremia	Due to enhanced antidiuretic hormone (ADH) action, it is more common in the elderly.
Teratogenicity	Increased incidence of minor fetal malformations.

Hypoglycemia (Option A), hypertension (Option B), and weight gain (Option C) are not commonly associated with carbamazepine use.

Solution for Question 8:

Correct Answer: D) Levetiracetam

Explanation:

The drug of choice for managing epilepsy during pregnancy due to its efficacy and safety profile is Levetiracetam.

Aspects	Description
Mechanism of Action	Levetiracetam binds to synaptic vesicular protein SV2A, modulating neurotransmitter release (e.g., glutamate, GABA) and inhibiting N-type calcium channels.
Uses	It is used for the management of partial seizures and myoclonic epilepsy. It is also utilised as adjunctive therapy for refractory partial seizures. Additionally, it serves as second-line adjunctive therapy for generalised tonic-clonic seizures and absence seizures, although it is not approved for children below 4 years.
Side effects	Sleepiness, dizziness, weakness, behavioural changes (rare) and potential driving impairment. It is generally well-tolerated and is not metabolised by CYP enzymes, resulting in minimal drug interactions.

Solution for Question 9:

Correct Answer: B) Lorazepam

Explanation:

- Lorazepam is the drug of choice for the management of status epilepticus due to its rapid onset of action and effectiveness in terminating seizures.
- It acts by enhancing the effects of gamma-aminobutyric acid (GABA) in the brain, thereby reducing neuronal excitability and terminating seizures.

Management of status epilepticus:

Position the head, provide oxygen, and monitor vital signs.

↓

Establish vascular access and check glucose levels.

↓

Administer benzodiazepines (preferably lorazepam IV).

↓

If seizures persist after 2 doses of benzodiazepines.

↓

Administer antiepileptic drugs (Fosphenytoin, Phenytoin, Levetiracetam, Valproic acid).

↓

If refractory status epilepticus suspected

↓

... (content truncated)

- Phenobarbital (Option A) and Valproate (Option D) are used in the management of status epilepticus but are not the first-line treatments.
- Intravenous administration of Lamotrigine (Option C) is contraindicated in emergency situations due to its association with Stevens-Johnson syndrome.

Solution for Question 10:

Correct Answer: C) 10 to 20 mcg/ml

Explanation:

- Phenytoin's therapeutic range refers to the optimal concentration of the drug in the bloodstream to achieve effective seizure control while minimising the risk of toxicity.
- The target range for maintaining therapeutic efficacy with phenytoin therapy is commonly accepted to be 10 to 20 mcg/ml.
- Phenytoin is metabolised in the liver primarily by CYP2C9 and CYP2C19. Over the therapeutic range, its kinetics shift from first-order to zero-order, leading to disproportionately high plasma concentrations with small dose increments.
- Monitoring plasma levels assists in dosage adjustments due to its capacity-limited metabolism.
- Therapeutic Drug Monitoring (TDM) involves measuring drug concentrations in the blood to ensure patients are within the therapeutic range and adjusting dosage as needed.

Solution for Question 11:

Correct Answer: C) Topiramate

Explanation:

- In the given case scenario, adding topiramate to the patient's existing regimen may help with seizure control and weight management, making it the most appropriate choice.
- Topiramate and Zonisamide are two anti-epileptic drugs (AEDs) that have been associated with weight loss. In contrast, many other AEDs, such as Valproic acid, Gabapentin (Option B), and Carbamazepine (Option A), are known to cause weight gain, possibly due to their effects on appetite stimulation and metabolic changes.
- In contrast, many other AEDs, such as Valproic acid, Gabapentin (Option B), and Carbamazepine (Option A), are known to cause weight gain, possibly due to their effects on appetite stimulation and metabolic changes.
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Category	Topiramate (Option C)	Zonisamide
Mechanism of Action	Acts through various mechanisms, including enhancing GABA-mediated inhibition and	

	inhibiting sodium/calcium channels and carbonic anhydrase.	
Indications	FDA-approved for partial-onset seizures, generalised tonic-clonic seizures, Lennox-Gastaut syndrome, and migraine prophylaxis.	FDA-approved for adjunctive treatment of partial-onset seizures in adults.
Weight Loss	Associated with significant weight loss due to suppression of appetite and increased energy expenditure.	
Anti-Obesity Drug	Combination with phentermine (Qsymia) is FDA-approved for chronic weight management in adults with BMI ≥ 30 or ≥ 27 with a weight-related comorbidity.	It is investigated as a potential anti-obesity medication but is not currently FDA-approved for this indication.

Solution for Question 12:

Correct Answer: C) Vigabatrin

Explanation:

- Vigabatrin is an antiepileptic drug used for infantile spasms and refractory complex partial seizures. It acts by irreversibly inhibiting GABA transaminase to increase GABA levels in the brain.
- Vigabatrin is associated with a significant risk of permanent peripheral visual field defects and retinal toxicity, necessitating regular ophthalmologic monitoring to detect and manage potential visual impairments early.
- Carbamazepine (Option A) and Valproic acid (Option B) can cause a few ophthalmic side effects, including visual disturbances such as diplopia and blurred vision, but not associated with permanent visual loss.

Among the given options, Levetiracetam (Option D) is considered to have a favourable safety profile with minimal ophthalmic side effects.

Solution for Question 13:

Correct Answer: C) Alprazolam

Explanation:

Alprazolam, a benzodiazepine, is the suitable choice for generalised anxiety disorder and panic attacks as it enhances GABA's effect by modulating GABA-A receptors, thus reducing anxiety symptoms effectively.

Drug	Mechanism of Action	Uses
Alprazolam (Benzodiazepine)(Option C)	Positive allosteric modulator of GABA-A receptors. Enhances the effect of GABA, leading to increased inhibitory neurotransmission.	Generalised anxiety disorder. Panic disorder. Short-term treatment of anxiety symptoms.
Zolpidem (Z-drug)(Option A)	Positive allosteric modulator of GABA-A receptors, with selectivity for $\alpha 1$ subunit-containing receptors. Enhances the effect of GABA, promoting sleep.	Short-term treatment of insomnia.
Gabapentin (Gabapentinoid)(Option B)	The exact mechanism is unclear, but it is believed to involve the modulation of voltage-gated calcium channels and indirect enhancement of GABA activity.	Postherpetic neuralgia. Neuropathic pain. Adjunctive therapy for partial seizures.
Topiramate (Antiepileptic)(Option D)	Multiple mechanisms, including enhancement of GABA activity and inhibition of glutamate activity. It also acts on voltage-gated sodium and calcium channels and inhibits carbonic anhydrase.	Epilepsy (partial and generalised seizures). Migraine prophylaxis.

Solution for Question 14:

Correct Answer: C) Acamprosate

Explanation:

Acamprosate, a glutamate receptor modulator, is the most appropriate choice for this patient with alcohol use disorder, as it helps maintain abstinence and reduces the risk of relapse by targeting the glutamate system.

Drugs acting on the Glutamate system:

Drug	Mechanism of Action	Uses
Acamprosate (Option C)	Modulates glutamate transmission, likely by acting as an antagonist at NMDA receptors and an agonist at GABAA receptors It helps restore the balance between excitatory and inhibitory neurotransmission disrupted by chronic alcohol use.	Maintenance of abstinence in alcohol dependence. Reduction of alcohol craving and relapse risk.
Memantine (Option A)	Non-competitive NMDA receptor antagonist. Blocks the effects of excessive glutamate activity.	Moderate to severe Alzheimer's disease
Riluzole (Option B)	Glutamate release inhibitor. Blocks voltage-gated sodium channels and reduces glutamate release.	Amyotrophic lateral sclerosis (ALS)

Topiramate (Option D) is primarily utilised as an antiepileptic and for migraine prophylaxis. Although it may reduce alcohol consumption in some cases, it lacks specific indications for maintaining abstinence from alcohol dependence but is not approved for this (OFF LABEL USE).

Solution for Question 15:

Correct Answer: B) CBD has been FDA-approved for the treatment of seizures associated with Lennox-Gastaut syndrome and Dravet syndrome.

Explanation:

- Cannabidiol (CBD) is a compound found in cannabis plants and is noted for its non-psychoactive nature, distinguishing it from THC (Tetrahydrocannabinol).
- Unlike THC (Tetrahydrocannabinol), which induces a "high," CBD does not cause intoxication or impairment, nor does it produce hallucinogenic effects or psychological dependence.

- While CBD has shown potential in preclinical and early clinical studies for managing chronic pain and inflammation, its primary FDA-approved use is for treating seizures associated with specific severe forms of epilepsy, such as Lennox-Gastaut syndrome , Infantile spasm with tuberous sclerosis and Dravet syndrome
 - CBD may influence the metabolism of certain drugs by inhibiting specific liver enzymes, such as Cytochrome P450. This could alter medication levels in the bloodstream and necessitate caution when used alongside other medications.
-

Antipsychotics and Antidepressants

1. A 35-year-old woman presents with persistent low mood, loss of interest, fatigue, difficulty concentrating, feelings of worthlessness, and occasional thoughts of self-harm for the past six months. She has a family history of depression. Diagnosed with major depressive disorder (MDD), she is started on paroxetine. Which class of drug does paroxetine fall under?

- A. Selective serotonin reuptake inhibitor
 - B. Serotonin and norepinephrine reuptake inhibitor
 - C. Tricyclic antidepressant
 - D. Atypical antidepressant
-

2. A 28-year-old man presents with excessive worry, restlessness, difficulty concentrating, muscle tension, and disturbed sleep for several months. He also experiences occasional panic attacks with palpitations, sweating, and a sense of impending doom. With a family history of anxiety disorders, he is diagnosed with Generalised anxiety disorder (GAD). Which of the following medications is most appropriate for initial treatment?

- A. Selective serotonin reuptake inhibitor (SSRI)
 - B. Benzodiazepine
 - C. Serotonin and norepinephrine reuptake inhibitor (SNRI)
 - D. Tricyclic antidepressant (TCA)
-

3. Which of the following is not true for SSRIs?

- A. Paroxetine is not safe in pregnancy

- B. Sertraline is safe to use with warfarin
 - C. Fluoxetine is the shortest-acting SSRI
 - D. Escitalopram is preferred in those with heart disease
-

4. A 45-year-old woman presents with widespread musculoskeletal pain, fatigue, and sleep disturbances for the past six months. She describes the pain as dull and achy, affecting various areas like the neck, shoulders, back, and hips, with heightened sensitivity at tender points. After ruling out other conditions, she is diagnosed with fibromyalgia. Which medication is most appropriate for initial treatment?

- A. Venlafaxine
 - B. Duloxetine
 - C. Vilazodone
 - D. Vortioxetine
-

5. A 9-year-old child presents with a history of involuntary bedwetting during sleep at least twice a week for the past 6 months. Physical examination and laboratory investigations reveal no abnormalities. The child has no daytime urinary symptoms or signs of urinary tract infection. After ruling out organic causes, what is the most appropriate pharmacological treatment for this condition?

- A. Amitriptyline
 - B. Amoxapine
 - C. Clomipramine
 - D. Imipramine
-

6. A 45-year-old woman with a history of major depressive disorder is prescribed amitriptyline for the management of her depression. She has been on the medication for two weeks. During a follow-up appointment, she reports experiencing constipation and difficulty urinating. Which of the following side effects is most likely associated with the use of amitriptyline in this patient?

- A. Diarrhea
 - B. Bradycardia
 - C. Urinary retention
 - D. Excessive sweating
-

7. Which of the following is false about MAO inhibitors?

- A. Use with sympathomimetics can cause hypertensive crisis
 - B. It is contraindicated with tyramine-rich food
 - C. Selegiline is a selective MAO-B inhibitor
 - D. Routinely used to treat depression
-

8. Which of the following atypical antidepressants has the maximum risk of seizures?

- A. Buspirone
 - B. Trazodone
 - C. Mirtazapine
 - D. Bupropion
-

9. A 30-year-old male presents with fever, altered mental status, tachycardia, rigidity, dilated pupils, clonus, and diarrhea after increasing his SSRI dose. He

reports that he recently started taking a new over-the-counter supplement for mood enhancement. The doctor prescribed him cyproheptadine. For which condition is cyproheptadine used as an antidote?

- A. Serotonin syndrome
 - B. Tricyclic antidepressant overdose
 - C. Rabbit syndrome
 - D. Cheese reaction
-

10. A 45-year-old man presents with complaints of persistent and intractable hiccups for the past 48 hours. He describes the hiccups as occurring every few seconds, significantly affecting his ability to eat, sleep, and carry out daily activities. He denies any recent changes in his diet or lifestyle and has no significant medical history. On examination, he appears anxious and distressed due to the hiccups. Which of the following medications is most appropriate for treating intractable hiccups in this patient?

- A. Thioridazine
 - B. Chlorpromazine
 - C. Prochlorperazine
 - D. Droperidol
-

11. A 70-year-old man with a 10-year history of Parkinson's disease presents to the clinic with complaints of visual hallucinations and delusional thinking, which have been progressively worsening over the past few months. He is started on an atypical antipsychotic to address his psychotic symptoms. Which of the following best describes the mechanism of action of the drug?

- A. Dopamine D2 receptor antagonism
- B. Serotonin 5-HT_{2A} receptor inverse agonism
- C. Dopamine D1 receptor partial agonism

D. Serotonin 5-HT_{1A} receptor agonism

12. A 42-year-old man has been diagnosed with schizophrenia for over 15 years. Despite multiple trials of antipsychotic medications, including risperidone, olanzapine, and quetiapine, he continues to experience severe and persistent psychotic symptoms. His psychiatrist decides to start clozapine due to the treatment-resistant nature of his schizophrenia. Which of the following does NOT best describe the drug?

- A. Drug choice for resistant mania
 - B. Weaker D₂ antagonism
 - C. Associated with decreased appetite
 - D. Wet pillow syndrome is a known side effect
-

13. Which of the following antipsychotic medications is considered the safest in pregnancy due to its minimal crossing of the placenta?

- A. Olanzapine
 - B. Risperidone
 - C. Aripiprazole
 - D. Quetiapine
-

14. A 40-year-old woman with a history of schizophrenia has been prescribed an antipsychotic medication for the past few months. She presents to the clinic with complaints of muscle stiffness, restlessness, and occasional tremors. During the examination, the physician noted her difficulty sitting still and observed her hands shaking intermittently. Which of the following symptoms is NOT a presentation associated with antipsychotic drugs?

- A. Tardive dyskinesia
 - B. Akathisia
 - C. Dystonia
 - D. Hyperprolactinemia
-

15. A 25-year-old man is brought to the emergency department due to severe agitation and aggressive behaviour. The patient has a history of bipolar disorder and is prescribed olanzapine, lithium, and fluoxetine. Temperature is 40.6 C (105.1 F), BP is 157/90 mm Hg. The patient requires several doses of antipsychotics to calm down. Examination shows diffuse rigidity in the upper and lower extremities. Which of the following most likely explains this patient's current symptoms?

- A. Drug-induced parkinsonism
 - B. Lithium toxicity
 - C. Neuroleptic malignant syndrome
 - D. Serotonin Syndrome
-

16. Which of the following antipsychotic medications is typically associated with a higher incidence of akathisia?

- A. Haloperidol
 - B. Clozapine
 - C. Olanzapine
 - D. Risperidone
-

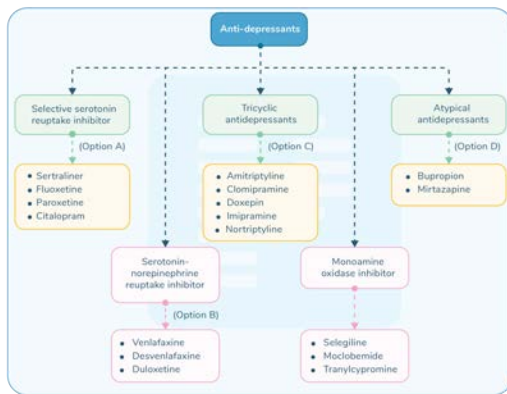
Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	3
Question 4	2
Question 5	4
Question 6	3
Question 7	4
Question 8	4
Question 9	1
Question 10	2
Question 11	2
Question 12	3
Question 13	4
Question 14	4
Question 15	3
Question 16	1

Solution for Question 1:

Correct Answer: A) Selective serotonin reuptake inhibitor

Explanation:



Solution for Question 2:

Correct Answer: A) Selective serotonin reuptake inhibitor (SSRI)

Explanation:

Selective serotonin reuptake inhibitors (SSRI) are first-line drugs in the treatment of Generalised anxiety disorder.

Mechanism of action:

SSRIs block serotonin reuptake in the brain, increasing its concentration in the synaptic cleft. This enhances serotonin signalling, which is thought to alleviate symptoms of depression and anxiety.

Uses:

- Major depressive disorder- drug of choice
- Generalised anxiety disorder
- Obsessive-compulsive disorder
- Premenstrual dysphoric disorder
- Post-traumatic stress disorder
- Bulimia nervosa
- Bipolar depression (using it in bipolar affective disorder can sometimes trigger mania episodes)

Side effects:

- Nausea, vomiting (Most common)
- Diarrhea
- Insomnia
- Anxiety during the initial weeks of treatment
- Sexual dysfunction (Reduced libido, Anorgasmia, Ejaculatory delay) - therefore can be used for the treatment of premature ejaculation
- Increased risk of suicide soon after starting drug
- Bleeding (thrombocytopenia)
- Loss of weight

Option B and C: Benzodiazepines and Serotonin and norepinephrine reuptake inhibitors are also used to treat anxiety but are not first-line drugs.

Benzodiazepines such as lorazepam and clonazepam are used in the acute phase of anxiety.

Option D: Tricyclic antidepressants are used off-label for treating anxiety and are not preferred.

Solution for Question 3:

Correct Answer: C) Fluoxetine is the shortest-acting SSRI

Explanation:

Fluoxetine is the longest-acting SSRI. Paroxetine is the shortest-acting SSRI.

Option A: Paroxetine is not safe in pregnancy. It is associated with an increased risk of heart malformation in newborns when taken in the first trimester. It can lead to persistent pulmonary hypertension in newborns, especially if used in the third trimester.

Paroxetine is drug of choice for premature ejaculation.

Option B: Sertraline is safe to use with warfarin. Sertraline and citalopram interact least with warfarin, whereas fluoxetine poses the highest risk of increasing the effects of warfarin.

Option D: Escitalopram is preferred in those with heart disease- this is due to evidence that exhibits escitalopram's ability to decrease cardiac events in the future among those who have preexisting heart disease.

Solution for Question 4:

Correct Answer: B) Duloxetine

Explanation:

- Duloxetine and milnacipram are used to treat symptoms of pain from fibromyalgia. They are classed as Serotonin and Norepinephrine Reuptake Inhibitors (SNRIs).
- Duloxetine is also used to treat anxiety, depression, ADHD, stress urinary incontinence, diabetic neuropathic pain and PTSD.
- SNRIs reduce fibromyalgia pain by modulating serotonin and norepinephrine levels in the central nervous system. By inhibiting their reuptake, SNRIs increase their concentrations in the synaptic cleft, enhancing neurotransmission. This helps regulate pain inhibition signalling pathways, reducing the pain experienced by individuals with fibromyalgia.

Option A: Venlafaxine is the shortest acting SNRI that is not approved for fibromyalgia treatment. It is approved for use in anxiety, depression, ADHD, and PTSD. Side effect of venlafaxine-hypertension

Option C and D: Vilazodone and vortioxetine are newer antidepressants that are approved for the treatment of major depressive disorder.

- Vilazodone is an SSRI and partial agonist at the 5HT 1A receptor and is faster acting than other SSRIs. It is used to treat depression, anxiety, panic disorder, and OCD.
 - Vortioxetine is a SERT (serotonin transporter) inhibitor and is classed as an atypical antidepressant. It is used to treat depression.
-

Solution for Question 5:

Correct Answer: D) Imipramine

Explanation:

Imipramine, a tricyclic antidepressant with anticholinergic properties, is effective in treating childhood nocturnal enuresis by reducing bedwetting episodes. However, desmopressin remains

the preferred drug of choice for this condition.

Option A: Amitriptyline is a TCA commonly used to treat depression and neuropathic pain. While it shares similarities with imipramine in terms of pharmacological class, it is not typically used as a first-line treatment for nocturnal enuresis. It can also be used for migraine prophylaxis.

Option B: Amoxapine is a TCA that inhibits norepinephrine and serotonin reuptake and acts as a dopamine receptor antagonist. It is used primarily for major depressive disorder and psychotic depression. It is not commonly used for nocturnal enuresis. Side effect of amoxapine - EPS

Option C: Clomipramine is another TCA, mainly used to treat obsessive-compulsive disorder (OCD) and depression. It is not typically used for bedwetting and can have several side effects, including sedation and anticholinergic effects.

Solution for Question 6:

Correct Answer: C) Urinary retention

Explanation:

Urinary retention is a common anticholinergic side effect of tricyclic antidepressants (TCAs) like amitriptyline.

- TCAs inhibit the action of acetylcholine, leading to reduced bladder contractility and consequent difficulty in initiating and maintaining urination.
- Diarrhea (Option A) is not a typical side effect of TCAs; instead, constipation is more common.
- Bradycardia (Option B) can occur with TCAs due to their anticholinergic effects, but it is less common than urinary retention. Excessive sweating (Option D) is also a possible side effect of TCAs but is less likely to occur than urinary retention.

Side effects of Tricyclic antidepressants:

Mechanism of side effect	Side effect
H1 receptor blockade	Sedation, weight gain

Anticholinergic action on muscarinic receptors	Hyperthermia Dry eyes/mouth Mydriasis (C/I in glaucoma) Urinary retention (option C) Constipation (Option A) Tachycardia
Increase in serotonin in the brain	Seizure threshold reduction
Alpha 1 adrenergic receptor antagonism	Postural hypotension
Sodium channel block	Arrhythmia/QT prolongation Bradycardia in heart block due to overdose (Option B)

Solution for Question 7:

Correct Answer: D) Routinely used to treat depression

Explanation:

MAO inhibitors are not routinely used to treat depression due to the better efficacy of other drugs and their extensive drug interaction profile.

Mechanism of Action:

MAO inhibitors inhibit the enzyme monoamine oxidase (MAO), which breaks down neurotransmitters like serotonin, dopamine, and norepinephrine in the brain. By inhibiting MAO, these drugs increase the levels of these neurotransmitters, helping to alleviate symptoms of depression and other mood disorders.

Types of MAO Inhibitors	Examples
Non-selective MAOIs	Phenelzine, Tranylcypromine
Selective MAO-A inhibitors	Moclobemide
Selective MAO-B inhibitors	Selegiline, Rasagiline (Option C)

Uses:

- Treatment-resistant depression
- Atypical depression
- Anxiety disorders

- Parkinson's disease (Selegiline) - Monoamine oxidase inhibitors (MAOIs), particularly selegiline and rasagiline, are used in Parkinson's disease for their neuroprotective effects and can help improve symptoms.

Side Effects:

- Hypertensive crisis: With consumption of tyramine-rich foods (e.g., aged cheeses, cured meats) or certain medications sympathomimetic drugs (e.g., amphetamines, decongestants) (Options A and B)
- Serotonin syndrome: Can result from combining MAOIs with other serotonergic drugs (e.g., SSRIs, SNRIs), causing agitation, confusion, rapid heart rate, and high BP.
- Orthostatic hypotension.

Solution for Question 8:

Correct Answer: D) Bupropion

Explanation:

All the drugs mentioned can increase the risk of seizures, but maximum with bupropion.

Atypical antidepressants:

Bupropion (Option D)	Norepinephrine dopamine reuptake inhibitor Used for smoking cessation and weight loss High risk of seizures
Trazodone (Option B)	Serotonin antagonist (5HT2 blocker) and reuptake inhibitor DOC-Depression + erectile dysfunction No sexual dysfunction side effects No nausea and vomiting side effects Can cause priapism
Mirtazapine, Mianserin (Option C)	Noradrenaline and selective serotonin antagonist Side effects: Somnolence, increased appetite, and weight gain Used in depression with insomnia
Buspirone, Gepirone (Option A)	5HT1a receptor partial agonist Used to treat generalised anxiety disorder

Tianeptine	Serotonin reuptake enhancer It also acts on opioid receptors and stimulates them Used to treat depression
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Solution for Question 9:

Correct Answer: A) Serotonin syndrome

Explanation:

Serotonin syndrome is treated with cyproheptadine, an antihistamine with antiserotonergic properties. Cyproheptadine blocks serotonin receptors, especially 5-HT_{2A} receptors, counteracting excessive serotonin activity. This helps alleviate symptoms like agitation, confusion, hyperthermia, and autonomic instability.

Option B: Tricyclic antidepressant overdose is managed with supportive care, gastric decontamination using activated charcoal, intravenous sodium bicarbonate to counteract cardiotoxic effects, benzodiazepines for seizures, and close monitoring of vital signs.

Option C: Rabbit syndrome is characterised by rapid, fine, rhythmic lip movements resembling chewing. It is treated by discontinuing the offending medication (e.g., antipsychotics or antidepressants) and using anticholinergic medications if symptoms are severe.

Option D: Cheese reaction (hypertensive crisis) occurs when tyramine-rich foods (e.g., aged cheeses, cured meats) are consumed by those taking MAO inhibitors. Treatment involves supportive care, BP monitoring, and symptomatic management. Intravenous medications like phentolamine or nitroprusside may be used to lower blood pressure rapidly. Avoiding tyramine-containing foods and medications interacting with MAO inhibitors is essential to prevent future crises.

Solution for Question 10:

Correct Answer: B) Chlorpromazine

Explanation:

Chlorpromazine, a typical antipsychotic medication, is commonly used to treat intractable hiccups due to its antiemetic and central nervous system depressant properties. It acts on the central nervous system to suppress the hiccup reflex.

Option A:

Thioridazine is an antipsychotic medication primarily used to treat schizophrenia. It is not typically indicated for hiccups. Additionally, thioridazine is associated with a risk of retinal detachment.

Option C: Prochlorperazine is primarily used as an antiemetic in the treatment of violent vomiting and migraine vomiting, not in treating hiccups.

Option D: Droperidol is an antiemetic and antipsychotic primarily used for nausea and vomiting. Droperidol, when combined with fentanyl, induces neuroleptic analgesia. It is not commonly used for treating hiccups. N2O if added to Neurolept analgesia forms Neurolept anaesthesia

Solution for Question 11:

Correct Answer: B) Serotonin 5-HT_{2A} receptor inverse agonism

Explanation:

The drug used herein is a novel antipsychotic, pimavanserin, and it's a serotonin 5-HT_{2A} inverse agonist specifically used to treat psychosis in Parkinson's disease.

PIMAVANSERIN		
Mechanism of Action	Clinical Use	Adverse effects
Pimavanserin is a novel antipsychotic drug. Primarily acts through inverse agonism of serotonin 5-HT _{2A} receptors. Does not appreciably bind to 5-HT _{2B} or other monoaminergic receptors.	Treats hallucinations and psychosis in Parkinson's disease (PD) patients Alleviates psychotic symptoms without worsening PD motor symptoms Lacks dopamine receptor antagonism, reducing the risk of worsening motor symptoms Suitable for PD patients intolerant to traditional antipsychotics (due to dopamine receptor blocking effects)	Carries a black-box warning for increased mortality in elderly patients. Essential to monitor QT interval due to the risk of QT prolongation and potential cardiac arrhythmias.

DOC for psychosis in a patient with parkinsonism- Pimavanserin

DOC for parkinsonism in a patient on antipsychotics- Clozapine as it primarily blocks D4 receptors rather than D2 receptors, minimizing the risk of exacerbating Parkinsonian symptoms.

Solution for Question 12:

Correct Answer: C)Associated with decreased appetite

Explanation:

Clozapine is linked to increased appetite due to blocking the 5-HT_{2C} receptor. There is a significant risk of weight gain, elevated blood sugar, and abnormal lipid levels.

Mechanism of Action	Atypical antipsychotic with weak dopamine D2 receptor antagonism. Exhibits potent antagonism at serotonin 5HT _{2A} receptors. Blocks 5HT _{2A} receptors more strongly than D2 receptors.
Uses of Clozapine	
Schizophrenia	DOC for treatment-resistant schizophrenia. Effective for schizophrenia with suicidality, polydipsia, or persistent impulsive aggression.
Mania (Option A)	Beneficial as adjunctive therapy or monotherapy for resistant mania.
Lower Risk of EPS (Option B)	Significantly lower risk of extrapyramidal symptoms (EPS) due to its weaker D2 antagonism compared to typical antipsychotics.
Other Uses	Used to treat antipsychotic-induced tardive dyskinesia.

Adverse effects of Clozapine:

Serious Adverse Effects of Clozapine	Description
Agranulocytosis	Rare but life-threatening decrease in white blood cells, requiring weekly monitoring. Discontinue if total leukocyte count < 3000/mm ³ or absolute neutrophil count < 2000/mm ³ . Avoid combining with carbamazepine.

	Monitoring of agranulocytosis: 1st 6 weeks - weekly; Next 6 months- Fortnightly; After 1 year- Monthly
Seizures	Dose-dependent seizure risk. Higher than other atypical antipsychotics.
Cardiac Effects	Risk of tachycardia, myocarditis (fatal, within first 6 wks) and dilated cardiomyopathy.
Non-serious Adverse effects of Clozapine	Description
Hypersalivation (Option D)	Known as "wet pillow syndrome" due to muscarinic receptor stimulation.
Metabolic Complications	Significant risk of weight gain, hyperglycemia, and dyslipidemia. Associated with increased appetite due to 5-HT _{2C} receptor blockade.
Cataracts	Higher risk than other atypical antipsychotics.
Effect in Pregnancy and Breastfeeding	Can cause decreased arousal and hypotonia in neonates. C/I during breastfeeding due to the risk of neutropenia, sedation, and seizures in infants.

Solution for Question 13:

Correct Answer: D) Quetiapine

Explanation:

Quetiapine is considered the safest due to the least placenta crossing. It is an atypical antipsychotic approved for schizophrenia, unipolar depression, acute mania and bipolar disorder.

Option A: Olanzapine

- Known for maximum placenta crossing, it is considered less safe and can cause neonatal EPS, sedation or withdrawal.
- Neonates exposed to olanzapine have higher NICU admission rates while sedating drugs like clozapine can cause decreased arousal and hypotonia ("floppy baby" syndrome).

Option B: Risperidone

- Associated with increased risk of stroke.

- An atypical antipsychotic approved for schizophrenia, autism, acute mania and bipolar disorder.
- Off-label use in anxiety disorders.
- Long-acting injectable form for bipolar I disorder maintenance treatment.

Option C: Aripiprazole

- Associated with the least weight gain.
- An atypical antipsychotic approved for schizophrenia, autism in children under 7 years old, and unipolar depression.
- Can also treat acute mania and is used for maintenance treatment in bipolar disorder.
- Treats both positive and negative symptoms of schizophrenia.

Solution for Question 14:

Correct Answer: D) Hyperprolactinemia

Explanation:

- Hyperprolactinemia is not an extrapyramidal symptom. This is due to the blockade of D2 receptors in the pituitary gland.
- Mechanism: Dopamine normally inhibits prolactin secretion via D2 receptors in the pituitary. Antipsychotics → Block D2 receptors → Hyperprolactinemia.
- Hyperprolactinemia is reversed upon discontinuation of the antipsychotic.

Risk of hyperprolactinemia by Antipsychotic Type:

Typical Antipsychotics	Atypical Antipsychotics
Haloperidol - highest risk Butyrophenones - potent D2 blockers with a high risk. Thioxanthenes -lower prolactin risk (least D2 blocking activity)	Risperidone and paliperidone - high risk (due to ■ P-glycoprotein affinity) Pimavanserin and clozapine - lowest risk.

Extrapyramidal Symptoms:

Condition	Onset	Treatment
Acute muscle Dystonia (Option C)	Within 1-5 days of starting antipsychotic	Administering antiparkinsonian agents such as diphenhydramine or benztropine intramuscularly.

Parkinsonism	Within 1 month	Benzhexol and benztropine (NOT Levodopa)
Akathisia (Option B) (Most common EPS)	5–60 days	Anti-anxiety medications DOC: Propranolol (C/I: Asthma) Diazepam
Rabbit Syndrome	Months or years	Antiparkinsonian agents. Amantadine is preferred due to less adverse cognitive effects.
Tardive Dyskinesia (Option A)	Late-onset adverse effect (after months or years) Occurs due to D2 receptor supersensitivity and can be permanent, even after discontinuing antipsychotic medication.	Stop AP drugs & start Clozapine. If symptoms persist, add Dopamine depleters (VMAT2 inhibitors) Valbenazine Tetrabenazine: DOC for Huntington's disease.

Solution for Question 15:

Correct Answer: C) Neuroleptic malignant syndrome

Explanation:

Given the clinical presentation of severe agitation, hyperthermia, **■**BP, diffuse muscle rigidity, and a history of antipsychotic use, the most likely explanation for the patient's symptoms is Neuroleptic malignant syndrome.

Key features describing NMS and Serotonin syndrome:

Feature	Neuroleptic Malignant Syndrome (NMS) (Option C)	Serotonin Syndrome (Option D)
Mechanism	Associated with high doses of potent D2 antagonists	Mediated by activation of 5HT _{2A} and 5HT _{1A} receptors (serotonergic agents)
Associated Drugs	Antipsychotic agents (greater risk with 1st generation than 2nd generation)	SSRIs, NE reuptake inhibitors, triptans, MAOIs.
Onset	1-3 days May persist for more than a week after the offending agent is	<1 day Occur when starting new or increasing antidepressant therapy

	discontinued	or combining an SSRI with an NEREuptake inhibitor or a triptan.
Common features	Altered mental status Sympathetic hyperactivity (eg, hyperthermia, hypertension, tachycardia, diaphoresis)	
Distinct features	Lead pipe rigidity, hyporeflexia, elevated enzymes (↑ CK), myoglobinuria	Clonus, hyperreflexia
Treatment	Cessation of antipsychotic, supportive care (e.g., aggressive cooling measures), DA agonist (e.g., bromocriptine), dantrolene, ECT	Cessation of drugs, benzodiazepines, supportive care, cyproheptadine (serotonin agonist) if symptoms persist. Avoid simultaneous serotonergic agents.

Option A: Drug-induced parkinsonism typically presents with bradykinesia, tremors, and rigidity but without severe agitation, high fever, and autonomic instability.

Option B: Lithium toxicity can cause symptoms like tremors, polyuria, polydipsia, confusion, and, in severe cases, seizures, but not typically high fever or severe rigidity.

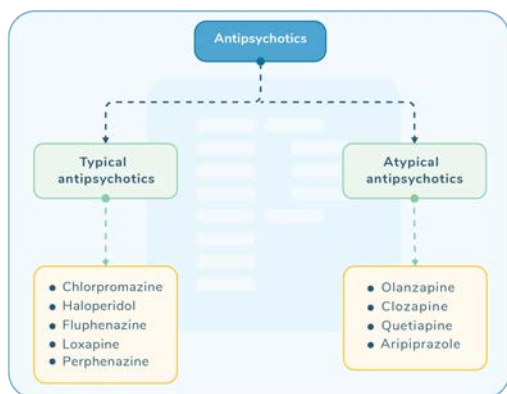
Solution for Question 16:

Correct Answer: A) Haloperidol

Explanation:

Haloperidol belongs to the class of first-generation antipsychotic medications, also known as typical antipsychotics. While effective in managing psychotic symptoms, haloperidol is notorious for its higher risk of causing extrapyramidal symptoms (EPS), which are movement disorders that can occur as side effects of antipsychotic medications. These include dystonia, akathisia, and tardive dyskinesia.

Atypical antipsychotics like clozapine (Option B), olanzapine (Option C), and risperidone (Option D) generally have a lower risk of causing these adverse effects.



Atypical antipsychotic

- maximum EPS- Lurasidone
- maximum Prolactin increase-Risperidone

Typical antipsychotic

- maximum EPS and Prolactin increase- Haloperidol

Opioids

1. Which action will not be seen by mu (μ) receptor stimulation?

- A. Strong analgesia
 - B. Euphoria
 - C. Diuresis
 - D. Respiratory depression
-

2. A 35-year-old construction worker presents to the emergency department after a crush injury to his lower leg from a collapsed beam. He is in severe pain, with swelling and deformity noted on examination. Vital signs are stable. What is the drug of choice for managing his pain?

- A. Baclofen
 - B. Ibuprofen
 - C. Morphine
 - D. Ketorolac
-

3. A 58-year-old male presents to the emergency department with severe chest pain radiating to his left arm and jaw. He appears diaphoretic and anxious. On examination, his blood pressure is 160/100 mmHg, heart rate is 110 bpm, and respiratory rate is 24 breaths per minute. An ECG reveals ST-segment elevation in leads II, III, and aVF. Which of the following medications is the most appropriate initial treatment for his severe pain?

- A. Meperidine
- B. Ibuprofen

- C. Nitroglycerin
 - D. Morphine
-

4. All of the following are true about the pharmacokinetics of morphine except?

- A. Morphine is primarily metabolised by the liver.
 - B. Morphine has a long half-life, typically around 2-4 hours.
 - C. Morphine undergoes extensive first-pass metabolism when administered orally.
 - D. Morphine is primarily eliminated unchanged through the kidneys.
-

5. Which of the following cough suppressants does not have an addictive potential?

- A. Codeine
 - B. Oxycodone
 - C. Pholcodine
 - D. Noscapine
-

6. A 32-year-old woman underwent an elective cesarean section at 39 weeks of gestation under spinal anaesthesia. Following the delivery and surgical procedure, she was transferred to the recovery room. Approximately 30 minutes later, she begins to experience intense shivering. Which of the following medications is most appropriate to treat this condition in this patient?

- A. Ondansetron
- B. Meperidine
- C. Midazolam
- D. Ketamine

7. A 35-year-old man presents to the clinic with complaints of muscle aches, insomnia, and intense cravings for opioids. He admits to using heroin for the past five years, often injecting multiple times a day. He has attempted to quit several times but has always relapsed due to severe withdrawal symptoms. He expresses a strong desire to stop using heroin and asks for medical help to manage his addiction. Which of the following is the most effective medication for the treatment?

- A. Naltrexone
 - B. Lofexidine
 - C. Methadone
 - D. Clonidine
-

8. Match the following drugs based on their receptor activity: Drug Receptor 1. Butorphanol a. Partial Mu Agonist + Kappa Antagonists 2. Nalfurafine b. Pure Antagonists 3. Buprenorphine c. Pure Kappa Agonist 4. Naloxone: d. Partial Kappa Agonist + Mu Antagonist

Drug	Receptor
1. Butorphanol	a. Partial Mu Agonist + Kappa Antagonists
2. Nalfurafine	b. Pure Antagonists
3. Buprenorphine	c. Pure Kappa Agonist
4. Naloxone:	d. Partial Kappa Agonist + Mu Antagonist

- A. 1:a, 2:d, 3:c, 4:b
 - B. 1:d, 2:c, 3:a, 4:b
 - C. 1:c, 2:d, 3:a, 4:b
 - D. 1:b, 2:c, 3:d, 4:a
-

9. A 26-year-old student was found unresponsive by his roommate after a party. A physical examination at the hospital revealed constricted pupils with a heart rate of 62 BPM, blood pressure of 90/60 mmHg, and a respiratory rate of 8 per minute. What is the most appropriate initial treatment for this patient's condition?

- A. Intravenous flumazenil
 - B. Intravenous naloxone
 - C. Gastric lavage with activated charcoal
 - D. Forced alkaline diuresis with sodium bicarbonate
-

**10. Which of the following statements regarding opioid withdrawal symptoms is false? It is a life-threatening condition resulting from chronic opioid dependence
Anxiety and irritability are common during early periods of opioid withdrawal.
Bradycardia and Sweating are typical symptoms of opioid withdrawal.
Buprenorphine is used to reduce withdrawal symptoms and cravings Naloxone is used to treat acute withdrawal symptoms and prevent future relapses.**

- A. 3 and 5
 - B. 2, 4, and 5
 - C. 4 and 5
 - D. 3, and 4
-

11. A 35-year-old male came to the clinic complaining of constipation for several days. A few weeks ago, he experienced diarrhoea after returning from Thailand, which was resolved after taking loperamide and ornidazole. What medication is best suited for relieving this patient's symptoms

- A. Bisacodyl
- B. Lactulose

C. Alvimopan

D. Psyllium

12. A 60-year-old male with a history of Parkinson's disease complained of pain after he underwent a laparoscopic inguinal hernia repair a few hours ago. Shortly after taking medication for pain relief, he became agitated and started sweating. Physical examination revealed tachycardia, hyperreflexia, and myoclonus in the legs. Which of the following drugs was most likely given to this patient?

A. Ibuprofen

B. Tramadol

C. Morphine

D. Oxycodone

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	4
Question 4	4
Question 5	4
Question 6	2
Question 7	3
Question 8	2
Question 9	2
Question 10	1
Question 11	3
Question 12	2

Solution for Question 1:

Correct Answer: C) Diuresis

Explanation:

Mu (μ) receptors are associated with strong analgesia, euphoria, and respiratory depression but not with diuresis. Diuresis is an action associated with kappa (κ) receptors.

Opioid receptors:

- They are G_i type of G-protein coupled receptors.
- Types: Mu (μ), kappa (κ), and delta (δ).

- Subtypes for kappa and delta receptors have been identified. Mu: $\mu 1$ and $\mu 2$. Kappa: $\kappa 1$ and $\kappa 2$.
- Mu: $\mu 1$ and $\mu 2$.
- Kappa: $\kappa 1$ and $\kappa 2$.
- Actions of three types of opioid receptors:
- Mu: $\mu 1$ and $\mu 2$.
- Kappa: $\kappa 1$ and $\kappa 2$.

Mu receptors	Kappa receptors	Delta receptors
High affinity for morphine & its congeners.	High affinity for ketocyclazocine & dynorphin A.	High affinity for Leu/metenkephalins.
Analgesia: Supraspinal ($\mu 1$) + spinal ($\mu 2$)	Analgesia: Spinal ($\kappa 1$) + Supraspinal ($\kappa 2$)	Analgesia: Spinal + affective of supraspinal)
Respiratory depression	Respiratory depression (minimal)	Respiratory depression
Sedation	Dysphoria	Affective behaviour
Euphoria	Miosis (minimal)	Reinforcing actions
Miosis	Sedation	Reduced GI motility
Muscular rigidity	Dependence	Proconvulsant
Reduced GI motility	Reduced GI motility	
Dependence (morphine type)		

Solution for Question 2:

Correct Answer: C) Morphine

Explanation:

Morphine is a potent opioid analgesic used for managing severe pain, including acute pain resulting from traumatic injuries like crush injuries.

It works by binding to opioid receptors in the brain and spinal cord, altering pain perception. It is the drug of choice for managing severe pain in an emergency.

Baclofen (Option A):

- It is a muscle relaxant used to treat muscle spasticity.

- Not typically used as primary pain management, especially for severe pain.
- Mechanism of action: Baclofen is an agonist for gamma-aminobutyric acid (GABA)_B receptors on pre and postsynaptic neurons in the central nervous system (CNS) and peripheral nervous system. Its action results in inhibition of the transmission of both mono and polysynaptic reflexes at the spinal cord, relaxing spasticity.

Ibuprofen (Option B):

- It is a (NSAID) used to relieve pain, reduce inflammation, and lower fever.
- It can be effective for mild to moderate pain
- It does not provide sufficient relief for severe pain.

Ketorolac (Option D):

- It is a (NSAID) with potent analgesic properties.
- It is often used for moderate to severe pain, but it is not as potent as opioids like morphine.
- Ketorolac is typically used when opioid analgesics are contraindicated or not preferred due to their side effects.

Solution for Question 3:

Correct Answer: D) Morphine

Explanation:

Morphine is the preferred choice for the management of severe pain in patients with acute myocardial infarction.

- It provides potent analgesia
- Vasodilatory effects ■ ↓ preload and afterload on the heart.
- ↓ Sympathetic activity (vagomimetic effect) ■ ↓ heart rate and ↓ blood pressure ■ ↓ O₂ demand.
- Additionally, It has sedative properties, which can be beneficial in calming an anxious patient experiencing severe pain and distress.

Meperidine (Option A) is an opioid analgesic, is not the preferred choice in acute myocardial infarction due to its potential to cause increased myocardial oxygen demand, tachycardia, and its short duration of action.

Ibuprofen (Option B) is a (NSAID) that is not indicated for the treatment of severe acute chest pain associated with myocardial infarction. Moreover, NSAIDs can increase the risk of cardiovascular events and should be avoided in this setting.

Nitroglycerin (Option C) is indicated for the relief of angina in patients with myocardial infarction, but it primarily acts as a vasodilator to reduce myocardial oxygen demand rather than directly targeting pain relief.

Solution for Question 4:

Correct Answer: D) Morphine is primarily eliminated unchanged through the kidneys.

Explanation:

- Morphine is primarily metabolized by the liver through glucuronidation to morphine-3-glucuronide (M3G) and morphine-6-glucuronide (M6G).
- Has a relatively long half-life, typically around 2-4 hours.
- Undergoes extensive first-pass metabolism when administered orally, resulting in reduced bioavailability.
- Morphine is metabolized in the liver and excreted primarily as metabolites through the kidneys. Only a small percentage is eliminated unchanged through the kidneys.
- Only a small percentage is eliminated unchanged through the kidneys.
- Only a small percentage is eliminated unchanged through the kidneys.

Solution for Question 5:

Correct Answer: D) Noscapine

Explanation:

- Codeine and oxycodone are opioid-based cough suppressants that have a high potential for addiction due to their narcotic properties. (Option A, B ruled out)

- Pholcodine, although less potent than codeine, is still an opioid derivative and carries a risk of addiction. (Option C ruled out)
- Noscapine is a non-narcotic cough suppressant derived from opium, but it lacks the addictive potential seen in opioid-based medications.

Antitussives (Cough suppressants):

Opioids	Non-opioids	Anti-histamines
Codeine. Ethylmorphine. Pholcodine. Oxycodone.	Noscapine Dextromethorphan.	Chlorpheniramine. Diphenhydramine. Promethazine.

Solution for Question 6:

Correct Answer: B) Meperidine

Explanation:

- Meperidine, also known as pethidine, is an opioid analgesic.
- Meperidine is considered the most effective drug for this specific purpose (DOC).
- It is thought to involve both opioid receptor activity and effects on the thermoregulatory center in the hypothalamus.

Ondansetron (Option A) (antiemetic), commonly used to prevent nausea and vomiting postoperatively or during chemotherapy. It acts as a selective serotonin 5-HT₃ receptor antagonist. Ondansetron may reduce post-anesthesia shivering; it is not the first-line treatment for this condition.

Midazolam (Option C) is a benzodiazepine used for sedation, anxiolysis, and amnesia. It can also have muscle relaxant properties. While it may reduce shivering in some cases, it is not used due to the risk of excessive sedation and respiratory depression, especially in the immediate postoperative period.

Ketamine (Option D) is a dissociative anaesthetic with analgesic properties. It can also have anti-shivering effects, likely due to its NMDA receptor antagonism and effects on central neurotransmitter systems. Because of its potent psychoactive effects and the potential for hallucinations, it is not considered better for the patient.

Solution for Question 7:

Correct Answer: C) Methadone

Explanation:

Methadone:

- Mechanism: It is a full opioid agonist that works by binding to the same receptors as other opioids, providing similar effects and thereby preventing withdrawal and reducing cravings.
- Use: It is highly effective for detoxification and long-term maintenance therapy.
- Advantages: Methadone has a long half-life, which allows for once-daily dosing.
- Effectiveness: It is considered one of the most effective medications for opioid addiction, especially for individuals with a long history of high-dose opioid use.

Naltrexone (Option A):

- Mechanism: It is an opioid antagonist that blocks the effects of opioids at their receptor sites.
- Use: It is used primarily for relapse prevention in opioid-dependent patients who have already undergone detoxification.
- Limitations: It is not effective for managing withdrawal symptoms and requires the patient to be opioid-free for a period before starting treatment.

Lofexidine (Option B):

- Mechanism: It is an alpha-2 adrenergic agonist, that helps to reduce the physical symptoms of opioid withdrawal.
- Use: It is specifically approved for the treatment of opioid withdrawal symptoms.
- Limitations: While it can ease withdrawal symptoms, it does not address cravings and is not used for long-term maintenance therapy.

Clonidine (Option D):

- Mechanism: It is an alpha-2 adrenergic agonist that helps to alleviate some of the physical symptoms of opioid withdrawal, such as anxiety, agitation, and muscle aches.
- Use: It is used off-label to manage withdrawal symptoms but does not address cravings or the underlying addiction.

Solution for Question 8:

Correct Answer: B) 1:d, 2:c, 3:a, 4:b

Explanation:

Mechanism of action of drugs acting on opioid receptors:

Receptor	Pure Kappa Agonist	Partial Kappa Agonist + Mu Antagonist	Partial Mu Agonist + Kappa Antagonists	Pure Antagonists (competitive block)
Drugs	Nalfurafine Fedotozine Eluxadoline Asimadoline	Pentazocine (Partial Mu agonist but acts as an antagonist) Nalorphine Butorphanol	Buprenorphine (binds mu receptor irreversibly) Dezocine	Naloxone Naltrexone (longest acting) Nalmefene (Liver safe) Naloxegol
Uses	1. Uremic pruritus in CKD (Nalfurafine) 2. Diarrhoea in IBS	1. Migraine (Intranasal Butorphanol)	1. Opioid deaddiction (Buprenorphine) 2. Analgesia (Dezocine)	1. Antidote for opioid poisoning (IV Naloxone) 2. Prevents relapse (Naltrexone) 3. Alcohol dependence (Nalmefene) 4. Morphine dependence, obesity (Naltrexone) 5. Opioid-induced constipation (Naloxol, methylnaltrexone)
S/E	Dysphoria or psychosis	Dysphoria or psychosis	Buprenorphine effects cannot be overcome by naloxone.	1. Naloxone: Hypertension, pulmonary oedema, arrhythmia 2. Naltrexone: Hepatotoxic
C/I		1. Opioid deaddiction (precipitate withdrawal) 2. Myocardial ischemia	Labor pain (neonatal respiratory depression)	

Solution for Question 9:

Correct Answer: B) Intravenous naloxone

Explanation:

Naloxone, a pure opioid antagonist, is the drug of choice for reversing opioid overdose in this patient, who presents with reduced consciousness, constricted pupils, and respiratory depression.

Opioid Overdose/ Poisoning	
Cause	Clinical, accidental or suicidal overdose
Symptoms	Unconsciousness/coma Miosis: Pin point pupil Respiratory depression Dry secretion in the eye/mouth Urinary retention Constipation Flaccid muscles Convulsions (in children)
Management	Acute overdose/poisoning: IV Naloxone (Antidote) ↓ Maintenance (Detoxification phase: To prevent withdrawal): Opioids: Buprenorphine, Methadone, Tramadol Non-Opioids- Clonidine, Lofexidine ↓ Abstinence (Prevent relapse): DOC: Oral Naltrexone

Flumazenil (Option A) is used to reverse benzodiazepine overdose, which includes shallow breathing, dilated pupils, and slow reflexes.

- Mechanism of action: Flumazenil is a specific benzodiazepine receptor antagonist that reverses most of the CNS effects of benzodiazepines. Flumazenil abolishes the hypnotic, psychomotor, cognitive and EEG effects of BZD's

Activated charcoal (Option C) is used for certain ingested toxins but is not effective in reversing opioid effects.

Forced alkaline diuresis (Option D) is recommended in barbiturate poisoning manifesting with coma, severe respiratory depression, and mild dilated pupils.

Solution for Question 10:

Correct Answer: A) 3 and 5

Explanation:

Opioid Withdrawal:

- A life-threatening condition resulting from opioid dependence.
- Caused by sudden cessation of opioid use or by the use of opioid partial agonists (buprenorphine) or antagonists (naloxone) by opioid-dependent patients.

Opioid withdrawal	
Causes	Abruptly stopping chronic use of opioids, including pain medication or illegal drug abuse.
Onset	6-12 hours: Short-acting opioids (heroin, fentanyl) 24-48 hours: Long-acting opioids (oxycodone)
Symptoms	W (M): Mydriasis I: Increased yawning (increase respiration), irritability T: Hyperthermia, Tachycardia H: Hyperventilation D: Diarrhea R: Rhinorrhea A: Anxiety, sweating W (M): Myalgia L: Lacrimation
Management	Deaddiction: Start on methadone (long-acting)/ buprenorphine (reduce craving) Methadone can be continued safely in pregnant women. Taper the dose Beta-blocker, Clonidine (symptomatic treatment) Naltrexone (Prevent relapse)

Solution for Question 11:

Correct Answer: C) Alvimopan

Explanation: This patient's history of diarrhoea treated with loperamide a few weeks ago indicates opioid-induced constipation, which is treated with Alvimopan, a peripherally acting μ -opioid receptor antagonist.

Opioid-induced constipation:

- MOA: Block peripheral opioid receptors in GIT, leaving CNS-mediated pain activity intact.
- Seen in chronic pain treatment, methadone maintenance, abdominal surgery
- Drugs Alvimopan: Mu opioid antagonist Oral naloxone blocks local receptors Methylnaltrexone Nalmefene Naloxol Naloxegol
- Alvimopan: Mu opioid antagonist
- Oral naloxone blocks local receptors
- Methylnaltrexone
- Nalmefene
- Naloxol
- Naloxegol
- Alvimopan: Mu opioid antagonist
- Oral naloxone blocks local receptors
- Methylnaltrexone
- Nalmefene
- Naloxol
- Naloxegol

Incorrect Options Explanation:

Bisacodyl (Option A) is a short-term stimulant laxative used to empty the bowel before surgical procedures but is not specifically indicated for opioid-induced constipation.

Lactulose (Option B) is an osmotic laxative used in the treatment of constipation and hepatic encephalopathy but not for opioid-induced constipation.

Psyllium husk (Option D) is a bulk-forming laxative used to relieve constipation and diarrhoea associated with intestinal problems but is not used for opioid-induced constipation.

Agent	Representative	Mechanism of action
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Bulk laxatives	Soluble (Psyllium) Insoluble (wheat bran)	Retaining water in stools, increasing
Stool softeners or wetting agents	Decussate sodium Decussate calcium	Promoting luminal water binding by detergent action and increasing stool bulk
Stimulant laxatives	Sodium picosulfate Bisacodyl	Increasing intestinal peristalsis by acting in myenteric plexus; decreasing large intestine water absorption
Osmotic agents	Polyethyleneglycol (PEG)	Osmotic water binding
Mixed (not classified)	Dried plums	Stool bulking and osmotic action

Solution for Question 12:

Correct Answer: B) Tramadol

Explanation:

Tramadol, an opioid analgesic used to treat moderate to severe pain, raises the risk of serotonin syndrome (agitation, sweating, tachycardia, hyperreflexia) if taken with MAO inhibitor (anti-Parkinson drug).

Tramadol:

- MOA: mu-opioid receptor agonist Inhibits reuptake of serotonin (5HT) and norepinephrine (NE)
- mu-opioid receptor agonist
- Inhibits reuptake of serotonin (5HT) and norepinephrine (NE)
- Use: Moderate pain: labour pain, post-op pain Post anaesthesia chills
- Moderate pain: labour pain, post-op pain
- Post anaesthesia chills
- Serotonin syndrome: In combination with MAO inhibitors (selegiline, etc.) →, elevated 5HT levels Symptoms: agitation, confusion, tachycardia, dilated pupils, muscle rigidity, and sweating. Treatment: Discontinuing the drug and supportive care

- In combination with MAO inhibitors (selegiline, etc.) →, elevated 5HT levels
- Symptoms: agitation, confusion, tachycardia, dilated pupils, muscle rigidity, and sweating.
- Treatment: Discontinuing the drug and supportive care
- Nausea and vomiting (activates the vomiting centre in the brain)
- Seizures
- mu-opioid receptor agonist
- Inhibits reuptake of serotonin (5HT) and norepinephrine (NE)
- Moderate pain: labour pain, post-op pain
- Post anaesthesia chills
- Serotonin syndrome: In combination with MAO inhibitors (selegiline, etc.) →, elevated 5HT levels Symptoms: agitation, confusion, tachycardia, dilated pupils, muscle rigidity, and sweating. Treatment: Discontinuing the drug and supportive care
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- Symptoms: agitation, confusion, tachycardia, dilated pupils, muscle rigidity, and sweating.
- Treatment: Discontinuing the drug and supportive care
- Nausea and vomiting (activates the vomiting centre in the brain)
- Seizures
- In combination with MAO inhibitors (selegiline, etc.) →, elevated 5HT levels
- Symptoms: agitation, confusion, tachycardia, dilated pupils, muscle rigidity, and sweating.
- Treatment: Discontinuing the drug and supportive care

Drugs avoided with MAO inhibitors: (Due to risk of Serotonin syndrome)

Tramadol

Fentanyl, Meperidine

Tricyclic depressants (amitriptyline)

SSRI (sertaline, citalopram etc.)

Dextromethorphan

Linezolid

MDMA, LSD, amphetamine

Food: soy products, red wine, cheese

Pethidine (causes seizure with MAO-)

Ibuprofen (Option A) is a non-steroidal anti-inflammatory drug (NSAID) and does not interact with MAOIs to cause serotonin syndrome.

Morphine (Option C), an opioid analgesic used in severe pain, causes respiratory depression and constipation and is not associated with serotonin syndrome.

Oxycodone (Option D) is an opioid analgesic but does not cause serotonin syndrome.

Parkinson's Disease and Other Disorders

1. A patient who was diagnosed with rheumatic heart disease now presents with irregular involuntary movements of the upper limbs and head alongside characteristic darting movement of the tongue. Which of the following drugs is least likely to be used in the management of this presentation?

- A. Trihexyphenidyl
 - B. Haloperidol
 - C. Sodium valproate
 - D. Carbamazepine
-

2. A 70-year-old male presents to the hospital with a 2-year history of tremors and rigidity. Physical examination revealed diminished facial expressions and dragging feet while walking. The tremors were reduced when he moved his hand. What is the mechanism of action used by the first-line and the most effective medication in managing this condition?

- A. Increasing dopamine release
 - B. Prevents degradation of dopamine
 - C. Acting as a dopamine precursor
 - D. Blocking acetylcholine receptors
-

3. A 60-year-old male with a history of glaucoma was recently diagnosed with primary Parkinson's disease and started treatment with syndopa (levodopa-carbidopa) therapy. The patient inquires about potential adverse events associated with the treatment. Which of the following statements is true? Nausea is the most common adverse effect due to the activation of D2 receptors inside the

brain. Dyskinesias and impulsiveness are common adverse effects associated with long-term levodopa use. Levodopa is contraindicated in this patient with a history of narrow-angle glaucoma. Long-term use of levodopa is associated with fluctuations in motor symptoms. Levodopa increases treatment efficacy as well as the risk of postural hypotension associated with carbidopa.

- A. 1, 3 and 5
 - B. 1 and 5
 - C. 2 and 5
 - D. 3 and 4
-

4. A patient who was diagnosed with Parkinson's disease and started on syndopa a few months ago now complains of worsening rigidity and dyskinesia in between brief periods of improvement. Which of the following is the least preferred add-on drug for this patient?

- A. Istradefylline
 - B. Entacapone
 - C. Apomorphine
 - D. Tolcapone
-

5. A patient was diagnosed with Parkinson's disease and started on drugs for the same 6 months ago. He now presents for a review with complaints of skin discolouration, as shown in the image below. Which of the following antiparkinson agents would have caused this presentation?

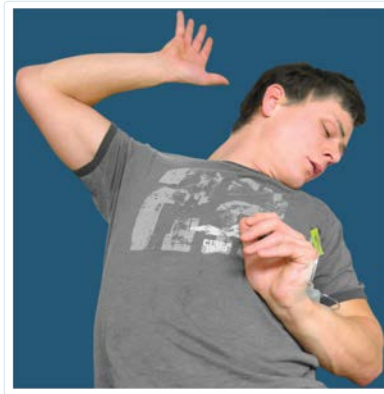


- A. Rasagiline
 - B. Trihexyphenidyl
 - C. Amantadine
 - D. Ropinirole
-

6. All of the following are true regarding non-ergot derivatives used in Parkinson's disease, except?

- A. Symptoms improve with pramipexole much quicker than with bromocriptine
 - B. Ropinirole can be used as an add-on drug during the on-off phase
 - C. The efficacy of apomorphine improves when combined with ondansetron
 - D. Ropinirole proves best in managing restless leg syndrome
-

7. A 55-year-old patient who was diagnosed with schizophrenia and started on risperidone 4 months ago now presents with resting tremors, muscular rigidity in the upper limbs and reduced pace in movements. Which of the following is the most appropriate drug of choice for this patient ?



- A. Rasagiline
 - B. Bromocriptine
 - C. Amantadine
 - D. Benzhexol
-

8. A patient who was diagnosed with rheumatic heart disease now presents with irregular involuntary movements of the upper limbs and head alongside characteristic darting movement of the tongue. Which of the following drugs is least likely to be used in the management of this presentation?

- A. Trihexyphenidyl
 - B. Haloperidol
 - C. Sodium valproate
 - D. Carbamazepine
-

9. A patient who presented with progressive muscle weakness and atrophy alongside dysarthria is diagnosed with Lou Gehrig's disease. All of the following drugs can be used to manage this condition except?

- A. Tizanidine

- B. Donepezil
 - C. Edaravone
 - D. Riluzole
-

10. Which of the following is the drug of choice in a child with Rett syndrome?

- A. Lonafarnib
 - B. Omaveloxolone
 - C. Trofinetide
 - D. Ocrelizumab
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	2
Question 4	4
Question 5	3
Question 6	3
Question 7	4
Question 8	1
Question 9	2
Question 10	3

Solution for Question 1:

Correct Answer: A) Trihexyphenidyl

Explanation:

The clinical presentation, along with the presence of rheumatic heart disease, is suggestive of Sydenham's chorea and anticholinergics like trihexyphenidyl will not be used in managing this.

Sydenham's chorea is characterised by unexpected, involuntary, and irregular movements of the upper limbs and head and characteristic darting movements of the tongue.

This can be due to dopaminergic excess in certain brain areas, and anticholinergics should not be given because central cholinergic excess will worsen symptoms.

Drugs used in the management of Sydenham's chorea:

- Haloperidol. (Option B ruled out)

- Sodium valproate. (Option C ruled out)
- Carbamazepine. (Option D ruled out)
- Corticosteroids (refractory cases).

Solution for Question 2:

Correct Answer: C) Acting as a dopamine precursor

Explanation:

The patient's 2-year history of resting hand tremors, muscle rigidity, and shuffling gait suggests primary Parkinson's disease. Levodopa, a dopamine precursor, is the treatment of choice for replenishing brain dopamine levels in this patient.

Primary Parkinson's disease:

- Idiopathic progressive neurodegenerative disease of dopaminergic neurons in the basal ganglia.
- Symptoms: rigidity, bradykinesia (slow movement), resting tremors, shuffling gait
- Treatment: Levodopa: DOC in Parkinson's disease > 65 years
- The drug is converted into dopamine inside the basal ganglia via dopa decarboxylase.
- Crosses the BBB and replenishes the dopamine level in the CNS
- Degraded via MAO-B and COMT enzymes.
- Alternative treatment options: Amantadine is an NMDA antagonist and increases dopamine release. (Option A). COMT inhibitors (Entacapone, Tolcapone) and MAO inhibitors (Rasagiline, Selegiline): Prevent dopamine degradation. (Option B) Benztropine: Reduces central cholinergic activity by blocking muscarinic receptors. (Option D)
- Amantadine is an NMDA antagonist and increases dopamine release. (Option A).
- COMT inhibitors (Entacapone, Tolcapone) and MAO inhibitors (Rasagiline, Selegiline): Prevent dopamine degradation. (Option B)
- Benztropine: Reduces central cholinergic activity by blocking muscarinic receptors. (Option D)
- Amantadine is an NMDA antagonist and increases dopamine release. (Option A).
- COMT inhibitors (Entacapone, Tolcapone) and MAO inhibitors (Rasagiline, Selegiline): Prevent dopamine degradation. (Option B)

- Benztropine: Reduces central cholinergic activity by blocking muscarinic receptors. (Option D)

Solution for Question 3:

Correct Answer: B) 1 and 5

Explanation:

Adverse events associated with Levodopa	
Side effects	1. Peripheral side effects: Nausea/ vomiting: Most common Due to D2 receptor activation in CTZ outside the blood-brain barrier. Treat with domperidone (Blocks D2 receptor) Postural hypotension (D1 receptor-induced vasodilation) Mydriasis (Stimulates alpha-1 receptor in the iris) Arrhythmia (Stimulates cardiac beta-1 receptor) Note: Combining Levodopa with peripheral dopa decarboxylase inhibitors (Carbidopa/Benserazide) prevents peripheral side effects. 2. CNS side effects: Dyskinesia: Treat with Amantadine, Levetiracetam Psychosis: Clozapine, Quetiapine Hypersexuality/impulsiveness ... (content truncated)
Contraindications	Angle-closure glaucoma (block trabecular outflow) Melanoma Peptic Ulcer Disease Psychosis In combination with vitamin B6/pyridoxine (increase peripheral conversion and side effects) In combination with MAO inhibitors (hypertensive crisis)

Solution for Question 4:

Correct Answer: D) Tolcapone

Explanation:

The above-mentioned scenario is suggestive of the on/off phenomenon seen in patients who are on syndopa, and tolcapone is the least preferred add-on drug for this presentation because it is potentially hepatotoxic.

When a patient has been taking syndopa for a while, the effect starts to wear off, as evidenced by an improvement in symptoms (ON) shortly after administration, followed by a setting of rigidity and dyskinesia (OFF). This is called the on/off phenomenon.

During this period, drugs that prolong the “on” phase or rescue the patients from the “off” phase should be added.

“ON” PHASE PROLONGATION	COMT inhibitors: Entacapone MAO-B inhibitors: Selegiline, rasagiline
RESCUE FROM “OFF” PHASE	Apomorphine, Istradefylline

Solution for Question 5:

Correct Answer: C) Amantadine

Explanation:

The skin discolouration given above, with a history of treatment for parkinsonism, is suggestive of livedo reticularis, which can occur as a side effect of using amantadine.

Livedo reticularis occurs due to the release of catecholamines, which result in peripheral vasoconstriction.

Amantadine:

- Amantadine acts primarily by promoting dopamine release in the brain and NMDA receptor antagonism.
- Amantadine can be used in milder cases or short courses to supplement levodopa for advanced cases.
- In the latter situation, it suppresses motor fluctuations and abnormal movements.
- A fixed dose of 100 mg BO is used (not titrated according to response).
- The effect of a single dose lasts 8- 12 hours.

- Other side effects: Insomnia. Restlessness. Confusion. Nightmares. Anticholinergic effects. Hallucinations (rare). Ankle oedema.
 - Insomnia.
 - Restlessness.
 - Confusion.
 - Nightmares.
 - Anticholinergic effects.
 - Hallucinations (rare).
 - Ankle oedema.
 - Insomnia.
 - Restlessness.
 - Confusion.
 - Nightmares.
 - Anticholinergic effects.
 - Hallucinations (rare).
 - Ankle oedema.
-

Solution for Question 6:

Correct Answer: C) The efficacy of apomorphine improves when combined with ondansetron

Explanation:

Apomorphine is contraindicated with ondansetron.

Apomorphine:

- This non-ergot derivative is an injectable dopamine agonist which acts predominantly on the D4 dopaminergic receptors.
- It causes profound hypotension and loss of consciousness when combined with serotonergic antiemetics (5HT3 antagonists like ondansetron) and hence is contraindicated to combine with them.
- Other side effects: QT prolongation.

Incorrect Options:

Ropinirole and pramipexole are other non-ergot derivatives that are selective dopaminergic D2/D3 receptor agonists.

Improvement in symptoms can be seen much quicker (as early as 1-2 weeks) when compared with bromocriptine (which may take months) (OPTION A)

Ropinirole is one of the drugs preferred as an add-on during the on/off phase (OPTION C) and is also approved by the FDA for use in restless leg syndrome (OPTION D)

Note: Dopamine agonists like bromocriptine and cabergoline are ergot derivatives.

Solution for Question 7:

Correct Answer: D) Benzhexol

Explanation:

The above-mentioned clinical vignette is suggestive of drug-induced parkinsonism, and central anticholinergics like benzhexol (trihexyphenidyl) are the only drugs of choice for this condition.

Drugs causing Parkinsonism:

- MOA: Dopaminergic antagonism.
- Drugs: Metoclopramide. Haloperidol. Domperidone. Prochlorperazine.
- Metoclopramide.
- Haloperidol.
- Domperidone.
- Prochlorperazine.
- Metoclopramide.
- Haloperidol.
- Domperidone.
- Prochlorperazine.

Central anticholinergics:

Patients with Parkinsonism have neurochemical imbalances in the brain, with a reduction in dopamine and an increase in acetylcholine levels together responsible for the symptoms.

These drugs have high central: peripheral anticholinergic action, hence used in parkinsonism. They include:

- Trihexyphenidyl (Benzhexol).
- Procyclidine.
- Biperiden,
- Orphenadrine.
- Promethazine.

Common side effects include urinary retention, acute confusion, blurring of vision (mydriasis), and euphoria (orphenadrine).

Solution for Question 8:

Correct Answer: A) Trihexyphenidyl

Explanation:

The clinical presentation, along with the presence of rheumatic heart disease, is suggestive of Sydenham's chorea and anticholinergics like trihexyphenidyl will not be used in managing this.

Sydenham's chorea is characterised by unexpected, involuntary, and irregular movements of the upper limbs and head and characteristic darting movements of the tongue.

This can be due to dopaminergic excess in certain brain areas, and anticholinergics should not be given because central cholinergic excess will worsen symptoms.

Drugs used in the management of Sydenham's chorea:

- Haloperidol.
- Sodium valproate.
- Carbamazepine.
- Corticosteroids (refractory cases).

Solution for Question 9:

Correct Answer: B) Donepezil

Explanation:

Donepezil will not be used in the management of Lou Gehrig's disease.

Lou Gehrig's disease/amyotrophic lateral sclerosis is an inherited motor neuron disorder affecting both upper and lower motor neurons, leading to progressive muscle weakness, muscular atrophy and dysarthria. Death usually occurs by respiratory failure.

Drugs used in the management of ALS:

Direct action on the nervous system	Symptomatic (relieve spasticity)
Riluzole. Edaravone.	Baclofen. Tizanidine. Clonazepam (should be used cautiously as it can cause respiratory depression in advanced ALS).

Toferson is a recently approved drug for the management of ALS. It is an antisense oligonucleotide that targets SOD1 mRNA to reduce the synthesis of SOD1 protein.

Solution for Question 10:

Correct Answer: C) Trofinetide

Explanation:

Trofinetide is the drug that is approved by the FDA to be used in Rett syndrome.

Rett syndrome is a rare inherited neurological disorder which is characterised by:

- Female predominance.
- It affects walking, talking and breathing at later stages.
- Extrapyrimal symptoms: flapping/grasping hands.
- Behavioural changes.

Incorrect Options:

Drug	Use
Lonafarnib	Progeria

Omaveloxolone	Friedreich's ataxia
Ocrelizumab	Multiple sclerosis

Previous Year Questions

1. A 35-year-old male presents to the emergency department with complaints of blurred vision, dizziness, and severe headache after ingesting an unknown quantity of homemade alcohol earlier in the day. On examination, he is disoriented. Laboratory tests reveal severe metabolic acidosis with a high anion gap. Which metabolite is primarily responsible for the toxicity in this case?

- A. Methanol
 - B. Formic acid
 - C. Formaldehyde
 - D. Acetic acid
-

2. What is the etiology of malignant neuroleptic syndrome?

- A. Antidepressants
 - B. Anxiolytics
 - C. Antipsychotics
 - D. Anti-epileptics
-

3. Which of the following drugs acts by inhibiting the catechol-O-methyl transferase enzyme?

- A. Amantadine
- B. Rotigotine
- C. Tolcapone
- D. Selegiline

4. A patient took a drug for migraine treatment following which there was numbness and his hand was as below. What is the possible drug that was taken?



- A. Dihydroergotamine
- B. Sumatriptan
- C. Butorphanol
- D. Aspirin

5. Which of the following is the disadvantage of oxcarbazepine over carbamazepine?

- A. Hepatotoxicity
- B. Stevens-Johnson syndrome
- C. Thrombocytopenia
- D. Dilutional hyponatremia

6. What is the appropriate antidote for Benzodiazepine overdose?

- A. Flumazenil
 - B. Flurazepam
 - C. Barbiturates
 - D. Buspirone
-

7. What is the preferred drug for the treatment of facial nerve palsy?

- A. Acyclovir
 - B. Ceftriaxone
 - C. Prednisolone
 - D. Naproxen
-

8. In the case of a newborn delivered by an HIV-positive woman who was not receiving treatment, what is the recommended initial course of treatment to reduce the risk of mother-to-child transmission of HIV?

- A. Nevirapine 6 months
 - B. Zidovudine 6 weeks
 - C. Nevirapine + Zidovudine + Lamivudine 6 weeks
 - D. Nevirapine 6 weeks
-

9. In the management of CMV retinitis, which drug is considered the drug of choice?

- A. Cotrimoxazole
- B. Valacyclovir

- C. Tenofovir
 - D. Ganciclovir
-

10. Which of the following is not classified as a selective serotonin reuptake inhibitor (SSRI) used in the treatment of depression?

- A. Fluoxetine
 - B. Sertraline
 - C. Trazadone
 - D. Escitalopram
-

11. After undergoing lithium therapy for a year, all of the following adverse effects are visible, except:

- A. Hypothyroidism
 - B. Tremors
 - C. Diabetes insipidus
 - D. Tardive dyskinesia
-

12. A child is brought to the clinic with a fever and has been experiencing continuous seizures for the last 30 minutes. The caregivers report no seizure-free intervals during this time. What is the most appropriate immediate action to manage this patient?

- A. Manage the fever and wait for seizure to subside
- B. Give midazolam immediately
- C. Give clobazam immediately

D. Manage the airway and wait for seizure to subside

13. A 22-year-old woman visits the clinic expressing her desire to conceive. She has a history of epilepsy and is seeking information regarding which antiepileptic medication carries the greatest risk of birth defects and is generally avoided during pregnancy.

- A. Valproate
 - B. Carbamazepine
 - C. Phenytoin
 - D. Lamotrigine
-

14. A chronic alcoholic patient, after stopping the intake of alcohol, developed whole-body tremors along with vomiting and visual and auditory hallucinations. Which among the following is used to prevent another episode of these symptoms?

- A. Chlordiazepoxide
 - B. Clonidine
 - C. Topiramate
 - D. Antabuse
-

15. Which of the following is the half-life of lithium?

- A. 8 hours
- B. 12 hours
- C. 24 hours
- D. 48 hours

16. What is the mechanism of action of Ethosuximide?

- A. Enhances GABAergic inhibition
 - B. Inhibits calcium channels in thalamic neurons
 - C. Blocks sodium channels
 - D. Increases serotonin levels
-

17. Which anti-epileptic drug is commonly associated with gum hypertrophy and dizziness as side effects?

- A. Carbamazepine
 - B. Phenytoin
 - C. Levetiracetam
 - D. Topiramate
-

18. Which medication could potentially cause constipation, blurred vision, and dry mouth in a 70-year-old patient with a history of schizophrenia receiving treatment for drug-induced Parkinson's disease?

- A. Lithium
 - B. Pramipexole
 - C. Benhexol
 - D. Selegiline
-

19. Which of the following drugs exacerbates myasthenia and is usually avoided in a patient with myasthenia gravis? 1. Lidocaine 2. Digene 3. Meropenem 4.

Phenytoin

- A. 1 & 2
 - B. 2 & 3
 - C. 1 & 3
 - D. 1 only
-

20. For a patient who has commenced lithium therapy, which of the subsequent adverse effects should they be instructed to be vigilant for in order to detect any signs of toxicity?

- A. Malaise and sore throat
 - B. Vertigo and Diarrhoea
 - C. Fever
 - D. Breathlessness
-

21. Which of the below statements is accurate regarding disulfiram?

- A. When used with alcohol, the patients feel drowsy, with chest discomfort and respiratory depression
 - B. It is an anticraving drug
 - C. It can be initiated even in minimally motivated person
 - D. Its action persists up to 2-3 days after stopping the drug
-

22. Antidote for alprazolam is:

- A. Protamine sulphate
 - B. Flumazenil
 - C. EDTA
 - D. BAL
-

23. A patient was treated with Tricyclic antidepressants for a major depressive episode in the hospital for 4 weeks. What is the main concern while considering his discharge from the hospital?

- A. Need for drug monitoring
 - B. ECG monitoring for arrhythmias
 - C. Suicidal risk
 - D. Sedation being the side effect
-

24. Which of the following drugs are used for absence seizures? A. Carbamazepine B. Valproate C. Ethosuximide D. Gabapentin

- A. A, B and C
 - B. B and C only
 - C. B only
 - D. A, B and D
-

25. Chlorpromazine acts on which of the following receptor? 1) D2 and 5-HT2 receptor 2) GABA and beta - adrenergic receptor 3) Muscarinic M1 and alpha - adrenergic receptors 4) H1 receptors

- A. 1, 3 and 4 are correct

- B. Only 2 is correct
 - C. 1 and 2 are correct
 - D. All are correct
-

26. Z track technique is used for?

- A. Administration of nicotine patches
 - B. Monitoring of carbamazepine therapy
 - C. Administration of long acting depot antipsychotics
 - D. Monitoring of lithium therapy
-

27. A bank employee suffering from depression came to AIIMS OPD. He was started on Escitalopram. Which of these adverse effects cannot be explained with escitalopram?

- A. Sialorrhea
 - B. Vivid dreams
 - C. Anorgasmia
 - D. Nausea
-

28. Which of the following recently approved medication is used to treat amyotrophic lateral sclerosis?

- A. Edaravone
- B. Piracetam
- C. Dabrafenib
- D. Trametinib

29. Choose the correctly matched pairs regarding the drugs used in schizophrenia:

1. D2 antagonism: Reduces positive symptoms 2. 5HT2A antagonism: Reduces negative Symptoms 3. 5HT1A agonism: Weight loss 4. Muscarinic antagonism: Reduces extrapyramidal symptoms

- A. 1, 3
 - B. 1, 4
 - C. 1, 2, 4
 - D. 1, 2, 3, 4
-

30. Which of the following is used as an antidote in opioid overdose?

- A. Naloxone
 - B. Flumazenil
 - C. Fomepizole
 - D. Pethidine
-

31. Which of the following anti-epileptic drug is least teratogenic during pregnancy?

- A. Phenytoin
 - B. Valproate
 - C. Carbamazepine
 - D. Levetiracetam
-

32. Why does a 35-year-old male patient with schizophrenia, who is currently receiving clozapine treatment, require frequent blood monitoring?

(or)

Why does a patient with schizophrenia, who is receiving clozapine treatment, require frequent blood monitoring?

- A. Agranulocytosis
 - B. Myocarditis
 - C. Cerebral bleed
 - D. Seizures
-

33. A patient who was diagnosed with epilepsy was put on retigabine. Now phenytoin is being added. Which of the following changes should be made to Retigabine?

- A. Increase the dose
 - B. Decrease the dose
 - C. Stop the drug
 - D. Stop retigabine and start on carbamazepine
-

34. What is the most common side effect of Lamotrigine?

- A. Diplopia
 - B. Tremors
 - C. Neutropenia
 - D. Rash
-

35. A psychiatrist prescribes lithium for a patient who is diagnosed with bipolar disorder. Which of the following statements is incorrect regarding lithium?

- A. Lithium is teratogenic if given to pregnant females
 - B. Lithium toxicity is exacerbated with thiazide
 - C. Lithium can decrease thyroid hormone levels
 - D. Hemodialysis is not useful in lithium overdose
-

36. A young male patient presents with ptosis and muscle weakness, which reportedly worsens in the evening and improves in the morning. This is relieved by neostigmine. What is the likely diagnosis?

- A. Myasthenia gravis
 - B. Huntington chorea
 - C. Amyotrophic lateral sclerosis
 - D. External ophthalmoplegia
-

37. A patient presented with dizziness, cool, clammy skin, pinpoint pupils with blue lips and fingernails, suffering from respiratory depression. The drug used to reverse the effects is?

- A. Naloxone
 - B. Atropine
 - C. Phentolamine
 - D. Physostigmine
-

38. A patient of schizophrenia is being treated with clozapine. For which rare but serious side effects should he be monitored?

- A. Agranulocytosis
 - B. Renal bleed
 - C. Hepatomegaly
 - D. Seizures
-

39. A 35-year male patient presents to surgery emergency with painful erection for past 7 hours. He has a history of mood disorder and was recently prescribed a medication by treating psychiatrist. Which is the likely offending drug?

- A. Mirtazapine
 - B. Venlafaxine
 - C. Trazodone
 - D. Tianeptine
-

40. Which anti-amoebic medication, when combined with alcohol, may have caused the lady to experience symptoms such as nausea, vomiting, and dizziness at a social gathering?

- A. Metronidazole
 - B. Nitazoxanide
 - C. Paromomycin
 - D. Diloxanide
-

41. Which antiepileptic drug has the lowest risk of causing birth defects during pregnancy?

- A. Valproate

- B. Phenytoin
 - C. Carbamazepine
 - D. Levetiracetam
-

42. After being prescribed an antipsychotic drug, the patient developed parkinsonism. Which of the following medications is suitable for treatment?

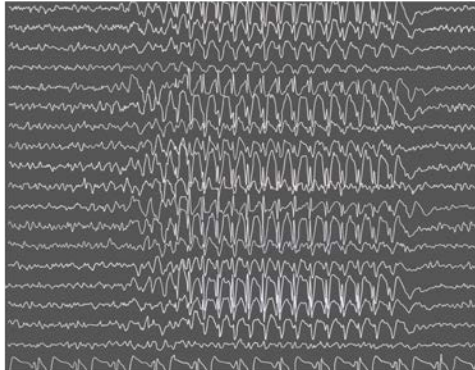
- A. Trihexyphenidyl
 - B. Entacapone
 - C. Rasagiline
 - D. Pramipexole
-

43. A female patient is on first-line pharmacological therapy for bipolar disorder for 6 months. She is fasting for a few days due to religious reasons and presented with seizures, coarse tremors, confusion, and weakness of limbs. Which of the following investigations should be done to help diagnose her condition?

- A. Serum potassium level
 - B. Serum lithium level
 - C. ECG
 - D. MRI
-

44. The parents of a 9-year-old girl have noticed their daughter often staring blankly, which is sometimes associated with chewing movements. There were also complaints of daydreaming and a decline in school performance. During the episodes, she did not have a problem with maintaining posture, nor were there generalized tonic and clonic movements. The child's EEG is given below. Which of

the following is a first-line agent to treat this condition?



- A. Carbamazepine
- B. Phenytoin
- C. Sodium valproate
- D. Clonazepam

45. A patient with symptoms of psychosis is prescribed aripiprazole. Which of the following best describes this drug, when compared to other antipsychotic drugs?

- A. No intrinsic activity, high affinity for D2 receptor, less extrapyramidal symptoms and more weight gain
- B. High intrinsic activity, no affinity for D2 receptor, more extrapyramidal symptoms and less weight gain
- C. Low intrinsic activity, high affinity for D2 receptor, less extrapyramidal symptoms and less weight gain
- D. High intrinsic activity, high affinity for D2 receptor, less extrapyramidal symptoms and less weight gain

46. Topiramate is used for:

- A. Treatment of acute migraine
 - B. Treatment of Lennox Gastaut syndrome
 - C. Prophylaxis of ADHD
 - D. Treatment of dementia
-

47. Which of the following is a novel anti-depressant?

- A. Lurasidone
 - B. Vilazodone
 - C. Asenapine
 - D. Blonanserin
-

48. Which of the following antidepressants causes the least sexual dysfunction?

- A. Fluoxetine
 - B. Venlafaxine
 - C. Mirtazapine
 - D. Imipramine
-

49. Which one of the following options refers to an opioid antitussive that acts centrally in the body?

- A. Ambroxol
- B. Diphenoxylate
- C. Levopropoxyphene
- D. Levorphanol

50. Which of the following is not an off-label use of risperidone?

- A. Obsessive Compulsive Disorder
 - B. Dementia
 - C. PTSD
 - D. Bipolar disorder
-

51. Which of the following anti-depressant drugs acts by inhibiting the reuptake of both serotonin and nor-adrenaline?

- A. Fluoxetine
 - B. Venlafaxine
 - C. Paroxetine
 - D. Dapoxetine
-

52. Which neurotransmitter is responsible for the condition observed in a 55-year-old male who presented with symptoms including rigidity, frequent falls, slowness in movements, tremors, and a short stepping gait upon entering the doctor's room for evaluation?

- A. GABA
 - B. Acetylcholine
 - C. Dopamine
 - D. Serotonin
-

53. Among the subsequent atypical antipsychotic medications, which one poses the greatest likelihood of galactorrhea?

- A. Iloperidone
 - B. Risperidone
 - C. Clozapine
 - D. Aripiprazole
-

54. Patient has pain in the facial nerve distribution while chewing and speaking. Which of the following drug should be given for the treatment?

- A. Lamotrigine
 - B. Carbamazepine
 - C. Phenobarbitone
 - D. Haloperidol
-

55. A patient presented to the casualty with signs of dehydration due to severe vomiting. The intern gave the patient a parenteral medication, following which the patient developed grimacing and abnormal posturing. What could have been the inciting drug of this reaction?

- A. Atropine
 - B. Ranitidine
 - C. Metoclopramide
 - D. Domperidone
-

56. The anti-epileptic drugs which can also be used for the treatment of neuropathic pain are: 1. Pregabalin 2. Carbamazepine 3. Lamotrigine 4. Gabapentin

- A. 1, 2 and 3
 - B. 1, 2 and 4
 - C. 1, 3 and 4
 - D. 2 and 4 only
-

57. A 25-year-old male presents with symptoms of decreased sleep, increased activity levels, and an elevated mood. He also has a family history of manic episodes. Which of the following treatments is most appropriate for his condition?

- A. Fluoxetine
 - B. Lithium
 - C. Carbamazepine
 - D. Lamotrigine
-

58. Anti-depressant drug which is used for smoking cessation is?

- A. Venlafaxine
 - B. Topiramate
 - C. Bupropion
 - D. Amitriptyline
-

59. Secretion of prolactin is inhibited by?

- A. Dopamine
- B. Nor-adrenaline

- C. Adrenaline
 - D. Serotonin
-

60. Which drug is most likely responsible for the visual disturbance and contraction of the visual field in a patient who was taking antiepileptics for seizure management?

- A. Phenobarbitone
 - B. Vigabatrin
 - C. Ethosuximide
 - D. Levetiracetam
-

61. A psychiatric patient was started on lithium for the management of bipolar disorder. To monitor the blood lithium levels, at what time interval should the blood sample be collected after the administration of lithium tablets?

- A. After 24 hours
 - B. After 12 hours
 - C. After 6 hours
 - D. Immediately after taking tablets
-

62. An elderly man presents with rigidity and tremors. On examination, he has blank facial expressions. Which of the following drugs can be used to manage this condition?

- A. Clozapine
- B. Donepezil
- C. Selegiline

D. Haloperidol

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	1
Question 5	4
Question 6	1
Question 7	3
Question 8	3
Question 9	4
Question 10	3
Question 11	4
Question 12	2
Question 13	1
Question 14	1
Question 15	3
Question 16	2
Question 17	2
Question 18	3
Question 19	1
Question 20	2
Question 21	1
Question 22	2

Question 23	3
Question 24	2
Question 25	1
Question 26	3
Question 27	1
Question 28	1
Question 29	3
Question 30	1
Question 31	4
Question 32	1
Question 33	1
Question 34	4
Question 35	4
Question 36	1
Question 37	1
Question 38	1
Question 39	3
Question 40	1
Question 41	4
Question 42	1
Question 43	2
Question 44	3
Question 45	3
Question 46	2
Question 47	2
Question 48	3

Question 49	3
Question 50	4
Question 51	2
Question 52	3
Question 53	2
Question 54	2
Question 55	3
Question 56	2
Question 57	2
Question 58	3
Question 59	1
Question 60	2
Question 61	2
Question 62	3

Solution for Question 1:

Answer: B) Formic acid

Explanation:

This patient's symptoms (blurred vision, dizziness, severe headache, disorientation) and high anion gap metabolic acidosis after ingesting homemade alcohol suggest methanol poisoning.

Methanol:

Adulteration in Country
liquor.
Solvent in industrial
processes.
Antifreeze.

Mechanism of Action	Methanol ■ Formaldehyde (oxidation by alcohol dehydrogenase). Formaldehyde ■ Formic acid (oxidation by aldehyde dehydrogenase). Formic Acid ■ Inhibition of Cytochrome c Oxidase ■ Cellular Hypoxia and Metabolic Acidosis.
Features of Toxicity	CNS: Headache, dizziness, confusion, ataxia, seizures Ocular: Blurred ("snowfield" vision). Gastrointestinal: Nausea, vomiting, fruity breath and abdominal pain Metabolic: Severe metabolic acidosis, elevated anion gap
Treatment	Mnemonic: SHAFE Sodium Bicarbonate to correct metabolic acidosis Hemodialysis to remove methanol and its metabolites Antidote: Fomepizole or Ethanol to inhibit ADH Folinic acid or folic acid to enhance formate metabolism.

- Methanol (Option A) is incorrect because methanol itself is not directly toxic; its metabolites (formic acid and formaldehyde) cause the toxicity.
- Formaldehyde (Option C) is incorrect because formaldehyde is an intermediate metabolite but is rapidly converted to formic acid, which is more toxic.
- Acetic Acid (Option D) is incorrect because acetic acid is the metabolic product of ethanol, not methanol, and does not cause significant toxicity.

Solution for Question 2:

Correct Option. C

- Antipsychotics: Malignant neuroleptic syndrome is primarily associated with the use of antipsychotic medications, which are used to treat psychiatric disorders such as schizophrenia and bipolar disorder. Malignant neuroleptic syndrome is a rare but potentially life-threatening adverse reaction to certain antipsychotic medications. It is characterized by symptoms such as severe muscle stiffness, high fever, altered mental status, autonomic dysfunction, and increased levels of creatine phosphokinase (CPK) in the blood. Prompt recognition and discontinuation of the offending medication are essential in managing malignant neuroleptic syndrome.

Incorrect options

Option A. Antidepressants: Antidepressants are medications used to treat various forms of depression. While certain antidepressants can have an impact on serotonin levels in the brain, they are not typically associated with causing the malignant neuroleptic syndrome. However, in rare cases, a condition known as serotonin syndrome can occur when taking certain antidepressants or combining them with other medications that increase serotonin levels. Serotonin syndrome shares some similarities with MNS but is distinct from it.

Option B. Anxiolytics: Anxiolytics, also known as anti-anxiety medications, are used to treat anxiety disorders and help manage symptoms of anxiety. They generally do not cause the malignant neuroleptic syndrome

Option D. Anti-epileptics: Anti-epileptic drugs are used to treat seizures and epilepsy. While some anti-epileptic medications may have certain side effects, they are not typically associated with causing malignant neuroleptic syndrome.

Solution for Question 3:

Correct Option C - Tolcapone:

- Tolcapone is a medication that acts by inhibiting the catechol-O-methyltransferase (COMT) enzyme. COMT is responsible for the breakdown of catecholamines, such as dopamine, norepinephrine, and epinephrine, in the central nervous system and peripheral tissues. By inhibiting COMT, tolcapone prolongs the action of these neurotransmitters, particularly dopamine, leading to increased dopamine levels and enhanced dopaminergic transmission.
- Therefore, among the options provided, tolcapone is the drug that acts by inhibiting the catechol-O-methyltransferase (COMT) enzyme.

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Therefore, among the options provided, tolcapone is the drug that acts by inhibiting the catechol-O-methyltransferase (COMT) enzyme.

Incorrect Options:

Option A - Amantadine: Amantadine is an antiviral and antiparkinsonian medication that primarily acts by increasing the release of dopamine and inhibiting its reuptake. It does not directly inhibit COMT.

Option B - Rotigotine: Rotigotine is a dopamine agonist used in the treatment of Parkinson's disease and restless legs syndrome. It stimulates dopamine receptors directly and does not inhibit COMT.

Option D - Selegiline: Selegiline is a selective monoamine oxidase-B (MAO-B) inhibitor, which increases dopamine levels by inhibiting the breakdown of dopamine by the MAO-B enzyme. It does not directly inhibit COMT.

Solution for Question 4:

Correct Option A - Dihydroergotamine:

- Based on the given information, the possible drug that was taken by the patient is Dihydroergotamine.
- Dihydroergotamine is a medication commonly used for the treatment of migraines. It belongs to a class of drugs called ergot alkaloids, which work by constricting blood vessels in the brain and reducing inflammation associated with migraines.
- One of the known side effects of Dihydroergotamine is peripheral vasoconstriction, which can lead to reduced blood flow to the extremities. This can manifest as numbness, tingling, and discoloration of the fingers, commonly referred to as Raynaud's phenomenon. Raynaud's phenomenon is characterized by color changes in the fingers (usually white, then blue, and finally red) in response to cold temperature or stress.
- Therefore, based on the symptoms of numbness and discoloration of the fingers, Dihydroergotamine is the drug most likely responsible for these effects in the patient.

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Therefore, based on the symptoms of numbness and discoloration of the fingers, Dihydroergotamine is the drug most likely responsible for these effects in the patient.

Incorrect Options:

Option B - Sumatriptan: Sumatriptan is another medication used for the treatment of migraines. While it can cause side effects such as tingling or numbness, it is less commonly associated with peripheral vasoconstriction and Raynaud's phenomenon compared to Dihydroergotamine.

Option C - Butorphanol: Butorphanol is an opioid analgesic used for the treatment of pain, including migraines. It is not typically associated with peripheral vasoconstriction and Raynaud's phenomenon.

Option D - Aspirin: Aspirin is a nonsteroidal anti-inflammatory drug (NSAID) commonly used for pain relief. It is not known to cause peripheral vasoconstriction or Raynaud's phenomenon.

Solution for Question 5:

Correct Option D - Dilutional hyponatremia:

Adverse effects of carbamazepine are:

- Hepatotoxicity
- Dilutional hyponatremia
- Steven Jones syndrome
- Agranulocytosis
- Thrombocytopenia
- Teratogenic

Oxcarbazepine has fewer side effects but it has a higher incidence of dilutional hyponatremia in patients that take this drug.

Incorrect Options:

Option A - Hepatotoxicity:

- This option is incorrect because hepatotoxicity is more commonly seen with carbamazepine use.

Option B - Stevens-Johnson syndrome:

- This option is incorrect because Stevens-Johnson syndrome although rare is more commonly seen with carbamazepine use.

Option C - Thrombocytopenia:

- This option is incorrect because thrombocytopenia is more commonly seen with carbamazepine use.

Solution for Question 6:

Correct Option A - Flumazenil:

- Flumazenil is the correct antidote for Benzodiazepine overdose. It is a specific antagonist that works by competitively binding to the same receptors in the brain that Benzodiazepines target, effectively reversing their effects. By blocking the receptors, flumazenil can rapidly reverse the sedative and respiratory-depressant effects of Benzodiazepines, potentially preventing life-threatening complications of overdose.

Incorrect Options:

Option B - Flurazepam: Flurazepam is not an antidote for Benzodiazepine overdose; it is actually a Benzodiazepine medication itself, used to treat insomnia. It would not counteract the effects of other Benzodiazepines.

Option C - Barbiturates: Barbiturates are a separate class of drugs that act as central nervous system depressants. They have a different mechanism of action compared to Benzodiazepines, and they are not considered antidotes for Benzodiazepine overdose.

Option D - Buspirone: Buspirone is an anxiolytic medication, but it does not function as an antidote for Benzodiazepine overdose. It has a distinct mechanism of action and is used for long-term treatment of anxiety disorders.

Solution for Question 7:

Correct Option C - Prednisolone:

- The drug of choice for facial nerve palsy, particularly Bell's palsy, is prednisolone (Option C). This medication helps reduce inflammation around the affected facial nerve, leading to improved nerve function and facial muscle recovery.

Incorrect Option:

Option A - Acyclovir: Acyclovir is an antiviral medication commonly used to treat viral infections caused by herpes viruses, such as herpes simplex and varicella-zoster. While facial nerve palsy can be caused by herpes zoster (Ramsay Hunt syndrome), acyclovir is not the primary drug of choice for the treatment of facial nerve palsy. The mainstay of treatment for this condition involves addressing inflammation and promoting nerve recovery.

Option B - Ceftriaxone: Ceftriaxone is a broad-spectrum antibiotic commonly used to treat various bacterial infections. However, bacterial infections are not a typical cause of facial nerve palsy, and antibiotics like ceftriaxone are not the primary treatment for this condition.

Option D - Naproxen: Naproxen is a nonsteroidal anti-inflammatory drug (NSAID) commonly used to relieve pain and inflammation. While NSAIDs can help manage pain associated with facial nerve palsy, they do not directly address the underlying inflammation of the facial nerve. Moreover, corticosteroids like prednisolone are more effective in reducing inflammation and improving outcomes in cases of facial nerve palsy.

Solution for Question 8:

Correct Option C - Nevirapine + Zidovudine + Lamivudine 6 weeks:

- The newborn in the given clinical scenario falls under the high-risk category, and the initiation of presumptive HIV therapy is indicated in this newborn.
- Presumptive HIV therapy in this case involves: Zidovudine + lamivudine + (either nevirapine or raltegravir) for 6 weeks. [Raltegravir is indicated in infants at risk of HIV-2 infection].

Incorrect Options:

Option A - Nevirapine 6 months: Nevirapine is an antiretroviral medication that can help prevent mother-to-child transmission of HIV. However, the recommended duration of nevirapine prophylaxis for newborns is typically 6 weeks, not 6 months. Prolonged use may expose the infant to unnecessary medication and potential side effects.

Option B - Zidovudine 6 weeks: Zidovudine is another antiretroviral drug used to prevent mother-to-child transmission of HIV. The standard regimen for newborns involves administering zidovudine for 6 weeks. This regimen has been shown to be effective in reducing the risk of transmission when combined with other preventive measures.

Option D - Nevirapine 6 weeks: Nevirapine alone is a commonly used antiretroviral medication for newborns at risk of HIV transmission. The recommended duration of nevirapine prophylaxis is generally 6 weeks. This option can be effective in reducing the risk of transmission when used as part of a comprehensive prevention strategy.

Solution for Question 9:

Correct Option D - Ganciclovir:

- Ganciclovir is the drug of choice for CMV retinitis. Ganciclovir and its oral prodrug valganciclovir are potent antiviral medications specifically designed to target CMV infections. They work by inhibiting the replication of CMV DNA. Ganciclovir can be administered intravenously, orally, or through intraocular implants to treat CMV retinitis effectively.

Incorrect Options:

Option A - Cotrimoxazole: Cotrimoxazole is not the drug of choice for CMV retinitis. It is an antibiotic combination of sulfamethoxazole and trimethoprim commonly used to treat various bacterial infections and some opportunistic infections, but it is not effective against cytomegalovirus (CMV) infections.

Option B - Valacyclovir: Valacyclovir is an antiviral medication primarily used for the treatment of herpes simplex and herpes zoster infections. However, it is not the drug of choice for CMV retinitis. While valacyclovir is effective against herpes viruses, it does not have the same potency against CMV.

Option C - Tenofovir: Tenofovir is an antiretroviral medication used in the treatment of HIV infection and prevention. While it can have activity against some other viruses, it is not typically used as a first-line treatment for CMV retinitis. Other medications like ganciclovir are more

effective against CMV.

Solution for Question 10:

Correct Option C - Trazodone:

- No, trazodone is not an SSRI. It is an atypical antidepressant that primarily functions as a serotonin antagonist and reuptake inhibitor (SARI). While it can be used to treat depression, it is not classified as an SSRI.

Incorrect Options:

Option A - Fluoxetine: Yes, fluoxetine is an SSRI (selective serotonin reuptake inhibitor) that is commonly prescribed for the management of depression. It works by increasing the levels of serotonin in the brain, which can help alleviate symptoms of depression.

Option B - Sertraline: Yes, sertraline is another SSRI that is frequently used to treat depression. Similar to other SSRIs, sertraline enhances serotonin levels in the brain, aiding in the reduction of depressive symptoms.

Option D - Escitalopram: Yes, escitalopram is an SSRI commonly prescribed for the management of depression. It functions by increasing the availability of serotonin in the brain, which can lead to an improvement in mood and reduction of depressive symptoms.

Solution for Question 11:

Correct Option D - Tardive dyskinesia:

- Tardive dyskinesia is not typically associated with lithium therapy. Tardive dyskinesia is a movement disorder that can occur as a side effect of certain medications, particularly antipsychotic medications. It involves involuntary and repetitive movements, often involving the face and tongue. This side effect is more commonly associated with antipsychotic drugs rather than lithium.

Incorrect Options:

Option A - Hypothyroidism: Lithium therapy is associated with the risk of hypothyroidism, which is an underactive thyroid gland. This adverse effect occurs due to the impact of lithium on thyroid function. Regular monitoring of thyroid function is often recommended during lithium treatment.

Option B - Tremors: Tremors, specifically fine hand tremors, are a well-known side effect of lithium therapy. They can occur as a result of the medication's effect on the central nervous system. These tremors are usually dose-dependent and can be managed by adjusting the dosage of lithium.

Option C - Diabetes insipidus: Lithium therapy can lead to the development of diabetes insipidus, which is a condition characterized by excessive thirst and production of large amounts of diluted urine. It occurs due to the impact of lithium on the kidneys' ability to concentrate urine. Regular monitoring of kidney function and hydration status is important during lithium treatment.

Solution for Question 12:

Correct Option B - Give midazolam immediately:

- Giving midazolam immediately is the most appropriate action in the case of continuous seizures. Midazolam is a fast-acting benzodiazepine that can help terminate the seizure activity.

Incorrect Options:

Option A - Manage the fever and wait for seizure to subside: Managing the fever and waiting for the seizure to subside might not be the best option in the case of continuous seizures, as this could lead to prolonged and potentially harmful seizures.

Option C - Give clobazam immediately: Giving clobazam immediately is not the first-line treatment for acute seizure management, especially in the case of continuous seizures.

Option D - Manage the airway and wait for seizure to subside: Managing the airway and waiting for the seizure to subside is important, but providing a medication like midazolam to stop the continuous seizure is usually the preferred initial approach.

Solution for Question 13:

Correct Option A:

- Valproate: Valproate has the highest teratogenic potential among the options provided and is not typically prescribed to pregnant women due to the increased risk of birth defects.

Incorrect options:

Option B. Carbamazepine: Carbamazepine has a lower teratogenic potential compared to valproate.

Option C. Phenytoin: Phenytoin has a lower teratogenic potential compared to valproate.

Option D. Lamotrigine: Lamotrigine can be given during pregnancy.

Solution for Question 14:

Correct Option A - Chlordiazepoxide:

- This medication belongs to the benzodiazepine class and is commonly used to manage alcohol withdrawal symptoms, including tremors, anxiety, agitation, and seizures. It acts by enhancing the inhibitory effects of gamma-aminobutyric acid (GABA) in the brain, thereby reducing the excitability of neurons and calming the central nervous system. It is long-acting and self-tapering because of active metabolites.

Incorrect Options:

Option B - Clonidine: While clonidine can be used to manage some symptoms of alcohol withdrawal, such as hypertension and agitation, it is not typically used as a first-line agent for preventing the full spectrum of withdrawal symptoms, including tremors, vomiting, and hallucinations.

Option C - Topiramate: Topiramate is an antiepileptic drug that has been studied for its potential role in reducing alcohol cravings and promoting abstinence in patients with alcohol dependence. However, it is not primarily used to manage acute alcohol withdrawal symptoms.

Option D - Antabuse: Disulfiram is a medication used to support the treatment of chronic alcoholism by producing an acute sensitivity to alcohol. It works by inhibiting the enzyme acetaldehyde dehydrogenase, leading to an accumulation of acetaldehyde in the body when

alcohol is consumed. However, it is not typically used to manage acute alcohol withdrawal symptoms.

Solution for Question 15:

Correct Option C -24 hours:

- The correct half-life of lithium is approximately 24 hours.

Incorrect Options:

Options A, B ,D:

- These are not the half-lives of Lithium
-

Solution for Question 16:

Correct Option B - Inhibits calcium channels in thalamic neurons:

- Ethosuximide exclusively blocks the T type of calcium channel in thalamic neurons.
- It is only used in absence seizures

Incorrect Options:

Options A, C, D:

- Enhancement of GABAergic inhibition, blockage of sodium channels, and Increase in serotonin levels are not the mechanism of action of Ethosuximide
-

Solution for Question 17:

Answer: B) Phenytoin

Explanation:

Phenytoin is a commonly used anti-epileptic drug (AED) that is well known for causing gum hypertrophy (gingival hyperplasia) and dizziness as side effects.

- Mechanism of Action: Blocks voltage-gated sodium channels, stabilising neuronal membranes and preventing seizures.
- Blocks voltage-gated sodium channels, stabilising neuronal membranes and preventing seizures.
- Blocks voltage-gated sodium channels, stabilising neuronal membranes and preventing seizures.

Adverse effects of Phenytoin include:

Carbamazepine (Option A) is incorrect because while it causes dizziness and ataxia, it is not associated with gum hypertrophy.

Levetiracetam (Option C) is incorrect because it commonly causes irritability and mood changes, but not gum hypertrophy.

Topiramate (Option D) is incorrect because it is associated with cognitive slowing, weight loss, and nephrolithiasis, but not gingival hyperplasia.

Solution for Question 18:

Correct Option C - Benzhexol:

- The above symptoms are likely due to anticholinergic toxicity.
- The drug that may be responsible is Benzhexol.
- Benzhexol, also known as trihexyphenidyl, is an anticholinergic medication commonly used to treat the extrapyramidal symptoms (EPS) associated with the use of antipsychotic drugs such as drug-induced Parkinson's disease. EPS includes symptoms such as tremors, rigidity, and bradykinesia. Benzhexol works by blocking the action of acetylcholine, a neurotransmitter in the brain that is involved in motor control.
- However, because benzhexol blocks acetylcholine, it can also cause a number of anticholinergic side effects, such as constipation, blurred vision, and dry mouth, which are consistent with the symptoms described in the scenario.

Incorrect Options:

Option A - Lithium: Lithium is a mood stabilizer that is commonly used to treat bipolar disorder. While it can cause a number of side effects, including tremors and polyuria, the symptoms described in the scenario are not typically associated with lithium use.

Option B - Pramipexole: Pramipexole is a dopamine agonist that is commonly used to treat Parkinson's disease. While it can cause a number of side effects, including constipation, the other symptoms described in the scenario are not typically associated with pramipexole use.

Option D - Selegiline: Selegiline is a selective monoamine oxidase B (MAO-B) inhibitor that is commonly used to treat Parkinson's disease. While it can cause a number of side effects, including dry mouth, the other symptoms described in the scenario are not typically associated with selegiline use.

Solution for Question 19:

Correct Option A - 1 & 2:

- Both Lidocaine and Digene can exacerbate myasthenia gravis and are usually avoided in patients with this condition.
- Myasthenia gravis is an autoimmune neuromuscular disorder characterized by muscle weakness and fatigue. It occurs due to the production of autoantibodies against the acetylcholine receptors at the neuromuscular junction, leading to impaired neuromuscular transmission.
- Lidocaine is a local anesthetic that can have a depolarizing effect on the neuromuscular junction, potentially worsening the symptoms of myasthenia gravis. It can interfere with the normal functioning of acetylcholine receptors, leading to increased weakness and muscle fatigue.
- Digene, also known as antacids or antacid formulations, can contain magnesium or aluminum salts. These salts can interfere with the absorption of certain medications, including pyridostigmine, which is commonly used to improve muscle strength in myasthenia gravis. Therefore, the use of Digene can potentially reduce the effectiveness of pyridostigmine and exacerbate symptoms in patients with myasthenia gravis.

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Incorrect Options:

Option B - 2 & 3:

- (3) Meropenem is a broad-spectrum antibiotic and is not specifically known to exacerbate myasthenia gravis. While it is important to consider potential drug interactions and side effects when prescribing antibiotics to patients with myasthenia gravis, meropenem itself is not known to worsen the condition.

(3) Meropenem is a broad-spectrum antibiotic and is not specifically known to exacerbate myasthenia gravis. While it is important to consider potential drug interactions and side effects when prescribing antibiotics to patients with myasthenia gravis, meropenem itself is not known to worsen the condition.

Option C - 1 & 3:

- (3) Meropenem is a broad-spectrum antibiotic and is not specifically known to exacerbate myasthenia gravis. While it is important to consider potential drug interactions and side effects when prescribing antibiotics to patients with myasthenia gravis, meropenem itself is not known to worsen the condition.

Option D - 1 only:

Solution for Question 20:

Correct Option B - Vertigo and Diarrhoea:

- Lithium is a medication commonly used for the treatment of bipolar disorder. While it can be effective in managing mood symptoms, it is important to monitor for potential side effects, including signs of lithium toxicity.
- One of the hallmark symptoms of lithium toxicity is neurologic and gastrointestinal disturbance. Vertigo, which is a sensation of spinning or dizziness, can occur as a result of lithium toxicity. Diarrhea is another common gastrointestinal symptom associated with lithium toxicity.
- Other signs and symptoms of lithium toxicity can include tremors, muscle weakness, lack of coordination, confusion, blurred vision, excessive thirst, increased urination, and persistent nausea.

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Other signs and symptoms of lithium toxicity can include tremors, muscle weakness, lack of coordination, confusion, blurred vision, excessive thirst, increased urination, and persistent nausea.

Incorrect Options:

Option A - Malaise and sore throat: While malaise (a general feeling of discomfort or uneasiness) and sore throat can be associated with certain conditions or infections, they are not specific symptoms of lithium toxicity. They are not commonly observed as side effects of lithium therapy.

Option C - Fever: Fever is not a typical symptom of lithium toxicity. If a patient on lithium therapy develops a fever, it is more likely to be due to an unrelated cause such as an infection.

Option D - Breathlessness: Breathlessness (shortness of breath or difficulty breathing) is not a known side effect or symptom of lithium toxicity. It is important to differentiate between potential respiratory issues unrelated to lithium therapy and the specific side effects associated with lithium toxicity.

Solution for Question 21:

Correct Option A - When used with alcohol, the patients feel drowsy, with chest discomfort and respiratory depression:

- When disulfiram is used with alcohol, patients may experience symptoms such as drowsiness, chest discomfort, and respiratory depression.
- Disulfiram is a medication used in the treatment of alcohol dependence. It works by inhibiting the enzyme acetaldehyde dehydrogenase, which is responsible for breaking down acetaldehyde, a toxic byproduct of alcohol metabolism. When a person taking disulfiram consumes alcohol, acetaldehyde accumulates in the body, leading to a range of unpleasant and potentially dangerous symptoms known as the disulfiram-alcohol reaction.
- Other symptoms may include flushing, nausea, vomiting, headache, blurred vision, confusion, and hypotension. Severe reactions can include cardiovascular collapse, myocardial infarction, acute congestive heart failure, unconsciousness, convulsions, and even death in rare cases.

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Other symptoms may include flushing, nausea, vomiting, headache, blurred vision, confusion, and hypotension. Severe reactions can include cardiovascular collapse, myocardial infarction, acute congestive heart failure, unconsciousness, convulsions, and even death in rare cases.

Incorrect Options:

Option B - It is an anti-craving drug: Disulfiram is not primarily considered an anticraving drug. It does not directly reduce the craving for alcohol or address the underlying psychological aspects of alcohol dependence. Its primary mechanism of action is to create an aversive reaction to alcohol intake, acting as a deterrent to drinking rather than directly targeting cravings.

Option C - It can be initiated even in minimally motivated person: Disulfiram is generally initiated in individuals who are highly motivated to quit drinking and who understand the potential consequences of consuming alcohol while on the medication. It requires a level of commitment and understanding from the individual as they need to be aware of the risks associated with the disulfiram-alcohol reaction. Therefore, it is typically not initiated in minimally motivated individuals.

Option D - Its action persists up to 2-3 days after stopping the drug: The effects of disulfiram do not persist for only 2-3 days after stopping the drug. Disulfiram effects can last up to 7-14 days. Therefore, patients taking disulfiram should be counseled about this to avoid untoward disulfiram-alcohol reactions.

Solution for Question 22:

Correct Option B - Flumazenil:

- Flumazenil is the correct antidote for alprazolam.
- It is a specific antagonist for benzodiazepines, including alprazolam.
- Flumazenil works by competitively binding to the benzodiazepine receptor sites in the brain, reversing the sedative and anxiolytic effects of benzodiazepines.
- It is used in cases of benzodiazepine overdose or in situations where the reversal of sedation or respiratory depression caused by benzodiazepines is necessary.

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It is used in cases of benzodiazepine overdose or in situations where the reversal of sedation or respiratory depression caused by benzodiazepines is necessary.

Incorrect Options:

Option A - Protamine sulfate: Protamine sulfate is not an antidote for alprazolam. It is a medication primarily used as an antidote for heparin, a blood-thinning medication.

Option C - EDTA: EDTA (Ethylene Diamine Tetraacetic Acid) is a chelating agent used to treat heavy metal poisoning. It is not an antidote for alprazolam.

Option D - BAL: BAL (British Anti-Lewisite) , also known as dimercaprol, is a chelating agent used as an antidote for heavy metal poisoning, specifically for arsenic, mercury, and lead. It is not an antidote for alprazolam.

Solution for Question 23:

Correct Option C - Suicidal risk:

- The main concern while considering the patient's discharge from the hospital after being treated with a tricyclic antidepressant (TCA) for a major depressive episode is the suicidal risk.
- Patients with major depressive episodes, especially when they have just started or changed their antidepressant medication, are at an increased risk of suicidal ideation and behavior. This risk is particularly significant in the initial weeks of treatment, including during hospitalization.
- It is crucial to assess the patient's suicidal risk thoroughly before considering discharge. The healthcare team should evaluate the patient's mood, suicidal thoughts, and overall mental state. The patient should be closely monitored for any signs of worsening depression or emergence of suicidal ideation. A comprehensive safety plan should be developed, including providing appropriate resources and ensuring the patient has access to support systems.

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Incorrect Options:

Option A - Need for drug monitoring: While some medications, including TCAs, may require therapeutic drug monitoring (TDM) in certain situations, routine drug monitoring is not the main concern when considering the patient's discharge. TDM is primarily used to assess drug levels and adjust dosages in specific cases where there is a narrow therapeutic index or potential drug interactions. However, for TCAs, routine TDM is not typically required.

Option B - ECG monitoring for arrhythmias: While TCAs can have cardiac side effects, such as QT prolongation and cardiac arrhythmias, routine ECG monitoring may be performed during the initial phase of TCA treatment, especially in the hospital setting. The main concern during discharge, however, is not primarily focused on ECG monitoring. If ECG abnormalities were detected during the hospital stay, appropriate management and monitoring would have been conducted at that time.

Option D - Sedation being the side effect: Sedation is indeed a side effect of TCAs, and it is important to assess and monitor the patient for sedation during treatment. However, the main concern while considering discharge is not solely focused on sedation. Other factors, such as the patient's overall clinical response, safety, and risk of suicide, are more critical considerations during the discharge planning process.

Solution for Question 24:

Correct Option B - B and C only:

- Valproate (option B) and Ethosuximide (option C) are the drugs commonly used for the treatment of absence seizures.
- Absence seizures, also known as petit mal seizures, are a type of generalized seizure characterized by a brief loss of consciousness or awareness. They typically last for a few seconds and are characterized by a sudden interruption of ongoing activities, blank staring, and subtle motor movements like blinking or lip smacking.
- Valproate is an antiepileptic drug that is effective in managing absence seizures. It works by increasing the levels of the inhibitory neurotransmitter gamma-aminobutyric acid (GABA) in the brain, which helps to control and reduce seizure activity.
- Ethosuximide is another antiepileptic drug that is specifically indicated for the treatment of absence seizures. It works by reducing the low-threshold calcium currents in neurons, which helps to suppress the abnormal electrical activity associated with absence seizures.

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Ethosuximide is another antiepileptic drug that is specifically indicated for the treatment of absence seizures. It works by reducing the low-threshold calcium currents in neurons, which helps to suppress the abnormal electrical activity associated with absence seizures.

Incorrect Options:

Option A - A, B and C: Carbamazepine is an antiepileptic drug that is mainly used to treat focal seizures and certain types of generalized seizures. However, it is not considered the first-line treatment for absence seizures. In fact, Carbamazepine may exacerbate absence seizures and is generally avoided in this specific seizure type.

Option C - B only is incorrect

Option D - A, B and D: Gabapentin is an antiepileptic drug that is primarily used to treat focal seizures and neuropathic pain. While it may have some efficacy in certain types of generalized seizures, it is not typically considered a first-line treatment for absence seizures.

Solution for Question 25:

Correct Option A - 1, 3 and 4 are correct

Chlorpromazine is an antipsychotic medication that acts on multiple receptors in the body. Let's examine each option:

- (1) D2 and 5-HT2 receptor: Chlorpromazine primarily acts as an antagonist at the D2 dopamine receptors and the 5-HT2 serotonin receptors. By blocking these receptors, chlorpromazine helps to reduce the symptoms of psychosis and has antipsychotic effects.
- (3) Muscarinic M1 and alpha-adrenergic receptors: Chlorpromazine also exhibits antagonistic effects at the muscarinic M1 receptors and the alpha-adrenergic receptors. Antagonism at muscarinic receptors can lead to side effects such as dry mouth, blurred vision, and constipation. Antagonism at alpha-adrenergic receptors can result in symptoms like orthostatic hypotension (low blood pressure when standing up).
- (4) H1 receptors: Chlorpromazine also acts as an antagonist at the H1 histamine receptors. Blockade of H1 receptors contributes to the sedative effects of chlorpromazine.

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(4) H1 receptors: Chlorpromazine also acts as an antagonist at the H1 histamine receptors. Blockade of H1 receptors contributes to the sedative effects of chlorpromazine.

Incorrect Options:

Option B - Only 2 is correct: Chlorpromazine does not primarily act on GABA and beta-adrenergic receptors. While it does have some interaction with these receptors, it is not the primary mechanism of action.

Option C - 1 and 2 are correct: While option 1 (D2 and 5-HT2 receptor) is correct, option 2 (GABA and beta-adrenergic receptor) is not accurate. Chlorpromazine's main actions are not related to GABA or beta-adrenergic receptors.

Option D - All are correct: While options 1 and 4 are correct, option 3 (Muscarinic M1 and alpha-adrenergic receptors) is inaccurate. Chlorpromazine does act on these receptors, but it is not the primary focus of its pharmacological effects.

Solution for Question 26:

Correct Option C - Administration of long-acting depot antipsychotics:

- The Z-track technique is a method used for the administration of certain medications, particularly long-acting depot formulations. It involves pulling the skin and underlying tissue to one side before injecting the medication into a deep muscle, and then releasing the skin after the injection is complete. This technique helps to create a zig-zag or "Z" path for the medication, preventing leakage or tracking of the medication along the injection site.
- The Z-track technique is specifically used for medications that may cause skin staining or irritation or have a potential for leakage. Long-acting depot antipsychotics, such as haloperidol decanoate or fluphenazine decanoate, are commonly administered using the

Z-track technique to ensure proper delivery of the medication into the muscle and minimizes skin irritation or discoloration.

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Incorrect Options:

Option A - Administration of nicotine patches: Nicotine patches are typically applied to the skin using a different technique, usually adhering to the manufacturer's instructions for application.

Option B - Monitoring of carbamazepine therapy: Carbamazepine therapy typically involves monitoring blood levels of the medication and adjusting the dosage as needed but does not require the Z-track technique for administration.

Option D - Monitoring of lithium therapy: Lithium therapy is usually orally administered. Z track is for IM injections.

Solution for Question 27:

Correct Option A - Sialorrhea:

- Sialorrhea (excessive salivation) cannot be explained with escitalopram.
- Escitalopram is a selective serotonin reuptake inhibitor (SSRI) commonly used for the treatment of depression and anxiety disorders. SSRI's can produce decreased saliva production and dryness of the mouth. Therefore sialorrhea would not be explained by escitalopram use..

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Incorrect Options:

Option B - Vivid dreams: Vivid dreams or increased dreaming can be a side effect of escitalopram and other SSRIs. Many individuals experience changes in dreaming patterns, such as more intense or vivid dreams, while taking these medications.

Option C - Anorgasmia: Anorgasmia refers to the difficulty or inability to achieve orgasm. It is a recognized side effect of SSRIs, including escitalopram. SSRIs can interfere with sexual function and may cause sexual side effects such as decreased libido, delayed ejaculation, or difficulty reaching orgasm.

Option D - Nausea: Nausea is a common side effect of escitalopram and other SSRIs, especially during the initial stages of treatment. It may occur in some individuals but usually improves over time or with dose adjustments.

Solution for Question 28:

Correct Option A - Edaravone:

- Edaravone is a drug that has been approved for the treatment of amyotrophic lateral sclerosis (ALS) in some countries. It is an antioxidant agent that helps to reduce oxidative stress, which is believed to contribute to the progression of ALS. By reducing oxidative damage, Edaravone aims to slow down the decline in physical function in ALS patients.

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Incorrect Options:

Option B - Piracetam: Piracetam is a nootropic agent that is commonly used for its cognitive-enhancing effects. It is not approved for the treatment of ALS. While some studies have explored the potential use of piracetam in ALS, the evidence for its effectiveness in treating the disease is limited and inconclusive.

Option C - Dabrafenib: Dabrafenib is a targeted therapy drug that is primarily used for the treatment of certain types of cancers, particularly melanoma. It works by inhibiting specific genetic mutations that drive the growth of cancer cells. Dabrafenib is not approved for the treatment of ALS and is not considered a standard therapy for the condition.

Option D - Trametinib: Trametinib is another targeted therapy drug that is primarily used for the treatment of certain types of cancers, such as melanoma with specific genetic mutations. Like Dabrafenib, Trametinib is not approved for the treatment of ALS and is not considered a standard therapy for the condition.

Solution for Question 29:

Correct Option C - 1, 2, 4:

(1) D2 antagonism - Reduces positive symptoms: Antipsychotics block dopamine receptors, specifically the D2 receptors, in the mesolimbic pathway. This decreases the dopamine activity in this pathway, reducing positive symptoms like hallucinations, delusions, and disordered thinking.

(2) 5HT2A antagonism - Reduces negative symptoms: Negative symptoms of schizophrenia refer to the absence or reduction of normal behavior, such as apathy, social withdrawal, and lack of motivation.

(3) 5HT1A agonism - Weight loss: This is an incorrect match. 5-HT1A receptor agonists are not used in the treatment of schizophrenia. 5-HT1A agonists like buspirone are used in the treatment of anxiety disorders and depression.

(4) Muscarinic antagonism - Reduces extrapyramidal symptoms: These include Parkinsonism, dystonia, and akathisia. Antipsychotics exert their effect by blocking dopamine receptors, which can lead to an imbalance between dopamine and acetylcholine. Muscarinic antagonism can reduce this imbalance and, thus, reduce EPS.

Incorrect Options:

- Options A, B and D are incorrect choice.

Solution for Question 30:

Correct option A: Naloxone

- Naloxone sodium is used as an antidote for opioid poisoning cases.
- It is a competitive blockers of mu/kappa/delta receptors and produces lacrimation, mydriasis, salivation and increases respiration.

Incorrect options:

Option B. Flumazenil is used as the antidote for Benzodiazepene poisoning.

Option C. Fomepizole is used as an antidote in case of methanol poisoning.

Option D. Pethidine itself is a kind of opioid used for relieving pain. The antidote for pethidine is Naloxone.

Solution for Question 31:

Correct option D: Levetiracetam-

- Levetiracetam does not interact with other drugs, is excreted through the kidney, and has rapid onset action. So it is used as the most preferred antiepileptic drug during pregnancy.

Incorrect options:

Option A. Exposure to Phenytoin can result in a complication called fetal hydantoin syndrome.

Option B. Exposure to Valproate can result in fetal Valproate syndrome.

Option C. Carbamazepine has teratogenic effects, such that it can cause developmental delay, growth retardation, and abnormal IQ.

Solution for Question 32:

Correct Option A - Agranulocytosis:

- Clozapine is an atypical antipsychotic medication commonly used in the treatment of schizophrenia.
- One of the most significant side effects of clozapine is agranulocytosis
- On starting clozapine, blood counts are monitored every week for 6 months, then every other week for 6 months, then monthly, if all neutrophil counts are adequate.
- Other side effects include metabolic syndrome, obesity, seizures, sedation, Type 2 DM.

Incorrect Options:

Option B - Myocarditis: While clozapine can potentially cause myocarditis, it is a rare side effect.

Option C - Cerebral bleed: Cerebral bleeding, or intracranial hemorrhage, is not a commonly reported side effect of clozapine nor does regular blood tests help in assessing risk or occurrence of one.

Option D - Seizures: Seizures can occur as a side effect of clozapine, particularly at higher doses.

Solution for Question 33:

Correct Option A - Increase the dose:

- Phenytoin is an enzyme inducer and increases the clearance of Retigabine. Therefore, the dose of Retigabine should be increased.

Phenytoin is an enzyme inducer and increases the clearance of Retigabine. Therefore, the dose of Retigabine should be increased.

Incorrect Options: Options B, C, and D are incorrect.

Solution for Question 34:

Correct Option D - Rash:

- Lamotrigine is an antiepileptic drug that is also used in the treatment of bipolar disorder. It works by stabilizing the activity of nerve cells in the brain that are involved in seizures and mood disorders.
- The most common side effect of lamotrigine, an antiepileptic and mood-stabilizing medication, is a rash. In clinical trials, approximately 10% of people taking lamotrigine experienced a rash, especially during the initial weeks of treatment. The rash can range from mild to severe, and in rare cases, it can be a sign of a serious allergic reaction such as Stevens-Johnson syndrome or toxic epidermal necrolysis, which requires immediate medical attention. Other less common side effects of lamotrigine may include dizziness, headache, double vision, nausea, vomiting, and somnolence (sleepiness).

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Incorrect Options:

Option A - Diplopia: Diplopia or double vision is an uncommon side effect of lamotrigine. It occurs in less than 1% of patients and usually resolves independently or with reduced drug dosage.

Option B - Tremors: Tremors are a rare side effect of lamotrigine. They occur in less than 1% of patients and may be related to the drug dosage. Tremors usually resolve with a reduction in the dose or discontinuation of the drug.

Option C - Neutropenia: Neutropenia, which is a low level of white blood cells called neutrophils, is a rare but potentially serious side effect of lamotrigine. It occurs in less than 1% of patients and can increase the risk of infection. Blood cell counts should be monitored in patients taking lamotrigine, especially during the first few months of treatment.

Solution for Question 35:

Correct Option D - Hemodialysis is not useful in lithium overdose:

- Hemodialysis is useful in lithium overdose. Hemodialysis can effectively treat severe lithium toxicity, as it can rapidly remove lithium from the bloodstream.

Hemodialysis is useful in lithium overdose. Hemodialysis can effectively treat severe lithium toxicity, as it can rapidly remove lithium from the bloodstream.

Incorrect Options:

Option A - Lithium is teratogenic if given to pregnant females: Lithium is known to cross the placenta, and exposure to lithium during pregnancy can increase the risk of fetal cardiac malformations. Women who are pregnant or planning to become pregnant should consult with their healthcare provider before taking lithium.

Option B - Lithium toxicity is exacerbated with thiazide: Thiazide diuretics, such as hydrochlorothiazide, can increase the serum levels of lithium by decreasing its renal clearance. This can lead to lithium toxicity, manifesting with symptoms such as tremors, confusion, and renal failure.

Option C - Lithium can decrease thyroid hormone levels: Lithium can cause hypothyroidism by decreasing the synthesis and secretion of thyroid hormones. Patients who are taking lithium should have their thyroid function monitored regularly.

Solution for Question 36:

Correct Option A - Myasthenia gravis:

- Myasthenia gravis is neuromuscular junction disease characterized by ptosis and muscle weakness that worsens in the evening and improves in the morning. Symptoms are relieved by anti-choline esterase such as neostigmine.

Myasthenia gravis is neuromuscular junction disease characterized by ptosis and muscle weakness that worsens in the evening and improves in the morning. Symptoms are relieved by anti-choline esterase such as neostigmine.

Incorrect Options:

Option B - Huntington chorea: Involuntary jerking movements and a decline in cognitive and psychiatric function characterize Huntington's chorea. No ptosis or improvement with neostigmine.

Option C - Amyotrophic lateral sclerosis: Amyotrophic lateral sclerosis is characterized by progressive muscle weakness and atrophy. It has a mix of upper neuron and lower neuron signs in its presentation. Though ptosis may be seen, No improvement with neostigmine is seen.

Option D - External ophthalmoplegia: External ophthalmoplegia is characterized by paralysis or weakness of the muscles that control eye movement resulting in diplopia. No improvement with neostigmine is seen.

Solution for Question 37:

Correct Option A - Naloxone:

- Naloxone is a medication used to reverse the effects of opioid overdose. The symptoms described in the patient, including respiratory depression, pinpoint pupils, and dizziness, are consistent with opioid overdose. Naloxone works by binding to the opioid receptors in the body, blocking the effects of opioids and rapidly reversing respiratory depression. Therefore, naloxone is the correct answer in this scenario.
- Therefore, in this scenario, the appropriate drug to reverse the effects is Naloxone, as the symptoms described are consistent with opioid overdose.

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Therefore, in this scenario, the appropriate drug to reverse the effects is Naloxone, as the symptoms described are consistent with opioid overdose.

Incorrect Options:

Option B - Atropine: Atropine is a medication that belongs to the class of anticholinergic drugs. It is primarily used to increase heart rate and treat certain types of bradycardia (slow heart rate). While atropine may be useful in specific situations, such as in the treatment of bradycardia or organophosphate poisoning, it is not the appropriate drug for reversing respiratory depression

caused by opioid overdose.

Option C - Phentolamine: Phentolamine is an alpha-adrenergic antagonist medication primarily used for the treatment of high blood pressure, certain types of vasospasm, and to counteract the effects of excessive levels of catecholamines. It is not indicated for reversing respiratory depression caused by opioid overdose.

Option D - Physostigmine: Physostigmine is a cholinesterase inhibitor that increases the levels of acetylcholine in the body. It is primarily used to reverse anticholinergic toxicity and certain types of drug-induced delirium. However, it is not indicated for the reversal of respiratory depression caused by opioid overdose.

Solution for Question 38:

Correct Option A - Agranulocytosis:

- Agranulocytosis is a potentially life-threatening condition characterized by a severe decrease in white blood cell count, specifically granulocytes. Clozapine has been associated with a risk of agranulocytosis, and therefore, regular monitoring of the absolute neutrophil count (ANC) is necessary. This involves frequent blood tests, especially during the initial stages of treatment. If agranulocytosis occurs, clozapine must be discontinued immediately.

Agranulocytosis is a potentially life-threatening condition characterized by a severe decrease in white blood cell count, specifically granulocytes. Clozapine has been associated with a risk of agranulocytosis, and therefore, regular monitoring of the absolute neutrophil count (ANC) is necessary. This involves frequent blood tests, especially during the initial stages of treatment. If agranulocytosis occurs, clozapine must be discontinued immediately.

Incorrect Options:

Option B - Renal bleed: Renal bleeding refers to bleeding within the kidneys or urinary tract. Renal bleeding is not a well-established side effect specifically associated with clozapine.

Option C - Hepatomegaly: Hepatomegaly refers to an enlargement of the liver. While clozapine treatment can cause mild elevations in liver enzymes, significant hepatomegaly is not a common side effect.

Option D - Seizures: Clozapine can lower the seizure threshold, increasing the risk of seizures, particularly at higher doses. Patients with a history of seizures or epilepsy may be more susceptible. If seizures occur, appropriate medical management and dose adjustments may be

necessary. However, this is not a serious complication at normal therapeutic levels.

Solution for Question 39:

Correct Option C - Trazodone:

- Trazodone is an antidepressant that belongs to the class of medications known as serotonin antagonist and reuptake inhibitors (SARIs). Priapism is a known side effect of trazodone, and it is more commonly associated with this medication compared to other antidepressants.
- The likely offending drug in this scenario is Trazodone

Trazodone is an antidepressant that belongs to the class of medications known as serotonin antagonist and reuptake inhibitors (SARIs). Priapism is a known side effect of trazodone, and it is more commonly associated with this medication compared to other antidepressants.

The likely offending drug in this scenario is Trazodone

Incorrect Options:

Option A - Mirtazapine: Mirtazapine is an antidepressant that belongs to the class of medications known as noradrenergic and specific serotonergic antidepressants (NASSAs). While it can cause priapism (prolonged and painful erection) as a rare side effect, it is not commonly associated with this condition.

Option B - Venlafaxine: Venlafaxine is an antidepressant that belongs to the class of medications known as selective serotonin and norepinephrine reuptake inhibitors (SNRIs). Priapism is a very rare side effect of venlafaxine, but it occurs less frequently compared to other medications.

Option D - Tianeptine: Tianeptine is an atypical antidepressant that has a different mechanism of action compared to other commonly prescribed antidepressants. Priapism is not a recognized side effect of tianeptine.

Solution for Question 40:

Correct Option A - Metronidazole:

- Metronidazole has a significant interaction with alcohol, which can result in a disulfiram-like reaction.
- This reaction occurs due to the inhibition of aldehyde dehydrogenase enzyme by metronidazole, leading to an accumulation of acetaldehyde when alcohol is consumed.
- Acetaldehyde is a toxic metabolite that causes unpleasant symptoms similar to a hangover, including nausea, vomiting, abdominal cramps, headache, and dizziness.

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Incorrect Options:

Option B - Nitazoxanide: Nitazoxanide is an anti-parasitic medication used to treat protozoal infections, including amoebiasis. It does not have a documented interaction with alcohol causing a disulfiram-like reaction.

Option C - Paromomycin: Paromomycin is an antibiotic used to treat intestinal infections, including amoebiasis. It does not have a documented interaction with alcohol causing a disulfiram-like reaction.

Option D - Diloxanide: Diloxanide is an anti-amoebic drug used to eliminate the cysts of *Entamoeba histolytica* from the intestinal tract after acute amoebic colitis. It does not have a documented interaction with alcohol causing a disulfiram-like reaction.

Solution for Question 41:

Correct Option D - Levetiracetam:

- Among the options given, levetiracetam is considered the least teratogenic antiepileptic drug in pregnancy. Teratogenicity refers to the ability of a drug to cause malformations or developmental abnormalities in a developing fetus.

Among the options given, levetiracetam is considered the least teratogenic antiepileptic drug in pregnancy. Teratogenicity refers to the ability of a drug to cause malformations or developmental abnormalities in a developing fetus.

Incorrect Options:

Option A - Valproate: Valproate is known to have a high teratogenic potential and is associated with an increased risk of major congenital malformations, including neural tube defects and facial abnormalities.

Option B & C - Phenytoin & Carbamazepine:

- Phenytoin & Carbamazepine are also associated with an increased risk of teratogenic effects. They have been linked to fetal hydantoin syndrome and fetal carbamazepine syndrome, respectively, which can cause a range of developmental abnormalities.

Phenytoin & Carbamazepine are also associated with an increased risk of teratogenic effects. They have been linked to fetal hydantoin syndrome and fetal carbamazepine syndrome, respectively, which can cause a range of developmental abnormalities.

Solution for Question 42:

Correct Option A - Trihexyphenidyl:

- Trihexyphenidyl can be given to a patient who developed parkinsonism as a side effect of antipsychotic medication.
- Parkinsonism refers to a group of symptoms similar to those seen in Parkinson's disease, including bradykinesia (slowness of movement), rigidity, tremors, and postural instability. It can be a side effect of certain antipsychotic medications, known as extrapyramidal symptoms (EPS).
- Trihexyphenidyl is an anticholinergic medication commonly used to treat parkinsonism and reduce the symptoms associated with EPS. It works by blocking the activity of acetylcholine, a neurotransmitter that is involved in motor control. By reducing the activity of acetylcholine, trihexyphenidyl helps to restore the balance of neurotransmitters in the brain and alleviate the parkinsonian symptoms caused by antipsychotic medication.

Trihexyphenidyl can be given to a patient who developed parkinsonism as a side effect of antipsychotic medication.

Parkinsonism refers to a group of symptoms similar to those seen in Parkinson's disease, including bradykinesia (slowness of movement), rigidity, tremors, and postural instability. It can be a side effect of certain antipsychotic medications, known as extrapyramidal symptoms (EPS).

Trihexyphenidyl is an anticholinergic medication commonly used to treat parkinsonism and reduce the symptoms associated with EPS. It works by blocking the activity of acetylcholine, a neurotransmitter that is involved in motor control. By reducing the activity of acetylcholine, trihexyphenidyl helps to restore the balance of neurotransmitters in the brain and alleviate the parkinsonian symptoms caused by antipsychotic medication.

Incorrect Options:

Option B, C & D - Entacapone, Rasagiline & Pramipexole:

- Entacapone, Rasagiline & Pramipexole, on the other hand, are medications used in the treatment of Parkinson's disease itself, and they are not typically used to manage parkinsonism caused by antipsychotic medication.

Entacapone, Rasagiline & Pramipexole, on the other hand, are medications used in the treatment of Parkinson's disease itself, and they are not typically used to manage parkinsonism caused by antipsychotic medication.

Solution for Question 43:

Correct Option B - Serum lithium level:

- In this scenario, given the patient's history of bipolar disorder and fasting, it is important to assess the serum lithium level to help diagnose her condition.
- Lithium is a commonly used first-line pharmacological treatment for bipolar disorder. It is essential to monitor lithium levels regularly to ensure that they are within the therapeutic range.
- Fasting can significantly affect the lithium levels in the blood, as decreased food intake and altered electrolyte balance can lead to changes in lithium absorption and excretion. Low serum lithium levels (below the therapeutic range) can increase the risk of relapse or breakthrough symptoms of bipolar disorder, while high levels can lead to lithium toxicity.
- The symptoms described by the patient, such as seizures, coarse tremors, confusion, and weakness of limbs, are consistent with lithium toxicity. Fasting can exacerbate this condition by altering lithium levels in the blood.

In this scenario, given the patient's history of bipolar disorder and fasting, it is important to assess the serum lithium level to help diagnose her condition.

Lithium is a commonly used first-line pharmacological treatment for bipolar disorder. It is essential to monitor lithium levels regularly to ensure that they are within the therapeutic range.

Fasting can significantly affect the lithium levels in the blood, as decreased food intake and altered electrolyte balance can lead to changes in lithium absorption and excretion. Low serum lithium levels (below the therapeutic range) can increase the risk of relapse or breakthrough symptoms of bipolar disorder, while high levels can lead to lithium toxicity.

The symptoms described by the patient, such as seizures, coarse tremors, confusion, and weakness of limbs, are consistent with lithium toxicity. Fasting can exacerbate this condition by altering lithium levels in the blood.

Incorrect Options:

Option A - Serum potassium level: Serum potassium level is not the preferred option

Option C - ECG: ECG is not the preferred answer.

Option D - MRI: MRI is not the preferred answer.

Solution for Question 44:

Correct Option C - Sodium valproate:

- In the above case, a 9 year old girl stares blankly and this is at times associated with chewing movements. This is accompanied with complaints of daydreaming and declining scholastic performance. The given EEG shows symmetric 3 Hz spike and wave pattern which is characteristic of absence seizures.
- The best anti-epileptic from amongst the given options is sodium valproate.
- Ethosuximide and sodium valproate are the only two drugs that are FDA approved as first line treatment for absence seizures and Ethosuximide is preferred over sodium valproate.

In the above case, a 9 year old girl stares blankly and this is at times associated with chewing movements. This is accompanied with complaints of daydreaming and declining scholastic performance. The given EEG shows symmetric 3 Hz spike and wave pattern which is characteristic of absence seizures.

The best anti-epileptic from amongst the given options is sodium valproate.

Ethosuximide and sodium valproate are the only two drugs that are FDA approved as first line treatment for absence seizures and Ethosuximide is preferred over sodium valproate.

Incorrect Options:

Option A - Carbamazepine: Carbamazepine is contraindicated in myoclonic and absence seizures.

Option B - Phenytoin: Phenytoin should not be used in myoclonic and absence seizures because they increase seizures.

Option D - Clonazepam: Clonazepam is not a first line drug for absence seizures and may be used for myoclonic seizures.

Solution for Question 45:

Correct Option C - Low intrinsic activity, high affinity for D2 receptor, less extrapyramidal symptoms and less weight gain:

- Aripiprazole has submaximal or low intrinsic activity and high affinity as it is a D2 receptor partial agonist. Partial agonists behave like an antagonist in the presence of an agonist and block the binding of agonists to the receptor resulting in inhibition of complete response.

Incorrect Options:

- Option A, B & D are not the correct option.

Solution for Question 46:

Correct Option B - Treatment of Lennox Gastaut syndrome:

- Lennox-Gastaut syndrome (LGS) is a rare and severe form of epilepsy that typically begins in early childhood and is characterized by seizures that are difficult to control. The seizures associated with LGS may involve different parts of the body and can cause falls and injury. There is no cure for LGS, and management focuses on controlling seizures and improving

quality of life.

- Topiramate is one of the medications that are commonly used in the management of LGS. It is an antiepileptic medication that works by reducing the activity of certain neurotransmitters in the brain, which helps to prevent seizures.
- Topiramate is usually taken orally and is often prescribed in combination with other medications to manage LGS. Topiramate work by modulating the activity of several neurotransmitters in the brain, including gamma-aminobutyric acid (GABA) and glutamate. GABA is an inhibitory neurotransmitter that helps to reduce the activity of neurons in the brain, while glutamate is an excitatory neurotransmitter that helps to increase the activity of neurons in the brain.

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Topiramate is usually taken orally and is often prescribed in combination with other medications to manage LGS. Topiramate work by modulating the activity of several neurotransmitters in the brain, including gamma-aminobutyric acid (GABA) and glutamate. GABA is an inhibitory neurotransmitter that helps to reduce the activity of neurons in the brain, while glutamate is an excitatory neurotransmitter that helps to increase the activity of neurons in the brain.

Incorrect Option:

Option A - Treatment of acute migraine:

- Triptans: These medications, including sumatriptan, rizatriptan, and eletriptan, are commonly used for the acute treatment of migraines. They work by constricting blood vessels and blocking pain pathways in the brain. Topiramate has been used for migraine prophylaxis.
- NSAIDs: Nonsteroidal anti-inflammatory drugs like ibuprofen, naproxen sodium, and aspirin can provide relief for mild to moderate migraines.

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NSAIDs: Nonsteroidal anti-inflammatory drugs like ibuprofen, naproxen sodium, and aspirin can provide relief for mild to moderate migraines.

Option C - Prophylaxis of ADHD:

- Attention deficit hyperactivity disorder (ADHD) is a condition that affects both children and adults, and it is characterized by symptoms such as impulsivity, hyperactivity, and inattention. While Topiramate may have some use in ADHD treatment, it is not a first-line treatment for ADHD prophylaxis.

Option D - Treatment of dementia:

- Dementia is a condition that affects memory, thinking, and behavior, and it is often associated with aging. While Topiramate may have some use in the management of dementia-related symptoms such as agitation and aggression, it is not a primary treatment for dementia.

Solution for Question 47:

Correct Option B - Vilazodone:

- It is a novel antidepressant medication that is used to treat major depressive disorder. It is a selective serotonin reuptake inhibitor (SSRI) and a partial agonist at the serotonin 5-HT_{1A} receptor. Vilazodone works by blocking the reuptake of serotonin, which helps to increase the amount of serotonin in the brain. It also activates the 5-HT_{1A} receptor, which is thought to enhance the antidepressant effect. Vilazodone was approved by the FDA in 2011 for the treatment of major depressive disorder.

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Incorrect Options:

Option A - Lurasidone: It is an atypical antipsychotic medication that is used to treat schizophrenia and bipolar disorder.

Option C - Asenapine: It is an atypical antipsychotic medication that is used to treat schizophrenia and bipolar disorder.

Option D - Blonanserin: It is an atypical antipsychotic medication that is used to treat schizophrenia. It works by blocking certain receptors in the brain, such as dopamine and serotonin receptors. Blonanserin is believed to block the activity of certain dopamine receptors, which helps to modify the balance of these chemicals in the brain.

Solution for Question 48:

Correct Option C- Mirtazapine:

- Sexual dysfunction is a common side effect of many antidepressant medications, and it can significantly impact a person's quality of life. While all antidepressants have the potential to cause sexual side effects, some are associated with a higher risk than others.
- Mirtazapine is an atypical antidepressant that works by enhancing the release of norepinephrine and serotonin. It has a different mechanism of action compared to selective serotonin reuptake inhibitors (SSRIs) like fluoxetine or serotonin-norepinephrine reuptake inhibitors (SNRIs) like venlafaxine.
- Mirtazapine is generally associated with a lower risk of sexual dysfunction compared to SSRIs and SNRIs. This is because mirtazapine does not directly inhibit the reuptake of serotonin, which is thought to play a role in sexual side effects. Instead, mirtazapine's mechanism of action primarily involves antagonism of certain serotonin and alpha-2 receptors, which can lead to increased release of norepinephrine and serotonin.

Sexual dysfunction is a common side effect of many antidepressant medications, and it can significantly impact a person's quality of life. While all antidepressants have the potential to cause sexual side effects, some are associated with a higher risk than others.

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Mirtazapine is generally associated with a lower risk of sexual dysfunction compared to SSRIs and SNRIs. This is because mirtazapine does not directly inhibit the reuptake of serotonin, which is thought to play a role in sexual side effects. Instead, mirtazapine's mechanism of action primarily involves antagonism of certain serotonin and alpha-2 receptors, which can lead to increased release of norepinephrine and serotonin.

Incorrect Options:

Option A - Fluoxetine: Fluoxetine is an SSRI that is commonly associated with sexual dysfunction as a side effect. It can cause various sexual side effects, including decreased libido, delayed ejaculation, and erectile dysfunction.

Option B - Venlafaxine: Venlafaxine is an SNRI that also carries a risk of sexual side effects. Similar to SSRIs, it can cause decreased libido, delayed orgasm, and other sexual dysfunctions.

Option D - Imipramine: Imipramine is a tricyclic antidepressant (TCA) that is known to have a higher incidence of sexual side effects compared to mirtazapine. TCAs, in general, have a higher risk of causing sexual dysfunction due to their effects on various neurotransmitters, including serotonin and norepinephrine.

Solution for Question 49:

Correct Option C - Levopropoxyphene:

- Levopropoxyphene is a medication that was used as a centrally acting opioid antitussive medication to treat cough. It works by suppressing the cough reflex in the brain.

Levopropoxyphene was withdrawn from the market in many countries due to its potential for abuse and serious side effects. It has high abuse potential and can lead to respiratory depression, cardiac arrhythmias, and other severe side effects.

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Levopropoxyphene was withdrawn from the market in many countries due to its potential for abuse and serious side effects. It has high abuse potential and can lead to respiratory depression, cardiac arrhythmias, and other severe side effects.

Incorrect Options:

Option A - Ambroxol: It is a medication that is used to treat respiratory diseases such as bronchitis, asthma, and COPD. Ambroxol works by breaking down and thinning mucus in the respiratory tract, making it easier to cough up. It is a mucolytic medication that is not related to opioids or antitussive medications.

Option B - Diphenoxylate: It is a medication that is used to treat diarrhea. Diphenoxylate works by slowing down the movement of the intestines, reducing the number of bowel movements and making the stool more solid. It is an opioid medication and can be abused, leading to addiction. Diphenoxylate is usually combined with atropine to discourage its abuse due to its unpleasant

side effects.

Option D - Levorphanol: It is a potent opioid medication that is used to treat severe pain. Levorphanol works by binding to opioid receptors in the brain and spinal cord, reducing the perception of pain. It has a higher potency than morphine and is used when other opioid medications are not effective or have intolerable side effects. Levorphanol is not an antitussive medication.

Solution for Question 50:

Correct Option D - Bipolar disorder:

- Risperidone is an antipsychotic medication that is approved by the FDA for the treatment of schizophrenia and bipolar disorder (both manic and mixed episodes). Risperidone works by blocking dopamine and serotonin receptors in the brain, which can help to reduce symptoms of psychosis and mania.
- Bipolar disorder is a mental health condition characterized by episodes of mania and depression. Mania is a state of elevated mood, energy, and activity levels, and may include symptoms such as grandiosity, reduced need for sleep, racing thoughts, and increased risk-taking behavior. Risperidone can be effective in treating manic episodes in bipolar disorder, and may also be used in combination with other medications to help stabilize mood and prevent future episodes.

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Bipolar disorder is a mental health condition characterized by episodes of mania and depression. Mania is a state of elevated mood, energy, and activity levels, and may include symptoms such as grandiosity, reduced need for sleep, racing thoughts, and increased risk-taking behavior. Risperidone can be effective in treating manic episodes in bipolar disorder, and may also be used in combination with other medications to help stabilize mood and prevent future episodes.

Incorrect Options:

Option A - OCD (Obsessive Compulsive Disorder): Risperidone is not specifically approved by the FDA for the treatment of OCD, but it has been used off-label for this condition. There is some

evidence to suggest that risperidone may be effective in reducing the symptoms of OCD, particularly in combination with other treatments.

Option B - Dementia: Risperidone is sometimes used off-label in the treatment of behavioral symptoms associated with dementia, such as aggression, agitation, and psychosis. However, this use has been associated with an increased risk of death, and is therefore generally not recommended.

Option C - PTSD (Post Traumatic Stress Disorder): Risperidone is not specifically approved for the treatment of PTSD, but it has been used off-label in some cases. Some studies have suggested that risperidone may be effective in reducing symptoms of PTSD, such as hyperarousal, avoidance, and re-experiencing.

Solution for Question 51:

Correct Option B - Venlafaxine:

- Venlafaxine belongs to serotonin noradrenaline reuptake inhibitors (SNRIs). Their mechanism of action includes inhibiting NA and 5HT reuptake, increasing brain-derived neurotrophic factors and no blockade of alpha, muscarinic and histamine receptors.

Venlafaxine belongs to serotonin noradrenaline reuptake inhibitors (SNRIs). Their mechanism of action includes inhibiting NA and 5HT reuptake, increasing brain-derived neurotrophic factors and no blockade of alpha, muscarinic and histamine receptors.

Incorrect Options:

Option A - Fluoxetine: This drug belongs to selective serotonin reuptake inhibitors (SSRIs) and is used in the treatment of depression. Their main action is by inhibiting the reuptake of serotonin increasing the BDNF. SSRIs don't block alpha, histamine, and muscarinic receptors.

Option C - Paroxetine: This drug belongs to selective serotonin reuptake inhibitors (SSRIs) and is used in the treatment of depression. Their main action is by inhibiting the reuptake of serotonin. It does not affect noradrenaline uptake.

Option D - Dapoxetine: This drug belongs to selective serotonin reuptake inhibitors (SSRIs) and is used in the treatment of depression. Their main action is by inhibiting the reuptake of serotonin. It does not affect noradrenaline uptake.

Solution for Question 52:

Correct Option C - Dopamine:

- The patient is exhibiting symptoms of parkinsonism.
- Reduced levels/activity of dopamine causes parkinsonism symptoms.
- Dopamine is a major neurotransmitter in basal ganglia, limbic system, CTZ and anterior pituitary

Incorrect Options:

Option A - GABA: It is an inhibitory neurotransmitter and is used to reduce anxiety, stress, depression. It also inhibits alpha cells and stimulates beta cells.

Option B - Acetylcholine: It is an excitatory neurotransmitter that carries messages from brain to other parts of the body. Excess of Ach can lead to muscle weakness, fatigue and cramps.

Option D - Serotonin: It is a neurotransmitter as well as a hormone. It is responsible for sleep wake cycle, mood and happiness.

Solution for Question 53:

Correct Option B - Risperidone:

- Risperidone is known to have a higher risk of causing galactorrhea compared to other atypical antipsychotics.

Risperidone is known to have a higher risk of causing galactorrhea compared to other atypical antipsychotics.

Incorrect Options:

Option A - Iloperidone:

- While iloperidone can potentially cause hyperprolactinemia (elevated levels of prolactin hormone), it is not commonly associated with a high risk of galactorrhea.

While iloperidone can potentially cause hyperprolactinemia (elevated levels of prolactin hormone), it is not commonly associated with a high risk of galactorrhea.

Option C - Clozapine:

Adverse effects

- Can produce Agranulocytosis (dose independent).
- Seizures (dose dependent).
- Myocarditis.
- Wet pillow syndrome- Causes Sialorrhea due to M4 agonism by Clozapine. The treatment is Atropine given locally
- Anticholinergic properties Risk of mydriasis Contraindicated in angle closure glaucoma
- Risk of mydriasis
- Contraindicated in angle closure glaucoma

Can produce Agranulocytosis (dose independent).

Seizures (dose dependent).

Myocarditis.

Wet pillow syndrome- Causes Sialorrhea due to M4 agonism by Clozapine. The treatment is Atropine given locally

Anticholinergic properties

- Risk of mydriasis
- Contraindicated in angle closure glaucoma

Risk of mydriasis

Contraindicated in angle closure glaucoma

Option D - Aripiprazole:

- Aripiprazole is known as a partial agonist of dopamine receptors and has a lower risk of causing hyperprolactinemia and subsequent galactorrhea compared to some other atypical antipsychotics.

Solution for Question 54:

Correct Option B - Carbamazepine:

- Carbamazepine is the drug of choice for the treatment of trigeminal neuralgia, which presents with severe, episodic facial pain in the distribution of the trigeminal nerve (cranial nerve V). The pain is often triggered by activities such as chewing, speaking, or touching the affected area.
- Therefore, among the options provided, carbamazepine is the most appropriate drug for the treatment of facial pain in the distribution of the facial nerve, specifically in the case of trigeminal neuralgia.

Carbamazepine is the drug of choice for the treatment of trigeminal neuralgia, which presents with severe, episodic facial pain in the distribution of the trigeminal nerve (cranial nerve V). The pain is often triggered by activities such as chewing, speaking, or touching the affected area.

Therefore, among the options provided, carbamazepine is the most appropriate drug for the treatment of facial pain in the distribution of the facial nerve, specifically in the case of trigeminal neuralgia.

Incorrect Options:

Option A - Lamotrigine: Lamotrigine is an anticonvulsant medication used in the treatment of epilepsy and bipolar disorder. While it has some efficacy in neuropathic pain, it is not considered a first-line treatment for trigeminal neuralgia.

Option C - Phenobarbital: Phenobarbital is a barbiturate that is primarily used as an anticonvulsant. It is not commonly used for the treatment of trigeminal neuralgia.

Option D - Haloperidol: Haloperidol is an antipsychotic medication primarily used in the treatment of psychotic disorders such as schizophrenia. It is not indicated for the treatment of trigeminal neuralgia.

Solution for Question 55:

Correct Option C - Metoclopramide:

- Metoclopramide, a dopamine receptor antagonist primarily used for gastrointestinal motility disorders and nausea, can induce extrapyramidal symptoms (EPS) due to its central blockade of dopamine receptors, particularly D2 receptors in the basal ganglia. This disruption in dopamine signaling can lead to dystonia, tremor, rigidity, and bradykinesia, with characteristic features including grimacing and abnormal posturing.

- EPS risk increases with higher doses or in susceptible individuals, necessitating close monitoring and consideration of alternative therapies in affected patients, with symptomatic management involving discontinuation of metoclopramide and, if necessary, the use of anticholinergic agents or benzodiazepines.

Metoclopramide, a dopamine receptor antagonist primarily used for gastrointestinal motility disorders and nausea, can induce extrapyramidal symptoms (EPS) due to its central blockade of dopamine receptors, particularly D2 receptors in the basal ganglia. This disruption in dopamine signaling can lead to dystonia, tremor, rigidity, and bradykinesia, with characteristic features including grimacing and abnormal posturing.

EPS risk increases with higher doses or in susceptible individuals, necessitating close monitoring and consideration of alternative therapies in affected patients, with symptomatic management involving discontinuation of metoclopramide and, if necessary, the use of anticholinergic agents or benzodiazepines.

Incorrect Options:

Option A - Atropine: Atropine is an anticholinergic medication that is used to treat various conditions, including bradycardia, certain types of poisoning, and as a premedication before surgery. It works by blocking the action of acetylcholine, a neurotransmitter in the parasympathetic nervous system. While atropine can cause side effects such as dry mouth, blurred vision, and increased heart rate, it is less commonly associated with grimacing and abnormal posturing.

Option B - Ranitidine: Ranitidine is a medication used to reduce the production of stomach acid and is commonly prescribed for conditions like gastric ulcers and gastroesophageal reflux disease (GERD). It belongs to the class of drugs known as H2 blockers. While ranitidine can cause side effects such as headache, dizziness, and gastrointestinal disturbances, it is not typically associated with grimacing and abnormal posturing.

Option D - Domperidone: Domperidone is another medication used to treat nausea and vomiting. It works by blocking dopamine receptors in the gut, which helps to alleviate symptoms. While domperidone is generally well-tolerated, it does not cross the Blood brain barrier and thus is less likely to cause EPS unlike metoclopramide.

Solution for Question 56:

Correct Option B - 1, 2 and 4:

The anti-epileptic drugs that can also be used for the treatment of neuropathic pain are:

- Pregabalin: Pregabalin is an anticonvulsant drug that is approved for the treatment of neuropathic pain, including diabetic neuropathy, postherpetic neuralgia, and fibromyalgia.
- Carbamazepine: Carbamazepine is an anticonvulsant drug that has been used for many years in the treatment of neuropathic pain, particularly trigeminal neuralgia.
- Gabapentin: Gabapentin is an anticonvulsant drug that is commonly used for the treatment of neuropathic pain, including diabetic neuropathy, postherpetic neuralgia, and peripheral neuropathy.

Pregabalin: Pregabalin is an anticonvulsant drug that is approved for the treatment of neuropathic pain, including diabetic neuropathy, postherpetic neuralgia, and fibromyalgia.

Carbamazepine: Carbamazepine is an anticonvulsant drug that has been used for many years in the treatment of neuropathic pain, particularly trigeminal neuralgia.

Gabapentin: Gabapentin is an anticonvulsant drug that is commonly used for the treatment of neuropathic pain, including diabetic neuropathy, postherpetic neuralgia, and peripheral neuropathy.

Incorrect Options:

Option A - 1, 2 and 3:

- Lamotrigine is an antiepileptic medication that is primarily used for the treatment of epilepsy. It belongs to the class of drugs known as anticonvulsants. Lamotrigine works by inhibiting the release of excitatory neurotransmitters, such as glutamate, in the brain, thereby reducing abnormal electrical activity and preventing seizures.
- Pregabalin, Carbamazepine, Gabapentin can help in reducing the neuropathic pain.

Lamotrigine is an antiepileptic medication that is primarily used for the treatment of epilepsy. It belongs to the class of drugs known as anticonvulsants. Lamotrigine works by inhibiting the release of excitatory neurotransmitters, such as glutamate, in the brain, thereby reducing abnormal electrical activity and preventing seizures.

Pregabalin, Carbamazepine, Gabapentin can help in reducing the neuropathic pain.

Option C - 1, 3 and 4:

- As mentioned above, Pregabalin, Carbamazepine, Gabapentin are the drugs which are anti-epileptic but can also reduce neuropathic pain. But lamotrigine doesn't show such properties, therefore this option is also incorrect.

As mentioned above, Pregabalin, Carbamazepine, Gabapentin are the drugs which are anti-epileptic but can also reduce neuropathic pain. But lamotrigine doesn't show such properties, therefore this option is also incorrect.

Option D - 2 and 4 only:

- Both Carbamazepine and Gabapentin helps in reduction of neuropathic pain but the options also have pregabalin which is why this option is incomplete as well as incorrect.

Both Carbamazepine and Gabapentin helps in reduction of neuropathic pain but the options also have pregabalin which is why this option is incomplete as well as incorrect.

Solution for Question 57:

Correct Option B - Lithium carbonate:

- Lithium carbonate is the drug of choice for long-term treatment of bipolar disorder, particularly for managing episodes of mania.
- It helps stabilize mood, reduce the frequency and intensity of manic episodes, and prevent relapses.

Lithium carbonate is the drug of choice for long-term treatment of bipolar disorder, particularly for managing episodes of mania.

It helps stabilize mood, reduce the frequency and intensity of manic episodes, and prevent relapses.

Incorrect Options:

Option A - Fluoxetine :Fluoxetine is a selective serotonin reuptake inhibitor (SSRI) commonly used to treat depression and anxiety disorders. It is not the first-line treatment for manic episodes and may exacerbate symptoms of mania.

Option C - Carbamazepine: Carbamazepine is not a first line drug

Option D - Lamotrigine: Used in bipolar depression

Solution for Question 58:

Correct Option C - Bupropion:

- Bupropion is an antidepressant medication that is also used as a smoking cessation aid. It helps reduce nicotine cravings and withdrawal symptoms, making it easier for individuals to quit smoking.

Bupropion is an antidepressant medication that is also used as a smoking cessation aid. It helps reduce nicotine cravings and withdrawal symptoms, making it easier for individuals to quit smoking.

Incorrect Options:

Option A - Venlafaxine: Venlafaxine is an antidepressant medication used primarily for the treatment of major depressive disorder, generalized anxiety disorder, and other psychiatric conditions. It is not specifically indicated for smoking cessation.

Option B - Topiramate: Topiramate is an anticonvulsant medication used for the treatment of epilepsy and prevention of migraine headaches. It is not commonly used as a smoking cessation aid.

Option D - Amitriptyline: Amitriptyline is a tricyclic antidepressant primarily used for the treatment of depression, neuropathic pain, and other conditions. It is not specifically indicated for smoking cessation.

Solution for Question 59:

Correct Option A - Dopamine:

- Prolactin secretion is primarily inhibited by dopamine, which is released from the hypothalamus. Dopamine acts on the lactotroph cells of the anterior pituitary gland to suppress the synthesis and secretion of prolactin.

Prolactin secretion is primarily inhibited by dopamine, which is released from the hypothalamus. Dopamine acts on the lactotroph cells of the anterior pituitary gland to suppress the synthesis and secretion of prolactin.

Incorrect Options:

Option B - Nor-adrenaline: Nor-adrenaline (norepinephrine) does not play a significant role in the direct inhibition of prolactin secretion.

Option C - Adrenaline: Adrenaline (epinephrine) does not have a direct inhibitory effect on prolactin secretion.

Option D - Serotonin: While serotonin can modulate prolactin secretion, its role is generally stimulatory rather than inhibitory. Serotonin stimulates prolactin release under certain physiological conditions, such as during pregnancy and nursing.

Solution for Question 60:

Correct Option B - Vigabatrin:

- Vigabatrin is an antiepileptic drug that is known to be associated with visual disturbances, including visual field defects. Specifically, it is associated with the development of concentric visual field constriction or "tunnel vision."
- Vigabatrin is primarily used in the treatment of epilepsy, particularly in patients with refractory seizures or those with infantile spasms. However, it carries a risk of causing visual field defects, which can range from mild to severe. The mechanism of this side effect is related to the drug's ability to irreversibly inhibit the enzyme GABA transaminase, leading to increased levels of gamma-aminobutyric acid (GABA) in the central nervous system.
- Visual field defects caused by vigabatrin are usually bilateral and symmetrical, and they can progress over time with continued use of the drug. Regular monitoring of visual fields is recommended for patients taking Vigabatrin to detect any changes and take appropriate action.

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Visual field defects caused by vigabatrin are usually bilateral and symmetrical, and they can progress over time with continued use of the drug. Regular monitoring of visual fields is recommended for patients taking Vigabatrin to detect any changes and take appropriate action.

Incorrect Options:

Option A - Phenobarbitone: Phenobarbital is an antiepileptic drug that is not commonly associated with visual disturbances or visual field defects as a side effect. It has a different mechanism of action and does not typically cause visual problems.

Option C - Ethosuximide: Ethosuximide is primarily used in the treatment of absence seizures. It is not typically associated with visual disturbances or visual field defects.

Option D - Levetiracetam: Levetiracetam is an antiepileptic drug that is generally well-tolerated, and visual disturbances are not commonly reported as side effects of this medication. While it can cause other side effects, such as drowsiness, headache, or mood changes, it is not known to cause visual field constriction.

Solution for Question 61:

Correct Option B - After 12 hours:

- Monitoring blood lithium levels is essential to ensure the therapeutic efficacy and safety of lithium therapy in patients with bipolar disorder. The timing of blood sample collection after the administration of lithium tablets is crucial to accurately assess the concentration of lithium in the bloodstream.
- After oral administration of lithium tablets, it takes time for the drug to be absorbed, distributed, and reach a steady state concentration in the blood. Therefore, blood samples for lithium level monitoring are typically collected after approximately 12 hours following the last dose of lithium.
- This time interval allows enough time for the drug to be fully absorbed and achieve steady-state levels in the bloodstream. By measuring lithium levels after 12 hours, healthcare providers can obtain an accurate representation of the drug's concentration and adjust the dosage if necessary.

Monitoring blood lithium levels is essential to ensure the therapeutic efficacy and safety of lithium therapy in patients with bipolar disorder. The timing of blood sample collection after the administration of lithium tablets is crucial to accurately assess the concentration of lithium in the bloodstream.

After oral administration of lithium tablets, it takes time for the drug to be absorbed, distributed, and reach a steady state concentration in the blood. Therefore, blood samples for lithium level

monitoring are typically collected after approximately 12 hours following the last dose of lithium.

This time interval allows enough time for the drug to be fully absorbed and achieve steady-state levels in the bloodstream. By measuring lithium levels after 12 hours, healthcare providers can obtain an accurate representation of the drug's concentration and adjust the dosage if necessary.

Incorrect Options:

Option A - After 24 hours: Waiting for 24 hours after the administration of lithium tablets may result in obtaining inaccurate lithium level measurements. By this time, the drug may have already been metabolized and eliminated from the body to a significant extent, making the blood sample less reflective of the therapeutic concentration.

Option C - After 6 hours: Taking a blood sample after only 6 hours may not provide an accurate representation of lithium levels. At this point, the drug may not have reached its peak concentration, and the sample may not reflect the steady-state levels necessary for proper monitoring.

Option D - Immediately after taking tablets: Collecting a blood sample immediately after taking lithium tablets would not accurately reflect the drug's concentration in the bloodstream. The drug needs time to be absorbed and distributed throughout the body before it reaches its peak concentration.

Solution for Question 62:

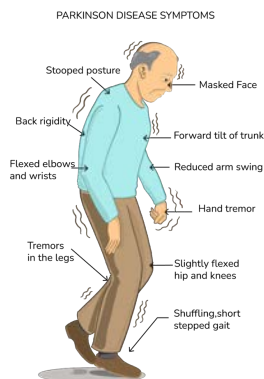
Correct Option C - Selegiline:

- This patient likely has Parkinsons disease
- Selegiline is a selective irreversible inhibitor of the monoamine oxidase-B (MAO-B) enzyme. In Parkinson's disease, the degeneration of dopaminergic neurons in the substantia nigra results in decreased dopamine levels in the striatum, leading to motor symptoms such as rigidity and tremors. By inhibiting the degradation of dopamine, Selegiline can increase dopamine levels in the brain, thereby improving the motor symptoms of Parkinson's disease.
- Selegiline is commonly used as adjunctive therapy with levodopa/carbidopa in the treatment of Parkinson's disease. It is especially useful in the early stages of Parkinson's disease, where the benefits of levodopa/carbidopa alone are insufficient to control symptoms.

This patient likely has Parkinsons disease

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Incorrect Options:

Option A - Clozapine: It is an atypical antipsychotic medication that is primarily used in the treatment of schizophrenia.

Option B - Donepezil: It is a drug used in the treatment of Alzheimer's disease, a condition characterized by progressive cognitive decline, memory loss, and difficulty with daily activities

Option D - Haloperidol: It is an antipsychotic medication commonly used to treat symptoms of psychosis such as hallucinations and delusion.

Antithyroid Drugs

1. What is the mechanism of action of Thiocyanate?

- A. Thyroid peroxidase inhibitor
 - B. Sodium Iodide symporter inhibitor (NIS)
 - C. Inhibit hormone release
 - D. Destruction of thyroid tissue
-

2. A 28-year-old woman with a history of hyperthyroidism is 4 weeks pregnant and presents for a prenatal visit. Which of the following is the DOC in the first trimester of pregnancy?

- A. Propylthiouracil
 - B. Carbimazole
 - C. Methimazole
 - D. I-131
-

3. A 6-month-old child is brought with the following defect over his scalp. Which of the following drugs, when used in the first trimester of pregnancy, is responsible for this condition?



[Image not available]

- A. Carbimazole
 - B. Radioactive I131
 - C. Iodine
 - D. Thiocyanate
-

4. Which of the following are the fastest-acting antithyroid drugs?

- A. Propylthiouracil
 - B. Sodium thiocyanate
 - C. Radioactive iodine
 - D. Potassium iodide
-

5. Which type of thyroid cancer does radioactive iodine treat?

- A. Papillary thyroid carcinoma
- B. Follicular thyroid carcinoma
- C. Medullary thyroid carcinoma
- D. Both A and B

6. A 38-year-old female with Graves' disease presents to the emergency department with severe agitation, palpitations, profuse sweating, and fever. On examination, she appears tremulous and diaphoretic with a heart rate of 160 bpm and a temperature of 103.1°F. Laboratory investigations reveal markedly elevated thyroid hormone levels. Which of the following is the drug of choice in this condition?

- A. Propranolol
- B. Methimazole
- C. Dexamethasone
- D. Amiodarone

7. A 67-year-old male with a long-standing history of hypothyroidism presents to the emergency department with altered mental status, hypothermia, bradycardia, and respiratory depression. His family reports increasing lethargy and confusion over the past week. He is unresponsive to stimuli. Labs reveal hyponatremia and elevated serum TSH levels. Which of the following is the DOC in this condition?

- A. Liothyronine (T3)
 - B. Levothyroxine (T4)
 - C. Teprotumumab
 - D. Cabozantinib
-

Correct Answers

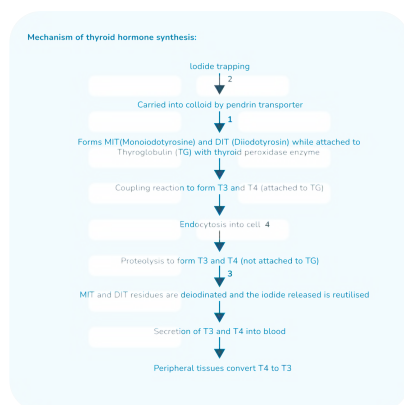
Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	4
Question 5	4
Question 6	1
Question 7	2

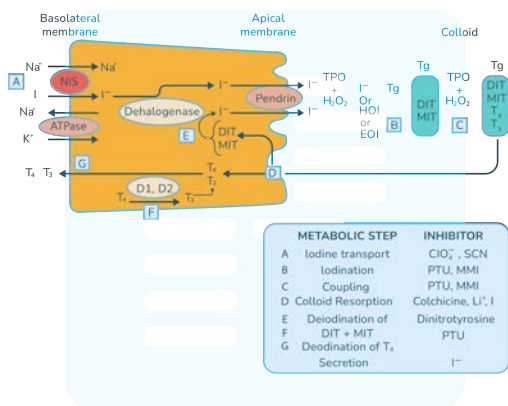
Solution for Question 1:

Correct Answer: B) Sodium Iodide symporter inhibitor (NIS)

EXPLANATION:

Thiocyanate inhibits iodide trapping by NIS into the thyroid cell and thus prevents the synthesis of thyroid hormone





Antithyroid drugs acting at each step include:

Inhibit hormone synthesis (Thioamides) (Thyroid peroxidase inhibitor) 1	Propylthiouracil Carbimazole Methimazole
Inhibit iodide trapping (Ionic inhibitors) (Sodium Iodide symporter inhibitors) 2	Thiocyanates (-SCN) Perchlorates (- ClO_4) Nitrates (- NO_3) Fluoroborates
Inhibit hormone release 3	Iodine Iodides of Na, K Organic iodide
Destroy thyroid tissue 4	Radioactive iodine I-131

Solution for Question 2:

Correct Answer: A) Propylthiouracil

EXPLANATION:

Propylthiouracil is the DOC in the first trimester of pregnancy since it has less placental transfer. It is contraindicated in 2nd and 3rd trimesters of pregnancy.

The uses of Propylthiouracil include

- Thyrotoxicosis- due to graves disease or toxic nodular goitre
- Drug of choice in the first trimester of pregnancy.

- Used alongside I-131 in patients with severe hyperthyroidism to prevent flare-ups from stored T4.

Mechanism of action:

- Propylthiouracil inhibits the production of new thyroid hormone in the thyroid gland.
- It acts by inhibiting the enzyme thyroid peroxidase, which usually converts iodide to an iodine molecule and incorporates the iodine molecule into amino acid tyrosine.
- Hence, DIT (diiodotyrosine) or MIT (monoiodotyrosine) does not get produced, which are the main constituents in the production of thyroxine (T4) and triiodothyronine (T3).
- Peripherally, it acts by inhibiting the conversion of T4 to T3.

The side effects include,

- Hepatotoxicity
- Hypothyroidism and goitre
- GI intolerance
- Skin rashes
- Joint pains

(Options B and C) Carbimazole and methimazole are used in the 2nd and 3rd trimesters of pregnancy; C/I in first trimester.

Radioactive I131 (Option D) is contraindicated in pregnancy.

Solution for Question 3:

Correct Answer: A) Carbimazole

EXPLANATION:

Carbimazole use during the first trimester of pregnancy causes the scalp formation defect known as aplasia cutis.

Carbimazole is a thyroid peroxidase inhibitor and is the prodrug of methimazole.

Its uses include

- DOC in hyperthyroidism and Graves disease.
- DOC in the second and third trimester of pregnancy.

- Used alongside I-131 in patients with severe hyperthyroidism to prevent flare-ups from stored T4.

The adverse effects include

- Teratogenic- Cutis aplasia, Esophageal atresia, Choanal atresia.
- Maculopapular rash
- Arthralgia
- Cholestatic jaundice

Differences between propylthiouracil and carbimazole:

Propylthiouracil	Carbimazole (Prodrug of Methimazole)
Less potent	5 x more potent
Highly plasma protein-bound	Less plasma protein bound
Less transferred across the placenta and in milk	Large amounts transferred across the placenta and in milk
Plasma t½ 1-2 hours; Acts for 4-8 hours	Plasma t½ 6-10 hours; Acts for 12-24 hours
Multiple (2-3) daily doses	Single daily dose
Inhibits peripheral T4 to T3 conversion	Does not inhibit T4 to T3 conversion
DOC in first trimester of pregnancy.(Less teratogenic)	DOC in 2nd and 3rd trimester of pregnancy.(Less hepatotoxic) C/I in first trimester.

Radioactive I-131 (Option 2): It is used to treat hyperthyroidism.

Contraindicated in pregnancy and <35 years.

Side effect: Permanent hypothyroidism.

Iodides (Option 3): it is used in thyroid storm and before surgery to reduce the size of thyroid.

Side effects: sialorrhea , sialadenitis.

Thiocyanate(Option 4): It is a Sodium Iodide symporter (Na I) inhibitor- inhibits iodine entry into the thyroid.

It is not specifically linked to aplasia cutis congenita.

Solution for Question 4:

Correct Answer: D) Potassium iodide

EXPLANATION:

Potassium iodide belongs to the oral iodide group, which are the fastest-acting thyroid inhibitors. Examples include potassium iodide and Lugol's iodine (5% iodine + 10% potassium iodide solution).

Table rendering failed

Propylthiouracil (Option 1): PTU inhibits the synthesis of thyroid hormones by blocking the enzyme thyroid peroxidase. It also prevents the conversion of T4 to T3 in peripheral tissues. While it starts to work within a few days, it may take several weeks to see the full effect. It is not the fastest-acting option.

Sodium thiocyanate (Option 2): It is used to ablate thyroid tissue and reduce hormone production. Its effects are not immediate

Radioactive iodine (Option 3): RAI is used to ablate thyroid tissue and reduce hormone production. Its effects are not immediate; it usually takes months for the thyroid hormone levels to drop significantly after administration. Therefore, it is not considered fast-acting.

Solution for Question 5:

Correct Answer: D) Both A and B

EXPLANATION:

Most papillary and follicular thyroid carcinomas are treated by surgery, radioiodine, and levothyroxine to suppress TSH. However, a small fraction progresses despite these therapies, and systemic therapy with kinase inhibitors may be appropriate in those cases.

- Radioactive iodine I-131 acts by destroying thyroid tissue (beta rays). It is used for the treatment of thyroid carcinoma, mainly follicular and papillary thyroid carcinoma.
- Radioactive iodine I-123 and I-125 are used for thyroid scans (gamma rays).

Table rendering failed

Option C: Medullary thyroid carcinoma is derived from parafollicular cells. Therefore, they are not responsive to radioactive iodine/TSH suppression.

Vandetanib, Cabozantinib are tyrosine kinase inhibitors given in medullary carcinoma of thyroid

Solution for Question 6:

Correct Answer: A) Propranolol

EXPLANATION:

Based on the given scenario, the patient is likely experiencing thyroid storm, an extreme manifestation of hyperthyroidism commonly seen in Graves' disease. Thyroid storm presents with severe agitation, palpitations, profuse sweating, and fever, often accompanied by markedly elevated thyroid hormone levels. Propranolol is the drug of choice in thyroid storm and works by inhibiting peripheral T4 to T3 conversion by inhibiting the 5' deiodinase enzyme.

Symptoms include,

- Fever of 104°F to 106°F with diaphoresis.
- CVS manifestations: Tachycardia, heart failure with pulmonary and peripheral edema, hypotension, arrhythmia, and death from cardiac arrest.
- CNS involvement includes agitation, delirium, anxiety, psychosis, or coma.
- Gastrointestinal symptoms include nausea, vomiting, diarrhoea, abdominal pain, intestinal obstruction, and acute hepatic failure.

Drugs inhibiting peripheral T4 to T3 conversion by inhibiting 5' deiodinase enzyme include,

- Beta-blockers: Propranolol
- Propylthiouracil
- Iodides
- Steroids: Dexamethasone
- Amiodarone

Management of thyroid storm:

- Large doses of propylthiouracil (500–1000 mg loading dose and 250 mg every 4 h) should be given orally or by nasogastric tube or per rectum; the drug's inhibitory action on T4 → T3 conversion makes it the antithyroid drug of choice.
- If not available, methimazole can be used in doses of 20 mg every 6 h.

- One hour after the first dose of propylthiouracil, stable iodide (5 drops SSKI every 6 h) is given to block thyroid hormone synthesis via the Wolff-Chaikoff effect (the delay allows the antithyroid drug to prevent the excess iodine from being incorporated into new hormone).
 - Propranolol should also be given to reduce tachycardia and other adrenergic manifestations (60–80 mg PO every 4 h, or 2 mg IV every 4 h). Although other β -adrenergic blockers can be used, high doses of propranolol decrease T4 $\rightarrow$ T3 conversion, and the doses can be easily adjusted.
 - Short-acting IV esmolol can be used to decrease heart rate while monitoring for signs of heart failure.
 - Additional therapeutic measures include glucocorticoids (e.g., hydrocortisone 300 mg IV bolus, then 100 mg every 8 h), antibiotics if infection is present, cholestyramine to sequester thyroid hormones, cooling, oxygen, and IV fluids.
-

Solution for Question 7:

Correct Answer: B) Levothyroxine (T4)

EXPLANATION;

The patient's condition suggests Myxedema coma, a rare and extreme complication of hypothyroidism with multiple organ abnormalities associated with altered sensorium, hypothermia, dry coarse skin with periorbital edema and bradycardia; it can be fatal. Hypothyroid patients exhibit multiple physiologic alterations to compensate for the deficiency of thyroid hormone. Intravenous levothyroxine (T4) is the DOC.

Other measures include,

- Warming the patient
- i.v. corticosteroids for adrenal insufficiency
- Ventilatory and cardiovascular support
- Correction of hyponatremia and hypoglycemia

Levothyroxine(T4)/liothyronine(T3) acts by binding to thyroid receptors. These thyroid receptors are nuclear receptors within the nucleus of cells.

The differences between T3 and T4 include,

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Note: Liotrix is a mixture of T4:T3 (4:1 mixture)

T3 and T4 exert their effect on thyroid-releasing hormone (TRH) by acting on transcription factors.

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Teprotumumab (Option C) is a monoclonal antibody used to treat thyroid eye disease. It works by inhibiting the insulin-like growth factor-1 receptor (IGF-1R). It is not used in the treatment of hypothyroidism or myxedema coma.

Cabozantinib and Vandetanib (Option D) are tyrosine kinase inhibitors used primarily to treat certain types of cancer, such as medullary thyroid cancer and renal cell carcinoma. They are not used for treating hypothyroidism or myxedema coma.

HPA Axis and Drugs acting on Bone and Osteoporosis

1. A 65-year-old male patient with a history of metastatic lung cancer presents to the emergency department with symptoms of nausea, vomiting, fatigue, and confusion. Blood tests reveal a significantly elevated serum calcium level. The patient is diagnosed with hypercalcemia of malignancy. Which of the following is the drug of choice for treating this condition?

- A. Cinacalcet
- B. Denosumab
- C. Zoledronic acid
- D. Calcitonin

2. A 30-year-old pregnant woman at 28 weeks of gestation presents to the emergency department with complaints of regular uterine contractions. A tocolytic agent is used to produce uterine relaxation. From the list of tocolytics given below, identify the mismatched drug and its mechanism of action.

- A. Terbutaline; β_2 selective antagonist
- B. Nifedipine; Calcium channel blocker
- C. Atosiban; Oxytocin receptor antagonist
- D. Magnesium sulfate; Competes with calcium for entry into the myometrium

3. Match the following uterine stimulant drugs with their mechanism of action
Drugs Mechanism of action
1. Oxytocin A. 15-Methyl PGF $_2$ alpha receptor stimulation
2. Carboprost B. Partial agonist at alpha-adrenergic and 5-HT $_2$ receptors
3. Methergine C. GPCR IP $_3$ /DAG pathway activation
4. Misoprostol D. PGE $_1$ receptor stimulation

Drugs	Mechanism of action
1. Oxytocin	A. 15-Methyl PGF2 alpha receptor stimulation
2. Carboprost	B. Partial agonist at alpha-adrenergic and 5-HT2 receptors
3. Methergine	C. GPCR IP3/DAG pathway activation
4. Misoprostol	D. PGE1 receptor stimulation

- A. 1-B, 2-A, 3-D, 4-C
- B. 1-D, 2-B, 3-A, 4-C
- C. 1-A, 2-C, 3-B, 4-D
- D. 1-C, 2-A, 3-B, 4-D

4. An 8-year-old female comes to the OPD due to the onset of menstrual bleeding. After investigation, a diagnosis of precocious puberty is established. The doctor prescribes her injections of Goserelin. How should the injection be administered for the management of precocious puberty?

- A. Intermittent weekly injection of Goserelin
- B. Continuous daily injection of Goserelin
- C. Single dose of Goserelin
- D. There is no specific method of administration of Goserelin

5. A 68-year-old man with difficulty in micturition and dull lower abdominal pain comes to the OPD. Investigations reveal prostate cancer, and the patient is prescribed a GnRH antagonist. Which of the following drugs is a GnRH antagonist?

- A. Goserelin
- B. Leuprolide
- C. Ganirelix

D. Finasteride

6. A 24-year-old woman comes to the OPD with complaints of galactorrhea and irregular menses. Investigation reveals a pituitary mass, and a diagnosis of prolactinoma was made. Which of the following is the drug of choice for prolactinoma?

- A. Cabergoline
 - B. Bromocriptine
 - C. Ropinirole
 - D. Pergolide
-

7. A 38-year-old man with acromegaly is started on octreotide injection to suppress growth hormone secretion. Which of the following is NOT true about octreotide?

- A. It is 40 times more potent than somatostatin in suppressing growth hormones.
 - B. Octreotide is preferred over Terlipressin for managing oesophageal variceal bleeding.
 - C. It is the drug of choice for Insulinoma.
 - D. It has a risk of developing gallbladder stones due to biliary stasis.
-

8. A 45-year-old man presents with enlarged hands and feet, coarse facial features, and elevated IGF-1 levels. He reports increasing shoe and ring size, excessive sweating, joint pain, and a deepening voice. An oral glucose tolerance test fails to suppress growth hormone, and an MRI shows a pituitary macroadenoma. Which drug targets the cause of these symptoms and biochemical abnormalities?

- A. Octreotide
- B. Bromocriptine

- C. Cabergoline
 - D. Pegvisomant
-

9. A 5-year-old boy visits a pediatric endocrinologist due to short stature. His height is below the 3rd percentile for his age, and his growth rate has been slower than normal. Which of the following medications would be most appropriate for treating dwarfism caused by a growth hormone deficiency?

- A. Mecasermin
 - B. Somatropin
 - C. IGF - 1
 - D. Sermorelin
-

10. Which phosphate binder is most appropriate for a 55-year-old man with end-stage renal disease (ESRD) on hemodialysis who has persistent hyperphosphatemia despite dietary phosphate restriction and has a history of coronary artery disease and coronary artery bypass grafting (CABG) 3 years ago?

- A. Vitamin D
 - B. Sevelamer
 - C. Teriparatide
 - D. Abaloparatide
-

11. A 65-year-old postmenopausal woman presents with a history of multiple fragility fractures. She has been previously treated for osteoporosis but continues to experience bone loss. Which medication would be most appropriate to stimulate bone formation in this patient?

- A. Bisphosphonates
 - B. Calcitonin
 - C. Denosumab
 - D. Teriparatide
-

12. A 58-year-old postmenopausal woman with a family history of osteoporosis is concerned about her risk of developing the condition. She is reluctant to start bisphosphonate therapy due to concerns about potential side effects. Which of the following medications would be the most appropriate choice for this patient?

- A. Raloxifene
 - B. Bazedoxifene
 - C. Tamoxifen
 - D. Lasofoxifene
-

13. How does pramlintide reduce blood glucose levels?

- A. Directly stimulating pancreatic beta cells to increase insulin secretion.
 - B. Inhibits the SGLT2 transporter in the kidneys, increasing glucose excretion.
 - C. It delays gastric emptying, suppresses postprandial glucagon secretion, and promotes satiety.
 - D. Activating PPAR-gamma receptors, improving insulin sensitivity in peripheral tissues.
-

14. Match the following monoclonal antibody;s with their respective mechanism of action. 1. Burosumab A. Inhibits RANKL, reducing osteoclast formation and activity. 2. Romosozumab B. Targets sclerostin, increasing bone formation and decreasing resorption. 3. Denosumab C. Inhibits fibroblast growth factor 23

(FGF23) 4. Rituximab D. Targets CD20 on B cells, leading to their depletion and affecting bone remodeling indirectly.

1. Burosumab	A. Inhibits RANKL, reducing osteoclast formation and activity.
2. Romosozumab	B. Targets sclerostin, increasing bone formation and decreasing resorption.
3. Denosumab	C. Inhibits fibroblast growth factor 23 (FGF23)
4. Rituximab	D. Targets CD20 on B cells, leading to their depletion and affecting bone remodeling indirectly.

- A. 1-C, 2-B, 3-A, 4-D
 - B. 1-A, 2-B, 3-C, 4-D
 - C. 1-B, 2-C, 3-D, 4-A
 - D. 1-C, 2-B, 3-D, 4-C
-

15. Which of the following drugs is contraindicated in Paget's disease?

- A. Alendronate
 - B. Teriparatide
 - C. Denosumab
 - D. Calcitonin
-

16. Which of the following drugs is NOT a form of Vitamin D?

- A. Cholecalciferol
 - B. Calcitriol
 - C. Alfacalcidol
 - D. Cinacalcet
-

17. Which of the following side effects is not associated with the use of bisphosphonates?

- A. Hypercalcemia
 - B. Esophagitis
 - C. Osteonecrosis of Jaw
 - D. Thrombophlebitis
-

18. A 65-year-old woman with a history of osteoporosis presents with severe bone pain and multiple vertebral fractures. She has previously undergone treatment that has failed. Which of the following bisphosphonates is most appropriate for this patient?

- A. Medronate
 - B. Pamidronate
 - C. Risedronate
 - D. Zoledronate
-

19. Which medication is considered the drug of choice for the treatment of osteoporosis?

- A. Teriparatide
 - B. Bisphosphonates
 - C. Denosumab
 - D. Calcitonin
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	1
Question 4	2
Question 5	3
Question 6	1
Question 7	2
Question 8	4
Question 9	4
Question 10	2
Question 11	4
Question 12	1
Question 13	3
Question 14	1
Question 15	2
Question 16	4
Question 17	1
Question 18	4
Question 19	2

Solution for Question 1:

Correct Answer: C) Zoledronic acid

Explanation:

Zoledronic acid is a bisphosphonate that is considered the drug of choice for the treatment of hypercalcemia of malignancy due to its potent ability to inhibit bone resorption and lower serum calcium levels effectively.

Management of hypercalcemia in malignancy:

Initial Treatment

- Intravenous hydration with normal saline to increase renal calcium excretion and glomerular filtration rate.
- Loop diuretics (e.g. furosemide) after initial volume expansion to enhance calciuresis.

Pharmacologic Therapy

- Calcitonin: Provides a rapid but short-lived reduction in serum calcium levels.
- Bisphosphonates (e.g. Zoledronic acid, Pamidronate): Potent inhibitors of osteoclast-mediated bone resorption, providing a more sustained reduction in calcium levels.
- Denosumab: A monoclonal antibody that inhibits osteoclast activity, effective in bisphosphonate-refractory hypercalcemia.
- Glucocorticoids: Used in calcitriol-mediated hypercalcemia to inhibit vitamin D conversion.

Refractory Hypercalcemia

- Denosumab can be used as an alternative for refractory or recurrent hypercalcemia despite treatment with bisphosphonates.
- Hemodialysis may be considered in patients with advanced kidney disease.

Solution for Question 2:

Correct Answer: A) Terbutaline; β_2 selective antagonist

Explanation:

Terbutaline is used as a tocolytic agent that produces uterine relaxation. It is a β_2 selective agonist and not an antagonist.

Uterine tocolytics :

Type	Drugs	Mechanism of action	Side effects
Adrenergic agonist (Option A)	Ritodrine Terbutaline Salbutamol Isoxsuprine	β_2 selective agonist resulting in myometrial relaxation.	Arrhythmia Hypotension Hyperglycemia Hyperinsulinemia Hyperkalemia
Calcium channel blocker (Option B)	Nifedipine Nicardipine Felodipine	Blocks L type calcium channels thus prevents calcium entry into the myometrium.	Tachyarrhythmia Hypotension Fetal hypoxia may occur
Oxytocin receptor antagonist (Option C)	Atosiban	Blocks oxytocin receptors in the uterus, thus preventing preterm contraction.	Nausea Vomiting Headache (Safe to use in cardiovascular disease)
Other drugs	MgSO ₄ (Option D)	Magnesium competes with calcium for entry into the myometrium, thus causing uterine relaxation.	Not used as a tocolytic due to the risk of fetal demise Used to prevent and treat seizures in pre-eclampsia and eclampsia.
Indomethacin	It is an NSAID that decreases prostaglandin formation, thus causing uterine relaxation.	Used in < 32 weeks POG, Premature closure of ductus arteriosus if used in >32 weeks. (in >32 weeks POG - Nifedipine is commonly used)	

Solution for Question 3:

Correct Answer: A) 1-B, 2-A, 3-D, 4-C

Explanation:

Drug	Mechanism of action	Uses	Side effects
------	---------------------	------	--------------

Oxytocin	GPCR activation ↓ IP3/DAG pathway ↓ Calcium influx into myometrial cells ↓ Uterine contraction	DOC in Induction and augmentation of labour. First line for postpartum haemorrhage. Other uses Uterine inertia Breast engorgement due to ineffective milk letdown reflex.	Injudicious use can lead to Uterine rupture Placental retention Fetal injury Large doses can lead to water intoxication.
Carboprost	15-Methyl PGF2 alpha receptor stimulation ↓ Uterine contraction	Used in postpartum haemorrhage that is resistant to oxytocin and methergine.	Nausea and vomiting Watery diarrhoea Flushing Hypotension Uterine cramps
Methergine	Partial agonists at alpha-adrenergic and 5-HT2 receptors cause an increase in uterine tone and contraction.	To prevent postpartum haemorrhage. To ensure normal involution of the uterus.	Vasospasm Angina Muscle cramps Paresthesia
Misoprostol	PGE1 receptor stimulation.	Cervical ripening	Nausea and vomiting Watery diarrhoea Flushing Hypotension Uterine cramps

Solution for Question 4:

Correct Answer: B) Continuous daily injection of Goserelin

Explanation:

- This patient has precocious puberty due to excessive gonadotropin production and is treated with daily continuous injections of Goserelin.
- Goserelin acts on the GnRH receptors in the pituitary gland, and its continuous elevation leads to a negative feedback mechanism, causing receptor downregulation. As a result, endogenous levels of LH and FSH decrease, which in turn lowers estrogen and progesterone levels.

GnRH elicits different responses based on the administration of the drug:

	Intermittent injection	Continuous injection
Mechanism of action	This is used when increased gonadotropin secretion is needed. As an agonist, its initial action stimulates the GnRH receptors, resulting in the release of gonadotropins.	This is preferred when decreased levels of gonadotropins or sex hormones are needed. Continuous stimulation results in a negative feedback mechanism on the receptors, leading to their downregulation.
Uses	DOC in Delayed puberty. Azoospermia (Male Infertility). Anovulation (Female Infertility).	DOC in Prostate carcinoma. DOC in Precocious puberty. Endometriosis.
Adverse effects	Increased size of the ovary may lead to pain in the abdomen. Rupture of the ovary: Hemoperitoneum.	Postmenopausal symptoms include hot flashes, Osteoporosis and Depression.

Solution for Question 5:

Correct Answer: C) Ganirelix

Explanation:

Ganirelix is a GnRH antagonist that inhibits gonadotropin release without initial stimulation.

	GnRH antagonist	GnRH agonist
Mechanism of action	GnRH antagonist ↓ Blocks the GnRH receptor in the pituitary ↓ Decreased FSH and LH levels ↓ Decreased Estrogen and Progesterone They have a lower risk of ovarian hyperstimulation. They offer complete suppression of endogenous gonadotropins.	GnRH agonist ↓ Initial stimulation of gonadotropin release ↓ Downregulation of GnRH receptors ↓ Suppression of gonadotropin release Recovery occurs after 2-3 months of the stoppage of medication.
Drugs	Cetrorelix Ganirelix Degarelix Abarelix Elagolix	Leuprolide (Option B) Goserelin (Option A) Histrelin Buserelin Triptorelin

Uses	Prostate cancer. Endometriosis. In Vitro Fertilization (Suppression of endogenous Gonadotropin before administration of exogenous hormones).	Delayed puberty Precocious puberty Anovulation Endometriosis
------	--	---

Option D: Finasteride is a 5-alpha reductase inhibitor that inhibits the conversion of testosterone to dihydrotestosterone (DHT). Although it has anti-androgenic activity, it is not used to treat prostate cancer; rather, it is used to treat benign prostatic hyperplasia (BPH) and androgenetic alopecia.

Solution for Question 6:

Correct Answer: A) Cabergoline

Explanation:

- Cabergoline, the drug of choice for prolactinoma, acts as a D2 receptor agonist, specifically targeting pituitary lactotroph D2 receptors to decrease prolactin secretion.
- Its extended half-life necessitates lower doses and enhances tolerability compared to other D2 agonists, leading to its widespread adoption over bromocriptine.

Dopaminergic agonists:

DRUG S	Mechanism of action	USES	SIDE EFFECTS
Cabergoline (Option A)	D2 agonist, selective for pituitary lactotrophs	Prolactinoma Acromegaly Post-abortion lactation suppression	Nausea and vomiting Postural hypotension Psychosis Ergotism. Peritoneal and pleural fibrosis

Bromocriptine (Option B)	D2 agonist D1 partial agonist Weak alpha blocker	Prolactinoma Parkinsonism Acromegaly Diabetes mellitus	Early Postural hypotension Constipation Nausea and vomiting Late Hallucination Psychosis Mental alteration Livedo reticularis
Ropinirole (Option C)	Selective D2/D3 agonist	Parkinson disease (oral drug) Restless leg syndrome (Now DOC for Restless Leg syndrome-Gabapentin)	Nausea and vomiting Postural hypotension Impulsive behaviour. Sexual hyperactivity
Pergolide (Option D)	D2 agonist	Parkinson disease	Valvular heart disease/Cardiac valve fibrosis. (thus no longer used)

Solution for Question 7:

Correct Answer: B) Octreotide is preferred over Terlipressin for managing oesophageal variceal bleeding.

Explanation:

Although octreotide can be used to manage oesophageal variceal bleeding, it is not preferred over terlipressin, which is more effective.

Octreotide:

- It is a somatostatin analogue, which is 40 times more potent than Somatostatin in the suppression of GH secretion. (Option A)
- Octreotide, a somatostatin analogue, has a significantly prolonged half-life of around 90 minutes compared to somatostatin's short half-life of 3-4 minutes, owing to enhanced stability and resistance to enzymatic degradation, enabling less frequent dosing and sustained therapeutic effects.

Mechanism of action	Uses
Reduces Growth hormone secretion due to inhibitory action on the pituitary.	Acromegaly

Reduces glucagon and insulin secretion. (Option C)	Glucagonoma / insulinoma
Reduces Gastrointestinal motility and secretion of secretory hormones such as CCK, VIP and serotonin.	Secretory diarrhoea in VIPoma, Carcinoid syndrome, Short bowel syndrome, Cancer-induced diarrhoea.
Reduces portal pressure by decreasing splanchnic blood flow. (But terlipressin is preferred over octreotide as it is more effective). (Option B)	Oesophageal variceal bleed.

Adverse effects of Octreotide:

- As it reduces secretory hormones such as CCK, it causes biliary stasis, thus increasing the risk of gallbladder stones. (Option D)
- Steatorrhea.
- Rarely hyperglycemia.

Solution for Question 8:

Correct Answer: D) Pegvisomant

Explanation:

Pegvisomant is a growth hormone receptor antagonist that directly blocks the action of excess growth hormone at the receptor level. By inhibiting growth hormone signalling, pegvisomant effectively reduces IGF-1 production (IGF 1 mainly produced by liver) and improves acromegaly symptoms and biochemical abnormalities.

Drug	Mechanism of Action	Primary Uses
Octreotide (Option A)	Inhibits growth hormone secretion by binding to somatostatin receptors.	First-line medical therapy for acromegaly.
Lanreotide		
Bromocriptine (Option B)	Reduces growth hormone secretion by binding to dopamine receptors.	Mixed prolactin and growth hormone-secreting adenomas. Bromocriptine is DOC for prolactinoma in pregnancy

Cabergoline (Option C)		
Pegvisomant (Option D)	Blocks the action of excess growth hormone at the receptor level.	Targeted therapy for acromegaly

Solution for Question 9:

Correct Answer: D) Sermorelin

Explanation:

Sermorelin is a synthetic analogue of growth hormone-releasing hormone (GHRH), naturally produced by the hypothalamus. It works by stimulating the pituitary gland to release growth hormone.

Drug	Mechanism of Action
Mecasermin (Option A)	Synthetic insulin-like growth factor-1 (IGF-1) receptor agonists promote growth by stimulating IGF-1 receptors.
Somatropin (Option B)	Somatropin is a Recombinant human growth hormone (rhGH) that stimulates growth by activating growth hormone receptors in target tissues.
IGF-1 (Option C)	Insulin-like growth factor-1 (IGF-1) promotes growth by binding to IGF-1 receptors, mediating the effects of growth hormone.
Sermorelin (Option D)	Growth hormone-releasing hormone (GHRH) analogue stimulates the release of endogenous growth hormone from the pituitary gland.

Solution for Question 10:

Correct Answer: B) Sevelamer

Explanation:

- Sevelamer carbonate is a non-calcium-based phosphate binder that effectively lowers serum phosphate levels without increasing the risk of hypercalcemia.
- It is used in patients with Chronic Kidney Disease (CKD) on hemodialysis to prevent hyperphosphatemia and its associated complications.
- Mechanism of Action: Sevelamer binds to dietary phosphate in the gut, preventing its absorption and thus reducing serum phosphorus levels. This decreases the stimulus for PTH secretion, helping to control secondary hyperparathyroidism in patients with chronic kidney disease on hemodialysis.

Drugs useful in Hyperphosphataemia:

Calcium based	Non-calcium based
Calcium Carbonate Calcium Acetate	Sevelamer Lanthanum Carbonate Sucroferric Oxyhydroxide

Solution for Question 11:

Correct Answer: D) Teriparatide

EXPLANATION:

- Teriparatide is a recombinant form of parathyroid hormone (PTH) that stimulates bone formation by promoting osteoblast activity.
- It enhances bone formation and improves bone density in patients with ongoing bone loss despite previous osteoporosis treatment.

Drugs used in Osteoporosis:

Pharmacological Agent	Mechanism of Action
Bisphosphonates (Option A)	Interfere with osteoclast activity, leading to decreased bone resorption.
Calcitonin (Option B)	Inhibits osteoclast activity, reducing bone resorption.
Denosumab (Option C)	It inhibits the RANK ligand, essential for osteoclast formation and function.
Estrogen/Raloxifene	Inhibit osteoclast activity and slow bone loss.

Cinacalcet	Indirectly reduces bone resorption by lowering PTH levels. (by stimulating CaSR receptors)
Teriparatide	A recombinant form of PTH stimulates osteoblast activity.
Strontium Ranelate	Dual action: It promotes bone formation by stimulating osteoblasts and decreases bone resorption by inhibiting osteoclast activity.

Solution for Question 12:

Correct Answer: A) Raloxifene

EXPLANATION:

- Raloxifene is a Selective Estrogen Receptor Modulator (SERM) that exhibits tissue-specific estrogen receptor agonist or antagonist effects.
- By acting as an estrogen agonist in bone, raloxifene increases bone mineral density (BMD) and improves bone strength, thereby reducing the risk of osteoporotic fractures, particularly vertebral fractures.
- Raloxifene is FDA-approved for both the prevention and treatment of postmenopausal osteoporosis, making it a suitable option for women at risk of developing osteoporosis or those who have already been diagnosed with the condition.
- Raloxifene is administered orally as a once-daily tablet.
- Adverse effects include hot flashes, leg cramps, venous thromboembolic events (VTE), and an increased risk of fatal stroke in postmenopausal women with coronary heart disease.

Bazedoxifene(Option B)

It is FDA-approved for the prevention of postmenopausal osteoporosis but not for treatment. While it increases BMD, its efficacy in reducing fracture risk is not as well-established as raloxifene.

Tamoxifen(Option C)

It is primarily used for the treatment and prevention of breast cancer recurrence. While it may have some beneficial effects on bone health, it is not FDA-approved for the treatment of osteoporosis.

Lasofloxifene(Option D)

It is not currently FDA-approved for the treatment of osteoporosis, although it has shown promise in increasing BMD and reducing fracture risk in clinical trials.

Solution for Question 13:

Correct Answer: C) It delays gastric emptying, suppresses postprandial glucagon secretion, and promotes satiety.

Explanation:

Pramlintide	
Type	Amylin analogue
MOA	Pramlintide ■ Mimics Amylin ■ Binds to Amylin Receptors ■ Delays Gastric Emptying ■ Reduces Postprandial Glucagon Secretion ■ Promotes Satiety ■ Decreased Food Intake ■ Decreased Postprandial Blood Glucose Levels.
Uses	Adjunct treatment in Type 1 and Type 2 diabetes.
Benefits	Improved postprandial (after meal) glucose control Reduced glycemic variability Potential weight loss due to reduced appetite
Side Effects	Nausea and Vomiting Headache Hypoglycemia (especially when used with insulin)
C/I	Gastroparesis (delayed gastric emptying) Hypoglycemia unawareness Poor adherence to insulin regimen
Special Points	Must be used with insulin but not mixed in the same syringe Requires education on proper dosing and hypoglycemia management.

Incorrect Options:

Option A: Sulfonylureas and meglitinides directly stimulate pancreatic beta cells to increase insulin secretion.

Option B: SGLT2 transporter is blocked by SGLT2 inhibitors, increasing glucose excretion.

Option D: Thiazolidinediones are PPAR-gamma agonists that stimulate PPAR-gamma receptors, increasing GLUT4 receptor expression, leading to reduced blood glucose and triglycerides.

Solution for Question 14:

Correct Answer: A) 1-C, 2-B, 3-A, 4-D

Explanation:

Burosumab	Inhibits fibroblast growth factor 23(FGF23)
Romosozumab	Targets sclerostin, increasing bone formation and decreasing resorption.
Denosumab	Inhibits RANKL, reducing osteoclast formation and activity.
Rituximab	Targets CD20 on B cells, leading to their depletion and affecting bone remodeling indirectly.

- Denosumab is preferred in patients with osteoporosis and chronic kidney disease because it is not eliminated by the kidneys and does not need dose adjustments for renal impairment.
- It effectively reduces fracture risk and increases bone density in postmenopausal women.

Monoclonal antibody	Description
Denosumab (Option C)	Mechanism of action: Denosumab → binds to RANKL (receptor activator of nuclear factor kappa-B ligand) → Prevents the interaction between RANKL and its receptor (RANK) on the surface of osteoclasts and osteoclast precursors → reduction in osteoclast formation, function, and survival → ↓ bone resorption.
Uses: Osteoporosis treatment in postmenopausal women at high fracture risk. Enhancing bone mass in men with osteoporosis. Treat bone loss due to hormone ablation therapy in men with prostate cancer. Hypercalcemia in malignancy. Treatment of Giant cell tumour.	
Burosumab (Option A)	Mechanism of action: Burosumab (monoclonal antibody) → binds to and inhibits fibroblast growth factor 23 (FGF23) → ↑ phosphate reabsorption in the kidney and ↑ production

	of active vitamin D.
Uses: Treatment of X-linked hypophosphatemia (XLH) (not osteoporosis)	
Romosozumab (Option B)	Mechanism of action: Romosozumab → inhibits sclerostin → ↑ bone formation and ↓ bone resorption.
Uses: Osteoporosis in postmenopausal women at high risk of fracture.	
Rituximab (Option D)	Mechanism of action: Rituximab → targets CD20 (a protein found on the surface of B-cells) → Destruction of B-cells.
Uses: B-cell non-Hodgkin's lymphoma, chronic lymphocytic leukaemia (CLL), and autoimmune diseases such as rheumatoid arthritis	

Solution for Question 15:

Correct Answer: B) Teriparatide

EXPLANATION:

Paget's disease of bone (or) Osteitis deformans is a chronic bone disorder characterised by the excessive breakdown and formation of bone tissue. This can lead to weakened bones that are misshapen and prone to fractures.

- Teriparatide is contraindicated in Paget's disease because it can exacerbate the condition by worsening the abnormal bone remodelling through its stimulation of bone formation.
- Paget's disease and hypercalcaemia are the contraindications of Teriparatide.

Options	Mechanism of Action	Uses	Adverse Effects
Alendronate (Bisphosphonates) (Option A)	Inhibits osteoclast mediated bone resorption	Osteoporosis Paget's disease Glucocorticoid-induced osteoporosis	Esophageal irritation Atypical fractures Osteonecrosis of the jaw
Denosumab (Option C)	Monoclonal antibody inhibits RANKL to prevent bone resorption	Osteoporosis Bone metastases Hypercalcaemia of malignancy	Hypocalcemia Infections Dermatitis

Calcitonin (Option D)	Inhibits osteoclast activity and bone resorption	Osteoporosis Paget's disease	Nasal irritation Flushing Nausea
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Solution for Question 16:

Correct Answer: D) Cinacalcet

EXPLANATION:

Cinacalcet is not a form of Vitamin D but a calcimimetic agent used to manage Secondary hyperparathyroidism.

	Vitamin D	Calcimimetic drugs
Exa mple s	Cholecalciferol Ergocalciferol Calciferol Doxercalciferol Alfacalcidol Maxacalcitol Paricalcitol	Cinacalcet Etelcalcetide Evocalcet
MOA	Increases intestinal absorption of calcium and phosphate. Promotes bone mineralisation by increasing calcium and phosphate levels in blood.	Modulates the calcium-sensing receptor (CaSR) activity on the parathyroid gland, leading to decreased secretion of PTH.
Uses	Nutritional rickets Vitamin D-deficiency rickets Familial hypophosphatemia Osteomalacia Osteoporosis Hypoparathyroidism	Secondary hyperparathyroidism in CKD patients. Hypercalcemia in parathyroid carcinoma.
Adve rse e ffects	Vitamin D is generally safe when taken in appropriate doses, but excessive intake can lead to various adverse effects due to increased calcium levels in the blood: Hypercalciuria, Increased risk of stones, Nausea, vomiting, constipation. Arrhythmias.	Hypocalcemia (principal adverse event) Potential development of adynamic bone disease if PTH level falls below 100 pg/mL. Potential increase in risk for upper gastrointestinal bleeding. Potential for QT interval prolongation.

Solution for Question 17:

Correct Answer: A) Hypercalcemia

EXPLANATION:

Hypercalcemia is not a side effect of bisphosphonates. Bisphosphonates are used to treat conditions like osteoporosis and are designed to reduce bone resorption, thereby lowering calcium levels in the blood.

Adverse effects of Bisphosphonates:

Short term	Long term
1. Upper GI: nausea, dyspepsia, abdominal pain, and gastritis	1. Osteonecrosis of jaw - Poor oral hygiene, invasive dental procedures or denture use, and prolonged exposure to high doses of IV bisphosphonates appear to increase the risk of ONJ development
2. Acute phase reactions associated with IV bisphosphonate therapy, it usually lasts 24 to 72 hours and is characterized by fever, myalgias, and arthralgias.	2. Atrial fibrillation
3. Severe musculoskeletal pain	3. Severe suppression of bone turnover
4. Hypocalcemia	4. Subtrochanteric femoral fractures.
5. Ocular inflammation-eg, uveitis, conjunctivitis, episcleritis, and scleritis (rare)	

Bisphosphonate	Adverse Effects
Etidronate	Interference with bone mineralisation, potential osteomalacia.
Pamidronate	Thrombophlebitis, bone pain, fever, leukopenia, flu-like symptom.
Alendronate	Esophagitis, gastric erosion, retrosternal pain, flatulence.
Ibandronate	Risk of flu-like symptoms with intravenous infusion.
Risedronate	Generally well-tolerated, potential for gastrointestinal upset.
Zoledronate	Severe hypocalcemia, nephrotoxicity, flu-like symptoms, osteonecrosis of the jaw.

- Oral Bisphosphonates causes esophagitis : Should be administered on an empty stomach with plenty of distilled water. Patient should not eat anything and should not lie down for 30 minutes - To avoid reflux esophagitis
- Bisphosphonates are associated with increased risk of Subtrochanteric femur fracture.
- Bisphosphonates are C/I in renal failure.
- Regular dental check-ups are advised to monitor for osteonecrosis of the jaw.

Solution for Question 18:

Correct Answer: D) Zoledronate

EXPLANATION:

- Zoledronate is a third-generation bisphosphonate known for its high potency and long-lasting efficacy after a single infusion.
- It is indicated for the treatment of osteoporosis and the prevention and treatment of skeletal-related events in patients with metastatic bone disease.

BISPHOSPHONATES:

1st generation	2nd generation	3rd generation
Etidronate Clodronate	Alendronate Risedronate Ibandronate	Zoledronic acid Pamidronate
Mechanism of Action	Direct inhibition of osteoclast activity and bone resorption.	
Frequency of administration	More potent drugs require less frequent administration Alendronate: Given once a week. Risedronate: Given once a week. Ibandronate: Given once a month. Zoledronate: Given once a year.	

Solution for Question 19:

Correct Answer: B) Bisphosphonates

EXPLANATION:

The patient's history of a pathological wrist fracture, combined with her T-scores (particularly the hip T-score of -2.5, which is classified as osteoporosis), indicates that she is at high risk for further fractures.

- Bisphosphonates are considered the drug of choice for the treatment of osteoporosis.
- They are highly effective in increasing bone mineral density (BMD) and reducing the risk of vertebral and non-vertebral fractures.
- The most commonly used bisphosphonates include Alendronate, Risedronate, Ibandronate, and Zoledronic acid.
- Bisphosphonates given maximum up to 10 yrs, beyond that it increase the risk of femur fracture.

Uses of Bisphosphonates:

Condition	Role of Bisphosphonates
Osteoporosis	Effective in preventing and treating postmenopausal osteoporosis, age-related, idiopathic, and steroid-induced osteoporosis.
Paget's Disease	Arrest osteolytic lesions, reduce bone pain, and improve secondary symptoms.
Hypercalcemia of Malignancy	Lower serum calcium levels, especially in severe cases.
Metastatic Bone Disease	Arrest osteolytic lesions and reduce bone pain.

Oral Hypoglycemic Drugs

1. Metformin is indicated in all of the following conditions except?

- A. Diabetic retinopathy
 - B. Polycystic ovary syndrome (PCOS)
 - C. Cushing Syndrome
 - D. Clozapine-induced glucose intolerance
-

2. How does pramlintide reduce blood glucose levels?

- A. Directly stimulating pancreatic beta cells to increase insulin secretion.
 - B. Inhibits the SGLT2 transporter in the kidneys, increasing glucose excretion.
 - C. It delays gastric emptying, suppresses postprandial glucagon secretion, and promotes satiety.
 - D. Activating PPAR-gamma receptors, improving insulin sensitivity in peripheral tissues.
-

3. Which of the following is the correct mechanism by which epalrestat acts?

- A. Inhibits the production of sorbitol
 - B. Increases expression of GLUT-4 receptors
 - C. Stimulates amylin receptors in the brain
 - D. Activation of AMP-dependent protein kinase
-

4. A patient who presented for an annual health checkup is found to have high blood glucose despite following lifestyle modifications. Which of the following is least likely to be used in managing this patient?

- A. Colesevelam
 - B. Bromocriptine
 - C. Indapamide
 - D. Telmisartan
-

5. A patient is recently diagnosed with type 2 diabetes mellitus and also found to have high triglyceride levels. Which of the following is the most appropriate drug for this patient?

- A. Fenofibrate
 - B. Ezetimibe
 - C. Icosapent ethyl
 - D. Saroglitazar
-

6. A patient who is a known diabetic and on metformin came up for a review, which showed no improvement in his glycemic profile. He has also been diagnosed with heart failure and is on drugs for the same. Which of the following is the most appropriate add-on drug for this patient?

- A. Repaglinide
 - B. Empagliflozin
 - C. Saxagliptin
 - D. Glipizide
-

7. A patient who is diagnosed with diabetes mellitus has been prescribed a drug which is a PPAR-gamma receptor agonist. All of the following are incorrect with respect to this group of drugs except?

- A. These drugs act by sensitizing skeletal muscles to insulin
 - B. These drugs reduce the production of adiponectin
 - C. The dose should be increased when combined with gemfibrozil
 - D. Play a supportive role in post-menopausal osteoporosis
-

8. Which of the following statements regarding alpha-glucosidase inhibitors is correct?

- A. Inhibits breakdown of complex carbohydrates
 - B. Weight gain is one of the common side effects
 - C. Acarbose enhances the absorption of digoxin leading to toxicity
 - D. Negligible risk to administer in patients with kidney disease
-

9. A 48-year-old diabetic patient was started on metformin. Before initiating therapy, his baseline serum creatinine was 0.9 mg/dL. At his annual follow-up, it has risen to 1.8 mg/dL. Considering this finding, what is the most appropriate next step in managing this patient?

- A. Continue metformin at the current dose and start sodium bicarbonate
 - B. Increase the dose of metformin
 - C. Discontinue metformin
 - D. Add another antidiabetic medication without changing the metformin dose
-

10. Which of the following classes of oral hypoglycemic drugs works by closing potassium channels in the pancreatic beta cells to increase insulin secretion?

- A. Rosiglitazone
 - B. Sitagliptin
 - C. Canagliflozin
 - D. Glimepiride
-

11. Which of the following statements correctly describes how metformin reduces blood glucose levels? Metformin stimulates the pancreatic receptors to increase insulin secretion. Metformin inhibits hepatic glucose production by inhibiting adenosine monophosphate kinase (AMPK). Metformin reduces both microvascular and macrovascular complications in type-2 diabetics. Metformin decreases lipid levels and increases insulin sensitivity in peripheral tissues. No effect on blood glucose levels during normoglycemic state.

- A. 3, 4, and 5
 - B. 1, 4 and 5
 - C. 1, 3 and 2
 - D. 2, 3 and 4
-

12. A 68-year-old male with type 2 diabetes mellitus managed on metformin presents to the clinic with increased blood glucose levels. Routine investigations revealed haemoglobin of 11 g/dL, serum creatinine of 2.3, and an estimated glomerular filtration rate (eGFR) of 30 mL/min/1.73 m². Which of the following additional medications can be safely prescribed to this patient?

- A. Dapagliflozin
- B. Saxagliptin
- C. Linagliptin

D. Exenatide

13. A 48-year-old patient with type 2 diabetes mellitus, previously managed with metformin, presents to the clinic with persistently elevated blood glucose levels. Considering the need to initiate a GLP-1 analogue, which options can be prescribed to accommodate the patient's request for oral medication?

- A. Exenatide
 - B. Semaglutide
 - C. Dulaglutide
 - D. Liraglutide
-

14. A 55-year-old male with a history of long-standing type 2 diabetes mellitus presents to the clinic with features suggestive of short bowel syndrome following abdominal surgical resection for bowel ischemia. He complains of chronic diarrhoea, malabsorption, and significant weight loss. His current diabetes management includes insulin therapy. Which of the following medications would be most appropriate to manage his diabetes in addition to addressing his condition?

- A. Metformin
 - B. Pramlintide
 - C. Sitagliptin
 - D. Teduglutide
-

15. Which of the following drugs has a mechanism of action similar to sulfonylureas?

- A. Repaglinide

- B. Pioglitazone
 - C. Acarbose
 - D. Metformin
-

16. All of the following adverse reactions are associated with chlorpropamide except?

- A. Hypoglycemia
 - B. Hyponatremia
 - C. Disulfiram like reaction
 - D. Hypertension
-

17. Which of the following sulfonylureas is considered the most potent?

- A. Tolbutamide
 - B. Chlorpropamide
 - C. Glimepiride
 - D. Glipizide
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	1
Question 4	3
Question 5	4
Question 6	2
Question 7	1
Question 8	1
Question 9	3
Question 10	4
Question 11	1
Question 12	3
Question 13	2
Question 14	4
Question 15	1
Question 16	4
Question 17	3

Solution for Question 1:

Correct Answer: A) Diabetic retinopathy

Diabetic retinopathy occurs due to neovascularization of the retinal blood vessels and is treated with intravitreal anti-VEGF injections (bevacizumab). Metformin is not used to treat diabetic retinopathy but to prevent macrovascular and microvascular problems, including diabetic retinopathy.

Uses of Metformin

First-line oral hypoglycemic for the treatment of type 2 diabetes mellitus.
 DOC: Prophylaxis in prediabetes or impaired glucose tolerance
 Metabolic syndrome
 DOC: Diabetes due to insulin resistance
 In PCOS: Metformin also promotes ovulation (Option B)
 Clozapine-induced DM/glucose intolerance (Option D)
 Cushing syndrome (Option C)
 Gestational diabetes

Solution for Question 2:

Correct Answer: C) It delays gastric emptying, suppresses postprandial glucagon secretion, and promotes satiety.

Explanation:

Pramlintide	
Type	Amylin analogue
MOA	Pramlintide ■ Mimics Amylin ■ Binds to Amylin Receptors ■ Delays Gastric Emptying ■ Reduces Postprandial Glucagon Secretion ■ Promotes Satiety ■ Decreased Food Intake ■ Decreased Postprandial Blood Glucose Levels.
Uses	Adjunct treatment in Type 1 and Type 2 diabetes.
Benefits	Improved postprandial (after meal) glucose control Reduced glycemic variability Potential weight loss due to reduced appetite
Side Effects	Nausea and Vomiting Headache Hypoglycemia (especially when used with insulin)
C/I	Gastroparesis (delayed gastric emptying) Hypoglycemia unawareness Poor adherence to insulin regimen

Special Points	It is a peptide drug, therefore given in injectable form only. Must be used with insulin but not mixed in the same syringe Requires education on proper dosing and hypoglycemia management.
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Option A: Sulfonylureas and meglitinides directly stimulate pancreatic beta cells to increase insulin secretion.

Option B: SGLT2 transporter is blocked by SGLT2 inhibitors, increasing glucose excretion.

Option D: Thiazolidinediones are PPAR-gamma agonists that stimulate PPAR-gamma receptors, increasing GLUT4 receptor expression and reducing blood glucose and triglycerides.

Solution for Question 3:

Correct Answer: A) Inhibits the production of sorbitol

Explanation:

- Epalrestat acts by reducing the production of sorbitol.
- It inhibits the enzyme aldose reductase, which is involved in the production of sorbitol.
- Accumulation of sorbitol in tissues, especially in nervous tissue, is the main pathophysiology behind the occurrence of neuropathy in diabetes.

Diabetic neuropathy:

- It occurs due to microvascular injury involving the vasa nervosum (vessels that supply the nerves).
- Management: Anticonvulsants: Carbamazepine, topiramate. Neurotrophic agents: Pregabalin, gabapentin. Antidepressants: Duloxetine, amitriptyline, nortriptyline. Antiarrhythmic: Mexiletine. Aldose reductase inhibitor: Epalrestat.
- Anticonvulsants: Carbamazepine, topiramate.
- Neurotrophic agents: Pregabalin, gabapentin.
- Antidepressants: Duloxetine, amitriptyline, nortriptyline.
- Antiarrhythmic: Mexiletine.
- Aldose reductase inhibitor: Epalrestat.

- Anticonvulsants: Carbamazepine, topiramate.
- Neurotrophic agents: Pregabalin, gabapentin.
- Antidepressants: Duloxetine, amitriptyline, nortriptyline.
- Antiarrhythmic: Mexiletine.
- Aldose reductase inhibitor: Epalrestat.

Drugs that improve insulin sensitivity, like PPAR-gamma agonists increase the expression of GLUT-4 receptors. (Option B)

Pramlintide (Option C) is an amylin analogue that acts on the corresponding receptors in the brain to reduce glucagon secretion from the pancreatic alpha cells.

Metformin acts by stimulating the enzyme ATP-dependant protein kinase. (Option D)

Solution for Question 4:

Correct Answer: C) Indapamide

Explanation:

- Indapamide (thiazide diuretic) will not be used to manage diabetes mellitus.
- Thiazide diuretics act by decreasing sodium reabsorption in the kidneys but will cause hyperglycemia and hyperuricemia. Hence, these drugs should not be used in patients with diabetes mellitus

Colesevelam (Option A) is a farnesoid X receptor antagonist, which is a bile acid binding resin. It helps in lowering blood cholesterol & blood sugar levels.

Bromocriptine (Option B) is a dopamine agonist which acts by hypothalamic control of hormones like GH & ACTH, thus reducing insulin resistance.

Telmisartan (Option D) is an angiotensin receptor blocker with partial PPAR-gamma agonist activity, hence improves insulin resistance by acting on the adipose tissue and skeletal muscle. It can be used in diabetes mellitus with hypertension.

Solution for Question 5:

Correct Answer: D) Saroglitazar

Explanation:

- The above patient is said to have diabetic dyslipidemia, and PPAR-alpha/gamma dual receptor agonists like saroglitazar are the drugs of choice in such patients.
 - Activation of PPAR-alpha receptors leads to the expression of genes involved in the uptake of fatty acids and their beta-oxidation, thus reducing triglyceride levels.
 - Activation of PPAR-gamma receptors leads to enhanced insulin sensitivity in adipose tissue and skeletal muscle, leading to reduced blood glucose levels.
 - Drugs: Saroglitazar. Aleglitazar. Muraglitazar. Tesaglitazar.
 - Saroglitazar.
 - Aleglitazar.
 - Muraglitazar.
 - Tesaglitazar.
 - Saroglitazar.
 - Aleglitazar.
 - Muraglitazar.
 - Tesaglitazar.
-

Solution for Question 6:

Correct Answer: B) Empagliflozin

Explanation:

- Diabetic patients with heart failure best benefit from adding a group of drugs called SGLT-2 (sodium-glucose cotransporter-2) inhibitors, and empagliflozin falls in this group.
- By inhibiting the SGLT-2 receptors in the proximal tubule of the kidneys, glucose (and sodium) reabsorption can be prevented.
- Since both sodium and glucose are osmotic molecules, increasing their excretion draws out more water, making these drugs effective in patients with heart failure.

- Drugs: Dapagliflozin. Empagliflozin. Canagliflozin. Ertugliflozin.
 - Dapagliflozin.
 - Empagliflozin.
 - Canagliflozin.
 - Ertugliflozin.
 - Dapagliflozin.
 - Empagliflozin.
 - Canagliflozin.
 - Ertugliflozin.
-

Solution for Question 7:

Correct Answer: A) These drugs act by sensitizing skeletal muscles to insulin

Explanation:

PPAR-gamma agonists/thiazolidinediones act on the PPAR (peroxisome proliferator-activated receptor) gamma in skeletal muscle and adipose tissue, sensitizing them to insulin and reducing blood glucose levels (primary action).

Activated PPAR-gamma receptors enhance the expression of insulin-dependent glucose transporter GLUT-4 in adipose tissue and skeletal muscle.

Secondary actions:

- Suppressing hepatic gluconeogenesis.
- Reduces fatty acid release.
- Promotes lipogenesis.

Drugs:

- Pioglitazone (available in India). Metabolised by CYP2C8 and CYP3A4. Given only in type 2 diabetes mellitus. Side effects: Plasma volume expansion, oedema, weight gain, headache, myalgia, mild anaemia, and increased risk of fractures (especially in elderly women). C/I: Liver disease and CHF.
- Metabolised by CYP2C8 and CYP3A4.
- Given only in type 2 diabetes mellitus.

- Side effects: Plasma volume expansion, oedema, weight gain, headache, myalgia, mild anaemia, and increased risk of fractures (especially in elderly women).
- Troglitazone (banned due to hepatotoxicity).
- Rosiglitazone (banned due to increased risk of MI/heart failure).
- Metabolised by CYP2C8 and CYP3A4.
- Given only in type 2 diabetes mellitus.
- Side effects: Plasma volume expansion, oedema, weight gain, headache, myalgia, mild anaemia, and increased risk of fractures (especially in elderly women).

Adiponectin (Option B) is released by the adipose tissue, enhancing insulin sensitivity.

Gemfibrozil (Option C) impairs the metabolism of these drugs, leading to potential toxicity; hence dose of the latter should be reduced almost to half. Rifampicin does the opposite by inducing the metabolism of these drugs.

Activating PPAR-gamma receptors in bones will lead to osteoclastic activation, resulting in bone resorption. Thus, these drugs can cause fractures, especially in those with osteoporosis (Option D).

Solution for Question 8:

Correct Answer: A) Inhibits breakdown of complex carbohydrates

Explanation:

Alpha-glucosidase inhibitors inhibit breakdown of complex carbohydrates, by inhibiting the enzyme alpha-glucosidase.

Alpha-glucosidase is an enzyme present in the brush border of the small intestine. It breaks down complex carbohydrates and enhances the absorption of the resulting simple carbohydrates. Hence, blood glucose levels can be controlled primarily by inhibiting this enzyme.

Since complex carbohydrates are made available to the gut after a meal, these drugs help control postprandial hyperglycemia.

Drugs: Acarbose, Voglibose, and Miglitol.

Weight gain is one of the common side effects (Option B) is an incorrect statement, as these drugs are weight and lipid-level neutral.

Acarbose enhances the absorption of digoxin leading to toxicity (Option C) is incorrect as Acarbose decreases the absorption of digoxin. Miglitol decreases the absorption of propranolol & ranitidine.

Negligible risk to administer in patients with kidney disease (Option D) is incorrect, these drugs are contraindicated to use in patients with stage 4 kidney disease.

Solution for Question 9:

Correct Answer: C) Discontinue metformin

Explanation:

Metformin is immediately discontinued in individuals with elevated serum creatinine levels (>1.5 mg/dL in males and >1.4 mg/dL in females), as it increases the risk of life-threatening lactic acidosis.

Complications associated with Metformin:

- Renal failure is common with metformin use.
- Plasma metformin levels are inversely related to GFR due to reduced drug clearance.
- Current guidelines suggest the safe use of metformin when GFR is more than 30 mL/min/1.73 m².
- Management: Assess renal function before starting metformin and monitor it annually. Discontinue if serum creatinine >1.5 mg/dL or eGFR < 30 mL/min/1.73 m². Discontinue pre-emptively before radiographic procedures using contrast dyes (till 48 hours after the procedure until renal function normalises) and hospitalization for severe illness (sepsis). Dose adjustment is needed for cationic drugs excreted by renal tubular secretion, such as cimetidine, furosemide, nifedipine, etc. It is indicated in patients with severe pulmonary disease, heart failure, liver disease, or chronic alcohol abuse.
- Assess renal function before starting metformin and monitor it annually.
- Discontinue if serum creatinine >1.5 mg/dL or eGFR < 30 mL/min/1.73 m².
- Discontinue pre-emptively before radiographic procedures using contrast dyes (till 48 hours after the procedure until renal function normalises) and hospitalization for severe illness (sepsis).
- Dose adjustment is needed for cationic drugs excreted by renal tubular secretion, such as cimetidine, furosemide, nifedipine, etc.

- It is indicated in patients with severe pulmonary disease, heart failure, liver disease, or chronic alcohol abuse.
- Assess renal function before starting metformin and monitor it annually.
- Discontinue if serum creatinine >1.5 mg/dL or eGFR < 30 mL/min/1.73 m².
- Discontinue pre-emptively before radiographic procedures using contrast dyes (till 48 hours after the procedure until renal function normalises) and hospitalization for severe illness (sepsis).
- Dose adjustment is needed for cationic drugs excreted by renal tubular secretion, such as cimetidine, furosemide, nifedipine, etc.
- It is indicated in patients with severe pulmonary disease, heart failure, liver disease, or chronic alcohol abuse.

(Option A) Intravenous sodium bicarbonate administration is the preferred treatment for metformin-associated lactic acidosis.

(Option B) Metformin can be given at a maximum 2550 mg/day dose in patients with increased blood glucose levels without demonstrating any adverse effects.

(Option D) If metformin monotherapy is not effective in controlling blood glucose levels in type 2 diabetics, the addition of sulfonylureas, GLP-1 receptor agonists, or DPP4 inhibitors may be beneficial.

Solution for Question 10:

Correct Answer: D) Glimepiride

Explanation:

Glimepiride (Sulfonylureas) functions by binding to the sulfonylurea receptor on pancreatic β Cells. This binding closes the ATP-sensitive K⁺ channels, leading to cell depolarisation. As a result, Ca²⁺ channels open, allowing Ca²⁺ influx, which triggers insulin secretion.

Sulfonylureas	
Site of Action	Pancreatic β cells
Uses	Type 2 Diabetes Mellitus
Mechanism of Action	Sulfonylureas ■ Bind to Receptors on Pancreatic β Cells ■ Closure of ATP-sensitive K ⁺ Channels ■ Membrane

	Depolarization ■ Opening of Voltage-Gated Ca^{2+} -Channels ■ Ca^{2+} Influx ■ Exocytosis of Insulin Granules ■ Increased Insulin Secretion ■ Lower Blood Glucose Levels.
Drugs	1st Generation Tolbutamide Chlorpropamide 2nd Generation Glibenclamide Glipizide Gliclazide Glimepiride

Rosiglitazone (Option A) (Thiazolidinediones) works by activating (PPAR- γ).

Sitagliptin (Option B) (DPP-4 inhibitors) inhibit the enzyme dipeptidyl peptidase-4.

Canagliflozin (Option C) (SGLT-2 inhibitors) works by inhibiting the sodium-glucose co-transporter 2 (SGLT-2) in proximal renal tubules.

Solution for Question 11:

Correct Answer: A) 3, 4, and 5

Explanation:

Unlike secretagogues, metformin does not stimulate insulin secretion from the pancreas; instead, it stimulates the AMPK enzyme, inhibiting hepatic glucose production and increasing peripheral insulin sensitivity. Further, it helps lower serum lipid levels and reduces both micro and macrovascular complications.

Metformin (Biguanides):

- First-line oral hypoglycemic for type-2 diabetes mellitus (DM).
- MOA: Stimulates the AMP-dependent protein kinase (AMPK) in the presence of insulin. It inhibits hepatic gluconeogenesis, fatty acid oxidation, and non-oxidative glucose metabolism and increases insulin sensitivity. Inhibits the mitochondrial glycerol phosphate dehydrogenase Reduces peripheral resistance to insulin in muscle and adipose tissues
- Stimulates the AMP-dependent protein kinase (AMPK) in the presence of insulin.
- It inhibits hepatic gluconeogenesis, fatty acid oxidation, and non-oxidative glucose metabolism and increases insulin sensitivity.

- Inhibits the mitochondrial glycerol phosphate dehydrogenase
- Reduces peripheral resistance to insulin in muscle and adipose tissues
- Stimulates the AMP-dependent protein kinase (AMPK) in the presence of insulin.
- It inhibits hepatic gluconeogenesis, fatty acid oxidation, and non-oxidative glucose metabolism and increases insulin sensitivity.
- Inhibits the mitochondrial glycerol phosphate dehydrogenase
- Reduces peripheral resistance to insulin in muscle and adipose tissues

Other beneficial effects:

- Weight loss in obese patients and weight neutrality in thin patients
- There is no risk of hypoglycemia (No effect during normoglycemic state)
- Reduction of lipogenesis (decrease triglyceride and LDL levels)
- Anti-ageing drug
- Decreases micro- and macrovascular complications of DM
Micro: diabetic neuropathy, nephropathy, retinopathy
Macro: MI and stroke
- Micro: diabetic neuropathy, nephropathy, retinopathy
- Macro: MI and stroke
- Micro: diabetic neuropathy, nephropathy, retinopathy
- Macro: MI and stroke
- Dose: 500 mg OD to 2550 mg/day (max)
- Side effects: Nausea/Vomiting/Diarrhoea
Megaloblastic anaemia: Reduces vitamin B12 absorption from GIT
Lactic acidosis: Avoid metformin in COPD, CHF, dehydration, and sepsis (hypoxia increases lactic acid production).
- Megaloblastic anaemia: Reduces vitamin B12 absorption from GIT
- Lactic acidosis: Avoid metformin in COPD, CHF, dehydration, and sepsis (hypoxia increases lactic acid production).
- Megaloblastic anaemia: Reduces vitamin B12 absorption from GIT
- Lactic acidosis: Avoid metformin in COPD, CHF, dehydration, and sepsis (hypoxia increases lactic acid production).

Solution for Question 12:

Correct Answer: C) Linagliptin

Explanation:

This patient should be started on linagliptin, a DPP-4 inhibitor that requires no dosage adjustment for renal impairment in this patient with type-2 diabetes and end-stage kidney disease (eGFR = 30 mL/min/1.73 m²).

Dipeptidyl peptidase-4 (DPP-4) inhibitors:

MOA	Increase Incretin Levels: Prevents degradation of incretin hormones (GLP-1 and GIP) Enhance Insulin Secretion from the pancreas in response to meals. (no risk of hypoglycemia) Reduce Glucagon Secretion Maintain neutral weight
Uses	Oral hypoglycemic drug (type-2 diabetes)
Drugs	Linagliptin: Safe in renal failure Teneligliptin: Safe in renal failure Vildagliptin: Hepatotoxic Alogliptin Sitagliptin Saxagliptin: (Option B) Metabolised by CYP3A4, it has more drug interactions. Requires dose adjustment in renal impairment. Risk of congestive heart failure.
Side effects	Nausea and vomiting (most common) Pancreatitis (lesser than GLP-1) Risk of pancreatic and thyroid cancer (lesser than GLP-1) Arthritis, Nasopharyngitis: Increase interleukins and cytokines levels (similar structure to CD26) Hypersensitivity reactions: Steven-Johnson syndrome, angioedema

Dapagliflozin (Option A), an SGLT-2 inhibitor, is not recommended for use to improve glycemic control in patients with type 2 diabetes mellitus whose eGFR is less than 45 mL/min/1.73 m².

Exenatide (Option C), a GLP-1 analogue, is avoided in patients with end-stage renal disease or ischemic renal failure due to hypovolemia.

Solution for Question 13:

Correct Answer: B) Semaglutide

Explanation:

Semaglutide is the only GLP analogue taken orally to manage elevated blood glucose levels in type 2 diabetics.

GLP-1 (glucagon-like peptide-1) analogues: (Drugs end with:- natide or - glutide)	
MOA	Enhance insulin secretion in response to food intake, which helps manage postprandial blood glucose levels. (No hypoglycemia risk) Suppress Glucagon Secretion Slow Gastric Emptying leads to reduced food intake. Promote satiety and help with weight loss.
Uses	Type-2 diabetes mellitus Anti-obesity drug: Liraglutide and Semaglutide
Drugs:	Injectables: Exenatide: Shortest acting (OPTION A) Lixisenatide: Albumin bound, longer acting Liraglutide (OPTION D) Dulaglutide (OPTION C) Albiglutide Injectable and Oral Route: Semaglutide: Longest-acting (OPTION B)
Side effects	Nausea and vomiting (most common) Gallstones Pancreatitis Risk of Pancreatic and thyroid cancer (medullary carcinoma of the thyroid): contraindicated in MEN syndrome

Solution for Question 14:

Correct Answer: D) Teduglutide

Explanation:

- Teduglutide is a (GLP-2) analogue that stimulates intestinal growth and enhances nutrient absorption in the small intestine.

- It is indicated for the treatment of short bowel syndrome in adult patients who are dependent on parenteral support. Additionally, it can help improve glycemic control by addressing the underlying malabsorption issues in this patient population.

Teduglutide	
Type	Peptide hormone (GLP-2 analogue)
Mechanism of Action	Teduglutide ■ Binds to GLP-2 Receptors on Intestinal Epithelial Cells ■ Activation of Guanylate Cyclase ■ Increased Production of cGMP ■ Activation of Protein Kinase G ■ Regulation of Tight Junctions and Epithelial Transporters ■ Enhanced Intestinal Absorption of Fluid and Nutrients ■ Improved Glycemic Control.
Uses	Treatment of short bowel syndrome (SBS) in adult patients who are dependent on parenteral support.
Benefits	Reduces dependence on parenteral nutrition Improves intestinal absorption and transit time Increases nutrient absorption and fluid balance
Side Effects	Nausea and Vomiting Headache Upper respiratory tract infection
Contraindications	Gastrointestinal obstruction or perforation Gastrointestinal cancer
Special Points	Requires subcutaneous injection Dosing and adjustment should be based on individual patient responses. Regular monitoring of nutritional status and fluid balance is essential.

- Metformin (Option A) is contraindicated in patients with renal impairment, which is common in short bowel syndrome due to fluid and electrolyte imbalances. Additionally, it primarily reduces hepatic glucose production and improves insulin sensitivity, which may not directly address the nutrient absorption issues in short bowel syndrome.
- Pramlintide (Option B) is an amylin analogue that slows gastric emptying and reduces postprandial glucagon secretion. It does not address the underlying issue of nutrient malabsorption in short bowel syndrome.
- Sitagliptin (Option C) (DPP-4 inhibitor) enhances incretin levels, increasing insulin and decreasing glucagon secretion. Therefore, it has no role in short bowel syndrome.

Solution for Question 15:

Correct Answer: A) Repaglinide

Explanation:

Repaglinide (Meglitinide analogue), like sulfonylureas, works by stimulating the release of insulin from Pancreatic β Cells. Both classes of drugs inhibit ATP-sensitive K^+ Channels in the β cell, leading to cell depolarisation, Ca^{2+} influx and subsequent insulin release.

Repaglinide	
Type	Meglitinide Analogue
Uses	Type 2 DM
Mechanism of Action	Meglitinide Analogue ■ Bind to Receptors on Pancreatic β Cells ■ Closure of ATP-sensitive K^+ Channels ■ Membrane Depolarization ■ Opening of Voltage-Gated Ca^{2+} Channels ■ Ca^{2+} Influx ■ Exocytosis of Insulin Granules ■ Increased Insulin Secretion.
Side Effects	Hypoglycemia Weight gain Headache Diarrhoea, Back pain cardiovascular events (e.g., myocardial ischemia) Liver enzyme abnormalities

Metformin (Option D): Primarily decreases hepatic glucose production and increases insulin sensitivity in peripheral tissues. It does not stimulate insulin release from pancreatic beta cells.

Pioglitazone (Option B): is a thiazolidinedione that improves insulin sensitivity by acting as an agonist for the (PPAR- γ). This results in increased glucose uptake in muscle and adipose tissue and decreased glucose production in the liver.

Acarbose (Option C): Is an alpha-glucosidase inhibitor that works by slowing down the digestion and absorption of carbohydrates in the small intestine.

Solution for Question 16:

Correct Answer: D) Hypertension

Explanation:

Chlorpropamide	
Class	Sulfonylurea (First Generation) Less commonly used
Side Effects	Hypoglycemia Gastrointestinal disturbances Weight gain Hyponatremia (Increased release of ADH) Photosensitivity Hepatic dysfunction Blood dyscrasias (Leukopenia, Thrombocytopenia, or Hemolytic anaemia) Allergic reaction (Itching, hives) Disulfiram like reaction
Drug Interactions	It may interact with NSAIDs, MAOIs, and Beta-blockers, potentially altering blood sugar control.
Contraindications	Type 1 diabetes Diabetic Ketoacidosis Severe kidney or liver disease Hypersensitivity to sulfonylureas

Solution for Question 17:

Correct Answer: C) Glimepiride

Explanation:

Glimepiride:

- Class: Sulfonylurea (Second Generation) Highly potent, requiring lower doses compared to other sulfonylureas. Administered once daily with the first main meal. Associated with a relatively lower risk of hypoglycemia
- Highly potent, requiring lower doses compared to other sulfonylureas.
- Administered once daily with the first main meal.
- Associated with a relatively lower risk of hypoglycemia
- Clinical Use: Used as Add-on therapy in type 2 DM when glycemic control is not achieved with metformin alone.

- Used as Add-on therapy in type 2 DM when glycemic control is not achieved with metformin alone.
 - Side effects: Hypoglycemia, weight gain, and occasional gastrointestinal disturbances.
 - Hypoglycemia, weight gain, and occasional gastrointestinal disturbances.
 - Highly potent, requiring lower doses compared to other sulfonylureas.
 - Administered once daily with the first main meal.
 - Associated with a relatively lower risk of hypoglycemia
 - Used as Add-on therapy in type 2 DM when glycemic control is not achieved with metformin alone.
 - Hypoglycemia, weight gain, and occasional gastrointestinal disturbances.
-

Parenteral Hypoglycemic Drugs

1. A 65-year-old male with a history of Type 2 diabetes mellitus and chronic kidney disease (CKD) stage 4. He is currently on oral hypoglycemic agents, but his glycemic control has been suboptimal. His endocrinologist decides to start him on insulin therapy. Which of the following precautions is most important to consider before initiating insulin therapy in this patient?

- A. Assessing liver function tests (LFTs)
- B. Evaluating renal function and adjusting the insulin dose accordingly
- C. Monitoring for signs of diabetic ketoacidosis (DKA)
- D. Assessing cardiac function to prevent hypoglycemia-related cardiac events

2. Match the different types of insulin with their respective onset and duration.

Types of Insulin Onset and Duration
1. Insulin aspart A. Onset: 1-2 hours, Duration: 24 hours or more
2. Regular insulin B. Onset: 15-30 minutes, Duration: 3-5 hours
3. NPH C. Onset: 30-60 minutes, Duration: 5-8 hours
4. Insulin detemir D. Onset: 1-2 hours, Duration: 12-18 hours

Types of Insulin	Onset and Duration
1. Insulin aspart	A. Onset: 1-2 hours, Duration: 24 hours or more
2. Regular insulin	B. Onset: 15-30 minutes, Duration: 3-5 hours
3. NPH	C. Onset: 30-60 minutes, Duration: 5-8 hours
4. Insulin detemir	D. Onset: 1-2 hours, Duration: 12-18 hours

- A. 1 - B, 2 - D, 3 - C, 4 - A
- B. 1 - C, 2 - B, 3 - A, 4 - D
- C. 1 - B, 2 - C, 3 - D, 4 - A
- D. 1 - A, 2 - B, 3 - C, 4 - D

3. Which of the following statements about the use of an insulin pump (continuous subcutaneous insulin infusion or CSII) is correct?

- A. Uses long-acting insulins to provide a constant basal infusion
 - B. Delivers insulin into the portal circulation
 - C. Uses only short-acting insulins
 - D. Not recommended in pronounced dawn phenomenon
-

4. Match the different types of insulin with their respective characteristics. Types of Insulin Characteristics

1. Insulin Lispro A. Proline at B28 replaced with aspartic acid

2. Insulin Aspart B. Glutamic acid at B29 and lysine at B3

3. Insulin Glulisine C. Complexed with zinc and protamine

4. NPH insulin D. Positions B28 and B29 modified

Types of Insulin	Characteristics
1. Insulin Lispro	A. Proline at B28 replaced with aspartic acid
2. Insulin Aspart	B. Glutamic acid at B29 and lysine at B3
3. Insulin Glulisine	C. Complexed with zinc and protamine
4. NPH insulin	D. Positions B28 and B29 modified

- A. 1 - D, 2 - A, 3 - B, 4 - C
 - B. 1 - B, 2 - D, 3 - A, 4 - C
 - C. 1 - A, 2 - C, 3 - D, 4 - B
 - D. 1 - C, 2 - D, 3 - B, 4 - A
-

5. Why is zinc used in the preparation of regular insulin?

- A. To stabilise insulin by maintaining a neutral pH solution
 - B. Forms complexes with insulin, allowing the gradual release of insulin monomers
 - C. Promotes rapid absorption of insulin after subcutaneous injection
 - D. Prevents insulin degradation in the bloodstream
-

6. A 54-year-old male with type 2 diabetes mellitus is currently managed on metformin and gliclazide. However, his diabetes is not well-controlled, and he requires insulin therapy. He has a busy work schedule and prefers a regimen that minimises the number of daily injections. Which of the following insulin regimens would be most appropriate for this patient?

- A. Insulin glargine once daily in the evening
 - B. A premixed combination of regular and NPH insulin twice daily
 - C. Insulin Aspart injected before each meal
 - D. CSII with regular insulin using an insulin pump
-

7. A 45-year-old man with a 10-year history of type 2 diabetes mellitus presents to the clinic for a follow-up appointment. He is currently on metformin and glipizide but has not achieved adequate glycemic control. His HbA1c is 8.2%, and he experiences frequent episodes of postprandial hyperglycemia. The physician decides to initiate a basal-bolus insulin regimen. Which of the following insulin regimens would be most appropriate for this patient?

- A. Insulin glargine at bedtime and insulin lispro before meals
 - B. Insulin detemir twice daily and insulin aspart before meals
 - C. Insulin degludec once daily and regular insulin before meals
 - D. NPH insulin twice daily and insulin glulisine before meals
-

8. A 54-year-old woman with type 2 diabetes mellitus presents to the clinic for a routine check-up. She has been managing her diabetes with a combination of diet, exercise, and insulin injections for the past 10 years. During the physical examination, the physician notes an area of subcutaneous fat atrophy at the site where the patient frequently administers her insulin injections. Which of the following is this finding most likely attributed to?

- A. Allergic reaction to insulin
 - B. Lipoatrophy caused by older insulin preparations
 - C. Lipohypertrophy caused by the lipogenic action of insulin
 - D. Somogyi effect
-

9. A 24-year-old male with a history of type 1 diabetes presents to the emergency department with symptoms of polyuria, polydipsia, nausea, vomiting, and abdominal pain. His blood glucose level is 450 mg/dL, arterial blood pH is 7.1, and serum bicarbonate is 12 mEq/L. Urine analysis shows high levels of ketones. He is diagnosed with diabetic ketoacidosis (DKA). Which of the following is the most appropriate initial insulin therapy for this patient?

- A. Subcutaneous injection of regular insulin
 - B. Intravenous infusion of regular insulin
 - C. Subcutaneous injection of insulin glargine
 - D. Intravenous infusion of insulin glargine.
-

10. Which of the following statements regarding Technosphere insulin (Afrezza), an inhaled insulin, is NOT TRUE?

- A. Absorbed from alveoli when administered using a breath-powered inhaler
- B. TI is a new dry powder formulation of recombinant human insulin

- C. Used to manage blood sugar after meals
 - D. Typically used alone without additional insulin
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	1
Question 5	2
Question 6	2
Question 7	1
Question 8	2
Question 9	2
Question 10	4

Solution for Question 1:

Correct Answer: B) Evaluating renal function and adjusting the insulin dose accordingly

Explanation:

- In patients with chronic kidney disease (CKD), the kidneys are less able to filter and metabolize insulin (by insulinase) → Longer insulin retention in the body → Lower insulin requirements.
- It is crucial to evaluate renal function and adjust the insulin dose appropriately before initiating therapy in CKD patients to avoid hypoglycemia.
- During dialysis, there may be a significant reduction in basal insulin requirements.
- Insulin administration with dialysate may necessitate a dose increase of up to 30% due to losses in tubing and dilution.

Solution for Question 2:

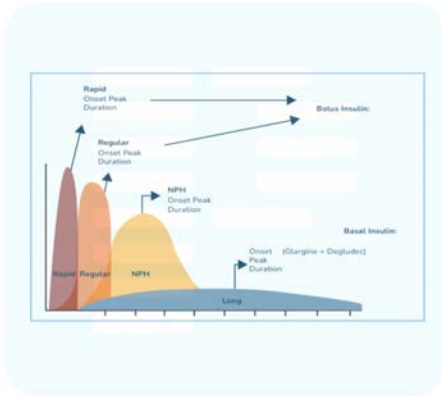
Correct Answer: C) 1 - B, 2 - C, 3 - D, 4 - A

Explanation:

Types of Insulin	Onset and Duration
1. Insulin aspart (Rapid acting insulin)	B. Onset: 15-30 minutes, Duration: 3-5 hours
2. Regular insulin (Short acting insulin)	C. Onset: 30-60 minutes, Duration: 5-8 hours
3. NPH (Intermediate acting insulin)	D. Onset: 1-2 hours, Duration: 12-18 hours
4. Insulin detemir (Long acting insulin)	A. Onset: 1-2 hours, Duration: 24 hours or more

Type	Preparation	Appearance	Onset	Peak (hr)	Duration (hr)	Can be mixed with
Rapid (ultra-short) acting	Insulin lispro	Clear	20-30 (mins)	1-1.5	3-5	Regular, NPH
Insulin aspart	20-30 (mins)	1-1.5	3-5			
Insulin glulisine	20-40 (mins)	1-2	3-5			
Short acting	Regular (soluble) insulin	Clear	30-60 (mins)	2-3	6-8	All preparations (except insulin glargine/detemir)
Intermediate acting	Insulin zinc suspension or Lente	Cloudy	1-2 (hrs)	8-10	12-20	Regular
Neutral Protamine Hagedorn (NPH) or isophane insulin	1-2 (hrs)	8-10	12-20			

Long acting	Insulin glargine	Clear	2-4 (hrs)		24	Cannot mix with other insulins
Insulin detemir	1-4 (hrs)		20-24			
Ultra-long acting	Insulin degludec	Clear	1-9 (hrs)		42	Cannot mix with other insulins



Solution for Question 3:

Correct Answer: C) Uses only short-acting insulins

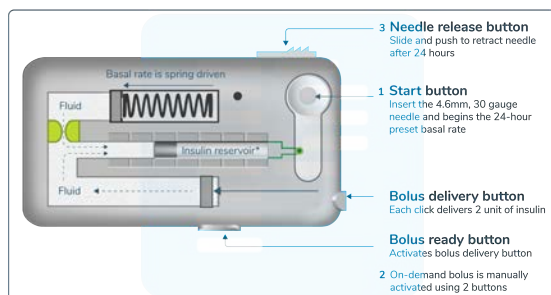
Explanation:

Insulin pumps use short acting insulin as it can adjust basal rates for different times of day, can program boluses for meals, and can improve glycemic control during exercise.

Route of Administration	Insulin Formulation	Uses	Advantages	Disadvantages
Subcutaneous (SC)	Short-acting insulin (regular, lispro, aspart, glulisine)	Long-term treatment of diabetes, prandial glucose control	Mimics physiological insulin release (analogues only), can be combined with long-acting insulins for basal-bolus regimen, lower risk of late postprandial	Requires multiple daily injections for optimal control (especially regular insulin), absorption can be affected by several factors

			hypoglycemia(analogues only)	
Intermediate-acting insulin (NPH)	Basal insulin coverage	Less expensive than long-acting analogues	Requires twice daily dosing, inconsistent absorption	
Long-acting insulin (glargine, detemir, degludec)	Basal insulin coverage, fasting glucose control	Once daily injection, smoother time-action profile (detemir, degludec), lower risk of nocturnal hypoglycemia (glargine, degludec), can be combined with short-acting insulins for basal-bolus regimen	Does not control mealtime glycemia, must be injected separately from other insulins (glargine)	
Intravenous (IV)	Short-acting insulin (regular)	DKA, hyperglycemia in hospitalised patients, situations where insulin requirements change rapidly	Rapid correction of metabolic abnormalities (DKA & hyperkalemia)	Long-acting insulin should not be administered intravenously
Insulin Pump (Continuous Subcutaneous Insulin Infusion)	Short-acting insulin (regular, analogues)	Provides both basal and prandial insulin	More physiological insulin replacement than injections, can adjust basal rates for different times of day, can program boluses for meals, and can improve glycemic control during exercise. Recommended in pronounced Dawn phenomenon.	Requires more patient responsibility than injections, risk of pump failure, cost, and infusion site infections.

INSULIN DELIVERY DEVICE

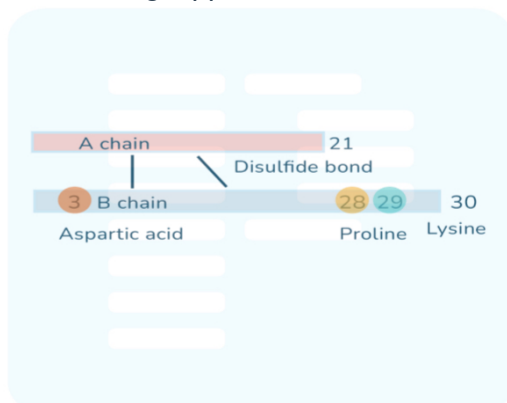


Solution for Question 4:

Correct Answer: A) 1 - D, 2 - A, 3 - B, 4 - C

Explanation:

- **Human Insulin Structure:** Human insulin is a two-chain polypeptide with 51 amino acids. The A chain has 21 amino acids, and the B chain has 30. Two disulfide bonds connect the A and B chains.
- **Synthesis:** Insulin is initially synthesised as preproinsulin, a single polypeptide chain with 110 amino acids. Preproinsulin has four domains: Signal peptide, B chain, C-peptide, and A chain.
- **Processing:** Preproinsulin is processed into proinsulin by removing the signal peptide and forming disulfide bonds between the A and B chains. Proinsulin is further processed into insulin and C-peptide (35 amino acids). C-peptide and insulin are stored in granules in the Golgi apparatus and secreted into the blood together.



Insulin Type	Chain Difference from Human Insulin
Rapid-acting insulin	
Insulin Lispro	Proline at B28 and lysine at B29 are reversed
Insulin Aspart	Aspartic acid replaces proline at B28
Insulin Glulisine	Lysine replaces asparagine at B23; glutamic acid replaces lysine at B29
Regular insulin	No difference (also called human physiological insulin)

Solution for Question 5:

Correct Answer: B) Forms complexes with insulin, allowing the gradual release of insulin monomers

Explanation:

- Regular insulin is a solution of unmodified insulin stabilised by a small amount of zinc.
- Insulin molecules self-aggregate to form hexamers around the zinc ions at injectable solution concentration.
- After subcutaneous injection, insulin monomers are gradually released by dilution, resulting in slow absorption.

Regular Insulin:

PROPERTY	REGULAR INSULIN
Appearance	Clear solution
Onset of action (subcutaneous)	30-45 minutes, 0.5-1 hours
Peak effect (subcutaneous)	2-3 hours
Effective duration (subcutaneous)	6-8 hours
Compatibility	Can be mixed with all preparations except insulin glargine and insulin detemir
Synthesis	Regular insulin is unmodified insulin
Route of administration	Subcutaneous or intravenous
Uses	Type 1 and type 2 diabetes; control prandial rise in blood glucose; acute correction of hyperglycemia; intravenous infusion for diabetic ketoacidosis and hyperglycemia in hospitalised patients

Solution for Question 6:

Correct Answer: B) A premixed combination of regular and NPH insulin twice daily

Explanation;

A premixed combination of regular and NPH insulin offers the convenience of twice-daily injections while addressing basal and prandial insulin needs.

- Mixtard insulin is a pre-mixed combination of regular insulin and isophane insulin.
- It is used in a split-mixed regimen, where the total daily dose is split and injected twice daily, before breakfast and dinner.
- Two formulations are available: A 30:70 mixture of regular and isophane insulin and a 50:50 mixture of regular and isophane insulin.
- Regular insulin has a rapid onset and short duration of action, while isophane insulin has a longer duration of action.

PROPERTY	INTERMEDIATE-ACTING INSULIN
Specific preparation	Insulin zinc suspension (Lente insulin), Neutral protamine Hagedorn (NPH) insulin, or isophane insulin
Appearance	Cloudy solution
Onset of action	1-2 hours
Peak effect	8-10 hours
Effective duration	12-20 hours (Lente), 12-20 hours (NPH)
Compatibility	Can be mixed with regular insulin
Disadvantages	The time course of absorption and intensity of action are relatively inconsistent
Major adverse effect	Hypoglycemia
Additional Information	Intermediate-acting insulin is formulated to prolong insulin absorption. It is often combined with regular insulin in a split-mixed regimen, subcutaneously twice daily, before breakfast and dinner. Lente - Insulin zinc suspension. NPH insulin is formulated by adding protamine to insulin in a quantity just sufficient to complex all the insulin molecules at a neutral pH.

Insulin glargine once daily (Option A) provides only basal insulin coverage and would necessitate additional injections of rapid-acting insulin before meals, making it less convenient.

Insulin aspart before each meal (Option C) only addresses mealtime glucose control and lacks basal coverage.

While effective, CSII (Option D) is more complex, requiring significant patient education and adherence.

Solution for Question 7:

Correct Answer: A) Insulin glargine at bedtime and insulin lispro before meals

Explanation:

Basal-Bolus Regimen:

- Involves long-acting insulin once daily for basal coverage and rapid-acting insulin before meals for post-prandial glucose control.
- Aimed at achieving round-the-clock euglycemia.
- Long-Acting Insulin: Eg., Insulin glargine Injected once daily, typically at bedtime. Provides smooth, peakless basal insulin coverage.
- Injected once daily, typically at bedtime.
- Provides smooth, peakless basal insulin coverage.
- Injected once daily, typically at bedtime.
- Provides smooth, peakless basal insulin coverage.
- Rapid-Acting Insulin: Eg., Insulin lispro or insulin aspart. Injected 0–20 minutes before meals. Provides postprandial insulin control.
- Injected 0–20 minutes before meals.
- Provides postprandial insulin control.
- Injected 0–20 minutes before meals.
- Provides postprandial insulin control.

Type of Insulin	Name of Insulin	Duration of Action	Properties
Long-acting	Glargine (Lantus)	12-24 hrs	Clear solution with a pH of 4.0 that stabilises the insulin hexamer; when injected into the neutral pH of the subcutaneous space, aggregation

			occurs, resulting in prolonged, predictable absorption from the injection site
	Detemir	6-10 hrs	Modified by the addition of a saturated fatty acid to the ε amino group of LysB29, yielding myristoylated insulin that binds to albumin via its fatty acid chain when injected subcutaneously
Ultra-long acting	Degludec	>24 hrs	Forms multi hexamers after injection subcutaneously and is associated with less severe hypoglycemia than glargine

Solution for Question 8:

Correct Answer: B) Lipoatrophy caused by older insulin preparations

Explanation:

This is a clinical scenario involving lipodystrophy, specifically lipoatrophy, as a potential side effect of long-term exogenous insulin use.

- Lipoatrophy is characterised by the loss of subcutaneous fat, most commonly observed at the site of repeated insulin injections. Older insulin preparations were more frequently associated with lipoatrophy as a side effect.
- The use of human insulin and highly purified insulin analogues has significantly reduced the occurrence of lipoatrophy.

Lipohypertrophy (Option C) involves an increase in subcutaneous fat at the injection site and is thought to be caused by the local lipogenic effects of high insulin concentrations.

Somogyi effect (Option D) is related to early morning hyperglycemia in diabetes. These conditions are not associated with lipoatrophy.

ADVERSE EFFECT	DESCRIPTION
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Hypoglycemia	The most frequent and serious reaction to exogenous insulin, especially in labile diabetes. Symptoms: Sweating, anxiety, palpitations, tremors (sympathetic stimulation); dizziness, headache, and behavioural changes (glucose deprivation). Severe cases: mental confusion, seizures, coma (blood glucose < 40 mg/dL).
Local Reactions	Swelling, erythema, and stinging, especially initially, at the injection site
Lipodystrophy	Rare; changes to subcutaneous fat at injection site, more common with older insulin. Caused by repetitive injections at the same site. Prevent by changing injection sites regularly.
Allergy	Very rare with human/purified insulins. Caused by contaminating proteins. Symptoms: Urticaria, angioedema, anaphylaxis.
Hypokalemia	Insulin moves potassium from the bloodstream into cells, leading to low blood potassium levels
Edema	Short-lived dependent edema due to sodium retention may occur when starting insulin therapy
Weight Gain	Modest weight gain with insulin treatment for type 1 and type 2 diabetes

Solution for Question 9:

Correct Answer: B) Intravenous infusion of regular insulin

Explanation:

The most appropriate initial insulin therapy for this patient with diabetic ketoacidosis (DKA) is intravenous infusion of regular insulin.

Acute complications of diabetes include:

- Diabetic ketoacidosis/DKA (Type 1 DM): Ketoacidosis of different grades generally occurs in insulin dependent diabetics, and is due to inadequate/no insulin replacement. Typically they are dehydrated, hyperventilating and have impaired consciousness.
- Hyperosmolar hyperglycemic state/HHS (Type 2 DM): This is characterized by high blood glucose (>600 mg/dl) and serum osmolality (>320 mOsm/L) along with deteriorating mental

status.

- Hyperkalemia

Condition	Insulin Use	Rationale
DKA	Intravenous (IV) regular insulin is used to correct metabolic abnormalities rapidly. Initial dose: Bolus of 0.1-0.2 U/kg, followed by a continuous infusion of 0.1 U/kg/hour. Adjustment: Double the infusion rate if no significant decrease in blood glucose after 2 hours. Maintenance: Once blood glucose reaches 300 mg/dL, reduce the infusion rate to 2-3 U/hour until consciousness is regained.	Regular insulin, the only type used for IV injections, promptly reverses the metabolic consequences of insulin deficiency. This method ensures a rapid blood glucose reduction, allows infusion rate adjustments based on patient response, and prevents hypoglycemia while providing ongoing metabolic support.
Hyperkalemia	Regular insulin IV is administered along with 50% dextrose	This approach helps shift potassium back into cells, lowering serum potassium levels
HHS	IV Regular insulin is preferred.	The general principles of treatment are the same as for ketoacidotic coma, except that faster fluid replacement is to be instituted.

Solution for Question 10:

Correct Answer: D) Typically used alone without additional insulin

Explanation:

Technosphere insulin, a new dry powder formulation of recombinant human insulin, has proven effective for both type 1 and type 2 diabetes patients. It is usually used alongside basal insulin injection.

Table rendering failed

Corticosteroids and Sex Hormones

1. A 35-year-old woman with severe rheumatoid arthritis has worsening joint pain, swelling, and stiffness despite taking methotrexate and an NSAID. The rheumatologist decides to add a corticosteroid to control inflammation. Which corticosteroid is the most appropriate, considering potency, duration, and side effects?

- A. Hydrocortisone
- B. Betamethasone
- C. Prednisolone
- D. Dexamethasone

2. A 28-year-old woman, pregnant with twins, is hospitalised at 30 weeks with PPRM. To reduce the risk of respiratory distress syndrome (RDS) in preterm infants, the obstetrician decides to administer corticosteroids. Which corticosteroid regimen is recommended to promote fetal lung maturity?

- A. Betamethasone 12 mg IM every 24 hours for 2 doses
- B. Dexamethasone 12 mg IM every 24 hours for 2 doses
- C. Hydrocortisone 250 mg intravenous (IV) every 6 hours for 4 doses
- D. Methylprednisolone 32 mg IM single dose

3. A 32-year-old woman with severe systemic lupus erythematosus (SLE) involving multiple organs presents with fatigue, joint pain, skin rashes, and signs of lupus nephritis. The rheumatologist plans high-dose corticosteroids to induce remission. What is the preferred dose of corticosteroid?

- A. Prednisone 60 mg daily

- B. Methylprednisolone 1 g intravenous (IV) for 3 days
 - C. Dexamethasone 40 mg daily
 - D. Hydrocortisone 200 mg IV every 6 hours
-

4. A 10-year-old boy with a widespread, itchy, erythematous rash on his arms, legs, and trunk has not found relief with over-the-counter hydrocortisone cream. Suspecting atopic dermatitis, the dermatologist prescribes a more potent topical steroid. Which topical steroid is most appropriate for this patient's condition and age?

- A. Hydrocortisone 0.5% cream
 - B. Triamcinolone 0.1% ointment.
 - C. Betamethasone 0.05% cream.
 - D. Clobetasol 0.05% ointment.
-

5. A 62-year-old man with type 2 diabetes and hypertension presents with severe nausea, vomiting, muscle cramps, fatigue, dizziness, dehydration, and blood pressure of 90/60 mmHg. Labs show hyponatremia, hyperkalemia, and metabolic acidosis. Which medication is most appropriate for his condition?

- A. Hydrocortisone
 - B. Fludrocortisone
 - C. Spironolactone
 - D. Dexamethasone
-

6. Which of the following is Diethylstilbestrol (DES) classified as?

- A. Synthetic and nonsteroidal estrogen

- B. Synthetic and steroidal estrogen
 - C. Natural and steroidal estrogen
 - D. Natural and nonsteroidal estrogen
-

7. A 54-year-old woman presents with a history of hormone receptor-positive breast cancer. She is prescribed tamoxifen as part of her adjuvant therapy regimen to reduce the risk of cancer recurrence. Which of the following best describes the mechanism of action of tamoxifen?

- A. It acts as a direct agonist on estrogen receptors in breast tissue.
 - B. It acts as an antagonist on estrogen receptors in breast tissue.
 - C. It inhibits aromatase enzyme activity.
 - D. It stimulates progesterone receptors in breast tissue.
-

8. A 32-year-old female presents with infertility. She has irregular menstrual cycles and is suspected of having anovulation. She was started on Clomiphene citrate. Which of the following best describes the mechanism of action of this drug?

- A. Binds to estrogen receptors, exerting agonistic effects in breast tissues.
 - B. Acts as a SERM, exerting antagonist effects in the pituitary.
 - C. Inhibits aromatase enzyme activity, reducing estrogen synthesis.
 - D. Stimulates gonadotropin secretion by blocking estrogenic feedback inhibition of the pituitary.
-

9. Which of the following drugs, whose half-life is 46 hours, primarily acts as an aromatase inhibitor?

- A. Clomiphene

- B. Tamoxifen
 - C. Letrozole
 - D. Fulvestrant
-

10. Which progestin type is characterized by both estrogenic and androgenic actions?

- A. Medroxyprogesterone acetate
 - B. Norethindrone
 - C. Levonorgestrel
 - D. Cyproterone
-

11. A 28-year-old woman with endometriosis comes to the clinic as her symptoms have not been relieved by first-line drugs. Which of the following drugs can be tried to provide relief to her symptoms?

- A. Nandrolone
 - B. Danazol
 - C. Methyltestosterone
 - D. Letrozole
-

12. A 70-year-old man who was diagnosed with metastatic prostatic cancer was started on combined androgen deprivation (CAD) therapy. Which of the following anti-androgenic drugs are used in combined androgen deprivation to prevent initial flare-ups due to initial LH and testosterone production due to GnRH agonist use?

- A. Flutamide
- B. Finasteride

- C. Ketoconazole
 - D. Cyproterone acetate
-

13. A 28-year-old man comes to the dermatologist with complaints of progressive thinning of hair for 3 years. His family has a history of male-pattern baldness. A diagnosis of androgenic alopecia was made, for which he was prescribed topical Minoxidil. What is the mechanism of action of this drug?

- A. Blockade of 5 alpha-reductase enzyme, thus preventing DHT formation.
 - B. Ornithine decarboxylase enzyme inhibitor, which is found in the hair follicles.
 - C. Blockade of androgenic receptors along with progestational activity.
 - D. K⁺ channel opening leads to vasodilation of microvessels around the hair follicles.
-

14. A 56-year-old woman complains of hot flashes that interfere with her daily activities and sleep. Considering her history of recurrent thromboembolism, which of the following drugs would be appropriate for the treatment of her post-menopausal hot flashes?

- A. Hormone replacement therapy with estrogen
 - B. Clonidine
 - C. Paroxetine
 - D. Bazedoxifene
-

15. A 48-year-old man comes to the OPD due to his inability to maintain a penile erection during sexual intercourse. He was prescribed Sildenafil to help with his erectile dysfunction. What is the mechanism of action of this drug?

- A. Inhibition of phosphodiesterase-5 enzyme, thus increasing cytosolic cGMP levels.

- B. Central alpha-2 agonist, reducing sympathetic outflow and causing vasodilation.
 - C. Inhibition of phosphodiesterase-3 enzyme, thus increasing cytosolic cAMP levels.
 - D. Peripheral alpha-blocker reduces vascular resistance and improve blood flow.
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	2
Question 5	2
Question 6	1
Question 7	2
Question 8	4
Question 9	3
Question 10	2
Question 11	2
Question 12	1
Question 13	4
Question 14	3
Question 15	1

Solution for Question 1:

Answer: C) Prednisolone

EXPLANATION:

- Prednisone is a synthetic glucocorticoid with intermediate potency and duration of action, making it a suitable choice for the long-term management of rheumatoid arthritis.

- It has potent anti-inflammatory and immunosuppressive effects, which can help control this patient's joint inflammation and disease activity.

Effects of corticosteroids on the body:

Duration of Action	Glucocorticoid	Potency	Effects/Uses
Short (8-12 hrs)	Hydrocortisone (Option A)	Low	Physiological replacement therapy, adrenal insufficiency, anti-inflammatory
Intermediate (12-36 hrs)	Cortisone	Anti-inflammatory, rheumatic disorders	
Prednisone	Intermediate	Anti-inflammatory, immunosuppressive, rheumatic disorders, asthma, autoimmune diseases.	
Prednisolone (Option C)			
Intermediate to Long	Methylprednisolone	Intermediate to High	Anti-inflammatory, immunosuppressive, severe inflammatory conditions, multiple sclerosis.
Long (36-72 hrs)	Dexamethasone (Option D)	High	Anti-inflammatory, immunosuppressive, severe inflammatory conditions, cerebral edema.
Betamethasone (Option B)			
Fludrocortisone	Mineralocorticoid activity, adrenal insufficiency, salt-wasting disorders.		

Solution for Question 2:

Answer: A) Betamethasone 12 mg IM every 24 hours for 2 doses

Explanation:

Antenatal corticosteroid administration has significantly reduced the risk of respiratory distress syndrome (RDS), intraventricular hemorrhage (IVH), and neonatal mortality in preterm infants.

- The optimal time for administration is between 24 hours and 7 days before delivery, allowing for sufficient time for the corticosteroids to stimulate surfactant production in the fetal lungs.
- The recommended corticosteroid regimen for promoting surfactant maturation in preterm pregnancies is

Betamethasone 12 mg IM every 24 hours for a total of 2 doses.
(OR)
Dexamethasone 6 mg IM every 12 hours for a total of 4 doses (Option B)

- Betamethasone and Dexamethasone are the preferred corticosteroids due to their potency, minimal mineralocorticoid activity, and ability to cross the placenta effectively.
- Hydrocortisone (Option C) and Methylprednisolone (Option D) single dose are not recommended regimens for promoting surfactant maturation in preterm pregnancies.

Solution for Question 3:

Answer: B) Methylprednisolone 1 g intravenous (IV) daily for 3 days

Explanation:

In severe cases of systemic lupus erythematosus (SLE) with multi-organ involvement, pulse therapy with high-dose intravenous corticosteroids is often used to induce remission and prevent further organ damage rapidly.

- The recommended regimen for Pulse therapy in this case would be methylprednisolone 1 gm intravenous (IV) daily for 3 consecutive days. This high-dose, short-term corticosteroid therapy aims to suppress autoimmune inflammation and prevent further organ damage quickly.

- Potential adverse effects associated with pulse therapy include hypertension, electrolyte imbalances (particularly hypokalemia), hyperglycemia, and an increased risk of infections.
- Close monitoring of blood pressure, electrolytes, and glucose levels is essential during and after pulse therapy.

Adverse effects of Corticosteroids:

Corticosteroids	Adverse Effects
Hydrocortisone (Option D)	Adrenal suppression, increased risk of infections, fluid retention, hypertension.
Cortisone	
Prednisone (Option A)	Weight gain, osteoporosis, diabetes mellitus, cushingoid features, skin changes.
Prednisolone	
Methylprednisolone (Option B)	Hypertension, hyperglycemia, electrolyte imbalances (hypokalemia), psychiatric disturbances
Dexamethasone (Option C)	Psychiatric disturbances, cushingoid features, increased risk of infections.
Betamethasone	
Fludrocortisone	Hypertension, edema, hypokalemia.

Solution for Question 4:

Answer: B) Triamcinolone 0.1% ointment

Explanation:

Triamcinolone 0.1% ointment is a mid-potency topical corticosteroid that would be an appropriate choice for treating this pediatric patient's widespread and moderately severe atopic dermatitis.

Topical Steroid	Typical Dose	Efficacy	Uses
Hydrocortisone (Option A)	0.5% - 2.5% cream/ointment	Mild	Mild inflammatory skin conditions in sensitive areas.

Triamcinolone (Option B)	0.025% - 0.1% cream/ointment	Moderate	Atopic dermatitis, contact dermatitis, psoriasis.
Betamethasone (Option C)	0.05% - 0.1% cream/ointment	High	Severe inflammatory skin conditions, refractory cases.
Clobetasol (Option D)	0.05% cream/ointment	Very high	Severe, recalcitrant cases, short-term use only.
Fluocinolone	0.025% - 0.2% cream/ointment	Moderate-high	Atopic dermatitis, psoriasis, lichen planus.
Mometasone	0.1% cream/ointment	Atopic dermatitis, eczema, psoriasis	
Fluticasone	0.005% - 0.05% cream/ointment		

Solution for Question 5:

Answer: A) Hydrocortisone

Explanation:

- The patient's hyponatremia, hyperkalemia, and metabolic acidosis are likely due to a lack of mineralocorticoid activity, resulting in impaired sodium reabsorption and potassium excretion in the kidneys.
- The clinical presentation and investigations indicate adrenal insufficiency, particularly mineralocorticoid deficiency.
- Hydrocortisone, a synthetic glucocorticoid, has both glucocorticoid and mineralocorticoid effects. Hence, it is the drug of choice in acute adrenal insufficiency. It can substitute for deficient endogenous cortisol, which maintains homeostasis.
- The recommended dose is 10-12 mg/m²/day, with an initial bolus of 100 mg followed by 200mg over 24 hours.

Fludrocortisone(Option B), a mineralocorticoid that can replace deficient aldosterone. It is also used in adrenal insufficiency, but synthetic cortisol must be given in acute emergency settings when hemodynamic instability occurs.

Spironolactone(Option C) is an aldosterone receptor antagonist used in treating hyperaldosteronism and heart failure but not appropriate for mineralocorticoid deficiency.

Dexamethasone(Option D) is a potent, long-acting synthetic glucocorticoid. However, it has only a glucocorticoid effect and no mineralocorticoid effect, so it is not the first-line agent.

Solution for Question 6:

Answer: A) Synthetic and nonsteroidal estrogen

Explanation:

Diethylstilbestrol (DES) is a synthetic and nonsteroidal estrogen.

Classification of estrogens:

Table rendering failed

Effects of Estrogen:

Aspect	Effect of Estrogen
Reproductive Organ Regulation	Governs growth and function of female reproductive organs and menstrual cycles.
Secondary Sexual Characteristics	Prompts development of female secondary sexual characteristics during puberty.
Bone Density Maintenance	It helps maintain bone density by inhibiting bone breakdown and promoting bone formation.
Cardiovascular	It improves cardiovascular health by regulating cholesterol levels.
Brain Function	Influences cognitive function, memory, mood regulation, and emotional well-being.
Skin and Hair Health	Contributes to skin thickness, elasticity, hydration, and hair growth.
Urinary Tract Integrity	It supports the integrity of urinary tract tissues and may prevent urinary incontinence.

Solution for Question 7:

Answer: B) It acts as an antagonist on estrogen receptors in breast tissue.

Explanation:

Tamoxifen, a Selective Estrogen Receptor Modulator (SERM), acts as an antagonist on estrogen receptors in breast tissue.

Selective Estrogen Receptor Modulators (SERMs) selectively target estrogen receptors in different tissues.

Aspect	Tamoxifen	Raloxifene
Mechanism of Action	Estrogen Receptor (ER) Antagonist ↓ Binds to ER ↓ Tissue-specific effects such as : Antagonist in breast tissue (prevents estrogen binding). Agonist in bone and cardiovascular tissues (estrogen-like effects)	
Clinical Use	Drug of choice for ER-positive breast cancer. Decrease the risk of contralateral breast cancer.	Prevention of breast cancer. Treatment of postmenopausal osteoporosis.
Side Effects	Increased risk of endometrial cancer. Thromboembolism, hot flashes, depression, hepatotoxicity, cataracts.	No increased risk of endometrial cancer. Hot flashes, depression. Thromboembolism are seen
Drug Interactions	It should not be combined with fluoxetine and other SSRIs. SSRIs inhibit the CYP2D6 enzyme, essential for converting tamoxifen into its active form, endoxifen. This interaction may lower endoxifen levels, diminishing tamoxifen's effectiveness against estrogen receptor-positive breast cancer.	No significant drug interactions.

Solution for Question 8:

Answer: D) Stimulates gonadotropin secretion by blocking estrogenic feedback inhibition of the pituitary.

Explanation:

Aspect	Clomiphene Citrate
Mechanism of Action	Blocking of estrogen receptors in hypothalamus by Clomiphene citrate ↓ Removal of estrogen feedback inhibition ↓ Increased GnRH secretion from the hypothalamus ↓ Increased FSH and LH secretion from the pituitary ↓ Stimulation of ovarian follicle growth ↓ Ovulation
Uses	Treatment of infertility due to failure of ovulation, often in cases of polycystic ovary syndrome (PCOS) or unexplained infertility. Used in assisted reproductive technology (ART) procedures. Treatment of oligozoospermia in men, where it increases gonadotropin secretion, promoting spermatogenesis and testosterone secretion.
Adverse Effects	Polycystic ovaries Multiple pregnancies Hot flashes Vertigo Allergic dermatitis Increased risk of ovarian tumour

Solution for Question 9:

Answer: C) Letrozole

Explanation:

Letrozole has a half-life of 46 hours and acts by inhibiting the Aromatase enzyme in estrogen synthesis.

Drug	Mechanism of Action	Use	Adverse Effects
Letrozole (Option C)	Inhibition of Aromatase Enzyme ↓ Reduction in Peripheral Estrogen Synthesis ↓ Decreased Estrogen Levels in circulation. The half-life of Letrozole is 46 hours	Breast cancer treatment Ovulation induction.	Hot flashes, joint pain, bone loss.
Clomiphene (Option A)	Blocks Estrogen Receptors in the Hypothalamus ↓ Prevents Negative Feedback Inhibition ↓ Increased Release of Gonadotropins (FSH and LH) from Pituitary ↓ Stimulation of Ovulation	Ovulation induction. Infertility in women.	Hot flashes, gastric upset, multiple pregnancies.
Tamoxifen (Option B)	Selective estrogen receptor modulator (SERM) Antagonist in breast tissue. Agonist in bone and cardiovascular tissues.	Breast cancer treatment and prophylaxis.	Endometrial hyperplasia, risk of endometrial carcinoma, and venous thromboembolism.
Fulvestrant (Option D)	Binding to Estrogen Receptors (ER) ↓ ER Degradation and Downregulation ↓ Inhibition of ER Signaling Pathways ↓ Decreased Estrogen	Advanced breast cancer treatment.	Injection site reactions, nausea, liver impairment.

Solution for Question 10:

Answer: B) Norethindrone

Explanation:

Norethindrone is a member of the 19-nortestosterone derivatives group, which includes compounds with weak estrogenic, androgenic, anabolic, and potent antiovaratory actions.

Progestins:

Drug	Actions	Side Effects
Medroxyprogesterone acetate	No significant estrogenic activity.	Nausea, breast tenderness, menstrual changes, headache, abdominal pain, fatigue, dizziness.
Levonorgestrel	Has minimal estrogenic activity.	
Norethindrone	Exhibits weak estrogenic and androgenic activity.	Similar to medroxyprogesterone acetate, but with the addition of possible androgenic side effects such as acne, hirsutism, and changes in libido.
Cyproterone	Has antiandrogenic activity: Blocks androgen receptors and inhibits testosterone production.	Menstrual changes, breast tenderness, weight changes, mood changes, decreased libido, erectile dysfunction, depression (rare but severe).

Solution for Question 11:

Answer: B) Danazol

Explanation:

Danazol is a testosterone receptor agonist with estrogen receptor antagonism action and glucocorticoid activity.

- The most prominent action is the suppression of gonadotropin release.
- It also suppresses gonadal function directly by inhibiting steroidogenic CYP enzymes. However, the aromatase needed for estrogen production is not affected.
- It is used to treat Endometriosis. Leiomyoma. Hereditary angioneurotic edema. Fibrocystic breast disease.

- Endometriosis.
- Leiomyoma.
- Hereditary angioneurotic edema.
- Fibrocystic breast disease.
- Endometriosis.
- Leiomyoma.
- Hereditary angioneurotic edema.
- Fibrocystic breast disease.
- Side effects: Elevation of hepatic enzymes. Females: Complete amenorrhoea occurs with higher doses. Androgenic effects include acne, hirsutism, decreased breast size and deepening of voice, Night sweats and hot flashes Males: Loss of libido
- Elevation of hepatic enzymes.
- Females: Complete amenorrhoea occurs with higher doses. Androgenic effects include acne, hirsutism, decreased breast size and deepening of voice, Night sweats and hot flashes
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- Androgenic effects include acne, hirsutism, decreased breast size and deepening of voice,
- Night sweats and hot flashes
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- Elevation of hepatic enzymes.
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- Night sweats and hot flashes
- Males: Loss of libido
- Complete amenorrhoea occurs with higher doses.
- Androgenic effects include acne, hirsutism, decreased breast size and deepening of voice,
- Night sweats and hot flashes

Other Androgen agonist agents:

Androgenic Drugs	Uses
Methyltestosterone (Option C)	Androgen replacement therapy in hypogonadism Hypopituitarism Testicular failure Idiopathic male infertility Delayed puberty in boys

Testosterone undecanoate	Cachexia in HIV or cancer patients.
Anabolic steroid Nandrolone (Option A) Oxymetholone Stanozolol	Bodybuilding Catabolic states such as acute illness, major surgery Malignancy associated anemia

Solution for Question 12:

Answer: A) Flutamide

Explanation:

Flutamide is a nonsteroidal androgen antagonist that is metabolised to an active metabolite, 2-hydroxyflutamide, which competitively blocks Androgen receptors on the pituitary and sex organs.

It also increases LH secretion by blocking negative feedback inhibition on the pituitary.

Anti androgenic Drugs	MOA	Uses	Side effects
Flutamide (Option A)	Non-steroidal androgen receptor blocker It also increases LH secretion.	CAB therapy for metastatic prostate cancer. Female hirsutism.	Hepatotoxic Gynecomastia Breast tenderness
Bicalutamide	Same as Flutamide but more potent and has a longer half-life	Metastatic prostate cancer	Less hepatotoxic Hot flashes Chills Oedema
Cyproterone acetate (Option D)	Weak androgen receptor blocker with progestational activity. It suppresses LH secretion.	Precocious puberty in boys Female hirsutism Oral contraception in combination with ethinylestradiol	Hepatotoxic
Ketoconazole (Option C)	Inhibits steroidogenic CYP450 enzyme, thus inhibiting steroid production	Antifungal agent	Antiandrogenic effects include gynecomastia and ↓ libido.

Finasteride (Option B)	5 alpha-reductase inhibitor thus reduces the conversion of testosterone to DHT.	Benign prostatic hyperplasia. Androgenic alopecia .	
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- Other drugs with anti-androgenic action are GnRH agonists, Cimetidine, and Spironolactone.

Solution for Question 13:

Answer: D) K⁺ channel opening leads to vasodilation of microvessels around the hair follicles.

Explanation:

- Minoxidil is a vasodilator used as a topical agent to treat Androgenic alopecia
- Mechanism of action: It acts by opening the K⁺ channel on the vascular smooth muscle of microvessels near the hair follicle, thus promoting hair growth.
- Adverse effects: Local irritation, burning sensation, and itching Rarely palpitation and dizziness occur Oral form can lead to hirsutism in females
- Local irritation, burning sensation, and itching
- Rarely palpitation and dizziness occur
- Oral form can lead to hirsutism in females
- Local irritation, burning sensation, and itching
- Rarely palpitation and dizziness occur
- Oral form can lead to hirsutism in females

Options	Description	
Finasteride /Dutasteride (Option A)	MOA	5α reductase enzyme inhibitor, which prevents the conversion of testosterone to dihydrotestosterone.
Uses	Benign prostatic hyperplasia. Androgenic alopecia (oral form and not topical as a topical agent)	
Eflornithine (Option B)	MOA	It inhibits the ornithine decarboxylase enzyme found in the hair follicle, thus slowing hair growth and serving as a hair depilator.

Uses	Topical agent to remove excessive facial hair in Hirsutism in females,	
Cyproterone acetate (Option C)	MOA	The weak blocker of androgenic receptors with progestational activity. It also suppresses LH secretion.
Uses	Female hirsutism. Precocious puberty in boys. Inappropriate sexual behavior in men. Oral contraception in combination with ethinylestradiol.	

Solution for Question 14:

Answer: C) Paroxetine

Explanation:

- Paroxetine, a selective serotonin reuptake inhibitor (SSRI), regulates mood and other functions.
- It reduces hot flashes by influencing thermoregulatory pathways in the hypothalamus.
- It is particularly suitable for women unable to use hormone replacement therapy.
- It is preferred over clonidine for treating hot flashes due to fewer side effects.

Hormone replacement therapy with estrogen(Option A)

- It is the first-line treatment for managing hot flashes in postmenopausal women. Effective for controlling symptoms like hot flashes, body aches, and vaginal dryness. Used to prevent postmenopausal osteoporosis, fractures, and cardiovascular events.
- Effective for controlling symptoms like hot flashes, body aches, and vaginal dryness.
- Used to prevent postmenopausal osteoporosis, fractures, and cardiovascular events.
- Contraindicated in: History of breast cancer or current breast cancer. History of endometrial cancer. History of recurrent thromboembolism due to increased risk of thromboembolic events.
- History of breast cancer or current breast cancer.
- History of endometrial cancer.
- History of recurrent thromboembolism due to increased risk of thromboembolic events.

- Effective for controlling symptoms like hot flashes, body aches, and vaginal dryness.
- Used to prevent postmenopausal osteoporosis, fractures, and cardiovascular events.
- History of breast cancer or current breast cancer.
- History of endometrial cancer.
- History of recurrent thromboembolism due to increased risk of thromboembolic events.

Clonidine(Option B)

- Clonidine, an alpha-2 adrenergic agonist, reduces sympathetic outflow to alleviate hot flashes by affecting the hypothalamus.
- It serves as a non-hormonal alternative for women unable to use estrogen, especially those with a history of hormone-sensitive breast cancers or recurrent thromboembolism.
- It is generally less effective than hormonal therapies and some non-hormonal options like SSRIs.
- Due to its side effects, including dry mouth, drowsiness, and hypotension, SSRIs like paroxetine are often preferred for their better tolerability and efficacy.

Bazedoxifene(Option D)

- Bazedoxifene, a selective estrogen receptor modulator, manages osteoporosis in postmenopausal women but may induce hot flashes in some cases.
- It is contraindicated in patients with a history of recurrent thromboembolism due to thromboembolic risk.

Solution for Question 15:

Answer: A) Inhibition of phosphodiesterase-5 enzyme, thus increasing cytosolic cGMP levels.

Explanation:

Sildenafil inhibits the phosphodiesterase-5 enzyme responsible for cGMP degradation.

Mechanism of action:

Sildenafil
↓
Inhibits phosphodiesterase-5 (PDE-5) enzyme
↓
Prevents cGMP degradation
↓
Increases cytosolic cGMP levels
↓
Prolongs activation of GMP-dependent protein kinase
↓
Inhibits myosin light chain kinase dephosphorylation
↓
Smooth muscle relaxation in the corpus cavernosum
... (content truncated)

- Other uses: Pulmonary artery hypertension
- Side effects:

Blue-green colour blindness	Nasal congestion
Non Arteritic ischaemic optic neuropathy (NAION)	Dyspepsia
Priapism	Hypotension
Flushing	

Central alpha-2 agonist, reducing sympathetic outflow and causing vasodilation.(Option B)

This mechanism is exhibited by Clonidine, which reduces blood pressure in hypertension, but it is not used due to the risk of withdrawal hypertension.

Inhibition of phosphodiesterase-3 enzyme, thus increasing cytosolic cAMP levels.(Option C)

Drugs acting by this mechanism include Amrinone, Milrinone (used as inotrope in heart failure), and Cilostazol (used in Peripheral vascular disease.)

Peripheral alpha-blocker reduces vascular resistance and improves blood flow(Option D)

Phentolamine acts by this mechanism, which is helpful in the intraoperative management of pheochromocytoma and hypertension control in clonidine withdrawal and cheese reaction.

Previous Year Questions

1. Which drug is preferred for management of diabetes in a patient with cardiac and renal failure?

- A. SGLT2 inhibitors
 - B. Pioglitazone
 - C. Metformin
 - D. Sulfonylureas
-

2. Which of the following can be given for long term management of obesity?

- A. Liraglutide
 - B. Metformin
 - C. Sibutramine
 - D. Fenfluramine
-

3. A patient presents to the emergency department with a blood pressure of 90/60 mmHg. There was a history of a local doctor prescribing him some drug for rheumatoid arthritis (RA) a few weeks ago, and he suddenly stopped taking that. On investigations, there was a serum potassium (K) level of 6.5 mEq/L, sodium (Na) level of 139 mEq/L, signs of dehydration, hypotension (90/40), and a blood glucose level of 40 mg/dL. Which of the following drugs should be used for the life-saving treatment of this patient?

- A. Hydrocortisone
- B. Sodium bicarbonate
- C. Insulin

D. Hypertonic saline

4. Which anti-diabetic drug is recently approved for the treatment of heart failure with low ejection fraction along with low urine output?

- A. Empagliflozin
 - B. Linagliptin
 - C. Rosiglitazone
 - D. Insulin degludec
-

5. All of the following drugs are used in thyroid storm except?

- A. Propranolol
 - B. Lugol's Iodine
 - C. Propylthiouracil
 - D. Aspirin
-

6. A young patient on treatment with drugs for MDR tuberculosis presents with weight gain, cold intolerance and lethargy. ON investigations, there was high levels of TSH and low levels of T3 and T4 in the blood. Which of the following is the likely drug responsible for these symptoms?

- A. Cycloserine
 - B. Ethionamide
 - C. Streptomycin
 - D. Pyrazinamide
-

7. Which medication is commonly used as preoperative management for pheochromocytoma to control blood pressure and prevent hypertensive crises?

- A. Captopril
 - B. Propranolol
 - C. Nifedipine
 - D. Phenoxybenzamine
-

8. How should the steroid therapy be administered for a 55-year-old male diagnosed with Addison's disease?

- A. Bolus dose of 100mg
 - B. Bolus of 100mg, followed by 200mg over 24 hours
 - C. Bolus of 200 mg, followed by 500 mg over 24 hours
 - D. Infusion of 500 mg over 24 hours
-

9. Which of the following adverse effects prevents the use of Teriparatide in a patient with Paget's disease?

- A. Osteonecrosis of jawbone
 - B. Osteosarcoma
 - C. Subtrochanteric fractures
 - D. Net loss of bone mass
-

10. Which of the following antidiabetic drugs can cause unexplained diarrhea?

- A. Metformin
 - B. Glimepiride
 - C. Pioglitazone
 - D. Sitagliptin
-

11. Which anti-diabetic drugs are approved for treatment in heart failure?

- A. SGLT2 inhibitor
 - B. DPP-4 inhibitor
 - C. GLP-1 inhibitor
 - D. Amylin analogues
-

12. A patient with type 2 diabetes has shown microalbuminuria in two separate urine analyses six months apart, with normal renal function test results. Which of the following is not useful in preventing renal damage progression?

- A. Hydrochlorothiazide
 - B. Finerenone
 - C. Canagliflozin
 - D. Dapagliflozin
-

13. A G2P1 delivered a preterm baby at 36 weeks. The baby was admitted to the NICU and given expressed breast milk. Unfortunately, the baby did not survive. What is used to reduce pain and engorgement of the lactating breast in this case?

- A. Pramipexole
- B. Chlorpromazine

- C. Sumatriptan
 - D. Cabergoline
-

14. Which of the following drugs increases the risk of UTI?

- A. Empagliflozin
 - B. Glimepiride
 - C. Metformin
 - D. Sitagliptin
-

15. Which drug is preferred for a 55-year-old with 150/100mm Hg BP and HbA1c 7.8%?

- A. Metformin
 - B. Sitagliptin
 - C. Glimepiride
 - D. Liraglutide
-

16. A 60-year-old female patient with type 2 diabetes mellitus has developed vaginal candidiasis. Which hypoglycemic agent should be avoided?

- A. Acarbose
 - B. Metformin
 - C. Gliflozins
 - D. Linagliptin
-

17. A diabetic patient suddenly fainted while exercising, and their blood sugar level was found to be 60 mg/dL. Which of the following antidiabetic drugs is the most likely cause?

- A. Sulfonylureas
 - B. Metformin
 - C. Acarbose
 - D. Pioglitazone
-

18. A 66-year-old man with a history of chronic kidney disease presents with poorly controlled diabetes mellitus. You are considering adjusting his diabetes medication regimen to improve glycemic control. Which of the following needs no dose modification in this patient?

- A. Repaglinide
 - B. Linagliptin
 - C. Glyburide
 - D. Saxagliptin
-

19. In the management of diabetes mellitus, lactic acidosis is caused by which of the following?

- A. Tolbutamide
 - B. Glipizide
 - C. Pioglitazone
 - D. Metformin
-

20. SGLT-2 inhibitors are recently used for type -2 diabetes mellitus. Which of the following are the rare but serious side effects? 1. Ketoacidosis 2. Urosepsis 3. Fournier gangrene 4. Angioedema

- A. 1, 2 & 4 are correct
 - B. 1, 2 & 3 are correct
 - C. Only 4 is correct
 - D. All are correct
-

21. In which of the following conditions is non-pulsatile dosing of GnRH agonist not used?

- A. Male infertility
 - B. Endometriosis
 - C. Precocious puberty
 - D. Prostate cancer
-

22. Which of the following statements accurately describes the effect of T3 and T4 on TRH?

- A. It activates phospholipase A
 - B. It acts on transcription factor
 - C. It acts by increasing cAMP
 - D. It is a tyrosine kinase receptor
-

23. Which one of the options listed below is classified as a Selective Estrogen Receptor Modular (SERM)?

- A. Estrogen
 - B. Metformin
 - C. Tamoxifen
 - D. Adriamycin
-

24. Which of the listed medications aids in the stimulation of bone formation?

- A. Calcitonin
 - B. Raloxifene
 - C. Bisphosphonates
 - D. Teriparatide
-

25. Purpose of adding zinc to insulin in commercial preparations for treatment of diabetes mellitus is:

- A. To make it long acting
 - B. To make it short acting
 - C. To make is fast acting
 - D. Addition of zinc make the insulin effective orally
-

26. Which of the options listed below is commonly employed as the initial treatment for postmenopausal osteoporosis?

- A. Estrogen
- B. Bisphosphonates

- C. Raloxifene
 - D. Combined oral contraceptives
-

27. A patient with a pituitary tumor that overproduced growth hormone underwent surgical removal of the tumor. The resection was found to be incomplete. What is the first-line treatment for this patient?

- A. Leuprolide
 - B. Goserelin
 - C. Nafarelin
 - D. Octreotide
-

28. Which medication, acting as a growth hormone receptor antagonist, is typically prescribed to address a condition marked by chin protrusion, excessive sweating, impaired glucose tolerance, and the enlargement of hands and feet in a middle-aged male patient?

- A. Octreotide
 - B. Pegvisomant
 - C. Cabergoline
 - D. Olcegepant
-

29. Which of the following is the most significant side-effect of tamoxifen?

- A. Cancer in the opposite breast
- B. Osteoporosis
- C. Endometrial hyperplasia

D. Infertility

30. What is the recommended duration for discontinuing tamoxifen treatment before a woman who has had a mastectomy for breast cancer can attempt to conceive?

- A. 1 month
 - B. 3 months
 - C. No need to stop
 - D. Can be stopped and conceived soon after stopping
-

31. Which enzymes does Finasteride inhibit in order to treat male-pattern baldness?

- A. Aromatase
 - B. 17-hydrolase
 - C. 5-alpha reductase
 - D. Phosphodiesterase
-

32. The half – life of letrozole is:

- A. 46 hours
 - B. 70 hours
 - C. 90 hours
 - D. 120 hours
-

33. A patient with diabetes was given an oral hypoglycemic drug which reduced her HbA1c from 7.6 to 6.7. On testing , the patient had high GIP and GLP1 levels with low glucagon levels. Which of the following drugs is possibly her medication?

- A. Liraglutide
 - B. Pramlintide
 - C. Voglibose
 - D. Sitagliptin
-

34. A female patient presented with vulvovaginal pruritis. On detailed history taking, it was found that she was on some antidiabetic drugs. Which antidiabetic drug can cause this condition as a side effect?

- A. Canagliflozin
 - B. Metformin
 - C. Linagliptin
 - D. Liraglutide
-

35. A 60-year-old postmenopausal female patient with a previous history of Colle's fracture came with complaints of lower backache. On evaluation, her T score was -2.5. Which of the following statements are false about the management of her condition?

- A. Teriparatide should be started before supplementing with bisphosphonates
 - B. Bisphosphonates are not given for more than a year
 - C. Calcium requirements is 1200 mg per day
 - D. Oral vitamin D3 is given along with oral calcium
-

36. What is the preferred medication for treating hyperthyroidism during the third trimester of pregnancy?

- A. Carbimazole
 - B. Propylthiouracil
 - C. Sodium iodide
 - D. Radioactive iodine
-

37. Mechanism of action of exenatide in diabetes mellitus is:

- A. It is an analogue of GLP and increases glucose dependent insulin secretion
 - B. It is DPP-4 inhibitor and result in decreased breakdown of GLP
 - C. It inhibits SGLT2 and cause glucosuria
 - D. It is amylin analogue and decrease glucagon
-

38. Which of the following drugs is not likely to cause Pulmonary fibrosis?

- A. Metformin
 - B. Methysergide
 - C. Bleomycin
 - D. Nitrofurantoin
-

39. A 45 year old obese lady with diabetes was managed with oral hypoglycemic agents. She now presents to the casualty with a hypoglycemic episode. Which of the following drugs is most likely responsible for her hypoglycemic episode?

- A. Metformin
- B. Pioglitazone

- C. Sitagliptin
 - D. Glimepiride
-

40. Among the given medications, which one does not necessitate any adjustment in dosage for individuals with chronic kidney disease?

- A. Linagliptin
 - B. Saxagliptin
 - C. Repaglinide
 - D. Glyburide
-

41. For the treatment of thyroid cancer, which radioactive iodine isotope among the options listed below is used?

- A. Iodine -125
 - B. Iodine-131
 - C. Iodine-127
 - D. Iodine-123
-

42. Which of the following medications is suitable for treating cryptorchidism?

- A. Pulsatile GnRH
 - B. Non-pulsatile GnRH
 - C. Testosterone
 - D. Estrogen
-

43. The best approach in the treatment of chronic SIADH is _____:

- A. Tolvaptan
 - B. Reducing fluid intake to <500 ml per day
 - C. Demeclocycline
 - D. Fludrocortisone
-

44. What is the prescribed dosage of hydrocortisone for adrenal insufficiency, measured in milligrams per square meter per day?

- A. 20
 - B. 25
 - C. 10
 - D. 5
-

45. Which of the following actions leads to a decrease in bone resorption and an increase in bone formation?

- A. Strontium ranelate
 - B. Ibandronate
 - C. Teriparatide
 - D. Calcitonin
-

46. Which of the options hinders the production of glucocorticoid?

- A. Mifepristone

- B. Flutamide
 - C. Finasteride
 - D. Metyrapone
-

47. Which of the following DMARD use requires regular monitoring of visual acuity and fundus examination?

- A. Methotrexate
 - B. Hydroxychloroquine
 - C. Sulfasalazine
 - D. Leflunomide
-

48.

- A. Transmembrane receptor linked to tyrosine kinase
 - B. Nuclear receptor
 - C. Ion channel receptor
 - D. G protein-coupled receptor
-

49. Shortest acting corticosteroid is?

- A. Dexamethasone
 - B. Hydrocortisone
 - C. Triamcinolone
 - D. Deflazacort
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	1
Question 5	4
Question 6	2
Question 7	4
Question 8	2
Question 9	2
Question 10	1
Question 11	1
Question 12	1
Question 13	4
Question 14	1
Question 15	4
Question 16	3
Question 17	1
Question 18	2
Question 19	4
Question 20	2
Question 21	1
Question 22	2

Question 23	3
Question 24	4
Question 25	1
Question 26	2
Question 27	4
Question 28	2
Question 29	3
Question 30	2
Question 31	3
Question 32	1
Question 33	4
Question 34	1
Question 35	2
Question 36	1
Question 37	1
Question 38	1
Question 39	4
Question 40	1
Question 41	2
Question 42	1
Question 43	1
Question 44	3
Question 45	1
Question 46	4
Question 47	2
Question 48	2

Solution for Question 1:

Answer: A) SGLT2 inhibitors

Explanation:

Sodium-glucose co-transporter 2 (SGLT2) inhibitors (e.g., empagliflozin, dapagliflozin) are preferred for diabetes management in patients with cardiac and renal failure due to their cardioprotective and renoprotective benefits.

- Cardiovascular Benefits: Reduce heart failure hospitalisation (especially in HF with reduced ejection fraction (HFrEF)). Lower major adverse cardiovascular events (MACE) in high-risk patients.
- Reduce heart failure hospitalisation (especially in HF with reduced ejection fraction (HFrEF)).
- Lower major adverse cardiovascular events (MACE) in high-risk patients.
- Renal Benefits: Slow progression of diabetic nephropathy. Reduce albuminuria and preserve glomerular function.
- Slow progression of diabetic nephropathy.
- Reduce albuminuria and preserve glomerular function.
- Reduce heart failure hospitalisation (especially in HF with reduced ejection fraction (HFrEF)).
- Lower major adverse cardiovascular events (MACE) in high-risk patients.
- Slow progression of diabetic nephropathy.
- Reduce albuminuria and preserve glomerular function.

Diabetic patients with heart failure best benefit from adding a group of drugs called SGLT-2 (sodium-glucose cotransporter-2) inhibitors, and empagliflozin falls in this group.

By inhibiting the SGLT-2 receptors in the proximal tubule of the kidneys, glucose (and sodium) reabsorption can be prevented.

Since both sodium and glucose are osmotic molecules, increasing their excretion draws out more water, making these drugs effective in patients with heart failure.

Drugs:

- Dapagliflozin.
- Empagliflozin.
- Canagliflozin.
- Ertugliflozin.

Side effects:

- Infections (vaginal thrush, balanoposthitis, Fournier's gangrene): Favour the growth of microbes in the lower urinary tract by providing an environment of excessive glucose.
- Euglycemic ketoacidosis is due to inhibiting glucose reabsorption and the body's preference for ketone bodies as an energy source after excess glucose excretion.
- Osteoporosis: Rarely due to loss of calcium ions upon inhibition of the SGLT-2 receptors.

Pioglitazone (Option B) is incorrect because pioglitazone (a thiazolidinedione, TZD) causes fluid retention, worsening heart failure.

Metformin (Option C) is incorrect because although it is the first-line drug for type 2 diabetes, it is contraindicated in severe renal failure (eGFR <30 mL/min/1.73m²) due to the risk of lactic acidosis.

Sulfonylureas (Option D) is incorrect because sulfonylureas (e.g., glibenclamide, glipizide) increase hypoglycemia risk, especially in renal impairment due to reduced clearance.

Solution for Question 2:

Correct Option A - Liraglutide:

- Liraglutide is a medication that is designed to help manage type 2 diabetes, but it has also been shown to promote weight loss. It works by mimicking the effects of a hormone called glucagon-like peptide-1 (GLP-1), which increases insulin secretion, slows down digestion, and reduces appetite. Liraglutide is typically administered as a daily injection and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Liraglutide include nausea, vomiting, diarrhea, and constipation. More serious side effects include pancreatitis and thyroid tumors, although these are rare.

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reduces appetite. Liraglutide is typically administered as a daily injection and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.

Common side effects of Liraglutide include nausea, vomiting, diarrhea, and constipation. More serious side effects include pancreatitis and thyroid tumors, although these are rare.

Incorrect Option:

Option B - Metformin:

- Metformin is a medication that is commonly used to manage type 2 diabetes, but it has also been shown to promote weight loss. It works by reducing glucose production in the liver and improving insulin sensitivity. Metformin is usually taken orally and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Metformin include nausea, vomiting, diarrhea, and stomach pain. More serious side effects include lactic acidosis, although this is rare.

Metformin is a medication that is commonly used to manage type 2 diabetes, but it has also been shown to promote weight loss. It works by reducing glucose production in the liver and improving insulin sensitivity. Metformin is usually taken orally and is usually prescribed along with a healthy diet and exercise to manage obesity in the long term.

Common side effects of Metformin include nausea, vomiting, diarrhea, and stomach pain. More serious side effects include lactic acidosis, although this is rare.

Option C - Sibutramine:

- Sibutramine is a medication that was previously used to manage obesity, but it has been discontinued due to safety concerns. It works by increasing the levels of certain neurotransmitters in the brain, which helps to reduce appetite and promote weight loss. Sibutramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Sibutramine included dry mouth, headache, constipation, and insomnia. More serious side effects included heart attacks and strokes, which is why it was taken off the market.

Sibutramine is a medication that was previously used to manage obesity, but it has been discontinued due to safety concerns. It works by increasing the levels of certain neurotransmitters in the brain, which helps to reduce appetite and promote weight loss. Sibutramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.

Common side effects of Sibutramine included dry mouth, headache, constipation, and insomnia. More serious side effects included heart attacks and strokes, which is why it was taken off the market.

Option D - Fenfluramine:

- Fenfluramine is a medication that was previously used to manage obesity, but it has also been discontinued due to safety concerns. It works by increasing serotonin levels in the brain, which helps reduce appetite and promote weight loss. Fenfluramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.
- Common side effects of Fenfluramine include dry mouth, headache, constipation, and insomnia. More serious side effects included heart valve damage, which is why it was taken off the market.

Fenfluramine is a medication that was previously used to manage obesity, but it has also been discontinued due to safety concerns. It works by increasing serotonin levels in the brain, which helps reduce appetite and promote weight loss. Fenfluramine was usually taken orally and was usually prescribed along with a healthy diet and exercise to manage obesity in the long term.

Common side effects of Fenfluramine include dry mouth, headache, constipation, and insomnia. More serious side effects included heart valve damage, which is why it was taken off the market.

Solution for Question 3:

Correct Option A - Hydrocortisone:

- The likely diagnosis in this patient is Hypothalamic - Pituitary - Adrenal Axis suppression due to exogenous corticosteroid use. The patient's current presentation is due to the abrupt cessation of steroids. Hydrocortisone is given as it has equal glucocorticoid and mineralocorticoid activity.

Incorrect Options:

Option B - Sodium bicarbonate: It has no role in the initial management of HPA axis suppression.

Option C - Insulin: Glucocorticoid deficiency causes hypoglycemia therefore treatment of choice is not insulin.

Option D - Hypertonic Saline: It has no role in the initial management of HPA axis suppression.

Solution for Question 4:

Correct Option A - Empagliflozin(SGLT- 2 inhibitors):

- SGLT- 2 inhibitors, inhibit reabsorb Na⁺ and water back, hence Na and water goes out which is good in chronic heart disease.

Other Drug used in decreasing mortality in chronic heart failure:

- Spironolactone
- SGLT- 2 inhibitors
- ACE inhibitors
- ARBs
- ARN(Combination of ARBs + sacubitril
- Beta Blockers
- For african americans - Hydralazine + Isosorbidedinitraat

Incorrect Options:

Option B - Linagliptin:

- Linagliptin is in a class of medications called dipeptidyl peptidase-4 (DPP-4) inhibitors. It is of no use in chronic heart failure.

Option C - Rosiglitazone:

- Rosiglitazone is a medication used in the management and treatment of type 2 diabetes mellitus. It is in the thiazolidinediones class of medications. It is of no use in chronic heart failure.

Option D - Insulin Degludec:

- This option is incorrect as it is of no use in chronic heart failure.

Solution for Question 5:

Correct Option D - Aspirin:

- Aspirin is of no use in thyroid storm. Aspirin displaces thyroxine from thyroxine-binding globulin and increases free T3 and T4 levels which can be dangerous and worsen thyroid

storm.

Incorrect Options:

Option A - Propranolol:

- Beta blockers are indicated to manage tachycardia in patients with thyroid storm.

Option B - Lugol's Iodine:

- Lugol's Iodine inhibits further release of thyroxine.

Option C - Propylthiouracil:

- It inhibits conversion of T4 to T3.

Solution for Question 6:

Correct Option B - Ethionamide:

- The given clinical picture points towards hypothyroidism the drug responsible in this scenario is Ethionamide.
- The mechanism of ethionamide-induced hypothyroidism is potentially caused by the structure of ethionamide compounds being chemically similar to thioamides, such as propylthiouracil (C₇H₈N₂S), which inhibits thyroid hormone synthesis.

Incorrect Options:

Option A - Cycloserine:

- This option is incorrect as cycloserine causes psychosis and seizures.

Option C - Streptomycin:

- This option is incorrect as it is ototoxic and is a teratogenic drug.

Option D - Pyrazinamide:

- This option is incorrect as its side effect is arthralgia and hyperuricemia.

Solution for Question 7:

Correct Option D - Phenoxybenzamine:

- Phenoxybenzamine is an alpha-adrenergic receptor antagonist and is the preferred choice for preoperative management of pheochromocytoma. By blocking alpha receptors, it prevents the vasoconstrictor effects of catecholamines, reducing the risk of hypertensive crises during surgery. It is initiated before surgery to ensure adequate alpha blockade and blood pressure control.

Incorrect Options:

Option A - Captopril: Captopril is an angiotensin-converting enzyme (ACE) inhibitor used to treat hypertension by blocking the conversion of angiotensin I to angiotensin II, a potent vasoconstrictor. However, it is not commonly used as the primary preoperative medication for pheochromocytoma. Other options are more effective in managing the condition.

Option B - Propranolol: Propranolol is a beta-blocker that blocks the effects of adrenaline and noradrenaline on beta receptors, leading to decreased heart rate and blood pressure. While it can be used in combination with other medications for preoperative management of pheochromocytoma, it is not the preferred initial choice due to the risk of unopposed alpha-adrenergic stimulation, which can worsen hypertensive crises.

Option C - Nifedipine: Nifedipine is a calcium channel blocker that relaxes blood vessels and reduces blood pressure. It can be used as part of the preoperative management of pheochromocytoma to control blood pressure. However, it is often used in combination with other medications, and it may not provide complete alpha-adrenergic blockade, which is crucial for managing pheochromocytoma.

Solution for Question 8:

Correct Option B - Bolus of 100mg, followed by 200mg over 24 hours:

- Bolus Dose (100mg): The initial bolus of 100mg aims to quickly elevate cortisol levels, providing an immediate response to address the acute adrenal insufficiency.
- Infusion (200mg over 24 hours): Following the bolus, a more extended infusion of 200mg over the next 24 hours ensures a sustained and continuous supply of cortisol. This helps maintain cortisol levels within the normal range and provides coverage throughout the day.

Solution for Question 9:

Correct Option B - Osteosarcoma:

Teriparatide is contraindicated in patients with Paget's disease due to the increased baseline risk of osteosarcoma associated with these patients.

- Although postmarketing surveillance has not established a definitive causal association between teriparatide use and osteosarcoma, it remains a serious concern due to findings in animal studies. Therefore, teriparatide carries a black-box warning for the potential risk of osteosarcoma.
- As a result, its use is limited to a maximum of 2 years and should be avoided in patients with conditions that increase the baseline risk for osteosarcoma, such as Paget disease of bone, unexplained elevations of alkaline phosphatase, open epiphyses, or prior radiation therapy involving the skeleton.

Incorrect Options:

- Options A, C, and D are not recognised adverse effects of Teriparatide.
-

Solution for Question 10:

Correct Option A - Metformin:

- The most common side effects (10%–25%) of metformin are gastrointestinal: nausea, indigestion, abdominal cramps or bloating, diarrhea, or some combination of these.
- Chronic treatment with metformin decreases bile acid absorption. The increased concentration of bile salts in the intestines, possibly caused by reduced absorption, may lead to diarrhea, which is often associated with metformin treatment.

Incorrect Options:

Option B - Glimepiride: Glimepiride, a sulfonylurea, is not commonly associated with diarrhea as a primary side effect. It can cause hypoglycemia and weight gain.

Option C - Pioglitazone: Pioglitazone can cause weight gain and fluid retention

Option D - Sitagliptin: Sitagliptin, a DPP-4 inhibitor, is generally well-tolerated and is not commonly associated with unexplained diarrhea.

Solution for Question 11:

Correct Option A - SGLT2 inhibitor:

- Dapagliflozin was the first SGLT2 inhibitor approved for treating heart failure in the U.S. in 2020. Its main effects include improved diuresis and preload reduction by increasing glucose excretion in the urine and sodium and water. It may also have off-target effects that improve cardiac function.

Incorrect Options:

- Options B, C, and D are primarily used for glycemic control alone. While GLP-1 agonists have shown some cardiovascular benefits in clinical trials, they are not approved for the treatment in heart failure.
-

Solution for Question 12:

Correct Option A - Hydrochlorothiazide:

- This is a thiazide diuretic commonly used to treat hypertension. While it helps control blood pressure, it is not specifically beneficial for preventing diabetic nephropathy progression.

Incorrect Options:

Option B - Finerenone: This is a non-steroidal mineralocorticoid receptor antagonist (MRA). It has been shown to have renal-protective effects in patients with diabetic nephropathy by reducing albuminuria and slowing the progression of kidney disease.

Option C - Canagliflozin: This is a sodium-glucose cotransporter 2 (SGLT2) inhibitor. SGLT2 inhibitors have been shown to reduce the progression of diabetic nephropathy and lower the risk

of renal events in patients with type 2 diabetes.

Option D - Dapagliflozin: This is another SGLT2 inhibitor with similar renal-protective benefits as canagliflozin.

Solution for Question 13:

Correct Option D - Cabergoline:

- Cabergoline is a potent agonist at D2 receptors and has a lower affinity for D1, 5HT, and α adrenergic receptors. It inhibits prolactin secretion, thereby suppressing lactation. It is commonly used in situations where lactation needs to be suppressed, such as in cases of fetal loss or when breastfeeding is not desired or feasible.

Incorrect Options:

Option A - Pramipexole: Pramipexole and ropinirole are widely used in the treatment of Parkinson's disease. They are agonists at all D2-like receptors but have the highest affinities for the D3 receptor subtype. However, these agents are less effective than L-DOPA in the early stages of PD treatment

Option B - Chlorpromazine: This antipsychotic medication can elevate prolactin levels, which contraindicates its use in this case.

Option C - Sumatriptan: This is a medication used to treat migraines. It does not have any effect on prolactin secretion or lactation.

Solution for Question 14:

Correct Option A - Empagliflozin:

- This is an SGLT2 inhibitor used to manage type 2 diabetes. SGLT2 inhibitors increase the amount of glucose excreted in the urine, which can create an environment conducive to bacterial growth, thereby increasing the risk of UTIs and genital infections.

Incorrect Options:

- Options B, C, and D do not generally increase the risk of UTI.
-

Solution for Question 15:

Correct Option D - Liraglutide:

- Given the cardiovascular benefits demonstrated by GLP-1 receptor agonists like liraglutide, dulaglutide, and semaglutide, the most suitable option from the provided list would be Liraglutide.
- GLP-1 receptor agonists have shown superiority over placebo in reducing the risk of cardiovascular events such as death from cardiovascular disease, stroke, or myocardial infarction. These medications provide additional cardiovascular protection beyond their glucose-lowering effects, making them particularly beneficial for patients with diabetes who have a high risk of cardiovascular events.

Incorrect Options:

- Options A, B, and C have not shown significant cardiovascular benefits as Liraglutide.
-

Solution for Question 16:

Correct Option C - Gliflozins:

- Gliflozins, also known as SGLT2 inhibitors, inhibit the sodium-glucose cotransporter 2 in the proximal renal tubules.
- This inhibition results in reduced reabsorption of glucose from the kidney, leading to increased glucose excretion in the urine (glycosuria).
- The presence of excess glucose in the urine creates an environment conducive to the growth of fungi, thereby increasing the risk of infections such as vaginal candidiasis.

Incorrect Options:

Options A, B, D:

- Acarbose, Metformin, and Linagliptin are not associated with an increased risk of infections
-

Solution for Question 17:

Correct Option A -Sulfonylurea:

- Sulfonylureas stimulate β -cells of the pancreas to secrete more insulin, which can lead to an increased risk of hypoglycemia.

Incorrect Options:

Options B, C, D:

- Metformin, Acarbose, and Pioglitazone are less likely to cause hypoglycemia.
-

Solution for Question 18:

Correct Option B- Linagliptin:

- Linagliptin is a dipeptidyl-peptidase-4 (DPP-4) inhibitor that does not require any dose adjustments in patients with chronic kidney disease because it is primarily eliminated through hepatic metabolism.
- Therefore, it can be safely used in CKD without the need for dose adjustments.

Incorrect Options:

Option A- Repaglinide: Repaglinide is a meglitinide antidiabetic drug that undergoes hepatic metabolism and has minimal renal excretion. Therefore, dose adjustment is still required in those with CKD.

Option C- Glyburide: Glyburide is a sulfonylurea drug that is predominantly metabolized in the liver and excreted in the urine. It requires renal dose adjustment in patients with CKD because its clearance is significantly affected by renal function, and accumulation can lead to hypoglycemia.

Option D- Saxagliptin: Saxagliptin is also a DPP-4 inhibitor; however, renal dose adjustment is required in patients with CKD due to its renal excretion. A lower dose is recommended in patients with reduced renal function.

Solution for Question 19:

Correct Option D - Metformin:

- Metformin is a widely used oral antidiabetic medication and is considered a first-line therapy for type 2 diabetes. It works by reducing glucose production in the liver, improving insulin sensitivity, and enhancing glucose uptake in peripheral tissues.
- Lactic acidosis is a well-known but rare side effect of metformin, occurring in less than 1 in 10,000 patients.
- The risk of developing lactic acidosis is higher in individuals with underlying kidney or liver disease, significant tissue hypoperfusion (e.g., heart failure), or excessive alcohol consumption. However, when used appropriately and with proper monitoring, the benefits of metformin in diabetes management generally outweigh the risk of lactic acidosis.

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The risk of developing lactic acidosis is higher in individuals with underlying kidney or liver disease, significant tissue hypoperfusion (e.g., heart failure), or excessive alcohol consumption. However, when used appropriately and with proper monitoring, the benefits of metformin in diabetes management generally outweigh the risk of lactic acidosis.

Incorrect Options:

Option A - Tolbutamide: Tolbutamide is a sulfonylurea medication used in the management of type 2 diabetes mellitus. It works by stimulating insulin release from the pancreas. Lactic acidosis is not a commonly reported side effect of tolbutamide.

Option B - Glipizide: Glipizide is another sulfonylurea medication used to lower blood sugar levels in individuals with type 2 diabetes. Like tolbutamide, lactic acidosis is not a commonly reported side effect of glipizide.

Option C - Pioglitazone: Pioglitazone is a thiazolidinedione medication used to improve insulin sensitivity in patients with type 2 diabetes. While pioglitazone has been associated with some rare cases of lactic acidosis, it is a relatively rare side effect and occurs in specific clinical scenarios, such as in patients with significant kidney or liver dysfunction.

Solution for Question 20:

Correct Option B - 1, 2 & 3 are correct:

SGLT-2 inhibitors are a class of medications used for the treatment of type 2 diabetes mellitus. While they are generally safe and well-tolerated, there are rare but serious side effects associated with their use.

1. Ketoacidosis: Ketoacidosis is a potentially life-threatening condition characterized by high levels of ketones in the blood. It is a rare side effect of SGLT-2 inhibitors but can occur. Patients using SGLT-2 inhibitors should be aware of the symptoms of ketoacidosis, such as excessive thirst, frequent urination, nausea, vomiting, abdominal pain, and fruity-smelling breath.

2. Urosepsis: Urosepsis refers to a severe infection that originates from the urinary tract and spreads to the bloodstream. While urosepsis is a rare side effect, there have been reports of an increased risk of urinary tract infections (UTIs) with the use of SGLT-2 inhibitors. If a UTI is not promptly treated, it can progress to urosepsis.

3. Fournier gangrene: Fournier gangrene is a rare but serious infection of the genital and perineal area. Although extremely rare, there have been reports of Fournier gangrene associated with the use of SGLT-2 inhibitors. Patients should be aware of symptoms such as severe pain, swelling, or redness in the genital or perineal area and seek immediate medical attention if these symptoms occur.

Incorrect Options:

Option A - 1, 2 & 4 are correct: While options 1 and 2 are correct, option 4 (Angioedema) is not a known rare but serious side effect of SGLT-2 inhibitors. Angioedema is more commonly associated with other classes of medications, such as ACE inhibitors.

Option C - Only 4 is correct: Option 4 (Angioedema) is not a known rare but serious side effect of SGLT-2 inhibitors. Therefore, this option is not accurate.

Option D - All are correct: While options 1 and 2 are correct, option 3 (Fournier gangrene) is also a rare but serious side effect associated with the use of SGLT-2 inhibitors. Therefore, this option

is not accurate.

Solution for Question 21:

Correct Option A - Male infertility:

- Non-pulsatile dosing of GnRH agonist is not used in male infertility because pulsatile release of GnRH is necessary to stimulate the secretion of luteinizing hormone (LH) and follicle-stimulating hormone (FSH), which are essential for spermatogenesis.

Non-pulsatile dosing of GnRH agonist is not used in male infertility because pulsatile release of GnRH is necessary to stimulate the secretion of luteinizing hormone (LH) and follicle-stimulating hormone (FSH), which are essential for spermatogenesis.

Incorrect Options:

Option B - Endometriosis: Non-pulsatile dosing of GnRH agonist is commonly used in endometriosis to suppress the production of estrogen and induce a hypoestrogenic state, which helps in reducing the symptoms and progression of the disease.

Option C - Precocious puberty: Non-pulsatile dosing of GnRH agonist is used in precocious puberty to suppress the production of gonadotropins, thereby delaying the onset of puberty and preventing early maturation of secondary sexual characteristics.

Option D - Prostate cancer: Non-pulsatile dosing of GnRH agonist is commonly used in prostate cancer to achieve and maintain medical castration by suppressing the production of gonadotropins and reducing testosterone levels, which helps in the treatment of hormone-sensitive prostate cancer.

Solution for Question 22:

Correct Option B - It acts on transcription factor:

- T3 and T4 exert their effects on TRH by acting on transcription factors. When T3 and T4 bind to their nuclear receptors, they act as ligands to activate gene transcription. This leads to the production of specific proteins, including transcription factors, which can then regulate the expression of genes involved in thyroid hormone feedback control.

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Incorrect Options:

Option A - It activates phospholipase A: T3 and T4, which are thyroid hormones, do not activate phospholipase A. Phospholipase A is an enzyme involved in the breakdown of phospholipids, and its activation is not directly associated with the actions of T3 and T4 on TRH (thyrotropin-releasing hormone).

Option C - It acts by increasing cAMP: The action of T3 and T4 on TRH does not involve the direct increase of cyclic adenosine monophosphate (cAMP). cAMP is a second messenger that is involved in various cellular signaling pathways, but it is not the primary mechanism through which T3 and T4 act on TRH.

Option D - It is a tyrosine kinase receptor: T3 and T4 do not act on TRH through a tyrosine kinase receptor. Tyrosine kinase receptors are a type of cell surface receptor involved in various signaling pathways, but they are not directly involved in the regulation of TRH by T3 and T4.

Solution for Question 23:

Correct Option C - Tamoxifen:

- Tamoxifen is a selective estrogen receptor modulator (SERM) used to treat breast cancer in men and women and as a prophylactic agent against breast cancer in women.
- It selectively inhibits or stimulates estrogen receptors depending on where it is located (hence selective modulator)

Tamoxifen is a selective estrogen receptor modulator (SERM) used to treat breast cancer in men and women and as a prophylactic agent against breast cancer in women.

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Incorrect Options:

Option A - Estrogen: Estrogen is a steroid hormone associated with the female reproductive organs and is responsible for developing female sexual characteristics. Estrogen or estradiol is the most common form of estrogen hormone for FDA-approved treatment as hormone replacement therapy in managing symptoms associated with menopause. It stimulates all estrogen receptors in all tissues.

Option B - Metformin: Metformin, a biguanide compound used for the treatment of diabetes, PCOS, and insulin resistance. Metformin can inhibit 17β -estradiol-stimulated cell proliferation and the expression of estrogen receptors in a dose-dependent manner in cancer cells.

Option D - Adriamycin: Adriamycin is used alone or with other drugs to treat many types of cancer, including leukemia, lymphoma, neuroblastoma, sarcoma, Wilms tumor, and certain cancers of the lung, breast, stomach, ovary, thyroid, and bladder.

Solution for Question 24:

Correct Option D - Teriparatide:

- Teriparatide is a synthetic form of parathyroid hormone (PTH), which is an essential hormone involved in regulating calcium and phosphate metabolism. Unlike other antiresorptive medications, such as bisphosphonates, teriparatide acts as an anabolic agent, meaning it promotes bone formation. It stimulates osteoblast activity, leading to increased bone formation and improved bone density. Teriparatide is primarily used in the treatment of severe osteoporosis in postmenopausal women and men at high risk of fractures.

Incorrect Options:

Option A - Calcitonin: Calcitonin is a hormone that regulates calcium and phosphate metabolism. While it has some modest antiresorptive effects, it primarily inhibits osteoclast activity, which reduces bone resorption. However, it does not have a significant role in promoting bone formation in the human body.

Option B - Raloxifene: Raloxifene is a selective estrogen receptor modulator (SERM) that mimics the beneficial effects of estrogen on bone density. It helps to prevent bone loss and reduce fracture risk by inhibiting bone resorption. However, raloxifene does not promote bone formation. It is used in the prevention and treatment of osteoporosis in postmenopausal women.

Option C - Bisphosphonates: Bisphosphonates are a class of medications commonly used to treat osteoporosis. They work by inhibiting bone resorption, thereby reducing bone loss and decreasing the risk of fractures. While bisphosphonates improve bone density by slowing down bone breakdown, they do not directly promote bone formation.

Solution for Question 25:

Correct Option A - To make it long acting:

- The purpose of adding zinc to insulin in commercial preparations for the treatment of diabetes mellitus is to make it long acting. Insulin is a hormone that regulates blood sugar levels, and its action can be modified by adding certain substances. Zinc is one such substance that can be added to insulin to prolong its duration of action.
- When zinc is added to insulin, it forms crystals that delay the absorption of insulin from the injection site into the bloodstream. This delayed absorption results in a slower and more prolonged release of insulin, leading to a long-acting effect. Long-acting insulin formulations are used to provide a steady basal level of insulin throughout the day, mimicking the body's natural insulin secretion patterns.

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Incorrect Options:

Option B - To make it short acting: Adding zinc to insulin does not make it short acting. Instead, it has the opposite effect of prolonging its duration of action.

Option C - To make it fast acting: Adding zinc to insulin does not make it fast acting. Fast-acting insulins, such as rapid-acting or short-acting insulins, are formulated differently to provide a rapid onset of action and shorter duration.

Option D - Addition of zinc makes the insulin effective orally: Adding zinc to insulin does not make it effective when taken orally. Insulin cannot be taken orally as it is broken down by digestive enzymes in the gastrointestinal tract. Insulin must be administered via subcutaneous injection or other suitable routes to be effective.

Solution for Question 26:

Correct Option B - Bisphosphonates:

- Bisphosphonates are typically used in the first-line management of postmenopausal osteoporosis. Postmenopausal osteoporosis is a condition characterized by a decrease in bone density and an increased risk of fractures that occurs after menopause. The goal of treatment is to prevent bone loss, maintain bone density, and reduce the risk of fractures. Bisphosphonates, such as alendronate, risedronate and ibandronate, are the most commonly prescribed medications for the first-line management of postmenopausal osteoporosis. They work by inhibiting bone resorption, reducing bone turnover, and increasing bone mineral density.
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Incorrect Options:

Option A - Estrogen: While estrogen was previously used as a first-line treatment for postmenopausal osteoporosis, its use has declined due to concerns about increased risks of cancer. Estrogen therapy is now typically reserved for women who require treatment for menopausal symptoms as well as osteoporosis.

Option C - Raloxifene: Raloxifene is a selective estrogen receptor modulator (SERM) that can be used to prevent and treat postmenopausal osteoporosis. It works by mimicking some of the beneficial effects of estrogen on bone density without some of the harmful effects on the breast and uterus. However, bisphosphonates are often preferred as first-line therapy.

Option D - Combined oral contraceptives: Combined oral contraceptives, which contain both estrogen and progestin, are not typically considered in treatment for postmenopausal osteoporosis.

Solution for Question 27:

Correct Option D - Octreotide:

- Octreotide is the first-line treatment for a patient with a pituitary tumor that overproduces growth hormone, particularly in cases where surgical resection was incomplete. Octreotide is a synthetic analog of somatostatin, a naturally occurring hormone that inhibits the release of growth hormone from the pituitary gland. Octreotide can be administered subcutaneously or intravenously and has a longer duration of action than natural somatostatin, making it an effective treatment to reduce the hypersecretion of growth hormone.

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intravenously and has a longer duration of action than natural somatostatin, making it an ineffective treatment to reduce the hypersecretion of growth hormone.

Incorrect Options:

Option A - Leuprolide: It is a medication that belongs to the class of drugs known as gonadotropin-releasing hormone (GnRH) agonists. It is commonly used to treat conditions such as prostate cancer, endometriosis, and uterine fibroids.

Option B - Goserelin: It is a medication that belongs to a class of drugs known as gonadotropin-releasing hormone (GnRH) agonists.

Option C - Nafarelin: It is a medication that also belongs to the class of drugs known as gonadotropin-releasing hormone (GnRH) agonists, like Leuprolide.

Solution for Question 28:

Correct Option B - Pegvisomant:

- The symptoms described in this question are consistent with acromegaly, which is caused by an overproduction of growth hormones.
- Acromegaly can result in excessive growth of soft tissues, bones, and organs, leading to a range of physical and metabolic changes.
- One medication that can be used to treat acromegaly is pegvisomant, which is a growth hormone receptor antagonist.
- Pegvisomant works by binding to the growth hormone receptor on target cells and blocking the action of growth hormone.
- This reduces the level of insulin-like growth factor 1 (IGF-1) in the blood, which is a downstream target of growth hormone and is responsible for many of the physical and metabolic changes seen in acromegaly.
- Pegvisomant is administered by subcutaneous injection, typically once daily.
- It is used in patients with acromegaly who have not responded to or cannot tolerate other medications, such as somatostatin analogs like octreotide or lanreotide.
- Pegvisomant can be used alone or in combination with other medications to control the level of growth hormone in the blood and alleviate symptoms associated with acromegaly.

- While pegvisomant is generally well-tolerated, it can cause some side effects, including injection site reactions, nausea, and liver function abnormalities.
- It is important for patients taking pegvisomant to be monitored regularly for IGF-1 levels and liver function, as well as for other potential complications of acromegaly, such as diabetes mellitus, hypertension, and sleep apnea.

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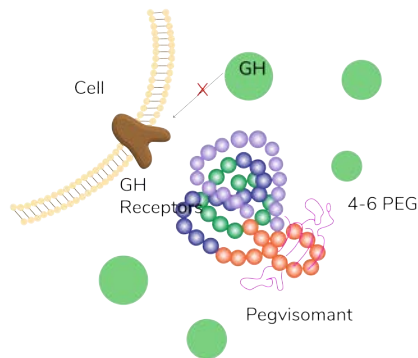
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It is important for patients taking pegvisomant to be monitored regularly for IGF-1 levels and liver function, as well as for other potential complications of acromegaly, such as diabetes mellitus, hypertension, and sleep apnea.



Incorrect Options:

Option A - Octreotide: It is a somatostatin analog that is commonly used to treat acromegaly by inhibiting the secretion of growth hormone (GH) from the pituitary gland.

Option C - Cabergoline: It is a dopamine receptor agonist that is often used to treat hyperprolactinemia.

Option D - Olcegepant: It is a calcitonin gene-related peptide (CGRP) receptor antagonist that is used for the treatment of migraine headaches.

Solution for Question 29:

Correct Option C - Endometrial hyperplasia:

- Tamoxifen is a selective estrogen receptor modulator (SERM) commonly used to treat breast cancer. While it has proven to be a valuable therapy for breast cancer patients, it also has potential side effects that must be considered.
- Endometrial hyperplasia is the most significant side-effect of tamoxifen. Tamoxifen is a partial agonist of the uterine's estrogen receptor, which can lead to an increased risk of endometrial hyperplasia, a precancerous condition. Endometrial hyperplasia is characterized by the abnormal growth of cells in the lining of the uterus and can lead to the development of endometrial cancer.

Tamoxifen is a selective estrogen receptor modulator (SERM) commonly used to treat breast cancer. While it has proven to be a valuable therapy for breast cancer patients, it also has potential side effects that must be considered.

Endometrial hyperplasia is the most significant side-effect of tamoxifen. Tamoxifen is a partial agonist of the uterine's estrogen receptor, which can lead to an increased risk of endometrial hyperplasia, a precancerous condition. Endometrial hyperplasia is characterized by the abnormal growth of cells in the lining of the uterus and can lead to the development of endometrial cancer.

Incorrect Options:

Option A - Cancer in the opposite breast: Tamoxifen is used to prevent the recurrence of breast cancer in the same and prevent the occurrence of a new cancer in the opposite breast.

Option B - Osteoporosis: Osteoporosis is not a significant side-effect of tamoxifen. Tamoxifen has a protective effect on bone density in postmenopausal women.

Option D - Infertility: Infertility is also not a significant side-effect of tamoxifen, as it does not affect fertility in women. In fact, tamoxifen has been used as a fertility treatment in women with breast cancer who wish to conceive after treatment. However, there is a risk of teratogenicity with use of tamoxifen and it is advised to stop the drug 3 months before attempting conception.

Solution for Question 30:

Correct Option B - 3 months:

- Tamoxifen is a selective estrogen receptor modulator (SERM) that is commonly used for the treatment of breast cancer.
- It works by blocking the estrogen receptors in breast tissue, which can prevent the growth and spread of cancer cells.
- The American Society of Clinical Oncology (ASCO) recommends that women who are taking tamoxifen should stop the medication at least 3 months before attempting to conceive.
- This allows time for the drug to be cleared from the body.

Incorrect Options:

- Option A, C & D: Refer above explanation

Solution for Question 31:

Correct Option C - 5-alpha reductase:

- Finasteride is a medication used to treat male-pattern baldness and benign prostatic hyperplasia (BPH).
- It acts by inhibiting the enzyme 5-alpha reductase, which converts testosterone to dihydrotestosterone (DHT), the hormone responsible for the growth of prostate cells and hair loss in men.

Finasteride is a medication used to treat male-pattern baldness and benign prostatic hyperplasia (BPH).

It acts by inhibiting the enzyme 5-alpha reductase, which converts testosterone to dihydrotestosterone (DHT), the hormone responsible for the growth of prostate cells and hair loss in men.

Incorrect Options:

Option A - Aromatase: Aromatase inhibitors are drugs like letrozole, anastrozole.

Option B - 17-hydrolase: 17-hydroxylase is an enzyme involved in the production of sex steroids, including testosterone. It is not targeted by finasteride.

Option D - Phosphodiesterase: Phosphodiesterase inhibitors, such as sildenafil, are used to treat erectile dysfunction by increasing blood flow to the penis. They do not affect 5-alpha reductase.

Solution for Question 32:

Correct Option A - 46 hours:

- The half-life of letrozole is 46 hours.
- Letrozole is an aromatase inhibitor drug used in the treatment of hormone receptor-positive breast cancer in postmenopausal women.
- It works by blocking the enzyme aromatase, which converts androgens into estrogen, leading to a reduction in estrogen levels.

The half-life of letrozole is 46 hours.

Letrozole is an aromatase inhibitor drug used in the treatment of hormone receptor-positive breast cancer in postmenopausal women.

It works by blocking the enzyme aromatase, which converts androgens into estrogen, leading to a reduction in estrogen levels.

Incorrect Options:

- Option B, C and D are incorrect. Refer to the explanation of the correct answer.

Option B, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 33:

Correct Option D - Sitagliptin:

- Sitagliptin is an oral medication classified as a dipeptidyl peptidase-4 (DPP-4) inhibitor. It works by inhibiting the enzyme DPP-4, which breaks down incretin hormones such as GLP-1 and GIP. By inhibiting DPP-4, sitagliptin increases the levels of GLP-1 and GIP, which stimulate insulin secretion and reduce glucagon secretion. This helps lower blood sugar levels. Sitagliptin is known to reduce HbA1c levels in patients with diabetes, making it a suitable option based on the provided information of high GIP and GLP1 levels along with low glucagon levels.
- Therefore, in this scenario, the most appropriate drug given to the patient would be Sitagliptin.

Sitagliptin is an oral medication classified as a dipeptidyl peptidase-4 (DPP-4) inhibitor. It works by inhibiting the enzyme DPP-4, which breaks down incretin hormones such as GLP-1 and GIP. By inhibiting DPP-4, sitagliptin increases the levels of GLP-1 and GIP, which stimulate insulin secretion and reduce glucagon secretion. This helps lower blood sugar levels. Sitagliptin is known to reduce HbA1c levels in patients with diabetes, making it a suitable option based on the provided information of high GIP and GLP1 levels along with low glucagon levels.

Therefore, in this scenario, the most appropriate drug given to the patient would be Sitagliptin.

Incorrect Options:

Options A - Liraglutide: Liraglutide is a medication in the class of glucagon-like peptide-1 (GLP-1) receptor agonists. GLP-1 receptor agonists help lower blood sugar levels by increasing insulin secretion, reducing glucagon secretion, slowing gastric emptying, and promoting satiety.

Liraglutide is effective in reducing HbA1c levels in patients with diabetes. Being a GLP-1 agonist, it would reduce the body's GLP-1 production. The information provided about high GIP (gastric inhibitory peptide) levels and low glucagon levels does not directly indicate the use of liraglutide. Therefore, it is not the most suitable option in this scenario.

Options B - Pramlintide: Pramlintide is an injectable medication used in the treatment of type 1 and type 2 diabetes. It is a synthetic analog of amylin, a hormone involved in glucose regulation. Pramlintide helps to lower blood sugar levels by reducing the secretion of glucagon, slowing gastric emptying, and promoting satiety. However, it has no effect on GLP-1 levels.

Options C - Voglibose: Voglibose is an oral medication that belongs to the class of alpha-glucosidase inhibitors. It works by inhibiting the enzymes in the intestines that break down carbohydrates, thereby delaying the absorption of glucose and reducing postprandial (after meal) blood sugar spikes. Voglibose does not directly affect GLP-1, GIP, or glucagon levels. Therefore, it may not be the most suitable option based on the information provided.

Solution for Question 34:

Correct Option A - Canagliflozin:

- Among the options provided, Canagliflozin is the antidiabetic drug that has been associated with vulvovaginal pruritus (itching) due to candidiasis. Canagliflozin belongs to the class of medications known as sodium-glucose co-transporter 2 (SGLT2) inhibitors. These drugs work by reducing glucose reabsorption in the kidneys, leading to increased glucose excretion in the urine.
- Vulvovaginal candidiasis is common due to the increased excretion of sugar in the urine of these individuals. The sugar promotes growth of pathogens like candida resulting in pruritus in the genital area, including the vulva and vagina.
- It can also result in UTI as well as Fournier's gangrene.

Among the options provided, Canagliflozin is the antidiabetic drug that has been associated with vulvovaginal pruritus (itching) due to candidiasis. Canagliflozin belongs to the class of medications known as sodium-glucose co-transporter 2 (SGLT2) inhibitors. These drugs work by reducing glucose reabsorption in the kidneys, leading to increased glucose excretion in the urine.

Vulvovaginal candidiasis is common due to the increased excretion of sugar in the urine of these individuals. The sugar promotes growth of pathogens like candida resulting in pruritus in the genital area, including the vulva and vagina.

It can also result in UTI as well as Fournier's gangrene.

Incorrect Options:

Option B, C & D - Metformin, Linagliptin & Liraglutide:

- On the other hand, are not typically associated with vulvovaginal candidiasis.

On the other hand, are not typically associated with vulvovaginal candidiasis.

Solution for Question 35:

Correct Option B - Bisphosphonates are not given for more than a year:

- Among the statements provided, the false statement about the management of the patient's condition is: "Bisphosphonates are not given for more than a year." Bisphosphonates are commonly used for the long-term management of osteoporosis. While the duration of treatment may vary depending on the individual's response and specific circumstances, bisphosphonates can be prescribed for several years to maintain bone density and reduce the risk of fractures.
- "Bisphosphonates are not given for more than a year."
- Bisphosphonates are commonly used for the long-term management of osteoporosis. While the duration of treatment may vary depending on the individual's response and specific circumstances, bisphosphonates can be prescribed for several years to maintain bone density and reduce the risk of fractures.

Among the statements provided, the false statement about the management of the patient's condition is:

- "Bisphosphonates are not given for more than a year."
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"Bisphosphonates are not given for more than a year."

Bisphosphonates are commonly used for the long-term management of osteoporosis. While the duration of treatment may vary depending on the individual's response and specific circumstances, bisphosphonates can be prescribed for several years to maintain bone density

and reduce the risk of fractures.

The other statements are true and aligned with general management principles for osteoporosis. Let's go through each statement to further clarify:

Incorrect Options:

Option A - Teriparatide should be started before supplementing with bisphosphonates: This statement is true. Teriparatide, a recombinant form of parathyroid hormone (PTH), is often considered as a first-line treatment option for severe osteoporosis. It stimulates new bone formation and is typically prescribed before bisphosphonates if the patient has severe osteoporosis or has experienced a recent fragility fracture.

Option C - Calcium requirement is 1200 mg per day: This statement is true. The recommended daily intake of calcium for postmenopausal women is generally 1200 mg. Adequate calcium intake is important for maintaining bone health and preventing osteoporosis.

Option D - Oral vitamin D3 is given along with oral calcium: This statement is true. Vitamin D plays a crucial role in calcium absorption and bone health. Postmenopausal women with osteoporosis often require supplemental vitamin D to ensure optimal calcium absorption. Vitamin D3 is the preferred form of vitamin D supplementation.

Solution for Question 36:

Correct Option A - Carbimazole:

- Drug of choice for Hyperthyroidism in 3rd trimester is Carbimazole
- Methimazole/Carbimazole can also be used in the third trimester of pregnancy.
- Carbimazole is a thyroid peroxidase inhibitor
- It has teratogenic potential in the first trimester and is switched over to propylthiouracil during this period. From the second trimester, it is advised to put the patient back on cabergoline due to the risk of propylthiouracil-induced hepatotoxicity.

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It has teratogenic potential in the first trimester and is switched over to propylthiouracil during this period. From the second trimester, it is advised to put the patient back on cabergoline due to the risk of propylthiouracil-induced hepatotoxicity.

Incorrect Options:

Option B - Propylthiouracil: From the second trimester, it is advised to put the patient back on cabergoline due to the risk of propylthiouracil-induced hepatotoxicity.

Option C - Sodium iodide: Sodium iodide is not used as a primary treatment for hyperthyroidism in pregnancy. It is not used for a long term management of hyperthyroidism, only short term like in the case of pre operative preparation.

Option D - Radioactive iodine: Radioactive iodine (RAI) is contraindicated during pregnancy as it can cross the placenta and harm the developing fetal thyroid gland. It is typically avoided in women who are pregnant or planning to become pregnant.

Solution for Question 37:

Correct Option A - It is an analogue of GLP and increases glucose-dependent insulin secretion:

- It is an analogue of GLP-1 released from the gut and increases glucose-dependent insulin secretion.
- Exenatide is a medication used in the treatment of type 2 diabetes mellitus. It belongs to the class of drugs known as incretin mimetics or GLP-1 receptor agonists. its mechanism of action:
- Exenatide is an analogue of glucagon-like peptide-1 (GLP-1), a hormone released from the gut in response to food intake. GLP-1 plays an important role in regulating blood sugar levels. Exenatide acts by mimicking the effects of GLP-1 on the body.

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Incorrect Options:

Option B - It is DPP-4 inhibitor and result in decreased breakdown of GLP:

- Drugs that cast as DPP4 inhibitors include- sitagliptin, vildagliptin etc.

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Option C - It inhibits SGLT2 and cause glucosuria:

- SGLT2 inhibitors examples include empagliflozin and Dapagliflozin

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Option D - It is amylin analogue and decrease glucagon:

- Amylin analogs include pramlintide

Amylin analogs include pramlintide

Solution for Question 38:

Correct Option A - Metformin:

- Metformin is a drug that is commonly used to treat type 2 diabetes.
- Increase insulin sensitivity(Biguanides)
- Mechanism: Inhibit mGPD which inhibits hepatic gluconeogenesis and the action of glucagon. Increase glycolysis, peripheral glucose uptake (increase insulin sensitivity).
- Adverse effects: GI upset, lactic acidosis (use with caution in renal insufficiency), vitamin B12 deficiency. Weight loss (often desired).
- It does Not cause Pulmonary fibrosis.

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Adverse effects: GI upset, lactic acidosis (use with caution in renal insufficiency), vitamin B12 deficiency. Weight loss (often desired).

It does Not cause Pulmonary fibrosis.

Incorrect Options:

Option B - Methysergide: Methysergide is a medication that was originally developed for the treatment of migraines and cluster headaches and can cause Pulmonary fibrosis.

Option C - Bleomycin: Bleomycin is a medication that is commonly used in the treatment of various types of cancers, including lymphomas, testicular cancer, and head and neck cancers and can cause Pulmonary fibrosis.

Option D - Nitrofurantoin: Nitrofurantoin is an antibiotic that is commonly used in the treatment of urinary tract infections and can cause Pulmonary fibrosis.

Solution for Question 39:

Correct Option D - Glimepiride:

- Glimepiride is the correct answer as it is a sulphonylurea and it causes hypoglycemia as it is an insulin secretagogue.

Glimepiride is the correct answer as it is a sulphonylurea and it causes hypoglycemia as it is an insulin secretagogue.

Incorrect Options:

Option A - Metformin: Metformin is not a direct insulin secretagogue and therefore it does not result in hypoglycemia as much as sulphonylureas.

Option B - Pioglitazone: Pioglitazone is not a direct insulin secretagogue and therefore it does not result in hypoglycemia as much as sulphonylureas.

Option C - Sitagliptin: Sitagliptin is a DPP-4 inhibitor and not a direct insulin secretagogue and therefore it does not result in hypoglycemia as much as sulphonylureas.

Solution for Question 40:

Correct Option A - Linagliptin:

- Linagliptin is an oral hypoglycemic drug belonging to DPP-4 inhibitors.
- MOA-increases the production of insulin and decreases the production of glucagon by the pancreas. It is excreted via the hepato-biliary route, hence dose adjustments are not needed

in chronic renal disease.

Linagliptin is an oral hypoglycemic drug belonging to DPP-4 inhibitors.

MOA-increases the production of insulin and decreases the production of glucagon by the pancreas. It is excreted via the hepato-biliary route, hence dose adjustments are not needed in chronic renal disease.

Incorrect Options:

Option B - Saxagliptin: Belongs to the class of DPP-4 inhibitors. Dose adjustment is needed if eGFR less than 50 mL/min/1.73 m² with the dose of 2.5 mg once daily.

Option C - Repaglinide: Belongs to the group of Megalitinides. It requires dose adjustment if CrCl is below 40ml/min

Option D - Glyburide: Belongs to Sulfonylureas. Act by binding to adenosine triphosphate-sensitive potassium channels in the beta cells of the pancreas leading to their inhibition. It alters the resting membrane potential of the cell, causing an influx of calcium and the stimulation of insulin secretion. Should be used with caution if CrCl is <50ml/min

Solution for Question 41:

Correct Option B - Iodine-131:

- Iodine-131 is used to deliver radiation directly to tissues during radioactive iodine therapy. It is used after the surgery for thyroid cancer to eliminate any remaining thyroid tissue and destroy remaining cancer cells. it can also be used for thyroid cancer that recurs after initial treatment. It is also used for treatment of hyperthyroidism.

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Incorrect Options:

- Option A, C & D: Refer above explanation

Option A, C & D: Refer above explanation

Solution for Question 42:

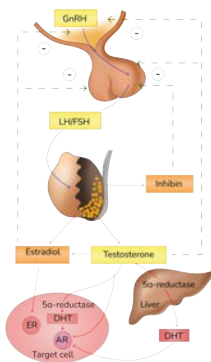
Correct Option A - Pulsatile GnRH:

- Synthetic GnRH injected i.v. (100 µg) induces prompt release of LH and FSH followed by elevation of gonadal steroid levels. It has a short plasma $t_{1/2}$ (4–8 min) due to rapid enzymatic degradation; has been used for testing pituitary gonadal axis in male as well as female hypogonadism. Only pulsatile exposure to GnRH induces FSH/LH secretion.
- This can stimulate the testes to descend.
- Undescended Testicles/Cryptorchidism is a condition wherein the testicles (or “testes”) fails to drop into the normal place in the scrotum during early development

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This can stimulate the testes to descend.

Undescended Testicles/Cryptorchidism is a condition wherein the testicles (or “testes”) fails to drop into the normal place in the scrotum during early development



Incorrect Options:

Option B - Non-pulsatile GnRH: Non-pulsatile GnRH continuous exposure to GnRH could desensitizes pituitary gonadotropes resulting in loss of Gonadotropin release, so it is not useful for treating hypogonadism.

Option C - Testosterone: Given for testicular failure(testis does not produce testosterone), hypopituitarism, primary hypogonadism.

Option D - Estrogen: Is used for treating delayed puberty, dysfunctional uterine bleeding and also as a contraceptive in combination with progesterone

Solution for Question 43:

Correct Option A - Tolvaptan:

- Tolvaptan is a selective vasopressin V2 receptor antagonist that helps to increase free water clearance and reduce the excessive water retention seen in SIADH (syndrome of inappropriate antidiuretic hormone secretion). It promotes water excretion without affecting sodium levels. Tolvaptan has been shown to be effective in treating chronic SIADH.

Tolvaptan is a selective vasopressin V2 receptor antagonist that helps to increase free water clearance and reduce the excessive water retention seen in SIADH (syndrome of inappropriate antidiuretic hormone secretion). It promotes water excretion without affecting sodium levels. Tolvaptan has been shown to be effective in treating chronic SIADH.

Incorrect Options:

Option B - Reducing fluid intake to <500 ml per day: While reducing fluid intake may help manage symptoms of fluid overload in some cases, it is not considered the primary treatment for chronic SIADH. Limiting fluid intake should be done cautiously and under medical supervision to prevent dehydration and electrolyte imbalances.

Option C - Demeclocycline: Demeclocycline is an antibiotic that has been used in the past for SIADH due to its ability to inhibit the action of antidiuretic hormone. However, it is no longer considered a first-line treatment due to concerns about its side effects, including nephrotoxicity and hepatotoxicity. Other treatment options, such as tolvaptan, have shown better efficacy and safety profiles.

Option D - Fludrocortisone: Fludrocortisone is a synthetic mineralocorticoid that acts on the kidneys to increase sodium reabsorption and fluid retention. It is not recommended as a first-line treatment for SIADH because it may worsen hyponatremia (low sodium levels) and fluid overload.

Solution for Question 44:

Correct Option C - 10:

- The replacement dose of hydrocortisone for adrenal insufficiency is typically based on the body surface area (BSA) of the individual.
- The recommended dose is approximately 10-12 mg/m²/day.

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The recommended dose is approximately 10-12 mg/m²/day.

Incorrect Options:

Option A - 20 mg/m²/day: This dosage is higher than the recommended replacement dose for hydrocortisone in adrenal insufficiency.

Option B - 25 mg/m²/day: This dosage is also higher than the recommended replacement dose.

Option D - 5 mg/m²/day: This dosage is lower than the recommended replacement dose and may not be sufficient to adequately manage adrenal insufficiency.

Solution for Question 45:

Correct Option A - Strontium ranelate:

- Strontium ranelate is a medication used for the treatment of osteoporosis.
- It exerts its effects by both decreasing bone resorption (the process of breaking down bone) and increasing bone formation.
- It enhances osteoblast activity, which is responsible for bone formation, while reducing osteoclast activity, which is responsible for bone resorption.
- This dual mechanism helps to improve bone density and strength.

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It exerts its effects by both decreasing bone resorption (the process of breaking down bone) and increasing bone formation.

It enhances osteoblast activity, which is responsible for bone formation, while reducing osteoclast activity, which is responsible for bone resorption.

This dual mechanism helps to improve bone density and strength.

Incorrect Options:

Option B - Ibandronate: Ibandronate is a bisphosphonate medication that primarily acts by inhibiting osteoclast activity and reducing bone resorption. It does not have a direct effect on bone formation.

Option C - Teriparatide: Teriparatide is a recombinant form of parathyroid hormone (PTH) and is used in the treatment of osteoporosis. It primarily stimulates bone formation by activating osteoblasts and increasing bone remodeling. However, it does not have a direct effect on reducing bone resorption.

Option D - Calcitonin: Calcitonin is a hormone that regulates calcium and phosphate metabolism. It can inhibit osteoclast activity and reduce bone resorption. However, it does not have a significant effect on promoting bone formation.

Solution for Question 46:

Correct Option D - Metyrapone:

- Metyrapone is a medication used to inhibit the synthesis of glucocorticoids in the adrenal glands.
- It acts by inhibiting the enzyme 11-beta-hydroxylase, which is involved in the conversion of 11-deoxycortisol to cortisol, a major glucocorticoid hormone.

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It acts by inhibiting the enzyme 11-beta-hydroxylase, which is involved in the conversion of 11-deoxycortisol to cortisol, a major glucocorticoid hormone.

Incorrect Options:

Option A - Mifepristone: Mifepristone, also known as RU-486, is a medication used for medical abortion and in some cases for the treatment of Cushing's syndrome. It acts as a progesterone and glucocorticoid receptor antagonist, but it does not directly inhibit the synthesis of glucocorticoids.

Option B - Flutamide: Flutamide is an antiandrogen medication primarily used in the treatment of prostate cancer. It does not inhibit the synthesis of glucocorticoids.

Option C - Finasteride: Finasteride is a medication primarily used for the treatment of benign prostatic hyperplasia (enlarged prostate) and male pattern hair loss. It inhibits the conversion of testosterone to dihydrotestosterone (DHT), but it does not directly inhibit the synthesis of glucocorticoids.

Solution for Question 47:

Correct Option B - Hydroxychloroquine:

- Hydroxychloroquine is an antimalarial drug commonly used as a Disease-Modifying Antirheumatic Drug (DMARD) in the treatment of rheumatoid arthritis and other autoimmune conditions. It is known to have potential ocular side effects, particularly on the retina. Therefore, regular monitoring of visual acuity and fundus examination is recommended to detect any changes in the eye.
- The reason for monitoring visual acuity and fundus examination in patients using hydroxychloroquine is to screen for retinal toxicity. Prolonged use of hydroxychloroquine can lead to damage to the retina, which can cause visual impairment or loss of vision.

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The reason for monitoring visual acuity and fundus examination in patients using hydroxychloroquine is to screen for retinal toxicity. Prolonged use of hydroxychloroquine can lead to damage to the retina, which can cause visual impairment or loss of vision.

Incorrect Options:

Option A, C & D: Methotrexate, sulfasalazine, & leflunomide:

- Methotrexate, sulfasalazine, & leflunomide are also DMARDs used in the treatment of rheumatoid arthritis and other autoimmune conditions. However, they do not typically require regular monitoring of visual acuity and fundus examination. These drugs have different potential side effects and monitoring requirements, but ocular toxicity is not commonly

associated with them.

- It's important for patients taking hydroxychloroquine to have regular eye examinations as part of their overall management to ensure early detection and intervention if retinal toxicity is detected. This helps in minimizing the risk of permanent vision loss.

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It's important for patients taking hydroxychloroquine to have regular eye examinations as part of their overall management to ensure early detection and intervention if retinal toxicity is detected. This helps in minimizing the risk of permanent vision loss.

Solution for Question 48:

Correct Option B - Nuclear receptor:

- Thyroid receptors are nuclear receptors within the nucleus of cells.

Thyroid receptors are nuclear receptors within the nucleus of cells.

Incorrect Options:

Option A - Transmembrane receptor linked to tyrosine kinase: This type of receptor is known as a receptor tyrosine kinase (RTK). RTKs are transmembrane proteins that span the cell membrane. They have an extracellular domain that binds specific ligands and an intracellular domain that possesses tyrosine kinase activity. Upon ligand binding, RTKs undergo dimerization and activation of their kinase domains, leading to the phosphorylation of tyrosine residues on the receptor itself and downstream signaling molecules. This phosphorylation initiates signaling cascades that regulate various cellular processes.

Option C - Ion channel receptor: Ion channel receptors are transmembrane proteins that form channels allowing the passage of ions across the cell membrane. When a ligand binds to these receptors, the channel opens or closes, regulating the flow of specific ions such as sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), or chloride (Cl^-). The opening or closing of ion channels can influence membrane potential and affect cellular excitability, neurotransmission, muscle contraction, and other physiological processes.

Option D - G protein-coupled receptor (GPCR): GPCRs, also known as seven-transmembrane receptors, are transmembrane proteins that traverse the cell membrane seven times. They are coupled to intracellular G proteins, which are involved in signal transduction pathways. Ligand binding to a GPCR induces conformational changes that activate the associated G protein, leading to the modulation of intracellular second messenger systems and downstream signaling cascades. GPCRs play a crucial role in regulating diverse physiological processes, including neurotransmission, hormone signaling, and sensory perception.

Solution for Question 49:

Correct Option B - Hydrocortisone:

- Hydrocortisone is the shortest acting corticosteroid among the options provided. It has a relatively short duration of action and requires multiple daily doses for optimal effect. With IV injection, peak effect is seen in 2-5 minutes.

Hydrocortisone is the shortest acting corticosteroid among the options provided. It has a relatively short duration of action and requires multiple daily doses for optimal effect. With IV injection, peak effect is seen in 2-5 minutes.

Incorrect Options:

Option A - Dexamethasone: Dexamethasone has a longer duration of action compared to hydrocortisone.

Option C - Triamcinolone: Triamcinolone has a longer duration of action compared to hydrocortisone.

Option D - Deflazacort: Deflazacort has a longer duration of action compared to hydrocortisone.

Cell Wall Synthesis Inhibitors

1. A patient is diagnosed with bilateral pneumonia that is caused by an extended-spectrum beta-lactamase-producing Klebsiella. Which of the following is the drug of choice in this patient?

- A. Amoxicillin-clavulanic acid
 - B. Ofloxacin-tinidazole
 - C. Ceftriaxone-doxycycline
 - D. Benzylpenicillin-sulbactam
-

2. All are true about colistin except?

- A. It is primarily used as a last-resort treatment for multidrug-resistant bacterial infections.
 - B. It is associated with potential nephrotoxic and neurotoxic side effects.
 - C. It is known for its efficacy against gram-negative bacteria.
 - D. It is primarily administered via oral route.
-

3. A 65-year-old male patient is admitted to the ICU for ventilator-associated pneumonia. He has been on mechanical ventilation for the past 48 hours due to respiratory failure secondary to severe sepsis. Cultures from his sputum reveal vancomycin-resistant organisms. What would be the preferred drug for this condition?

- A. Cefepime
- B. Meropenem
- C. Linezolid
- D. Daptomycin

4. A 65-year-old female patient presents to the emergency department with severe abdominal pain, watery diarrhoea, and fever. She has a history of recent antibiotic use for a respiratory infection. Physical examination reveals abdominal tenderness and distension. A colonoscopy showed the presence of pseudomembranes. Based on these findings, what is the best treatment choice for this patient?

- A. Ciprofloxacin
 - B. Metronidazole
 - C. Vancomycin
 - D. Azithromycin
-

5. A 45-year-old male patient with a history of Staphylococcus aureus infection is being treated with antibiotics. During his treatment, he develops a sudden onset of flushing, redness of the upper body, and pruritus (itching). Which drugs are most likely responsible for the patient's condition?



- A. Vancomycin
 - B. Aztreonam
 - C. Ciprofloxacin
 - D. Ceftriaxone
-

6. All are true regarding aztreonam except?

- A. Used primarily to treat Gram-negative bacterial infections.
 - B. It can be administered both orally and intravenously.
 - C. It has a low cross-reactivity with penicillins, making it suitable for patients with penicillin allergies.
 - D. Primarily undergoes renal excretion and requires dose adjustment in patients with renal impairment.
-

7. A 45-year-old male patient residing in Delhi presents to the emergency department with fever, chills, hypotension, and altered mental status. Blood cultures are positive for a gram-negative bacillus. The patient is diagnosed with sepsis, and subsequent laboratory testing reveals that the causative organism produces New Delhi metallo-beta-lactamase (NDM-1). Given this information, which of the following is the best treatment choice?

- A. Meropenem
 - B. Piperacillin-tazobactam
 - C. Ceftazidime-avibactam
 - D. Amoxicillin-clavulanate
-

8. A patient initially admitted to the intensive care unit with respiratory failure was escalated to invasive ventilation. Few days later, he developed a fever along with basal crackles and patchy consolidation can be seen in chest X-ray. Which of the following is the drug of choice for this patient?

- A. Cefoperazone-sulbactam
- B. Imipenem-cilastatin
- C. Piperacillin-tazobactam
- D. Amoxicillin-clavulanate

9. A patient who initially presented with symptoms of staphylococcal scalded skin syndrome is found to be methicillin-resistant. The report also said vancomycin would also be ineffective in this organism. Which of the following cephalosporins can most likely be used effectively in this patient?

- A. Cefetamet pivoxil
 - B. Cefuroxime axetil
 - C. Cefpodoxime proxetil
 - D. Ceftaroline fosamil
-

10. Which of the following is a 4th generation cephalosporin?

- A. Ceftaroline
 - B. Cefnidir
 - C. Cefpirome
 - D. Ceftament
-

11. All of the following drugs correctly match their mechanism of action. 1. Cycloserine a. Inhibits enzyme bactoprenol dephosphorylase 2. Fosfomycin b. Inhibits enzyme transpeptidase 3. Monobactam c. Inhibits enzyme enol pyruvate transferase 4. Bacitracin d. Inhibits enzyme alanine racemase and ligase

1. Cycloserine	a. Inhibits enzyme bactoprenol dephosphorylase
2. Fosfomycin	b. Inhibits enzyme transpeptidase
3. Monobactam	c. Inhibits enzyme enol pyruvate transferase
4. Bacitracin	d. Inhibits enzyme alanine racemase and ligase

- A. 1:d , 2:c , 3:b , 4:a
 - B. 1:b , 2:c , 3:d , 4:a
 - C. 1:d , 2:a , 3:b , 4:c
 - D. 1:d , 2:b , 3:a , 4:c
-

12. A patient who was admitted with suspected osteomyelitis is found to be caused by *Staphylococcus aureus* that shows resistance to methicillin. Which of the following is least likely to be the drug of choice for this patient?

- A. Teicoplanin
 - B. Chloramphenicol
 - C. Linezolid
 - D. Vancomycin
-

13. All of the following are true regarding penicillinase-resistant penicillins except?

- A. Dicloxacillin falls in this group of drugs.
 - B. Methicillin is preferred in patients with renal disease.
 - C. Not resistant to penicillinase of gram-negative bacteria.
 - D. Concomitant use of contraceptive pills is contraindicated with nafcillin.
-

14. All of the following statements are incorrect about extended spectrum penicillins except?

- A. Ampicillin is one of the preferred drugs of choice in typhoid fever.
- B. The biliary concentration of ampicillin is low and hence not used in cholecystitis.
- C. Piperacillin is more potent than carbenicillin in pseudomonas infections.

D. Carbenicillin is the preferred drug in this group in heart failure patients.

15. A 30-year-old male recently began treatment for painless lesions in his palms and soles and presented to the emergency department with fever, chills, and myalgias. These symptoms developed a few hours after receiving his first dose of benzathine penicillin G. On examination, he appears flushed (image) with tachycardia and hypotension. Based on the findings, match the most likely diagnosis and management of this condition. 1. Anaphylactic reaction A. Continue medication and start antipyretics 2. Serum Sickness B. Administer epinephrine 3. Jarisch-Herxheimer reaction C. Discontinue and switch to alternate medication 4. Steven-Johnson syndrome D. Discontinue medication and admit to ICU

1. Anaphylactic reaction	A. Continue medication and start antipyretics
2. Serum Sickness	B. Administer epinephrine
3. Jarisch-Herxheimer reaction	C. Discontinue and switch to alternate medication
4. Steven-Johnson syndrome	D. Discontinue medication and admit to ICU

- A. 3 and B
 - B. 2 and C
 - C. 3 and A
 - D. 4 and D
-

16. A 50-year-old male presents to the hospital with headaches, difficulty coordinating his movements, and memory loss over the past six months. On physical examination, his pupils are irregular and do not constrict to light but constrict during accommodation. He had a history of painless genital ulcer, which was resolved on its own a few years ago. What is the drug of choice used for this condition in this patient?

- A. A single dose of 2.4 million IU penicillin G

- B. Once weekly dose of 2.4 million IU penicillin G
 - C. Once weekly dose of 24 million IU penicillin G
 - D. Continues dose of 24 million IU penicillin G
-

17. Which of the following statements regarding penicillins is true? Acid-stable penicillins are destroyed by gastric acid and must be administered parenterally. Penicillin G is an acid-stable penicillin that should be given in high doses if administered orally. Acid-stable penicillins are avoided with food and antacids. Acid labile penicillins include amoxicillin, ampicillin and oxacillin.

- A. 1, 3 and 4
 - B. 2 and 3
 - C. 3 and 4
 - D. 3 only
-

18. A 30-year-old female presents to the clinic with a sore throat and fever for three days. Previous medical history includes breathing issues and rashes caused by an unknown medicine used to treat syphilis one year ago. A throat swab confirms the streptococcal infection. What is the best treatment option for this patient?

- A. Amoxicillin
 - B. Ceftriaxone
 - C. Erythromycin
 - D. Penicillin V
-

19. A 19-year-old college student arrives at the emergency room complaining of photophobia, high fever and severe headache. A physical examination reveals pain

when extending the legs and stiffness on neck flexion. Following a lumbar puncture, the cerebrospinal fluid was cloudy, with high protein, low glucose and raised white blood cell count. What is the most appropriate treatment of choice for this patient?

- A. Intravenous Acyclovir
 - B. Intravenous Vancomycin
 - C. Intravenous Ceftriaxone
 - D. Intravenous Amphotericin
-

20. Penicillin is helpful for the treatment of all of the following conditions except?

- A. Streptococcal pharyngitis
 - B. Neurosyphilis
 - C. MRSA pneumonia
 - D. Leptospirosis
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	3
Question 4	3
Question 5	1
Question 6	2
Question 7	3
Question 8	2
Question 9	4
Question 10	3
Question 11	1
Question 12	2
Question 13	2
Question 14	3
Question 15	3
Question 16	4
Question 17	4
Question 18	3
Question 19	3
Question 20	3

Solution for Question 1:

Correct answer: A) Amoxicillin-clavulanic acid

Explanation:

Amoxicillin-clavulanic acid would be the ideal option for this patient.

Extended-spectrum beta-lactamase is an enzyme that provides the bacteria with resistance to most penicillins, cephalosporins and even the monobactam aztreonam except carbapenems which are effective against ESBL-producing bacteria.

Beta-lactamase inhibitors have shown effectiveness against such enzyme-producing bacteria. These drugs should be used appropriately with other penicillins or cephalosporins to act against such beta-lactamase-producing organisms like *Klebsiella* or *Enterococcus*.

The table below shows the beta-lactamase inhibitors along with their combinations with penicillins/beta-lactams.

Amoxicillin, ticarcillin	Clavulanic acid
Ampicillin, cefoperazone	Sulbactam
Piperacillin, ceftolozane	Tazobactam
Meropenem	Vaborbactam
Imipenem	Relebactam
Ceftazidime	Avibactam

Solution for Question 2:

Correct Answer: D) It is primarily administered via oral route.

Explanation:

Colistin	
Type	Polypeptide antibiotic

Pharmacokinetics	Administration: IV or via inhalation Distribution: Poor penetration into tissues, primarily stays in the blood Metabolism: Minimal metabolism Excretion: Primarily excreted unchanged in urine
Mechanism of Action	Colistin ■ Binds to Lipopolysaccharides (LPS) in Gram-negative Bacterial Cell Walls ■ Disrupts Membrane Integrity ■ Increases Permeability of Cell Membrane ■ Leakage of Intracellular Contents ■ Bacterial Cell Death
Uses	Multidrug-resistant Gram-negative bacterial infections (Option C) Pseudomonas Acinetobacter Klebsiella pneumoniae Escherichia coli Ventilator-associated pneumonia (VAP) Catheter-associated UTI
Side Effects	Nephrotoxicity (Option B) Neurotoxicity (paresthesias, dizziness, confusion) (Option B) Respiratory depression (with systemic use) Hypersensitivity reactions (rare) Neuromuscular blockade (rare) Superinfection (especially with prolonged use) (rare)
Special Points	Due to its toxicity profile and risk of resistance development, it is considered a last-line option for treating multidrug-resistant Gram-negative infections (Option A). Inhalation treats respiratory infections, particularly in cystic fibrosis patients with Pseudomonas aeruginosa colonisation.

Solution for Question 3:

Correct Answer: C) Linezolid

Explanation:

Linezolid is an oxazolidinone antibiotic that has activity against vancomycin-resistant organisms, including Vancomycin-resistant Staphylococcus aureus (VRSA) and vancomycin-resistant Enterococcus (VRE).

It is often considered the preferred choice for infections caused by vancomycin-resistant organisms causing lung infections/ pneumonia.

Linezolid	
Type/Class	Oxazolidinone antibiotic
Pharmacokinetics	Administration: Oral or IV Absorption: Well-absorbed orally, with bioavailability of approximately 100% Distribution: Widely distributed in body tissues, including the lungs, skin, and bone Metabolism: Minimal metabolism Excretion: Excreted unchanged in urine
Mechanism of Action	Linezolid ■ Binds to the 50S Ribosomal Subunit ■ Inhibits Initiation of Protein Synthesis ■ Blocks Elongation of Peptide Chains ■ Inhibits Bacterial Protein Synthesis ■ Bacteriostatic or Bactericidal Activity.
Uses	Methicillin-resistant Staphylococcus aureus (MRSA) Vancomycin-resistant Staphylococcus aureus (VRSA) Vancomycin-resistant Enterococcus (VRE) Streptococcus pneumoniae Complicated skin and skin structure infections Nosocomial pneumonia (including those caused by MRSA)
Side Effects	Nausea and vomiting Diarrhea Rash Thrombocytopenia Peripheral neuropathy (with prolonged use) Serotonin syndrome Lactic acidosis (rare)

- Daptomycin is DOC for VRSA in skin infections (Option D)
- Linezolid is DOC for VRSA in lung infections.

Solution for Question 4:

Correct Answer: C) Vancomycin

Explanation:

The clinical scenario of a patient with severe abdominal pain, watery diarrhoea, fever with a history of recent antibiotic use, and colonoscopy showing pseudomembranes is suggestive of pseudomembranous colitis. Among the given options, vancomycin is the best drug to manage this condition.

Pseudomembranous enterocolitis:

- Cause: Clostridium difficile & antibiotics. Clindamycin. 3rd generation cephalosporins. Ampicillin. Fluoroquinolones.
- Clindamycin.
- 3rd generation cephalosporins.
- Ampicillin.
- Fluoroquinolones.
- The antibiotics kill the gut flora, leading to Clostridium difficile superinfection.
- Management: Fidaxomicin. Vancomycin. Metronidazole.
- Fidaxomicin.
- Vancomycin.
- Metronidazole.
- Clindamycin.
- 3rd generation cephalosporins.
- Ampicillin.
- Fluoroquinolones.
- Fidaxomicin.
- Vancomycin.
- Metronidazole.

(Option A and D ruled out)

- For an initial episode of pseudomembranous colitis, fidaxomicin is preferred due to its safety profile and higher beneficial effects. In resource-limited settings where fidaxomicin is unavailable, vancomycin can be used. (Option C)
- In resource-limited settings where fidaxomicin is unavailable, vancomycin can be used. (Option C)
- Metronidazole was previously used as a first-line treatment for pseudomembranous colitis; however, it is no longer the preferred treatment due to increasing resistance and higher recurrence rates than vancomycin. (Option B)
- In resource-limited settings where fidaxomicin is unavailable, vancomycin can be used. (Option C)

Other uses of vancomycin:

- Methicillin-resistant *Staphylococcus aureus*.
- Vancomycin is not effective against *PSEUDOMONAS*

Solution for Question 5:

Correct answer: A) Vancomycin

Explanation:

- Red Man Syndrome is a well-known adverse reaction to vancomycin.
- Vancomycin is a glycopeptide antibiotic with the potential to release histamine by direct action on mast cells through rapid intravenous injection, causing chills, fever, urticaria, and intense flushing, known as Red man syndrome.
- It can be ameliorated by administering vancomycin more slowly, sometimes with premedication with histamine blockers. Red man syndrome is not observed with teicoplanin.
- Other side effects of vancomycin:

Common	Rare
Nephrotoxicity (especially with prolonged use or high doses). Ototoxicity (especially with rapid infusion or high doses). Thrombophlebitis at IV site.	Neutropenia. Anaphylaxis.

Solution for Question 6:

Correct answer: B) It can be administered both orally and intravenously.

Explanation:

Aztreonam is administered intravenously (IV) or intramuscularly (IM), not orally.

Aztreonam	
Type	Monobactam Antibiotic

Pharmacokinetics	Absorption: Administered IV or IM Distribution: Widely distributed in the body, including CSF Metabolism: Minimal metabolism Excretion: Primarily excreted unchanged in urine, requires renal dose adjustment in patients with renal impairment. (Option D)
Mechanism of Action	Aztreonam ■ Binds to Penicillin-Binding Proteins (PBPs) ■ Inhibits Peptidoglycan Cross-linking ■ Disruption of Bacterial Cell Wall Synthesis ■ Cell Lysis ■ Bacterial Death Resistant to beta-lactamases, but susceptible to metallo-beta-lactamases and extended-spectrum beta-lactamases (ESBLs).
Uses	Treatment of Gram-negative bacterial infections (e.g. <i>Salmonella typhi</i>, <i>Pseudomonas</i>), not effective against Gram-positive bacteria or Anaerobes. (Option A) Urinary tract infections (UTIs) Lower respiratory tract infections Septicemia Skin and skin structure infections Intra-abdominal infections Gynecological infections It is particularly useful for patients allergic to penicillins or cephalosporins.
Side Effects	Common: - Rash - Nausea and Vomiting - Diarrhea Serious: - Hypersensitivity reactions (anaphylaxis) <i>Clostridium difficile</i>-associated diarrhoea - Hepatotoxicity - Neutropenia and thrombocytopenia
Special Points	It can be used in patients with penicillin allergy due to low cross-reactivity, except for Ceftazidime. (Option C) Often combined with beta-lactamase inhibitors (Avibactam) to expand the spectrum.

Solution for Question 7:

Correct answer: C) Ceftazidime-avibactam

Explanation:

Metallobactamase-producing bacteria produce an enzyme called metalloβ-lactamase, which can break down and make certain antibiotics ineffective.

These bacteria can cause severe infections and are often resistant to multiple antibiotics, making them challenging to treat.

Ceftazidime-avibactam is a combination of a third-generation cephalosporin and a non-β-lactam β-lactamase inhibitor. Avibactam can inhibit NDM-1, making this combination effective against NDM-1-producing organisms.

Meropenem (Option A) is a broad-spectrum carbapenem antibiotic. However, it is generally ineffective against organisms producing NDM-1, as these bacteria can hydrolyse carbapenems and render them useless.

Piperacillin-tazobactam (Option B) is a combination of a broad-spectrum penicillin and a β-lactamase inhibitor. It is ineffective against organisms producing NDM-1, as NDM-1 can hydrolyse a wide range of β-lactams, including those protected by β-lactamase inhibitors.

Amoxicillin-clavulanate (Option D) combines a penicillin antibiotic and a β-lactamase inhibitor. NDM-1 can also hydrolyse this combination.

Solution for Question 8:

Correct answer: B) Imipenem-cilastatin

Explanation:

The above clinical vignette suggests ventilator-associated pneumonia (VAP), and carbapenems like imipenem combined with cilastatin are the most likely drug of choice.

Carbapenems:

- Imipenem (with cilastatin)
- Meropenem
- Faropenem
- Doripenem
- Ertapenem

- Uses of carbapenems: Used in nosocomial infections. DOC in *Acinetobacter baumannii* (Ventilator-associated pneumonia)(most common cause of VAP nowadays) *Klebsiella* *Burkholderia* *Serratia* *Proteus* *Pseudomonas*
 - Used in nosocomial infections.
 - DOC in *Acinetobacter baumannii* (Ventilator-associated pneumonia)(most common cause of VAP nowadays) *Klebsiella* *Burkholderia* *Serratia* *Proteus* *Pseudomonas*
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 - *Burkholderia*
 - *Serratia*
 - *Proteus*
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 - *Burkholderia*
 - *Serratia*
 - *Proteus*
 - *Pseudomonas*
 - *Acinetobacter baumannii* (Ventilator-associated pneumonia)(most common cause of VAP nowadays)
 - *Klebsiella*
 - *Burkholderia*
 - *Serratia*
 - *Proteus*
 - *Pseudomonas*
-

Solution for Question 9:

Correct answer: D) Ceftaroline fosamil

Explanation:

Fifth-generation cephalosporins like ceftaroline fosamil and ceftobiprole medocaril are found to be effective against MRSA.

Uses of cephalosporins:

First generation	ENT, upper respiratory & cutaneous infections Cephazolin is the DOC for surgical site infection prophylaxis
Second generation	Mixed aerobic-anaerobic infections (Only cephalosporin class effective against anaerobes)
Third generation	Ceftriaxone/Cefotaxime: empirical therapy for meningitis Ceftazidime: Pseudomonas meningitis STI : Gonorrhea, chancroid, syphilis Enteric fever Patients with neutropenia
Fourth generation	Nosocomial infections (intensive care borne)
Fifth generation	Penicillin resistant community acquired and hospital acquired pneumonia by S.pneumoniae Penicillinase producing S.aureus infections MRSA

Cephalosporin effective against pseudomonas: All P containing cephalosporins: Cefpirome
Cefepime, Cefoperazone, and Ceftazidime (most effective)

Solution for Question 10:

Correct answer: C) Cefpirome

Explanation:

Cefpirome is a 4th generation cephalosporin.

Cephalosporins are a group of antibiotics derived from cephalosporin-C, which is obtained from the fungus Cephalosporium. They are chemically related to penicillins.

Classification of cephalosporins:

They are classified into 5 generations.

1st gen	2nd gen	3rd gen	4th gen	5th gen
Cephazolin (Option B ruled out) Cephalexin Cefadroxil	Cefuroxime Cefoxitin Cefaclor Cefprozil	Cefixime (Option A ruled out) Cefotaxime Ceftizoxime Ceftriaxone Ceftazidime Cefpodoxime proxetil Cefdinir (Option D ruled out) Ceftibuten Ceftamet pivoxil	Cefepime Cefpirome (Option C)	Ceftaroline Ceftobiprole

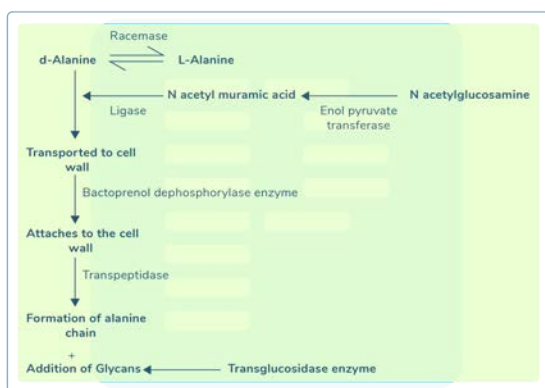
Solution for Question 11:

Correct answer: A) 1:d , 2:c , 3:b , 4:a

Explanation:

Cell wall synthesis inhibitors:

- The bacterial outer cell wall comprises peptidoglycan chains, chains of amino acid (alanine) connected by sugars or glycans.
- Cell wall synthesis:



Cell wall synthesis inhibitors			
MOA	Drugs	Uses	Side effects

Inhibits transpeptidase (penicillin-binding protein) and prevents cross-linking	Beta-lactam drugs 1. Penicillin 2. Cephalosporin 3. Carbapenems 4. Monobactam (Aztrenam)	MSSA infections Perioperative prophylaxis Syphilis Meningitis Community-acquired pneumonia GI and GU infection Aztrenam in penicillin allergy. Ertapenam in ESBL, P. aeruginosa, and Acinetobacter	1. Hypersensitivity 2. Cross hypersensitivity (except monobactams)
Inhibits enol pyruvate transferase and block GAG synthesis	Fosfomycin	Enterococcal UTI (Foley's catheter) Transurethral procedures Safe in pregnancy	1. Hematuria 2. Urinary discomfort 3. Neuropsychiatric: Psychosis, seizures
D-alanine analogue inhibits alanine racemase and ligase (Bacteriostatic)	Cycloserine	Multidrug resistant TB	1. Neurotoxicity 2. Psychosis 3. Depression (Suicide risk)
Inhibits bactoprenol dephosphorylase enzyme	Bacitracin	Topical cream in gram-positive cocci (Staphylococcus and Streptococcus) skin infection Oral: Enterococcal infection	1. Nephrotoxic (IV) 2. Bone marrow suppression
Inhibition of transglucosidase enzyme	Vancomycin	Pseudomembranous colitis(oral) MRSA E. faecalis	1. Nephrotoxicity 2. Ototoxicity 3. Red man syndrome

• Mnemonic: BBC Getting Famous

Solution for Question 12:

Correct answer: B) Chloramphenicol

Explanation:

The scenario shows osteomyelitis caused by methicillin-resistant staphylococcus aureus (MRSA) and chloramphenicol is not used in infections caused by them.

Drugs used against MRSA:

- Vancomycin (Option D)
- Linezolid (Option C)
- Teicoplanin (Option A)
- Tedizolid
- Spectinomycin
- Daptomycin
- Mupirocin

Solution for Question 13:

Correct answer: B) Methicillin is preferred in patients with renal disease.

Explanation:

The increasing incidence of hematuria, albuminuria and interstitial nephritis as adverse effects of being on methicillin is one of the major reasons why the drug is no longer used widely.

This group of drugs protects the beta-lactam ring of the drug from staphylococcal penicillinase. Their only indication is to use them in penicillinase-producing *Staphylococcus*, which is not methicillin-resistant.

Penicillinase-resistant penicillins	Side effects
Cloxacillin (oral)	Coagulation defect/bleeding
Oxacillin (oral)	Hepatotoxicity
Nafcillin (IV)	Neutropenia
Dicloxacillin (oral) (Option A)	Hepatotoxicity

This group of drugs is not resistant to penicillinase produced by Gram-negative bacteria and hence cannot be used in such cases. (Option C)

Nafcillin is a known inducer of cytochrome P450 enzymes which render drugs like oral contraceptive pills ineffective and hence concomitant use of these drugs is contraindicated. (Option D)

Solution for Question 14:

Correct answer: C) Piperacillin is more potent than carbenicillin in pseudomonas infections.

Explanation:

Piperacillin is almost 8 times more potent than carbenicillin in treating Pseudomonas infections

Extended-spectrum penicillins act against a variety of Gram-negative bacteria in addition to its basic nature of acting against Gram-positive bacteria. They include:

- Aminopenicillins: Ampicillin. Bacampicillin. Amoxicillin.
- Ampicillin.
- Bacampicillin.
- Amoxicillin.
- Carboxypenicillins: Carbenicillin. Ticarcillin.
- Carbenicillin.
- Ticarcillin.
- Ureidopenicillins: Piperacillin. Mezlocillin.
- Piperacillin.
- Mezlocillin.
- Ampicillin.
- Bacampicillin.
- Amoxicillin.
- Carbenicillin.
- Ticarcillin.
- Piperacillin.
- Mezlocillin.

Ampicillin is now not used in typhoid fever because of the emergence of resistance. (Option A)

Ampicillin is partly excreted in bile and also reabsorbed to enter the entero-hepatic circulation.

Hence it can be used in cholecystitis because it presents in high concentrations in bile. (Option B)

Carbenicillin when used in its salt format will retain water and hence subjectively not preferred in patients with heart or renal failure. (Option D)

Solution for Question 15:

Correct answer: C) 3 and A

Explanation:

This patient is experiencing a Jarisch-Herxheimer reaction, a typical response to antibiotic treatment for syphilis, characterised by fever, chills, myalgias and worsening of syphilitic rash, which usually resolves within 24 hours without discontinuing penicillin therapy. Supportive treatment includes antipyretics and corticosteroids.

Jarisch-Herxheimer reaction:

- Host reaction is due to the release of spirochetal proteins (hypersensitivity).
- Common in patients treated for secondary syphilis, leptospirosis, and Lyme disease.
- Symptoms: Chills, fever, headache, and myalgias occur several hours after the first dose. It is worsening of syphilitic rash (image) and lymphadenopathy.
- Chills, fever, headache, and myalgias occur several hours after the first dose.
- It is worsening of syphilitic rash (image) and lymphadenopathy.
- Management: Self-limited: resolves within 24 hours. Supportive care: Antipyretics, hydration and corticosteroids. Continue penicillin therapy.
- Self-limited: resolves within 24 hours.
- Supportive care: Antipyretics, hydration and corticosteroids.
- Continue penicillin therapy.
- Chills, fever, headache, and myalgias occur several hours after the first dose.
- It is worsening of syphilitic rash (image) and lymphadenopathy.
- Self-limited: resolves within 24 hours.
- Supportive care: Antipyretics, hydration and corticosteroids.
- Continue penicillin therapy.

Solution for Question 16:

Correct answer: D) Continues dose of 24 million IU penicillin G

Explanation:

This patient's history of untreated painless genital ulcer (chancre) presenting with progressive neurological symptoms along with examination revealing Argyll Robertson pupils suggest neurosyphilis as the most likely diagnosis, which should be treated with a continuous intravenous infusion of 18–24 IU of intravenous benzathine penicillin G.

Syphilis:

- Sexually transmitted disease caused by *Treponema pallidum*.
- Classification:

Stages of Syphilis	Treatment	
Primary syphilis	Chancre (painless genital ulcer)	Single IM Benzathine penicillin 2.4 million IU (Option A)
Secondary syphilis	Spread in blood: Rash involving palms and soles, uveitis	Single IM Benzathine penicillin 2.4 million IU
Early latent syphilis	Dormant for < 1 year	Single IM Benzathine penicillin 2.4 million IU
Late latent syphilis	Dormant for > 1 year	Weekly IM Benzathine penicillin 2.4 million IU for three weeks (Option B)
Tertiary syphilis	Activation after one year and damages the organs (gummas)	Weekly IM Benzathine penicillin 2.4 million IU for three weeks (Option B)
Damage to heart: Cardiovascular syphilis (aortic aneurysm, arteritis) Damage to the brain: Neurosyphilis (tabes dorsalis, ataxia, dementia, Argyll Robertson pupils)	Continues IV Aqueous Penicillin G 18-24 IU for 10-14 days. (Option D)	

Solution for Question 17:

Correct answer: D) 3 only

Explanation:

Acid Labile Penicillin	Acid Stable Penicillin
Penicillin G is degraded in the stomach by HCl Route: Parenteral If oral necessary: given in high doses	Resistant to gastric HCl Route: Oral Drugs: Mnemonic: VODCAA Penicillin V (No effect on meningococcus) Oxacillin Dicloxacillin, Flucloxacillin Cloxacillin Ampicillin Amoxicillin Avoid food and antacids except for amoxicillin Given 30 mins before food or >2 hrs after food

Solution for Question 18:

Correct answer: C) Erythromycin

Explanation:

This patient with a history of anaphylaxis (trouble breathing and rash) after starting medication to treat sore throat suggests penicillin hypersensitivity reaction and should be prescribed erythromycin due to the risk of cross-reactivity with similar drugs like amoxicillin, ceftriaxone, and penicillin V.

Penicillin Hypersensitivity:

- Incidence: 4-5%
- This is due to the product breakdown called "major determinant" or the penicilloyl moiety.
- It may occur at any dose and in the absence of a previous exposure.
- Allergic testing: Intradermal 6-aminopenicillins acid (benzylpenicilloyl polylysine).
- Types: Type 1 (immediate): Sudden death due to anaphylaxis (Purpuric eruption, hypotension, bronchoconstriction and GI symptoms). Type 2: Hemolytic anaemia (Detected through Coombs test), neutropenia. Type 3: Serum sickness (presents after one week and persists after discontinuation). Type 4 (Delayed type): Steven-Johnson syndrome, toxic epidermal necrolysis.
- Type 1 (immediate): Sudden death due to anaphylaxis (Purpuric eruption, hypotension, bronchoconstriction and GI symptoms).
- Type 2: Hemolytic anaemia (Detected through Coombs test), neutropenia.

- Type 3: Serum sickness (presents after one week and persists after discontinuation).
- Type 4 (Delayed type): Steven-Johnson syndrome, toxic epidermal necrolysis.
- Treatment: Discontinue the medication and avoid similar classes of antibiotics (avoid all beta-lactams but monobactam is safe as it has a different structure that lacks the characteristic beta-lactam ring configuration. This structural difference significantly reduces the risk of cross-reactivity with other beta-lactam antibiotics. (Options A, B and D ruled out).
DOC: Macrolides (Erythromycin, Azithromycin). Alternate: Clindamycin
Desensitisation in ICU setting: In patients who must receive the drug
- Discontinue the medication and avoid similar classes of antibiotics (avoid all beta-lactams but monobactam is safe as it has a different structure that lacks the characteristic beta-lactam ring configuration. This structural difference significantly reduces the risk of cross-reactivity with other beta-lactam antibiotics. (Options A, B and D ruled out).
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- DOC: Macrolides (Erythromycin, Azithromycin).
- Alternate: Clindamycin
- Desensitisation in ICU setting: In patients who must receive the drug

Solution for Question 19:

Correct answer: C) Intravenous Ceftriaxone

Explanation:

The young student has classical symptoms of bacterial meningitis (severe headache, high fever, photophobia, and positive Kernig's sign), and CSF findings of bacterial meningitis suggest meningococcal meningitis as the most likely diagnosis. This should be treated with ceftriaxone, a broad-spectrum third-generation cephalosporin with good CSF penetration.

Meningococcal Meningitis	
Causative organism	Neisseria meningitidis
Treatment	DOC: IV Ceftriaxone (Third generation cephalosporin)
Prophylaxis	Adult: Ciprofloxacin (earlier Rifampicin) Prevents nasal colonisation Children and Pregnant women: Ceftriaxone injection (as Ciprofloxacin (FQ) causes cartilage damage in children and fetus).

Intravenous acyclovir (Option A) is recommended for VZV-associated and HSV meningitis, showing normal glucose, increased protein, and lymphocytic pleocytosis on CSF examination.

Intravenous vancomycin (Option B) is advised for resistant cases of healthcare-associated meningitis and ventriculitis but not as a first-line treatment for bacterial meningitis.

Intravenous amphotericin B (Option D) plus flucytosine is used for cryptococcal meningitis in HIV patients, showing elevated opening pressure, increased protein, low glucose, and slight lymphocytic predominance on CSF examination.

Solution for Question 20:

Correct answer: C) MRSA pneumonia

Explanation:

Vancomycin, a glycopeptide antibiotic, is the drug of choice for Methicillin-resistant *Staphylococcus aureus* (MRSA) infection.

Penicillins	
-------------	--

Classification	Natural Penicillins Penicillin G/Benzylpenicillin/Crystalline penicillin First penicillin Parenteral route only Allergic potential Narrow spectrum Resistance exhibited by beta-lactamase enzyme-producing organisms Penicillin V: Enteral route Penicillinase resistant penicillin Methicillin, cloxacillin, nafcillin, oxacillin, and dicloxacillin Aminopenicillins: Ampicillin, amoxicillin. Anti-pseudomonal penicillins: Piperacillin, ticarcillin
Pharmacokinetics	Poor CNS penetration (except Penicillin G and nafcillin) Poor eye penetration Poor prostate penetration
Uses (SMART Player GAYLE)	S: Syphilis: DOC for neurosyphilis (Option B ruled out) M: Meningococcal meningitis A: Actinomycosis R: Rat bite fever: Streptobacillus, Spirillosis T: Tetanus Player: Pasteurella multocida G: Gas gangrene (Myonecrosis): Clostridium perfringens A: Anthrax Y: Yaws L: Leptospirosis (Option D ruled out) E: Every species: Streptococcus, Staphylococcus if sensitive (Option A ruled out)

Treatment of syphilis:

1° syphilis/ 2° syphilis/ Early latent : Benzathine penicillin single depot injection i/m 2.4 million IU

LATE latent: Benzathine penicillin G- three doses of 2.4 million units IM at one-week intervals.

Neuro syphilis: Aqueous Penicillin G continuous infusion i/v

Protein Synthesis Inhibitors

1. Match the following antibiotics with their appropriate mechanism of action.

Drugs **Mechanism of action**
1 Linezolid A. Blocks 50s ribosomal subunit; interferes with translocation
2 Tetracycline B. Block 23s subunit on 50s ribosome, interferes with formylmethionine-tRNA 70S complex formation
3 Erythromycin C. Blocks 30s ribosomal subunit; causes misreading of mRNA
4 Streptomycin D. Blocks 30S mRNA-ribosome; inhibits attachment of aminoacyl-t-RNA to the acceptor (A) site.

Drugs	Mechanism of action
1 Linezolid	A. Blocks 50s ribosomal subunit; interferes with translocation
2 Tetracycline	B. Block 23s subunit on 50s ribosome, interferes with formylmethionine-tRNA 70S complex formation
3 Erythromycin	C. Blocks 30s ribosomal subunit; causes misreading of mRNA
4 Streptomycin	D. Blocks 30S mRNA-ribosome; inhibits attachment of aminoacyl-t-RNA to the acceptor (A) site.

- A. 1-B, 2-D, 3-C, 4-A
- B. 1-B, 2-D, 3-A, 4-C
- C. 1-D, 2-A, 3-B, 4-C
- D. 1-C, 2-B, 3-A, 4-D

2. A drug sensitivity test of a patient's sputum sample shows the presence of vancomycin-resistant MRSA. A narrow-spectrum protein synthesis inhibitor antibiotic that inhibits the formation of formylmethionine-tRNA 70S initiation complex was started for the patient. Which drug was most likely used?

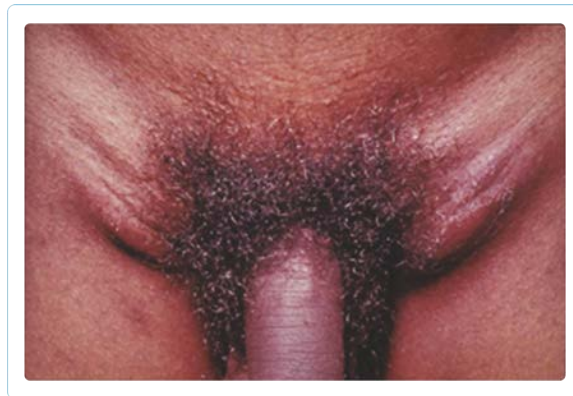
- A. Azithromycin
- B. Linezolid

- C. Chloramphenicol
 - D. Clindamycin
-

3. Which of the following statements is NOT true regarding Chloramphenicol?

- A. It is metabolised via hepatic conjugation
 - B. It can cause dose-dependent marrow suppression
 - C. It can be safely used in neonates
 - D. Resistance occurs due to plasmid-mediated inactivation by bacteria.
-

4. A 22-year-old man with a history of multiple sexual partners comes to the clinic with an inguinal swelling. Physical examination findings are given below. What would be the drug of choice for this patient?



- A. Penicillin G
 - B. Doxycycline
 - C. Cefixime
 - D. Acyclovir
-

5. A 26-year-old pregnant woman was treated with an antibiotic for chlamydia during her 6th month of pregnancy. Nine months later, she delivered a healthy baby boy. As the child grew, he developed teeth resembling those in the provided image and had a short stature. Which antibiotic was most likely used?



- A. Tetracyclines
- B. Amikacin
- C. Chloramphenicol
- D. Sulphonamide

6. Which of the following antimicrobials is safe for use in neonates?

- A. Erythromycin
- B. Chloramphenicol
- C. Azithromycin
- D. Sulfamethoxazole

7. Which of the following antibiotics is commonly used in triple therapy to treat H. pylori infection?

- A. Erythromycin
 - B. Doxycycline
 - C. Azithromycin
 - D. Clarithromycin
-

8. Which of the following is true regarding Erythromycin?

- A. Primarily eliminated through renal excretion.
 - B. Erythromycin is stable in an acidic environment.
 - C. Have a large volume of distribution, allowing for extensive tissue penetration.
 - D. Highly lipid-soluble, allowing for excellent penetration into the central nervous system
-

9. A 45-year-old woman presents with severe diarrhoea and abdominal pain after recently being treated with an antibiotic for a skin infection. Stool tests confirm the presence of *Clostridium difficile*. Which of the following antibiotics is associated with the highest risk of causing pseudomembranous colitis in this clinical scenario?

- A. Erythromycin
 - B. Tetracycline
 - C. Clindamycin
 - D. Ciprofloxacin
-

10. A 65-year-old patient with renal failure presents with a suspected Gram-positive bacterial infection. The patient's creatinine clearance is significantly reduced. Which antibiotic would be the most appropriate choice for this patient?

- A. Vancomycin

- B. Ciprofloxacin
 - C. Gentamicin
 - D. Linezolid
-

11. A 68-year-old woman with diabetes and chronic kidney disease is hospitalised for a severe diabetic foot infection, likely due to MRSA. The doctor is considering a Streptogramin antibiotic. Which of the following statements about Streptogramins is correct?

- A. Inhibit bacterial protein synthesis by binding to the 50S ribosomal subunit.
 - B. Cover both Gram-positive and Gram-negative bacteria.
 - C. First-line therapy for community-acquired pneumonia.
 - D. The most common adverse effect is nephrotoxicity.
-

12. A 55-year-old diabetic man presents with a severe UTI, showing high fever, chills, and dysuria. Urine culture reveals E. coli resistant to aminoglycosides and fluoroquinolones. What mechanism likely contributes to E. coli's resistance to aminoglycosides?

- A. Inactivation of the drug by bacterial enzymes
 - B. Alteration of the bacterial ribosomal binding site
 - C. Decreased uptake of the drug into bacterial cells
 - D. Formation of biofilms on mucosal surfaces
-

13. Which adverse effects are most commonly associated with aminoglycosides?

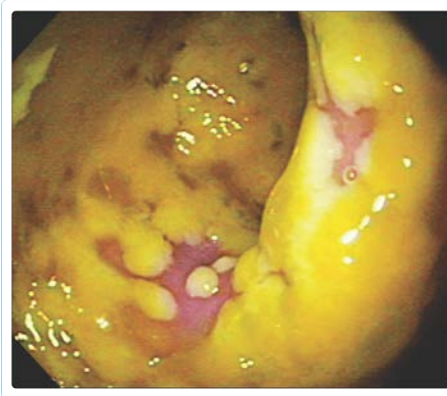
- A. Hepatotoxicity and hypoglycemia
- B. Nephrotoxicity and ototoxicity

- C. Cardiotoxicity and neutropenia
 - D. Pulmonary toxicity and hyperkalemia
-

14. Which antibiotics will be least helpful in treating *Enterococcus faecium* infection?

- A. Ampicillin
 - B. Linezolid
 - C. Daptomycin
 - D. Tigecycline
-

15. A 72-year-old woman with COPD develops severe diarrhoea, abdominal cramps, and fever two weeks after being treated with ceftriaxone and clindamycin. Stool testing confirms *Clostridioides difficile* infection. An endoscopic image shows pseudomembranous colitis. Which antibiotic is the most appropriate for treating her condition?



- A. Metronidazole
- B. Rifaximin
- C. Fidaxomicin

D. Clindamycin

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	2
Question 5	1
Question 6	3
Question 7	4
Question 8	3
Question 9	3
Question 10	4
Question 11	1
Question 12	1
Question 13	2
Question 14	1
Question 15	3

Solution for Question 1:

Correct Answer: B) 1- B, 2-D, 3-A, 4-C

Explanation:

Drugs	Ribosomal subunit blocked	Effect
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Linezolid	Block 23s subunit on 50s ribosome.	Interferes with formylmethionine-tRNA A 70S initiation complex formation, thus preventing protein synthesis.
Tetracycline /tigecycline Doxycycline Demeclocycline Minocycline	Blocks 30S mRNA - ribosome.	Inhibits attachment of aminoacyl-t-RNA to the acceptor (A) site.
Aminoglycosides Streptomycin Amikacin	Misreading of mRNA Inhibit polysome formation (Attachment of ribosomes on mRNA) Inhibits initiation of protein synthesis	
Chloramphenicol	Blocks 50s ribosomal subunit.	Inhibits transfer of the elongating peptide chain to the newly attached aminoacyl-tRNA.
Macrolides Erythromycin Azithromycin	Interferes with translocation.	

Solution for Question 2:

Correct Answer: B) Linezolid

Explanation:

In the above scenario, the patient's sample shows the presence of vancomycin-resistant MRSA, a multi-drug-resistant strain for whom linezolid is reserved.

- Linezolid is a narrow-spectrum protein synthesis inhibitor that targets only gram-positive bacteria.
- Mechanism of action: Inhibition of the formation of the formylmethionine-tRNA 70S initiation complex, thus preventing the initiation of protein synthesis.

Spectrum	Antibiotics	Uses
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Broad spectrum Effective against a wide range of bacteria, including both Gram-positive and Gram-negative organisms.	Chloramphenicol (Option C)	Pyogenic meningitis (Second line) Endophthalmitis Anaerobic infections
Tetracycline	Chlamydia Bacillus anthrax Psittacosis	
Clindamycin (Option D)	Ance Anaerobic infections Drug-resistant Malaria Toxoplasmosis	
Intermediate spectrum Typically, they cover Gram-positive or Gram-negative bacteria, but not both extensively.	Macrolides Azithromycin (Option A) Erythromycin	Atypical pneumonia Whooping cough Chlamydia
Narrow spectrum Effective against a limited range of bacteria.	Aminoglycoside Streptomycin Amikacin	Tuberculosis Tularemia Neonatal sepsis
Linezolid (Option B)	Vancomycin-resistant MRSA. VRE.	
Streptogramin	MRSA Penicillin-resistant streptococcus pneumonia.	

Solution for Question 3:

Correct Answer: C) It can be safely used in neonates

Explanation:

Chloramphenicol should be avoided in neonates due to their immature liver, which leads to toxicity. This results in "grey baby syndrome," characterised by hypotonia, distended abdomen, irregular respiration, ashen grey cyanosis, elevated lactic acid levels, cardiovascular collapse, and potentially death.

Category	Description
Mechanism of Action	Inhibits protein synthesis by blocking the 50s ribosomal subunit, preventing peptide chain elongation.

Absorption & Distribution	Rapidly absorbed, has a high volume of distribution (1 L/kg), has good CSF penetration, and freely penetrates bodily cavities.
Excretion	Excreted via hepatic glucuronidation. (Option A)
Drug Interactions	Inhibits metabolism of warfarin, cyclophosphamide, and phenytoin. Enhanced metabolism by phenobarbitone, phenytoin, and rifampin.
Resistance Mechanisms	Plasmid-mediated inactivation by bacteria decreases penetration into bacteria. (Option D)
Uses	Second line for pyogenic meningitis, endophthalmitis, anaerobic infections
Side Effects	Dose-dependent marrow suppression. (Option B) Dose-independent idiosyncratic reaction. Hypersensitivity. Grey baby syndrome.

Solution for Question 4:

Correct Answer: B) Doxycycline

Explanation:

Lymphogranuloma venereum is caused by *Chlamydia trachomatis*, and the drug of choice is Doxycycline + Azithromycin.

Doxycycline is a member of the tetracycline group. It is a protein synthesis inhibitor that blocks 30S mRNA-ribosome, thus inhibiting the attachment of aminoacyl-t-RNA to the acceptor (A) site.

Tetracyclines:

Features	Tetracycline Oxytetracycline	Demeclocycline	Doxycycline Minocycline
Potency	Low	Intermediate	High
Plasma protein binding	Low	High	high
Half-life	6 - 12 hours	10 -16 hours	18- 24 hours

Alteration of gut flora	Altered	Moderate alteration	Least alteration
Phototoxicity	Low	Highest	High
Other toxicity	Tooth discoloration is less marked	Diabetes insipidus	Doxycycline has low nephrotoxicity. Minocycline has Ototoxicity.
Uses of Tetracyclines	Chlamydial Infections: Urethritis, Endocervicitis, Lymphogranuloma venereum, granuloma inguinale. Atypical Pneumonia: Mycoplasma pneumonia, Psittacosis. Rickettsial Infections. Pleurodesis: Doxycycline for pneumothorax. H. pylori Infection: Part of quadruple therapy. Anthrax, Plague, Borrelia.		

Penicillin G (Option A) is a beta-lactam antibiotic that inhibits bacterial cell wall synthesis and is used to treat syphilis.

Cefixime (Option C) is a third-generation cephalosporin that inhibits bacterial cell wall synthesis and is used to treat gonorrhoea.

Acyclovir (Option D) is an anti-viral agent that inhibits viral DNA polymerase, which is used to treat herpes infection.

Solution for Question 5:

Correct Answer: A) Tetracyclines

Explanation:

- Tetracyclines are contraindicated in pregnancy, lactation, and children due to the risk of calcium-tetracycline chelation, which can lead to permanent yellowish discolouration of teeth and inhibited bone growth.

- It should also be avoided with dairy products as they reduce drug absorption.
- Contraindication of tetracycline : Pregnancy Children During lactation Renal failure (Except Doxyxline)
- Pregnancy
- Children
- During lactation
- Renal failure (Except Doxyxline)
- Adverse effects of tetracycline are:
- Pregnancy
- Children
- During lactation
- Renal failure (Except Doxyxline)

Acute hepatic necrosis: seen in pregnancy Fanconi syndrome Photosensitivity Permanent discolouration of teeth (due to in-utero exposure) Delayed growth	Diabetes insipidus Rarely, ↑ ICP in children/pseudotumor cerebri Vestibulotoxic (minocycline) Exfoliative dermatitis (hypersensitivity reaction)
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Amikacin (Option B) is also contraindicated during pregnancy due to potential ototoxicity and nephrotoxicity, but it does not affect bone growth or tooth colour.

Chloramphenicol (Option C) is contraindicated in pregnancy due to the risk of 'grey baby syndrome', which can lead to severe symptoms like hypotonia, distended abdomen, cyanosis, and cardiovascular collapse.

Sulfonamide (Option D) is a sulfa drug that displaces bilirubin from albumin, leading to unconjugated hyperbilirubinemia, which leads to kernicterus. Thus, it should be avoided during pregnancy and in neonates.

Solution for Question 6:

Correct Answer: C) Azithromycin

Explanation:

Azithromycin is considered safer for use in neonates. It is often used in neonates to treat infections such as chlamydia conjunctivitis and pneumonia.

Azithromycin	
Mechanism of Action	Azithromycin inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit, interfering with the elongation of the polypeptide chain and ultimately halting bacterial growth.
Pharmacokinetics	Excellent intracellular penetration, allowing it to effectively target intracellular pathogens. This intracellular accumulation contributes to its prolonged half-life. Once-daily dosing Hepatic clearance does not cross the blood-brain barrier.
Common Uses	Respiratory Infections (e.g., pneumonia (atypical), bronchitis) Genitourinary Infections (e.g., Chlamydia trachomatis) Skin and Soft Tissue Infections (e.g., cellulitis) Other Infections (e.g., otitis media, pharyngitis)
Side Effects	Mild gastric upset Abdominal pain Headache Dizziness

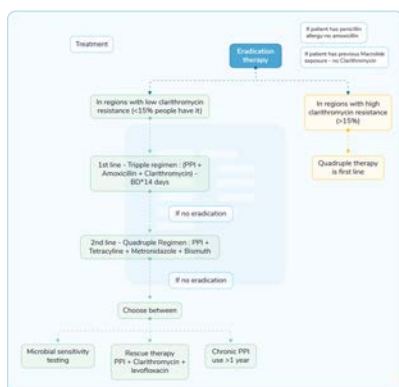
Solution for Question 7:

Correct Answer: D) Clarithromycin

Explanation:

The triple therapy regimen for H. pylori you provided is as follows:

- Lansoprazole: 30 mg
- Amoxicillin: 1000 mg
- Clarithromycin: 500 mg



Aspect	Clarithromycin
Mechanism of Action	Inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit.
Uses	Upper and lower respiratory tract infections, otitis media, whooping cough, atypical pneumonia, H. pylori eradication, MAC infection in AIDS patients, other atypical mycobacterial diseases, and leprosy.
Side Effects	Gastric intolerance, reversible hearing loss (at high doses), hepatic dysfunction, rhabdomyolysis (rare), and CYP3A4 inhibition lead to potential drug interactions.

Solution for Question 8:

Correct Answer: C) Have a large volume of distribution, allowing for extensive tissue penetration.

Explanation:

Erythromycin has a large volume of distribution and can penetrate various tissues, cells, and abscesses.

Category	Erythromycin
Mechanism of action	Binds to the 50S ribosomal subunit of bacteria and interferes with the translocation process during protein synthesis, thereby inhibiting the elongation of the peptide chain.
Pharmacokinetics	Widely distributed in the body, it enters cells, abscesses, and crosses serous membranes and the placenta. Primarily hepatic metabolism and biliary excretion, with minor renal excretion. Does not penetrate the CNS.

	Erythromycin is not stable in an acidic environment and is therefore given as enteric-coated tablets.
Uses	Drug of choice (DOC) for atypical pneumonia caused by <i>Mycoplasma pneumoniae</i> , and whooping cough. As an alternative to Penicillin, it is helpful in infections such as streptococcal pharyngitis, tonsillitis, and mastoiditis. Symptomatic relief in diabetic gastroparesis.
Adverse Effects	Epigastric pain, nausea, anorexia, diarrhoea. High doses may cause reversible hearing impairment. Hypersensitivity. CYP3A4 inhibition, Hypertrophic pyloric stenosis, Worsening of myasthenia gravis, Cholestatic jaundice.

Solution for Question 9:

Correct Answer: C) Clindamycin

Explanation:

The antibiotic most commonly implicated in causing pseudomembranous colitis is Clindamycin.

Clindamycin	
Mechanism of Action	Inhibits protein synthesis by binding to the 50S ribosome.
Spectrum of Activity	Effective against gram-positive cocci, <i>Clostridium</i> spp., <i>Nocardia</i> , <i>Actinomyces</i> , <i>Toxoplasma</i>
Use	Anaerobic and mixed infections, especially those caused by <i>Bacteroides fragilis</i> , can lead to conditions like abdominal and pelvic abscesses, lung abscesses, septic abortion, and infections from penetrating injuries.
Adverse Effects	Rashes, urticaria, abdominal pain. Liver damage Thrombophlebitis at the injection site. Major concern: Pseudomembranous enterocolitis due to <i>Clostridium difficile</i> superinfection, potentially fatal.

Erythromycin (Option A), Tetracycline (Option B), and Ciprofloxacin (Option D) are less commonly associated with pseudomembranous colitis than clindamycin.

Solution for Question 10:

Correct Answer: D) Linezolid

Explanation:

Linezolid is the preferred choice for this patient due to its hepatic metabolism and non-renal excretion, which enhances safety in individuals with renal impair

Linezolid	
Mechanism of Action	It inhibits bacterial protein synthesis by binding to the 23S ribosomal RNA of the P site of 50S ribosomal subunit. It interferes with the formation of the ternary N-formylmethionyl-tRNA (tRNA ^f -Met)-70S initiation complex, preventing the initiation of protein synthesis.
Spectrum of Activity	Gram-positive cocci (aerobic and anaerobic): MRSA, VRSA, VRE, Strep. viridans, Strep. pneumoniae Other sensitive bacteria: Clostridia, B. fragilis, M. tuberculosis.
Common Uses	Uncomplicated and complicated skin and soft tissue infections Hospital-acquired pneumonia (HAPs) Bacteremia.
Side Effects	Mild abdominal pain, nausea and diarrhoea. Chorioretinitis Optic neuritis Bone marrow suppression (monitor platelet count) Caution is advised when used concurrently with MAO inhibitors due to the risk of serotonin syndrome. Serotonin syndrome

Solution for Question 11:

Correct Answer: A) Streptogramins inhibit bacterial protein synthesis through binding to the 50S ribosomal subunit.

Explanation:

Streptogramins	
Mechanism of Action	A mixture of Quinpristine and Dalfopristine in 30:70 ratio. Inhibit bacterial protein synthesis by binding to the 50S ribosomal subunit, preventing the formation of functional ribosomes.
Spectrum of Activity	Narrow spectrum, primarily effective against Gram-positive bacteria, including multidrug-resistant pathogens like MRSA and VRE.
Clinical Use	Not commonly used as first-line therapy for community-acquired pneumonia. Reserved for infections caused by multidrug-resistant Gram-positive pathogens, especially in healthcare-associated settings.
Adverse Effects	Common adverse effects are nausea, vomiting, diarrhoea, headache Nephrotoxicity is not a common adverse effect.

Solution for Question 12:

Correct Answer: A) Inactivation of the drug by bacterial enzymes

Explanation:

- Aminoglycoside resistance typically involves modification by bacterial enzymes like acetyltransferases, phosphotransferases, and nucleotidyltransferases.
- These enzymes alter aminoglycosides, preventing their binding to bacterial ribosomes and blocking protein synthesis.
- This resistance mechanism is common in multidrug-resistant bacteria, including those causing urinary tract infections (UTIs).

Aminoglycosides	
Mechanism of action	Inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit, causing misreading of mRNA.
Drugs	Streptomycin, Gentamicin, Tobramycin, Amikacin, Neomycin, Netilmicin
Spectrum	Primarily Gram-negative aerobic bacteria; limited Gram-positive coverage. Includes Pseudomonas

	aeruginosa, Enterobacteriaceae.
Uses	Serious Gram-negative infections (e.g., sepsis, pneumonia) Tuberculosis (Streptomycin) Endocarditis (Gentamicin in combination therapy)
Resistance mechanism	Enzymatic inactivation by acetyltransferases, phosphotransferases, and nucleotidyltransferases. Alteration of the 30S ribosomal binding site.

Solution for Question 13:

Correct Answer: B) Nephrotoxicity and ototoxicity

Explanation:

Adverse Effect	Mechanism of Action	Reversibility	Risk Order
Ototoxicity	Damages hair cells (neurons projecting into endolymph). Cochlear damage leads to high-frequency hearing loss. Vestibular damage leads to imbalance, ataxia, vertigo.	Irreversible.	Neomycin > Amikacin > Kanamycin.
Nephrotoxicity	Damages tubular cells of nephron (PCT, DCT), leading to renal tubular acidosis.	Reversible.	Neomycin > Gentamicin > Streptomycin.
Neuromuscular Blockade	Inhibits release of ACh and blocks Nm receptor, leading to muscle paralysis Contraindicated in Myasthenia gravis	Partially reversible with calcium salts.	Neomycin > Streptomycin > Tobramycin.

Solution for Question 14:

Correct Answer: A) Ampicillin

Explanation:

Ampicillin inhibits cell wall synthesis by binding to PBPs. However, it is least effective against *E. faecium* strains due to resistance caused by altered PBPs.

Drugs useful in *Enterococcus* infection:

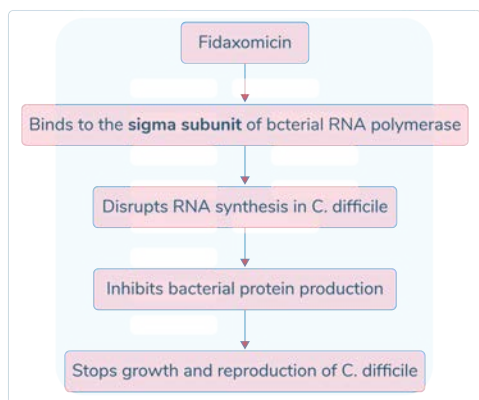
Drug	Mechanism of Action (MOA)
Vancomycin	Inhibits cell wall synthesis by binding to the D-alanyl-D-alanine terminus of cell wall precursor units.
Linezolid (Option B)	Inhibits protein synthesis by binding to the 50S ribosomal subunit.
Daptomycin (Option C)	Causes rapid depolarisation of bacterial membrane potential, inhibiting protein, DNA, and RNA synthesis.
Nitrofurantoin	Reduced by bacterial enzymes to reactive intermediates, altering or inactivating bacterial ribosomal proteins.
Tigecycline (Option D)	Inhibits protein synthesis by binding to the 30S ribosomal subunit.

Solution for Question 15:

Correct Answer: C) Fidaxomicin

Explanation:

- Fidaxomicin is the best choice for treating severe pseudomembranous enterocolitis caused by *Clostridioides difficile*.
- It specifically targets *C. difficile*, minimising disruption to the gut microbiome and reducing recurrence risk compared to broad-spectrum antibiotics like metronidazole or vancomycin.
- Mechanism of action:



Antimetabolites and Fluroquinolones

1. Which of the following statements correctly describes nitroimidazoles?
Metronidazole inhibits bacterial DNA synthesis by interacting directly with DNA. Metronidazole is a prodrug converted to its active form by the bacterial nitroreductase enzyme. Secnidazole is one of the shortest-acting nitroimidazoles. Metronidazole inhibits the PFOR enzyme, which damages DNA in protozoa. Resistance occurs through the activation of nitric oxide free radicals.

- A. 1, 2 and 4
- B. 2 and 4
- C. 1, 2 and 3
- D. 1 and 3

2. A 30-year-old man presents to the clinic with a history of abdominal pain and loose stools for the past two weeks but no h/o fever. He has recently returned from a vacation to Nepal. Physical examination reveals tenderness in the right lower quadrant of the abdomen. Stool testing shows mucoid stool with elevated white cell counts. What is the drug of choice for treating this condition?

- A. Ciprofloxacin
- B. Metronidazole
- C. Albendazole
- D. Doxycycline

3. A 25-year-old man presented to the clinic with foul-smelling diarrhoea and abdominal cramps for the past week. A stool examination revealed giardia cysts. After starting appropriate treatment, she returned a few days later, complaining of

severe nausea, vomiting, headaches, and flushing after the farewell party. Which of the following is not a side effect associated with the medication?

- A. Urine discolouration
 - B. Metallic taste
 - C. Peripheral neuropathy
 - D. Tendon rupture
-

4. A 35-year-old HIV-positive male presents to the clinic with a two-week history of profuse, watery diarrhoea and weight loss. His CD4 count is 150 cells/mm³, and stool examination reveals oocysts. What is the drug of choice used for treating this patient's condition?

- A. Benznidazole
 - B. Nifurtimox
 - C. Nitazoxanide
 - D. Paromomycin
-

5. Which of the following statements is true regarding bacterial resistance against nitroimidazoles?

- A. Increase efflux pump activity to remove nitroimidazoles from the cell.
 - B. Alter the structure of ribosomal subunits to prevent nitroimidazole binding.
 - C. Produce enzymes that inactivate nitroimidazoles before they can enter the cell.
 - D. Reduce the expression of nitroreductase enzymes needed to activate nitroimidazoles.
-

6. Rearrange the following steps in the correct sequence based on the mechanism of action of fluoroquinolones: 1. Inhibits DNA gyrase and topoisomerase IV. 2. DNA

replication and transcription are hindered. 3. Bacterial cell division is disrupted. 4. Bacterial DNA supercoiling and strand separation are affected. 5. Bacterial cell death occurs.

- A. 1, 4, 2, 3, 5
 - B. 2, 4, 1, 3, 5
 - C. 1, 2, 4, 3, 5
 - D. 1, 4, 3, 2, 5
-

7. Which of the following statements about second-generation fluoroquinolones is false?

- A. They have a broader spectrum of activity compared to first-generation fluoroquinolones.
 - B. They inhibit bacterial DNA gyrase and topoisomerase IV.
 - C. They have a shorter half-life compared to first-generation fluoroquinolones.
 - D. They can cause QT prolongation as a side effect.
-

8. A 35-year-old male presents with flu-like symptoms, including fever, malaise, and cough. Upon further questioning, he reports recent contact with animal hides during his work at a tannery. Chest X-ray revealed a widened mediastinum. Blood cultures are drawn, and the patient is to be started on antibiotics. Which of the following is the most appropriate treatment choice for this patient?

- A. Ciprofloxacin
 - B. Amoxicillin
 - C. Doxycycline
 - D. Vancomycin
-

9. A 65-year-old male presents to the emergency department with a 3-day history of productive cough, fever, and shortness of breath. Chest X-ray reveals consolidation in the right lower lobe consistent with community-acquired pneumonia. What is the most appropriate initial antibiotic therapy for this patient?

- A. Amoxicillin
 - B. Azithromycin
 - C. Levofloxacin
 - D. Ceftriaxone
-

10. Which of the following is the longest-acting fluoroquinolone?

- A. Sparfloxacin
 - B. Pefloxacin
 - C. Moxifloxacin
 - D. Delafloxacin
-

11. Which of the side effects is not associated with fluoroquinolones?

- A. Tendon rupture
 - B. QT interval prolongation
 - C. Photosensitivity
 - D. Alopecia
-

12. A 25-year-old female has been complaining of dysuria, increased frequency of urination, and mild lower abdominal discomfort for the past three days. She reports no fever, chills, or flank pain, and she has no known drug allergies. A urine dipstick test in the clinic shows positive nitrites and leukocyte esterase, indicating a

possible mild, uncomplicated UTI. What is the most appropriate initial antibiotic treatment for this patient?

- A. Amoxicillin
 - B. Ciprofloxacin
 - C. Nitrofurantoin
 - D. Cotrimoxazole
-

13. Choose the correct statements regarding nitrofurantoin: Is primarily used for the treatment of systemic infections. Works by damaging bacterial DNA and ribosomal proteins. Is well absorbed from the gastrointestinal tract and reaches high concentrations in urine. Is safe for use in patients with significant renal impairment.

- A. 1 and 3
 - B. 2 and 4
 - C. 2 and 3
 - D. 1 and 4
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	4
Question 4	3
Question 5	4
Question 6	1
Question 7	3
Question 8	1
Question 9	3
Question 10	1
Question 11	4
Question 12	4
Question 13	3

Solution for Question 1:

Correct Answer: A) 1, 2 and 4

Explanation:

Statements 1, 2 and 4 are correct regarding nitroimidazoles

Nitroimidazoles	
MOA	Prodrug is converted to active form through nitro group reduction inside the bacteria(Statement 2) → formation

	of nitric oxide free radicals → causes DNA damage→inhibits replication(Statement 1) Inhibits PFOR enzyme (pyruvate ferritin oxido reductase) in protozoa → causing DNA damage(Statement 4)
Drugs	Metronidazole Ornidazole Tinidazole Satranidazole Secnidazole: longest acting drug Other: Benznidazole, Nifurtimox, Nitazoxanide

Solution for Question 2:

Correct Answer: B) Metronidazole

Explanation:

The patient presents with symptoms (abdominal pain, diarrhoea, and abdominal tenderness) consistent with amoebiasis, a parasitic infection caused by *E. histolytica*. Metronidazole is the drug of choice. The history of travel to a region with poor sanitation (Nepal), mucoid stools, and elevated white cell counts supports this diagnosis.

Metronidazole	
Coverage	Broad spectrum anaerobes (<i>Bacteroides</i> , <i>Clostridium</i> , <i>Fusobacterium</i> , <i>Peptococcus</i> , etc.)
Uses	Giardiasis (in HIV-positive patients) <i>Entamoeba histolytica</i> → Amebic dysentery & amebic liver abscess <i>Trichomonas vaginalis</i> Bacterial Vaginosis <i>Dracunculus</i> (guinea worm) <i>Gardnerella</i> / bacterial vaginosis Empiric intra-abdominal and soft tissue anaerobic infections <i>H. pylori</i> gastritis (combined with proton pump inhibitor) Prophylaxis for colorectal surgery Fulminant <i>C. difficile</i> infection (IV metronidazole + oral vancomycin) Mild to moderate Crohn's disease

Solution for Question 3:

Correct Answer: D) Tendon rupture

Explanation:

The clinical findings of the patient (foul-smelling stool, abdominal cramps), along with the stool examination, indicate giardiasis which is most likely treated with metronidazole. The patient experienced a disulfiram-like reaction after drinking alcohol during the party. Common side effects include a metallic taste, neurotoxicity (peripheral neuropathy), and dark urine, while tendon rupture is a side effect of fluoroquinolones (e.g., ciprofloxacin).

Adverse effects of Metronidazole	
Side effects	Metallic taste, intensely bitter (Option B) Headache, nausea, dry mouth (most common) Vomiting, diarrhoea, and abdominal distress Dysuria, cystitis, and a sense of pelvic pressure Dizziness, vertigo, Neurotoxicity: Peripheral neuropathy, numbness, paresthesias, ataxia (Option C) Brown color urine (Option A) Urticaria, flushing, and pruritus (rare cause of Stevens-Johnson syndrome)
Drug Interaction	Inhibitor of CYP enzymes Disulfiram-like effect: abdominal distress, vomiting, flushing, headache or psychosis with alcoholic beverages during or within 3 days of therapy. Precipitate lithium toxicity Prothrombin time prolongation with warfarin (anticoagulant).
Contraindications	CNS diseases Severe hepatic disease First trimester of pregnancy

Solution for Question 4:

Correct Answer: C) Nitazoxanide

Explanation:

Nitazoxanide is the drug of choice for *Cryptosporidium parvum*, a parasitic infection that causes severe diarrhoea and infectious oocysts, particularly in immunocompromised individuals, such as those with HIV/AIDS.

Nitazoxanide:

- MOA: Inhibit PFOR enzyme-dependent electron transfer. Inhibits growth of parasitic oocytes and trophozoites.
- Inhibit PFOR enzyme-dependent electron transfer.
- Inhibits growth of parasitic oocytes and trophozoites.
- Uses: Giardiasis resistant to metronidazole Cryptosporidiosis (DOC) Safe in pregnancy
- Giardiasis resistant to metronidazole
- Cryptosporidiosis (DOC)
- Safe in pregnancy
- Side effect: Green urine.
- Inhibit PFOR enzyme-dependent electron transfer.
- Inhibits growth of parasitic oocytes and trophozoites.
- Giardiasis resistant to metronidazole
- Cryptosporidiosis (DOC)
- Safe in pregnancy

Solution for Question 5:

Correct Answer: D) Reduce the expression of nitroreductase enzymes needed to activate nitroimidazoles.

Explanation:

Bacteria reduces the expression of nitroreductase enzyme to resist nitroimidazole

Resistance in Nitroimidazoles

1. Reduced nitroreductase enzyme expression (Option D)
↓

Nitroimidazoles (such as metronidazole) cannot activate in their active form.



It prevents drug activation and stops the formation of the reactive nitroso group within the bacterial cell.

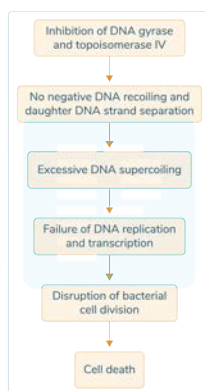
2. Lower levels of PFOR and ferredoxin. (less common)

Solution for Question 6:

Correct answer: A) 1, 4, 2, 3, 5

Explanation:

Mechanism of action of fluoroquinolones:



Solution for Question 7:

Correct Answer: C) They have a shorter half-life compared to first-generation fluoroquinolones.

Explanation:

Second-generation fluoroquinolones typically have a longer half-life than first-generation fluoroquinolones and require less frequent dosing.

	First Generation	Second Generation
--	------------------	-------------------

Drugs Mnemonic: New COP has Small Lite Machine Gun	Norfloxacin Ciprofloxacin Ofloxacin Pefloxacin	Sparfloxacin Levofloxacin Moxifloxacin Gemifloxacin
Pharmacokinetics	Good oral absorption Moderate half-life	Good oral absorption Longer half-life (Option C is the false statement) Better tissue penetration
Mechanism of Action	Inhibit bacterial DNA gyrase	Inhibit bacterial DNA gyrase and topoisomerase IV (Option B)
Spectrum	Effective against Gram-negative bacteria	Broader spectrum, including Gram (+ve), Gram (-ve) and Atypical pathogens. (Option A)
Uses	Urinary tract infections Prostatitis	Respiratory infections (e.g. Pneumonia) Skin infections Complicated UTIs
Side Effects	Nausea and Vomiting Diarrhea Dizziness	QT prolongation (Option D) Tendonitis CNS effects Phototoxicity Peripheral neuropathy
Contraindications	Pregnancy Children Myasthenia gravis	Same as first generation Patients with cardiac arrhythmias

Solution for Question 8:

Correct Answer: A) Ciprofloxacin

Explanation:

The given clinical vignette suggests anthrax and ciprofloxacin, which is a first-generation fluoroquinolone that is effective against it. It is the first-line treatment for anthrax, particularly in cases of inhalational anthrax or systemic dissemination.

Ciprofloxacin

Pharmacokinetics	Good oral absorption Bioavailability ~70%, Moderate half-life (~4-5 hours) Widely distributed in tissues and body fluids Renal excretion
Uses	Urinary tract infections Prostatitis Pseudomonas aeruginosa (DOC) Anthrax (Bacillus anthracis) (DOC) Gastrointestinal infections (e.g., traveller's diarrhoea) Bone and joint infections
Side Effects	Gastrointestinal disturbances (nausea, diarrhoea). CNS effects (headache, dizziness). Tendonitis and tendon rupture. Photosensitivity.
Contraindications	Pregnancy Children (due to potential for cartilage damage) Myasthenia gravis (can exacerbate muscle weakness)

Solution for Question 9:

Correct Answer: C) Levofloxacin

Explanation:

Levofloxacin is a Second Generation Fluoroquinolone Antibiotic with excellent coverage against typical and atypical pathogens commonly associated with community-acquired pneumonia. It is recommended as initial therapy for patients with moderate to severe pneumonia who require hospitalisation.

Levofloxacin	
Pharmacokinetics	Good oral absorption Bioavailability ~99% Longer half-life (~6-8 hours) Widely distributed in tissues and body fluids Renal excretion.

Uses	Community-acquired pneumonia (DOC) Acute bacterial sinusitis Chronic bronchitis exacerbation Second-line drug for TB Urinary tract infections Prostatitis Skin and soft tissue infections Effective against atypical pathogens (e.g., Mycoplasma, Chlamydia, Legionella)
Side Effects	Gastrointestinal disturbances (nausea, diarrhoea) CNS effects (headache, dizziness) Tendonitis and Tendon rupture. QT interval prolongation Photosensitivity
Contraindications	Pregnancy and lactation. Children and adolescents (due to potential for cartilage damage) Myasthenia gravis (can exacerbate muscle weakness) Patients with a history of tendon disorders Cardiac arrhythmias (due to risk of QT prolongation)

Solution for Question 10:

Correct answer: A) Sparfloxacin

Explanation:

Sparfloxacin is considered the longest-acting fluoroquinolone (Half-life: 19 Hours), but it is not used due to safety concerns, particularly related to cardiac arrhythmias.

Solution for Question 11:

Correct Answer: D) Alopecia

Explanation:

Alopecia is not an adverse effect of fluoroquinolones

Adverse effects of fluoroquinolones:

- Gastrointestinal disturbances (nausea, diarrhoea).
- CNS effects (headache, dizziness).
- Tendonitis and tendon rupture.(Option A)
- QT interval prolongation.(Option B)
- Photosensitivity.(Option C)
- Potential for exacerbation of myasthenia gravis.

Solution for Question 12:

Correct Answer: D) Cotrimoxazole

Explanation:

Cotrimoxazole, a combination of trimethoprim and sulfamethoxazole, is an effective and commonly used treatment for uncomplicated UTIs. It is effective against a broad range of uropathogens, including many strains of *E. coli*.

Cotrimoxazole is generally well-tolerated and has a favourable resistance profile, making it an appropriate choice for this patient.

Cotrimoxazole	
Type	Competitive Inhibitor of DHPS (PABA analogues)
Mechanism of Action	Trimethoprim-sulfamethoxazole (cotrimoxazole) ■ Inhibit Dihydrofolate Reductase and Dihydropteroate Synthase ■ Disruption of Folate Synthesis ■ Inhibition of DNA, RNA, and Protein Synthesis ■ Impaired Bacterial Cell Growth and Division ■ Accumulation of Cellular Damage ■ Bacterial Death.
Uses	Urinary tract infections (DOC for mild cases) Respiratory tract infections GIT Infections Prevention and treatment of <i>Pneumocystis jirovecii</i> pneumonia in immunocompromised patients
Side Effects	Nausea and Vomiting Diarrhoea Rash

	Hypersensitivity reactions (including Stevens-Johnson syndrome and toxic epidermal necrolysis) Blood dyscrasias (e.g., neutropenia, thrombocytopenia)
Special Points	Contraindicated in patients with folate deficiency Use with caution in pregnant women, infants, and individuals with G6PD deficiency Increased risk of hyperkalemia with potassium-sparing diuretics

Solution for Question 13:

Correct Answer: C) 2 and 3

Explanation:

Nitrofurantoin	
Type	Urinary Antiseptic
Pharmacokinetics	Absorption: Rapidly absorbed from the GI tract (Food Increases Absorption)(Statement 3) Distribution: High concentration in urine(Statement 3) Metabolism: Minimal hepatic metabolism Excretion: Excreted unchanged in urine, not recommended for use in patients with significant renal impairment (CrCl < 60 mL/min).
Mechanism of Action	Nitrofurantoin ■ Bacterial Reduction to Reactive Intermediates ■ Damage to Bacterial DNA and Ribosomal Proteins(Statement 2) ■ Disruption of DNA Replication, Transcription, and Translation ■ Inhibition of Bacterial Cell Growth and Division ■ Bacterial Death
Uses	Treatment of uncomplicated urinary tract infections (UTIs) Prophylaxis for recurrent UTIs Not effective for pyelonephritis or prostatitis
Side Effects	Pulmonary reactions (acute and chronic) Brown Urine Hemolysis in G6PD deficiency Hepatotoxicity Peripheral neuropathy

Antiviral Drugs

1. A 45-year-old male patient presents to the clinic with fatigue, jaundice, and abdominal discomfort. Laboratory tests reveal elevated liver enzymes and positive Hepatitis B surface antigen (HBsAg). Further evaluation shows HBsAg +ve, Anti-HBC IgG +ve, HBeAg +ve and evidence of liver inflammation on biopsy. What should be the drug of choice for the treatment of this patient?

- A. Lamivudine
- B. Interferon-alpha
- C. Tenofovir
- D. Adefovir

2. A 65-year-old male patient with a history of hypertension and type 2 DM presents with significant shortness of breath, a persistent cough, and a high fever. His oxygen saturation is 85% on room air. After a thorough examination and necessary tests, he is diagnosed with COVID-19 pneumonia. Which of the following treatment measures has been shown to offer the maximum reduction in mortality for this patient?

- A. Hydroxychloroquine
- B. Convalescent Plasma
- C. Dexamethasone
- D. Remdesivir

3. Which of the following drugs is generally not used for topical HSV infections?

- A. Acyclovir

- B. Trifluridine
 - C. Ganciclovir
 - D. Docosanol
-

4. Choose the correct statements regarding cidofovir. 1. It is primarily used in the treatment of CMV retinitis. 2. It can be administered orally. 3. It is commonly used as a first-line treatment for influenza infections. 4. Requires probenecid co-administration.

- A. 1 and 4
 - B. 2 and 3
 - C. 1 and 2
 - D. 1 and 3
-

5. A 60-year-old male with a history of HIV infection presents to the clinic with complaints of blurred vision, floaters, and pain in his left eye. On fundus examination, following appearance is seen. Which of the following drugs is most appropriate to manage this condition?



- A. Acyclovir

- B. Valganciclovir
 - C. Oseltamivir
 - D. Ribavirin
-

6. A 45-year-old male presents with painful vesicular lesions on his lips and genital area. He has been on acyclovir therapy for HSV infection, but there has been no improvement in his symptoms. Given the concern for acyclovir-resistant HSV, which of the following drugs is most appropriate to manage this condition?

- A. Valacyclovir
 - B. Famciclovir
 - C. Foscarnet
 - D. Oseltamivir
-

7. A 22-year-old male presents to the physician with painful ulcers and a burning sensation over his genital area. Physical examination reveals a fluid-filled vesicular lesion with a red erythematous base (image). All of the following are true regarding the drug of choice (DOC) for this condition, except?



- A. It is an antiviral drug that works as a nucleotide analogue, preventing viral replication.

- B. It is applied topically, intravenously, or orally.
 - C. It should be taken with plenty of water.
 - D. It is also used to treat Epstein Barr virus (EBV) and varicella-zoster virus (VZV)
-

8. A 30-year-old woman arrives at the clinic with a high-grade fever, myalgia, and a dry cough that has persisted for the last 36 hours. Her family doctor diagnosed her with a flu infection and recommended baloxavir. Which of the following correctly describes the drug's mechanism of action?

- A. Inhibits neuraminidase, preventing the release of new viral particles
 - B. Inhibits viral RNA polymerase, preventing viral replication
 - C. Inhibits M2 ion channels, preventing viral uncoating
 - D. Inhibits viral protease, preventing viral maturation
-

9. A 23-year-old male presents to the hospital with a high-grade fever, chest pain and dry cough for the last two days. A rapid influenza test confirmed the diagnosis. Which of the following is the treatment of choice for this patient's condition?

- A. Reassurance
 - B. Oseltamivir
 - C. Amantidine
 - D. Zanamivir
-

10. A 2-year-old child presents to the hospital with persistent cough, wheezing, and difficulty breathing. A viral panel is conducted, and the child is diagnosed with respiratory syncytial virus (RSV). What is the drug of choice for this condition?

- A. Lenecapavir

- B. Ribavirin
 - C. Interferon-alpha
 - D. Palivizumab
-

11. A 47-year-old female with chronic hepatitis C, who is currently on sofosbuvir, presents with a persistent cough, weight loss, and fever. Chest X-ray revealed a cavitary lesion with hilar lymphadenopathy in the right lower lobe. The patient was started on an appropriate treatment regimen. Which of the following complications is likely to be seen in this case?

- A. Decreased efficacy of sofosbuvir
 - B. Increased risk of hepatotoxicity
 - C. Increased risk of peripheral neuropathy
 - D. Increased risk of fatal cardiac arrest
-

12. A 38-year-old man with a history of drug abuse presents to the clinic with jaundice and right upper quadrant pain. Laboratory tests confirm the presence of anti-HCV antibodies. The gastroenterologist advises starting NS5A and NS5B inhibitor combination therapy. Which of the following correctly describes the mechanism of action of the drugs used in this patient? Inhibits viral protease, preventing the cleavage of viral polyproteins Inhibits viral RNA polymerase, halting viral RNA synthesis Inhibits the viral protein essential for viral RNA replication and assembly Inhibits viral exit through degradation of viral RNA

- A. 1 and 2
 - B. 2 and 3
 - C. 3 and 4
 - D. 1 and 3
-

13. A 32-year-old man presents to the clinic with persistent fatigue, weight loss, and recurrent infections. After testing, he is diagnosed with HIV. The physician discusses initiating antiretroviral therapy (ART) with drugs that inhibit nucleoside reverse transcriptase enzyme. Which of the following drugs does not belong to that class?

- A. Abacavir.
 - B. Tenofovir.
 - C. Stavudine.
 - D. Didanosine.
-

14. A 35-year-old male patient who is HIV-positive presents with a persistent cough, night sweats, weight loss, and fever. Sputum samples confirm the diagnosis of tuberculosis. His CD4 count is 150 cells/mm³, and he is not currently on antiretroviral therapy. What should be the next step in the management of this patient?

- A. Start antiretroviral therapy (ART) immediately
 - B. Initiate anti-tuberculosis treatment first and defer ART
 - C. Start both anti-tuberculosis treatment and ART simultaneously
 - D. Administer prophylactic treatment for opportunistic infections before starting any TB or HIV treatment
-

15. An intern presents to the occupational health department after sustaining a needle stick injury from a suspected HIV-positive patient. In this case, what is the regimen of choice for post-exposure prophylaxis (PEP)?

- A. Zidovudine + Lamivudine + Ritonavir
- B. Emtricitabine + Tenofovir + Atazanavir

- C. Tenofovir + Emtricitabine + Raltegravir
 - D. Tenofovir + Lamivudine + Dolutegravir
-

16. A 30-year-old patient presents to the clinic with symptoms suggestive of HIV infection. Upon testing, the diagnosis of HIV is confirmed. What is the regimen of choice for initiating antiretroviral therapy for this patient?

- A. Abacavir + Lamivudine + Dolutegravir.
 - B. Tenofovir + Lamivudine + Efavirenz.
 - C. Tenofovir + Lamivudine + Dolutegravir.
 - D. Abacavir + Lamivudine + Lopinavir.
-

17. Which of the following is a CCR5 inhibitor used in the treatment of HIV?

- A. Enfuvirtide
 - B. Fostemsavir
 - C. Ibalizumab
 - D. Maraviroc
-

18. A 42-year-old patient with HIV is being treated with darunavir as part of their antiretroviral therapy. To enhance the effectiveness of darunavir, the physician adds another medication. Which of the following drugs can be used for this purpose?

- A. Atazanavir
- B. Efavirenz
- C. Cobicistat

D. Lamivudine

19. A 35-year-old patient on anti-HIV therapy presents to the emergency department with complaints of severe flank pain, nausea, and vomiting. Laboratory tests reveal hematuria, and a CT scan shows evidence of nephrolithiasis. Which of the following protease inhibitors is most likely responsible for these symptoms?

- A. Indinavir
 - B. Ritonavir
 - C. Atazanavir
 - D. Darunavir
-

20. Which of the following statements regarding the properties of protease inhibitors are true? Nelfinavir is an aspartyl protease inhibitor that prevents viral protein assembly in the cell. Lipodystrophy is a common side effect seen with the majority of protease inhibitors. Ritonavir boosts plasma levels of lopinavir by inhibiting the CYP2C9 enzyme. Ritonavir is associated with a higher risk of intracranial haemorrhage. Cross-resistance among other protease inhibitors is rare.

- A. Statements 1 and 3 are correct.
 - B. Statements 1 and 2 are correct.
 - C. Statements 1, 3, and 4 are correct.
 - D. All statements are correct
-

21. Match the following integrase inhibitors with their unique characteristics: 1. Cabotegravir a. Associated with rhabdomyolysis 2. Dolutegravir b. Intramuscular administered every 3 months 3. Raltegravir c. Safe in pregnancy and widely used in India 4. Elvitegravir d. Used in combination with Cobicistat

1. Cabotegravir	a. Associated with rhabdomyolysis
2. Dolutegravir	b. Intramuscular administered every 3 months
3. Raltegravir	c. Safe in pregnancy and widely used in India
4. Elvitegravir	d. Used in combination with Cobicistat

- A. 1:b, 2:c, 3:a, 4:d
- B. 1:c, 2:b, 3:a, 4:d
- C. 1:c, 2:b, 3:d, 4:a
- D. 1:b, 2:c, 3:d, 4:a

22. A 30-year-old HIV-1 positive primigravida who has been adherent to her antiretroviral therapy with a consistently low viral load delivers a full-term baby boy. What is the most appropriate drug of choice to reduce the risk of HIV transmission in this neonate?

- A. Zidovudine
- B. Efavirenz
- C. Nevirapine
- D. Nevirapine + Zidovudine

23. All of the following are true about tenofovir except

- A. It is an NRTI (nucleotide reverse transcriptase inhibitor) that requires phosphorylation activation.
- B. Tenofovir disoproxil is a shorter-acting salt with a high risk of acute renal failure.
- C. It is used in the treatment of both HIV and Hepatitis B.
- D. It is not recommended for children under 10 years or those with pre-existing renal disease.

24. A 5-year-old child diagnosed with HIV is scheduled to begin therapy with nucleoside reverse transcriptase inhibitors (NRTIs). His baseline laboratory tests reveal a haemoglobin level of 8.8 g/dL. Which of the following drugs should be avoided in this patient?

- A. Abacavir
 - B. Lamivudine
 - C. Zidovudine
 - D. Emtricitabine
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	1
Question 5	2
Question 6	3
Question 7	1
Question 8	2
Question 9	2
Question 10	2
Question 11	1
Question 12	2
Question 13	2
Question 14	2
Question 15	4
Question 16	3
Question 17	4
Question 18	3
Question 19	1
Question 20	2
Question 21	1
Question 22	3

Question 23	1
Question 24	3

Solution for Question 1:

Correct Answer: C) Tenofovir

Explanation:

Tenofovir is a potent antiviral agent with a high barrier to resistance, making it a preferred first-line treatment for chronic Hepatitis B. It suppresses viral replication, reduces liver inflammation, and has a favourable safety profile.

TDF is widely recommended due to its efficacy and low rate of resistance.

First-Line Treatments:

These drugs are preferred because they have a high genetic barrier to resistance, potent antiviral activity, and are generally well-tolerated. They effectively suppress the hepatitis B virus and are less likely to result in drug resistance.

Second-Line Treatments:

Lamivudine (Option A) is used to treat Hepatitis B. Its long-term use is associated with a high rate of viral resistance.

Interferon-alpha (Option B) can be used to treat chronic Hepatitis B. However, its use is limited by significant side effects and lower efficacy. It is often reserved for select cases where oral antiviral agents are contraindicated or not preferred.

Adefovir (Option D) is an antiviral medication that can be used to treat chronic Hepatitis B. It is generally considered a second-line agent because it is less potent than TDF and entecavir and has a higher risk of nephrotoxicity.

Solution for Question 2:

Correct Answer: C) Dexamethasone

Explanation:

The given clinical vignette suggests severe COVID-19 and dexamethasone has been shown to reduce mortality in moderate to severe cases.

Severe COVID-19 (Rx- admission in ICU/HDU):

- Respiratory rate >30/min.
- Breathlessness (+ve).
- SpO₂ <90% on room air.

Dexamethasone (Option C) has been shown to reduce mortality in patients with severe COVID-19 who require supplemental oxygen or mechanical ventilation. This is likely due to its anti-inflammatory effects, which help reduce the cytokine storm and lung damage associated with severe COVID-19.

Drugs used to treat COVID-19:

Remdesivir (Option D)	Inhibits RNA-dependent RNA polymerase
Favipiravir	
Molnupiravir	Causes errors in the viral RNA during replication
Nirmatrelvir	Protease inhibitor
Bebtelovimab	Monoclonal antibodies against spike protein of SARS-CoV2
Bamlanivimab	
Sotrovimab	
Tocilizumab	IL-6 inhibitor
Baricitinib	Janus kinase inhibitor
Corticosteroids	Methylprednisolone/dexamethasone. Reduction of mortality and inflammation.
Heparin	Anti-coagulation and prevention of thrombotic complications.

Hydroxychloroquine and convalescent plasma therapy (Option A&B) are no longer used for the treatment of COVID-19, as per revised guidelines.

Solution for Question 3:

Correct Answer: C) Ganciclovir

Explanation:

Drugs used for topical HSV infection:

Mnemonic- VITACD.

Drug	MOA	Use
Nucleoside analogues		
Vidarabine	Viral DNA polymerase Inhibition.	Topical treatment of HSV infections
Idoxuridine	Inhibits viral DNA synthesis by competing with thymidine triphosphate.	
Trifluridine (OptionB)	Herpes simplex keratitis	
Acyclovir (DOC) (Option A)	Viral DNA polymerase Inhibition.	Genital herpes, cold sores (HSV-1, HSV-2)
Cidofovir	Viral DNA polymerase Inhibition.	Off-label topical use for certain HSV infections
Alcohol		
Docosanol (Option D)	Inhibits fusion between the plasma membrane and the herpes simplex virus	Cold sores (HSV-1)

Common side effects: Local irritation, burning sensation, and stinging.

Solution for Question 4:

Correct Answer: A) 1 and 4

Explanation:

Cidofovir	
It is a monophosphate nucleotide analogue of cytidine.	
Pharmacokinetics	Administered intravenously (Statement 2- Ruled Out) Excreted unchanged by kidneys A long half-life allows for once-weekly dosing
Mechanism of Action	Penetrates into infected cells ■ Converted to active form ■ Inhibits viral DNA polymerase ■ Incorporation into viral DNA ■ Terminates DNA chain elongation.
Uses	Treatment of cytomegalovirus (CMV) retinitis in immunocompromised patients (Statement 1). Off-label use in treating other viral infections, such as adenovirus, herpesviruses (HSV, VZV), and papillomavirus infections (Statement 3- Ruled Out)
Side Effects	Nephrotoxicity (Administer with Probenecid to reduce toxicity) (Statement 4). Neutropenia. Hypotension. Ocular effects (including uveitis and iritis). Gastrointestinal disturbances (nausea, diarrhoea).

Solution for Question 5:

Correct Answer: B) Valganciclovir

Explanation:

The fundoscopy image shows Pizza Pie appearance, characteristic of CMV Retinitis. Valganciclovir is an oral prodrug of ganciclovir specifically used to treat CMV infections. It effectively manages CMV retinitis and other CMV-related conditions, making it the appropriate choice for this patient.

Valganciclovir	
Class	Nucleoside analogue
Pharmacokinetics	Oral administration. Converted to ganciclovir (Prodrug). Renal clearance.
MOA	Conversion to Ganciclovir ■ Incorporation into Viral DNA ■ Inhibition of Viral DNA Polymerase ■ Prevention

	of Viral DNA Synthesis ■ Decreased Viral Replication
Uses	CMV retinitis CMV prophylaxis in transplant patients
Side Effects	Neutropenia Thrombocytopenia Anemia Fever

Solution for Question 6:

Correct Answer: C) Foscarnet

Explanation:

Foscarnet is an antiviral drug that does not require activation by viral thymidine kinase (as Acyclovir) and works by directly inhibiting viral DNA polymerase. This makes it effective against Acyclovir-resistant HSV infections.

Drugs used in Acyclovir-resistant herpes Infection:

	Foscarnet	Cidofovir
Class	Antiviral	Antiviral
Pharmacokinetics	IV administration Renal clearance	IV administration Renal clearance
MOA	Inhibits viral DNA polymerase directly	Inhibits viral DNA polymerase Incorporates into viral DNA
Uses	HSV infections resistant to acyclovir CMV retinitis	Acyclovir-resistant HSV infections CMV retinitis Respiratory papillomatosis
Side Effects	Nephrotoxicity Electrolyte imbalances Seizures Anemia	Nephrotoxicity Ocular hypotony Neutropenia

Valacyclovir (Option A) is a prodrug of acyclovir, and it shares the exact mechanism of action. Therefore, it would not be effective in managing acyclovir-resistant HSV infections.

Famciclovir (Option C) is another antiviral medication used to treat herpesvirus infections. It is a prodrug of penciclovir and shares a similar action mechanism with acyclovir. Hence, it is

generally not effective against acyclovir-resistant HSV.

Oseltamivir (Option D) is an antiviral medication that treats and prevents influenza A & B. It is ineffective against herpes simplex virus.

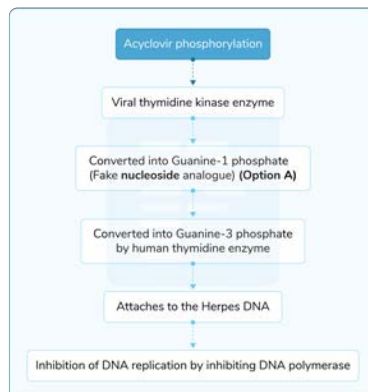
Solution for Question 7:

Correct Answer: A) It is an antiviral drug that works as a nucleotide analogue, preventing viral replication.

Explanation:

Acyclovir is the drug of choice for treating genital herpes simplex virus presented with painful fluid-filled vesicles with a red erythematous base associated with burning and itchy sensation.

Acyclovir:



- Uses: (Option D) Herpes simplex virus-1 and 2 Genital herpes. Mucocutaneous herpes simplex (use IV). Herpes labialis, Keratoconjunctivitis (use topical). HSV encephalitis (use IV). Varicella zoster virus (VZV). Epstein Barr virus.
- Herpes simplex virus-1 and 2 Genital herpes. Mucocutaneous herpes simplex (use IV). Herpes labialis, Keratoconjunctivitis (use topical). HSV encephalitis (use IV).
- Genital herpes.
- Mucocutaneous herpes simplex (use IV).
- Herpes labialis, Keratoconjunctivitis (use topical).
- HSV encephalitis (use IV).
- Varicella zoster virus (VZV).

- Epstein Barr virus.
- Route: Oral, IV, topical (Option B)
- Congeners: Valaciclovir: most effective for VZV. Famciclovir.
- Valaciclovir: most effective for VZV.
- Famciclovir.
- Resistance: Associated with viral thymidine kinase mutation and AIDS. Foscarnet and Cidofovir (do not require thymidine kinase activation).
- Associated with viral thymidine kinase mutation and AIDS.
- Foscarnet and Cidofovir (do not require thymidine kinase activation).
- Side effects: Neurotoxicity: seizures, confusion, hallucinations. Nephrotoxicity: Obstructive renal failure (crystalluria). Contraindicated with cyclosporine and probenecid. Administer plenty of IV fluid and water (Option C). Thrombocytopenia.
- Neurotoxicity: seizures, confusion, hallucinations.
- Nephrotoxicity: Obstructive renal failure (crystalluria). Contraindicated with cyclosporine and probenecid. Administer plenty of IV fluid and water (Option C).
- Contraindicated with cyclosporine and probenecid.
- Administer plenty of IV fluid and water (Option C).
- Thrombocytopenia.
- Herpes simplex virus-1 and 2 Genital herpes. Mucocutaneous herpes simplex (use IV). Herpes labialis, Keratoconjunctivitis (use topical). HSV encephalitis (use IV).
- Genital herpes.
- Mucocutaneous herpes simplex (use IV).
- Herpes labialis, Keratoconjunctivitis (use topical).
- HSV encephalitis (use IV).
- Varicella zoster virus (VZV).
- Epstein Barr virus.
- Genital herpes.
- Mucocutaneous herpes simplex (use IV).
- Herpes labialis, Keratoconjunctivitis (use topical).
- HSV encephalitis (use IV).
- Valaciclovir: most effective for VZV.
- Famciclovir.
- Associated with viral thymidine kinase mutation and AIDS.
- Foscarnet and Cidofovir (do not require thymidine kinase activation).

- Neurotoxicity: seizures, confusion, hallucinations.
 - Nephrotoxicity: Obstructive renal failure (crystalluria). Contraindicated with cyclosporine and probenecid. Administer plenty of IV fluid and water (Option C).
 - Contraindicated with cyclosporine and probenecid.
 - Administer plenty of IV fluid and water (Option C).
 - Thrombocytopenia.
 - Contraindicated with cyclosporine and probenecid.
 - Administer plenty of IV fluid and water (Option C).
-

Solution for Question 8:

Correct Answer: B) Inhibits viral RNA polymerase, preventing viral replication

Explanation:

Baloxavir:

- MOA: Inhibits the cap-dependent endonuclease of RNA polymerase → Inhibits initiation of mRNA synthesis → Prevents viral replication.
 - Use: Acute uncomplicated influenza A and B in > 12 years of age who is symptomatic for < 48 hours. Post-exposure prophylaxis in > 12 years of age
 - Acute uncomplicated influenza A and B in > 12 years of age who is symptomatic for < 48 hours.
 - Post-exposure prophylaxis in > 12 years of age
 - Route: Single oral dose.
 - Side effect: Diarrhoea.
 - Avoid co-administration with dairy products, antacids, or divalent supplements (calcium, iron, zinc).
 - Acute uncomplicated influenza A and B in > 12 years of age who is symptomatic for < 48 hours.
 - Post-exposure prophylaxis in > 12 years of age
-

Solution for Question 9:

Correct Answer: B) Oseltamivir

Explanation:

The preferred treatment for influenza is oral oseltamivir, which should be started within 48 hours of symptom onset for five days to minimise the duration and severity of the illness.

Management of Influenza (flu virus)		
Class	M-Protein inhibitors	Neuraminidase inhibitors
MOA	Blocks protein synthesis and virion assembly	Blocks viral release
Drug and Uses	1. Amantadine (Option C) DOC in influenza + parkinsons disease 2. Rimantidine	1. Oseltamavir (Tamiflu) Oral DOC: Influenza A and B 2. Zanamavir (Option D): Inhalational DOC: Resistant influenza A and B C/I in asthma and COPD 3. IV Peramavir 4. Lanimavir
Side effects	High resistance Ankle oedema Livedo reticularis	Nausea/vomiting Bronchospasm (zanamavir)

Reassurance (Option A) is not recommended as early treatment initiation within 48 hours of symptom onset reduces the duration of symptoms and the severity

Solution for Question 10:

Correct Answer: B) Ribavirin

Explanation:

Ribavirin is the drug of choice for the treatment of RSV.

Ribavirin:

- MOA: Guanine nucleotide synthesis inhibitor → Inhibition of RNA-dependent RNA polymerase.
- Uses: Active against all RNA viruses. Respiratory syncytial virus. Asthma-like illness: Bronchiolitis obliterans in newborn. Route: Inhalational. DOC for RSV prophylaxis: Palivizumab (Option D) HCV: Interferon alpha + Ribavirin (Life long treatment). Lassa fever. Haemorrhagic fever. Severe influenza (Intravenous route).
- Respiratory syncytial virus. Asthma-like illness: Bronchiolitis obliterans in newborn. Route: Inhalational. DOC for RSV prophylaxis: Palivizumab (Option D)
- Asthma-like illness: Bronchiolitis obliterans in newborn.
- Route: Inhalational.
- DOC for RSV prophylaxis: Palivizumab (Option D)
- HCV: Interferon alpha + Ribavirin (Life long treatment).
- Lassa fever.
- Haemorrhagic fever.
- Severe influenza (Intravenous route).
- Side effects: Aplastic anaemia.
- Respiratory syncytial virus. Asthma-like illness: Bronchiolitis obliterans in newborn. Route: Inhalational. DOC for RSV prophylaxis: Palivizumab (Option D)
- Asthma-like illness: Bronchiolitis obliterans in newborn.
- Route: Inhalational.
- DOC for RSV prophylaxis: Palivizumab (Option D)
- HCV: Interferon alpha + Ribavirin (Life long treatment).
- Lassa fever.
- Haemorrhagic fever.
- Severe influenza (Intravenous route).
- Asthma-like illness: Bronchiolitis obliterans in newborn.
- Route: Inhalational.
- DOC for RSV prophylaxis: Palivizumab (Option D)

Lenecapavir (Option A) is a capsid inhibitor used to treat HIV infection.

Interferon-beta (Option C) is used to treat papillomavirus, not RSV infection.

Solution for Question 11:

Correct Answer: A) Decreased efficacy of sofosbuvir

Explanation:

Anti-tubercular therapy was most likely initiated for this patient, who was receiving sofosbuvir treatment for chronic hepatitis and presenting with signs and symptoms of pulmonary tuberculosis (persistent cough, fever, and cavitary lesions on X-ray).

Rifampicin induces the P-glycoprotein pump (P-gp), which increases sofosbuvir excretion and decreases sofosbuvir's efficacy in this patient.

Increased risk of hepatotoxicity (Option B) is seen with voxilaprevir (NS3/4A Inhibitor) and ethinyl estradiol-containing hormonal contraception.

Increased risk of peripheral neuropathy (Option C) is seen with the use of isoniazid in slow acetylators.

Increased risk of fatal cardiac arrest (Option D) bradycardia is seen with concurrent use of sofosbuvir and amiodarone.

Solution for Question 12:

Correct Answer: B) 2 and 3

Explanation:

Regimen for Hepatitis C management:

- Sofosbuvir + Velpatasvir: Oral for 12 weeks (DOC)
- Sofosbuvir + Velpatasvir + Voxilaprevir: Oral for 12 weeks
- Glecaprevir + Pibrentasvir: Oral for 8 weeks

Management of Hepatitis C				
Directly acting antivirals (DAA): Drug of choice	Interferon alpha			
Type	NS3/4A Inhibitors	NS5A Inhibitors	NS5B Inhibitor	

MOA	Functions as a protease inhibitor, cleave functional protein	Inhibit viral replication and assembly	RNA polymerase inhibitor	Produce viral RNA degrading enzyme and inhibits exit of the virus
Drugs	Ends with -previr- 1. Simeprevir 2. Boceprevir 3. Voxilaprevir 4. Telaprevir	Ends with -Asvir- 1. Ledipasvir 2. Ombitasvir 3. Velpatasvir 4. Elbasvir 5. Daclatasvir	Ends with -bruvir- 1. Dasabuvir 2. Sofosbuvir	Pegylated interferon α Used in Hepatitis C, B and D DOC for Hepatitis B & D coinfection Kaposi Sarcoma CML Hepatitis induced thrombocytopenia
Adverse events	Avoid use with ethinyl estradiol (hepatotoxicity)	Avoid with antacids and potent P-gp inducers	Avoid use with amiodarone (severe bradycardia)	1. BM suppression 2. Hypo/ hyperthyroidism 3. Flu-like syndrome 4. Nephrotoxicity

Solution for Question 13:

Correct Answer: B) Tenofovir

Explanation:

Tenofovir is a nucleotide reverse transcriptase inhibitor.

Nucleoside reverse transcriptase inhibitors (NRTIs)	
MOA	Requires phosphorylation to convert into fake active nucleotides Competitive inhibition of reverse transcriptase (RNA-dependent DNA polymerase) enzyme Incorporated into the viral DNA chain during replication, leading to chain termination
Use	Active against HIV1 and HIV2
Pharmacokinetics	All excreted by the kidney except Zidovudine and Abacavir Metabolised by non-microsomal enzymes
Drugs	Mnemonic: DAZZELS D - Didanosine (Option D ruled out)

A - Abacavir (Option A ruled out) Z - Zidovudine Z - Zalcitabine E - Emtricitabine L - Lamivudine S - Stavudine (Option C ruled out)

Solution for Question 14:

Correct Answer: B) Initiate anti-tuberculosis treatment first and defer ART

Explanation:

Anti-tubercular therapy must be started first, and ART should be started after 2 weeks. This approach prioritises the immediate control of the TB infection while planning for the timely initiation of ART within 2-8 weeks.

This balances the need to manage both conditions effectively and reduces the risk of complications such as Immune Reconstitution Inflammatory Syndrome (IRIS).

Initiation of ART first or anti-tubercular drugs with ART simultaneously can potentially result in IRIS. (Options A and C)

Prophylactic treatment for opportunistic infections (e.g., Pneumocystis pneumonia) is important in HIV patients, especially those with low CD4 counts. However, in the presence of active TB, initiating specific TB treatment takes precedence. (Option D)

Solution for Question 15:

Correct Answer: D) Tenofovir + Lamivudine + Dolutegravir

Explanation:

This combination of Tenofovir+ Lamivudine+dolutegravir (Option D) is widely recommended as the first-line regimen for PEP due to its effectiveness and tolerability.

A 4-week regimen of at least 2 ARV drugs (mostly 2 RTIs) has been found to reduce the risk of HIV infection significantly if started within 72 hours after exposure.

Solution for Question 16:

Correct Answer: C) Tenofovir + Lamivudine + Dolutegravir

Explanation:

- The first-line regimen is tenofovir (TDF 300 mg) + Lamivudine (3TC 300 mg) + Dolutegravir (DTG 50 mg) regimen (TLD), which is a fixed-dose combination given in a single pill once a day (at a fixed time every day as per the patient's convenience).
- This regimen provides a harmonized treatment approach for all adults, adolescents, and pregnant women with HIV-1, HIV-2, prior nevirapine exposure, and those co-infected with TB or hepatitis.
- It's a simple, potent, and well-tolerated option that facilitates decentralized service delivery, simplifies the supply chain, and reduces monitoring requirements.
- Tenofovir and lamivudine are both NRTIs. Dolutegravir is an integrase inhibitor with high efficacy, good tolerability, and a high barrier to resistance.

Abacavir + Lamivudine + Dolutegravir (Option A)

The regimen of choice for:

- Children aged 6-10 years and body weight of 20-30kg.
- People living with HIV <30kg.
- HIV patients with high serum creatinine values.

Tenofovir+ Lamivudine + Efavirenz (Option B)

- The regimen of choice for women of childbearing age group who do not wish to take dolutegravir.

Abacavir + Lamivudine + Lopinavir (Option D)

- The regimen of choice for children aged <6 years and body weight of <20kg.
-

Solution for Question 17:

Correct Answer: D) Maraviroc

Explanation:

Maraviroc is a CCR5 inhibitor designed to block the CCR5 receptor on the surface of T cells. By doing so, it prevents HIV from entering these cells, thereby inhibiting the infection process.

The other options listed are not CCR5 inhibitors but belong to different classes of HIV treatment.

Drugs	Mechanism of Action
Entry Inhibitors	Block the HIV virus from entering the host cell.
CCR5 Inhibitor (Maraviroc) (Option D)	Bind to the CCR5 co-receptor on the host cell membrane. Only attachment inhibitors active against both HIV 1&2
Fusion Inhibitors (Enfuvirtide) (Option A)	Prevent the fusion of the HIV virus with the host cell membrane.
Attachment Inhibitors (Fostemsavir) (Option B)	Bind to the gp120 subunit of the HIV envelope, preventing attachment to the CD4 receptor.
Post-Attachment Inhibitors (Ibalizumab) (Option C)	Bind to CD4 receptors, preventing conformational changes needed for HIV entry.

Solution for Question 18:

Correct Answer: C) Cobicistat

Explanation:

- Cobicistat (Option C) is a pharmacokinetic enhancer specifically used to boost the effectiveness of certain protease inhibitors, including darunavir.
- It works by inhibiting cytochrome CYP3A4, which increases the plasma concentration of darunavir, thereby enhancing its antiviral activity.
- Low-dose ritonavir can also be used as a booster combined with lopinavir and darunavir.

Atazanavir, efavirenz, and lamivudine (Option A, B&D) are also anti-HIV drugs but are not used as boosters.

Solution for Question 19:

Correct Answer: A) Indinavir

Explanation:

Indinavir (Option A) is a protease inhibitor associated with developing nephrolithiasis (Kidney stones - Radiolucent on X-Ray / may be radiolucent on CT as well) due to its poor solubility in urine.

- It can lead to the formation of crystals in the urine, causing flank pain, hematuria, and potential obstruction.
- Indinavir also causes hyperbilirubinemia and jaundice

Ritonavir, atazanavir, and darunavir (Option B,C&D) are protease inhibitors but are not associated with nephrolithiasis. The table below shows their side effects.

Ritonavir	Gastrointestinal disturbances. Hyperlipidemia. Hepatotoxicity.
Atazanavir	Indirect hyperbilirubinemia (jaundice).
Darunavir	Rash. Liver function abnormalities. Gastrointestinal disturbances.

Solution for Question 20:

Correct Answer: B) Statements 1 and 2 are correct.

Explanation:

Protease inhibitors (PI): (-navir-)	
MOA	Inhibits the assembly/packaging of viral proteins Competitively inhibits the action of HIV aspartyl protease
Pharmacokinetics	All are CYP3A4 inhibitors: Except Nelfinavir (CYP2C9) Increases toxicity of other drugs Simvastatin: Risk of myopathy Midazolam: Causes respiratory depression
Uses	HIV treatment and 2nd line Post-exposure prophylaxis
Drugs: RAFT ISLAND	Ritonavir Maximum CYP inhibition → inhibits metabolism of other PI Low-dose ritonavir boosts other protease inhibitors (Lopinavir) Amprenavir Fos-amprenavir Tipranavir Indinavir Saquinavir (least CYP inhibitor) Lopinavir Atazanavir (no dyslipidemia) Nelfinavir Darunavir (maximum bioavailability, taken with food)
Resistance	P-glycoprotein efflux transporter Cross-resistance with other protease inhibitors.

Solution for Question 21:

Correct Answer: A) 1:b, 2:c, 3:a, 4:d

Explanation:

Integrase Inhibitors:

MOA	Prevent the formation of covalent bonds between host and viral DNA. Inhibit DNA called strand transfer.
Drugs	(ends with the word -gravir): BCDER Bictegravir Cabotegravir Dolutegravir (most used in India and safe during pregnancy) Elvitegravir (used in combination with cobicistat)

	Raltegravir
Route	All oral drugs except Cabotegravir (IM, once every 3 months)
Metabolism	CYP3A4 substrates Rifampicin reduces the effect, hence increasing the dose of the integrase inhibitor
Uses	Treatment and post-exposure prophylaxis of HIV Dolutegravir and Lamivudine: First line in treatment-naïve HIV patients
Side effects	Rashes Steven Johnson syndrome: Epidermis of the skin is damaged Raltegravir: Rhabdomyolysis, DRESS Bictegravir: Increase metformin concentration, hepatotoxicity Avoid administration with divalent cations (magnesium- or aluminium-containing antacids, iron supplements)

Solution for Question 22:

Correct Answer: C) Nevirapine

Explanation:

In low-risk infants whose mothers are well-adherent to antiretroviral therapy (ART) and have a proven low viral load, nevirapine is the treatment of choice for neonatal HIV-1 prophylaxis.

Non-nucleoside reverse transcriptase inhibitor (NNRTI)- Safe in pregnancy			
Drug	Zidovudine (Option A)	Nevirapine (NVP)	Efavirenz (Option B)
Metabolism	No CYP enzyme interaction	CYP3A4 inducer	Mixed (inducer and inhibitor)

Use	1. Used for neonatal prophylaxis in: Mother with HIV-2 or HIV-1+HIV-2 co-infection (viral load <1000 copies/ml) Mother received NVP during past pregnancy Mother on PI-based ART regimen.	1. First line for HIV-1 neonatal prophylaxis in low-risk infants (viral load <1000 copies/ml) 2. Treatment in adults and children	Treatment of HIV-1 in adults & children > 3 months
Side effects	Erythrocytic macrocytosis Insomnia Nail hyperpigmentation Skeletal myopathy Hepatotoxicity	Hepatotoxicity, Rash, pruritus Steven-Johnson syndrome Lower plasma levels of coadministered CYP3A4 substrates Methadone withdrawal Contraceptive failure	CNS effects Insomnia, Depression Psychosis, dysphoria Peripheral neuropathy Increase cholesterol Lower levels of efavirenz with phenobarbital, phenytoin, and carbamazepine.

Dual ARV Prophylaxis (NVP+Zidovudine) (Option D) is recommended for high-risk infants born to HIV-positive mothers who have high viral load (>1000 copies/ml), are not on ART, or are newly diagnosed HIV-positive (within 6 weeks of delivery).

Non-nucleoside reverse transcriptase inhibitor (NNRTI) Drugs (NEED RTI)

- Nevirapine
- Efavirenz
- Etravirine
- Delavirdine
- Doravirine
- Rilpivirine

Solution for Question 23:

Correct Answer: A) It is an NRTI (nucleotide reverse transcriptase inhibitor) that requires phosphorylation activation.

Explanation:

Tenofovir (Nucleotide analogue RTI)	
MOA	No phosphorylation required (Option A) Competitively inhibits viral reverse transcriptase Incorporates into HIV DNA, causing chain termination (due to incomplete ribose ring)
Types	Tenofovir disoproxil: short-acting, more side effects (Option B) Tenofovir alafenamide: long-acting
Uses	DOC for HIV DOC for Hepatitis B virus + HIV (Option C) HIV preexposure prophylaxis (in combination with emtricitabine)
Side effect	Nephrotoxic (Acute renal failure) Regular creatinine and phosphorus monitoring. Avoid with NSAIDs Fanconi syndrome (causes damage to PCT of nephron) Leads to phosphate loss Hypophosphatemic rickets, osteoporosis Hepatitis B exacerbation with abrupt discontinuation
Contraindication(Option D)	Renal disease (High creatinine) Replace tenofovir with Zidovudine or Abacavir Child < 10 years Weight < 30 kg

Solution for Question 24:

Correct Answer: C) Zidovudine

Explanation:

Zidovudine, a NRTI is known for its hematologic side effects, which include anaemia and neutropenia.

- 2nd line drug for anemic children is Abacavir

Side Effects of
NRTIs

Common side effect	Bone marrow suppression Alopecia Myopathy Pancreatitis Peripheral neuropathy Hepatic steatosis Lipodystrophy: Obese, type 2 DM Lactic acidosis: Inhibition of human mitochondrial respiration
Zidovudine (Option C)	Bone marrow suppression (maximum risk): anaemia, leukopenia Contraindicated: Severe anaemia if Hb <9 gm/dl Treatment: Erythropoietin injections Erythrocytic macrocytosis Nail hyperpigmentation Alopecia, myopathy (due to depletion of mitochondrial DNA) Hepatotoxicity and steatosis Insomnia
Didanosine	Pancreatitis (maximum risk)
Stavudine	Maximum risk for: Peripheral neuropathy Hepatic steatosis Lipodystrophy Lactic acidosis
Abacavir (Option A)	Steven-Johnson syndrome (fatal hypersensitivity) with HLAB 5701 (Contraindicated if tested positive) Fever, abdominal pain, and rash within 6 weeks of starting abacavir necessitates immediate discontinuation Myocardial infarction
Emtricitabine (Option D)	Skin hyperpigmentation in sun-exposed areas Exacerbation of hepatitis B with abrupt discontinuation
Lamivudine (Option B)	Exacerbation of hepatitis B with abrupt discontinuation (Safest NRTI)
Never combined NRTIs	Increased toxicity: Stavudine, Didanosine and Zalcitabine Cross-resistance: Lamivudine and Emtricitabine Zidovudine and Stavudine

Anti-Parasitic and Anti Protozoal Drugs

1. A 26-year-old male patient presents to the OPD. He is travelling to Bangladesh for a period of 12 weeks for work. What is the drug of choice for malaria prophylaxis for this patient?

- A. Primaquine
- B. Doxycycline
- C. Chloroquine
- D. Mefloquine

2. Match the following diseases with their respective drugs of choice

DISEASES	DRUGS OF CHOICE
1. Sleeping Sickness (West African Type) - Early Stage	A. Fexinidazole
2. Sleeping Sickness (West African Type) - Late Stage	B. Suramin
3. Sleeping Sickness (East African Type) - Early Stage	C. Nifurtimox-eflornithine combination
4. Sleeping Sickness (East African Type) - Late Stage	D. Melarsoprol
5. Chagas Disease	E. Benznidazole
6. Pneumocystis jirovecii	F. Cotrimoxazole (TMP/SMX)

Table rendering failed

- A. 1-A, 2-C, 3-B, 4-D, 5-E, 6-F
- B. 1-D, 2-A, 3-E, 4-C, 5-F, 6-B
- C. 1-A, 2-B, 3-C, 4-D, 5-E, 6-F
- D. 1-E, 2-C, 3-F, 4-D, 5-B, 6-A

3. Match the following diseases with their respective drug of choice for treatment.

Diseases **Drugs of Choice** **Giardiasis** **Toxoplasma gondii** **Balantidium coli** **Naegleria**
Pneumocystis pneumonia **A. Metronidazole or Tinidazole** **B. Amphotericin B** **C.**
Tetracycline **D. Sulfadiazine + Pyrimethamine** **E. Cotrimoxazole**

Diseases	Drugs of Choice
Giardiasis	A. Metronidazole or Tinidazole
Toxoplasma gondii	B. Amphotericin B
Balantidium coli	C. Tetracycline
Naegleria	D. Sulfadiazine + Pyrimethamine
Pneumocystis pneumonia	E. Cotrimoxazole

- A. 1-D, 2-A, 3-B, 4-E, 5-C
- B. 1-A, 2-D, 3-C, 4-B, 5-E
- C. 1-A, 2-B, 3-C, 4-E, 5-D
- D. 1-E, 2-B, 3-C, 4-D, 5-A

4. A 42-year-old male patient presents to the clinic with a chronic, non-healing ulcer on his arm. The ulcer started as a small, painless papule and gradually enlarged over the past three months. A biopsy of the lesion shows amastigotes of Leishmania species, confirming a diagnosis of cutaneous leishmaniasis. Which of the following is the drug of choice for treating this cutaneous lesion?

- A. Liposomal Amphotericin B
- B. Pentavalent antimonials
- C. Miltefosine
- D. Paromomycin

5. A 28-year-old female presents with intermittent abdominal pain, diarrhea, and flatulence for two weeks after consuming local street food. Examination reveals mild lower abdominal tenderness. Stool tests show Entamoeba histolytica cysts,

confirming luminal amebiasis. What is the drug of choice for treating luminal amebiasis in this patient?

- A. Metronidazole
 - B. Tinidazole
 - C. Paromomycin
 - D. Quiniodochlor
-

6. A 45-year-old male presents with symptoms of severe malaria. He complains of high fever, chills, profuse sweating, altered mental status, and jaundice. Laboratory investigations reveal thrombocytopenia, elevated serum creatinine, and metabolic acidosis. A blood smear confirms the presence of Plasmodium falciparum infection. Which of the following antimalarial medications is the most appropriate initial treatment for this patient's complicated malaria?

- A. Chloroquine
 - B. Artemether
 - C. Quinine
 - D. Artesunate
-

7. A 28-year-old male traveller presents to the clinic for malaria prophylaxis before his trip to a region where Plasmodium vivax is endemic. He has a history of G6PD deficiency. Which of the following antimalarial medications should be avoided due to the risk of hemolysis in individuals with G6PD deficiency?

- A. Quinine
 - B. Primaquine
 - C. Proguanil
 - D. Mefloquine
-

8. A 35-year-old male traveller returns from a recent trip to a malaria-endemic region. He presents to the clinic with complaints of blurred vision and halos around lights. He reports taking chloroquine prophylaxis during his trip. Which of the following ophthalmic side effects is most likely associated with his antimalarial medication?

- A. Optic neuritis
 - B. Corneal deposits
 - C. Retinal detachment
 - D. Glaucoma
-

9. Which of the following antimalarials is safe to use in all trimesters?

- A. Primaquine
 - B. Artemisinin
 - C. Clindamycin
 - D. Doxycycline
-

10. Which of the following drug combinations is used to treat malaria infections due to multiple plasmodium species?

- A. Quinine + Doxycycline
 - B. Chloroquine + Primaquine
 - C. Parenteral ACT + Oral ACT
 - D. ACT+ Primaquine
-

11. Which of the following mechanisms is characteristic of albendazole (benzimidazole-class) anti-helminthic drugs?

- A. Inhibition of acetylcholinesterase
 - B. Activation of GABA receptors
 - C. Inhibition of microtubule synthesis
 - D. Blockade of nicotinic acetylcholine receptors
-

12. Which of the following drugs is used for an effective radical cure in malarial fever?

- A. Artemisinin
 - B. Mefloquine
 - C. Atovaquone
 - D. Primaquine
-

13. Match the following stages of malaria with the corresponding drugs used
Stages of malaria Drugs
1. Sporozoites A. Tafenoquine, Primaquine
2. Hepatic schizogony B. Mosquirix vaccine
3. Gametocytes C. Proguanil, Primaquine
4. Hypnozoites D. Only primaquine

Stages of malaria	Drugs
1. Sporozoites	A. Tafenoquine, Primaquine
2. Hepatic schizogony	B. Mosquirix vaccine
3. Gametocytes	C. Proguanil, Primaquine
4. Hypnozoites	D. Only primaquine

- A. 1-c, 2-d, 3-a, 4-b
- B. 1-d, 2-a, 3-b, 4-c

C. 1-a, 2-b, 3-c, 4-d

D. 1-b, 2-c, 3-d, 4-a

14. A 35-year-old male patient is admitted to the hospital with high fever, chills, and severe headache. Upon examination and laboratory tests, he is diagnosed with *P.falciparum* malaria. The patient's medical history reveals that he has schizophrenia and is currently on antipsychotic medication. Considering the patient's medical history, which of the following antimalarial treatments is contraindicated for this patient?

A. Artemether-lumefantrine

B. Quinine

C. Mefloquine

D. Atovaquone-proguanil

15. A 45-year-old male patient presents with complaints of intense itching, especially at night, and a rash that has developed over the past few weeks. On examination, multiple small, red papules and burrows are noted in the interdigital spaces of his hands, wrists, and waistline. The patient mentions that his wife and children are also experiencing similar symptoms. Which of the following is the most appropriate treatment for this patient's condition?

A. Lindane

B. Permethrin

C. Sulfur

D. Dicophane

16. Match the following diseases to their respective drug of choice for treatment.

DISEASES DRUG OF CHOICE 1. Diphyllbothriasis A. Albendazole 2.

Neurocysticercosis B. Praziquantel 3. Lymphatic Filariasis C. Diethylcarbamazine (DEC) 4. Liver Fluke Infection D. Triclabendazole

DISEASES	DRUG OF CHOICE
1. Diphyllbothriasis	A. Albendazole
2. Neurocysticercosis	B. Praziquantel
3. Lymphatic Filariasis	C. Diethylcarbamazine (DEC)
4. Liver Fluke Infection	D. Triclabendazole

A. 1-B, 2-A, 3-C, 4-D

B. 1-C, 2-B, 3-D, 4-A

C. 1-D, 2-C, 3-B, 4-A

D. 1-A, 2-D, 3-B, 4-C

17. A 30-year-old woman visits a tropical medicine clinic after returning from a humanitarian trip to sub-Saharan Africa. She reports experiencing intense itching and skin changes. Blood tests reveal the presence of *Loa loa* microfilariae at a level of 40,000 microfilariae/mL of blood. Which of the following medications is contraindicated in this patient?

A. Ivermectin

B. Pyrantel pamoate

C. Piperazine

D. Niclosamide

18. Which of the following drugs is contraindicated in ocular cysticercosis?

A. Levamisole

- B. Oxamniquine
 - C. Praziquantel
 - D. Metrifonate
-

19. An 18-month-old child presents with complaints of recurrent abdominal discomfort, irritability, and poor weight gain. Upon examination, the child appears pale with no signs of acute distress. Stool examination reveals the presence of ova and parasites. Which of the following medications is the most appropriate for the treatment of this child's condition?

- A. Mebendazole
 - B. Pyrantel pamoate
 - C. Albendazole
 - D. Praziquantel
-

20. Which of the following is false regarding the benzimidazole class of antihelminths?

- A. Acts by inhibiting microtubule synthesis
 - B. High bioavailability
 - C. Acts by reducing glucose transport
 - D. Contraindicated in those less than 1 year of age
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	2
Question 4	1
Question 5	3
Question 6	4
Question 7	4
Question 8	2
Question 9	3
Question 10	4
Question 11	3
Question 12	4
Question 13	4
Question 14	3
Question 15	2
Question 16	1
Question 17	1
Question 18	3
Question 19	3
Question 20	2

Solution for Question 1:

Correct Answer: D) Mefloquine

Explanation:

Mefloquine is the DOC for travelers going to malaria endemic regions for a period longer than 6 weeks.

Causal prophylaxis	Suppressive prophylaxis
The pre-erythrocytic phase (in liver) is the target for this prophylaxis.	The schizonts seen in the erythrocytic phase are the target for this prophylaxis
Before the onset of symptoms	The exoerythrocytic phase continues, but clinical disease does not appear
Targets the cause of malaria.	Given to travellers going to countries with malaria
Drugs used include Primaquine, Proguanil, Doxycycline	Drugs used include Chloroquine, Doxycycline, Mefloquine, Proguanil
	Non-endemic countries: Chloroquine started 2 weeks before travel and continued for 4 weeks after return (600mg weekly). Endemic countries (chloroquine-resistant): Travel < 6 weeks: Doxycycline started 1 day before travel and continued for 4 weeks after return (100 mg daily). Travel > 6 weeks: Mefloquine started 2 weeks before travel and continued for 4 weeks after return (250 mg weekly).

Solution for Question 2:

Correct Answer: A) 1-A, 2-C, 3-B, 4-D, 5-E, 6-F

Explanation:

Disease	Early Stage Treatment	Late Stage Treatment	Drug of Choice (DOC)	Also Used
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Sleeping sickness: West African Type (<i>T. brucei gambiense</i>)	Fexinidazole	Nifurtimox-eflornithine combination therapy (NECT)	Fexinidazole	Pentamidine (in children < 6 yrs old or < 20 kg)
East African Type (<i>T. brucei rhodesiense</i>)	Suramin	Melarsoprol	Suramin	Melarsoprol
Chagas Disease (<i>T. cruzi</i> infection)			Benznidazole	Nifurtimox
Pneumocystis jirovecii			Cotrimoxazole (TMP/SMX)	

Fexinidazole is the recommended front-line treatment in Europe and Africa for adults with early-stage *T. brucei gambiense* and for less severe late-stage disease with CNS involvement.

Solution for Question 3:

Correct Answer: B) 1-A, 2-D, 3-C, 4-B, 5-E

Explanation:

Drug	Uses	Pregnancy
Metronidazole	Treats amebiasis, giardiasis, trichomoniasis & other anaerobic infections. 1st line drug in all forms of amebiasis Part of combination therapy for <i>H. pylori</i> infection.	Crosses placenta and enters breast milk, caution in pregnancy
Sulfadiazine & Pyrimethamine	Along with folic acid used as DOC for <i>Toxoplasma</i> encephalitis. Treats congenital toxoplasmosis in infants and mothers.	Used for intermittent preventive treatment in pregnancy (IPTp) during the second and third trimesters. Teratogenic at high doses.
Tetracycline	DOC in <i>Balantidium coli</i>	Not recommended in pregnancy

Amphotericin B	Liposomal amphotericin B is the drug of choice for Leishmania donovani. Naegleria infections.	Drug of choice for pregnant and breastfeeding women with visceral leishmaniasis
Cotrimoxazole	DOC for both the treatment and prevention of Pneumocystis jirovecii infection, particularly in HIV-positive individuals. DOC in Cyclospora & Isospora infections - Diarrhea in HIV-positive individuals	

Solution for Question 4:

Correct Answer: A) Liposomal Amphotericin B

Explanation:

Liposomal Amphotericin B is the drug of choice for cutaneous leishmaniasis

LEISHMANIASIS			
Drug	Route	Duration	Notes
Liposomal Amphotericin B (Option A)	IV		Drug of choice for Kala-azar, especially in pregnancy and children. 10 mg/kg single dose Drug of choice for visceral and cutaneous leishmaniasis
Miltefosine (Option C ruled out)	Oral	30 days	Only oral drug for Kala-azar. C/I in pregnancy (teratogenic) and children. Drug of choice for Post Kala-azar dermal leishmaniasis (PKDL)
Paromomycin (Option D ruled out)	IM	30 days	Used in combination with miltefosine for 30 days (Lower efficacy than miltefosine)

Amphotericin B deoxycholate	IV	30 d ays	Second-line drug for visceral leishmaniasis according to the NVBDCP. Effective for antimony-resistant cases and for immunocompromised patients.
Sodium stibogluconate (Pentavalent antimonial) (Option B ruled out)	IV	30 d ays	Least preferred treatment for Kala-azar due to resistance. Initially made for kala-azar, also known as pentavalent antimonial. Contains antimony (Sb), which is toxic to Leishmania. S/E: QT prolongation Pancreatitis Nephrotoxic

Solution for Question 5:

Correct Answer: C) Paromomycin

Explanation:

Paromomycin is the drug of choice for treating luminal amebiasis. It is an aminoglycoside antibiotic not absorbed from the gastrointestinal tract, effectively eradicating *Entamoeba histolytica* cysts and trophozoites in the intestinal lumen without affecting the systemic tissues.

Table rendering failed

Solution for Question 6:

Correct Answer: D) Artesunate

Explanation:

According to current guidelines, Artesunate is the preferred initial treatment for severe or complicated malaria caused by *Plasmodium falciparum*. It has been shown to reduce parasite burden and mortality in severe malaria cases rapidly. Other options like quinine and artemether are also effective, but artesunate is generally favoured due to its faster onset of action and superior efficacy in severe malaria. Chloroquine is not recommended for the treatment of severe malaria.

Complicated malaria is a life-threatening form of malaria caused by *P. falciparum*. Treatment is given parenterally.

- DOC: IV Artemisinin
- 2nd: IM artemether/artether
- 3rd: IV Quinine in 5% dextrose (quinine is hypoglycaemic)

Solution for Question 7:

Correct Answer: D) Mefloquine

Explanation:

Mefloquine is safe to use in G6PD deficiency. Its other side effects include severe neurological and psychiatric side effects such as vivid dreams, psychosis, and seizures. It is metabolised by CYP3A4, which is induced by drugs like rifampin.

Glucose-6-phosphate dehydrogenase (G6PD) deficiency impairs RBCs' ability to withstand oxidative stress. Certain medications, like antimalarials, can trigger hemolysis in individuals with G6PD deficiency by inducing oxidative stress within rRBCs.

Antimalarials safe in G6PD deficiency	Antimalarials NOT safe in G6PD deficiency
Mefloquine (safest)(Option D) Chloroquine	Quinine (Option A ruled out) Primaquine (Option B ruled out) Proguanil (Option C ruled out)

Solution for Question 8:

Correct Answer: B) Corneal deposits

Explanation:

Corneal deposits are a recognised ophthalmic side effect of chloroquine therapy for malaria prophylaxis or treatment. These deposits can lead to symptoms such as blurred vision and halos around lights. Optic neuritis, retinal detachment, glaucoma, and macular edema are not typically associated with chloroquine use for malaria.

CHLOROQUINE	
MOA	Chloroquine accumulates in acidic compartments of parasite-infected cells, disrupting heme polymerisation and inhibiting parasite growth and replication.
Resistance	Due to mutations in genes encoding transporters involved in chloroquine uptake and efflux, reducing chloroquine accumulation within parasites.
Uses	Malaria Treatment and Prophylaxis Amoebiasis Lepra reaction Giardiasis Autoimmune Diseases (RA, SLE)
S/E	Ophthalmic: Bull's eye maculopathy, anterior sub capsular cataract, vortex keratopathy, corneal deposits Gastrointestinal: Nausea, vomiting, diarrhea Skin: Itching, rash, photosensitivity Neuropsychiatric: Agitation, confusion, hallucinations Cardiac: QT prolongation, cardiomyopathy (rare)

Solution for Question 9:

Correct Answer: C) Clindamycin

Explanation:

Clindamycin is safe to use in all trimesters of pregnancy.

Safe in all trimesters	Contraindicated in 1st trimester only	Contraindicated in all trimesters
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Chloroquine Quinine Clindamycin (Option C)	Artesunate combination therapy (ACT) (Option B)	Primaquine (Option A) Doxycycline (Option D)
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Treatment:

P. Vivax/ovale	P. falciparum	Complicated malaria
Primaquine contraindicated. It is not cured; relapse is treated after pregnancy.	Chloroquine resistance is seen. 1st trimester: Quinine + clindamycin 2nd trimester onwards: ACT	Life-threatening to the mother All life-saving measures are taken despite teratogenicity IV Artesunate is used regardless of trimester

Solution for Question 10:

Correct Answer: D) ACT+ Primaquine

Explanation:

ACT (artemisinin-based combination therapy) + Primaquine is used to treat mixed malarial infections.

The National Vector Borne Diseases Control Programme (NVBDPC) provides guidelines for the early diagnosis and prompt, complete, and effective treatment of malaria.

NVBDPC guidelines:

Scenario	Treatment
Uncomplicated P. Vivax malaria	Chloroquine 25 mg/kg over 3 days (10-10-5) + Primaquine oral 0.25 mg/kg for 14 days
Complicated malaria	Parenteral ACT-Artesunate i.v. given for 2 days, followed by a full course of oral ACT. OR Parenteral quinine i.v. given for 2 days, and oral quinine given for 5 days. Add on oral doxycycline
Mixed infection (P. vivax/ovale/falciparum)	ACT+ Primaquine for 14 days
Pregnancy	1st trimester: Parenteral quinine 2nd and 3rd trimester: Parenteral artemisinin

Solution for Question 11:

Correct Answer: C) Inhibition of microtubule synthesis

Explanation:

Albendazoles (Benzimidazoles) act by inhibition of microtubule synthesis

Anthelmintics and their mechanism of action:

Microtubule inhibitors (Option C)	Spastic paralysis	Flaccid paralysis
Benzimidazole group: Albendazole Mebendazole Triclabendazole Thiabendazole	Praziquantel: Calcium influx Levamisole: Nicotinic muscle receptor stimulation Pyrantel pamoate: muscle depolarisation	Piperazine: Opens GABA-A Cl ⁻ channel Ivermectin: Opens glutamate Cl ⁻ channel

Solution for Question 12:

Correct Answer: D) Primaquine

Explanation:

Primaquine is the DOC for effective radical cure in malarial fever.

Clinical Cure	Radical Cure
Erythrocytic schizonts are killed to prevent an episode of malarial fever.	Hypnozoites(dormant stage) are killed to prevent relapse of malarial fever.
Seen in all plasmodium species	Seen in P. Ovale and P. Vivax
All drugs are effective except primaquine	DOC: Primaquine and Tafenoquine

Solution for Question 13:

Correct Answer: D) 1-b, 2-c, 3-d, 4-a

Explanation:

Stages of malaria	Drugs acting on them
Hepatic schizogony (pre-erythrocytic phase) Causal prophylaxis	Primaquine Proguanil Doxycycline
Erythrocytic schizogony Clinical cure	All drugs except primaquine
Hypnozoites(dormant form) Radical cure	Primaquine Tafenoquine
Gametocytes	Primaquine
Sporozoites	Mosquirix vaccine RTS,S/AS01 vaccine Given i.m. in children Sporozoites are coated in HbSAg, leading to a T-cell response against the sporozoites.

Solution for Question 14:

Correct Answer: C) Mefloquine

Explanation:

Mefloquine is C/I in patients with a history of psychiatric disorders such as schizophrenia. Mefloquine can exacerbate psychiatric symptoms, potentially leading to severe psychiatric reactions, including anxiety, paranoia, depression, hallucinations, and even suicidal thoughts.

The choice of antimalarial drug depends on the species of Plasmodium causing the infection, the severity of the infection, drug resistance patterns in the area, and individual patient factors such as pregnancy and age.

ANTI-MALARIAL			
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DRUGS			
Drug Class	Mechanism of Action	Drugs & Indications	Notes
Chloroquine Group	Inhibits the conversion of hemoglobin to hemozoin in parasites	Chloroquine: Treatment of malaria (clinical cure), lepra reaction, atopic dermatitis Amodiaquine and Piperaquine: Treatment of malaria.	Chloroquine can cause ocular toxicity, including bull's eye maculopathy, anterior subcapsular cataracts, and vortex keratopathy.
Artemisinin Group	Forms free radicals that damage parasite proteins and DNA	Artesunate: Treatment of malaria (clinical cure), including complicated malaria (IV administration). Artemether, Arteether, and Arterolane: Treatment of malaria (clinical cure).	Artemisinin is obtained from the Chinese herb <i>Artesuna annua</i> .
Quinoline	Blocks potassium channels in parasites, leading to paralysis	Quinine: Treatment of malaria (clinical cure), including complicated malaria (IV administration). Quinidine: Treatment of malaria (clinical cure) and used as a Class IA antiarrhythmic drug.	Quinine is obtained from the cinchona plant and can cause ototoxicity (cinchonism), hypoglycemia, QT prolongation, hypotension, and diarrhea.
Primaquine	Disrupts the parasite's electron transport chain	Radical cure of malaria caused by <i>P. vivax</i> and <i>P. ovale</i> (targets hypnozoites).	Primaquine is not effective against erythrocytic schizogony and, therefore not used for clinical cure.
Antifolates	Inhibit dihydrofolate reductase, essential for parasite folate synthesis	Pyrimethamine and Sulfadoxine: Treatment of malaria (clinical cure), typically in combination with artesunate. Proguanil: Treatment of malaria (clinical cure) and prophylaxis. Sulfadoxine: Treatment of malaria Typically in combination with artesunate.	These drugs should be combined with folic acid (tetrahydrofolate) to prevent megaloblastic anemia.
Antibiotics	Various mechanisms targeting parasite protein synthesis	Doxycycline: Treatment of malaria (clinical cure and causal prophylaxis), pneumocystis pneumonia (in combination with primaquine).	Doxycycline is C/I in pregnancy and children <8 years old.

		Clindamycin: Treatment of malaria (clinical cure), pneumocystis pneumonia (in combination with primaquine), toxoplasmosis.	
Others		Mefloquine: Treatment of malaria (clinical cure and suppressive prophylaxis), particularly chloroquine-resistant strains. Lumefantrine: Treatment of malaria (clinical cure), particularly with artemether in northeastern Indian states. Tafenoquine: Radical cure of malaria caused by P. vivax and P. ovale. Atovaquone: Treatment of malaria (clinical cure) and pneumocystis pneumonia (in combination with dapsone).	Mefloquine has a high risk of neuropsychiatric symptoms and is C/I in cerebral malaria. Lumefantrine can cause QT prolongation.

Solution for Question 15:

Correct Answer: B) Permethrin

Explanation:

The clinical scenario describes a patient with classic symptoms of scabies, A highly contagious parasitic infection caused by *Sarcoptes scabiei* that presents as an intensely itchy rash, often affecting the hands, wrists, and genitals. Permethrin is the most efficacious and convenient drug for treating scabies.

DRUGS FOR SCABIES	
5 % Permethrin (1st line)	A pyrethroid insecticide that is highly effective, with nearly 100% cure rates in scabies after a single application.
Lindane (2nd line)	A broad-spectrum insecticide, but less effective than permethrin and has a higher risk of systemic toxicity,

	particularly in children.
Benzyl Benzoate (2nd line)	Minimally absorbed through the skin and has low systemic toxicity but can cause neurological symptoms in children.
Crotamiton (2nd line)	Less prone to cause skin irritation and has low systemic toxicity but has produced lower cure rates in scabies.
Sulfur	The oldest scabicide but has several disadvantages, including messy application, unpleasant odour, and the need for repeated applications.
Dicophane (DDT)	Can be absorbed through the skin and cause systemic toxicity, rarely used due to potential side effects and safer alternatives.
Ivermectin	An oral medication that is highly effective in treating scabies. Not to be given to children <5 yrs old, pregnant, or breastfeeding women.

Solution for Question 16:

Correct Answer: A) 1-B, 2-A, 3-C, 4-D

Explanation:

Condition	Treatment	Other Information
Cestodes	Praziquantel	DOC for all cestodes. C/I in ocular cysticercosis
Niclosamide	2nd line drug for cestodes	
Albendazole	DOC for Echinococcus (Dog tapeworm) - Causes Hydatid cyst	
Neurocysticercosis	Albendazole	First, corticosteroids are given to prevent inflammation, followed by Albendazole to kill the worms. Corticosteroids and anti-epileptic drugs (Phenytoin) are then continued for six months. Starting with Albendazole alone can trigger an inflammatory reaction

Nematodes	Albendazole	DOC for all Nematodes, except Strongyloides, Onchocerca volvulus, Wuchereria, Brugia malayi/timori, Loa-loa, Dracunculus, and Microsporidium
Mebendazole		
Ivermectin	DOC for Strongyloides (Thread worm) and Onchocerca volvulus (causes river blindness); Also used for Scabies.	
DEC (Diethylcarbamazine)	DOC for Wuchereria, Brugia malayi/timori, and Loa-loa; Add Ivermectin or Albendazole to DEC.	
Metronidazole	DOC for Dracunculus (Guinea worm)	
Pyrantel pamoate	2nd line drug for Nematode infection	
Piperazine	Used for treatment of Ascariasis and Enterobius	
Levamisole	2nd line drug for treatment of Ascariasis; Given in the treatment of steroid-resistant nephrotic syndrome	
Trematodes	Praziquantel	DOC for all trematodes; For Schistosoma hematobium: Praziquantel + Metrifonate; For Schistosoma mansoni: Praziquantel + Oxamniquine
Triclabendazole	DOC for Fasciola hepatica (Liver fluke)	
Microsporidium	Albendazole	May cause Keratitis, which should be treated with eye drops - Topical fumagillin

Solution for Question 17:

Correct Answer: A) Ivermectin

Explanation:

The patient exhibits signs and symptoms indicative of loiasis.

- High microfilarial load is detected.
- Ivermectin is contraindicated for *Loa loa* microfilaremia levels $\geq 30,000$ microfilariae/mL of blood.
- Risk of severe and potentially fatal encephalopathies with Ivermectin use in such cases.

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Drug	Mechanism of Action	Clinical Uses	Other Information
Pyra ntel Pam oate (Opti on B)	Depolarisation of muscle by opening non-selective cation channels in worms, leading to spastic paralysis.	2nd line drug for Nematodes	
Piper azin e (Opti on C)	Opens - GABA-A Cl channel , hyperpolarizes muscle & causes flaccid paralysis.	To treat Ascariasis and Enterobius	C/I in renal disease and epilepsy. Safe in pregnancy.
Niclo sami de (Opti on D)	Calcium influx causing spastic muscle paralysis	2nd line drug for cestodes. Used to treat intestinal T. solium infections, especially with suspected neurocysticercosis.	C/I in ocular cysticercosis as it causes eye inflammation leading to blindness

Solution for Question 18:

Correct Answer: C) Praziquantel

Explanation:

Praziquantel is C/I in ocular cysticercosis as it can trigger an inflammatory response, potentially leading to vision loss.

PRAZIQUANTEL

MOA	Causes calcium ion influx into the tapeworm, resulting in paralysis, detachment from the intestine, and death. At higher concentrations, it breaks down the tapeworm's outer layer.
Pharmacokinetics	Extensively metabolised in the liver ($\Rightarrow$ less bioavailability)
Uses	Highly effective for all human tapeworm infections, especially for <i>T. solium</i> cysticercosis, where larvae invade tissues like the brain, muscles, and skin.
Adverse effects	In neurocysticercosis, killing larval cysts in the brain can trigger inflammation, leading to seizures and ■ intracranial pressure. C/I in ocular cysticercosis due to the risk of intense inflammatory response, tissue damage, and vision loss

Levamisole (Option A)

- Levamisole is an antiparasitic drug that is primarily used to treat ascariasis (2nd line drug) and ancylostomiasis.
- MOA: Stimulates Nm receptor causing spastic paralysis.
- A low dose of levamisole is an immunostimulant; High dose: Immunosuppression $\rightarrow$ bone marrow suppression (agranulocytosis).
- Used in the treatment of steroid-resistant nephrotic syndrome.

Oxamniquine (Option B) is used primarily to treat schistosomiasis caused by *Schistosoma mansoni* but is ineffective for *S. haematobium* and *S. japonicum* infections.

Metrifonate (Option D) is used primarily to treat infections caused by *Schistosoma haematobium*.

Solution for Question 19:

Correct Answer: C) Albendazole

Explanation:

Albendazole is a broad-spectrum benzimidazole anthelmintic medication commonly used to treat parasitic infections, is the treatment of choice in pediatric patients with evidence of parasitic infection and symptoms such as recurrent abdominal discomfort and poor weight gain.

- It is used for mass deworming and treating infections including those caused by intestinal worms such as nematode infection, cysticercosis, cutaneous larva migrans, toxocariasis and echinococcosis.
- It is C/I in 1st trimester of pregnancy and in age <1 year.

Mebendazole (Option A) is contraindicated in this age group. It is approved for use in >2 years of age.

Pyrantel pamoate (Option B) is a second-line drug used to treat intestinal nematode infections.

Praziquantel (Option D) primarily treats infections caused by tapeworms and flukes (except *Fasciola hepatica*, where DOC is Triclabendazole). It is not indicated for treating common intestinal worms seen in this age group.

Solution for Question 20:

Correct Answer: B) It has high bioavailability

Explanation:

Benzimidazoles have low bioavailability

Benzimidazoles (Albendazole, Mebendazole, Triclabendazole)	
MOA	Microtubule inhibition and subsequent disruption of cell division (Option A) Reduced glucose transport (Option C)
Pharmacokinetics	Low bioavailability (Option B is the false statement) Increased absorption when taken with fatty meals
Side effects	Anorexia, nausea/vomiting, dizziness Hypersensitivity Aplastic anemia Hepatotoxicity
Pregnancy	Contraindicated in 1st trimester Given in 2nd and 3rd trimesters (Only in severe cases)
Children	Albendazole: C/I in <1 year of age (Option D) Mebendazole: C/I in <2 years of age

Dose for deworming	Albendazole 1-2 years: 200mg oral tablet >2 years: 400mg oral tablet Mebendazole >2 years: 100mg oral tablet
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Antifungal Drugs

1. Which of the following drugs acts by inhibiting 1,3-beta-glucan synthase?

- A. Natamycin
 - B. Fluconazole
 - C. Caspofungin
 - D. Terbinafine
-

2. Which of the following is NOT true regarding Amphotericin B?

- A. Acts by binding to ergosterol in fungal cell membranes.
 - B. Liposomal AMB is associated with lower nephrotoxicity than conventional AMB.
 - C. AMB is absorbed orally, making it suitable for treatment with systemic mycoses.
 - D. Drug of choice for the treatment of cryptococcal meningitis.
-

3. Which drug belongs to the triazole group of antifungals known for its broad spectrum against yeast and moulds, including mucormycosis?

- A. Miconazole
 - B. Posaconazole
 - C. Clotrimazole
 - D. Ketoconazole
-

4. All of the following azoles are primarily excreted through bile except?

- A. Ketoconazole
- B. Itraconazole
- C. Posaconazole
- D. Fluconazole

5. A 65-year-old woman visits the dermatology clinic with a persistent and intensely itchy rash on her arms, legs, and trunk. She mentions that over-the-counter antihistamines have provided minimal relief. Which azole drug with antipruritic effects would be most suitable for managing her condition?



- A. Fluconazole
- B. Ketoconazole
- C. Itraconazole
- D. Voriconazole

6. Match the following diseases with their respective drugs of choice: Diseases
 Drugs of Choice Fungal Keratitis a. Natamycin Seborrheic Dermatitis b.
 Ketoconazole Mucormycosis c. Amphotericin B Sporotrichosis d. Itraconazole

Diseases	Drugs of Choice
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Fungal Keratitis	a. Natamycin
Seborrheic Dermatitis	b. Ketoconazole
Mucormycosis	c. Amphotericin B
Sporotrichosis	d. Itraconazole

- A. 1-a, 2-b, 3-c, 4-d
- B. 1-c, 2-a, 3-d, 4-b
- C. 1-a, 2-b, 3-d, 4-c
- D. 1-d, 2-c, 3-b, 4-a

7. A kidney transplant patient receiving cyclosporine develops a new fungal infection and is prescribed ketoconazole for treatment. What is the potential consequence of adding ketoconazole to the patient's regimen?

- A. Decreased effectiveness of cyclosporine
- B. Elevated risk of toxicity with cyclosporine
- C. No significant interaction between ketoconazole and cyclosporine
- D. Enhanced elimination of cyclosporine

8. Match the following antifungal drugs with their properties, uses, and specific adverse effects: 1.Fluconazole a. Preferred for Candida infections; Can cause QT prolongation 2.Itraconazole b. Preferred for Aspergillus infections; Can cause visual disturbances 3.Voriconazole c. Preferred for mucormycosis; Can cause hepatotoxicity 4.Posaconazole d. Preferred for prophylaxis in immunocompromised patients; Can cause gastrointestinal upset

1.Fluconazole	a. Preferred for Candida infections; Can cause QT prolongation
2.Itraconazole	b. Preferred for Aspergillus infections; Can cause visual disturbances

3.Voriconazole	c. Preferred for mucormycosis; Can cause hepatotoxicity
4.Posaconazole	d. Preferred for prophylaxis in immunocompromised patients; Can cause gastrointestinal upset

- A. 1-d, 2-c, 3-b, 4-a
- B. 1-a, 2-b, 3-c, 4-d
- C. 1-b, 2-c, 3-a, 4-d
- D. 1-c, 2-d, 3-b, 4-a

9. A 23-year-old man comes to the dermatology OPD. After examination, a diagnosis of onychomycosis was made, and he was started on oral terbinafine. What is the mechanism of action of this drug?



- A. Inhibition of thymidylate synthase
- B. Inhibition of lanosterol 14-methylase
- C. Inhibition of squalene epoxidase
- D. Formation of pore in ergosterol membrane

10. An 18-year-old high school student presents to the dermatology outpatient clinic with symptoms of tinea pedis and tinea capitis. The dermatologist opts for an antifungal agent that inhibits microtubules within fungal cells, thereby preventing mitosis. Which drug was most likely prescribed?

- A. Ciclopirox olamine
 - B. Griseofulvin
 - C. Nystatin
 - D. Caspofungin
-

11. A 65-year-old diabetic patient is admitted to the ICU with meningitis. He reported contact with pigeon droppings two weeks prior. India ink staining of his CSF sample confirmed *Cryptococcus neoformans* infection. What is the most appropriate drug combination for treating this fungal infection?

- A. Caspofungin + Flucytosine
 - B. Amphotericin B + Flucytosine
 - C. Voriconazole + Caspofungin
 - D. Amphotericin B + Posaconazole
-

12. Match the following superficial fungal infection with the appropriate drug for treatment. FUNGAL INFECTION TOPICAL DRUG
1. Vulvo-vaginal candidiasis a. Griseofulvin
2. Tinea capitis b. Miconazole
3. Tinea versicolor c. Clotrimazole
4. Seborrheic dermatitis d. Ketoconazole

FUNGAL INFECTION	TOPICAL DRUG
1. Vulvo-vaginal candidiasis	a. Griseofulvin
2. Tinea capitis	b. Miconazole
3. Tinea versicolor	c. Clotrimazole
4. Seborrheic dermatitis	d. Ketoconazole

A. 1-c, 2-b, 3-d, 4-a

B. 1-a, 2-c, 3-b, 4-d

C. 1-c, 2-a, 3-b, 4-d

D. 1-d, 2-b, 3-a, 4-c

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	2
Question 4	4
Question 5	2
Question 6	2
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	2
Question 12	3

Solution for Question 1:

Correct Answer: C) Caspofungin

Explanation:

Caspofungin, an echinocandin antifungal, inhibits 1,3-beta-glucan synthase, disrupting fungal cell wall synthesis and causing cell lysis and death.

- Cell wall inhibiting antifungals: Act by inhibiting the enzyme 1,3-beta-glucan synthase, essential for fungal cell wall formation. Caspofungin Micafungin Anidulafungin
- Caspofungin
- Micafungin

- Anidulafungin
- Cell membrane inhibiting antifungals:
- Caspofungin
- Micafungin
- Anidulafungin

Drugs	Mechanism of action
Natamycin (Option A)	Binds to ergosterol in the fungal cell membrane, disrupting membrane integrity.
Terbinafine (Option D)	It inhibits squalene epoxidase, reduces ergosterol synthesis, and affects cell membrane integrity.
Amphotericin B	Binds to ergosterol, forming pores in the fungal cell membrane, leading to cell death.
Fluconazole, Itraconazole, Ketoconazole (Option B)	Inhibits ergosterol synthesis, affecting cell membrane function.

Solution for Question 2:

Correct answer: C) AMB is absorbed orally, making it suitable for treatment with systemic mycoses.

Explanation:

Amphotericin B (AMB) is not absorbed orally, it is administered intravenously (IV) to achieve therapeutic levels in the bloodstream and treat systemic fungal infections.

Aspect	Amphotericin B
Mechanism of Action	Binds to ergosterol in fungal cell membranes (Option A), forming pores that disrupt membrane integrity, leading to leakage of cellular contents and fungal cell death.
Preparation	Conventional Amphotericin B (c-AMB) Amphotericin B lipid complex (ABLC) Liposomal Amphotericin B (L-AMB)
Use	Cryptococcal meningitis (Option D), mucormycosis, coccidioidal meningitis.
Adverse Effects	Nephrotoxicity: Most notable with conventional Amphotericin B (c-AMB); less with liposomal

formulations (Option B). Fever, chills, and hypotension can be managed with pre-treatment (NSAIDs, antihistamines). Anemia (requires erythropoietin treatment). Renal tubular acidosis leads to electrolyte imbalances (hypomagnesemia, hypokalemia). Arachnoiditis, when used intrathecally

Solution for Question 3:

Correct Answer: B) Posaconazole

Explanation:

Posaconazole is the triazole antifungal known for its broad spectrum against yeast and moulds, including mucormycosis.

Table rendering failed

Aspect	Posaconazole
Mechanism of Action	Inhibits fungal sterol 14 α -demethylase, disrupting ergosterol synthesis in fungal cell membranes.
Spectrum of Activity	Broad spectrum against various fungi, including <i>Aspergillus</i> spp., <i>Candida</i> spp., and Mucorales (effective against mucormycosis).
Adverse Effects	Common: Nausea, abdominal pain, diarrhoea, headache, dizziness. Rare: Anaemia, neutropenia, cardiac arrhythmias, visual disturbances.
Route of Administration	Absorption of oral suspension is enhanced by low pH and fatty foods.

Solution for Question 4:

Correct answer: D) Fluconazole

Explanation:

Fluconazole is primarily excreted through urine. It has a high renal clearance and is eliminated largely unchanged in the urine, making it contraindicated in renal disease.

Pharmacokinetics of Azoles:

- Absorption: Oral Absorption: Azoles generally have good oral bioavailability, but this varies among agents. Some, like itraconazole and posaconazole, absorb better in acidic environments or with fatty food. Topical Absorption: Clotrimazole and ketoconazole penetrate skin layers effectively, treating superficial fungal infections.
- Oral Absorption: Azoles generally have good oral bioavailability, but this varies among agents. Some, like itraconazole and posaconazole, absorb better in acidic environments or with fatty food.
- Topical Absorption: Clotrimazole and ketoconazole penetrate skin layers effectively, treating superficial fungal infections.
- Oral Absorption: Azoles generally have good oral bioavailability, but this varies among agents. Some, like itraconazole and posaconazole, absorb better in acidic environments or with fatty food.
- Topical Absorption: Clotrimazole and ketoconazole penetrate skin layers effectively, treating superficial fungal infections.
- Distribution: Azoles distribute widely into tissues and body fluids, including skin, nails, CSF, and eyes. However, CNS penetration is limited, except for fluconazole.
- Metabolism: Azoles are metabolized in the liver via CYP3A4, forming inactive metabolites. Ketoconazole and Itraconazole inhibit CYP3A4, influencing drug interactions.
- Azoles are metabolized in the liver via CYP3A4, forming inactive metabolites.
- Ketoconazole and Itraconazole inhibit CYP3A4, influencing drug interactions.
- Azoles are metabolized in the liver via CYP3A4, forming inactive metabolites.
- Ketoconazole and Itraconazole inhibit CYP3A4, influencing drug interactions.

Excretion: Primarily excreted via hepatic metabolism, except fluconazole, which is excreted unchanged in urine.

Solution for Question 5:

Correct answer: B) Ketoconazole

Explanation:

- Ketoconazole, an azole antifungal, possesses anti-inflammatory and antipruritic properties, making it well-suited for managing dermatological conditions associated with itching.
- Fluconazole, Itraconazole, Voriconazole, and Posaconazole are other azole drugs that lack significant anti-pruritic properties.

Solution for Question 6:

Correct Answer: B) 1-c, 2-a, 3-d, 4-b

Explanation:

Disease	Preferred antifungals	Mechanism of Action
Fungal Keratitis	Natamycin	Disrupts fungal cell membrane integrity.
Seborrheic Dermatitis	Ketoconazole (topical and oral)	Inhibits fungal cytochrome P450-dependent enzyme.
Mucormycosis	Amphotericin B	Disrupts fungal cell membrane integrity.
Sporotrichosis	Amphotericin B (induction), Itraconazole (maintenance)	Inhibits ergosterol synthesis, essential for fungal membranes.

Solution for Question 7:

Correct Answer: B) Elevated risk of toxicity with cyclosporine.

Explanation:

- Ketoconazole is a potent inhibitor of the CYP3A4 enzyme system, which extensively metabolises cyclosporine.

- Inhibiting CYP3A4 with ketoconazole reduces cyclosporine metabolism, increasing blood levels of cyclosporine.
- Elevated cyclosporine levels can result in toxicity, including nephrotoxicity (kidney damage), hepatotoxicity (liver damage), and neurotoxicity (nervous system damage).

Adverse Effects of Ketoconazole:

Gastrointestinal	Nausea, vomiting, abdominal pain, diarrhoea.
Hepatic	Elevated liver enzymes, hepatotoxicity (rare but serious)
Endocrine	Gynecomastia, decreased libido, menstrual irregularities
Skin	Rash, photosensitivity
Neurological	Headache, dizziness
Others	Adrenal insufficiency (with prolonged use at high doses)

Solution for Question 8:

Correct Answer: B) 1-a, 2-b, 3-c, 4-d

Explanation:

	Fluconazole	Itraconazole	Voriconazole	Posaconazole
Mechanism of Action	Inhibits fungal cytochrome P450-dependent enzyme lanosterol 14 α -demethylase, leading to inhibition of ergosterol synthesis.			
Pharmacokinetics	Well-absorbed orally. Metabolised by CYP2C19. Excreted in bile except fluconazole.			
Uses	Candida infections	Aspergillus infecti	Mucormyc	Prophylaxis in immunocompromise

		ons	osis	d patients
Adverse Effects	QT prolongation	Visual disturbances	Hepatotoxicity	Gastrointestinal upset

Solution for Question 9:

Correct Answer: C) Inhibition of squalene epoxidase

Explanation:

Aspects	Terbinafine
Mechanism of action	Competitive inhibition of squalene epoxidase (Option C) ↓ Accumulation of squalene within fungal cells ↓ Inhibition of ergosterol synthesis ↓ Disruption of fungal cell membrane integrity ↓ Fungicidal action
Features	It has a high affinity for sebum and thus accumulates in sebum, stratum corneum and nails.
Uses	Topical: Used for treating tinea infections (ringworm) and pityriasis versicolor (fungal skin infection). Oral: DOC for onychomycosis (fungal infection of the nails), tinea capitis (scalp ringworm), and extensive fungal lesions. It is also used in oromucosal candidiasis as an alternative to fluconazole.
Side effects	Taste disturbance and rash; topical application causes dryness, itching and irritation.

Inhibition of thymidylate synthase (Option A)

- Flucytosine (5-FC) converts into 5-fluorodeoxyuridylic acid within the fungal cell and acts as an inhibitor of thymidylate synthase.
- It is a narrow-spectrum fungistatic agent active towards *Cryptococcus neoformans* and *chromoblastomyces*.

Inhibition of lanosterol 14-methylase (Option B)

- Azoles inhibit fungal cytochrome P450 enzyme - lanosterol 15-methylase, which results in abnormal membrane formation.\
- It is a broad-spectrum antifungal agent that is active against dermatophytes, candida and other deep fungal infections (except mucormycosis)

Formation of pore in ergosterol membrane (Option D)

- Amphotericin B has lipophilic and lipophobic sides, forming micropores in the ergosterol membrane of the fungal wall.
- It is a broad-spectrum fungicidal agent effective against Candida, Histoplasma, Sporothrix, Blastomycosis, and Leishmania.

Solution for Question 10:

Correct Answer: B) Griseofulvin

Explanation:

Aspects	Griseofulvin
MOA	Inhibition of microtubules within the fungal cells, thus inhibiting mitosis. It results in the formation of multi-nucleated and stunted fungal hyphae.
Features	Keratinophilic thus gets deposited in high concentration in Stratum corneum, hair and nails Plasma half-life is 24 hours
Uses	DOC in Superficial fungal infections such as Tinea. (T. capitis- most effective; T.curis , T.unguium, T.corporis- less effective; T. versicolor – it is not effective)
Side effects	Headache Neuritis Hepatitis Photosensitivity Disulfiram like reaction

Solution for Question 11:

Correct Answer: B) Amphotericin B + Flucytosine

Explanation:

The treatment of choice for cryptococcal meningitis combines IV Amphotericin B and IV Flucytosine.

	Mechanism of action	Uses	Side effects
Amphotericin B (Polyene antibiotic molecule)	Forms micropores in the ergosterol membrane.	Cryptococcus neoformans Leishmania	Nephrotoxicity Allergic reaction Arachnoiditis Anemia
Flucytosine (pyrimidine antimetabolite)	Inhibitor of thymidylate synthase	Cryptococcus neoformans Chromoblastomyces.	Dose-dependent marrow suppression Enteritis.

Caspofungin + Flucytosine (Option A) is used in the treatment of systemic candidiasis in an immunocompromised patient

Voriconazole + Caspofungin (Option C) is used to treat invasive aspergillosis.

Amphotericin B + Posaconazole (Option D) is used in the treatment of mucormycosis

Solution for Question 12:

Correct Answer: C) 1-c, 2-a, 3-b, 4-d

Explanation:

Topical antifungals:

Drug	Mechanism of action	Topical Uses
Imidazole Oxiconazole Clotrimazole Miconazole Sertaconazole Ketoconazole	Inhibition of lanosterol 14-methylase	Vulvovaginal candidiasis (Clotrimazole) Tinea infections Tinea versicolor (Miconazole) Onychomycosis Dandruff- Seborrheic dermatitis caused by Malassezia furfur

		(ketoconazole)
Triazole Fluconazole Itraconazole		
Griseofulvin	Inhibition of microtubules within the fungal cells, thus inhibiting mitosis.	Cases covering large area Onychomycosis Tinea capitis Athletes foot Pityriasis versicolor - not effective
Allylamine Terbinafine (topical and oral) Butenafine	Competitive inhibition of squalene epoxidase results in squalene accumulation within the cells.	Tinea infection and pityriasis versicolor
Other topical agents: Ciclopirox olamine, Undecylenic acid, Benzoic acid		

Properties of Anti-microbial Agents

1. A 30-year-old male patient presents with a fever and signs of acute hemolytic anemia. He has a medical history significant for glucose-6-phosphate dehydrogenase (G6PD) deficiency. Laboratory investigations reveal hemoglobinuria and a decreased hemoglobin level. The patient is diagnosed with a hemolytic crisis secondary to G6PD deficiency. Which of the following antibiotics is safest to administer to this patient?

- A. Ciprofloxacin
- B. Azithromycin
- C. Metronidazole
- D. Nitrofurantoin

2. A 28-year-old pregnant woman at 32 weeks gestation presents with symptoms suggestive of a urinary tract infection (UTI). She has a past medical history of penicillin allergy but is otherwise healthy. Considering her pregnancy status, which of the following antibiotics would be contraindicated due to potential adverse effects on the fetus?

- A. Erythromycin
- B. Cephalexin
- C. Amoxicillin
- D. Ciprofloxacin

3. Which of the following are not safe to use in severe renal disease?

- A. Linezolid

- B. Rifampicin
 - C. Tobramycin
 - D. Cefoperazone
-

4. Which of the following statements regarding Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) accurately distinguishes bactericidal from bacteriostatic drugs?

- A. Bactericidal drugs have a narrower range between MIC and MBC than bacteriostatic drugs
 - B. Bacteriostatic drugs show a higher MIC to MBC ratio than bactericidal drugs
 - C. Both drug types have similar MIC but different MBC values
 - D. Bactericidal drugs have higher MIC but lower MBC values than bacteriostatic drugs
-

5. What is the Post-Antibiotic Effect (PAE)?

- A. The ability of antibiotics to continue killing bacteria post-administration
 - B. The period during which bacteria develop resistance post-antibiotic exposure
 - C. The ability of antibiotics to suppress bacterial growth beyond MIC
 - D. Extended inhibition of bacterial regrowth post-antibiotic clearance
-

6. Which statement is true about acquired resistance?

- A. It depends on both the microorganism and the drug
- B. It depends only on the microorganism
- C. It depends only on the drug
- D. It is independent of the microorganism and the drug

7. The transfer of antimicrobial resistance through a virus called bacteriophage is called?

- A. Conjugation
 - B. Transduction
 - C. Transformation
 - D. Multistep mutation
-

8. Bacteria lacking a cell wall, such as Mycoplasma species, primarily exhibit resistance through which mechanism?

- A. Drug Efflux
 - B. Drug Influx
 - C. Enzyme Inactivation
 - D. Mutation in Target
-

9. When combining a bactericidal and bacteriostatic drug, what is the most likely outcome?

- A. Drug antagonism
 - B. Drug synergism
 - C. Both Drug antagonism and drug synergism
 - D. Neither Drug antagonism nor drug synergism
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	1
Question 5	4
Question 6	1
Question 7	2
Question 8	2
Question 9	1

Solution for Question 1:

Correct Answer: B) Azithromycin

Explanation:

Among the listed antibiotics, azithromycin, a macrolide, is generally considered safe for patients with G6PD deficiency.

Safe in G6PD deficiency	Unsafe in G6PD deficiency
Mnemonic: PCM Penicillins Cephalosporins Macrolides (Option B) Mefloquine	Mnemonic: MAD FAN Can Metronidazole (Option C ruled out) Antimalarials- except mefloquine Dapsone Fluoroquinolones (Option A ruled out) Aspirin Nitro drugs- Nitrofurantoin (Option D ruled out) Chloramphenicol

Solution for Question 2:

Correct Answer: D) Ciprofloxacin

Explanation:

Ciprofloxacin (fluoroquinolone) is contraindicated in pregnancy due to its association with potential adverse effects on fetal bone and cartilage.

Erythromycin, Cephalexin, and Amoxicillin are generally considered safe to use in pregnancy and are commonly prescribed for the treatment of UTIs during pregnancy, with appropriate consideration of maternal and fetal factors.

Safe in pregnancy	Unsafe in pregnancy
Mnemonic: PCM Penicillins (Option C ruled out) Cephalosporins (Option B ruled out) Macrolides (Option A ruled out) Mefloquine	Mnemonic: FATS Fluoroquinolones: Fetal bone and cartilage damage due to chelating effect (Option D) Aminoglycosides: Ototoxic to the fetus Tetracycline Sulphonamides: Fetal bone and cartilage damage due to chelating effect, kernicterus in 3rd trimester

Solution for Question 3:

Correct Answer: C) Tobramycin

Explanation:

Tobramycin (aminoglycoside) can pose risks in individuals with kidney disease due to its potential for nephrotoxicity. In people with compromised kidney function, tobramycin may not be effectively cleared from the body, leading to its accumulation and increased toxicity.

Renal safe antibiotics

Drugs that are excreted in bile are safe to use in renal disease (No dose reduction required)

Mnemonic: Totally Contraindicated In RENAL Failure Disease

Tigecycline

Cefoperazone (Option D ruled out)

Isoniazid

Rifampicin (Option B ruled out)

Erythromycin

Nafcillin

Azithromycin

Linezolid (Option A ruled out)

Fluoroquinolones- Pefloxacin, Moxifloxacin, Trovafloxacin

Doxycycline

Solution for Question 4:

Correct Answer: A) Bactericidal drugs have a narrower range between MIC and MBC than bacteriostatic drugs

Explanation:

Bactericidal drugs often have a smaller difference between the MIC (the concentration required to inhibit bacterial growth) and MBC (the concentration required to kill bacteria) than bacteriostatic drugs. This indicates a more potent killing effect at concentrations close to those needed to inhibit growth.

Bacteriostatic agents inhibit bacterial growth, typically measured by the Minimum Inhibitory Concentration (MIC), while bactericidal agents kill bacteria, usually measured by the Minimum Bactericidal Concentration (MBC). In other words, bacteriostatic prevents further growth, while bactericidal directly eliminates bacteria.

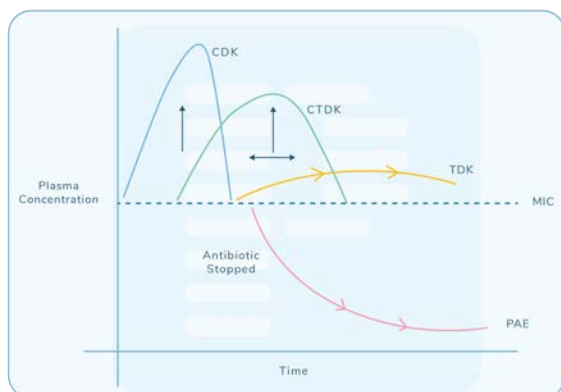
Bacteriostatic	Bactericidal
MIC<<<MBC Bacterial growth is restricted but not killed The remaining bacteria are killed by the immunity of the body Examples- Tetracyclines, macrolides	MIC = MBC Bacteria are completely killed Examples- Beta-lactams, Fluoroquinolones, aminoglycosides

Solution for Question 5:

Correct Answer: D) Extended inhibition of bacterial regrowth post-antibiotic clearance

Explanation:

The Post-Antibiotic Effect (PAE) refers to the continued suppression of bacterial growth or a delay in bacterial regrowth after a period of exposure to antibiotics, even after the drugs have been eliminated from the body.



Concentration-dependent killing (CDK)	Represents the dose of antimicrobials above MIC that can kill the microbe It is a single high-dose Examples- Aminoglycosides and metronidazole
Time-dependent killing (TDK)	Represents the duration over a longer period of time for antimicrobials to kill the bacteria It is multiple daily doses. Examples- Penicillin, Azithromycin
CTDK (combination of CDK & TDK)	CTDK is also known as area-dependent killing. Examples- Vancomycin, Fluoroquinolones
Post Antibiotic Effect (PAE)	Prolonged suppression of bacterial growth after antibiotic exposure, even when drug levels fall below MIC, due to the body's immune response. Advantage: Duration of therapy can be reduced. E.g. In TB, Rifampicin is given once a day due to its CDK property, but in leprosy, it is given once a month to utilise its PAE effectively.

Solution for Question 6:

Correct Answer: A) It depends on both the microorganism and the drug

Explanation:

Resistance is the unresponsiveness of a microorganism to an antimicrobial agent (AMA). It depends on both the microorganism and the drug. There are 2 types:

Natural resistance	Acquired resistance
Some microbes lack the metabolic process or target site the particular drug affects	Development of resistance by an organism (sensitive before) due to the use of an AMA over a period of time
e.g. gram-negative bacilli are unaffected by penicillin G aerobic organisms are not affected by metronidazole anaerobic bacteria are not inhibited by aminoglycoside antibiotics	The development of resistance is dependent on the microorganism as well as on the drug. Some bacteria rapidly acquire resistance, e.g. staphylococci, coliforms, tubercle bacilli, gonococci
This does not pose a significant clinical problem	This can happen with any microbe and is a major clinical problem
	They are of 2 types Mutation Gene transfer

Solution for Question 7:

Correct Answer: B) Transduction

Explanation:

Transduction is the transfer of a gene carrying antimicrobial resistance through the agency of a bacteriophage. It is a type of gene transfer.

Mutation	Gene transfer
It is a stable and heritable genetic change that occurs spontaneously and randomly among microorganisms	The gene that causes resistance is passed from one organism to another
Vertical transfer of resistance; relatively slow and usually of lower grade	Horizontal transfer of resistance; rapid spread; high-level resistance to several antibiotics
There are 2 types Single step: A single gene mutation confers high resistance, emerging rapidly (e.g., enterococci to	There are 3 types Conjugation: One bacteria acquires a resistance gene from another bacteria via plasmid (e.g., MDR TB,

streptomycin)

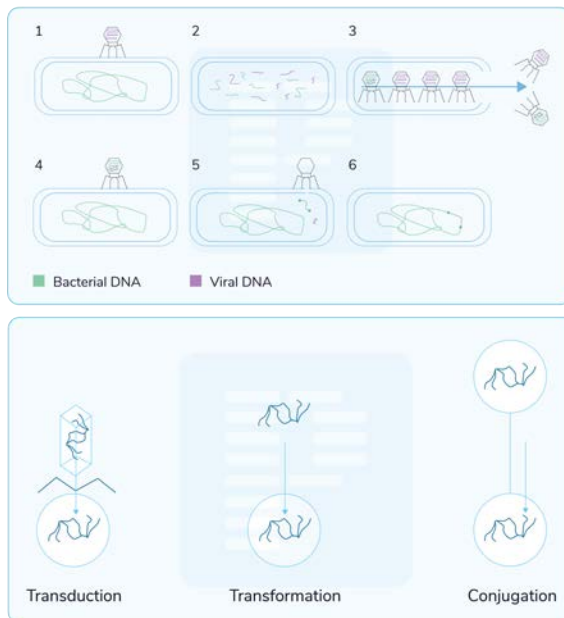
Multistep: Several gene modifications are involved; sensitivity decreases gradually stepwise. (e.g., resistance to erythromycin, tetracyclines and chloramphenicol) (Option D)

typhoid)(Option A)

Transduction: Bacteria acquire resistance genes from a bacteriophage (e.g., MRSA)

Transformation-Bacteria acquire genes directly from the environment (e.g., pneumococci, gonococci)(Option C)

TRANSDUCTION



Solution for Question 8:

Correct Answer: B) Drug Influx

Explanation:

Mycoplasma species, lacking a cell wall, resist antibiotics primarily by preventing the uptake of antimicrobial agents (AMAs) into the cell (Drug influx)

There are 4 main causes of antimicrobial resistance.

	Description	Examples
Drug Ef flux (Option A)	Bacteria possess efflux pumps to expel toxic substances and antimicrobial agents (AMAs)	Antimalarials (chloroquine), Tetracyclines, Tigecycline

Drug In flux (Option B)	Natural mechanisms in bacteria prevent the uptake of AMAs	Mycoplasma (resistant to β -lactams and glycopeptides due to lack of cell wall)
Enzyme inactivation (Option C)	Bacterial enzymes like β -lactamases inactivate β -lactam drugs by hydrolysing a specific site in their ring structure	β -lactams (penicillins, cephalosporins), Aminoglycosides, Chloramphenicol
Mutation in target (Option D)	Bacteria modify components of their cell to resist AMAs	β -lactam resistance in gram-positive bacteria via altered penicillin-binding proteins (PBPs), Fluoroquinolones (DNA gyrase mutation), Vancomycin (van genes altering peptidoglycan precursors)

Solution for Question 9:

Correct Answer: A) Drug antagonism

Explanation:

Table rendering failed

Anti-mycobacterial drugs

1. Which of the following medications primarily exerts its therapeutic effect by targeting mycobacterial ATP synthase?

- A. Bedaquiline
 - B. Levofloxacin
 - C. Linezolid
 - D. Thiacetazone
-

2. A 42-year-old male patient with a history of TB presents for a follow-up appointment. He has been compliant with his anti-TB medications. However, his recent sputum culture results indicate the persistence of TB bacilli. Which of the following mechanisms of resistance is not seen with isoniazid use?

- A. KatG gene mutation
 - B. KasA gene mutation
 - C. InhA gene overexpression
 - D. PncA gene mutation
-

3. A 35-year-old male currently on treatment for leprosy presents to the emergency department with a sudden onset of new lesions and enlargement of existing lesions. They appear red, swollen and tender to the touch. What is the type of hypersensitivity reaction seen in this condition?

- A. Type 1 hypersensitivity reaction
- B. Type 2 hypersensitivity reaction
- C. Type 3 hypersensitivity reaction

D. Type 4 hypersensitivity reaction

4. What is the drug of choice for post-exposure prophylaxis in leprosy?

- A. Rifampin 600 mg
 - B. Clofazimine 300 mg
 - C. Dapsone 100 mg
 - D. Clarithromycin 500 mg
-

5. A 40-year-old male patient presents with hypopigmented skin lesions (<5 in number) and numbness in her extremities. Examination reveals the presence of nerve thickening in the ulnar area. What is the duration of multidrug therapy in this patient?

- A. 3 months
 - B. 6 months
 - C. 9 months
 - D. 12 months
-

6. Which of the following drugs is contraindicated in patients with G6PD deficiency?

- A. Dapsone
 - B. Rifampin
 - C. Clarithromycin
 - D. Clofazimine
-

7. A 35-year-old female patient presents to the dermatology clinic with multiple hypopigmented skin lesions and numbness in her extremities. A diagnosis is made after a thorough examination and testing. Which among the following drugs used to treat this condition is bacteriostatic?

- A. Ofloxacin
 - B. Moxifloxacin
 - C. Dapsone
 - D. Rifampin
-

8. A 55-year-old male with a history of advanced HIV/AIDS presents for a routine follow-up. His CD4 count is 50 cells/mm³, and he has been adherent to antiretroviral therapy for the past 6 months. Upon review of his medical history, it is noted that he had a previous episode of disseminated Mycobacterium avium complex (MAC) infection. Which of the following is the most appropriate prophylactic regimen for this patient?

- A. Ethambutol monotherapy
 - B. Clarithromycin monotherapy
 - C. Azithromycin monotherapy
 - D. Rifabutin plus azithromycin combination therapy
-

9. A 42-year-old male with multidrug-resistant tuberculosis presents for initiation of treatment. Given his MDR-TB, the clinician decides to initiate oral bedaquiline therapy. Which of the following statements regarding oral bedaquiline therapy is true?

- A. Recommended in patients with fluoroquinolone resistance
- B. Contraindicated with HIV co-infection

- C. No risk of QT prolongation
 - D. Regular liver function tests required
-

10. A 35-year-old patient presents with a history of pulmonary tuberculosis that has failed to respond to standard first-line treatment. Despite completing the initial course of therapy, the patient's symptoms have worsened, and sputum cultures confirm the presence of multidrug-resistant tuberculosis. Which of the following medications is most appropriate for managing this patient's condition?

- A. Bedaquiline
 - B. Rifampicin
 - C. Levofloxacin
 - D. Isoniazid
-

11. A 40-year-old patient diagnosed with multidrug-resistant tuberculosis (MDR-TB) is prescribed delamanid as part of the treatment regimen. The patient asks the physician how delamanid works to treat tuberculosis. Which of the following explanations best describes the mechanism of action of delamanid?

- A. Inhibits DNA gyrase and topoisomerase IV enzymes in bacteria
 - B. Inhibits mycolic acid synthesis in the cell wall of M.tuberculosis
 - C. Disrupts bacterial protein synthesis by binding to the 23S ribosomal RNA
 - D. Inhibits the synthesis of arabinogalactan in the cell wall of M.tuberculosis
-

12. A 50-year-old male with multidrug-resistant tuberculosis (MDR-TB) presents with generalised weakness, cold intolerance, weight gain despite reduced appetite, decreased libido, and difficulty maintaining erections. Lab tests show elevated liver enzymes and TSH. Which medication is likely causing these effects?

- A. Ethambutol can cause side effects such as optic neuritis (vision changes), rash, and gastrointestinal disturbances.
 - B. Isoniazid
 - C. Ethionamide
 - D. Levofloxacin
-

13. Which of the following statements is false?

- A. Vitamin B6 is given to treat Cycloserine induced toxicity
 - B. Clofazimine causes urine discolouration
 - C. Cycloserine causes suicidal ideation
 - D. Clofazimine can cause constipation
-

14. A 58-year-old woman with a history of cirrhosis presents to the clinic with fever, night sweats, and a persistent cough. After evaluation, she is diagnosed with pulmonary tuberculosis (TB). Given her liver disease, which of the following antitubercular drugs needs to be avoided?

- A. Streptomycin
 - B. Rifampin
 - C. Pyrazinamide
 - D. Ethambutol
-

15. A 28-year-old HIV-positive male patient presents to the clinic reporting a recent contact with a sputum-positive TB patient. He has no active symptoms of TB. Which of the following is the recommended prophylactic regimen for TB in this patient?

- A. Isoniazid and pyrazinamide combination therapy
 - B. Isoniazid and ethambutol combination therapy
 - C. Isoniazid and streptomycin combination therapy
 - D. Isoniazid and rifapentine combination therapy
-

16. A patient is newly diagnosed with TB and is found to be HIV positive. Which of the following ART drugs do not require dose-adjustment with Rifampin?

- A. Nevirapine
 - B. Dolutegravir
 - C. Ritonavir
 - D. Efavirenz
-

17. Which of the following drugs is not used in TB patients with liver disease but is considered safe in renal disease?

- A. Streptomycin
 - B. Fluoroquinolone
 - C. Ethambutol
 - D. Rifampin
-

18. A 56-year-old male patient with newly diagnosed pulmonary TB begins treatment with ATT. After several weeks of therapy, he complains of difficulty distinguishing colors and experiencing blurred vision. On examination, he exhibits decreased visual acuity and bilateral central scotomas. Which of the following drugs is responsible for this condition?

- A. Rifampin

- B. Pyrazinamide
 - C. Streptomycin
 - D. Ethambutol
-

19. Which of the following drugs is not used in the continuation phase of Drug susceptible TB treatment?

- A. Isoniazid
 - B. Rifampin
 - C. Pyrazinamide
 - D. Ethambutol
-

20. A 45-year-old male with pulmonary tuberculosis presents with acute pain and swelling in his right big toe after 6 weeks of antitubercular therapy. On examination, his right big toe is tender, erythematous, and warm to touch. Lab tests reveal elevated serum uric acid levels. Which of the following antitubercular drugs is most likely responsible for his current presentation?

- A. Isoniazid
 - B. Rifampicin
 - C. Ethambutol
 - D. Pyrazinamide
-

21. Which rifamycin drug is specifically utilized in the treatment of hepatic encephalopathy?

- A. Rifampicin
- B. Rifapentine

- C. Rifabutin
 - D. Rifaximin
-

22. A 60-year-old male with a history of atrial fibrillation is on warfarin therapy to prevent thromboembolic events. He is diagnosed with active pulmonary tuberculosis and started on antitubercular therapy. However, upon follow-up, his international normalized ratio (INR) is found to be subtherapeutic despite stable warfarin dosing. Which of the following antitubercular drugs is most likely responsible for the decreased efficacy of warfarin?

- A. Isoniazid
 - B. Ethambutol
 - C. Pyrazinamide
 - D. Rifampicin
-

23. A 40-year-old man presents with a persistent cough, fever, and night sweats for the past two weeks. He has a history of close contact with a family member diagnosed with active tuberculosis. A sputum sample is collected and tested using the Cartridge-Based Nucleic Acid Amplification Test (CBNAAT), which detects the genetic material of *Mycobacterium tuberculosis* and mutations associated with drug resistance. Which of the following antitubercular drugs is most likely to be affected based on this result?

- A. Isoniazid
 - B. Ethambutol
 - C. Pyrazinamide
 - D. Rifampicin
-

24. Which of the following is not a symptom of isoniazid toxicity?

- A. Hepatotoxicity
 - B. Systemic lupus erythematosus
 - C. Optic neuritis
 - D. Renal failure
-

25. A 55-year-old female patient with newly diagnosed TB is prescribed a regimen as part of her treatment plan. The healthcare provider emphasizes the importance of monitoring for neurotoxicity as a potential adverse effect associated with this therapy. What dose of Vitamin B6 is given to prevent neurotoxicity with the use of this drug?

- A. 25-50 mg/day
 - B. 200 mg/day
 - C. 100 mg/day
 - D. 10 mg/day
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	3
Question 4	1
Question 5	2
Question 6	1
Question 7	3
Question 8	3
Question 9	4
Question 10	1
Question 11	2
Question 12	3
Question 13	4
Question 14	3
Question 15	4
Question 16	4
Question 17	4
Question 18	4
Question 19	3
Question 20	4
Question 21	4
Question 22	4

Question 23	4
Question 24	4
Question 25	4

Solution for Question 1:

Correct Answer: A) Bedaquiline

Explanation:

Bedaquiline Inhibits mycobacterial ATP synthase, a crucial enzyme involved in the energy production of Mycobacterium tuberculosis. By disrupting this process, bedaquiline effectively kills the bacteria.

- Cross-resistance is seen with clofazimine.
- It is metabolised by CYP3A4. Hence, rifampicin, a CYP3A4 inducer, reduces its efficacy.
- Side effects: Nausea and diarrhea (common side effects) QT prolongation
- Nausea and diarrhea (common side effects)
- QT prolongation
- The BPaL regimen is a combination of drugs used to treat drug-resistant tuberculosis (TB): Bedaquiline (B) Pretomanid (Pa) Linezolid (L)
- Bedaquiline (B)
- Pretomanid (Pa)
- Linezolid (L)
- Nausea and diarrhea (common side effects)
- QT prolongation
- Bedaquiline (B)
- Pretomanid (Pa)
- Linezolid (L)

Levofloxacin (Option B) is a fluoroquinolone antibiotic that inhibits DNA gyrase and topoisomerase IV enzymes in bacteria. It is safe to use in liver failure, unlike moxifloxacin.

Linezolid (Option C) inhibits bacteria's protein synthesis by binding to 23S ribosomal RNA of the 50S subunit. Side effects include chorioretinitis, optic neuritis, MAO inhibition, and bone marrow suppression.

Thiacetazone (Option D) acts by inhibiting mycolic acid synthesis in the cell wall of *Mycobacterium tuberculosis*.

Solution for Question 2:

Correct Answer: D) PncA gene mutation

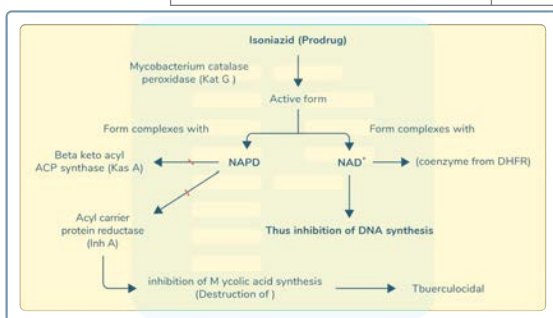
Explanation:

PncA gene mutation causes resistance to Rifampin, not isoniazid.

Isoniazid inhibits the synthesis of mycolic acids, which are fatty acid components of the mycobacterial cell wall.

The mechanisms of resistance with Isoniazid are given below.

Mechanism of resistance	
KatG gene mutation (Option A)	INH is a prodrug, converted to the active form by catalase-peroxidase produced by the KATg gene. The mutated catalase enzyme does not form the active drug
KasA gene mutation (Option B)	Component of cell wall mycolic acid
InhA gene overexpression (Option C)	Causes cross-resistance to Ethionamide (second-line drug)



Solution for Question 3:

Correct Answer: C) Type 3 hypersensitivity reaction

Explanation:

The patient is most likely to have a type 2 lepra reaction, which is a type 3 hypersensitivity reaction

Type 1 lepra reaction	Type 2 lepra reaction
aka Reversal reaction	aka Erythema nodosum leprosum
Delayed type 4 hypersensitivity reaction	Type 3 hypersensitivity reaction involving immune complex deposition
Cutaneous ulceration, multiple nerve involvement with swollen, painful, and tender nerves occur suddenly	Abrupt onset; existing lesions enlarge, become red, swollen, and painful. Several new lesions may appear.
Treatment Don't stop MDT Clofazimine or DOC: Corticosteroids (Prednisone) In mild hypersensitivity: Chloroquine	Treatment Don't stop MDT Clofazimine or DOC: Corticosteroids (Prednisone) or Thalidomide (corticosteroid resistant cases) In mild hypersensitivity: Chloroquine

Solution for Question 4:

Correct Answer: A) Rifampin 600 mg

Explanation:

Rifampin 600 mg single dose is the drug of choice for post-exposure prophylaxis for leprosy.

It is given in

- Healthcare professionals
- Contacts of index patients

Age group	Dose of Rifampicin
≥ 15 years	600 mg
10–14 years	450 mg
6–9 years (Weight ≥20 Kg)	300 mg

2–5 years (Weight ≤20 Kg)	10-15 mg/Kg
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Leprosy vaccines:

BCG (Bacillus Calmette Guérin)	MIP (Mycobacterium indicus pranii)
Live vaccine	Killed vaccine
Cannot be used in immunocompromised patients	Can be used in immunocompromised patients
Used for TB but also protects against leprosy	Used as an adjunct for leprosy PEP along with Rifampin

Solution for Question 5:

Correct Answer: B) 6 months

Explanation:

The duration of treatment in this patient with paucibacillary M.leprae infection is 6 months.

CHARACTERISTIC		PAUCIBACILLARY		MULTIBACILLARY	
Skin lesions		1-5 lesions		>5 lesions	
Nerve involvement		No nerve		≥1 nerve involved	
Slit skin smear		Negative		Positive	

		Treatment duration	Dapsone	Clofazimine	Rifampicin
Paucibacillary (PB)	Adult	6 months	100 mg daily	300 mg once a month & 50 mg daily	600 mg once a month
Child (9-14 yrs)	6 months	50 mg daily	150 mg once a month & 50 mg alternate day	450 mg once a month	
Multibacillary (MB)	Adult	12 months	100 mg daily	300 mg once a month & 50 mg daily	600 mg once a month
Child (9-14 yrs)	12 months	50 mg daily	150 mg once a month & 50 mg alternate day	450 mg once a month	

Single lesion paucibacillary:

- One skin lesion with no nerve involvement.
- In India: Treatment is similar to paucibacillary (Rifampicin, Dapsone, Clofazimine for 6 months)
- According to WHO: ROM therapy Rifampicin (600mg), Ofloxacin (400mg), Minocycline (100mg) - Single dose. For a child, half of these adult doses are administered.
- Rifampicin (600mg), Ofloxacin (400mg), Minocycline (100mg) - Single dose.
- For a child, half of these adult doses are administered.
- Rifampicin (600mg), Ofloxacin (400mg), Minocycline (100mg) - Single dose.
- For a child, half of these adult doses are administered.

Drug-resistant leprosy Regimen		
	Intensive Phase (6 months)	Continuation Phase (18 months)
Rifampicin-resistant cases: 2 second line + 1 first line (Clofazimine)	Ofloxacin + Minocycline + Clofazimine (or) Ofloxacin + Clarithromycin + Clofazimine	Ofloxacin + Minocycline + Clofazimine (or) Ofloxacin + Clofazimine
Rifampicin & ofloxacin resistant cases	Clarithromycin + Minocycline + Clofazimine	(Clarithromycin or Minocycline) + Clofazimine

Solution for Question 6:

Correct Answer: A) Dapsone

Explanation:

Dapsone is contraindicated in patients with G6PD deficiency due to hemolysis of RBCs.

Dapsone	
MOA	Inhibits folic acid synthesis by inhibiting PABA incorporation into folic acid by folate synthase.
Pharmacokinetics	Dapsone is bacteriostatic It undergoes acetylation and glucuronide and sulfate conjugation in the liver Excreted in the urine Plasma $\frac{1}{2}$ life: >24 hours

Uses	First-line for M.leprae Chloroquine-resistant malaria Toxoplasmosis prophylaxis Pneumocystis jirovecii infection prophylaxis Dermatitis herpetiformis
Side effects	Hemolytic anemia - Hemolysis in patients with G6PD deficiency (Antidote for hemolytic anemia: Methylene Blue) Methaemoglobinaemia Paresthesias Neuropsychiatric symptoms

Solution for Question 7:

Correct Answer: C) Dapsone

Explanation:

Dapsone is the first-line drug used to treat leprosy and is bacteriostatic.

Ofloxacin, Moxifloxacin and Rifampin are bactericidal.

Table rendering failed

Duration of treatment:

- Multibacillary Leprosy (MB): 12 months
- Paucibacillary Leprosy (PB): 6 months

Solution for Question 8:

Correct Answer: C) Azithromycin monotherapy

Explanation:

Azithromycin is the preferred agent for MAC prophylaxis due to its efficacy and ease of administration.

Mycobacterium avium complex is an opportunistic infection that occurs in HIV-positive individuals in the form of pneumonia. It occurs when CD4 count is less than 50/mm³.

Prophylaxis:

- Azithromycin weekly or
- Rifabutin daily

Treatment:

- Clarithromycin (Most effective) or Azithromycin + Ethambutol + Rifabutin

Duration: 12 months

Solution for Question 9:

Correct Answer: D) Regular liver function tests required

Explanation:

9-month-all oral bedaquiline therapy regimen:

- The WHO recommends using a 9-month all-oral regimen (shorter regimen) instead of longer 18-month regimens for patients with MDR/RR-TB and in patients without resistance to fluoroquinolones. (Option A ruled out)
- Given when the BPaL-M regimen is unavailable.
- This regimen involves the administration of oral bedaquiline once daily, in combination with other oral anti-TB drugs.
- Bedaquiline can be used in patients with HIV co-infection (Option B ruled out)
- Regular monitoring is crucial throughout the treatment period with 9 months of all oral bedaquiline therapy, including assessments of cardiac function (due to the risk of QT prolongation associated with bedaquiline) (Option C ruled out) and liver function (Option D) (due to potential hepatotoxicity).

9-month-all oral bedaquiline therapy regimen	
Medications	Duration

Initial Phase: Bedaquiline (6 months) + Levofloxacin/Moxifloxacin + Ethionamide + Ethambutol + Isoniazid (high dose) + Pyrazinamide + Clofazimine Alternative for Ethionamide: Linezolid (2 months)	4-6 months (extend to 6 months if sputum smear positive at 4 months)
Continuation Phase: Levofloxacin/Moxifloxacin + Clofazimine + Ethambutol + Pyrazinamide	5 months

Solution for Question 10:

Correct Answer: A) Bedaquiline

Explanation:

MDR-TB is resistant to at least isoniazid and rifampicin, which are the two most potent first-line anti-TB drugs. In such cases, alternative medications must be used.

Bedaquiline is a new medication specifically developed for the treatment of MDR-TB. It is recommended for use in combination with other anti-TB drugs for the treatment of MDR-TB.

Monodrug resistance	MDR	Pre-XDR
Isoniazid resistance: RZE+Levofloxacin for 8 months Rifampicin resistance: treat as MDR TB	It is isoniazid + rifampicin resistance BPAL-M is most preferred (Bedaquiline + pretomanid + linezolid + moxifloxacin) (Option A) All oral bedaquiline regimen is more commonly used throughout India	It is additional resistance to levofloxacin and moxifloxacin BPAL regimen is used if available All oral bedaquiline regimen used of BPAL are not available

Levofloxacin (Option C) although effective against TB, may not be sufficient for MDR-TB treatment alone.

Isoniazid (Option D) and Rifampicin (Option B) are first-line drugs and are ineffective against MDR-TB.

Solution for Question 11:

Correct Answer: B) Inhibits mycolic acid synthesis in the cell wall of M.tuberculosis

Explanation:

Delamanid and pretomanid are bicyclic nitroimidazoles that inhibit mycolic acid synthesis in cell wall synthesis of mycobacterium tuberculosis. They are part of the Group C regimen for MDR TB.

	Mechanism of action	Side effects
Delamanid	Inhibits the synthesis of mycolic acid by nitric oxide production, an essential component of the cell wall of M.tuberculosis	Nausea, vomiting, abdominal pain, headache, and elevated liver enzymes. QT prolongation, which can increase the risk of cardiac arrhythmias Monitoring of liver function and ECG is recommended during treatment
Pretomanid	Inhibiting the synthesis of mycolic acid by disrupting the synthesis of fatty acids in the cell wall of M.tuberculosis	Nausea, vomiting, and diarrhea Peripheral neuropathy Hepatotoxicity Acne

Inhibits DNA gyrase and topoisomerase IV enzymes in bacteria (Option A) is the mechanism of action of fluoroquinolones, such as ciprofloxacin and moxifloxacin.

Disrupts bacterial protein synthesis by binding to the 23S ribosomal RNA (Option C) describes the mechanism of action of macrolides, such as erythromycin and azithromycin.

Inhibits the synthesis of arabinogalactan in the cell wall of M. tuberculosis (Option D) describes the mechanism of action of ethambutol.

Solution for Question 12:

Correct Answer: C) Ethionamide

Explanation:

The patient's symptoms strongly suggest hypothyroidism and possible liver dysfunction.

Ethionamide, used to treat multidrug-resistant tuberculosis (MDR-TB), can cause hepatotoxicity and disrupt thyroid function, leading to these adverse effects. It is part of group C drugs in MDR TB treatment.

Ethionamide-MOA: It acts similarly to isoniazid, a first-line anti-TB drug. Ethionamide works by inhibiting the synthesis of mycolic acids, which are essential components of the bacterial cell wall of *Mycobacterium tuberculosis*.

Side effects:

- Nausea, vomiting, and loss of appetite
- Hepatotoxicity
- Neurological symptoms include peripheral neuropathy, dizziness, and confusion. These are treated with pyridoxine (Vitamin B6).
- Erectile dysfunction
- Hypothyroidism

Ethambutol (Option A) can cause side effects such as optic neuritis (vision changes) and rash,

Isoniazid (Option B) is a first-line anti-TB drug known to cause hepatotoxicity, but it is not the correct answer in this scenario.

Levofloxacin (Option D) is not typically associated with gastrointestinal symptoms or hepatotoxicity to the extent seen with ethionamide.

Solution for Question 13:

Correct Answer: D) Clofazimine can cause constipation

Explanation:

Clofazimine is an antimycobacterial agent primarily used to treat leprosy and MDR TB. It does not typically cause constipation but rather gastrointestinal disturbances such as nausea, vomiting, and diarrhea.

	Clofazimine	Cycloserine
M O A	Binds to bacterial DNA, inhibiting replication/transcription. It also has anti-inflammatory properties.	Inhibits bacterial cell wall synthesis by blocking alanine racemase.
S / E	Nausea, vomiting, diarrhea, discolouration of skin and body secretions	Neurotoxicity (treated with Vitamin B6), dizziness, drowsiness, headache, confusion, psychosis, suicidal ideation, vomiting, diarrhea,

		and risk of seizures
--	--	----------------------

Solution for Question 14:

Correct Answer: C) Pyrazinamide

Explanation:

The Hepatotoxic drugs, in order of their toxicity, include Pyrazinamide > Isoniazid > Rifampin.

Name of drug	Isoniazid	Rifampin (Option B)	Pyrazinamide (Option C)	Ethambutol (Option D)	Streptomycin (Option A)
MOA	Inhibition of synthesis of mycolic acids (cell wall). It also inhibits mycobacterium DHFR, thereby interfering with nucleic acid synthesis.	Interrupts RNA synthesis by binding to mycobacterial DNA-dependent RNA polymerase	Blocks mycolic acid synthesis by getting converted into pyrazinoic acid	Inhibits arabinosyl transferases involved in mycolic acid incorporation in the cell wall	Inhibits 30S ribosomal protein synthesis
ROA	Oral	Oral	Oral	Oral	Intramuscular
Hepatotoxicity	Yes	Yes	Yes	No	No
Cellularity	Intracellular and extracellular	Intracellular and extracellular	Intracellular	Extracellular	Extracellular
Bactericidal	Bactericidal	Bactericidal	Bactericidal	Bacteriostatic	Bactericidal

/ Ba cte rio sta tic					
Ba cte ria in hib ited	Rapidly dividing M.TB	Slow growers	Slow growers	Rapidly dividing M.TB	Rapidly dividing M.TB

Solution for Question 15:

Correct Answer: D) Isoniazid and rifapentine combination therapy

Explanation:

Isoniazid and rifapentine combination therapy is used for TB prophylaxis in high-risk patients.

TB prophylaxis is indicated in the following high-risk patient groups:

- Contacts of open cases who show recent Mantoux conversion.
- Children with a sputum-positive TB patient in the family.
- Neonate of tubercular mother.
- Patients of leukaemia, diabetes, silicosis or those receiving immunosuppressant medication or those on long-term corticosteroid therapy.
- HIV-infected contacts of sputum-positive cases.

The regimens include one of the below options:

- Isoniazid (5 mg/kg) daily for 6 months (or)
- Rifampin (10 mg/kg) daily for 4 months (or)
- Isoniazid (5 mg/kg) + Rifampin (10 mg/kg) daily for 3 months (or)
- Once a week, Rifapentine (10 mg/kg) + high dose Isoniazid (15 mg/kg) weekly for 3 months

Solution for Question 16:

Correct Answer: D) Efavirenz

Explanation:

Efavirenz is an NNRTI that is not influenced by the activation of the CYP enzyme by Rifampin.

- HIV-positive individuals face an 8-fold increased risk of developing TB and are prone to more severe and potentially fatal extrapulmonary TB.
- To manage co-infection, ATT is initiated first, followed by ART after 2 weeks to mitigate the risk of Immune Reconstitution Inflammatory Syndrome (IRIS), an exaggerated inflammatory response.
- Rifampin, a potent CYP inducer in ATT, accelerates the metabolism of certain ART drugs like nevirapine (NNRTI), dolutegravir (integrase inhibitor), and protease inhibitors (e.g., Ritonavir, Indinavir, lopinavir), reducing their effectiveness. However, nucleoside reverse transcriptase inhibitors (NRTIs) and efavirenz can be safely combined with Rifampin.
- Rifabutin, a weaker enzyme inducer, is often used as an alternative to mitigate drug interactions in ART.
- All HIV-positive TB patients should also receive cotrimoxazole preventive therapy to guard against *Pneumocystis jirovecii* infection.

Solution for Question 17:

Correct Answer: D) Rifampin

Explanation:

Rifampin is NOT used in TB patients with liver disease since it is highly hepatotoxic but is considered safe in renal disease due to its biliary excretion.

Table rendering failed

Solution for Question 18:

Correct Answer: D) Ethambutol

Explanation:

Ethambutol is a first-line anti-tubercular drug that causes retrobulbar neuritis, decreased visual acuity, and color blindness.

Ethambutol	
MOA	Inhibits arabinosyl transferases involved in arabinogalactan synthesis, thereby interfering with mycolic acid synthesis in cell wall
Mechanism of resistance	Mutation of embAB genes, which encodes the enzyme arabinosyl transferase, reducing its affinity for ethambutol
Uses	In the treatment of Mycobacterium tuberculosis Mycobacterium avium complex (MAC) Mycobacterium kansasii
Side effects	Painless loss of vision Field defects due to retrobulbar neuritis Loss of visual acuity Red-green color blindness (Yellow vision- Digoxin; Blue vision- Sildenafil) Nausea, rashes, fever

Solution for Question 19:

Correct Answer: C) Pyrazinamide

Explanation:

Pyrazinamide is not used in the continuation phase of drug-susceptible TB treatment.

Patients who have been newly diagnosed or previously treated for TB are considered drug-susceptible.

	Pulmonary	Extrapulmonary
Intensive phase	Isoniazid	Isoniazid

	Rifampin Pyrazinamide Ethambutol	Rifampin Pyrazinamide Ethambutol
For 2 months	For 2 months	
Continuation phase	Isoniazid Rifampin Ethambutol	Isoniazid Rifampin Ethambutol
For 4 months	For 7-10 months	

Solution for Question 20:

Correct Answer: D) Pyrazinamide

Explanation:

Pyrazinamide increases serum uric acid levels and predisposes patients to gouty arthritis. It achieves this by inhibiting renal tubular secretion of uric acid, leading to hyperuricemia. The patient's symptoms of acute gouty arthritis and elevated serum uric acid levels are consistent with pyrazinamide-induced gout.

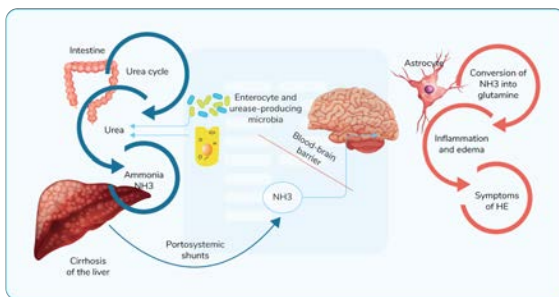
Pyrazinamide	
Drug type	Prodrug
Activation	Converted to pyrazinoic acid inside the bacterial cell by pyrazinamidase
Gene involved in activation	pncA gene encodes for pyrazinamidase, which converts pyrazinamide into its active form
Mechanism of action	Pyrazinoic acid accumulates in an acidic medium and likely inhibits mycolic acid synthesis by interacting with a different fatty acid synthase than INH. It also disrupts the mycobacterial cell membrane and transport functions.
Mechanism of resistance	Resistance to pyrazinamide develops rapidly if used alone and is mostly due to mutation in the pncA gene
Uses	Only to treat tuberculosis
Side effects	Most hepatotoxic Most potent at increasing uric acid levels

Solution for Question 21:

Correct Answer: D) Rifaximin

Explanation:

Rifaximin is a rifamycin drug used to treat hepatic encephalopathy by reducing ammonia production and absorption in the gut. Due to its minimal absorption from the gastrointestinal tract, it has no significant systemic effects.



Rifamycin group of drugs:

Rifampicin (or Rifampin)	Used in TB, leprosy Strong CYP inducer Food decreases absorption, so taken on an empty stomach
Rifapentine	Used for TB prophylaxis Strong CYP inducer
Rifabutin	Used in MAC Less potent CYP inducer Side effects: Pseudo-jaundice (Orange discoloration of the skin) Panuveitis
Rifaximin	It is unabsorbable, so it is used to treat gut infections Diarrhea predominant IBS, Traveller's diarrhea To treat hepatic encephalopathy No effect on CYP

Solution for Question 22:

Correct Answer: D) Rifampicin

Explanation:

- Rifampicin induces the hepatic cytochrome P450 enzyme system, including CYP2C9 and CYP3A4, which are involved in the metabolism of warfarin.
- Enhanced metabolism of warfarin due to rifampicin can lead to decreased plasma concentrations of warfarin, resulting in subtherapeutic levels and reduced anticoagulant efficacy. Therefore, rifampicin is the antitubercular drug most likely responsible for the interaction with warfarin in this clinical scenario.

Adverse effects of Rifampicin:

- Hepatotoxicity: Risk increases in patients with liver disease and is dose-dependent. Developing jaundice requires discontinuing rifampin, but it is typically reversible.
- Red-orange discoloration of body fluids like sweat, tears, urine, and saliva (Temporary and harmless)
- Cutaneous: Flushing, itching with a rash (especially on the face and scalp), redness, and watery eyes.
- Hypersensitivity reactions: Thrombotic thrombocytopenic purpura, acute renal failure
- Flu-like syndrome: Chills, fever, headache, malaise, and bone pain.
- Gastrointestinal: Abdominal cramps, nausea, vomiting, and diarrhea.

Isoniazid (Option A), Ethambutol (Option B), and Pyrazinamide (Option C) do not significantly affect the metabolism of warfarin.

Solution for Question 23:

Correct Answer: D) Rifampicin

Explanation:

The Cartridge-Based Nucleic Acid Amplification Test (CBNAAT) is commonly used for the rapid detection of Mycobacterium tuberculosis and mutations associated with drug resistance to rifampicin. Rifampicin resistance is often the first indicator of multidrug-resistant tuberculosis

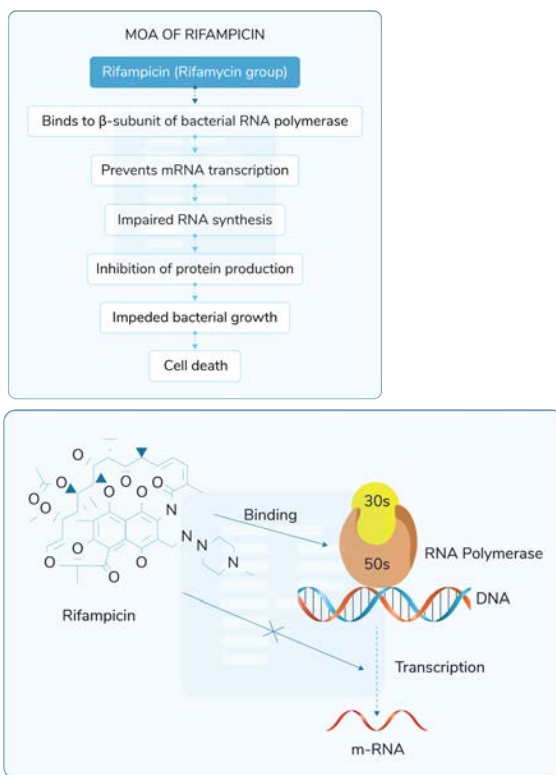
(MDR-TB).

CBNAAT typically provides results within 2 hours after the sample is processed.

Indications for CBNAAT:

- HIV-positive individuals
- Immunocompromised patients
- Recent contacts of TB cases
- Healthcare workers exposed to TB

MOA:



Resistance: Rifampicin resistance occurs due to mutations in the *rpoB* gene, which encodes RNA polymerase's β -subunit. These mutations alter rifampicin's binding site, diminishing its effectiveness in inhibiting RNA synthesis. Resistance to other tuberculosis drugs like isoniazid often coincides with resistance to rifampicin, causing MDR-TB and extensively drug-resistant TB (XDR-TB).

Uses: (Mnemonic- TB LAMBDAs)

- TB
- Leprosy

- Anthrax
 - Meningitis prophylaxis from H. influenzae
 - Brucellosis
 - Diphtheria (resistant to erythromycin)
 - All Staphylococcal infections
-

Solution for Question 24:

Correct Answer: D) Renal failure

Explanation:

Renal failure is not a common manifestation of isoniazid toxicity.

Acute Isoniazid toxicity presents as seizures (unresponsive to anticonvulsants), high anion gap metabolic acidosis, and coma.

Features of Isoniazid toxicity:

- Neurotoxicity: INH inhibits the activation of pyridoxine into pyridoxal phosphate, which impairs GABA synthesis. This can cause peripheral neuritis, paresthesias, numbness, hallucinations, psychosis, euphoria, transient memory loss, and rarely convulsions.
- Optic neuritis: More pronounced with ethambutol than isoniazid (Option B)
- Hepatotoxicity (Option A)
- Sideroblastic anemia
- Systemic Lupus Erythematosus/Drug-induced lupus erythematosus (Option C)
- Arthralgia
- Gynecomastia

Treat with vitamin B6

- If the isoniazid dose is known, gram to gram equivalent is given.
 - If the isoniazid dose is unknown, 70mg/kg up to 5g maximum is given.
-

Solution for Question 25:

Correct Answer: D) 10 mg/day

Explanation:

10mg of Vit B6 is given prophylactically to prevent neurotoxicity with the use of Isoniazid (INH) as part of the anti-tubercular drug regimen.

The side effects and uses of INH are given below

Table rendering failed

25-50 mg/day (Option A) is the prophylactic dose given in pregnancy and children.

100 mg/day (Option C) is the dose given for isoniazid toxicity.

Previous Year Questions

1. Which drug amongst the options has a tendency to accumulate in the cornea?

- A. Leflunomide
 - B. Chloroquine
 - C. Methotrexate
 - D. Sulfasalazine
-

2. In an individual traveling to an area with endemic malaria, which of the following is the preferred chemoprophylaxis option?

- A. Chloroquine
 - B. Quinine
 - C. Lumefantrine
 - D. Pyrimethamine
-

3. What is the clinical application of tafenoquine among the options provided?

- A. Radical cure of *Plasmodium vivax*
 - B. Prophylaxis of malaria in pregnancy
 - C. Treatment of severe *falciparum* malaria
 - D. Treatment of malaria in children <2 years
-

4. Identify the incorrect statement in relation to antiviral medications:

- A. Ombitasvir prevents HCV protein synthesis
 - B. Imiquimod used in treatment of condyloma acuminata
 - C. Simprevir inhibits protease of hepatitis C
 - D. Oseltamivir is used for swine flu
-

5. Assertion: Larger doses of acyclovir are recommended for treatment of genital herpes in HIV positive patients Reason: Frequent recurrences of genital herpes is seen in HIV patients

- A. Both assertion and reason are true and reason is correct explanation for the assertion
 - B. Both assertion and reason are true and reason is not the correct explanation for the assertion
 - C. Assertion is true, but the reason is false
 - D. Assertion is false, but the reason is true
 - E. Both assertion and reason are false
-

6. In a patient with severe COVID-19, which of the following options is unlikely to provide significant assistance?

- A. Low molecular weight heparin
 - B. Oxygen
 - C. Favipiravir
 - D. Steroids
-

7. Which of the options listed below is classified as a second-line medication for treating tuberculosis?

- A. Streptomycin
 - B. Ethionamide
 - C. Acyclovir
 - D. Vancomycin
-

8. Which topical antifungal exhibits both anti-inflammatory properties and provides relief from itching?

- A. Ketoconazole
 - B. Sertaconazole
 - C. Luliconazole
 - D. Terbinafine
-

9. What is the drug that decreases the death rate in patients with COVID-19 among the options provided? A. Dexamethasone, B. Remdesivir, C. Favipravir

- A. B and C
 - B. A and B
 - C. Only A
 - D. A, B, C
-

10. Which of the following anti-tubercular drug is not bactericidal?

- A. Isoniazid
- B. Rifapentin
- C. Thioacetazone
- D. Kanamycin

11. Against which of the following is Voriconazole ineffective?

- A. *Candida albicans*
 - B. Mucormycosis
 - C. *Candida tropicalis*
 - D. Aspergillosis
-

12. Which of the options listed below acts as a hindrance to the process of RNA synthesis?

- A. 5-Fluorouracil
 - B. Rifampicin
 - C. Nitrofurantoin
 - D. Novobiocin
-

13. Praziquantel is utilized to treat which medical condition?

- A. Strongyloidiasis
 - B. Trichomoniasis
 - C. Schistosomiasis
 - D. Rhinosporidiosis
-

14. A 60-year-old male came to the emergency department after sustaining a fall. X-ray showed a fractured neck of the femur and was operated on for the same. On

postoperative day 2, he experienced a sharp, severe, constant, burning pain over his right back. On examination, the lesion is as seen below. The area is tender to light touch with a rash over the area. What is the most appropriate treatment for this patient?



- A. Amitriptyline
- B. Topical corticosteroids
- C. Valacyclovir
- D. Mupirocin

15. What should patients with bacterial vaginosis and trichomoniasis on metronidazole avoid consuming?

- A. Grapefruit juice
- B. Alcohol
- C. Benzodiazepine
- D. MAOI/SSRI

16. Which medication among digoxin, potassium chloride, clarithromycin, atenolol, and triamterene is responsible for the increased serum digoxin levels in a patient

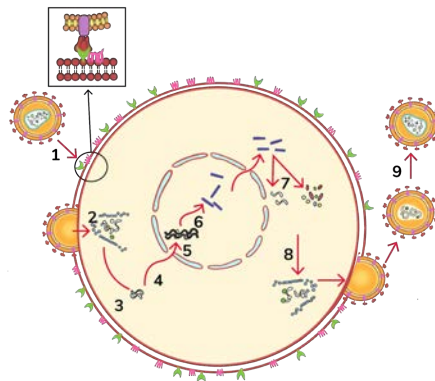
who developed atrial fibrillation with controlled ventricular rate?

- A. Potassium chloride
 - B. Clarithromycin
 - C. Atenolol
 - D. Triamterene
-

17. Why was the patient advised to use an additional barrier method of contraception after being diagnosed with tuberculosis and starting antitubercular therapy while already on oral contraceptive pills?

- A. Teratogenic
 - B. Failure
 - C. Rifampicin decreases OCP metabolism
 - D. Rifampicin induces OCP metabolism and cause failure
-

18. The image below shows the life cycle of a virus. Which of the following drugs acts at step 5?



- A. Raltegravir
- B. Enfuvirtide

- C. Tenofovir
 - D. Zidovudine
-

19. Which medication is most likely responsible for the development of hyperpigmentation in the palms and soles of a patient who has been receiving combined active antiretroviral therapy (cART) for 5 months, and is a known case of HIV?

- A. Nelfinavir
 - B. Emtricitabine
 - C. Abacavir
 - D. Zidovudine
-

20. What is the recommended medication for a diabetic patient with rhinitis and facial swelling, where an intranasal biopsy showed the presence of broad-based aseptate hyphae with right angle branching and evidence of vascular invasion?

- A. Fluconazole
 - B. Amphotericin B
 - C. Ketoconazole
 - D. Griseofulvin
-

21. In the treatment of bacterial vaginosis (BV), which antibiotic is not recommended by the Centers for Disease Control and Prevention (CDC) guidelines?

- A. Metronidazole
- B. Tinidazole

- C. Clindamycin
 - D. Erythromycin
-

22. Which cellular process does Bedaquiline primarily target to exert its therapeutic effect, and how does this mechanism contribute to its antimycobacterial activity?

- A. ATP synthesis inhibition
 - B. Mycolic acid synthesis inhibition
 - C. Protein inhibition
 - D. Inhibiting reverse transcriptase enzyme
-

23. Patient was taking ampicillin. Now he is experiencing diarrhea. Drug of choice to manage this condition?

- A. Oral metronidazole
 - B. Oral Fidaxomicin
 - C. Oral ciprofloxacin
 - D. Bezlotoxumab
-

24. Which drug shows intrinsic resistance?

- A. Fluconazole - Candida glabrata and Candida krusei
 - B. Amphotericin B - candida glabrata
 - C. Voriconazole - aspergillus fumigatus
 - D. Itraconazole - sporothrix
-

25. Which drug would be considered safe for use in a pregnant woman at 20 weeks gestation presenting with fever and dysuria and preliminarily diagnosed with cystitis?

- A. Ciprofloxacin
 - B. Gentamicin
 - C. Cotrimoxazole
 - D. Amoxicillin
-

26. Which of the following drug dosages is incorrect for the treatment of leprosy in adults?

- A. Dapsone 100 mg
 - B. Clofazimine 50 mg
 - C. Clofazimine 300 mg
 - D. Rifampicin 450 mg
-

27. A TB patient with HIV begins ART and experiences deterioration in health after 2 months. What is the likely cause?

- A. Development of drug resistance to TB drugs
 - B. Drug-drug interactions between TB and ART medications
 - C. Immunological changes in the body
 - D. Development of drug resistance to ART
-

28. A patient with penicillin allergy presents with fever, burning micturition, and hearing problems accompanied by dizziness. Which drug is the most appropriate for the management?

- A. Azithromycin
 - B. Amikacin
 - C. Ciprofloxacin
 - D. Ampicillin-Clavulanic acid
-

29. Which anti-tubercular drug is associated with mutations in the KATG gene?

- A. Rifampicin
 - B. Isoniazid
 - C. Moxifloxacin
 - D. Ethambutol
-

30. A 55-year-old patient is receiving a combination antiretroviral therapy including Zidovudine, Lamivudine, and Nevirapine for HIV management. Which of the following drugs should be avoided due to potential drug interactions with this regimen?

- A. Rifampicin
 - B. Streptomycin
 - C. INH
 - D. Ethambutol
-

31. What is the inclusion criterion for starting bedaquiline in a patient presenting with multidrug-resistant (MDR) or rifampicin-resistant (RR) tuberculosis?

- A. Resistance to both rifampicin and isoniazid with mutations in the inhA and katG genes
 - B. Resistance to rifampicin but sensitivity to fluoroquinolones
 - C. Resistance to rifampicin and sensitivity to isoniazid
 - D. Resistance to both rifampicin and fluoroquinolones
-

32. Which drug can be used to treat both HIV and Hepatitis B?

- A. Dolutegravir
 - B. Tenofovir
 - C. Zidovudine
 - D. Ritonavir
-

33. Which of the following is the preferred treatment for Plasmodium falciparum malaria?

- A. Chloroquine
 - B. Artemisinin-based combination therapy (ACT)
 - C. Primaquine
 - D. Doxycycline
-

34. Which of the following anti-tubercular drugs is associated with visual disturbances?

- A. Rifampicin
- B. Streptomycin
- C. Pyrazinamide
- D. Ethambutol

35. Which of the following is false about voriconazole?

- A. It should be taken on an empty stomach
 - B. It requires therapeutic drug monitoring
 - C. It should not be co-administered with pyrazinamide
 - D. It requires a loading dose.
-

36. Which among the following anti-leprosy drugs can cause skin hyperpigmentation?

- A. Clofazimine
 - B. Rifampicin
 - C. Dapsone
 - D. Ofloxacin
-

37. Which is the incorrectly matched option in respect of the mechanism of action of the antimicrobial drugs?

- A. Linezolid: Binds to 30S ribosome and inhibits protein synthesis
 - B. Rifampicin: Inhibits DNA dependent RNA polymerase
 - C. Aztreonam: Inhibit cell wall synthesis
 - D. Daptomycin: Cause depolarization of membranes
-

38. What could be the likely cause for shifting the initial treatment plan of starting on ampicillin to ceftriaxone in a 14-month-old diagnosed with Hemophilus influenza meningitis?

- A. Hemophilus alters penicillin binding protein sites, hence resistant to ampicillin
 - B. Easier to administer IV ceftriaxone than IV ampicillin
 - C. Cotrimoxazole is the drug of choice, but patient is allergic to sulfa drugs
 - D. Hemophilus produces beta lactamase, hence resistant to ampicillin
-

39. Which of the following drugs, when administered, would enhance the bioavailability of saquinavir?

- A. Ritonavir
 - B. Vitamin C
 - C. Cimetidine
 - D. Ganciclovir
-

40. In fungi, where does amphotericin B primarily exert its effects?

- A. Nucleic acid
 - B. Ribosome
 - C. Cell membrane
 - D. Cytoplasm
-

41. Which antiviral drug possesses dual antiviral activity against both HIV and HBV?

- A. Enfuvirtide

- B. Abacavir
 - C. Tenofovir
 - D. Ritonavir
-

42. Which of the following statements regarding foscarnet is false?

- A. It is used for resistant CMV
 - B. Renal toxicity is seen
 - C. It is activated by viral thymidine kinase
 - D. Regular monitoring of serum electrolytes is required
 - E. It can cause genital ulceration
-

43. Which of the following statements is not true about the management of a nurse who experienced an accidental needle prick from an HIV-infected needle?

- A. Zidovudine is used as monotherapy in post exposure prophylaxis
 - B. Washing hands with soap and water is advised
 - C. Baseline viral markers of healthcare personnel should be done at the time of prick
 - D. Follow up viral markers of healthcare personnel should be measure at 6 weeks
 - E. Tenofovir + Lamivudine + Efavirenz for a period of 4 weeks
-

44. In which medical conditions is Tafenoquine used?

- A. Radical cure of plasmodium
- B. Prophylaxis of malaria in pregnancy
- C. Treatment of severe falciparum malaria

D. Treatment of malaria in children <2 years

45. What is the recommended medication for treating a 23-year-old female who is experiencing burning pain while urinating and has been diagnosed with a co-infection of chlamydia and gonococcus?

- A. Azithromycin
 - B. Ceftriaxone
 - C. Fluconazole
 - D. Clindamycin
-

46. What is the preferred medication for preventing meningococcal meningitis in a pregnant woman following exposure to a confirmed case?

- A. Ciprofloxacin
 - B. Ceftriaxone
 - C. Cefalexin
 - D. Rifabutin
-

47. What is the aerosol medication that is administered to children for the treatment of RSV infection?

- A. Indinavir
 - B. Amantadine
 - C. Ribavirin
 - D. Tenofovir
-

48. A patient with fever and eschar. What will be the treatment?



- A. Doxycycline
 - B. Doxycycline + Azithromycin
 - C. Ceftriaxone + Doxycycline
 - D. Ceftriaxone
-

49. Which of the following is a novel anti-depressant?

- A. Lurasidone
 - B. Vilazodone
 - C. Asenapine
 - D. Blonanserin
-

50. What is the mechanism of action of cyclosporine?

- A. Inhibition of calcineurin
- B. AMP kinase stimulation

- C. IMP dehydrogenase inhibition
 - D. Dihydro-orotate dehydrogenase inhibition
-

51. What is the mechanism of resistance by lactamases to β -lactam antibiotics?

- A. Alteration of penicillin binding protein
 - B. Degradation of antibiotics
 - C. Drug efflux
 - D. Alternation in 50s ribosome structure
-

52. Which one of the following medications should not be recommended in conjunction with theophylline?

- A. Erythromycin
 - B. Cefotaxime
 - C. Cotrimoxazole
 - D. Amoxicillin
-

53. Which of the following options does not involve the use of antimicrobial combinations?

- A. Uncomplicated Gonorrhea
 - B. Tuberculosis
 - C. Intra abdominal infections
 - D. Malaria
-

54. An example of a bacteriostatic drug is _____:

- A. Aminoglycoside
 - B. Vancomycin
 - C. Metronidazole
 - D. Linezolid
-

55. Which of the following carbapenems has increased risk of causing seizures?

- A. Meropenem
 - B. Imipenem
 - C. Etrapanem
 - D. Doripenem
-

56. What is the mechanism of triazoles?

- A. Interferes with DNA and RNA synthesis
 - B. Inhibition of ergosterol synthesis
 - C. Forms pores in fungal membranes
 - D. Inhibits epoxidation of squalene in fungi
-

57. Which of the following fluoroquinolones has maximum bioavailability?

- A. Ciprofloxacin
- B. Levofloxacin

- C. Gatifloxacin
 - D. Norfloxacin
-

58. What is the mechanism of action of oseltamivir and zanamivir?

- A. DNA polymerase inhibition
 - B. Protein synthesis inhibition
 - C. Nucleotide analogue
 - D. Neuraminidase inhibition
-

59. What is the drug of choice for diphtheria carrier state?

- A. Amoxicillin
 - B. Erythromycin
 - C. Rifampicin
 - D. Tetracycline
-

60. A 32-year-old man who had traveled to Assam recently presents with fever, chills, seizures, and altered sensorium. On examination, his blood pressure was 70/50 mm of Hg and labored breathing was present. A blood smear was sent which revealed plasmodium falciparum. Which of the following is the drug of choice in this condition?

- A. Chloroquine
- B. Quinine
- C. Intravenous artesunate
- D. Artemether

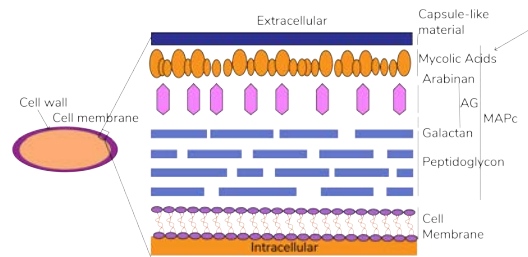
61. Patients receiving which of the following anti-microbial drugs are least likely to develop seizures?

- A. Beta-lactam compounds
 - B. Quinolones
 - C. Macrolides
 - D. Isoniazid
-

62. A tuberculosis patient on anti-tubercular treatment presents with a tingling sensation and paresthesia in the lower limbs. He is not a diabetic and occasionally consumes alcohol. Which of the following vitamins must be supplemented to this patient?

- A. Vitamin B6
 - B. Vitamin B12
 - C. Vitamin B2
 - D. Vitamin B3
-

63. A patient was diagnosed with pulmonary tuberculosis and was started on first time antitubercular drugs. Which of the following drug acts by inhibiting the synthesis of the marked layer shown in the image?



- A. Isoniazid
- B. Rifampicin
- C. Pyrazinamide
- D. Ethambutol

64. What could be the likely cause for the change in treatment plan from ampicillin to ceftriaxone in a 14-month-old diagnosed with Hemophilus influenza meningitis?

- A. Hemophilus alters penicillin binding protein sites, hence resistant to ampicillin
- B. Easier to administer IV ceftriaxone than IV ampicillin
- C. Cotrimoxazole is the drug of choice, but patient is allergic to sulfa drugs
- D. Hemophilus produces beta lactamase, hence resistant to ampicillin

65. A woman presenting with symptoms of urinary tract infection was prescribed a drug that causes tendon rupture and arthropathy. What is the mechanism of action of the drug?

- A. DNA gyrase inhibition
- B. Ribosomal inhibition
- C. Cell wall synthesis
- D. Inhibition of folic acid synthesis

66. What are the mechanisms of action of doxycycline?

- A. Cell wall synthesis inhibitor
 - B. Protein synthesis inhibitor
 - C. DNA gyrase inhibitor
 - D. Leucyl-tRNA synthetase inhibitor
-

67. Which of the following fluoroquinolone is not given in liver disease?

- A. Ofloxacin
 - B. Levofloxacin
 - C. Pefloxacin
 - D. Delafloxacin
-

68. A diabetic patient presents with a fungal sinus and peri-orbital region infection with significant visual impairment, as shown below. The most appropriate drug for the treatment of this patient is:



- A. Itraconazole
 - B. Liposomal amphotericin B
 - C. Ketoconazole
 - D. Broad spectrum antibiotics
-

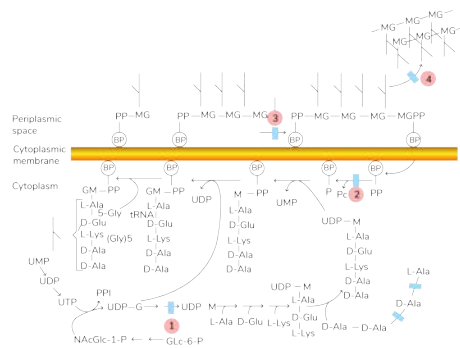
69. What is the medication that can be prescribed to treat the white patchy lesions on the tongue and buccal mucosa of a patient with bronchial asthma who is using inhalational steroids?

- A. Clotrimazole
 - B. Griseofulvin
 - C. Terbinafine
 - D. Flucytosine
-

70. For vancomycin dosage, which nomogram scale is employed?

- A. Hallstead scale
 - B. Salazar scale
 - C. Rumack Matthew nomogram
 - D. Matzke scale
-

71. The given diagram shows the steps in bacterial cell wall formation. Identify the site of action of beta – lactam antibiotics?



- A. 1
- B. 2
- C. 3
- D. 4

72. A known asthmatic patient with a fever and sore throat was prescribed an antibiotic. Two days later he presents with a maculopapular rash. Which of the following drugs might have caused this?

- A. Prednisolone
- B. Salbutamol
- C. Fexofenadine
- D. Amoxicillin

73. A 48 year old male patient presents to the OPD with complaints of cough, fever, night sweats and weight loss since the last 3 months. He has a history of HIV and is on antiretroviral medication. His current medication is Tenofovir 150mg, Lamivudine 300 mg and Dolutegravir 50 mg once daily at night. He is diagnosed with TB and started on antitubercular therapy. What is the dose adjustment required in this case?

- A. No dose adjustment required.
 - B. Add Tenofovir 150mg in the morning
 - C. Add Dolutegravir 50 mg in the morning
 - D. Add Lamivudine 300 mg in the morning
-

74. What is the required dose adjustment in a person who started anti-tubercular therapy and is already on ART therapy with Tenofovir 200 mg, Dolutegravir 100 mg at night, and Lamivudine 200 mg?

- A. Add tenofovir 200 mg in the morning
 - B. Add dolutegravir 100 mg in the morning
 - C. No dose adjustment
 - D. Add lamivudine 200 mg in the morning
-

75. What is ECG change associated with the use of anti-tubercular drug bedaquiline?

- A. QT interval prolongation
 - B. PR interval prolongation
 - C. QRS widening
 - D. ST-segment elevation
-

76. What is the probable cause for hypotension, flushing, and nausea in a patient using anti-amoebic drugs?

- A. Delirium tremens
- B. Alcohol withdrawal

- C. Disulfiram like reaction
 - D. Hallucinosiis
-

77. What is the mechanism of action of remdesivir?

- A. RNA-dependent DNA polymerase inhibition
 - B. RNA-dependent RNA polymerase inhibition
 - C. Fusion inhibition
 - D. Entry inhibition
-

78. Which of the following is true about tedizolid?

- A. Active against gram positive organisms
 - B. Poor oral bioavailability
 - C. Active against anaerobes
 - D. Causes peripheral neuropathy
-

**79. Assertion: In a patient admitted in hospital for community – acquired pneumonia, combination therapy of beta – lactams and azithromycin is given.
Reason: This combination covers Gram – positive organisms and anaerobes**

- A. Both assertion and reason are true and the reason is the correct explanation for the assertion
- B. Both assertion and reason are true, but the reason is not the correct explanation for the assertion
- C. Assertion is true, but the reason is false
- D. Both assertion and reason are false



- A. 1
- B. 2
- C. 3
- D. 4

83. Match the following bacteria with the drug of choice used for each. : A. Bacteroides 1. Vancomycin B. MRSA 2. Metronidazole C. Hemophilus 3. Meropenem D. Klebsiella 4. Penicillin

A. Bacteroides	1. Vancomycin
B. MRSA	2. Metronidazole
C. Hemophilus	3. Meropenem
D. Klebsiella	4. Penicillin

- A. A-1, B-3, C-4, D-2
- B. A-2, B-1, C-4, D-3
- C. A-3, B-4, C-1, D-2
- D. A-4, B-1, C-2, D-3

84. Which of the following is the drug of choice for scrub typhus?

- A. Doxycycline
- B. Azithromycin
- C. Chloramphenicol
- D. Ciprofloxacin

85. Which drug from the options provided is known as a protein synthesis inhibitor?

- A. Linezolid
 - B. Nevirapine
 - C. Indinavir
 - D. Ceftriazone
-

86. Drug of choice for nasal carriers of MRSA is:

- A. Vancomycin
 - B. Teicoplanin
 - C. Mupirocin
 - D. Linezolid
-

87. Which of the following is gametocidal to all the species causing malaria?

- A. Primaquine
 - B. Chloroquine
 - C. Quinine
 - D. None
-

88. What is the drug of choice for scrub typhus during pregnancy and otherwise respectively?

- A. Chloramphenicol, Doxycycline

- B. Doxycycline, Doxycycline
 - C. Doxycycline, Azithromycin
 - D. Azithromycin, Doxycycline
-

89. Which of the following drugs could potentially be responsible for the patient's development of nephrolithiasis?

- A. Ketoconazole
 - B. Indinavir
 - C. Lamivudine
 - D. Zidovudine
-

90. Identify the incorrect pairing of antifungals with their respective mechanisms of action.:

- A. Ketoconazole: Lanosterol-14-demethylase inhibition
 - B. Flucytosine: Inhibits microtubule synthesis and thus hinders mitosis
 - C. Terbinafine: Squalene epoxidase inhibition
 - D. Amphotericin B: Sequesters ergosterol
-

91. A female presented with burning micturition and symptoms of urinary tract infection (UTI). Which of the following fluoroquinolones is least likely to be effective?

- A. Ofloxacin
- B. Ciprofloxacin
- C. Moxifloxacin

D. Levofloxacin

92. A female patient presented with burning micturition. A diagnosis of cystitis was made. Which of the following drug will use for treatment?

- A. Amoxicillin
 - B. TMP-SMX
 - C. Ciprofloxacin
 - D. Doxycycline
-

93. Antibiotic of choice for UTI in pregnancy?

- A. Amoxicillin
 - B. Azithromycin
 - C. Clarithromycin
 - D. Ceftriaxone
-

94. Which of the following antimicrobials is effective against an organism producing extended-spectrum beta-lactamases?

- A. Amoxicillin – Clavulanic acid
 - B. Penicillin
 - C. Piperacillin – Tazobactam
 - D. Ceftriaxone
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	1
Question 5	1
Question 6	3
Question 7	2
Question 8	2
Question 9	3
Question 10	3
Question 11	2
Question 12	2
Question 13	3
Question 14	3
Question 15	2
Question 16	2
Question 17	4
Question 18	1
Question 19	2
Question 20	2
Question 21	4
Question 22	1

Question 23	2
Question 24	1
Question 25	4
Question 26	4
Question 27	3
Question 28	3
Question 29	2
Question 30	1
Question 31	1
Question 32	2
Question 33	2
Question 34	4
Question 35	3
Question 36	1
Question 37	1
Question 38	4
Question 39	1
Question 40	3
Question 41	3
Question 42	3
Question 43	1
Question 44	1
Question 45	1
Question 46	2
Question 47	3
Question 48	1

Question 49	2
Question 50	1
Question 51	2
Question 52	1
Question 53	1
Question 54	4
Question 55	2
Question 56	2
Question 57	2
Question 58	4
Question 59	2
Question 60	3
Question 61	3
Question 62	1
Question 63	1
Question 64	4
Question 65	1
Question 66	2
Question 67	3
Question 68	2
Question 69	1
Question 70	4
Question 71	4
Question 72	4
Question 73	3
Question 74	2

Question 75	1
Question 76	3
Question 77	2
Question 78	1
Question 79	3
Question 80	1
Question 81	1
Question 82	4
Question 83	2
Question 84	1
Question 85	1
Question 86	3
Question 87	1
Question 88	4
Question 89	2
Question 90	2
Question 91	3
Question 92	3
Question 93	4
Question 94	3

Solution for Question 1:

Correct Option B: Chloroquine

- Among the options provided, chloroquine is the drug that can get deposited in the cornea.

- Chloroquine is an antimalarial medication that is also used in the treatment of certain autoimmune diseases such as rheumatoid arthritis and lupus. Prolonged or high-dose use of chloroquine can lead to a condition known as chloroquine keratopathy. This condition is characterized by the deposition of the drug in the cornea, specifically in the corneal epithelium and endothelium.
- The cornea is the transparent outermost layer of the eye, and the deposition of chloroquine in this tissue can lead to various ocular side effects. These include corneal opacities, blurred vision, halos around lights, and changes in color vision. In severe cases, chloroquine keratopathy can affect visual acuity and lead to permanent visual impairment.

Incorrect options:

Option A: Leflunomide- Do not deposit in cornea

Option C: Methotrexate - Do not deposit in cornea

Option D: Sulfasalazine - Do not deposit in cornea

Solution for Question 2:

Correct option:

Option A: Chloroquine

- Due to sensitive strains of all 4 species, chloroquine is very efficient in treating malaria.
- A 250 mg pill of chloroquine phosphate contains 150 mg of the base.
- The recommended dosage is CHLOROQUINE 1 g (600 mg base) at 0 followed by 0.5 g at 6, 24, and 48 hr. Prophylaxis is another usage for it; 300 mg base weekly. The WHO has condensed the chloroquine regimen to 1 g (600 mg dosage) at 0 hours, 24 hours, and 48 hours (2-2-1 regimen).
- Regions where there are strains that are sensitive to chloroquine
- Areas with chloroquine-resistant *P. falciparum*, such as India, should take two tablets each week, starting one week before leaving the endemic area and continuing for four weeks after leaving.

Incorrect options:

Options B

Quinine is used in uncomplicated, complicated, and cerebral malaria.

Option C

Lumefantrine is used in the treatment of MDR falciparum malaria.

Option D

Pyrimethamine was given as chemoprophylaxis before but is not used now.

Solution for Question 3:

Correct Option A.

Tafenoquine is an antimalarial medication used for the radical cure (prevention of relapse) of Plasmodium vivax malaria. It is effective in eliminating the dormant liver stages (hypnozoites) of the parasite, which are responsible for causing relapses of the disease. Tafenoquine is administered as a single dose in combination with a blood schizonticide, typically chloroquine.

Incorrect Options:

Option B. Prophylaxis of malaria in pregnancy: Tafenoquine is not recommended for prophylaxis of malaria in pregnancy due to limited safety data.

Option C. Treatment of severe falciparum malaria: Tafenoquine is not indicated for the treatment of severe falciparum malaria. It is primarily used for Plasmodium vivax malaria.

Option D. Treatment of malaria in children <2 years: Tafenoquine is not approved for use in children younger than 18 years of age.

Solution for Question 4:

Correct Option A.

Ombitasvir is not involved in the prevention of HCV (Hepatitis C Virus) protein synthesis.

Ombitasvir is actually a direct-acting antiviral agent that is part of a combination therapy used for

the treatment of chronic hepatitis C infection. It specifically targets the HCV NS5A protein, which is involved in viral replication and assembly.

Incorrect Options:

Option B. Imiquimod used in the treatment of condyloma acuminata: Imiquimod is an immune response modifier that is used topically in the treatment of certain viral infections, including genital warts (condyloma acuminata). It stimulates the body's immune system to produce interferon and other cytokines, which help in the elimination of the viral infection.

Option C. Simeprevir inhibits the protease of hepatitis C: Simeprevir is a direct-acting antiviral agent that inhibits the protease enzyme of the hepatitis C virus. It is used in combination with other antiviral medications for the treatment of chronic hepatitis C infection.

Option D. Oseltamivir is used for swine flu: Oseltamivir is an antiviral medication that is commonly used for the treatment and prevention of influenza (flu) caused by influenza A and B viruses, including the H1N1 strain (commonly known as swine flu).

Solution for Question 5:

Correct Option A.

- Assertion: Larger doses of acyclovir are recommended for the treatment of genital herpes in HIV-positive patients. This assertion is true because HIV-positive patients tend to experience more frequent recurrences of genital herpes.
- Reason: Frequent recurrences of genital herpes are seen in HIV patients. This reason is also true and provides a valid explanation for the assertion. The compromised immune system in HIV-positive individuals can lead to more frequent reactivation of the herpes virus, requiring higher doses of acyclovir or other antiviral medications to effectively manage the infection.
- Since both the assertion and reason are true and the reason correctly explains the assertion, option A is the correct answer.

Incorrect Options:

Option B. Both assertion and reason are true, but the reason is not the correct explanation for the assertion: This option is incorrect because the reason provided is indeed the correct explanation for the assertion.

Option C. Assertion is true, but the reason is false: This option is incorrect because both the assertion and reason are true.

Option D. Assertion is false, but the reason is true: This option is incorrect because the assertion is true, as HIV-positive patients do require larger doses of acyclovir for genital herpes treatment.

Option E. Both assertion and reason are false: This option is incorrect because both the assertion and reason are true.

Solution for Question 6:

Correct Option C:

- Favipiravir is an antiviral medication that has been investigated for the treatment of various viral infections, including influenza and COVID-19. However, its effectiveness in treating severe cases of COVID-19 is still uncertain, and current evidence is limited.

Incorrect Options:

Option A: Low molecular weight heparin: Severe COVID-19 can be associated with an increased risk of blood clots and thromboembolic events. Low molecular weight heparin (LMWH) is commonly used as a prophylactic or therapeutic anticoagulant in hospitalized patients with severe COVID-19 to prevent or treat these complications.

Option B: Oxygen: Severe COVID-19 can cause respiratory distress and hypoxemia. Supplemental oxygen therapy is a crucial supportive measure in managing severe cases, helping to maintain adequate oxygenation and alleviate respiratory distress.

Option D: Steroids: Steroids, such as dexamethasone, have shown beneficial effects in reducing mortality and improving outcomes in hospitalized patients with severe COVID-19. They have anti-inflammatory properties and help suppress an exaggerated immune response, known as cytokine storm, which can contribute to severe lung injury.

Solution for Question 7:

Correct Option B:

- Ethionamide is a second-line anti-tubercular drug used in the treatment of tuberculosis (TB). It is typically prescribed when the first-line drugs, such as isoniazid and rifampicin, are ineffective or when the TB infection is resistant to these medications. Ethionamide is classified as a second-line drug due to its use in cases of drug-resistant TB.
- Ethionamide works by inhibiting the synthesis of mycolic acids, which are crucial components of the bacterial cell wall in *Mycobacterium tuberculosis*, the bacterium that causes TB. By targeting the cell wall synthesis, ethionamide helps to suppress the growth and replication of the bacteria.

Incorrect Options:

Option A: Streptomycin: Streptomycin is a first-line anti-tubercular drug used in the initial treatment of TB. It is typically administered in combination with other first-line drugs and is effective against susceptible strains of *M. tuberculosis*.

Option C: Acyclovir: Acyclovir is an antiviral medication used to treat viral infections, particularly those caused by the herpes simplex virus (HSV). It is not used as an anti-tubercular drug.

Option D: Vancomycin: Vancomycin is an antibiotic primarily used to treat infections caused by Gram-positive bacteria, including methicillin-resistant *Staphylococcus aureus* (MRSA). It is not an anti-tubercular drug and is not commonly used in the treatment of TB.

Solution for Question 8:

Correct Option B:

- Sertaconazole is a topical antifungal medication that belongs to the azole class of antifungals. It is used to treat various fungal infections, including dermatophyte infections (such as ringworm) and yeast infections (such as candidiasis).
- Sertaconazole has both anti-inflammatory and anti-pruritic (anti-itch) effects in addition to its antifungal activity. It works by inhibiting the synthesis of ergosterol, an essential component of fungal cell membranes, thereby disrupting fungal growth and replication.
- The anti-inflammatory effects of sertaconazole help reduce redness, swelling, and other signs of inflammation associated with fungal infections. The anti-pruritic action helps relieve itching, which is a common symptom of many fungal infections.

Incorrect Options:

Option A: Ketoconazole: Ketoconazole is an antifungal medication that belongs to the azole class as well. While it is effective against various fungal infections, it does not have inherent anti-inflammatory or anti-pruritic effects. Its primary mode of action is the inhibition of ergosterol synthesis in fungal cells.

Option C: Luliconazole: Luliconazole is another azole antifungal used topically to treat fungal infections of the skin. Similar to ketoconazole, it does not have direct anti-inflammatory or anti-pruritic effects. Its primary mechanism of action is the inhibition of fungal ergosterol synthesis.

Option D: Terbinafine: Terbinafine is an allylamine antifungal medication used topically and orally for the treatment of fungal infections, especially dermatophyte infections. It primarily works by inhibiting the synthesis of ergosterol in fungal cells. Terbinafine does not possess intrinsic anti-inflammatory or anti-pruritic properties.

Solution for Question 9:

Correct Option C:

Among the options provided, dexamethasone (option A) is the drug that has been shown to reduce mortality in COVID-19 patients. Dexamethasone is a corticosteroid that has been widely studied in the treatment of COVID-19 and has demonstrated beneficial effects, particularly in severe cases. It has been shown to reduce inflammation and improve outcomes in patients requiring supplemental oxygen or mechanical ventilation.

Incorrect Options:

Option A. B and C:

Option B: Remdesivir has been authorized for emergency use in certain situations for the treatment of COVID-19. However, its impact on mortality reduction is not as pronounced as that of dexamethasone. Remdesivir primarily aims to reduce the duration of hospitalization and improve clinical recovery but does not significantly impact overall mortality rates.

Option C: Favipiravir is an antiviral medication that has shown some promise in the treatment of COVID-19. However, there is currently limited evidence regarding its effect on mortality reduction. More research is needed to establish its definitive role in COVID-19 management.

Option B. A and B

Option D. A, B, C is incorrect too

Solution for Question 10:

Correct Option C - Thioacetazone:

- Thioacetazone is an anti-tubercular drug that exhibits bacteriostatic activity rather than bactericidal activity. Bacteriostatic drugs inhibit the growth and reproduction of bacteria, but they do not directly kill them. Thioacetazone is primarily used in the treatment of drug-sensitive tuberculosis (TB) in combination with other anti-tubercular drugs.

Thioacetazone is an anti-tubercular drug that exhibits bacteriostatic activity rather than bactericidal activity. Bacteriostatic drugs inhibit the growth and reproduction of bacteria, but they do not directly kill them. Thioacetazone is primarily used in the treatment of drug-sensitive tuberculosis (TB) in combination with other anti-tubercular drugs.

Incorrect Options:

Option A - Isoniazid: Isoniazid is a first-line anti-tubercular drug that exhibits bactericidal activity. It works by inhibiting the synthesis of mycolic acids, an essential component of the bacterial cell wall in *Mycobacterium tuberculosis*. Isoniazid kills actively dividing tubercular bacteria.

Option B - Rifapentin: Rifapentin is a rifamycin derivative and a bactericidal anti-tubercular drug. It works by inhibiting bacterial RNA synthesis and has potent activity against *M. tuberculosis*. Rifapentin is often used in the treatment of drug-susceptible TB and as part of the regimen for latent TB infection.

Option D - Kanamycin: Kanamycin is an aminoglycoside antibiotic that exhibits bactericidal activity against a broad range of bacteria, including *Mycobacterium tuberculosis*. It works by binding to the bacterial ribosome and inhibiting protein synthesis, leading to the death of the bacteria.

Solution for Question 11:

Correct Option B: Mucormycosis

- Voriconazole is not effective against Mucormycosis.
- Voriconazole is an antifungal medication that belongs to the azole class of drugs. It is commonly used for the treatment of various fungal infections, including *Candida albicans*, *Candida tropicalis*, and Aspergillosis. Voriconazole works by inhibiting the synthesis of ergosterol, an essential component of the fungal cell membrane.
- However, voriconazole is not effective against Mucormycosis, which is caused by fungi of the order Mucorales. Mucormycosis is a serious and often life-threatening fungal infection that primarily affects individuals with compromised immune systems, such as those with poorly controlled diabetes, organ transplant recipients, or individuals receiving immunosuppressive therapy. The treatment of choice for Mucormycosis is typically amphotericin B, either alone or in combination with surgical debridement to remove infected tissue.

Incorrect Options:

Option A, C and D: Voriconazole is effective against *Candida albicans*, *Candida tropicalis*, and Aspergillosis

Solution for Question 12:

Correct Option B: Rifampicin

- Among the options provided, Rifampicin is an inhibitor of RNA synthesis.
- Rifampicin is a broad-spectrum antibiotic that is primarily used in the treatment of tuberculosis and other bacterial infections. It works by inhibiting bacterial RNA polymerase, which is responsible for synthesizing RNA during transcription. By blocking RNA polymerase, Rifampicin effectively inhibits RNA synthesis in bacteria.

Incorrect options:

Option A: 5-Fluorouracil is an antimetabolite that interferes with DNA synthesis, not RNA synthesis.

Option C: Nitrofurantoin is an antibiotic primarily used for urinary tract infections and does not directly inhibit RNA synthesis.

Option D: Novobiocin is a bacteriostatic antibiotic that acts by inhibiting bacterial DNA gyrase, which is involved in DNA replication and not RNA synthesis.

Solution for Question 13:

Correct Option C

- Schistosomiasis: Praziquantel is the drug of choice for the treatment of schistosomiasis, a parasitic infection caused by trematodes of the genus *Schistosoma*.

Incorrect options:

Option A. Strongyloidiasis: Although Praziquantel can be used for the treatment of certain parasitic infections, it is not the drug of choice for strongyloidiasis.

Option B. Trichomoniasis: Praziquantel is not used for the treatment of trichomoniasis. Metronidazole is the preferred medication for this sexually transmitted infection.

Option D. Rhinosporidiosis: Praziquantel is not the drug of choice for rhinosporidiosis, a chronic fungal infection. Surgical excision is the primary treatment modality for this condition.

Solution for Question 14:

Correct Option C: Valacyclovir

- The above clinical condition is suggestive of herpes zoster rash, and the most appropriate treatment is valacyclovir.

Incorrect options:

Option A: Amitriptyline, a tricyclic antidepressant indicated for neuropathic pain, can be given for post-herpetic neuralgia.

Option B: topical corticosteroids have no role in herpes zoster.

Option D: Mupirocin, an antibiotic, is used to treat impetigo as well as other skin infections caused by bacteria. It is not effective against fungal or viral infections.

Solution for Question 15:

Correct Option B:

- Patients with bacterial vaginosis and trichomoniasis who are taking metronidazole should avoid consuming alcohol.
- Metronidazole is an antibiotic commonly used to treat bacterial infections, including bacterial vaginosis and trichomoniasis. When metronidazole is taken along with alcohol, a disulfiram-like reaction can occur. This reaction leads to symptoms such as flushing, nausea, vomiting, headache, rapid heart rate, and in some cases, severe discomfort. These symptoms can be quite unpleasant and may discourage patients from completing their antibiotic treatment.
- It is essential to adhere to the instructions provided by the healthcare professional and abstain from alcohol consumption while taking metronidazole and for at least 24 to 48 hours after completing the course of treatment. This time frame may vary depending on the specific instructions given by the healthcare provider.

Incorrect Options :

Option A: Grapefruit juice: Grapefruit juice does not interact directly with metronidazole and is generally considered safe to consume.

Option C: Benzodiazepines: Benzodiazepines are a class of medications used to treat anxiety, insomnia, and other conditions. There are no specific interactions between metronidazole and benzodiazepines that would necessitate avoiding them altogether. However, it is always important to follow the guidance of the prescribing healthcare professional.

Option D: MAOIs/SSRIs: Monoamine oxidase inhibitors (MAOIs) and selective serotonin reuptake inhibitors (SSRIs) are types of antidepressant medications. While there are no significant interactions between metronidazole and these antidepressants, it is important to consult with a healthcare professional regarding the specific medications being taken to ensure there are no contraindications or interactions.

Solution for Question 16:

Correct Option B:

- In the given scenario, the medication most likely responsible for the raised serum digoxin levels and the development of atrial fibrillation is clarithromycin.
- Clarithromycin is an antibiotic that belongs to the macrolide class. It can inhibit the metabolism of digoxin, leading to increased serum digoxin levels. When clarithromycin is taken concomitantly with digoxin, it can interfere with the hepatic metabolism of digoxin through inhibition of the P-glycoprotein efflux transporter system, resulting in higher digoxin concentrations in the blood.
- Elevated digoxin levels can increase the risk of toxic effects, including the development or exacerbation of cardiac arrhythmias such as atrial fibrillation. Atrial fibrillation is a known side effect of digoxin toxicity.

Incorrect Options:

Option A: Potassium chloride: Potassium chloride is a supplement used to correct potassium deficiencies and is not known to directly interact with digoxin metabolism.

Option C: Atenolol: Atenolol is a beta-blocker used to treat hypertension and other cardiovascular conditions. While it can interact with digoxin, it typically reduces its effects rather than increasing its levels.

Option D: Triamterene: Triamterene is a potassium-sparing diuretic that can interact with digoxin, potentially leading to increased serum digoxin levels. However, in this case, clarithromycin is a more likely candidate for the increased digoxin levels.

Solution for Question 17:

Correct Option D:

- The reason for advising the use of an additional barrier method of contraception in a female patient on oral contraceptive pills (OCP) who is starting antitubercular therapy (ATT) is that rifampicin, one of the medications commonly used in ATT, can interact with oral

contraceptives and reduce their effectiveness.

- Rifampicin is known to induce the metabolism of many drugs, including oral contraceptives. It increases the activity of certain enzymes in the liver that metabolize medications, including the hormones in OCPs. This increased metabolism can lead to lower hormone levels in the body, potentially reducing the effectiveness of the oral contraceptives and increasing the risk of contraceptive failure.
- Therefore, using an additional barrier method of contraception, such as condoms, is advised to provide extra protection against unintended pregnancy during the course of ATT when rifampicin is being taken. This is important to ensure adequate contraception and prevent unwanted pregnancies while undergoing tuberculosis treatment

Incorrect Options:

Option A: Teratogenic Rifampicin is not Teratogenic

Option B failure: This is incorrect option

Option C: Rifampicin decreases OCP metabolism : This statement is wrong instead rifampicin increases it.

Solution for Question 18:

Correct Option A - Raltegravir:

- The life cycle of a virus involves several steps, including entry, replication, assembly, and release.
- In the given image, step 5 represents the integration of the viral DNA into the host genome. This step is crucial for the virus to establish a persistent infection.
- Raltegravir belongs to a class of antiviral drugs called integrase inhibitors. It inhibits the integration of viral DNA into the host genome by targeting the viral integrase enzyme. Therefore, raltegravir acts at step 5

Incorrect Options:

Option B - Enfuvirtide: Enfuvirtide belongs to a class of antiviral drugs called fusion inhibitors. It works by preventing the fusion of the viral and host cell membranes, which occurs at step 2 of the viral life cycle.

Option C - Tenofovir: Tenofovir belongs to a class of antiviral drugs called nucleoside reverse transcriptase inhibitors. It inhibits the reverse transcription of the viral RNA into DNA, which occurs at step 3 of the viral life cycle. Therefore, tenofovir does not act at step 5.

Option D - Zidovudine : Zidovudine drug also belongs to the class of nucleoside reverse transcriptase inhibitors. It inhibits the reverse transcription of viral RNA into DNA, which occurs at step 3 of the viral life cycle.

Solution for Question 19:

Correct Option B - Emtricitabine:

- The most likely drug causing hyperpigmentation in the palms and soles in a patient who is on combined active antiretroviral therapy (cART) is Emtricitabine.
- Emtricitabine is a nucleoside reverse transcriptase inhibitor (NRTI) that is commonly used in the treatment of HIV.
- It has been associated with the development of hyperpigmentation in some patients.
- The hyperpigmentation is usually seen on the palms and soles and may be accompanied by nail discoloration.
- This side effect is thought to be due to the drug's effects on melanin synthesis.

Incorrect Options:

Option A - Nelfinavir: Nelfinavir is a protease inhibitor used in the treatment of HIV. It works by inhibiting the protease enzyme, which is required for the production of mature, infectious viral particles. While nelfinavir can cause several side effects such as diarrhea, nausea, and abdominal pain, hyperpigmentation is not a known side effect of this drug.

Option C - Abacavir: Abacavir is a nucleoside reverse transcriptase inhibitor (NRTI) used in the treatment of HIV. Like emtricitabine, it can cause a hypersensitivity reaction that presents with skin rash, fever, and other symptoms. However, hyperpigmentation is not a known side effect of abacavir.

Option D - Zidovudine: Zidovudine is another NRTI used in the treatment of HIV. It works by inhibiting the reverse transcriptase enzyme, which is required for the replication of the virus. While zidovudine can cause several side effects such as nausea, headache, and fatigue, hyperpigmentation is not a known side effect of this drug. Therefore, zidovudine is an incorrect option.

Solution for Question 20:

Correct Option B:

- Amphotericin B is the drug of choice for mucormycosis because it has broad-spectrum antifungal activity and is effective against many different types of fungi, including the ones that cause mucormycosis.
- It works by binding to the fungal cell membrane and disrupting its structure, leading to cell death. Additionally, it is able to penetrate into tissues and reach the site of infection, which is important in the treatment of mucormycosis, which often involves deep tissue invasion. While Amphotericin B has some side effects, such as fever, chills, and kidney toxicity, it remains the first-line treatment for mucormycosis due to its effectiveness against the infection.

Incorrect Options:

Option A: Fluconazole: It is a triazole antifungal agent that is effective against many fungal species, including some forms of *Candida*, *Cryptococcus*, and *Histoplasma*.

Option C: Ketoconazole: It is a broad-spectrum antifungal medication that is effective against many types of fungi, including *Candida* species.

Option D: Griseofulvin: It is an antifungal drug that is mainly used to treat superficial fungal infections of the skin, hair, and nails.

Solution for Question 21:

Correct Option D - Erythromycin:

- Erythromycin is not typically recommended for the treatment of bacterial vaginosis. While it is an antibiotic, it is not as effective against the specific bacteria that contribute to BV compared to metronidazole, tinidazole, or clindamycin. Therefore, according to CDC guidelines, erythromycin is not a preferred or alternative treatment option for bacterial vaginosis.

Incorrect Options:

Option A - Metronidazole: Metronidazole is commonly used as a first-line treatment for bacterial vaginosis. It has been shown to effectively target the anaerobic bacteria that contribute to BV. CDC guidelines recommend oral metronidazole (500 mg twice daily for 7 days) or metronidazole gel (0.75%, once daily for 5 days) as a preferred treatment option.

Option B - Tinidazole: Tinidazole is another antibiotic that is effective against the bacteria associated with bacterial vaginosis. It is generally considered an alternative to metronidazole, especially for patients who cannot tolerate metronidazole. CDC guidelines suggest a single 2 g dose of oral tinidazole as an alternative treatment option.

Option C - Clindamycin: Clindamycin, in the form of both oral and vaginal preparations, is also used for the treatment of bacterial vaginosis. It has been shown to be effective against the condition by suppressing the overgrowth of certain bacteria. CDC guidelines recommend oral clindamycin (300 mg twice daily for 7 days) or clindamycin vaginal cream (2%, once daily for 7 days) as alternative treatment options.

Solution for Question 22:

Correct Option A - ATP synthesis inhibition:

- Bedaquiline targets the ATP synthase enzyme of *Mycobacterium tuberculosis*, inhibiting its activity. ATP synthase is essential for the production of ATP, the main energy currency of cells. By inhibiting ATP synthesis, Bedaquiline disrupts the energy production of the bacterium, leading to its growth inhibition.

Incorrect Options:

Option B - Mycolic acid synthesis inhibition: Mycolic acids are crucial components of the mycobacterial cell wall, contributing to its impermeability and resistance to the host immune response. However, Bedaquiline's primary mechanism of action is not directly related to inhibiting mycolic acid synthesis. Instead, it targets ATP synthesis as mentioned in option A.

Option C - Protein inhibition: Bedaquiline's mechanism of action is not primarily related to protein inhibition. Its main target is the ATP synthase enzyme, which affects energy production in the bacterium, as explained in option A.

Option D - Inhibiting reverse transcriptase enzyme: This mechanism is not relevant to Bedaquiline's action. Reverse transcriptase is an enzyme found in retroviruses like HIV, and

Bedaquiline is not used to treat viral infections but rather for tuberculosis.

Solution for Question 23:

Correct Option B - Oral Fidaxomicin:

- Antibiotics usually kill normal gut flora leading to multiplication of pathogenic bacteria (one of the most common among them is *C. Difficile* which further causes pseudomembranous colitis, risk of it is also increased by usage of ampicillin, clindamycin and 3rd generation cephalosporins)
- Drug of choice for this condition is Oral Fidaxomicin.

Incorrect Options:

Option A - Oral metronidazole:

- This option is incorrect as Oral metronidazole is not the drug of choice for this condition. It is the 2nd drug of choice.

Option C - Oral ciprofloxacin:

- This option is incorrect as Oral ciprofloxacin is not the drug of choice for this condition.

Option D - Bezlotoxumab:

- This option is incorrect as Bezlotoxumab is not the drug of choice for this condition.
-

Solution for Question 24:

Correct Option A - Fluconazole - *Candida glabrata* and *Candida krusei*:

- Fluconazole - does not work on *Candida glabrata* and *Candida krusei*
- Resistance is mainly by Efflux pump overexpression ABC and MFS transporters
- ABC and MFS transporters
- ABC and MFS transporters

Incorrect Options:

Option B - Amphotericin B - candida glabrata:

- This statement is incorrect as amphotericin B is effective against *Candida glabrata*.

Option C - Voriconazole - aspergillus fumigatus:

- This statement is incorrect as voriconazole is effective against *Aspergillus fumigatus*.

Option D - Itraconazole - sporothrix:

- This option is incorrect as itraconazole is effective against *Sporothrix*.

Solution for Question 25:

Correct Option D.

- Amoxicillin: Amoxicillin is considered safe to use during pregnancy for the treatment of urinary tract infections, including cystitis.

Incorrect options:

Option A. Ciprofloxacin: Amoxicillin is preferred.

Option B. Gentamicin: Gentamicin is an aminoglycoside antibiotic that is not commonly used for the treatment of cystitis. It is generally reserved for severe infections and may have potential risks during pregnancy.

Option C. Cotrimoxazole: Cotrimoxazole (trimethoprim/sulfamethoxazole) is an antibiotic that is generally avoided during pregnancy, particularly during the first trimester, due to potential risks to the developing fetus.

Solution for Question 26:

Answer: D) Rifampicin 450 mg

Explanation:

For leprosy treatment in adults, the correct dosage of Rifampicin is 600 mg monthly, not 450 mg.

The World Health Organization (WHO) Multi-Drug Therapy (MDT) regimen for adults is:

- Paucibacillary (PB) leprosy (6 months MDT): Rifampicin 600 mg once monthly (supervised)
Dapsone 100 mg daily (self-administered)
- Rifampicin 600 mg once monthly (supervised)
- Dapsone 100 mg daily (self-administered)
- Multibacillary (MB) leprosy (12 months MDT): Rifampicin 600 mg once monthly (supervised) Clofazimine 300 mg once monthly (supervised) + 50 mg daily (self-administered) Dapsone 100 mg daily (self-administered)
- Rifampicin 600 mg once monthly (supervised)
- Clofazimine 300 mg once monthly (supervised) + 50 mg daily (self-administered)
- Dapsone 100 mg daily (self-administered)
- Rifampicin 600 mg once monthly (supervised)
- Dapsone 100 mg daily (self-administered)
- Rifampicin 600 mg once monthly (supervised)
- Clofazimine 300 mg once monthly (supervised) + 50 mg daily (self-administered)
- Dapsone 100 mg daily (self-administered)

Rifampicin 450 mg is incorrect because the standard dose for adults is 600 mg once monthly. However, 450 mg is used in children (10 mg/kg).

Characteristic		Paucibacillary		Multibacillary	
Skin lesions		1-5 lesions		>5 lesions	
Nerve involvement		No nerve		≥1 nerve involved	
Slit skin smear		Negative		Positive	
		Treatment duration	Dapsone	Clofazimine	Rifampicin
Paucibacillary (PB)	Adult	6 months	100 mg daily	300 mg once a month & 50 mg daily	600 mg once a month
Child (9-14 yrs)	6 months	50 mg daily	150 mg once a month & 50 mg alternate day	450 mg once a month	
Multibacillary (MB)	Adult	12 months	100 mg daily	300 mg once a month & 50 mg daily	600 mg once a month

Child (9-14 yrs)	12 months	50 mg daily	150 mg once a month & 50 mg alternate day	450 mg once a month	
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Single lesion paucibacillary:

- One skin lesion with no nerve involvement.
- In India: Treatment is similar to paucibacillary (Rifampicin, Dapsone, Clofazimine for 6 months)
- According to WHO: ROM therapy Rifampicin (600mg), Ofloxacin (400mg), Minocycline (100mg) - Single dose. For a child, half of these adult doses are administered.
- Rifampicin (600mg), Ofloxacin (400mg), Minocycline (100mg) - Single dose.
- For a child, half of these adult doses are administered.
- Rifampicin (600mg), Ofloxacin (400mg), Minocycline (100mg) - Single dose.
- For a child, half of these adult doses are administered.

Drug-resistant leprosy Regimen		
	Intensive Phase (6 months)	Continuation Phase (18 months)
Rifampicin-resistant cases: 2 second line + 1 first line (Clofazimine)	Ofloxacin + Minocycline + Clofazimine (or) Ofloxacin + Clarithromycin + Clofazimine	Ofloxacin + Minocycline + Clofazimine (or) Ofloxacin + Clofazimine
Rifampicin & ofloxacin resistant cases	Clarithromycin + Minocycline + Clofazimine	(Clarithromycin or Minocycline) + Clofazimine

- Dapsone 100 mg (Option A): Correct for both PB and MB leprosy.
- Clofazimine 50 mg (Option B): Correct as a daily dose for MB leprosy.
- Clofazimine 300 mg (Option C): Correct as a monthly supervised dose for MB leprosy.

Solution for Question 27:

Answer: C) Immunological changes in the body

Explanation:

The deterioration of health after starting ART in an HIV-TB co-infected patient is most likely due to Immune Reconstitution Inflammatory Syndrome (IRIS), which is caused by immunological

changes in response to ART.

- Mechanism: ART restores CD4+ T-cell function and improves the immune response. This leads to a stronger inflammatory response against pre-existing TB infection, worsening symptoms. IRIS typically occurs 2–12 weeks after ART initiation.
- ART restores CD4+ T-cell function and improves the immune response.
- This leads to a stronger inflammatory response against pre-existing TB infection, worsening symptoms.
- IRIS typically occurs 2–12 weeks after ART initiation.
- Clinical Presentation: Worsening TB symptoms (fever, lymphadenopathy, worsening respiratory symptoms). New inflammatory lesions (e.g., enlarging TB lymph nodes, worsening pleural effusions). No evidence of drug resistance.
- Worsening TB symptoms (fever, lymphadenopathy, worsening respiratory symptoms).
- New inflammatory lesions (e.g., enlarging TB lymph nodes, worsening pleural effusions).
- No evidence of drug resistance.
- Management: Continue both TB treatment and ART. NSAIDs or corticosteroids may be used for severe cases.
- Continue both TB treatment and ART.
- NSAIDs or corticosteroids may be used for severe cases.
- ART restores CD4+ T-cell function and improves the immune response.
- This leads to a stronger inflammatory response against pre-existing TB infection, worsening symptoms.
- IRIS typically occurs 2–12 weeks after ART initiation.
- Worsening TB symptoms (fever, lymphadenopathy, worsening respiratory symptoms).
- New inflammatory lesions (e.g., enlarging TB lymph nodes, worsening pleural effusions).
- No evidence of drug resistance.
- Continue both TB treatment and ART.
- NSAIDs or corticosteroids may be used for severe cases.
- HIV-positive individuals face an 8-fold increased risk of developing TB and are prone to more severe and potentially fatal extrapulmonary TB.
- To manage co-infection, ATT is initiated first, followed by ART after 2 weeks to mitigate the risk of Immune Reconstitution Inflammatory Syndrome (IRIS), an exaggerated inflammatory response.
- Rifampin, a potent CYP inducer in ATT, accelerates the metabolism of certain ART drugs like nevirapine (NNRTI), dolutegravir (integrase inhibitor), and protease inhibitors (e.g., Ritonavir, Indinavir, lopinavir), reducing their effectiveness. However, nucleoside reverse

transcriptase inhibitors (NRTIs) and efavirenz can be safely combined with Rifampin.

- Rifabutin, a weaker enzyme inducer, is often used as an alternative to mitigate drug interactions in ART.
- All HIV-positive TB patients should also receive cotrimoxazole preventive therapy to guard against *Pneumocystis jirovecii* infection.
- Development of drug resistance to TB drugs (Option A) is incorrect because TB drug resistance develops over months to years, whereas IRIS occurs within weeks of ART initiation.
- Drug-drug interactions between TB and ART medications (Option B) is incorrect because although interactions (e.g., rifampicin reducing ART levels) are common, they usually lead to virologic failure, not an inflammatory worsening.
- Development of drug resistance to ART (Option D) is incorrect because ART resistance presents with viral load rebound and worsening HIV, not a paradoxical inflammatory response.

Solution for Question 28:

Answer: C) Ciprofloxacin

Explanation:

The patient presents with fever, burning micturition (suggestive of a urinary tract infection, UTI), and hearing problems with dizziness (suggestive of ototoxicity). Given the penicillin allergy, the most appropriate alternative is Ciprofloxacin (a fluoroquinolone), which is:

- Effective against common UTI pathogens (*E. coli*, *Klebsiella*, *Proteus*, *Pseudomonas*).
- Safe in penicillin-allergic patients.
- Less likely to exacerbate hearing loss compared to aminoglycosides.

Ciprofloxacin	
Pharmacokinetics	Good oral absorption Bioavailability ~70%, Moderate half-life (~4-5 hours) Widely distributed in tissues and body fluids Renal excretion

Uses	Urinary tract infections Prostatitis Pseudomonas aeruginosa (DOC) Anthrax (Bacillus anthracis) (DOC) Gastrointestinal infections (e.g., traveller's diarrhoea) Bone and joint infections
Side Effects	Gastrointestinal disturbances (nausea, diarrhoea). CNS effects (headache, dizziness). Tendonitis and tendon rupture. Photosensitivity.
Contraindications	Pregnancy Children (due to potential for cartilage damage) Myasthenia gravis (can exacerbate muscle weakness)

- Azithromycin (Option A) is incorrect because macrolides are not first-line for UTIs, as they have poor urinary penetration and limited coverage against Gram-negative uropathogens.
- Amikacin (Option B) is incorrect because although aminoglycosides are effective against UTIs, they are highly ototoxic and nephrotoxic, which is risky given the patient's hearing problems and dizziness.
- Ampicillin-Clavulanic Acid (Option D) is incorrect because the patient has a penicillin allergy, making this option unsuitable.

Solution for Question 29:

Answer: B) Isoniazid

Explanation:

Isoniazid (INH) resistance in *Mycobacterium tuberculosis* is commonly associated with mutations in the KATG gene, which encodes catalase-peroxidase.

- Mechanism of Action: Isoniazid is a prodrug that requires activation by KatG (catalase-peroxidase enzyme). Activated INH inhibits mycolic acid synthesis, disrupting the mycobacterial cell wall.
- Isoniazid is a prodrug that requires activation by KatG (catalase-peroxidase enzyme).
- Activated INH inhibits mycolic acid synthesis, disrupting the mycobacterial cell wall.
- Mechanism of Resistance: Mutations in KATG (especially Ser315Thr mutation) prevent isoniazid activation, leading to drug resistance. Other mutations involve inhA gene (affecting the target enzyme enoyl-ACP reductase), leading to low-level resistance.

- Mutations in KATG (especially Ser315Thr mutation) prevent isoniazid activation, leading to drug resistance.
- Other mutations involve inhA gene (affecting the target enzyme enoyl-ACP reductase), leading to low-level resistance.
- Isoniazid is a prodrug that requires activation by KatG (catalase-peroxidase enzyme).
- Activated INH inhibits mycolic acid synthesis, disrupting the mycobacterial cell wall.
- Mutations in KATG (especially Ser315Thr mutation) prevent isoniazid activation, leading to drug resistance.
- Other mutations involve inhA gene (affecting the target enzyme enoyl-ACP reductase), leading to low-level resistance.

The mechanisms of resistance with Isoniazid are given below.

Mechanism of resistance	
KatG gene mutation	INH is a prodrug, converted to the active form by catalase-peroxidase produced by the KATg gene. The mutated catalase enzyme does not form the active drug
KasA gene mutation	Component of cell wall mycolic acid
InhA gene overexpression	Causes cross-resistance to Ethionamide (second-line drug)



Rifampicin (Option A) is incorrect because resistance is due to rpoB gene mutations, affecting RNA polymerase.

Moxifloxacin (Option C) is incorrect because fluoroquinolone resistance is linked to mutations in gyrA and gyrB genes, which encode DNA gyrase.

Ethambutol (Option D) is incorrect because resistance involves mutations in the embB gene, which encodes arabinosyl transferase, affecting cell wall synthesis.

Solution for Question 30:

Correct Option A - Rifampicin:

- Rifampicin is a potent inducer of the cytochrome P450 enzymes, particularly CYP3A4, which can significantly lower the plasma concentrations of Nevirapine. Nevirapine is metabolized by the CYP3A4 enzyme, and induction of this enzyme by rifampicin can lead to subtherapeutic levels of Nevirapine, reducing its effectiveness in managing HIV. This interaction can lead to treatment failure and the development of drug resistance. Therefore, rifampicin should be avoided or used with extreme caution, and alternative antiretroviral drugs or anti-tuberculosis medications should be considered.

Incorrect Options:

Option B - Streptomycin:

- Streptomycin is an aminoglycoside antibiotic used primarily for the treatment of tuberculosis. It does not have significant interactions with the antiretroviral drugs Zidovudine, Lamivudine, or Nevirapine. Therefore, it can be used safely in combination with this regimen without the risk of significant drug interactions.

Option C - INH (Isoniazid):

- Isoniazid (INH) is commonly used in the treatment and prophylaxis of tuberculosis. It does not significantly interact with Zidovudine, Lamivudine, or Nevirapine, and can generally be used safely in patients receiving this combination antiretroviral therapy. There may be some concern about hepatotoxicity when used with other hepatotoxic drugs, but this does not constitute a direct contraindication.

Option D - Ethambutol:

- Ethambutol is another first-line anti-tuberculosis medication. It has minimal drug interactions with antiretroviral drugs, including Zidovudine, Lamivudine, and Nevirapine. Ethambutol can be used safely in combination with this HIV regimen without significant concerns for drug interactions.
-

Solution for Question 31:

Correct Option A - Resistance to Both Rifampicin and Isoniazid with Mutations in the inhA and katG Genes:

- Bedaquiline is typically used for treating multidrug-resistant (MDR) tuberculosis, which includes cases with resistance to both rifampicin and isoniazid. This includes strains with mutations in the inhA and katG genes, which are associated with isoniazid resistance. Bedaquiline is considered in such cases when other second-line drugs are not effective or suitable, providing a vital treatment option for MDR-TB.

Incorrect Options:

Option B - Resistance to Rifampicin but Sensitivity to Fluoroquinolones:

- While resistance to rifampicin is a key factor in MDR-TB, the inclusion criterion for starting bedaquiline is more specific. Sensitivity to fluoroquinolones alone does not meet the criteria for bedaquiline use; it is usually considered in the broader context of multidrug-resistant TB, including resistance to both rifampicin and isoniazid.

Option C - Resistance to Rifampicin and Sensitivity to Isoniazid:

- Resistance to rifampicin and sensitivity to isoniazid would be classified as rifampicin-resistant TB rather than MDR-TB. Bedaquiline is typically used for MDR-TB, which involves resistance to both rifampicin and isoniazid, not just one.

Option D - Resistance to Both Rifampicin and Fluoroquinolones:

- Resistance to both rifampicin and fluoroquinolones suggests a more severe form of drug-resistant TB, potentially extensively drug-resistant (XDR-TB). Bedaquiline may still be used in such cases, but the initial criterion for starting it is based on MDR-TB, which includes resistance to both rifampicin and isoniazid.

Solution for Question 32:

Correct Option B -Tenofovir:

- Tenofovir is a nucleotide reverse transcriptase inhibitor (NtRTI) is the drug of choice used in the treatment of both HIV and Hepatitis B infections.

Incorrect Options:

Options A, C, D:

- Dolutegravir, Zidovudine, and Ritonavir can not be used for both HIV and Hepatitis B infections.
-

Solution for Question 33:

Correct Option B - Artemisinin-based combination therapy (ACT):

- Artemisinin-based combination therapy (ACT) is the preferred treatment for *Plasmodium falciparum* malaria.

Incorrect Options:

Option A - Chloroquine

- *Plasmodium falciparum* is resistant to chloroquine, making it ineffective for treatment.

Option C - Primaquine

- Primaquine is used as a gametocidal agent in *Plasmodium falciparum* malaria to prevent transmission but not for the treatment of acute infection.

Option D - Doxycycline

- Doxycycline is given as traveler's prophylaxis for malaria.
-

Solution for Question 34:

Correct Option D - Ethambutol:

- Ethambutol is the anti-tubercular drug most closely associated with visual disturbances, particularly optic neuritis, color blindness, and reduced visual acuity.

Incorrect Options:

Option A – Rifampicin: The side effects of rifampicin include hepatotoxicity, flu-like symptoms, and red-orange discoloration of urine.

Option B – Streptomycin: The side effects of streptomycin include ototoxicity and nephrotoxicity.

Option C – Pyrazinamide: Pyrazinamide is associated with a maximum risk of hepatotoxicity and hyperuricemia.

Solution for Question 35:

Correct Option C - It should not be co-administered with pyrazinamide:

- There is no known significant interaction between voriconazole and pyrazinamide. Voriconazole's interactions primarily involve drugs metabolized by CYP2C19, CYP2C9, and CYP3A4.

Incorrect Options:

Option A - It should be taken on an empty stomach: Because high-fat meals can reduce voriconazole bioavailability, the oral drug should be administered either 1 hour before or 1 hour after meals.

Option B - It requires therapeutic drug monitoring: Therapeutic drug monitoring is vital for maintaining target serum concentrations between 1 and 5 mg/L, particularly for children due to their unpredictable plasma levels of voriconazole.

Option D - It requires a loading dose: Voriconazole typically requires a loading dose to quickly reach therapeutic levels.

Solution for Question 36:

Correct Option A - Clofazimine:

- Clofazimine is a fat-soluble riminophenazine dye used in the treatment of leprosy. Prolonged treatment with clofazimine leads to extensive bioaccumulation and precipitation

throughout the body, forming crystal-like drug inclusions (CLDIs).

- Common side effects of the drug include gastrointestinal problems, ichthyosis, and discoloration of skin, eyes, and bodily secretions. This discoloration, which can lead to skin hyperpigmentation, is one of the more distressing or stigmatizing effects of the drug. It can also prolong the QT interval on an ECG.

Incorrect Options:

- Options B, C, and D are not typically associated with discoloration of skin or hyperpigmentation.

Solution for Question 37:

Correct Option A - Linezolid: Binds to 30S ribosome and inhibits protein synthesis:

- Protein synthesis inhibitors bind either to 30 s or 50 s
- Drugs binding to 30s are - Tetracycline and aminoglycoside
- Drugs binding to 50s are - Chloramphenicol, Linezolid and clindamycin

Incorrect Options:

Option B - Rifampicin: Inhibits DNA-dependent RNA polymerase:

- This option is correct as Rifampicin works by inhibiting DNA-dependent RNA polymerase.

Option C - Aztreonam: Inhibit cell wall synthesis:

- This Option is correct as Aztreonam works by inhibiting cell wall synthesis.

Option D - Daptomycin: Cause depolarization of membranes:

- This option is correct as Daptomycin causes depolarization of membranes.

Solution for Question 38:

Correct Choice: d

Hemophilus produces beta-lactamase, hence resistant to ampicillin: This statement is likely the reason for switching from ampicillin to ceftriaxone. Hemophilus influenzae is known to produce beta-lactamase, an enzyme that can break down and inactivate beta-lactam antibiotics such as ampicillin. Ceftriaxone, on the other hand, is a third-generation cephalosporin antibiotic that is not as susceptible to beta-lactamase degradation and can be effective against Hemophilus influenzae infections.

Incorrect Choices:

Option a. Hemophilus alters penicillin binding protein sites, hence resistant to ampicillin: This statement is not completely accurate. While some strains of Hemophilus influenzae may have altered penicillin binding protein sites and be resistant to ampicillin, this is not the only mechanism of resistance. In addition, not all strains of Hemophilus influenzae are resistant to ampicillin.

Option b. Easier to administer IV ceftriaxone than IV ampicillin: While ceftriaxone can be administered once a day, which may be more convenient than the multiple daily doses required for ampicillin, ease of administration is not the primary reason for switching from ampicillin to ceftriaxone.

Option c. Cotrimoxazole is the drug of choice, but the patient is allergic to sulfa drugs: Cotrimoxazole (trimethoprim-sulfamethoxazole) is an antibiotic that is often used to treat Hemophilus influenzae infections. However, if the patient is allergic to sulfa drugs, cotrimoxazole may not be a suitable treatment option.

Solution for Question 39:

Correct Choice: a

Ritonavir: This medication is a protease inhibitor that is used to treat HIV. When used in combination with saquinavir, ritonavir can also be used to boost the levels of saquinavir in the blood. Ritonavir can inhibit the activity of an enzyme called cytochrome P450, which is responsible for breaking down saquinavir in the liver. By inhibiting the activity of this enzyme, ritonavir can increase the bioavailability of saquinavir, leading to higher levels of the medication in the blood and improved antiviral activity.

Incorrect Choices:

Option b. Vitamin C: Vitamin C is an essential nutrient that is involved in many important functions in the body, including immune function, collagen production, and wound healing. However, it is not known to have any significant effect on the bioavailability of saquinavir.

Option c. Cimetidine: Cimetidine is a medication that is used to reduce stomach acid production and is sometimes used to treat acid reflux and ulcers. It can also inhibit the activity of cytochrome P450, which is involved in the metabolism of many medications, including saquinavir. However, current evidence suggests that cimetidine has only a modest effect on the bioavailability of saquinavir and may not be a reliable method for increasing saquinavir levels.

Option d. Ganciclovir: Ganciclovir is an antiviral medication that is used to treat herpes viruses and cytomegalovirus (CMV) infections. It is not known to have any significant effect on the bioavailability of saquinavir, as it works through a different mechanism of action compared to saquinavir.

Solution for Question 40:

Correct Option: C: Cell membrane

- Amphotericin B primarily acts on the cell membrane of fungi.
- It binds to ergosterol, a component of the fungal cell membrane, forming pores and disrupting the integrity and function of the membrane.
- This leads to leakage of intracellular contents, altered ion balance, and ultimately, cell death.
- By targeting the cell membrane, amphotericin B exhibits its antifungal activity.

Incorrect Options:

Option A: Nucleic acid- Amphotericin B does not directly target nucleic acids in fungi. Its mechanism of action is primarily related to the disruption of the cell membrane.

Option B: Ribosome- Amphotericin B does not target the ribosomes of fungi. Ribosomes are involved in protein synthesis, and their inhibition is the mechanism of action of certain antibiotics, but not amphotericin B.

Option D: Cytoplasm- Amphotericin B does not act within the cytoplasm of fungi. Its primary site of action is the cell membrane, as mentioned above.

Solution for Question 41:

Correct Option C - Tenofovir:

- Tenofovir is a nucleotide reverse transcriptase inhibitor (NRTI) that is effective against both HIV and HBV.
- It inhibits the reverse transcriptase enzyme, which is essential for the replication of both viruses.
- Tenofovir is available in two main formulations: tenofovir disoproxil fumarate (TDF) and tenofovir alafenamide (TAF).
- Both are used in the treatment of HIV infection as well as chronic hepatitis B infection.
- TDF and TAF are included in various combination antiretroviral therapy (ART) regimens for HIV/AIDS, providing dual antiviral activity against both viruses.

Incorrect Options: Options A, B, and D are incorrect.

Solution for Question 42:

Correct Option is C: It is activated by viral thymidine kinase:

- Foscarnet is an antiviral medication used for the treatment of certain viral infections, including resistant cytomegalovirus (CMV) infections. It works by directly inhibiting viral DNA polymerase without requiring activation by viral thymidine kinase. Therefore, option C is incorrect.

Incorrect Options:

Option A: It is used for resistant CMV: Foscarnet is indicated for the treatment of CMV retinitis in immunocompromised patients, particularly those who have failed or are intolerant to ganciclovir therapy.

Option B: Renal toxicity is seen: Foscarnet can cause renal toxicity, including electrolyte imbalances, impaired renal function, and nephrotoxicity. Regular monitoring of renal function and

serum electrolytes is necessary during foscarnet therapy.

Option D: Regular monitoring of serum electrolytes is required: As mentioned above, foscarnet can cause electrolyte imbalances, so regular monitoring of serum electrolytes, including calcium, magnesium, potassium, and phosphate, is important.

Option E: It can cause genital ulceration: Foscarnet has been associated with the development of genital ulcerations, which can be a side effect of the medication.

Solution for Question 43:

Correct Option is A: False: Zidovudine is not used as monotherapy in post-exposure prophylaxis.

- Zidovudine (also known as AZT) is not used as monotherapy in post-exposure prophylaxis (PEP) for HIV. PEP typically involves using a combination of antiretroviral drugs to reduce the risk of HIV transmission after an accidental exposure. Zidovudine is commonly used as part of a multi-drug regimen in PEP protocols.

Incorrect Options:

Option B: Washing hands with soap and water is advised after an accidental needle prick to minimize the risk of infection.

Option C: Baseline viral markers of healthcare personnel should be done at the time of the needle prick to establish their current HIV status.

Option D: Follow-up viral markers of healthcare personnel should be measured at 6 weeks after the needle prick to monitor for any possible seroconversion or HIV infection.

Option E: Tenofovir + Lamivudine + Efavirenz is a commonly used combination of antiretroviral drugs for post-exposure prophylaxis (PEP). This combination is typically taken for a period of 4 weeks to reduce the risk of HIV transmission.

Solution for Question 44:

Correct Option is A: Tafenoquine is used in the radical cure of Plasmodium infections, which means it is used to eliminate both the blood-stage and dormant liver-stage parasites in patients with malaria.

Incorrect Options:

Option B: Prophylaxis of malaria in pregnancy typically involves the use of other antimalarial medications such as chloroquine or mefloquine.

Option C: Severe falciparum malaria is usually treated with intravenous antimalarial medications such as quinine or artesunate.

Option D: Tafenoquine is not approved for use in children under 18 years of age.

Solution for Question 45:

Correct Option: A.

- In the case of co-infection with Chlamydia trachomatis and Neisseria gonorrhoeae, the drug of choice for treatment is Azithromycin. Azithromycin is an antibiotic that is effective against both Chlamydia and Gonococcus. It is a macrolide antibiotic that inhibits bacterial protein synthesis, leading to the eradication of the bacteria causing the infection.

Incorrect Options:

Option B. Ceftriaxone: Ceftriaxone is primarily used for the treatment of Neisseria gonorrhoeae infection. While it is effective against Gonococcus, it is not sufficient to treat Chlamydia. Therefore, it is important to use a combination therapy with Azithromycin to cover both organisms in cases of co-infection.

Option C. Fluconazole: Fluconazole is an antifungal medication used to treat fungal infections. It is not effective against bacterial infections like Chlamydia and Gonococcus.

Option D. Clindamycin: Clindamycin is an antibiotic primarily used for treating anaerobic bacterial infections. It is not the drug of choice for treating Chlamydia and Gonococcus.

Solution for Question 46:

Correct Option: B

- Ceftriaxone is the drug of choice for prophylaxis of meningococcal meningitis in a pregnant female after contact with a case. Meningococcal meningitis is a severe bacterial infection caused by the bacterium *Neisseria meningitidis*. It can be transmitted through respiratory droplets and close contact with an infected individual.
- Ceftriaxone is a third-generation cephalosporin antibiotic that has excellent activity against *Neisseria meningitidis*. It is highly effective in eradicating the bacteria from the nasopharynx, thus preventing the development of meningitis in individuals who have been exposed to the bacterium.

Incorrect Options:

Option A) Ciprofloxacin: Ciprofloxacin is a fluoroquinolone antibiotic that has activity against *Neisseria meningitidis*. However, it is generally not recommended for prophylaxis in pregnant females due to potential adverse effects on the fetus, such as cartilage development issues.

Option C) Cephalexin: Cephalexin is a first-generation cephalosporin antibiotic that has limited activity against *Neisseria meningitidis*. It is not the drug of choice for prophylaxis in this situation.

Option D) Rifabutin: Rifabutin is an antibiotic primarily used in the treatment of tuberculosis and certain *Mycobacterium avium* complex infections. It does not provide adequate coverage against *Neisseria meningitidis* and is not indicated for prophylaxis in this case.

Solution for Question 47:

Correct Option C.

- Ribavirin: Ribavirin is an antiviral medication that can be administered via aerosol for the treatment of respiratory syncytial virus (RSV) infection in children. It helps reduce the severity of symptoms and improve outcomes in RSV-infected infants and young children.

Incorrect options:

Option A. Indinavir: Indinavir is an antiretroviral medication used for the treatment of HIV infection. It is not indicated for the treatment of RSV infection.

Option B. Amantadine: Amantadine is an antiviral medication primarily used for the prevention and treatment of influenza A virus infection. It does not have activity against RSV.

Option D. Tenofovir: Tenofovir is an antiretroviral medication used in the treatment of HIV and hepatitis B virus infection. It is not indicated for the treatment of RSV infection.

Solution for Question 48:

Correct Option A - Doxycycline:

The given clinical picture is suggestive of mild to moderate scrub typhus:

- In mild to moderate scrub typhus only Doxycycline is given.
- The given picture of eschar is the site of the bite.
- It is given through oral route.

Incorrect Options:

Option B - Doxycycline + Azithromycin:

- This option is incorrect as Doxycycline + Azithromycin (Dual Therapy) is given in severe scrub typhus which presents with high grade fever and body pain along with end-organ damage.

Option C - Ceftriaxone + Doxycycline:

- This combination is not preferred in scrub typhus.

Option D - Ceftriaxone:

- This option is incorrect because Doxycycline is a preferred drug in this case.
-

Solution for Question 49:

Correct Option B - Vilazodone:

- It is a novel antidepressant medication that is used to treat major depressive disorder. It is a selective serotonin reuptake inhibitor (SSRI) and a partial agonist at the serotonin 5-HT_{1A} receptor. Vilazodone works by blocking the reuptake of serotonin, which helps to increase the amount of serotonin in the brain. It also activates the 5-HT_{1A} receptor, which is thought to enhance the antidepressant effect. Vilazodone was approved by the FDA in 2011 for the treatment of major depressive disorder.

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Incorrect Options:

Option A - Lurasidone: It is an atypical antipsychotic medication that is used to treat schizophrenia and bipolar disorder. It works by modifying the balance of certain chemicals in the brain, such as dopamine and serotonin. Lurasidone is believed to block the activity of certain dopamine receptors and enhance the activity of specific serotonin receptors. It was approved by the FDA in 2010 for the treatment of schizophrenia, and in 2013 for the treatment of bipolar depression.

Option C - Asenapine: It is an atypical antipsychotic medication that is used to treat schizophrenia and bipolar disorder. It works by blocking certain receptors in the brain, such as dopamine and serotonin receptors. Asenapine is believed to block the activity of certain dopamine and serotonin receptors, which helps to modify the balance of these chemicals in the brain. It was approved by the FDA in 2009 for the treatment of schizophrenia, and in 2010 for the treatment of bipolar disorder.

Option D - Blonanserin: It is an atypical antipsychotic medication that is used to treat schizophrenia. It works by blocking certain receptors in the brain, such as dopamine and serotonin receptors. Blonanserin is believed to block the activity of certain dopamine receptors, which helps to modify the balance of these chemicals in the brain. It was approved by the FDA in Japan in 2008 for the treatment of schizophrenia.

Solution for Question 50:

Correct Option A - Inhibition of calcineurin:

- Cyclosporine is an immunosuppressive medication that works by inhibiting the activity of calcineurin. Calcineurin is a phosphatase enzyme that is involved in T-cell activation, a crucial component of the immune system's response to foreign substances and pathogens.
- When T-cells are activated, they release cytokines, signaling molecules that can attract and activate other immune cells. Cyclosporine works by binding to cyclophilin, a protein that is expressed in T-cells, and this complex then binds to and inhibits calcineurin. This results in a reduction of cytokine production and T-cell proliferation.
- The reduction of cytokine production and T-cell proliferation leads to immunosuppression and is the basis for the clinical use of cyclosporine as an immunosuppressive medication. Cyclosporine is commonly used to prevent organ transplant rejection and to treat autoimmune diseases such as rheumatoid arthritis and psoriasis.

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Incorrect Option:

Option B - AMP kinase stimulation: AMP kinase (AMPK) is an enzyme that plays a key role in regulating cellular energy metabolism. It is activated when cellular energy levels are low, and it promotes the production of ATP, the primary energy source for cells. Cyclosporine is not known to stimulate the activity of AMP kinase. In fact, some research suggests that cyclosporine may inhibit the activity of AMPK, although the mechanisms underlying this effect are not well understood.

Option C - IMP dehydrogenase inhibition: Inosine monophosphate dehydrogenase (IMPDH) is an enzyme that is involved in the synthesis of guanosine nucleotides, which are building blocks of DNA and RNA. Cyclosporine is not known to inhibit the activity of IMPDH. However, another immunosuppressive medication called mycophenolate mofetil (MMF) does inhibit IMPDH, and it is often used in combination with cyclosporine to treat transplant rejection.

Option D - Dihydro-orotate dehydrogenase inhibition: Dihydro-orotate dehydrogenase (DHODH) is an enzyme that is involved in the synthesis of pyrimidine nucleotides, which are also building blocks of DNA and RNA. Cyclosporine is not known to inhibit the activity of DHODH.

Solution for Question 51:

Correct Option B - Degradation of antibiotics:

- The mechanism of resistance by lactamases to β -lactam antibiotics is degradation of antibiotics.
- Lactamases, also known as β -lactamases, are enzymes produced by certain bacteria that are capable of inactivating β -lactam antibiotics, such as penicillins, cephalosporins, and carbapenems. These enzymes hydrolyze the β -lactam ring, which is a key structural component of β -lactam antibiotics, rendering them inactive.
- The degradation of β -lactam antibiotics by lactamases prevents the antibiotics from binding to their target site, which is usually penicillin-binding proteins (PBPs) located on the bacterial cell wall. The β -lactam ring of the antibiotic is essential for inhibiting the activity of PBPs, which are involved in cell wall synthesis. By hydrolyzing the β -lactam ring, lactamases prevent the antibiotic from binding to PBPs and exerting its bactericidal or bacteriostatic effects.
- It's important to note that while lactamases are a common mechanism of resistance, bacteria can employ multiple mechanisms to resist β -lactam antibiotics.

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It's important to note that while lactamases are a common mechanism of resistance, bacteria can employ multiple mechanisms to resist β -lactam antibiotics.

Incorrect Options:

- Option A, C & D is not the correct answer.

Option A, C & D is not the correct answer.

Solution for Question 52:

Correct Option A - Erythromycin:

- Among the options provided, erythromycin should not be prescribed with theophylline due to a significant drug interaction.
- Erythromycin is an antibiotic that can inhibit the metabolism of theophylline by blocking the activity of an enzyme called cytochrome P450 3A4 (CYP3A4). Theophylline is primarily metabolized by CYP3A4, and when its metabolism is inhibited by erythromycin, the levels of theophylline in the blood can increase significantly.
- This interaction can lead to theophylline toxicity, which can cause symptoms such as nausea, vomiting, rapid heartbeat, tremors, and seizures. Therefore, it is generally recommended to avoid concomitant use of erythromycin and theophylline.

Among the options provided, erythromycin should not be prescribed with theophylline due to a significant drug interaction.

Erythromycin is an antibiotic that can inhibit the metabolism of theophylline by blocking the activity of an enzyme called cytochrome P450 3A4 (CYP3A4). Theophylline is primarily metabolized by CYP3A4, and when its metabolism is inhibited by erythromycin, the levels of theophylline in the blood can increase significantly.

This interaction can lead to theophylline toxicity, which can cause symptoms such as nausea, vomiting, rapid heartbeat, tremors, and seizures. Therefore, it is generally recommended to avoid concomitant use of erythromycin and theophylline.

Incorrect Options:

Option B - Cefotaxime: Cefotaxime do not have significant interactions with theophylline and can be prescribed concurrently.

Option C - Cotrimoxazole: Cotrimoxazole (trimethoprim-sulfamethoxazole) do not have significant interactions with theophylline and can be prescribed concurrently.

Option D - Amoxicillin: Amoxicillin, on the other hand, does not have significant interactions with theophylline and can be prescribed concurrently.

Solution for Question 53:

Correct Option A - Uncomplicated Gonorrhea:

- Gonorrhea is a sexually transmitted bacterial infection caused by the *Neisseria gonorrhoeae* bacterium. It primarily affects the genital tract, but can also infect the rectum, throat, and eyes. Gonorrhea is usually treated with a single agent such as ceftriaxone, azithromycin, or doxycycline.
- Antimicrobial combinations are not commonly used to treat gonorrhea because the bacteria are usually sensitive to a single antibiotic, antibiotics used for treating gonorrhea typically have high efficacy and low resistance rates when used as a single agent. In fact, the World Health Organization (WHO) recommends using a single agent as first-line treatment for uncomplicated gonorrhea. However, in some cases, combination therapy may be used to treat coexisting infections or to address concerns about antibiotic resistance.
- In recent years, there has been growing concern about the development of antibiotic-resistant strains of gonorrhea.

Gonorrhea is a sexually transmitted bacterial infection caused by the *Neisseria gonorrhoeae* bacterium. It primarily affects the genital tract, but can also infect the rectum, throat, and eyes. Gonorrhea is usually treated with a single agent such as ceftriaxone, azithromycin, or doxycycline.

Antimicrobial combinations are not commonly used to treat gonorrhea because the bacteria are usually sensitive to a single antibiotic, antibiotics used for treating gonorrhea typically have high efficacy and low resistance rates when used as a single agent. In fact, the World Health Organization (WHO) recommends using a single agent as first-line treatment for uncomplicated gonorrhea. However, in some cases, combination therapy may be used to treat coexisting infections or to address concerns about antibiotic resistance.

In recent years, there has been growing concern about the development of antibiotic-resistant strains of gonorrhea.

Incorrect Options:

Option B - Tuberculosis: Tuberculosis is a bacterial infection that is typically treated with a combination of antibiotics. This is because TB bacteria can develop resistance to single antibiotics if used alone. The standard treatment for active TB includes a combination of isoniazid, rifampin, pyrazinamide, and ethambutol, taken for several months.

Option C - Intra abdominal infections: Antimicrobial combinations are commonly used to treat intra-abdominal infections, which can be caused by a variety of bacteria. The combination of antibiotics used depends on the type of infection and the bacteria causing it. For example, in cases of complicated intra-abdominal infections, a combination of piperacillin-tazobactam plus metronidazole or cefotaxime plus metronidazole may be used.

Option D - Malaria: Antimicrobial combinations are commonly used to treat malaria, especially in areas where drug resistance is high. This is because using a single antimalarial drug can lead to the development of drug-resistant strains of the malaria parasite. Examples of antimalarial combinations include atovaquone-proguanil, artemether-lumefantrine, and mefloquine-artesunate.

Solution for Question 54:

Correct Option D - Linezolid:

- Linezolid is an example of a bacteriostatic drug.
 - It inhibits bacterial protein synthesis by binding to the 50S subunit of the bacterial ribosome, preventing the formation of functional ribosomes.
 - This inhibits bacterial growth and replication without necessarily killing the bacteria.
- Therefore, linezolid is a bacteriostatic drug.

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It inhibits bacterial protein synthesis by binding to the 50S subunit of the bacterial ribosome, preventing the formation of functional ribosomes.

This inhibits bacterial growth and replication without necessarily killing the bacteria. Therefore, linezolid is a bacteriostatic drug.

Incorrect Options:

Option A - Aminoglycoside: Aminoglycosides such as gentamicin and amikacin are examples of bactericidal drugs. They work by inhibiting bacterial protein synthesis and killing the bacteria, rather than just inhibiting their growth. Therefore, aminoglycosides are not bacteriostatic drugs.

Option B - Vancomycin: Vancomycin is primarily a bactericidal drug. It interferes with bacterial cell wall synthesis, leading to cell lysis and bacterial death. Therefore, vancomycin is not a bacteriostatic drug.

Option C - Metronidazole: Metronidazole is primarily a bactericidal drug. It acts by disrupting the DNA structure and inhibiting nucleic acid synthesis in susceptible bacteria, leading to their death. Therefore, metronidazole is not a bacteriostatic drug.

Solution for Question 55:

Correct Option B - Imipenem:

- Imipenem is a carbapenem antibiotic that has a higher risk of causing seizures compared to other carbapenems. It can lower the seizure threshold, especially at higher doses or in patients with underlying risk factors for seizures.

Imipenem is a carbapenem antibiotic that has a higher risk of causing seizures compared to other carbapenems. It can lower the seizure threshold, especially at higher doses or in patients with underlying risk factors for seizures.

Incorrect Options:

Option A - Meropenem: Meropenem is a carbapenem antibiotic that is generally associated with a lower risk of causing seizures compared to imipenem.

Option C - Etrapanem: Etrapanem is not a commonly used carbapenem. It has similar properties to other carbapenems but is less frequently prescribed.

Option D - Doripenem: Doripenem is a carbapenem antibiotic that is generally associated with a lower risk of causing seizures compared to imipenem

Solution for Question 56:

Option B - Inhibition of ergosterol synthesis:

- Triazoles are a class of antifungal medications that work by inhibiting the synthesis of ergosterol, a vital component of the fungal cell membrane. Ergosterol is the fungal equivalent of cholesterol in human cell membranes. Disruption of ergosterol synthesis impairs the integrity and function of the fungal cell membrane, leading to cell death and inhibition of fungal growth.

Triazoles are a class of antifungal medications that work by inhibiting the synthesis of ergosterol, a vital component of the fungal cell membrane. Ergosterol is the fungal equivalent of cholesterol in human cell membranes. Disruption of ergosterol synthesis impairs the integrity and function of the fungal cell membrane, leading to cell death and inhibition of fungal growth.

Incorrect Options:

Option A - Interferes with DNA and RNA synthesis: Some antifungal medications, such as nucleoside analogs, do interfere with DNA and RNA synthesis in fungi, but this is not the mechanism of action of triazoles.

Option C - Forms pores in fungal membranes: This is the mechanism of action of polyenes, not triazoles. Polyenes like amphotericin B create pores in fungal membranes, disrupting their integrity.

Option D - Inhibits epoxidation of squalene in fungi: This describes the mechanism of allylamines, another class of antifungal drugs. Allylamines inhibit the conversion of squalene to ergosterol, but this is different from the mechanism of action of triazoles, which directly inhibit ergosterol synthesis.

Solution for Question 57:

Correct Option B - Levofloxacin.:

- Levofloxacin is a fluoroquinolone antibiotic that exhibits high oral bioavailability. It is the L-isomer of ofloxacin and has a longer half-life compared to other fluoroquinolones. Due to its excellent bioavailability, it is widely used for various infections.

Levofloxacin is a fluoroquinolone antibiotic that exhibits high oral bioavailability. It is the L-isomer of ofloxacin and has a longer half-life compared to other fluoroquinolones. Due to its excellent bioavailability, it is widely used for various infections.

Incorrect Options:

Option A - Ciprofloxacin: Ciprofloxacin is a commonly used fluoroquinolone antibiotic. While it has good bioavailability, it is not the fluoroquinolone with the maximum bioavailability.

Option C - Gatifloxacin: Gatifloxacin is another fluoroquinolone antibiotic. While it has good bioavailability, it is not the fluoroquinolone with the maximum bioavailability.

Option D - Norfloxacin: Norfloxacin is an older fluoroquinolone antibiotic. It has lower bioavailability compared to some of the newer fluoroquinolones and is typically used for urinary tract infections.

Solution for Question 58:

Correct Option D - Neuraminidase inhibition.:

- Both oseltamivir and zanamivir are neuraminidase inhibitors. Neuraminidase is an enzyme found on the surface of influenza viruses. By inhibiting neuraminidase, these drugs prevent the release of newly formed viral particles from infected cells, reducing the spread of the virus within the body. This mechanism of action helps to decrease the severity and duration of influenza symptoms and can be effective in treating and preventing influenza infections.

Both oseltamivir and zanamivir are neuraminidase inhibitors. Neuraminidase is an enzyme found on the surface of influenza viruses. By inhibiting neuraminidase, these drugs prevent the release of newly formed viral particles from infected cells, reducing the spread of the virus within the body. This mechanism of action helps to decrease the severity and duration of influenza symptoms and can be effective in treating and preventing influenza infections.

Incorrect Options:

Option A - DNA polymerase inhibition: Oseltamivir and zanamivir do not inhibit DNA polymerase. DNA polymerase is an enzyme involved in DNA replication and is targeted by antiviral drugs used to treat viral infections such as herpes or hepatitis B.

Option B - Protein synthesis inhibition: Oseltamivir and zanamivir do not inhibit protein synthesis. Protein synthesis inhibitors are antibiotics that target bacterial ribosomes and interfere with the process of protein production.

Option C - Nucleotide analogue: Oseltamivir and zanamivir are not nucleotide analogues. Nucleotide analogues are compounds that mimic the structure of nucleotides and can be

incorporated into viral DNA or RNA during replication, leading to chain termination and inhibition of viral replication.

Solution for Question 59:

Correct Option B - Erythromycin:

- Erythromycin is the drug of choice for the diphtheria carrier state. It is effective against the *Corynebacterium diphtheriae* bacterium responsible for diphtheria. Erythromycin is typically used to treat carriers to prevent the spread of the disease to others, as carriers can still transmit the bacteria even if they do not exhibit symptoms of the disease.

Erythromycin is the drug of choice for the diphtheria carrier state. It is effective against the *Corynebacterium diphtheriae* bacterium responsible for diphtheria. Erythromycin is typically used to treat carriers to prevent the spread of the disease to others, as carriers can still transmit the bacteria even if they do not exhibit symptoms of the disease.

Incorrect Options

Option A - Amoxicillin: Amoxicillin is not the drug of choice for the diphtheria carrier state. While amoxicillin is an antibiotic commonly used to treat bacterial infections, it is not the preferred treatment for diphtheria carriers.

Option C - Rifampicin: Rifampicin is not the drug of choice for the diphtheria carrier state. rifampicin is a broad-spectrum antibiotic used to treat various bacterial infections.

Option D - Tetracycline: Tetracycline is not the drug of choice for the diphtheria carrier state. Although tetracycline is an antibiotic with a broad spectrum of activity against various bacteria, including some *Corynebacterium* species, it is not the recommended treatment for diphtheria carriers.

Solution for Question 60:

Correct Option C - Intravenous artesunate:

- *P. falciparum* causes complicated malaria with loss of consciousness
- It is a life-threatening condition. Injectable drugs are given in life-threatening conditions and in emergencies
- Intravenous artesunate is the preferred and recommended treatment for severe *falciparum* malaria.

P. falciparum causes complicated malaria with loss of consciousness

It is a life-threatening condition. Injectable drugs are given in life-threatening conditions and in emergencies

Intravenous artesunate is the preferred and recommended treatment for severe *falciparum* malaria.

Incorrect Options:

Option A - Chloroquine: Effective for Rheumatoid arthritis, Extraintestinal amebiasis, Discoid lupus erythematosus, Lepa reaction (type II lepra reaction), Mononucleosis, Photogenic reaction, Giardiasis.

Option B - Quinine: Quinine is given via the intravenous route in the treatment of complicated malaria or cerebral malaria. Quinine is also given via the oral route in combination with clindamycin to treat malaria in pregnancy (first trimester of pregnancy) in *Plasmodium falciparum*.

Option D - Artemether: Artemether is another artemisinin derivative used in the treatment of malaria, but intravenous artesunate is preferred for severe *falciparum* malaria due to its more rapid onset of action.

Solution for Question 61:

Correct Option C - Macrolides:

- Macrolides carry a certain level of risk from typical adverse effects like nausea, vomiting, abdominal pain, and diarrhea to more specific side effects like causing IHPS in children but they are generally not associated with a risk of seizures.

Macrolides carry a certain level of risk from typical adverse effects like nausea, vomiting, abdominal pain, and diarrhea to more specific side effects like causing IHPS in children but they are generally not associated with a risk of seizures.

Incorrect Options:

Option A - Beta-lactam compounds: Adverse reactions associated with beta-lactamase inhibitors include gastrointestinal side effects, such as diarrhea, nausea, and constipation; nervous system effects such as headaches, insomnia, and seizures; hematological effects such as impaired platelet function; allergic reactions including anaphylaxis; pain at the injection site; and dermatologic side effects including Stevens-Johnson syndrome, toxic epidermal necrolysis, and drug-induced eosinophilia and systemic symptoms.

Option B - Quinolones: The most frequent side effects are gastrointestinal reactions (nausea, dyspepsia, vomiting) and CNS reactions such as dizziness, insomnia and headache. Many of the more severe CNS reactions seem to be due to metabolic interaction with theophylline, especially when enoxacin is used.

Option D - Isoniazid: The most common symptoms include hives, difficulty breathing, and swelling in your face or throat) or a severe skin reaction (fever, sore throat, burning in your eyes, skin pain, red or purple skin rash that spreads and causes blistering and peeling). symptoms may include: skin rash, fever, swollen glands, flu-like symptoms, muscle aches, severe weakness, unusual bruising, or yellowing of your skin or eyes and a seizure (convulsions).

Solution for Question 62:

Correct Option A - Vitamin B6:

- Vitamin B6 (pyridoxine) should be supplemented to this patient. Tingling sensation and paresthesia in the lower limbs are symptoms of peripheral neuropathy, which is a common side effect of isoniazid, one of the first-line drugs used in the treatment of tuberculosis. Isoniazid causes a deficiency of vitamin B6 by inhibiting its conversion from pyridoxine, leading to neuropathy. Therefore, supplementation of vitamin B6 is recommended to prevent or treat this side effect.

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Incorrect Options:

Option B - Vitamin B12: Vitamin B12 deficiency can cause neurological symptoms, including tingling sensation and paresthesia in the lower limbs, but this patient is on anti-tubercular treatment, which includes isoniazid.

Option C - Vitamin B2: Vitamin B2 (riboflavin) is not directly associated with the nervous system, but it is involved in energy metabolism and has a role in the production of red blood cells.

Option D - Vitamin B3: Vitamin B3 (niacin) deficiency is rare and typically only occurs in cases of severe malnutrition.

Solution for Question 63:

Correct Option A - Isoniazid:

- The marked layer in the image is the mycolic acid layer, which is a characteristic feature of the cell wall of Mycobacterium species.
- Isoniazid is a first-line antitubercular drug that is commonly used in the treatment of TB. It works by targeting the synthesis of mycolic acids.
- Isoniazid inhibits the synthesis of mycolic acids by targeting the enzyme enoyl-ACP reductase, which is involved in the synthesis of long-chain fatty acids
- By inhibiting this enzyme, isoniazid disrupts the synthesis of mycolic acids, leading to the loss of structural integrity and permeability barrier of the mycobacterial cell wall. This, in turn, leads to the death of the bacterial cell.

The marked layer in the image is the mycolic acid layer, which is a characteristic feature of the cell wall of Mycobacterium species.

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Isoniazid inhibits the synthesis of mycolic acids by targeting the enzyme enoyl-ACP reductase, which is involved in the synthesis of long-chain fatty acids

By inhibiting this enzyme, isoniazid disrupts the synthesis of mycolic acids, leading to the loss of structural integrity and permeability barrier of the mycobacterial cell wall. This, in turn, leads to the death of the bacterial cell.

Incorrect Options:

Option B - Rifampicin: Rifampicin is another first-line antitubercular drug that works by inhibiting the RNA polymerase enzyme in the bacteria, thereby preventing the transcription of bacterial DNA.

Option C - Pyrazinamide: Pyrazinamide is not the correct answer because it targets pyrazinamide disrupts the bacterial membrane potential and reduces the pH of the bacterial cytoplasm, leading to bactericidal effects. While pyrazinamide can be effective against *Mycobacterium tuberculosis*, it does not specifically target the mycolic acid layer of the cell wall.

Option D - Ethambutol: Ethambutol is also not the correct answer because it targets the arabinogalactan layer. Ethambutol inhibits the enzyme arabinosyl transferase, which is involved in the synthesis of arabinogalactan. Arabinogalactan is a polysaccharide that links the mycolic acid layer to the peptidoglycan layer in the cell wall.

Solution for Question 64:

Correct Option D - Hemophilus produces beta lactamase, hence resistant to ampicillin:

- Hemophilus produces beta-lactamase, hence resistant to ampicillin: This statement is likely the reason for switching from ampicillin to ceftriaxone. Hemophilus influenzae is known to produce beta-lactamase, an enzyme that can break down and inactivate beta-lactam antibiotics such as ampicillin. Ceftriaxone, on the other hand, is a third-generation cephalosporin antibiotic that is not as susceptible to beta-lactamase degradation and can be effective against Hemophilus influenzae infections.

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Incorrect Option:

Option A - Hemophilus alters penicillin binding protein sites, hence resistant to ampicillin: This statement is not completely accurate. While some strains of Hemophilus influenzae may have altered penicillin binding protein sites and be resistant to ampicillin, this is not the only mechanism of resistance. In addition, not all strains of Hemophilus influenzae are resistant to ampicillin.

Option B - Easier to administer IV ceftriaxone than IV ampicillin: While ceftriaxone can be administered once a day, which may be more convenient than the multiple daily doses required for ampicillin, ease of administration is not the primary reason for switching from ampicillin to ceftriaxone.

Option C - Cotrimoxazole is the drug of choice, but the patient is allergic to sulfa drugs: Cotrimoxazole (trimethoprim-sulfamethoxazole) is an antibiotic that is often used to treat *Hemophilus influenzae* infections. However, if the patient is allergic to sulfa drugs, cotrimoxazole may not be a suitable treatment option.

Solution for Question 65:

Correct Option A - DNA gyrase inhibition:

- Fluoroquinolones are a class of antibiotics that target bacterial DNA gyrase, a type II topoisomerase enzyme that plays a crucial role in DNA replication and repair.
- By inhibiting DNA gyrase, fluoroquinolones prevent the bacteria from synthesizing and replicating their DNA, leading to bacterial cell death.
- Examples of fluoroquinolones include ciprofloxacin, levofloxacin, and moxifloxacin, which are commonly used for the treatment of urinary tract infections, respiratory tract infections, and other bacterial infections.

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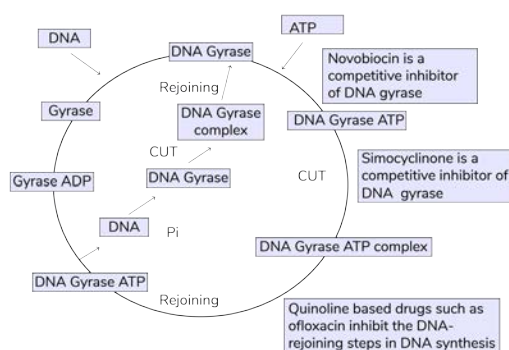
- However, fluoroquinolones are also known to have a rare but potentially serious side effect of causing tendon rupture and arthropathy. This is because DNA gyrase plays a role in the synthesis of collagen, the primary structural protein in tendons, ligaments, and other connective tissues. By inhibiting DNA gyrase, fluoroquinolones can disrupt collagen synthesis, leading to weakened and damaged connective tissues in the tendons and joints. This can result in symptoms such as joint pain, swelling, and stiffness, as well as tendonitis.

and tendon rupture.

- Therefore, in the case of a woman presenting with symptoms of urinary tract infection who has been prescribed a drug that causes tendon rupture and arthropathy, the likely mechanism of action of the drug is DNA gyrase inhibition, which is characteristic of fluoroquinolones.

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Therefore, in the case of a woman presenting with symptoms of urinary tract infection who has been prescribed a drug that causes tendon rupture and arthropathy, the likely mechanism of action of the drug is DNA gyrase inhibition, which is characteristic of fluoroquinolones.



Incorrect Options:

Option B - Ribosomal inhibition: Ribosomal inhibition is a mechanism of action of some antibiotics, such as macrolides and aminoglycosides.

Option C - Cell wall synthesis: Cell wall synthesis inhibition is another mechanism of action of certain antibiotics, such as beta-lactams (e.g. penicillins and cephalosporins).

Option D - Inhibition of folic acid synthesis: Inhibition of folic acid synthesis is a mechanism of action of certain antibiotics, such as sulfonamides and trimethoprim.

Solution for Question 66:

Correct Option B - Protein synthesis inhibitor:

- Doxycycline acts by inhibiting protein synthesis and is a tetracycline.
- Tetracyclines act by inhibiting protein synthesis by binding to the 30s subunit of 70s ribosomes.
- Aminoglycosides too act by binding to the 30s subunit of 70s ribosome.

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Tetracyclines act by inhibiting protein synthesis by binding to the 30s subunit of 70s ribosomes.

Aminoglycosides too act by binding to the 30s subunit of 70s ribosome.

Incorrect Options:

Option A - Cell wall synthesis inhibitor: Drugs that inhibit cell wall synthesis include cefotetan and oxacillin and are beta lactam antibiotics.

Option C - DNA gyrase inhibitor: DNA gyrase inhibitors include ciprofloxacin which is a fluoroquinolone.

Option D - Leucyl-tRNA synthetase inhibitor: Leucyl-tRNA synthetase inhibitor includes tavaborole. It inhibits fungal leucyl-tRNA gyrase and hence inhibits fungal protein.

Solution for Question 67:

Correct Option C - Pefloxacin:

- FQs undergoing bile elimination are: Pefloxacin Moxifloxacin Trovafloxacin.
- Pefloxacin
- Moxifloxacin
- Trovafloxacin.
- These are safe in renal failure.
- These are contraindicated in hepatic failure as they are eliminated by bile. (Trova > Moxi > Peflox).

FQs undergoing bile elimination are:

- Pefloxacin

- Moxifloxacin
- Trovafloxacin.

Pefloxacin

Moxifloxacin

Trovafloxacin.

These are safe in renal failure.

These are contraindicated in hepatic failure as they are eliminated by bile. (Trova > Moxi > Peflox).

Incorrect Options:

Option A - Ofloxacin:

- It is used for the treatment of Leprosy, TB, and diarrhoea.
- Pharmacokinetics: This drug has ~100% Bioavailability when used orally.
- This drug and normally fluoroquinolones have the increase risk of seizures.

It is used for the treatment of Leprosy, TB, and diarrhoea.

Pharmacokinetics: This drug has ~100% Bioavailability when used orally.

This drug and normally fluoroquinolones have the increase risk of seizures.

Option B - Levofloxacin:

- It is used for the treatment of respiratory infections, and TB.
- Pharmacokinetics: This drug has approximately ~100% BA when used orally.

It is used for the treatment of respiratory infections, and TB.

Pharmacokinetics: This drug has approximately ~100% BA when used orally.

Option D - Delafloxacin:

- It is used to treat ABSSSI (acute bacterial skin and skin structure infection).

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Solution for Question 68:

Correct Option B - Liposomal amphotericin B:

- The most appropriate drug for the treatment of fungal infection in a diabetic patient with peri-orbital region involvement and significant visual impairment is liposomal amphotericin B.
- Fungal infections are common in diabetic patients due to hyperglycemia, decreased immunity, and impaired blood supply to tissues. Fungal infections can be superficial or systemic, and their treatment depends on the type and severity of the infection.
- Amphotericin B is a broad-spectrum antifungal drug that is used to treat severe and life-threatening fungal infections. It is particularly effective against invasive fungal infections such as aspergillosis, candidiasis, and mucormycosis, which are more common in immunocompromised patients such as those with diabetes. Liposomal amphotericin B is a newer formulation of amphotericin B that has a lower incidence of adverse effects such as nephrotoxicity and infusion-related reactions.

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Incorrect Options:

Option A - Itraconazole: Itraconazole is a triazole antifungal drug that is used to treat superficial and systemic fungal infections. It is particularly effective against dermatophyte infections such as tinea corporis, tinea cruris, and tinea pedis. It is less effective against invasive fungal infections such as aspergillosis and mucormycosis.

Option C - Ketoconazole: Ketoconazole is an imidazole antifungal drug that is used to treat superficial fungal infections such as dermatophytosis, candidiasis, and pityriasis versicolor. It is not effective against invasive fungal infections.

Option D - Broad spectrum antibiotics: Broad-spectrum antibiotics are ineffective against fungal infections and may even worsen the infection by killing beneficial bacteria and allowing fungal overgrowth.

Solution for Question 69:

Correct Option A - Clotrimazole:

- The white patchy lesions observed on the tongue and buccal mucosa are likely oral thrush, a fungal infection caused by *Candida* species. Inhalational steroids can suppress the immune system, increasing the risk of developing fungal infections.
- Clotrimazole is an antifungal medication used to treat oral thrush, among other fungal infections. It works by inhibiting the growth and reproduction of fungi, allowing the body's immune system to fight off the infection.

The white patchy lesions observed on the tongue and buccal mucosa are likely oral thrush, a fungal infection caused by *Candida* species. Inhalational steroids can suppress the immune system, increasing the risk of developing fungal infections.

Clotrimazole is an antifungal medication used to treat oral thrush, among other fungal infections. It works by inhibiting the growth and reproduction of fungi, allowing the body's immune system to fight off the infection.

Incorrect Options:

Option B - Griseofulvin: Griseofulvin is an antifungal medication for fungal skin, hair, and nail infections. However, it is ineffective against *Candida* species, which is the likely cause of the patient's oral thrush.

Option C - Terbinafine: Terbinafine is another antifungal medication used to treat fungal infections of the skin, hair, and nails, but it is also ineffective against *Candida* species.

Option D - Flucytosine: Flucytosine is an antifungal medication used to treat severe fungal infections, such as those caused by *Cryptococcus* and *Candida* species. However, it is not the first-line treatment for oral thrush and is typically reserved for more severe cases.

Solution for Question 70:

Correct Options D - Matzke scale:

- The Matzke scale is used for vancomycin dosage. It is a scale used to estimate renal function in patients with renal impairment, which can be important for adjusting drug dosages, including vancomycin.

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Incorrect Options:

Option A - Hallstead scale: The Hallstead scale is not specifically used for vancomycin dosage. It is a scale used to estimate the severity of alcohol withdrawal symptoms.

Option B - Salazar scale: The Salazar scale is not used for vancomycin dosage. It is a formula used to estimate creatinine clearance in children.

Option C - Rumack Matthew nomogram: The Rumack Matthew nomogram is a tool used to assess the risk of hepatotoxicity (liver toxicity) following an acute acetaminophen (paracetamol) overdose. It helps determine the need for treatment with the antidote N-acetylcysteine. It is not used for vancomycin dosage.

Solution for Question 71:

Correct Option D - 4:

- Beta-lactam antibiotics work by inhibiting the action of enzymes involved in cell wall synthesis, weakening the cell wall and making it more susceptible to osmotic pressure.
- The given diagram shows the steps in bacterial cell wall formation. The various steps involved in bacterial cell wall formation are: UDP-N-acetylglucosamine (UDP-NAG) and UDP-N-acetylmuramic acid (UDP-NAM) are synthesized in the cytoplasm of the bacterial cell. UDP-NAM is linked to a lipid carrier molecule called bactoprenol phosphate to form lipid II. Lipid II is flipped across the cytoplasmic membrane to the outer surface, where it is incorporated into the growing cell wall. The peptide chains of adjacent lipid II molecules are cross-linked by enzymes called transpeptidases, also known as penicillin-binding proteins (PBPs).
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- The peptide chains of adjacent lipid II molecules are cross-linked by enzymes called transpeptidases, also known as penicillin-binding proteins (PBPs).
- Based on the above steps, the site of action of beta-lactam antibiotics is option 4, which is the site where transpeptidases or PBPs are involved in cross-linking the peptide chains of adjacent lipid II molecules.
- Beta-lactam antibiotics, such as penicillins, cephalosporins, and carbapenems, work by irreversibly binding to the active site of PBPs, thereby inhibiting their function. This leads to a weakened cell wall and eventual cell death.

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Based on the above steps, the site of action of beta-lactam antibiotics is option 4, which is the site where transpeptidases or PBPs are involved in cross-linking the peptide chains of adjacent lipid II molecules.

Beta-lactam antibiotics, such as penicillins, cephalosporins, and carbapenems, work by irreversibly binding to the active site of PBPs, thereby inhibiting their function. This leads to a weakened cell wall and eventual cell death.

Incorrect Options:

- Options A, B & C are incorrect choices.
-

Solution for Question 72:

Correct Option D - Amoxicillin:

- Amoxicillin might have caused the maculopapular rash in the patient.
- Amoxicillin is a commonly prescribed antibiotic, and one of the known adverse reactions associated with its use is the development of a maculopapular rash. This type of rash is a delayed hypersensitivity reaction and is considered a common manifestation of an allergic response to amoxicillin. The rash typically appears a few days after starting the medication and presents as small, flat, reddened lesions that may be raised or have a papular appearance. It can be widespread and may involve different areas of the body.

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Incorrect Options:

Option A - Prednisolone: Prednisolone is a corticosteroid medication that is frequently used in the treatment of asthma and various other inflammatory conditions. While corticosteroids can cause side effects, such as skin thinning and increased vulnerability to skin infections, a maculopapular rash is not a typical reaction to prednisolone.

Option B - Salbutamol: Salbutamol is a bronchodilator medication commonly used in the management of asthma symptoms. It works by relaxing the airway muscles and facilitating better airflow. Salbutamol is not known to cause a maculopapular rash as an adverse reaction.

Option C - Fexofenadine: Fexofenadine is an antihistamine medication used to relieve allergy symptoms, including sneezing, itching, runny nose, and watery eyes. It is generally well-tolerated and does not commonly cause a maculopapular rash as a side effect.

Solution for Question 73:

Correct Option B - Add Dolutegravir 50 mg in the morning:

- The dose of Dolutegravir(Integrase inhibitor) has to be increased in this patient. Rifampicin, an Antitubercular drug is a potent CYP inducer and therefore, the effect of Dolutegravir is reduced.

Incorrect Options:

Option A - No dose adjustment required: Dose adjustment is required as rifampicin is a CYP inducer

Option B- Add Tenofovir 150mg in the morning: Tenofovir, an NRTI can be given safely with Rifampicin

Option D - Add Lamivudine 300 mg in the morning: Lamivudine, an NRTI can be given safely with Rifampicin.

Solution for Question 74:

Correct Option B - Add dolutegravir 100mg in the morning:

- Dolutegravir blocks the catalytic activity of the HIV-encoded integrase of HIV-1 and HIV-2, preventing the integration of viral DNA into the host chromosome.
- Dolutegravir is susceptible to decreases in plasma concentrations caused by inducers of drug-metabolizing enzymes.
- The dose of Dolutegravir needs to be increased to twice daily when given with enzyme inducers like rifampicin, efavirenz, and ritonavir.

Incorrect Options:

- Options A, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 75:

Correct Option A - QT interval prolongation:

- Bedaquiline is an anti-tubercular drug most effective in resistant TB.
- It inhibits mycobacterial ATP synthase, thus reducing energy production inside TB bacteria.
- QT prolongation is the most characteristic side effect of Bedaquiline.
- The most common side effect of Bedaquiline is nausea.

Incorrect Options:

Option B - PR interval prolongation: PR interval is increased by drugs like Procainamide, Propranolol, and Esmolol.

Option C - QRS widening: QRS widening is caused by class 1 antiarrhythmic drugs.

Option D - ST-segment elevation: ST-segment elevation is caused by sodium channel blockers.

Solution for Question 76:

Correct Option C - Disulfiram-like reaction:

- Metronidazole is used in the treatment of amoebic colitis and liver abscesses.
- It has a disulfiram-like action which can cause nausea, vomiting, abdominal pain, chest pain, flushing, and hypotension.
- Drugs causing disulfiram-like action and contraindicated in alcoholics: Chlorpropamide, Cephalosporins (Cefoperazone, Cefamandole, Cefotetan, Cefoxitin), Griseofulvin, Metronidazole, Procarbazine.
- Chlorpropamide
- Cephalosporins (Cefoperazone, Cefamandole, Cefotetan, Cefoxitin)
- Griseofulvin
- Metronidazole
- Procarbazine
- Chlorpropamide
- Cephalosporins (Cefoperazone, Cefamandole, Cefotetan, Cefoxitin)

- Griseofulvin
- Metronidazole
- Procarbazine

Incorrect Options:

Option A - Delirium tremens:

- Delirium tremens is a severe form of alcohol withdrawal and characterized by hallucinations, delirium, tachycardia, and a potentially fatal fever or cardiac arrhythmia

Option B - Alcohol withdrawal:

- Alcohol withdrawal is characterized by alcohol craving, tremor, sleep disturbance, tachycardia, hypertension, sweating, etc.,

Option D - Hallucinoses:

- Hallucinoses is characterized by hallucinations and caused by drugs like amphetamine, cocaine, fentanyl, etc.,

Solution for Question 77:

Correct Option B - RNA-dependent RNA polymerase inhibition:

- The mechanism of action of remdesivir is RNA-dependent RNA polymerase inhibition. This is a unique mechanism of action that targets the replication of RNA viruses, such as coronavirus. RNA-dependent RNA polymerase is an enzyme that is essential for the replication of RNA viruses, and it is responsible for synthesizing new copies of viral RNA in infected cells.
- Remdesivir is a nucleotide analog that can interfere with the activity of RNA-dependent RNA polymerase. Specifically, it can be incorporated into new RNA strands during replication, which can result in premature termination of the replication process. This can prevent the virus from replicating and spreading in the body, which can help to reduce the severity of the infection.
- Remdesivir has been shown to be effective against a variety of RNA viruses, including coronaviruses, Ebola virus, and respiratory syncytial virus.

The mechanism of action of remdesivir is RNA-dependent RNA polymerase inhibition. This is a unique mechanism of action that targets the replication of RNA viruses, such as coronavirus.

RNA-dependent RNA polymerase is an enzyme that is essential for the replication of RNAviruses, and it is responsible for synthesizing new copies of viral RNA in infected cells.

Remdesivir is a nucleotide analog that can interfere with the activity of RNA-dependent RNA polymerase. Specifically, it can be incorporated into new RNA strands during replication, which can result in premature termination of the replication process. This can prevent the virus from replicating and spreading in the body, which can help to reduce the severity of the infection.

Remdesivir has been shown to be effective against a variety of RNA viruses, including coronaviruses, Ebola virus, and respiratory syncytial virus.

Incorrect Options:

Option A - RNA-dependent DNA polymerase inhibition: This mechanism of action is typically associated with antiviral medications used to treat retroviruses, such as HIV. It involves blocking the activity of the reverse transcriptase enzyme, which is responsible for converting viral RNA into DNA. This conversion allows the virus to integrate its DNA into the host cell's genome, where it can replicate and cause disease. Remdesivir does not act through this mechanism.

Option C - Fusion inhibition: This mechanism of action is typically associated with antiviral medications used to treat viruses such as HIV and hepatitis C. It involves blocking the fusion of the virus with the host cell, which is necessary for the virus to enter the cell and replicate. Remdesivir does not act through this mechanism.

Option D - Entry inhibition: This mechanism of action involves blocking the ability of the virus to enter the host cell, which is necessary for the virus to replicate and cause disease. This mechanism is used by some antiviral medications to treat viruses such as HIV and hepatitis C. Remdesivir does not act through this mechanism.

Solution for Question 78:

Correct Option A - Tedizolid is active against gram-positive organisms:

- Tedizolid is an antibiotic that belongs to the oxazolidinone class. It exhibits activity against a wide range of gram-positive bacteria, including methicillin-resistant *Staphylococcus aureus* (MRSA) and vancomycin-resistant *Enterococcus faecalis* (VRE).

Incorrect Options:

Option B - Poor oral bioavailability: Tedizolid has good oral bioavailability, which means it can be effectively absorbed when taken orally.

Option C - Active against anaerobes: Tedizolid primarily targets gram-positive organisms and does not have significant activity against anaerobes.

Option D - Causes peripheral neuropathy: Peripheral neuropathy is not a known adverse effect of tedizolid.

Solution for Question 79:

Correct Option C - Assertion is true, but the reason is false:

- The assertion states that in a patient admitted to the hospital for community-acquired pneumonia, combination therapy of beta-lactams and azithromycin is given. This assertion is true, as this combination therapy is a commonly recommended approach for treating community-acquired pneumonia.
- The reason provided is that this combination covers Gram-positive organisms and anaerobes. However, this reason is false. Beta-lactams, such as penicillin or cephalosporins, primarily target Gram-positive bacteria, while azithromycin has broader coverage against Gram-negative bacteria and atypical pathogens. It does not specifically target anaerobes. Therefore, the reason is incorrect and does not provide a valid explanation for the assertion.

Incorrect Options:

Option A - Both assertion and reason are true, and the reason is the correct explanation for the assertion: This option is incorrect because the reason provided is false.

Option B - Both assertion and reason are true, but the reason is not the correct explanation for the assertion: This option is incorrect because the reason provided is false and does not explain the assertion.

Option D - Both assertion and reason are false: This option is incorrect because the assertion is true, but the reason is false.

Solution for Question 80:

Correct Option A - Lincosamides:

- Clostridium difficile-associated diarrhea (CDAD) occurs maximally due to the use of lincosamides. Lincosamides are a class of antibiotics that includes drugs like clindamycin. These antibiotics have been strongly associated with the development of CDAD. They disrupt the normal gut microbiota, allowing Clostridium difficile, a bacteria that can cause diarrhea, to overgrow and produce toxins.

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Incorrect Options:

Option B - Tetracyclines: While tetracyclines can also be associated with the development of CDAD, lincosamides have a higher association.

Option C - Carbapenems: Carbapenems, a broad-spectrum antibiotic class, are not typically associated with an increased risk of CDAD. However, they can disrupt the normal gut microbiota and lead to other gastrointestinal side effects.

Option D - Macrolides: Macrolides, such as erythromycin or azithromycin, have a lower risk of causing CDAD compared to lincosamides. They are not commonly associated with CDAD, although they can have other gastrointestinal side effects.

Solution for Question 81:

Correct Option A - Ethambutol:

- Ethambutol is an antimycobacterial drug commonly used in the treatment of tuberculosis. While it is effective in combating tuberculosis, it is also associated with ophthalmic toxicity as a potential adverse effect. Ethambutol can cause optic neuritis or retrobulbar neuritis, which involves inflammation of the optic nerve. This can lead to visual disturbances, blurred vision, and changes in color vision. Ocular toxicity from ethambutol is dose-related and more likely

at higher doses or with prolonged use. Regular monitoring of visual acuity, color vision, and visual fields is important during ethambutol therapy to detect any ocular toxicity.

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Incorrect Options:

Option B - Isoniazid: Isoniazid is a primary medication used in the treatment of tuberculosis. It is generally well-tolerated, but it can have adverse effects such as hepatotoxicity (liver damage), peripheral neuropathy (nerve damage in the limbs), and hypersensitivity reactions. However, it is not specifically associated with ophthalmic toxicity.

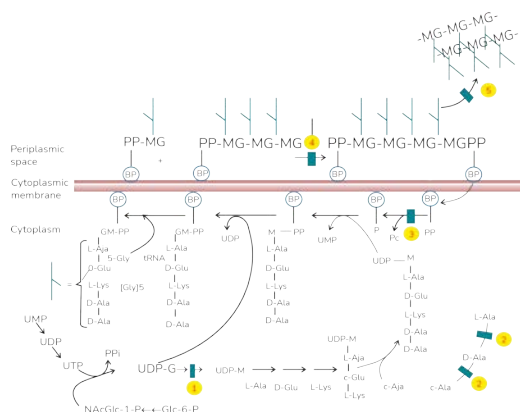
Option C - Kanamycin: Kanamycin is an aminoglycoside antibiotic occasionally used in the treatment of tuberculosis, particularly in cases of drug-resistant tuberculosis. While aminoglycosides can have various side effects, ophthalmic toxicity is not a commonly reported adverse effect associated with kanamycin use.

Option D - Rifampin: Rifampin is another primary medication used in the treatment of tuberculosis. It is generally well-tolerated, but it can have side effects such as hepatotoxicity, gastrointestinal disturbances, and flu-like symptoms. Ophthalmic toxicity is not commonly associated with rifampin use.

Solution for Question 82:

Correct Option D - 4 (Vancomycin):

- Binds to the D-alanyl-D-alanine portion of peptidoglycan, preventing its incorporation into the growing cell wall. This will prevent further elongation of peptidoglycan and cross linking.
- here in this diagram, step 4 is marked by point of action of vancomycin.



Incorrect Options:

Option A - 1 (Fosfomycin): Inhibits the enzyme enolpyruvyl transferase, which is involved in the synthesis of peptidoglycan precursors. This will prevent formation of UDP-N-aceylmuramic acid. that is precursor of N-aceylmuramic acid

Option B - 2 (Cycloserine): Inhibits the formation of peptidoglycan cross-links by interfering with the synthesis of peptidoglycan. This will inhibit alanine racemase that converts L-alanile to D-alanile

Option C - 3 (Bacitracin): Blocks the dephosphorylation of the lipid carrier, preventing the recycling of the carrier required for peptidoglycan synthesis.

Solution for Question 83:

Correct Option B - A-2, B-1, C-4, D-3:

- **Bacteroides (option A):** The drug of choice for Bacteroides infections is metronidazole (option 2). Metronidazole has excellent anaerobic coverage and is effective against Bacteroides species.
- **MRSA (Methicillin-resistant Staphylococcus aureus) (option B):** The drug of choice for MRSA infections is vancomycin (option 1). Vancomycin is an antibiotic that is active against MRSA and is commonly used to treat infections caused by this resistant strain of Staphylococcus aureus.
- **Hemophilus (option C):** Hemophilus infections are commonly treated with penicillin (option 4). Penicillin is effective against Hemophilus influenzae, the bacterium responsible for Hemophilus infections.

- *Klebsiella* (option D): The drug of choice for *Klebsiella* infections is meropenem (option 3). Meropenem is a broad-spectrum antibiotic that covers many Gram-negative bacteria, including *Klebsiella* species.

Bacteroides (option A): The drug of choice for *Bacteroides* infections is metronidazole (option 2). Metronidazole has excellent anaerobic coverage and is effective against *Bacteroides* species.

MRSA (Methicillin-resistant *Staphylococcus aureus*) (option B): The drug of choice for MRSA infections is vancomycin (option 1). Vancomycin is an antibiotic that is active against MRSA and is commonly used to treat infections caused by this resistant strain of *Staphylococcus aureus*.

Hemophilus (option C): *Hemophilus* infections are commonly treated with penicillin (option 4). Penicillin is effective against *Hemophilus influenzae*, the bacterium responsible for *Hemophilus* infections.

Klebsiella (option D): The drug of choice for *Klebsiella* infections is meropenem (option 3). Meropenem is a broad-spectrum antibiotic that covers many Gram-negative bacteria, including *Klebsiella* species.

Incorrect Options:

Option A - A-1, B-3, C-4, D-2: This option incorrectly matches *Bacteroides* with vancomycin. Vancomycin is not the drug of choice for *Bacteroides* infections; metronidazole is the preferred agent.

Option C - A-3, B-4, C-1, D-2: This option incorrectly matches *Hemophilus* with meropenem. Penicillin is the drug of choice for *Hemophilus* infections, not meropenem.

Option D - A-4, B-1, C-2, D-3: This option incorrectly matches *Klebsiella* with metronidazole. Meropenem, not metronidazole, is the preferred drug for *Klebsiella* infections.

Solution for Question 84:

Correct Option A - Doxycycline:

- This is the drug of choice for scrub typhus.
- Doxycycline is effective against *Orientia tsutsugamushi* and is commonly used as the first-line treatment.

This is the drug of choice for scrub typhus.

Doxycycline is effective against *Orientia tsutsugamushi* and is commonly used as the first-line treatment.

Incorrect Options:

Option B - Azithromycin: Azithromycin can be an alternative treatment option for scrub typhus, especially in cases where doxycycline is contraindicated or not tolerated. However, doxycycline is generally preferred due to its established efficacy.

Option C - Chloramphenicol: Chloramphenicol was previously considered the drug of choice for scrub typhus. However, due to its potential side effects and the availability of safer and more effective alternatives, such as doxycycline, it is no longer recommended as the first-line treatment.

Option D - Ciprofloxacin: Ciprofloxacin is not typically used as the first-line treatment for scrub typhus. Other antibiotics, such as doxycycline and azithromycin, are more commonly recommended.

Solution for Question 85:

Correct Option A - Linezolid:

- Linezolid is a protein synthesis inhibitor. It belongs to a class of antibiotics known as oxazolidinones. Linezolid works by binding to the bacterial 23S ribosomal RNA of the 50S subunit, which prevents the formation of a functional 70S initiation complex. This ultimately inhibits bacterial protein synthesis.

Incorrect Options:

Option B - Nevirapine: Nevirapine is an antiretroviral medication used in the treatment of HIV/AIDS. It belongs to the class of drugs called non-nucleoside reverse transcriptase inhibitors (NNRTIs). Nevirapine inhibits the reverse transcriptase enzyme, which is necessary for the replication of the HIV virus, but it does not directly inhibit protein synthesis.

Option C - Indinavir: Indinavir is an antiretroviral medication used in the treatment of HIV/AIDS. It belongs to the class of drugs called protease inhibitors. Indinavir works by inhibiting the protease enzyme, which is required for the replication of the HIV virus, but it does not directly inhibit protein synthesis.

Option D - Ceftriaxone: Ceftriaxone is a broad-spectrum antibiotic belonging to the class of drugs called cephalosporins. It inhibits bacterial cell wall synthesis by binding to penicillin-binding proteins, leading to bacterial cell lysis. Ceftriaxone does not directly inhibit protein synthesis.

Solution for Question 86:

Correct Option C - Mupirocin:

- Mupirocin is the drug of choice for nasal carriers of methicillin-resistant *Staphylococcus aureus* (MRSA). It is used topically in the nostrils to eliminate the colonization of MRSA.

Incorrect Options:

Option A - Vancomycin: Vancomycin is an antibiotic used to treat systemic infections caused by MRSA. It is not specifically indicated for nasal carriers.

Option B - Teicoplanin: Teicoplanin is another antibiotic with activity against MRSA. Like Vancomycin, it is not the preferred choice for nasal carriers.

Option D - Linezolid: Linezolid is an antibiotic used for the treatment of infections caused by MRSA. However, Mupirocin is the specific drug of choice for nasal carriers in this context

Solution for Question 87:

Correct Option A - Primaquine:

- Primaquine is gametocidal to all species causing malaria. It is effective against both the asexual blood stages and the sexual forms (gametocytes) of the malaria parasite. It is primarily used for the treatment of *Plasmodium vivax* and *Plasmodium ovale* malaria.

Incorrect Options:

Option B - Chloroquine: Chloroquine is primarily effective against the asexual blood stages of malaria parasites. It is commonly used for the treatment and prevention of malaria caused by *Plasmodium falciparum* and *Plasmodium vivax*. However, it is not gametocidal to all species

causing malaria.

Option C - Quinine: Quinine is an older antimalarial drug that is primarily effective against the asexual blood stages of malaria parasites. It is used for the treatment of severe malaria caused by *Plasmodium falciparum*. Like chloroquine, it is not gametocidal to all species causing malaria.

Solution for Question 88:

Correct Option D - Azithromycin, Doxycycline:

- Azithromycin is the drug of choice for treating scrub typhus in pregnant women. It is considered safe for use during pregnancy and has been shown to effectively treat the infection. Doxycycline, on the other hand, is the preferred choice for non-pregnant individuals due to its efficacy

Azithromycin is the drug of choice for treating scrub typhus in pregnant women. It is considered safe for use during pregnancy and has been shown to effectively treat the infection. Doxycycline, on the other hand, is the preferred choice for non-pregnant individuals due to its efficacy

Incorrect Options:

Option A - Chloramphenicol, Doxycycline: Chloramphenicol is an antibiotic that can be used for the treatment of scrub typhus. However, it is generally not the first-line choice due to its potential side effects, including a rare but serious condition called gray baby syndrome. Doxycycline is a tetracycline antibiotic that is highly effective against the bacteria that cause scrub typhus. It is commonly used in non-pregnant individuals for the treatment of scrub typhus.

Option B - Doxycycline, Doxycycline: Doxycycline is the drug of choice for treating scrub typhus in non-pregnant individuals. It is a broad-spectrum antibiotic that is effective against the bacteria causing scrub typhus. It is commonly used due to its efficacy and relatively low risk of adverse effects.

Option C - Doxycycline, Azithromycin: Doxycycline, as mentioned earlier, is effective against scrub typhus in non-pregnant individuals. Azithromycin is an alternative option that can be used for the treatment of scrub typhus in pregnant women. It is a macrolide antibiotic that is generally considered safe during pregnancy.

Solution for Question 89:

Correct Option B - Indinavir:

- Indinavir is a protease inhibitor used in the treatment of HIV/AIDS. While it is effective in suppressing the HIV virus, it has been associated with the development of nephrolithiasis. Indinavir crystals can precipitate in the urine, leading to kidney stone formation. Nephrolithiasis is a known side effect of indinavir therapy. However, central obesity, hyperlipidemia, and insulin resistance are not typically associated with indinavir use.

Indinavir is a protease inhibitor used in the treatment of HIV/AIDS. While it is effective in suppressing the HIV virus, it has been associated with the development of nephrolithiasis. Indinavir crystals can precipitate in the urine, leading to kidney stone formation. Nephrolithiasis is a known side effect of indinavir therapy. However, central obesity, hyperlipidemia, and insulin resistance are not typically associated with indinavir use.

Incorrect Options:

Option A - Ketoconazole: Ketoconazole is an antifungal medication and is not typically associated with nephrolithiasis. Common side effects of ketoconazole may include gastrointestinal upset, liver function abnormalities, and skin reactions.

Option C - Lamivudine: Lamivudine is a nucleoside reverse transcriptase inhibitor (NRTI) used in the treatment of HIV/AIDS. Lamivudine is generally well-tolerated. Common side effects of lamivudine may include gastrointestinal symptoms, headache, and fatigue.

Option D - Zidovudine: Zidovudine, also known as AZT, is an NRTI used in the treatment of HIV/AIDS. Zidovudine has been associated with various metabolic abnormalities. It can cause mitochondrial toxicity, which may result in fat redistribution, including central obesity. Zidovudine is also known to cause hyperlipidemia and insulin resistance.

Solution for Question 90:

Correct Option B - Flucytosine: Inhibits microtubule synthesis and thus hinders mitosis:

- The correct mechanism of action of Flucytosine is different. Flucytosine is an antifungal medication that is converted into 5-fluorouracil within fungal cells. It interferes with fungal RNA and protein synthesis by inhibiting fungal DNA and RNA synthesis. It is primarily used

in combination with other antifungal drugs for the treatment of systemic fungal infections.

Incorrect Options:

Option A - Ketoconazole: Lanosterol-14-demethylase inhibition:

- Lanosterol-14-demethylase inhibition Ketoconazole is an antifungal medication that works by inhibiting the enzyme lanosterol-14-demethylase. This enzyme is responsible for converting lanosterol into ergosterol, which is an essential component of fungal cell membranes. By inhibiting this enzyme, ketoconazole disrupts the synthesis of ergosterol and leads to the accumulation of abnormal fungal cell membranes, ultimately causing fungal cell death.

Option C - Terbinafine: Squalene epoxidase inhibition:

- Terbinafine is an antifungal drug that inhibits the enzyme squalene epoxidase. This enzyme is involved in the synthesis of ergosterol, an essential component of fungal cell membranes. By inhibiting squalene epoxidase, terbinafine disrupts ergosterol synthesis and leads to the accumulation of squalene, causing fungal cell death.

Option D - Amphotericin B: Sequesters ergosterol:

- Sequesters ergosterol Amphotericin B is a broad-spectrum antifungal medication that binds to ergosterol, a key component of fungal cell membranes. It forms a complex with ergosterol, which results in the formation of pores or channels in the fungal cell membrane. This disruption of the cell membrane integrity leads to the leakage of intracellular contents and ultimately results in fungal cell death.

Solution for Question 91:

Correct Option C - Moxifloxacin:

- Moxifloxacin is less likely to be effective in treating urinary tract infections (UTIs) caused by Enterococcus species. While it is a broad-spectrum fluoroquinolone antibiotic that is effective against many bacterial pathogens, including those commonly associated with UTIs, its coverage against Enterococcus species is limited.

Moxifloxacin is less likely to be effective in treating urinary tract infections (UTIs) caused by Enterococcus species. While it is a broad-spectrum fluoroquinolone antibiotic that is effective against many bacterial pathogens, including those commonly associated with UTIs, its coverage against Enterococcus species is limited.

Incorrect Options:

Option A, B & D - Ofloxacin, Ciprofloxacin and Levofloxacin:

- Ofloxacin, ciprofloxacin, and levofloxacin are fluoroquinolone antibiotics that have broader coverage against a wide range of pathogens, including Enterococcus species. They are commonly used in the treatment of UTIs and have demonstrated efficacy against both gram-negative and gram-positive bacteria, including Enterococcus.
- Enterococcus species are a common cause of urinary tract infections, particularly in healthcare-associated and complicated UTIs. Infections caused by Enterococcus can be challenging to treat due to their intrinsic resistance to many antibiotics. While some fluoroquinolones have activity against Enterococcus, such as ofloxacin, ciprofloxacin, and levofloxacin, moxifloxacin has limited effectiveness against this pathogen.
- Therefore, in the given scenario, if the suspected UTI is caused by Enterococcus, moxifloxacin would be the least effective choice among the listed fluoroquinolones. The selection of an appropriate antibiotic should be based on the specific pathogens involved, their susceptibility patterns, and individual patient factors.

Ofloxacin, ciprofloxacin, and levofloxacin are fluoroquinolone antibiotics that have broader coverage against a wide range of pathogens, including Enterococcus species. They are commonly used in the treatment of UTIs and have demonstrated efficacy against both gram-negative and gram-positive bacteria, including Enterococcus.

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Therefore, in the given scenario, if the suspected UTI is caused by Enterococcus, moxifloxacin would be the least effective choice among the listed fluoroquinolones. The selection of an appropriate antibiotic should be based on the specific pathogens involved, their susceptibility patterns, and individual patient factors.

Solution for Question 92:

Correct Option C - Ciprofloxacin:

- The correct answer for the treatment of cystitis in this case would be: Ciprofloxacin.
- Cystitis is commonly caused by a bacterial infection, and the choice of antibiotic depends on the suspected or identified bacteria and its susceptibility pattern. In the case of uncomplicated cystitis, the most common pathogen is Escherichia coli. Ciprofloxacin is a fluoroquinolone antibiotic that has excellent activity against E. coli and is often recommended as a first-line treatment for uncomplicated cystitis.

The correct answer for the treatment of cystitis in this case would be: Ciprofloxacin.

Cystitis is commonly caused by a bacterial infection, and the choice of antibiotic depends on the suspected or identified bacteria and its susceptibility pattern. In the case of uncomplicated cystitis, the most common pathogen is Escherichia coli. Ciprofloxacin is a fluoroquinolone antibiotic that has excellent activity against E. coli and is often recommended as a first-line treatment for uncomplicated cystitis.

Incorrect Options:

Option A - Amoxicillin: While amoxicillin is effective against many bacterial infections, it is not the preferred choice for treating uncomplicated cystitis. It may be used in certain cases, but it is not the first-line treatment due to increasing antibiotic resistance.

Option B - TMP-SMX: TMP-SMX (Trimethoprim-sulfamethoxazole) is another commonly used antibiotic for cystitis. It is effective against E. coli, but its use can be limited by bacterial resistance in some regions. However, it is still considered a first-line treatment option in many cases.

Option D - Doxycycline: Doxycycline is a broad-spectrum antibiotic that is effective against various bacterial infections, including urinary tract infections. However, it is not typically the first-line choice for uncomplicated cystitis unless there is a specific indication or resistance pattern.

Solution for Question 93:

Correct Option D - Ceftriaxone:

- The antibiotic of choice for treating urinary tract infections (UTIs) in pregnancy is Ceftriaxone.
- UTIs are a common concern during pregnancy due to hormonal changes and anatomical factors that can increase the risk of urinary tract infections. When treating UTIs in pregnancy,

certain antibiotics are preferred to ensure the safety of both the mother and the developing fetus.

- Ceftriaxone is the preferred antibiotic for UTIs in pregnancy due to its effectiveness against common pathogens that cause UTIs, including *Escherichia coli*. It is a broad-spectrum antibiotic that is generally safe for use during pregnancy.

The antibiotic of choice for treating urinary tract infections (UTIs) in pregnancy is Ceftriaxone.

UTIs are a common concern during pregnancy due to hormonal changes and anatomical factors that can increase the risk of urinary tract infections. When treating UTIs in pregnancy, certain antibiotics are preferred to ensure the safety of both the mother and the developing fetus.

Ceftriaxone is the preferred antibiotic for UTIs in pregnancy due to its effectiveness against common pathogens that cause UTIs, including *Escherichia coli*. It is a broad-spectrum antibiotic that is generally safe for use during pregnancy.

Incorrect Options:

Option A - Amoxicillin:

- Amoxicillin is an antibiotic commonly used for various infections.
- This antibiotic may be used if the bacterial strain is known to be sensitive.
- It is less commonly used due to rising resistance rates.

Option B - Azithromycin:

- Azithromycin is an antibiotic often used for respiratory tract infections, but it is not typically recommended as a first-line treatment for UTIs.
- Its spectrum of activity is not as focused on the common bacteria that cause UTIs.

Option C - Clarithromycin:

- Clarithromycin is primarily used for respiratory tract infections and is not typically recommended as a first-line treatment for UTIs.
- It may have limited effectiveness against the specific bacteria that commonly cause UTIs.

Solution for Question 94:

Correct Option C - Piperacillin – Tazobactam:

- Extended-spectrum beta-lactamases (ESBLs) are enzymes produced by certain bacteria that confer resistance to a broad range of beta-lactam antibiotics, including penicillins and cephalosporins.
- Piperacillin-Tazobactam: This combination antibiotic consists of piperacillin, a penicillin, and tazobactam, a beta-lactamase inhibitor. Tazobactam inhibits the activity of beta-lactamases, including ESBLs, thereby restoring the efficacy of piperacillin against ESBL-producing bacteria.

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Piperacillin-Tazobactam: This combination antibiotic consists of piperacillin, a penicillin, and tazobactam, a beta-lactamase inhibitor. Tazobactam inhibits the activity of beta-lactamases, including ESBLs, thereby restoring the efficacy of piperacillin against ESBL-producing bacteria.

Incorrect Options:

Option A - Amoxicillin-Clavulanic acid: This combination antibiotic consists of amoxicillin, a penicillin, and clavulanic acid, a beta-lactamase inhibitor. While it is effective against some beta-lactamase-producing organisms, it may not be effective against ESBL-producing bacteria, as ESBLs can still hydrolyze the amoxicillin component.

Option B - Penicillin: Penicillin antibiotics are not effective against bacteria producing ESBLs, as these enzymes can hydrolyze and inactivate penicillins.

Option D - Ceftriaxone: Ceftriaxone is a cephalosporin antibiotic and is not effective against ESBL-producing bacteria. ESBLs can hydrolyze and inactivate cephalosporins, including ceftriaxone.

Respiratory Pharmacology

1. A 12-year-old girl with a known history of cystic fibrosis (CF) presents with a persistent cough and increased sputum production over the past two weeks. Her pulmonary function tests show a significant decline in FEV1 compared to her baseline. Sputum culture reveals *Pseudomonas aeruginosa*. She has been compliant with her routine medications, including a CFTR modulator, mucolytics, and pancreatic enzyme replacement therapy. Which of the following pharmacological adjustments is the most appropriate next step in managing this patient's current clinical condition?

- A. Add inhaled tobramycin to her current regimen
- B. Switch her CFTR modulator to a different one
- C. Increase the dose of her mucolytic agent
- D. Start intravenous ceftazidime and tobramycin

2. Which of the following statements is false regarding the mechanism of action of theophylline?

- A. Inhibit phosphodiesterase 4
- B. Block A1 receptors
- C. Activation of NF kappa B
- D. Cause histone acetylation

3. A 45-year-old female patient presents with progressive dyspnea on exertion and fatigue. She has a history of systemic sclerosis and is diagnosed with pulmonary arterial hypertension (PAH) based on right heart catheterization, which shows elevated pulmonary arterial pressures. The patient is initiated on inhaled nitric

oxide (NO) therapy as part of her treatment plan. Which of the following statements best describes the mechanism by which inhaled nitric oxide improves her symptoms?

- A. Nitric oxide acts by reducing systemic blood pressure, thereby decreasing the workload on the left ventricle
 - B. Nitric oxide decreases the production of inflammatory cytokines, thereby reducing pulmonary artery inflammation and fibrosis.
 - C. Nitric oxide causes selective pulmonary vasodilation, decreasing pulmonary vascular resistance and alleviating right ventricular strain.
 - D. Nitric oxide increases the proliferation of endothelial cells in pulmonary arteries, promoting vascular remodeling and reducing hypertension.
-

4. A 45-year-old female patient with a history of asthma and nasal polyps presents with wheezing, shortness of breath, and chest tightness. She reports that her symptoms began shortly after taking aspirin for a headache. On examination, she has diffuse wheezing and reduced breath sounds bilaterally. She is given supplemental oxygen, and nebulized bronchodilators are started.

- A. Inhibition of cyclooxygenase-1 (COX-1) leading to increased leukotriene production
 - B. Direct stimulation of the beta-2 adrenergic receptors causing bronchoconstriction
 - C. Inhibition of lipoxygenase leading to decreased leukotriene production
 - D. Blocking of the muscarinic receptors in the bronchial smooth muscle
-

5. During a diagnostic workup for a patient presenting with symptoms of suspected pulmonary arterial hypertension (PAH), a vaso-reactive testing procedure is planned. The patient undergoes right heart catheterization, and baseline hemodynamic measurements are recorded. Subsequently, a vasoactive medication is administered directly into the pulmonary circulation through the catheter. What outcome of vaso-reactive testing in PAH would indicate a positive response?

- A. A positive response indicates a decrease in mean pulmonary artery pressure of at least 10 mmHg
- B. A positive response indicates a reduction in pulmonary vascular resistance of at least 10%
- C. A negative response indicates a favourable response to vasodilator medications.
- D. Calcium channel blockers are contraindicated in positive responders.

6. A 45-year-old female presents to the cardiology clinic with complaints of increasing shortness of breath. She was diagnosed with pulmonary artery hypertension. Which of the following correctly describes the mechanism of action with respect to the drug? Drug Mechanism of Action 1. Bosentan A. Prostaglandin I-2 agonist 2. Selexipag B. Prostacyclin (PGI-2) analogue 3. Riociguat C. Endothelin receptor antagonist 4. Epoprostenol D. Soluble guanylate cyclase stimulator

Drug	Mechanism of Action
1. Bosentan	A. Prostaglandin I-2 agonist
2. Selexipag	B. Prostacyclin (PGI-2) analogue
3. Riociguat	C. Endothelin receptor antagonist
4. Epoprostenol	D. Soluble guanylate cyclase stimulator

- A. 1-C, 2-A, 3-D, 4-B
- B. 1-B, 2-D, 3-C, 4-A
- C. 1-D, 2-B, 3-A, 4-C
- D. 1-A, 2-C, 3-B, 4-D

7. Which of the following best describes the role of corticosteroids in asthma management?

- A. Corticosteroids are not bronchodilators

- B. Corticosteroids are only used for severe asthma attacks.
 - C. Corticosteroids are limited to maintenance therapy for chronic asthma.
 - D. Corticosteroids are used in both acute exacerbations and chronic management of asthma, with their administration route varying by use case.
-

8. A 12-year-old child with a history of asthma presents with symptoms of coughing, wheezing, and shortness of breath, particularly during physical activities. The child has been using a short-acting beta-agonist inhaler as needed but continues to experience frequent asthma symptoms. The physician decides to add a medication that does not cause bronchodilation but restricts the inflammatory mediators implicated in asthma. Which medication is being referred to in this scenario?

- A. Sodium cromoglycate
 - B. Salmeterol
 - C. Magnesium sulfate
 - D. Theophylline
-

9. A 45-year-old female with a history of asthma presents with acute shortness of breath and wheezing. Despite initial treatment with inhaled bronchodilators and systemic corticosteroids, her symptoms persist. Which of the following can be used to manage this presentation?

- A. Magnesium sulfate
 - B. N-acetylcysteine
 - C. Bromhexine
 - D. Montelukast
-

10. A 65-year-old male with a history of COPD on medication presented to the ER with complaints of headache and chest pain. His ECG showed ventricular arrhythmia, and he subsequently experienced a seizure. He had recently begun taking the anti-gout medication allopurinol. What medication is he likely using for COPD?

- A. Theophylline
 - B. Albuterol
 - C. Ipratropium
 - D. Fluticasone
-

11. A 60-year-old patient has symptoms of a productive cough and difficulty clearing thick mucus from their airways. The patient has a history of COPD and frequently experiences episodes of breathing difficulties. Which of the following drugs can be used to reduce the viscosity of the mucus, thus increasing its clearance in this patient?

- A. Albuterol
 - B. Dextromethorphan
 - C. Bromhexine
 - D. Guaifenesin
-

12. A 65-year-old male with a long history of chronic obstructive pulmonary disease (COPD) is prescribed an anticholinergic to help manage his symptoms. At his follow-up appointment, he reports experiencing increased shortness of breath and wheezing shortly after using his inhaler. Which of the following causes paradoxical bronchoconstriction?

- A. Tiotropium
- B. Ipratropium
- C. Oxitropium

D. Umeclidinium

13. A 68-year-old male with a history of chronic obstructive pulmonary disease (COPD) presents to the clinic for a routine follow-up. He reports experiencing increased shortness of breath and wheezing, particularly during physical activity. He is on albuterol used as needed, and the physician decides to add ipratropium. Which of the following is the mechanism of action of the latter?

- A. β_2 receptor agonist
 - B. M2 receptor agonist
 - C. M2 receptor antagonist
 - D. M3 receptor antagonist
-

14. A 30-year-old male with a history of asthma complains of wheezing, shortness of breath, and nighttime awakenings occurring about twice a week, with each episode lasting more than a day. His current medication regimen includes a SABA. Pulmonary function tests show a FEV1 of 70% predicted. Which is the most appropriate pharmacological management for this patient's condition?

- A. Low-dose inhaled corticosteroids
 - B. High-dose inhaled corticosteroids
 - C. Long-acting beta-agonists and low-dose inhaled corticosteroids
 - D. Oral corticosteroids
-

15. A 25-year-old asthmatic presents with severe respiratory distress and is unresponsive to her albuterol inhaler. She has a respiratory rate of 32, oxygen saturation of 88%, and audible wheezing. Initial treatment with high-dose nebulized albuterol and systemic corticosteroids has shown only minimal improvement.

What is the most appropriate next step in managing her condition?

- A. Repeat
 - B. Aminophylline
 - C. Ipratropium
 - D. Magnesium sulfate
-

16. A 24-year-old male presents with shortness of breath, wheezing, and chest tightness. He has a known history of asthma. On examination, he appears distressed, with an RR of 30/min, HR of 110/min, and a SpO₂ of 92% on room air. Auscultation reveals widespread wheezing. Which of the following is the most appropriate initial pharmacological management for this presentation?

- A. Oral Prednisone 40 mg
 - B. Inhaled Salbutamol 2.5 mg via nebulizer
 - C. Inhaled Fluticasone 100 mcg
 - D. Montelukast 10 mg orally
-

17. A 42-year-old female patient with a history of severe asthma presents with frequent exacerbations despite being on corticosteroids. She is about to get started on omalizumab. Which of the following mechanisms best describes the action of omalizumab?

- A. Inhibiting leukotriene synthesis
 - B. Blocking the activity of TNF-alpha
 - C. Neutralizing free IgE in circulation
 - D. Suppressing eosinophil activation
-

18. A 12-year-old female patient with cystic fibrosis (CF) presents to the clinic with increased cough and thick sputum production, causing shortness of breath and fatigue. Which of the following medications would be the most effective for this patient?

- A. Pancrelipase
 - B. Dornase alfa
 - C. Cevimeline
 - D. Lumacaftor
-

19. A 30-year-old female presented with complaints of a persistent dry cough for the past week and an antitussive was prescribed. Two days later, the patient returned with nausea, vomiting, and constipation. Which centrally acting antitussive medication is most likely responsible for these adverse effects in this patient?

- A. Codeine
 - B. Dextromethorphan
 - C. Noscapine
 - D. Promethazine
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	3
Question 4	1
Question 5	1
Question 6	1
Question 7	4
Question 8	1
Question 9	1
Question 10	1
Question 11	3
Question 12	2
Question 13	4
Question 14	3
Question 15	4
Question 16	2
Question 17	3
Question 18	2
Question 19	1

Solution for Question 1:

Correct Answer: D) Start intravenous ceftazidime and tobramycin

Explanation:

Here, the significant decline in pulmonary function and the presence of *Pseudomonas aeruginosa* suggest a severe exacerbation that likely requires intravenous antibiotics for effective management. Intravenous antibiotics, such as ceftazidime and tobramycin, are used for severe or refractory infections.

Inhaled tobramycin specifically targets *Pseudomonas aeruginosa* in the lungs and is effective in treating pulmonary exacerbations in CF patients. This can be a suitable option if the exacerbation is not severe (Option A).

Changing the CFTR modulator might be considered for long-term management if her current one is ineffective, but it is not an immediate intervention for acute exacerbation due to *Pseudomonas* infection (Option B).

Increasing mucolytic dosage may help in mucus clearance but is not a targeted approach for a *Pseudomonas* infection (Option C).

Solution for Question 2:

Correct Answer: C) Activation of NF kappa B

Explanation:

Theophylline acts by inhibiting the NF kappa B rather than activating it.

Mechanism of action of Theophylline:

Mnemonic- PANCHI

- P- Phosphodiesterase 4 inhibitor (Option A) Inhibition of PDEs, → increased cellular cAMP and cyclic GMP levels → Bronchodilation Important PDE isoenzymes in smooth muscle relaxation include PDE3(primary action), PDE4, and PDE5.
- Inhibition of PDEs, → increased cellular cAMP and cyclic GMP levels → Bronchodilation
- Important PDE isoenzymes in smooth muscle relaxation include PDE3(primary action), PDE4, and PDE5.

- A- Blocks Adenosine (A1) receptor (Option B) Adenosine induces bronchoconstriction in asthmatic patients by releasing histamine and leukotrienes. Theophylline acts as an antagonist to adenosine receptors → bronchodilation. Blocking adenosine A1 receptors might lead to serious side effects, such as cardiac arrhythmias and seizures.
- Adenosine induces bronchoconstriction in asthmatic patients by releasing histamine and leukotrienes.
- Theophylline acts as an antagonist to adenosine receptors → bronchodilation.
- Blocking adenosine A1 receptors might lead to serious side effects, such as cardiac arrhythmias and seizures.
- N- Inhibit NF kappa B Theophylline inhibits the movement of the proinflammatory transcription factor NF-κB into the nucleus → reducing the expression of inflammatory genes in asthma and COPD.
- Theophylline inhibits the movement of the proinflammatory transcription factor NF-κB into the nucleus → reducing the expression of inflammatory genes in asthma and COPD.
- C- Calcium channel blockade
- H- Causes Histone acetylation (Option D) → Decrease the inflammatory activity of IL-5, IL-4.
- I- Reduces the Inflammation.
- Inhibition of PDEs, → increased cellular cAMP and cyclic GMP levels → Bronchodilation
- Important PDE isoenzymes in smooth muscle relaxation include PDE3(primary action), PDE4, and PDE5.
- Adenosine induces bronchoconstriction in asthmatic patients by releasing histamine and leukotrienes.
- Theophylline acts as an antagonist to adenosine receptors → bronchodilation.
- Blocking adenosine A1 receptors might lead to serious side effects, such as cardiac arrhythmias and seizures.
- Theophylline inhibits the movement of the proinflammatory transcription factor NF-κB into the nucleus → reducing the expression of inflammatory genes in asthma and COPD.

Solution for Question 3:

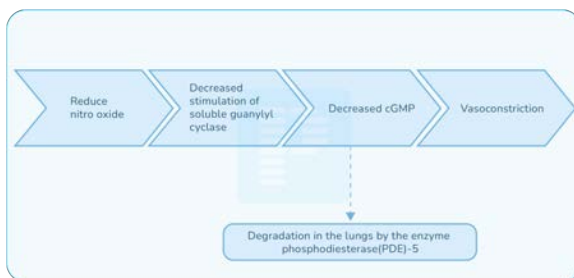
Correct Answer: C) Nitric oxide causes selective pulmonary vasodilation, decreasing pulmonary vascular resistance and alleviating right ventricular strain.

Explanation:

Inhaled nitric oxide acts as a selective pulmonary vasodilator, which helps to reduce pulmonary vascular resistance.

- This reduction in resistance lowers the pressure in the pulmonary arteries, thereby decreasing the afterload on the right ventricle.
- This is crucial in PAH, where increased pulmonary pressures can lead to right ventricular hypertrophy and failure.
- By decreasing this strain, nitric oxide improves the patient's symptoms and cardiac function.

Role of Nitric Oxide in PAH:



Nitric oxide administered via inhalation does not significantly affect systemic blood pressure; its primary action is localized to the pulmonary vasculature (Option A).

While nitric oxide may have some anti-inflammatory effects, its primary role in PAH is through vasodilation rather than reducing inflammation and fibrosis (Option B).

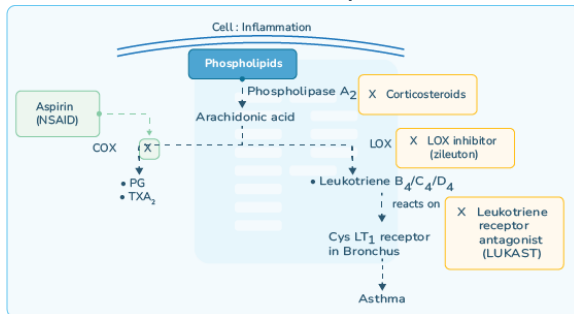
Nitric oxide does not increase endothelial cell proliferation in a way that promotes beneficial vascular remodeling; its therapeutic effect is mainly through vasodilation of the pulmonary arteries (Option D).

Solution for Question 4:

Correct Answer: A) Inhibition of cyclooxygenase-1 (COX-1) leading to increased leukotriene production

Explanation: Based on the history and Clinical findings, the patient developed aspirin-induced asthma, which is primarily due to the inhibition of the COX-1 enzyme.

- This inhibition reduces the production of prostaglandins, which are protective in the airways, and shifts the arachidonic acid metabolism toward the lipoxygenase pathway, leading to increased production of leukotrienes.
- Leukotrienes are potent bronchoconstrictors and contribute to the symptoms of asthma.



Beta-2 adrenergic receptors, when stimulated, cause bronchodilation, not bronchoconstriction. Beta-2 agonists are used as bronchodilators in asthma treatment (Option B).

Inhibition of lipoxygenase would decrease leukotriene production, leading to bronchodilation and relief from asthma symptoms, which is the opposite of what occurs in aspirin-induced asthma (Option C).

Blocking muscarinic receptors in the bronchial smooth muscle would lead to bronchodilation, not bronchoconstriction. Muscarinic antagonists are used as bronchodilators in the treatment of chronic obstructive pulmonary disease (COPD) and asthma (Option D).

Solution for Question 5:

Correct Answer: A) A positive response indicates a decrease in mean pulmonary artery pressure of at least 10 mmHg

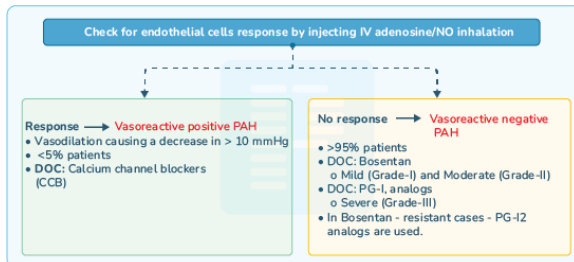
Explanation:

A positive response during vasoactive testing indicates a decrease in mean pulmonary artery pressure of at least 10 mmHg or a reduction in pulmonary vascular resistance of at least 20%. (Option B)

- Vaso-reactive testing assesses pulmonary vascular response to vasodilators in PAH.
- It involves right heart catheterization to measure baseline pulmonary artery pressure and vascular resistance.

- Vasodilators are then administered directly into the pulmonary circulation.
- Hemodynamic measurements are repeated post-administration to evaluate the response.

Vasoactive testing:



A negative response indicates non-reactivity to the vasoactive drug (Option C)

Positive responders are actually considered for long-term calcium blocker therapy (Option D).

Solution for Question 6:

Correct Answer: A) 1-C, 2-A, 3-D, 4-B

Explanation: The drugs used in PAH are as follows: Mnemonic : PEPSI Coca Cola

Mechanism of action	Drugs
PDE 5 Inhibitors	Sildenafil (Viagra), Tadalafil
Endothelin Receptor Antagonists	Bosentan, Sitaxsentan, Ambrisentan
PG-I2 Analogs	Treprostinil, Iloprost, Epoprostenol
Soluble Guanylyl Cyclase Activators	Riociguat
I-2 prostaglandin agonist	Selexipag
Continuous NO Inhalation	Nitric oxide
Calcium Channel Blockers	Amlodipine, Nifedipine

Solution for Question 7:

Correct Answer : D) Corticosteroids play a critical role in both acute exacerbations and chronic management of asthma, with their administration route varying by use case.

Explanation:

Corticosteroids are integral to both acute and chronic asthma management.

- Inhaled corticosteroids are used in long-term maintenance therapy for chronic asthma, helping to control persistent inflammation.
- For acute exacerbations which are not controlled by inhalational corticosteroids and bronchodilators, systemic corticosteroids (oral or IV) are used to rapidly reduce inflammation despite the fact that corticosteroids are not bronchodilators.

Route	Examples with use
Inhalational Corticosteroids	DOC- Chronic or persistent asthma Eg: Beclomethasone Fluticasone-Most Potent Mometasone Triamcinolone Budesonide- Most commonly used Ciclesonide
Oral Corticosteroids	Recommended for refractory asthma Eg: Prednisolone
IV Corticosteroids	Recommended for status asthmaticus Eg: Hydrocortisone Dexamethasone

Solution for Question 8:

Correct Answer: A) Sodium cromoglycate

Explanation:

Sodium cromoglycate is a mast cell stabilizer that inhibits degranulation of mast cells (as well as other inflammatory cells). Release of mediators of asthma-like histamine, LTs, PAF, interleukins, etc. is restricted. It does not cause bronchodilation.

Salmeterol (Option B) is a beta 2 sympathomimetic that causes bronchodilation and relieves symptoms of asthma

Magnesium sulfate (Option C) blocks calcium channels which leads to bronchodilation. It may be used in acute exacerbations of asthma but not for long-term management.

Theophylline (Option D) inhibits PDE3 and PDE 4, resulting in high cAMP levels inside smooth muscle cells resulting in bronchodilation.

Solution for Question 9:

Correct Answer: A) Magnesium sulfate

Explanation :

Magnesium sulfate blocks calcium channels, which leads to bronchodilation. Therefore, it may be used in asthma attacks unresolved by administration of other drugs.

Mnemonic for the uses of magnesium sulfate: TABLETS

- Torsades De Pointes (DOC for arrhythmia)
- Antacid (Magnesium hydroxide)
- Barium poisoning and Calcium poisoning antidote
- Laxative effect (Treatment of constipation)
- Eclampsia (Drug of choice)
- Tocolytic agent (Relaxes the uterus)
- Status asthmaticus

N-acetylcysteine (Option B) & bromhexine (Option C) are mucolytic agents used to thin mucus secretions. Both drugs do not have a role in treating severe asthma exacerbations.

Montelukast (Option D) is a leukotriene receptor antagonist used for long-term asthma management. It is not used for acute exacerbations because it does not provide rapid relief of bronchospasm.

Solution for Question 10:

Correct Answer: A) Theophylline

Explanation :

- Theophylline has a narrow therapeutic index.
- It is metabolized by the CYP1A2 enzyme. Thus, enzyme inhibitors (eg. allopurinol) increase blood levels of theophylline causing toxicity.

Drugs reducing the effect of theophylline	Drugs increasing the effect of theophylline
CYP enzyme inducers Drugs include Griseofulvin Phenobarbitone Phenytoin Rifampicin Smoking and alcohol	CYP inhibitors Drugs include Ciprofloxacin Omeprazole Cimetidine Allopurinol Erythromycin

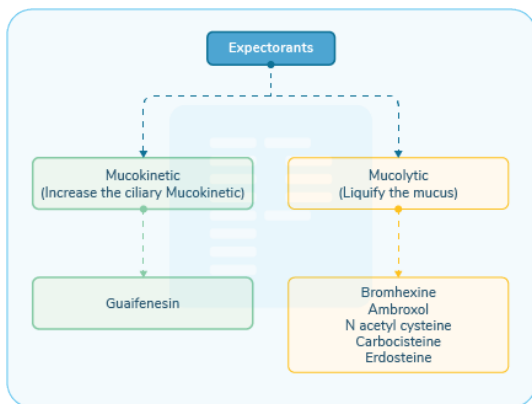
Solution for Question 11:

Correct Answer: C) Bromhexine

Explanation:

Bromhexine is a mucolytic agent that helps to break down and reduce the viscosity of mucus, facilitating its removal by coughing.

This action can be beneficial in conditions like COPD, where mucus plugs can obstruct airways and exacerbate breathing difficulties.



- Albuterol (Option A) is a bronchodilator.
- Dextromethorphan (Option B) is an antitussive.
- Guaifenesin (Option D) is a mucokinetic.

None of these directly address the need to reduce mucus viscosity and enhance mucus clearance through coughing.

Solution for Question 12:

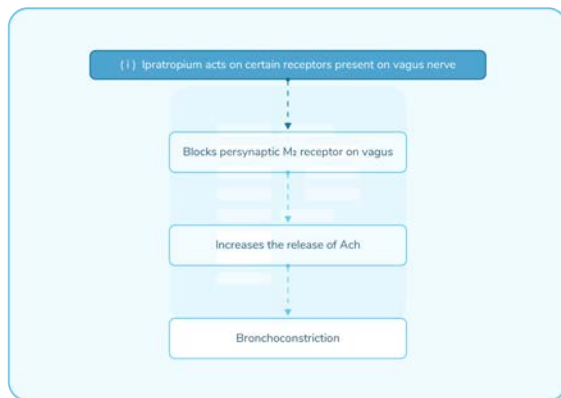
Correct Answer: B) Ipratropium

Explanation:

The occurrence of wheezing and increased shortness of breath after using inhaled anticholinergic in a patient with COPD suggests paradoxical bronchoconstriction. Ipratropium bromide can result in such a presentation.

Mechanism of Bronchoconstriction:

- Nebulizer: Due to hypotonic solution and antibacterial additives (benzalkonium chloride, EDTA)
- Metered-dose inhaler (MDI): Possible due to blockade of prejunctional M2 receptors, increasing ACh release



Tiotropium bromide and other LAMAs have no similar reports of paradoxical bronchoconstriction (Option A, C & D).

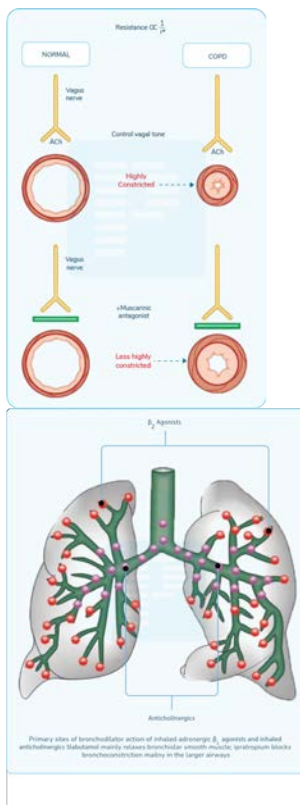
Solution for Question 13:

Correct Answer : D) M3 receptor antagonist

Explanation:

Ipratropium bromide is an anticholinergic that inhibits the M3 receptors on the vagus nerve supplying the bronchioles of the lung to cause bronchodilation.

- The bronchodilation effect is minimal in normal airways.
- The effect is more significant in chronic obstructive pulmonary disease (COPD) patients.
- COPD patients have structurally narrowed airways.
- Higher airway resistance in COPD due to the inverse relationship between airway resistance and the fourth power of the airway radius.



- Inhaled adrenergic beta 2 agonists and inhaled anticholinergics act on different primary sites in the respiratory system.
- Salbutamol primarily relaxes the smooth muscle in the bronchioles.
- Ipratropium mainly prevents bronchoconstriction in the larger airways.

Ipratropium does not act on β_2 or M2 receptors (Options A, B & C).

Solution for Question 14:

Correct Answer: C) Long-acting beta-agonists and low-dose inhaled corticosteroids

Explanation:

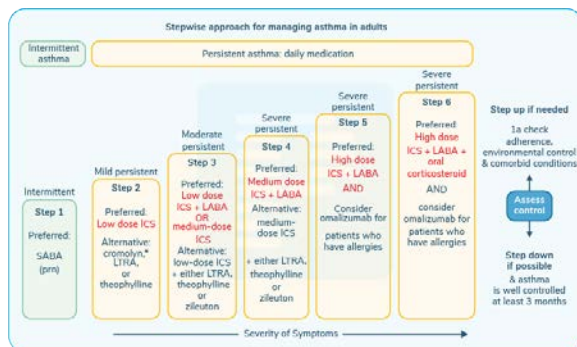
The above Clinical vignette suggests the presence of moderate persistent asthma in this patient.

Categories of Asthma:

Asthma classification	Signs and symptoms
-----------------------	--------------------

Mild intermittent	Mild symptoms up to 2 days a week and up to 2 nights a month
Mild persistent	Symptoms >2 times a week but not >1 time in a single day
Moderate persistent	Symptoms every day and >1 time/ week at night
Severe persistent	Symptoms throughout the day on most days and also frequent awakenings at night

Step-up therapy for Chronic/Persistent asthma:



Solution for Question 15:

CorrectAnswer : D) Magnesium sulfate

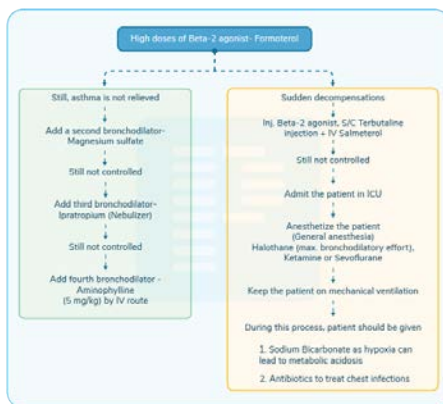
Explanation :

The above Clinical scenario suggests a classic presentation of Status asthmaticus.

Status asthmaticus :

- Defined as severe asthma that does not respond to repeated beta-agonist treatments (inhaled albuterol, levalbuterol, or subcutaneous epinephrine)
- Considered a medical emergency
- Requires immediate recognition and treatment

Management :



Solution for Question 16:

Correct Answer: B) Inhaled Salbutamol 2.5 mg via nebuliser

Explanation:

In the management of an acute asthma exacerbation, the first-line treatment is a short-acting beta-agonist (SABA) such as salbutamol (albuterol).

- This medication works to relax bronchial smooth muscle, leading to rapid improvement in airway obstruction and relief of symptoms.
- It acts as quickly as 5 minutes within the administration, and the effect lasts for around 2-4 hours.

Oral corticosteroids (eg. prednisolone) (Option A) are not the first-line treatment for immediate relief of symptoms, instead they are typically given after initial stabilization with a SABA.

Inhaled corticosteroids (eg. fluticasone) and leukotriene receptor antagonists (eg. montelukast) (Options C & D) are used for long-term management and prevention of asthma symptoms and are not effective for immediate relief in an acute exacerbation.

Solution for Question 17:

Correct Answer: C) Neutralizing free IgE in circulation

Explanation :

Omalizumab is a humanized monoclonal antibody that targets and neutralizes free IgE in circulation. It prevents the activation of mast cells and other inflammatory cells, reducing the release of mediators that cause bronchoconstriction upon antigen challenge.

Options A, B, and D do not accurately describe the mechanism of action of Omalizumab in asthma treatment.

Anti-inflammatory drugs in Asthma :

Mast cell stabilizers	Inhibit the release of inflammatory mediators (-CROM-)
LOX inhibitor	Drug- Zileuton
Leukotriene receptor antagonist (LRA)	• Blocks CYS LT1 receptor of leukotriene • Drugs-(-Lukast-)
Corticosteroids	• Has broad mechanism of action • Other Name- Broad spectrum anti inflammatory drugs • DOC-Chronic or Persistent asthma
Monoclonal antibodies	• Blocks IgE antibody- Omalizumab • Blocks IL-5-Mepolizumab, Reslizumab, • Blocks IL-5 receptor- Benralizumab • Blocks IL-4-Dupilumab

Solution for Question 18:

Correct Answer: B) Dornase alfa

Explanation:

Dornase alfa is an enzyme that reduces mucus viscosity by cleaving extracellular DNA in the airway secretions.

Management of Cystic fibrosis:

Symptom	Treatment
Dry eyes, mouth (Xerophthalmia, xerostomia)	Pilocarpine Cevimeline (Option C)

Hard mucus	Mucolytics - N acetylcysteine Erdosteine Carbocysteine Dornase alpha (Degrades DNA) Hypertonic saline
Recurrent pneumonia (Pseudomonas)	Inhalational antibiotics - Tobramycin Amikacin Colistin Aztreonam
Chronic pancreatitis	Pancreatic enzyme replacement (Pancrezyme)
Disease-modifying therapy	Lumacaftor (Option D) Ivacaftor Elexacaftor

Pancrelipase (Option A) is a pancreatic enzyme replacement therapy to aid digestion.

Solution for Question 19:

Correct Answer: A) Codeine

Explanation:

The patient presents with symptoms suggestive of codeine-induced adverse effects, including nausea, vomiting, and constipation. Opioids commonly cause gastrointestinal adverse effects due to agonistic activity on their receptors in the gastrointestinal tract.

Dextromethorphan (Option B), noscapine (Option C) and promethazine (Option D) are less likely to cause gastrointestinal adverse effects compared to codeine.

Centrally acting Antitussives:

Class	Drugs	Mechanism	Adverse effects
Opioids	Morphine Hydromorphone Oxycodone Codeine Hydrocodone Oxycodone Pholcodine	Suppress cough centre	Codeine - Constipation Respiratory depression Drowsiness

Non Opioids	Dextromethorphan Noscapine	Dextromethorphan- Acts on glutamate receptor-NMDA antagonist. Noscapine-Sigma agonist	Dextromethorphan- Dizziness Nausea Drowsiness At high doses- Hallucinations Ataxia
Antihistaminic drugs	Cetirizine Promethazine	H1 receptor antagonists	Drowsiness Sedation Somnolence Fatigue Headache

Previous Year Questions

1. Which of the following statements is incorrect regarding Samter's triad?

- A. It is only triggered by aspirin.
 - B. Other NSAIDs with COX-1 blocking ability can also trigger symptoms.
 - C. It is characterized by asthma, nasal polyps, and aspirin sensitivity.
 - D. It involves overproduction of leukotrienes.
-

2. A patient presents with a cough and thick sputum. What should be the drug given to this patient?

- A. Bromhexine
 - B. Codeine
 - C. Dextromethorphan
 - D. Benzonatate
-

3. Which one of the options listed below is classified as a centrally acting opioid antitussive?

- A. Ambroxol
 - B. Diphenoxylate
 - C. Levopropoxyphene
 - D. Levorphanol
-

4. Which of the following agents is not used in the management of bronchial asthma?

- A. Phosphodiesterase inhibitors
 - B. Steroids
 - C. Beta-agonists
 - D. Cholinergic drugs
-

5. A group of people are traveling to the mountains, and a girl starts complaining of mountain sickness. What is the drug of choice?

- A. Promethazine
 - B. Acetazolamide
 - C. Dimenhydrinate
 - D. Thiazide
-

6. Inhaled Nitric oxide is approved for treatment of:

- A. Labile hypertension
 - B. Pulmonary hypertension
 - C. Treatment of resistant migraine
 - D. Resistant Parkinson's disease
-

7. What is the most common drawback of the given device?



- A. Bacterial contamination
 - B. Does not provide multiple doses
 - C. Drug deposition in the oropharynx
 - D. Not portable
-

8. If codeine is coming to the market in combination with dextromethorphan as an anti-tussive, this combination_____:

- A. Should not be used due to risk of addiction
 - B. Can be used as both have different mechanism of action
 - C. Should not be used as it is irrational
 - D. Can be used to treat cough effectively
-

9. Mechanism of action of zileuton:

- A. Cyclooxygenase inhibitor
- B. 5-lipoxygenase inhibitor
- C. 5-phosphodiesterase inhibitor
- D. Prevents mast cell degranulation

10. A male patient with chronic obstructive pulmonary disease (COPD) was prescribed theophylline. He noticed that his urine output had increased the following day. This action of the drug is mediated through which of the following receptors?

- A. Interleukin – 10
 - B. Histone deacetylase
 - C. Adenosine A1
 - D. Beta 2 adrenergic receptors
-

11. Which of the following is not a side effect of salbutamol?

- A. Hypoglycemia
 - B. Hypokalemia
 - C. Tremors
 - D. Tachycardia
-

12. Which one of the following medications utilized in the immediate treatment of asthma does not exert its effects by widening the bronchial airways?

- A. Salbutamol
 - B. Ipratropium
 - C. Deriphylline
 - D. Hydrocortisone
-

13. Which of the following drugs is not considered safe for a pregnant woman diagnosed with extensively drug-resistant TB?

- A. Amoxicillin
 - B. Clofazimine
 - C. Kanamycin
 - D. Linezolid
-

14. A patient who is a known case of chronic obstructive pulmonary disease who is on inhaled salmeterol therapy presents to the outpatient department with complaints that there is no improvement. Which of the following drugs can be added to his regimen?

- A. Theophylline
 - B. Azithromycin
 - C. Oral glucocorticoids
 - D. Ipratropium
-

15. Theophylline causes diuresis by:

- A. Adenosine agonism
 - B. PDE 4 inhibition
 - C. Beta 2 agonism
 - D. PDE 3 inhibition
-

16. In a patient with interstitial lung disease, which of the following antiarrhythmic medications should not be used due to contraindication?

- A. Amiodarone
 - B. Sotalol
 - C. Quinidine
 - D. Lignocaine
-

17. Which anti-asthma drug should be avoided with Erythromycin?

- A. Salbutamol
 - B. Theophylline
 - C. Terbutaline
 - D. Ipratropium
-

18. A patient comes to you with increased cough, increased breathlessness, and decreased exercise capacity. Chest X-ray shows pulmonary fibrosis. Which drug can be administered in the given condition?

- A. Bortezomib
 - B. Roflumilast
 - C. Imatinib
 - D. Pirfenidone
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	3
Question 4	4
Question 5	2
Question 6	2
Question 7	3
Question 8	3
Question 9	2
Question 10	3
Question 11	1
Question 12	4
Question 13	3
Question 14	4
Question 15	4
Question 16	1
Question 17	2
Question 18	4

Solution for Question 1:

Correct Option A - It is only triggered by aspirin (Incorrect Statement):

- This statement is incorrect because Samter's triad, also known as aspirin-exacerbated respiratory disease (AERD), can be triggered not only by aspirin but also by other nonsteroidal anti-inflammatory drugs (NSAIDs) that inhibit cyclooxygenase-1 (COX-1). These include ibuprofen, naproxen, and others that block COX-1. Therefore, patients with Samter's triad must avoid all COX-1 inhibiting NSAIDs, not just aspirin.

Incorrect Options:

Option B - Other NSAIDs with COX-1 blocking ability can also trigger symptoms (Correct Statement):

- This is a correct statement. In Samter's triad, the use of NSAIDs that inhibit COX-1 can trigger symptoms like asthma exacerbation, nasal congestion, and other respiratory issues due to their impact on the arachidonic acid pathway, leading to increased production of leukotrienes.

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Option C - It is characterized by asthma, nasal polyps, and aspirin sensitivity (Correct Statement):

- This is a correct statement. Samter's triad is defined by the presence of three key features: asthma, chronic rhinosinusitis with nasal polyps, and sensitivity to aspirin or other COX-1 inhibiting NSAIDs.

Option D - It involves overproduction of leukotrienes (Correct Statement):

- Explanation: This is a correct statement. In Samter's triad, the inhibition of COX-1 by aspirin or NSAIDs leads to the overproduction of leukotrienes, which are inflammatory mediators that contribute to the symptoms of asthma, nasal polyps, and other respiratory issues.

Solution for Question 2:

Correct Option A - Bromhexine:

- This drug is a mucolytic agent that helps to thin and loosen mucus in the airways, making it easier to cough up and clear from the respiratory tract. It is particularly useful for patients with thick sputum.

Incorrect Options:

- Options B, C, and D are cough suppressants and are not recommended in case of thick sputum.

Solution for Question 3:

Correct Choice: C

- Levopropoxyphene is a medication that was used as a centrally acting opioid antitussive medication to treat cough. It works by suppressing the cough reflex in the brain. Levopropoxyphene was withdrawn from the market in many countries due to its potential for abuse and serious side effects. It has high abuse potential and can lead to respiratory depression, cardiac arrhythmias, and other severe side effects.

Incorrect Choices:

Option A. Ambroxol: It is a medication that is used to treat respiratory diseases such as bronchitis, asthma, and COPD. Ambroxol works by breaking down and thinning mucus in the respiratory tract, making it easier to cough up. It is a mucolytic medication that is not related to opioids or antitussive medications.

Option B. Diphenoxylate: It is a medication that is used to treat diarrhea. Diphenoxylate works by slowing down the movement of the intestines, reducing the number of bowel movements and making the stool more solid. It is an opioid medication and can be abused, leading to addiction. Diphenoxylate is usually combined with atropine to discourage its abuse due to its unpleasant side effects.

Option D. Levorphanol: It is a potent opioid medication that is used to treat severe pain. Levorphanol works by binding to opioid receptors in the brain and spinal cord, reducing the perception of pain. It has a higher potency than morphine and is used when other opioid medications are not effective or have intolerable side effects. Levorphanol is not an antitussive medication.

Solution for Question 4:

Correct Option D - Cholinergic drugs:

- Cholinergic drugs are not typically used in the management of bronchial asthma. Cholinergic stimulation can lead to further bronchoconstriction which would be detrimental to a patient with asthma.

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Incorrect Options:

Option A - Phosphodiesterase inhibitors: Phosphodiesterase inhibitors, such as theophylline, were previously used in the management of bronchial asthma. However, their use has declined in recent years due to the availability of more effective and safer medications.

Phosphodiesterase inhibitors work by relaxing the smooth muscles in the airways, which can help improve airflow and relieve symptoms of asthma. While they may not be commonly used, they can still have a benefit with use in asthma.

Option B - Steroids: Steroids, such as inhaled corticosteroids, are a cornerstone of asthma management. They work by reducing inflammation in the airways, which helps prevent and control asthma symptoms. Inhaled corticosteroids are considered the most effective long-term control medication for asthma and are commonly prescribed to reduce airway inflammation and prevent asthma exacerbations.

Option C - Beta-agonists: Beta-agonists, such as short-acting beta-agonists (SABAs) like albuterol and long-acting beta-agonists (LABAs), are bronchodilators commonly used in the treatment of asthma. They work by relaxing the smooth muscles in the airways, leading to bronchodilation and improved airflow. SABAs are typically used as quick-relief medications for acute asthma symptoms, while LABAs are used as part of long-term control therapy in combination with inhaled corticosteroids.

Solution for Question 5:

Correct Option B - Acetazolamide:

- Acetazolamide is a carbonic anhydrase inhibitor which acts on PCT and leads to $\text{Na}^+/\text{HCO}_3^-$ excretion in urine.

- It is the drug of choice in mountain sickness and taken orally.
- In mountain sickness, at high altitudes due lesser oxygen and hyperventilation causes CO₂ washout and respiratory alkalosis (alkaline blood suppresses CNS).
- When Acetazolamide is taken it produces metabolic acidosis and reduces hyperventilation.

Incorrect Options:

- Options A, C, and D are incorrect.

Solution for Question 6:

Correct Option B - Pulmonary hypertension:

- Pulmonary hypertension is a type of high blood pressure that affects the pulmonary arteries, which are the blood vessels that carry blood from the heart to the lungs. In pulmonary hypertension, these arteries become narrowed or blocked, making it harder for blood to flow through them. This can lead to shortness of breath, fatigue, chest pain, and other symptoms.
- Inhaled nitric oxide is a gas that is used to treat pulmonary hypertension because it acts as a potent vasodilator, meaning it relaxes the smooth muscle in the walls of the pulmonary arteries and improves blood flow. Nitric oxide works by activating an enzyme called guanylate cyclase, which converts guanosine triphosphate (GTP) to cyclic guanosine monophosphate (cGMP). cGMP then causes relaxation of the smooth muscle cells in the pulmonary arteries, leading to vasodilation.
- Inhaled nitric oxide is administered via a face mask or tracheal tube to patients who are in the hospital or other medical setting. It is considered a relatively safe treatment, although it can have side effects such as headache, nausea, or vomiting. It is also relatively expensive and requires careful monitoring to ensure that the patient is receiving the correct dose.

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Incorrect Options:

Option A - Labile hypertension: Labile hypertension is a type of high blood pressure that is characterized by episodic or fluctuating increases in blood pressure. It is also sometimes called "white coat hypertension" because it may be triggered by the stress of being in a medical setting or interacting with healthcare providers. Treatment for labile hypertension may involve lifestyle changes such as reducing salt intake and increasing physical activity, as well as medications such as beta-blockers and diuretics.

Option C - Treatment of resistant migraine: Resistant migraine is a type of migraine headache that does not respond to standard treatments such as pain relievers or triptans. It can be a challenging condition to manage, and treatment may involve a combination of medications, lifestyle modifications, and alternative therapies such as acupuncture or biofeedback. Inhaled nitric oxide is not approved for the treatment of resistant migraine.

Option D - Resistant Parkinson's disease: Parkinson's disease is a neurodegenerative disorder that affects movement and coordination. It is caused by the loss of dopamine-producing neurons in the brain. Treatment for Parkinson's disease may involve medications such as levodopa or dopamine agonists, as well as lifestyle modifications such as exercise and physical therapy. Inhaled nitric oxide is not approved for the treatment of resistant Parkinson's disease

Solution for Question 7:

Correct Option C - Drug deposition in the oropharynx:

- Metered dose inhalers (MDIs) are commonly used for delivering medication directly into the respiratory system to treat conditions such as asthma and chronic obstructive pulmonary disease (COPD). However, one of the drawbacks of MDIs is that a significant portion of the medication can deposit in the oropharynx (back of the throat) instead of reaching the intended target in the lungs. This can reduce the effectiveness of the medication and may lead to inadequate symptom relief.

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Incorrect Options:

Option A - Bacterial contamination: While bacterial contamination is a concern in medical devices, it is not the most common drawback specifically associated with metered dose inhalers. Proper cleaning and maintenance can help mitigate the risk of bacterial contamination.

Option B - Does not provide multiple doses: Metered dose inhalers are designed to provide multiple doses of medication. They contain a canister that holds a certain number of doses, and each actuation of the inhaler delivers a measured dose of medication. However, it is important to keep track of the remaining doses to ensure that the inhaler is not empty when needed.

Option D - Not portable: Metered dose inhalers are compact and portable, making them convenient for use in various settings. They are designed to be easily carried and used by individuals on the go. Therefore, "not portable" is an incorrect option regarding the drawbacks of metered dose inhalers

Solution for Question 8:

Correct Option C - Should not be used as it is irrational:

- Codeine and dextromethorphan are both antitussive medications used to suppress cough. However, combining them in a single product is considered irrational for several reasons.
- First, both codeine and dextromethorphan have similar mechanisms of action, as they act on the cough reflex in the brain. They both suppress cough by affecting the central nervous system. Therefore, combining them does not provide any additional therapeutic benefit.
- Second, codeine is an opioid medication with potential for abuse and addiction. It carries a risk of respiratory depression, sedation, and other opioid-related side effects. Dextromethorphan, on the other hand, is a non-opioid medication that does not carry the same risk of addiction or respiratory depression. Combining codeine with dextromethorphan increases the risk of potential opioid-related adverse effects without providing any significant clinical advantage.

Codeine and dextromethorphan are both antitussive medications used to suppress cough. However, combining them in a single product is considered irrational for several reasons.

First, both codeine and dextromethorphan have similar mechanisms of action, as they act on the cough reflex in the brain. They both suppress cough by affecting the central nervous system. Therefore, combining them does not provide any additional therapeutic benefit.

Second, codeine is an opioid medication with potential for abuse and addiction. It carries a risk of respiratory depression, sedation, and other opioid-related side effects. Dextromethorphan, on the other hand, is a non-opioid medication that does not carry the same risk of addiction or respiratory depression. Combining codeine with dextromethorphan increases the risk of potential opioid-related adverse effects without providing any significant clinical advantage.

Incorrect Options:

Option A - Should not be used due to risk of addiction: This option suggests that the combination of codeine and dextromethorphan should not be used due to the risk of addiction. While it is true that codeine, being an opioid, carries a risk of addiction, dextromethorphan does not have the same addictive potential. Combining these drugs does not increase the risk of addiction.

Option B - Can be used as both have different mechanisms of action: This option suggests that the combination can be used because codeine and dextromethorphan have different mechanisms of action. However, this statement is incorrect as both medications act on the cough reflex in the brain and suppress cough in a similar manner. Combining them does not provide any additional therapeutic benefit.

Option D - Can be used to treat cough effectively: This option suggests that the combination can effectively treat cough. However, as mentioned earlier, combining codeine and dextromethorphan does not provide any added benefit compared to using a single-entity dextromethorphan preparation. The combination is considered irrational, and safer alternatives exist for treating cough effectively.

Solution for Question 9:

Correct Option B - 5-lipoxygenase inhibitor:

- Zileuton is a 5-lipoxygenase inhibitor. 5-lipoxygenase is an enzyme involved in the metabolism of arachidonic acid to leukotrienes. By inhibiting 5-lipoxygenase, zileuton helps reduce the production of leukotrienes, which are potent mediators of inflammation in

conditions like asthma. Zileuton is primarily used for the maintenance treatment of asthma and is particularly effective in reducing bronchoconstriction and airway inflammation.

Zileuton is a 5-lipoxygenase inhibitor. 5-lipoxygenase is an enzyme involved in the metabolism of arachidonic acid to leukotrienes. By inhibiting 5-lipoxygenase, zileuton helps reduce the production of leukotrienes, which are potent mediators of inflammation in conditions like asthma. Zileuton is primarily used for the maintenance treatment of asthma and is particularly effective in reducing bronchoconstriction and airway inflammation.

Incorrect Options:

Option A - Cyclooxygenase inhibitor: Cyclooxygenase (COX) inhibitors, such as nonsteroidal anti-inflammatory drugs (NSAIDs), work by inhibiting the activity of the COX enzymes involved in the production of prostaglandins and thromboxanes. These drugs are commonly used to reduce pain, inflammation, and fever. Zileuton does not inhibit cyclooxygenase.

Option C - 5-phosphodiesterase inhibitor: 5-phosphodiesterase inhibitors are a class of drugs that inhibit the activity of the enzyme phosphodiesterase, which breaks down cyclic nucleotides like cyclic adenosine monophosphate (cAMP) and cyclic guanosine monophosphate (cGMP). Drugs in this class, such as sildenafil (Viagra), are primarily used for conditions like erectile dysfunction and pulmonary arterial hypertension. Zileuton does not inhibit 5-phosphodiesterase.

Option D - Prevents mast cell degranulation: Mast cell degranulation refers to the release of inflammatory mediators, such as histamine and cytokines, from mast cells. While mast cell stabilizers, like cromolyn sodium, are used to prevent mast cell degranulation, zileuton does not directly prevent mast cell degranulation. It primarily acts as a 5-lipoxygenase inhibitor to reduce leukotriene production.

Solution for Question 10:

Correct Option C - Adenosine A1:

- The increased urine output observed in the patient with COPD after taking theophylline is likely due to the drug's action on adenosine receptors. Theophylline is a non-selective adenosine receptor antagonist, meaning it blocks the action of adenosine at all types of adenosine receptors, including A1, A2A, A2B, and A3 receptors.
- Adenosine A1 receptors are found in the kidneys and are involved in regulating water and electrolyte balance. By blocking adenosine A1 receptors, theophylline can lead to increased

urine production and diuresis. Therefore, the answer is Adenosine A1 receptors.

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Incorrect Options:

Option A - Interleukin-10: IL-10 is an anti-inflammatory cytokine that is involved in regulating immune responses and reducing inflammation. While IL-10 has many important roles in the body, it is not directly involved in regulating urine production or diuresis.

Option B - Histone deacetylase: Histone deacetylases (HDACs) are a family of enzymes that are involved in regulating gene expression by removing acetyl groups from histone proteins.

Option D - Beta 2 adrenergic receptors: Beta 2 adrenergic receptors are a subtype of adrenergic receptors that are primarily found in smooth muscle, such as in the airways of the lungs. They are not present in the kidneys.

Solution for Question 11:

Correct Option A - Hypoglycemia:

- Hypoglycemia, which refers to low blood sugar levels, is not a known side effect of salbutamol. Salbutamol is primarily used as a bronchodilator to relieve symptoms of asthma and other respiratory conditions. It works by relaxing the smooth muscles in the airways, which helps improve breathing. While salbutamol can affect glucose metabolism, it typically does not cause significant changes in blood sugar levels.

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Incorrect Options:

Option B - Hypokalemia: Hypokalemia refers to low levels of potassium in the blood. Salbutamol is a beta-2 adrenergic agonist, and its use can lead to an intracellular shift of potassium, potentially causing transient hypokalemia. This occurs due to the stimulation of beta-2 adrenergic receptors, which can activate the sodium-potassium pump and cause the movement of potassium into cells. However, significant hypokalemia is more commonly associated with higher doses or prolonged use of salbutamol. It can also be used to treat hyperkalemia.

Option C - Tremors: Tremors, or involuntary shaking of the hands, is a known side effect of salbutamol. It is a result of the drug's stimulation of beta-2 adrenergic receptors in the muscles. While tremors can be bothersome, they are generally temporary and resolve once the drug is metabolized and cleared from the body.

Option D - Tachycardia: Tachycardia, which refers to a rapid heart rate, is a common side effect of salbutamol. Salbutamol stimulates beta-2 adrenergic receptors in the heart, leading to an increased heart rate. This effect is desired in the context of bronchodilation but can be a concern in individuals with pre-existing heart conditions. It is important to monitor heart rate during salbutamol use, especially in individuals with cardiovascular issues.

Solution for Question 12:

Correct Option D - Hydrocortisone:

- Steroids produce annexin-1, which inhibits Phospholipase A2 (PLA₂). PLA₂ through a series of reactions is responsible for inflammation which causes bronchoconstriction during an asthmatic attack. Thus steroids have an anti-inflammatory effect. This effect causes reduction in bronchoconstriction, it does not actively cause bronchodilation.

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Incorrect Options:

Option A - Salbutamol: Salbutamol is a short acting beta agonist. It works only for 4-5 hours. So, the patient needs to take four times a day. Known to cause tremors, tachycardia, and tolerance(3T) after repeated use due to β -2 receptor desensitization.

Option B - Ipratropium: It is a Short-acting Muscarinic Antagonist mainly used in the management of COPD than bronchial asthma. It causes bronchodilation.

Option C - Deriphylline: Deriphylline derived from methylxanthine (a purine derivative) .It is a smooth muscle relaxant and bronchodilator. Aids in bronchial dilation, diuretic, cardiac and CNS stimulant.

Solution for Question 13:

Correct Option C - Kanamycin:

- Kanamycin is an aminoglycoside. Aminoglycosides are contraindicated in pregnancy due to ototoxicity in the fetus, so kanamycin cannot be used in the above case.

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Other antibiotics that are unsafe in pregnancy include:

- Tetracycline
- Chloramphenicol
- Metronidazole in first trimester
- Fluoroquinolones

Tetracycline

Chloramphenicol

Metronidazole in first trimester

Fluoroquinolones

Incorrect Options:

Option A - Amoxicillin: Amoxicillin belongs to the penicillin group of antibiotics and is safe in pregnancy.

Option B - Clofazimine: Clofazimine is safe in pregnancy.

Option D - Linezolid: Linezolid is safe in pregnancy.

Solution for Question 14:

Correct Option D - Ipratropium:

- In the above case, the patient is on a bronchodilator salmeterol. The next best step is to add another bronchodilator like ipratropium which is a muscarinic antagonist.

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Incorrect Options:

Option A - Theophylline: Theophylline produces improvements in expiratory flow rates but is not the next best step. If LABA is not sufficient patient should have a LAMA added to their therapy.

Option B - Azithromycin: the patient has no fever or any symptoms that will prompt the use of an antibiotic like azithromycin.

Option C - Oral glucocorticoids: Oral glucocorticoids are not the next best step.

Solution for Question 15:

Correct Option D - PDE 3 inhibition:

- Theophylline acts as a non-selective phosphodiesterase (PDE) inhibitor. It inhibits the enzyme PDE 3, which is primarily found in cardiac and smooth muscle cells. PDE 3 inhibition leads to increased levels of cyclic adenosine monophosphate (cAMP) within these cells. Elevated cAMP levels then trigger various downstream effects, including smooth muscle relaxation and increased myocardial contractility.
- In the context of diuresis, PDE 3 inhibition by theophylline leads to increased cAMP levels in the renal tubules, specifically in the distal convoluted tubules and collecting ducts. This increase in cAMP promotes water and sodium excretion, resulting in diuresis.

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Incorrect Options:

Option A - Adenosine agonism: Theophylline does not agonize adenosine receptors. Instead, it antagonizes adenosine receptors, which can lead to various effects, such as bronchodilation and central nervous system stimulation as well as diuresis.

Option B - PDE 4 inhibition: While theophylline does inhibit phosphodiesterase (PDE), it is not primarily through PDE 4 inhibition. Theophylline has a non-selective inhibitory effect on PDE enzymes, which leads to increased cAMP levels in cells. However, its diuretic effects are not primarily mediated by PDE 4 inhibition.

Option C - Beta 2 agonism: Theophylline does not act as a beta 2 agonist. Beta 2 agonists, such as albuterol, are commonly used for their bronchodilatory effects and are distinct from the mechanisms of action of theophylline.

Solution for Question 16:

Correct Option A - Amiodarone:

- Amiodarone is contraindicated in a patient with interstitial lung disease.
- Amiodarone is an antiarrhythmic medication that is effective in treating various types of cardiac arrhythmias. However, it has been associated with the development or exacerbation of interstitial lung disease, a condition characterized by inflammation and scarring of lung tissue. Interstitial lung disease can cause symptoms such as shortness of breath, cough, and decreased lung function.
- Due to the risk of pulmonary toxicity, including interstitial lung disease, amiodarone is generally contraindicated in patients with pre-existing interstitial lung disease. If interstitial lung disease develops or worsens during amiodarone therapy, the medication should be discontinued.
- Patients on amiodarone should undergo screening as well as regular follow up PFT's.

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Patients on amiodarone should undergo screening as well as regular follow up PFT's.

Incorrect Options:

Option B, C & D - Sotalol, Quinidine and Lignocaine:

- Sotalol, quinidine, and lignocaine are not specifically contraindicated in patients with interstitial lung disease, although caution may still be required and individual patient factors should be considered when prescribing these medications.

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Solution for Question 17:

Correct Option B - Theophylline:

- Among the options provided, the anti-asthma drug that is typically avoided with erythromycin is Theophylline. Erythromycin is known to inhibit the metabolism of theophylline, leading to an increase in its blood levels. This can result in theophylline toxicity, which may manifest as symptoms such as nausea, vomiting, tremors, rapid heartbeat, and even seizures.
- Theophylline is a bronchodilator used in the management of asthma and other respiratory conditions. It works by relaxing the smooth muscles in the airways, thereby improving breathing. As it is metabolised by cytochrome P450 and erythromycin is a Microsomal enzyme (cytochrome P450) inhibitors causing increased theophylline level..

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Incorrect Options:

Option A - Salbutamol: There is no significant interaction between salbutamol (also known as albuterol) and erythromycin. Salbutamol is a short-acting beta-agonist bronchodilator used for the relief of acute asthma symptoms.

Option C - Terbutaline: Like salbutamol, terbutaline is a beta-agonist bronchodilator. It does not have a documented interaction with erythromycin that requires avoidance.

Option D - Ipratropium: Ipratropium is an anticholinergic bronchodilator used for the treatment of asthma and chronic obstructive pulmonary disease (COPD). It does not have a significant interaction with erythromycin.

Solution for Question 18:

Correct Option D - Pirfenidone:

- Pirfenidone is the correct answer, as it is an antifibrotic agent approved for treating idiopathic pulmonary fibrosis (IPF)

Pirfenidone is the correct answer, as it is an antifibrotic agent approved for treating idiopathic pulmonary fibrosis (IPF)

Incorrect Options:

Option A - Bortezomib: Bortezomib is a proteasome inhibitor for treating multiple myeloma and mantle cell lymphoma.

Option B - Roflumilast: Roflumilast is a phosphodiesterase-4 inhibitor used to treat chronic obstructive pulmonary disease (COPD).

Option C - Imatinib: Imatinib is a tyrosine kinase inhibitor used to treat chronic myeloid leukemia (CML) and gastrointestinal stromal tumors (GISTs).

Drugs in Acid Peptic Disease

1. A 58-year-old male with a history of chronic kidney disease presents with worsening epigastric pain and frequent heartburn. Given his renal impairment, which class of medications for Peptic ulcer disease should be avoided, and which specific agent would provide the most optimal relief without exacerbating his comorbid conditions as well as for long-term use?

- A. Avoid proton pump inhibitors; use ranitidine.
 - B. Avoid antacids containing magnesium; use omeprazole.
 - C. Avoid H₂ receptor antagonists; use sucralfate.
 - D. Avoid proton pump inhibitors; use calcium carbonate.
-

2. A 30-year-old primigravida presents with complaints of recurrent heartburn and epigastric discomfort. She has a history of closed-angle glaucoma, which is currently well-managed with medication. Considering her medical history, which combination of acid-suppressant drugs should be avoided?

- A. PPIs only
 - B. H₂ blockers only
 - C. Anticholinergics only
 - D. Anticholinergics and Prostaglandin analogues
-

3. A 42-year-old man presents with heartburn and epigastric pain, diagnosed with peptic ulcer disease. The doctor opts for a PPI, choosing between Ilaprazole, Lansoprazole, Rabeprazole and Omeprazole. Which of these medications requires the lowest dosage to achieve equivalent therapeutic effects?

- A. Ilaprazole

- B. Lansoprazole
 - C. Rabeprazole
 - D. Omeprazole
-

4. A 68-year-old female patient with a history of myocardial infarction is currently on dual antiplatelet therapy, including clopidogrel. She visited a government hospital where an intern prescribed her omeprazole to be taken in the morning, not knowing that omeprazole is contraindicated with clopidogrel. Which Cytochrome P450 enzyme is implicated with respect to the reduced effectiveness of clopidogrel when taking omeprazole?

- A. CYP2C9
 - B. CYP2C19
 - C. CYP3A4
 - D. CYP2D6
-

5. Which of the following is incorrect with respect to the adverse effects of chronic use of proton pump inhibitors (PPI)?

- A. Use of PPI is associated with reduced absorption of B12, Mg, and Iron.
 - B. Long-term use of PPI is associated with an increased risk of bone fractures.
 - C. Increased risk of pseudomembranous enterocolitis is associated with long-term use of PPI.
 - D. Reduced serum gastrin levels can occur due to long-term use of PPI, causing ECL hyperplasia and atrophic gastritis.
-

6. A 38-year-old male with a history of frequent episodes of acidity was prescribed an H₂ receptor antagonist. During a follow-up visit, he reports loss of libido and erectile dysfunction. Laboratory tests reveal elevated prolactin levels. Which of the following is the most likely cause of these symptoms?

- A. Ranitidine
 - B. Famotidine
 - C. Nizatidine
 - D. Cimetidine
-

7. A 32-year-old woman seeks advice on medication to prevent gastric ulcers. She has a history of rheumatoid arthritis and has been on long-term NSAID therapy. Her medical history reveals no contraindications to the use of prostaglandin analogues. The physician considers prescribing Misoprostol for ulcer prevention. Which of the following is the most appropriate counseling point regarding the use of Misoprostol in this patient?

- A. Misoprostol should be taken on an empty stomach to enhance absorption.
 - B. Misoprostol is contraindicated in women of childbearing potential unless they are using effective contraception.
 - C. Misoprostol can be safely used in patients with a history of allergy to prostaglandins.
 - D. Misoprostol works by inhibiting the H₂ receptors in the stomach lining to reduce acid secretion.
-

8. Which of the following statements is true regarding the use of Bismuth compounds in ulcer treatment?

- A. They inhibit the proton pump in gastric parietal cells.
- B. They form a protective layer over the ulcer base, shielding it from acid and pepsin.
- C. They primarily act as antacids.
- D. They are H₂ receptor antagonists.

9. Which of the following clinical features is characteristic of Milk-Alkali Syndrome?

- A. Hyperkalemia
 - B. Metabolic acidosis
 - C. Hypercalcemia
 - D. Hypophosphatemia
-

10. A 68 y/o female presents to the emergency department with severe abdominal pain, bloating and difficulty breathing. She has a H/O CKD and GERD. The patient admits to using sodium bicarbonate antacids frequently to manage her GERD symptoms. On examination, her abdomen is distended and she exhibits signs of peritonitis. An abdominal X-ray reveals free air under the diaphragm. Which of the following is the most likely cause of the patient's symptoms?

- A. Hyperkalemia from excessive sodium bicarbonate use.
 - B. Excessive gas production from sodium bicarbonate neutralising stomach acid.
 - C. Hypocalcemia-induced muscle spasms causing bowel rupture.
 - D. Hypermagnesemia from concomitant use of magnesium-containing antacids.
-

11. A 25 y/o female, a competitive gymnast presents with complaints of heartburn. She has a history of regular consumption of dairy-based calcium supplements for the past 5 years to help with her athletic lifestyle. She insists the doctor prescribe her calcium-based antacids. The doctor suggests against it and explains to her the reason. Why did the doctor suggest against it?

- A. Risk of developing diarrhea
- B. Risk of developing osteoporosis

- C. Risk of developing milk-alkali syndrome
 - D. Risk of developing hypertension and edema
-

12. A 72-year-old woman with a history of CHF (congestive heart failure) presents to the clinic complaining of heartburn, chest pain, and regurgitation of food after meals for the past week. Her CHF symptoms are currently well-controlled with medication. On examination, there is no evidence of angina pectoris. Which medication would be MOST APPROPRIATE for this patient's symptoms?

- A. Magaldrate
 - B. Sodium Bicarbonate
 - C. Aspirin
 - D. Metoprolol
-

13. A 5y/o girl is brought to the emergency department by her parents with complaints of persistent vomiting, confusion, and lethargy. The parents mentioned that she had a viral illness two weeks prior, during which she was given a combination of medications to alleviate her symptoms. On examination, the child is found to be drowsy, with signs of hepatomegaly. Laboratory tests reveal elevated liver enzymes and ammonia levels. Which of the following medications is most likely associated with the child's current presentation?

- A. Paracetamol
 - B. Amoxicillin
 - C. Bismuth subsalicylate
 - D. Ranitidine
-

14. A 45-year-old male patient presents with symptoms of peptic ulcer disease and is diagnosed with H. pylori infection. He is allergic to penicillin. As part of his initial treatment, you decide to start him on triple therapy for H. pylori eradication. Which of the following drugs are used in Triple therapy in this patient?

- A. PPI + Clarithromycin + Metronidazole for 14 days
 - B. PPI + Clarithromycin + Amoxicillin for 14 days
 - C. PPI + Levofloxacin + Clarithromycin for 14 days
 - D. PPI + Clarithromycin + Metronidazole for 7 days
-

15. Which of the following is the most preferred treatment regimen for H pylori in India?

- A. Triple therapy
 - B. Quadruple therapy
 - C. Sequential therapy
 - D. Hybrid therapy
-

16. A 50-year-old patient with a history of gastroesophageal reflux disease (GERD) reports persistent symptoms despite lifestyle modifications and dietary changes. The physician decides to add alginic acid to the patient's treatment regimen. What is the mechanism of action of alginic acid in the treatment of GERD?

- A. Makes a neutral covering over HCL
 - B. Bursts bubbles in the gastrointestinal tract
 - C. Increases prostaglandin synthesis
 - D. Polymerizes with proteins to form a protective covering
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	1
Question 4	2
Question 5	4
Question 6	4
Question 7	2
Question 8	2
Question 9	3
Question 10	2
Question 11	3
Question 12	1
Question 13	3
Question 14	1
Question 15	2
Question 16	1

Solution for Question 1:

Correct Answer: B) Avoid antacids containing magnesium; use omeprazole.

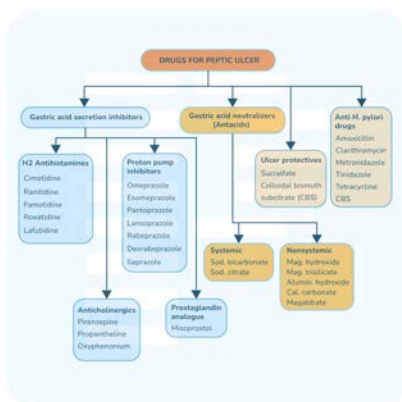
Explanation:

Antacids containing magnesium (e.g. Magnesium hydroxide and Magaldrate) should be avoided in patients with renal impairment because they can lead to hypermagnesemia. Omeprazole, a proton pump inhibitor, is generally safe for use in patients with chronic kidney disease and is effective for treating acid peptic disease. This is the most appropriate option.

Avoid proton pump inhibitors; use ranitidine (Option A) :Proton pump inhibitors (PPIs) are the DOC for acid peptic disease and are not contraindicated in renal impairment. Ranitidine, an H2 receptor antagonist, was commonly used for acid-related conditions but has been withdrawn from the market due to carcinogenic risk.

Avoid H2 receptor antagonists; use sucralfate (Option C): H2 receptor antagonists can generally be used in patients with renal impairment. Sucralfate is a mucosal protectant that can provide symptomatic relief by coating ulcers and protecting them from stomach acid. However, it is less effective than PPIs for acid suppression.

Avoid proton pump inhibitors; use calcium carbonate (Option D): While calcium carbonate (a type of antacid) can provide symptomatic relief, it may lead to hypercalcemia and potential renal complications. On long term use, Calcium ions diffuse into the gastric mucosa and increase HCl production directly by parietal cells as well as by releasing gastrin. Acid rebound occurs. Avoiding PPIs is unnecessary in renal impairment.



Gastric acid secretion inhibitors:

- M1 blocker-Pirenzepine, Telenzepine
- K-CAB blocker- Vonoprazan

Solution for Question 2:

Correct answer: D) Anticholinergics and Prostaglandin analogues

Explanation:

- Anticholinergics can cause pupillary dilation, leading to an increase in intraocular pressure, which can precipitate an acute angle closure glaucoma attack.
- Prostaglandin analogues can cause uterine contractions and are contraindicated in pregnancy due to the risk of abortion or premature labor.

Classification of acid suppressant drugs with mechanism of action:

Acid Suppressant Drugs	Mechanism of Action
Proton Pump Inhibitors (PPIs) (Option A)	Irreversibly inhibit the H ⁺ /K ⁺ ATPase pump in the gastric parietal cells, leading to sustained suppression of gastric acid secretion.
H ₂ Blockers (Option B)	Competitively antagonizes histamine H ₂ receptors on the surface of gastric parietal cells, thereby diminishing the stimulatory effect of histamine on the production of gastric acid.
Prostaglandin Analogues	Stimulate prostaglandin E ₂ receptors on gastric epithelial cells, promoting the secretion of mucus and bicarbonate while inhibiting gastric acid secretion.
Anticholinergics (Option C)	Act as competitive antagonists at M ₃ muscarinic receptors in the stomach, inhibiting the stimulatory effects of acetylcholine on gastric acid secretion and motility.

Solution for Question 3:

Correct Answer: A) Ilaprazole

Explanation:

- Potency of PPIs - Dex Rabeprazole > Ilaprazole > Lansoprazole > Rabeprazole > Omeprazole > Pantoprazole
- All PPIs have equal efficacy.

Route of Administration:

- Oral: All PPIs
- Injectable: IV/IM (Pantoprazole, Es-omeprazole, Rabeprazole)

Properties:

- Acid production reduction persists for 2 to 4 days even after stopping the drug and resumes later.
- Duration of action is typically 2 to 4 days.
- Omeprazole is the fastest acting PPI.
- PPIs are acid labile; Enteric coating can protect from acid degradation.
- They should be taken on empty stomach when only ~10% of proton pumps are active, followed 1 hour later by a meal to activate the H⁺/K⁺/ATPase at a time when plasma concentration of the PPI is maximal.

Metabolism and Interactions:

- All PPIs are metabolised by the CYP2C19 enzyme.
- Omeprazole is the maximum CYP inhibitor.
- Rabeprazole has the lowest CYP inhibition, followed by Pantoprazole.
- Omeprazole should never be given with Clopidogrel due to potential interactions. Omeprazole inhibits the enzyme CYP2C19, thus preventing the conversion of Clopidogrel to its active metabolite.

Solution for Question 4:

Correct Answer: B) CYP2C19

Explanation-

- Omeprazole inhibits the enzyme CYP2C19, which is crucial for converting clopidogrel into its active metabolite. This interaction reduces the effectiveness of clopidogrel, potentially decreasing its ability to prevent blood clots in patients who are at risk, such as those with a history of myocardial infarction.
- If a patient requires both clopidogrel and a PPI, it is recommended to use pantoprazole or rabeprazole to reduce the risk of interaction.
- Omeprazole reduces the metabolism of antidepressants and antipsychotics, increasing the chances of toxicity. Alongside, Omeprazole also blocks the activation of Tamoxifen.
- Omeprazole also inhibits other CYP enzymes like CYP2C9, CYP2D6 & CYP3A4. However, these enzymes are not involved in metabolic activation of Clopidogrel. (Option A, C and D

ruled out)

Solution for Question 5:

Correct answer: D) Reduced serum gastrin levels can occur due to long-term use of PPI, causing ECL hyperplasia and atrophic gastritis.

Explanation:

- Chronic use of PPIs can lead to hypergastrinemia, which can further result in: ECL hyperplasia Fundic gland polyposis Atrophic gastritis
- ECL hyperplasia
- Fundic gland polyposis
- Atrophic gastritis
- ECL hyperplasia
- Fundic gland polyposis
- Atrophic gastritis
- Hypergastrinemia is detected by measuring serum gastrin levels. This condition may predispose individuals to rebound hypersecretion of gastric acid upon discontinuation of therapy and may also promote the growth of gastrointestinal tumours.

Other side effects of PPI use include:

- Gastrointestinal upset (nausea, constipation, diarrhoea).
 - Increased microbial growth leads to a higher risk of hospital-acquired pneumonia, community-acquired *Clostridium difficile* infection, spontaneous bacterial peritonitis and pseudomembranous enterocolitis (Option C ruled out).
 - Increased risk of bone fractures due to hypocalcemia (reduced absorption of calcium) (Option B ruled out).
-

Solution for Question 6:

Correct Answer: D) Cimetidine

Explanation-

Cimetidine, an H₂ receptor antagonist, can cause anti-androgenic effects, leading to loss of libido and erectile dysfunction. It can also increase prolactin levels, contributing to these symptoms. These side effects are more commonly associated with cimetidine than with other H₂ receptor antagonists like Famotidine.

Side effects associated with H₂ Blockers:

- Gastrointestinal: Nausea, vomiting, diarrhoea and constipation.
- Neurological: Headache and dizziness.
- Endocrine and Reproductive: Gynecomastia, erectile dysfunction, loss of libido and elevated prolactin levels.
- Hematologic: Thrombocytopenia and agranulocytosis.
- Hepatic: Elevated Liver Enzymes and hepatotoxicity.
- Renal: Interstitial Nephritis

Drugs causing gynecomastia:

- D: Dopamine antagonists (e.g., antipsychotics like risperidone, and metoclopramide)
- I: Isoniazid (antitubercular drug)
- S: Spironolactone (a potassium-sparing diuretic)
- C: Cimetidine (an H₂ receptor antagonist)
- O: Oestrogens (including some medications like hormone replacement therapy)

Solution for Question 7:

Correct Answer: B) Misoprostol is contraindicated in women of childbearing potential unless they are using effective contraception.

Explanation:

Misoprostol is contraindicated for reducing the risk of NSAID-induced ulcers in women of childbearing potential unless the patient is at high risk of complications from gastric ulcers associated with the use of the NSAID.

It is also completely contraindicated during pregnancy because it can increase uterine contractility and abort a pregnancy.

Misoprostol is a longer-acting synthetic PGE1 derivative that acts on EP receptors (EP2, EP3, and EP4) to increase mucin and bicarbonate production and decrease HCl production. (Option D)

- It has a shorter duration of action.
- Taking misoprostol with food may potentially reduce the incidence of systemic side effects by lowering the early high peak plasma concentrations of misoprostol acid. (Option A)
- Recommended dose: 200 µg four times daily.
- Primary indication: prevention and treatment of NSAID-associated gastric ulcer and bleeding. But PPI is the DOC.
- Adverse effects seen with misoprostol use: Diarrhea, abdominal cramps, uterine bleeding, abortion.
- Contraindicated in Pregnancy, Inflammatory bowel disease (IBD), previous episodes of allergy to prostaglandins (Option C).

Solution for Question 8:

Correct answer: B) They form a protective layer over the ulcer base, shielding it from acid and pepsin.

Explanation:

Colloidal bismuth compounds act as ulcer protective agents, which at acidic pH (<5) precipitate mucus glycoprotein, forming a protective layer over the ulcer base.

Ulcer Protective Agents:

	Bismuth	Sucralfate
Properties	Water-soluble, but precipitates when pH <5.	Basic aluminium salt of sulfated sucrose. Polymerises at pH <4, forming a sticky gel-like consistency.
Mechanism of Action	Increases production of gastric mucosal prostaglandin E, mucus, and bicarbonate. Precipitate mucus glycoproteins and coat the ulcer base. Detach and inhibit <i>Helicobacter pylori</i> directly.	Precipitates surface proteins at the ulcer base, creating a physical barrier against acid, pepsin, and bile. Does not neutralise acid but delays gastric emptying. Potentially augments gastric mucosal prostaglandin synthesis for

		additional protection.
Dosage and Administration	120 mg taken half an hour before the three major meals and at bedtime for 4-8 weeks.	1 gram taken on an empty stomach one hour before the three major meals and at bedtime for 4-8 weeks.
Side Effects	Diarrhoea Headachy Dizziness Blackening of the tongue, dentures, and stools.	Constipation Hypophosphatemia Dry mouth Nausea
Indications	Gastritis and non-ulcer dyspepsia associated with <i>H. pylori</i> . Occasionally used as part of a triple-drug regimen.	Bile reflux, gastritis and prophylaxis of stress ulcers.

Solution for Question 9:

Correct Answer: C) Hypercalcemia

Explanation:

- Milk-Alkali Syndrome occurs when milk was prescribed with absorbable alkali like CaCO_3 for peptic ulcer in the past.
- Nowadays, it may be seen when patients self-medicate with OTC Calcium supplements.
- Symptoms: Headache, anorexia, weakness, abdominal discomfort.
- It is characterised by a triad of hypercalcemia, metabolic alkalosis, and acute kidney injury.

Hyperkalemia, Metabolic acidosis, and Hypophosphatemia are not seen in Milk Alkali Syndrome. (Options A,B, and D)

Therefore, the correct option is hypercalcemia, reflecting the primary biochemical disturbance seen in this syndrome.

Solution for Question 10:

Correct answer: B) Excessive gas production from sodium bicarbonate neutralising stomach acid.

Explanation:

The patient's symptoms, clinical signs of peritonitis and X-ray findings suggest bowel perforation in the patient.

- Sodium bicarbonate antacids neutralise stomach acid, producing CO₂ gas as a byproduct.
- Excessive gas can lead to significant gastric and intestinal distension in patients who use these antacids frequently or in large amounts.
- This increased intraluminal pressure can cause discomfort, and bloating, and, in severe cases, lead to perforation of the bowel.
- This is particularly risky in individuals with predisposing conditions like CKD, where the handling of excess sodium and bicarbonate might be compromised, exacerbating the problem.

Hyperkalemia is not caused by sodium bicarbonate use (Option A ruled out).

Hypocalcemia-induced muscle spasms causing bowel rupture (Option C) is a rare but severe complication associated with hypoparathyroidism and is not associated with Sodium bicarbonate use.

Hypermagnesemia from concomitant use of magnesium-containing antacids (Option D) is not related to sodium bicarbonate but to magnesium-containing antacids.

Solution for Question 11:

Correct answer: C) Risk of developing milk-alkali syndrome

Explanation:

- The patient is already on dairy-based calcium supplements for a prolonged period. Adding calcium-based antacids would put her at risk of developing Milk Alkali syndrome.
- Milk Alkali Syndrome is a recognized risk with excessive intake of dairy and absorbable alkali (Like CaCO₃- a calcium-based antacid), leading to hypercalcemia, alkalosis, and renal insufficiency.
- Symptoms can include nausea, vomiting, confusion, and in severe cases, kidney failure.

Side effects of other antacids

Calcium-based antacids typically cause constipation rather than diarrhea. Diarrhea is more commonly associated with magnesium-based antacids. (Option A ruled out)

Calcium is essential for bone health, and calcium-based antacids could actually help prevent osteoporosis by supplementing calcium. Osteoporosis is usually a side effect associated with aluminum-based antacids. (Option B ruled out)

Calcium-based antacids are not typically associated with causing hypertension or edema. These conditions are more commonly related to sodium-based antacids. Hence should not be given to patients with CHF. (Option D ruled out)

Solution for Question 12:

Correct answer: A) Magaldrate

Explanation:

The patient's symptoms of heartburn, chest pain, and regurgitation of food after meals are suggestive of GERD.

- Magaldrate is a nonsystemic antacid and is a combination of two antacids: magnesium hydroxide and aluminum hydroxide and can be safely used in patients with heart disease.
- Magnesium hydroxide reacts quickly with stomach acid to neutralize it, providing rapid relief from heartburn. However, it causes diarrhea as a side effect.
- Aluminum hydroxide works slower but provides longer-lasting relief. It also coats the esophagus, protecting it from further irritation by stomach acid. However, it causes constipation as a side effect.
- Hence, Magaldrate is a fixed combination of the two and works by counteracting each other's adverse effects and hence giving optimum results.

Sodium Bicarbonate (Option B) is a high sodium antacid, which can worsen fluid retention in CHF patients causing hypertension and edema. It is usually contraindicated in patients with heart disease.

Aspirin (NSAID) (Option C) can worsen heartburn symptoms and should be avoided.

Metoprolol (Option D) is a beta-blocker used to treat high blood pressure and angina pectoris, but it is not typically used for GERD.

Solution for Question 13:

Correct Answer: C) Bismuth Subsalicylate

Explanation:

- The patient's symptoms, clinical presentation, and lab values suggest she has Reye's Syndrome.
- Reye's syndrome is a rare but serious condition that causes swelling in the liver and brain. It most commonly affects children and teenagers recovering from a viral infection, such as influenza or chickenpox. The use of salicylates during such viral infections has been strongly associated with the development of Reye's syndrome.
- Bismuth subsalicylate, found in some over-the-counter medications for stomach upset (such as Pepto-Bismol), contains salicylate, which can lead to Reye's syndrome when used in children recovering from viral infections.
- Other commonly used salicylates include: Aspirin: an NSAID Methyl Salicylate : found in topical relief creams and ointments
- Aspirin: an NSAID
- Methyl Salicylate : found in topical relief creams and ointments
- Other side effects of Bismuth include: Black tongue and black faeces- when bismuth salicylate comes in contact with sulfur compounds in mouth and GIT, it forms Bismuth sulfide which is a black compound. Bone: Osteoporosis
- Black tongue and black faeces- when bismuth salicylate comes in contact with sulfur compounds in mouth and GIT, it forms Bismuth sulfide which is a black compound.
- Bone: Osteoporosis
- Aspirin: an NSAID
- Methyl Salicylate : found in topical relief creams and ointments
- Black tongue and black faeces- when bismuth salicylate comes in contact with sulfur compounds in mouth and GIT, it forms Bismuth sulfide which is a black compound.
- Bone: Osteoporosis

Paracetamol (Option A) is widely used as an analgesic and antipyretic. While an overdose of it can lead to acute liver failure, it is not associated with Reye's syndrome.

Amoxicillin (Option B) is an antibiotic commonly used for URIs and is not associated with Reye's syndrome.

Ranitidine (Option D) is an H₂ receptor antagonist used to reduce stomach acid production. It is not linked to Reye's syndrome.

Solution for Question 14:

Correct Answer: A) PPI + Clarithromycin + Metronidazole for 14 days

Explanation

This patient has been diagnosed with H. Pylori and needs to be started on triple therapy. He is also allergic to penicillin so Amoxicillin is avoided, hence he has to be started on PPI + Clarithromycin + Metronidazole for 14 days.

Antibiotics used against H. Pylori:	Generation
Clarithromycin Amoxicillin Metronidazole	1st generation
Tetracycline Levofloxacin	2nd generation
Rifabutin Furazolidone	3rd generation

Guidelines for treatment of H. Pylori:

Anti-H. pylori regimens:	Drugs used	No. of days:
Triple therapy: PCA (first line therapy used)	PPI + Clarithromycin + Amoxicillin (Option B)	BD for 14 days
Triple therapy: PCM (in case of penicillin allergy or resistance)	PPI + Clarithromycin + Metronidazole	BD for 14 days
Levofloxacin-based triple therapy:	PPI + Levofloxacin + Clarithromycin (PAL) (Option C)	BD for 14 days

PPI + Tetracycline + Amoxicillin for 14 days (Option D) are not used in combination, instead PPI + Amoxicillin followed by PPI + Metronidazole + Levofloxacin for 14 days (PA → PML) is used as a new concomitant therapy.

Solution for Question 15:

Correct answer: B) Quadruple therapy

Explanation:

- Quadruple therapy (tetracycline, metronidazole, bismuth, PPI) is effective against metronidazole resistance with a success rate of 90%+ for 10-14 days in India.
- Due to the risk of development of Clarithromycin resistance in India.

Anti-H.pylori regimens:	Drugs used	No. of days:
Triple therapy: PCM (in case of penicillin allergy or resistance)	PPI + Clarithromycin + Metronidazole	BD for 14 days
Triple therapy: PCA	PPI + Clarithromycin + Amoxicillin	BD for 14 days
Quadruple therapy (In clarithromycin resistance)	PPI + Bismuth salts + Metronidazole + Tetracycline	BD for 14 days
Sequential therapy	PPI + Amoxicillin (for 7 days) → PPI + Clarithromycin + Metronidazole (for 7 days)	BD For 14 days
Hybrid therapy	PPI + Amoxicillin (for 7 days) → PPI + Clarithromycin + Amoxicillin + Metronidazole (for 7 days)	BD For 14 days

Solution for Question 16:

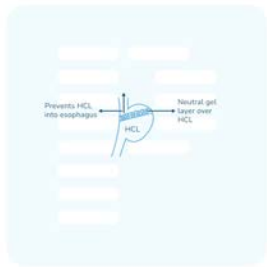
Correct answer: A) Makes a neutral covering over HCL

Explanation:

This patient with a history of GERD has failed lifestyle modifications and dietary changes, so the physician decides to prescribe Alginic acid.

Mechanism of action of Alginic acid in GERD:

- Alginates treat GERD by forming a mechanical barrier.
- They create a gel in the presence of gastric acid.
- This gel forms a raft that localizes to the acid pocket in the proximal stomach.
- This prevents the acid from getting displaced to the esophagus
- Hence helping patients with symptoms of GERD



Bursts bubbles in the gastrointestinal tract (Option B): This is the mechanism of action of surfactant-like molecules e.g; Simethicone and dimethicone

Increases prostaglandin synthesis (Option C): This is the mechanism of action of Bismuth salts.

Polymerizes with proteins to form a protective covering (Option D): This is the mechanism of action of Sucralfate which is ulcer protective.

Laxatives and Anti-Diarrheal Drugs

1. A 45-year-old male patient is scheduled for a colonoscopy. Bowel cleaning is required as part of the preparation. Which of the following laxatives would be most appropriate for this purpose?

- A. Bisacodyl
 - B. Lactulose
 - C. Polyethylene glycol (PEG)
 - D. Docusate sodium
-

2. Which of the following statements is false regarding the mechanism of action of bulk laxatives?

- A. Bulk laxatives mainly act by fermenting laxatives by gut bacteria, which produces short-chain fatty acids and increases bacterial mass in the stool.
 - B. Short-chain fatty acids have a prokinetic effect, and bacterial mass increases stool volume.
 - C. The more fermentable the laxative, the more the stool bulk and the faster the transit.
 - D. Dietary fibres such as lignin resist fermentation, and fibres such as pectins are readily fermented.
-

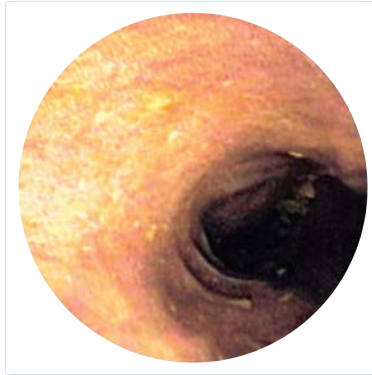
3. A 70-year-old male presents with severe abdominal distension and discomfort. He reports that he has not had a bowel movement for over a week and has been experiencing nausea and vomiting. Physical examination reveals a significantly distended abdomen with absent bowel sounds. An abdominal X-ray shows a megacolon. Which of the following treatments is contraindicated in this patient due to the condition of atonic colon?

- A. Intravenous fluids for rehydration & parenteral nutrition
 - B. Nasogastric decompression to relieve distension
 - C. Bulk-forming laxatives to increase stool bulk
 - D. Antibiotic therapy for suspected infection
-

4. A 45-year-old woman with a history of chronic constipation visits her primary care physician. She has been prescribed bisacodyl, a stimulant laxative, to help with her bowel movements. The physician advises her to take the medication before bedtime. What is the BEST explanation for prescribing bisacodyl before bedtime?

- A. Bisacodyl takes 6-8 hours to reach the small bowel, thus needs to be given at bedtime for morning evacuation.
 - B. Bisacodyl has an associated sedative effect, thus needs to be prescribed before sleep.
 - C. The drug requires hydrolysis by endogenous esterases in the bowel for activation, which takes 6–10 hours on oral dose.
 - D. Bisacodyl mechanism of action involves the process of fermentation, which needs 6-8 hours to take place.
-

5. A 55-year-old woman with a history of chronic constipation has been using over-the-counter laxatives for several months. During a routine colonoscopy, the gastroenterologist observes the following image. Which of the following is the implicated reason for this condition?



- A. Bisacodyl
- B. Isabgol
- C. Castor Oil
- D. Senna

6. A 45-year-old male presents with complaints of chronic indigestion, bloating and delayed gastric emptying. After a thorough evaluation, the physician prescribes a prokinetic agent to improve gastrointestinal motility. Which of the following statements about prokinetic agents is false?

- A. Metoclopramide acts as a D2 antagonist and has serotonergic activity, enhancing GI motility.
- B. Domperidone primarily acts as a peripheral D2 receptor antagonist, facilitating gastric emptying and increasing LES tone.
- C. Erythromycin acts as a motilin receptor agonist, mimicking the action of motilin to stimulate gastric contractions.
- D. Cholecystokinin acts on the CCK-A receptor and stimulates stomach contraction, promoting motility.

7. A 45-year-old woman presents with complaints of chronic GERD and occasional delayed gastric emptying. After a thorough evaluation, the physician decides to prescribe her Mosapride. Which of the following statements regarding Mosapride is false?

- A. Mosapride is a selective 5-HT₄ receptor agonist.
 - B. Mosapride enhances gastric motility by increasing acetylcholine release.
 - C. Mosapride is commonly used to treat gastroesophageal reflux disease (GERD).
 - D. Mosapride is a Motilin receptor stimulant.
-

8. A 23-year-old female presents with complaints of severe nausea and vomiting for the past three days. She has a history of migraine headaches and recently started taking metoclopramide to manage her symptoms. On examination, the patient appears anxious and has abnormal facial movements, including grimacing and involuntary movements of her neck and shoulders. Which of the following is the most likely diagnosis for this patient's symptoms?

- A. Serotonin syndrome
 - B. Acute dystonia
 - C. Neuroleptic malignant syndrome
 - D. Allergic reaction
-

9. A 55-year-old male with a history of cirrhosis secondary to chronic hepatitis C presents with altered mental status. He is disoriented to time and place and has asterixis on physical examination. His family reports that he has been non-compliant with his medications, including lactulose. Laboratory tests reveal elevated serum ammonia levels. Which of the following is the primary mechanism by which lactulose helps in the management of hepatic encephalopathy in this patient?

- A. Lactulose reduces serum ammonia levels by promoting its excretion through urine.

- B. Lactulose acts as a sedative to reduce agitation and improve mental status.
 - C. Lactulose acidifies the colonic contents, converting ammonia to ammonium, which is less readily absorbed.
 - D. Lactulose directly binds to ammonia in the bloodstream, facilitating its elimination.
-

10. A 28-year-old pregnant woman at 24 weeks of gestation presents with complaints of constipation for the past two weeks. She is concerned about using medications that might affect her pregnancy. The obstetrician discusses the use of stool softeners to alleviate her symptoms. Which of the following is the most appropriate stool softener to recommend for this patient?

- A. Bisacodyl
 - B. Docusate sodium
 - C. Magnesium citrate
 - D. Senna
-

11. A 40-year-old female presents with a chronic history of watery diarrhoea for the past 4 months, up to 10 times daily. She reports significant weight loss despite maintaining a good appetite. Blood tests reveal mild hypokalemia. Stool analysis shows no faecal pathogens. Abdominal imaging identifies a small pancreatic mass. Further hormonal testing demonstrates elevated vasoactive intestinal peptide (VIP) levels. Which of the following medications is most likely to be beneficial for controlling the patient's diarrhoea?

- A. Loperamide
 - B. Metronidazole
 - C. Octreotide
 - D. Diphenoxylate with Atropine
-

12. A 32-year-old woman presents to the clinic with complaints of cramping abdominal pain and irregular bowel movements, including episodes of both diarrhea and constipation. She mentions occasional mucus in her stool but denies any recent travel, bloody stools, or weight loss. She also reports feeling increasingly stressed due to work deadlines. Her stool culture and other lab results are found to be normal. Which of the following would not be an ideal treatment option for her?

- A. Psyllium
 - B. Dicyclomine
 - C. Alosetron
 - D. Rifaximin
-

13. A 32-year-old male presents with a 6-month history of chronic diarrhea, cramping abdominal pain, and bloating. He reports that his symptoms are exacerbated by stress. There is no h/o rectal bleeding, weight loss, or mucous in the stools. On physical exam, his abdomen is mildly distended with diffuse tenderness but no masses. His lab values are normal. His doctor suspects he has diarrhoea-dominant IBS (IBS-D). Which of the following treatments should be avoided in this patient?

- A. Loperamide
 - B. Alosetron
 - C. Hyoscyamine
 - D. Fluoxetine
-

14. A 38-year-old female presents with abdominal pain, bloating, and diarrhoea for the past week. She mentions that she was recently treated with a 7-day course of amoxicillin for a sinus infection. On further questioning, she reports no fever or

blood in the stool. A stool sample is negative for common pathogens and parasites. The physician suspects antibiotic-associated diarrhoea. Which of the following medications is most appropriate next best to alleviate the patient's symptoms?

- A. Metronidazole
 - B. Lactobacillus
 - C. Rifaximin
 - D. Loperamide
-

15. A 45-year-old female with breast cancer is undergoing chemotherapy with cisplatin. Despite receiving antiemetics during the initial 24 hours, she experiences nausea and vomiting 48 hours after chemotherapy. Which of the following drugs below is most effective in preventing delayed emesis in this patient?

- A. Dronabinol
 - B. Domperidone
 - C. Dexamethasone
 - D. Aprepitant
-

Correct Answers

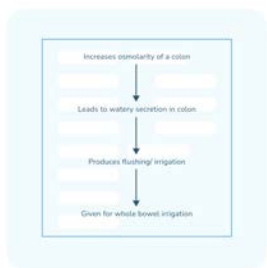
Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	3
Question 5	4
Question 6	4
Question 7	4
Question 8	2
Question 9	3
Question 10	2
Question 11	3
Question 12	4
Question 13	2
Question 14	2
Question 15	4

Solution for Question 1:

Correct Answer: C) Polyethylene glycol (PEG)

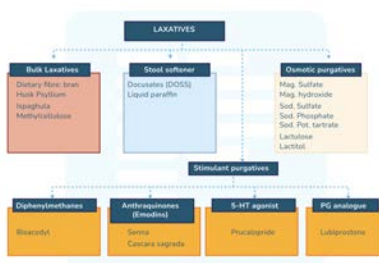
Explanation:

Polyethylene glycol (PEG) is the drug of choice for rapid bowel evacuation before surgery, ultrasound, or X-ray.



This results in rapid and complete bowel cleansing, crucial for procedures like colonoscopy.

- For Bowel Evacuation, the preferred agents are Saline Purgatives (sodium sulphate, sodium picosulphate, Sodium Phosphate) and Osmotic Laxatives (Lactulose, PEG).
- Saline purgatives are the fastest-acting agents used in Bowel evacuation.



Bisacodyl (Option A) is a stimulant laxative that is used in patients of Chronic constipation. It is not used in rapid bowel evacuation due to its slow action and incomplete evacuation.

Lactulose (Option B) is an osmotic laxative, but it is not routinely preferred for bowel evacuation due to its slow action. It is the laxative of choice for patients with hepatic encephalopathy or liver disease because it helps reduce ammonia levels in the blood.

Docusate sodium (Option D) is a stool softener that helps to moisten the stool, making it easier to pass. It is not effective for the rapid and complete evacuation of the bowel needed before procedures like a colonoscopy. Docusates are the laxative of choice in Pregnancy.

Solution for Question 2:

Correct Answer: C) More fermentable the laxative, the more the stool bulk and faster the transit.

Explanation:

The more fermentable the laxative, the less bulky the stool and the slower the transit. The reason behind this is that poorly fermentable fibres (Lignin, Cellulose) absorb more water, resulting in more stool volume.

Fibre is that part of food that resists enzymatic digestion and reaches the colon largely unchanged.

Colonic bacteria ferment fibre to varying degrees, depending on its chemical nature and water solubility.

Fermentation of fibre has two important effects:

- It produces short-chain fatty acids, which are trophic for colonic epithelium and have a prokinetic effect.
- It increases bacterial mass, which results in increased stool volume.

Fibers have varying degrees of fermentation, which is summarised in the table:-

TABLE 54-3 ■ PROPERTIES OF DIFFERENT DIETARY FIBERS		
TYPE OF FIBER	WATER SOLUBILITY	% FERMENTED
Nonpolysaccharides		
Lignin	Poor	0
Cellulose	Poor	15
Noncellulose polysaccharides		
Hemicellulose	Good	56-67
Mucilages and gums	Good	85-95
Pectins	Good	90-95

In general, insoluble, poorly fermentable fibers, such as lignin, are most effective in increasing stool bulk and transit.

Bulk Laxatives such as Bran, Psyllium husk, & Methylcellulose contain more Lignin, which aids in their effectiveness.

Fruits and vegetables contain more pectins and hemicellulose, which is readily fermentable, and has a reduced effect on increasing stool volume.

Solution for Question 3:

Correct Answer: C) Bulk-forming laxatives to increase stool bulk

Explanation:

- In the case of paralytic ileus, where the sensory nerves of the colon are not functional (atonic colon or megacolon), the use of bulk-forming laxatives is an absolute contraindication.
 - These laxatives increase the stool's bulk, requiring active peristalsis to move through the intestines.
 - In a non-functional, atonic colon, this can lead to further distension, increased pressure, and the risk of perforation. Instead, treatment should focus on relieving distension, rehydration, correcting electrolytes, and managing any underlying causes or infections.
 - The rest of the options, intravenous fluids for rehydration & parenteral nutrition, nasogastric decompression to relieve distension and antibiotic therapy for suspected infection are components of the treatment protocol for Paralytic Ileus. (Option A, B and D ruled out)
-

Solution for Question 4:

Correct Answer: C) The drug requires hydrolysis by endogenous esterases in the bowel for activation, which occurs after 6–10 hours on oral dose.

Explanation:

Bisacodyl and other Stimulant laxatives require hydrolysis by endogenous esterases in the bowel for activation, so the Bisacodyl effects are seen after 6–10 hours of an oral dose.

However, Bisacodyl can also be given via rectal suppositories which act within 15-60 mins.

- Bisacodyl acts by causing low grade inflammation in small and large bowel, stimulating intestinal motility and promoting accumulation of water and electrolytes. Hence, there's no relation with the drug reaching the colon to act. (Option A ruled out)
 - Bisacodyl does not have any sedative or sleep inducing effect. (Option B ruled out)
 - Bisacodyl does not involve fermentation. It is activated by the process of hydrolysis by endogenous esterases. (Option D rule out)
-

Solution for Question 5:

Correct Answer: D) Senna

Explanation:

The Image shown is of Melanosis Coli, a benign condition where dark pigmentation of the colonic mucosa is seen on colonoscopy, often associated with the long-term use of anthraquinone-containing laxatives like Senna and Cascara.

Senna & Cascara:

- Herbal Laxatives, which are a type of Stimulant laxatives.
- It causes low-grade inflammation in the small and large bowel, stimulating intestinal motility and promoting the accumulation of water and electrolytes.
- Other side effects of Senna/Cascara besides Melanosis Coli are nausea, vomiting and abdominal cramping.

Contraindications:

- Pregnancy (Increased the risk of abortion due to contractions)
- Intestinal Obstruction (can cause rupture of the intestine)
- Before Colonoscopy (can interfere with colonoscopy findings)

Solution for Question 6:

Correct Answer: D) Cholecystokinin acts on the CCK-A receptor and stimulates stomach contraction, promoting motility.

Explanation:

Cholecystokinin acts on the CCK A receptor to inhibit stomach contraction and not stimulate it.

Prokinetic agents:

Class of Drug	Drugs	Uses
Dopamine (D2) Receptor Antagonists	Metoclopramide, Domperidone, Levosulpiride	GERD, gastroparesis, dyspepsia, antiemetic. (Option A and B ruled out)

5-HT4 agonists	Cisapride, Tegaserod (banned due to QT prolongation), Mosapride, Prucalopride, Cinitapride, Metoclopramide.	GERD, gastroparesis, dyspepsia.
Motilin receptor agonists	Erythromycin	In Acute Upper GI haemorrhage to promote gastric emptying of blood before endoscopy. (Option C ruled out)
CCK - A (CCK -1) inhibitors	Dexlorglumide	GERD, IBS, functional dyspepsia, constipation and gastroparesis.
↑ ACh levels	Neostigmine	Postoperative paralytic ileus

Solution for Question 7:

Correct Answer: D) Mosapride is a Motilin receptor stimulant.

Explanation:

Mosapride is a selective 5-HT4 receptor agonist that increases ACh release from enteric neurons, enhancing GI motility. It is used in conditions like GERD and functional dyspepsia.

Erythromycin acts on motilin receptors, stimulating gastric motility, and is sometimes used in gastroparesis.

Mechanism of Action of Prokinetic Agents:

Prokinetic Agent	Primary Receptor(s) Targeted	Mechanism of Action
Mosapride	5-HT4 receptors	Agonist of 5-HT4 receptors; increases release of acetylcholine, enhancing gastric motility.
Metoclopramide	D2 receptors, 5-HT4 receptors, 5-HT3 receptors	Antagonist of D2 and 5-HT3 receptors; agonist of 5-HT4 receptors; enhances gastric motility and antiemetic effect.
Domperidone	D2 receptors	An antagonist of D2 receptors increases acetylcholine release, enhancing gastric motility.

Erythromycin	Motilin receptors	Agonist of motilin receptors; mimics motilin, stimulating gastrointestinal motility.
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Solution for Question 8:

Correct Answer: B) Acute dystonia

Explanation:

Acute dystonia is an extrapyramidal symptom characterised by sudden, involuntary muscle contractions, often affecting the face, neck and shoulders.

This condition is a known side effect of metoclopramide, especially in younger patients.

- Metoclopramide is a dopamine (D2) antagonist, and it is used for nausea, vomiting, and gastroparesis but carries a risk of extrapyramidal symptoms since it significantly crosses BBB (Blood brain-barrier).
- In contrast, domperidone, another (D2) antagonist, has a lower risk of causing extrapyramidal symptoms because it does not cross the BBB as readily as metoclopramide.

Therefore, domperidone is often preferred in patients who are at higher risk for these side effects.

Solution for Question 9:

Correct Answer: C) Lactulose acidifies the colonic contents, converting ammonia to ammonium, which is less readily absorbed.

Explanation:

Hepatic encephalopathy (HE) is a neuropsychiatric syndrome resulting from liver dysfunction and portosystemic shunting, leading to the accumulation of neurotoxins such as ammonia. Elevated ammonia levels contribute significantly to the pathogenesis of HE.

Lactulose is a non-absorbable disaccharide used in the management of HE. Its primary mechanism involves:

- Lactulose is metabolized by colonic bacteria to lactic acid and acetic acid, leading to a decrease in colonic pH.
- The acidic environment converts ammonia (NH_3) to ammonium (NH_4^+), which is not readily absorbed by the gut.
- This conversion reduces the overall absorption of ammonia into the bloodstream, thereby lowering serum ammonia levels and improving symptoms of HE.

Gut Sterilization:

- The use of antibiotics to reduce the production of ammonia by gut bacteria
- Drugs Used: Rifaximin.

Solution for Question 10:

Correct Answer: B) Docusate sodium

Explanation:

Docusate sodium is a stool softener, which

- Emulsifies colonic contents
- Increases water penetration into feces, making the stool softer and easier to pass.
- Laxative of choice during pregnancy.
- Enhances absorption of nonabsorbable drugs (e.g., liquid paraffin should not be combined)



Bisacodyl and Senna are stimulant laxatives - risk of abdominal cramping - not the first choice in pregnancy. (Options A and D ruled out)

Magnesium citrate is an osmotic laxative - might cause electrolyte imbalances - not recommended as a first-line treatment in pregnancy. (Option C ruled out)

Solution for Question 11:

Correct Answer: C) Octreotide

Explanation:

- The patient's presentation of chronic watery diarrhoea, weight loss, and hypokalemia is suggestive of a secretory diarrhoea syndrome.
- The presence of a pancreatic mass on imaging, along with elevated VIP levels, points towards a VIPoma (vasoactive intestinal peptide tumour).
- Octreotide is a somatostatin analogue that helps control the excessive secretion of VIP, improving diarrhoea.

Loperamide (Option A) is an opioid analog used for symptomatic relief of acute diarrhoea but is ineffective for chronic secretory diarrhoea.

Metronidazole (Option B) is an antibiotic effective against certain bacteria and parasites that can cause diarrhea. However, the negative stool cultures rule out this option.

Diphenoxylate with Atropine (Option D) : (Lomotil) Similar to Loperamide, Lomotil is an ant motility agent that provides symptomatic relief but doesn't address the underlying cause of diarrhea in VIPoma.

Solution for Question 12:

Correct Answer: D) Rifaximin

Explanation:

The patient's symptoms of cramping abdominal pain, bloating, and irregular bowel movements, combined with a history of stress and normal lab results, indicate a likely diagnosis of Irritable Bowel Syndrome (IBS).

Rifaximin is an antibiotic, it's not a typical first-line treatment for IBS. It may be used in specific cases where diarrhea due to bacterial overgrowth is suspected, however the stool culture results of this patient did not show any bacteria or pathogens, hence the use of rifaximin here would be unwarranted.

Psyllium (Option A) is a bulk laxative that helps IBS by

- Softening the stools in constipation & reducing bowel movement frequency in diarrhoea.
- Acting as a prebiotic, promoting the growth of beneficial gut bacteria.

Dicyclomine (Option B) helps abdominal pain and cramping in IBS by:

Blocking the action of Ach on muscarinic receptors in the smooth muscle of the GI tract. Ach is a neurotransmitter that stimulates muscle contractions. By inhibiting its action, dicyclomine reduces involuntary muscle spasms.

Alosetron (Option C) helps with the frequency and urgency of bowel movements in IBS by:

By blocking serotonin (5-HT₃) receptors in the GI tract. Serotonin plays a key role in regulating bowel function, including motility, secretion, and sensation. By antagonising these receptors, alosetron can reduce the effects of serotonin, which are often heightened in IBS-D patients.

Drugs used in IBS

Constipation	Diarrhea	Abdominal	Gas or bloating
Psyllium/ Bulk/ methylcellulose (Bulk Laxatives)	Antimotility: Loperamide ↓ Antispasmodic	Anticholinergics (SMR) Dicyclomine Hyoscyine ↓ (TCA) Tricyclic	Prokinetics ↓ Dysbiosis ↓ Rifaximin
Prokinetic (pharmacologic)	Cholestyramine	antidepressants	Low FODMAP diet
↓ Chloride secretor Lubiprostone Linaclotide	↓ Alosetron in females	↓ SSRI/fluoxetine	
↓ Osmotic Laxatives (PEG, MG/OL, sorbitol, lactulose)			

Solution for Question 13:

Correct Answer: B) Alosetron

Explanation:

- Alosetron is a 5-HT₃ antagonist that has been approved for the treatment of severe IBS-D in women who have not responded to conventional therapy. However, it is not approved for use in male patients due to a lack of efficacy and serious adverse effects, such as ischemic colitis and severe constipation.
- Drugs for IBS that are not contraindicated in males:

Loperamide (Option A) is an antimotility drug that reduces intestinal peristalsis hence helping to relieve the diarrhea symptoms in patients with IBS- D.

Hyoscyamine (Option C) (Levsin): is an antispasmodic. It relieves cramping and pain by reducing intestinal muscle spasms.

Fluoxetine (Option D) is an antidepressant (SSRI) that can help some patients by modulating gut-brain interaction and reducing anxiety related to IBS.

Solution for Question 14:

Correct Answer: B) Lactobacillus

Explanation:

Lactobacillus probiotics can help restore the balance of normal gut flora that may have been disrupted by antibiotic treatment. Probiotics promote the growth of beneficial bacteria in the gut, potentially alleviating symptoms of antibiotic-associated diarrhoea by re-establishing healthy microbiota.

Other normal gut flora include:

Bacteria

- Bifidobacterium
- Streptomyces

Fungi

- Saccharomyces

Metronidazole (Option A) is an antibiotic used to treat specific infections, including *C. difficile*, but is not indicated for restoring gut flora balance.

Rifaximin (Option C) is another antibiotic used for certain GI conditions like traveler's diarrhea and IBS involving bacterial diarrhea but does not restore normal flora.

Loperamide (Option D) is used to treat diarrhea, it does not address the underlying imbalance of gut flora.

Solution for Question 15:

Correct Answer: D) Aprepitant

Explanation:

This patient's history and clinical symptoms suggest delayed chemotherapy-induced vomiting (CINV). Among the given options, aprepitant is the most effective drug for delayed CINV.

- Aprepitant is an NK1 receptor antagonist that prevents delayed phases of CINV (CINV > 24hrs) when standard anti-emetics are ineffective.
- It blocks substance P, a neuropeptide associated with vomiting.
- Other NK1 receptor antagonists include Fosapripitant (injectable) Netupitant. Rolapitant (longest half-life of 180 hours).
- Fosapripitant (injectable)
- Netupitant.
- Rolapitant (longest half-life of 180 hours).
- Fosapripitant (injectable)
- Netupitant.
- Rolapitant (longest half-life of 180 hours).

Antiemetic drugs:

	Examples	Uses	Side effects
5-HT ₃ receptor antagonists	Ondansetron (shortest-acting, weak 5-HT ₄ antagonist) Palonosetron (longest-acting +	Chemotherapy-induced nausea and vomiting Early onset: Ondansetron Late onset: Palonosetron	Headache QT prolongation Maximum: Dolasetron Minimum: Palonosetron

	most potent) Granisetron (can be given intradermal)		Hypotension
Centrally acting D2 antagonists	Metoclopramide	Chemotherapy-induced nausea and vomiting. Gastroparesis. Paralytic ileus. Bowel cleansing.	Extra-pyramidal symptoms (acute dystonia). Severe hyperprolactinemia.
Domperidone (Option B)	Mild-moderate vomiting. GERD.	Minor EPS features. Mild hyperprolactinemia.	
Neurokinin-1 inhibitors	Aprepitant Fosaprepitant Rolapitant Netupitant	Late-onset nausea and vomiting (DOC) Post-operative nausea and vomiting.	Rolapitant: Neutropenia. Hiccups. Decreased appetite. Dizziness.
Cannabinoids	Dronabinol (Option A) Nabilone	Chemotherapy-induced nausea and vomiting (refractory cases). Appetite stimulant (can be used in HIV and anorexia patients).	Hypotension. Conjunctival injection (bloodshot eyes)
Antihistamines	Cyclizine Meclizine Promethazine Diphenhydramine	Motion sickness	Sedation.
Doxylamine	Morning sickness (given with pyridoxine)	Dry mouth Drowsiness Dry mouth Constipation	
Muscarinic receptor antagonists	Hyoscine (scopolamine)	Motion sickness (DOC, available as a transdermal patch)	Dry mouth. Visual disturbances. Drowsiness.
Glucocorticoids (Option D) (dexamethasone) are used as an adjunct in the management of chemotherapy-induced nausea and vomiting. M/C used drugs for delayed emesis associated with cancer chemotherapy: Ondansetron (5-HT ₃ antagonist) + Aprepitant			

(NK-1 inhibitor) +Dexamethasone.			
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GIT Miscellaneous

1. All of the following are contraindications of emetics except?

- A. Corrosive ingestion
 - B. Unconscious patient
 - C. CNS stimulant toxicity
 - D. Parkinson's disease
-

2. A 25-year-old woman is just being shifted to the recovery room after an appendectomy. She feels nauseous. Which of the following is the most appropriate antiemetic for this patient?

- A. Ondansetron
 - B. Aprepitant
 - C. Domperidone
 - D. Hyoscine
-

3. A 35-year-old woman has a history of nausea and vomiting, especially during her journeys. Which of the following would be the most appropriate drug for this patient to prevent the same during her next journey?

- A. Diphenhydramine
 - B. Hyoscine
 - C. Ondansetron
 - D. Metoclopramide
-

4. A 55-year-old woman is undergoing chemotherapy for breast cancer. Despite taking her prescribed antiemetic medication, she experiences significant nausea and vomiting starting 24 hours after her chemotherapy sessions and lasting for several days. Which of the following is the 1st line drug used to relieve this patient's chemotherapy-induced nausea and vomiting?

- A. Ondansetron
 - B. Dronabinol
 - C. Fezolinetant
 - D. Aprepitant
-

5. A 70-year-old man with Parkinson's disease has recently started treatment with levodopa. He reports experiencing significant nausea and vomiting since beginning the medication. Which of the following antiemetics is appropriate for this patient's condition?

- A. Metoclopramide
 - B. Prochlorperazine
 - C. Chlorpromazine
 - D. Domperidone
-

6. A 64-year-old male patient with a history of lung cancer undergoing chemotherapy comes to the clinic with complaints of dizziness and red eyes for 4 days. He complained of chemotherapy-induced nausea and vomiting 5 days ago, for which he was prescribed an antiemetic. Which of the following antiemetics would have led to his current presentation?

- A. Ondansetron
- B. Metoclopramide

- C. Aprepitant
 - D. Dranabinol
-

7. A 25-year-old female comes to the clinic with complaints of vomiting associated with her migraine headache for the past 2 days. Which of the following can be effective in this patient?

- A. Lorazepam
 - B. Nabilone
 - C. Prochlorperazine
 - D. Dexamethasone
-

8. A 45-year-old female patient presents with mild to moderate pain in the upper right quadrant of her abdomen. Upon ultrasound examination, she is diagnosed with cholesterol gallstones. The physician prescribes ursodeoxycholic acid. Which of the following is the mechanism of action of this drug?

- A. Decreases cholesterol absorption from the intestine
 - B. Decreases cholesterol synthesis in the liver
 - C. Increases cholesterol breakdown
 - D. Increases cholesterol uptake by adipocytes
-

9. A 45-year-old female patient presents to the clinic with symptoms of severe malabsorption following an extensive intestinal resection due to Crohn's disease. She has been diagnosed with short bowel syndrome and is currently receiving parenteral nutrition. Which of the following medications is not typically used to treat short bowel syndrome?

- A. Growth hormone
 - B. Octreotide
 - C. Teduglutide
 - D. Ursodeoxycholic acid
-

10. A 45-year-old male presents to the clinic with complaints of chronic abdominal pain and significant weight loss over the past year. He reports frequent, bulky and foul-smelling stools. His medical history is significant for alcohol abuse over the past 20 years. Laboratory investigations reveal normal LFT, and imaging studies show calcifications in the pancreas. Considering the patient's clinical presentation and medical history, which of the following is the most appropriate treatment?

- A. Lipase only
 - B. Amylase only
 - C. Lipase and trypsin
 - D. Proteinase and amylase
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	2
Question 4	4
Question 5	4
Question 6	4
Question 7	3
Question 8	1
Question 9	4
Question 10	3

Solution for Question 1:

Correct answer: D) Parkinson's disease

Explanation:

Emetics are used in Parkinson's disease

Emetics induce vomiting. There are two drugs which are used to induce vomiting:

- Syrup of Ipecac (oral preparation) used in oral poisoning
- Apomorphine (given as injection- I/M or S/C) used in Parkinson's disease (Option D) and erectile dysfunction (D2 agonist in the hypothalamus → ■ sexual activity)

MECHANISM OF ACTION: Activates D2 receptors in the CTZ area.

CONTRAINDICATIONS OF EMETICS IN POISONING :

- Corrosive poisoning (acids/ alkali) → ■risk of perforation (Option A ruled out)
- CNS Stimulant (cocaine/ amphetamines) → ■risk of seizures (Option C ruled out)
- Unconscious patient → ■risk of aspiration pneumonia (Option B ruled out)

Solution for Question 2:

Correct answer: A) Ondansetron

Explanation:

The drug of choice for postoperative nausea and vomiting is ondansetron.

CLASSIFICATION OF ANTIEMETICS :

Drug class	Examples	Uses
Anticholinergic (M1 receptor)	Hyoscine (scopolamine) (Option D)	DOC: motion sickness (Option D ruled out)
Antihistamine (H1 receptor)	Diphenhydramine Promethazine Cyclizine Meclizine	Motion sickness
5HT3 antagonist	Ondansetron Palonosetron Dolasetron Tropisetron Granisetron	DOC: early onset chemo-induced vomiting DOC: Postoperative and postradiotherapy-induced vomiting (Option A)
D2 blocker	Metoclopramide Domperidone	Mild to moderate vomiting GERD (Option C ruled out)
NK1 inhibitor	Aprepitant Fosaprepitant Netupitant Rolapitant	DOC: late-onset chemotherapy-induced vomiting (Option B ruled out)

Solution for Question 3:

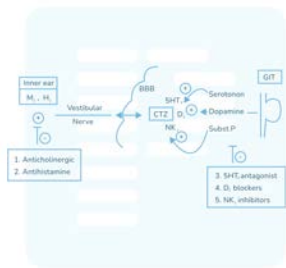
Correct answer: B) Hyoscine

Explanation:

The above scenario is suggestive of motion sickness, and the drug of choice for the same is hyoscine/ scopolamine, which is an anticholinergic (M1 receptor antagonist)

Mechanism of action of anticholinergics and antihistaminic drugs :

- M1 receptor and H1 receptors are present in the inner ear labyrinth and they get activated when there is movement (during journeys) in the semi-circular canal → activate the vestibular nerve → stimulates CTZ in the brain → nausea and vomiting.
- M1 and H1 receptor blockers block this effect → ■ stimulation of the CTZ



DRUGS USED FOR MOTION SICKNESS :

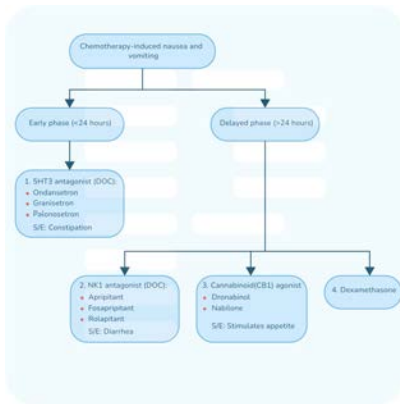
- Anticholinergics : Scopolamine Used as a transdermal patch. Use 4 to 5 hours before travel.
- Used as a transdermal patch.
- Use 4 to 5 hours before travel.
- Antihistamines : Diphenhydramine, Promethazine, and Cyclizine.
- Used as a transdermal patch.
- Use 4 to 5 hours before travel.

Solution for Question 4:

Correct answer: D) Aprepitant

Explanation:

This patient is undergoing chemotherapy for breast cancer, after 24 hours of therapy she complained of nausea and vomiting for several days. This suggests a delayed phase in chemotherapy-induced nausea and vomiting (CINV). The DOC for the same is an NK1 receptor antagonist (e.g; Aprepitant)



Fezolinetant is an NK3 receptor antagonist used to treat postmenopausal vasomotor syndrome (Option C).

Solution for Question 5:

Correct answer: D) Domperidone

Explanation:

The above scenario suggests levodopa-induced nausea and vomiting, and the drug of choice for the same is domperidone, which is a dopamine receptor (D2) antagonist.

D2 receptor antagonists :

- Domperidone : Treats levodopa-induced nausea and vomiting. It does not cross the blood-brain barrier (vs metoclopramide) and hence does not interfere with the therapeutic effects of levodopa, but inhibits the D2 receptors on CTZ → decreasing vomiting.
- Treats levodopa-induced nausea and vomiting.
- It does not cross the blood-brain barrier (vs metoclopramide) and hence does not interfere with the therapeutic effects of levodopa, but inhibits the D2 receptors on CTZ → decreasing vomiting.

- Metoclopramide : 5 HT4 agonist + D2 antagonist + raised Ach Crosses the blood-brain barrier and interferes with the therapeutic effects of levodopa, hence not preferred in levodopa-induced vomiting (Option A ruled out) Causes extrapyramidal side effects.
- Crosses the blood-brain barrier and interferes with the therapeutic effects of levodopa, hence not preferred in levodopa-induced vomiting (Option A ruled out)
- Causes extrapyramidal side effects.
- Antipsychotics : Prochlorperazine (Option B)- DOC for violent vomiting Chlorpromazine (Option C)
- Prochlorperazine (Option B)- DOC for violent vomiting
- Chlorpromazine (Option C)
- Treats levodopa-induced nausea and vomiting.
- It does not cross the blood-brain barrier (vs metoclopramide) and hence does not interfere with the therapeutic effects of levodopa, but inhibits the D2 receptors on CTZ → decreasing vomiting.
- Crosses the blood-brain barrier and interferes with the therapeutic effects of levodopa, hence not preferred in levodopa-induced vomiting (Option A ruled out)
- Causes extrapyramidal side effects.
- Prochlorperazine (Option B)- DOC for violent vomiting
- Chlorpromazine (Option C)

Solution for Question 6:

Correct answer: D) Dronabinol

Explanation:

Dronabinol causes hypotension (leading to dizziness) and bloodshot eyes.

Mechanism of action: CB1 agonist (Cannabinoid receptor)

Uses:

- Chemotherapy-induced nausea and vomiting
- Appetite stimulation.

ADVERSE EFFECTS OF DRONABINOL :

- CNS effects : Exacerbation of mania, depression, and schizophrenia Dizziness Fatigue Cognitive impairment
- Exacerbation of mania, depression, and schizophrenia
- Dizziness
- Fatigue
- Cognitive impairment
- Cardiovascular effects : Tachycardia Hypotension or hypertension
- Tachycardia
- Hypotension or hypertension
- Conjunctival congestion
- Occasional abdominal pain, nausea, and vomiting are potentially indicative of cannabinoid hyperemesis syndrome.
- Exacerbation of mania, depression, and schizophrenia
- Dizziness
- Fatigue
- Cognitive impairment
- Tachycardia
- Hypotension or hypertension

Drug	Side effect
Ondansetron	Constipation, QT interval prolongation, headache (Option A ruled out)
Metoclopramide	Extrapyramidal side effects (by crossing the blood brain barrier) (Option B ruled out)
Aprepitant	QT interval prolongation (Option C ruled out)

Solution for Question 7:

Correct answer: C) Prochlorperazine

Explanation:

The drug of choice for migraine-induced vomiting is an antipsychotic such as prochlorperazine and chlorpromazine.

Antipsychotics :

- They have antihistamine and anticholinergic properties and hence can be used in migraine-induced nausea and vomiting.
- They inhibit D2 receptors in CTZ, which is the mechanism behind preventing nausea and vomiting.
- They are also indicated in post-chemotherapy, postoperative nausea and vomiting and also in motion sickness.

Lorazepam (Option A): The sedative, amnesic and anxiolytic properties of lorazepam help be used as an anticipatory medication in patients with terminal diseases.

Nabilone (Option B) which is a CB1 receptor agonist, is used prophylactically for post-chemotherapy-induced nausea and vomiting.

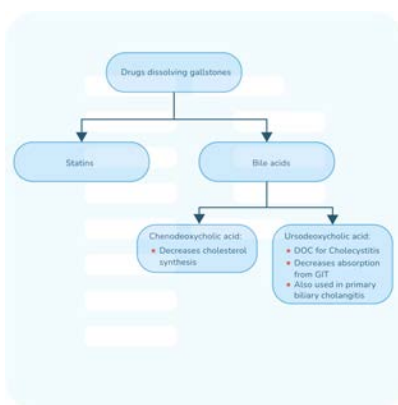
Dexamethasone (Option D) (Corticosteroids) reduce inflammation → reduce the inflammatory mediators like serotonin, dopamine, and substance P; mainly used in chemotherapy-induced nausea and vomiting.

Solution for Question 8:

Correct answer: A) Decreases cholesterol absorption from the intestine

Explanation:

Ursodeoxycholic acid acts by decreasing cholesterol absorption from the intestine.



- Ursodeoxycholic acid is a naturally occurring hydrophilic bile acid.

- When taken orally, it Alters relative concentrations of bile acids by reducing its absorption from the intestine Decreases biliary lipid secretion Reduces cholesterol content in bile, making it less likely to form gallstones
- Alters relative concentrations of bile acids by reducing its absorption from the intestine
- Decreases biliary lipid secretion
- Reduces cholesterol content in bile, making it less likely to form gallstones
- It is also cytoprotective in cholestatic liver diseases like primary sclerosing cholangitis.
- Alters relative concentrations of bile acids by reducing its absorption from the intestine
- Decreases biliary lipid secretion
- Reduces cholesterol content in bile, making it less likely to form gallstones

Decreased cholesterol synthesis in the liver (Option B) is the mechanism of action of chenodeoxycholic acid and statins.

Solution for Question 9:

Correct answer: D) Ursodeoxycholic acid

Explanation:

Ursodeoxycholic acid is not used in short bowel syndrome but instead used in patients with biliary colic symptoms.

SHORT BOWEL SYNDROME :

- Short bowel syndrome (SBS) is a difficult and frequently debilitating condition characterised by malabsorption and chronic diarrhoea. It occurs when a large portion of the intestine is removed.
- 3 new drugs for the treatment of short bowel syndrome :

Growth hormone (Option A): → increases production of IGF-1 → stimulates growth of bowel cells → enhances nutrient absorption.

Octreotide (Option B): → decreases intestinal secretions → decreases malabsorptive diarrhoea,

Teduglutide (Option D): it is a recombinant GLP2 analogue → increases the production of IGF-1 → stimulates gut epithelial proliferation.

Solution for Question 10:

Correct answer: C) Lipase and trypsin

Explanation:

This patient with steatorrhoea due to chronic pancreatitis needs to be treated with pancreatic enzyme replacement.

Indications for pancreatic enzyme replacement :

- Cystic fibrosis,
- Chronic pancreatitis,
- Post pancreatectomy.

Drug used : pancrelipase = Lipase + trypsin (lipase>trypsin)

Previous Year Questions

1. In paracetamol poisoning, N-acetylcysteine is administered as an antidote. How does it act to prevent toxicity?

- A. Inhibits the toxin
 - B. Removes the toxin
 - C. Restores glutathione levels
 - D. Neutralizes liver enzymes
-

2. A 25-year-old woman presents to the emergency department 6 hours after ingesting a large quantity of paracetamol tablets. What is the most likely progression of paracetamol poisoning in this case?

- A. Increase in liver enzymes within 24 hours
 - B. Liver failure within 24 hours
 - C. Paracetamol achieves peak concentration within 4 hours
 - D. Recovery within 72 hours
-

3. Diphenoxylate is used to treat secretory diarrhea, and it is often combined with atropine. What is the primary reason for this combination therapy?

- A. To correct the side effects of diphenoxylate
 - B. To enhance the antidiarrheal efficacy of diphenoxylate
 - C. To provide an additional antidiarrheal effect
 - D. To reduce the potential for abuse of diphenoxylate
-

4. What is the most appropriate treatment for a patient presenting with diarrhea and a flask-shaped ulcer on a colonoscopy?

- A. Metronidazole
 - B. Ciprofloxacin
 - C. Vancomycin
 - D. Ampicillin
-

5. A patient underwent gastrointestinal surgery and had her jejunum resected. Which of the following drugs should be added to her treatment?

- A. Liraglutide
 - B. Teduglutide
 - C. Exenatide
 - D. Semaglutide
-

6. Which of the following drugs is used in GERD?

- A. Ondansetron
 - B. Domperidone
 - C. Loperamide
 - D. Diphenoxylate
-

7. Which of the following is the drug of choice (DOC) for treating morning sickness in pregnant women?

- A. Pyridoxine (Vitamin B6)
 - B. Domperidone
 - C. Ondansetron
 - D. Metoclopramide
-

8. A patient undergoing chemotherapy was given an antiemetic, after which he developed symptoms like acute dystonia, bradykinesia, and tremors. Which of the following drugs would have caused these symptoms?

- A. Ondansetron
 - B. Metoclopramide
 - C. Meclizine
 - D. Scopolamine
-

9. Proton pump inhibitors suppress gastric acid secretion by acting on:

- A. H⁺/Cl⁻ ATPase pump
 - B. H⁺/K⁺ ATPase pump
 - C. Na⁺/K⁺ ATPase pump
 - D. Na⁺/K⁺/Cl⁻ ATPase pump
-

10. Which is not used in treatment in Wilson's disease?

- A. Zinc
- B. Penicillamine
- C. Calcium citrate
- D. Trientine

11. Prucalopride is a _____:

- A. 5HT_{2a} agonist
 - B. 5HT_{2b} agonist
 - C. 5HT₄ antagonist
 - D. 5HT₄ agonist
-

12. A pediatric patient was prescribed erythromycin for the prevention of recurrent rheumatic fever in a rural health center. If the patient complained of a history of constipation, which medication should be avoided to prevent the development of any arrhythmia?

- A. Lubiprostone
 - B. Streptomycin
 - C. Ebastine
 - D. Cisapride
-

13. Which of these drugs stimulates PPAR-alpha?

- A. Gemfibrozil
 - B. Ezetimibe
 - C. Colestipol
 - D. Simvastatin
-

14. Treatment for Zollinger-Ellison syndrome is:

- A. Cimetidine
 - B. Omeprazole
 - C. Misoprostol
 - D. Aluminum hydroxide
-

15. A 35-year-old female comes to the OPD for well-care visits. The patient is completely well and eats a balanced diet. She never smokes or drinks alcohol and has no significant past medical history. On examination, her vitals are normal. The patient tells the physician that next week she will go on a trip to an island and is worried about motion sickness. Which of the following drugs may be considered first-line therapy for this patient's prophylaxis of motion sickness?

- A. Promethazine
 - B. Ondansetron
 - C. Metoclopramide
 - D. Domperidone
-

16. Which of the following conditions can be treated with pirenzepine?

- A. Asthma
 - B. Peptic ulcer disease
 - C. Glaucoma
 - D. Hypertension
-

17. A colonoscopy of an elderly patient revealed brownish pigmented lesions as shown below. He admitted to chronic laxative abuse. Which of the following

laxatives is most likely to cause this condition?



- A. Bran
- B. Psyllium
- C. Senna
- D. Methylcellulose

18. True statement about teduglutide is:

- A. It is a GLP-2 analogue
- B. It's half life is 6-8 hours
- C. It is administered orally
- D. It is recommended for patients with colorectal cancer

19. Most effective antiemetic for chemotherapy-induced vomiting is

_____.

- A. Domperidone
- B. Tegaserod
- C. Granisetron

D. Doxylamine

20. What is the most frequently observed adverse effect of orlistat among the options provided?

- A. Acute kidney injury
 - B. Osteoporosis
 - C. Steatorrhea
 - D. Colorectal cancer
-

21. Which one among the following options is not categorized as a prokinetic agent?

- A. 5HT4 agonist
 - B. Dopamine antagonist
 - C. Motilin agonists
 - D. Diphenylmethane
-

22. Which of the following drugs are used to treat chemotherapy-induced nausea and vomiting?

- A. Doxylamine + domperidone + aprepitant
 - B. Prochlorperazine + metoclopramide + domperidone
 - C. Dexamethasone + metoclopramide + domperidone
 - D. Granisteron + dexamethasone + aprepitant
-

23. Which drug is most appropriate for treating diarrhea in a patient treated for colorectal carcinoma with 5-fluorouracil?

- A. Ciprofloxacin
 - B. Loperamide
 - C. Atropine
 - D. Ornidazole
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	4
Question 4	1
Question 5	2
Question 6	2
Question 7	1
Question 8	2
Question 9	2
Question 10	3
Question 11	4
Question 12	4
Question 13	1
Question 14	2
Question 15	1
Question 16	2
Question 17	3
Question 18	1
Question 19	3
Question 20	3
Question 21	4
Question 22	4

Solution for Question 1:

Answer: C) Restores glutathione levels

Explanation:

N-acetylcysteine (NAC) is the antidote for paracetamol (acetaminophen) poisoning and primarily works by restoring glutathione levels to prevent liver damage.

- Mechanism of Paracetamol Toxicity: Paracetamol is metabolized by the liver into: Non-toxic metabolites (via sulfation & glucuronidation). A toxic metabolite, NAPQI (N-acetyl-p-benzoquinone imine), via CYP450. Normally, NAPQI is detoxified by glutathione (GSH). In overdose, glutathione is depleted, allowing NAPQI to accumulate → hepatic necrosis.
- Paracetamol is metabolized by the liver into: Non-toxic metabolites (via sulfation & glucuronidation). A toxic metabolite, NAPQI (N-acetyl-p-benzoquinone imine), via CYP450.
- Non-toxic metabolites (via sulfation & glucuronidation).
- A toxic metabolite, NAPQI (N-acetyl-p-benzoquinone imine), via CYP450.
- Normally, NAPQI is detoxified by glutathione (GSH).
- In overdose, glutathione is depleted, allowing NAPQI to accumulate → hepatic necrosis.
- How NAC Works: Acts as a glutathione precursor, replenishing GSH stores. Enhances non-toxic sulfation pathway for paracetamol metabolism. Directly binds NAPQI in high doses, reducing toxicity.
- Acts as a glutathione precursor, replenishing GSH stores.
- Enhances non-toxic sulfation pathway for paracetamol metabolism.
- Directly binds NAPQI in high doses, reducing toxicity.
- Paracetamol is metabolized by the liver into: Non-toxic metabolites (via sulfation & glucuronidation). A toxic metabolite, NAPQI (N-acetyl-p-benzoquinone imine), via CYP450.
- Non-toxic metabolites (via sulfation & glucuronidation).
- A toxic metabolite, NAPQI (N-acetyl-p-benzoquinone imine), via CYP450.
- Normally, NAPQI is detoxified by glutathione (GSH).
- In overdose, glutathione is depleted, allowing NAPQI to accumulate → hepatic necrosis.
- Non-toxic metabolites (via sulfation & glucuronidation).

- A toxic metabolite, NAPQI (N-acetyl-p-benzoquinone imine), via CYP450.
- Acts as a glutathione precursor, replenishing GSH stores.
- Enhances non-toxic sulfation pathway for paracetamol metabolism.
- Directly binds NAPQI in high doses, reducing toxicity.

Management of Paracetamol poisoning	
Management Aspect	Details
Initial Treatment	Gastric Lavage Administer activated charcoal or cholestyramine (effective within 30 minutes of ingestion)
N-acetylcysteine (NAC) Therapy	Initiate within 8 hours of ingestion; may be partially effective up to 24–36 hours post-overdose. Administration: IV: 140 mg/kg loading dose over 1 hour, followed by 70 mg/kg every 4 hours for 15–20 doses.
Mechanism of Action	NAC acts as a sulfhydryl donor, replenishing glutathione. Neutralizes toxic metabolites and prevents binding to cell proteins, reducing the severity of hepatic necrosis.
Liver Transplantation	Considered in cases of hepatic failure (jaundice, coagulopathy, confusion). King's College Criteria is used to predict the need for a liver transplant in patients with paracetamol-induced acute liver failure : Arterial blood pH < 7.30 Or all of the following: Prothrombin time (PT) > 100 seconds (INR > 6.5) Serum creatinine > 300 µmol/L Grade III or IV hepatic encephalopathy.

- Inhibits the toxin (Option A) is incorrect because NAC does not inhibit NAPQI formation, it helps detoxify it.
- Removes the toxin (Option B) is incorrect because NAC does not directly remove paracetamol or NAPQI from the body.
- Neutralizes liver enzymes (Option D) is incorrect because NAC does not neutralize liver enzymes, it restores glutathione to prevent hepatic injury.

Solution for Question 2:

Answer: A) Increase in liver enzymes within 24 hours

Explanation:

Paracetamol (acetaminophen) poisoning follows a predictable progression based on the time since ingestion. The first 24 hours are usually asymptomatic or present with mild symptoms, but a rise in liver enzymes (AST, ALT) occurs as hepatotoxicity begins.

Metabolism of Acetaminophen
Acetaminophen is mainly metabolized via phase II reactions into harmless sulfate and glucuronide metabolites. A small portion undergoes phase I metabolism by CYP2E1, forming the hepatotoxic metabolite N-acetyl-p-benzoquinone-imine (NAPQI). Detoxification of NAPQI NAPQI is neutralized by binding to glutathione, forming water-soluble mercapturic acid, which is excreted via the kidneys. Glutathione Depletion When excessive NAPQI is produced, or glutathione is depleted, NAPQI binds to hepatocyte macromolecules, forming acetaminophen-protein adducts, causing hepatocyte necrosis.
Risk Factors
Chronic alcohol use (induces CYP2E1, depletes glutathione). Starvation. Prior use of mixed-function oxidase-inducing drugs (e.g., phenobarbital, INH). Hepatitis C infection increases the risk of liver injury in overdose.
Clinical features
Early Symptoms (4-12 hrs): Nausea, vomiting, diarrhea, abdominal pain, shock. Delayed Symptoms (24-48 hrs): Hepatic injury becomes apparent. Coagulopathy and encephalopathy. Maximal damage seen 3–5 days post-ingestion with aminotransferase levels >10,000 IU/L. Renal failure and myocardial injury may occur.

Stages of acute acetaminophen poisoning		
Stage	Time Post-Ingestion	Description
I	0–24 hours	Nausea, vomiting, and anorexia.
II	24–72 hours	Right upper quadrant abdominal pain; elevated liver enzymes (AST, ALT), and sometimes increased bilirubin and INR if poisoning is severe.
III	72–96 hours	Vomiting, signs of liver failure; peak levels of AST, ALT, bilirubin, and INR. Possible renal failure and pancreatitis.

IV	> 5 days	Recovery of liver function or progression to multiple organ failure.
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- Liver failure within 24 hours (Option B) is incorrect because severe liver failure typically occurs at 72–96 hours, not within the first 24 hours.
- Paracetamol achieves peak concentration within 4 hours (Option C) is incorrect because while paracetamol reaches peak plasma levels in 30 minutes to 2 hours, this does not define the course of toxicity.
- Recovery within 72 hours (Option D) is incorrect because severe toxicity peaks at 72–96 hours, with either recovery or progression to liver failure afterward.

Solution for Question 3:

Correct Option D - To reduce the potential for abuse of diphenoxylate:

- Diphenoxylate is an opioid derivative with antidiarrheal properties. It can potentially be abused due to its opioid-like effects, especially at higher doses. To mitigate the risk of abuse, diphenoxylate is combined with atropine, an anticholinergic agent. The addition of atropine causes unpleasant side effects (such as dry mouth, blurred vision, and tachycardia) at higher doses, which discourages abuse of the medication.

Incorrect Options:

Option A - To correct the side effects of diphenoxylate:

- Atropine is not added to correct side effects of diphenoxylate. In fact, atropine can add its own set of side effects. The primary purpose of atropine in this combination is to deter abuse, not to alleviate side effects of diphenoxylate.

Option B - To enhance the antidiarrheal efficacy of diphenoxylate:

- While atropine has some anticholinergic effects that might contribute to reducing diarrhea, its primary role in this combination is not to enhance diphenoxylate's antidiarrheal efficacy, but rather to deter abuse.

Option C - To provide an additional antidiarrheal effect:

- Although atropine may provide some additional antidiarrheal effects due to its anticholinergic properties, this is not the main reason for its combination with diphenoxylate. The primary reason is to reduce the potential for abuse of diphenoxylate.

Solution for Question 4:

Correct Option A -Metronidazole:

- The patient's clinical presentation and colonoscopic findings are suggestive of amebic dysentery. Metronidazole is the drug of choice (DOC) for treating amebic dysentery. The standard regimen is 400 mg three times a day for a week. Additionally, luminal amebicidal drugs are added to eradicate the cysts within the intestines.

Incorrect Options:

Options B, C, D:

- Ciprofloxacin, Vancomycin, and Ampicillin are not the drugs used for amebic dysentery
-

Solution for Question 5:

Correct Option B - Teduglutide:

- Teduglutide is a GLP-2 analogue which causes growth of intestinal cells. It increases absorptive power of intestine and is approved for short bowel syndrome treatment.

Incorrect Options:

Options A, C, D:

- Liraglutide, Exenatide, and Semaglutide are not the drugs used for Short bowel syndrome
-

Solution for Question 6:

Correct Option B – Domperidone:

- The primary treatment of GERD includes proton pump inhibitors and prokinetic agents. Domperidone is a prokinetic agent used in the treatment of GERD. It helps to close the lower esophageal sphincter (LES), thereby reducing reflux of stomach contents into the esophagus.

Incorrect Options:

Options A, C, and D:

- Ondansetron, Loperamide, and Diphenoxylate are not the drugs used in GERD.
-

Solution for Question 7:

Correct Option A – Pyridoxine (Vitamin B6):

- The drug of choice for treating morning sickness in pregnant women is Pyridoxine (Vitamin B6) and Doxylamine.

Incorrect Options:

Options B, C, D:

- Domperidone, Ondansetron, and Metoclopramide are antiemetic drugs, but the drug of choice is Pyridoxine
-

Solution for Question 8:

Correct Option B - Metoclopramide:

- Metoclopramide is the drug that is most likely to cause the symptoms described in the given scenario.
- Metoclopramide is an antiemetic medication that is often used to treat nausea and vomiting associated with chemotherapy.
- It can cause acute dystonia (sudden, involuntary muscle contractions), bradykinesia (slowness of movement) and tremors. These symptoms are known as extrapyramidal side effects and are related to the drug's dopamine receptor antagonism in the brain.

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Incorrect Options:

Option A - Ondansetron: Ondansetron works by blocking serotonin receptors, not dopamine receptors. Hence, extrapyramidal side effects are not seen.

Option C - Meclizine: Meclizine is an antihistamine medication that is used to treat motion sickness and is not associated with extrapyramidal side effects.

Option D - Scopolamine: Scopolamine is an anticholinergic medication. Hence, extrapyramidal side effects are not seen

Solution for Question 9:

Correct Option B - H⁺/K⁺ ATPase pump:

- This option refers to the hydrogen-potassium ATPase pump, also known as the proton pump, which is the correct answer. Proton pump inhibitors (PPIs) specifically target and inhibit the H⁺/K⁺ ATPase pump located in the parietal cells of the stomach. By binding irreversibly to this pump, PPIs prevent the secretion of gastric acid into the stomach. This leads to a significant reduction in gastric acid production, providing relief from acid-related conditions such as gastroesophageal reflux disease (GERD) and peptic ulcers.

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Incorrect Options:

Option A - H⁺/Cl⁻ ATPase pump: proton pump inhibitors (PPIs) do not act on a H⁺/Cl⁻ ATPase pump.

Option C - Na⁺/K⁺ ATPase pump: The Na⁺/K⁺ ATPase pump is not directly involved in gastric acid secretion. It is an enzyme found in the cell membranes of many types of cells. This is the target pump for cardiac glycosides like digoxin.

Option D - Na⁺/K⁺/Cl⁻ ATPase pump: There is no Na⁺/K⁺/Cl⁻ ATPase pump in the body.

Solution for Question 10:

Correct Option C - Calcium citrate:

- Wilson's disease is a genetic disorder characterized by the abnormal accumulation of copper in various organs, primarily the liver and brain. The mainstay of treatment for Wilson's disease involves the use of medications that help reduce copper levels in the body. These medications include:

Wilson's disease is a genetic disorder characterized by the abnormal accumulation of copper in various organs, primarily the liver and brain. The mainstay of treatment for Wilson's disease involves the use of medications that help reduce copper levels in the body. These medications include:

Incorrect Options:

Option A - Zinc: Zinc acts by blocking the absorption of copper from the diet and promotes the excretion of copper in the stool. It is commonly used as a first-line therapy for Wilson's disease.

Option B - Penicillamine: Penicillamine is a chelating agent that binds to copper and facilitates its removal from the body through urine. It is also used in the treatment of Wilson's disease, especially in patients who cannot tolerate or do not respond to zinc therapy.

Option D - Trientine: Trientine is another chelating agent used in the treatment of Wilson's disease. It works similarly to penicillamine by binding to copper and promoting its elimination from the body.

Solution for Question 11:

Correct Option D - 5HT4 agonist:

- Prucalopride is a 5HT4 agonist. It selectively activates the 5HT4 receptors in the gastrointestinal tract. By binding to and stimulating these receptors, prucalopride promotes increased gut motility and enhances the movement of stool. This helps to alleviate chronic constipation and improve bowel function.

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Incorrect Options:

Option A - 5HT2a agonist: Prucalopride does not primarily target the 5HT2a receptors. Activation of the 5HT2a receptors is associated with various effects, including vasoconstriction, hallucinations, and changes in mood and cognition. Prucalopride's mechanism of action is primarily through 5HT4 receptor activation.

Option B - 5HT2b agonist: Prucalopride does not primarily target the 5HT2b receptors. Activation of the 5HT2b receptors is associated with effects such as cardiac valve fibrosis. Prucalopride's therapeutic action is not mediated through 5HT2b receptor activation.

Option C - 5HT4 antagonist: Prucalopride is a 5HT4 agonist, not an antagonist. As mentioned earlier, prucalopride stimulates the 5HT4 receptors, leading to enhanced gastrointestinal motility and improved bowel function.

Solution for Question 12:

Correct Option D - Cisapride:

- Cisapride is a prokinetic medication that was commonly used to treat gastrointestinal disorders, including constipation and gastroesophageal reflux disease (GERD). However, it has been withdrawn from the market in many countries due to its potential to cause serious cardiac arrhythmias such as Qt prolongation and torsades de pointes.

- In the given scenario, the patient has a history of constipation, and erythromycin has been prescribed for the prevention of recurrent rheumatic fever. Erythromycin is an antibiotic that is sometimes used as a prophylactic treatment for rheumatic fever. However, erythromycin can interact with cisapride and increase the risk of cardiac arrhythmias. This is because erythromycin can inhibit the metabolism of cisapride, leading to increased levels of cisapride in the body.

Cisapride is a prokinetic medication that was commonly used to treat gastrointestinal disorders, including constipation and gastroesophageal reflux disease (GERD). However, it has been withdrawn from the market in many countries due to its potential to cause serious cardiac arrhythmias such as Qt prolongation and torsades de pointes.

In the given scenario, the patient has a history of constipation, and erythromycin has been prescribed for the prevention of recurrent rheumatic fever. Erythromycin is an antibiotic that is sometimes used as a prophylactic treatment for rheumatic fever. However, erythromycin can interact with cisapride and increase the risk of cardiac arrhythmias. This is because erythromycin can inhibit the metabolism of cisapride, leading to increased levels of cisapride in the body.

Incorrect Options:

Option A - Lubiprostone: Lubiprostone is a medication used for the treatment of chronic idiopathic constipation and irritable bowel syndrome with constipation. It works by increasing fluid secretion in the intestines, promoting bowel movements. Lubiprostone is not associated with the risk of cardiac arrhythmias.

Option B - Streptomycin: Streptomycin is an antibiotic from the aminoglycoside class. It is primarily used to treat tuberculosis and certain other bacterial infections. While streptomycin can have side effects, including effects on the auditory and vestibular systems, it is not known to cause arrhythmias.

Option C - Ebastine: Ebastine is an antihistamine medication used for the treatment of allergic rhinitis and chronic idiopathic urticaria. It is generally well-tolerated and does not have a significant risk of causing arrhythmias.

Solution for Question 13:

Correct Option A - Gemfibrozil:

- Gemfibrozil is a fibric acid derivative that acts as a PPAR-alpha agonist.

- By activating PPAR-alpha leads to decreased triglyceride levels

Gemfibrozil is a fibric acid derivative that acts as a PPAR-alpha agonist.

By activating PPAR-alpha leads to decreased triglyceride levels

Incorrect Options:

Option B - Ezetimibe: Ezetimibe is a cholesterol absorption inhibitor that blocks dietary cholesterol absorption in the small intestine

Option C - Colestipol: Colestipol is a bile acid sequestrant that binds to bile acids in the intestine, preventing their reabsorption and promoting their excretion in the feces.

Option D - Simvastatin: Simvastatin is an HMG-CoA reductase inhibitor (statin) that works by inhibiting the production of cholesterol in the liver

Solution for Question 14:

Correct Option B - Omeprazole:

- Omeprazole is a proton pump inhibitor (PPI) and is the drug of choice for the treatment of Zollinger-Ellison syndrome. PPIs work by irreversibly inhibiting the gastric proton pump, the final step in gastric acid secretion. By reducing stomach acid production, omeprazole helps alleviate the symptoms and complications associated with Zollinger-Ellison syndrome. Other PPIs such as pantoprazole, lansoprazole, and esomeprazole may also be used.

Omeprazole is a proton pump inhibitor (PPI) and is the drug of choice for the treatment of Zollinger-Ellison syndrome. PPIs work by irreversibly inhibiting the gastric proton pump, the final step in gastric acid secretion. By reducing stomach acid production, omeprazole helps alleviate the symptoms and complications associated with Zollinger-Ellison syndrome. Other PPIs such as pantoprazole, lansoprazole, and esomeprazole may also be used.

Incorrect Options:

Option A - Cimetidine: Cimetidine is an H₂ receptor antagonist that reduces the production of stomach acid. While it can provide symptomatic relief for conditions like gastroesophageal reflux disease (GERD) and peptic ulcers, it is not typically the first-line treatment for Zollinger-Ellison syndrome. Zollinger-Ellison syndrome is a rare condition characterized by the overproduction of gastrin, which leads to excessive stomach acid secretion. Therefore, more potent acid-suppressing medications are usually needed.

Option C - Misoprostol: Misoprostol is a prostaglandin analog that has cytoprotective properties and can help prevent gastric ulcers caused by nonsteroidal anti-inflammatory drugs (NSAIDs). While it may be used for certain conditions involving gastric ulcers, it is not typically considered a primary treatment for Zollinger-Ellison syndrome.

Option D - Aluminum hydroxide: Aluminum hydroxide is an antacid that can neutralize stomach acid. It provides temporary relief for symptoms related to excessive stomach acids, such as heartburn and indigestion. However, it is not a specific treatment for Zollinger-Ellison syndrome, as it does not address the underlying cause of increased acid production. Excessive use of Aluminum hydroxide can cause constipation as a side effect.

Solution for Question 15:

Correct Option A - Promethazine:

- Promethazine is commonly used as a first-line therapy for prophylaxis of motion sickness. It belongs to a class of medications called antihistamines, which can help alleviate symptoms of motion sickness, such as nausea, vomiting, and dizziness.
- Motion sickness arises from a mismatch between sensory inputs from the visual, vestibular, and proprioceptive systems, resulting in a mismatch in the brain's perception of motion and spatial orientation.
- Histamine receptors play a pivotal role in modulating the body's response to motion stimuli. By antagonizing these receptors, antihistamines mitigate the neural signals responsible for triggering nausea and vomiting, thereby ameliorating symptoms of motion sickness.

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Motion sickness arises from a mismatch between sensory inputs from the visual, vestibular, and proprioceptive systems, resulting in a mismatch in the brain's perception of motion and spatial orientation.

Histamine receptors play a pivotal role in modulating the body's response to motion stimuli. By antagonizing these receptors, antihistamines mitigate the neural signals responsible for triggering nausea and vomiting, thereby ameliorating symptoms of motion sickness.

Incorrect Options:

Option B - Ondansetron: Ondansetron is a medication primarily used for the prevention and treatment of nausea and vomiting caused by chemotherapy or surgery. While it can be effective in managing nausea and vomiting, it is not typically considered a first-line therapy for motion sickness. Antihistamines like promethazine are generally preferred in this context.

Option C - Metoclopramide: Metoclopramide is a medication that helps to relieve symptoms of nausea and vomiting by increasing the movements or contractions of the stomach and intestines. While it can be effective in certain situations, it is not commonly used as a first-line therapy for motion sickness. Antihistamines, such as promethazine, are typically preferred due to their specific mechanism of action.

Option D - Domperidone: Domperidone is another medication used for the management of nausea and vomiting. It works by blocking dopamine receptors in the gut, which can help reduce the symptoms. However, similar to metoclopramide, it is not commonly used as a first-line therapy for motion sickness. Antihistamines are generally the preferred choice.

Solution for Question 16:

Correct Option B - Peptic ulcer disease:

- Pirenzepine is an anticholinergic medication that can be used in the treatment of peptic ulcer disease.
- It works by selectively blocking the M1 receptors of gastric parietal cells, which reduces gastric acid secretion.
- By reducing acid production, pirenzepine helps in the healing and prevention of ulcers in the stomach and duodenum.

Pirenzepine is an anticholinergic medication that can be used in the treatment of peptic ulcer disease.

It works by selectively blocking the M1 receptors of gastric parietal cells, which reduces gastric acid secretion.

By reducing acid production, pirenzepine helps in the healing and prevention of ulcers in the stomach and duodenum.

Incorrect Options:

Option A - Asthma: Pirenzepine is not indicated for the treatment of asthma. Asthma is a chronic respiratory condition characterized by airway inflammation and bronchoconstriction. The primary treatment for asthma involves bronchodilators, anti-inflammatory medications, and in some cases, corticosteroids. Antimuscarinic drugs used in asthma target M3 receptors which are found in the bronchioles.

Option C - Glaucoma: Pirenzepine is not used in the treatment of glaucoma. Glaucoma is an eye condition characterized by increased intraocular pressure, which can lead to optic nerve damage. The mainstay of treatment for glaucoma involves medications that lower intraocular pressure, such as prostaglandin analogs, beta-blockers, and carbonic anhydrase inhibitors. Anticholinergics are not used in treatment of glaucoma as they cause mydriasis and potentially worsen the condition by blocking aqueous humour outflow

Option D - Hypertension: Pirenzepine is not indicated for the treatment of hypertension. Hypertension, or high blood pressure, is typically managed with antihypertensive medications such as diuretics, beta-blockers, ACE inhibitors, or calcium channel blockers.

Solution for Question 17:

Correct Option C - Senna:

- Senna is a stimulant laxative commonly used for the treatment of constipation.
- It contains anthraquinone derivatives that can cause pigment deposition in the colon, resulting in melanosis coli.

Senna is a stimulant laxative commonly used for the treatment of constipation.

It contains anthraquinone derivatives that can cause pigment deposition in the colon, resulting in melanosis coli.

Incorrect Options:

Option A - Bran: Bran is a high-fiber substance derived from the outer layer of cereal grains. It is commonly used as a natural laxative. Bran itself does not typically cause melanosis coli, which is characterized by brownish pigmented lesions in the colon.

Option B - Psyllium: Psyllium is a soluble fiber commonly used as a bulk-forming laxative. It absorbs water and adds bulk to the stool, promoting regular bowel movements. Psyllium is not known to cause melanosis coli.

Option D - Methylcellulose: Methylcellulose is a bulk-forming laxative that absorbs water and forms a gel-like substance, adding bulk to the stool. It does not typically cause melanosis coli. Thus, it is not the likely cause of the condition described.

Solution for Question 18:

Correct Option A - It is a GLP-2 analogue:

- Teduglutide is a synthetic analogue of glucagon-like peptide-2 (GLP-2). GLP-2 is a naturally occurring hormone that stimulates intestinal growth and enhances nutrient absorption. Teduglutide acts similarly to GLP-2 and is used for the treatment of short bowel syndrome.

Teduglutide is a synthetic analogue of glucagon-like peptide-2 (GLP-2). GLP-2 is a naturally occurring hormone that stimulates intestinal growth and enhances nutrient absorption. Teduglutide acts similarly to GLP-2 and is used for the treatment of short bowel syndrome.

Incorrect Option:

Option B - Its half-life is 6-8 hours: The half-life of teduglutide is approximately 3-5 hours, not 6-8 hours.

Option C - It is administered orally: Teduglutide is not administered orally. It is given as a subcutaneous injection.

Option D - It is recommended for patients with colorectal cancer: Teduglutide is not indicated or recommended for the treatment of colorectal cancer. It is specifically used for the treatment of short bowel syndrome, a condition characterized by a significant reduction in functional small intestine length.

Solution for Question 19:

Correct Option C - Granisetron:

- Granisetron, along with ondansetron, belongs to a class of medications called 5-HT₃ receptor antagonists. These medications are highly effective in preventing and treating chemotherapy-induced vomiting. Granisetron works by blocking serotonin (5-HT₃) receptors in the brain and gut, thereby reducing nausea and vomiting. It is frequently used in combination with other antiemetic drugs in chemotherapy regimens.

Granisetron, along with ondansetron, belongs to a class of medications called 5-HT₃ receptor antagonists. These medications are highly effective in preventing and treating chemotherapy-induced vomiting. Granisetron works by blocking serotonin (5-HT₃) receptors in the brain and gut, thereby reducing nausea and vomiting. It is frequently used in combination with other antiemetic drugs in chemotherapy regimens.

Incorrect Options:

Option A - Domperidone: Domperidone is an antiemetic medication primarily used to treat nausea and vomiting associated with gastrointestinal disorders. It is not considered a first-line treatment for chemotherapy-induced vomiting.

Option B - Tegaserod: Tegaserod is a medication used to treat irritable bowel syndrome with constipation but is not commonly used as an antiemetic for chemotherapy-induced vomiting.

Option D - Doxylamine: Doxylamine is an antihistamine that can be used for the management of morning sickness.

Solution for Question 20:

Correct Option C - Steatorrhea:

- The most common side effect of orlistat which is an anti-obesity drug is steatorrhea.
- It forms a covalent bond with lipases and reduces fat absorption.
- They can also prevent the absorption of fat soluble vitamins like vit. A, D, E, K.

The most common side effect of orlistat which is an anti-obesity drug is steatorrhea.

It forms a covalent bond with lipases and reduces fat absorption.

They can also prevent the absorption of fat soluble vitamins like vit. A, D, E, K.

Incorrect Options:

Option A - Acute kidney injury: Renal injury may be a side effect but it is not the most common one and is rarely acute. It is due to the unabsorbed fat binding to calcium resulting in renal stones that can lead to oxalate nephropathy.

Option B - Osteoporosis: Osteoporosis is caused due to vit D deficiency and calcium. It is not the most common side effect.

Option D - Colorectal cancer: Colorectal cancer is not a side effect.

Solution for Question 21:

Correct Option D - Diphenylmethane:

- Diphenylmethane derivatives like hydroxyzine are not prokinetic agents, they are antihistaminics used to treat allergic disorders.
- Prokinetics are medications that enhance gastrointestinal motility and promote the movement of food through the digestive system. They are commonly used in the treatment of conditions such as gastroesophageal reflux disease (GERD), gastroparesis, and functional dyspepsia. The main classes of prokinetic agents include 5HT4 agonists, dopamine antagonists, and motilin agonists.

Diphenylmethane derivatives like hydroxyzine are not prokinetic agents, they are antihistaminics used to treat allergic disorders.

Prokinetics are medications that enhance gastrointestinal motility and promote the movement of food through the digestive system. They are commonly used in the treatment of conditions such as gastroesophageal reflux disease (GERD), gastroparesis, and functional dyspepsia. The main classes of prokinetic agents include 5HT4 agonists, dopamine antagonists, and motilin agonists.

Incorrect Options:

Option A - 5HT4 agonist: 5HT4 agonists (e.g., cisapride) stimulate the 5HT4 receptors in the gastrointestinal tract, leading to increased release of acetylcholine and enhancement of gut motility.

Option B - Dopamine antagonists: Dopamine antagonists (E.g. Metoclopramide, Domperidone) block dopamine receptors in the gastrointestinal tract, reducing inhibitory dopamine effects and enhancing gastrointestinal motility.

Option C - Motilin agonists: Motilin agonists (e.g., erythromycin) mimic the action of the hormone motilin, which promotes gastric emptying and enhances gastrointestinal motility.

Solution for Question 22:

Correct Option D - Granisetron + dexamethasone + aprepitant:

Chemotherapy can stimulate the release of certain chemicals, such as serotonin and substance P, which can trigger the vomiting center in the brain.

First line therapies include any of or a combination of

1. Granisetron: Granisetron or Ondansetron are serotonin 5-HT₃ receptor antagonists. Chemotherapy can trigger the release of serotonin, which stimulates these receptors in the gut and the brain, leading to nausea and vomiting. By blocking these receptors, granisetron helps to reduce the sensation of nausea and prevent vomiting.
2. Dexamethasone: Dexamethasone is a corticosteroid with anti-inflammatory and immunosuppressive properties. It is believed to work by reducing inflammation in the gastrointestinal tract and suppressing the release of certain chemicals in the brain that trigger nausea and vomiting.
3. Aprepitant: Aprepitant is a substance P/neurokinin 1 (NK1) receptor antagonist. Substance P is a neurotransmitter involved in triggering nausea and vomiting. By blocking the NK1 receptors, aprepitant helps to inhibit the action of substance P, thereby reducing the likelihood of experiencing nausea and vomiting.

These medications are often used together because they target different pathways involved in the development of CINV, providing a more comprehensive approach to prevention and management. This combination therapy is particularly effective in preventing both acute and delayed CINV associated with chemotherapy.

Incorrect Options:

Option A - Doxylamine + domperidone + aprepitant: Doxylamine is primarily an antihistamine used for morning sickness in pregnant women and is not commonly used to treat chemotherapy-induced nausea and vomiting. Domperidone is a medication used to treat nausea and vomiting but is not the first-line treatment for chemotherapy-induced nausea and vomiting.

Option B - Prochlorperazine + metoclopramide + domperidone: Prochlorperazine is an antipsychotic medication used to treat schizophrenia and other psychiatric disorders. While it can be used to treat nausea and vomiting, it is not a first-line treatment for chemotherapy-induced nausea and vomiting.

Option C - Dexamethasone + metoclopramide + domperidone: Dexamethasone and metoclopramide can be used to treat chemotherapy-induced nausea and vomiting; they are not first-line treatments. Additionally, the addition of domperidone is not typically recommended for the treatment of chemotherapy-induced nausea and vomiting.

Solution for Question 23:

Correct Option B -Loperamide:

- 5-fluorouracil is an antimetabolite chemotherapy drug that can cause gastrointestinal toxicity, including diarrhea, as a side effect. Loperamide is an antidiarrheal agent that decreases intestinal motility and increases water absorption and electrolytes in the intestine, forming more solid stools. Therefore, it is the most appropriate drug to be used in this case.

5-fluorouracil is an antimetabolite chemotherapy drug that can cause gastrointestinal toxicity, including diarrhea, as a side effect. Loperamide is an antidiarrheal agent that decreases intestinal motility and increases water absorption and electrolytes in the intestine, forming more solid stools. Therefore, it is the most appropriate drug to be used in this case.

Incorrect Options:

Option A - Ciprofloxacin: Ciprofloxacin is an antibiotic that is used to treat bacterial infections including diarrhea causing pathogens. However in this given case, the cause is most likely due to the GI toxic effects of 5-FU. Therefore use of an antibiotic would be inappropriate.

Option C - Atropine: Atropine is an anticholinergic agent that can be used to treat dysentery and diverticulitis.

Option D - Ornidazole: Ornidazole is an antimicrobial drug used to treat protozoal and anaerobic bacterial infections. However in this given case, the cause is most likely due to the GI toxic effects of 5-FU. Therefore use of an antibiotic would be inappropriate.

Anti Platelets and Pro Platelets, Thrombolytics and Anti Thrombolytics

1. A 70-year-old woman came with complaints of sudden onset of pain, swelling, and redness in her left calf. She has been experiencing difficulty walking and standing for the past two days. She has a history of hypertension and high cholesterol, which is being managed with diet and medication. On giving which of the following drugs will we need to monitor her clotting time for?

- A. Aspirin
- B. Heparin
- C. Alteplase
- D. Apixaban

2. A 58-year-old female patient with a history of atrial fibrillation visits for routine follow-up. She has been prescribed an antiplatelet drug to reduce her risk of stroke. Her medication regimen includes clopidogrel, which she has been taking for the past six months without any adverse effects. Which receptor is primarily targeted by clopidogrel to inhibit platelet aggregation?

- A. P2Y₁₂ receptor
- B. GP IIb/IIIa receptor
- C. COX-1 enzyme
- D. Thrombin receptor

3. A 65-year-old male presents with acute chest pain suggestive of MI. He has a history of hypertension, hyperlipidemia, and type 2 diabetes mellitus. Which of the following is the most appropriate initial dose of aspirin for this patient?

- A. 75 mg orally
 - B. 150 mg orally
 - C. 325 mg orally
 - D. 500 mg orally
-

4. A 72-year-old female patient with a history of atrial fibrillation and coronary artery disease is scheduled for elective hip replacement surgery in two weeks. She has been taking low-dose aspirin (75 mg daily) for secondary prevention of MI and stroke. Which of the following is the most appropriate management regarding her aspirin therapy before surgery?

- A. Continue aspirin at the same dose through the surgery.
 - B. Stop aspirin immediately and resume after the surgery.
 - C. Stop aspirin 7-10 days before surgery and resume after the surgery.
 - D. Switch from aspirin to clopidogrel until after the surgery.
-

5. A 65-year-old male with a history of hypertension and type 2 diabetes presents with chest pain and is diagnosed with acute coronary syndrome. The cardiology team decides to initiate antiplatelet therapy. Which of the following ADP receptor blockers would be the best choice if the team wants an agent with reversible action?

- A. Vorapaxar
 - B. Prasugrel
 - C. Ticagrelor
 - D. Ticlopidine
-

6. A 58-year-old man presents with a sudden onset of severe chest pain radiating to his left arm. He is diaphoretic and appears anxious. His medical history includes hypertension and hyperlipidemia. An ECG shows ST-segment elevation in the anterior leads, and laboratory tests reveal elevated troponin levels. The cardiologist considers initiating therapy with prasugrel. Which of the following conditions is a contraindication for the use of prasugrel in this patient?

- A. Hypertension
 - B. Hyperlipidemia
 - C. History of ischemic stroke
 - D. Diabetes mellitus
-

7. A 60-year-old male with a history of stable angina is currently on a regimen that includes aspirin and nitrates. His cardiologist decides to add a new medication to his treatment plan to prevent thrombotic events further. After a few weeks, the patient reports increased episodes of chest pain, particularly during physical exertion. The cardiologist suspects that it might be coronary steal syndrome. Which of the following medications is most likely responsible?

- A. Pentoxifylline
 - B. Dipyridamole
 - C. Cilostazol
 - D. Aspirin
-

8. A 55-year-old male with a history of myocardial infarction is being evaluated for ongoing chest pain and shortness of breath. The patient's current medication regimen includes aspirin, a beta-blocker, and a statin. The cardiologist considers adding a Phosphodiesterase 3 (PDE-3) inhibitor to further manage the patient's symptoms but recalls that certain PDE-3 inhibitors are contraindicated in patients with myocardial infarction. Which is the correct reason for PDE-3 inhibitors being contraindicated in patients with a history of myocardial infarction?

- A. They increase the risk of bleeding.
 - B. They cause severe hypotension.
 - C. They may inhibit PDE3 in the heart, leading to increased cardiac stimulation and oxygen demand.
 - D. They can cause bradycardia and heart block.
-

9. A 65-year-old male patient with a history of chronic kidney disease presents to the emergency department with severe chest pain. An electrocardiogram (ECG) confirms an acute myocardial infarction (MI). The cardiologist decides to proceed with percutaneous coronary intervention (PCI) and considers using a glycoprotein IIb/IIIa inhibitor to prevent further thrombotic events during the procedure. Given the patient's renal condition, which glycoprotein IIb/IIIa inhibitor is most appropriate for this patient?

- A. Eptifibatide
 - B. Abciximab
 - C. Tirofiban
 - D. Aspirin
-

10. Which of the following statements about newer antiplatelet drugs is incorrect?

- A. Vorapaxar is a PAR (Protease Activating Receptor) inhibitor.
 - B. Blocking PAR receptors on platelets prevents the attachment of thrombin, leading to no platelet aggregation.
 - C. Dazoxiben is a thromboxane A₂ synthase inhibitor.
 - D. Dazoxiben is recently approved as an add-on drug to aspirin in myocardial infarction (MI).
-

11. A 45-year-old female presents to the clinic with a history of easy bruising, petechiae, and prolonged bleeding from minor cuts. Laboratory tests reveal a significantly low platelet count, and she is diagnosed with idiopathic thrombocytopenic purpura (ITP). The physician decides to start treatment to increase her platelet count. Which of the following pro-platelet drugs is most appropriate for treating this patient's condition?

- A. Vorapaxar
 - B. Oprelvekin
 - C. Romiplostim
 - D. Abciximab
-

12. Which of the following drugs is approved for immune thrombocytopenia (ITP) ?

- A. Fostamatinib
 - B. Bosutinib
 - C. Oprelvekin
 - D. Erlotinib
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	3
Question 4	3
Question 5	3
Question 6	3
Question 7	2
Question 8	3
Question 9	2
Question 10	4
Question 11	3
Question 12	1

Solution for Question 1:

Correct answer: B) Heparin

Explanation:

- When a patient is started on Heparin, an anticoagulant, it is crucial to monitor their clotting time to ensure the dosage is effective and safe. This is usually done by measuring the prothrombin time (PT) and activated partial thromboplastin time (aPTT).
- Based on her symptoms and physical exam findings, she likely has deep vein thrombosis (DVT) in her left leg.

- Hence, an anticoagulant like Heparin should be started immediately to prevent the clot from growing or breaking off and travelling to the lungs (pulmonary embolism).
- Lepirudin (used in Heparin-Induced Thrombocytopenia) also requires aPTT monitoring

Aspirin (Option A): is an antiplatelet drug that does not require regular monitoring of clotting time. When given, it requires monitoring of bleeding time.

Alteplase (Options C): is a thrombolytic agents used to dissolve clots but monitoring of clotting time is not typically part of their standard administration protocol. Instead, they require careful monitoring of the patient's overall condition and signs of bleeding. Further, when given, its action can be monitored using D- Dimer to detect the degradation product of fibrin.

Apixaban (Option D) is an oral direct factor Xa inhibitor that prevents thrombin generation and clot formation, with no routine monitoring of clotting time required, though renal function and bleeding risk should be periodically assessed.

	Action	Monitorings
Antiplatelets	Prevent the formation of platelet-rich thrombus	- Effect is monitored by bleeding time - Increase in bleeding time - Antiplatelet toxicity
Anticoagulants	Prevent the formation of fibrin-rich thrombus	- Increases clotting time - PT (Prothrombin time) - aPTT (Activated Partial Thromboplastin Time)
Fibrinolytic (Thrombolytic)	- Degrade the deposited fibrin - Cane remove or dissolve a clot	- Degradation product of fibrin - High amount indicates more fibrin - Degradation

Solution for Question 2:

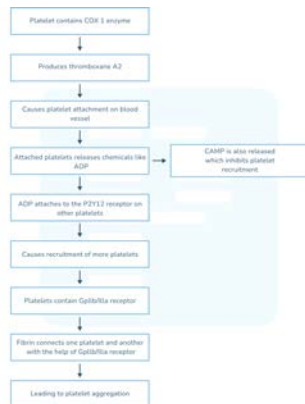
Correct answer: A) P2Y12 receptor

Explanation:

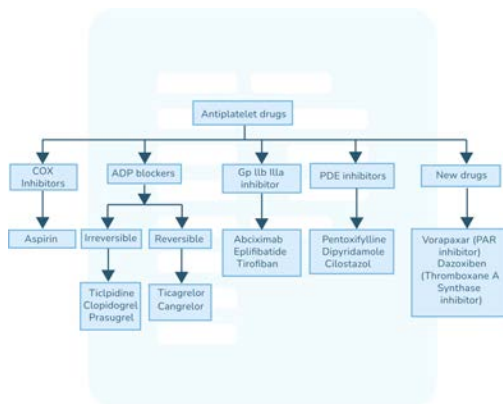
Clopidogrel is an antiplatelet drug that it is used for primary prophylaxis of MI and stroke, specifically targets the P2Y12 receptor on the surface of platelets.

By inhibiting this receptor, clopidogrel prevents ADP from binding, thereby reducing platelet activation and aggregation, which helps to lower the risk of thrombotic events such as stroke in patients with atrial fibrillation.

Normal Action of Platelets:



Antiplatelet drugs are grouped into different categories based on the receptor on which they act:
(Option B and C ruled out)



Solution for Question 3:

Correct answer: C) 325 mg orally

Explanation:

- In acute myocardial infarction, the recommended initial dose of aspirin is 160-325 mg, administered orally as soon as possible.
- This dose helps to quickly and significantly inhibit platelet thromboxane synthesis which leads to reduced platelet attachment.
- Dosage of Aspirin:

75 mg (Option A) is more suitable for long-term secondary prevention

150 mg (Option B) is not commonly used as an initial dose in this context.

500 mg (Option D) is not recommended as it exceeds the effective dose range without additional benefit.

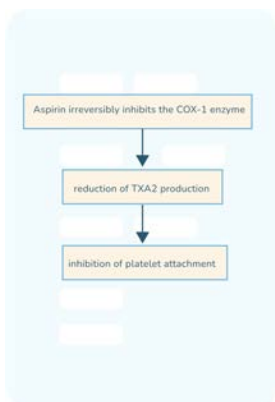
Solution for Question 4:

Correct answer: C) Stop aspirin 7-10 days before surgery and resume after the surgery.

Explanation:

The patient is scheduled for elective surgery, which carries a risk of bleeding. Stopping aspirin 7-10 days before surgery (Option C) is recommended to allow time for new platelets to form (since platelets have a life span of 7-10 days), thus reducing bleeding risk while maintaining cardiovascular protection.

Mechanism of Action of Aspirin:



Clinical Uses and Dosage:

- Aspirin is the drug of choice for MI (Myocardial infarction) Stroke Buerger's disease
- MI (Myocardial infarction)
- Stroke
- Buerger's disease
- Aspirin is used in low doses (typically 75 mg daily) for long-term prophylaxis against MI and stroke in patients with cardiovascular disease or atrial fibrillation.

- Higher doses of aspirin (160-325 mg) are used in the initial management of acute myocardial infarction (MI).
- MI (Myocardial infarction)
- Stroke
- Buerger's disease

Precautions:

Pre-surgical Considerations:

- To minimise the risk of perioperative bleeding, it is generally recommended to stop aspirin 7-10 days before elective surgery to allow regeneration of platelets with normal function.
- In its place, Low Molecular Weight Heparin (Ex, Enoxaparin) 5000U s.c. twice daily is used.
- Switching to clopidogrel (Option D) is not appropriate as it also affects platelet function and can increase bleeding risk.

Solution for Question 5:

Correct answer: C) Ticagrelor

Explanation:

Ticagrelor is a reversible antiplatelet agent that falls under ADP receptor blocker.

ADP receptor blockers are a class of antiplatelet agents that inhibit the P2Y₁₂ receptor on platelets, thus preventing platelet aggregation.

- These drugs are commonly used in the management of acute coronary syndrome and other cardiovascular conditions to reduce the risk of thrombotic events.
- The ADP receptor blockers can be categorized into reversible and irreversible agents, which differ in their duration of action and the binding nature to the P2Y₁₂ receptor.
- Irreversible blockers - permanently bind to the P2Y₁₂ receptor on the platelet surface. Once a platelet is affected, it remains inhibited for its lifespan.
- Reversible blockers - bind to the P2Y₁₂ receptor in a reversible manner, allowing for a quicker offset of action if needed.

Vorapaxar (Option A) (PAR 1 blocker) also has reversible action.

Irreversible blockers
(prodrugs)

Reversible blockers (active drugs)

Ticlopidine.(Option D) Clopidogrel Prasugrel.(Option B)	Ticagrelor. Cangrelor: IV drug (used during percutaneous transluminal coronary angiography- PTCA).
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Clinical Uses of ADP receptor blockers: ACS, PCI, stroke prevention.

Solution for Question 6:

Correct answer: C) History of ischemic stroke

Explanation:

From the given history, the patient likely has developed acute MI.

- ADP receptor blockers, such as clopidogrel, prasugrel, and ticagrelor, are antiplatelet agents used to reduce the risk of thrombotic events in patients with acute coronary syndromes, including MI. Inhibit platelet aggregation by blocking the ADP receptors on platelet cell membranes, thereby preventing clot formation. While ticlopidine and clopidogrel can be given in both MI and stroke, prasugrel, ticagrelor, and cangrelor are used only in MI and are contraindicated in patients with a history stroke due to the increased risk of severe bleeding (risk of hemorrhagic stroke).
- Inhibit platelet aggregation by blocking the ADP receptors on platelet cell membranes, thereby preventing clot formation.
- While ticlopidine and clopidogrel can be given in both MI and stroke, prasugrel, ticagrelor, and cangrelor are used only in MI and are contraindicated in patients with a history stroke due to the increased risk of severe bleeding (risk of hemorrhagic stroke).
- Inhibit platelet aggregation by blocking the ADP receptors on platelet cell membranes, thereby preventing clot formation.
- While ticlopidine and clopidogrel can be given in both MI and stroke, prasugrel, ticagrelor, and cangrelor are used only in MI and are contraindicated in patients with a history stroke due to the increased risk of severe bleeding (risk of hemorrhagic stroke).

Hypertension, hyperlipidemia, and diabetes mellitus (Option A,B, and D) are not contraindications for ADP receptor blockers. Many patients with acute coronary syndrome also have hypertension, hyperlipidemia, and diabetes and ADP receptor blockers can be safely used in these patients.

Solution for Question 7:

Correct answer: B) Dipyridamole

Explanation:

- Dipyridamole, a PDE-3 Inhibitor, is known to cause coronary steal syndrome, which can exacerbate angina by dilating the relatively constricted arterioles of the healthy myocardium, leading to redistribution of blood flow away from the ischemic myocardium.
- Dipyridamole and other PDE-3 inhibitors (Pentoxifylline, Cilostazol) act by blocking PDE-3 Receptors on platelets thereby increasing cAMP and inhibiting platelet recruitment. However, Dipyridamole is approved for secondary prevention of stroke when it is combined with aspirin.

PDE-3 Inhibitors

Drug	Common uses	Notes
Pentoxifylline (Option A)	Used as an add-on drug to aspirin in cases of Stroke and Buerger's disease	Improves blood circulation (rheological agent) Increases elasticity of RBCs
Cilostazol (Option C)	-	
Dipyridamole	Dipyridamole is approved for secondary prevention of stroke when it is combined with aspirin	

Solution for Question 8:

Correct answer: C) They may inhibit PDE3 in the heart, leading to increased cardiac stimulation and oxygen demand.

Explanation:

PDE-3 inhibitors increase cAMP levels, which can lead to increased cardiac stimulation. In patients with a history of myocardial infarction, this increased cardiac stimulation can elevate heart rate and oxygen demand, potentially worsening the patient's condition and increasing the risk of adverse cardiac events.

- While PDE-3 inhibitors do have some effects on blood platelets and may be used to prevent clot formation, the primary concern in myocardial infarction patients is not an increased risk of bleeding but rather the increased cardiac workload and oxygen demand. (Option A)
- PDE-3 inhibitors can cause hypotension as a side effect due to their vasodilatory effects, but this is not the primary reason they are contraindicated in patients with a history of myocardial infarction. (Option B)
- PDE-3 inhibitors are more likely to cause tachycardia (increased heart rate) rather than bradycardia (decreased heart rate). Heart block is also not a typical concern with PDE-3 inhibitors. (Option D)

Solution for Question 9:

Correct answer: B) Abciximab

Explanation:

- Abciximab is a monoclonal antibody that inhibits glycoprotein IIb/IIIa. It is safe in renal failure as it is not renally eliminated and does not require dosage adjustment in Renal patients.

GpIIb/IIIa inhibitors			
Drug	Structure	Common Uses	Route
Abciximab	Monoclonal antibody	Acute attack of MI Stroke Stent replacement (Percutaneous transluminal coronary angioplasty)	i.v.
Eptifibatide (Option A)	Peptide molecule		

Tirofiban (Option C)	Non-peptide molecule		
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Eptifibatide and Tirofiban are renally cleared and requires dose adjustment in patients with renal impairment.

Also, Aspirin (Option D) is an antiplatelet agent commonly used in the management of acute coronary syndromes and for secondary prevention of thrombotic events. However, it is not a glycoprotein IIb/IIIa inhibitor.

Solution for Question 10:

Correct answer: D) Dazoxiben is recently approved as an add-on drug to aspirin in myocardial infarction (MI).

Explanation:

Vorapaxar, not Dazoxiben, has been recently approved as an add-on drug to aspirin in MI.

Dazoxiben is still under trial for MI and has not been approved as an add-on drug.

Newer Anti-platelet Drugs:

Drug Name	Drug Type	Mechanism of Action	Approval Status
Vorapaxar	PAR (Protease Activating Receptor) inhibitor (Option A)	Blocks PAR receptor on platelets, preventing thrombin attachment and platelet aggregation (Option B)	Recently approved as an add-on drug to aspirin in MI
Dazoxiben	Thromboxane A2 synthase inhibitor (Option C)	Arachidonic Acid (via COX-1 enzyme) ↓ Prostaglandin G2 (via Thromboxane A2 Synthase) ↓ Thromboxane A2 (TxA2) Dazoxiben blocks Thromboxane A2 Synthase	Under trial for MI

Solution for Question 11:

Correct answer: C) Romiplostim

Explanation:

Romiplostim is a recombinant form of thrombopoietin given as a subcutaneous injection specifically for the treatment of idiopathic thrombocytopenic purpura (ITP). It acts by increasing platelet production by mimicking thrombopoietin & stimulating megakaryocytes in the bone marrow.

Treatment options for ITP:

- Observation for mild cases.
- Corticosteroids and IVIG for moderate to severe cases.
- Splenectomy for refractory ITP or when other treatments fail.

Pro-Platelet Drugs			
Drug Name	Drug Type	Route	Uses
Romiplostim	Recombinant form of thrombopoietin	Subcutaneous injection	Treatment of idiopathic thrombocytopenic purpura (ITP)
Eltrombopag	Thrombopoietin receptor agonist	Oral	Treatment of ITP and chronic liver disease
Oprelvekin	Recombinant form of IL-11	Subcutaneous injection	DOC in chemotherapy-induced thrombocytopenia

Vorapaxar (Option A): is an antiplatelet drug that inhibits the PAR-1 receptor, used to reduce thrombotic cardiovascular events, and is not indicated for increasing platelet count in ITP.

Oprelvekin (Option B): is a recombinant form of interleukin-11 (IL-11) used for the prevention of severe thrombocytopenia during chemotherapy. It is not the primary choice for ITP.

Abciximab (Option D): is a glycoprotein IIb/IIIa inhibitor used in acute coronary syndromes and during PCI to prevent platelet aggregation, and is not indicated for treating ITP.

Solution for Question 12:

Correct answer: A) Fostamatinib

Explanation:

Fostamatinib is approved for Chronic immune thrombocytopenia

Drug	Class	Use
Fostamatinib	BTK inhibitor	Chronic immune thrombocytopenia
Bosutinib (Option B)	Tyrosine kinase inhibitor	Chronic, accelerated, or blast-phase Ph+ CML with resistance or intolerance to prior therapy
Oprelvekin (Option C)	Thrombopoietic growth factor (recombinant IL-11)	Patients undergoing chemotherapy for non-myeloid malignancies with severe thrombocytopenia
Erlotinib (Option D)	EGFR inhibitor	Patients with advanced or metastatic Non-Small Cell Lung Cancer after failure of platinum-based chemotherapy treatment Newly diagnosed Non-Small Cell Lung Cancer patients with EGFR mutations In combination with gemcitabine for the treatment of patients with locally advanced, unresectable, or metastatic pancreatic cancer

Oral and Parenteral Anticoagulants

1. A 72-year-old woman with a history of atrial fibrillation and a recent deep vein thrombosis (DVT) is admitted to the hospital. The physician decides to initiate anticoagulation therapy with warfarin to prevent further thromboembolic complications. Which of the following best describes the mechanism of action of warfarin?

- A. Inhibits the synthesis of fibrin
- B. Inhibits platelet aggregation
- C. Inhibits the synthesis of vitamin K-dependent clotting factors
- D. Directly binds to and inactivates thrombin

2. A 58-year-old woman with a history of mitral valve replacement (on warfarin) and trigeminal neuralgia (on carbamazepine) has been prescribed clarithromycin for a chest infection. Her regular medications include amlodipine and atorvastatin. Her INR is 1.3 (target > 3 for mechanical heart valve). Which drug most likely interacted with warfarin to give this INR?

- A. Amlodipine
- B. Atorvastatin
- C. Clarithromycin
- D. Carbamazepine

3. An 82-year-old man presents to the emergency department with a sudden onset of severe headache and altered mental status. His CT Scan reveals an intracranial haemorrhage. He has a history of atrial fibrillation and has been taking warfarin. His INR is 5.8. Which of the following is the most appropriate treatment?

- A. Fresh Frozen Plasma
 - B. Tranexamic acid
 - C. Vitamin K
 - D. Prothrombin complex concentrate
-

4. A 68-year-old patient is being evaluated for the initiation of warfarin therapy for various indications. Which of the following indications for warfarin therapy requires the highest target INR range?

- A. Atrial fibrillation (non-valvular)
 - B. Deep vein thrombosis (DVT) prophylaxis
 - C. Mechanical heart valve replacement
 - D. Myocardial infarction (MI) prevention
-

5. A 72-year-old man has been taking Warfarin 7mg daily for the treatment of left arm DVT for the past 2 years. He also takes furosemide 40mg for HTN. He is usually compliant with his medications. What is the most important adverse effect the patient should be careful with?

- A. Peripheral Neuropathy
 - B. Phototoxicity
 - C. Headache
 - D. Constipation
-

6. Which oral anticoagulant is known for inhibiting activated Factor Xa and is used for prophylaxis and treatment of deep vein thrombosis (DVT)?

- A. Warfarin

- B. Dabigatran
 - C. Apixaban
 - D. Ethylbiscoumacetate
-

7. Which form of Vitamin K is used as an antidote for coumarin anticoagulant overdose, such as warfarin?

- A. Vitamin K1
 - B. Vitamin K2
 - C. Vitamin K3
 - D. Vitamin K4
-

8. Which drugs exert their anticoagulant effect primarily by binding to antithrombin III (AT-III) to inhibit factor Xa selectively?

- A. Lepirudin
 - B. Dabigatran
 - C. Fondaparinux
 - D. Heparin
-

9. Which of the following statements is not true regarding the features of various types of heparin?

- A. ULMW heparin has the shortest half-life.
- B. LMW heparin is primarily excreted through urine.
- C. HMW heparin requires monitoring of aPTT.
- D. Fondaparinux is a synthetic pentasaccharide heparin

10. A 65-year-old man who was admitted to the ICU due to septicemia started showing features of disseminated intravascular coagulation. He was then started on a parenteral anticoagulant injection. Which of the following anticoagulants was most probably used in this patient?

- A. Fondaparinux
 - B. Enoxaparin
 - C. Unfractionated heparin
 - D. All of the above
-

11. A 48-year-old man who was recently diagnosed with unstable angina was started on heparin. After 2 days, he developed 'Heparin-induced thrombocytopenia'. Which of the following types of heparin is not associated with HIT?

- A. Unfractionated heparin
 - B. Enoxaparin
 - C. Dalteparin
 - D. Fondaparinux
-

12. Which anticoagulant is preferred for treating patients with heparin-induced thrombocytopenia (HIT) due to its specific indication, effectiveness in avoiding cross-reactivity with heparin antibodies, and safety profile?

- A. Warfarin
- B. Fondaparinux
- C. Argatroban

D. Enoxaparin

13. A 28-year-old pregnant female who is in her first trimester was diagnosed with antiphospholipid syndrome. As a prophylaxis for thrombosis and risk of miscarriage, anticoagulant therapy was started. Which of the following anticoagulant agents is unsafe to use during the first trimester of pregnancy?

- A. Fondaparinux
 - B. Warfarin
 - C. Dalteparin
 - D. Unfractionated heparin
-

14. A 78-year-old male patient presents to the emergency department with features of acute deep vein thrombosis. The patient has a history of atrial fibrillation and chronic kidney disease. He is currently on anticoagulant therapy. Which set of anticoagulants is safe to use in this patient?

- A. Warfarin and Fondaparinux
 - B. Warfarin and unfractionated heparin
 - C. Unfractionated heparin and Dabigatran
 - D. Rivaroxaban and Fondaparinux
-

15. A 45-year-old man with atrial fibrillation who is on warfarin started having nosebleeds, gum bleeding and hematuria. His PT/INR was 6.2, and a diagnosis of warfarin toxicity was made. Immediate intervention with an antidote for warfarin was given. Match the following anticoagulants with their antidotes. Anticoagulants ANTIDOTE 1. Warfarin A. Protamine Sulfate 2. Heparin B. Andexanet Alfa 3. Dabigatran C. Vitamin K 4. Apixaban D. Idarucizumab

Anticoagulants	ANTIDOTE
1. Warfarin	A. Protamine Sulfate
2. Heparin	B. Andexanet Alfa
3. Dabigatran	C. Vitamin K
4. Apixaban	D. Idarucizumab

- A. 1-B, 2-C, 3-A, 4-D
 - B. 1-C, 2-A, 3-D, 4-B
 - C. 1-C, 2-D, 3-B, 4-A
 - D. 1-B, 2-A, 3-C, 4-D
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	4
Question 4	3
Question 5	3
Question 6	3
Question 7	1
Question 8	3
Question 9	1
Question 10	3
Question 11	4
Question 12	3
Question 13	2
Question 14	2
Question 15	2

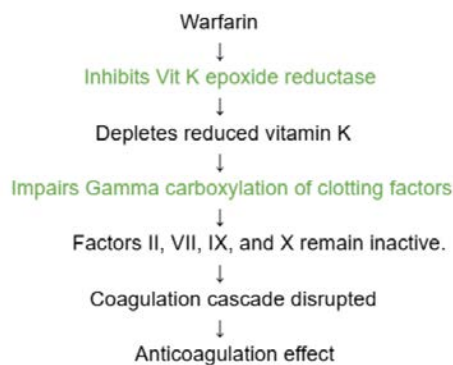
Solution for Question 1:

Correct answer: C) Inhibits the synthesis of vitamin K-dependent clotting factors

Explanation:

By inhibiting the synthesis of vitamin K-dependent clotting factors, warfarin reduces the risk of clot formation and helps prevent adverse events like stroke or recurrent DVT.

Mechanism of action:



Solution for Question 2:

Correct answer: D) Carbamazepine

Explanation:

Carbamazepine is a cytochrome P450 enzyme inducer. It will decrease the anticoagulant effects of warfarin (↓ INR).

Pharmacokinetics of Warfarin:

- Oral administration, zero-order kinetics.
- Excreted in bile, safe in renal failure.
- High protein binding (99% albumin): Not removed by hemodialysis.

Examples:

Enzyme Inhibitors (go to SICKFACES.COM)

Enzyme Inducers (want to CRAP GPS)

Sodium Valproate Isoniazid Cimetidine Ketoconazole Fluconazole Alcohol (Binge drinking) Chloramphenicol Erythromycin Sulfonamides Ciprofloxacin Omeprazole Metronidazole	Carbamazepine Rifampin Alcohol (Chronic use) Phenytoin Griseofulvin Phenobarbital Sulphonylureas
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Solution for Question 3:

Correct answer: D) Carbamazepine

Explanation:

- An intracranial haemorrhage in a patient on warfarin is considered a major bleed, necessitating the rapid reversal of anticoagulation.
- Prothrombin complex concentrate (PCC) is a concentrated source of clotting factors.
- PCC acts faster than fresh frozen plasma and vitamin K to restore the clotting factors and reduce the INR.

Warfarin and high INR management:

Major bleeding	Stop warfarin. Administer prothrombin complex concentrate. Administer intravenous vitamin K1. If prothrombin complex concentrate is not available, give fresh frozen plasma.
INR > 8 with minor or no bleeding	Stop warfarin. Administer intravenous or oral vitamin K1. Restart warfarin when INR is < 5.
INR 5 - 8 with minor bleeding	
INR 5 - 8 with no bleeding	Stop warfarin. Restart warfarin when INR is < 5.
INR above target range but < 5	Reduce or omit one or two doses of warfarin Measure INR in 2-3 days

Solution for Question 4:

Correct answer: C) Mechanical heart valve replacement

Explanation:

For patients with mechanical heart valve replacements, warfarin therapy requires a higher target INR range of 3.0 to 3.5. This higher intensity of anticoagulation is necessary to prevent thromboembolic complications associated with mechanical valves.

Use	Target INR Range
Atrial Fibrillation	2.0 - 2.5
Venous Thromboembolism (VTE) Prophylaxis	
Bioprosthetic Heart Valves (first 3 months)	
Post-Myocardial Infarction	
Treatment of DVT, PE, TIA	2.5 - 3.0
Mechanical Prosthetic Heart Valves	3.0 - 3.5
Recurrent Thromboembolism	

Solution for Question 5:

Correct answer: C) Headache

Explanation:

The most important adverse effect of warfarin is headache, which may result from intracranial haemorrhage. Patients should be counselled to report any unusual or persistent headaches while taking Warfarin. HTN is also a risk factor for intracranial haemorrhage.

Adverse Effects:

- Bleeding: Ecchymosis (bruising), Epistaxis, Hematuria, Gastrointestinal bleeding, Intracranial or other internal haemorrhages, which can be fatal.
- Frequent gastrointestinal disturbances
- Oral ulceration
- Dermatitis
- Urticaria (hives)
- Alopecia (hair loss)

Contraindications for Warfarin:

- Bleeding disorders.
- Severe hypertension (risk of cerebral haemorrhage), threatened abortion, piles, gastrointestinal ulcers (risk of aggravated bleeding).
- Subacute bacterial endocarditis (risk of embolism), large malignancies (risk of bleeding in the central necrosed area of the tumour), tuberculosis (risk of hemoptysis).
- Ocular and neurosurgery, lumbar puncture.
- Chronic alcoholics, cirrhosis, renal failure.
- Aspirin and other antiplatelet drugs should be used very cautiously during warfarin therapy.
- Early Pregnancy: Increases birth defects, especially skeletal abnormalities; can produce fetal warfarin syndrome (stippled epiphysis, eye socket, hand bones, and growth retardation).
- Later in Pregnancy: Can cause CNS defects, fetal haemorrhage, fetal death, and accentuates neonatal hypoprothrombinemia.

Solution for Question 6:

Correct answer: C) Apixaban

Explanation:

Apixaban, an oral anticoagulant, directly inhibits activated Factor Xa, thereby reducing clot formation by preventing the conversion of prothrombin to thrombin.

NOVEL ORAL ANTICOAGULANTS:

Category	Direct Factor Xa Inhibitors	Direct Thrombin Inhibitors
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Mechanism of action	Inhibit Factor Xa directly	Inhibit thrombin (Factor IIa) directly
Drugs	Apixaban, Rivaroxaban, Edoxaban, Betrixaban	Dabigatran, Ximelagatran
Uses	Prophylaxis of venous thromboembolism following knee or hip replacement surgery. Prophylaxis of stroke in patients with atrial fibrillation. Treatment of deep vein thrombosis and pulmonary embolism.	
Advantages of novel anticoagulants over warfarin	Rapid onset and offset of action. No need for routine laboratory monitoring. Lower risk of bleeding. Fewer drug interactions.	

Solution for Question 7:

Correct answer: A) Vitamin K1

Explanation:

Vitamin K1 (phytonadione) is an antidote for coumarin anticoagulant overdose, such as warfarin, by promoting the synthesis of clotting factors in the liver.

VITAMIN K:

Forms	Use	Adverse Effects
Vitamin K1 (Phytonadione)	Antidote for coumarin anticoagulant overdose (e.g., warfarin). Prophylaxis and treatment of Vitamin K deficiency bleeding. Newborns to prevent hemorrhagic disease of the newborn.	Severe anaphylactoid reactions on IV injection.
Vitamin K2 (Menaquinone) (Option B)	Bone metabolism and Calcium regulation in bones and arteries. Used in osteoporosis.	Generally considered safe. Very few reported adverse effects.

Vitamin K3 (Menadione) (Option C)	Not commonly used for anticoagulant reversal.	Hemolysis (especially in G6PD deficiency). Risk of kernicterus in newborns.
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Solution for Question 8:

Correct answer: C) Fondaparinux

Explanation:

Fondaparinux is a synthetic pentasaccharide that exerts its anticoagulant effect by binding to antithrombin III (AT-III) and selectively inhibiting factor Xa through enhanced AT-III activity. It does not directly inhibit thrombin (factor IIa), making it a specific factor Xa inhibitor.

Type	Mechanism of Action	Drugs
Direct Thrombin Inhibitors	Inhibit thrombin (factor IIa) directly without cofactors.	Hirudin derivatives: Hirudin, Lepirudin.
Non-hirudin (Synthetic): Argatroban, Bivalirudin.		
Indirect Thrombin Inhibitors	Enhance antithrombin III activity to inhibit thrombin indirectly.	Unfractionated Heparin: Heparin Long chain of molecule bind to both factor 2 and 10
Low Molecular Weight Heparin (LMWH): Enoxaparin, Dalteparin, Tinzaparin Long chain of molecule binds to factor 10 > 2		
Ultra Low Molecular Weight Heparin (ULMWH): Fondaparinux, Idraparinux, Danaparoid (not commonly used) Long chain of molecule bind selectively to factor 2		
Synthetic Direct Factor Xa Inhibitors	Bind directly to factor Xa to inhibit its activity.	Fondaparinux, Idraparinux

Solution for Question 9:

Correct answer: A) ULMW heparin has the shortest half-life.

Explanation:

ULMW (Ultra-low molecular weight) heparin has the longest half-life among the three types (HMW, LMW, and ULMW).

Type of Heparin	Mechanism of Action	Features
High Molecular Weight (HMW) Heparin (unfractionated)	Binds to antithrombin III, inhibits thrombin and factor Xa.	Derived from porcine intestine or bovine lung. Requires monitoring of aPTT Shorter half-life. Safe in renal failure
Low Molecular Weight (LMW) Enoxaparin, Dalteparin	Preferentially inhibits factor Xa.	Derived from LMW fragments of unfractionated heparin. Primarily renal excretion Monitoring of anti-factor Xa levels.
Ultra Low Molecular Weight (ULMW) Fondaparinux	Selectively inhibits factor Xa.	Synthetic pentasaccharide Longer half-life. No routine monitoring is needed for coagulation parameters.

Solution for Question 10:

Correct answer: C) Unfractionated heparin

Explanation:

Unfractionated heparin is the drug of choice for Disseminated intravascular coagulation (DIC), as it has

- Rapid onset of action is crucial during the management of DIC.
- A short half-life helps in quick cessation of action if needed.
- Toxicity can be easily reversed with protamine sulfate.

- The dose can be titrated by monitoring aPTT.

	Uses	Contraindication
HMWH Unfractionated heparin (Option C)	Disseminated intravascular coagulation Hemodialysis Atrial fibrillation Maintain patency of cannulae and shunts.	Any major bleed Thrombocytopenia Heparin-induced thrombocytopenia Cerebrovascular accident Aneurysm Recent CNS surgery or ophthalmic surgery Severe renal impairment (except for unfractionated heparin) GIT ulcer Hemophilia Severe uncontrolled hypertension Bacterial endocarditis Hypersensitivity to heparin Large malignancies ... (content truncated)
LMWH Dalteparin Enoxaparin (Option B) Tinzaparin	Prophylaxis of Deep vein thrombosis & Pulmonary embolism Treatment of Acute coronary syndrome & Antiphospholipid syndrome.	
ULMWH Fondaparinux (Option A) Idraparinux		

Solution for Question 11:

Correct answer: D) Fondaparinux

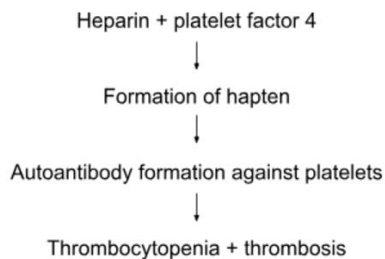
Explanation:

Fondaparinux is a synthetic ultra-low molecular weight heparin that is not associated with heparin-induced thrombocytopenia.

Adverse effects of heparin:

- Hemorrhage due to heparin toxicity 1st sign is hematuria, which can affect the GIT, CVS and joints. Antidote - Protamine sulfate (most effective against UFH and least effective against ULMWH).

- 1st sign is hematuria, which can affect the GIT, CVS and joints.
- Antidote - Protamine sulfate (most effective against UFH and least effective against ULMWH).
- Heparin-induced thrombocytopenia
- 1st sign is hematuria, which can affect the GIT, CVS and joints.
- Antidote - Protamine sulfate (most effective against UFH and least effective against ULMWH).



- Serotonin assay shows elevated levels of serotonin (released from platelet)
- Management - discontinue heparin + start on direct thrombin inhibitor (argatroban, lepirudin)
- All types of heparin except for Fondaparinux cause HIT

Other adverse effects of heparin:

- Hyperkalemia (due to aldosterone synthesis inhibition)
- Osteoporosis
- Transient and reversible alopecia
- Hypersensitivity reaction (rigor, chills, fever and anaphylaxis)

Solution for Question 12:

Correct answer: C) Argatroban

Explanation:

- Argatroban is a direct thrombin inhibitor that binds directly to the active site of thrombin, inhibiting its activity.
- Unlike heparin, which enhances the activity of antithrombin III (AT-III), argatroban does not require AT-III for its anticoagulant effect.

- This makes it effective in patients with heparin-induced thrombocytopenia (HIT), where antibodies to heparin-platelet factor 4 complexes cause immune-mediated thrombosis.

Aspect	Argatroban
Mechanism of Action (MOA)	It is a direct thrombin inhibitor that binds reversibly to the catalytic site of thrombin, inhibiting its activity. It prevents thrombin-mediated conversion of fibrinogen to fibrin and inhibits platelet activation and aggregation.
Adverse Effects	Bleeding (ranging from minor to severe haemorrhage). Hypersensitivity reactions. Elevated liver enzymes. Local irritation at the injection site.
Uses	Prophylaxis and treatment of thrombosis in patients at risk of HIT.
Contraindications	Active major bleeding Hypersensitivity to Argatroban. Severe hepatic impairment. Concomitant use with strong CYP3A4 inhibitors.

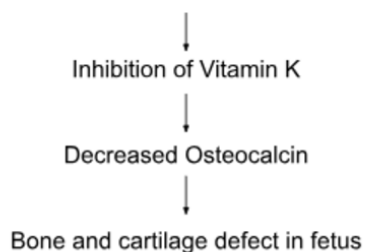
Solution for Question 13:

Correct answer: B) Warfarin

Explanation:

- Warfarin is contraindicated in the first trimester of pregnancy as it is teratogenic, leading to Congenital warfarin syndrome.

Warfarin crosses placenta and enters the fetus



- If used early in pregnancy Hypoplasia of the nose (saddle nose), eye sockets. Hypoplasia of limbs. Growth retardation.
- Hypoplasia of the nose (saddle nose), eye sockets.

- Hypoplasia of limbs.
- Growth retardation.
- Hypoplasia of the nose (saddle nose), eye sockets.
- Hypoplasia of limbs.
- Growth retardation.
- And if used later in pregnancy, it can lead to CNS defects. Fetal haemorrhage. Neonatal hypoprothrombinemia.
- CNS defects.
- Fetal haemorrhage.
- Neonatal hypoprothrombinemia.
- CNS defects.
- Fetal haemorrhage.
- Neonatal hypoprothrombinemia.
- Heparin is safe to use during pregnancy since it does not cross the placenta, and it is employed in conditions such as antiphospholipid syndrome, deep vein thrombosis, and pulmonary embolism.
- Novel oral anticoagulants (NOAC) such as Dabigatran and Apixaban are also not preferred during pregnancy or lactation.

Gestational period	Anticoagulants preferred
0-12 weeks	LMW/ULMW heparin
12-36 weeks	LMW/ULMW heparin As organogenesis is completed, Warfarin may be used only in cases of patients with mechanical prosthetic valves
36 weeks till delivery	LMW/ULMW heparin
During delivery	No anticoagulant is used due to the risk of postpartum hemorrhage
During lactation	Warfarin is preferred over heparin

Solution for Question 14:

Correct answer: B) Warfarin and unfractionated heparin

Explanation:

Warfarin and unfractionated heparin are safe for patients with chronic kidney disease.

Drug	Excretion	Safety in renal disease
Warfarin (Vitamin K antagonist)	Hepatic excretion	Safe
Unfractionated heparin (High molecular weight heparin)	Reticuloendothelial cells and hepatic excretion	Safe
Fondaparinux (Ultra-low molecular weight heparin)	Renal excretion	Unsafe
Dabigatran (Direct thrombin inhibitor)	Renal excretion	Unsafe
Rivaroxaban (Direct factor Xa inhibitor)	Renal and hepatic excretion	Unsafe

Solution for Question 15:

Correct answer: B) 1-C, 2-A, 3-D, 4-B

Explanation:

Anticoagulant	Antidote	Mechanism
Warfarin	Vitamin K	Vitamin K helps reverse the warfarin-induced inhibition of Vitamin K epoxide reductase, resulting in ↑ carboxylation and activation of clotting factors II, VII, IX, and X.
Heparin	Protamine Sulfate	It neutralises and prevents the binding of heparin and antithrombin III.
Dabigatran	Idarucizumab	It is a monoclonal antibody that binds to dabigatran and neutralises its anticoagulant activity.

Apixaban	Andexanet Alfa	It acts as a decoy receptor for apixaban, binds to apixaban, and reverses its anticoagulant effects.
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Hematinics, Iron, and Chelators

1. Which of the following features of iron dextran is NOT true?

- A. It can be administered intravenously (IV) or intramuscularly (IM).
 - B. It is useful for patients with chronic gastrointestinal blood loss.
 - C. Staining of subcutaneous fat can occur with IM injections
 - D. Allergy testing is not required before administration.
-

2. A 55-year-old male patient with CKD presents complaining of persistent fatigue and weakness. Labs reveal anaemia. After assessing the patient's medical history and conducting further investigations, the physician decides to prescribe an erythropoiesis-stimulating agent to manage the anaemia associated with CKD. Which of the following is the longest-acting erythropoietin analogue?

- A. Epoetin alfa
 - B. Darbepoetin
 - C. Peginesatide
 - D. Daprodustat
-

3. A 45-year-old male patient with non-Hodgkin's lymphoma is undergoing high-dose chemotherapy. Post-chemotherapy, he develops severe neutropenia. His oncologist decides to use a hematopoietic drug to stimulate neutrophil production. Which of the following treatments is most appropriate for managing his condition?

- A. Erythropoietin
- B. Vitamin B12 and folic acid
- C. Filgrastim

D. Plerixafor

4. A 60-year-old male patient with chronic hepatitis C is scheduled to undergo a liver biopsy. He has a history of thrombocytopenia, with a platelet count of 45,000/ μ L, which poses a risk for bleeding during the procedure. His hepatologist decides to use a pharmacological agent to increase his platelet count before the surgery. Which of the following treatments is most appropriate to achieve this goal?

- A. Romiplostim
 - B. Eltrombopag
 - C. Erythropoietin
 - D. IVIG (Intravenous Immunoglobulin)
-

5. A 35-year-old female presents with symptoms of fatigue, pallor, and shortness of breath. Laboratory tests reveal a haemoglobin level of 9 g/dL and microcytic hypochromic red blood cells, consistent with iron deficiency anemia. She is prescribed oral ferrous sulfate. What is the earliest sign that indicates a positive response to iron therapy?

- A. An increase in serum ferritin levels
 - B. A rise in haemoglobin levels
 - C. An increase in reticulocyte count
 - D. Improvement in symptoms of fatigue
-

6. Match the diseases with their respective chelator of choice for treatment:
Poisoning Chelators
1. Bismuth poisoning A. BAL (Dimercaprol)
2. Copper poisoning B. Penicillamine
3. Iron poisoning in thalassemia C. Calcium Disodium

EDTA 4. Cadmium poisoning D. Deferoxamine 5. Acute iron poisoning E. Deferiprone

Poisoning	Chelators
1. Bismuth poisoning	A. BAL (Dimercaprol)
2. Copper poisoning	B. Penicillamine
3. Iron poisoning in thalassemia	C. Calcium Disodium EDTA
4. Cadmium poisoning	D. Deferoxamine
5. Acute iron poisoning	E. Deferiprone

- A. 1-A, 2-B, 3-E, 4-C, 5-D
 - B. 1-D, 2-C, 3-E, 4-B, 5-A
 - C. 1-B, 2-D, 3-E, 4-A, 5-C
 - D. 1-A, 2-C, 3-E, 4-D, 5-B
-

7. Which of the following does not reduce oral iron absorption?

- A. Tea
 - B. Coffee
 - C. Pantoprazole
 - D. Amla
-

8. Which of the following statements on thrombolytics is false?

- A. Activates plasminogen to plasmin
- B. Acts via tissue plasminogen activator
- C. Contraindicated in pregnancy
- D. Can be used in NSTEMI

9. Which of the following statements regarding Epsilon Amino-Caproic Acid (EACA) is true?

- A. It is an antidote for fibrinolytic agents
 - B. Promotes clot dissolution
 - C. Activates plasminogen
 - D. It is an example of a thrombolytic agent
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	3
Question 3	3
Question 4	2
Question 5	3
Question 6	1
Question 7	4
Question 8	4
Question 9	1

Solution for Question 1:

Correct answer: D) Allergy testing is not required before administration.

Explanation:

- Allergy testing needed before iron dextran due to severe adverse reactions risk.
- More prone to hypersensitivity reactions than newer formulations.
- Not recommended for patients with rheumatoid arthritis or connective tissue diseases.

Indications for parenteral iron therapy:

- Intolerance to oral iron (e.g., severe GI upset)
- Inability to absorb oral iron (e.g., malabsorption syndromes)
- Non-compliance with oral iron therapy
- Severe iron deficiency anemia with chronic bleeding
- Need for rapid iron repletion (e.g., CKD patients on erythropoietin therapy)

Parenteral Iron Formulation & ROA	Anaphylaxis Risk	Uses	Disadvantages
Iron dextran (IV, IM)	Yes	Useful for patients with chronic gastrointestinal blood loss.	Subcutaneous fat staining with IM injections (Z-track technique helps prevent staining) Allergy testing required before administration Not preferred in rheumatoid arthritis or connective tissue diseases
Iron sorbitol citrate (IM)	Yes		
Iron sucrose (IV)	No	Safer than iron dextran. Indicated for anemia in patients with kidney disease.	Total dose IV infusion not possible Reports of kidney damage exist Avoid oral iron concurrently and for 5 days after the last injection.
Ferric gluconate (IV)	No	About 80% is delivered to transferrin within 24 hours Safer than iron dextran	

Parenteral Iron Dose Calculation:

The Ganzoni formula can be used to calculate the total iron dose needed for a patient:

Total iron dose = Weight (kg) x [15 - Current Hb (g/dL)] x 2.4 + 500 (to replenish stores)

Where: Hb deficit = 15 - Current Hb (g/dL)

Solution for Question 2:

Correct answer: C) Peginesatide

Explanation: Peginesatide is the longest acting erythropoietin analogue.

Hematinics that increase the RBC production:

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Solution for Question 3:

Correct answer: C) Filgrastim

Explanation: Filgrastim (Recombinant human G-CSF) treats severe neutropenia post-autologous hematopoietic stem cell transplantation and high-dose cancer chemotherapy.

Drug Name	Mechanism of Action	Clinical Uses
G-CSF (Glycoprotein with 175 amino acids)	Stimulates colony forming unit (CFU) granulocytes to increase neutrophil production. Enhances phagocytic and cytotoxic activities of neutrophils	Treats severe neutropenia after autologous hematopoietic stem cell transplantation and high-dose cancer chemotherapy Treats severe congenital neutropenias
Filgrastim (Recombinant human G-CSF) (Option C)	Same use as G-CSF Treats neutropenia in patients with AIDS receiving zidovudine Used in patients undergoing peripheral blood stem cells collection for stem cell transplantation	
PEG-Filgrastim (Longer acting pegylated form of G-CSF)	Prevention of chemotherapy-induced neutropenia	
Sargramostim (Recombinant human GM-CSF)	Stimulates myelopoiesis	Autologous marrow transplantation treatment Mobilizes CD34+ progenitor cells for peripheral blood stem cell collection post-myeloablative chemotherapy Treats neutropenia in patients on intensive cancer chemotherapy Supports myelopoiesis in cyclic neutropenia, myelodysplasia, aplastic anemia, and AIDS-associated neutropenia

Molgramostim	Targets GM-CSF receptors on bone marrow precursor cells	Increases granulocytes and monocytes Approved for chemotherapy-induced neutropenia
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Plerixafor: Mobilizes hematopoietic stem cells to peripheral blood (used with G-CSF, not alone for neutropenia)

Solution for Question 4:

Correct answer: B) Eltrombopag

Explanation: Eltrombopag is a thrombopoietin receptor agonist that stimulates platelet production and is specifically approved for use in patients with chronic hepatitis C-associated thrombocytopenia to increase platelet counts and reduce the risk of bleeding.

- Eltrombopag is preferred over Romiplostim in this scenario

Drug Name & ROA	Mechanism of Action	Clinical Use
Thrombopoietin (SC)	Stimulates platelet production: Binds to thrombopoietin receptors on bone marrow megakaryocytes	Used for ITP treatment: Abandoned due to antibody development
Romiplostim (SC) (Option A)	Thrombopoietin receptor agonist: Mimics effects of thrombopoietin	Treats ITP and chronic liver disease Approved for liver disease patients undergoing surgery
Eltrombopag (Oral) (Option B)		
Fostamatinib (Oral)	Tyrosine kinase inhibitor; Reduces inflammation, indirectly increases platelet production	New drug for the treatment of ITP.
Oprelvekin -IL-11 (SC)	Recombinant IL-11; Same MOA as thrombopoietin	Treats chemotherapy-induced thrombocytopenia For non-myeloid malignancy patients undergoing chemotherapy with severe thrombocytopenia

IVIg (Option D) can be used to increase platelet counts in certain conditions like ITP, but it is not a first-line treatment for thrombocytopenia associated with chronic hepatitis C and has a more

transient effect.

Solution for Question 5:

Correct answer: C) An increase in reticulocyte count

Explanation:

- Early response to iron therapy: Increased reticulocyte count within days to a week.
- Reticulocytes rise as bone marrow produces more red blood cells.

Oral Iron	
Characteristic	Information
Factors Increasing Absorption	Best in acidic environment, take on empty stomach Taking iron with vitamin C increases absorption
Factors Decreasing Absorption	PPIs and antacids lowers absorption Tetracycline binds to iron, reducing absorption Food, tea, coffee contain phytates and tannins that decrease iron absorption Calcium reduces iron absorption by forming insoluble complexes with non-heme iron in the intestine,
Side Effects	Nausea, vomiting, diarrhea, constipation Black stools and tongue indicate regular iron intake
Indications for Parenteral Iron	Severe GI irritation or poor compliance: Start parenteral iron
Iron Content of Preparations	Ferrous sulfate (most common, DOC for iron deficiency anemia): 32% iron. Ferrous gluconate: 12% iron. Ferrous fumarate (maximum iron content): 33% iron. Ferrous sulfate hydrate: 20% iron. Ferrous ascorbate - new preparation containing Vitamin C (↑ iron absorption)
Earliest Response to Therapy	Increase in reticulocyte count: Earliest response to iron therapy Hemoglobin levels increase by 0.5-1% per week in microcytic anemia Oral and parenteral iron equally effective for treating iron deficiency anemia

Solution for Question 6:

Correct answer: A) 1-A, 2-B, 3-E, 4-C, 5-D

Explanation:

- Chelators form stable complexes with metallic ions, used mainly in heavy metal poisonings.
- They bind to toxic metals, preventing interaction with essential biomolecules.
- Chelation involves ring structure formation within the molecule.

Effective chelating agents possess several key properties:

- Higher affinity for toxic metals: Must outcompete body ligands for binding.
- Greater affinity for toxic metals than calcium: Crucial due to calcium abundance in plasma.
- Appropriate distribution: Aligns with target metal distribution in the body.
- Water solubility: Essential for effective chelation and excretion.

Chelator	Uses	Other Information
Dimercaprol (BAL)	Arsenic poisoning Mercury poisoning Gold poisoning Bismuth poisoning Nickel poisoning Antimony poisoning Lead poisoning (as adjuvant to Calcium Disodium EDTA) Wilson's disease (as adjuvant to Penicillamine)	C/I in iron and cadmium poisoning because the complexes are toxic.
Calcium Disodium EDTA	Lead poisoning Zinc poisoning Copper poisoning Manganese poisoning Cadmium poisoning Radioactive metal poisoning	DOC for Manganese, lead, Cadmium poisoning Not effective in mercury poisoning due to mercury's tight binding and localization in deep tissues (inaccessible) Doesn't enter brain or CSF, removes toxic metals from accessible sites only.

Penicillamine	Wilson's disease Copper poisoning Mercury poisoning Lead poisoning (as adjuvant to Calcium Disodium EDTA) Cystinuria and cystine stones Scleroderma	DOC for copper poisoning Alternative drug to dimercaprol/succimer for mercury poisoning.
Desferoxamine	Acute iron poisoning Transfusion siderosis	Given parenterally. Not preferred in chronic iron overload (Deferiprone is preferred)
Deferiprone	Iron poisoning Transfusion siderosis in thalassemia patients	Oral drug. Less effective than desferoxamine for iron overload due to repeated blood transfusion.
Dimercaptosuccinic acid (Succimer)	Arsenic poisoning Mercury poisoning Lead poisoning	May chelate some essential metals like copper and zinc.

Solution for Question 7:

Correct answer: D) Amla

Explanation:

- Amla (Vitamin C): Vitamin C (ascorbic acid) enhances iron absorption by reducing ferric iron (Fe^{3+}) to ferrous iron (Fe^{2+}), which is more readily absorbed in the gastrointestinal tract. This makes Amla (Vitamin C) beneficial for iron absorption.
- Non-heme iron in food is primarily found in the ferric state (Fe^{3+}), which needs to be converted to the ferrous state (Fe^{2+}) to be absorbed effectively.
- Iron salts have low solubility at the neutral pH of the small intestine, but in the stomach's acidic environment, iron dissolves and attaches to a protein carrier.
- Vitamin C releases iron from the carrier, creates soluble iron-ascorbate complexes, and converts ferric iron (Fe^{3+}) to the more absorbable ferrous iron (Fe^{2+}).

Tea (Option A) contains tannins that bind to iron, forming insoluble complexes that cannot be absorbed by the body. Thus, tea reduces iron absorption.

Coffee (Option B) contains polyphenols that inhibit iron absorption by binding to iron and forming insoluble complexes.

Pantoprazole (Option C) is a proton pump inhibitor (PPI) that reduces stomach acidity. An acidic environment in the stomach is crucial for iron absorption, and by reducing stomach acid, pantoprazole decreases iron absorption.

Solution for Question 8:

Correct answer: D) Can be used in NSTEMI

Explanation:

- Thrombolytics are contraindicated in NSTEMI, it is used only in STEMI

Fibrinolytic Drugs (Thrombolytics):

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Solution for Question 9:

Correct answer: A) It is an antidote for fibrinolytic agents

Explanation:

- Epsilon Amino-Caproic Acid (EACA) is used as an antidote for fibrinolytic agents.

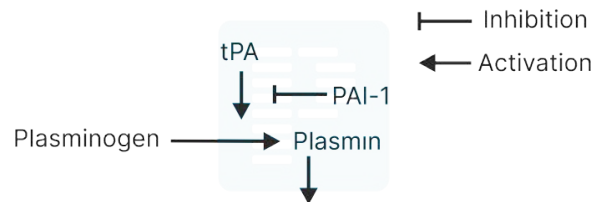
Antifibrinolytic (Option D) drugs:

Mechanism	Inhibits plasminogen activation (Option C)→plasmin not synthesized→fibrin not lysed→prevents clot dissolution (Option B)
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Examples and its uses	Epsilon Amino-Caproic Acid (EACA) Antidote for fibrinolytic agents Hyperplasmaemic states with excessive fibrinolysis Adjunctive in hemophiliacs for bleeding control (e.g., tooth extraction, prostatectomy, trauma) Tranexamic Acid Prevention/control of bleeding due to fibrinolytic drugs Cardiopulmonary bypass surgery Tonsillectomy, prostatic surgery, tooth extraction in hemophiliacs Menorrhagia, recurrent epistaxis, hyphema due to ocular trauma, peptic ulcer
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Previous Year Questions

1. Which drug acts on the rounded up enzyme?



- A. Aminocapric acid
- B. Tenecteplase
- C. Streptokinase
- D. Apixaban

2. What is the mechanism of action (MOA) of Rivaroxaban?

- A. Inhibition of thrombin (Factor IIa)
- B. Inhibition of Factor Xa
- C. Activation of antithrombin III
- D. Inhibition of platelet aggregation

3. Which among the following anticoagulants requires aPTT monitoring?

- A. Lepirudin
- B. Enoxaparin
- C. Dabigatran
- D. Apixaban

4. What is the mode of action of the medication administered to a patient with hyperlipidemia who subsequently develops gallstones?

- A. Inhibits HMG CoA reductase
 - B. Activates PPAR alpha
 - C. Binds to deoxycholic acid
 - D. Decrease VLDL
-

5. Which of the statements below correctly describes pegylated filgrastim?

- A. Duration of action is shorter than normal filgrastim
 - B. Added advantage is that it can be taken orally
 - C. It should not be administered in sickle cell patients
 - D. It is not effective in the treatment of severe neutropenia
-

6. In a 35-year-old female with sickle cell disease who presents with unprovoked pain in her extremities and torso, thromboembolic events are suspected. What is the initial medication that should be administered for prophylaxis of thromboembolic events in this patient?

- A. LMWH
 - B. Warfarin
 - C. Aspirin
 - D. Dabigatran
-

7. A patient presenting with purpura is diagnosed with isolated thrombocytopenia. Tyrosine kinase inhibitor used in isolated thrombocytopenia is?

- A. Oprelvekin
 - B. Fostamatinib
 - C. Eltrombopag
 - D. Romiplostim
-

8. In patients with indications for warfarin usage, what is the recommended range of INR in a patient with mechanical heart valves?

- A. 1.0-1.5
 - B. 2.0-3.0
 - C. 2.5-3.5
 - D. 3.5-4.5
-

9. For the action of heparin as an anticoagulant, availability of which of the following chemical substances is needed?

- A. Thrombin
 - B. Plasmin
 - C. Antithrombin – III
 - D. Thrombomodulin
-

10. A first-trimester pregnant female has DVT and needs anticoagulation. Which of the following anticoagulants is preferred?

- A. Enoxaparin
 - B. Clopidogrel
 - C. Betrixaban
 - D. Warfarin
-

11. A child accidentally took a bottle full of iron tablets. Which drug is used as antidote for poisoning?

- A. Luspatercept
 - B. Deferoxamine
 - C. Deferiprone
 - D. DMSA
-

12. A patient with history of ischemic stroke was started on clopidogrel. However, she had another attack of stroke after 6 months. Which of the following could be responsible for the failure of clopidogrel in this patient?

- A. Reduced activity of CYP2C19
 - B. Downregulation of CYP2D6
 - C. Upregulation of CYP1A1
 - D. Downregulation of CYP2E1
-

13. LMWH is given by which route?

- A. Intravenous

- B. Subcutaneous
 - C. Intramuscular
 - D. Intradermal
-

14. What is the antidote for fibrinolytic therapy?

- A. Vitamin K
 - B. Dalteparin
 - C. EACA
 - D. Alteptase
-

15. Which drug directly inhibits clotting factor Xa among the options provided?

- A. Apixaban
 - B. Argatroban
 - C. Fondaparinux
 - D. Aspirin
-

16. Which of the following is used in acute mountain sickness?

- A. Dexamethasone
 - B. Digoxin
 - C. Diltiazem
 - D. Acetazolamide
-

17. A patient with deep vein thrombosis was started on a new drug. After 2 days, he presented with the given finding. Which of the following drugs is implicated in causing the above condition?



- A. Warfarin
 - B. Heparin
 - C. Dabigatran
 - D. Rivarox
-

18. To which class of drugs does Apixaban belong in?

- A. Antifibrinolytic
 - B. Factor Xa inhibitors
 - C. Oral direct thrombin inhibitor
 - D. Parenteral direct thrombin inhibitor
-

19. Which of the following statements accurately describes prasugrel?

- A. It acts as an P2Y₁₂ agonist
- B. It is 10 times less potent than clopidogrel

- C. It is a pro drug
 - D. Omeprazole interferes in its activation to an active metabolite
-

20. Which drugs require monitoring of the coagulation profile during therapy?

- A. Fondaparinux
 - B. Enoxaparin
 - C. Dabigatran
 - D. Lepirudin
-

21. Which one of the following drugs inhibits neutrophil recruitment?

- A. Montelukast
 - B. Sodium cromolyn
 - C. Colchicine
 - D. Febuxostat
-

22. Which of the following is an oral factor Xa inhibitor?

- A. Bivalirudin
 - B. Dabigatran
 - C. Rivaroxaban
 - D. Enoxaparin
-

23. What is the suggested follow-up treatment for a patient post valve replacement?

- A. Warfarin
 - B. ACE inhibitors
 - C. Beta blockers
 - D. Thiazides
-

24. Which of the options listed below is an oral medication that directly inhibits thrombin?

- A. Dabigatran
 - B. Rivaroxaban
 - C. Warfarin
 - D. Lepirudin
-

25. A young boy with a history of multiple blood transfusions presents with bone pain for the past 6 months. A peripheral smear showed sickle- shaped RBCs. Which of the following drugs are not used in the treatment of his condition?

- A. Bebtelovimab
 - B. Voxelotor
 - C. Hydroxyurea
 - D. L-glutamine
-

26. Which of the following statements about statins is incorrect?

- A. Although HMG-CoA reductase inhibitors substantially reduce the risk of cardiovascular events, there is mild increase in lipoprotein a (Lpa) levels.

- B. These drugs should not be stopped even in severe conditions like injury, surgery etc.
 - C. They can be given with verapamil and other enzyme inhibitors
 - D. With the long term use, there is slight increase in the incidence of type 2 diabetes mellitus.
-

27. Which one of the following medications can be administered to this patient undergoing chemotherapy for the management of decreased platelet counts observed after the last cycle of treatment?

- A. Oprelvekin (IL-11)
 - B. Filgrastim
 - C. Erythropoietin
 - D. Amifostine
-

28. What is the mechanism of action of dicumarol?

- A. Activates antithrombin III .
 - B. Inhibits vitamin K dependant Clotting factors
 - C. Inhibits tissue plasminogen activator
 - D. Inhibitor of factor Xa
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	2
Question 5	3
Question 6	1
Question 7	2
Question 8	3
Question 9	3
Question 10	1
Question 11	2
Question 12	1
Question 13	2
Question 14	3
Question 15	1
Question 16	4
Question 17	1
Question 18	2
Question 19	3
Question 20	4
Question 21	3
Question 22	3

Question 23	1
Question 24	1
Question 25	1
Question 26	3
Question 27	1
Question 28	2

Solution for Question 1:

Correct Option A - Aminocaproic Acid:

- Aminocaproic acid is an antifibrinolytic agent that works by inhibiting plasminogen activation, thereby preventing the conversion of plasminogen to plasmin, which is responsible for fibrin clot breakdown. PAI-1 (Plasminogen Activator Inhibitor-1) inhibits tissue plasminogen activator (tPA), reducing plasmin formation. Aminocaproic acid effectively counteracts excessive fibrinolysis and stabilizes clots by preventing plasmin formation, indirectly related to the action of PAI-1.

Incorrect Options:

Option B - Tenecteplase:

- Tenecteplase is a fibrinolytic (thrombolytic) agent that works by directly activating plasminogen to plasmin, which breaks down fibrin clots. Tenecteplase does not act on PAI-1; instead, its action would be opposed by PAI-1 because PAI-1 inhibits tissue plasminogen activator (tPA), reducing the activation of plasminogen.

Option C - Streptokinase:

- Streptokinase is another fibrinolytic agent that works by forming a complex with plasminogen, leading to the formation of plasmin and subsequent breakdown of clots. Like tenecteplase, streptokinase's action is not directly on PAI-1, but rather on plasminogen, and its activity can be inhibited by PAI-1.

Option D - Apixaban:

- Apixaban is an anticoagulant that works by directly inhibiting Factor Xa, an essential enzyme in the coagulation cascade. Apixaban does not interact with the fibrinolytic system or PAI-1, and its mechanism of action is unrelated to PAI-1 or plasminogen activation.

Solution for Question 2:

Correct Option B - Inhibition of factor Xa:

- The mechanism of action of Rivaroxaban is by inhibition of factor Xa.

Incorrect options

Options A, C, D:

- Inhibition of thrombin (factor IIa), activation of antithrombin III, and inhibition of platelet aggregation are not the mechanism of action of Rivaroxaban
-

Solution for Question 3:

Correct Option A - Lepirudin:

- Lepirudin is a recombinant form of hirudin. Desirudin and lepirudin are medications used to treat thrombosis in patients with heparin-induced thrombocytopenia. They are eliminated by the kidneys, with a half-life of about 2 hours after subcutaneous administration and approximately 10 minutes after intravenous infusion. Caution should be exercised when using these drugs in patients with decreased renal function, and daily monitoring of serum creatinine and aPTT levels is recommended.

Incorrect Options:

Option B - Enoxaparin: This is a low molecular weight heparin (LMWH). LMWH usually produces a predictable anticoagulant response, so routine monitoring is not needed. However, patients with renal impairment, obesity, pregnancy, and children may require monitoring using an anti-factor Xa assay due to potential variations in the drug's effects.

Option C - Dabigatran: While dabigatran prolongs the aPTT, routine monitoring is not required due to its predictable anticoagulant response at fixed doses. Thrombin time is too sensitive for monitoring dabigatran, and a diluted thrombin time assay has been developed for quantifying plasma dabigatran concentrations, but these are not routinely used.

Option D - Apixaban: This is an oral direct factor Xa inhibitor. Apixaban is given in fixed doses and does not require routine coagulation monitoring. The drugs affect the PT more than the aPTT, but the PT does not reliably measure their anticoagulant activity. Anti-factor Xa assays can measure drug levels. It's important to assess renal function at least yearly in patients taking oral factor Xa inhibitors or more frequently in patients with renal dysfunction.

Solution for Question 4:

Correct Option B - Activates PPAR alpha:

- Gallstones are typically formed due to the precipitation of cholesterol in bile, which can occur due to a variety of factors, including excess cholesterol production or secretion, decreased bile acid secretion, and altered motility of the gallbladder.
- In the context of hyperlipidemia, drugs that decrease cholesterol synthesis or increase cholesterol excretion can alter the composition of bile and increase the risk of gallstone formation.
- Fibrates, such as fenofibrate and gemfibrozil, are a class of drugs commonly used to treat hyperlipidemia. They work by activating peroxisome proliferator-activated receptors (PPARs), which are transcription factors involved in lipid and glucose metabolism.
- PPAR activation leads to increased expression of enzymes involved in fatty acid oxidation and decreased expression of enzymes involved in triglyceride synthesis and VLDL secretion.
- They reduce bile acid excretion and increase incidence of gallstones.

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PPAR activation leads to increased expression of enzymes involved in fatty acid oxidation and decreased expression of enzymes involved in triglyceride synthesis and VLDL secretion.

They reduce bile acid excretion and increase incidence of gallstones.

Incorrect Options:

Option A - Inhibits HMG CoA reductase: Statins work by inhibiting HMG CoA reductase, which is the rate-limiting enzyme in cholesterol synthesis. Major side effects are myopathy and derangement of LFTs. Multiple studies show a decreased risk for gallstones in patients on statins

Option C - Binds to deoxycholic acid: Deoxycholic acid is a bile acid that is involved in the emulsification and absorption of dietary fat. Bile acid sequestrants bind to bile acids in the GI contents and prevent their reabsorption. They do not have an increased risk of gallstones associated with them.

Option D - Decrease VLDL: Decreasing VLDL is the mechanism of action of some lipid-lowering drugs, such as niacin and omega-3 fatty acids. While decreasing VLDL can have beneficial effects on lipid metabolism, it is not directly related to the formation of gallstones.

Solution for Question 5:

Correct Option C: Pegylated-filgrastim should not be administered in sickle cell patients.

Option C. It should not be administered in sickle cell patients: Pegylated-filgrastim is contraindicated in sickle cell patients. It can potentially stimulate the production of sickled red blood cells, leading to vaso-occlusive crises and worsening of sickle cell disease symptoms.

Incorrect options:

Option A. Duration of action is shorter than normal filgrastim: Pegylated-filgrastim has a longer duration of action compared to normal filgrastim. The addition of polyethylene glycol (PEG) to filgrastim prolongs its half-life, resulting in sustained release and a longer duration of action.

Option B. Added advantage is that it can be taken orally: Pegylated-filgrastim is not available in an oral form. It is typically administered via subcutaneous injection.

Option D. It is not effective in the treatment of severe neutropenia: Pegylated-filgrastim is effective in the treatment of severe neutropenia. It stimulates the production of neutrophils and helps reduce the duration and severity of neutropenia, which is a condition characterized by a low neutrophil count.

Solution for Question 6:

Correct Option A - LMWH:

- LMWH (Low Molecular Weight Heparin): LMWH, such as enoxaparin, is often used as the initial medication for prophylaxis of thromboembolic events in patients with sickle cell disease. It has both preventive and immediate anticoagulant effects, making it suitable for both long-term prophylaxis and initial management of acute thromboembolic events.

Incorrect Options:

Option B - Warfarin: Warfarin is an oral anticoagulant medication used for long-term prophylaxis against blood clots. However, it takes some time to reach its full anticoagulant effect and is not typically the first choice for immediate management of acute events.

Option C - Aspirin: Aspirin is an antiplatelet medication that helps prevent the formation of blood clots, but it may not be the first-line choice for thromboembolic event prophylaxis in patients with sickle cell disease. It is more commonly used in other conditions, such as cardiovascular disease.

Option D - Dabigatran: Dabigatran is one of the newer oral anticoagulant medications used for long-term prophylaxis. It is not usually the initial choice for immediate management of acute thromboembolic events.

Solution for Question 7:

Correct Option B - Fostamatinib:

- It is a Tyrosine kinase inhibitor used in the management of chronic idiopathic thrombocytopenic purpura. The first line of drugs used to manage this condition is steroids. Fostamatinib is only used in patients who do not respond to first-line therapy.

Option A - Oprelvekin:

- This option is incorrect because it is an IL 11 analog and it is used in thrombocytopenia induced due to anti-cancer drug therapy.

Option C - Eltrombopag:

- It is incorrect as it is a thrombopoietin receptor agonist used in idiopathic thrombocytopenic purpura.

Option D - Romiplostim:

- It is incorrect as it is a thrombopoietin receptor agonist used in idiopathic thrombocytopenic purpura.
-

Solution for Question 8:

Correct Option C:

- The recommended range of INR in a patient with a mechanical heart valve is 2.5 to 3.5.
- Warfarin therapy is monitored using prothrombin time as it is sensitive to reduction in levels of prothrombin, factor VII, and factor X. INR helps to standardize the reporting of prothrombin time.

Incorrect Options:

Option A: This range is too low for the beneficial effects of warfarin.

Option B: On regular warfarin therapy, the INR range maintained is 2 to 3.

Option D: INR more than 3.5 is not advised as it raises the risk of a bleed.

Solution for Question 9:

Correct Option C - Antithrombin-III:

- Antithrombin-III is a natural anticoagulant protein present in the bloodstream. Heparin acts as a cofactor to AT-III, enhancing its anticoagulant activity. The availability of AT-III is crucial for heparin to exert its anticoagulant effect by inhibiting thrombin and other clotting factors.

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Incorrect Options:

Option A - Thrombin: Thrombin is a key enzyme in the blood coagulation cascade that converts fibrinogen to fibrin, leading to the formation of blood clots. While heparin indirectly inhibits thrombin by enhancing the activity of antithrombin-III, the availability of thrombin itself is not mandatory for heparin's action as an anticoagulant.

Option B - Plasmin: Plasmin is an enzyme responsible for fibrinolysis, which is the breakdown of blood clots. It is involved in the dissolution of fibrin clots by cleaving fibrin into smaller fragments. Heparin does not directly affect plasmin activity, so the availability of plasmin is not mandatory for heparin's anticoagulant action.

Option D - Thrombomodulin: Thrombomodulin is a glycoprotein present on the surface of endothelial cells lining blood vessels. It plays a role in regulating blood coagulation by binding to thrombin and converting its activity from a procoagulant to an anticoagulant. While thrombomodulin is important for the regulation of thrombin activity, it is not directly related to heparin's anticoagulant action.

Solution for Question 10:

Correct Option A - Enoxaparin:

- Enoxaparin is a low molecular weight heparin that inhibits clotting factors 10 and 2 (10>>2).
- It can be used safely in pregnancy.
- Anticoagulants in pregnancy:

Gestational period	Drug of choice
0-12 weeks	LMW/ULMW Heparin Contraindication- Warfarin
12-36 weeks	Warfarin > Heparin (LMW/ULMW)
36 weeks to 1 day before delivery	LMW/ULMW Heparin
During delivery	No anticoagulant drugs are recommended Increases postpartum hemorrhage
Lactation	Warfarin (preferred), LMW/ULMW Heparin

Incorrect Options:

Option B - Clopidogrel: Clopidogrel is an antiplatelet drug and is not used in DVT.

Option C - Betrixaban: Betrixaban is a direct factor Xa inhibitor. It is not used in pregnancy.

Option D - Warfarin: Warfarin is contraindicated in the 1st trimester of pregnancy as it causes fetal warfarin syndrome.

Solution for Question 11:

Correct Option B - Deferoxamine:

- The correct antidote for iron poisoning is Deferoxamine.
- Iron poisoning can occur when a person ingests a toxic amount of iron tablets or other iron-containing substances. Iron toxicity can be particularly dangerous in children, as it can lead to severe symptoms and potentially life-threatening complications.
- Deferoxamine is a chelating agent that is used as the antidote for iron poisoning. It works by binding to iron in the body and forming a stable complex that can be excreted through urine and feces. Deferoxamine helps to remove excess iron from the body and prevent its toxic effects.
- Therefore, in the case of iron poisoning, the appropriate antidote is Deferoxamine. It is important to seek immediate medical attention in such cases to ensure proper management and administration of the antidote.

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Therefore, in the case of iron poisoning, the appropriate antidote is Deferoxamine. It is important to seek immediate medical attention in such cases to ensure proper management and administration of the antidote.

Incorrect Options:

Option A - Luspatercept: Luspatercept is a medication used for the treatment of anemia in certain blood disorders and is not indicated for the treatment of iron poisoning.

Option C - Deferiprone: Deferiprone is an oral chelating agent used in the treatment of iron overload disorders such as thalassemia. While it can also chelate iron, Deferoxamine is the preferred antidote for acute iron poisoning.

Option D - DMSA: DMSA (Dimercaptosuccinic acid) is another chelating agent used for heavy metal poisoning, but it is not the preferred antidote for iron poisoning. It is more commonly used for lead or mercury poisoning.

Solution for Question 12:

Correct Option A - Reduced activity of CYP2C19:

- Clopidogrel is a prodrug that requires activation by CYP2C19 to become an active antiplatelet medication. CYP2C19 is an enzyme found in the liver and is responsible for converting clopidogrel into its active form. If the activity of CYP2C19 is decreased, as can occur due to certain genetic polymorphisms or drug interactions, the activation of clopidogrel may be reduced, leading to decreased antiplatelet effects and possible therapeutic failure.
- In the case of a patient with reduced CYP2C19 activity (poor metabolizer phenotype), clopidogrel may not be adequately converted to its active form. This can lead to decreased effectiveness of the drug in inhibiting platelet aggregation and preventing blood clots. As a result, patients with a reduced CYP2C19 function may be at a higher risk of experiencing recurrent strokes or other cardiovascular events despite taking clopidogrel.
- It's important to note that CYP2C19 is just one of the factors that can influence the response to clopidogrel. Other factors, such as non-genetic factors (e.g., drug interactions, patient compliance) involved in clopidogrel's mechanism of action, can also contribute to treatment failure.

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Incorrect Options:

Option B - Downregulation of CYP2D6: CYP2D6 is not involved in the activation of clopidogrel, so downregulation of this enzyme is unlikely to be responsible for the failure of clopidogrel in this patient. CYP2D6 is responsible for the metabolism of many other medications, including antidepressants and antipsychotics, so downregulation of this enzyme may affect the efficacy of those medications.

Option C - Upregulation of CYP1A1: CYP1A1 is an enzyme involved in the metabolism of many drugs, but it is not directly involved in the activation of clopidogrel. Upregulation of this enzyme may affect the metabolism of other medications, but it is unlikely to be responsible for the failure of clopidogrel in this patient.

Option D - Downregulation of CYP2E1: CYP2E1 is not involved in the activation of clopidogrel, so downregulation of this enzyme is also unlikely to be responsible for the failure of clopidogrel in this patient. CYP2E1 is involved in the metabolism of many other medications, including alcohol and acetaminophen, so downregulation of this enzyme may affect the efficacy or toxicity of those medications.

Solution for Question 13:

Correct Option B - Subcutaneous:

- LMWH (Low Molecular Weight Heparin) is typically administered via the subcutaneous route. Subcutaneous administration involves injecting the medication into the fatty tissue layer just beneath the skin. This route allows for slow and sustained absorption of the medication into the bloodstream. LMWH is specifically formulated for subcutaneous use and is available in pre-filled syringes or cartridges.

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bloodstream. LMWH is specifically formulated for subcutaneous use and is available in pre-filled syringes or cartridges.

Incorrect Options:

Option A - Intravenous (IV): LMWH is not typically administered intravenously. Regular heparin can be given intravenously.

Option C - Intramuscular (IM): LMWH is not administered intramuscularly as it can cause pain, hematoma formation, and unpredictable absorption.

Option D - Intradermal (ID): LMWH is not administered intradermally as it is not intended for local effects and may not be well absorbed through the skin.

Solution for Question 14:

Correct Option C - EACA:

- Fibrinolytic therapy involves the use of drugs that promote the breakdown of blood clots, such as alteplase (recombinant tissue plasminogen activator, or tPA). However, in some cases, it may be necessary to reverse or neutralize the effects of fibrinolytic therapy, particularly if there is excessive bleeding or other complications.
- The antidote for fibrinolytic therapy is ϵ -Aminocaproic acid (EACA). EACA is an antifibrinolytic agent that works by inhibiting the activation of plasminogen to plasmin, thereby preventing the breakdown of fibrin clots. It can be administered intravenously or orally to counteract the effects of fibrinolytic drugs and help control bleeding.

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Incorrect Options:

Option A - Vitamin K: Vitamin K is an antidote for certain anticoagulants like warfarin, but it is not used to reverse the effects of fibrinolytic therapy, which involves breaking down clots rather than preventing their formation. The correct antidote for fibrinolytic therapy is typically antifibrinolytic agents like tranexamic acid or ϵ -aminocaproic acid (EACA).

Option B - Dalteparin: Dalteparin is a low molecular weight heparin anticoagulant and is not used as an antidote for fibrinolytic therapy. It works by inhibiting the formation of blood clots but does not reverse the effects of fibrinolytic drugs.

Option D - Alteplase: Alteplase, also known as tPA, is a fibrinolytic agent used to dissolve blood clots. It is not an antidote for fibrinolytic therapy but rather the drug that requires an antidote if needed to reverse its effects.

Solution for Question 15:

Correct Option A - Apixaban:

- Apixaban is a direct inhibitor of clotting factor Xa. It belongs to a class of medications known as direct oral anticoagulants (DOACs) or non-vitamin K antagonist oral anticoagulants (NOACs). Apixaban specifically inhibits factor Xa, which is a crucial enzyme involved in the coagulation cascade. By inhibiting factor Xa, apixaban prevents the formation of thrombin and subsequent blood clot formation.

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Incorrect Options:

Option B - Argatroban: Argatroban is a direct thrombin inhibitor, not a direct inhibitor of clotting factor Xa. It acts by directly inhibiting thrombin, another key enzyme involved in the coagulation pathway.

Option C - Fondaparinux: Fondaparinux is an indirect inhibitor of clotting factor Xa. It selectively binds to antithrombin III, which in turn inhibits factor Xa. Although it indirectly inhibits factor Xa, it does not directly inhibit the enzyme itself.

Option D - Aspirin: Aspirin is not a direct inhibitor of clotting factor Xa. It belongs to the class of medications known as antiplatelet drugs. Aspirin exerts its antiplatelet effects by irreversibly inhibiting cyclooxygenase enzymes, leading to decreased synthesis of thromboxane A₂, a potent platelet aggregator.

Solution for Question 16:

Correct Option D - Acetazolamide:

- Acetazolamide is a medication used for the prevention and treatment of acute mountain sickness (AMS). It is a carbonic anhydrase inhibitor that works by promoting renal excretion of bicarbonate, leading to a mild metabolic acidosis.
- This acidosis stimulates the respiratory center in the brain, increasing ventilation and improving oxygenation. By aiding in the adjustment to higher altitudes, acetazolamide can help reduce the symptoms of AMS, such as headache, nausea, and dizziness.

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This acidosis stimulates the respiratory center in the brain, increasing ventilation and improving oxygenation. By aiding in the adjustment to higher altitudes, acetazolamide can help reduce the symptoms of AMS, such as headache, nausea, and dizziness.

Incorrect Options:

Option A - Dexamethasone: Dexamethasone is a corticosteroid medication with anti-inflammatory and immunosuppressive properties. While it is not the first-line treatment for acute mountain sickness (AMS), it may be used in severe cases or as part of a comprehensive treatment plan for high-altitude cerebral edema (HACE). HACE is a potentially life-threatening condition characterized by brain swelling at high altitudes. Dexamethasone helps reduce inflammation and swelling in the brain, helping to alleviate symptoms and stabilize the condition.

Option B - Digoxin: Digoxin is a medication from the cardiac glycoside class commonly used to treat heart failure and certain heart rhythm disorders, such as atrial fibrillation. It is not indicated for the treatment of acute mountain sickness. Digoxin works by increasing the force of contractions in the heart and slowing down the heart rate.

Option C - Diltiazem: Diltiazem is a calcium channel blocker primarily used for the treatment of hypertension (high blood pressure) and certain heart rhythm disorders. It is not specifically indicated for acute mountain sickness. Diltiazem works by relaxing blood vessels and reducing the workload on the heart by blocking the entry of calcium ions into cells.

Solution for Question 17:

Correct Option A - Warfarin:

- Warfarin is an anticoagulant medication that is commonly used in the treatment and prevention of blood clots, including deep vein thrombosis (DVT) and pulmonary embolism (PE).
- Warfarin works by interfering with the synthesis of vitamin K-dependent clotting factors in the liver.
- When we start warfarin for initial 1 to 2 days there is an increased risk of clotting Known as Hypercoagulation or dermal vascular necrosis.
- It is also known as Purple toe syndrome(mainly seen in genetic deficiency of protein C)

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It is also known as Purple toe syndrome(mainly seen in genetic deficiency of protein C)

Incorrect Options:

Option B - Heparin: Heparin is an anticoagulant medication that is commonly used in the treatment and prevention of blood clots, including deep vein thrombosis (DVT) and pulmonary embolism (PE).

- Its side effects are Heparin induced thrombocytopenia not the above given condition.
- DOC for heparin induced thrombocytopenia is Direct thrombin inhibitor(Dabigatran)

Its side effects are Heparin induced thrombocytopenia not the above given condition.

DOC for heparin induced thrombocytopenia is Direct thrombin inhibitor(Dabigatran)

Option C - Dabigatran: Dabigatran is an anticoagulant medication.

- It is a Thrombin II a inhibitor(direct thrombin inhibitor).
- It is effective in Venous thrombosis and indicated in DVT and Pulmonary embolism.
- It does not cause the above given condition.

It is a Thrombin II a inhibitor(direct thrombin inhibitor).

It is effective in Venous thrombosis and indicated in DVT and Pulmonary embolism.

It does not cause the above given condition.

Option D - Rivaroxaban: Rivaroxaban is an anticoagulant medication.

- It is direct Factor X a inhibitor
- Oral Anticoagulant
- Do not require monitoring
- Andexanet alfa is an antidote for Factor Xa inhibitor overdose.

It is direct Factor X a inhibitor

Oral Anticoagulant

Do not require monitoring

Andexanet alfa is an antidote for Factor Xa inhibitor overdose.

Solution for Question 18:

Correct Option B - Factor Xa inhibitors:

- Apixaban belongs to the class of drugs known as Factor Xa inhibitors.

Apixaban belongs to the class of drugs known as Factor Xa inhibitors.

Incorrect Options:

Option A - Antifibrinolytic: Antifibrinolytic drugs work by promoting clot stability and preventing the breakdown of blood clots. Examples of antifibrinolytic drugs include tranexamic acid and aminocaproic acid. They have a different mechanism of action compared to apixaban.

Option C - Oral direct thrombin inhibitor: Oral direct thrombin inhibitors, such as dabigatran, directly inhibit the activity of thrombin, which is a key enzyme involved in the clotting cascade. These drugs act upstream in the clotting process compared to Factor Xa inhibitors like apixaban.

Option D - Parenteral direct thrombin inhibitor: Parenteral direct thrombin inhibitors are administered through injection or infusion and directly inhibit thrombin.

Solution for Question 19:

Correct Option C - It is a pro drug:

- Prasugrel is a thienopyridine antiplatelet agent that is administered in its inactive prodrug form.
- It requires enzymatic conversion in the liver to its active metabolite, which then irreversibly inhibits the P2Y₁₂ adenosine diphosphate (ADP) receptor on platelets, preventing platelet activation and aggregation.

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It requires enzymatic conversion in the liver to its active metabolite, which then irreversibly inhibits the P2Y₁₂ adenosine diphosphate (ADP) receptor on platelets, preventing platelet activation and aggregation.

Incorrect Options:

Option A - It acts as a P2Y₁₂ agonist: Prasugrel acts as a P2Y₁₂ antagonist, not an agonist. It blocks the P2Y₁₂ receptor on platelets, leading to inhibition of platelet activation.

Option B - It is 10 times less potent than clopidogrel: Prasugrel is considered to be more potent than clopidogrel. It has a faster onset of action, greater inhibition of platelet aggregation, and reduced inter-individual variability compared to clopidogrel.

Option D - Omeprazole interferes in its activation to an active metabolite: Omeprazole, a proton pump inhibitor, is known to inhibit the activation of clopidogrel to its active metabolite. However, there is no significant interaction between prasugrel and omeprazole that interferes with its activation.

Solution for Question 20:

Correct Option D - Lepirudin:

Coagulation profile monitoring is needed in therapy with lepirudin.

- Lepirudin is a direct thrombin inhibitor that is mainly used as an anticoagulant during heparin-induced thrombocytopenia.
- Coagulation profile monitoring, including activated partial thromboplastin time (aPTT), is necessary during lepirudin therapy to ensure appropriate anticoagulation.

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Incorrect Options:

Option A - Fondaparinux: Fondaparinux is a synthetic anticoagulant that specifically inhibits factor Xa. It does not require routine coagulation profile monitoring.

Option B - Enoxaparin: Enoxaparin is a low molecular weight heparin that also inhibits factor Xa. While regular coagulation profile monitoring is not necessary, periodic monitoring of platelet count is recommended.

Option C - Dabigatran: Dabigatran is an oral direct thrombin inhibitor. It does not require routine coagulation profile monitoring, as it has a predictable anticoagulant effect.

Solution for Question 21:

Correct Option C - Colchicine:

- It is an anti-inflammatory medication that works by disrupting microtubules in neutrophils, which reduces their ability to migrate to sites of inflammation and thus inhibits neutrophil recruitment. It's commonly used to manage gout and can be helpful in some inflammatory conditions.

It is an anti-inflammatory medication that works by disrupting microtubules in neutrophils, which reduces their ability to migrate to sites of inflammation and thus inhibits neutrophil recruitment.

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Incorrect Options:

Option A - Montelukast: Montelukast is a leukotriene receptor antagonist used primarily in the management of asthma and allergic rhinitis. While it has anti-inflammatory properties, it doesn't directly inhibit neutrophil recruitment.

Option B - Sodium cromolyn: Sodium cromolyn is a mast cell stabilizer that is used for the prevention and management of asthma, allergic rhinitis, and other allergic conditions. It helps prevent the release of inflammatory mediators, but it does not specifically inhibit neutrophil recruitment.

Option D - Febuxostat: Febuxostat is a xanthine oxidase inhibitor used to treat hyperuricemia and prevent gout. It doesn't have a direct effect on neutrophil recruitment.

Solution for Question 22:

Correct Option C - Rivaroxaban:

- Rivaroxaban is an oral factor Xa inhibitor. It works by inhibiting factor Xa, which is an important component in the coagulation cascade. Rivaroxaban is used for various indications, including prevention and treatment of deep vein thrombosis, pulmonary embolism, and prevention of stroke and systemic embolism in patients with atrial fibrillation.

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Incorrect Options:

Option A - Bivalirudin: It is a direct thrombin inhibitor and is administered intravenously. Bivalirudin is primarily used as an anticoagulant during certain cardiac procedures.

Option B - Dabigatran: It is a direct thrombin inhibitor like bivalirudin, but it is available in an oral formulation. Dabigatran is commonly used for stroke prevention in atrial fibrillation and the treatment of deep vein thrombosis and pulmonary embolism.

Option D - Enoxaparin: It is a low molecular weight heparin (LMWH) that is administered subcutaneously. Enoxaparin acts by enhancing the activity of antithrombin III, thereby inhibiting

multiple clotting factors and preventing blood clots.

Solution for Question 23:

Correct Option A - Warfarin

- Among the options provided, a patient who has undergone valve replacement surgery will typically require follow-up treatment with warfarin. Warfarin is an oral anticoagulant medication that works by inhibiting the synthesis of vitamin K-dependent clotting factors in the liver. It helps to prevent the formation of blood clots and reduce the risk of thromboembolic events, such as strokes and deep vein thrombosis.
- After valve replacement surgery, especially with mechanical heart valves, the risk of blood clot formation on the valve surface is increased. Therefore, long-term anticoagulation therapy with warfarin is often recommended to maintain adequate blood-thinning and prevent clot-related complications.

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After valve replacement surgery, especially with mechanical heart valves, the risk of blood clot formation on the valve surface is increased. Therefore, long-term anticoagulation therapy with warfarin is often recommended to maintain adequate blood-thinning and prevent clot-related complications.

Incorrect Options:

Option B - ACE inhibitors: ACE inhibitors are a class of medications used primarily to manage hypertension and heart failure. While they have important roles in the treatment of certain cardiac conditions, they are not typically the primary follow-up treatment after valve replacement surgery.

Option C - Beta blockers: Beta blockers are commonly used in the management of various cardiovascular conditions, such as hypertension, angina, and certain arrhythmias. While beta blockers may be prescribed as part of the post-operative care following valve replacement surgery, they are not specifically indicated as the primary follow-up treatment.

Option D - Thiazides: Thiazides are a type of diuretic medication used to treat hypertension and edema. They are not typically the primary follow-up treatment after valve replacement surgery.

Solution for Question 24:

Correct Option A - Dabigatran:

- Dabigatran is an oral direct thrombin inhibitor and is the correct answer to the question. It belongs to a class of medications called non-vitamin K antagonist oral anticoagulants (NOACs) or direct oral anticoagulants (DOACs). Dabigatran directly inhibits the activity of thrombin, which is a key enzyme involved in the clotting cascade. By inhibiting thrombin, dabigatran prevents the formation of fibrin and the subsequent development of blood clots.

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Incorrect Options:

Option B - Rivaroxaban: Rivaroxaban is an oral anticoagulant that belongs to the class of medications known as factor Xa inhibitors. It works by directly inhibiting factor Xa, another enzyme involved in the clotting cascade.

Option C - Warfarin: Warfarin is an oral anticoagulant that works by inhibiting the synthesis of certain clotting factors in the liver. It is not a direct thrombin inhibitor.

Option D - Lepirudin: Lepirudin is an intravenous direct thrombin inhibitor that is used as an anticoagulant in patients with heparin-induced thrombocytopenia. It is not administered orally.

Solution for Question 25:

Correct Option A - Bebtelovimab:

- The drug not used in treating sickle cell disease (SCD) in the given scenario is Bebtelovimab.
- Bebtelovimab is a monoclonal antibody that targets a protein called P-selectin, which is involved in the adhesion of white blood cells to the inner lining of blood vessels.
- It is used for the treatment of COVID-19 and reduces risk of hospitalisation and death.

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It is used for the treatment of COVID-19 and reduces risk of hospitalisation and death.

Incorrect Options:

- Options B, C and D can be used to treat patients with sickle cell disease

Options B, C and D can be used to treat patients with sickle cell disease

Solution for Question 26:

Correct Option C - They can be given with verapamil and other enzyme inhibitors:

- Statins are a class of drugs that lower cholesterol levels by inhibiting the enzyme HMG-CoA reductase, which is responsible for the synthesis of cholesterol in the liver.
- They are widely used to reduce the risk of cardiovascular events such as heart attack and stroke.
- Statins should not be given with verapamil is cytochrome 3A4 enzyme inhibitors as they can increase the levels of statins in blood leading to increase risk of myopathy

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They are widely used to reduce the risk of cardiovascular events such as heart attack and stroke.

Statins should not be given with verapamil is cytochrome 3A4 enzyme inhibitors as they can increase the levels of statins in blood leading to increase risk of myopathy

Incorrect Options:

Option A - Although HMG-CoA reductase inhibitors substantially reduce the risk of cardiovascular events, there is mild increase in lipoprotein a (Lpa) levels: It is a true statement regarding statins. It is well established that statins substantially reduce the risk of cardiovascular events. However, there is a mild increase in lipoprotein a (Lpa) levels, which is a risk factor for cardiovascular disease.

Option B - These drugs should not be stopped even in severe conditions like injury, surgery etc: It is a true statement regarding statins. Statins should not be stopped abruptly, even in severe conditions such as injury or surgery, as it can increase the risk of cardiovascular events.

Option D - With the long term use, there is slight increase in the incidence of type 2 diabetes mellitus: It is a true statement regarding statins. With long-term use, there is a slight increase in the incidence of type 2 diabetes mellitus. However, the benefits of statins in reducing the risk of cardiovascular events outweigh this risk.

Solution for Question 27:

Correct Option A - Oprelvekin (IL-11):

- Oprelvekin (IL-11) is a medication used to stimulate platelet production in patients who have undergone chemotherapy. It works by stimulating the production of megakaryocytes, which are the cells that produce platelets. Oprelvekin is approved for use in patients with chemotherapy-induced thrombocytopenia (CIT), a condition characterized by a low platelet count caused by chemotherapy.

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Incorrect Options:

Option B - Filgrastim: Filgrastim is a medication used to stimulate the production of white blood cells in patients undergoing chemotherapy. It works by stimulating the bone marrow to produce more white blood cells, which can help to prevent infections. While low platelet counts can be a side effect of chemotherapy, filgrastim is not indicated for treating thrombocytopenia.

Option C - Erythropoietin: Erythropoietin is a medication that stimulates red blood cell production in patients with anemia. It is not indicated for the treatment of thrombocytopenia.

Option D - Amifostine is a medication used to protect healthy cells from the toxic effects of chemotherapy. It does not stimulate the production of platelets and is not indicated for the treatment of thrombocytopenia.

Solution for Question 28:

Correct Option B - Inhibits vitamin K dependant Clotting factors:

- Dicumarol is an anticoagulant drug that inhibits the production of vitamin K-dependent clotting factors, including factors II, VII, IX, and X.
- It acts by inhibiting enzyme vit k VKOR

Dicumarol is an anticoagulant drug that inhibits the production of vitamin K-dependent clotting factors, including factors II, VII, IX, and X.

It acts by inhibiting enzyme vit k VKOR

Incorrect Options:

Option A - Activates antithrombin III: Dicumarol does not activate antithrombin III, nor does it inhibit the action of vitamin K-dependent clotting factors by this mechanism.

Option C - Inhibits tissue plasminogen activator: Dicumarol does not inhibit the action of the tissue plasminogen activator, which is involved in fibrinolysis.

Option D - Inhibitor of factor Xa: Dicumarol does not directly inhibit factor Xa. Instead, it indirectly inhibits the production of factor Xa by inhibiting the production of vitamin K-dependent clotting factors.

Cytotoxic drugs

1. A 65-year-old male presents with a recent diagnosis of advanced non-small cell lung cancer (NSCLC). The oncologist decides to initiate treatment with pemetrexed. The patient has a history of COPD and hypertension, which are well-controlled with medication. Which of the following best describes the mechanism of action of pemetrexed in the treatment of non-small cell lung cancer (NSCLC)?

- A. Inhibits thymidylate synthase, dihydrofolate reductase, and glycinamide ribonucleotide formyltransferase, thereby disrupting DNA and RNA synthesis.
 - B. Binds to tubulin, inhibiting microtubule formation and causing cell cycle arrest in the M phase.
 - C. Acts as an alkylating agent, causing cross-linking of DNA strands and preventing DNA replication.
 - D. Inhibits topoisomerase II, leading to DNA strand breaks and apoptosis.
-

2. Which of the following is true about Eribulin?

- A. It is less affected by the PGP efflux pump
 - B. Used in colorectal cancer
 - C. Inhibits depolymerization of microtubule
 - D. Forms toxic metabolite called SN-38 causing GIT damage
-

3. A 55-year-old male patient undergoing treatment for testicular cancer develops pruritic, erythematous bands on his arms and back. What is the mechanism of action of the drug that causes this cutaneous presentation?



- A. Generates free radicals and induces DNA cleavage via deoxyribose ring damage
 - B. Inhibits ribonucleotide reductase
 - C. Prevents polymerization and disrupts microtubule formation
 - D. Inhibits topoisomerase II
-

4. A 62-year-old female patient, receiving treatment with daunorubicin for breast cancer, presents with symptoms of fatigue, shortness of breath, and chest pain. An echocardiogram shows decreased left ventricular ejection fraction. Which of the following agents should be administered to prevent further cardiac damage?

- A. Penicillamine
 - B. Dexrazoxane
 - C. Ethylenediaminetetraacetic acid (EDTA)
 - D. Dimercaprol
-

5. A 45 y/o female was recently diagnosed with acute promyelocytic leukemia (APML). She has been undergoing treatment for the same for the past 1 month. She now presents with sudden dyspnea, tachypnea & dry cough. O/E cyanosis is present. The doctor suspects ARDS. Her CBC showed elevated WBC counts as well. Which of the following drugs is responsible for the patient's condition?

- A. All-trans retinoic acid (ATRA)
 - B. Methotrexate
 - C. Cyclophosphamide
 - D. Azathioprine
-

6. A 30 y/o male came with h/o testicular cancer. He underwent chemotherapy 6 months ago for the same. He now presents with progressive dyspnea and a dry cough for the past 3 months. Pulmonary function tests reveal restrictive lung disease with decreased lung volumes. Chest Xray shows diffuse basilar infiltrates and Pulmonary fibrosis. Which of the following chemotherapy drugs is most likely responsible for the patient's condition?

- A. Cisplatin
 - B. Etoposide
 - C. Bleomycin
 - D. Methotrexate
-

7. A 25-year-old male patient with a known history of Crigler-Najjar syndrome type I is diagnosed with metastatic colon cancer. The oncology team is planning his treatment regimen, which includes various anticancer drugs. Which of the following anti-cancer drugs is not contraindicated in this patient?

- A. Irinotecan
 - B. Topotecan
 - C. Etoposide
 - D. 5 Fluro-Uracil
-

8. A 55 y/o woman presents with a new diagnosis of metastatic breast cancer. Her oncologist decides to initiate treatment with a drug that inhibits microtubules. Considering her condition and the available microtubule- inhibiting drugs, the oncologist discusses the options, including vincristine, vinblastine, paclitaxel, and docetaxel. Which of the following statements is true for microtubule-inhibiting anti-cancer drugs?

- A. Vincristine & vinblastine are both vinca alkaloids that inhibit tubulin polymerization, thereby preventing the formation of microtubules.
- B. Vincristine & vinblastine are both equally neurotoxic.
- C. Paclitaxel and docetaxel both destabilize microtubules by inhibiting tubulin assembly, leading to mitotic spindle dysfunction.
- D. Paclitaxel and docetaxel are both contraindicated in liver disease

9. Which of the following purine analogues is considered the drug of choice (DOC) for treating hairy cell leukemia?

- A. Thioguanine
- B. Cladribine
- C. Fludarabine
- D. Mercaptopurine

10. Match the following conditions and the pyrimidine analogues which are used as drugs of choice. a) Pancreatic cancer 1. Cytosine arabinoside b) Acute Myeloid Leukemia (AML) 2. Azacitidine c) Myelodysplastic syndrome 3. 5-Fluorouracil d) Colorectal Cancer 4. Gemcitabine

a) Pancreatic cancer	1. Cytosine arabinoside
b) Acute Myeloid Leukemia (AML)	2. Azacitidine
c) Myelodysplastic syndrome	3. 5-Fluorouracil

- A. a) - 4; b) - 1; c) - 2; d) - 3
B. a) - 3; b) - 4; c) - 1; d) - 2
C. a) - 1; b) - 2; c) - 3; d) - 4
D. a) - 2; b) - 3; c) - 4; d) - 1
-

11. A 55-year-old woman with metastatic breast cancer is undergoing chemotherapy. She presents with complaints of redness, swelling, and pain on her palms and soles. She describes a tingling sensation that has progressed to a burning pain over the past week. Upon physical examination, there is erythema and mild swelling of the affected areas, with some peeling of the skin as shown below. Her current chemotherapy regimen includes capecitabine and docetaxel. Which of the following is the most likely diagnosis for this patient's symptoms?



- A. Erythromelalgia
B. Erythema multiforme
C. Hand-foot syndrome (palmar-plantar erythrodysesthesia)
D. Stevens-Johnson syndrome
-

12. A 55-year-old patient with colorectal cancer is being treated with 5-fluorouracil (5-FU) as part of their chemotherapy regimen. Which of the following best describes the mechanism by which 5-fluorouracil exerts its anti-cancer effects in this patient?

- A. 5-Fluorouracil inhibits DNA synthesis by incorporating into the DNA strand and causing chain termination.
 - B. 5-Fluorouracil inhibits RNA polymerase, preventing the transcription of mRNA.
 - C. 5-Fluorouracil is converted into fluorodeoxyuridine monophosphate (FdUMP), which inhibits thymidylate synthase, leading to decreased thymidine production and inhibition of DNA synthesis.
 - D. 5-Fluorouracil activates topoisomerase I, causing breaks in DNA strands.
-

13. A 55-year-old woman with a history of rheumatoid arthritis under medication presents with complaints of fatigue, oral ulcers, and a non-productive cough. On examination, she appears pale, and reveals a temp of 100.9°F. Laboratory investigations show the following: Pancytopenia, elevated BUN and creatinine levels and elevated transaminases. Which of the following is the most likely diagnosis?

- A. Systemic lupus erythematosus (SLE)
 - B. Methotrexate toxicity
 - C. Felty's syndrome
 - D. Acute viral hepatitis
-

14. A 35-year-old female presents with vaginal bleeding, abdominal pain, and a high serum β -hCG level. Imaging studies and histopathology confirm a diagnosis of choriocarcinoma. The oncologist decides to start treatment with a chemotherapeutic agent. Which of the following drugs is most appropriate for this patient?

- A. Cisplatin

- B. Methotrexate
 - C. Doxorubicin
 - D. Vincristine
-

15. A 65 y/o male is undergoing chemotherapy for the treatment of non-Hodgkin's lymphoma. During his treatment, he developed pancytopenia, characterized by anemia, leukopenia, and thrombocytopenia. The oncologist reviews the patient's medication list to determine which drug might be responsible for the patient's lab results. All of the following drugs could be responsible for the patient's condition except?

- A. Cyclophosphamide
 - B. Methotrexate
 - C. Doxorubicin
 - D. Vincristine
-

16. A 25-year-old male presents with a history of recurrent episodes of severe pain in his limbs and chest, especially during cold weather or after vigorous physical activity. His medical history reveals multiple hospital admissions for similar episodes, diagnosed as vaso-occlusive crises. Laboratory investigations confirm the presence of hemoglobin S. His physician decides to start him on a medication that helps to reduce the frequency of painful crises and the need for blood transfusions. Which of the following is the most appropriate medication for this patient's condition?

- A. Methotrexate
 - B. Hydroxyurea
 - C. Imatinib
 - D. Anagrelide
-

17. A 60-year-old male patient is diagnosed with locally advanced squamous cell carcinoma of the head and neck. His oncologist recommends a combination of radiation therapy and chemotherapy. The patient expresses concern about the potential damage to his surrounding normal tissues from the radiation. To address his concerns, the oncologist considers using both a radio-sensitizer and a radio-protector. Which of the following combinations of agents is most appropriate for this patient's treatment plan?

- A. Cisplatin and amifostine
 - B. Doxorubicin and vitamin C
 - C. Metformin and hyperbaric oxygen
 - D. Gemcitabine and sodium selenite
-

18. A 55-year-old male patient, who has been diagnosed with metastatic testicular cancer, has been receiving cisplatin as part of his chemotherapy regimen. After his third cycle, he reports experiencing fatigue, decreased urine output, and mild swelling in his ankles. Blood tests reveal an elevated serum creatinine level and reduced glomerular filtration rate (GFR). Which of the following is the most appropriate management step?

- A. Increase the cisplatin dosage to overcome resistance.
 - B. Switch to another platinum-based drug such as carboplatin, which has a similar nephrotoxicity profile.
 - C. Administer i.v. hydration and consider using amifostine as a nephroprotective agent.
 - D. Discontinue all chemotherapy and initiate dialysis immediately.
-

19. As a pharmaceutical researcher, you are tasked with summarizing the primary uses of platinum-based chemotherapy agents. Which of the following statements is incorrect with respect to uses of cisplatin and carboplatin?

- A. Carboplatin is used for the treatment of retinoblastoma along with bleomycin and vincristine.
- B. Platinum compounds are used in bone tumors such as osteosarcoma.
- C. Platinum compounds are combined with 5- fluorouracil & radiosensitizers for the treatment of gastrointestinal tract cancers and head & neck cancers.
- D. Paclitaxel, when combined with cisplatin/carboplatin, produces complete resolution in the patients of ovarian cancer.
-

20. Match the following drugs with their primary uses: Drug Primary Use

1. Sargramostim A. Granulocyte- CSF used in Stem cell extraction

2. Melphalan B. Granulocyte- Macrophage CSF used in Stem cell extraction

3. Plerixafor C. Used in myeloablation for bone marrow transplantation

4. Filgrastim D. Blocks CXCR4 receptor for stem cell mobilization

Drug	Primary Use
1. Sargramostim	A. Granulocyte- CSF used in Stem cell extraction
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3. Plerixafor	C. Used in myeloablation for bone marrow transplantation
4. Filgrastim	D. Blocks CXCR4 receptor for stem cell mobilization

- A. 1-B, 2-C, 3-A, 4-D
- B. 1-B, 2-C, 3-D, 4-A
- C. 1-A, 2-D, 3-C, 4-B
- D. 1-A, 2-C, 3-D, 4-B
-

21. Match the following cytotoxic drugs with their primary uses: Cytotoxic Drugs Primary Use

1. Temozolomide A. Treatment of Hodgkin's lymphoma

2. Dacarbazine B. DOC of glioma tumors

3. Trabectedin C. Organophosphate used in

myeloablative procedures 4. Thiotepa D. Used in cisplatin-resistant ovarian cancer cases 5. Altretamine E. Treatment of leiomyosarcoma

Cytotoxic Drugs	Primary Use
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4. Thiotepa	D. Used in cisplatin-resistant ovarian cancer cases
5. Altretamine	E. Treatment of leiomyosarcoma

- A. 1-B, 2-A, 3-E, 4-C, 5-D
 - B. 1-A, 2-B, 3-E, 4-D, 5-C
 - C. 1-B, 2-E, 3-A, 4-D, 5-C
 - D. 1-E, 2-C, 3-B, 4-A, 5-D
-

22. A 60-year-old female with a history of chronic myeloid leukemia (CML) is being treated with Busulfan. She presents to the clinic with complaints of abdominal pain and swelling, fatigue, and yellowing of the skin. Additionally, the patient reports experiencing episodes of seizures since starting busulfan therapy. Upon examination, the patient has hepatomegaly and jaundice. Liver function tests indicate elevated liver enzymes. Which of the following statements is incorrect for the next steps?

- A. The patient's should be started on defibrotide as the symptoms are likely due to Budd-Chiari syndrome (Veno-Occlusive Disease)
 - B. Prompt administration of warfarin to reduce chances of another episode of VOD.
 - C. Patients should be started on antiepileptics such as phenytoin to prevent seizures.
 - D. Adjust the dose of busulfan based on plasma drug-level monitoring.
-

23. A 55-year-old male with a history of non-Hodgkin's lymphoma is receiving chemotherapy, including cyclophosphamide. After a few weeks of treatment, he presents with symptoms of urinary frequency, urgency, and dysuria. Urinalysis reveals hematuria. The patient's physician suspects hemorrhagic cystitis. Which of the following statements is false regarding the use of cyclophosphamide?

- A. Hemorrhagic cystitis is primarily caused by the direct cytotoxic effect of cyclophosphamide on the bladder epithelium.
 - B. Ifosfamide, a modification of cyclophosphamide, is more likely to cause hemorrhagic cystitis than cyclophosphamide itself.
 - C. Cyclophosphamide is metabolized into 4-hydroxy cyclophosphamide in the liver by CYP2B6.
 - D. Hemorrhagic cystitis is treated by MESNA (Mercapto-Ethane-Sulfonate-Sodium,) which neutralizes toxic effects in the bladder.
-

24. A 65 y/o male presents with complaints of fatigue, lower back pain, and recurrent infections over the past few months. He also reports recent weight loss and increased thirst. O/E, pallor is noted. His lab investigations revealed anemia, hypercalcemia, elevated serum creatinine, hypoproteinemia, and M Spike in SPEP test. His urinalysis also showed Bence Jones proteins. Which of the following nitrogen mustards should be used in treating this patient?

- A. Cyclophosphamide
 - B. Mechlorethamine
 - C. Chlorambucil
 - D. Melphalan
-

25. A 55 y/o male presents with stage 3 lung cancer. The oncologist recommends treatment with a chemotherapy regimen that includes a cytotoxic drug known to be most effective during the S phase of the cell cycle. Which of the following cytotoxic drugs did he most likely recommend?

- A. Vincristine
 - B. Cisplatin
 - C. 5- Fluorouracil
 - D. Bleomycin
-

26. A 50 y/o male was diagnosed with Hodgkin's lymphoma two years ago. He underwent chemotherapy for the same. Recently, the patient has presented with symptoms of fatigue, easy bruising, and recurrent infections. Laboratory tests and bone marrow biopsy confirmed a diagnosis of AML. Which of the following drugs used in the treatment of this patient's Hodgkin's lymphoma is most likely responsible for the development of his condition?

- A. Doxorubicin
 - B. Vincristine
 - C. Bleomycin
 - D. Procarbazine
-

27. A 45 y/o male with AML has been undergoing intensive chemotherapy. Three days after starting treatment, he developed a sudden onset of weakness, nausea, and muscle cramps. Laboratory tests reveal hyperkalemia, hyperuricemia, hyperphosphatemia, and hypocalcemia. His urine output has decreased significantly over the past 24 hours. Which of the following is not a treatment option for this patient?

- A. Aggressive intravenous hydration
 - B. Allopurinol
 - C. Rasburicase
 - D. Sodium bicarbonate
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	2
Question 5	1
Question 6	3
Question 7	4
Question 8	1
Question 9	2
Question 10	1
Question 11	3
Question 12	3
Question 13	2
Question 14	2
Question 15	4
Question 16	2
Question 17	1
Question 18	3
Question 19	1
Question 20	2
Question 21	1
Question 22	2

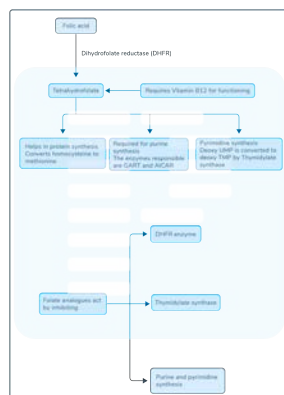
Question 23	1
Question 24	4
Question 25	3
Question 26	4
Question 27	4

Solution for Question 1:

Correct Answer: A) Inhibits thymidylate synthase, dihydrofolate reductase, and glycinamide ribonucleotide formyl transferase, thereby disrupting DNA and RNA synthesis.

Explanation:

Pemetrexed is a folate analogue used in the treatment of various cancers, including non-small cell lung cancer (NSCLC) and malignant pleural mesothelioma. It works by inhibiting several key enzymes involved in folate metabolism and nucleotide synthesis which disrupts DNA and RNA synthesis, leading to the inhibition of cancer cell growth and proliferation.



Trials for the treatment regimens of non-small cell lung cancer (NSCLC):

- JMDB Trial
- PARAMOUNT Trial
- LUX-Lung Trials

Mechanism of Action of Folate Analogues:

Folate Analogue	Mechanism of Action	Uses
Pemetrexed	Inhibition of Thymidylate synthase Dihydrofolate reductase Glycinamide ribonucleotide formyltransferase (GARFT) → depletion of nucleotide pools essential for DNA synthesis and repair.	Mesothelioma Non-small cell lung cancer (NSCLC) In combination with cisplatin for malignant pleural mesothelioma.
Methotrexate	Inhibition of dihydrofolate reductase - prevents the formation of THF. Disrupts DNA synthesis and cell proliferation.	Acute lymphoblastic leukemia (ALL) Non-Hodgkin lymphoma Osteosarcoma Gestational trophoblastic neoplasia Rheumatoid arthritis Psoriasis.
Raltitrexed	Inhibition of thymidylate synthase - interfering with DNA synthesis.	Colorectal cancer.
Pralatrexate	Competitive inhibitor of dihydrofolate reductase thymidylate synthase, → disrupting folate metabolism and inhibiting DNA synthesis.	Relapsed or refractory peripheral T-cell lymphoma (PTCL).

Solution for Question 2:

Correct answer: A) It is less affected by the PGP efflux pump

Explanation:

Eribulin less affected by the PGP efflux pump than other microtubule disruptors, making it effective in drug-resistant tumors that overexpress PGP.

- Eribulin is used to treat drug-resistant metastatic breast cancer and liposarcoma. (Option B Ruled out)
- Eribulin works by binding to the vinca site on β -tubulin and inhibiting polymerization of the microtubule assembly. (Option C Ruled out)
- Irinotecan, a Topoisomerase I inhibitor, forms a toxic metabolite called SN-38 causing GIT damage. (Option D Ruled out)

Solution for Question 3:

Correct Answer: A) Generates free radicals and induces DNA cleavage via deoxyribose ring damage

Explanation:

Bleomycin is commonly used in the treatment of testicular cancer. It is a key component of the chemotherapy regimens BEP (bleomycin, etoposide, and cisplatin)

- Bleomycin is known to cause a variety of cutaneous toxicities, including a rare side effect called flagellate pigmentation which presents as pruritic, erythematous bands on the arms, back, scalp, and hands and can be treated with topical corticosteroids.
 - The mechanism of action of bleomycin is it binds to DNA to generate free radicals and induce DNA cleavage via deoxyribose ring damage in the G2 phase.
 - Hydroxyurea inhibits ribonucleotide reductase (Option B), blocking DNA synthesis
 - Vinca alkaloids bind to tubulin, preventing polymerization and disrupting microtubule formation (Option C)
 - Anthracyclines Inhibit topoisomerase II (Option D) acting on the G2 and S phase
-

Solution for Question 4:

Correct Answer: B) Dexrazoxane

Explanation

- The mechanism of cardiotoxicity by Daunorubicin is believed to be partly due to cardiac lipid peroxidation, which is promoted by Fe^{2+} causing Cardiomyopathy.
- Dexrazoxane is a cardioprotective, iron-chelating agent. It is used to prevent cardiotoxicity caused by the Anthracyclines like Daunorubicin and Doxorubicin.
- Dexrazoxane may also prevent anthracycline-derived DNA damage by interacting with topoisomerase II in cardiac cells.

Penicillamine (Option A) is used for copper chelation and certain conditions like rheumatoid arthritis

Dimercaprol (Option D) is used for arsenic and other heavy metal poisonings

EDTA (Option C) is primarily used for lead poisoning.

None of these agents provide the specific cardioprotective effects that dexrazoxane does against anthracycline-induced damage.

Solution for Question 5:

Correct Answer: A) All-trans retinoic acid (ATRA)

Explanation

ATRA is the mainstay of treatment for APML.

The patient's elevated WBCs and presence of ARDS post-chemotherapy for her APML suggest she has developed Leukocyte maturation syndrome.

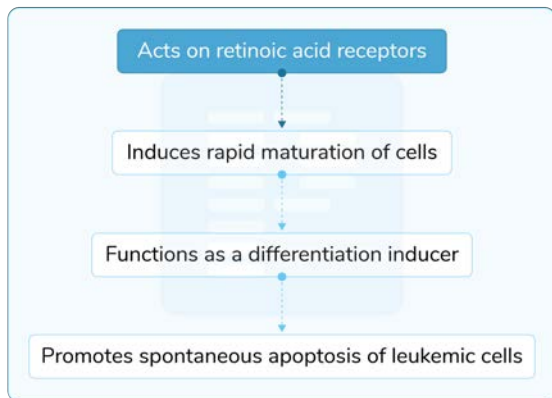
While highly effective in inducing remission in APML patients, it can lead to a complication known as leukocytic maturation syndrome, also called retinoic acid syndrome (RAS) which is characterized by highly elevated leukocyte counts (sometimes even 50,000 to 100,000) and ARDS.

RAS typically occurs within the first month of ATRA therapy and is managed by temporarily discontinuing ATRA and administering dexamethasone.

Other side effects of ATRA include:

- Diabetes
- Dyslipidemia
- Infertility

Mechanism of Action of All-Trans Retinoic Acid (ATRA) in APML:



Methotrexate and Cyclophosphamide (Option B&C) is used to treat several leukemias, however, it is not typically associated with leukocytic maturation syndrome (retinoic acid syndrome)

Azathioprine (Option D) is an immunosuppressive drug used in the treatment of various autoimmune diseases and in preventing organ rejection in transplant recipients. While azathioprine can cause bone marrow suppression and increase the risk of infections, it is not associated with leukocyte maturation syndrome.

Solution for Question 6:

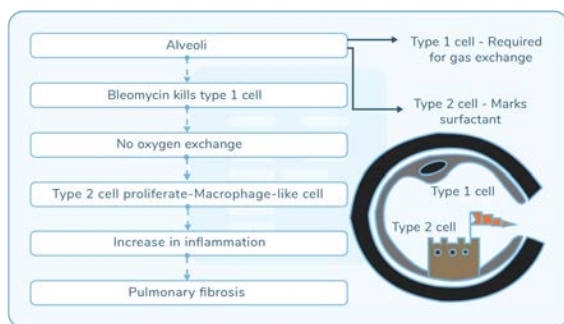
Correct Answer: C) Bleomycin

Explanation:

The patient's signs, symptoms, and imaging studies suggest that he most likely has pulmonary fibrosis.

This could be a result of the side effects of the drug bleomycin used in his chemotherapy regimen (BEP- Bleomycin, Etoposide, Cisplatin)

Bleomycin is an antitumor antibiotic that is well-known for its association with pulmonary toxicity, including pulmonary fibrosis which occurs as follows:



Cisplatin (Option A) is a platinum-based chemotherapy drug commonly used in the treatment of testicular cancer.

Etoposide (Option B) is a topoisomerase II inhibitor used in the treatment of testicular cancer among other malignancies. Pulmonary fibrosis is not a recognized adverse effect of cisplatin and Etoposide

Methotrexate (Option D) is a folate antagonist commonly used in the treatment of various malignancies, but it is not a standard component of chemotherapy regimens for testicular cancer. Pulmonary toxicity can occur with methotrexate, presenting as pneumonitis rather than fibrosis, and it is associated with high-dose or prolonged therapy used in conditions like rheumatoid arthritis or psoriasis.

Solution for Question 7:

Correct Answer: D) 5- Fluorouracil

Explanation:

5-fluorouracil (5-FU) is not contraindicated in Crigler-Najjar syndrome.

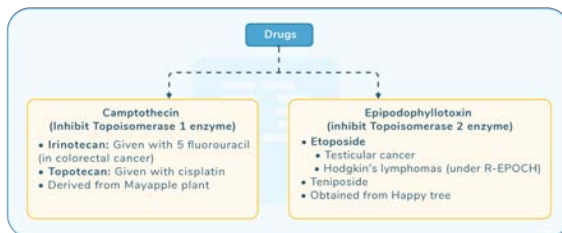
- Crigler-Najjar syndrome is primarily associated with a deficiency in the enzyme uridine diphosphate glucuronosyltransferase 1A1 (UGT1A1), which is responsible for the glucuronidation and metabolism of certain drugs.
- 5-FU does not undergo significant metabolism via the UGT1A1 pathway.
- Instead, it is primarily metabolized by dihydropyrimidine dehydrogenase (DPD) and is excreted via the kidneys.
- Therefore, 5-fluorouracil is generally considered safe for use in patients with Crigler-Najjar syndrome.

Irinotecan & topotecan (Option A) & (Option B) are topoisomerase 1 inhibitors.

- They are metabolized by the enzyme UGT1A1.
- Therefore, it is contraindicated in patients with Crigler-Najjar syndrome due to deficiency of UGT1A1.

Etoposide (Option C) is a topoisomerase 2 inhibitor used in testicular cancers. Like irinotecan and topotecan, it is also metabolized by the enzyme UGT1A1. Hence, it is contraindicated in patients with Crigler-Najjar syndrome.

Topoisomerase inhibitors:



Solution for Question 8:

Correct Answer: A) Vincristine & vinblastine are both vinca alkaloids that inhibit tubulin polymerization, thereby preventing the formation of microtubules.

Explanation:

Both vinblastine and vincristine are vinca alkaloids.

- These drugs bind to tubulin and inhibit its polymerization into microtubules, which prevents the formation of the mitotic spindle.
- The cell division is hence arrested at metaphase, effectively halting the proliferation of cancer cells.

Vinca Alkaloids and Taxanes:

Table rendering failed

Vincristine is more neurotoxic than vinblastine (Option B):

- Both of them disrupt microtubule dynamics by inhibiting tubulin polymerization, but vincristine does so more potently in neuronal cells.

- Microtubules are essential for axonal transport, a process critical for the movement of organelles, proteins, and other materials within neurons.
- Vincristine's stronger disruption of this process leads to greater neurotoxicity.

Both docetaxel and paclitaxel stabilize microtubules (Option C):

They enhance the polymerization of tubulin and over-stabilize the microtubules, preventing their disassembly, which disrupts the normal dynamic equilibrium required for mitotic spindle function.

Only Paclitaxel is contraindicated in liver disease whereas docetaxel is not (Option D):

- Paclitaxel is extensively metabolized by the liver, primarily through the cytochrome P450 enzyme system, specifically CYP2C8 and CYP3A4.
- In patients with liver disease, the metabolic capacity of the liver is significantly impaired leading to decreased clearance of paclitaxel from the body. This results in higher systemic concentrations of the drug, increasing the risk of toxicity.
- In contrast, docetaxel is a drug that gets excreted mainly via the kidneys, hence it is contraindicated in kidney disease.

Solution for Question 9:

Correct Answer: B) Cladribine

Explanation

Purine analogues are the DOC in various neoplastic conditions due to the following MOA:

- Enter the cell.
- Converted to fake nucleosides by HGPRTase enzyme (Hypoxanthine guanine phosphoribosyltransferase).
- Incorporated into DNA.
- Inhibits DNA polymerase enzyme.
- Inhibits DNA synthesis.

Purine Analogues and their uses:

Purine Analogue	Uses
Cladribine	Hairy cell leukemia (DOC), multiple sclerosis

Fludarabine (Option C -Ruled out)	Chronic lymphocytic leukemia (CLL) [DOC], low-grade non-Hodgkin's lymphoma
Mercaptopurine (Option D -Ruled out)	Acute lymphoblastic leukemia (ALL) maintenance, inflammatory bowel disease (off-label)
Thioguanine (Option A- Ruled out)	Acute myeloid leukemia (AML)
Pentostatin	Hairy cell leukemia
Azathioprine	Organ transplantation, rheumatoid arthritis, IBD (DOC) and SLE (DOC)
Nelarabine	T-cell acute lymphoblastic leukemia (T-ALL), T-cell lymphoblastic lymphoma (T-LBL) [DOC]
Clofarabine	Pediatric ALL (DOC)

Solution for Question 10:

Correct Answer: A) a) - 4; b) - 1; c) - 2; d) - 3

Explanation:

Drug of Choices for given conditions are:

Pancreatic cancer - Gemcitabine

Acute Myeloid Leukemia (AML) - Cytosine arabinoside

Myelodysplastic syndrome - Azacitidine

Colorectal Cancer - 5- Fluorouracil

Pyrimidine Analogues with uses:

Pyrimidine Analogue	Use(s)
5-Fluorouracil	Colorectal cancer, breast cancer, other solid tumors
Cytarabine (Cytosine arabinoside)	Acute myeloid leukemia (AML), acute lymphocytic leukemia (ALL), non-Hodgkin lymphoma
Gemcitabine	Pancreatic cancer (DOC), non-small cell lung cancer, bladder cancer
Azacitidine and Decitabine	Myelodysplastic syndromes (DOC), AML

Capecitabine	Colorectal cancer, breast cancer, gastric cancer
--------------	--

Solution for Question 11:

Correct Answer: C) Hand-foot syndrome (palmar-plantar erythrodysesthesia)

Explanation:

This patient on chemotherapy with capecitabine and docetaxel has developed redness, swelling, and pain on her palms and soles and tingling sensation that has progressed to a burning pain. There is also mild swelling with some peeling of the skin. These symptoms point towards hand-foot syndrome.

Hand-foot syndrome, also known as palmar-plantar erythrodysesthesia or acral erythema:

- Occurs due to the leakage of the chemotherapeutic drugs into the capillaries of the hands and feet, causing damage to the surrounding tissues.
- Initially presents with numbness, tingling, or burning pain in the palms and soles.
- Accompanied by sharply demarcated erythema (redness) which may or may not include edema (swelling), cracking, or desquamation (peeling skin).
- Blistering and ulceration seen in advanced cases
- Lateral parts and distal fat pads of the palms are often affected before the soles of the feet.



Drugs causing Hand-foot syndrome:

- 5- Fluorouracil
- Capecitabine (maximum)
- Docetaxel

- Doxorubicin (including pegylated liposomal doxorubicin)

Preventive management of Hand-foot syndrome: Use of pyridoxine (vitamin B6) has been recommended.

Erythromelalgia (Option A):

- Characterized by episodes of red, hot, and painful extremities.
- Usually not associated with chemotherapy and does not typically cause skin peeling. Episodic presentation rather than continuous, which contrasts with the progressive and persistent symptoms seen in HFS.
- Erythromelalgia can be caused by genetic mutations (primary erythromelalgia), myeloproliferative disorders, autoimmune diseases, neurological conditions, infections, vascular disorders, and environmental factors

Erythema multiforme (Option B):

- Hypersensitivity reaction often triggered by infections, medications, or other factors.
- Characterized by the appearance of target-like lesions, primarily on the extremities, but can also affect mucous membranes.
- It is less commonly associated with chemotherapy.

Stevens-Johnson syndrome (SJS) (Option D):

- Presents with widespread skin necrosis and mucous membrane involvement, often starting with flu-like symptoms followed by a painful red or purplish rash that blisters and causes the skin to shed.
- SJS is much more severe and extensive compared to hand-foot syndrome and involves more than just the hands and feet.

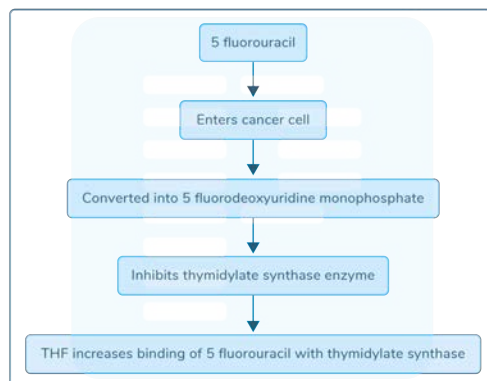


Solution for Question 12:

Correct Answer: C) 5-Fluorouracil is converted into fluorodeoxyuridine monophosphate (FdUMP), which inhibits thymidylate synthase, leading to decreased thymidine production and inhibition of DNA synthesis.

Explanation:

5-Fluorouracil inhibits thymidylate synthase after conversion to FdUMP, leading to decreased thymidine production and inhibition of DNA synthesis. It is always combined with THF to increase the effectiveness since THF increases binding of 5 fluorouracil with thymidylate synthase.



Uses of 5-FU:

- Given IV for all GIT cancer along with platin
- Given in breast cancer
- Topical application on skin: For treatment of Mycosis Fungoides (Sezary Syndrome) Basal Cell Carcinoma Actinic Keratosis
- Mycosis Fungoides (Sezary Syndrome)
- Basal Cell Carcinoma
- Actinic Keratosis
- Capecitabine is an oral pro-drug approved for breast cancer treatment
- Colorectal cancer FOLFOX Regimen: Folinic acid , Fluorouracil (5-FU), Oxaliplatin
- FOLFIRINOX Regimen: Folinic acid, Fluorouracil (5-FU), Irinotecan, Oxaliplatin
- FOLFOX Regimen: Folinic acid , Fluorouracil (5-FU), Oxaliplatin
- FOLFIRINOX Regimen: Folinic acid, Fluorouracil (5-FU), Irinotecan, Oxaliplatin
- Mycosis Fungoides (Sezary Syndrome)
- Basal Cell Carcinoma

- Actinic Keratosis
 - FOLFOX Regimen: Folinic acid , Fluorouracil (5-FU), Oxaliplatin
 - FOLFIRINOX Regimen: Folinic acid, Fluorouracil (5-FU), Irinotecan, Oxaliplatin
-

Solution for Question 13:

Correct Answer: B) Methotrexate toxicity

Explanation:

The patient's presentation of fatigue, oral ulcers, fever, and pancytopenia along with elevated BUN and creatinine levels suggest methotrexate toxicity.

Methotrexate Toxicity and Side Effects:

- Bone Marrow Suppression- Risk of spontaneous hemorrhage, life-threatening infection
- Megaloblastic anemia
- Alopecia and dermatitis
- Pneumonitis
- Nephrotoxicity (high doses)
- Defective oogenesis or spermatogenesis, abortion, and teratogenesis
- Hepatotoxic (Cirrhosis)
- Intrathecal Administration of MTX Meningismus Inflammatory Response in CSF Seizures, coma, and rare cases of death.
- Meningismus
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Management of Methotrexate Toxicity:

1. Stop methotrexate immediately.
2. Administer leucovorin: It is advised to monitor methotrexate levels to establish the correct dosage and duration for leucovorin rescue therapy. (Note: Leucovorin cannot reverse

neurotoxicity)

3. Administer glucarpidase if serum methotrexate concentration is significantly elevated or persistent elevation with renal dysfunction.

4. Hydration and alkalinization: IV fluids with sodium bicarbonate to alkalinize urine and promote methotrexate excretion.

5. Hemodialysis

6. Supportive Care: Blood transfusions for significant anemia, Antibiotics for infections due to neutropenia, Platelet transfusions for significant thrombocytopenia.

- SLE (Option A) can present with fatigue, oral ulcers, fever, and hematological abnormalities (like anemia and leukopenia). However, SLE often has more specific findings such as a malar rash, photosensitivity, arthritis, serositis, and renal involvement (e.g., proteinuria).
 - However, SLE often has more specific findings such as a malar rash, photosensitivity, arthritis, serositis, and renal involvement (e.g., proteinuria).
 - Felty's syndrome (Option C) is characterized by the triad of rheumatoid arthritis, splenomegaly, and neutropenia.
 - Acute Viral Hepatitis (Option D) Although the patient has elevated liver enzymes and fever, the presence of oral ulcers, pancytopenia, and renal impairment are not typically associated with acute viral hepatitis.
 - However, SLE often has more specific findings such as a malar rash, photosensitivity, arthritis, serositis, and renal involvement (e.g., proteinuria).
-

Solution for Question 14:

Correct Answer: B) Methotrexate

Explanation

Methotrexate has been shown to be curative and is the drug of choice for the treatment of choriocarcinoma.

Mechanism of Action:

- Methotrexate is a folate antagonist that inhibits dihydrofolate reductase (DHFR) and thymidylate synthase, leading to inhibition of DNA synthesis. This action is most effective during the S phase of the cell cycle, where DNA replication occurs.

- Methotrexate (MTX) inhibits AICAR transformylase, disrupting purine synthesis. This leads to the accumulation of AICAR, which is converted to adenosine. Elevated adenosine levels exert anti-inflammatory effects by modulating immune responses, inhibiting lymphocyte activation, and reducing pro-inflammatory cytokine production.

Efficacy in Choriocarcinoma: Methotrexate has been shown to be curative in choriocarcinoma. The typical dosage for choriocarcinoma is 15-30 mg/day for 5 days orally or 20-40 mg/m² BSA i.m. or i.v. twice weekly.

Other Uses:

Uses	Anticancer drug (high dose 100 mg/day)	Immunosuppressive drug (low dose 30 to 70 mg / week)
	• Choriocarcinoma (DOC) • Osteosarcoma (HD-MTx) • Meningeal leukemia • (intrathecal methotrexate) • Breast cancer • Mycosis fungoides • ALL tumor • Head and neck cancer	• Organ transplantation • Psoriasis • Erythrodermic psoriasis • Psoriatic arthritis (DOC) • Wegner's granulomatosis • Sarcoidosis • Crohn's disease • Rheumatoid arthritis (DOC) • Ectopic Pregnancy • Abortion • Multiple sclerosis • (less common)

The other options, while effective for other cancers, are not used for treating choriocarcinoma.

Solution for Question 15:

Correct Answer: D) Vincristine

Explanation:

The patient's pancytopenia is probably due to bone marrow suppression caused by his anti-cancer drugs.

- Vincristine, a vinca alkaloid, specifically targets cells during the M phase (mitosis).
- It binds to tubulin and inhibits microtubule formation.
- This disruption primarily affects cells with high mitotic activity but has a preferential effect on neural tissues due to their microtubule-dependent processes and has very little effect on bone marrow cells.

Other anticancer drugs which do not cause bone marrow suppression include:

- Asparaginase
- Bleomycin

- Chlorambucil

Cyclophosphamide (Option A) is an alkylating agent very well known to cause bone marrow suppression.

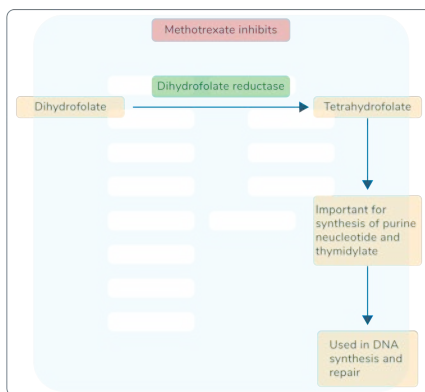
MOA of Alkylating agents causing cell death:



Methotrexate (Option B) is a folate analog that inhibits dihydrofolate reductase (DHFR) needed for DNA synthesis and repair, causing bone marrow suppression.

Doxorubicin (Option C) causes bone marrow suppression by DNA Intercalation, Topoisomerase II Inhibition, and free radical formation causing DNA damage.

MOA:



Solution for Question 16:

Correct Answer: B) Hydroxyurea

Explanation:

This patient most likely has sickle cell disease and is experiencing painful vaso-occlusive crises. Hydroxyurea is the drug of choice in this scenario and it works by increasing fetal hemoglobin (HbF) levels, which helps reduce the frequency of sickle cell crises.

Mechanism of Action: Hydroxyurea blocks the conversion of ribonucleotides to deoxyribonucleotides by inhibiting the enzyme ribonucleoside diphosphate reductase (RNR), thereby interfering with DNA synthesis.

Hydroxyurea treats sickle cell anemia by:

Other Uses of Hydroxyurea:

- Chronic myeloid leukemia,
- Psoriasis,
- Polycythemia vera,
- Occasionally used in some solid tumors as a radiosensitizer before radiotherapy.

Dosage of Hydroxyurea:

- Daily dose: 20-30 mg/kg
- Twice weekly dose: 80 mg/kg

Methotrexate (Option A): Methotrexate is primarily used for rheumatoid arthritis and certain types of cancer. Methotrexate inhibits dihydrofolate reductase, blocking DNA synthesis by preventing the conversion of dihydrofolate to tetrahydrofolate.

Imatinib (Option C): Imatinib is used for chronic myeloid leukemia and gastrointestinal stromal tumors. Imatinib is a tyrosine kinase inhibitor that specifically targets the BCR-ABL fusion protein in CML and other cancers, blocking signaling pathways that promote cell proliferation and survival.

Anagrelide (Option D): Anagrelide is used for thrombocythemia. Anagrelide is a selective phosphodiesterase III inhibitor that reduces platelet production by inhibiting megakaryocyte maturation, thereby lowering elevated platelet counts in conditions like essential thrombocythemia.

Solution for Question 17:

Correct Answer: A) Cisplatin and amifostine

Explanation:

In the given clinical scenario, the combination of cisplatin and amifostine is appropriate. Cisplatin will enhance the radiation's impact on the squamous cell carcinoma, increasing the likelihood of tumor cell death. Meanwhile, amifostine will protect the patient's normal tissues from radiation-induced damage, thus addressing the patient's concerns about potential harm to surrounding healthy tissues.

Radio-sensitizers:

- Agents that enhance the effectiveness of radiation therapy by making cancer cells more susceptible to the damage caused by radiation.
- Cisplatin is a commonly used radio-sensitizer, particularly in head and neck cancers. It works by forming DNA adducts that interfere with DNA repair mechanisms, thus increasing the likelihood of radiation-induced cell death in tumor cells.
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Radio-protectors:

- Substances used to protect normal tissues from the harmful effects of radiation.
- Amifostine is a well-known radio-protector that selectively protects normal tissues without shielding the tumor cells. It acts by scavenging free radicals and providing thiol groups that repair radiation-induced damage in normal cells.
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Radiosensitizers	Radio-protectors
• Makes cancer cells prone to radiotherapy • Drugs • Cisplatin • Hydroxyurea • 5 Fluorouracil • Gemcitabine	• Protects normal organs from radiotherapy • Drugs • Vitamin A, C, E • Amifostine • N-acetylcysteine (Liver protector) • Antidote for paracetamol poisoning • Cough suppressants

Doxorubicin and vitamin C (Option B):

- Doxorubicin has some radiosensitizing properties but is not typically used in head and neck cancers for this purpose.
- Vitamin C is not a recognized radio-sensitizer.

Metformin and hyperbaric oxygen (Option C):

- Metformin has some evidence of radiosensitizing effects, especially in certain cancer types, but it is not a standard agent for this purpose.
- Hyperbaric oxygen is more commonly used as a radio-sensitizer but is not combined with Metformin in standard practice.

Gemcitabine and sodium selenite (Option D):

- Gemcitabine can act as a radiosensitizer in some cancers but is not standard for head and neck cancers.
- Sodium selenite is not a recognized radio-protector.

Solution for Question 18:

Correct Answer: C) Administer i.v. hydration and consider using amifostine as a nephroprotective agent.

Explanation:

This patient with metastatic testicular cancer, who has been receiving cisplatin and reports of decreased urine output, and mild swelling in his ankles with elevated serum creatinine level and reduced GFR likely developed cisplatin-induced nephrotoxicity.

Cisplatin is known for its nephrotoxic effects, which can lead to acute kidney injury.

The management of cisplatin-induced nephrotoxicity includes

- Adequate i.v. fluids - Chloride diuresis (Cl inactivates cisplatin)
- Mannitol - diuresis of inactivated cisplatin
- Use of nephroprotective agents like IV Amifostine (by scavenging free radicals and reducing oxidative stress)

Increasing the cisplatin dosage could exacerbate the kidney damage.(Option A)

Switching to carboplatin is not ideal, as carboplatin can also cause nephrotoxicity, although to a lesser extent than cisplatin.(Option B)

Discontinuing all chemotherapy and initiating dialysis is too extreme without trying other nephroprotective measures first.(Option D)

Solution for Question 19:

Correct Answer: A) Carboplatin is used for the treatment of retinoblastoma along with bleomycin and vincristine.

Explanation: Carboplatin is used in retinoblastoma as a part of VEC regimen, of which bleomycin is not a component.

VEC regimen for retinoblastoma- Vincristine, Etoposide, Carboplatin.

Also, paclitaxel, when used with cisplatin or carboplatin induces complete responses in the majority of patients with carcinoma of the ovary.

Uses of Cisplatin and Carboplatin:

Gastrointestinal Tract (GIT) Cancers:	Esophagus, stomach, colorectal, anal cancers Combined with 5- fluorouracil and radiosensitizers
Head and Neck Cancers: (Option C)	Combined with 5- fluorouracil and radiosensitizers
Urogenital Cancers:	Cervical Cancer Ovarian Cancer (Option D) Testicular Cancer (BEP regimen: Bleomycin, Etoposide, Platin- Cisplatin/Carboplatin)
Retinoblastoma:	VEC regimen (Vincristine, Etoposide, Carboplatin)
Other Uses:	Osteosarcoma (Option B) Lung Cancer

Paclitaxel (Option D): It inhibits microtubule disassembly during cell division, leading to cell cycle arrest and apoptosis. Cisplatin/carboplatin cross-links DNA strands, preventing DNA replication and transcription, which also leads to cell death. Thus, they act synergistically to produce complete resolution of ovarian cancer.

Solution for Question 20:

Correct Answer:- B) 1-B, 2-C, 3-D, 4-A

Explanation:

Sargramostim

- Indications: Accelerates bone marrow recovery after chemotherapy or transplant; treats severe neutropenia.
- MOA: GM-CSF that stimulates the production and activation of granulocytes and macrophages.

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Melphalan

- Indications: Treats multiple myeloma; used as a conditioning regimen before stem cell transplant.
- MOA: Alkylating agent that forms DNA cross-links, preventing replication and causing cell death.

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Plerixafor

- Indications: Mobilizes stem cells for collection in lymphoma or multiple myeloma patients.
- MOA: CXCR4 antagonist that disrupts the binding of stem cells to bone marrow, facilitating their release into the bloodstream.

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MOA: CXCR4 antagonist that disrupts the binding of stem cells to bone marrow, facilitating their release into the bloodstream.

Filgrastim

- Indications: Prevents or treats neutropenia from chemotherapy or bone marrow disorders.
- MOA: G-CSF that stimulates neutrophil production and function in the bone marrow.

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MOA: G-CSF that stimulates neutrophil production and function in the bone marrow.

Drugs for Myeloablation	Drugs for stem cell extraction
- Given for bone marrow transplantation - Drugs - Busulfan - Melphalan - Cyclophosphamide - Thiotepa	- G-CSF: Filgrastim - GM-CSF: Sargramostim - Plerixafor: Blocks CXCR 4 receptor (holds stem cells inside the bone marrow.)

Solution for Question 21:

Correct Answer: A) 1-B, 2-A, 3-E, 4-C, 5-D

Explanation:

Miscellaneous Alkylating agents used as Cytotoxic drugs

Thiotepa - BBB penetration is high - DOC for Glioma tumors • Astrocytoma • Glioblastoma multiforme - Resistant Glioma tumors are given - Previously used in Melanoma
Busulfan - Used in the treatment of Hodgkin's lymphoma - Given under the ABVD regimen
Procarbazine - Treatment of Hodgkin's lymphoma - Was given under the MOPP regimen - Side effects • Male infertility (Azoospermia) • Depression - Disulfiram like reaction • Procarbazine should be avoided in alcoholics
Flucytosine Ornithine decarboxylase used in myelodysplastic procedure
Alkylating - Used in the treatment of Ovarian cancer (DOC: cisplatin) - Given in cisplatin resistant cases
Taxanes Used in the treatment of Neuroblastoma
Lutetide Used in the treatment of lung cancer resistant to cisplatin

Alkylating agents are chemotherapy drugs that attach alkyl groups to DNA, causing cross-linking, disrupting replication, and leading to cell death.

Solution for Question 22:

Correct Answer: B) Prompt administration of warfarin to reduce chances of another episode of VOD.

Explanation:

The patient is a typical case of Budd-Chiari like Syndrome, a type of Veno Occlusive Disease (VOD) commonly seen in the patients on busulfan. Warfarin is an anticoagulant and is not used for the treatment of VOD. Therefore, this statement is incorrect.

Side effects of Busulfan includes:-

- Bone marrow suppression (maximum risk)
- Pulmonary fibrosis
- Seizures
- Budd-Chiari like syndrome
- Tumor lysis syndrome

The patient's should be started on defibrotide as the symptoms are likely due to Budd-Chiari syndrome (Veno-Occlusive Disease) (Option A):

The management of VOD focuses on use of defibrotide which is an antithrombotic agent as the treatment of choice.

Patients should be started on antiepileptics such as phenytoin to prevent seizures (Option C):

Patients on Busulfan, if experiencing episodes of seizures, are to be started on phenytoin, which is the frequently used anti-epileptic for patients on Busulfan. However, Phenytoin also induces the enzymes that metabolize busulfan, reducing its plasma concentration by about 20% thus reducing its side effects.

Adjust the dose of busulfan based on plasma drug-level monitoring (Option D):

Adjusting the dose of Busulfan based on plasma drug-level monitoring is frequently done to balance therapeutic efficacy and minimize toxicity, especially since Busulfan clearance can vary significantly among patients.

Solution for Question 23:

Correct Answer: A) Hemorrhagic cystitis is primarily caused by the direct cytotoxic effect of cyclophosphamide on the bladder epithelium.

Explanation:

Hemorrhagic cystitis is not caused by the direct cytotoxic effect of cyclophosphamide itself. Instead, it is caused by a metabolite of cyclophosphamide, acrolein, which is excreted in the

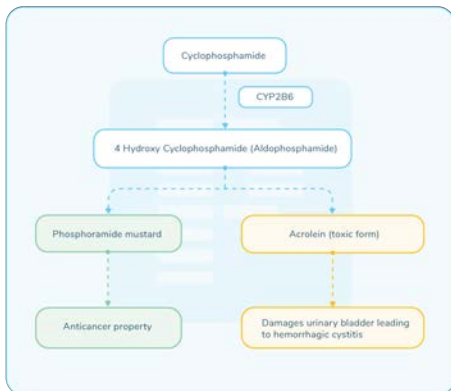
urine and irritates the bladder lining. It is characterized by painful hematuria & increased urinary frequency & urgency.

Side effects of cyclophosphamide includes:

- Ototoxicity
- Hemorrhagic cystitis

Cyclophosphamide is metabolized into 4-hydroxy cyclophosphamide in the liver by CYP2B6 (Option C):

Cyclophosphamide is metabolized in the liver by CYP2B6 as follows:



Ifosfamide, a modification of cyclophosphamide, is more likely to cause hemorrhagic cystitis than cyclophosphamide itself (Option B):

Ifosfamide, a modification of cyclophosphamide, is more likely to cause hemorrhagic cystitis than cyclophosphamide, by producing higher levels of the same toxic metabolite, acrolein.

Hemorrhagic cystitis is treated by MESNA (Mercapto-Ethane-Sulfonate-Sodium), which neutralizes toxic effects in the bladder (Option D):

Hemorrhagic cystitis due to Cyclophosphamide/Ifosfamide is prevented by Oral MESNA and treated by i.v. MESNA, which neutralizes the toxic effects of Acrolein.

Solution for Question 24:

Correct Answer: D) Melphalan

Explanation:

The patient's symptoms, clinical signs, and lab investigations confirm the diagnosis of multiple myeloma

- Melphalan is a nitrogen mustard derivative specifically used to treat multiple myeloma.
- It works by alkylating DNA and RNA, leading to the disruption of cancer cell replication and eventually cell death.
- It is often used in combination with other drugs: Melphalan + Thalidomide + Dexamethasone
- It is also given for myeloablative procedure

Cyclophosphamide (Option A), mechlorethamine (Option B) and chlorambucil (Option C): They are also nitrogen mustard derivatives used in oncology, but they are not the first line of choice for multiple myeloma.

Nitrogen mustard derivatives and its uses:

Nitrogen mustards	Uses
Melphalan (Option D)	Tx of Multiple Myeloma Myeloablative procedures
Mechlorethamine (Option B ruled out)	Tx of Hodgkin's lymphoma (ABVD & MOPP Regimen) Tx of cutaneous T-cell lymphoma
Cyclophosphamide (Option A ruled out)	DOC in Burkitt's lymphoma Tx of other cancers eg. Breast cancer Ovarian cancer Neuroblastoma Non-Hodgkin's lymphoma Immunosuppressant in Wegener's Granulomatosis (DOC) Acute cerebral vasculitis Rheumatoid arthritis Steroid dependent nephrotic syndrome
Chlorambucil (Option C ruled out)	2nd line Tx of CLL
Busulfan	Tx of CML and AML in resistant cases

Solution for Question 25:

Correct Answer: C) 5- Fluorouracil

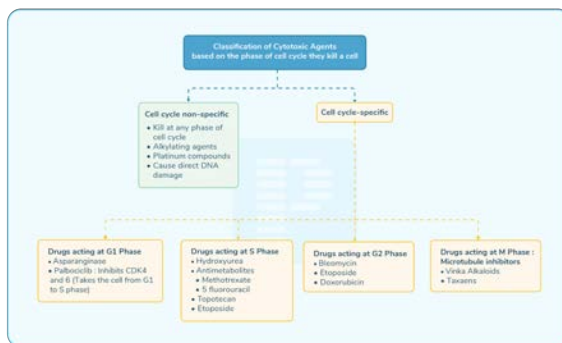
Explanation:

Fluorouracil primarily acts on the S phase by inhibiting thymidylate synthase, an enzyme essential for DNA synthesis.

Vincristine (Option A): Acts during the M phase (mitosis) by disrupting microtubule assembly.

Cisplatin (Option B): Acts during all phases of the cell cycle by forming DNA adducts and inhibiting DNA repair.

Bleomycin (Option D): Acts during the G2 phase of cell cycle by binding to the DNA and producing free radicals that damage the DNA.



Solution for Question 26:

Correct Answer: D) Procarbazine

Explanation:

The patient has likely developed secondary cancer (AML) due to the use of procarbazine.

- Procarbazine has the maximum risk of causing secondary AML and myelodysplastic syndrome (t-MDS) due to their mechanism of action, which involves DNA damage and mutagenesis.

Other drugs causing secondary cancer

- Etoposide: Causes early cancer (within <3yrs)
- Mitoxantrone
- Cisplatin: Causes chronic cancer (after >3yrs)

Doxorubicin (Option A): It is an anthracycline antibiotic that is commonly used in the ABVD regimen for Hodgkin's lymphoma. It is mainly known for its cardiotoxicity, which can lead to heart failure.

Vincristine (Option B): It is a vinca alkaloid used in the Stanford V and BEACOPP regimens for Hodgkin's lymphoma. Its major side effects include neurotoxicity, which can cause peripheral neuropathy and myelosuppression.

Bleomycin (Option C): It is included in the ABVD, BEACOPP, and Stanford V regimens. The primary side effect of bleomycin is pulmonary toxicity, which can lead to pulmonary fibrosis. It is also associated with skin changes and fever.

Solution for Question 27:

Correct Answer: D) Sodium bicarbonate

Explanation:

The patient's symptoms and lab values suggest that he has tumor lysis syndrome.

Tumor lysis syndrome occurs due to the rapid killing of cancerous cells in the blood. It is characterized by :

- Hyperkalemia
- Hyperuricemia
- Hyperphosphatemia
- Hypocalcemia

Intrinsic tumor - related risk factors	Patient - related risk factors
High risk cancers - Burkitt lymphoma, ALL and AML (with WBC > 1 lakh), Diffuse large B cell lymphoma, etc.	Pre-treatment Hyperuricemia or Hyperphosphatemia.
High proliferation Rate.	Pre-existing nephropathy / oliguria.
Large tumor burden (> 10cm).	Inadequate hydration.
Chemosensitivity.	

Sodium bicarbonate worsens the condition by increasing the precipitation of calcium phosphate in kidneys:

- Sodium bicarbonate is an alkaline substance. When administered, it can increase the pH of the urine, making it more alkaline.
- Calcium phosphate has decreased solubility in alkaline conditions. As the urine pH rises, the solubility of calcium phosphate decreases, leading to a higher risk of precipitation.
- Hence, sodium bicarbonate is not given to patients with tumor lysis syndrome.

Aggressive intravenous hydration (Option A): It is essential to help flush out the kidneys and reduce the risk of uric acid and calcium phosphate crystal deposition.

Allopurinol (Option B): It is used to inhibit the production of uric acid, thus preventing hyperuricemia present in tumor lysis syndrome.

Rasburicase (Option C): It is an enzyme that helps break down uric acid to a more soluble form (allantoin), which is easier to excrete.

Targeted Therapy and Monoclonal Antibodies

1. A 45 y/o male presents with fatigue, unintentional weight loss, and left upper quadrant abdominal discomfort. Physical examination shows splenomegaly. Lab tests show leukocytosis. The peripheral blood smear shows increased basophils, myelocytes, and segmented neutrophils. Bone marrow biopsy confirms hypercellularity with increased myeloid precursors and the presence of the Philadelphia chromosome (BCR-ABL fusion gene). Which of the following drugs would be your DOC to treat the patient?

- A. Ponatinib
 - B. Imatinib
 - C. Nilotinib
 - D. Dasatinib
-

2. A 60-year-old female patient diagnosed with a gastrointestinal stromal tumor (GIST) has been started on first-line treatment. Genetic testing reveals a mutation in the c-KIT gene. After several months, the patient develops resistance to the initial therapy. The oncologist considers alternative medications. Which of the following drugs is specifically known for selectively inhibiting the Platelet-Derived Growth Factor Receptor (PDGFR) rather than multiple tyrosine kinase receptors?

- A. Imatinib
 - B. Sunitinib
 - C. Ripretinib
 - D. Regorafenib
-

3. Which of the following conditions is not commonly treated with Bruton's Tyrosine Kinase (BTK) inhibitors?

- A. Mantle Cell Lymphoma
- B. Marginal Zone Lymphoma
- C. Chronic Lymphocytic Leukemia
- D. Hodgkin's Lymphoma

4. Match the following drugs to their appropriate mutation targets in cancer treatment: 1. Erlotinib A. HER2 (+)ve Breast Cancer 2. Osimertinib B. HER1 (EGFR) mutation in Non-Small Cell Lung Cancer 3. Lapatinib C. T790M mutation in EGFR receptor 4. Neratinib D. Trastuzumab-resistant HER2 (+)ve Breast Cancer

1. Erlotinib	A. HER2 (+)ve Breast Cancer
2. Osimertinib	B. HER1 (EGFR) mutation in Non-Small Cell Lung Cancer
3. Lapatinib	C. T790M mutation in EGFR receptor
4. Neratinib	D. Trastuzumab-resistant HER2 (+)ve Breast Cancer

- A. 1-C, 2-B, 3-D, 4-A
- B. 1-B, 2-C, 3-A, 4-D
- C. 1-C, 2-B, 3-A, 4-D
- D. 1-B, 2-C, 3-D, 4-A

5. Match the drugs & their clinical use: Drug Clinical Use 1. Pazopanib A. Cholangiocarcinoma 2. Midostaurin B. Urothelial Carcinoma 3. Erdafitinib C. Acute Myeloid Leukemia (AML) with FLT-3 mutation 4. Infigratinib D. Renal Cell Carcinoma (RCC)

Drug	Clinical Use
------	--------------

1. Pazopanib	A. Cholangiocarcinoma
2. Midostaurin	B. Urothelial Carcinoma
3. Erdafitinib	C. Acute Myeloid Leukemia (AML) with FLT-3 mutation
4. Infigratinib	D. Renal Cell Carcinoma (RCC)

- A. 1-D, 2-C, 3-B, 4-A
- B. 1-D, 2-C, 3-A, 4-B
- C. 1-C, 2-D, 3-A, 4-B
- D. 1-C, 2-D, 3-B, 4-A

6. A 58-year-old male patient diagnosed with malignant melanoma has not responded to initial therapies. His oncologist is considering targeted therapy options. Which of the following inhibitors can be used in malignant melanoma?

- A. Sirolimus
- B. Binimetinib
- C. Idelalisib
- D. Adagrasib

7. A 52-year-old female presents with a diagnosis of invasive ductal carcinoma of the breast. The tumour measures 3 cm in diameter and has been staged as T2N1M0. Immunohistochemical testing reveals that the cancer cells overexpress the HER2 protein and are ER-negative. The patient has no significant comorbidities and has been scheduled for surgery. Which of the following drugs is most suitable for this patient as part of her adjuvant chemotherapy regimen

- A. Tamoxifen
- B. Cetuximab
- C. Trastuzumab

D. Bevacizumab

8. A 60-year-old male with a significant smoking history presents with a persistent cough, dyspnea, weight loss, and fatigue. A Chest CT reveals a suspicious mass in the right upper lobe. A biopsy of the mass shows adenocarcinoma. Further molecular testing reveals high PD-L1 expression (>50%) and no actionable mutations in EGFR, ALK, or ROS1. Which of the following drugs would you choose as the first line of treatment in this case?

- A. Erlotinib
 - B. Bevacizumab
 - C. Pembrolizumab
 - D. Ipilimumab
-

9. A 45-year-old man presents to the dermatology clinic with chronic, thick, silvery-white, scaly plaques on his elbows, knees, and lower back. He reports that the plaques have been present for several years and have gradually worsened. He has tried various OTC creams with minimal relief. The plaques are extensive on examination, and the patient rates his itching and discomfort as moderate to severe. Which of the following drugs is NOT relevant for this patient's treatment?

- A. Ixekizumab
 - B. Secukinumab
 - C. Etanercept
 - D. Mogamulizumab
-

10. A 65-year-old male with hypertension and CKD presents with gross hematuria and flank pain. Imaging studies reveal a 7 cm mass in the left kidney concerning

renal cell carcinoma. He undergoes a radical nephrectomy with pathological confirmation of clear cell RCC, pT3aN0M0 (stage III). Due to his advanced stage, you are considering adjuvant therapy with Aldesleukin. Which of the following is a potential mechanism of action of aldesleukin in treating RCC?

- A. Induction of angiogenesis
 - B. Activation of T cells
 - C. Inhibition of vascular endothelial growth factor (VEGF)
 - D. Direct cytotoxicity against tumor cells
-

11. A 65-year-old male with refractory multiple myeloma has undergone multiple lines of treatment including proteasome inhibitors, immunomodulatory drugs, and autologous stem cell transplantation. Despite these treatments, his disease has progressed. His oncologist decided to start him on a novel therapy that targets B-cell maturation antigen (BCMA) and delivers a cytotoxic agent to myeloma cells. Which drug is the patient likely being prescribed?

- A. Daratumumab
 - B. Elotuzumab
 - C. Belantamab Vedotin
 - D. Ixekizumab
-

12. A 58-year-old female with metastatic melanoma has experienced disease progression after treatment with a BRAF inhibitor and an immune checkpoint inhibitor targeting CTLA-4. Her oncologist now recommends a monoclonal antibody Pembrolizumab, that targets the PD-1 receptor to help reinvigorate T-cell activity against the tumor. Pembrolizumab can be used in the treatment of all of the following, except?

- A. Non-small cell lung cancer
- B. Breast cancer

- C. Renal Cell carcinoma
 - D. Multiple myeloma
-

13. A 48-year-old woman with newly diagnosed HER2-positive invasive ductal carcinoma is started on a neoadjuvant therapy that includes a monoclonal antibody. This antibody specifically binds to the extracellular domain of the HER2 receptor, blocking the activation of downstream signaling pathways involved in cell growth and survival. Additionally, it enhances antibody-dependent cellular cytotoxicity (ADCC), leading to improved immune-mediated destruction of tumor cells. Despite the initial efficacy of the treatment, her oncologist monitors her closely for potential cardiotoxicity, a known side effect of this therapy. Which monoclonal antibody is the patient most likely receiving?

- A. Pertuzumab
 - B. Trastuzumab
 - C. Cetuximab
 - D. Bevacizumab
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	4
Question 4	4
Question 5	1
Question 6	2
Question 7	3
Question 8	3
Question 9	4
Question 10	2
Question 11	3
Question 12	4
Question 13	2

Solution for Question 1:

Correct Answer: B) Imatinib

Explanation:

The clinical scenario given, describes a patient who has CML.

Chronic Myeloid Leukemia (CML) is a myeloproliferative disorder characterized by the presence of the Philadelphia chromosome, resulting from a reciprocal translocation between chromosomes 9 and 22 [t(9;22)(q34;q11)], which generates the BCR-ABL fusion gene. This fusion gene encodes a constitutively active tyrosine kinase that drives the uncontrolled

proliferation of myeloid cells

Imatinib binds to the ABL kinase portion of the BCR-ABL tyrosine kinase receptor, inhibiting its attachment and preventing cell division. By inhibiting this aberrant tyrosine kinase activity, Imatinib selectively targets the underlying cause of CML, leading to inhibition of leukemic cell proliferation and induction of apoptosis.

Ponatinib (Option A), Nilotinib (Option C) & Dasatinib (Option D), and Bosutinib can also be used for treating CML but neither of them is DOC for CML.

- Dasatinib and Nilotinib are used in Imatinib-resistant cases.
- Ponatinib is used in Dasatinib and Nilotinib-resistant cases. In further resistant cases, Omacetaxine.

Generation	Drugs
1st Generation	Erlotinib Gefitinib
2nd Generation	Afatinib Dacomitinib
3rd Generation	Osimertinib
4th Generation	Mobocertinib

Solution for Question 2:

Correct Answer: C) Ripretinib

Explanation:

Ripretinib is a selective inhibitor of Platelet-Derived Growth Factor Receptor (PDGFR) & C-KIT, while the rest of the drugs block multiple types of Tyrosine kinase receptors (BCR-ABL, PDGFR, C-KIT).

In GIST, there is a mutation in either of the 2 receptors, PDGFR Alpha or C-KIT leading to this tumor.

Newer drugs such as Ripretinib & Avapritinib selectively block PDGFR Alpha & C-KIT Receptors.

Drugs used in GIST with their mechanism of action:

MOA	Drugs
-----	-------

Multiple TK Inhibitor (BCR-ABL, PDGFR, C-KIT)	Sunitinib, Imatinib, Regorafenib (Option A,B,D)
Selective Inhibitor of PDGFR & C-KIT	Repretinib, Avapritinib

Solution for Question 3:

Correct Answer: D) Hodgkin's Lymphoma

Explanation:

BTK inhibitors include drugs like Ibrutinib, Acalabrutinib, and Zanubrutinib are primarily used to treat various types of non-Hodgkin lymphomas and leukemias like:

- Mantle Cell Lymphoma (Option A)
- Marginal Zone Lymphoma (Option B)
- Chronic Lymphocytic Leukemia (Option C)
- Small Lymphocytic Leukemia
- Waldenstrom Macrogammaglobulinemia

They are not used for treating Hodgkin's Lymphoma■.

Solution for Question 4:

Correct Answer: D) 1-B, 2-C, 3-D, 4-A

Explanation:

Erlotinib is used for treating EGFR (HER1) positive non-small cell lung cancer.

Osimertinib is used for treating non-small cell lung cancer with the T790M mutation in the EGFR receptor.

Lapatinib is given in cases of trastuzumab-resistant HER2 positive breast cancer.

Neratinib is used for treating HER2 positive breast cancer■.

Mutation Target	Drug	Generation/Type	Explanation
HER1 (EGFR) - Non-Small Cell Lung Cancer	Erlotinib, Gefitinib	1st Generation	Given in EGFR (+)ve lung cancer, pancreatic cancer.
Afatinib, Dacomitinib	2nd Generation	Given in resistance of Erlotinib & Gefitinib.	
Osimertinib	3rd Generation	Given in resistant cases of Afatinib & Dacomitinib. Given in T790M mutation in EGFR receptor.	
Mobocertinib	4th Generation	Given in resistant cases of Osimertinib.	
Cetuximab, Panitumumab, Necitumumab	MAB Approved	Blocks EGFR.	
HER2+ve - Breast Cancer	Lapatinib		Given in Trastuzumab-resistant cases.
Neratinib, Tucatinib		Used in HER2 (+) ve breast cancer.	

Solution for Question 5:

Correct Answer: A) 1-D, 2-C, 3-B, 4-A

Explanation:

- Pazopanib - VEGFR Inhibitor- Renal Cell Carcinoma (RCC)
- Midostaurin - VEGFR Inhibitor- Acute Myeloid Leukemia (AML) with FLT-3 mutation
- Erdafitinib - FGFR Inhibitor- Urothelial Carcinoma
- Infigratinib - FGFR Inhibitor- Cholangiocarcinoma

VEGFR Inhibitors and uses:

Drug Name	Clinical Use
Pazopanib, Axitinib, Tivozanib	Renal Cell Carcinoma (RCC)
Sorafenib, Lenvatinib	Renal Cell Carcinoma (RCC), Hepatocellular Carcinoma (HCC)

Sunitinib, Regorafenib	Renal Cell Carcinoma (RCC), Gastrointestinal Stromal Tumors (GIST)
Vandetanib, Cabozantinib	Medullary Carcinoma of the Thyroid
Midostaurin, Gilteritinib	Acute Myeloid Leukemia (AML) with FLT-3 mutation

FGFR Inhibitors and uses:

Drug Name	Clinical Use
Erdafitinib	Urothelial Carcinoma
Pemigatinib, Infigratinib	Cholangiocarcinoma

Solution for Question 6:

Correct Answer: B) Binimetinib

Explanation:

Binimetinib is a MEK inhibitor, used specifically in malignant melanoma. MEK Inhibitors belong to the class of Protein Kinase Signaling Inhibitors.

Protein Kinase Signaling Inhibitors:

Inhibitor Type	Drugs	Uses
RAS Inhibitors	Sotorasib, Adagrasib (Option D Ruled out)	NSCLC with KRAS mutation
B-RAF Inhibitors	Vemurafenib, Dabrafenib, Encorafenib	Malignant Melanoma with B-RAF mutation
MEK Inhibitors	Trametinib, Cobimetinib, Binimetinib (Option B), Selumetinib	Malignant Melanoma
PI3K Inhibitors	Idelalisib (Option C Ruled out), Copanlisib, Duvelisib, Umbralisib, Alpelisib	B Cell Lymphoma (CLL, SLL, FL)
mTOR Inhibitors	Sirolimus (Option A Ruled out), Temsirolimus, Everolimus	RCC (Renal Cell Carcinoma), Breast Cancer, PNET (Pancreatic Neuroendocrine Tumors)

Solution for Question 7:

Correct Answer: C) Trastuzumab

Explanation:

- Trastuzumab (Option C) is a monoclonal antibody targeting the HER2 protein, which is overexpressed in HER2-positive breast cancers. This patient has HER2 +ve, ER -ve stage II breast cancer.
- It binds to the extracellular domain IV of HER2, reduces HER2 signaling, and can induce antibody-dependent, immune cell-mediated cytotoxicity, inhibiting cancer cell proliferation.
- It is specifically used to treat HER2-positive breast cancer, both in early-stage and metastatic settings.

Tamoxifen (Option A):

- A selective estrogen receptor modulator (SERM), binds to estrogen receptors on breast cancer cells, thereby blocking the effects of estrogen, which can promote the growth of breast cancer cells.
- It is primarily used for estrogen receptor-positive (ER+) breast cancer.

Cetuximab (Option B):

- It is a monoclonal antibody that targets the epidermal growth factor receptor (EGFR) and thus prevents cell proliferation.
- It treats certain types of colorectal and head and neck cancers that overexpress EGFR rather than HER2.

Bevacizumab (Option D):

- It is a monoclonal antibody that targets vascular endothelial growth factor (VEGF), inhibiting angiogenesis.
- It is used for metastatic colorectal cancer, non-small cell lung cancer, renal cell carcinoma, and glioblastoma. It is not usually associated with the treatment of HER2 +ve breast cancer.

Important MABs in tumours:

MAB	Uses
Trastuzumab Pertuzumab	• HER2 +ve Breast Cancer
ADO-Trastuzumab	• Trastuzumab resistant HER2+ve Breast cancer
Panitumumab Necitumumab	• Non-small cell lung cancer • Metastatic Head & Neck cancer • Metastatic colorectal cancer
Cetuximab	• Non-small cell lung cancer • Metastatic Head & Neck cancer • Metastatic colorectal cancer
Rituximab	• Rheumatoid Arthritis • L: Lupus erythematosus • I: Idiopathic thrombocytopenic purpura • A: Autoimmune hemolytic anemia • N: Non-Hodgkin lymphoma • C: CLL
Isatuximab	• Multiple Myeloma
Alemtuzumab	• CLL • Multiple sclerosis
Mogamulizumab	• Mycosis fungoides • Sezary syndrome

Solution for Question 8:

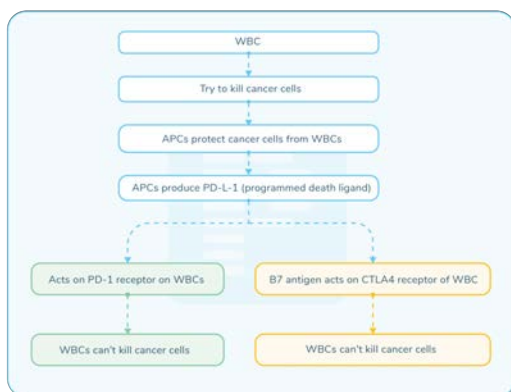
Correct Answer: C) Pembrolizumab

Explanation:

Pembrolizumab is an immune checkpoint inhibitor that targets the PD-1 receptor on T-cells. The patient's symptoms, CT findings, molecular testing, and biopsy tests of adenocarcinoma (the most common subtype of NSCLC) results are typical presentations for a patient with non-small cell lung carcinoma (NSCLC).

- Here, the patient has high PD-L1 expression (>50%).
- Pembrolizumab is particularly effective in NSCLC patients with high PD-L1 expression ($\geq 50\%$) and is approved as a first-line treatment for these patients when there are no actionable mutations in EGFR, ALK, or ROS1.

MOA of Immune Checkpoint Inhibitors:



Erlotinib (Option A): It is a tyrosine kinase inhibitor (TKI) that targets the EGFR. These mutations are more common in non-smokers.

Bevacizumab (Option B): It is a monoclonal antibody that inhibits VEGF, thereby inhibiting angiogenesis. It is used in combination with chemotherapy for certain types of cancer, including NSCLC.

Ipilimumab (Option D): It is a CTLA-4 inhibitor. By blocking CTLA-4, ipilimumab enhances T-cell activation and proliferation, promoting an immune response against cancer cells.

Mnemonic	Drugs	Uses
New	Nivolumab	• NSCLC, Melanoma, Hodgkin lymphoma
Tumour	Tremelimumab	• NSCLC, Melanoma, Hodgkin lymphoma
Drugs	Durvalumab	• Endometrial cancer
Acting	Avelumab	• Urothelial cancers
At	Atezolizumab	
Immune	Ipilimumab	• NSCLC, Melanoma
Check	Cepilimumab	
Point	Pembrolizumab	• Endometrial cancer • > 30 cancers

Immune Checkpoint Inhibitors and their uses:

Blocks CTLA4 receptor	Blocks PD-L1 receptor	Blocks PD-1 receptor
Ipilimumab Tremelimumab	Atezolizumab Avelumab Durvalumab	Nivolumab Cepilimumab Pembrolizumab

Solution for Question 9:

Correct Answer: D) Mogamulizumab

Explanation:

This patient's clinical findings are characteristic of Plaque psoriasis. Mogamulizumab is unsuitable for treating plaque psoriasis.

Mogamulizumab:

- It is a monoclonal antibody that targets CCR4, which is expressed on certain T-cells, including malignant T-cells in cutaneous T-cell lymphoma (CTCL), where it is used as a primary agent for treatment.
- It is mainly used as a treatment for relapsed mycosis fungoides & Sezary syndrome.
- Mogamulizumab is unsuitable for treating plaque psoriasis because it does not target the mechanisms underlying psoriasis inflammation, such as TNF-alpha or IL-17A.

Ixekizumab (Option A) & Secukinumab (Option B) are monoclonal antibodies that specifically target interleukin-17A (IL-17A), a pro-inflammatory cytokine involved in the pathogenesis of psoriasis. They are indicated for treating moderate to severe plaque psoriasis in adults.

Etanercept (Option C) is a soluble TNF-alpha receptor fusion protein that binds to and neutralizes TNF-alpha, a key cytokine involved in the inflammatory process in psoriasis. It is indicated for treating moderate to severe plaque psoriasis, psoriatic arthritis, and other autoimmune conditions.

MABs used in Plaque Psoriasis

Drug	Target	Uses
Ixekizumab Secukinumab Brodalumab	• IL-17	• Plaque psoriasis
Tildrakizumab Guselkumab	• IL-23	• Plaque psoriasis
Ustekinumab	• IL-12 & IL-23	• Plaque psoriasis • IBD (UC/CD)
Efalizumab	• CD11a (LFA) • Leukocyte function antigen	• Plaque psoriasis
Alefacept	• CD2 (LFA)	• Plaque psoriasis

Solution for Question 10:

Correct Answer: B) Activation of T cells

Explanation:

- Aldesleukin (IL-2) stimulates the proliferation of activated T cells and the secretion of cytokines from NK cells and monocytes. IL-2 stimulation increases cytotoxic killing by T cells and NK cells.
- In the context of renal cell carcinoma (RCC), it enhances immune responses against tumour cells, leading to their destruction.

Other Interleukin used in cancer:

Denileukin Diftitox is an IL2 linked with diphtheria toxin and is used in cutaneous T-cell lymphomas.

Induction of angiogenesis (Option A): Aldesleukin does not induce angiogenesis; rather, it enhances the immune response against tumours.

Inhibition of vascular endothelial growth factor (VEGF) (Option C): Aldesleukin does not directly inhibit VEGF.

Direct cytotoxicity against tumour cells (Option D): Aldesleukin primarily works through immune-mediated mechanisms rather than direct cytotoxic effects.

Solution for Question 11:

Correct Answer: C) Belantamab Vedotin

Explanation:

Belantamab Vedotin is an armed monoclonal antibody or an antibody-toxin conjugate approved for the treatment of refractory multiple myeloma.

Drug	Target	Approved for
Gemtuzumab Ozogamicin	· CD33 + Calicheamicin	Relapsed AML
Brentuximab Vedotin	· CD30 + Vedotin (microtubule cleaver)	· Hodgkin's lymphoma (refractory) · Large B-cell lymphoma
Belantamab Vedotin	· BCMA+ Vedotin (microtubule cleaver)	· Refractory multiple myeloma

Polatuzumab Vedotin	· CD19 + Vedotin (microtubule cleaver)	· Relapsed AML
Moxetumomab Pasudotox	· CD22 + Linked to pseudomonas toxin A	· Hairy cell leukemia

Daratumumab (Option A): Acts by blocking CD38 and is used in the treatment of multiple myeloma.

Elotuzumab (Option B): Acts by blocking CD319 and is used in the treatment of multiple myeloma.

Ixekizumab (Option D): Acts by blocking IL-17 and is used in the treatment of plaque psoriasis.

Solution for Question 12:

Correct Answer: D) Multiple myeloma

Explanation:

Pembrolizumab is an immune checkpoint inhibitor that targets the PD-1 receptor on T-cells. It has various clinical uses which can be summarized using the following mnemonic.

Pembrolizumab uses: (Mnemonic)

- P: Pulmonary cancer/ NSCLC (Tested in Keynote trial) (Option A)
- E: Endometrial cancer
- M: Melanoma
- B: Breast cancer (Option B)
- R: Renal cell carcinoma (Option C)
- O: Oesophagus, stomach cancer
- L: Lynch Syndrome/Mismatch DNA
- I: Intestinal cancer/Colorectal
- Z: Neck & head cancer
- U: Urothelial + genital cancer
- M: Merkel cell cancer, mediastinal cancer
- A: Abnormal microsatellite instability cancer
- B: B cell cancer (large)

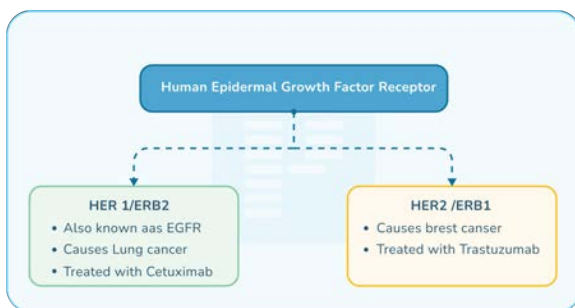
Solution for Question 13:

Correct Answer: B) Trastuzumab

Explanation:

Trastuzumab is a humanized IgG1 monoclonal antibody that binds to the extracellular domain IV of HER2, reduces HER2 signaling, and can induce antibody-dependent, immune cell-mediated cytotoxicity.

Mechanism of action: Blocks HER2/Neu receptor



Therapeutic Uses:

- Trastuzumab is approved for HER2-overexpressing breast and gastric cancer.
- It is used in combination with cytotoxic chemotherapeutics such as taxanes as initial treatment or as a single agent following relapse of disease after cytotoxic chemotherapy.

Side effects of Trastuzumab:

- Cardiotoxic
- Should be avoided with Doxorubicin (>30%)

Previous Year Questions

1. What is basiliximab?

- A. IL-1 receptor antagonist
 - B. Anti-CD3 antibody
 - C. IL-2 receptor antagonist
 - D. TNF inhibitor
-

2. Which chemotherapeutic medication may lead to renal tubular damage in a patient?

- A. Cisplatin
 - B. Streptozocin
 - C. Methysergide
 - D. Cyclophosphamide
-

3. Which of the following is true about lithium?

- A. It can cause postural tremors at therapeutic dosage
 - B. It is not absorbed through gut
 - C. It is used for absent seizure treatment
 - D. Lithium is not teratogenic
-

4. A new drug approved for Rett's syndrome is?

- A. Trofinetide
 - B. Folinic acid
 - C. L-carnitine
 - D. Naltrexone
-

5. What treatment options are available for breast cancer that is HER-2 positive and resistant to trastuzumab?

- A. Lapatinib
 - B. Sorafenib
 - C. Vemurafenib
 - D. Erlotinib
-

6. What is the mechanism of action of thalidomide?

- A. Angiogenesis inhibitor
 - B. Inhibits leukotrienes
 - C. Prevents folic acid synthesis in bacteria
 - D. Inhibits factor Xa
-

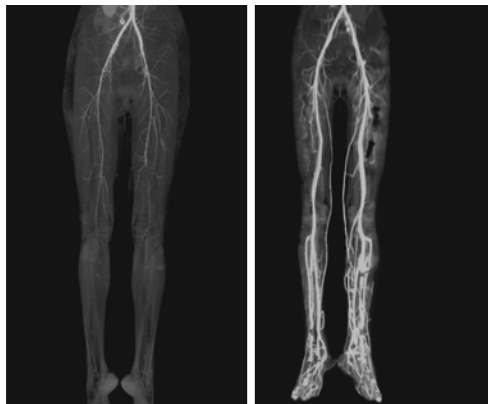
7. Which of the medications listed below is considered safe for the treatment of pregnancy-induced hypertension?

- A. Lisinopril
- B. Propranolol
- C. Alpha methyldopa
- D. Losartan

8. Category A, B, C, D, X in drug classification are indicative of which of the following?

- A. Safety in pregnancy
 - B. Dose adjustment in renal failure
 - C. Cost ratio from cheap to expensive
 - D. As per safety or therapeutic index
-

9. An HIV-positive woman on ART was prescribed ergotamine for a migraine attack four days ago. She presented with complaints that she has been unable to feel her legs, from the mid-thigh to her toes, for the past two days. The angiographic image is given below. What is the likely diagnosis?



- A. Atherosclerosis
 - B. Ergotism
 - C. Polyradicular neuropathy
 - D. Herniated disc
-

10. Nivolumab is used as checkpoint inhibitor in

- A. Retinoblastoma
 - B. Medulloblastoma
 - C. Pleuropulmonaryblastoma
 - D. Hodgkin's lymphoma
-

11. Which drug should not be administered to a pregnant woman?

- A. Erythromycin
 - B. Amoxicillin
 - C. Ceftriaxone
 - D. Streptomycin
-

12. What is the mechanism of action of methyl dopa?

- A. Blockade of beta adrenergic receptors
 - B. Agonism of beta 2 receptors
 - C. Agonism of alpha 2 receptors
 - D. Blockade of alpha 2 receptors
-

13. Which corticosteroid is recommended for promoting the maturation of fetal lungs in preterm labor?

- A. Hydrocortisone

- B. Triamcinolone
 - C. Methylprednisolone
 - D. Betamethasone
-

14. Which of the following medications limit the accessibility of thymidylate?

- A. Rituximab
 - B. Cisplatin
 - C. 5-fluorouracil
 - D. Methotrexate
-

15. Which of the following medications must only be dispensed with a prescription from a licensed medical practitioner?

- A. Schedule M drugs
 - B. Schedule H drugs
 - C. Schedule J Drugs
 - D. Schedule X drugs
-

16. An infant had a serum bilirubin level of 33 mg/dL. The mother might have taken which of the following drugs during the third trimester for this to occur?

- A. Ampicillin
- B. Co-trimoxazole
- C. Azithromycin
- D. Chloramphenicol

17. What is the term used to describe drugs that claim to contain a certain amount of a specific ingredient on their packaging, but actually have a lower amount in reality?

- A. Misbranded drug
 - B. Adulterated drugs
 - C. Spurious drugs
 - D. Counterfeit drugs
-

18. Which of the following teratogenic effects can be caused by the use of lithium during pregnancy?

- A. Cardiac defects
 - B. Facial defects
 - C. CNS defects
 - D. Urogenital defects
-

19. Which among the following is an HER-1 inhibitor?

- A. Panitumumab
 - B. Crizotinib
 - C. Alectinib
 - D. Lorlatinib
-

20. Which of the following anti-cancer medications is used for the treatment of Gastrointestinal Stromal Tumors (GIST) and Chronic Myeloid Leukemia (CML)?

- A. Imatinib
 - B. Methotrexate
 - C. Doxorubicin
 - D. Cisplatin
-

21. What is the primary mechanism of action of methotrexate?

- A. Inhibits DNA synthesis by blocking thymidylate synthase
 - B. Inhibits dihydrofolate reductase, leading to decreased DNA synthesis
 - C. Inhibits RNA synthesis by blocking RNA polymerase
 - D. Interferes with microtubule formation during cell division
-

22. Which of the following drugs used in the treatment of Multiple Myeloma is associated with an increased risk of developing Herpes zoster infection?

- A. Bortezomib
 - B. Daratumumab
 - C. Lomitapide
 - D. Lenalidomide
-

23. Aprepitant, a drug used to prevent late-onset chemotherapy-induced nausea and vomiting, acts through which of the following mechanisms?

- A. NK-1 antagonist
- B. NK-1 agonist

- C. NK-3 antagonist
 - D. NK-2 antagonist
-

24. A 60-year-old patient undergoing treatment with Mycophenolate mofetil (MMF) presents with a serum calcium level of 13.5 mg/dL. He also has myasthenia gravis. What is the most likely cause of the hypercalcemia in this patient?

- A. Drug-induced hypercalcemia
 - B. Primary hyperparathyroidism
 - C. Small cell lung cancer
 - D. Thymoma
-

25. Which adverse effect of Bleomycin is exacerbated by radiation exposure?

- A. Pulmonary toxicity
 - B. Gastric toxicity
 - C. Neural toxicity
 - D. Hepatotoxicity
-

26. What is the key difference between Cisplatin and Carboplatin?

- A. Cisplatin is less neurotoxic than Carboplatin
 - B. Cisplatin causes fewer gastric issues compared to Carboplatin
 - C. Carboplatin is more neurotoxic than Cisplatin
 - D. Cisplatin has more nephrotoxic side effects compared to Carboplatin
-

27. In Carcinoma cervix, which chemotherapeutic agent is used?

- A. Paclitaxel
 - B. Cyclophosphamide
 - C. Cisplatin
 - D. Topotecan
-

28. Which of the following is the most common side effect induced by cisplatin chemotherapy?

- A. GI symptoms
 - B. Neuropathy
 - C. Cardiotoxicity
 - D. Myopathy
-

29. Which of the following drugs is used to prevent painful vaso-occlusive crisis and prevent long-term complications in sickle cell anemia?

- A. Methionine
 - B. Hydroxyurea
 - C. Methotrexate + Leucovorin
 - D. Eltrombopag
-

30. Which drug can be administered in the next cycle of chemotherapy to prevent neutropenia in a patient who has received anti-cancer therapy?

- A. Filgrastim
 - B. Prednisolone
 - C. Vitamin B12
 - D. Folic acid
-

31. How does methotrexate work in osteosarcoma?

- A. Analog of UMP
 - B. Competitive inhibitor of DHFR
 - C. Blocks purine synthesis
 - D. Blocks pyrimidine synthesis
-

32. Which interleukin (IL) is suppressed by tocilizumab among the options provided?

- A. IL-6
 - B. IL-2
 - C. IL-4
 - D. IL-5
-

33. A patient on anti-cancer therapy developed an infection. Blood analysis revealed neutropenia. Which of the following drug is likely to be effective in this patient?

- A. Romiplostim
- B. Oprelvekin
- C. Filgrastim

D. Epoetin alfa

34. Which of the following drugs is associated with untoward side effects of renal tubular damage?

- A. Cisplatin
 - B. Streptozocin
 - C. Methysergide
 - D. Cyclophosphamide
-

35. Which among the following drugs is the new FDA approved immune check point inhibitor for carcinoma endometrium?

- A. Nivolumab
 - B. Pembrolizumab
 - C. Ipilimumab
 - D. Transtuzumab
-

36. Please match the drugs with their respective targets of action. A. Trastuzumab 1. BCR-ABL tyrosine kinase B. Infliximab 2. mTOR C. Sirolimus 3. TNF alpha D. Imatinib 4. HER2/neu

A. Trastuzumab	1. BCR-ABL tyrosine kinase
B. Infliximab	2. mTOR
C. Sirolimus	3. TNF alpha
D. Imatinib	4. HER2/neu

- A. A-2, B-3, C-1, D-4
 - B. A-4, B-3, C-1, D-2
 - C. A-3, B-4, C-2, D-1
 - D. A-4, B-3, C-2, D-1
-

37. Which of the following is not an immune checkpoint inhibitor?

- A. Nivolumab
 - B. Pembrolizumab
 - C. Atezolimumab
 - D. Cetuximab
-

38. KEYNOTE 189 was a clinical trial involving an immune checkpoint inhibitor. It is found to improve survival rates with which of the following?

- A. Pembrolizumab with chemotherapy given for NSCLC (Non-small cell lung cancer)
 - B. Nivolumab with chemotherapy given for NSCLC
 - C. Only Pembrolizumab for NSCLC
 - D. Only nivolumab for NSCLC
-

39. Which is most cardiotoxic anticancer drug?

- A. Anthracyclines
- B. Cyclophosphamide
- C. Tamoxifen
- D. Imatinib

40. Which of the following anti-CD 20 monoclonal antibodies is used in cancer therapy?

- A. Methotrexate
 - B. Rituximab
 - C. Cisplatin
 - D. 5-Fluorouracil
-

41. Monoclonal antibody used in cancer?

- A. Rituximab
 - B. Cisplatin
 - C. 5-Fluorouracil
 - D. Methotrexate
-

42. Which one of the following substances acts as a suppressor of DNA synthesis?

- A. 6-mercaptopurine
 - B. Mitomycin
 - C. Actinomycin
 - D. Asparaginase
-

43. What medication is most appropriate for treating the condition described in a 66-year-old male who is experiencing urinary difficulties, back pain, an elevated

PSA blood test, two radiolucent lesions in the bony pelvis on X-ray, and enlarged lymph nodes in the lower abdomen on MRI?

- A. Anastrozole
 - B. Desogestrel
 - C. Goserelin
 - D. Oxandrolone
-

44. What medication is administered to prevent chemotherapy-induced nausea and vomiting in a 45-year-old woman with lung adenocarcinoma who has experienced these symptoms during her previous treatment visits?

- A. Metoclopramide
 - B. Ropinirole
 - C. Domperidone
 - D. Palonosetron
-

45. In a patient diagnosed with breast cancer, who is 53 years old and also has heart failure, which medication should be avoided?

- A. Methotrexate
 - B. Epirubicin
 - C. Cyclophosphamide
 - D. Cisplatin
-

46. A patient on anti cancer therapy developed infection. Blood analysis revealed a total leucocyte count of 2000/mm³. Which of the following drug is likely to be effective in this patient?

- A. Romiplostim
 - B. Oprelvekin
 - C. Filgrastim
 - D. Erythropoietin
-

47. A 40 year old man was treated for a high-grade soft-tissue sarcoma with radiotherapy. Since he was not responding, he was shifted to chemotherapy. After initiating chemotherapy, he developed erythematous rashes over the areas where radiotherapy was administered earlier. Which of the following drugs causes this condition?

- A. Vincristine
 - B. Doxorubicin
 - C. Cyclophosphamide
 - D. 6-mercaptopurine
-

48. Which of the following drugs causes flagellate pigmentation?

- A. Bleomycin
 - B. Doxorubicin
 - C. Vincristine
 - D. Minocycline
-

49. Which is the anticoagulant of choice for the management of cancer-associated thromboembolism?

- A. Unfractionated Heparin

- B. Warfarin
 - C. Low molecular weight heparin
 - D. Dabigatran
-

50. Which of the following is not an alkylating agent:

- A. Melphalan
 - B. Cyclophosphamide
 - C. Busulfan
 - D. 5-Fluorouracil
-

51. A boy presents with pallor and lymphadenopathy, the X-ray of his skull is given below. Which of the following drug is not used for the management of the suspected diagnosis?



- A. Cyclosporine
- B. Methotrexate
- C. Vinblastine
- D. Vincristine

52. What is the specific monoclonal antibody utilized for cancer treatment?

- A. Omalizumab
 - B. Denosumab
 - C. Rituximab
 - D. Methotrexate
-

53. The Nobel Prize in Physiology or Medicine, 2018 was awarded to James P. Allison and Tasuku Honjo for which of the following discoveries?

- A. Discovery of mechanisms of autophagy
 - B. Discoveries of molecular mechanisms controlling the circadian rhythm
 - C. Discovery of cancer therapy by inhibition of negative immune regulation
 - D. The discovery that matures cells can be reprogrammed to become pluripotent
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	1
Question 4	1
Question 5	1
Question 6	1
Question 7	3
Question 8	1
Question 9	2
Question 10	4
Question 11	4
Question 12	3
Question 13	4
Question 14	3
Question 15	2
Question 16	4
Question 17	1
Question 18	1
Question 19	1
Question 20	1
Question 21	2
Question 22	1

Question 23	1
Question 24	4
Question 25	1
Question 26	4
Question 27	3
Question 28	1
Question 29	2
Question 30	1
Question 31	2
Question 32	1
Question 33	3
Question 34	1
Question 35	2
Question 36	4
Question 37	4
Question 38	1
Question 39	1
Question 40	2
Question 41	1
Question 42	1
Question 43	3
Question 44	4
Question 45	2
Question 46	3
Question 47	2
Question 48	1

Question 49	3
Question 50	4
Question 51	4
Question 52	3
Question 53	3

Solution for Question 1:

Correct option: C: IL-2 receptor antagonist

- Basiliximab is an immunosuppressive medication that acts as an IL-2 receptor antagonist.
- It specifically binds to the alpha chain of the IL-2 receptor, which is expressed on activated T lymphocytes.
- By blocking this receptor, basiliximab inhibits IL-2 signaling and prevents T-cell activation and proliferation.
- It is primarily used as part of immunosuppressive therapy in kidney transplantation to reduce the risk of organ rejection.

Incorrect Options:

Option A: IL-1 receptor antagonist- IL-1 receptor antagonists are a class of drugs that specifically block the activity of interleukin-1 (IL-1), a pro-inflammatory cytokine. Basiliximab does not act as an IL-1 receptor antagonist.

Option B: Anti-CD3 antibody- Anti-CD3 antibodies are monoclonal antibodies that target the CD3 antigen on T lymphocytes. They are used in certain immune-mediated disorders and organ transplantation to modulate T-cell function. Basiliximab is not an anti-CD3 antibody.

Option D: TNF inhibitor- Tumor necrosis factor (TNF) inhibitors are a class of drugs that block the activity of TNF, a pro-inflammatory cytokine involved in various immune-mediated diseases. Basiliximab does not have TNF inhibitory properties.

Solution for Question 2:

Correct Option A - Cisplatin:

- Cisplatin is a chemotherapy drug that is well-known for its nephrotoxic (kidney-damaging) effects. It is widely used in the treatment of various cancers, including testicular, ovarian, and bladder cancer. The nephrotoxicity of cisplatin is primarily due to its accumulation in the renal tubules, leading to cellular damage and impairment of renal function.

Incorrect Options:

Option B - Streptozocin: Streptozocin, is not typically associated with renal tubular damage. It is used in the treatment of pancreatic neuroendocrine tumors.

Option C - Methysergide: Methysergide, is not a common chemotherapeutic medication. It is primarily used for the prevention of vascular headaches.

Option D - Cyclophosphamide: Cyclophosphamide, can lead to hemorrhagic cystitis, but it is less commonly associated with renal tubular damage compared to Cisplatin.

Solution for Question 3:

Correct options: A-It can cause postural tremors at therapeutic dosage

This statement is correct. Postural tremors can be a side effect of lithium treatment.

Incorrect options:

b) It is not absorbed through the gut: This statement is incorrect. Lithium is readily absorbed in the gastrointestinal tract after oral administration.

c) It is used for absent seizure treatment: This statement is incorrect. Lithium is not commonly used for the treatment of absent seizures (also known as absence seizures). It is primarily indicated for bipolar disorder and certain other psychiatric conditions. Drugs used for absence seizures include valproate, lamotrigine, and ethosuximide (especially in < 2 years of age).

d) Lithium is not teratogenic: Lithium is a medication commonly used for the treatment of bipolar disorder. However, it is known to be teratogenic, meaning it can cause harm to a developing fetus if taken during pregnancy. Lithium exposure during the first trimester has been associated with an increased risk of cardiac malformations, particularly Ebstein's anomaly, in the fetus.

Solution for Question 4:

Among the options provided, the drug approved for Rett syndrome is Trofinetide.

Correct Options A: Trofinetide

- Trofinetide is a synthetic peptide used to treat Rett syndrome. Trofinetide is a synthetic analog of glycine-proline-glutamate, a naturally occurring peptide. The exact mechanism by which trofinetide exerts therapeutic effects is unknown; trofinetide may exhibit anti-inflammatory and trophic effects.
- Trofinetide has shown promising results in clinical trials, with improvements observed in certain aspects of Rett syndrome symptoms.

Incorrect Options:

Options B: Folinic acid: Folinic acid, also known as leucovorin, serves multiple roles in medicine. It is utilized alongside 5-FU in treating different cancers. Additionally, it acts as an antidote to folic acid antagonists like methotrexate. Folinic acid serves as both an antidote and a modulating agent for chemotherapy, providing support for this class of medications. It is not specifically approved for the treatment of Rett syndrome.

Options C: L-carnitine: L-carnitine is a naturally occurring substance that plays a role in energy metabolism. It is commonly used as a dietary supplement and may benefit certain medical conditions, particularly those related to mitochondrial dysfunction. However, L-carnitine is not specifically approved for the treatment of Rett syndrome.

Options D: Naltrexone: Naltrexone is a medication primarily used to treat opioid addiction and alcohol dependence. It works by blocking the effects of opioids in the brain. It is not helpful for the treatment of Rett syndrome.

Solution for Question 5:

Correct Option A: Lapatinib

- The treatment for HER-2 positive trastuzumab-resistant breast cancer can include various options, and the choice of treatment depends on several factors. However, among the options provided, the correct answer is Lapatinib.
- Lapatinib: Lapatinib is an oral tyrosine kinase inhibitor that targets both HER-2 and EGFR (epidermal growth factor receptor). It is approved for the treatment of HER-2 positive breast cancer, either as monotherapy or in combination with trastuzumab. Lapatinib can be considered as a treatment option in patients with trastuzumab-resistant breast cancer.

Incorrect options

Option B: Sorafenib: Sorafenib is a multi-kinase inhibitor primarily used for the treatment of advanced kidney cancer and liver cancer. While it has shown activity against various kinases involved in tumor growth and angiogenesis, it is not a standard treatment for trastuzumab-resistant breast cancer.

Option C: Vemurafenib: Vemurafenib is a targeted therapy used for the treatment of metastatic melanoma with a specific mutation (BRAF V600E). It is not indicated or effective in the treatment of HER-2 positive breast cancer.

Option D: Erlotinib: Erlotinib is an EGFR tyrosine kinase inhibitor primarily used for the treatment of non-small cell lung cancer (NSCLC) and pancreatic cancer. It does not have a specific indication for the treatment of HER-2 positive breast cancer, especially in the context of trastuzumab resistance.

Solution for Question 6:

Correct Option A: Angiogenesis inhibitor

- Thalidomide is known for its ability to inhibit angiogenesis, which is the formation of new blood vessels. It acts by inhibiting the production of vascular endothelial growth factor (VEGF) and basic fibroblast growth factor (bFGF), both of which play crucial roles in angiogenesis.

Incorrect Options

Option B: Inhibits leukotrienes: Thalidomide does not directly inhibit leukotrienes. Leukotrienes are inflammatory mediators derived from arachidonic acid metabolism, and their synthesis is inhibited by drugs like leukotriene receptor antagonists or 5-lipoxygenase inhibitors, e.g., Montelukast.

Option C: Prevents folic acid synthesis in bacteria: Thalidomide does not target folic acid synthesis in bacteria. It does not have antibacterial properties and does not interfere with the synthesis of folic acid in bacteria.

Option D: Inhibits factor Xa: Thalidomide does not inhibit factor Xa, which is a coagulation factor involved in the blood clotting cascade. Anticoagulant medications like direct factor Xa inhibitors (e.g., rivaroxaban, apixaban) or indirect factor Xa inhibitors (e.g., heparin, fondaparinux) specifically target factor Xa for anticoagulant effects.

Solution for Question 7:

Correct Option C- Alpha methyldopa:

Incorrect Options:

Option A- Lisinopril: Lisinopril is an ACE inhibitor and is avoided in pregnancy.

Option B- Propranolol: Propranolol is a non selective beta blocker which should be avoided in pregnancy.

Option D- Losartan: Losartan is an ARB and is avoided in pregnancy.

Solution for Question 8:

Correct Option A:

The categories A, B, C, D, and X in drug classification are used to indicate the safety of a medication in pregnancy, with Category A being the safest and Category X being the most dangerous.

Category A: Adequate and well-controlled studies have failed to demonstrate a risk to the fetus in the first trimester of pregnancy, and there is no evidence of risk in later trimesters.

Category B: Animal reproduction studies have failed to demonstrate a risk to the fetus, but there are no adequate and well-controlled studies in pregnant women. Or, animal reproduction studies

have shown an adverse effect on the fetus, but adequate and well-controlled studies in pregnant women have failed to demonstrate a risk to the fetus in any trimester.

Category C: Animal reproduction studies have shown an adverse effect on the fetus, and there are no adequate and well-controlled studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks.

Category D: There is positive evidence of human fetal risk based on adverse reaction data from investigational or marketing experience or studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks.

Category X: Studies in animals or humans have demonstrated fetal abnormalities, and/or there is positive evidence of human fetal risk based on adverse reaction data from investigational or marketing experience, and the risks involved in use of the drug in pregnant women clearly outweigh potential benefits.

Incorrect Option:

Option B. Dose adjustment in renal failure: This is not correct. Dose adjustment in renal failure typically depends on the drug's pharmacokinetics and how it is metabolized and eliminated from the body. Medications that are renally excreted may require dose adjustment in patients with impaired kidney function.

Option C. Cost ratio from cheap to expensive: This is not correct. The categories A, B, C, D, and X do not provide any information about the cost of a medication. Cost information is typically provided separately.

Option D. Safety or therapeutic index: This is not correct. The categories A, B, C, D, and X are used to indicate the safety of a medication in pregnancy, and do not provide information on the therapeutic index or overall safety of a medication in other populations.

Solution for Question 9:

Correct option B: Ergotism:

- Ergotism is a condition caused by the ingestion of ergot-contaminated grains or the use of ergot-derived medications such as ergotamine.
- Ergotamine is a medication commonly used for the treatment of migraines.

- It can lead to constriction of blood vessels, which can cause symptoms such as numbness, tingling, and pain in the extremities.
- In severe cases, it can lead to gangrene and limb loss.

Incorrect options:

Option A: Atherosclerosis: Atherosclerosis is a chronic condition that develops over time and is usually associated with risk factors such as high blood pressure, high cholesterol, and smoking.

Option C: Polyradicular neuropathy:

- This is a condition where multiple nerves in the body become damaged, leading to symptoms such as pain, weakness, and numbness.
- It can be caused by a variety of factors, including infections, autoimmune disorders, and exposure to toxins.
- While this is a possibility, the recent use of ergotamine makes ergotism a more likely diagnosis.

Option D: Herniated disc: It is a condition where the soft inner portion of a spinal disc bulges out and puts pressure on surrounding nerves.

Solution for Question 10:

Correct Choice: d

Hodgkin's lymphoma is a type of cancer that affects the lymphatic system, which is part of the immune system. Nivolumab is a checkpoint inhibitor that targets the programmed cell death protein 1 (PD-1) receptor on T cells, which can help to overcome the tumor's ability to evade the immune system.

Nivolumab is a monoclonal antibody that targets the programmed cell death protein 1 (PD-1) receptor on T cells. PD-1 is a checkpoint protein that is expressed on the surface of T cells and helps to regulate the immune response by inhibiting T cell activation and proliferation. PD-1 binds to two ligands, PD-L1 and PD-L2, which are expressed on the surface of some cancer cells and other immune cells. When PD-1 binds to these ligands, it sends a signal to the T cell to "turn off" and stop attacking the cancer cell.

By blocking the PD-1 checkpoint, nivolumab enhances the anti-tumor activity of T cells and promotes an immune response against cancer cells. This mechanism of action is different from

traditional chemotherapy and radiation therapy, which work by directly killing cancer cells. Nivolumab is considered a type of immunotherapy that harnesses the immune system to fight cancer.

Incorrect Choices:

Option a. Retinoblastoma - The treatment of retinoblastoma depends on the size and location of the tumor, as well as the extent of the disease. The most common treatments for retinoblastoma include surgery, radiation therapy, chemotherapy, or a combination of these approaches. Some common chemotherapy drugs used to treat retinoblastoma include vincristine, carboplatin, etoposide, and cyclophosphamide.

Option b. Medulloblastoma - Treatment for medulloblastoma may involve surgery, radiation therapy, chemotherapy, or a combination of these approaches. Some common chemotherapy drugs used to treat medulloblastoma include cisplatin, carboplatin, vincristine, and cyclophosphamide. Radiation therapy is also commonly used in the treatment of medulloblastoma.

Option c. Pleuropulmonaryblastoma - Treatment for pleuropulmonaryblastoma may involve surgery, chemotherapy, radiation therapy, or a combination of these approaches. Some common chemotherapy drugs used to treat pleuropulmonaryblastoma include cisplatin, carboplatin, etoposide, and ifosfamide.

Solution for Question 11:

Correct Option: D

- Streptomycin is contraindicated in pregnant females due to its potential to cause harm to the developing fetus. Streptomycin is an antibiotic from the aminoglycoside class and has been associated with adverse effects on the fetal auditory and vestibular systems. It can cross the placenta and reach the developing fetus, potentially leading to hearing loss, balance problems, and other developmental abnormalities. Therefore, it is generally avoided during pregnancy unless the potential benefits outweigh the risks and no alternative treatment options are available.

Incorrect Option:

Option A) Erythromycin: Erythromycin is an antibiotic that is generally considered safe for use during pregnancy. It belongs to the macrolide class of antibiotics and has been widely used for

the treatment of various infections in pregnant women without significant evidence of harm to the fetus.

Option B) Amoxicillin: Amoxicillin is a commonly used antibiotic from the penicillin class and is generally considered safe for use during pregnancy. It has a long history of use in pregnant women with bacterial infections and has not been associated with significant risks to the fetus.

Option C) Ceftriaxone: Ceftriaxone is a broad-spectrum antibiotic from the cephalosporin class. It is generally considered safe for use during pregnancy and is often prescribed for the treatment of various infections in pregnant women.

Solution for Question 12:

Correct Option: C

- Methyldopa is an antihypertensive medication commonly used in the management of hypertension during pregnancy. Its mechanism of action involves agonism of alpha 2 adrenergic receptors. Methyldopa is converted in the body into methylnorepinephrine, which acts as a false neurotransmitter. Methylnorepinephrine stimulates alpha 2 receptors in the central nervous system (CNS), leading to a decrease in sympathetic outflow from the CNS. This ultimately results in reduced peripheral vascular resistance and lowered blood pressure.

Incorrect Options:

Option A) Blockade of beta adrenergic receptors: Methyldopa does not block beta adrenergic receptors. Beta blockers, on the other hand, are a separate class of antihypertensive medications that block beta adrenergic receptors to reduce blood pressure.

Option B) Agonism of beta 2 receptors: Methyldopa does not directly agonize beta 2 receptors. Agonists of beta 2 receptors, such as albuterol, are commonly used in the treatment of asthma and chronic obstructive pulmonary disease (COPD).

Option D) Blockade of alpha 2 receptors: Methyldopa does not block alpha 2 receptors. Blocking alpha 2 receptors would result in increased sympathetic outflow and potentially raise blood pressure. Medications that block alpha receptors, like alpha blockers, are used for different therapeutic purposes, such as the treatment of benign prostatic hyperplasia (BPH) or alpha-adrenergic antagonist poisoning.

Solution for Question 13:

Correct Option D: Betamethasone

- Betamethasone is the corticosteroid indicated for the stimulation of fetal lung maturation in preterm labor.
- Betamethasone belongs to the class of glucocorticoids and has potent anti-inflammatory and immunosuppressive properties.
- It is commonly used in obstetrics to enhance fetal lung maturity when preterm delivery is anticipated.
- Administering betamethasone to the pregnant woman helps promote the production of surfactant in the fetal lungs, which is crucial for the proper functioning of the lungs and prevention of respiratory distress syndrome (RDS) in preterm infants.

Incorrect Options:

Option A: Hydrocortisone is a corticosteroid that has a wide range of uses, but it is not specifically indicated for fetal lung maturation in preterm labor.

Option B: Triamcinolone, another corticosteroid, is primarily used for its anti-inflammatory properties and is not the preferred choice for fetal lung maturation.

Option C: Methylprednisolone- Methylprednisolone is a corticosteroid that is used for various medical conditions, including inflammatory and autoimmune disorders. However, it is not the corticosteroid of choice for the stimulation of fetal lung maturation in preterm labor.

Solution for Question 14:

Correct Option C - 5-fluorouracil:

- 5-fluorouracil (5-FU) is an antimetabolite chemotherapy drug that is used in the treatment of various cancers, including colorectal cancer and breast cancer. It exerts its anticancer effects by inhibiting thymidylate synthase, an enzyme that plays a crucial role in the synthesis of thymidine, a building block of DNA. By inhibiting thymidylate synthase, 5-FU reduces the

availability of thymidine, thereby interfering with DNA synthesis and inhibiting the growth of cancer cells.

Incorrect Options:

Option A - Rituximab: Rituximab is a monoclonal antibody used in the treatment of certain types of cancer, such as non-Hodgkin lymphoma, and autoimmune diseases like rheumatoid arthritis. It does not inhibit thymidylate synthesis.

Option B - Cisplatin: Cisplatin is a chemotherapy drug used to treat various cancers, including testicular, ovarian, bladder, and lung cancer. It works by forming covalent bonds with DNA, causing cross-links and inhibiting DNA synthesis, but it does not directly inhibit thymidylate synthesis.

Option D - Methotrexate: Methotrexate is an antimetabolite drug used in cancer chemotherapy and to treat autoimmune diseases like rheumatoid arthritis. It inhibits dihydrofolate reductase, which reduces the availability of tetrahydrofolate, an essential cofactor for thymidylate synthesis. However, it is not primarily known for restricting thymidylate availability.

Solution for Question 15:

Correct Option B. Schedule H drugs

- Schedule H drugs are drugs that are required to be sold only on prescription from a registered medical practitioner. These drugs have a higher potential for misuse or abuse and may pose risks if used without proper medical supervision. Examples of Schedule H drugs include antibiotics, opioids, sedatives, and certain other prescription medications.

Incorrect Options

Option A. Schedule M drugs: Schedule M refers to drugs that are manufactured and marketed in India. These drugs are required to comply with Good Manufacturing Practices (GMP) guidelines. Schedule M does not specifically regulate the sale or prescription requirements of these drugs. However, it is important to note that the sale of all drugs, including Schedule M drugs, should comply with the applicable drug regulations and requirements, which may include prescription-only status for certain medications.

Option C. Schedule J drugs: Schedule J refers to a category of drugs known as "Strengthening Schedule J Drugs." These drugs are considered essential medicines for public health and are used to treat various diseases and conditions. The sale and prescription requirements for

Schedule J drugs may vary depending on the specific drug and its formulation. Some Schedule J drugs may be available over-the-counter (OTC) or sold with certain restrictions, while others may require a prescription.

Option D. Schedule X drugs: Schedule X drugs are a category of drugs that have a high potential for abuse and addiction. These drugs are tightly regulated and are subject to strict prescription and dispensing requirements. Schedule X drugs include narcotics, psychotropic substances, and certain other drugs that are considered dangerous and are controlled under the Narcotic Drugs and Psychotropic Substances (NDPS) Act, of 1985. These drugs can only be prescribed and dispensed by registered medical practitioners with specific licenses.

Solution for Question 16:

Option D. Chloramphenicol: Chloramphenicol is an antibiotic known to inhibit bilirubin glucuronidation in the liver. By interfering with this metabolic pathway, chloramphenicol can increase serum bilirubin levels and potentially lead to neonatal hyperbilirubinemia. Therefore, if the mother took chloramphenicol during the third trimester, it could be a possible cause of the high serum bilirubin level in the infant.

Incorrect Options:

Option A. Ampicillin: Ampicillin is a broad-spectrum antibiotic commonly used during pregnancy to treat bacterial infections. It is not known to have a direct impact on bilirubin metabolism or cause significant changes in serum bilirubin levels. Therefore, it is unlikely to be the cause of the high serum bilirubin level in the infant.

Option: B. Co-trimoxazole is a combination antibiotic used to treat various bacterial infections. Similar to ampicillin, co-trimoxazole is not known to directly affect bilirubin metabolism. It is therefore unlikely to be the cause of the high serum bilirubin level.

Option C. Azithromycin: Azithromycin is an antibiotic commonly prescribed for respiratory tract infections and other bacterial infections. It has not been reported to have a direct impact on bilirubin metabolism. Therefore, it is unlikely to be the cause of the high serum bilirubin level.

Solution for Question 17:

Correct Option A

- A misbranded drug refers to a medication whose labeling or packaging does not comply with the legal requirements or regulations set by the regulatory authorities. This could include inaccurate or misleading information on the label, improper instructions for use, or failure to meet labeling standards.

Incorrect Options

Option B. Adulterated drugs:

- Adulterated drugs are medications that have been tampered with or contaminated. They may contain impurities, substandard ingredients, or other substances that are not supposed to be present in the drug. Adulteration can occur during the manufacturing, packaging, or distribution process, compromising the quality and safety of the drug.

Option C. Spurious drugs:

- Spurious drugs are counterfeit medications that are intentionally and fraudulently labeled with the name of a genuine drug, but they do not contain the correct active ingredients or may contain harmful substances. These drugs are intended to deceive consumers and may pose serious risks to health.

Option D. Counterfeit drugs:

- Counterfeit drugs are medications that are intentionally and fraudulently produced or distributed with the intent to deceive. They may mimic the appearance, packaging, or labeling of genuine drugs, but they do not meet the quality standards or contain the correct active ingredients. Counterfeit drugs can be ineffective, unsafe, or even harmful to individuals who unknowingly use them.
-

Solution for Question 18:

Correct Option A

- Lithium exposure during pregnancy has been linked to an increased risk of cardiac defects in the developing fetus. These defects may involve structural abnormalities of the heart, such as atrial and ventricular septal defects, as well as abnormalities in the heart's electrical

conduction system.

Incorrect Option

Option B. Facial defects:

- Some studies have suggested a potential association between lithium use during pregnancy and an increased risk of facial defects in the fetus. These defects may include cleft lip or palate, although the evidence is not conclusive.

Option C. CNS defects:

- Central nervous system (CNS) defects have also been reported in infants exposed to lithium in utero. These defects may include malformations of the brain and spinal cord, neural tube defects, and developmental delays. However, the specific risks and extent of these defects are still being studied.

Option D. Urogenital defects:

- Lithium use during pregnancy has been associated with an increased risk of urogenital defects in the developing fetus. These defects may involve abnormalities in the urinary system, reproductive organs, or both.

Solution for Question 19:

Correct Option A - Panitumumab:

- Panitumumab is a human IgG2κ monoclonal antibody that binds to the extracellular domain III of the epidermal growth factor receptor (EGFR or HER-1). By binding to this receptor, panitumumab prevents ligand-dependent signaling, effectively inhibiting the EGFR pathway, which is often upregulated in certain types of cancer.

Incorrect Options:

- Options B, C, and D are not inhibitors of EGFR or HER-1. They inhibit tyrosine kinase receptors.

Option B - Crizotinib: Crizotinib is an orally bioavailable inhibitor of several receptor tyrosine kinases, including ROS1, HGFR/cMET, and ALK. It is Primarily used for non-small cell lung cancer (NSCLC) that is ALK-positive or ROS1-positive.

Option C - Alectinib: Alectinib is also an orally bioavailable inhibitor of ALK kinase activity. It shows a higher selectivity for ALK than crizotinib and is active against mutant forms of ALK

found in tumors resistant to crizotinib.

Option D - Lorlatinib: Lorlatinib is a third-generation ALK inhibitor that targets ALK and ROS1.

Solution for Question 20:

Correct Option A – Imatinib:

- Imatinib is a drug of choice- in the management of Chronic Myeloid Leukemia.
- It is also used for the treatment of GIST (Gastrointestinal stromal tumors).

Incorrect Options:

Options B, C, D:

- Methotrexate, Doxorubicin, and Cisplatin are not used for the treatment of both Gastrointestinal Stromal Tumors and Chronic Myeloid Leukemia
-

Solution for Question 21:

Correct Option B - Inhibits dihydrofolate reductase, leading to decreased DNA synthesis:

- Methotrexate is an antimetabolite and antifolate drug that primarily inhibits the enzyme dihydrofolate reductase (DHFR).
- DHFR is responsible for converting dihydrofolate (DHF) into tetrahydrofolate (THF).
- THF is essential for the synthesis of purine nucleotides and thymidylate, which are crucial for DNA synthesis and repair.
- By inhibiting DHFR, methotrexate decreases the production of THF, leading to a reduction in DNA synthesis.

Incorrect Options:

Options A, C, D:

- These are not the mechanism of action of Methotrexate.

Solution for Question 22:

Correct Option A - Bortezomib:

- Mechanism of Action: Bortezomib is a proteasome inhibitor used in the treatment of multiple myeloma and mantle cell lymphoma.
- Association with Herpes Zoster: Bortezomib has been associated with an increased risk of reactivation of the varicella-zoster virus (VZV), leading to Herpes zoster infection. This is due to the immunosuppressive effects of bortezomib, which can impair the immune system's ability to keep latent infections in check.
- Prophylaxis: Patients receiving bortezomib are often prescribed antiviral prophylaxis (e.g., acyclovir) to reduce the risk of Herpes zoster reactivation.

Incorrect Options:

Option B - Daratumumab:

- Mechanism of Action: Daratumumab is a monoclonal antibody that targets CD38 on plasma cells, used in the treatment of multiple myeloma.
- Infection Risk: While daratumumab is associated with an increased risk of respiratory infections due to its immunomodulatory effects, it is not specifically linked to a higher risk of Herpes zoster infection.

Option C - Lomitapide:

- Mechanism of Action: Lomitapide is a microsomal triglyceride transfer protein (MTP) inhibitor used to treat homozygous familial hypercholesterolemia, not multiple myeloma.
- Relevance: Lomitapide is not associated with an increased risk of Herpes zoster, nor is it relevant in the context of multiple myeloma treatment.

Option D - Lenalidomide:

- Mechanism of Action: Lenalidomide is an immunomodulatory drug used in multiple myeloma.
- Infection Risk: Although lenalidomide is associated with an increased risk of neutropenia and infections, it is not specifically associated with an increased risk of Herpes zoster infection compared to bortezomib.

Solution for Question 23:

Correct Option A - NK-1 Antagonist:

- Mechanism of Action: Aprepitant is a selective antagonist of the neurokinin-1 (NK-1) receptor. By blocking the NK-1 receptors, aprepitant inhibits the action of substance P, a neuropeptide involved in the emetic pathway. This helps to prevent nausea and vomiting, particularly the late-onset type associated with chemotherapy.
- Use: It is effective in managing delayed chemotherapy-induced nausea and vomiting (CINV) when used in combination with other antiemetic agents.

Incorrect Options:

Option B - NK-1 Agonist:

- This is incorrect because aprepitant is not an agonist but an antagonist of the NK-1 receptor. An NK-1 agonist would stimulate the receptor, potentially increasing the risk of nausea and vomiting rather than preventing it.

Option C - NK-3 Antagonist:

- Aprepitant does not act on the NK-3 receptors. NK-3 receptors are another subtype of neurokinin receptors, and antagonists targeting these receptors are not used in the treatment of chemotherapy-induced nausea and vomiting.

Option D - NK-2 Antagonist:

- Aprepitant is not an NK-2 antagonist. NK-2 receptors are another subtype involved in the emetic response, but aprepitant specifically targets the NK-1 receptors. NK-2 antagonists are not the primary agents used for preventing chemotherapy-induced nausea and vomiting.
-

Solution for Question 24:

Correct Option D - Thymoma:

- Thymoma is a tumor originating from the thymus gland and is frequently associated with myasthenia gravis. Hypercalcemia can occur in patients with thymoma due to the production of parathyroid hormone-related peptide (PTHrP) by the tumor or associated paraneoplastic syndromes. Given the patient's history of myasthenia gravis, which is strongly associated with thymoma, the presence of hypercalcemia points to thymoma as the most likely cause.

Incorrect Options:

Option A - Drug-induced Hypercalcemia:

- Mycophenolate mofetil (MMF) is an immunosuppressant that is not typically associated with hypercalcemia. Drug-induced hypercalcemia is less likely in this scenario, particularly since MMF does not have hypercalcemia as a known side effect.

Option B - Primary Hyperparathyroidism:

- Primary hyperparathyroidism is a common cause of hypercalcemia and would be a strong consideration in many patients. However, the association between myasthenia gravis and thymoma, along with the presence of hypercalcemia, makes thymoma a more likely diagnosis in this specific context.

Option C - Small Cell Lung Cancer:

- Small cell lung cancer is more commonly associated with ectopic hormone production, such as ACTH or ADH, rather than PTHrP, which leads to hypercalcemia. Additionally, there is no direct link between myasthenia gravis and small cell lung cancer. Therefore, small cell lung cancer is an unlikely cause in this patient.

Solution for Question 25:

Answer: A) Pulmonary toxicity

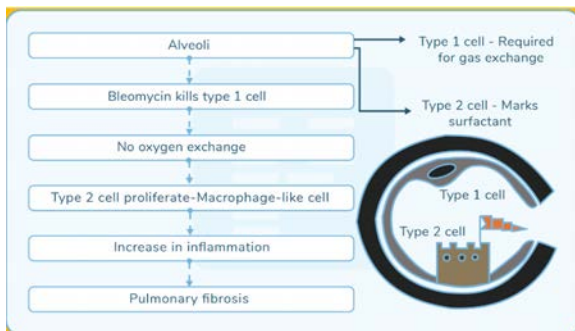
Explanation:

Bleomycin-induced pulmonary toxicity is the most significant adverse effect, and it is exacerbated by radiation exposure (a phenomenon known as radiation recall pneumonitis).

- Mechanism: Bleomycin generates free radicals, leading to oxidative damage and fibrosis in the lungs. The lungs have low bleomycin hydrolase activity, making them more susceptible to damage. Radiation exposure further increases oxidative stress and inflammatory responses, worsening lung injury.

- Bleomycin generates free radicals, leading to oxidative damage and fibrosis in the lungs.
- The lungs have low bleomycin hydrolase activity, making them more susceptible to damage.
- Radiation exposure further increases oxidative stress and inflammatory responses, worsening lung injury.
- Risk Factors: Cumulative bleomycin dose >400 units Concurrent oxygen therapy (risk of oxygen toxicity) Prior or concurrent radiation therapy Older age and renal impairment
- Cumulative bleomycin dose >400 units
- Concurrent oxygen therapy (risk of oxygen toxicity)
- Prior or concurrent radiation therapy
- Older age and renal impairment
- Clinical Features: Progressive dyspnoea, non-productive cough Interstitial pneumonitis → pulmonary fibrosis Restrictive lung disease with ↓ DLCO (diffusing capacity for carbon monoxide)
- Progressive dyspnoea, non-productive cough
- Interstitial pneumonitis → pulmonary fibrosis
- Restrictive lung disease with ↓ DLCO (diffusing capacity for carbon monoxide)
- Bleomycin generates free radicals, leading to oxidative damage and fibrosis in the lungs.
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- Radiation exposure further increases oxidative stress and inflammatory responses, worsening lung injury.
- Cumulative bleomycin dose >400 units
- Concurrent oxygen therapy (risk of oxygen toxicity)
- Prior or concurrent radiation therapy
- Older age and renal impairment
- Progressive dyspnoea, non-productive cough
- Interstitial pneumonitis → pulmonary fibrosis
- Restrictive lung disease with ↓ DLCO (diffusing capacity for carbon monoxide)

Bleomycin is an antitumor antibiotic that is well-known for its association with pulmonary toxicity, including pulmonary fibrosis which occurs as follows:



Gastric toxicity (Option B) is incorrect because bleomycin does not significantly affect the gastrointestinal tract.

Neural toxicity (Option C) is incorrect because bleomycin does not cause neurotoxicity (unlike vincristine, which affects nerves).

Hepatotoxicity (Option D) is incorrect because bleomycin has minimal liver toxicity compared to other chemotherapy agents.

Solution for Question 26:

Answer: D) Cisplatin has more nephrotoxic side effects compared to Carboplatin

Explanation:

Cisplatin and Carboplatin are platinum-based chemotherapy agents, but Cisplatin is more nephrotoxic due to its accumulation in renal tubular cells.

- Key Differences: Cisplatin: More nephrotoxic → Causes acute tubular necrosis (ATN) and requires aggressive hydration with mannitol or amifostine to reduce damage. More neurotoxic and ototoxic → Causes peripheral neuropathy and sensorineural hearing loss (ototoxicity). More severe nausea and vomiting. Carboplatin: Less nephrotoxic and does not require aggressive hydration. Less neurotoxic compared to cisplatin. More myelosuppressive (causes thrombocytopenia).
- Cisplatin: More nephrotoxic → Causes acute tubular necrosis (ATN) and requires aggressive hydration with mannitol or amifostine to reduce damage. More neurotoxic and ototoxic → Causes peripheral neuropathy and sensorineural hearing loss (ototoxicity). More severe nausea and vomiting.

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- More neurotoxic and ototoxic → Causes peripheral neuropathy and sensorineural hearing loss (ototoxicity).
- More severe nausea and vomiting.
- Carboplatin: Less nephrotoxic and does not require aggressive hydration. Less neurotoxic compared to cisplatin. More myelosuppressive (causes thrombocytopenia).
- Less nephrotoxic and does not require aggressive hydration.
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- More nephrotoxic → Causes acute tubular necrosis (ATN) and requires aggressive hydration with mannitol or amifostine to reduce damage.
- More neurotoxic and ototoxic → Causes peripheral neuropathy and sensorineural hearing loss (ototoxicity).
- More severe nausea and vomiting.
- Less nephrotoxic and does not require aggressive hydration.
- Less neurotoxic compared to cisplatin.
- More myelosuppressive (causes thrombocytopenia).

Cisplatin is known for its nephrotoxic effects, which can lead to acute kidney injury.

The management of cisplatin-induced nephrotoxicity includes

- Adequate i.v. fluids - Chloride diuresis (Cl inactivates cisplatin)

- Mannitol - diuresis of inactivated cisplatin
 - Use of nephroprotective agents like amifostine (only i.v.)
 - Cisplatin is less neurotoxic than Carboplatin (Option A): Incorrect because Cisplatin is more neurotoxic (causing peripheral neuropathy).
 - Cisplatin causes fewer gastric issues compared to Carboplatin (Option B): Incorrect because Cisplatin causes more nausea and vomiting (highly emetogenic).
 - Carboplatin is more neurotoxic than Cisplatin (Option C): Incorrect because Cisplatin is more neurotoxic than Carboplatin.
-

Solution for Question 27:

Correct Option C - Cisplatin:

- Cisplatin is a platinum-containing chemotherapy drug that is commonly used in the treatment of Carcinoma cervix, particularly in combination with other drugs or radiation therapy. It works by interfering with the DNA in cancer cells, preventing their growth and division. Cisplatin is often a key component in the treatment regimen for cervical cancer.

Incorrect Options:

Option A - Paclitaxel: Paclitaxel is a chemotherapy drug often used in the treatment of various cancers, including ovarian, breast, and lung cancers. While it may be used in combination with other drugs for certain gynecological cancers, it is not typically the primary chemotherapeutic agent used for Carcinoma cervix.

Option B - Cyclophosphamide: Cyclophosphamide is a chemotherapy drug used to treat a variety of cancers, including some types of lymphoma and leukemia, as well as autoimmune disorders. It is not the primary chemotherapeutic agent used for Carcinoma cervix.

Option D - Topotecan: Topotecan is a chemotherapy drug that is used in the treatment of various cancers, including ovarian and lung cancers. While it might be used in certain cases of recurrent or advanced cervical cancer, it is not typically the first-line chemotherapeutic agent of choice.

Solution for Question 28:

Correct Option A - GI symptoms:

- Gastrointestinal (GI) symptoms are the most common side effects associated with cisplatin treatment. These symptoms may include nausea, vomiting, and loss of appetite. Cisplatin can irritate the stomach lining and trigger the body's vomiting reflex, leading to these discomforting GI effects.

Incorrect Options:

Option B - Neuropathy: Neuropathy, characterized by nerve damage and resulting in symptoms like numbness, tingling, and weakness, is a known side effect of cisplatin. However, it is not the most common side effect.

Option C - Cardiotoxicity: While cisplatin can have cardiovascular side effects, such as abnormal heart rhythms and changes in blood pressure, cardiotoxicity is not typically the most common side effect of cisplatin.

Option D - Myopathy: Myopathy refers to muscle-related side effects, including muscle weakness or pain. Although cisplatin can lead to myopathy, it is not the most common side effect.

Solution for Question 29:

Correct Option B - Hydroxyurea:

Hydroxyurea, a myelosuppressive agent, is the only effective drug proven to reduce the frequency of painful episodes. It raises the level of HbF and the hemoglobin level.

- It usually decreases the rate of painful episodes by 50 %. Its mechanisms of action for HbF induction remain incompletely understood. Current evidence suggests that several potential mechanisms of action by hydroxyurea may be relevant for patients with SCA, which together lead not only to HbF induction but also to additional benefits.
- The most important mechanism of action is inhibition of ribonucleotide reductase (RR), the enzyme involved in transforming ribonucleosides into deoxyribonucleosides that serve as building blocks for DNA synthesis.

Incorrect Options:

Option A - Methionine:

- This option is incorrect as this drug is of no use in preventing painful vaso-occlusive crises and preventing long-term complications in sickle cell anemia.

Option C - Methotrexate + Leucovorin:

- This option is incorrect as this drug is of no use in preventing painful vaso-occlusive crises and preventing long-term complications in sickle cell anemia.

Option D - Eltrombopag:

- This option is incorrect as this drug is of no use in preventing painful vaso-occlusive crises and preventing long-term complications in sickle cell anemia.

Solution for Question 30:

Correct Option: A

- Filgrastim is a granulocyte colony-stimulating factor (G-CSF) that can be used to prevent neutropenia in patients undergoing chemotherapy. Neutropenia is a condition characterized by a low level of neutrophils, a type of white blood cell that plays a crucial role in fighting off infections. Chemotherapy drugs can suppress the production of neutrophils, leading to an increased risk of infections.
- Filgrastim works by stimulating the production of neutrophils in the bone marrow, helping to prevent or reduce the severity and duration of neutropenia. By increasing the neutrophil count, it enhances the patient's ability to fight off infections.

Incorrect Options :

Option B. Prednisolone: Prednisolone is a corticosteroid that has anti-inflammatory and immunosuppressive effects. While it may be used in certain cases to manage inflammation or immune-related complications, it is not specifically indicated for the prevention of neutropenia.

Option C. Vitamin B12: Vitamin B12 is a nutrient that plays a role in the production of red blood cells and the maintenance of nerve cells. While it is essential for overall health, it does not have a direct effect on preventing neutropenia.

Option D. Folic acid: Folic acid is a B-vitamin that is important for the synthesis and repair of DNA and red blood cell production. Similar to vitamin B12, it does not have a direct effect on preventing neutropenia.

Solution for Question 31:

Correct Option B - Competitive inhibitor of DHFR:

- The mechanism of action of methotrexate in osteosarcoma involves being a competitive inhibitor of dihydrofolate reductase (DHFR).
- Methotrexate is a folate analog that interferes with the synthesis of DNA, RNA, and proteins. It specifically inhibits the enzyme dihydrofolate reductase, which plays a critical role in the folate metabolism pathway.
- By inhibiting DHFR, methotrexate prevents the conversion of dihydrofolate (DHF) to tetrahydrofolate (THF), which is necessary for the synthesis of purines and pyrimidines. This inhibition leads to decreased availability of nucleotides required for DNA and RNA synthesis.
- In osteosarcoma, methotrexate is often used as part of combination therapy to target and kill cancer cells. By interfering with nucleotide synthesis, methotrexate inhibits the growth and proliferation of osteosarcoma cells, helping to control the progression of the disease.
- It's important to note that methotrexate is typically used as part of a multimodal treatment approach for osteosarcoma, and the specific treatment plan should be determined by a medical oncologist based on the individual patient's condition and other factors.

The mechanism of action of methotrexate in osteosarcoma involves being a competitive inhibitor of dihydrofolate reductase (DHFR).

Methotrexate is a folate analog that interferes with the synthesis of DNA, RNA, and proteins. It specifically inhibits the enzyme dihydrofolate reductase, which plays a critical role in the folate metabolism pathway.

By inhibiting DHFR, methotrexate prevents the conversion of dihydrofolate (DHF) to tetrahydrofolate (THF), which is necessary for the synthesis of purines and pyrimidines. This inhibition leads to decreased availability of nucleotides required for DNA and RNA synthesis.

In osteosarcoma, methotrexate is often used as part of combination therapy to target and kill cancer cells. By interfering with nucleotide synthesis, methotrexate inhibits the growth and proliferation of osteosarcoma cells, helping to control the progression of the disease.

It's important to note that methotrexate is typically used as part of a multimodal treatment approach for osteosarcoma, and the specific treatment plan should be determined by a medical oncologist based on the individual patient's condition and other factors.

Incorrect Options:

Options A, C and D are incorrect.

Solution for Question 32:

Correct Option A - IL-6:

- Tocilizumab inhibits interleukin-6 (IL-6).
- Tocilizumab is a monoclonal antibody that specifically targets and blocks the IL-6 receptor. IL-6 is a pro-inflammatory cytokine that plays a role in immune responses and inflammatory processes. It is involved in various conditions, including rheumatoid arthritis, giant cell arteritis, and cytokine release syndrome.
- By inhibiting the IL-6 receptor, tocilizumab can reduce the activity of IL-6 and modulate the inflammatory response. This can help alleviate symptoms and improve outcomes in conditions where IL-6 is overactive.

Tocilizumab inhibits interleukin-6 (IL-6).

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By inhibiting the IL-6 receptor, tocilizumab can reduce the activity of IL-6 and modulate the inflammatory response. This can help alleviate symptoms and improve outcomes in conditions where IL-6 is overactive.

Incorrect Options:

Option B - IL-2: IL-2 is involved in the activation and proliferation of T-cells and is not directly inhibited by tocilizumab.

Option C - IL-4: IL-4 is an important cytokine involved in immune responses, particularly in allergic reactions and Th2 immune responses. Tocilizumab does not target or inhibit IL-4.

Option D - IL-5: IL-5 is primarily associated with eosinophil activation and recruitment. Tocilizumab does not directly inhibit IL-5.

Solution for Question 33:

Correct Option C - Filgrastim:

- Recombinant human G-CSF(Granulocyte colony stimulating factor)
- Increases neutrophil production
- Administered subcutaneously or slow IV infusion

Recombinant human G-CSF(Granulocyte colony stimulating factor)

Increases neutrophil production

Administered subcutaneously or slow IV infusion

Incorrect Options:

Option A - Romiplostim: Romiplostim thrombopoietin receptor agonist . Romiplostim stimulates the production of platelets and is used to treat thrombocytopenia.

Option B - Oprelvekin: Oprelvekin stimulates the production of megakaryocytes and is also used to treat thrombocytopenia.

Option D - Epoetin alfa: Epoetin alfa is used to stimulate red blood cell production and is ineffective in treating neutropenia.

Solution for Question 34:

Correct Option A - Cisplatin:

- Cisplatin is a potent chemotherapy agent used to treat various cancers, including testicular, ovarian, bladder, and lung cancers.
- Renal tubular damage is a well-known side effect of cisplatin. Cisplatin is taken up by the proximal tubular cells in the kidney, where it accumulates and causes the tubular cells' oxidative stress, inflammation, and apoptosis. This can lead to acute kidney injury, electrolyte imbalances, and chronic kidney disease.

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Renal tubular damage is a well-known side effect of cisplatin. Cisplatin is taken up by the proximal tubular cells in the kidney, where it accumulates and causes the tubular cells' oxidative stress, inflammation, and apoptosis. This can lead to acute kidney injury, electrolyte imbalances, and chronic kidney disease.

Incorrect Options:

Option B - Streptozocin: Streptozocin is an alkylating agent used to treat pancreatic islet cell carcinoma. It causes diabetes mellitus due to destruction of beta cells of pancreas. It can cause kidney damage, but this is typically due to its effects on the glomerulus rather than the renal tubules.

Option C - Methysergide: Methysergide is a serotonin antagonist that is used to treat migraines. It is associated with retroperitoneal fibrosis not renal tubular damage.

Option D - Cyclophosphamide: Cyclophosphamide is an alkylating agent that is used to treat a variety of cancers and autoimmune diseases. It causes hemorrhagic cystitis.

Solution for Question 35:

Correct Option B - Pembrolizumab:

- Pembrolizumab is a monoclonal antibody that targets the programmed cell death protein 1 (PD-1) receptor on T cells.
- It is a type of immunotherapy known as an immune checkpoint inhibitor, which works by blocking the PD-1 receptor, thereby enabling the immune system to recognize and attack cancer cells.
- It was approved by the FDA in 2019 for the treatment of advanced or recurrent endometrial cancer that is microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR).

Pembrolizumab is a monoclonal antibody that targets the programmed cell death protein 1 (PD-1) receptor on T cells.

It is a type of immunotherapy known as an immune checkpoint inhibitor, which works by blocking the PD-1 receptor, thereby enabling the immune system to recognize and attack cancer cells.

It was approved by the FDA in 2019 for the treatment of advanced or recurrent endometrial cancer that is microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR).

Incorrect Options:

Option A & C - Nivolumab & Ipilimumab: Nivolumab and Ipilimumab are other immune checkpoint inhibitors that the FDA has approved for the treatment of various types of cancer, including melanoma and lung cancer, but they have not been specifically approved for the treatment of endometrial cancer.

Option D - Trastuzumab: Trastuzumab, on the other hand, is not an immune checkpoint inhibitor, but rather a monoclonal antibody that targets the human epidermal growth factor receptor 2 (HER2). It is used to treat HER2-positive breast and gastric cancer, but not endometrial cancer.

Solution for Question 36:

Correct Option D - A-4, B-3, C-2, D-1:

The correct pairs are as follows:

A. Trastuzumab	4. HER2/neu
B. Infliximab	3. TNF alpha
C. Sirolimus	2. mTOR
D. Imatinib	1. BCR-ABL tyrosine kinase

Solution for Question 37:

Correct Option D - Cetuximab:

- Cetuximab is not an immune checkpoint inhibitor, but rather a monoclonal antibody that targets the epidermal growth factor receptor (EGFR) on the surface of cancer cells.
- It is used to treat certain types of cancer, including colorectal and head and neck.

Incorrect Options:

Option A - Nivolumab: Nivolumab is an immune checkpoint inhibitor.

Option B - Pembrolizumab: Pembrolizumab is an immune checkpoint inhibitor.

Option C - Atezolimumab: Atezolimumab is an immune checkpoint inhibitor.

Solution for Question 38:

Correct Options A - Pembrolizumab with chemotherapy given for NSCLC(Non-small cell lung cancer):

- The KEYNOTE -189 trial investigated the use of pembrolizumab and chemotherapy in the treatment of non-small cell lung cancer (NSCLC).

The KEYNOTE -189 trial investigated the use of pembrolizumab and chemotherapy in the treatment of non-small cell lung cancer (NSCLC).

Incorrect Options:

Option B - Nivolumab with chemo given for NSCLC: The KEYNOTE-189 trial specifically focused on pembrolizumab and not nivolumab.

Option C - Only Pembrolizumab for NSCLC: The KEYNOTE-189 trial investigated the combination of pembrolizumab with chemotherapy, not pembrolizumab alone, to treat NSCLC. The study had two treatment arms: one receiving pembrolizumab plus chemotherapy and the other receiving a placebo alongside chemotherapy.

Option D - Only Nivolumab for NSCLC: The KEYNOTE-189 trial did not evaluate nivolumab for the treatment of NSCLC.

Solution for Question 39:

Correct Option A - Anthracyclines:

- Daunorubicin and doxorubicin are anthracyclines. Epirubicin, idarubicin, and mitoxantrone are other anthracycline analogs. Doxorubicin's cardiotoxicity is primarily associated with its potential to cause damage to the heart muscle, leading to a condition called cardiomyopathy.

- It can result in symptoms such as shortness of breath, fatigue, chest pain, and fluid retention.

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It can result in symptoms such as shortness of breath, fatigue, chest pain, and fluid retention.

Incorrect Options:

Option B - Cyclophosphamide: Cyclophosphamide belongs to alkylating agents

Option C - Tamoxifen: Tamoxifen is a selective estrogen receptor modulator

Option D - Imatinib: Imatinib is a tyrosine kinase inhibitor.

- All these drugs are less cardiotoxic than anthracyclines.

Solution for Question 40:

Correct Option B - Rituximab:

- Rituximab is an anti-CD 20 monoclonal antibody used in cancer treatments.
- It blocks the CD20 receptor, which is a marker of B-lymphocyte.

Incorrect options:

Option A. Methotrexate is an antimetabolite. It is also a chemotherapy agent and immune suppressant.

Option C. Cisplatin is an antineoplastic agent.

Option D. 5-Fluorouracil is a cytotoxic drug.

Solution for Question 41:

Correct Option A - Rituximab:

- Rituximab is a monoclonal antibody used in the treatment of certain types of cancer. It specifically targets and binds to a protein called CD20, which is found on the surface of B-lymphocytes. Rituximab works by promoting the destruction of cancerous B-cells and modulating the immune response against cancer.
- Rituximab is primarily used in the treatment of B-cell non-Hodgkin lymphoma (NHL), including follicular lymphoma, diffuse large B-cell lymphoma, and chronic lymphocytic leukemia (CLL). It is also utilized in the management of certain autoimmune disorders, such as rheumatoid arthritis and pemphigus vulgaris.

Rituximab is a monoclonal antibody used in the treatment of certain types of cancer. It specifically targets and binds to a protein called CD20, which is found on the surface of B-lymphocytes. Rituximab works by promoting the destruction of cancerous B-cells and modulating the immune response against cancer.

Rituximab is primarily used in the treatment of B-cell non-Hodgkin lymphoma (NHL), including follicular lymphoma, diffuse large B-cell lymphoma, and chronic lymphocytic leukemia (CLL). It is also utilized in the management of certain autoimmune disorders, such as rheumatoid arthritis and pemphigus vulgaris.

Incorrect Options:

Option B - Cisplatin: Cisplatin is a platinum-based chemotherapy drug that is not a monoclonal antibody. It works by damaging the DNA in cancer cells, leading to their death.

Option C - 5-Fluorouracil: 5-Fluorouracil is a chemotherapy drug that inhibits the synthesis of DNA and RNA, leading to the destruction of cancer cells. It is not a monoclonal antibody.

Option D - Methotrexate: Methotrexate is a chemotherapy drug that interferes with DNA synthesis and cell division. It is used in the treatment of various cancers, as well as autoimmune conditions. However, it is not a monoclonal antibody.

Solution for Question 42:

Correct Option A - 6-mercaptopurine:

- Among the options provided, 6-mercaptopurine, also known as 6-MP, is an inhibitor of DNA synthesis. It is a medication that belongs to the class of drugs called antimetabolites. 6-MP interferes with the synthesis of DNA by inhibiting the enzyme hypoxanthine-guanine phosphoribosyltransferase (HGPRT), which is involved in the synthesis of DNA and RNA.

Among the options provided, 6-mercaptopurine, also known as 6-MP, is an inhibitor of DNA synthesis. It is a medication that belongs to the class of drugs called antimetabolites. 6-MP interferes with the synthesis of DNA by inhibiting the enzyme hypoxanthine-guanine phosphoribosyltransferase (HGPRT), which is involved in the synthesis of DNA and RNA.

Incorrect Options:

Option B - Mitomycin: Mitomycin inhibits DNA synthesis by binding to DNA and preventing its replication.

Option C - Actinomycin: Actinomycin inhibits DNA synthesis by binding to DNA and preventing its replication.

Option D - Asparaginase: Asparaginase is an enzyme that depletes the amino acid asparagine. It does not directly inhibit DNA synthesis but affects protein synthesis by depleting asparagine, an essential amino acid for the growth of some cancer cells.

Solution for Question 43:

Correct Option C - Goserelin:

- The patient complains of difficulty in urination, back pain, and an elevated PSA blood test. It also says that on x-ray, two radiolucent lesions were found in the bony pelvis and MRI revealed several enlarged lymph nodes. The features suggest that this is a case of prostate cancer and the drug used is Goserelin which is a GnRH agonist.

The patient complains of difficulty in urination, back pain, and an elevated PSA blood test. It also says that on x-ray, two radiolucent lesions were found in the bony pelvis and MRI revealed several enlarged lymph nodes. The features suggest that this is a case of prostate cancer and the drug used is Goserelin which is a GnRH agonist.

Incorrect Options:

Option A - Anastrozole: Anastrozole is used for the treatment of breast cancer and is an aromatase inhibitor.

Option B - Desogestrel: Desogestrel is used as oral contraceptive and is a synthetic progestin without androgenic activity.

Option D - Oxandrolone: Oxandrolone is used for the treatment of bone pain in patients with osteoporosis and is a synthetic derivative of testosterone.

Solution for Question 44:

Correct Option D - Palonosetron:

- In cases of vomiting due to chemotherapy, palonosetron is used.
- Palonosetron is a 5HT₃ receptor antagonist that blocks serotonin in vagal terminals and centrally in the chemoreceptor trigger zone(CTZ).
- It has a higher receptor affinity as compared to the first-generation 5HT₃ antagonists making it more potent.

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Incorrect Options:

Option A - Metoclopramide: Metoclopramide is an antiemetic agent and it is a 5HT₄ agonist and a D₂ blocker.

Option B - Ropinirole: Ropinirole is the drug of choice for restless leg syndrome and is an anti parkinsonian drug.

Option C - Domperidone: Domperidone is not used in such cases.

Solution for Question 45:

Correct Option B - Epirubicin:

- Epirubicin is avoided as it is cardiotoxic and hence is not used in this patient as she already has a history of heart failure. Epirubicin acts by interacting with DNA directly and it affects its transcription and replication. It causes irreversible cardiotoxicity and cardiomyopathy due to

free radicals and is dose dependent.

Incorrect Options:

Option A - Methotrexate: Methotrexate is not cardiotoxic.

Option C - Cyclophosphamide: Cyclophosphamide is not cardiotoxic.

Option D - Cisplatin: Cisplatin is not cardiotoxic.

Solution for Question 46:

Correct Option C - Filgrastim:

- In a patient on anti-cancer therapy who develops infection with a low total leukocyte count of 2000/mm³, the most likely effective drug would be Filgrastim.
- Filgrastim is a granulocyte colony-stimulating factor (G-CSF) that stimulates the production and maturation of neutrophils, a type of white blood cell responsible for fighting bacterial infections. It is commonly used to treat neutropenia, which is a low level of neutrophils in the blood.
- Therefore, in this case, Filgrastim would be the most appropriate and effective drug for increasing the leukocyte count and aiding in the treatment of infection.

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Therefore, in this case, Filgrastim would be the most appropriate and effective drug for increasing the leukocyte count and aiding in the treatment of infection.

Incorrect Options:

Option A & B: Romiplostim & Oprelvekin are not indicated for the treatment of infection or low leukocyte count. Romiplostim is a thrombopoietin receptor agonist used to increase platelet production in patients with thrombocytopenia, a low platelet count. Oprelvekin is a recombinant interleukin-11 used to stimulate platelet production in patients with severe thrombocytopenia.

Option D - Erythropoietin: Erythropoietin is a hormone that stimulates the production of red blood cells and is used to treat anemia, not low leukocyte count or infection.

Solution for Question 47:

Correct Option B - Doxorubicin:

- Doxorubicin is an anthracycline chemotherapy drug commonly used in the treatment of various cancers, including soft-tissue sarcomas. It works by interfering with the replication of DNA and inhibiting the activity of topoisomerase II, an enzyme involved in DNA synthesis.
- One of the known side effects of Doxorubicin is radiation recall dermatitis, which presents as erythematous rashes or skin reactions in areas previously exposed to radiation therapy. This phenomenon occurs when chemotherapy drugs exacerbate or "recall" the inflammatory reaction caused by prior radiation exposure.
- Therefore, among the options given, Doxorubicin is the drug most likely responsible for the development of erythematous rashes over the areas previously treated with radiotherapy.

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Therefore, among the options given, Doxorubicin is the drug most likely responsible for the development of erythematous rashes over the areas previously treated with radiotherapy.

Incorrect Options:

Option A - Vincristine: Vincristine is a vinca alkaloid chemotherapy drug that acts by disrupting the formation of microtubules in cells. It is not typically associated with radiation recall dermatitis.

Option C - Cyclophosphamide: Cyclophosphamide is an alkylating agent commonly used in chemotherapy. It is not known to cause radiation recall dermatitis.

Option D - 6-mercaptopurine: 6-mercaptopurine is an antimetabolite chemotherapy drug used to treat various types of cancer. It does not typically cause radiation recall dermatitis.

Solution for Question 48:

Correct Option A - Bleomycin:

- Bleomycin is an anticancer medication used in the treatment of various malignancies, including lymphomas, testicular cancer, and squamous cell carcinomas. One of the known side effects of Bleomycin is flagellate pigmentation, which refers to a distinctive linear pattern of pigmentation resembling whip-like streaks on the skin. It is an uncommon side effect but can occur with prolonged use of the drug.

Bleomycin is an anticancer medication used in the treatment of various malignancies, including lymphomas, testicular cancer, and squamous cell carcinomas. One of the known side effects of Bleomycin is flagellate pigmentation, which refers to a distinctive linear pattern of pigmentation resembling whip-like streaks on the skin. It is an uncommon side effect but can occur with prolonged use of the drug.

Incorrect Options:

Option B - Doxorubicin: Doxorubicin is an anthracycline chemotherapy drug used to treat various types of cancer, including breast cancer, lung cancer, and leukemia. While doxorubicin can cause various side effects, including hair loss and discoloration of nails, it is not associated with flagellate pigmentation.

Option C - Vincristine: Vincristine is a chemotherapy drug commonly used in the treatment of leukemia, lymphoma, and solid tumors. It is not known to cause flagellate pigmentation. However, vincristine can cause other side effects, such as peripheral neuropathy, constipation, and hair loss.

Option D - Minocycline: Minocycline is an antibiotic in the tetracycline class, commonly used to treat bacterial infections, including acne. Prolonged use of minocycline has been associated with a side effect known as minocycline-induced pigmentation. This pigmentation can be present in various types, including blue-gray pigmentation of the skin, mucous membranes, and nails. However, it does not specifically cause flagellate pigmentation.

Solution for Question 49:

Correct Option C - Low molecular weight heparin:

- LMWH is derived from unfractionated heparin but has a smaller molecular size, which allows for more predictable anticoagulant effects and longer half-life. LMWH, such as enoxaparin or dalteparin, is typically administered subcutaneously once or twice daily and does not require routine monitoring. It has demonstrated efficacy and safety in preventing and treating venous thromboembolism in cancer patients and is considered the anticoagulant of choice in this population.

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Incorrect Options:

Option A - Unfractionated Heparin: Unfractionated Heparin UFH requires continuous intravenous infusion or subcutaneous injections and necessitates monitoring of the activated partial thromboplastin time (aPTT) to adjust the dose. While UFH can be used in cancer-associated thromboembolism, it is generally less preferred compared to low molecular weight heparin (LMWH) due to the need for frequent monitoring and a higher risk of heparin-induced thrombocytopenia.

Option B - Warfarin: Warfarin is an oral vitamin K antagonist that inhibits the production of clotting factors in the liver. It requires regular monitoring of the international normalized ratio (INR) to maintain the desired level of anticoagulation. While warfarin is commonly used for long-term anticoagulation in various conditions, its use in cancer-associated thromboembolism is limited due to drug interactions, variable response, and the need for careful dose adjustment. Additionally, cancer patients may have an increased bleeding risk, which can complicate the use of warfarin.

Option D - Dabigatran: Dabigatran is a direct oral anticoagulant (DOAC) that directly inhibits thrombin, a key enzyme in the blood clotting process. While DOACs have gained popularity in various clinical settings, including non-valvular atrial fibrillation and venous thromboembolism, their use in cancer-associated thromboembolism is limited. The evidence for DOACs in this specific population is limited, and they may have interactions with certain cancer treatments.

Solution for Question 50:

Correct Option D - 5-Fluorouracil:

- 5-Fluorouracil (5-FU) is not an alkylating agent. It is a fluoropyrimidine antimetabolite that interferes with DNA and RNA synthesis. 5-FU is converted to its active form within cells and inhibits the enzyme thymidylate synthase, which is necessary for the synthesis of DNA. By inhibiting DNA synthesis, 5-FU disrupts cell division and growth. It is commonly used in the treatment of various solid tumors, including colorectal, breast, and gastrointestinal cancers.

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Incorrect Options:

Option A - Melphalan: Melphalan is an alkylating agent. It belongs to the class of nitrogen mustards and works by forming covalent bonds with DNA, leading to cross-linking and damage to the DNA strands. This ultimately inhibits cell division and growth. Melphalan is commonly used in the treatment of multiple myeloma and certain types of cancer.

Option B - Cyclophosphamide: Cyclophosphamide is an alkylating agent. It is a prodrug that is metabolized in the body to its active form, phosphoramidate mustard. Like other alkylating agents, cyclophosphamide forms covalent bonds with DNA, leading to DNA damage and disruption of cell division. It is used in the treatment of various cancers, including lymphomas, leukemia, and solid tumors.

Option C - Busulfan: Busulfan is an alkylating agent. It is used in the treatment of chronic myelogenous leukemia (CML) and as part of conditioning regimens for stem cell transplantation. Busulfan works by forming DNA cross-links and inhibiting DNA synthesis, leading to the suppression of abnormal cell growth.

Solution for Question 51:

Correct Option D - Vincristine:

- Langerhans cell histiocytosis (LCH) is a rare disorder characterized by the abnormal proliferation of Langerhans cells, a type of immune cell. The disease can affect various organs, including the bone, skin, and lymph nodes. Based on the presentation of pallor, lymphadenopathy, and the X-ray showing punched-out lesions, LCH is a possible diagnosis.
- The management of Langerhans cell histiocytosis typically involves a multidisciplinary approach and may include the use of various treatment modalities depending on the extent and severity of the disease.
- Vincristine is not typically used as a primary treatment for Langerhans cell histiocytosis. While vincristine is used in the treatment of certain cancers, it is not considered a standard treatment option for LCH.

Langerhans cell histiocytosis (LCH) is a rare disorder characterized by the abnormal proliferation of Langerhans cells, a type of immune cell. The disease can affect various organs, including the bone, skin, and lymph nodes. Based on the presentation of pallor, lymphadenopathy, and the X-ray showing punched-out lesions, LCH is a possible diagnosis.

The management of Langerhans cell histiocytosis typically involves a multidisciplinary approach and may include the use of various treatment modalities depending on the extent and severity of the disease.

Vincristine is not typically used as a primary treatment for Langerhans cell histiocytosis. While vincristine is used in the treatment of certain cancers, it is not considered a standard treatment option for LCH.

Incorrect Options:

Option A - Cyclosporine: Cyclosporine is an immunosuppressant medication that can be used in the treatment of LCH, especially in cases involving multi-system or severe disease.

Option B - Methotrexate: Methotrexate, a chemotherapy agent, is commonly used in the treatment of LCH. It can help control the proliferation of Langerhans cells and reduce the inflammatory response associated with the disease.

Option C - Vinblastine: Vinblastine is a chemotherapy medication that is often used as a first-line treatment for LCH, particularly in cases involving multi-system disease or high-risk organ involvement.

Solution for Question 52:

Correct Option C - Rituximab:

- Rituximab, an anti-CD20 monoclonal antibody, is used in the management of lymphoproliferative and autoimmune conditions. Currently, it has been approved for use by the FDA under the following conditions. CD20-positive B-cell Non-Hodgkin's Lymphoma (NHL) Chronic Lymphocytic Leukemia Rheumatoid Arthritis Pemphigus Vulgaris
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CD20-positive B-cell Non-Hodgkin's Lymphoma (NHL)

Chronic Lymphocytic Leukemia

Rheumatoid Arthritis

Pemphigus Vulgaris

Incorrect Options:

Option A - Omalizumab:

- Omalizumab is a recombinant, humanized, monoclonal antibody against human immunoglobulin E (IgE). It is used in Urticaria and Atopic Dermatitis.

Option B - Denosumab:

- Denosumab is a fully human monoclonal antibody that binds the cytokine RANKL (receptor activator of NF κ B ligand), an essential factor initiating bone turnover. RANKL inhibition blocks osteoclast maturation, function and survival, thus reducing bone resorption.
- It is used for the treatment of Osteoporosis.

It is used for the treatment of Osteoporosis.

Option D - Methotrexate:

- Methotrexate is an anti-metabolite, not a monoclonal antibody. It is most commonly used in chemotherapy and immunosuppressant in auto-immune diseases.

Solution for Question 53:

Correct Option C - Discovery of cancer therapy by inhibition of negative immune regulation:

- Discovery of cancer therapy by inhibition of negative immune regulation: The Nobel Prize in Physiology or Medicine, 2018 was awarded to James P. Allison and Tasuku Honjo for their discovery of cancer therapy by inhibition of negative immune regulation. Their work led to the development of immune checkpoint inhibitors, which are drugs that can block the interaction between checkpoint proteins and their receptors, unleashing the immune system's ability to attack cancer cells.

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Incorrect Options:

Option A - Discovery of mechanisms of autophagy: The Nobel Prize in Physiology or Medicine, 2016 was awarded to Yoshinori Ohsumi for his discoveries of mechanisms of autophagy.

Option B - Discoveries of molecular mechanisms controlling the circadian rhythm: The Nobel Prize in Physiology or Medicine, 2017 was awarded to Jeffrey C. Hall, Michael Rosbash, and Michael W. Young for their discoveries of molecular mechanisms controlling the circadian rhythm.

Option D - Discovery that mature cells can be reprogrammed to become pluripotent: The Nobel Prize in Physiology or Medicine, 2012 was awarded to Sir John B. Gurdon and Shinya Yamanaka for their discovery that mature cells can be reprogrammed to become pluripotent.

Anaesthetic Pharmacology

1. A 45-year-old male patient with a history of hyperthyroidism is diagnosed with a brain tumor and is scheduled for neurosurgery. Thiopentone has been chosen as the induction agent for this surgery. Which of the following is an incorrect statement about Thiopentone?

- A. Thiopentone decreases intracranial pressure, making it suitable for neurosurgical procedures
- B. Porphyria is a relative contraindication for thiopentone
- C. The mechanism for termination of action is by redistribution from brain to fat cells
- D. Thiopentone has antithyroid properties and thus is an agent of choice for hyperthyroid patients

2. A 45 year old male undergoes a minor surgical procedure for lipoma excision under local anesthesia with bupivacaine. Shortly after administration, he complains of tinnitus, dizziness and a metallic taste in his mouth. What is the most appropriate treatment for this patient's condition?

- A. Intravenous atropine
- B. Intravenous epinephrine
- C. 20% lipid emulsion therapy
- D. Intravenous sodium bicarbonate

3. A 55-year-old male patient with a history of chronic kidney disease and liver cirrhosis is scheduled for elective abdominal surgery. The anesthesiologist must choose an appropriate muscle relaxant for intubation and intraoperative muscle relaxation. Considering the patient's comorbidities, the anesthesiologist is

evaluating the use of cisatracurium instead of atracurium. Which of the following statements is true regarding Atracurium?

- A. Metabolised by Hoffman elimination only
 - B. Dose-dependent histamine release
 - C. Preferred muscle relaxant in liver/ kidney disease
 - D. Made of only one isomer
-

4. A 40-year-old male patient is brought to the emergency department for an urgent intubation due to respiratory failure. The anesthesiologist decides to use succinylcholine to facilitate rapid sequence intubation. Which of the following statements is false regarding succinylcholine?

- A. Toxicity causes diaphragm paralysis
 - B. Toxicity causes hyperkalemia
 - C. Contraindicated in muscular dystrophy
 - D. Drug of choice in stroke and hemiplegia
-

5. A 32-year-old female undergoing elective laparoscopic cholecystectomy develops muscle fasciculations followed by flaccid paralysis after a neuromuscular blocking agent is administered. Despite neostigmine administration, the muscle relaxation persists. Which muscle relaxant was administered?

- A. Suxamethonium
 - B. Pancuronium
 - C. Pipecuronium
 - D. Vecuronium
-

6. Which of the following local anaesthetics is used to diagnose pseudocholinesterase deficiency?

- A. Ropivacaine
 - B. Bupivacaine
 - C. Etidocaine
 - D. Dibucaine
-

7. In the context of differential sensitivity of nerve fibers to local anaesthetics, which of the following sequences correctly represents the order in which different types of nerve fibres are blocked, starting from the most sensitive to the least sensitive?

- A. Type A alpha, Type A beta, Type A gamma, Type A delta, Type C
 - B. Type C, Type A delta, Type A gamma, Type A beta, Type A alpha
 - C. Type A delta, Type A alpha, Type A beta, Type A gamma, Type C
 - D. Type C, Type A gamma, Type A delta, Type A beta, Type A alpha
-

8. A 12-year-old patient was sedated for more than 48 hours following status epilepticus. He developed refractory bradycardia, hyperlipidemia, and renal failure. Arterial blood gas (ABG) analysis showed metabolic acidosis. Which of the following sedatives is most likely responsible for these symptoms?

- A. Midazolam
 - B. Propofol
 - C. Dexmedetomidine
 - D. Fentanyl
-

9. A 40-year-old patient is administered an intermediate-acting amide as a local anesthetic during a minor surgical procedure on the upper extremity. Shortly after administration, the patient develops cyanosis and complains of shortness of breath. Which of the following is most likely the local anesthetic based on the above symptoms?

- A. Prilocaine
 - B. Mepivacaine
 - C. Lidocaine
 - D. Bupivacaine
-

10. A 34-year-old woman is scheduled for a laparoscopic cholecystectomy. During the preoperative evaluation, the anesthesiologist explains the different types of anaesthetics that will be used and points at a cylinder having a blue body and white shoulder, along with a fast-acting IV drug. What are these anaesthetic agents?

- A. Nitrous oxide + Oxygen; Propofol
 - B. Cyclopropane; Fentanyl
 - C. Carbon dioxide; Ketamine
 - D. Nitrous oxide + Oxygen; Midazolam
-

11. Which of the following statements about isoflurane is correct?

- A. Isoflurane primarily causes a drop in blood pressure through cardiac depression
 - B. Isoflurane maintains cardiac output while causing vasodilation
 - C. Isoflurane is known to provoke seizures, making it unsuitable for patients undergoing neurosurgery
 - D. Isoflurane causes significant metabolic byproducts, leading to renal and hepatic toxicity.
-

12. A 45-year-old male patient with a history of asthma presents to the operating room for elective surgery. Halothane is used as the primary anaesthetic agent. During the induction phase, the patient suddenly develops ventricular arrhythmias. What is the most likely mechanism underlying the ventricular arrhythmias induced by halothane in this patient?

- A. Direct myocardial depression
 - B. Inhibition of catecholamine release
 - C. Sensitization of the myocardium to catecholamines
 - D. Potentiation of calcium influx into cardiac myocytes
-

13. A 45-year-old male is undergoing surgery for an abdominal hernia repair. Cyclopropane is chosen as the anaesthetic agent. During the procedure, electrocautery is used to control bleeding. Due to concerns about cyclopropane's high flammability, the anesthesiologist decides to discontinue its administration. The patient experiences a drop in blood pressure. What is the most likely cause of the patient's hypotension?

- A. An equipment malfunction resulted in inadequate delivery of oxygen during surgery
 - B. Sudden withdrawal of cyclopropane led to a sharp decrease in sympathetic nervous system activity
 - C. The patient experienced a common side effect of the electrocautery
 - D. The patient had an allergic reaction to cyclopropane
-

14. Entonox, used for analgesia during painful procedures, exhibits the Poynting effect. Which of the following statements best describes the Poynting effect in Entonox?

- A. It causes the individual properties of nitrogen and oxygen to intensify when combined

- B. It results in the production of a gas mixture that behaves differently from its individual components
 - C. It increases the likelihood of adverse effects when administering Entonox
 - D. It leads to the rapid separation of nitrogen and oxygen at high pressures
-

15. A 40-year-old male with a history of substance abuse presents for a wisdom tooth extraction. The patient is known to abuse nitrous oxide recreationally. His medical history is otherwise unremarkable. The anesthesiologist is considering the use of nitrous oxide for sedation. Which of the following complications should the anesthesiologist be most concerned about for this patient?

- A. Malignant Hyperthermia
 - B. Hepatotoxicity
 - C. Vitamin B 12 Deficiency
 - D. Renal toxicity
-

16. A 45-year-old male presents to the emergency department with sudden onset sharp chest pain and shortness of breath. On examination, he has decreased breath sounds on the left side, and chest X-ray confirms left-sided pneumothorax. The patient is scheduled for an urgent chest tube insertion under general anaesthesia. Which of the following inhaled anaesthetics is contraindicated in this patient?

- A. Sevoflurane
 - B. Isoflurane
 - C. Desflurane
 - D. Nitrous oxide
-

17. Match each inhalation agent with its corresponding blood/gas partition coefficient: Inhaled anaesthetic B/G partition coefficient a) Sevoflurane 1) 0.47 b) Isoflurane 2) 0.68 c) Desflurane 3) 1.4 d) Nitrous oxide 4) 0.42

Inhaled anaesthetic	B/G partition coefficient
a) Sevoflurane	1) 0.47
b) Isoflurane	2) 0.68
c) Desflurane	3) 1.4
d) Nitrous oxide	4) 0.42

- A. a-4, b-1, c-3, d-2
 - B. a-3, b-4, c-1, d-2
 - C. a-2, b-3, c-4, d-1
 - D. a-3, b-2, c-1, d-4
-

18. A 25-year-old male presents to the emergency department with acute onset right lower quadrant abdominal pain, nausea, and vomiting. On examination, he has right lower quadrant rebound tenderness and guarding. The doctor decides to take him up for an emergency surgery. Which inhalation anaesthetic would be best for induction in this patient based on its MAC values and blood gas solubility?

- A. Nitrous oxide
 - B. Sevoflurane
 - C. Desflurane
 - D. Halothane
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	2
Question 4	4
Question 5	1
Question 6	4
Question 7	2
Question 8	2
Question 9	1
Question 10	1
Question 11	2
Question 12	3
Question 13	2
Question 14	2
Question 15	3
Question 16	4
Question 17	3
Question 18	2

Solution for Question 1:

Correct answer: B) Porphyria is a relative contraindication for thiopentone

Explanation:

Porphyria is not a relative contraindication but an absolute contraindication for the use of Thiopentone due to the risk of inducing acute porphyric crisis.

Contraindications for thiopentone:

Table rendering failed

Thiopentone decreases intracranial pressure, making it suitable for neurosurgical procedures (Option A): Thiopentone, an intravenous barbiturate, is used as an induction agent of choice for neurosurgeries for the following reasons:

- Causes maximum decrease in ICP
- Anticonvulsant properties
- Decreases cerebral metabolic rate
- Maximum neuroprotective among IV anaesthetics

The mechanism for termination of action is by redistribution from brain to fat cells (Option C):

The primary mechanism for the termination of its action is redistribution from the brain to fat and muscle tissues.

Mechanism of Termination	Redistribution from the brain to fat and muscle tissues
Metabolism	Metabolised in the liver by hydroxylation
Excretion	Excreted by the kidneys

Thiopentone has antithyroid properties and thus is an agent of choice for hyperthyroid patients (Option D): Thiopentone has antithyroid properties, making it an induction agent of choice for hyperthyroid patients.

All i/v anaesthetics cause :

- Decreased Cerebral metabolic O₂ demand
- Decreased ICP and CBF
- Decreased Respiratory drive
- Myocardial depression

Exceptions KETAMINE: Sympathomimetic drug, increase all of the above ETOMIDATE: Cardio-stable

Solution for Question 2:

Correct answer: C) 20% lipid emulsion therapy

Explanation:

- 20% lipid emulsion therapy is the most appropriate treatment for bupivacaine toxicity.
- Lipid emulsion acts as a "lipid sink", sequestering the lipophilic bupivacaine, thus reducing its concentration in tissues and mitigating toxicity.
- Bupivacaine is a long-acting local anesthetic.
- Toxicity often results from overdose or accidental intravascular injection, leading to serious and potentially life-threatening complications.

Table rendering failed

- Intravenous atropine (Option A) and Intravenous epinephrine (Option B) may be used for bradycardia and cardiac arrest, but they are not the primary treatment for bupivacaine toxicity.
 - Intravenous sodium bicarbonate (Option D) can be helpful in managing acidosis, which may exacerbate toxicity, but it is not the primary antidote.
-

Solution for Question 3:

Correct answer: B) Dose-dependent histamine release

Explanation:

Differences between Atracurium and Cisatracurium:

Atracurium	Cisatracurium
Mixture of >10 isomers Metabolized by Hoffmann elimination and plasma esterase Dose-dependent histamine release → allergy (Option B) Should be avoided in patients with kidney/ liver disease, as its toxic metabolite laudanosine gets accumulated → CNS toxicity → seizures	Made of one isomer (Option D) 100% of the drug is metabolized by Hofmann elimination (Option A) No histamine release Safe in patients with kidney/ liver disease (Option C).

Solution for Question 4:

Correct answer: D) Drug of choice in stroke and hemiplegia

Explanation:

Succinylcholine:

- It is a depolarising neuromuscular blocking agent
- It is the fastest and shortest-acting neuromuscular blocking agent
- Causes initial fasciculations
- It is metabolized by pseudocholinesterase.
- Its actions cannot be reversed by neostigmine.

Toxicity of succinylcholine:	Diaphragm paralysis → Scoline apnea (Option A) Hyperkalemia (Option B) Due to potassium release → cardiac arrest Increased in trauma, burns, rhabdomyolysis Malignant hyperthermia Treated by Dantrolene Histamine release SA node partial agonist leads to bradycardia , junctional rhythm - treatment: i/v atropine
Contraindications:	Muscle dystrophy (Option C) Neurological disease Hemiplegia Stroke (Option D) Contraindicated in head injury, glaucoma Increases ICP, IOP

Solution for Question 5:

Correct answer: A) Suxamethonium

Explanation:

This patient is administered a neuromuscular blocking agent during intubation for muscle relaxation and develops muscle fasciculations followed by flaccid paralysis, and neostigmine

does not reverse the neuromuscular blockade. These findings suggest that the drug administered is a depolarizing blocking agent such as Suxamethonium/ Succinylcholine.

Difference between depolarizing and non-depolarizing blockers:

	Depolarizing Blockers	Non-Depolarizing Blockers
Drugs	Suxamethonium/ succinylcholine	Curonium, curium, curare
MOA	Phase1 Opening of Nm receptor (like Ach) Initial muscle contraction-fasciculation Depolarizing SMR Enters into phase 2 Desensitization of the Nm receptor → flaccid paralysis of the muscle	Competitively block Nm receptor of Ach They are non-depolarizing and cause flaccid paralysis
Role of Neostigmine	Neostigmine cannot reverse the effect of succinylcholine	For reversal of NDMR: increase Ach o Neostigmine o Edrophonium
Initial Fasciculations	Present	Absent
Postoperative myalgia	Present	Absent
Cold Temperature	Increase the effect	Decrease the effect
Muscle sensitivity	1st to paralyze: Neck muscle Last to paralyze: Diaphragm (Most resistant, 1st to recover)	1st: Fingers Last: Diaphragm

Solution for Question 6:

Correct answer: D) Dibucaine

Explanation:

To diagnose pseudocholinesterase deficiency, the local anesthetic used is dibucaine.

Dibucaine test to diagnose pseudocholinesterase deficiency:

- A laboratory analysis for pseudocholinesterase deficiency involves taking a plasma sample from the patient and subjecting it to a qualitative test to assess the activity of the pseudocholinesterase enzyme.
- Dibucaine is a long-acting & highly potent local anesthetic that inhibits about 80% of pseudocholinesterase (PChE) activity in normal individuals.
- In heterozygotes (with 1 variant allele), PChE activity is inhibited by approximately 40-60%.
- In homozygotes (with 2 variant alleles), PChE activity is inhibited by around 20%.
- The percentage of inhibition is known as the dibucaine number.
- A lower dibucaine number indicates a longer duration of residual neuromuscular blockade after administration of succinylcholine (SCh) or mivacurium.

Solution for Question 7:

Correct answer: B) Type C, Type A delta, Type A gamma, Type A beta, Type A alpha

Explanation:

- Local anaesthetics block nerve fibres in a specific order based on their diameter and myelination.
- B fibres are the most sensitive and are blocked first, followed by small unmyelinated C fibres and small myelinated A-delta fibres.
- Subsequently, A gamma, A beta, and finally, the large, heavily myelinated A alpha fibres are blocked.

CLASSIFICATION	ANATOMIC LOCATION	MYELIN	DIAMETER (µm)	CONDUCTION VELOCITY (m/s)	FUNCTION	CRITICAL SENSITIVITY TO BLOCK
A Fibers						
Aα	Afferent to and efferent from muscles and joints	Yes	6-22	10-85	Motor and proprioception	+
Aβ						++
Aγ	Efferent to muscle spindles	Yes	3-6	15-35	Muscle and tone	++
Aδ	Sensory roots and afferent peripheral nerves	Yes	1-4	5-25	Pain, temperature, touch	+++
B Fibers						
	Preganglionic sympathetic	Yes	<3	3-15	Vasomotor, visceromotor, sudomotor, pilomotor	++++
C Fibers						
Sympathetic	Postganglionic sympathetic	No	0.3-1.3	0.7-1.3	Vasomotor, visceromotor, sudomotor, pilomotor	++++
Dorsal root	Sensory roots and afferent peripheral nerves	No	0.4-1.2	0.1-2	Pain, temperature, touch	++++

Solution for Question 8:

Correct answer: B) Propofol

Explanation:

The patient's presentation is indicative of Propofol Infusion Syndrome (PRIS), which can occur with prolonged propofol sedation. PRIS is characterized by symptoms such as refractory bradycardia, hyperlipidemia, renal failure, and metabolic acidosis.

- PRIS typically occurs when a dose of >4 mg/kg/hr of propofol is given for >48 hours. Additional features of PRIS include: Hyperlipidemia (first sign to get detected) Refractory bradycardia and asystole Acute renal failure Rhabdomyolysis Severe metabolic acidosis Green-coloured urine can also be seen.
- Hyperlipidemia (first sign to get detected)
- Refractory bradycardia and asystole
- Acute renal failure
- Rhabdomyolysis
- Severe metabolic acidosis
- Green-coloured urine can also be seen.
- Recognizing these signs is crucial for diagnosing and managing PRIS, as this is potentially fatal.
- Hyperlipidemia (first sign to get detected)
- Refractory bradycardia and asystole
- Acute renal failure
- Rhabdomyolysis
- Severe metabolic acidosis
- Green-coloured urine can also be seen.

Midazolam (Option A): While midazolam can cause prolonged sedation and respiratory depression, it is not typically associated with hyperlipidemia, metabolic acidosis, or renal failure.

Dexmedetomidine (Option B): It can cause bradycardia, but it does not cause hyperlipidemia, metabolic acidosis, or renal failure.

Fentanyl (Option D): It can cause respiratory depression and hypotension but is not associated with hyperlipidemia, metabolic acidosis, or renal failure.

Solution for Question 9:

Correct answer: A) Prilocaine

Explanation:

- Prilocaine, a local anaesthetic, can cause methemoglobinemia, where haemoglobin is oxidised to methemoglobin, reducing blood's oxygen-carrying capacity.
- This leads to symptoms like cyanosis and shortness of breath. Benzocaine also causes methemoglobinemia
- Methemoglobinemia typically develops after administering prilocaine at a dose of 8 mg/kg. Methemoglobinemia can be treated with intravenous methylene blue at a dose of 1–2 mg/kg.

Duration of Action	Local Anaesthetic
Ultra Short	Chloroprocaine (shortest)
Short	Procaine
Intermediate	Lidocaine, Prilocaine, Mepivacaine
Long	Bupivacaine, Ropivacaine, Etidocaine, Tetracaine, Dibucaine (longest)

	Amides	Esters
Physical property	Stable solution	Unstable, structurally similar to PABA.
Metabolism	By liver	Esterase enzyme
Exceptions	Artikane by esterase	Cocaine in liver
Examples	Prilocaine, lignocaine	Procaine

Solution for Question 10:

Correct answer: A) Nitrous oxide + Oxygen; Propofol

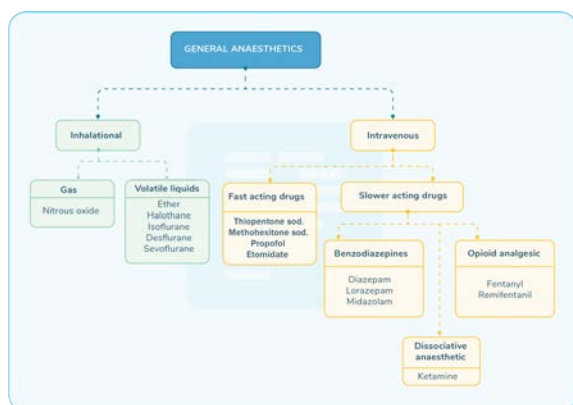
Explanation:

- The colour coding of entonox (Nitrous Oxide + Oxygen) is Blue body + white shoulder.
- The fast-acting IV anaesthetic drug refers to Propofol, as the IV anaesthetics mentioned in other options are slow-acting.

The colour coding of gases used in anaesthesia is as follows:

Gas	Colour Coding	Pin Index Safety System
Air	Grey body + black/white shoulder	1,5
Oxygen	Black body + white shoulder	2,5
Nitrous Oxide	Blue	3,5
Entonox (Nitrous Oxide + Oxygen) (Option A)	Blue body + white shoulder	7
Carbon Dioxide (>7.5%)	Grey	1,6
Carbon Dioxide (<7.5%)	Grey	2,6
Cyclopropane	Orange	3,6

The classification of general anaesthetics is as follows:



Solution for Question 11:

Correct answer: B) Isoflurane maintains cardiac output while causing vasodilation

Explanation:

Isoflurane is particularly beneficial in patients with heart failure because it maintains cardiac output while causing vasodilation and reducing BP, unlike halothane, which primarily causes a drop in BP through cardiac depression.

Effects of Isoflurane:

System	Effects
CVS	Decrease in BP - due to vasodilation (Option A) CO is well maintained - suitable for patients with myocardial insufficiency (Option B) HR is increased due to reflex tachycardia Dilates coronaries - possibility of coronary steal syndrome
RS	Respiratory depression Increased secretions
Uterus and Skeletal Muscles	Muscle Relaxation
Ocular	Pupils do not dilate and light reflex is not lost.
Metabolism	Negligible - No renal or hepatotoxicity

- It is also noted for its minimal metabolism, reducing the risk of renal and hepatic toxicity (Option D).
- It does not provoke seizures (Option C), making it suitable for neurosurgical procedures (whereas sevoflurane provokes seizures and is contraindicated in neurosurgical procedures).
- Agent of choice in liver failure.

All inhalational anaesthetic agents cause :

- Decreased Respiratory drive
- Decreased cerebral metabolic O₂ demand
- Increased CBF and ICP

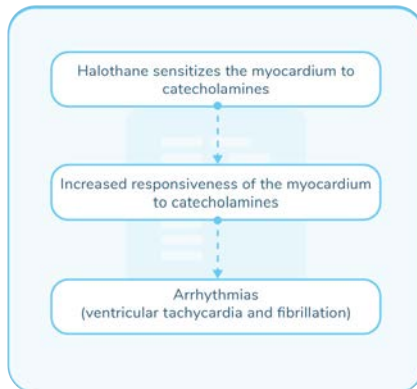
Solution for Question 12:

Correct answer: C) Sensitization of the myocardium to catecholamines

Explanation:

The patient's sudden development of ventricular arrhythmias during halothane anesthesia is most likely due to the sensitisation of the myocardium to catecholamines rather than direct

myocardial depression or inhibition of catecholamine release.



Effects of Halothane:

System	Effects
CVS	Decrease in BP due to cardiac depression CO is reduced with deepening anesthesia HR is reduced Sensitize the heart to the arrhythmogenic action of adrenaline/noradrenaline
RS	Respiratory depression Bronchodilation - DOC for asthmatics
GI/GU	Inhibits intestinal and uterine contractions ↓BP → ↓GFR → ↓ urine formation
Metabolism	By liver - Hepatotoxic

Halothane can depress myocardial contractility by reducing intracellular calcium concentration, but this is not the primary mechanism for halothane-induced arrhythmias.

Solution for Question 13:

Correct answer: B) Sudden withdrawal of cyclopropane led to a sharp decrease in sympathetic nervous system activity

Explanation:

- Cyclopropane is used as an inhalational general anaesthetic agent that increases blood pressure by stimulating the sympathetic nervous system.

- Sudden discontinuation of cyclopropane can result in a rapid decrease in sympathetic activity, causing hypotension. This phenomenon is known as cyclopropane shock.

Solution for Question 14:

Correct answer: B) It results in the production of a gas mixture that behaves differently from its individual components

Explanation:

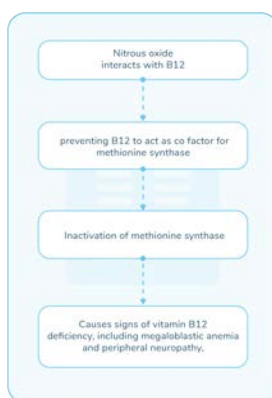
Entonox, a blend of nitrogen oxide (N₂O) and oxygen in a 50:50 ratio, serves as an analgesic in both adults and children. Also used in labor.

- Its production involves passing oxygen through liquid nitrogen oxide (N₂O), where the gases mutually dissolve, resulting in a gas mixture that behaves as a single gas differently from its individual components, known as the "Poynting effect."
- Entonox is presented as a gas in blue cylinders with white and blue checked shoulders.

Solution for Question 15:

Correct answer: C) Vitamin B 12 Deficiency

Explanation:



Recreational abuse of nitrous oxide can lead to vitamin B12 deficiency, causing:

- Peripheral neuropathy
- Myelopathy
- Cognitive impairment

Malignant Hyperthermia (Option A): It is a rare but life-threatening complication typically associated with volatile anaesthetics such as halothane and depolarising muscle relaxants like succinylcholine, not with nitrous oxide.

Hepatotoxicity (Option B): It is a significant side effect of halothane and is not seen as associated with nitrous oxide.

The mechanism of halothane hepatotoxicity is by metabolic pathway and immune-mediated pathway.

Metabolic pathway:

- Halothane is metabolized primarily in the liver by the cytochrome P450 enzyme system, into trifluoroacetic acid (TFA) producing reactive intermediates, such as trifluoroacetyl chloride, which can bind to hepatic proteins, forming neoantigens.

Immune-Mediated Response:

- The neoantigens formed by the binding of reactive intermediates to liver proteins can trigger an immune response.
- In susceptible individuals, this can lead to hepatic inflammation and injury known as halothane hepatitis.

Renal toxicity (Option D): It is seen with anaesthetics such as methoxyflurane and halothane and not with nitrous oxide.

Solution for Question 16:

Correct answer: D) Nitrous oxide

Explanation:

Nitrous oxide, because of its low B/G partition coefficients, will enter body cavities rapidly, thereby increasing the volume or pressure in this cavity.

- It is contraindicated in patients with pneumothorax because it can rapidly diffuse into the chest cavity, causing expansion of the pneumothorax and potentially worsening the patient's condition.
- It can also cause expansion of other air collections such as an obstructed middle ear, air embolus, intraocular air bubble, and obstructed bowel loop.

Sevoflurane (Option A), Isoflurane (Option B), and Desflurane (Option D): They are not associated with the expansion of air-filled spaces and can be used in patients with pneumothorax.

Solution for Question 17:

Correct answer: C) a-2, b-3, c-4, d-1

Explanation:

The blood/gas partition coefficient is a measure of how soluble gas is in the blood. Lower values indicate faster induction and recovery times and helps minimize respiratory depression.

- Desflurane has the lowest B/G coefficient (0.42). It is the fastest-acting flurane. For a patient with COPD, minimizing time spent under deep anesthesia is crucial, hence desflurane is preferred for rapid onset and offset of anesthesia.
- It is the fastest-acting flurane.
- For a patient with COPD, minimizing time spent under deep anesthesia is crucial, hence desflurane is preferred for rapid onset and offset of anesthesia.
- Sevoflurane: B/G partition coefficient is 0.68.
- Isoflurane: B/G partition coefficient is 1.4.
- Nitrous Oxide: B/G partition coefficient is 0.47.
- It is the fastest-acting flurane.
- For a patient with COPD, minimizing time spent under deep anesthesia is crucial, hence desflurane is preferred for rapid onset and offset of anesthesia.

Physical and anaesthetic properties of inhalational anaesthetics						
Anaesthetic	Boiling point (°C)	Inflammability	Irritancy (odour)	Oil: Gas partition coefficient*	Blood : Gas partition coefficient*	MAC (%)
1. Ether	35	Infl. + Exflo.	+++ (Pungent)	65	12.1	1.9
Halothane	50	Non inflammable	- (Pleasant)	224	2.3	0.75
Isosflurane	48	Non inflammable	+ (Unpleasant)	99	1.4	1.2
Desflurane	24	Non inflammable	+ (Unpleasant)	19	0.42	6.0
Sevoflurane	59	Non inflammable	- (Pleasant)	50	0.68	2.0
Nitrous oxide	Gas	Non inflammable	-	1.4	0.47	105

Solution for Question 18:

Correct answer: B) Sevoflurane

Explanation:

- Sevoflurane has a moderate MAC value (2%) and is also preferred for its additional benefit of rapid induction and recovery times due to its relatively low blood-gas solubility.
- The patient's complaints and clinical features point towards the suspicion of Appendicitis. Hence, he is possibly being taken up for an emergency laparoscopic appendicectomy.
- COMPOUND A of sevoflurane causes renal toxicity

MAC Definition:

The concentration of the general anaesthetic in the alveoli of the lung that produces unresponsiveness to the surgical incision in 50% of patients.

Nitrous oxide (Option A): It has a very high MAC value (105%) and is not potent enough to be used as a sole anaesthetic agent.

Desflurane (Option C): It has a MAC value of around 6%, so it is less potent than sevoflurane. However, it has a very rapid onset and offset, making it suitable for short procedures and outpatient surgeries.

Halothane (Option D): It has a very low MAC value (0.75%), indicating high potency, but it is associated with significant side effects, including hepatotoxicity, and is less commonly used today.

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Sevoflurane	59	Non inflammable	- (Pleasant)	50	0.68	2.0
Nitrous oxide	Gas	Non inflammable	-	1.4	0.47	105